S0 · Reviewed final Figure 4 interpretation · Interpretation and alternative framings · A size advantage in heading persistence can coexist with multiple active regions.

A size advantage in heading persistence can coexist with multiple active regions.

overall interpretation: H1 is inconclusive, H2 fails the declared overall gate and H3 is inconclusive in both arms and all seven views, with forced states shown separately. No model satisfies all H1 requirements (three required); only Holmes meets the main H2 model criterion (two required). The multiple-region predictor has Spearman +0.091 and 0/4 successful sign folds; count has−0.389 and 3/4, with one unavailable fold each. Neither reaches the required negative association magnitude 0.6. No eligible distinct H3 mode pair satisfies the complete corroboration/fingerprint conjunction. The chosen main figure instead reports the robust f = 0.05 observation in Wave-pinning, Holmes and Spring, with the full four-model overview alongside it. This is a conditional polarity result, not support for universal noise buffering or a validated multiplicity predictor.

retained qualifications: Finite-window bounds, unavailable comparisons and unbounded intervals remain explicit. Across all 185 conditions, each arm has 156 passing,17 failing and 12 inconclusive descriptive window checks. For the 52 preselected extended conditions, each view has 50 passing and 2 inconclusive horizon checks; the original-cohort check has the same counts. These are sensitivity qualifications, not additional hypothesis gates or equivalence proofs. At the selected input, all three-model size-ratio CIs remain above 1 in all seven views; selected endpoint persistence/regime window checks pass. Native regimes and fixed-physical smoothed/downsampled imaging proxies stay distinct. Global-heading physical mobility is a proxy, literal single-region site MSD is separate, and neither establishes cell migration. Historical Spring sampling failures and original/repaired/E4 Gaussian-lineage accounting remain in the supplement. The selected f = 0.05 conditions were not preselected for E4 extension. Both selector tables reuse the same common-horizon observations here; they are one cohort, rather than independent long-horizon confirmation.

Reviewed final Figure 4 interpretation. Reviewed interpretation remains distinct from raw hypothesis truth and its window/horizon qualifications. All deciding criterion records and complete memo text remain in the page CSV. Operational angular mobility and geometric conversion are not resolved long-lag persistence or measured migration. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_reviewed_interpretation_v3_20261020_001_S0_memo_01_01.csv

S0 · Reviewed final Figure 4 interpretation · Interpretation and alternative framings · Wave-pinning

Wave-pinning

pattern: At f = 0.05, the large/small heading-persistence ratio is 4.44 (95% CI 3.20–9.07). The primary optical large-cell multiple-region time fraction is 60.1% (51.1–69.3%), versus 3.4% in small cells. Size-dependent heading persistence therefore coexists with multiple observed regions.

numeric shortfalls: H1 has 5 supported gains among 11 evaluable inputs, but no complete model pass. Its f½ CIs overlap: common small[0.0001,0.03], large[0.0001,0.1]; longest small[0.0001,0.02], large[0.0001,0.05]. H2 has no qualified reversal/drop range. H3 shows the multiple-region class, without a qualified distinct second class.

qualifications: The f = 0.05 size advantage survives all seven views and both arms. Late-quarter persistence is bounded, with an unbounded CI endpoint. Its direct geometry-normalized persistence CI includes 1; the global-heading physical-mobility proxy increases and is a separate short-lag estimand.

Audited deciding record:
Wave-pinning · H1 model · inconclusive
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: inconclusive; reason: Every evaluable ratio CI must exclude1 above; undefined or missing
qualification cannot pass; evaluable inputs: 11; gain inputs: 5; reversal inputs: none observed; unknown forced qualification: no, criterion: H1 no
supported reversal; status: criterion met; reason: All designed comparisons remain in the coverage table; reversal inputs: none observed, criterion:
H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction;
small ci: 0.0001, 0.03; large ci: 0.0001, 0.1

Audited deciding record:
Wave-pinning · H2 model · inconclusive
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged

Reviewed final Figure 4 interpretation. Reviewed interpretation remains distinct from raw hypothesis truth and its window/horizon qualifications. All deciding criterion records and complete memo text remain in the page CSV. Operational angular mobility and geometric conversion are not resolved long-lag persistence or measured migration. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_reviewed_interpretation_v3_20261020_002_S0_memo_02_01.csv

S0 · Reviewed final Figure 4 interpretation · Interpretation and alternative framings · Wave-pinning

reversal inputs: none observed
peak drop ranges: none observed
evaluable inputs: 11

Reviewed final Figure 4 interpretation. Reviewed interpretation remains distinct from raw hypothesis truth and its window/horizon qualifications. All deciding criterion records and complete memo text remain in the page CSV. Operational angular mobility and geometric conversion are not resolved long-lag persistence or measured migration. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_reviewed_interpretation_v3_20261020_003_S0_memo_02_02.csv

S0 · Reviewed final Figure 4 interpretation · Interpretation and alternative framings · Goryachev

Goryachev

pattern: Non-forced weak-input heading persistence remains censored. At f = 0.05 the large-cell native low-contrast fraction is 90.0%, while usable-pair coverage for persistence is 0.131, below 0.5. Those observations stay visible, with the operational forced-state flag.

numeric shortfalls: H1 has 0 supported direct-ratio gains among 5 evaluable non-forced inputs and unbounded overlapping f½ CIs. H2 has no qualifying non-forced range. Goryachev supplies the candidate contrast-loss class for H3, but forcing/coverage qualification and the full distinct-class fingerprint conjunction do not pass.

qualifications: Forced-inclusive results remain separate. A numerical ratio computed from the remaining valid pairs does not establish an eligible persistent heading. The unmatched f = 0.015 designed endpoint and its missing comparison remain explicit.

Audited deciding record:
Goryachev · H1 model · inconclusive
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: inconclusive; reason: Every evaluable ratio CI must exclude1 above; undefined or missing
qualification cannot pass; evaluable inputs: 5; gain inputs: 0; reversal inputs: none observed; unknown forced qualification: no, criterion: H1 no
supported reversal; status: criterion met; reason: All designed comparisons remain in the coverage table; reversal inputs: none observed, criterion:
H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction;
small ci: 0.0001, unavailable; large ci: 0.0001, unavailable

Audited deciding record:
Goryachev · H2 model · inconclusive
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: none observed
peak drop ranges: none observed

Reviewed final Figure 4 interpretation. Reviewed interpretation remains distinct from raw hypothesis truth and its window/horizon qualifications. All deciding criterion records and complete memo text remain in the page CSV. Operational angular mobility and geometric conversion are not resolved long-lag persistence or measured migration. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_reviewed_interpretation_v3_20261020_004_S0_memo_03_01.csv

S0 · Reviewed final Figure 4 interpretation · Interpretation and alternative framings · Goryachev

evaluable inputs: 5

Reviewed final Figure 4 interpretation. Reviewed interpretation remains distinct from raw hypothesis truth and its window/horizon qualifications. All deciding criterion records and complete memo text remain in the page CSV. Operational angular mobility and geometric conversion are not resolved long-lag persistence or measured migration. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_reviewed_interpretation_v3_20261020_005_S0_memo_03_02.csv

S0 · Reviewed final Figure 4 interpretation · Interpretation and alternative framings · Holmes 3

Holmes 3

pattern: At f = 0.05 the heading-persistence ratio is 2.74 (95% CI 2.20–3.47); the primary optical large-cell multiple-region fraction is 83.0% (79.5–85.9%), versus 9.8% in small cells. The same multiple-region class appears under strong input.

numeric shortfalls: H1 has 4 supported gains among 11 evaluable inputs; f½ CIs overlap (common small[0.0001,0.02], large[0.0001,0.03]; longest both[0.0001,0.02]). H2 passes for this model in main:2.228-fold drop from f = 0.03 to 0.3. Overall requires at least 2 models, and the primary count predictor ρ = −0.389 falls short of−0.6 by 0.211.

qualifications: The conditional H2 model result becomes inconclusive in the second-half and late-quarter views. The f = 0.05 size advantage itself survives all seven views; its late-quarter upper bound is unbounded. This distinction remains visible rather than redefining the hypothesis after the window checks.

Audited deciding record:
Holmes 3 · H1 model · inconclusive
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: inconclusive; reason: Every evaluable ratio CI must exclude1 above; undefined or missing
qualification cannot pass; evaluable inputs: 11; gain inputs: 4; reversal inputs: none observed; unknown forced qualification: no, criterion: H1 no
supported reversal; status: criterion met; reason: All designed comparisons remain in the coverage table; reversal inputs: none observed, criterion:
H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction;
small ci: 0.0001, 0.02; large ci: 0.0001, 0.03

Audited deciding record:
Holmes 3 · H2 model · criterion met
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: none observed

Reviewed final Figure 4 interpretation. Reviewed interpretation remains distinct from raw hypothesis truth and its window/horizon qualifications. All deciding criterion records and complete memo text remain in the page CSV. Operational angular mobility and geometric conversion are not resolved long-lag persistence or measured migration. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_reviewed_interpretation_v3_20261020_006_S0_memo_04_01.csv

S0 · Reviewed final Figure 4 interpretation · Interpretation and alternative framings · Holmes 3

peak drop ranges: earlier f: 0.03; later f: 0.3; fold drop: 2.228; observation bound certified: yes, earlier f: 0.05; later f: 0.3; fold drop: 2.202;
observation bound certified: yes
evaluable inputs: 11

Reviewed final Figure 4 interpretation. Reviewed interpretation remains distinct from raw hypothesis truth and its window/horizon qualifications. All deciding criterion records and complete memo text remain in the page CSV. Operational angular mobility and geometric conversion are not resolved long-lag persistence or measured migration. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_reviewed_interpretation_v3_20261020_007_S0_memo_04_02.csv

S0 · Reviewed final Figure 4 interpretation · Interpretation and alternative framings · Spring

Spring

pattern: At f = 0.05 the persistence ratio is 1.167 (95% CI 1.093–1.251), while the primary optical large-cell multiple-region fraction is 53.0% (50.4–55.3%), versus 0.08% in small cells. The count increase is therefore compatible with a modest positive heading-persistence size effect.

numeric shortfalls: H1 has 7 supported gains among 10 evaluable inputs, with overlapping f½ CIs: small[0.0001,0.003], large[0.0001,0.01]. The f = 1 reversal, ratio 0.877 (0.785–0.976), is operationally forced and cannot support the non-forced H2 call. H3 corroborates the same multiple-region class, not an eligible distinct second class.

qualifications: Use the measured changing circumference: the actual geometric ratio at f = 0.05 is 8.534, rather than the resting-size reference 9. Historical sampled-duration/cadence failures remain explicit; successful native replay does not establish all region-lifetime or duration convergence.

Audited deciding record:
Spring · H1 model · inconclusive
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: inconclusive; reason: Every evaluable ratio CI must exclude1 above; undefined or missing
qualification cannot pass; evaluable inputs: 10; gain inputs: 7; reversal inputs: none observed; unknown forced qualification: no, criterion: H1 no
supported reversal; status: criterion met; reason: All designed comparisons remain in the coverage table; reversal inputs: none observed, criterion:
H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction;
small ci: 0.0001, 0.003; large ci: 0.0001, 0.01

Audited deciding record:
Spring · H2 model · inconclusive
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged

Reviewed final Figure 4 interpretation. Reviewed interpretation remains distinct from raw hypothesis truth and its window/horizon qualifications. All deciding criterion records and complete memo text remain in the page CSV. Operational angular mobility and geometric conversion are not resolved long-lag persistence or measured migration. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_reviewed_interpretation_v3_20261020_008_S0_memo_05_01.csv

S0 · Reviewed final Figure 4 interpretation · Interpretation and alternative framings · Spring

reversal inputs: none observed
peak drop ranges: none observed
evaluable inputs: 10

Reviewed final Figure 4 interpretation. Reviewed interpretation remains distinct from raw hypothesis truth and its window/horizon qualifications. All deciding criterion records and complete memo text remain in the page CSV. Operational angular mobility and geometric conversion are not resolved long-lag persistence or measured migration. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_reviewed_interpretation_v3_20261020_009_S0_memo_05_02.csv

S0 · Reviewed final Figure 4 interpretation · Interpretation and alternative framings · Persistence and global-heading mobility respond differently to enlargement.

Persistence and global-heading mobility respond differently to enlargement.

supporting panels: S4/S5 geometry and mobility, S12 cadence checks, and the same f = 0.05 small/large native comparisons. At that input, global-heading-derived physical-mobility ratios are 1.845,3.606 and 6.356 in Wave-pinning, Holmes and Spring, each withCI above 1, alongside persistence ratios above 1.

weaknesses: The global-heading conversion is not one tracked site diffusivity. Short-lag slopes are cadence-sensitive; Wave-pinning direct excess-persistence CI includes the geometric reference, so the two source estimands cannot be collapsed into a strict common geometric claim.

further runs: Save finer scalar output and test converged short-lag slopes at the same fixed inputs; extend qualified single-region tracks to compare literal site MSD with the heading-derived proxy. No new integrations are included in this delivery.

Reviewed final Figure 4 interpretation. Reviewed interpretation remains distinct from raw hypothesis truth and its window/horizon qualifications. All deciding criterion records and complete memo text remain in the page CSV. Operational angular mobility and geometric conversion are not resolved long-lag persistence or measured migration. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_reviewed_interpretation_v3_20261020_010_S0_memo_06_01.csv

S0 · Reviewed final Figure 4 interpretation · Interpretation and alternative framings · Model-dependent loss of a single persistent polarity pattern.

Model-dependent loss of a single persistent polarity pattern.

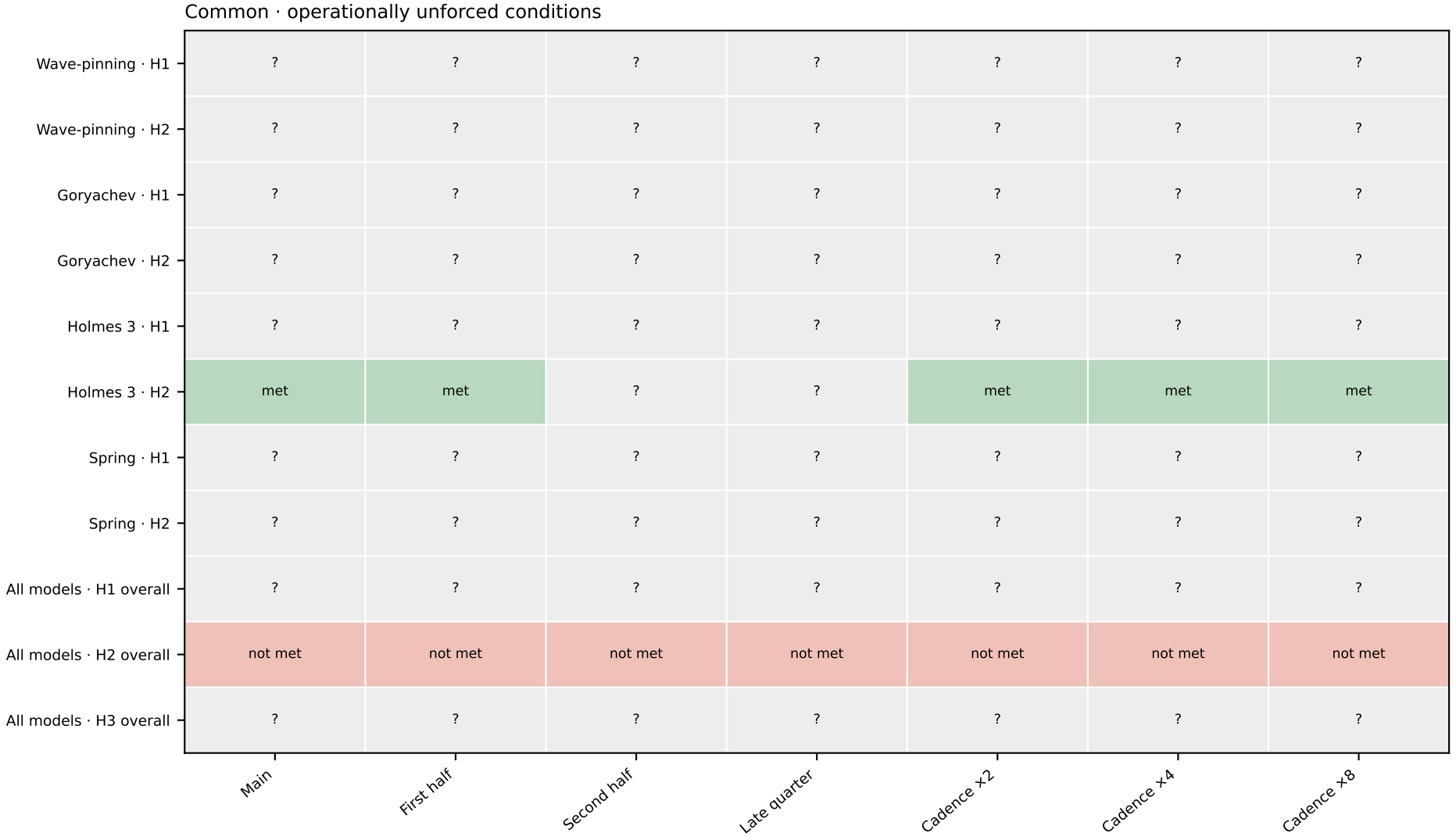
supporting panels: Full S2 regime/coverage maps, S7 fixed-mean sigma sweeps, S9 optical fingerprints and S0 H3 component/pair tables; the all-model main overview can lead this framing.

weaknesses: Only the multiple-region class is qualified and corroborated across the eligible models. Goryachev contrast loss is forced/undercovered, and no distinct eligible pair passes the full fingerprint test. A descriptive difference cannot be promoted to a confirmed mechanism-specific failure law.

further runs: Sample additional fixed-mean Goryachev transition inputs with mean controls and sigma contrasts to seek an abnormal state with adequate heading coverage, then predeclare and retest the distinct-class fingerprints. Longer weak-input runs would also tighten the currently overlapping noise-tolerance bounds.

Reviewed final Figure 4 interpretation. Reviewed interpretation remains distinct from raw hypothesis truth and its window/horizon qualifications. All deciding criterion records and complete memo text remain in the page CSV. Operational angular mobility and geometric conversion are not resolved long-lag persistence or measured migration. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_reviewed_interpretation_v3_20261020_011_S0_memo_07_01.csv

S0 · Completed audited Figure 4 criteria and qualifications · Declared hypothesis criteria across every view



Green: criterion met. Red: declared criterion not met. ?: inconclusive.
Raw criterion truth is shown here. A7 window, horizon and seed-cohort qualifications are separate S12 evidence.
An unmet criterion does not establish its biological opposite; no main framing is selected by this renderer.

Completed audited Figure 4 criteria and qualifications. Both forcing views are retained. All seven declared windows/cadences are displayed; missing work is refused. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_001_S0_common_primary.csv

S0 · Completed audited Figure 4 criteria and qualifications · Main-view criterion components · common · operationally unforced conditions

Wave-pinning · H1 model · inconclusive
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: inconclusive; reason: Every evaluable ratio CI must exclude1 above; undefined or missing qualification cannot pass; evaluable inputs: 11; gain inputs: 5; reversal inputs: none observed; unknown forced qualification: no, criterion: H1 no supported reversal; status: criterion met; reason: All designed comparisons remain in the coverage table; reversal inputs: none observed, criterion: H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction; small ci: 0.0001, 0.03; large ci: 0.0001, 0.1

Wave-pinning · H2 model · inconclusive
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: none observed
peak drop ranges: none observed
evaluable inputs: 11

Goryachev · H1 model · inconclusive
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: inconclusive; reason: Every evaluable ratio CI must exclude1 above; undefined or missing qualification cannot pass; evaluable inputs: 5; gain inputs: 0; reversal inputs: none observed; unknown forced qualification: no, criterion: H1 no supported reversal; status: criterion met; reason: All designed comparisons remain in the coverage table; reversal inputs: none observed, criterion: H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction; small ci: 0.0001, unavailable; large ci: 0.0001, unavailable

Goryachev · H2 model · inconclusive
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: none observed
peak drop ranges: none observed
evaluable inputs: 5

Holmes 3 · H1 model · inconclusive
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: inconclusive; reason: Every evaluable ratio CI must exclude1 above; undefined or missing qualification cannot pass; evaluable inputs: 11; gain inputs: 4; reversal inputs: none observed; unknown forced qualification: no, criterion: H1 no supported reversal; status: criterion met; reason: All designed comparisons remain in the coverage table; reversal inputs: none observed, criterion: H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction; small ci: 0.0001, 0.02; large ci: 0.0001, 0.03

Holmes 3 · H2 model · criterion met
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: none observed
peak drop ranges: earlier f: 0.03; later f: 0.3; fold drop: 2.228; observation bound certified: yes, earlier f: 0.05; later f: 0.3; fold drop: 2.202; observation bound certified: yes
evaluable inputs: 11

Completed audited Figure 4 criteria and qualifications. Criterion components are copied from the audited final tables, without recomputing truth. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_002_S0_common_primary_details_01.csv

S0 · Completed audited Figure 4 criteria and qualifications · Main-view criterion components · common · operationally unforced conditions

Spring · H1 model · inconclusive
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: inconclusive; reason: Every evaluable ratio CI must exclude1 above; undefined or missing qualification cannot pass; evaluable inputs: 10; gain inputs: 7; reversal inputs: none observed; unknown forced qualification: no, criterion: H1 no supported reversal; status: criterion met; reason: All designed comparisons remain in the coverage table; reversal inputs: none observed, criterion: H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction; small ci: 0.0001, 0.003; large ci: 0.0001, 0.01

Spring · H2 model · inconclusive
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: none observed
peak drop ranges: none observed
evaluable inputs: 10

all · H1 overall · inconclusive
reason: Predeclared model count
passes: 0
inconclusive: 4
required: 3
total: 4

all · H2 overall · criterion not met
reason: All required criteria are reported separately
components: criterion: H2 model count; status: inconclusive; reason: Predeclared model count; passes: 1; inconclusive: 3; required: 2; total: 4, criterion: H2 A6 predictor gate; status: criterion not met; reason: Missing folds are not successes; incomplete coverage is distinguished from evaluated disagreement; features: criterion: multiple front fraction; status: criterion not met; reason: Only the two frozen direct-tau negative-association predictors can support H2; pooled spearman: 0.09091; successful folds: 0; unavailable folds: 1; planned folds: 4, criterion: region count mean; status: criterion not met; reason: Only the two frozen direct-tau negative-association predictors can support H2; pooled spearman: -0.389; successful folds: 3; unavailable folds: 1; planned folds: 4

all · H3 overall · inconclusive
reason: At least two different models must show distinct corroborated, non-forced failure modes with both seed fingerprints separated
All 65 preset distinct-model/mode pairs are displayed separately on H3 pair pages, including every inconclusive or unmet component.

Completed audited Figure 4 criteria and qualifications. Criterion components are copied from the audited final tables, without recomputing truth. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_003_S0_common_primary_details_02.csv

S0 · Completed audited Figure 4 criteria and qualifications · Declared hypothesis criteria across every view

	Main	First half	Second half	Late quarter	Cadence x2	Cadence x4	Cadence x8
Wave-pinning · H1	?	?	?	?	?	?	?
Wave-pinning · H2	?	?	?	?	?	?	?
Goryachev · H1	?	?	?	?	?	?	?
Goryachev · H2	?	?	?	?	?	?	?
Holmes 3 · H1	?	?	?	?	?	?	?
Holmes 3 · H2	met	met	met	?	met	met	met
Spring · H1	not met	?	not met	?	not met	not met	?
Spring · H2	met	?	met	?	met	met	?
All models · H1 overall	?	?	?	?	?	?	?
All models · H2 overall	not met	not met	not met	not met	not met	not met	not met
All models · H3 overall	?	?	?	?	?	?	?

Green: criterion met. Red: declared criterion not met. ?: inconclusive.
Raw criterion truth is shown here. A7 window, horizon and seed-cohort qualifications are separate S12 evidence.
An unmet criterion does not establish its biological opposite; no main framing is selected by this renderer.

Completed audited Figure 4 criteria and qualifications. Both forcing views are retained. All seven declared windows/cadences are displayed; missing work is refused. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_004_S0_common_inclusive.csv

S0 · Completed audited Figure 4 criteria and qualifications · Main-view criterion components · common · all conditions including operationally forced

Wave-pinning · H1 model · inconclusive
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: inconclusive; reason: Every evaluable ratio CI must exclude1 above; undefined or missing qualification cannot pass; evaluable inputs: 12; gain inputs: 6; reversal inputs: none observed; unknown forced qualification: no, criterion: H1 no supported reversal; status: criterion met; reason: All designed comparisons remain in the coverage table; reversal inputs: none observed, criterion: H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction; small ci: 0.0001, 0.03; large ci: 0.0001, 0.1

Wave-pinning · H2 model · inconclusive
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: none observed
peak drop ranges: none observed
evaluable inputs: 12

Goryachev · H1 model · inconclusive
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: inconclusive; reason: Every evaluable ratio CI must exclude1 above; undefined or missing qualification cannot pass; evaluable inputs: 7; gain inputs: 2; reversal inputs: none observed; unknown forced qualification: no, criterion: H1 no supported reversal; status: criterion met; reason: All designed comparisons remain in the coverage table; reversal inputs: none observed, criterion: H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction; small ci: 0.0001, 0.01; large ci: 0.0001, unavailable

Goryachev · H2 model · inconclusive
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: none observed
peak drop ranges: none observed
evaluable inputs: 7

Holmes 3 · H1 model · inconclusive
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: inconclusive; reason: Every evaluable ratio CI must exclude1 above; undefined or missing qualification cannot pass; evaluable inputs: 12; gain inputs: 5; reversal inputs: none observed; unknown forced qualification: no, criterion: H1 no supported reversal; status: criterion met; reason: All designed comparisons remain in the coverage table; reversal inputs: none observed, criterion: H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction; small ci: 0.0001, 0.02; large ci: 0.0001, 0.03

Holmes 3 · H2 model · criterion met
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: none observed
peak drop ranges: earlier f: 0.03; later f: 0.3; fold drop: 2.228; observation bound certified: yes, earlier f: 0.05; later f: 0.3; fold drop: 2.202; observation bound certified: yes
evaluable inputs: 12

Completed audited Figure 4 criteria and qualifications. Criterion components are copied from the audited final tables, without recomputing truth. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_005_S0_common_inclusive_details_01.csv

S0 · Completed audited Figure 4 criteria and qualifications · Main-view criterion components · common · all conditions including operationally forced

Spring · H1 model · criterion not met
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: criterion not met; reason: Every evaluable ratio CI must exclude1 above; undefined or missing qualification cannot pass; evaluable inputs: 12; gain inputs: 7; reversal inputs: 1; unknown forced qualification: no, criterion: H1 no supported reversal; status: criterion not met; reason: All designed comparisons remain in the coverage table; reversal inputs: 1, criterion: H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction; small ci: 0.0001, 0.003; large ci: 0.0001, 0.01

Spring · H2 model · criterion met
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: 1
peak drop ranges: none observed
evaluable inputs: 12

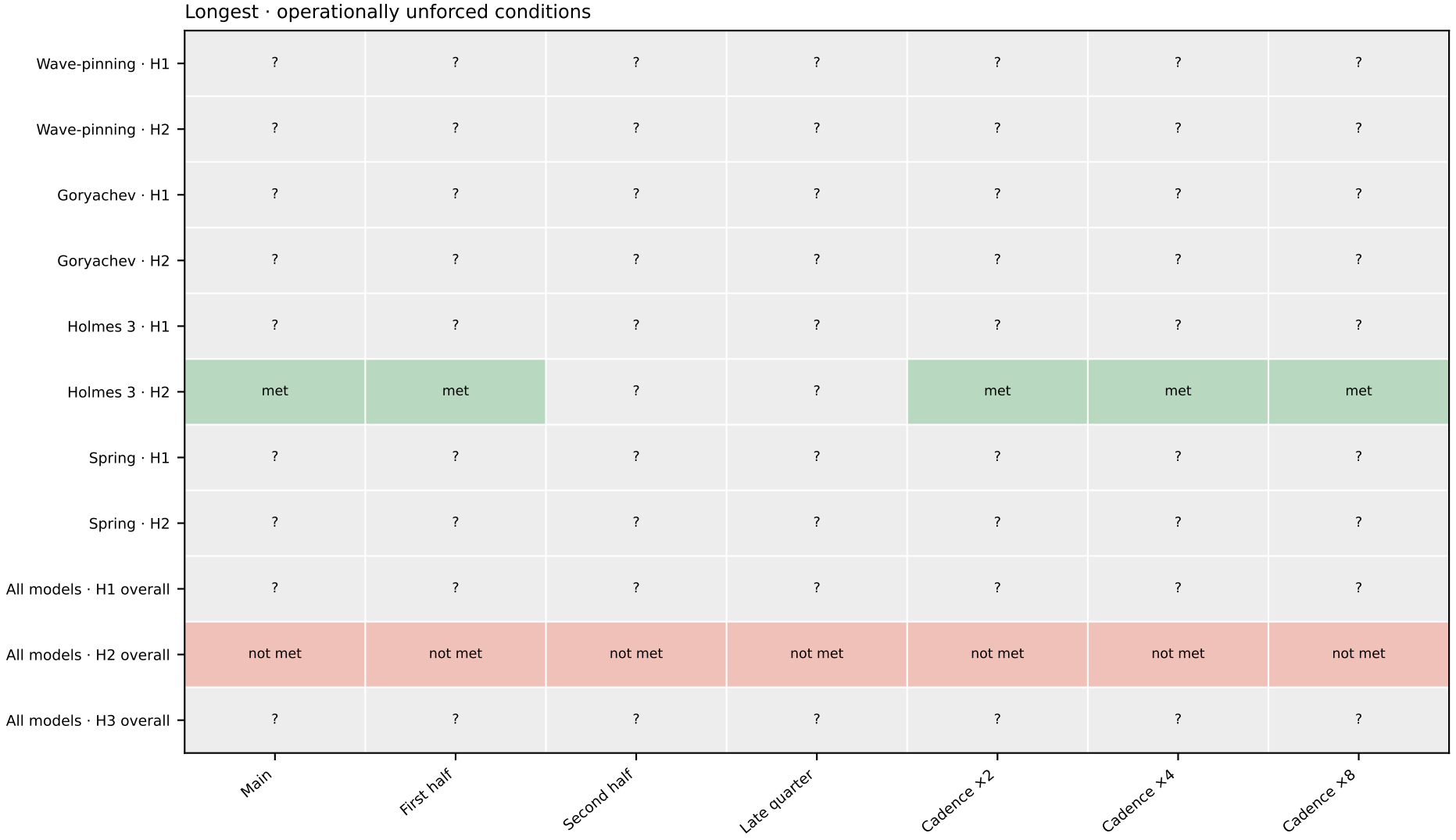
all · H1 overall · inconclusive
reason: Predeclared model count
passes: 0
inconclusive: 3
required: 3
total: 4

all · H2 overall · criterion not met
reason: All required criteria are reported separately
components: criterion: H2 model count; status: criterion met; reason: Predeclared model count; passes: 2; inconclusive: 2; required: 2; total: 4, criterion: H2 A6 predictor gate; status: criterion not met; reason: Missing folds are not successes; incomplete coverage is distinguished from evaluated disagreement; features: criterion: multiple front fraction; status: criterion not met; reason: Only the two frozen direct-tau negative-association predictors can support H2; pooled spearman: -0.02143; successful folds: 0; unavailable folds: 1; planned folds: 4, criterion: region count mean; status: criterion not met; reason: Only the two frozen direct-tau negative-association predictors can support H2; pooled spearman: -0.3416; successful folds: 3; unavailable folds: 1; planned folds: 4

all · H3 overall · inconclusive
reason: At least two different models must show distinct corroborated, non-forced failure modes with both seed fingerprints separated
All 65 preset distinct-model/mode pairs are displayed separately on H3 pair pages, including every inconclusive or unmet component.

Completed audited Figure 4 criteria and qualifications. Criterion components are copied from the audited final tables, without recomputing truth. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_006_S0_common_inclusive_details_02.csv

S0 · Completed audited Figure 4 criteria and qualifications · Declared hypothesis criteria across every view



Green: criterion met. Red: declared criterion not met. ?: inconclusive.
Raw criterion truth is shown here. A7 window, horizon and seed-cohort qualifications are separate S12 evidence.
An unmet criterion does not establish its biological opposite; no main framing is selected by this renderer.

Completed audited Figure 4 criteria and qualifications. Both forcing views are retained. All seven declared windows/cadences are displayed; missing work is refused. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_007_S0_longest_primary.csv

S0 · Completed audited Figure 4 criteria and qualifications · Main-view criterion components · longest · operationally unforced conditions

Wave-pinning · H1 model · inconclusive
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: inconclusive; reason: Every evaluable ratio CI must exclude1 above; undefined or missing qualification cannot pass; evaluable inputs: 11; gain inputs: 5; reversal inputs: none observed; unknown forced qualification: no, criterion: H1 no supported reversal; status: criterion met; reason: All designed comparisons remain in the coverage table; reversal inputs: none observed, criterion: H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction; small ci: 0.0001, 0.02; large ci: 0.0001, 0.05

Wave-pinning · H2 model · inconclusive
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: none observed
peak drop ranges: none observed
evaluable inputs: 11

Goryachev · H1 model · inconclusive
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: inconclusive; reason: Every evaluable ratio CI must exclude1 above; undefined or missing qualification cannot pass; evaluable inputs: 5; gain inputs: 0; reversal inputs: none observed; unknown forced qualification: no, criterion: H1 no supported reversal; status: criterion met; reason: All designed comparisons remain in the coverage table; reversal inputs: none observed, criterion: H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction; small ci: 0.0001, unavailable; large ci: 0.0001, unavailable

Goryachev · H2 model · inconclusive
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: none observed
peak drop ranges: none observed
evaluable inputs: 5

Holmes 3 · H1 model · inconclusive
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: inconclusive; reason: Every evaluable ratio CI must exclude1 above; undefined or missing qualification cannot pass; evaluable inputs: 11; gain inputs: 4; reversal inputs: none observed; unknown forced qualification: no, criterion: H1 no supported reversal; status: criterion met; reason: All designed comparisons remain in the coverage table; reversal inputs: none observed, criterion: H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction; small ci: 0.0001, 0.02; large ci: 0.0001, 0.02

Holmes 3 · H2 model · criterion met
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: none observed
peak drop ranges: earlier f: 0.03; later f: 0.3; fold drop: 2.228; observation bound certified: yes, earlier f: 0.05; later f: 0.3; fold drop: 2.202; observation bound certified: yes
evaluable inputs: 11

Completed audited Figure 4 criteria and qualifications. Criterion components are copied from the audited final tables, without recomputing truth. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_008_S0_longest_primary_details_01.csv

S0 · Completed audited Figure 4 criteria and qualifications · Main-view criterion components · longest · operationally unforced conditions

Spring · H1 model · inconclusive
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: inconclusive; reason: Every evaluable ratio CI must exclude1 above; undefined or missing qualification cannot pass; evaluable inputs: 10; gain inputs: 7; reversal inputs: none observed; unknown forced qualification: no, criterion: H1 no supported reversal; status: criterion met; reason: All designed comparisons remain in the coverage table; reversal inputs: none observed, criterion: H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction; small ci: 0.0001, 0.003; large ci: 0.0001, 0.01

Spring · H2 model · inconclusive
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: none observed
peak drop ranges: none observed
evaluable inputs: 10

all · H1 overall · inconclusive
reason: Predeclared model count
passes: 0
inconclusive: 4
required: 3
total: 4

all · H2 overall · criterion not met
reason: All required criteria are reported separately
components: criterion: H2 model count; status: inconclusive; reason: Predeclared model count; passes: 1; inconclusive: 3; required: 2; total: 4, criterion: H2 A6 predictor gate; status: criterion not met; reason: Missing folds are not successes; incomplete coverage is distinguished from evaluated disagreement; features: criterion: multiple front fraction; status: criterion not met; reason: Only the two frozen direct-tau negative-association predictors can support H2; pooled spearman: 0.09091; successful folds: 0; unavailable folds: 1; planned folds: 4, criterion: region count mean; status: criterion not met; reason: Only the two frozen direct-tau negative-association predictors can support H2; pooled spearman: -0.389; successful folds: 3; unavailable folds: 1; planned folds: 4

all · H3 overall · inconclusive
reason: At least two different models must show distinct corroborated, non-forced failure modes with both seed fingerprints separated
All 65 preset distinct-model/mode pairs are displayed separately on H3 pair pages, including every inconclusive or unmet component.

Completed audited Figure 4 criteria and qualifications. Criterion components are copied from the audited final tables, without recomputing truth. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_009_S0_longest_primary_details_02.csv

S0 · Completed audited Figure 4 criteria and qualifications · Declared hypothesis criteria across every view

	Main	First half	Second half	Late quarter	Cadence x2	Cadence x4	Cadence x8
Wave-pinning · H1	?	?	?	?	?	?	?
Wave-pinning · H2	?	?	?	?	?	?	?
Goryachev · H1	?	?	?	?	?	?	?
Goryachev · H2	?	?	?	?	?	?	?
Holmes 3 · H1	?	?	?	?	?	?	?
Holmes 3 · H2	met	met	met	?	met	met	met
Spring · H1	not met	?	not met	?	not met	not met	?
Spring · H2	met	?	met	?	met	met	?
All models · H1 overall	?	?	?	?	?	?	?
All models · H2 overall	not met	not met	not met	not met	not met	not met	not met
All models · H3 overall	?	?	?	?	?	?	?

Green: criterion met. Red: declared criterion not met. ?: inconclusive.
Raw criterion truth is shown here. A7 window, horizon and seed-cohort qualifications are separate S12 evidence.
An unmet criterion does not establish its biological opposite; no main framing is selected by this renderer.

Completed audited Figure 4 criteria and qualifications. Both forcing views are retained. All seven declared windows/cadences are displayed; missing work is refused. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_010_S0_longest_inclusive.csv

S0 · Completed audited Figure 4 criteria and qualifications · Main-view criterion components · longest · all conditions including operationally forced

Wave-pinning · H1 model · inconclusive
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: inconclusive; reason: Every evaluable ratio CI must exclude1 above; undefined or missing qualification cannot pass; evaluable inputs: 12; gain inputs: 6; reversal inputs: none observed; unknown forced qualification: no, criterion: H1 no supported reversal; status: criterion met; reason: All designed comparisons remain in the coverage table; reversal inputs: none observed, criterion: H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction; small ci: 0.0001, 0.02; large ci: 0.0001, 0.05

Wave-pinning · H2 model · inconclusive
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: none observed
peak drop ranges: none observed
evaluable inputs: 12

Goryachev · H1 model · inconclusive
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: inconclusive; reason: Every evaluable ratio CI must exclude1 above; undefined or missing qualification cannot pass; evaluable inputs: 7; gain inputs: 2; reversal inputs: none observed; unknown forced qualification: no, criterion: H1 no supported reversal; status: criterion met; reason: All designed comparisons remain in the coverage table; reversal inputs: none observed, criterion: H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction; small ci: 0.0001, 0.01; large ci: 0.0001, unavailable

Goryachev · H2 model · inconclusive
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: none observed
peak drop ranges: none observed
evaluable inputs: 7

Holmes 3 · H1 model · inconclusive
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: inconclusive; reason: Every evaluable ratio CI must exclude1 above; undefined or missing qualification cannot pass; evaluable inputs: 12; gain inputs: 5; reversal inputs: none observed; unknown forced qualification: no, criterion: H1 no supported reversal; status: criterion met; reason: All designed comparisons remain in the coverage table; reversal inputs: none observed, criterion: H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction; small ci: 0.0001, 0.02; large ci: 0.0001, 0.02

Holmes 3 · H2 model · criterion met
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: none observed
peak drop ranges: earlier f: 0.03; later f: 0.3; fold drop: 2.228; observation bound certified: yes, earlier f: 0.05; later f: 0.3; fold drop: 2.202; observation bound certified: yes
evaluable inputs: 12

Completed audited Figure 4 criteria and qualifications. Criterion components are copied from the audited final tables, without recomputing truth. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_011_S0_longest_inclusive_details_01.csv

S0 · Completed audited Figure 4 criteria and qualifications · Main-view criterion components · longest · all conditions including operationally forced

Spring · H1 model · criterion not met
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: criterion not met; reason: Every evaluable ratio CI must exclude1 above; undefined or missing qualification cannot pass; evaluable inputs: 12; gain inputs: 7; reversal inputs: 1; unknown forced qualification: no, criterion: H1 no supported reversal; status: criterion not met; reason: All designed comparisons remain in the coverage table; reversal inputs: 1, criterion: H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction; small ci: 0.0001, 0.003; large ci: 0.0001, 0.01

Spring · H2 model · criterion met
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: 1
peak drop ranges: none observed
evaluable inputs: 12

all · H1 overall · inconclusive
reason: Predeclared model count
passes: 0
inconclusive: 3
required: 3
total: 4

all · H2 overall · criterion not met
reason: All required criteria are reported separately
components: criterion: H2 model count; status: criterion met; reason: Predeclared model count; passes: 2; inconclusive: 2; required: 2; total: 4, criterion: H2 A6 predictor gate; status: criterion not met; reason: Missing folds are not successes; incomplete coverage is distinguished from evaluated disagreement; features: criterion: multiple front fraction; status: criterion not met; reason: Only the two frozen direct-tau negative-association predictors can support H2; pooled spearman: -0.02143; successful folds: 0; unavailable folds: 1; planned folds: 4, criterion: region count mean; status: criterion not met; reason: Only the two frozen direct-tau negative-association predictors can support H2; pooled spearman: -0.3416; successful folds: 3; unavailable folds: 1; planned folds: 4

all · H3 overall · inconclusive
reason: At least two different models must show distinct corroborated, non-forced failure modes with both seed fingerprints separated
All 65 preset distinct-model/mode pairs are displayed separately on H3 pair pages, including every inconclusive or unmet component.

Completed audited Figure 4 criteria and qualifications. Criterion components are copied from the audited final tables, without recomputing truth. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_012_S0_longest_inclusive_details_02.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · main · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0009372 [0, 0.002187] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0171 [0.007002, 0.03033] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.9893 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.2347 [0.124, 0.3757] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2696, 2696] resolved	[8000, ∞] right censored	S 0.9795 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.6389 [0.5518, 0.7217] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[505.5, 505.5] resolved	[8000, ∞] right censored	S 0.9425 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9545 [0.9444, 0.9647] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[127.9, 127.9] resolved	[8000, ∞] right censored	S 0.9218 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9863 [0.9853, 0.9872] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	inconclusive	inconclusive	[83.26, 83.26] resolved	[8000, ∞] right censored	S 0.9328 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.9613 [0.9603, 0.9622] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_013_S0_H3_candidates_common_main_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · main · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	inconclusive	[474.5, 474.5] resolved	[8000, ∞] right censored	S 0.2376 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.07688 [0.06599, 0.08612] S low 0.856 [0.8388, 0.8749] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[564.4, 564.4] resolved	[8000, ∞] right censored	S 0.1314 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.0624 [0.05545, 0.06938] S low 0.9004 [0.8883, 0.9136] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[734.3, 734.3] resolved	[8000, ∞] right censored	S 0.05307 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.02684 [0.02106, 0.03265] S low 0.9579 [0.9502, 0.9669] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1257, 1257] resolved	[8000, ∞] right censored	S 0.01543 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.004842 [0.001874, 0.00781] S low 0.9925 [0.9875, 0.9975] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_014_S0_H3_candidates_common_main_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · main · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[8000, ∞] right censored	S 0 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0003905 [0, 0.001119] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.9789 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.1968 [0.0342, 0.3847] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3901, 3901] resolved	[8000, ∞] right censored	S 0.9798 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.4833 [0.3906, 0.5751] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2012, 2012] resolved	[8000, ∞] right censored	S 0.985 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.7162 [0.6475, 0.7784] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[986.5, 986.5] resolved	[8000, ∞] right censored	S 0.9754 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.8569 [0.8322, 0.8799] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[388.1, 388.1] resolved	[8000, ∞] right censored	S 0.9626 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9196 [0.901, 0.9378] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[103.8, 103.8] resolved	[8000, ∞] right censored	S 0.964 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9311 [0.9288, 0.9335] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_015_S0_H3_candidates_common_main_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · main · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	inconclusive	inconclusive	[51.37, 51.37] resolved	[8000, ∞] right censored	S 0.9693 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.8552 [0.8485, 0.8619] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.000781 [7.81e-05, 0.001744] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[533.4, 533.4] resolved	[800, ∞] right censored	S 0.9981 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.02353 [0.006378, 0.04493] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[146, 146] resolved	[800, ∞] right censored	S 0.9921 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.1367 [0.1076, 0.1706] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[37.92, 37.92] resolved	[800, ∞] right censored	S 0.9845 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.3529 [0.3182, 0.3847] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.18, 18.18] resolved	[800, ∞] right censored	S 0.9828 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.4992 [0.4707, 0.531] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.65, 10.65] resolved	[800, ∞] right censored	S 0.9792 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.6586 [0.6433, 0.6732] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.794, 6.794] resolved	[800, ∞] right censored	S 0.9745 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7892 [0.7759, 0.8008] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	inconclusive	inconclusive	[4.964, 4.964] resolved	[800, ∞] right censored	S 0.9746 / W 1 H 3200 / 3200 95% bound check: ?	S multi 0.8399 [0.8372, 0.8425] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_016_S0_H3_candidates_common_main_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · main · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	inconclusive	inconclusive	[4.121, 4.121] resolved	[800, ∞] right censored	S 0.9903 / W 1 H 3200 / 3200 95% bound check: ?	S multi 0.768 [0.7476, 0.7885] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_017_S0_H3_candidates_common_main_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · main · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.011 [1.002, 1.023] N 12/12; valid 1	1.005 [1.004, 1.007] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8679, low 0; forced mean no	1.212 [1.09, 1.34] N 12/12; valid 1	1.009 [1.007, 1.01] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.668 [1.558, 1.771] N 12/12; valid 1	1.009 [1.006, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_018_S0_H3_fingerprints_common_main_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · main · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3857, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9133, low 0; forced mean no	2.309 [2.248, 2.367] N 12/12; valid 1	1.024 [1.017, 1.031] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.482 [2.376, 2.593] N 2/2; valid 1	1.124 [1.11, 1.139] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.507 [2.432, 2.583] N 2/2; valid 1	1.25 [1.233, 1.268] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1.008 [1.007, 1.01] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.03 [1.029, 1.031] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.11, 1.121] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9581 [0.936, 0.9784] N 12/12; valid 1	1.484 [1.393, 1.597] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7913 [0.7772, 0.8073] N 12/12; valid 1	3.345 [3.282, 3.41] N 12/12; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_019_S0_H3_fingerprints_common_main_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · main · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.01544; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1406; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.7685 [0.7559, 0.7812] N 12/12; valid 1	2.616 [2.548, 2.681] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.03	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check single front; H 32000 multi 0.008175, low 0; forced mean yes	1.454 [1.386, 1.527] N 12/12; valid 1	0.4761 [0.4656, 0.4867] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.05	not declared	not declared	not declared	1.348 [1.277, 1.415] N 12/12; valid 1	0.4608 [0.45, 0.4716] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.1	not declared	not declared	not declared	1.206 [1.133, 1.282] N 12/12; valid 1	0.4371 [0.4272, 0.4482] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.3	not declared	not declared	not declared	0.654 [0.6049, 0.7068] N 2/2; valid 1	0.4247 [0.4159, 0.4336] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=1	not declared	not declared	not declared	0.07067 [0.05517, 0.08696] N 2/2; valid 1	0.4228 [0.3957, 0.4538] N 2/2; valid 1	count unavailable contrast unavailable
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_020_S0_H3_fingerprints_common_main_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · main · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.008] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.194 [1.022, 1.375] N 12/12; valid 1	1.013 [1.012, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.509 [1.389, 1.617] N 12/12; valid 1	1.029 [1.026, 1.031] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.01148, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8567, low 0; forced mean no	1.771 [1.657, 1.891] N 12/12; valid 1	1.049 [1.045, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.012 [1.942, 2.085] N 12/12; valid 1	1.076 [1.07, 1.083] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7005, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8086, low 0; forced mean no	2.11 [2.05, 2.176] N 12/12; valid 1	1.134 [1.123, 1.146] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_021_S0_H3_fingerprints_common_main_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · main · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.354 [2.32, 2.389] N 2/2; valid 1	1.305 [1.285, 1.325] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.215 [2.078, 2.359] N 2/2; valid 1	1.692 [1.633, 1.755] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.021 [1.004, 1.043] N 12/12; valid 1	1.007 [1.005, 1.009] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.121 [1.094, 1.153] N 12/12; valid 1	1.001 [0.9976, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.298 [1.265, 1.33] N 12/12; valid 1	1 [0.9938, 1.006] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_022_S0_H3_fingerprints_common_main_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · main · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1727, low 0; forced mean no	no eligible check multiple fronts; H 3200 multi 0.5114, low 0; forced mean no	1.434 [1.397, 1.473] N 12/12; valid 1	0.9973 [0.9902, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.567 [1.538, 1.592] N 12/12; valid 1	1.012 [1, 1.024] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5599, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.744, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6043, low 0; forced mean no	1.701 [1.68, 1.722] N 12/12; valid 1	1.033 [1.018, 1.05] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.762 [1.704, 1.821] N 2/2; valid 1	1.172 [1.093, 1.26] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=1	not declared	not declared	not declared	1.759 [1.752, 1.766] N 2/2; valid 1	1.25 [1.225, 1.276] N 2/2; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_023_S0_H3_fingerprints_common_main_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_024_S0_H3_pairs_common_main_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_025_S0_H3_pairs_common_main_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_026_S0_H3_pairs_common_main_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_027_S0_H3_pairs_common_main_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_028_S0_H3_pairs_common_main_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_029_S0_H3_pairs_common_main_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_030_S0_H3_pairs_common_main_primary_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_031_S0_H3_pairs_common_main_primary_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_032_S0_H3_pairs_common_main_primary_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · main · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0009372 [0, 0.002187] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0171 [0.007002, 0.03033] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.9893 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.2347 [0.124, 0.3757] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2696, 2696] resolved	[8000, ∞] right censored	S 0.9795 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.6389 [0.5518, 0.7217] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[505.5, 505.5] resolved	[8000, ∞] right censored	S 0.9425 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9545 [0.9444, 0.9647] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[127.9, 127.9] resolved	[8000, ∞] right censored	S 0.9218 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9863 [0.9853, 0.9872] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	criterion met	inconclusive	[83.26, 83.26] resolved	[8000, ∞] right censored	S 0.9328 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9613 [0.9603, 0.9622] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_033_S0_H3_candidates_common_main_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · main · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	criterion met	[474.5, 474.5] resolved	[8000, ∞] right censored	S 0.2376 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.07688 [0.06599, 0.08612] S low 0.856 [0.8388, 0.8749] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[564.4, 564.4] resolved	[8000, ∞] right censored	S 0.1314 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.0624 [0.05545, 0.06938] S low 0.9004 [0.8883, 0.9136] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[734.3, 734.3] resolved	[8000, ∞] right censored	S 0.05307 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.02684 [0.02106, 0.03265] S low 0.9579 [0.9502, 0.9669] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1257, 1257] resolved	[8000, ∞] right censored	S 0.01543 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.004842 [0.001874, 0.00781] S low 0.9925 [0.9875, 0.9975] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_034_S0_H3_candidates_common_main_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · main · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[8000, ∞] right censored	S 0 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0003905 [0, 0.001119] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.9789 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.1968 [0.0342, 0.3847] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3901, 3901] resolved	[8000, ∞] right censored	S 0.9798 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.4833 [0.3906, 0.5751] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2012, 2012] resolved	[8000, ∞] right censored	S 0.985 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.7162 [0.6475, 0.7784] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[986.5, 986.5] resolved	[8000, ∞] right censored	S 0.9754 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.8569 [0.8322, 0.8799] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[388.1, 388.1] resolved	[8000, ∞] right censored	S 0.9626 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9196 [0.901, 0.9378] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[103.8, 103.8] resolved	[8000, ∞] right censored	S 0.964 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9311 [0.9288, 0.9335] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_035_S0_H3_candidates_common_main_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · main · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	criterion met	inconclusive	[51.37, 51.37] resolved	[8000, ∞] right censored	S 0.9693 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.8552 [0.8485, 0.8619] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.000781 [7.81e-05, 0.001744] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[533.4, 533.4] resolved	[800, ∞] right censored	S 0.9981 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.02353 [0.006378, 0.04493] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[146, 146] resolved	[800, ∞] right censored	S 0.9921 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.1367 [0.1076, 0.1706] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[37.92, 37.92] resolved	[800, ∞] right censored	S 0.9845 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.3529 [0.3182, 0.3847] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.18, 18.18] resolved	[800, ∞] right censored	S 0.9828 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.4992 [0.4707, 0.531] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.65, 10.65] resolved	[800, ∞] right censored	S 0.9792 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.6586 [0.6433, 0.6732] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.794, 6.794] resolved	[800, ∞] right censored	S 0.9745 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7892 [0.7759, 0.8008] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	criterion met	inconclusive	[4.964, 4.964] resolved	[800, ∞] right censored	S 0.9746 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.8399 [0.8372, 0.8425] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_036_S0_H3_candidates_common_main_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · main · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	criterion met	inconclusive	[4.121, 4.121] resolved	[800, ∞] right censored	S 0.9903 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.768 [0.7476, 0.7885] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_037_S0_H3_candidates_common_main_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · main · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.011 [1.002, 1.023] N 12/12; valid 1	1.005 [1.004, 1.007] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8679, low 0; forced mean no	1.212 [1.09, 1.34] N 12/12; valid 1	1.009 [1.007, 1.01] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.668 [1.558, 1.771] N 12/12; valid 1	1.009 [1.006, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_038_S0_H3_fingerprints_common_main_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · main · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3857, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9133, low 0; forced mean no	2.309 [2.248, 2.367] N 12/12; valid 1	1.024 [1.017, 1.031] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.482 [2.376, 2.593] N 2/2; valid 1	1.124 [1.11, 1.139] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.507 [2.432, 2.583] N 2/2; valid 1	1.25 [1.233, 1.268] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1.008 [1.007, 1.01] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.03 [1.029, 1.031] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.11, 1.121] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9581 [0.936, 0.9784] N 12/12; valid 1	1.484 [1.393, 1.597] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7913 [0.7772, 0.8073] N 12/12; valid 1	3.345 [3.282, 3.41] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_039_S0_H3_fingerprints_common_main_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · main · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.01544; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1406; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.7685 [0.7559, 0.7812] N 12/12; valid 1	2.616 [2.548, 2.681] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.03	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	single front; H 32000 qualified yes; same no multi 0.008175, low 0; forced mean yes	1.454 [1.386, 1.527] N 12/12; valid 1	0.4761 [0.4656, 0.4867] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.05	not declared	not declared	not declared	1.348 [1.277, 1.415] N 12/12; valid 1	0.4608 [0.45, 0.4716] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.1	not declared	not declared	not declared	1.206 [1.133, 1.282] N 12/12; valid 1	0.4371 [0.4272, 0.4482] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.3	not declared	not declared	not declared	0.654 [0.6049, 0.7068] N 2/2; valid 1	0.4247 [0.4159, 0.4336] N 2/2; valid 1	count yes contrast yes
Goryachev f=1	not declared	not declared	not declared	0.07067 [0.05517, 0.08696] N 2/2; valid 1	0.4228 [0.3957, 0.4538] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_040_S0_H3_fingerprints_common_main_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · main · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.008] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.194 [1.022, 1.375] N 12/12; valid 1	1.013 [1.012, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.509 [1.389, 1.617] N 12/12; valid 1	1.029 [1.026, 1.031] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.01148, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8567, low 0; forced mean no	1.771 [1.657, 1.891] N 12/12; valid 1	1.049 [1.045, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.012 [1.942, 2.085] N 12/12; valid 1	1.076 [1.07, 1.083] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7005, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8086, low 0; forced mean no	2.11 [2.05, 2.176] N 12/12; valid 1	1.134 [1.123, 1.146] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_041_S0_H3_fingerprints_common_main_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · main · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.354 [2.32, 2.389] N 2/2; valid 1	1.305 [1.285, 1.325] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.215 [2.078, 2.359] N 2/2; valid 1	1.692 [1.633, 1.755] N 2/2; valid 1	count yes contrast yes
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.021 [1.004, 1.043] N 12/12; valid 1	1.007 [1.005, 1.009] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.121 [1.094, 1.153] N 12/12; valid 1	1.001 [0.9976, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.298 [1.265, 1.33] N 12/12; valid 1	1 [0.9938, 1.006] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_042_S0_H3_fingerprints_common_main_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · main · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1727, low 0; forced mean no	no eligible check multiple fronts; H 3200 multi 0.5114, low 0; forced mean no	1.434 [1.397, 1.473] N 12/12; valid 1	0.9973 [0.9902, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.567 [1.538, 1.592] N 12/12; valid 1	1.012 [1, 1.024] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5599, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.744, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6043, low 0; forced mean no	1.701 [1.68, 1.722] N 12/12; valid 1	1.033 [1.018, 1.05] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.762 [1.704, 1.821] N 2/2; valid 1	1.172 [1.093, 1.26] N 2/2; valid 1	count yes contrast yes
Spring f=1	not declared	not declared	not declared	1.759 [1.752, 1.766] N 2/2; valid 1	1.25 [1.225, 1.276] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_043_S0_H3_fingerprints_common_main_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / criterion met	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_044_S0_H3_pairs_common_main_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_045_S0_H3_pairs_common_main_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_046_S0_H3_pairs_common_main_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_047_S0_H3_pairs_common_main_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_048_S0_H3_pairs_common_main_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_049_S0_H3_pairs_common_main_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_050_S0_H3_pairs_common_main_inclusive_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_051_S0_H3_pairs_common_main_inclusive_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_052_S0_H3_pairs_common_main_inclusive_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis first half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0002082 [0, 0.0006246] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.02441 [0.006141, 0.0519] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 0.9964 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.2057 [0.0673, 0.3891] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	inconclusive	inconclusive	[2704, 2704] resolved	[4000, ∞] right censored	S 0.9964 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.6192 [0.5258, 0.7165] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[487, 487] resolved	[4000, ∞] right censored	S 0.9483 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9579 [0.946, 0.9681] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[132.3, 132.3] resolved	[4000, ∞] right censored	S 0.9308 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.985 [0.985, 0.985] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	inconclusive	inconclusive	[77.8, 77.8] resolved	[4000, ∞] right censored	S 0.9445 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.9622 [0.9594, 0.965] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_053_S0_H3_candidates_common_analysis_first_half_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis first half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	inconclusive	[540.8, 540.8] resolved	[4000, ∞] right censored	S 0.2132 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.06949 [0.05278, 0.08584] S low 0.8661 [0.844, 0.8911] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[606, 606] resolved	[4000, ∞] right censored	S 0.1347 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.05569 [0.04679, 0.06673] S low 0.9029 [0.8896, 0.9143] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[748.5, 748.5] resolved	[4000, ∞] right censored	S 0.05051 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.0266 [0.01764, 0.03618] S low 0.9619 [0.9517, 0.9717] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[0, ∞] unavailable lag gap	[4000, ∞] right censored	S 0.0003448 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.00812 [0.003748, 0.01249] S low 0.9866 [0.9781, 0.995] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_054_S0_H3_candidates_common_analysis_first_half_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis first half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[4000, ∞] right censored	S 0 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 0.9932 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.1835 [0.03965, 0.375] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	inconclusive	inconclusive	[3999, 3999] resolved	[4000, ∞] right censored	S 0.998 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.367 [0.2444, 0.4962] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[1827, 1827] resolved	[4000, ∞] right censored	S 0.9906 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.7034 [0.6057, 0.7941] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[975.5, 975.5] resolved	[4000, ∞] right censored	S 0.9642 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.8384 [0.7979, 0.8804] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[397.5, 397.5] resolved	[4000, ∞] right censored	S 0.9622 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9261 [0.9032, 0.9492] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[98.02, 98.02] resolved	[4000, ∞] right censored	S 0.9469 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9341 [0.9288, 0.9394] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_055_S0_H3_candidates_common_analysis_first_half_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis first half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	inconclusive	inconclusive	[49.28, 49.28] resolved	[4000, ∞] right censored	S 0.9661 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.8695 [0.8682, 0.8707] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[400, ∞] right censored	[400, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[400, ∞] right censored	[400, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[400, ∞] right censored	[400, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.0009369 [0, 0.002655] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[347.3, 347.3] resolved	[400, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.02056 [0.002342, 0.05127] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[134.2, 134.2] resolved	[400, ∞] right censored	S 0.9921 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.1433 [0.09447, 0.193] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[34.54, 34.54] resolved	[400, ∞] right censored	S 0.9857 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.3674 [0.3309, 0.4097] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.22, 18.22] resolved	[400, ∞] right censored	S 0.9847 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.4844 [0.4473, 0.5205] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.31, 10.31] resolved	[400, ∞] right censored	S 0.9774 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.6649 [0.6391, 0.6876] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.701, 6.701] resolved	[400, ∞] right censored	S 0.975 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7977 [0.7819, 0.814] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	inconclusive	inconclusive	[4.789, 4.789] resolved	[400, ∞] right censored	S 0.9742 / W 1 H 3200 / 3200 95% bound check: ?	S multi 0.8267 [0.8226, 0.8307] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_056_S0_H3_candidates_common_analysis_first_half_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis first half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	inconclusive	inconclusive	[4.337, 4.337] resolved	[400, ∞] right censored	S 0.9893 / W 1 H 3200 / 3200 95% bound check: ?	S multi 0.7561 [0.727, 0.7851] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_057_S0_H3_candidates_common_analysis_first_half_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis first half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.018 [1.002, 1.042] N 12/12; valid 1	1.004 [1.003, 1.006] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8815, low 0; forced mean no	1.178 [1.031, 1.372] N 12/12; valid 1	1.007 [1.003, 1.011] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.651 [1.505, 1.778] N 12/12; valid 1	1.012 [1.007, 1.017] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_058_S0_H3_fingerprints_common_analysis_first_half_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis first half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3779, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9141, low 0; forced mean no	2.309 [2.253, 2.364] N 12/12; valid 1	1.023 [1.014, 1.033] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.506 [2.388, 2.633] N 2/2; valid 1	1.112 [1.078, 1.147] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.459 [2.368, 2.554] N 2/2; valid 1	1.286 [1.259, 1.314] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1.009 [1.007, 1.01] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.03 [1.029, 1.032] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.112 [1.103, 1.121] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9412 [0.9057, 0.9772] N 12/12; valid 1	1.593 [1.418, 1.838] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7942 [0.7712, 0.8197] N 12/12; valid 1	3.393 [3.313, 3.462] N 12/12; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_059_S0_H3_fingerprints_common_analysis_first_half_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis first half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0.08333; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.7682 [0.7542, 0.7843] N 12/12; valid 1	2.601 [2.511, 2.698] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.03	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check single front; H 32000 multi 0.01228, low 0; forced mean yes	1.429 [1.342, 1.51] N 12/12; valid 1	0.4705 [0.4524, 0.4893] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.05	not declared	not declared	not declared	1.381 [1.271, 1.485] N 12/12; valid 1	0.4573 [0.4425, 0.4726] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.1	not declared	not declared	not declared	1.219 [1.139, 1.304] N 12/12; valid 1	0.4295 [0.4176, 0.4423] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.3	not declared	not declared	not declared	0.6434 [0.6143, 0.6742] N 2/2; valid 1	0.4216 [0.4195, 0.4237] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=1	not declared	not declared	not declared	0.04676 [0, 0.09559] N 2/2; valid 1	0.4164 [0.3826, 0.4562] N 2/2; valid 1	count unavailable contrast unavailable
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_060_S0_H3_fingerprints_common_analysis_first_half_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis first half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.004 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.009] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.179 [1.032, 1.363] N 12/12; valid 1	1.012 [1.011, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.351 [1.211, 1.499] N 12/12; valid 1	1.029 [1.025, 1.033] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8559, low 0; forced mean no	1.771 [1.622, 1.937] N 12/12; valid 1	1.049 [1.044, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.019 [1.911, 2.12] N 12/12; valid 1	1.073 [1.066, 1.08] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7088, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8021, low 0; forced mean no	2.151 [2.065, 2.239] N 12/12; valid 1	1.135 [1.119, 1.15] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_061_S0_H3_fingerprints_common_analysis_first_half_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis first half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.496 [2.397, 2.6] N 2/2; valid 1	1.265 [1.254, 1.276] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.185 [2.056, 2.315] N 2/2; valid 1	1.6 [1.54, 1.663] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1.001] N 2/2; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1.001 [1, 1.003] N 12/12; valid 1	1.002 [1.001, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.019 [1.002, 1.049] N 12/12; valid 1	1.005 [1.002, 1.007] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.128 [1.082, 1.177] N 12/12; valid 1	1.001 [0.995, 1.006] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.312 [1.274, 1.352] N 12/12; valid 1	1 [0.9924, 1.009] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_062_S0_H3_fingerprints_common_analysis_first_half_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis first half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1632, low 0; forced mean no	no eligible check single front; H 3200 multi 0.4919, low 0; forced mean no	1.418 [1.371, 1.465] N 12/12; valid 1	1.003 [0.9936, 1.013] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.569 [1.521, 1.612] N 12/12; valid 1	1.006 [0.989, 1.022] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5564, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7449, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.5985, low 0; forced mean no	1.7 [1.675, 1.726] N 12/12; valid 1	1.039 [1.017, 1.061] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.738 [1.693, 1.782] N 2/2; valid 1	1.17 [1.097, 1.253] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=1	not declared	not declared	not declared	1.723 [1.693, 1.752] N 2/2; valid 1	1.271 [1.236, 1.306] N 2/2; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_063_S0_H3_fingerprints_common_analysis_first_half_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 1.429 [1.342, 1.51] N 12/12; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4705 [0.4524, 0.4893] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 1.381 [1.271, 1.485] N 12/12; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4573 [0.4425, 0.4726] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 1.219 [1.139, 1.304] N 12/12; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4295 [0.4176, 0.4423] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 0.6434 [0.6143, 0.6742] N 2/2; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4216 [0.4195, 0.4237] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 0.04676 [0, 0.09559] N 2/2; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4164 [0.3826, 0.4562] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 1.429 [1.342, 1.51] N 12/12; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4705 [0.4524, 0.4893] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 1.381 [1.271, 1.485] N 12/12; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4573 [0.4425, 0.4726] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 1.219 [1.139, 1.304] N 12/12; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4295 [0.4176, 0.4423] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_064_S0_H3_pairs_common_analysis_first_half_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 0.6434 [0.6143, 0.6742] N 2/2; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4216 [0.4195, 0.4237] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 0.04676 [0, 0.09559] N 2/2; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4164 [0.3826, 0.4562] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 1.429 [1.342, 1.51] N 12/12; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4705 [0.4524, 0.4893] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 1.381 [1.271, 1.485] N 12/12; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4573 [0.4425, 0.4726] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 1.219 [1.139, 1.304] N 12/12; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4295 [0.4176, 0.4423] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 0.6434 [0.6143, 0.6742] N 2/2; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4216 [0.4195, 0.4237] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 0.04676 [0, 0.09559] N 2/2; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4164 [0.3826, 0.4562] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 1.429 [1.342, 1.51] N 12/12; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4705 [0.4524, 0.4893] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_065_S0_H3_pairs_common_analysis_first_half_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 1.381 [1.271, 1.485] N 12/12; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4573 [0.4425, 0.4726] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 1.219 [1.139, 1.304] N 12/12; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4295 [0.4176, 0.4423] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 0.6434 [0.6143, 0.6742] N 2/2; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4216 [0.4195, 0.4237] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 0.04676 [0, 0.09559] N 2/2; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4164 [0.3826, 0.4562] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_066_S0_H3_pairs_common_analysis_first_half_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_067_S0_H3_pairs_common_analysis_first_half_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_068_S0_H3_pairs_common_analysis_first_half_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_069_S0_H3_pairs_common_analysis_first_half_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_070_S0_H3_pairs_common_analysis_first_half_primary_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_071_S0_H3_pairs_common_analysis_first_half_primary_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_072_S0_H3_pairs_common_analysis_first_half_primary_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis first half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0002082 [0, 0.0006246] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.02441 [0.006141, 0.0519] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 0.9964 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.2057 [0.0673, 0.3891] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	inconclusive	inconclusive	[2704, 2704] resolved	[4000, ∞] right censored	S 0.9964 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.6192 [0.5258, 0.7165] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[487, 487] resolved	[4000, ∞] right censored	S 0.9483 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9579 [0.946, 0.9681] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[132.3, 132.3] resolved	[4000, ∞] right censored	S 0.9308 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.985 [0.985, 0.985] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	criterion met	inconclusive	[77.8, 77.8] resolved	[4000, ∞] right censored	S 0.9445 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9622 [0.9594, 0.965] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_073_S0_H3_candidates_common_analysis_first_half_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis first half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	criterion met	[540.8, 540.8] resolved	[4000, ∞] right censored	S 0.2132 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.06949 [0.05278, 0.08584] S low 0.8661 [0.844, 0.8911] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[606, 606] resolved	[4000, ∞] right censored	S 0.1347 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.05569 [0.04679, 0.06673] S low 0.9029 [0.8896, 0.9143] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[748.5, 748.5] resolved	[4000, ∞] right censored	S 0.05051 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.0266 [0.01764, 0.03618] S low 0.9619 [0.9517, 0.9717] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[0, ∞] unavailable lag gap	[4000, ∞] right censored	S 0.0003448 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.00812 [0.003748, 0.01249] S low 0.9866 [0.9781, 0.995] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_074_S0_H3_candidates_common_analysis_first_half_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis first half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[4000, ∞] right censored	S 0 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 0.9932 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.1835 [0.03965, 0.375] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	inconclusive	inconclusive	[3999, 3999] resolved	[4000, ∞] right censored	S 0.998 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.367 [0.2444, 0.4962] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[1827, 1827] resolved	[4000, ∞] right censored	S 0.9906 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.7034 [0.6057, 0.7941] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[975.5, 975.5] resolved	[4000, ∞] right censored	S 0.9642 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.8384 [0.7979, 0.8804] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[397.5, 397.5] resolved	[4000, ∞] right censored	S 0.9622 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9261 [0.9032, 0.9492] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[98.02, 98.02] resolved	[4000, ∞] right censored	S 0.9469 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9341 [0.9288, 0.9394] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_075_S0_H3_candidates_common_analysis_first_half_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis first half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	criterion met	inconclusive	[49.28, 49.28] resolved	[4000, ∞] right censored	S 0.9661 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.8695 [0.8682, 0.8707] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[400, ∞] right censored	[400, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[400, ∞] right censored	[400, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[400, ∞] right censored	[400, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.0009369 [0, 0.002655] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[347.3, 347.3] resolved	[400, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.02056 [0.002342, 0.05127] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[134.2, 134.2] resolved	[400, ∞] right censored	S 0.9921 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.1433 [0.09447, 0.193] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[34.54, 34.54] resolved	[400, ∞] right censored	S 0.9857 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.3674 [0.3309, 0.4097] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.22, 18.22] resolved	[400, ∞] right censored	S 0.9847 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.4844 [0.4473, 0.5205] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.31, 10.31] resolved	[400, ∞] right censored	S 0.9774 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.6649 [0.6391, 0.6876] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.701, 6.701] resolved	[400, ∞] right censored	S 0.975 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7977 [0.7819, 0.814] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	criterion met	inconclusive	[4.789, 4.789] resolved	[400, ∞] right censored	S 0.9742 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.8267 [0.8226, 0.8307] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_076_S0_H3_candidates_common_analysis_first_half_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis first half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	criterion met	inconclusive	[4.337, 4.337] resolved	[400, ∞] right censored	S 0.9893 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7561 [0.727, 0.7851] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_077_S0_H3_candidates_common_analysis_first_half_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis first half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.018 [1.002, 1.042] N 12/12; valid 1	1.004 [1.003, 1.006] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8815, low 0; forced mean no	1.178 [1.031, 1.372] N 12/12; valid 1	1.007 [1.003, 1.011] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.651 [1.505, 1.778] N 12/12; valid 1	1.012 [1.007, 1.017] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_078_S0_H3_fingerprints_common_analysis_first_half_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis first half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3779, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9141, low 0; forced mean no	2.309 [2.253, 2.364] N 12/12; valid 1	1.023 [1.014, 1.033] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.506 [2.388, 2.633] N 2/2; valid 1	1.112 [1.078, 1.147] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.459 [2.368, 2.554] N 2/2; valid 1	1.286 [1.259, 1.314] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1.009 [1.007, 1.01] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.03 [1.029, 1.032] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.112 [1.103, 1.121] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9412 [0.9057, 0.9772] N 12/12; valid 1	1.593 [1.418, 1.838] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7942 [0.7712, 0.8197] N 12/12; valid 1	3.393 [3.313, 3.462] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_079_S0_H3_fingerprints_common_analysis_first_half_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis first half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0.08333; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.7682 [0.7542, 0.7843] N 12/12; valid 1	2.601 [2.511, 2.698] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.03	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	single front; H 32000 qualified yes; same no multi 0.01228, low 0; forced mean yes	1.429 [1.342, 1.51] N 12/12; valid 1	0.4705 [0.4524, 0.4893] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.05	not declared	not declared	not declared	1.381 [1.271, 1.485] N 12/12; valid 1	0.4573 [0.4425, 0.4726] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.1	not declared	not declared	not declared	1.219 [1.139, 1.304] N 12/12; valid 1	0.4295 [0.4176, 0.4423] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.3	not declared	not declared	not declared	0.6434 [0.6143, 0.6742] N 2/2; valid 1	0.4216 [0.4195, 0.4237] N 2/2; valid 1	count yes contrast yes
Goryachev f=1	not declared	not declared	not declared	0.04676 [0, 0.09559] N 2/2; valid 1	0.4164 [0.3826, 0.4562] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_080_S0_H3_fingerprints_common_analysis_first_half_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis first half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.004 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.009] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.179 [1.032, 1.363] N 12/12; valid 1	1.012 [1.011, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.351 [1.211, 1.499] N 12/12; valid 1	1.029 [1.025, 1.033] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8559, low 0; forced mean no	1.771 [1.622, 1.937] N 12/12; valid 1	1.049 [1.044, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.019 [1.911, 2.12] N 12/12; valid 1	1.073 [1.066, 1.08] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7088, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8021, low 0; forced mean no	2.151 [2.065, 2.239] N 12/12; valid 1	1.135 [1.119, 1.15] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_081_S0_H3_fingerprints_common_analysis_first_half_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis first half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.496 [2.397, 2.6] N 2/2; valid 1	1.265 [1.254, 1.276] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.185 [2.056, 2.315] N 2/2; valid 1	1.6 [1.54, 1.663] N 2/2; valid 1	count yes contrast yes
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1.001] N 2/2; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1.001 [1, 1.003] N 12/12; valid 1	1.002 [1.001, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.019 [1.002, 1.049] N 12/12; valid 1	1.005 [1.002, 1.007] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.128 [1.082, 1.177] N 12/12; valid 1	1.001 [0.995, 1.006] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.312 [1.274, 1.352] N 12/12; valid 1	1 [0.9924, 1.009] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_082_S0_H3_fingerprints_common_analysis_first_half_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis first half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1632, low 0; forced mean no	no eligible check single front; H 3200 multi 0.4919, low 0; forced mean no	1.418 [1.371, 1.465] N 12/12; valid 1	1.003 [0.9936, 1.013] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.569 [1.521, 1.612] N 12/12; valid 1	1.006 [0.989, 1.022] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5564, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7449, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.5985, low 0; forced mean no	1.7 [1.675, 1.726] N 12/12; valid 1	1.039 [1.017, 1.061] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.738 [1.693, 1.782] N 2/2; valid 1	1.17 [1.097, 1.253] N 2/2; valid 1	count yes contrast yes
Spring f=1	not declared	not declared	not declared	1.723 [1.693, 1.752] N 2/2; valid 1	1.271 [1.236, 1.306] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_083_S0_H3_fingerprints_common_analysis_first_half_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 1.429 [1.342, 1.51] N 12/12; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4705 [0.4524, 0.4893] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 1.381 [1.271, 1.485] N 12/12; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4573 [0.4425, 0.4726] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 1.219 [1.139, 1.304] N 12/12; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4295 [0.4176, 0.4423] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 0.6434 [0.6143, 0.6742] N 2/2; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4216 [0.4195, 0.4237] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 0.04676 [0, 0.09559] N 2/2; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4164 [0.3826, 0.4562] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / criterion met	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 1.429 [1.342, 1.51] N 12/12; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4705 [0.4524, 0.4893] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 1.381 [1.271, 1.485] N 12/12; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4573 [0.4425, 0.4726] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 1.219 [1.139, 1.304] N 12/12; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4295 [0.4176, 0.4423] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_084_S0_H3_pairs_common_analysis_first_half_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 0.6434 [0.6143, 0.6742] N 2/2; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4216 [0.4195, 0.4237] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 0.04676 [0, 0.09559] N 2/2; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4164 [0.3826, 0.4562] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 1.429 [1.342, 1.51] N 12/12; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4705 [0.4524, 0.4893] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 1.381 [1.271, 1.485] N 12/12; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4573 [0.4425, 0.4726] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 1.219 [1.139, 1.304] N 12/12; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4295 [0.4176, 0.4423] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 0.6434 [0.6143, 0.6742] N 2/2; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4216 [0.4195, 0.4237] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 0.04676 [0, 0.09559] N 2/2; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4164 [0.3826, 0.4562] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 1.429 [1.342, 1.51] N 12/12; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4705 [0.4524, 0.4893] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_085_S0_H3_pairs_common_analysis_first_half_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 1.381 [1.271, 1.485] N 12/12; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4573 [0.4425, 0.4726] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 1.219 [1.139, 1.304] N 12/12; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4295 [0.4176, 0.4423] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 0.6434 [0.6143, 0.6742] N 2/2; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4216 [0.4195, 0.4237] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 0.04676 [0, 0.09559] N 2/2; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4164 [0.3826, 0.4562] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_086_S0_H3_pairs_common_analysis_first_half_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_087_S0_H3_pairs_common_analysis_first_half_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_088_S0_H3_pairs_common_analysis_first_half_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_089_S0_H3_pairs_common_analysis_first_half_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_090_S0_H3_pairs_common_analysis_first_half_inclusive_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_091_S0_H3_pairs_common_analysis_first_half_inclusive_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_092_S0_H3_pairs_common_analysis_first_half_inclusive_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis second half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.001666 [0, 0.003956] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.009786 [0.003748, 0.01713] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 0.9823 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.2636 [0.1409, 0.4316] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	inconclusive	inconclusive	[2591, 2591] resolved	[4000, ∞] right censored	S 0.9641 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.6587 [0.5212, 0.7791] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[519.6, 519.6] resolved	[4000, ∞] right censored	S 0.9356 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9512 [0.9327, 0.9693] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[120.8, 120.8] resolved	[4000, ∞] right censored	S 0.9141 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9875 [0.9856, 0.9894] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	inconclusive	inconclusive	[89.93, 89.93] resolved	[4000, ∞] right censored	S 0.921 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.9603 [0.9557, 0.965] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_093_S0_H3_candidates_common_analysis_second_half_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis second half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	inconclusive	[445.8, 445.8] resolved	[4000, ∞] right censored	S 0.261 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.08432 [0.07131, 0.09593] S low 0.8457 [0.8243, 0.8687] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[513, 513] resolved	[4000, ∞] right censored	S 0.1159 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.06907 [0.05767, 0.08078] S low 0.898 [0.8794, 0.9162] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[778.2, 778.2] resolved	[4000, ∞] right censored	S 0.05398 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.02712 [0.01973, 0.03409] S low 0.9539 [0.9411, 0.9668] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1169, 1169] resolved	[4000, ∞] right censored	S 0.02487 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.001562 [0, 0.003123] S low 0.9984 [0.9969, 1] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_094_S0_H3_candidates_common_analysis_second_half_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis second half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[4000, ∞] right censored	S 0 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0007808 [0, 0.002238] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 0.9646 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.2101 [0.01353, 0.4176] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	multiple fronts	inconclusive	inconclusive	[3164, 3164] resolved	[4000, ∞] right censored	S 0.9536 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.5997 [0.4581, 0.7432] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	inconclusive	criterion met	[2436, 2436] resolved	[4000, ∞] right censored	S 0.9774 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.729 [0.6299, 0.8188] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[932.3, 932.3] resolved	[4000, ∞] right censored	S 0.9844 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.8754 [0.8284, 0.9147] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[382.3, 382.3] resolved	[4000, ∞] right censored	S 0.9625 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9132 [0.8915, 0.9359] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[110.7, 110.7] resolved	[4000, ∞] right censored	S 0.9823 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9279 [0.9269, 0.9288] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_095_S0_H3_candidates_common_analysis_second_half_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis second half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	inconclusive	inconclusive	[53.4, 53.4] resolved	[4000, ∞] right censored	S 0.972 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.8407 [0.8289, 0.8526] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[400, ∞] right censored	[400, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[400, ∞] right censored	[400, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[400, ∞] right censored	[400, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.0006246 [0, 0.00177] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[400, ∞] right censored	[400, ∞] right censored	S 0.995 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.02649 [0.00406, 0.05945] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[199, 199] resolved	[400, ∞] right censored	S 0.9903 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.13 [0.08468, 0.182] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[42.25, 42.25] resolved	[400, ∞] right censored	S 0.9823 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.3384 [0.2924, 0.388] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	multiple fronts	criterion met	criterion met	[18.06, 18.06] resolved	[400, ∞] right censored	S 0.9805 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.5139 [0.4664, 0.562] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[11.07, 11.07] resolved	[400, ∞] right censored	S 0.9809 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.6522 [0.6354, 0.669] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.896, 6.896] resolved	[400, ∞] right censored	S 0.9739 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7807 [0.759, 0.8005] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	inconclusive	inconclusive	[5.19, 5.19] resolved	[400, ∞] right censored	S 0.9749 / W 1 H 3200 / 3200 95% bound check: ?	S multi 0.8532 [0.8438, 0.8626] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_096_S0_H3_candidates_common_analysis_second_half_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis second half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	inconclusive	inconclusive	[3.881, 3.881] resolved	[400, ∞] right censored	S 0.9912 / W 1 H 3200 / 3200 95% bound check: ?	S multi 0.7801 [0.7683, 0.792] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_097_S0_H3_candidates_common_analysis_second_half_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis second half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1.001 [1, 1.002] N 12/12; valid 1	1.003 [1.001, 1.004] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.004 [1, 1.009] N 12/12; valid 1	1.006 [1.005, 1.008] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8543, low 0; forced mean no	1.246 [1.112, 1.416] N 12/12; valid 1	1.01 [1.007, 1.014] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.685 [1.497, 1.846] N 12/12; valid 1	1.007 [1.001, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_098_S0_H3_fingerprints_common_analysis_second_half_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis second half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3933, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9126, low 0; forced mean no	2.31 [2.228, 2.389] N 12/12; valid 1	1.025 [1.017, 1.035] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.459 [2.364, 2.555] N 2/2; valid 1	1.137 [1.132, 1.143] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.556 [2.481, 2.633] N 2/2; valid 1	1.216 [1.208, 1.224] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1.008 [1.007, 1.009] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.03 [1.029, 1.032] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.119 [1.114, 1.124] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9756 [0.9592, 0.9901] N 12/12; valid 1	1.39 [1.33, 1.457] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7884 [0.7767, 0.8] N 12/12; valid 1	3.299 [3.213, 3.39] N 12/12; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_099_S0_H3_fingerprints_common_analysis_second_half_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis second half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.03087; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1978; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.7687 [0.7458, 0.7905] N 12/12; valid 1	2.631 [2.551, 2.704] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.03	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check single front; H 32000 multi 0.00406, low 0; forced mean yes	1.478 [1.369, 1.585] N 12/12; valid 1	0.4817 [0.4616, 0.5027] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.05	not declared	not declared	not declared	1.316 [1.238, 1.397] N 12/12; valid 1	0.4644 [0.4506, 0.4779] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.1	not declared	not declared	not declared	1.193 [1.074, 1.325] N 12/12; valid 1	0.4449 [0.4296, 0.4626] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.3	not declared	not declared	not declared	0.6643 [0.5959, 0.7388] N 2/2; valid 1	0.4279 [0.4122, 0.4438] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=1	not declared	not declared	not declared	0.09375 [0.07432, 0.1143] N 2/2; valid 1	0.4292 [0.4084, 0.4522] N 2/2; valid 1	count unavailable contrast unavailable
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_100_S0_H3_fingerprints_common_analysis_second_half_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis second half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.009] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.208 [1.002, 1.431] N 12/12; valid 1	1.014 [1.013, 1.016] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.669 [1.476, 1.851] N 12/12; valid 1	1.028 [1.024, 1.033] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.02295, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8576, low 0; forced mean no	1.77 [1.621, 1.919] N 12/12; valid 1	1.049 [1.042, 1.057] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.004 [1.889, 2.119] N 12/12; valid 1	1.079 [1.071, 1.088] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.6923, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8151, low 0; forced mean no	2.07 [1.996, 2.152] N 12/12; valid 1	1.134 [1.12, 1.148] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_101_S0_H3_fingerprints_common_analysis_second_half_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis second half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.218 [2.101, 2.339] N 2/2; valid 1	1.347 [1.313, 1.381] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.245 [2.098, 2.405] N 2/2; valid 1	1.787 [1.703, 1.878] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [0.9999, 1.001] N 2/2; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.004 [1.003, 1.006] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.023 [1.002, 1.056] N 12/12; valid 1	1.009 [1.005, 1.012] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.113 [1.068, 1.163] N 12/12; valid 1	1.001 [0.9955, 1.006] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.284 [1.242, 1.334] N 12/12; valid 1	0.9997 [0.9923, 1.007] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_102_S0_H3_fingerprints_common_analysis_second_half_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis second half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	single front; H 3200 qualified yes; same no multi 0, low 0; forced mean no	single front; H 3200 qualified yes; same no multi 0.1823, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.5307, low 0; forced mean no	1.449 [1.399, 1.503] N 12/12; valid 1	0.9915 [0.9815, 1] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.564 [1.538, 1.588] N 12/12; valid 1	1.019 [0.9997, 1.038] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5634, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7431, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.61, low 0; forced mean no	1.702 [1.667, 1.739] N 12/12; valid 1	1.027 [1, 1.053] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.787 [1.716, 1.86] N 2/2; valid 1	1.173 [1.09, 1.268] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=1	not declared	not declared	not declared	1.796 [1.762, 1.83] N 2/2; valid 1	1.23 [1.172, 1.291] N 2/2; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_103_S0_H3_fingerprints_common_analysis_second_half_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 1.478 [1.369, 1.585] N 12/12; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4817 [0.4616, 0.5027] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 1.316 [1.238, 1.397] N 12/12; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4644 [0.4506, 0.4779] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 1.193 [1.074, 1.325] N 12/12; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4449 [0.4296, 0.4626] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 0.6643 [0.5959, 0.7388] N 2/2; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4279 [0.4122, 0.4438] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 0.09375 [0.07432, 0.1143] N 2/2; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4292 [0.4084, 0.4522] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 1.478 [1.369, 1.585] N 12/12; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4817 [0.4616, 0.5027] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 1.316 [1.238, 1.397] N 12/12; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4644 [0.4506, 0.4779] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 1.193 [1.074, 1.325] N 12/12; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4449 [0.4296, 0.4626] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_104_S0_H3_pairs_common_analysis_second_half_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 0.6643 [0.5959, 0.7388] N 2/2; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4279 [0.4122, 0.4438] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 0.09375 [0.07432, 0.1143] N 2/2; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4292 [0.4084, 0.4522] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 1.478 [1.369, 1.585] N 12/12; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4817 [0.4616, 0.5027] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 1.316 [1.238, 1.397] N 12/12; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4644 [0.4506, 0.4779] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 1.193 [1.074, 1.325] N 12/12; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4449 [0.4296, 0.4626] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 0.6643 [0.5959, 0.7388] N 2/2; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4279 [0.4122, 0.4438] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 0.09375 [0.07432, 0.1143] N 2/2; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4292 [0.4084, 0.4522] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 1.478 [1.369, 1.585] N 12/12; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4817 [0.4616, 0.5027] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_105_S0_H3_pairs_common_analysis_second_half_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 1.316 [1.238, 1.397] N 12/12; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4644 [0.4506, 0.4779] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 1.193 [1.074, 1.325] N 12/12; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4449 [0.4296, 0.4626] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 0.6643 [0.5959, 0.7388] N 2/2; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4279 [0.4122, 0.4438] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 0.09375 [0.07432, 0.1143] N 2/2; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4292 [0.4084, 0.4522] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_106_S0_H3_pairs_common_analysis_second_half_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_107_S0_H3_pairs_common_analysis_second_half_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_108_S0_H3_pairs_common_analysis_second_half_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_109_S0_H3_pairs_common_analysis_second_half_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_110_S0_H3_pairs_common_analysis_second_half_primary_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_111_S0_H3_pairs_common_analysis_second_half_primary_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_112_S0_H3_pairs_common_analysis_second_half_primary_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_113_S0_H3_pairs_common_analysis_second_half_primary_010.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis second half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.001666 [0, 0.003956] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.009786 [0.003748, 0.01713] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 0.9823 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.2636 [0.1409, 0.4316] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	inconclusive	inconclusive	[2591, 2591] resolved	[4000, ∞] right censored	S 0.9641 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.6587 [0.5212, 0.7791] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[519.6, 519.6] resolved	[4000, ∞] right censored	S 0.9356 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9512 [0.9327, 0.9693] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[120.8, 120.8] resolved	[4000, ∞] right censored	S 0.9141 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9875 [0.9856, 0.9894] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	criterion met	inconclusive	[89.93, 89.93] resolved	[4000, ∞] right censored	S 0.921 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9603 [0.9557, 0.965] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_114_S0_H3_candidates_common_analysis_second_half_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis second half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	criterion met	[445.8, 445.8] resolved	[4000, ∞] right censored	S 0.261 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.08432 [0.07131, 0.09593] S low 0.8457 [0.8243, 0.8687] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[513, 513] resolved	[4000, ∞] right censored	S 0.1159 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.06907 [0.05767, 0.08078] S low 0.898 [0.8794, 0.9162] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[778.2, 778.2] resolved	[4000, ∞] right censored	S 0.05398 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.02712 [0.01973, 0.03409] S low 0.9539 [0.9411, 0.9668] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1169, 1169] resolved	[4000, ∞] right censored	S 0.02487 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.001562 [0, 0.003123] S low 0.9984 [0.9969, 1] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_115_S0_H3_candidates_common_analysis_second_half_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis second half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[4000, ∞] right censored	S 0 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0007808 [0, 0.002238] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 0.9646 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.2101 [0.01353, 0.4176] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	multiple fronts	inconclusive	inconclusive	[3164, 3164] resolved	[4000, ∞] right censored	S 0.9536 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.5997 [0.4581, 0.7432] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	inconclusive	criterion met	[2436, 2436] resolved	[4000, ∞] right censored	S 0.9774 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.729 [0.6299, 0.8188] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[932.3, 932.3] resolved	[4000, ∞] right censored	S 0.9844 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.8754 [0.8284, 0.9147] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[382.3, 382.3] resolved	[4000, ∞] right censored	S 0.9625 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9132 [0.8915, 0.9359] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[110.7, 110.7] resolved	[4000, ∞] right censored	S 0.9823 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9279 [0.9269, 0.9288] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_116_S0_H3_candidates_common_analysis_second_half_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis second half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	criterion met	inconclusive	[53.4, 53.4] resolved	[4000, ∞] right censored	S 0.972 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.8407 [0.8289, 0.8526] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[400, ∞] right censored	[400, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[400, ∞] right censored	[400, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[400, ∞] right censored	[400, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.0006246 [0, 0.00177] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[400, ∞] right censored	[400, ∞] right censored	S 0.995 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.02649 [0.00406, 0.05945] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[199, 199] resolved	[400, ∞] right censored	S 0.9903 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.13 [0.08468, 0.182] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[42.25, 42.25] resolved	[400, ∞] right censored	S 0.9823 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.3384 [0.2924, 0.388] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	multiple fronts	criterion met	criterion met	[18.06, 18.06] resolved	[400, ∞] right censored	S 0.9805 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.5139 [0.4664, 0.562] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[11.07, 11.07] resolved	[400, ∞] right censored	S 0.9809 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.6522 [0.6354, 0.669] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.896, 6.896] resolved	[400, ∞] right censored	S 0.9739 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7807 [0.759, 0.8005] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	criterion met	inconclusive	[5.19, 5.19] resolved	[400, ∞] right censored	S 0.9749 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.8532 [0.8438, 0.8626] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_117_S0_H3_candidates_common_analysis_second_half_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis second half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	criterion met	inconclusive	[3.881, 3.881] resolved	[400, ∞] right censored	S 0.9912 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7801 [0.7683, 0.792] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_118_S0_H3_candidates_common_analysis_second_half_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis second half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1.001 [1, 1.002] N 12/12; valid 1	1.003 [1.001, 1.004] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.004 [1, 1.009] N 12/12; valid 1	1.006 [1.005, 1.008] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8543, low 0; forced mean no	1.246 [1.112, 1.416] N 12/12; valid 1	1.01 [1.007, 1.014] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.685 [1.497, 1.846] N 12/12; valid 1	1.007 [1.001, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_119_S0_H3_fingerprints_common_analysis_second_half_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis second half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3933, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9126, low 0; forced mean no	2.31 [2.228, 2.389] N 12/12; valid 1	1.025 [1.017, 1.035] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.459 [2.364, 2.555] N 2/2; valid 1	1.137 [1.132, 1.143] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.556 [2.481, 2.633] N 2/2; valid 1	1.216 [1.208, 1.224] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1.008 [1.007, 1.009] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.03 [1.029, 1.032] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.119 [1.114, 1.124] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9756 [0.9592, 0.9901] N 12/12; valid 1	1.39 [1.33, 1.457] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7884 [0.7767, 0.8] N 12/12; valid 1	3.299 [3.213, 3.39] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_120_S0_H3_fingerprints_common_analysis_second_half_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis second half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.03087; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1978; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.7687 [0.7458, 0.7905] N 12/12; valid 1	2.631 [2.551, 2.704] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.03	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	single front; H 32000 qualified yes; same no multi 0.00406, low 0; forced mean yes	1.478 [1.369, 1.585] N 12/12; valid 1	0.4817 [0.4616, 0.5027] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.05	not declared	not declared	not declared	1.316 [1.238, 1.397] N 12/12; valid 1	0.4644 [0.4506, 0.4779] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.1	not declared	not declared	not declared	1.193 [1.074, 1.325] N 12/12; valid 1	0.4449 [0.4296, 0.4626] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.3	not declared	not declared	not declared	0.6643 [0.5959, 0.7388] N 2/2; valid 1	0.4279 [0.4122, 0.4438] N 2/2; valid 1	count yes contrast yes
Goryachev f=1	not declared	not declared	not declared	0.09375 [0.07432, 0.1143] N 2/2; valid 1	0.4292 [0.4084, 0.4522] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_121_S0_H3_fingerprints_common_analysis_second_half_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis second half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.009] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.208 [1.002, 1.431] N 12/12; valid 1	1.014 [1.013, 1.016] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.669 [1.476, 1.851] N 12/12; valid 1	1.028 [1.024, 1.033] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.02295, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8576, low 0; forced mean no	1.77 [1.621, 1.919] N 12/12; valid 1	1.049 [1.042, 1.057] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.004 [1.889, 2.119] N 12/12; valid 1	1.079 [1.071, 1.088] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.6923, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8151, low 0; forced mean no	2.07 [1.996, 2.152] N 12/12; valid 1	1.134 [1.12, 1.148] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_122_S0_H3_fingerprints_common_analysis_second_half_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis second half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.218 [2.101, 2.339] N 2/2; valid 1	1.347 [1.313, 1.381] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.245 [2.098, 2.405] N 2/2; valid 1	1.787 [1.703, 1.878] N 2/2; valid 1	count yes contrast yes
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [0.9999, 1.001] N 2/2; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.004 [1.003, 1.006] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.023 [1.002, 1.056] N 12/12; valid 1	1.009 [1.005, 1.012] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.113 [1.068, 1.163] N 12/12; valid 1	1.001 [0.9955, 1.006] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.284 [1.242, 1.334] N 12/12; valid 1	0.9997 [0.9923, 1.007] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_123_S0_H3_fingerprints_common_analysis_second_half_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis second half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	single front; H 3200 qualified yes; same no multi 0, low 0; forced mean no	single front; H 3200 qualified yes; same no multi 0.1823, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.5307, low 0; forced mean no	1.449 [1.399, 1.503] N 12/12; valid 1	0.9915 [0.9815, 1] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.564 [1.538, 1.588] N 12/12; valid 1	1.019 [0.9997, 1.038] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5634, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7431, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.61, low 0; forced mean no	1.702 [1.667, 1.739] N 12/12; valid 1	1.027 [1, 1.053] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.787 [1.716, 1.86] N 2/2; valid 1	1.173 [1.09, 1.268] N 2/2; valid 1	count yes contrast yes
Spring f=1	not declared	not declared	not declared	1.796 [1.762, 1.83] N 2/2; valid 1	1.23 [1.172, 1.291] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_124_S0_H3_fingerprints_common_analysis_second_half_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 1.478 [1.369, 1.585] N 12/12; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4817 [0.4616, 0.5027] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 1.316 [1.238, 1.397] N 12/12; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4644 [0.4506, 0.4779] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 1.193 [1.074, 1.325] N 12/12; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4449 [0.4296, 0.4626] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 0.6643 [0.5959, 0.7388] N 2/2; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4279 [0.4122, 0.4438] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 0.09375 [0.07432, 0.1143] N 2/2; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4292 [0.4084, 0.4522] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / criterion met	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 1.478 [1.369, 1.585] N 12/12; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4817 [0.4616, 0.5027] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 1.316 [1.238, 1.397] N 12/12; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4644 [0.4506, 0.4779] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 1.193 [1.074, 1.325] N 12/12; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4449 [0.4296, 0.4626] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_125_S0_H3_pairs_common_analysis_second_half_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 0.6643 [0.5959, 0.7388] N 2/2; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4279 [0.4122, 0.4438] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 0.09375 [0.07432, 0.1143] N 2/2; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4292 [0.4084, 0.4522] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 1.478 [1.369, 1.585] N 12/12; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4817 [0.4616, 0.5027] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 1.316 [1.238, 1.397] N 12/12; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4644 [0.4506, 0.4779] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 1.193 [1.074, 1.325] N 12/12; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4449 [0.4296, 0.4626] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 0.6643 [0.5959, 0.7388] N 2/2; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4279 [0.4122, 0.4438] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 0.09375 [0.07432, 0.1143] N 2/2; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4292 [0.4084, 0.4522] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 1.478 [1.369, 1.585] N 12/12; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4817 [0.4616, 0.5027] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_126_S0_H3_pairs_common_analysis_second_half_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 1.316 [1.238, 1.397] N 12/12; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4644 [0.4506, 0.4779] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 1.193 [1.074, 1.325] N 12/12; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4449 [0.4296, 0.4626] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 0.6643 [0.5959, 0.7388] N 2/2; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4279 [0.4122, 0.4438] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 0.09375 [0.07432, 0.1143] N 2/2; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4292 [0.4084, 0.4522] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: criterion met / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: criterion met / criterion met	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_127_S0_H3_pairs_common_analysis_second_half_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_128_S0_H3_pairs_common_analysis_second_half_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_129_S0_H3_pairs_common_analysis_second_half_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_130_S0_H3_pairs_common_analysis_second_half_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_131_S0_H3_pairs_common_analysis_second_half_inclusive_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_132_S0_H3_pairs_common_analysis_second_half_inclusive_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_133_S0_H3_pairs_common_analysis_second_half_inclusive_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_134_S0_H3_pairs_common_analysis_second_half_inclusive_010.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · late quarter · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.002081 [0, 0.006242] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.009155 [0.001144, 0.01987] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 0.9757 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.2917 [0.1244, 0.5021] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 0.9902 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.6283 [0.4561, 0.7684] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[523.6, 523.6] resolved	[2000, ∞] right censored	S 0.9389 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9597 [0.9421, 0.9774] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[113.6, 113.6] resolved	[2000, ∞] right censored	S 0.9101 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.985 [0.9825, 0.9875] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	inconclusive	inconclusive	[90.81, 90.81] resolved	[2000, ∞] right censored	S 0.9227 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.9638 [0.9563, 0.9713] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_135_S0_H3_candidates_common_late_quarter_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · late quarter · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	inconclusive	[450, 450] resolved	[2000, ∞] right censored	S 0.2955 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.0748 [0.0594, 0.09124] S low 0.8537 [0.8408, 0.8707] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[428, 428] resolved	[2000, ∞] right censored	S 0.128 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.0671 [0.05452, 0.07886] S low 0.9087 [0.892, 0.926] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[978, 978] resolved	[2000, ∞] right censored	S 0.05124 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.03069 [0.01987, 0.04203] S low 0.9511 [0.9319, 0.9681] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[2000, ∞] right censored	S 0 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_136_S0_H3_candidates_common_late_quarter_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · late quarter · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[2000, ∞] right censored	S 0 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 0.9919 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.162 [0.01007, 0.3558] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	multiple fronts	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 0.9537 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.5145 [0.3285, 0.706] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	inconclusive	criterion met	[2000, ∞] right censored	[2000, ∞] right censored	S 0.9877 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.6821 [0.5204, 0.8242] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	inconclusive	inconclusive	[1347, 1347] resolved	[2000, ∞] right censored	S 0.9783 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.8719 [0.8317, 0.9134] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[367.8, 367.8] resolved	[2000, ∞] right censored	S 0.9576 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9181 [0.8925, 0.944] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[97.62, 97.62] resolved	[2000, ∞] right censored	S 0.9923 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9295 [0.9201, 0.9388] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_137_S0_H3_candidates_common_late_quarter_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · late quarter · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	inconclusive	inconclusive	[58.16, 58.16] resolved	[2000, ∞] right censored	S 0.9759 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.8102 [0.7928, 0.8277] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[200, ∞] right censored	[200, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[200, ∞] right censored	[200, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[200, ∞] right censored	[200, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.00104 [0, 0.003121] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[200, ∞] right censored	[200, ∞] right censored	S 0.99 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.04359 [0.00437, 0.1033] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	inconclusive	inconclusive	[179.1, 179.1] resolved	[200, ∞] right censored	S 0.9928 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.1256 [0.0723, 0.1927] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[48.74, 48.74] resolved	[200, ∞] right censored	S 0.9833 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.3343 [0.27, 0.4022] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	multiple fronts	criterion met	criterion met	[18.09, 18.09] resolved	[200, ∞] right censored	S 0.9776 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.533 [0.4707, 0.6013] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[11.29, 11.29] resolved	[200, ∞] right censored	S 0.9828 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.6471 [0.6185, 0.6756] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.988, 6.988] resolved	[200, ∞] right censored	S 0.9724 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7762 [0.7507, 0.7998] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	inconclusive	inconclusive	[5.513, 5.513] resolved	[200, ∞] right censored	S 0.9747 / W 1 H 3200 / 3200 95% bound check: ?	S multi 0.829 [0.7903, 0.8677] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_138_S0_H3_candidates_common_late_quarter_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · late quarter · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	inconclusive	inconclusive	[3.991, 3.991] resolved	[200, ∞] right censored	S 0.9924 / W 1 H 3200 / 3200 95% bound check: ?	S multi 0.7709 [0.7191, 0.8227] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_139_S0_H3_candidates_common_late_quarter_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · late quarter · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [0.9999, 1] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [0.9999, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1.002 [1, 1.005] N 12/12; valid 1	1.002 [1.001, 1.004] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.005 [1, 1.012] N 12/12; valid 1	1.008 [1.005, 1.011] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8688, low 0; forced mean no	1.29 [1.103, 1.521] N 12/12; valid 1	1.011 [1.006, 1.015] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.641 [1.428, 1.853] N 12/12; valid 1	1.008 [0.9981, 1.018] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_140_S0_H3_fingerprints_common_late_quarter_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · late quarter · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3682, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9189, low 0; forced mean no	2.312 [2.243, 2.38] N 12/12; valid 1	1.027 [1.017, 1.04] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.442 [2.348, 2.54] N 2/2; valid 1	1.143 [1.112, 1.175] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.617 [2.484, 2.75] N 2/2; valid 1	1.187 [1.123, 1.253] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1.008 [1.007, 1.01] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.03 [1.028, 1.033] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.107, 1.125] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9756 [0.9419, 1] N 12/12; valid 1	1.39 [1.313, 1.511] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7926 [0.7663, 0.8219] N 12/12; valid 1	3.225 [3.093, 3.362] N 12/12; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_141_S0_H3_fingerprints_common_late_quarter_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · late quarter · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.06169; forced mean no	no eligible check single front; H 32000 multi 0, low 0.2398; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.7782 [0.7472, 0.8097] N 12/12; valid 1	2.615 [2.542, 2.694] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.03	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check single front; H 32000 multi 0.007074, low 0; forced mean yes	1.452 [1.346, 1.561] N 12/12; valid 1	0.472 [0.4473, 0.4987] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.05	not declared	not declared	not declared	1.317 [1.203, 1.428] N 12/12; valid 1	0.4675 [0.45, 0.4849] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.1	not declared	not declared	not declared	1.191 [1.04, 1.369] N 12/12; valid 1	0.4325 [0.4136, 0.4551] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.3	not declared	not declared	not declared	0.5441 [0.4932, 0.6032] N 2/2; valid 1	0.4432 [0.4413, 0.4451] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=1	not declared	not declared	not declared	0.03546 [0.02778, 0.04348] N 2/2; valid 1	0.4206 [0.3952, 0.4478] N 2/2; valid 1	count unavailable contrast unavailable
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_142_S0_H3_fingerprints_common_late_quarter_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · late quarter · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.004 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.009 [1.008, 1.01] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.161 [0.9983, 1.358] N 12/12; valid 1	1.015 [1.012, 1.018] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.614 [1.355, 1.89] N 12/12; valid 1	1.028 [1.022, 1.035] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8859, low 0; forced mean no	1.678 [1.477, 1.875] N 12/12; valid 1	1.05 [1.041, 1.059] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	1.975 [1.841, 2.108] N 12/12; valid 1	1.081 [1.069, 1.093] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7209, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8092, low 0; forced mean no	2.112 [2.008, 2.218] N 12/12; valid 1	1.132 [1.114, 1.153] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_143_S0_H3_fingerprints_common_late_quarter_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · late quarter · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.363 [2.228, 2.5] N 2/2; valid 1	1.379 [1.31, 1.45] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.164 [2.06, 2.279] N 2/2; valid 1	1.827 [1.787, 1.867] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.003 [1, 1.006] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.038 [1, 1.107] N 12/12; valid 1	1.009 [1.003, 1.014] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.107 [1.053, 1.173] N 12/12; valid 1	1.004 [0.9956, 1.012] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.27 [1.203, 1.352] N 12/12; valid 1	0.9951 [0.9836, 1.007] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_144_S0_H3_fingerprints_common_late_quarter_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · late quarter · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	single front; H 3200 qualified yes; same no multi 0, low 0; forced mean no	single front; H 3200 qualified yes; same no multi 0.1828, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.5309, low 0; forced mean no	1.473 [1.403, 1.553] N 12/12; valid 1	0.9857 [0.9721, 0.9981] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.548 [1.522, 1.573] N 12/12; valid 1	1.025 [1.001, 1.048] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5624, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7423, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6064, low 0; forced mean no	1.677 [1.64, 1.716] N 12/12; valid 1	1.023 [0.9881, 1.059] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.743 [1.686, 1.8] N 2/2; valid 1	1.153 [1.086, 1.221] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=1	not declared	not declared	not declared	1.752 [1.686, 1.82] N 2/2; valid 1	1.277 [1.102, 1.491] N 2/2; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_145_S0_H3_fingerprints_common_late_quarter_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 1.452 [1.346, 1.561] N 12/12; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.472 [0.4473, 0.4987] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 1.317 [1.203, 1.428] N 12/12; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.4675 [0.45, 0.4849] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 1.191 [1.04, 1.369] N 12/12; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.4325 [0.4136, 0.4551] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 0.5441 [0.4932, 0.6032] N 2/2; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.4432 [0.4413, 0.4451] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 0.03546 [0.02778, 0.04348] N 2/2; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.4206 [0.3952, 0.4478] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 1.452 [1.346, 1.561] N 12/12; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.472 [0.4473, 0.4987] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 1.317 [1.203, 1.428] N 12/12; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.4675 [0.45, 0.4849] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 1.191 [1.04, 1.369] N 12/12; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.4325 [0.4136, 0.4551] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_146_S0_H3_pairs_common_late_quarter_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 0.5441 [0.4932, 0.6032] N 2/2; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.4432 [0.4413, 0.4451] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 0.03546 [0.02778, 0.04348] N 2/2; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.4206 [0.3952, 0.4478] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 1.452 [1.346, 1.561] N 12/12; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.472 [0.4473, 0.4987] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 1.317 [1.203, 1.428] N 12/12; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.4675 [0.45, 0.4849] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 1.191 [1.04, 1.369] N 12/12; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.4325 [0.4136, 0.4551] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 0.5441 [0.4932, 0.6032] N 2/2; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.4432 [0.4413, 0.4451] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 0.03546 [0.02778, 0.04348] N 2/2; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.4206 [0.3952, 0.4478] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 1.452 [1.346, 1.561] N 12/12; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.472 [0.4473, 0.4987] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_147_S0_H3_pairs_common_late_quarter_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 1.317 [1.203, 1.428] N 12/12; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.4675 [0.45, 0.4849] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 1.191 [1.04, 1.369] N 12/12; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.4325 [0.4136, 0.4551] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 0.5441 [0.4932, 0.6032] N 2/2; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.4432 [0.4413, 0.4451] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 0.03546 [0.02778, 0.04348] N 2/2; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.4206 [0.3952, 0.4478] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_148_S0_H3_pairs_common_late_quarter_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_149_S0_H3_pairs_common_late_quarter_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_150_S0_H3_pairs_common_late_quarter_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_151_S0_H3_pairs_common_late_quarter_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_152_S0_H3_pairs_common_late_quarter_primary_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_153_S0_H3_pairs_common_late_quarter_primary_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_154_S0_H3_pairs_common_late_quarter_primary_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_155_S0_H3_pairs_common_late_quarter_primary_010.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · late quarter · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.002081 [0, 0.006242] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.009155 [0.001144, 0.01987] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 0.9757 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.2917 [0.1244, 0.5021] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 0.9902 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.6283 [0.4561, 0.7684] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[523.6, 523.6] resolved	[2000, ∞] right censored	S 0.9389 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9597 [0.9421, 0.9774] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[113.6, 113.6] resolved	[2000, ∞] right censored	S 0.9101 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.985 [0.9825, 0.9875] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	criterion met	inconclusive	[90.81, 90.81] resolved	[2000, ∞] right censored	S 0.9227 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9638 [0.9563, 0.9713] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_156_S0_H3_candidates_common_late_quarter_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · late quarter · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	criterion met	[450, 450] resolved	[2000, ∞] right censored	S 0.2955 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.0748 [0.0594, 0.09124] S low 0.8537 [0.8408, 0.8707] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[428, 428] resolved	[2000, ∞] right censored	S 0.128 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.0671 [0.05452, 0.07886] S low 0.9087 [0.892, 0.926] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[978, 978] resolved	[2000, ∞] right censored	S 0.05124 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.03069 [0.01987, 0.04203] S low 0.9511 [0.9319, 0.9681] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[2000, ∞] right censored	S 0 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_157_S0_H3_candidates_common_late_quarter_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · late quarter · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[2000, ∞] right censored	S 0 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 0.9919 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.162 [0.01007, 0.3558] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	multiple fronts	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 0.9537 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.5145 [0.3285, 0.706] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	inconclusive	criterion met	[2000, ∞] right censored	[2000, ∞] right censored	S 0.9877 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.6821 [0.5204, 0.8242] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	inconclusive	inconclusive	[1347, 1347] resolved	[2000, ∞] right censored	S 0.9783 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.8719 [0.8317, 0.9134] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[367.8, 367.8] resolved	[2000, ∞] right censored	S 0.9576 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9181 [0.8925, 0.944] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[97.62, 97.62] resolved	[2000, ∞] right censored	S 0.9923 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9295 [0.9201, 0.9388] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_158_S0_H3_candidates_common_late_quarter_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · late quarter · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	criterion met	inconclusive	[58.16, 58.16] resolved	[2000, ∞] right censored	S 0.9759 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.8102 [0.7928, 0.8277] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[200, ∞] right censored	[200, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[200, ∞] right censored	[200, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[200, ∞] right censored	[200, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.00104 [0, 0.003121] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[200, ∞] right censored	[200, ∞] right censored	S 0.99 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.04359 [0.00437, 0.1033] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	inconclusive	inconclusive	[179.1, 179.1] resolved	[200, ∞] right censored	S 0.9928 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.1256 [0.0723, 0.1927] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[48.74, 48.74] resolved	[200, ∞] right censored	S 0.9833 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.3343 [0.27, 0.4022] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	multiple fronts	criterion met	criterion met	[18.09, 18.09] resolved	[200, ∞] right censored	S 0.9776 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.533 [0.4707, 0.6013] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[11.29, 11.29] resolved	[200, ∞] right censored	S 0.9828 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.6471 [0.6185, 0.6756] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.988, 6.988] resolved	[200, ∞] right censored	S 0.9724 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7762 [0.7507, 0.7998] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	criterion met	inconclusive	[5.513, 5.513] resolved	[200, ∞] right censored	S 0.9747 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.829 [0.7903, 0.8677] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_159_S0_H3_candidates_common_late_quarter_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · late quarter · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	criterion met	inconclusive	[3.991, 3.991] resolved	[200, ∞] right censored	S 0.9924 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7709 [0.7191, 0.8227] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_160_S0_H3_candidates_common_late_quarter_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · late quarter · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [0.9999, 1] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [0.9999, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1.002 [1, 1.005] N 12/12; valid 1	1.002 [1.001, 1.004] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.005 [1, 1.012] N 12/12; valid 1	1.008 [1.005, 1.011] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8688, low 0; forced mean no	1.29 [1.103, 1.521] N 12/12; valid 1	1.011 [1.006, 1.015] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.641 [1.428, 1.853] N 12/12; valid 1	1.008 [0.9981, 1.018] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_161_S0_H3_fingerprints_common_late_quarter_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · late quarter · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3682, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9189, low 0; forced mean no	2.312 [2.243, 2.38] N 12/12; valid 1	1.027 [1.017, 1.04] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.442 [2.348, 2.54] N 2/2; valid 1	1.143 [1.112, 1.175] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.617 [2.484, 2.75] N 2/2; valid 1	1.187 [1.123, 1.253] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1.008 [1.007, 1.01] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.03 [1.028, 1.033] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.107, 1.125] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9756 [0.9419, 1] N 12/12; valid 1	1.39 [1.313, 1.511] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7926 [0.7663, 0.8219] N 12/12; valid 1	3.225 [3.093, 3.362] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_162_S0_H3_fingerprints_common_late_quarter_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · late quarter · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.06169; forced mean no	no eligible check single front; H 32000 multi 0, low 0.2398; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.7782 [0.7472, 0.8097] N 12/12; valid 1	2.615 [2.542, 2.694] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.03	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	single front; H 32000 qualified yes; same no multi 0.007074, low 0; forced mean yes	1.452 [1.346, 1.561] N 12/12; valid 1	0.472 [0.4473, 0.4987] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.05	not declared	not declared	not declared	1.317 [1.203, 1.428] N 12/12; valid 1	0.4675 [0.45, 0.4849] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.1	not declared	not declared	not declared	1.191 [1.04, 1.369] N 12/12; valid 1	0.4325 [0.4136, 0.4551] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.3	not declared	not declared	not declared	0.5441 [0.4932, 0.6032] N 2/2; valid 1	0.4432 [0.4413, 0.4451] N 2/2; valid 1	count yes contrast yes
Goryachev f=1	not declared	not declared	not declared	0.03546 [0.02778, 0.04348] N 2/2; valid 1	0.4206 [0.3952, 0.4478] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_163_S0_H3_fingerprints_common_late_quarter_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · late quarter · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.004 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.009 [1.008, 1.01] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.161 [0.9983, 1.358] N 12/12; valid 1	1.015 [1.012, 1.018] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.614 [1.355, 1.89] N 12/12; valid 1	1.028 [1.022, 1.035] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8859, low 0; forced mean no	1.678 [1.477, 1.875] N 12/12; valid 1	1.05 [1.041, 1.059] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	1.975 [1.841, 2.108] N 12/12; valid 1	1.081 [1.069, 1.093] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7209, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8092, low 0; forced mean no	2.112 [2.008, 2.218] N 12/12; valid 1	1.132 [1.114, 1.153] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_164_S0_H3_fingerprints_common_late_quarter_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · late quarter · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.363 [2.228, 2.5] N 2/2; valid 1	1.379 [1.31, 1.45] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.164 [2.06, 2.279] N 2/2; valid 1	1.827 [1.787, 1.867] N 2/2; valid 1	count yes contrast yes
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.003 [1, 1.006] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.038 [1, 1.107] N 12/12; valid 1	1.009 [1.003, 1.014] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.107 [1.053, 1.173] N 12/12; valid 1	1.004 [0.9956, 1.012] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.27 [1.203, 1.352] N 12/12; valid 1	0.9951 [0.9836, 1.007] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_165_S0_H3_fingerprints_common_late_quarter_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · late quarter · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	single front; H 3200 qualified yes; same no multi 0, low 0; forced mean no	single front; H 3200 qualified yes; same no multi 0.1828, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.5309, low 0; forced mean no	1.473 [1.403, 1.553] N 12/12; valid 1	0.9857 [0.9721, 0.9981] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.548 [1.522, 1.573] N 12/12; valid 1	1.025 [1.001, 1.048] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5624, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7423, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6064, low 0; forced mean no	1.677 [1.64, 1.716] N 12/12; valid 1	1.023 [0.9881, 1.059] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.743 [1.686, 1.8] N 2/2; valid 1	1.153 [1.086, 1.221] N 2/2; valid 1	count yes contrast yes
Spring f=1	not declared	not declared	not declared	1.752 [1.686, 1.82] N 2/2; valid 1	1.277 [1.102, 1.491] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_166_S0_H3_fingerprints_common_late_quarter_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 1.452 [1.346, 1.561] N 12/12; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.472 [0.4473, 0.4987] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 1.317 [1.203, 1.428] N 12/12; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.4675 [0.45, 0.4849] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 1.191 [1.04, 1.369] N 12/12; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.4325 [0.4136, 0.4551] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 0.5441 [0.4932, 0.6032] N 2/2; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.4432 [0.4413, 0.4451] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 0.03546 [0.02778, 0.04348] N 2/2; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.4206 [0.3952, 0.4478] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / criterion met	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 1.452 [1.346, 1.561] N 12/12; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.472 [0.4473, 0.4987] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 1.317 [1.203, 1.428] N 12/12; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.4675 [0.45, 0.4849] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 1.191 [1.04, 1.369] N 12/12; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.4325 [0.4136, 0.4551] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_167_S0_H3_pairs_common_late_quarter_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 0.5441 [0.4932, 0.6032] N 2/2; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.4432 [0.4413, 0.4451] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 0.03546 [0.02778, 0.04348] N 2/2; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.4206 [0.3952, 0.4478] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 1.452 [1.346, 1.561] N 12/12; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.472 [0.4473, 0.4987] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 1.317 [1.203, 1.428] N 12/12; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.4675 [0.45, 0.4849] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 1.191 [1.04, 1.369] N 12/12; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.4325 [0.4136, 0.4551] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 0.5441 [0.4932, 0.6032] N 2/2; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.4432 [0.4413, 0.4451] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 0.03546 [0.02778, 0.04348] N 2/2; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.4206 [0.3952, 0.4478] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 1.452 [1.346, 1.561] N 12/12; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.472 [0.4473, 0.4987] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_168_S0_H3_pairs_common_late_quarter_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 1.317 [1.203, 1.428] N 12/12; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.4675 [0.45, 0.4849] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 1.191 [1.04, 1.369] N 12/12; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.4325 [0.4136, 0.4551] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 0.5441 [0.4932, 0.6032] N 2/2; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.4432 [0.4413, 0.4451] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 0.03546 [0.02778, 0.04348] N 2/2; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.4206 [0.3952, 0.4478] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: criterion met / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: criterion met / criterion met	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: criterion met / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_169_S0_H3_pairs_common_late_quarter_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_170_S0_H3_pairs_common_late_quarter_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_171_S0_H3_pairs_common_late_quarter_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_172_S0_H3_pairs_common_late_quarter_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_173_S0_H3_pairs_common_late_quarter_inclusive_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_174_S0_H3_pairs_common_late_quarter_inclusive_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_175_S0_H3_pairs_common_late_quarter_inclusive_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_176_S0_H3_pairs_common_late_quarter_inclusive_010.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x2 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0009369 [0, 0.002186] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.01712 [0.007027, 0.0304] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.9892 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.235 [0.1245, 0.3759] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2696, 2696] resolved	[8000, ∞] right censored	S 0.9798 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.6387 [0.5516, 0.7211] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[505.9, 505.9] resolved	[8000, ∞] right censored	S 0.9422 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9548 [0.9448, 0.9647] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[128.3, 128.3] resolved	[8000, ∞] right censored	S 0.9207 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9863 [0.9844, 0.9881] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	inconclusive	inconclusive	[83.27, 83.27] resolved	[8000, ∞] right censored	S 0.9319 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.9606 [0.9582, 0.9631] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_177_S0_H3_candidates_common_cadence_x2_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x2 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	inconclusive	[475.4, 475.4] resolved	[8000, ∞] right censored	S 0.2373 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.07594 [0.06543, 0.08484] S low 0.8565 [0.8397, 0.8749] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[563.6, 563.6] resolved	[8000, ∞] right censored	S 0.131 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.06283 [0.05564, 0.07001] S low 0.9004 [0.8879, 0.9136] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[733.2, 733.2] resolved	[8000, ∞] right censored	S 0.05288 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.0266 [0.02082, 0.03206] S low 0.9579 [0.9503, 0.9669] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1251, 1251] resolved	[8000, ∞] right censored	S 0.01525 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.004997 [0.001874, 0.00812] S low 0.9925 [0.9875, 0.9975] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_178_S0_H3_candidates_common_cadence_x2_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x2 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[8000, ∞] right censored	S 0 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0004164 [0, 0.001145] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.979 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.1969 [0.03435, 0.3848] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3903, 3903] resolved	[8000, ∞] right censored	S 0.9796 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.4837 [0.3914, 0.5751] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2012, 2012] resolved	[8000, ∞] right censored	S 0.985 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.7161 [0.6474, 0.7783] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[986.9, 986.9] resolved	[8000, ∞] right censored	S 0.9749 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.857 [0.832, 0.8798] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[388.5, 388.5] resolved	[8000, ∞] right censored	S 0.9627 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.919 [0.9007, 0.9373] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[104, 104] resolved	[8000, ∞] right censored	S 0.9616 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9335 [0.9332, 0.9338] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_179_S0_H3_candidates_common_cadence_x2_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x2 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	inconclusive	inconclusive	[51.19, 51.19] resolved	[8000, ∞] right censored	S 0.9708 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.852 [0.8463, 0.8576] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.0007287 [5.205e-05, 0.001666] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[533.5, 533.5] resolved	[800, ∞] right censored	S 0.9981 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.02363 [0.006558, 0.04497] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[145.8, 145.8] resolved	[800, ∞] right censored	S 0.9922 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.1366 [0.108, 0.1706] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[37.9, 37.9] resolved	[800, ∞] right censored	S 0.9847 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.3533 [0.3188, 0.3849] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.18, 18.18] resolved	[800, ∞] right censored	S 0.9823 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.4987 [0.4698, 0.5309] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.66, 10.66] resolved	[800, ∞] right censored	S 0.9803 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.658 [0.6436, 0.6719] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.808, 6.808] resolved	[800, ∞] right censored	S 0.9744 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7891 [0.7747, 0.8021] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	inconclusive	inconclusive	[4.953, 4.953] resolved	[800, ∞] right censored	S 0.9749 / W 1 H 3200 / 3200 95% bound check: ?	S multi 0.842 [0.8407, 0.8432] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_180_S0_H3_candidates_common_cadence_x2_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x2 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	inconclusive	inconclusive	[4.101, 4.101] resolved	[800, ∞] right censored	S 0.9887 / W 1 H 3200 / 3200 95% bound check: ?	S multi 0.7633 [0.7395, 0.787] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_181_S0_H3_candidates_common_cadence_x2_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x2 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.011 [1.002, 1.023] N 12/12; valid 1	1.005 [1.004, 1.007] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8682, low 0; forced mean no	1.212 [1.09, 1.34] N 12/12; valid 1	1.009 [1.007, 1.01] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.668 [1.558, 1.771] N 12/12; valid 1	1.009 [1.006, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_182_S0_H3_fingerprints_common_cadence_x2_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x2 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3856, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9124, low 0; forced mean no	2.309 [2.248, 2.367] N 12/12; valid 1	1.024 [1.017, 1.031] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.482 [2.376, 2.593] N 2/2; valid 1	1.124 [1.11, 1.139] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.507 [2.432, 2.583] N 2/2; valid 1	1.25 [1.233, 1.268] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1.008 [1.007, 1.01] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.03 [1.029, 1.031] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.11, 1.121] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9581 [0.936, 0.9784] N 12/12; valid 1	1.484 [1.393, 1.597] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7913 [0.7772, 0.8073] N 12/12; valid 1	3.345 [3.282, 3.41] N 12/12; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_183_S0_H3_fingerprints_common_cadence_x2_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x2 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.01546; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1406; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.7685 [0.7559, 0.7812] N 12/12; valid 1	2.616 [2.548, 2.681] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.03	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check single front; H 32000 multi 0.008224, low 0; forced mean yes	1.454 [1.386, 1.527] N 12/12; valid 1	0.4761 [0.4656, 0.4867] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.05	not declared	not declared	not declared	1.348 [1.277, 1.415] N 12/12; valid 1	0.4608 [0.45, 0.4716] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.1	not declared	not declared	not declared	1.206 [1.133, 1.282] N 12/12; valid 1	0.4371 [0.4272, 0.4482] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.3	not declared	not declared	not declared	0.654 [0.6049, 0.7068] N 2/2; valid 1	0.4247 [0.4159, 0.4336] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=1	not declared	not declared	not declared	0.07067 [0.05517, 0.08696] N 2/2; valid 1	0.4228 [0.3957, 0.4538] N 2/2; valid 1	count unavailable contrast unavailable
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_184_S0_H3_fingerprints_common_cadence_x2_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x2 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.008] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.194 [1.022, 1.375] N 12/12; valid 1	1.013 [1.012, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.509 [1.389, 1.617] N 12/12; valid 1	1.029 [1.026, 1.031] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.01145, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8576, low 0; forced mean no	1.771 [1.657, 1.891] N 12/12; valid 1	1.049 [1.045, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.012 [1.942, 2.085] N 12/12; valid 1	1.076 [1.07, 1.083] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7007, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8082, low 0; forced mean no	2.11 [2.05, 2.176] N 12/12; valid 1	1.134 [1.123, 1.146] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_185_S0_H3_fingerprints_common_cadence_x2_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x2 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.354 [2.32, 2.389] N 2/2; valid 1	1.305 [1.285, 1.325] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.215 [2.078, 2.359] N 2/2; valid 1	1.692 [1.633, 1.755] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.021 [1.004, 1.043] N 12/12; valid 1	1.007 [1.005, 1.009] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.121 [1.094, 1.153] N 12/12; valid 1	1.001 [0.9976, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.298 [1.265, 1.33] N 12/12; valid 1	1 [0.9938, 1.006] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_186_S0_H3_fingerprints_common_cadence_x2_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x2 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1726, low 0; forced mean no	no eligible check multiple fronts; H 3200 multi 0.5119, low 0; forced mean no	1.434 [1.397, 1.473] N 12/12; valid 1	0.9973 [0.9902, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.567 [1.538, 1.592] N 12/12; valid 1	1.012 [1, 1.024] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5596, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7444, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6049, low 0; forced mean no	1.701 [1.68, 1.722] N 12/12; valid 1	1.033 [1.018, 1.05] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.762 [1.704, 1.821] N 2/2; valid 1	1.172 [1.093, 1.26] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=1	not declared	not declared	not declared	1.759 [1.752, 1.766] N 2/2; valid 1	1.25 [1.225, 1.276] N 2/2; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_187_S0_H3_fingerprints_common_cadence_x2_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_188_S0_H3_pairs_common_cadence_x2_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_189_S0_H3_pairs_common_cadence_x2_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_190_S0_H3_pairs_common_cadence_x2_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_191_S0_H3_pairs_common_cadence_x2_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_192_S0_H3_pairs_common_cadence_x2_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_193_S0_H3_pairs_common_cadence_x2_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_194_S0_H3_pairs_common_cadence_x2_primary_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_195_S0_H3_pairs_common_cadence_x2_primary_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_196_S0_H3_pairs_common_cadence_x2_primary_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x2 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0009369 [0, 0.002186] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.01712 [0.007027, 0.0304] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.9892 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.235 [0.1245, 0.3759] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2696, 2696] resolved	[8000, ∞] right censored	S 0.9798 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.6387 [0.5516, 0.7211] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[505.9, 505.9] resolved	[8000, ∞] right censored	S 0.9422 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9548 [0.9448, 0.9647] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[128.3, 128.3] resolved	[8000, ∞] right censored	S 0.9207 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9863 [0.9844, 0.9881] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	criterion met	inconclusive	[83.27, 83.27] resolved	[8000, ∞] right censored	S 0.9319 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9606 [0.9582, 0.9631] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_197_S0_H3_candidates_common_cadence_x2_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x2 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	criterion met	[475.4, 475.4] resolved	[8000, ∞] right censored	S 0.2373 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.07594 [0.06543, 0.08484] S low 0.8565 [0.8397, 0.8749] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[563.6, 563.6] resolved	[8000, ∞] right censored	S 0.131 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.06283 [0.05564, 0.07001] S low 0.9004 [0.8879, 0.9136] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[733.2, 733.2] resolved	[8000, ∞] right censored	S 0.05288 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.0266 [0.02082, 0.03206] S low 0.9579 [0.9503, 0.9669] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1251, 1251] resolved	[8000, ∞] right censored	S 0.01525 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.004997 [0.001874, 0.00812] S low 0.9925 [0.9875, 0.9975] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_198_S0_H3_candidates_common_cadence_x2_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x2 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[8000, ∞] right censored	S 0 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0004164 [0, 0.001145] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.979 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.1969 [0.03435, 0.3848] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3903, 3903] resolved	[8000, ∞] right censored	S 0.9796 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.4837 [0.3914, 0.5751] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2012, 2012] resolved	[8000, ∞] right censored	S 0.985 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.7161 [0.6474, 0.7783] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[986.9, 986.9] resolved	[8000, ∞] right censored	S 0.9749 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.857 [0.832, 0.8798] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[388.5, 388.5] resolved	[8000, ∞] right censored	S 0.9627 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.919 [0.9007, 0.9373] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[104, 104] resolved	[8000, ∞] right censored	S 0.9616 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9335 [0.9332, 0.9338] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_199_S0_H3_candidates_common_cadence_x2_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x2 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	criterion met	inconclusive	[51.19, 51.19] resolved	[8000, ∞] right censored	S 0.9708 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.852 [0.8463, 0.8576] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.0007287 [5.205e-05, 0.001666] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[533.5, 533.5] resolved	[800, ∞] right censored	S 0.9981 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.02363 [0.006558, 0.04497] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[145.8, 145.8] resolved	[800, ∞] right censored	S 0.9922 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.1366 [0.108, 0.1706] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[37.9, 37.9] resolved	[800, ∞] right censored	S 0.9847 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.3533 [0.3188, 0.3849] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.18, 18.18] resolved	[800, ∞] right censored	S 0.9823 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.4987 [0.4698, 0.5309] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.66, 10.66] resolved	[800, ∞] right censored	S 0.9803 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.658 [0.6436, 0.6719] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.808, 6.808] resolved	[800, ∞] right censored	S 0.9744 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7891 [0.7747, 0.8021] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	criterion met	inconclusive	[4.953, 4.953] resolved	[800, ∞] right censored	S 0.9749 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.842 [0.8407, 0.8432] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_200_S0_H3_candidates_common_cadence_x2_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x2 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	criterion met	inconclusive	[4.101, 4.101] resolved	[800, ∞] right censored	S 0.9887 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7633 [0.7395, 0.787] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_201_S0_H3_candidates_common_cadence_x2_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x2 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.011 [1.002, 1.023] N 12/12; valid 1	1.005 [1.004, 1.007] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8682, low 0; forced mean no	1.212 [1.09, 1.34] N 12/12; valid 1	1.009 [1.007, 1.01] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.668 [1.558, 1.771] N 12/12; valid 1	1.009 [1.006, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_202_S0_H3_fingerprints_common_cadence_x2_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x2 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3856, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9124, low 0; forced mean no	2.309 [2.248, 2.367] N 12/12; valid 1	1.024 [1.017, 1.031] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.482 [2.376, 2.593] N 2/2; valid 1	1.124 [1.11, 1.139] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.507 [2.432, 2.583] N 2/2; valid 1	1.25 [1.233, 1.268] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1.008 [1.007, 1.01] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.03 [1.029, 1.031] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.11, 1.121] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9581 [0.936, 0.9784] N 12/12; valid 1	1.484 [1.393, 1.597] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7913 [0.7772, 0.8073] N 12/12; valid 1	3.345 [3.282, 3.41] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_203_S0_H3_fingerprints_common_cadence_x2_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x2 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.01546; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1406; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.7685 [0.7559, 0.7812] N 12/12; valid 1	2.616 [2.548, 2.681] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.03	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	single front; H 32000 qualified yes; same no multi 0.008224, low 0; forced mean yes	1.454 [1.386, 1.527] N 12/12; valid 1	0.4761 [0.4656, 0.4867] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.05	not declared	not declared	not declared	1.348 [1.277, 1.415] N 12/12; valid 1	0.4608 [0.45, 0.4716] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.1	not declared	not declared	not declared	1.206 [1.133, 1.282] N 12/12; valid 1	0.4371 [0.4272, 0.4482] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.3	not declared	not declared	not declared	0.654 [0.6049, 0.7068] N 2/2; valid 1	0.4247 [0.4159, 0.4336] N 2/2; valid 1	count yes contrast yes
Goryachev f=1	not declared	not declared	not declared	0.07067 [0.05517, 0.08696] N 2/2; valid 1	0.4228 [0.3957, 0.4538] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_204_S0_H3_fingerprints_common_cadence_x2_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x2 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.008] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.194 [1.022, 1.375] N 12/12; valid 1	1.013 [1.012, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.509 [1.389, 1.617] N 12/12; valid 1	1.029 [1.026, 1.031] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.01145, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8576, low 0; forced mean no	1.771 [1.657, 1.891] N 12/12; valid 1	1.049 [1.045, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.012 [1.942, 2.085] N 12/12; valid 1	1.076 [1.07, 1.083] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7007, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8082, low 0; forced mean no	2.11 [2.05, 2.176] N 12/12; valid 1	1.134 [1.123, 1.146] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_205_S0_H3_fingerprints_common_cadence_x2_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x2 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.354 [2.32, 2.389] N 2/2; valid 1	1.305 [1.285, 1.325] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.215 [2.078, 2.359] N 2/2; valid 1	1.692 [1.633, 1.755] N 2/2; valid 1	count yes contrast yes
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.021 [1.004, 1.043] N 12/12; valid 1	1.007 [1.005, 1.009] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.121 [1.094, 1.153] N 12/12; valid 1	1.001 [0.9976, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.298 [1.265, 1.33] N 12/12; valid 1	1 [0.9938, 1.006] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_206_S0_H3_fingerprints_common_cadence_x2_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x2 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1726, low 0; forced mean no	no eligible check multiple fronts; H 3200 multi 0.5119, low 0; forced mean no	1.434 [1.397, 1.473] N 12/12; valid 1	0.9973 [0.9902, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.567 [1.538, 1.592] N 12/12; valid 1	1.012 [1, 1.024] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5596, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7444, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6049, low 0; forced mean no	1.701 [1.68, 1.722] N 12/12; valid 1	1.033 [1.018, 1.05] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.762 [1.704, 1.821] N 2/2; valid 1	1.172 [1.093, 1.26] N 2/2; valid 1	count yes contrast yes
Spring f=1	not declared	not declared	not declared	1.759 [1.752, 1.766] N 2/2; valid 1	1.25 [1.225, 1.276] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_207_S0_H3_fingerprints_common_cadence_x2_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / criterion met	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_208_S0_H3_pairs_common_cadence_x2_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_209_S0_H3_pairs_common_cadence_x2_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_210_S0_H3_pairs_common_cadence_x2_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_211_S0_H3_pairs_common_cadence_x2_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_212_S0_H3_pairs_common_cadence_x2_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_213_S0_H3_pairs_common_cadence_x2_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_214_S0_H3_pairs_common_cadence_x2_inclusive_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_215_S0_H3_pairs_common_cadence_x2_inclusive_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_216_S0_H3_pairs_common_cadence_x2_inclusive_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x4 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0009363 [0, 0.002185] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.01737 [0.007176, 0.03059] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.989 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.2348 [0.1243, 0.3759] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2696, 2696] resolved	[8000, ∞] right censored	S 0.9804 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.6395 [0.5528, 0.7213] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[504.7, 504.7] resolved	[8000, ∞] right censored	S 0.9445 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.954 [0.9441, 0.964] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[128.6, 128.6] resolved	[8000, ∞] right censored	S 0.9195 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9863 [0.985, 0.9875] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	inconclusive	inconclusive	[84, 84] resolved	[8000, ∞] right censored	S 0.9341 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.9638 [0.96, 0.9675] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_217_S0_H3_candidates_common_cadence_x4_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x4 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	inconclusive	[478, 478] resolved	[8000, ∞] right censored	S 0.2386 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.07563 [0.06471, 0.0849] S low 0.8562 [0.8387, 0.8753] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[563.3, 563.3] resolved	[8000, ∞] right censored	S 0.13 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.06149 [0.05379, 0.06898] S low 0.9012 [0.8878, 0.915] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[736.8, 736.8] resolved	[8000, ∞] right censored	S 0.05301 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.02643 [0.0205, 0.03215] S low 0.9581 [0.9501, 0.9672] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1264, 1264] resolved	[8000, ∞] right censored	S 0.01493 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.005618 [0.002497, 0.008739] S low 0.9919 [0.9863, 0.9975] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_218_S0_H3_candidates_common_cadence_x4_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x4 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[8000, ∞] right censored	S 0 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0004161 [0, 0.001248] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.9788 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.1964 [0.03391, 0.3844] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3907, 3907] resolved	[8000, ∞] right censored	S 0.9785 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.4842 [0.3916, 0.5754] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2013, 2013] resolved	[8000, ∞] right censored	S 0.9844 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.7165 [0.6475, 0.7785] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[985.7, 985.7] resolved	[8000, ∞] right censored	S 0.975 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.8579 [0.8328, 0.8807] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[390.9, 390.9] resolved	[8000, ∞] right censored	S 0.9621 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9184 [0.9004, 0.9365] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[104.4, 104.4] resolved	[8000, ∞] right censored	S 0.9597 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9332 [0.9313, 0.9351] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_219_S0_H3_candidates_common_cadence_x4_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x4 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	inconclusive	inconclusive	[52.68, 52.68] resolved	[8000, ∞] right censored	S 0.9712 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.8508 [0.8452, 0.8564] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.0006242 [0, 0.001457] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[533.5, 533.5] resolved	[800, ∞] right censored	S 0.998 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.02362 [0.006656, 0.04484] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[145.8, 145.8] resolved	[800, ∞] right censored	S 0.9919 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.1363 [0.1071, 0.1705] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[37.89, 37.89] resolved	[800, ∞] right censored	S 0.9848 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.3548 [0.3199, 0.3867] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.15, 18.15] resolved	[800, ∞] right censored	S 0.983 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.4988 [0.4698, 0.5304] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.72, 10.72] resolved	[800, ∞] right censored	S 0.9789 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.6572 [0.6426, 0.6712] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.891, 6.891] resolved	[800, ∞] right censored	S 0.9746 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7887 [0.7737, 0.8026] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	inconclusive	inconclusive	[5.058, 5.058] resolved	[800, ∞] right censored	S 0.9724 / W 1 H 3200 / 3200 95% bound check: ?	S multi 0.8452 [0.8427, 0.8477] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_220_S0_H3_candidates_common_cadence_x4_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x4 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	inconclusive	inconclusive	[4.068, 4.068] resolved	[800, ∞] right censored	S 0.9862 / W 1 H 3200 / 3200 95% bound check: ?	S multi 0.7665 [0.7403, 0.7928] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_221_S0_H3_candidates_common_cadence_x4_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x4 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.011 [1.002, 1.023] N 12/12; valid 1	1.005 [1.004, 1.007] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8685, low 0; forced mean no	1.212 [1.09, 1.34] N 12/12; valid 1	1.009 [1.007, 1.01] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.668 [1.558, 1.771] N 12/12; valid 1	1.009 [1.006, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_222_S0_H3_fingerprints_common_cadence_x4_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x4 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3855, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9143, low 0; forced mean no	2.309 [2.248, 2.367] N 12/12; valid 1	1.024 [1.017, 1.031] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.482 [2.376, 2.593] N 2/2; valid 1	1.124 [1.11, 1.139] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.507 [2.432, 2.583] N 2/2; valid 1	1.25 [1.233, 1.268] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1.008 [1.007, 1.01] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.03 [1.029, 1.031] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.11, 1.121] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9581 [0.936, 0.9784] N 12/12; valid 1	1.484 [1.393, 1.597] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7913 [0.7772, 0.8073] N 12/12; valid 1	3.345 [3.282, 3.41] N 12/12; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_223_S0_H3_fingerprints_common_cadence_x4_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x4 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.0155; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1407; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.7685 [0.7559, 0.7812] N 12/12; valid 1	2.616 [2.548, 2.681] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.03	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check single front; H 32000 multi 0.008219, low 0; forced mean yes	1.454 [1.386, 1.527] N 12/12; valid 1	0.4761 [0.4656, 0.4867] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.05	not declared	not declared	not declared	1.348 [1.277, 1.415] N 12/12; valid 1	0.4608 [0.45, 0.4716] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.1	not declared	not declared	not declared	1.206 [1.133, 1.282] N 12/12; valid 1	0.4371 [0.4272, 0.4482] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.3	not declared	not declared	not declared	0.654 [0.6049, 0.7068] N 2/2; valid 1	0.4247 [0.4159, 0.4336] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=1	not declared	not declared	not declared	0.07067 [0.05517, 0.08696] N 2/2; valid 1	0.4228 [0.3957, 0.4538] N 2/2; valid 1	count unavailable contrast unavailable
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_224_S0_H3_fingerprints_common_cadence_x4_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x4 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.008] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.194 [1.022, 1.375] N 12/12; valid 1	1.013 [1.012, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.509 [1.389, 1.617] N 12/12; valid 1	1.029 [1.026, 1.031] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.01144, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8553, low 0; forced mean no	1.771 [1.657, 1.891] N 12/12; valid 1	1.049 [1.045, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.012 [1.942, 2.085] N 12/12; valid 1	1.076 [1.07, 1.083] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7007, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8081, low 0; forced mean no	2.11 [2.05, 2.176] N 12/12; valid 1	1.134 [1.123, 1.146] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_225_S0_H3_fingerprints_common_cadence_x4_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x4 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.354 [2.32, 2.389] N 2/2; valid 1	1.305 [1.285, 1.325] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.215 [2.078, 2.359] N 2/2; valid 1	1.692 [1.633, 1.755] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.021 [1.004, 1.043] N 12/12; valid 1	1.007 [1.005, 1.009] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.121 [1.094, 1.153] N 12/12; valid 1	1.001 [0.9976, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.298 [1.265, 1.33] N 12/12; valid 1	1 [0.9938, 1.006] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_226_S0_H3_fingerprints_common_cadence_x4_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x4 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1728, low 0; forced mean no	no eligible check multiple fronts; H 3200 multi 0.5112, low 0; forced mean no	1.434 [1.397, 1.473] N 12/12; valid 1	0.9973 [0.9902, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.567 [1.538, 1.592] N 12/12; valid 1	1.012 [1, 1.024] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5605, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.743, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6073, low 0; forced mean no	1.701 [1.68, 1.722] N 12/12; valid 1	1.033 [1.018, 1.05] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.762 [1.704, 1.821] N 2/2; valid 1	1.172 [1.093, 1.26] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=1	not declared	not declared	not declared	1.759 [1.752, 1.766] N 2/2; valid 1	1.25 [1.225, 1.276] N 2/2; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_227_S0_H3_fingerprints_common_cadence_x4_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_228_S0_H3_pairs_common_cadence_x4_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_229_S0_H3_pairs_common_cadence_x4_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_230_S0_H3_pairs_common_cadence_x4_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_231_S0_H3_pairs_common_cadence_x4_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_232_S0_H3_pairs_common_cadence_x4_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_233_S0_H3_pairs_common_cadence_x4_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_234_S0_H3_pairs_common_cadence_x4_primary_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_235_S0_H3_pairs_common_cadence_x4_primary_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_236_S0_H3_pairs_common_cadence_x4_primary_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x4 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0009363 [0, 0.002185] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.01737 [0.007176, 0.03059] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.989 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.2348 [0.1243, 0.3759] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2696, 2696] resolved	[8000, ∞] right censored	S 0.9804 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.6395 [0.5528, 0.7213] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[504.7, 504.7] resolved	[8000, ∞] right censored	S 0.9445 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.954 [0.9441, 0.964] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[128.6, 128.6] resolved	[8000, ∞] right censored	S 0.9195 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9863 [0.985, 0.9875] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	criterion met	inconclusive	[84, 84] resolved	[8000, ∞] right censored	S 0.9341 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9638 [0.96, 0.9675] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_237_S0_H3_candidates_common_cadence_x4_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x4 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	criterion met	[478, 478] resolved	[8000, ∞] right censored	S 0.2386 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.07563 [0.06471, 0.0849] S low 0.8562 [0.8387, 0.8753] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[563.3, 563.3] resolved	[8000, ∞] right censored	S 0.13 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.06149 [0.05379, 0.06898] S low 0.9012 [0.8878, 0.915] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[736.8, 736.8] resolved	[8000, ∞] right censored	S 0.05301 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.02643 [0.0205, 0.03215] S low 0.9581 [0.9501, 0.9672] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1264, 1264] resolved	[8000, ∞] right censored	S 0.01493 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.005618 [0.002497, 0.008739] S low 0.9919 [0.9863, 0.9975] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_238_S0_H3_candidates_common_cadence_x4_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x4 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[8000, ∞] right censored	S 0 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0004161 [0, 0.001248] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.9788 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.1964 [0.03391, 0.3844] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3907, 3907] resolved	[8000, ∞] right censored	S 0.9785 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.4842 [0.3916, 0.5754] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2013, 2013] resolved	[8000, ∞] right censored	S 0.9844 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.7165 [0.6475, 0.7785] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[985.7, 985.7] resolved	[8000, ∞] right censored	S 0.975 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.8579 [0.8328, 0.8807] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[390.9, 390.9] resolved	[8000, ∞] right censored	S 0.9621 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9184 [0.9004, 0.9365] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[104.4, 104.4] resolved	[8000, ∞] right censored	S 0.9597 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9332 [0.9313, 0.9351] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_239_S0_H3_candidates_common_cadence_x4_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x4 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	criterion met	inconclusive	[52.68, 52.68] resolved	[8000, ∞] right censored	S 0.9712 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.8508 [0.8452, 0.8564] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.0006242 [0, 0.001457] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[533.5, 533.5] resolved	[800, ∞] right censored	S 0.998 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.02362 [0.006656, 0.04484] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[145.8, 145.8] resolved	[800, ∞] right censored	S 0.9919 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.1363 [0.1071, 0.1705] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[37.89, 37.89] resolved	[800, ∞] right censored	S 0.9848 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.3548 [0.3199, 0.3867] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.15, 18.15] resolved	[800, ∞] right censored	S 0.983 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.4988 [0.4698, 0.5304] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.72, 10.72] resolved	[800, ∞] right censored	S 0.9789 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.6572 [0.6426, 0.6712] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.891, 6.891] resolved	[800, ∞] right censored	S 0.9746 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7887 [0.7737, 0.8026] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	criterion met	inconclusive	[5.058, 5.058] resolved	[800, ∞] right censored	S 0.9724 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.8452 [0.8427, 0.8477] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_240_S0_H3_candidates_common_cadence_x4_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x4 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	criterion met	inconclusive	[4.068, 4.068] resolved	[800, ∞] right censored	S 0.9862 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7665 [0.7403, 0.7928] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_241_S0_H3_candidates_common_cadence_x4_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x4 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.011 [1.002, 1.023] N 12/12; valid 1	1.005 [1.004, 1.007] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8685, low 0; forced mean no	1.212 [1.09, 1.34] N 12/12; valid 1	1.009 [1.007, 1.01] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.668 [1.558, 1.771] N 12/12; valid 1	1.009 [1.006, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_242_S0_H3_fingerprints_common_cadence_x4_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x4 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3855, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9143, low 0; forced mean no	2.309 [2.248, 2.367] N 12/12; valid 1	1.024 [1.017, 1.031] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.482 [2.376, 2.593] N 2/2; valid 1	1.124 [1.11, 1.139] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.507 [2.432, 2.583] N 2/2; valid 1	1.25 [1.233, 1.268] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1.008 [1.007, 1.01] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.03 [1.029, 1.031] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.11, 1.121] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9581 [0.936, 0.9784] N 12/12; valid 1	1.484 [1.393, 1.597] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7913 [0.7772, 0.8073] N 12/12; valid 1	3.345 [3.282, 3.41] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_243_S0_H3_fingerprints_common_cadence_x4_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x4 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.0155; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1407; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.7685 [0.7559, 0.7812] N 12/12; valid 1	2.616 [2.548, 2.681] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.03	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	single front; H 32000 qualified yes; same no multi 0.008219, low 0; forced mean yes	1.454 [1.386, 1.527] N 12/12; valid 1	0.4761 [0.4656, 0.4867] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.05	not declared	not declared	not declared	1.348 [1.277, 1.415] N 12/12; valid 1	0.4608 [0.45, 0.4716] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.1	not declared	not declared	not declared	1.206 [1.133, 1.282] N 12/12; valid 1	0.4371 [0.4272, 0.4482] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.3	not declared	not declared	not declared	0.654 [0.6049, 0.7068] N 2/2; valid 1	0.4247 [0.4159, 0.4336] N 2/2; valid 1	count yes contrast yes
Goryachev f=1	not declared	not declared	not declared	0.07067 [0.05517, 0.08696] N 2/2; valid 1	0.4228 [0.3957, 0.4538] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_244_S0_H3_fingerprints_common_cadence_x4_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x4 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.008] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.194 [1.022, 1.375] N 12/12; valid 1	1.013 [1.012, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.509 [1.389, 1.617] N 12/12; valid 1	1.029 [1.026, 1.031] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.01144, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8553, low 0; forced mean no	1.771 [1.657, 1.891] N 12/12; valid 1	1.049 [1.045, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.012 [1.942, 2.085] N 12/12; valid 1	1.076 [1.07, 1.083] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7007, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8081, low 0; forced mean no	2.11 [2.05, 2.176] N 12/12; valid 1	1.134 [1.123, 1.146] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_245_S0_H3_fingerprints_common_cadence_x4_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x4 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.354 [2.32, 2.389] N 2/2; valid 1	1.305 [1.285, 1.325] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.215 [2.078, 2.359] N 2/2; valid 1	1.692 [1.633, 1.755] N 2/2; valid 1	count yes contrast yes
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.021 [1.004, 1.043] N 12/12; valid 1	1.007 [1.005, 1.009] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.121 [1.094, 1.153] N 12/12; valid 1	1.001 [0.9976, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.298 [1.265, 1.33] N 12/12; valid 1	1 [0.9938, 1.006] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_246_S0_H3_fingerprints_common_cadence_x4_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x4 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1728, low 0; forced mean no	no eligible check multiple fronts; H 3200 multi 0.5112, low 0; forced mean no	1.434 [1.397, 1.473] N 12/12; valid 1	0.9973 [0.9902, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.567 [1.538, 1.592] N 12/12; valid 1	1.012 [1, 1.024] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5605, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.743, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6073, low 0; forced mean no	1.701 [1.68, 1.722] N 12/12; valid 1	1.033 [1.018, 1.05] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.762 [1.704, 1.821] N 2/2; valid 1	1.172 [1.093, 1.26] N 2/2; valid 1	count yes contrast yes
Spring f=1	not declared	not declared	not declared	1.759 [1.752, 1.766] N 2/2; valid 1	1.25 [1.225, 1.276] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_247_S0_H3_fingerprints_common_cadence_x4_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / criterion met	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_248_S0_H3_pairs_common_cadence_x4_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_249_S0_H3_pairs_common_cadence_x4_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_250_S0_H3_pairs_common_cadence_x4_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_251_S0_H3_pairs_common_cadence_x4_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_252_S0_H3_pairs_common_cadence_x4_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_253_S0_H3_pairs_common_cadence_x4_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_254_S0_H3_pairs_common_cadence_x4_inclusive_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_255_S0_H3_pairs_common_cadence_x4_inclusive_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_256_S0_H3_pairs_common_cadence_x4_inclusive_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x8 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.001247 [0, 0.002702] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.01808 [0.008105, 0.03096] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.9888 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.2344 [0.1236, 0.3753] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2702, 2702] resolved	[8000, ∞] right censored	S 0.98 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.6394 [0.5521, 0.7211] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[504.9, 504.9] resolved	[8000, ∞] right censored	S 0.9444 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.953 [0.9426, 0.9634] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[130.7, 130.7] resolved	[8000, ∞] right censored	S 0.9131 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.99 [0.985, 0.995] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	inconclusive	inconclusive	[82.42, 82.42] resolved	[8000, ∞] right censored	S 0.9286 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.9663 [0.9626, 0.9701] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_257_S0_H3_candidates_common_cadence_x8_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x8 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	inconclusive	[479.3, 479.3] resolved	[8000, ∞] right censored	S 0.2363 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.07461 [0.06234, 0.08541] S low 0.8585 [0.841, 0.8786] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[556.9, 556.9] resolved	[8000, ∞] right censored	S 0.1307 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.06234 [0.05486, 0.07004] S low 0.9009 [0.8872, 0.9156] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[767.4, 767.4] resolved	[8000, ∞] right censored	S 0.05192 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.02702 [0.02078, 0.03304] S low 0.9572 [0.9493, 0.9663] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1269, 1269] resolved	[8000, ∞] right censored	S 0.01491 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.004988 [0.002494, 0.007481] S low 0.9925 [0.9875, 0.9975] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_258_S0_H3_candidates_common_cadence_x8_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x8 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[8000, ∞] right censored	S 0 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0004156 [0, 0.001247] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.9797 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.1966 [0.0349, 0.3847] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3906, 3906] resolved	[8000, ∞] right censored	S 0.9783 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.4823 [0.3896, 0.5738] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2015, 2015] resolved	[8000, ∞] right censored	S 0.9841 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.718 [0.6498, 0.7791] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[981.7, 981.7] resolved	[8000, ∞] right censored	S 0.9783 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.8603 [0.8356, 0.8832] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[392.1, 392.1] resolved	[8000, ∞] right censored	S 0.9591 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9175 [0.899, 0.936] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[104.9, 104.9] resolved	[8000, ∞] right censored	S 0.9648 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9302 [0.9252, 0.9352] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_259_S0_H3_candidates_common_cadence_x8_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x8 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	inconclusive	inconclusive	[0, 80] unresolved below two cadences	[8000, ∞] right censored	S 0.9724 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.8354 [0.8204, 0.8504] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.0008313 [0, 0.00187] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[534.3, 534.3] resolved	[800, ∞] right censored	S 0.9978 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.02328 [0.006234, 0.04489] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[146.2, 146.2] resolved	[800, ∞] right censored	S 0.9913 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.1357 [0.1062, 0.1704] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[37.83, 37.83] resolved	[800, ∞] right censored	S 0.984 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.3554 [0.3215, 0.3874] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.25, 18.25] resolved	[800, ∞] right censored	S 0.9838 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.4985 [0.4701, 0.5299] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.83, 10.83] resolved	[800, ∞] right censored	S 0.9799 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.6586 [0.6424, 0.6735] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[0, 8] unresolved below two cadences	[800, ∞] right censored	S 0.9745 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.787 [0.7695, 0.802] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	inconclusive	inconclusive	[0, 8] unresolved below two cadences	[800, ∞] right censored	S 0.9599 / W 1 H 3200 / 3200 95% bound check: ?	S multi 0.8628 [0.8454, 0.8803] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_260_S0_H3_candidates_common_cadence_x8_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x8 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	inconclusive	inconclusive	[0, 8] unresolved below two cadences	[800, ∞] right censored	S 0.9925 / W 1 H 3200 / 3200 95% bound check: ?	S multi 0.7793 [0.7456, 0.813] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_261_S0_H3_candidates_common_cadence_x8_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x8 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.011 [1.002, 1.023] N 12/12; valid 1	1.005 [1.004, 1.007] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.868, low 0; forced mean no	1.212 [1.09, 1.34] N 12/12; valid 1	1.009 [1.007, 1.01] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.668 [1.558, 1.771] N 12/12; valid 1	1.009 [1.006, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_262_S0_H3_fingerprints_common_cadence_x8_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x8 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3859, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9092, low 0; forced mean no	2.309 [2.248, 2.367] N 12/12; valid 1	1.024 [1.017, 1.031] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.482 [2.376, 2.593] N 2/2; valid 1	1.124 [1.11, 1.139] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.507 [2.432, 2.583] N 2/2; valid 1	1.25 [1.233, 1.268] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1.008 [1.007, 1.01] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.03 [1.029, 1.031] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.11, 1.121] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9581 [0.936, 0.9784] N 12/12; valid 1	1.484 [1.393, 1.597] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7913 [0.7772, 0.8073] N 12/12; valid 1	3.345 [3.282, 3.41] N 12/12; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_263_S0_H3_fingerprints_common_cadence_x8_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x8 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.01559; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1407; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.7685 [0.7559, 0.7812] N 12/12; valid 1	2.616 [2.548, 2.681] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.03	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check single front; H 32000 multi 0.008105, low 0; forced mean yes	1.454 [1.386, 1.527] N 12/12; valid 1	0.4761 [0.4656, 0.4867] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.05	not declared	not declared	not declared	1.348 [1.277, 1.415] N 12/12; valid 1	0.4608 [0.45, 0.4716] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.1	not declared	not declared	not declared	1.206 [1.133, 1.282] N 12/12; valid 1	0.4371 [0.4272, 0.4482] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.3	not declared	not declared	not declared	0.654 [0.6049, 0.7068] N 2/2; valid 1	0.4247 [0.4159, 0.4336] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=1	not declared	not declared	not declared	0.07067 [0.05517, 0.08696] N 2/2; valid 1	0.4228 [0.3957, 0.4538] N 2/2; valid 1	count unavailable contrast unavailable
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_264_S0_H3_fingerprints_common_cadence_x8_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x8 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.008] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.194 [1.022, 1.375] N 12/12; valid 1	1.013 [1.012, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.509 [1.389, 1.617] N 12/12; valid 1	1.029 [1.026, 1.031] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.01143, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8543, low 0; forced mean no	1.771 [1.657, 1.891] N 12/12; valid 1	1.049 [1.045, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.012 [1.942, 2.085] N 12/12; valid 1	1.076 [1.07, 1.083] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7014, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8103, low 0; forced mean no	2.11 [2.05, 2.176] N 12/12; valid 1	1.134 [1.123, 1.146] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_265_S0_H3_fingerprints_common_cadence_x8_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x8 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.354 [2.32, 2.389] N 2/2; valid 1	1.305 [1.285, 1.325] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.215 [2.078, 2.359] N 2/2; valid 1	1.692 [1.633, 1.755] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.021 [1.004, 1.043] N 12/12; valid 1	1.007 [1.005, 1.009] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.121 [1.094, 1.153] N 12/12; valid 1	1.001 [0.9976, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.298 [1.265, 1.33] N 12/12; valid 1	1 [0.9938, 1.006] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_266_S0_H3_fingerprints_common_cadence_x8_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x8 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1727, low 0; forced mean no	no eligible check multiple fronts; H 3200 multi 0.5106, low 0; forced mean no	1.434 [1.397, 1.473] N 12/12; valid 1	0.9973 [0.9902, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.567 [1.538, 1.592] N 12/12; valid 1	1.012 [1, 1.024] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5576, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7423, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6027, low 0; forced mean no	1.701 [1.68, 1.722] N 12/12; valid 1	1.033 [1.018, 1.05] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.762 [1.704, 1.821] N 2/2; valid 1	1.172 [1.093, 1.26] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=1	not declared	not declared	not declared	1.759 [1.752, 1.766] N 2/2; valid 1	1.25 [1.225, 1.276] N 2/2; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_267_S0_H3_fingerprints_common_cadence_x8_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_268_S0_H3_pairs_common_cadence_x8_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_269_S0_H3_pairs_common_cadence_x8_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_270_S0_H3_pairs_common_cadence_x8_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_271_S0_H3_pairs_common_cadence_x8_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_272_S0_H3_pairs_common_cadence_x8_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_273_S0_H3_pairs_common_cadence_x8_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_274_S0_H3_pairs_common_cadence_x8_primary_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_275_S0_H3_pairs_common_cadence_x8_primary_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_276_S0_H3_pairs_common_cadence_x8_primary_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x8 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.001247 [0, 0.002702] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.01808 [0.008105, 0.03096] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.9888 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.2344 [0.1236, 0.3753] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2702, 2702] resolved	[8000, ∞] right censored	S 0.98 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.6394 [0.5521, 0.7211] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[504.9, 504.9] resolved	[8000, ∞] right censored	S 0.9444 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.953 [0.9426, 0.9634] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[130.7, 130.7] resolved	[8000, ∞] right censored	S 0.9131 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.99 [0.985, 0.995] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	criterion met	inconclusive	[82.42, 82.42] resolved	[8000, ∞] right censored	S 0.9286 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9663 [0.9626, 0.9701] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_277_S0_H3_candidates_common_cadence_x8_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x8 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	criterion met	[479.3, 479.3] resolved	[8000, ∞] right censored	S 0.2363 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.07461 [0.06234, 0.08541] S low 0.8585 [0.841, 0.8786] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[556.9, 556.9] resolved	[8000, ∞] right censored	S 0.1307 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.06234 [0.05486, 0.07004] S low 0.9009 [0.8872, 0.9156] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[767.4, 767.4] resolved	[8000, ∞] right censored	S 0.05192 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.02702 [0.02078, 0.03304] S low 0.9572 [0.9493, 0.9663] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1269, 1269] resolved	[8000, ∞] right censored	S 0.01491 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.004988 [0.002494, 0.007481] S low 0.9925 [0.9875, 0.9975] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_278_S0_H3_candidates_common_cadence_x8_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x8 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[8000, ∞] right censored	S 0 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0004156 [0, 0.001247] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.9797 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.1966 [0.0349, 0.3847] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3906, 3906] resolved	[8000, ∞] right censored	S 0.9783 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.4823 [0.3896, 0.5738] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2015, 2015] resolved	[8000, ∞] right censored	S 0.9841 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.718 [0.6498, 0.7791] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[981.7, 981.7] resolved	[8000, ∞] right censored	S 0.9783 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.8603 [0.8356, 0.8832] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[392.1, 392.1] resolved	[8000, ∞] right censored	S 0.9591 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9175 [0.899, 0.936] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[104.9, 104.9] resolved	[8000, ∞] right censored	S 0.9648 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9302 [0.9252, 0.9352] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_279_S0_H3_candidates_common_cadence_x8_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x8 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	criterion met	inconclusive	[0, 80] unresolved below two cadences	[8000, ∞] right censored	S 0.9724 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.8354 [0.8204, 0.8504] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.0008313 [0, 0.00187] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[534.3, 534.3] resolved	[800, ∞] right censored	S 0.9978 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.02328 [0.006234, 0.04489] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[146.2, 146.2] resolved	[800, ∞] right censored	S 0.9913 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.1357 [0.1062, 0.1704] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[37.83, 37.83] resolved	[800, ∞] right censored	S 0.984 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.3554 [0.3215, 0.3874] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.25, 18.25] resolved	[800, ∞] right censored	S 0.9838 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.4985 [0.4701, 0.5299] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.83, 10.83] resolved	[800, ∞] right censored	S 0.9799 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.6586 [0.6424, 0.6735] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[0, 8] unresolved below two cadences	[800, ∞] right censored	S 0.9745 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.787 [0.7695, 0.802] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	criterion met	inconclusive	[0, 8] unresolved below two cadences	[800, ∞] right censored	S 0.9599 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.8628 [0.8454, 0.8803] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_280_S0_H3_candidates_common_cadence_x8_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x8 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	criterion met	inconclusive	[0, 8] unresolved below two cadences	[800, ∞] right censored	S 0.9925 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7793 [0.7456, 0.813] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_281_S0_H3_candidates_common_cadence_x8_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x8 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.011 [1.002, 1.023] N 12/12; valid 1	1.005 [1.004, 1.007] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.868, low 0; forced mean no	1.212 [1.09, 1.34] N 12/12; valid 1	1.009 [1.007, 1.01] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.668 [1.558, 1.771] N 12/12; valid 1	1.009 [1.006, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_282_S0_H3_fingerprints_common_cadence_x8_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x8 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3859, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9092, low 0; forced mean no	2.309 [2.248, 2.367] N 12/12; valid 1	1.024 [1.017, 1.031] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.482 [2.376, 2.593] N 2/2; valid 1	1.124 [1.11, 1.139] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.507 [2.432, 2.583] N 2/2; valid 1	1.25 [1.233, 1.268] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1.008 [1.007, 1.01] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.03 [1.029, 1.031] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.11, 1.121] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9581 [0.936, 0.9784] N 12/12; valid 1	1.484 [1.393, 1.597] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7913 [0.7772, 0.8073] N 12/12; valid 1	3.345 [3.282, 3.41] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_283_S0_H3_fingerprints_common_cadence_x8_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x8 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.01559; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1407; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.7685 [0.7559, 0.7812] N 12/12; valid 1	2.616 [2.548, 2.681] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.03	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	single front; H 32000 qualified yes; same no multi 0.008105, low 0; forced mean yes	1.454 [1.386, 1.527] N 12/12; valid 1	0.4761 [0.4656, 0.4867] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.05	not declared	not declared	not declared	1.348 [1.277, 1.415] N 12/12; valid 1	0.4608 [0.45, 0.4716] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.1	not declared	not declared	not declared	1.206 [1.133, 1.282] N 12/12; valid 1	0.4371 [0.4272, 0.4482] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.3	not declared	not declared	not declared	0.654 [0.6049, 0.7068] N 2/2; valid 1	0.4247 [0.4159, 0.4336] N 2/2; valid 1	count yes contrast yes
Goryachev f=1	not declared	not declared	not declared	0.07067 [0.05517, 0.08696] N 2/2; valid 1	0.4228 [0.3957, 0.4538] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_284_S0_H3_fingerprints_common_cadence_x8_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x8 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.008] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.194 [1.022, 1.375] N 12/12; valid 1	1.013 [1.012, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.509 [1.389, 1.617] N 12/12; valid 1	1.029 [1.026, 1.031] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.01143, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8543, low 0; forced mean no	1.771 [1.657, 1.891] N 12/12; valid 1	1.049 [1.045, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.012 [1.942, 2.085] N 12/12; valid 1	1.076 [1.07, 1.083] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7014, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8103, low 0; forced mean no	2.11 [2.05, 2.176] N 12/12; valid 1	1.134 [1.123, 1.146] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_285_S0_H3_fingerprints_common_cadence_x8_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x8 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.354 [2.32, 2.389] N 2/2; valid 1	1.305 [1.285, 1.325] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.215 [2.078, 2.359] N 2/2; valid 1	1.692 [1.633, 1.755] N 2/2; valid 1	count yes contrast yes
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.021 [1.004, 1.043] N 12/12; valid 1	1.007 [1.005, 1.009] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.121 [1.094, 1.153] N 12/12; valid 1	1.001 [0.9976, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.298 [1.265, 1.33] N 12/12; valid 1	1 [0.9938, 1.006] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_286_S0_H3_fingerprints_common_cadence_x8_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x8 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1727, low 0; forced mean no	no eligible check multiple fronts; H 3200 multi 0.5106, low 0; forced mean no	1.434 [1.397, 1.473] N 12/12; valid 1	0.9973 [0.9902, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.567 [1.538, 1.592] N 12/12; valid 1	1.012 [1, 1.024] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5576, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7423, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6027, low 0; forced mean no	1.701 [1.68, 1.722] N 12/12; valid 1	1.033 [1.018, 1.05] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.762 [1.704, 1.821] N 2/2; valid 1	1.172 [1.093, 1.26] N 2/2; valid 1	count yes contrast yes
Spring f=1	not declared	not declared	not declared	1.759 [1.752, 1.766] N 2/2; valid 1	1.25 [1.225, 1.276] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_287_S0_H3_fingerprints_common_cadence_x8_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / criterion met	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_288_S0_H3_pairs_common_cadence_x8_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_289_S0_H3_pairs_common_cadence_x8_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_290_S0_H3_pairs_common_cadence_x8_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_291_S0_H3_pairs_common_cadence_x8_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_292_S0_H3_pairs_common_cadence_x8_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_293_S0_H3_pairs_common_cadence_x8_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_294_S0_H3_pairs_common_cadence_x8_inclusive_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_295_S0_H3_pairs_common_cadence_x8_inclusive_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_296_S0_H3_pairs_common_cadence_x8_inclusive_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · main · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.001146 [0.0002929, 0.00222] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.0171 [0.007002, 0.03033] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9893 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.2347 [0.124, 0.3757] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2696, 2696] resolved	[3.2e+04, ∞] right censored	S 0.9795 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.6389 [0.5518, 0.7217] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[505.5, 505.5] resolved	[3.2e+04, ∞] right censored	S 0.9425 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9545 [0.9444, 0.9647] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[127.9, 127.9] resolved	[3.2e+04, ∞] right censored	S 0.9218 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9863 [0.9853, 0.9872] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	inconclusive	inconclusive	[83.26, 83.26] resolved	[3.2e+04, ∞] right censored	S 0.9328 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.9613 [0.9603, 0.9622] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_297_S0_H3_candidates_longest_main_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · main · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9574 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.003893 [0, 0.01091] S low 0.03678 [0, 0.1046] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	inconclusive	[474.5, 474.5] resolved	[3.2e+04, ∞] right censored	S 0.2376 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.07688 [0.06599, 0.08612] S low 0.856 [0.8388, 0.8749] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[564.4, 564.4] resolved	[3.2e+04, ∞] right censored	S 0.1314 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.0624 [0.05545, 0.06938] S low 0.9004 [0.8883, 0.9136] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[734.3, 734.3] resolved	[3.2e+04, ∞] right censored	S 0.05307 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.02684 [0.02106, 0.03265] S low 0.9579 [0.9502, 0.9669] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1257, 1257] resolved	[3.2e+04, ∞] right censored	S 0.01543 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.004842 [0.001874, 0.00781] S low 0.9925 [0.9875, 0.9975] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_298_S0_H3_candidates_longest_main_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · main · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[3.2e+04, ∞] right censored	S 0 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.0001367 [0, 0.0004101] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.0003905 [0, 0.001119] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9789 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.1968 [0.0342, 0.3847] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3901, 3901] resolved	[3.2e+04, ∞] right censored	S 0.9798 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.4833 [0.3906, 0.5751] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2012, 2012] resolved	[3.2e+04, ∞] right censored	S 0.985 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.7162 [0.6475, 0.7784] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[986.5, 986.5] resolved	[3.2e+04, ∞] right censored	S 0.9754 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.8569 [0.8322, 0.8799] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[388.1, 388.1] resolved	[3.2e+04, ∞] right censored	S 0.9626 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9196 [0.901, 0.9378] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[103.8, 103.8] resolved	[3.2e+04, ∞] right censored	S 0.964 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9311 [0.9288, 0.9335] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_299_S0_H3_candidates_longest_main_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · main · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	inconclusive	inconclusive	[51.37, 51.37] resolved	[3.2e+04, ∞] right censored	S 0.9693 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.8552 [0.8485, 0.8619] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[3200, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[3200, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[800, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.000781 [7.81e-05, 0.001744] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	criterion met	inconclusive	[533.4, 533.4] resolved	[3200, ∞] right censored	S 0.9981 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.02353 [0.006378, 0.04493] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[146, 146] resolved	[3200, ∞] right censored	S 0.9921 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.1367 [0.1076, 0.1706] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[37.92, 37.92] resolved	[3200, ∞] right censored	S 0.9845 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.3529 [0.3182, 0.3847] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.18, 18.18] resolved	[3200, ∞] right censored	S 0.9828 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.4992 [0.4707, 0.531] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.65, 10.65] resolved	[3200, ∞] right censored	S 0.9792 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.6586 [0.6433, 0.6732] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.794, 6.794] resolved	[3200, ∞] right censored	S 0.9745 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7892 [0.7759, 0.8008] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	inconclusive	inconclusive	[4.964, 4.964] resolved	[3200, ∞] right censored	S 0.9746 / W 1 H 3200 / 12800 95% bound check: ?	S multi 0.8399 [0.8372, 0.8425] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_300_S0_H3_candidates_longest_main_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · main · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	inconclusive	inconclusive	[4.121, 4.121] resolved	[3200, ∞] right censored	S 0.9903 / W 1 H 3200 / 12800 95% bound check: ?	S multi 0.768 [0.7476, 0.7885] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_301_S0_H3_candidates_longest_main_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · main · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1.001 [1, 1.002] N 12/12; valid 1	1.002 [1.002, 1.003] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.011 [1.002, 1.023] N 12/12; valid 1	1.005 [1.004, 1.007] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8679, low 0; forced mean no	1.212 [1.09, 1.34] N 12/12; valid 1	1.009 [1.007, 1.01] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.668 [1.558, 1.771] N 12/12; valid 1	1.009 [1.006, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_302_S0_H3_fingerprints_longest_main_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · main · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3857, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9133, low 0; forced mean no	2.309 [2.248, 2.367] N 12/12; valid 1	1.024 [1.017, 1.031] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.482 [2.376, 2.593] N 2/2; valid 1	1.124 [1.11, 1.139] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.507 [2.432, 2.583] N 2/2; valid 1	1.25 [1.233, 1.268] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.008, 1.008] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.029 [1.029, 1.03] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.113, 1.117] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9581 [0.936, 0.9784] N 12/12; valid 1	1.484 [1.393, 1.597] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7913 [0.7772, 0.8073] N 12/12; valid 1	3.342 [3.279, 3.408] N 12/12; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_303_S0_H3_fingerprints_longest_main_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · main · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.01544; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1406; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.8003 [0.7627, 0.8608] N 12/12; valid 1	2.556 [2.371, 2.688] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.03	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check single front; H 32000 multi 0.008175, low 0; forced mean yes	1.454 [1.386, 1.527] N 12/12; valid 1	0.4761 [0.4656, 0.4867] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.05	not declared	not declared	not declared	1.348 [1.277, 1.415] N 12/12; valid 1	0.4608 [0.45, 0.4716] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.1	not declared	not declared	not declared	1.206 [1.133, 1.282] N 12/12; valid 1	0.4371 [0.4272, 0.4482] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.3	not declared	not declared	not declared	0.654 [0.6049, 0.7068] N 2/2; valid 1	0.4247 [0.4159, 0.4336] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=1	not declared	not declared	not declared	0.07134 [0.05671, 0.08629] N 12/2; valid 1	0.4328 [0.4281, 0.4376] N 12/2; valid 1	count unavailable contrast unavailable
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_304_S0_H3_fingerprints_longest_main_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · main · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.008] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.194 [1.022, 1.375] N 12/12; valid 1	1.013 [1.012, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.509 [1.389, 1.617] N 12/12; valid 1	1.029 [1.026, 1.031] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.01148, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8567, low 0; forced mean no	1.771 [1.657, 1.891] N 12/12; valid 1	1.049 [1.045, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.012 [1.942, 2.085] N 12/12; valid 1	1.076 [1.07, 1.083] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7005, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8086, low 0; forced mean no	2.11 [2.05, 2.176] N 12/12; valid 1	1.134 [1.123, 1.146] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_305_S0_H3_fingerprints_longest_main_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · main · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.354 [2.32, 2.389] N 2/2; valid 1	1.305 [1.285, 1.325] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.215 [2.078, 2.359] N 2/2; valid 1	1.692 [1.633, 1.755] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.021 [1.004, 1.043] N 12/12; valid 1	1.007 [1.005, 1.009] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.121 [1.094, 1.153] N 12/12; valid 1	1.001 [0.9976, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.298 [1.265, 1.33] N 12/12; valid 1	1 [0.9938, 1.006] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_306_S0_H3_fingerprints_longest_main_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · main · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1727, low 0; forced mean no	no eligible check multiple fronts; H 3200 multi 0.5114, low 0; forced mean no	1.434 [1.397, 1.473] N 12/12; valid 1	0.9973 [0.9902, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.567 [1.538, 1.592] N 12/12; valid 1	1.012 [1, 1.024] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5599, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.744, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6043, low 0; forced mean no	1.701 [1.68, 1.722] N 12/12; valid 1	1.033 [1.018, 1.05] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.762 [1.704, 1.821] N 2/2; valid 1	1.172 [1.093, 1.26] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=1	not declared	not declared	not declared	1.759 [1.752, 1.766] N 2/2; valid 1	1.25 [1.225, 1.276] N 2/2; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_307_S0_H3_fingerprints_longest_main_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_308_S0_H3_pairs_longest_main_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_309_S0_H3_pairs_longest_main_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_310_S0_H3_pairs_longest_main_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_311_S0_H3_pairs_longest_main_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_312_S0_H3_pairs_longest_main_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_313_S0_H3_pairs_longest_main_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_314_S0_H3_pairs_longest_main_primary_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_315_S0_H3_pairs_longest_main_primary_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_316_S0_H3_pairs_longest_main_primary_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · main · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.001146 [0.0002929, 0.00222] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.0171 [0.007002, 0.03033] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9893 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.2347 [0.124, 0.3757] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2696, 2696] resolved	[3.2e+04, ∞] right censored	S 0.9795 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.6389 [0.5518, 0.7217] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[505.5, 505.5] resolved	[3.2e+04, ∞] right censored	S 0.9425 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9545 [0.9444, 0.9647] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[127.9, 127.9] resolved	[3.2e+04, ∞] right censored	S 0.9218 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9863 [0.9853, 0.9872] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	criterion met	inconclusive	[83.26, 83.26] resolved	[3.2e+04, ∞] right censored	S 0.9328 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9613 [0.9603, 0.9622] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_317_S0_H3_candidates_longest_main_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · main · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9574 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.003893 [0, 0.01091] S low 0.03678 [0, 0.1046] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	criterion met	[474.5, 474.5] resolved	[3.2e+04, ∞] right censored	S 0.2376 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.07688 [0.06599, 0.08612] S low 0.856 [0.8388, 0.8749] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[564.4, 564.4] resolved	[3.2e+04, ∞] right censored	S 0.1314 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.0624 [0.05545, 0.06938] S low 0.9004 [0.8883, 0.9136] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[734.3, 734.3] resolved	[3.2e+04, ∞] right censored	S 0.05307 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.02684 [0.02106, 0.03265] S low 0.9579 [0.9502, 0.9669] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1257, 1257] resolved	[3.2e+04, ∞] right censored	S 0.01543 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.004842 [0.001874, 0.00781] S low 0.9925 [0.9875, 0.9975] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_318_S0_H3_candidates_longest_main_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · main · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[3.2e+04, ∞] right censored	S 0 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.0001367 [0, 0.0004101] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.0003905 [0, 0.001119] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9789 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.1968 [0.0342, 0.3847] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3901, 3901] resolved	[3.2e+04, ∞] right censored	S 0.9798 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.4833 [0.3906, 0.5751] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2012, 2012] resolved	[3.2e+04, ∞] right censored	S 0.985 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.7162 [0.6475, 0.7784] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[986.5, 986.5] resolved	[3.2e+04, ∞] right censored	S 0.9754 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.8569 [0.8322, 0.8799] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[388.1, 388.1] resolved	[3.2e+04, ∞] right censored	S 0.9626 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9196 [0.901, 0.9378] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[103.8, 103.8] resolved	[3.2e+04, ∞] right censored	S 0.964 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9311 [0.9288, 0.9335] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_319_S0_H3_candidates_longest_main_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · main · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	criterion met	inconclusive	[51.37, 51.37] resolved	[3.2e+04, ∞] right censored	S 0.9693 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.8552 [0.8485, 0.8619] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[3200, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[3200, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[800, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.000781 [7.81e-05, 0.001744] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	criterion met	inconclusive	[533.4, 533.4] resolved	[3200, ∞] right censored	S 0.9981 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.02353 [0.006378, 0.04493] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[146, 146] resolved	[3200, ∞] right censored	S 0.9921 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.1367 [0.1076, 0.1706] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[37.92, 37.92] resolved	[3200, ∞] right censored	S 0.9845 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.3529 [0.3182, 0.3847] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.18, 18.18] resolved	[3200, ∞] right censored	S 0.9828 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.4992 [0.4707, 0.531] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.65, 10.65] resolved	[3200, ∞] right censored	S 0.9792 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.6586 [0.6433, 0.6732] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.794, 6.794] resolved	[3200, ∞] right censored	S 0.9745 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7892 [0.7759, 0.8008] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	criterion met	inconclusive	[4.964, 4.964] resolved	[3200, ∞] right censored	S 0.9746 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.8399 [0.8372, 0.8425] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_320_S0_H3_candidates_longest_main_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · main · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	criterion met	inconclusive	[4.121, 4.121] resolved	[3200, ∞] right censored	S 0.9903 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.768 [0.7476, 0.7885] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_321_S0_H3_candidates_longest_main_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · main · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1.001 [1, 1.002] N 12/12; valid 1	1.002 [1.002, 1.003] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.011 [1.002, 1.023] N 12/12; valid 1	1.005 [1.004, 1.007] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8679, low 0; forced mean no	1.212 [1.09, 1.34] N 12/12; valid 1	1.009 [1.007, 1.01] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.668 [1.558, 1.771] N 12/12; valid 1	1.009 [1.006, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_322_S0_H3_fingerprints_longest_main_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · main · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3857, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9133, low 0; forced mean no	2.309 [2.248, 2.367] N 12/12; valid 1	1.024 [1.017, 1.031] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.482 [2.376, 2.593] N 2/2; valid 1	1.124 [1.11, 1.139] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.507 [2.432, 2.583] N 2/2; valid 1	1.25 [1.233, 1.268] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.008, 1.008] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.029 [1.029, 1.03] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.113, 1.117] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9581 [0.936, 0.9784] N 12/12; valid 1	1.484 [1.393, 1.597] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7913 [0.7772, 0.8073] N 12/12; valid 1	3.342 [3.279, 3.408] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_323_S0_H3_fingerprints_longest_main_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · main · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.01544; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1406; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.8003 [0.7627, 0.8608] N 12/12; valid 1	2.556 [2.371, 2.688] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.03	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	single front; H 32000 qualified yes; same no multi 0.008175, low 0; forced mean yes	1.454 [1.386, 1.527] N 12/12; valid 1	0.4761 [0.4656, 0.4867] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.05	not declared	not declared	not declared	1.348 [1.277, 1.415] N 12/12; valid 1	0.4608 [0.45, 0.4716] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.1	not declared	not declared	not declared	1.206 [1.133, 1.282] N 12/12; valid 1	0.4371 [0.4272, 0.4482] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.3	not declared	not declared	not declared	0.654 [0.6049, 0.7068] N 2/2; valid 1	0.4247 [0.4159, 0.4336] N 2/2; valid 1	count yes contrast yes
Goryachev f=1	not declared	not declared	not declared	0.07134 [0.05671, 0.08629] N 12/2; valid 1	0.4328 [0.4281, 0.4376] N 12/2; valid 1	count yes contrast yes
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_324_S0_H3_fingerprints_longest_main_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · main · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.008] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.194 [1.022, 1.375] N 12/12; valid 1	1.013 [1.012, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.509 [1.389, 1.617] N 12/12; valid 1	1.029 [1.026, 1.031] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.01148, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8567, low 0; forced mean no	1.771 [1.657, 1.891] N 12/12; valid 1	1.049 [1.045, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.012 [1.942, 2.085] N 12/12; valid 1	1.076 [1.07, 1.083] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7005, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8086, low 0; forced mean no	2.11 [2.05, 2.176] N 12/12; valid 1	1.134 [1.123, 1.146] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_325_S0_H3_fingerprints_longest_main_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · main · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.354 [2.32, 2.389] N 2/2; valid 1	1.305 [1.285, 1.325] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.215 [2.078, 2.359] N 2/2; valid 1	1.692 [1.633, 1.755] N 2/2; valid 1	count yes contrast yes
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.021 [1.004, 1.043] N 12/12; valid 1	1.007 [1.005, 1.009] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.121 [1.094, 1.153] N 12/12; valid 1	1.001 [0.9976, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.298 [1.265, 1.33] N 12/12; valid 1	1 [0.9938, 1.006] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_326_S0_H3_fingerprints_longest_main_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · main · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1727, low 0; forced mean no	no eligible check multiple fronts; H 3200 multi 0.5114, low 0; forced mean no	1.434 [1.397, 1.473] N 12/12; valid 1	0.9973 [0.9902, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.567 [1.538, 1.592] N 12/12; valid 1	1.012 [1, 1.024] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5599, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.744, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6043, low 0; forced mean no	1.701 [1.68, 1.722] N 12/12; valid 1	1.033 [1.018, 1.05] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.762 [1.704, 1.821] N 2/2; valid 1	1.172 [1.093, 1.26] N 2/2; valid 1	count yes contrast yes
Spring f=1	not declared	not declared	not declared	1.759 [1.752, 1.766] N 2/2; valid 1	1.25 [1.225, 1.276] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_327_S0_H3_fingerprints_longest_main_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / criterion met	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_328_S0_H3_pairs_longest_main_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_329_S0_H3_pairs_longest_main_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_330_S0_H3_pairs_longest_main_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_331_S0_H3_pairs_longest_main_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_332_S0_H3_pairs_longest_main_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_333_S0_H3_pairs_longest_main_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_334_S0_H3_pairs_longest_main_inclusive_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_335_S0_H3_pairs_longest_main_inclusive_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_336_S0_H3_pairs_longest_main_inclusive_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis first half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.001146 [2.604e-05, 0.002956] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.02441 [0.006141, 0.0519] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 0.9964 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.2057 [0.0673, 0.3891] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2704, 2704] resolved	[1.6e+04, ∞] right censored	S 0.9964 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.6192 [0.5258, 0.7165] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[487, 487] resolved	[1.6e+04, ∞] right censored	S 0.9483 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9579 [0.946, 0.9681] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[132.3, 132.3] resolved	[1.6e+04, ∞] right censored	S 0.9308 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.985 [0.985, 0.985] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	inconclusive	inconclusive	[77.8, 77.8] resolved	[1.6e+04, ∞] right censored	S 0.9445 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.9622 [0.9594, 0.965] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_337_S0_H3_candidates_longest_analysis_first_half_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis first half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	inconclusive	[540.8, 540.8] resolved	[1.6e+04, ∞] right censored	S 0.2132 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.06949 [0.05278, 0.08584] S low 0.8661 [0.844, 0.8911] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[606, 606] resolved	[1.6e+04, ∞] right censored	S 0.1347 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.05569 [0.04679, 0.06673] S low 0.9029 [0.8896, 0.9143] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[748.5, 748.5] resolved	[1.6e+04, ∞] right censored	S 0.05051 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.0266 [0.01764, 0.03618] S low 0.9619 [0.9517, 0.9717] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[0, ∞] unavailable lag gap	[1.6e+04, ∞] right censored	S 0.0003448 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.00812 [0.003748, 0.01249] S low 0.9866 [0.9781, 0.995] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_338_S0_H3_candidates_longest_analysis_first_half_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis first half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[1.6e+04, ∞] right censored	S 0 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.0002734 [0, 0.0008202] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 0.9932 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.1835 [0.03965, 0.375] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3999, 3999] resolved	[1.6e+04, ∞] right censored	S 0.998 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.367 [0.2444, 0.4962] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[1827, 1827] resolved	[1.6e+04, ∞] right censored	S 0.9906 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.7034 [0.6057, 0.7941] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[975.5, 975.5] resolved	[1.6e+04, ∞] right censored	S 0.9642 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.8384 [0.7979, 0.8804] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[397.5, 397.5] resolved	[1.6e+04, ∞] right censored	S 0.9622 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9261 [0.9032, 0.9492] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[98.02, 98.02] resolved	[1.6e+04, ∞] right censored	S 0.9469 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9341 [0.9288, 0.9394] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_339_S0_H3_candidates_longest_analysis_first_half_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis first half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	inconclusive	inconclusive	[49.28, 49.28] resolved	[1.6e+04, ∞] right censored	S 0.9661 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.8695 [0.8682, 0.8707] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[1600, ∞] right censored	[1600, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[1600, ∞] right censored	[1600, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[400, ∞] right censored	[1600, ∞] right censored	S 1 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.0009369 [0, 0.002655] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	criterion met	inconclusive	[347.3, 347.3] resolved	[1600, ∞] right censored	S 1 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.02056 [0.002342, 0.05127] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[134.2, 134.2] resolved	[1600, ∞] right censored	S 0.9921 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.1433 [0.09447, 0.193] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[34.54, 34.54] resolved	[1600, ∞] right censored	S 0.9857 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.3674 [0.3309, 0.4097] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.22, 18.22] resolved	[1600, ∞] right censored	S 0.9847 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.4844 [0.4473, 0.5205] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.31, 10.31] resolved	[1600, ∞] right censored	S 0.9774 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.6649 [0.6391, 0.6876] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.701, 6.701] resolved	[1600, ∞] right censored	S 0.975 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7977 [0.7819, 0.814] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	inconclusive	inconclusive	[4.789, 4.789] resolved	[1600, ∞] right censored	S 0.9742 / W 1 H 3200 / 12800 95% bound check: ?	S multi 0.8267 [0.8226, 0.8307] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_340_S0_H3_candidates_longest_analysis_first_half_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis first half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	inconclusive	inconclusive	[4.337, 4.337] resolved	[1600, ∞] right censored	S 0.9893 / W 1 H 3200 / 12800 95% bound check: ?	S multi 0.7561 [0.727, 0.7851] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_341_S0_H3_candidates_longest_analysis_first_half_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis first half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1.001 [1, 1.002] N 12/12; valid 1	1.002 [1.001, 1.003] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.018 [1.002, 1.042] N 12/12; valid 1	1.004 [1.003, 1.006] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8815, low 0; forced mean no	1.178 [1.031, 1.372] N 12/12; valid 1	1.007 [1.003, 1.011] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.651 [1.505, 1.778] N 12/12; valid 1	1.012 [1.007, 1.017] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_342_S0_H3_fingerprints_longest_analysis_first_half_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis first half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3779, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9141, low 0; forced mean no	2.309 [2.253, 2.364] N 12/12; valid 1	1.023 [1.014, 1.033] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.506 [2.388, 2.633] N 2/2; valid 1	1.112 [1.078, 1.147] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.459 [2.368, 2.554] N 2/2; valid 1	1.286 [1.259, 1.314] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.008, 1.009] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.029 [1.029, 1.03] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.116 [1.113, 1.119] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9412 [0.9057, 0.9772] N 12/12; valid 1	1.593 [1.418, 1.838] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7942 [0.7712, 0.8197] N 12/12; valid 1	3.39 [3.312, 3.459] N 12/12; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_343_S0_H3_fingerprints_longest_analysis_first_half_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis first half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0.08333; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.7682 [0.7542, 0.7843] N 12/12; valid 1	2.646 [2.553, 2.746] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.03	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check single front; H 32000 multi 0.01228, low 0; forced mean yes	1.429 [1.342, 1.51] N 12/12; valid 1	0.4705 [0.4524, 0.4893] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.05	not declared	not declared	not declared	1.381 [1.271, 1.485] N 12/12; valid 1	0.4573 [0.4425, 0.4726] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.1	not declared	not declared	not declared	1.219 [1.139, 1.304] N 12/12; valid 1	0.4295 [0.4176, 0.4423] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.3	not declared	not declared	not declared	0.6434 [0.6143, 0.6742] N 2/2; valid 1	0.4216 [0.4195, 0.4237] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=1	not declared	not declared	not declared	0.04648 [0, 0.09419] N 12/2; valid 1	0.4256 [0.4173, 0.4334] N 12/2; valid 1	count unavailable contrast unavailable
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_344_S0_H3_fingerprints_longest_analysis_first_half_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis first half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.009] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.179 [1.032, 1.363] N 12/12; valid 1	1.012 [1.011, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.351 [1.211, 1.499] N 12/12; valid 1	1.029 [1.025, 1.033] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8559, low 0; forced mean no	1.771 [1.622, 1.937] N 12/12; valid 1	1.049 [1.044, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.019 [1.911, 2.12] N 12/12; valid 1	1.073 [1.066, 1.08] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7088, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8021, low 0; forced mean no	2.151 [2.065, 2.239] N 12/12; valid 1	1.135 [1.119, 1.15] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_345_S0_H3_fingerprints_longest_analysis_first_half_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis first half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.496 [2.397, 2.6] N 2/2; valid 1	1.265 [1.254, 1.276] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.185 [2.056, 2.315] N 2/2; valid 1	1.6 [1.54, 1.663] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1.001 [1, 1.003] N 12/12; valid 1	1.002 [1.001, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.019 [1.002, 1.049] N 12/12; valid 1	1.005 [1.002, 1.007] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.128 [1.082, 1.177] N 12/12; valid 1	1.001 [0.995, 1.006] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.312 [1.274, 1.352] N 12/12; valid 1	1 [0.9924, 1.009] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_346_S0_H3_fingerprints_longest_analysis_first_half_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis first half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1632, low 0; forced mean no	no eligible check single front; H 3200 multi 0.4919, low 0; forced mean no	1.418 [1.371, 1.465] N 12/12; valid 1	1.003 [0.9936, 1.013] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.569 [1.521, 1.612] N 12/12; valid 1	1.006 [0.989, 1.022] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5564, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7449, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.5985, low 0; forced mean no	1.7 [1.675, 1.726] N 12/12; valid 1	1.039 [1.017, 1.061] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.738 [1.693, 1.782] N 2/2; valid 1	1.17 [1.097, 1.253] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=1	not declared	not declared	not declared	1.723 [1.693, 1.752] N 2/2; valid 1	1.271 [1.236, 1.306] N 2/2; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_347_S0_H3_fingerprints_longest_analysis_first_half_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 1.429 [1.342, 1.51] N 12/12; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4705 [0.4524, 0.4893] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 1.381 [1.271, 1.485] N 12/12; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4573 [0.4425, 0.4726] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 1.219 [1.139, 1.304] N 12/12; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4295 [0.4176, 0.4423] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 0.6434 [0.6143, 0.6742] N 2/2; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4216 [0.4195, 0.4237] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 0.04648 [0, 0.09419] N 12/2; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4256 [0.4173, 0.4334] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 1.429 [1.342, 1.51] N 12/12; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4705 [0.4524, 0.4893] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 1.381 [1.271, 1.485] N 12/12; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4573 [0.4425, 0.4726] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 1.219 [1.139, 1.304] N 12/12; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4295 [0.4176, 0.4423] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_348_S0_H3_pairs_longest_analysis_first_half_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 0.6434 [0.6143, 0.6742] N 2/2; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4216 [0.4195, 0.4237] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 0.04648 [0, 0.09419] N 12/2; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4256 [0.4173, 0.4334] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 1.429 [1.342, 1.51] N 12/12; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4705 [0.4524, 0.4893] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 1.381 [1.271, 1.485] N 12/12; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4573 [0.4425, 0.4726] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 1.219 [1.139, 1.304] N 12/12; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4295 [0.4176, 0.4423] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 0.6434 [0.6143, 0.6742] N 2/2; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4216 [0.4195, 0.4237] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 0.04648 [0, 0.09419] N 12/2; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4256 [0.4173, 0.4334] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 1.429 [1.342, 1.51] N 12/12; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4705 [0.4524, 0.4893] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_349_S0_H3_pairs_longest_analysis_first_half_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 1.381 [1.271, 1.485] N 12/12; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4573 [0.4425, 0.4726] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 1.219 [1.139, 1.304] N 12/12; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4295 [0.4176, 0.4423] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 0.6434 [0.6143, 0.6742] N 2/2; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4216 [0.4195, 0.4237] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 0.04648 [0, 0.09419] N 12/2; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4256 [0.4173, 0.4334] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_350_S0_H3_pairs_longest_analysis_first_half_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_351_S0_H3_pairs_longest_analysis_first_half_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_352_S0_H3_pairs_longest_analysis_first_half_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_353_S0_H3_pairs_longest_analysis_first_half_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_354_S0_H3_pairs_longest_analysis_first_half_primary_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_355_S0_H3_pairs_longest_analysis_first_half_primary_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_356_S0_H3_pairs_longest_analysis_first_half_primary_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis first half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.001146 [2.604e-05, 0.002956] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.02441 [0.006141, 0.0519] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 0.9964 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.2057 [0.0673, 0.3891] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2704, 2704] resolved	[1.6e+04, ∞] right censored	S 0.9964 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.6192 [0.5258, 0.7165] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[487, 487] resolved	[1.6e+04, ∞] right censored	S 0.9483 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9579 [0.946, 0.9681] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[132.3, 132.3] resolved	[1.6e+04, ∞] right censored	S 0.9308 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.985 [0.985, 0.985] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	criterion met	inconclusive	[77.8, 77.8] resolved	[1.6e+04, ∞] right censored	S 0.9445 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9622 [0.9594, 0.965] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_357_S0_H3_candidates_longest_analysis_first_half_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis first half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	criterion met	[540.8, 540.8] resolved	[1.6e+04, ∞] right censored	S 0.2132 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.06949 [0.05278, 0.08584] S low 0.8661 [0.844, 0.8911] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[606, 606] resolved	[1.6e+04, ∞] right censored	S 0.1347 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.05569 [0.04679, 0.06673] S low 0.9029 [0.8896, 0.9143] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[748.5, 748.5] resolved	[1.6e+04, ∞] right censored	S 0.05051 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.0266 [0.01764, 0.03618] S low 0.9619 [0.9517, 0.9717] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[0, ∞] unavailable lag gap	[1.6e+04, ∞] right censored	S 0.0003448 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.00812 [0.003748, 0.01249] S low 0.9866 [0.9781, 0.995] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_358_S0_H3_candidates_longest_analysis_first_half_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis first half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[1.6e+04, ∞] right censored	S 0 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.0002734 [0, 0.0008202] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 0.9932 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.1835 [0.03965, 0.375] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3999, 3999] resolved	[1.6e+04, ∞] right censored	S 0.998 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.367 [0.2444, 0.4962] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[1827, 1827] resolved	[1.6e+04, ∞] right censored	S 0.9906 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.7034 [0.6057, 0.7941] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[975.5, 975.5] resolved	[1.6e+04, ∞] right censored	S 0.9642 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.8384 [0.7979, 0.8804] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[397.5, 397.5] resolved	[1.6e+04, ∞] right censored	S 0.9622 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9261 [0.9032, 0.9492] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[98.02, 98.02] resolved	[1.6e+04, ∞] right censored	S 0.9469 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9341 [0.9288, 0.9394] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_359_S0_H3_candidates_longest_analysis_first_half_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis first half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	criterion met	inconclusive	[49.28, 49.28] resolved	[1.6e+04, ∞] right censored	S 0.9661 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.8695 [0.8682, 0.8707] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[1600, ∞] right censored	[1600, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[1600, ∞] right censored	[1600, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[400, ∞] right censored	[1600, ∞] right censored	S 1 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.0009369 [0, 0.002655] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	criterion met	inconclusive	[347.3, 347.3] resolved	[1600, ∞] right censored	S 1 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.02056 [0.002342, 0.05127] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[134.2, 134.2] resolved	[1600, ∞] right censored	S 0.9921 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.1433 [0.09447, 0.193] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[34.54, 34.54] resolved	[1600, ∞] right censored	S 0.9857 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.3674 [0.3309, 0.4097] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.22, 18.22] resolved	[1600, ∞] right censored	S 0.9847 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.4844 [0.4473, 0.5205] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.31, 10.31] resolved	[1600, ∞] right censored	S 0.9774 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.6649 [0.6391, 0.6876] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.701, 6.701] resolved	[1600, ∞] right censored	S 0.975 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7977 [0.7819, 0.814] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	criterion met	inconclusive	[4.789, 4.789] resolved	[1600, ∞] right censored	S 0.9742 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.8267 [0.8226, 0.8307] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_360_S0_H3_candidates_longest_analysis_first_half_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis first half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	criterion met	inconclusive	[4.337, 4.337] resolved	[1600, ∞] right censored	S 0.9893 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7561 [0.727, 0.7851] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_361_S0_H3_candidates_longest_analysis_first_half_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis first half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1.001 [1, 1.002] N 12/12; valid 1	1.002 [1.001, 1.003] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.018 [1.002, 1.042] N 12/12; valid 1	1.004 [1.003, 1.006] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8815, low 0; forced mean no	1.178 [1.031, 1.372] N 12/12; valid 1	1.007 [1.003, 1.011] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.651 [1.505, 1.778] N 12/12; valid 1	1.012 [1.007, 1.017] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_362_S0_H3_fingerprints_longest_analysis_first_half_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis first half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3779, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9141, low 0; forced mean no	2.309 [2.253, 2.364] N 12/12; valid 1	1.023 [1.014, 1.033] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.506 [2.388, 2.633] N 2/2; valid 1	1.112 [1.078, 1.147] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.459 [2.368, 2.554] N 2/2; valid 1	1.286 [1.259, 1.314] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.008, 1.009] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.029 [1.029, 1.03] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.116 [1.113, 1.119] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9412 [0.9057, 0.9772] N 12/12; valid 1	1.593 [1.418, 1.838] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7942 [0.7712, 0.8197] N 12/12; valid 1	3.39 [3.312, 3.459] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_363_S0_H3_fingerprints_longest_analysis_first_half_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis first half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0.08333; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.7682 [0.7542, 0.7843] N 12/12; valid 1	2.646 [2.553, 2.746] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.03	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	single front; H 32000 qualified yes; same no multi 0.01228, low 0; forced mean yes	1.429 [1.342, 1.51] N 12/12; valid 1	0.4705 [0.4524, 0.4893] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.05	not declared	not declared	not declared	1.381 [1.271, 1.485] N 12/12; valid 1	0.4573 [0.4425, 0.4726] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.1	not declared	not declared	not declared	1.219 [1.139, 1.304] N 12/12; valid 1	0.4295 [0.4176, 0.4423] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.3	not declared	not declared	not declared	0.6434 [0.6143, 0.6742] N 2/2; valid 1	0.4216 [0.4195, 0.4237] N 2/2; valid 1	count yes contrast yes
Goryachev f=1	not declared	not declared	not declared	0.04648 [0, 0.09419] N 12/2; valid 1	0.4256 [0.4173, 0.4334] N 12/2; valid 1	count yes contrast yes
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_364_S0_H3_fingerprints_longest_analysis_first_half_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis first half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.009] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.179 [1.032, 1.363] N 12/12; valid 1	1.012 [1.011, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.351 [1.211, 1.499] N 12/12; valid 1	1.029 [1.025, 1.033] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8559, low 0; forced mean no	1.771 [1.622, 1.937] N 12/12; valid 1	1.049 [1.044, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.019 [1.911, 2.12] N 12/12; valid 1	1.073 [1.066, 1.08] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7088, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8021, low 0; forced mean no	2.151 [2.065, 2.239] N 12/12; valid 1	1.135 [1.119, 1.15] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_365_S0_H3_fingerprints_longest_analysis_first_half_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis first half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.496 [2.397, 2.6] N 2/2; valid 1	1.265 [1.254, 1.276] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.185 [2.056, 2.315] N 2/2; valid 1	1.6 [1.54, 1.663] N 2/2; valid 1	count yes contrast yes
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1.001 [1, 1.003] N 12/12; valid 1	1.002 [1.001, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.019 [1.002, 1.049] N 12/12; valid 1	1.005 [1.002, 1.007] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.128 [1.082, 1.177] N 12/12; valid 1	1.001 [0.995, 1.006] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.312 [1.274, 1.352] N 12/12; valid 1	1 [0.9924, 1.009] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_366_S0_H3_fingerprints_longest_analysis_first_half_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis first half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1632, low 0; forced mean no	no eligible check single front; H 3200 multi 0.4919, low 0; forced mean no	1.418 [1.371, 1.465] N 12/12; valid 1	1.003 [0.9936, 1.013] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.569 [1.521, 1.612] N 12/12; valid 1	1.006 [0.989, 1.022] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5564, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7449, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.5985, low 0; forced mean no	1.7 [1.675, 1.726] N 12/12; valid 1	1.039 [1.017, 1.061] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.738 [1.693, 1.782] N 2/2; valid 1	1.17 [1.097, 1.253] N 2/2; valid 1	count yes contrast yes
Spring f=1	not declared	not declared	not declared	1.723 [1.693, 1.752] N 2/2; valid 1	1.271 [1.236, 1.306] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_367_S0_H3_fingerprints_longest_analysis_first_half_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 1.429 [1.342, 1.51] N 12/12; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4705 [0.4524, 0.4893] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 1.381 [1.271, 1.485] N 12/12; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4573 [0.4425, 0.4726] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 1.219 [1.139, 1.304] N 12/12; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4295 [0.4176, 0.4423] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 0.6434 [0.6143, 0.6742] N 2/2; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4216 [0.4195, 0.4237] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 0.04648 [0, 0.09419] N 12/2; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4256 [0.4173, 0.4334] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / criterion met	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 1.429 [1.342, 1.51] N 12/12; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4705 [0.4524, 0.4893] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 1.381 [1.271, 1.485] N 12/12; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4573 [0.4425, 0.4726] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 1.219 [1.139, 1.304] N 12/12; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4295 [0.4176, 0.4423] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_368_S0_H3_pairs_longest_analysis_first_half_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 0.6434 [0.6143, 0.6742] N 2/2; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4216 [0.4195, 0.4237] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 0.04648 [0, 0.09419] N 12/2; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4256 [0.4173, 0.4334] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 1.429 [1.342, 1.51] N 12/12; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4705 [0.4524, 0.4893] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 1.381 [1.271, 1.485] N 12/12; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4573 [0.4425, 0.4726] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 1.219 [1.139, 1.304] N 12/12; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4295 [0.4176, 0.4423] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 0.6434 [0.6143, 0.6742] N 2/2; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4216 [0.4195, 0.4237] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 0.04648 [0, 0.09419] N 12/2; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4256 [0.4173, 0.4334] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 1.429 [1.342, 1.51] N 12/12; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4705 [0.4524, 0.4893] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_369_S0_H3_pairs_longest_analysis_first_half_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 1.381 [1.271, 1.485] N 12/12; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4573 [0.4425, 0.4726] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 1.219 [1.139, 1.304] N 12/12; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4295 [0.4176, 0.4423] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 0.6434 [0.6143, 0.6742] N 2/2; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4216 [0.4195, 0.4237] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 0.04648 [0, 0.09419] N 12/2; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4256 [0.4173, 0.4334] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_370_S0_H3_pairs_longest_analysis_first_half_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_371_S0_H3_pairs_longest_analysis_first_half_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_372_S0_H3_pairs_longest_analysis_first_half_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_373_S0_H3_pairs_longest_analysis_first_half_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_374_S0_H3_pairs_longest_analysis_first_half_inclusive_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_375_S0_H3_pairs_longest_analysis_first_half_inclusive_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_376_S0_H3_pairs_longest_analysis_first_half_inclusive_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis second half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.001146 [7.811e-05, 0.002591] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.009786 [0.003748, 0.01713] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 0.9823 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.2636 [0.1409, 0.4316] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2591, 2591] resolved	[1.6e+04, ∞] right censored	S 0.9641 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.6587 [0.5212, 0.7791] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[519.6, 519.6] resolved	[1.6e+04, ∞] right censored	S 0.9356 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9512 [0.9327, 0.9693] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[120.8, 120.8] resolved	[1.6e+04, ∞] right censored	S 0.9141 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9875 [0.9856, 0.9894] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	inconclusive	inconclusive	[89.93, 89.93] resolved	[1.6e+04, ∞] right censored	S 0.921 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.9603 [0.9557, 0.965] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_377_S0_H3_candidates_longest_analysis_second_half_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis second half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 0.9282 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.007785 [0, 0.02182] S low 0.07356 [0, 0.2091] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	inconclusive	[445.8, 445.8] resolved	[1.6e+04, ∞] right censored	S 0.261 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.08432 [0.07131, 0.09593] S low 0.8457 [0.8243, 0.8687] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[513, 513] resolved	[1.6e+04, ∞] right censored	S 0.1159 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.06907 [0.05767, 0.08078] S low 0.898 [0.8794, 0.9162] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[778.2, 778.2] resolved	[1.6e+04, ∞] right censored	S 0.05398 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.02712 [0.01973, 0.03409] S low 0.9539 [0.9411, 0.9668] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1169, 1169] resolved	[1.6e+04, ∞] right censored	S 0.02487 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.001562 [0, 0.003123] S low 0.9984 [0.9969, 1] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_378_S0_H3_candidates_longest_analysis_second_half_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis second half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[1.6e+04, ∞] right censored	S 0 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.0007808 [0, 0.002238] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 0.9646 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.2101 [0.01353, 0.4176] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	multiple fronts	criterion met	inconclusive	[3164, 3164] resolved	[1.6e+04, ∞] right censored	S 0.9536 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.5997 [0.4581, 0.7432] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2436, 2436] resolved	[1.6e+04, ∞] right censored	S 0.9774 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.729 [0.6299, 0.8188] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[932.3, 932.3] resolved	[1.6e+04, ∞] right censored	S 0.9844 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.8754 [0.8284, 0.9147] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[382.3, 382.3] resolved	[1.6e+04, ∞] right censored	S 0.9625 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9132 [0.8915, 0.9359] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[110.7, 110.7] resolved	[1.6e+04, ∞] right censored	S 0.9823 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9279 [0.9269, 0.9288] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_379_S0_H3_candidates_longest_analysis_second_half_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis second half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	inconclusive	inconclusive	[53.4, 53.4] resolved	[1.6e+04, ∞] right censored	S 0.972 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.8407 [0.8289, 0.8526] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[1600, ∞] right censored	[1600, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[1600, ∞] right censored	[1600, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[400, ∞] right censored	[1600, ∞] right censored	S 1 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.0006246 [0, 0.00177] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[400, ∞] right censored	[1600, ∞] right censored	S 0.995 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.02649 [0.00406, 0.05945] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[199, 199] resolved	[1600, ∞] right censored	S 0.9903 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.13 [0.08468, 0.182] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[42.25, 42.25] resolved	[1600, ∞] right censored	S 0.9823 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.3384 [0.2924, 0.388] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	multiple fronts	criterion met	criterion met	[18.06, 18.06] resolved	[1600, ∞] right censored	S 0.9805 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.5139 [0.4664, 0.562] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[11.07, 11.07] resolved	[1600, ∞] right censored	S 0.9809 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.6522 [0.6354, 0.669] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.896, 6.896] resolved	[1600, ∞] right censored	S 0.9739 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7807 [0.759, 0.8005] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	inconclusive	inconclusive	[5.19, 5.19] resolved	[1600, ∞] right censored	S 0.9749 / W 1 H 3200 / 12800 95% bound check: ?	S multi 0.8532 [0.8438, 0.8626] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_380_S0_H3_candidates_longest_analysis_second_half_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis second half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	inconclusive	inconclusive	[3.881, 3.881] resolved	[1600, ∞] right censored	S 0.9912 / W 1 H 3200 / 12800 95% bound check: ?	S multi 0.7801 [0.7683, 0.792] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_381_S0_H3_candidates_longest_analysis_second_half_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis second half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1.001 [1, 1.002] N 12/12; valid 1	1.003 [1.002, 1.003] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.004 [1, 1.009] N 12/12; valid 1	1.006 [1.005, 1.008] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8543, low 0; forced mean no	1.246 [1.112, 1.416] N 12/12; valid 1	1.01 [1.007, 1.014] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.685 [1.497, 1.846] N 12/12; valid 1	1.007 [1.001, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_382_S0_H3_fingerprints_longest_analysis_second_half_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis second half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3933, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9126, low 0; forced mean no	2.31 [2.228, 2.389] N 12/12; valid 1	1.025 [1.017, 1.035] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.459 [2.364, 2.555] N 2/2; valid 1	1.137 [1.132, 1.143] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.556 [2.481, 2.633] N 2/2; valid 1	1.216 [1.208, 1.224] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.008, 1.008] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.029 [1.029, 1.03] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.114 [1.111, 1.117] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9756 [0.9592, 0.9901] N 12/12; valid 1	1.39 [1.33, 1.457] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7884 [0.7767, 0.8] N 12/12; valid 1	3.296 [3.212, 3.387] N 12/12; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_383_S0_H3_fingerprints_longest_analysis_second_half_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis second half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.03087; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1978; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.8323 [0.7592, 0.954] N 12/12; valid 1	2.465 [2.105, 2.697] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.03	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check single front; H 32000 multi 0.00406, low 0; forced mean yes	1.478 [1.369, 1.585] N 12/12; valid 1	0.4817 [0.4616, 0.5027] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.05	not declared	not declared	not declared	1.316 [1.238, 1.397] N 12/12; valid 1	0.4644 [0.4506, 0.4779] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.1	not declared	not declared	not declared	1.193 [1.074, 1.325] N 12/12; valid 1	0.4449 [0.4296, 0.4626] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.3	not declared	not declared	not declared	0.6643 [0.5959, 0.7388] N 2/2; valid 1	0.4279 [0.4122, 0.4438] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=1	not declared	not declared	not declared	0.0961 [0.07751, 0.1151] N 12/2; valid 1	0.44 [0.4345, 0.4466] N 12/2; valid 1	count unavailable contrast unavailable
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_384_S0_H3_fingerprints_longest_analysis_second_half_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis second half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.009] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.208 [1.002, 1.431] N 12/12; valid 1	1.014 [1.013, 1.016] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.669 [1.476, 1.851] N 12/12; valid 1	1.028 [1.024, 1.033] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.02295, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8576, low 0; forced mean no	1.77 [1.621, 1.919] N 12/12; valid 1	1.049 [1.042, 1.057] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.004 [1.889, 2.119] N 12/12; valid 1	1.079 [1.071, 1.088] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.6923, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8151, low 0; forced mean no	2.07 [1.996, 2.152] N 12/12; valid 1	1.134 [1.12, 1.148] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_385_S0_H3_fingerprints_longest_analysis_second_half_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis second half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.218 [2.101, 2.339] N 2/2; valid 1	1.347 [1.313, 1.381] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.245 [2.098, 2.405] N 2/2; valid 1	1.787 [1.703, 1.878] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.004 [1.003, 1.006] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.023 [1.002, 1.056] N 12/12; valid 1	1.009 [1.005, 1.012] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.113 [1.068, 1.163] N 12/12; valid 1	1.001 [0.9955, 1.006] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.284 [1.242, 1.334] N 12/12; valid 1	0.9997 [0.9923, 1.007] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_386_S0_H3_fingerprints_longest_analysis_second_half_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis second half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	single front; H 3200 qualified yes; same no multi 0, low 0; forced mean no	single front; H 3200 qualified yes; same no multi 0.1823, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.5307, low 0; forced mean no	1.449 [1.399, 1.503] N 12/12; valid 1	0.9915 [0.9815, 1] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.564 [1.538, 1.588] N 12/12; valid 1	1.019 [0.9997, 1.038] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5634, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7431, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.61, low 0; forced mean no	1.702 [1.667, 1.739] N 12/12; valid 1	1.027 [1, 1.053] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.787 [1.716, 1.86] N 2/2; valid 1	1.173 [1.09, 1.268] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=1	not declared	not declared	not declared	1.796 [1.762, 1.83] N 2/2; valid 1	1.23 [1.172, 1.291] N 2/2; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_387_S0_H3_fingerprints_longest_analysis_second_half_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 1.478 [1.369, 1.585] N 12/12; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4817 [0.4616, 0.5027] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 1.316 [1.238, 1.397] N 12/12; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4644 [0.4506, 0.4779] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 1.193 [1.074, 1.325] N 12/12; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4449 [0.4296, 0.4626] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 0.6643 [0.5959, 0.7388] N 2/2; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4279 [0.4122, 0.4438] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 0.0961 [0.07751, 0.1151] N 12/2; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.44 [0.4345, 0.4466] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 1.478 [1.369, 1.585] N 12/12; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4817 [0.4616, 0.5027] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 1.316 [1.238, 1.397] N 12/12; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4644 [0.4506, 0.4779] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 1.193 [1.074, 1.325] N 12/12; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4449 [0.4296, 0.4626] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_388_S0_H3_pairs_longest_analysis_second_half_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 0.6643 [0.5959, 0.7388] N 2/2; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4279 [0.4122, 0.4438] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 0.0961 [0.07751, 0.1151] N 12/2; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.44 [0.4345, 0.4466] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 1.478 [1.369, 1.585] N 12/12; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4817 [0.4616, 0.5027] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 1.316 [1.238, 1.397] N 12/12; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4644 [0.4506, 0.4779] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 1.193 [1.074, 1.325] N 12/12; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4449 [0.4296, 0.4626] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 0.6643 [0.5959, 0.7388] N 2/2; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4279 [0.4122, 0.4438] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 0.0961 [0.07751, 0.1151] N 12/2; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.44 [0.4345, 0.4466] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 1.478 [1.369, 1.585] N 12/12; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4817 [0.4616, 0.5027] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_389_S0_H3_pairs_longest_analysis_second_half_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 1.316 [1.238, 1.397] N 12/12; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4644 [0.4506, 0.4779] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 1.193 [1.074, 1.325] N 12/12; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4449 [0.4296, 0.4626] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 0.6643 [0.5959, 0.7388] N 2/2; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4279 [0.4122, 0.4438] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 0.0961 [0.07751, 0.1151] N 12/2; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.44 [0.4345, 0.4466] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_390_S0_H3_pairs_longest_analysis_second_half_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_391_S0_H3_pairs_longest_analysis_second_half_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_392_S0_H3_pairs_longest_analysis_second_half_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_393_S0_H3_pairs_longest_analysis_second_half_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_394_S0_H3_pairs_longest_analysis_second_half_primary_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_395_S0_H3_pairs_longest_analysis_second_half_primary_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_396_S0_H3_pairs_longest_analysis_second_half_primary_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_397_S0_H3_pairs_longest_analysis_second_half_primary_010.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis second half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.001146 [7.811e-05, 0.002591] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.009786 [0.003748, 0.01713] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 0.9823 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.2636 [0.1409, 0.4316] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2591, 2591] resolved	[1.6e+04, ∞] right censored	S 0.9641 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.6587 [0.5212, 0.7791] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[519.6, 519.6] resolved	[1.6e+04, ∞] right censored	S 0.9356 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9512 [0.9327, 0.9693] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[120.8, 120.8] resolved	[1.6e+04, ∞] right censored	S 0.9141 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9875 [0.9856, 0.9894] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	criterion met	inconclusive	[89.93, 89.93] resolved	[1.6e+04, ∞] right censored	S 0.921 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9603 [0.9557, 0.965] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_398_S0_H3_candidates_longest_analysis_second_half_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis second half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 0.9282 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.007785 [0, 0.02182] S low 0.07356 [0, 0.2091] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	criterion met	[445.8, 445.8] resolved	[1.6e+04, ∞] right censored	S 0.261 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.08432 [0.07131, 0.09593] S low 0.8457 [0.8243, 0.8687] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[513, 513] resolved	[1.6e+04, ∞] right censored	S 0.1159 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.06907 [0.05767, 0.08078] S low 0.898 [0.8794, 0.9162] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[778.2, 778.2] resolved	[1.6e+04, ∞] right censored	S 0.05398 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.02712 [0.01973, 0.03409] S low 0.9539 [0.9411, 0.9668] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1169, 1169] resolved	[1.6e+04, ∞] right censored	S 0.02487 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.001562 [0, 0.003123] S low 0.9984 [0.9969, 1] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_399_S0_H3_candidates_longest_analysis_second_half_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis second half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[1.6e+04, ∞] right censored	S 0 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.0007808 [0, 0.002238] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 0.9646 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.2101 [0.01353, 0.4176] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	multiple fronts	criterion met	inconclusive	[3164, 3164] resolved	[1.6e+04, ∞] right censored	S 0.9536 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.5997 [0.4581, 0.7432] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2436, 2436] resolved	[1.6e+04, ∞] right censored	S 0.9774 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.729 [0.6299, 0.8188] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[932.3, 932.3] resolved	[1.6e+04, ∞] right censored	S 0.9844 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.8754 [0.8284, 0.9147] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[382.3, 382.3] resolved	[1.6e+04, ∞] right censored	S 0.9625 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9132 [0.8915, 0.9359] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[110.7, 110.7] resolved	[1.6e+04, ∞] right censored	S 0.9823 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9279 [0.9269, 0.9288] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_400_S0_H3_candidates_longest_analysis_second_half_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis second half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	criterion met	inconclusive	[53.4, 53.4] resolved	[1.6e+04, ∞] right censored	S 0.972 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.8407 [0.8289, 0.8526] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[1600, ∞] right censored	[1600, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[1600, ∞] right censored	[1600, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[400, ∞] right censored	[1600, ∞] right censored	S 1 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.0006246 [0, 0.00177] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[400, ∞] right censored	[1600, ∞] right censored	S 0.995 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.02649 [0.00406, 0.05945] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[199, 199] resolved	[1600, ∞] right censored	S 0.9903 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.13 [0.08468, 0.182] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[42.25, 42.25] resolved	[1600, ∞] right censored	S 0.9823 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.3384 [0.2924, 0.388] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	multiple fronts	criterion met	criterion met	[18.06, 18.06] resolved	[1600, ∞] right censored	S 0.9805 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.5139 [0.4664, 0.562] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[11.07, 11.07] resolved	[1600, ∞] right censored	S 0.9809 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.6522 [0.6354, 0.669] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.896, 6.896] resolved	[1600, ∞] right censored	S 0.9739 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7807 [0.759, 0.8005] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	criterion met	inconclusive	[5.19, 5.19] resolved	[1600, ∞] right censored	S 0.9749 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.8532 [0.8438, 0.8626] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_401_S0_H3_candidates_longest_analysis_second_half_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis second half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	criterion met	inconclusive	[3.881, 3.881] resolved	[1600, ∞] right censored	S 0.9912 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7801 [0.7683, 0.792] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_402_S0_H3_candidates_longest_analysis_second_half_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis second half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1.001 [1, 1.002] N 12/12; valid 1	1.003 [1.002, 1.003] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.004 [1, 1.009] N 12/12; valid 1	1.006 [1.005, 1.008] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8543, low 0; forced mean no	1.246 [1.112, 1.416] N 12/12; valid 1	1.01 [1.007, 1.014] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.685 [1.497, 1.846] N 12/12; valid 1	1.007 [1.001, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_403_S0_H3_fingerprints_longest_analysis_second_half_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis second half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3933, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9126, low 0; forced mean no	2.31 [2.228, 2.389] N 12/12; valid 1	1.025 [1.017, 1.035] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.459 [2.364, 2.555] N 2/2; valid 1	1.137 [1.132, 1.143] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.556 [2.481, 2.633] N 2/2; valid 1	1.216 [1.208, 1.224] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.008, 1.008] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.029 [1.029, 1.03] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.114 [1.111, 1.117] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9756 [0.9592, 0.9901] N 12/12; valid 1	1.39 [1.33, 1.457] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7884 [0.7767, 0.8] N 12/12; valid 1	3.296 [3.212, 3.387] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_404_S0_H3_fingerprints_longest_analysis_second_half_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis second half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.03087; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1978; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.8323 [0.7592, 0.954] N 12/12; valid 1	2.465 [2.105, 2.697] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.03	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	single front; H 32000 qualified yes; same no multi 0.00406, low 0; forced mean yes	1.478 [1.369, 1.585] N 12/12; valid 1	0.4817 [0.4616, 0.5027] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.05	not declared	not declared	not declared	1.316 [1.238, 1.397] N 12/12; valid 1	0.4644 [0.4506, 0.4779] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.1	not declared	not declared	not declared	1.193 [1.074, 1.325] N 12/12; valid 1	0.4449 [0.4296, 0.4626] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.3	not declared	not declared	not declared	0.6643 [0.5959, 0.7388] N 2/2; valid 1	0.4279 [0.4122, 0.4438] N 2/2; valid 1	count yes contrast yes
Goryachev f=1	not declared	not declared	not declared	0.0961 [0.07751, 0.1151] N 12/2; valid 1	0.44 [0.4345, 0.4466] N 12/2; valid 1	count yes contrast yes
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_405_S0_H3_fingerprints_longest_analysis_second_half_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis second half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.009] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.208 [1.002, 1.431] N 12/12; valid 1	1.014 [1.013, 1.016] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.669 [1.476, 1.851] N 12/12; valid 1	1.028 [1.024, 1.033] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.02295, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8576, low 0; forced mean no	1.77 [1.621, 1.919] N 12/12; valid 1	1.049 [1.042, 1.057] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.004 [1.889, 2.119] N 12/12; valid 1	1.079 [1.071, 1.088] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.6923, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8151, low 0; forced mean no	2.07 [1.996, 2.152] N 12/12; valid 1	1.134 [1.12, 1.148] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_406_S0_H3_fingerprints_longest_analysis_second_half_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis second half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.218 [2.101, 2.339] N 2/2; valid 1	1.347 [1.313, 1.381] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.245 [2.098, 2.405] N 2/2; valid 1	1.787 [1.703, 1.878] N 2/2; valid 1	count yes contrast yes
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.004 [1.003, 1.006] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.023 [1.002, 1.056] N 12/12; valid 1	1.009 [1.005, 1.012] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.113 [1.068, 1.163] N 12/12; valid 1	1.001 [0.9955, 1.006] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.284 [1.242, 1.334] N 12/12; valid 1	0.9997 [0.9923, 1.007] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_407_S0_H3_fingerprints_longest_analysis_second_half_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis second half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	single front; H 3200 qualified yes; same no multi 0, low 0; forced mean no	single front; H 3200 qualified yes; same no multi 0.1823, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.5307, low 0; forced mean no	1.449 [1.399, 1.503] N 12/12; valid 1	0.9915 [0.9815, 1] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.564 [1.538, 1.588] N 12/12; valid 1	1.019 [0.9997, 1.038] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5634, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7431, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.61, low 0; forced mean no	1.702 [1.667, 1.739] N 12/12; valid 1	1.027 [1, 1.053] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.787 [1.716, 1.86] N 2/2; valid 1	1.173 [1.09, 1.268] N 2/2; valid 1	count yes contrast yes
Spring f=1	not declared	not declared	not declared	1.796 [1.762, 1.83] N 2/2; valid 1	1.23 [1.172, 1.291] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_408_S0_H3_fingerprints_longest_analysis_second_half_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 1.478 [1.369, 1.585] N 12/12; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4817 [0.4616, 0.5027] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 1.316 [1.238, 1.397] N 12/12; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4644 [0.4506, 0.4779] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 1.193 [1.074, 1.325] N 12/12; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4449 [0.4296, 0.4626] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 0.6643 [0.5959, 0.7388] N 2/2; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4279 [0.4122, 0.4438] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 0.0961 [0.07751, 0.1151] N 12/2; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.44 [0.4345, 0.4466] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / criterion met	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 1.478 [1.369, 1.585] N 12/12; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4817 [0.4616, 0.5027] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 1.316 [1.238, 1.397] N 12/12; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4644 [0.4506, 0.4779] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 1.193 [1.074, 1.325] N 12/12; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4449 [0.4296, 0.4626] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_409_S0_H3_pairs_longest_analysis_second_half_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 0.6643 [0.5959, 0.7388] N 2/2; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4279 [0.4122, 0.4438] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 0.0961 [0.07751, 0.1151] N 12/2; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.44 [0.4345, 0.4466] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 1.478 [1.369, 1.585] N 12/12; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4817 [0.4616, 0.5027] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 1.316 [1.238, 1.397] N 12/12; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4644 [0.4506, 0.4779] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 1.193 [1.074, 1.325] N 12/12; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4449 [0.4296, 0.4626] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 0.6643 [0.5959, 0.7388] N 2/2; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4279 [0.4122, 0.4438] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 0.0961 [0.07751, 0.1151] N 12/2; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.44 [0.4345, 0.4466] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 1.478 [1.369, 1.585] N 12/12; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4817 [0.4616, 0.5027] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_410_S0_H3_pairs_longest_analysis_second_half_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 1.316 [1.238, 1.397] N 12/12; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4644 [0.4506, 0.4779] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 1.193 [1.074, 1.325] N 12/12; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4449 [0.4296, 0.4626] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 0.6643 [0.5959, 0.7388] N 2/2; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4279 [0.4122, 0.4438] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 0.0961 [0.07751, 0.1151] N 12/2; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.44 [0.4345, 0.4466] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_411_S0_H3_pairs_longest_analysis_second_half_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_412_S0_H3_pairs_longest_analysis_second_half_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_413_S0_H3_pairs_longest_analysis_second_half_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_414_S0_H3_pairs_longest_analysis_second_half_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_415_S0_H3_pairs_longest_analysis_second_half_inclusive_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_416_S0_H3_pairs_longest_analysis_second_half_inclusive_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_417_S0_H3_pairs_longest_analysis_second_half_inclusive_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_418_S0_H3_pairs_longest_analysis_second_half_inclusive_010.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · late quarter · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[2000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.0004686 [0, 0.00125] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[2000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.009155 [0.001144, 0.01987] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[2000, ∞] right censored	[8000, ∞] right censored	S 0.9757 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.2917 [0.1244, 0.5021] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	inconclusive	inconclusive	[2000, ∞] right censored	[8000, ∞] right censored	S 0.9902 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.6283 [0.4561, 0.7684] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[523.6, 523.6] resolved	[8000, ∞] right censored	S 0.9389 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9597 [0.9421, 0.9774] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[113.6, 113.6] resolved	[8000, ∞] right censored	S 0.9101 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.985 [0.9825, 0.9875] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	inconclusive	inconclusive	[90.81, 90.81] resolved	[8000, ∞] right censored	S 0.9227 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.9638 [0.9563, 0.9713] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_419_S0_H3_candidates_longest_late_quarter_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · late quarter · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[2000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.9147 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.008148 [0, 0.02031] S low 0.09125 [0, 0.2506] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	inconclusive	[450, 450] resolved	[8000, ∞] right censored	S 0.2955 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.0748 [0.0594, 0.09124] S low 0.8537 [0.8408, 0.8707] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[428, 428] resolved	[8000, ∞] right censored	S 0.128 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.0671 [0.05452, 0.07886] S low 0.9087 [0.892, 0.926] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[978, 978] resolved	[8000, ∞] right censored	S 0.05124 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.03069 [0.01987, 0.04203] S low 0.9511 [0.9319, 0.9681] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[8000, ∞] right censored	S 0 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_420_S0_H3_candidates_longest_late_quarter_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · late quarter · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[8000, ∞] right censored	S 0 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[2000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[2000, ∞] right censored	[8000, ∞] right censored	S 0.9919 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.162 [0.01007, 0.3558] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	multiple fronts	inconclusive	inconclusive	[2000, ∞] right censored	[8000, ∞] right censored	S 0.9537 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.5145 [0.3285, 0.706] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	inconclusive	criterion met	[2000, ∞] right censored	[8000, ∞] right censored	S 0.9877 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.6821 [0.5204, 0.8242] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[1347, 1347] resolved	[8000, ∞] right censored	S 0.9783 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.8719 [0.8317, 0.9134] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[367.8, 367.8] resolved	[8000, ∞] right censored	S 0.9576 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9181 [0.8925, 0.944] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[97.62, 97.62] resolved	[8000, ∞] right censored	S 0.9923 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9295 [0.9201, 0.9388] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_421_S0_H3_candidates_longest_late_quarter_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · late quarter · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	inconclusive	inconclusive	[58.16, 58.16] resolved	[8000, ∞] right censored	S 0.9759 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.8102 [0.7928, 0.8277] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[200, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.00104 [0, 0.003121] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[200, ∞] right censored	[800, ∞] right censored	S 0.99 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.04359 [0.00437, 0.1033] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[179.1, 179.1] resolved	[800, ∞] right censored	S 0.9928 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.1256 [0.0723, 0.1927] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[48.74, 48.74] resolved	[800, ∞] right censored	S 0.9833 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.3343 [0.27, 0.4022] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	multiple fronts	criterion met	criterion met	[18.09, 18.09] resolved	[800, ∞] right censored	S 0.9776 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.533 [0.4707, 0.6013] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[11.29, 11.29] resolved	[800, ∞] right censored	S 0.9828 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.6471 [0.6185, 0.6756] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.988, 6.988] resolved	[800, ∞] right censored	S 0.9724 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7762 [0.7507, 0.7998] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	inconclusive	inconclusive	[5.513, 5.513] resolved	[800, ∞] right censored	S 0.9747 / W 1 H 3200 / 12800 95% bound check: ?	S multi 0.829 [0.7903, 0.8677] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_422_S0_H3_candidates_longest_late_quarter_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · late quarter · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	inconclusive	inconclusive	[3.991, 3.991] resolved	[800, ∞] right censored	S 0.9924 / W 1 H 3200 / 12800 95% bound check: ?	S multi 0.7709 [0.7191, 0.8227] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_423_S0_H3_candidates_longest_late_quarter_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · late quarter · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.003] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.005 [1, 1.012] N 12/12; valid 1	1.008 [1.005, 1.011] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8688, low 0; forced mean no	1.29 [1.103, 1.521] N 12/12; valid 1	1.011 [1.006, 1.015] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.641 [1.428, 1.853] N 12/12; valid 1	1.008 [0.9981, 1.018] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_424_S0_H3_fingerprints_longest_late_quarter_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · late quarter · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3682, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9189, low 0; forced mean no	2.312 [2.243, 2.38] N 12/12; valid 1	1.027 [1.017, 1.04] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.442 [2.348, 2.54] N 2/2; valid 1	1.143 [1.112, 1.175] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.617 [2.484, 2.75] N 2/2; valid 1	1.187 [1.123, 1.253] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.008, 1.008] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.029 [1.028, 1.03] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.11, 1.12] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9756 [0.9419, 1] N 12/12; valid 1	1.39 [1.313, 1.511] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7926 [0.7663, 0.8219] N 12/12; valid 1	3.221 [3.09, 3.355] N 12/12; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_425_S0_H3_fingerprints_longest_late_quarter_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · late quarter · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.06169; forced mean no	no eligible check single front; H 32000 multi 0, low 0.2398; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.8531 [0.7673, 0.976] N 12/12; valid 1	2.315 [1.928, 2.588] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.03	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check single front; H 32000 multi 0.007074, low 0; forced mean yes	1.452 [1.346, 1.561] N 12/12; valid 1	0.472 [0.4473, 0.4987] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.05	not declared	not declared	not declared	1.317 [1.203, 1.428] N 12/12; valid 1	0.4675 [0.45, 0.4849] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.1	not declared	not declared	not declared	1.191 [1.04, 1.369] N 12/12; valid 1	0.4325 [0.4136, 0.4551] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.3	not declared	not declared	not declared	0.5441 [0.4932, 0.6032] N 2/2; valid 1	0.4432 [0.4413, 0.4451] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=1	not declared	not declared	not declared	0.03468 [0.02741, 0.04213] N 12/2; valid 1	0.4227 [0.4056, 0.4392] N 12/2; valid 1	count unavailable contrast unavailable
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_426_S0_H3_fingerprints_longest_late_quarter_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · late quarter · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.004 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.009 [1.008, 1.01] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.161 [0.9983, 1.358] N 12/12; valid 1	1.015 [1.012, 1.018] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.614 [1.355, 1.89] N 12/12; valid 1	1.028 [1.022, 1.035] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8859, low 0; forced mean no	1.678 [1.477, 1.875] N 12/12; valid 1	1.05 [1.041, 1.059] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	1.975 [1.841, 2.108] N 12/12; valid 1	1.081 [1.069, 1.093] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7209, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8092, low 0; forced mean no	2.112 [2.008, 2.218] N 12/12; valid 1	1.132 [1.114, 1.153] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_427_S0_H3_fingerprints_longest_late_quarter_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · late quarter · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.363 [2.228, 2.5] N 2/2; valid 1	1.379 [1.31, 1.45] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.164 [2.06, 2.279] N 2/2; valid 1	1.827 [1.787, 1.867] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.003 [1, 1.006] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.038 [1, 1.107] N 12/12; valid 1	1.009 [1.003, 1.014] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.107 [1.053, 1.173] N 12/12; valid 1	1.004 [0.9956, 1.012] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.27 [1.203, 1.352] N 12/12; valid 1	0.9951 [0.9836, 1.007] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_428_S0_H3_fingerprints_longest_late_quarter_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · late quarter · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	single front; H 3200 qualified yes; same no multi 0, low 0; forced mean no	single front; H 3200 qualified yes; same no multi 0.1828, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.5309, low 0; forced mean no	1.473 [1.403, 1.553] N 12/12; valid 1	0.9857 [0.9721, 0.9981] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.548 [1.522, 1.573] N 12/12; valid 1	1.025 [1.001, 1.048] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5624, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7423, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6064, low 0; forced mean no	1.677 [1.64, 1.716] N 12/12; valid 1	1.023 [0.9881, 1.059] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.743 [1.686, 1.8] N 2/2; valid 1	1.153 [1.086, 1.221] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=1	not declared	not declared	not declared	1.752 [1.686, 1.82] N 2/2; valid 1	1.277 [1.102, 1.491] N 2/2; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_429_S0_H3_fingerprints_longest_late_quarter_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 1.452 [1.346, 1.561] N 12/12; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.472 [0.4473, 0.4987] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 1.317 [1.203, 1.428] N 12/12; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.4675 [0.45, 0.4849] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 1.191 [1.04, 1.369] N 12/12; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.4325 [0.4136, 0.4551] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 0.5441 [0.4932, 0.6032] N 2/2; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.4432 [0.4413, 0.4451] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 0.03468 [0.02741, 0.04213] N 12/2; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.4227 [0.4056, 0.4392] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 1.452 [1.346, 1.561] N 12/12; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.472 [0.4473, 0.4987] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 1.317 [1.203, 1.428] N 12/12; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.4675 [0.45, 0.4849] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 1.191 [1.04, 1.369] N 12/12; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.4325 [0.4136, 0.4551] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_430_S0_H3_pairs_longest_late_quarter_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 0.5441 [0.4932, 0.6032] N 2/2; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.4432 [0.4413, 0.4451] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 0.03468 [0.02741, 0.04213] N 12/2; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.4227 [0.4056, 0.4392] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 1.452 [1.346, 1.561] N 12/12; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.472 [0.4473, 0.4987] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 1.317 [1.203, 1.428] N 12/12; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.4675 [0.45, 0.4849] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 1.191 [1.04, 1.369] N 12/12; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.4325 [0.4136, 0.4551] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 0.5441 [0.4932, 0.6032] N 2/2; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.4432 [0.4413, 0.4451] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 0.03468 [0.02741, 0.04213] N 12/2; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.4227 [0.4056, 0.4392] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 1.452 [1.346, 1.561] N 12/12; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.472 [0.4473, 0.4987] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_431_S0_H3_pairs_longest_late_quarter_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 1.317 [1.203, 1.428] N 12/12; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.4675 [0.45, 0.4849] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 1.191 [1.04, 1.369] N 12/12; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.4325 [0.4136, 0.4551] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 0.5441 [0.4932, 0.6032] N 2/2; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.4432 [0.4413, 0.4451] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 0.03468 [0.02741, 0.04213] N 12/2; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.4227 [0.4056, 0.4392] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_432_S0_H3_pairs_longest_late_quarter_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_433_S0_H3_pairs_longest_late_quarter_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_434_S0_H3_pairs_longest_late_quarter_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_435_S0_H3_pairs_longest_late_quarter_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_436_S0_H3_pairs_longest_late_quarter_primary_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_437_S0_H3_pairs_longest_late_quarter_primary_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_438_S0_H3_pairs_longest_late_quarter_primary_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_439_S0_H3_pairs_longest_late_quarter_primary_010.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · late quarter · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[2000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.0004686 [0, 0.00125] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[2000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.009155 [0.001144, 0.01987] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[2000, ∞] right censored	[8000, ∞] right censored	S 0.9757 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.2917 [0.1244, 0.5021] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	inconclusive	inconclusive	[2000, ∞] right censored	[8000, ∞] right censored	S 0.9902 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.6283 [0.4561, 0.7684] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[523.6, 523.6] resolved	[8000, ∞] right censored	S 0.9389 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9597 [0.9421, 0.9774] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[113.6, 113.6] resolved	[8000, ∞] right censored	S 0.9101 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.985 [0.9825, 0.9875] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	criterion met	inconclusive	[90.81, 90.81] resolved	[8000, ∞] right censored	S 0.9227 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9638 [0.9563, 0.9713] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_440_S0_H3_candidates_longest_late_quarter_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · late quarter · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[2000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.9147 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.008148 [0, 0.02031] S low 0.09125 [0, 0.2506] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	criterion met	[450, 450] resolved	[8000, ∞] right censored	S 0.2955 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.0748 [0.0594, 0.09124] S low 0.8537 [0.8408, 0.8707] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[428, 428] resolved	[8000, ∞] right censored	S 0.128 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.0671 [0.05452, 0.07886] S low 0.9087 [0.892, 0.926] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[978, 978] resolved	[8000, ∞] right censored	S 0.05124 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.03069 [0.01987, 0.04203] S low 0.9511 [0.9319, 0.9681] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[8000, ∞] right censored	S 0 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_441_S0_H3_candidates_longest_late_quarter_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · late quarter · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[8000, ∞] right censored	S 0 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[2000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[2000, ∞] right censored	[8000, ∞] right censored	S 0.9919 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.162 [0.01007, 0.3558] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	multiple fronts	inconclusive	inconclusive	[2000, ∞] right censored	[8000, ∞] right censored	S 0.9537 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.5145 [0.3285, 0.706] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	inconclusive	criterion met	[2000, ∞] right censored	[8000, ∞] right censored	S 0.9877 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.6821 [0.5204, 0.8242] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[1347, 1347] resolved	[8000, ∞] right censored	S 0.9783 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.8719 [0.8317, 0.9134] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[367.8, 367.8] resolved	[8000, ∞] right censored	S 0.9576 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9181 [0.8925, 0.944] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[97.62, 97.62] resolved	[8000, ∞] right censored	S 0.9923 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9295 [0.9201, 0.9388] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_442_S0_H3_candidates_longest_late_quarter_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · late quarter · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	criterion met	inconclusive	[58.16, 58.16] resolved	[8000, ∞] right censored	S 0.9759 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.8102 [0.7928, 0.8277] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[200, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.00104 [0, 0.003121] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[200, ∞] right censored	[800, ∞] right censored	S 0.99 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.04359 [0.00437, 0.1033] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[179.1, 179.1] resolved	[800, ∞] right censored	S 0.9928 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.1256 [0.0723, 0.1927] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[48.74, 48.74] resolved	[800, ∞] right censored	S 0.9833 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.3343 [0.27, 0.4022] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	multiple fronts	criterion met	criterion met	[18.09, 18.09] resolved	[800, ∞] right censored	S 0.9776 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.533 [0.4707, 0.6013] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[11.29, 11.29] resolved	[800, ∞] right censored	S 0.9828 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.6471 [0.6185, 0.6756] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.988, 6.988] resolved	[800, ∞] right censored	S 0.9724 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7762 [0.7507, 0.7998] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	criterion met	inconclusive	[5.513, 5.513] resolved	[800, ∞] right censored	S 0.9747 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.829 [0.7903, 0.8677] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_443_S0_H3_candidates_longest_late_quarter_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · late quarter · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	criterion met	inconclusive	[3.991, 3.991] resolved	[800, ∞] right censored	S 0.9924 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7709 [0.7191, 0.8227] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_444_S0_H3_candidates_longest_late_quarter_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · late quarter · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.003] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.005 [1, 1.012] N 12/12; valid 1	1.008 [1.005, 1.011] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8688, low 0; forced mean no	1.29 [1.103, 1.521] N 12/12; valid 1	1.011 [1.006, 1.015] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.641 [1.428, 1.853] N 12/12; valid 1	1.008 [0.9981, 1.018] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_445_S0_H3_fingerprints_longest_late_quarter_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · late quarter · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3682, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9189, low 0; forced mean no	2.312 [2.243, 2.38] N 12/12; valid 1	1.027 [1.017, 1.04] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.442 [2.348, 2.54] N 2/2; valid 1	1.143 [1.112, 1.175] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.617 [2.484, 2.75] N 2/2; valid 1	1.187 [1.123, 1.253] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.008, 1.008] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.029 [1.028, 1.03] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.11, 1.12] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9756 [0.9419, 1] N 12/12; valid 1	1.39 [1.313, 1.511] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7926 [0.7663, 0.8219] N 12/12; valid 1	3.221 [3.09, 3.355] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_446_S0_H3_fingerprints_longest_late_quarter_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · late quarter · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.06169; forced mean no	no eligible check single front; H 32000 multi 0, low 0.2398; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.8531 [0.7673, 0.976] N 12/12; valid 1	2.315 [1.928, 2.588] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.03	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	single front; H 32000 qualified yes; same no multi 0.007074, low 0; forced mean yes	1.452 [1.346, 1.561] N 12/12; valid 1	0.472 [0.4473, 0.4987] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.05	not declared	not declared	not declared	1.317 [1.203, 1.428] N 12/12; valid 1	0.4675 [0.45, 0.4849] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.1	not declared	not declared	not declared	1.191 [1.04, 1.369] N 12/12; valid 1	0.4325 [0.4136, 0.4551] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.3	not declared	not declared	not declared	0.5441 [0.4932, 0.6032] N 2/2; valid 1	0.4432 [0.4413, 0.4451] N 2/2; valid 1	count yes contrast yes
Goryachev f=1	not declared	not declared	not declared	0.03468 [0.02741, 0.04213] N 12/2; valid 1	0.4227 [0.4056, 0.4392] N 12/2; valid 1	count yes contrast yes
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_447_S0_H3_fingerprints_longest_late_quarter_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · late quarter · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.004 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.009 [1.008, 1.01] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.161 [0.9983, 1.358] N 12/12; valid 1	1.015 [1.012, 1.018] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.614 [1.355, 1.89] N 12/12; valid 1	1.028 [1.022, 1.035] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8859, low 0; forced mean no	1.678 [1.477, 1.875] N 12/12; valid 1	1.05 [1.041, 1.059] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	1.975 [1.841, 2.108] N 12/12; valid 1	1.081 [1.069, 1.093] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7209, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8092, low 0; forced mean no	2.112 [2.008, 2.218] N 12/12; valid 1	1.132 [1.114, 1.153] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_448_S0_H3_fingerprints_longest_late_quarter_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · late quarter · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.363 [2.228, 2.5] N 2/2; valid 1	1.379 [1.31, 1.45] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.164 [2.06, 2.279] N 2/2; valid 1	1.827 [1.787, 1.867] N 2/2; valid 1	count yes contrast yes
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.003 [1, 1.006] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.038 [1, 1.107] N 12/12; valid 1	1.009 [1.003, 1.014] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.107 [1.053, 1.173] N 12/12; valid 1	1.004 [0.9956, 1.012] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.27 [1.203, 1.352] N 12/12; valid 1	0.9951 [0.9836, 1.007] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_449_S0_H3_fingerprints_longest_late_quarter_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · late quarter · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	single front; H 3200 qualified yes; same no multi 0, low 0; forced mean no	single front; H 3200 qualified yes; same no multi 0.1828, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.5309, low 0; forced mean no	1.473 [1.403, 1.553] N 12/12; valid 1	0.9857 [0.9721, 0.9981] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.548 [1.522, 1.573] N 12/12; valid 1	1.025 [1.001, 1.048] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5624, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7423, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6064, low 0; forced mean no	1.677 [1.64, 1.716] N 12/12; valid 1	1.023 [0.9881, 1.059] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.743 [1.686, 1.8] N 2/2; valid 1	1.153 [1.086, 1.221] N 2/2; valid 1	count yes contrast yes
Spring f=1	not declared	not declared	not declared	1.752 [1.686, 1.82] N 2/2; valid 1	1.277 [1.102, 1.491] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_450_S0_H3_fingerprints_longest_late_quarter_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 1.452 [1.346, 1.561] N 12/12; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.472 [0.4473, 0.4987] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 1.317 [1.203, 1.428] N 12/12; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.4675 [0.45, 0.4849] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 1.191 [1.04, 1.369] N 12/12; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.4325 [0.4136, 0.4551] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 0.5441 [0.4932, 0.6032] N 2/2; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.4432 [0.4413, 0.4451] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 0.03468 [0.02741, 0.04213] N 12/2; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.4227 [0.4056, 0.4392] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / criterion met	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 1.452 [1.346, 1.561] N 12/12; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.472 [0.4473, 0.4987] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 1.317 [1.203, 1.428] N 12/12; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.4675 [0.45, 0.4849] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 1.191 [1.04, 1.369] N 12/12; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.4325 [0.4136, 0.4551] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_451_S0_H3_pairs_longest_late_quarter_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 0.5441 [0.4932, 0.6032] N 2/2; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.4432 [0.4413, 0.4451] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 0.03468 [0.02741, 0.04213] N 12/2; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.4227 [0.4056, 0.4392] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 1.452 [1.346, 1.561] N 12/12; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.472 [0.4473, 0.4987] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 1.317 [1.203, 1.428] N 12/12; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.4675 [0.45, 0.4849] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 1.191 [1.04, 1.369] N 12/12; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.4325 [0.4136, 0.4551] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 0.5441 [0.4932, 0.6032] N 2/2; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.4432 [0.4413, 0.4451] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 0.03468 [0.02741, 0.04213] N 12/2; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.4227 [0.4056, 0.4392] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 1.452 [1.346, 1.561] N 12/12; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.472 [0.4473, 0.4987] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_452_S0_H3_pairs_longest_late_quarter_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 1.317 [1.203, 1.428] N 12/12; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.4675 [0.45, 0.4849] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 1.191 [1.04, 1.369] N 12/12; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.4325 [0.4136, 0.4551] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 0.5441 [0.4932, 0.6032] N 2/2; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.4432 [0.4413, 0.4451] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 0.03468 [0.02741, 0.04213] N 12/2; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.4227 [0.4056, 0.4392] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: criterion met / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: criterion met / criterion met	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_453_S0_H3_pairs_longest_late_quarter_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_454_S0_H3_pairs_longest_late_quarter_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_455_S0_H3_pairs_longest_late_quarter_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_456_S0_H3_pairs_longest_late_quarter_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_457_S0_H3_pairs_longest_late_quarter_inclusive_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_458_S0_H3_pairs_longest_late_quarter_inclusive_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_459_S0_H3_pairs_longest_late_quarter_inclusive_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_460_S0_H3_pairs_longest_late_quarter_inclusive_010.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x2 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.001159 [0.0002991, 0.002239] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.01712 [0.007027, 0.0304] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9892 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.235 [0.1245, 0.3759] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2696, 2696] resolved	[3.2e+04, ∞] right censored	S 0.9798 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.6387 [0.5516, 0.7211] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[505.9, 505.9] resolved	[3.2e+04, ∞] right censored	S 0.9422 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9548 [0.9448, 0.9647] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[128.3, 128.3] resolved	[3.2e+04, ∞] right censored	S 0.9207 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9863 [0.9844, 0.9881] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	inconclusive	inconclusive	[83.27, 83.27] resolved	[3.2e+04, ∞] right censored	S 0.9319 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.9606 [0.9582, 0.9631] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_461_S0_H3_candidates_longest_cadence_x2_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x2 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9573 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.00401 [0, 0.01127] S low 0.03671 [0, 0.1044] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	inconclusive	[475.4, 475.4] resolved	[3.2e+04, ∞] right censored	S 0.2373 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.07594 [0.06543, 0.08484] S low 0.8565 [0.8397, 0.8749] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[563.6, 563.6] resolved	[3.2e+04, ∞] right censored	S 0.131 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.06283 [0.05564, 0.07001] S low 0.9004 [0.8879, 0.9136] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[733.2, 733.2] resolved	[3.2e+04, ∞] right censored	S 0.05288 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.0266 [0.02082, 0.03206] S low 0.9579 [0.9503, 0.9669] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1251, 1251] resolved	[3.2e+04, ∞] right censored	S 0.01525 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.004997 [0.001874, 0.00812] S low 0.9925 [0.9875, 0.9975] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_462_S0_H3_candidates_longest_cadence_x2_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x2 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[3.2e+04, ∞] right censored	S 0 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.0001302 [0, 0.0003906] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.0004164 [0, 0.001145] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 0.979 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.1969 [0.03435, 0.3848] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3903, 3903] resolved	[3.2e+04, ∞] right censored	S 0.9796 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.4837 [0.3914, 0.5751] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2012, 2012] resolved	[3.2e+04, ∞] right censored	S 0.985 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.7161 [0.6474, 0.7783] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[986.9, 986.9] resolved	[3.2e+04, ∞] right censored	S 0.9749 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.857 [0.832, 0.8798] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[388.5, 388.5] resolved	[3.2e+04, ∞] right censored	S 0.9627 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.919 [0.9007, 0.9373] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[104, 104] resolved	[3.2e+04, ∞] right censored	S 0.9616 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9335 [0.9332, 0.9338] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_463_S0_H3_candidates_longest_cadence_x2_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x2 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	inconclusive	inconclusive	[51.19, 51.19] resolved	[3.2e+04, ∞] right censored	S 0.9708 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.852 [0.8463, 0.8576] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[3200, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[3200, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[800, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.0007287 [5.205e-05, 0.001666] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	criterion met	inconclusive	[533.5, 533.5] resolved	[3200, ∞] right censored	S 0.9981 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.02363 [0.006558, 0.04497] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[145.8, 145.8] resolved	[3200, ∞] right censored	S 0.9922 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.1366 [0.108, 0.1706] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[37.9, 37.9] resolved	[3200, ∞] right censored	S 0.9847 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.3533 [0.3188, 0.3849] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.18, 18.18] resolved	[3200, ∞] right censored	S 0.9823 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.4987 [0.4698, 0.5309] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.66, 10.66] resolved	[3200, ∞] right censored	S 0.9803 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.658 [0.6436, 0.6719] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.808, 6.808] resolved	[3200, ∞] right censored	S 0.9744 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7891 [0.7747, 0.8021] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	inconclusive	inconclusive	[4.953, 4.953] resolved	[3200, ∞] right censored	S 0.9749 / W 1 H 3200 / 12800 95% bound check: ?	S multi 0.842 [0.8407, 0.8432] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_464_S0_H3_candidates_longest_cadence_x2_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x2 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	inconclusive	inconclusive	[4.101, 4.101] resolved	[3200, ∞] right censored	S 0.9887 / W 1 H 3200 / 12800 95% bound check: ?	S multi 0.7633 [0.7395, 0.787] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_465_S0_H3_candidates_longest_cadence_x2_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x2 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1.001 [1, 1.002] N 12/12; valid 1	1.002 [1.002, 1.003] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.011 [1.002, 1.023] N 12/12; valid 1	1.005 [1.004, 1.007] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8682, low 0; forced mean no	1.212 [1.09, 1.34] N 12/12; valid 1	1.009 [1.007, 1.01] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.668 [1.558, 1.771] N 12/12; valid 1	1.009 [1.006, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_466_S0_H3_fingerprints_longest_cadence_x2_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x2 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3856, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9124, low 0; forced mean no	2.309 [2.248, 2.367] N 12/12; valid 1	1.024 [1.017, 1.031] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.482 [2.376, 2.593] N 2/2; valid 1	1.124 [1.11, 1.139] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.507 [2.432, 2.583] N 2/2; valid 1	1.25 [1.233, 1.268] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.008, 1.008] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.029 [1.029, 1.03] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.113, 1.117] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9581 [0.936, 0.9784] N 12/12; valid 1	1.484 [1.393, 1.597] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7913 [0.7772, 0.8073] N 12/12; valid 1	3.342 [3.279, 3.408] N 12/12; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_467_S0_H3_fingerprints_longest_cadence_x2_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x2 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.01546; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1406; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.8003 [0.7627, 0.8608] N 12/12; valid 1	2.556 [2.371, 2.688] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.03	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check single front; H 32000 multi 0.008224, low 0; forced mean yes	1.454 [1.386, 1.527] N 12/12; valid 1	0.4761 [0.4656, 0.4867] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.05	not declared	not declared	not declared	1.348 [1.277, 1.415] N 12/12; valid 1	0.4608 [0.45, 0.4716] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.1	not declared	not declared	not declared	1.206 [1.133, 1.282] N 12/12; valid 1	0.4371 [0.4272, 0.4482] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.3	not declared	not declared	not declared	0.654 [0.6049, 0.7068] N 2/2; valid 1	0.4247 [0.4159, 0.4336] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=1	not declared	not declared	not declared	0.07134 [0.05671, 0.08629] N 12/2; valid 1	0.4328 [0.4281, 0.4376] N 12/2; valid 1	count unavailable contrast unavailable
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_468_S0_H3_fingerprints_longest_cadence_x2_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x2 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.008] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.194 [1.022, 1.375] N 12/12; valid 1	1.013 [1.012, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.509 [1.389, 1.617] N 12/12; valid 1	1.029 [1.026, 1.031] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.01145, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8576, low 0; forced mean no	1.771 [1.657, 1.891] N 12/12; valid 1	1.049 [1.045, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.012 [1.942, 2.085] N 12/12; valid 1	1.076 [1.07, 1.083] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7007, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8082, low 0; forced mean no	2.11 [2.05, 2.176] N 12/12; valid 1	1.134 [1.123, 1.146] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_469_S0_H3_fingerprints_longest_cadence_x2_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x2 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.354 [2.32, 2.389] N 2/2; valid 1	1.305 [1.285, 1.325] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.215 [2.078, 2.359] N 2/2; valid 1	1.692 [1.633, 1.755] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.021 [1.004, 1.043] N 12/12; valid 1	1.007 [1.005, 1.009] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.121 [1.094, 1.153] N 12/12; valid 1	1.001 [0.9976, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.298 [1.265, 1.33] N 12/12; valid 1	1 [0.9938, 1.006] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_470_S0_H3_fingerprints_longest_cadence_x2_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x2 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1726, low 0; forced mean no	no eligible check multiple fronts; H 3200 multi 0.5119, low 0; forced mean no	1.434 [1.397, 1.473] N 12/12; valid 1	0.9973 [0.9902, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.567 [1.538, 1.592] N 12/12; valid 1	1.012 [1, 1.024] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5596, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7444, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6049, low 0; forced mean no	1.701 [1.68, 1.722] N 12/12; valid 1	1.033 [1.018, 1.05] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.762 [1.704, 1.821] N 2/2; valid 1	1.172 [1.093, 1.26] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=1	not declared	not declared	not declared	1.759 [1.752, 1.766] N 2/2; valid 1	1.25 [1.225, 1.276] N 2/2; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_471_S0_H3_fingerprints_longest_cadence_x2_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_472_S0_H3_pairs_longest_cadence_x2_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_473_S0_H3_pairs_longest_cadence_x2_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_474_S0_H3_pairs_longest_cadence_x2_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_475_S0_H3_pairs_longest_cadence_x2_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_476_S0_H3_pairs_longest_cadence_x2_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_477_S0_H3_pairs_longest_cadence_x2_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_478_S0_H3_pairs_longest_cadence_x2_primary_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_479_S0_H3_pairs_longest_cadence_x2_primary_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_480_S0_H3_pairs_longest_cadence_x2_primary_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x2 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.001159 [0.0002991, 0.002239] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.01712 [0.007027, 0.0304] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9892 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.235 [0.1245, 0.3759] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2696, 2696] resolved	[3.2e+04, ∞] right censored	S 0.9798 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.6387 [0.5516, 0.7211] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[505.9, 505.9] resolved	[3.2e+04, ∞] right censored	S 0.9422 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9548 [0.9448, 0.9647] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[128.3, 128.3] resolved	[3.2e+04, ∞] right censored	S 0.9207 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9863 [0.9844, 0.9881] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	criterion met	inconclusive	[83.27, 83.27] resolved	[3.2e+04, ∞] right censored	S 0.9319 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9606 [0.9582, 0.9631] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_481_S0_H3_candidates_longest_cadence_x2_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x2 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9573 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.00401 [0, 0.01127] S low 0.03671 [0, 0.1044] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	criterion met	[475.4, 475.4] resolved	[3.2e+04, ∞] right censored	S 0.2373 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.07594 [0.06543, 0.08484] S low 0.8565 [0.8397, 0.8749] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[563.6, 563.6] resolved	[3.2e+04, ∞] right censored	S 0.131 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.06283 [0.05564, 0.07001] S low 0.9004 [0.8879, 0.9136] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[733.2, 733.2] resolved	[3.2e+04, ∞] right censored	S 0.05288 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.0266 [0.02082, 0.03206] S low 0.9579 [0.9503, 0.9669] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1251, 1251] resolved	[3.2e+04, ∞] right censored	S 0.01525 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.004997 [0.001874, 0.00812] S low 0.9925 [0.9875, 0.9975] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_482_S0_H3_candidates_longest_cadence_x2_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x2 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[3.2e+04, ∞] right censored	S 0 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.0001302 [0, 0.0003906] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.0004164 [0, 0.001145] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 0.979 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.1969 [0.03435, 0.3848] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3903, 3903] resolved	[3.2e+04, ∞] right censored	S 0.9796 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.4837 [0.3914, 0.5751] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2012, 2012] resolved	[3.2e+04, ∞] right censored	S 0.985 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.7161 [0.6474, 0.7783] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[986.9, 986.9] resolved	[3.2e+04, ∞] right censored	S 0.9749 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.857 [0.832, 0.8798] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[388.5, 388.5] resolved	[3.2e+04, ∞] right censored	S 0.9627 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.919 [0.9007, 0.9373] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[104, 104] resolved	[3.2e+04, ∞] right censored	S 0.9616 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9335 [0.9332, 0.9338] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_483_S0_H3_candidates_longest_cadence_x2_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x2 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	criterion met	inconclusive	[51.19, 51.19] resolved	[3.2e+04, ∞] right censored	S 0.9708 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.852 [0.8463, 0.8576] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[3200, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[3200, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[800, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.0007287 [5.205e-05, 0.001666] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	criterion met	inconclusive	[533.5, 533.5] resolved	[3200, ∞] right censored	S 0.9981 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.02363 [0.006558, 0.04497] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[145.8, 145.8] resolved	[3200, ∞] right censored	S 0.9922 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.1366 [0.108, 0.1706] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[37.9, 37.9] resolved	[3200, ∞] right censored	S 0.9847 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.3533 [0.3188, 0.3849] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.18, 18.18] resolved	[3200, ∞] right censored	S 0.9823 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.4987 [0.4698, 0.5309] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.66, 10.66] resolved	[3200, ∞] right censored	S 0.9803 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.658 [0.6436, 0.6719] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.808, 6.808] resolved	[3200, ∞] right censored	S 0.9744 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7891 [0.7747, 0.8021] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	criterion met	inconclusive	[4.953, 4.953] resolved	[3200, ∞] right censored	S 0.9749 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.842 [0.8407, 0.8432] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_484_S0_H3_candidates_longest_cadence_x2_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x2 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	criterion met	inconclusive	[4.101, 4.101] resolved	[3200, ∞] right censored	S 0.9887 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7633 [0.7395, 0.787] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_485_S0_H3_candidates_longest_cadence_x2_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x2 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1.001 [1, 1.002] N 12/12; valid 1	1.002 [1.002, 1.003] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.011 [1.002, 1.023] N 12/12; valid 1	1.005 [1.004, 1.007] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8682, low 0; forced mean no	1.212 [1.09, 1.34] N 12/12; valid 1	1.009 [1.007, 1.01] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.668 [1.558, 1.771] N 12/12; valid 1	1.009 [1.006, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_486_S0_H3_fingerprints_longest_cadence_x2_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x2 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3856, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9124, low 0; forced mean no	2.309 [2.248, 2.367] N 12/12; valid 1	1.024 [1.017, 1.031] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.482 [2.376, 2.593] N 2/2; valid 1	1.124 [1.11, 1.139] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.507 [2.432, 2.583] N 2/2; valid 1	1.25 [1.233, 1.268] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.008, 1.008] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.029 [1.029, 1.03] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.113, 1.117] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9581 [0.936, 0.9784] N 12/12; valid 1	1.484 [1.393, 1.597] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7913 [0.7772, 0.8073] N 12/12; valid 1	3.342 [3.279, 3.408] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_487_S0_H3_fingerprints_longest_cadence_x2_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x2 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.01546; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1406; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.8003 [0.7627, 0.8608] N 12/12; valid 1	2.556 [2.371, 2.688] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.03	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	single front; H 32000 qualified yes; same no multi 0.008224, low 0; forced mean yes	1.454 [1.386, 1.527] N 12/12; valid 1	0.4761 [0.4656, 0.4867] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.05	not declared	not declared	not declared	1.348 [1.277, 1.415] N 12/12; valid 1	0.4608 [0.45, 0.4716] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.1	not declared	not declared	not declared	1.206 [1.133, 1.282] N 12/12; valid 1	0.4371 [0.4272, 0.4482] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.3	not declared	not declared	not declared	0.654 [0.6049, 0.7068] N 2/2; valid 1	0.4247 [0.4159, 0.4336] N 2/2; valid 1	count yes contrast yes
Goryachev f=1	not declared	not declared	not declared	0.07134 [0.05671, 0.08629] N 12/2; valid 1	0.4328 [0.4281, 0.4376] N 12/2; valid 1	count yes contrast yes
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_488_S0_H3_fingerprints_longest_cadence_x2_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x2 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.008] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.194 [1.022, 1.375] N 12/12; valid 1	1.013 [1.012, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.509 [1.389, 1.617] N 12/12; valid 1	1.029 [1.026, 1.031] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.01145, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8576, low 0; forced mean no	1.771 [1.657, 1.891] N 12/12; valid 1	1.049 [1.045, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.012 [1.942, 2.085] N 12/12; valid 1	1.076 [1.07, 1.083] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7007, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8082, low 0; forced mean no	2.11 [2.05, 2.176] N 12/12; valid 1	1.134 [1.123, 1.146] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_489_S0_H3_fingerprints_longest_cadence_x2_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x2 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.354 [2.32, 2.389] N 2/2; valid 1	1.305 [1.285, 1.325] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.215 [2.078, 2.359] N 2/2; valid 1	1.692 [1.633, 1.755] N 2/2; valid 1	count yes contrast yes
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.021 [1.004, 1.043] N 12/12; valid 1	1.007 [1.005, 1.009] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.121 [1.094, 1.153] N 12/12; valid 1	1.001 [0.9976, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.298 [1.265, 1.33] N 12/12; valid 1	1 [0.9938, 1.006] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_490_S0_H3_fingerprints_longest_cadence_x2_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x2 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1726, low 0; forced mean no	no eligible check multiple fronts; H 3200 multi 0.5119, low 0; forced mean no	1.434 [1.397, 1.473] N 12/12; valid 1	0.9973 [0.9902, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.567 [1.538, 1.592] N 12/12; valid 1	1.012 [1, 1.024] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5596, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7444, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6049, low 0; forced mean no	1.701 [1.68, 1.722] N 12/12; valid 1	1.033 [1.018, 1.05] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.762 [1.704, 1.821] N 2/2; valid 1	1.172 [1.093, 1.26] N 2/2; valid 1	count yes contrast yes
Spring f=1	not declared	not declared	not declared	1.759 [1.752, 1.766] N 2/2; valid 1	1.25 [1.225, 1.276] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_491_S0_H3_fingerprints_longest_cadence_x2_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / criterion met	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_492_S0_H3_pairs_longest_cadence_x2_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_493_S0_H3_pairs_longest_cadence_x2_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_494_S0_H3_pairs_longest_cadence_x2_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_495_S0_H3_pairs_longest_cadence_x2_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_496_S0_H3_pairs_longest_cadence_x2_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_497_S0_H3_pairs_longest_cadence_x2_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_498_S0_H3_pairs_longest_cadence_x2_inclusive_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_499_S0_H3_pairs_longest_cadence_x2_inclusive_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_500_S0_H3_pairs_longest_cadence_x2_inclusive_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x4 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.001172 [0.0002603, 0.002292] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.01737 [0.007176, 0.03059] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 0.989 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.2348 [0.1243, 0.3759] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2696, 2696] resolved	[3.2e+04, ∞] right censored	S 0.9804 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.6395 [0.5528, 0.7213] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[504.7, 504.7] resolved	[3.2e+04, ∞] right censored	S 0.9445 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.954 [0.9441, 0.964] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[128.6, 128.6] resolved	[3.2e+04, ∞] right censored	S 0.9195 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9863 [0.985, 0.9875] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	inconclusive	inconclusive	[84, 84] resolved	[3.2e+04, ∞] right censored	S 0.9341 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.9638 [0.96, 0.9675] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_501_S0_H3_candidates_longest_cadence_x4_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x4 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9575 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.004035 [0, 0.01135] S low 0.03676 [0, 0.1045] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	inconclusive	[478, 478] resolved	[3.2e+04, ∞] right censored	S 0.2386 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.07563 [0.06471, 0.0849] S low 0.8562 [0.8387, 0.8753] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[563.3, 563.3] resolved	[3.2e+04, ∞] right censored	S 0.13 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.06149 [0.05379, 0.06898] S low 0.9012 [0.8878, 0.915] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[736.8, 736.8] resolved	[3.2e+04, ∞] right censored	S 0.05301 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.02643 [0.0205, 0.03215] S low 0.9581 [0.9501, 0.9672] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1264, 1264] resolved	[3.2e+04, ∞] right censored	S 0.01493 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.005618 [0.002497, 0.008739] S low 0.9919 [0.9863, 0.9975] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_502_S0_H3_candidates_longest_cadence_x4_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x4 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[3.2e+04, ∞] right censored	S 0 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.0001302 [0, 0.0003905] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.0004161 [0, 0.001248] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9788 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.1964 [0.03391, 0.3844] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3907, 3907] resolved	[3.2e+04, ∞] right censored	S 0.9785 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.4842 [0.3916, 0.5754] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2013, 2013] resolved	[3.2e+04, ∞] right censored	S 0.9844 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.7165 [0.6475, 0.7785] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[985.7, 985.7] resolved	[3.2e+04, ∞] right censored	S 0.975 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.8579 [0.8328, 0.8807] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[390.9, 390.9] resolved	[3.2e+04, ∞] right censored	S 0.9621 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9184 [0.9004, 0.9365] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[104.4, 104.4] resolved	[3.2e+04, ∞] right censored	S 0.9597 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9332 [0.9313, 0.9351] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_503_S0_H3_candidates_longest_cadence_x4_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x4 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	inconclusive	inconclusive	[52.68, 52.68] resolved	[3.2e+04, ∞] right censored	S 0.9712 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.8508 [0.8452, 0.8564] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[3200, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[3200, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[800, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.0006242 [0, 0.001457] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	criterion met	inconclusive	[533.5, 533.5] resolved	[3200, ∞] right censored	S 0.998 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.02362 [0.006656, 0.04484] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[145.8, 145.8] resolved	[3200, ∞] right censored	S 0.9919 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.1363 [0.1071, 0.1705] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[37.89, 37.89] resolved	[3200, ∞] right censored	S 0.9848 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.3548 [0.3199, 0.3867] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.15, 18.15] resolved	[3200, ∞] right censored	S 0.983 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.4988 [0.4698, 0.5304] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.72, 10.72] resolved	[3200, ∞] right censored	S 0.9789 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.6572 [0.6426, 0.6712] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.891, 6.891] resolved	[3200, ∞] right censored	S 0.9746 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7887 [0.7737, 0.8026] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	inconclusive	inconclusive	[5.058, 5.058] resolved	[3200, ∞] right censored	S 0.9724 / W 1 H 3200 / 12800 95% bound check: ?	S multi 0.8452 [0.8427, 0.8477] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_504_S0_H3_candidates_longest_cadence_x4_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x4 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	inconclusive	inconclusive	[4.068, 4.068] resolved	[3200, ∞] right censored	S 0.9862 / W 1 H 3200 / 12800 95% bound check: ?	S multi 0.7665 [0.7403, 0.7928] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_505_S0_H3_candidates_longest_cadence_x4_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x4 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1.001 [1, 1.002] N 12/12; valid 1	1.002 [1.002, 1.003] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.011 [1.002, 1.023] N 12/12; valid 1	1.005 [1.004, 1.007] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8685, low 0; forced mean no	1.212 [1.09, 1.34] N 12/12; valid 1	1.009 [1.007, 1.01] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.668 [1.558, 1.771] N 12/12; valid 1	1.009 [1.006, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_506_S0_H3_fingerprints_longest_cadence_x4_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x4 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3855, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9143, low 0; forced mean no	2.309 [2.248, 2.367] N 12/12; valid 1	1.024 [1.017, 1.031] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.482 [2.376, 2.593] N 2/2; valid 1	1.124 [1.11, 1.139] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.507 [2.432, 2.583] N 2/2; valid 1	1.25 [1.233, 1.268] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.008, 1.008] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.029 [1.029, 1.03] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.113, 1.117] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9581 [0.936, 0.9784] N 12/12; valid 1	1.484 [1.393, 1.597] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7913 [0.7772, 0.8073] N 12/12; valid 1	3.342 [3.279, 3.408] N 12/12; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_507_S0_H3_fingerprints_longest_cadence_x4_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x4 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.0155; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1407; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.8003 [0.7627, 0.8608] N 12/12; valid 1	2.556 [2.371, 2.688] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.03	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check single front; H 32000 multi 0.008219, low 0; forced mean yes	1.454 [1.386, 1.527] N 12/12; valid 1	0.4761 [0.4656, 0.4867] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.05	not declared	not declared	not declared	1.348 [1.277, 1.415] N 12/12; valid 1	0.4608 [0.45, 0.4716] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.1	not declared	not declared	not declared	1.206 [1.133, 1.282] N 12/12; valid 1	0.4371 [0.4272, 0.4482] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.3	not declared	not declared	not declared	0.654 [0.6049, 0.7068] N 2/2; valid 1	0.4247 [0.4159, 0.4336] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=1	not declared	not declared	not declared	0.07134 [0.05671, 0.08629] N 12/2; valid 1	0.4328 [0.4281, 0.4376] N 12/2; valid 1	count unavailable contrast unavailable
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_508_S0_H3_fingerprints_longest_cadence_x4_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x4 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.008] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.194 [1.022, 1.375] N 12/12; valid 1	1.013 [1.012, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.509 [1.389, 1.617] N 12/12; valid 1	1.029 [1.026, 1.031] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.01144, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8553, low 0; forced mean no	1.771 [1.657, 1.891] N 12/12; valid 1	1.049 [1.045, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.012 [1.942, 2.085] N 12/12; valid 1	1.076 [1.07, 1.083] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7007, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8081, low 0; forced mean no	2.11 [2.05, 2.176] N 12/12; valid 1	1.134 [1.123, 1.146] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_509_S0_H3_fingerprints_longest_cadence_x4_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x4 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.354 [2.32, 2.389] N 2/2; valid 1	1.305 [1.285, 1.325] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.215 [2.078, 2.359] N 2/2; valid 1	1.692 [1.633, 1.755] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.021 [1.004, 1.043] N 12/12; valid 1	1.007 [1.005, 1.009] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.121 [1.094, 1.153] N 12/12; valid 1	1.001 [0.9976, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.298 [1.265, 1.33] N 12/12; valid 1	1 [0.9938, 1.006] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_510_S0_H3_fingerprints_longest_cadence_x4_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x4 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1728, low 0; forced mean no	no eligible check multiple fronts; H 3200 multi 0.5112, low 0; forced mean no	1.434 [1.397, 1.473] N 12/12; valid 1	0.9973 [0.9902, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.567 [1.538, 1.592] N 12/12; valid 1	1.012 [1, 1.024] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5605, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.743, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6073, low 0; forced mean no	1.701 [1.68, 1.722] N 12/12; valid 1	1.033 [1.018, 1.05] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.762 [1.704, 1.821] N 2/2; valid 1	1.172 [1.093, 1.26] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=1	not declared	not declared	not declared	1.759 [1.752, 1.766] N 2/2; valid 1	1.25 [1.225, 1.276] N 2/2; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_511_S0_H3_fingerprints_longest_cadence_x4_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_512_S0_H3_pairs_longest_cadence_x4_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_513_S0_H3_pairs_longest_cadence_x4_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_514_S0_H3_pairs_longest_cadence_x4_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_515_S0_H3_pairs_longest_cadence_x4_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_516_S0_H3_pairs_longest_cadence_x4_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_517_S0_H3_pairs_longest_cadence_x4_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_518_S0_H3_pairs_longest_cadence_x4_primary_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_519_S0_H3_pairs_longest_cadence_x4_primary_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_520_S0_H3_pairs_longest_cadence_x4_primary_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x4 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.001172 [0.0002603, 0.002292] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.01737 [0.007176, 0.03059] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 0.989 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.2348 [0.1243, 0.3759] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2696, 2696] resolved	[3.2e+04, ∞] right censored	S 0.9804 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.6395 [0.5528, 0.7213] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[504.7, 504.7] resolved	[3.2e+04, ∞] right censored	S 0.9445 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.954 [0.9441, 0.964] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[128.6, 128.6] resolved	[3.2e+04, ∞] right censored	S 0.9195 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9863 [0.985, 0.9875] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	criterion met	inconclusive	[84, 84] resolved	[3.2e+04, ∞] right censored	S 0.9341 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9638 [0.96, 0.9675] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_521_S0_H3_candidates_longest_cadence_x4_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x4 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9575 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.004035 [0, 0.01135] S low 0.03676 [0, 0.1045] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	criterion met	[478, 478] resolved	[3.2e+04, ∞] right censored	S 0.2386 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.07563 [0.06471, 0.0849] S low 0.8562 [0.8387, 0.8753] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[563.3, 563.3] resolved	[3.2e+04, ∞] right censored	S 0.13 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.06149 [0.05379, 0.06898] S low 0.9012 [0.8878, 0.915] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[736.8, 736.8] resolved	[3.2e+04, ∞] right censored	S 0.05301 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.02643 [0.0205, 0.03215] S low 0.9581 [0.9501, 0.9672] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1264, 1264] resolved	[3.2e+04, ∞] right censored	S 0.01493 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.005618 [0.002497, 0.008739] S low 0.9919 [0.9863, 0.9975] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_522_S0_H3_candidates_longest_cadence_x4_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x4 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[3.2e+04, ∞] right censored	S 0 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.0001302 [0, 0.0003905] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.0004161 [0, 0.001248] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9788 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.1964 [0.03391, 0.3844] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3907, 3907] resolved	[3.2e+04, ∞] right censored	S 0.9785 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.4842 [0.3916, 0.5754] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2013, 2013] resolved	[3.2e+04, ∞] right censored	S 0.9844 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.7165 [0.6475, 0.7785] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[985.7, 985.7] resolved	[3.2e+04, ∞] right censored	S 0.975 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.8579 [0.8328, 0.8807] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[390.9, 390.9] resolved	[3.2e+04, ∞] right censored	S 0.9621 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9184 [0.9004, 0.9365] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[104.4, 104.4] resolved	[3.2e+04, ∞] right censored	S 0.9597 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9332 [0.9313, 0.9351] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_523_S0_H3_candidates_longest_cadence_x4_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x4 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	criterion met	inconclusive	[52.68, 52.68] resolved	[3.2e+04, ∞] right censored	S 0.9712 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.8508 [0.8452, 0.8564] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[3200, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[3200, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[800, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.0006242 [0, 0.001457] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	criterion met	inconclusive	[533.5, 533.5] resolved	[3200, ∞] right censored	S 0.998 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.02362 [0.006656, 0.04484] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[145.8, 145.8] resolved	[3200, ∞] right censored	S 0.9919 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.1363 [0.1071, 0.1705] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[37.89, 37.89] resolved	[3200, ∞] right censored	S 0.9848 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.3548 [0.3199, 0.3867] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.15, 18.15] resolved	[3200, ∞] right censored	S 0.983 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.4988 [0.4698, 0.5304] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.72, 10.72] resolved	[3200, ∞] right censored	S 0.9789 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.6572 [0.6426, 0.6712] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.891, 6.891] resolved	[3200, ∞] right censored	S 0.9746 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7887 [0.7737, 0.8026] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	criterion met	inconclusive	[5.058, 5.058] resolved	[3200, ∞] right censored	S 0.9724 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.8452 [0.8427, 0.8477] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_524_S0_H3_candidates_longest_cadence_x4_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x4 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	criterion met	inconclusive	[4.068, 4.068] resolved	[3200, ∞] right censored	S 0.9862 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7665 [0.7403, 0.7928] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_525_S0_H3_candidates_longest_cadence_x4_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x4 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1.001 [1, 1.002] N 12/12; valid 1	1.002 [1.002, 1.003] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.011 [1.002, 1.023] N 12/12; valid 1	1.005 [1.004, 1.007] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8685, low 0; forced mean no	1.212 [1.09, 1.34] N 12/12; valid 1	1.009 [1.007, 1.01] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.668 [1.558, 1.771] N 12/12; valid 1	1.009 [1.006, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_526_S0_H3_fingerprints_longest_cadence_x4_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x4 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3855, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9143, low 0; forced mean no	2.309 [2.248, 2.367] N 12/12; valid 1	1.024 [1.017, 1.031] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.482 [2.376, 2.593] N 2/2; valid 1	1.124 [1.11, 1.139] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.507 [2.432, 2.583] N 2/2; valid 1	1.25 [1.233, 1.268] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.008, 1.008] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.029 [1.029, 1.03] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.113, 1.117] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9581 [0.936, 0.9784] N 12/12; valid 1	1.484 [1.393, 1.597] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7913 [0.7772, 0.8073] N 12/12; valid 1	3.342 [3.279, 3.408] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_527_S0_H3_fingerprints_longest_cadence_x4_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x4 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.0155; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1407; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.8003 [0.7627, 0.8608] N 12/12; valid 1	2.556 [2.371, 2.688] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.03	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	single front; H 32000 qualified yes; same no multi 0.008219, low 0; forced mean yes	1.454 [1.386, 1.527] N 12/12; valid 1	0.4761 [0.4656, 0.4867] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.05	not declared	not declared	not declared	1.348 [1.277, 1.415] N 12/12; valid 1	0.4608 [0.45, 0.4716] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.1	not declared	not declared	not declared	1.206 [1.133, 1.282] N 12/12; valid 1	0.4371 [0.4272, 0.4482] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.3	not declared	not declared	not declared	0.654 [0.6049, 0.7068] N 2/2; valid 1	0.4247 [0.4159, 0.4336] N 2/2; valid 1	count yes contrast yes
Goryachev f=1	not declared	not declared	not declared	0.07134 [0.05671, 0.08629] N 12/2; valid 1	0.4328 [0.4281, 0.4376] N 12/2; valid 1	count yes contrast yes
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_528_S0_H3_fingerprints_longest_cadence_x4_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x4 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.008] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.194 [1.022, 1.375] N 12/12; valid 1	1.013 [1.012, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.509 [1.389, 1.617] N 12/12; valid 1	1.029 [1.026, 1.031] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.01144, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8553, low 0; forced mean no	1.771 [1.657, 1.891] N 12/12; valid 1	1.049 [1.045, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.012 [1.942, 2.085] N 12/12; valid 1	1.076 [1.07, 1.083] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7007, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8081, low 0; forced mean no	2.11 [2.05, 2.176] N 12/12; valid 1	1.134 [1.123, 1.146] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_529_S0_H3_fingerprints_longest_cadence_x4_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x4 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.354 [2.32, 2.389] N 2/2; valid 1	1.305 [1.285, 1.325] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.215 [2.078, 2.359] N 2/2; valid 1	1.692 [1.633, 1.755] N 2/2; valid 1	count yes contrast yes
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.021 [1.004, 1.043] N 12/12; valid 1	1.007 [1.005, 1.009] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.121 [1.094, 1.153] N 12/12; valid 1	1.001 [0.9976, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.298 [1.265, 1.33] N 12/12; valid 1	1 [0.9938, 1.006] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_530_S0_H3_fingerprints_longest_cadence_x4_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x4 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1728, low 0; forced mean no	no eligible check multiple fronts; H 3200 multi 0.5112, low 0; forced mean no	1.434 [1.397, 1.473] N 12/12; valid 1	0.9973 [0.9902, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.567 [1.538, 1.592] N 12/12; valid 1	1.012 [1, 1.024] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5605, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.743, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6073, low 0; forced mean no	1.701 [1.68, 1.722] N 12/12; valid 1	1.033 [1.018, 1.05] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.762 [1.704, 1.821] N 2/2; valid 1	1.172 [1.093, 1.26] N 2/2; valid 1	count yes contrast yes
Spring f=1	not declared	not declared	not declared	1.759 [1.752, 1.766] N 2/2; valid 1	1.25 [1.225, 1.276] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_531_S0_H3_fingerprints_longest_cadence_x4_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / criterion met	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_532_S0_H3_pairs_longest_cadence_x4_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_533_S0_H3_pairs_longest_cadence_x4_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_534_S0_H3_pairs_longest_cadence_x4_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_535_S0_H3_pairs_longest_cadence_x4_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_536_S0_H3_pairs_longest_cadence_x4_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_537_S0_H3_pairs_longest_cadence_x4_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_538_S0_H3_pairs_longest_cadence_x4_inclusive_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_539_S0_H3_pairs_longest_cadence_x4_inclusive_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_540_S0_H3_pairs_longest_cadence_x4_inclusive_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x8 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.001093 [0.0002082, 0.002186] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.01808 [0.008105, 0.03096] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9888 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.2344 [0.1236, 0.3753] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2702, 2702] resolved	[3.2e+04, ∞] right censored	S 0.98 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.6394 [0.5521, 0.7211] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[504.9, 504.9] resolved	[3.2e+04, ∞] right censored	S 0.9444 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.953 [0.9426, 0.9634] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[130.7, 130.7] resolved	[3.2e+04, ∞] right censored	S 0.9131 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.99 [0.985, 0.995] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	inconclusive	inconclusive	[82.42, 82.42] resolved	[3.2e+04, ∞] right censored	S 0.9286 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.9663 [0.9626, 0.9701] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_541_S0_H3_candidates_longest_cadence_x8_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x8 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9577 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.00406 [0, 0.0114] S low 0.0368 [0, 0.1046] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	inconclusive	[479.3, 479.3] resolved	[3.2e+04, ∞] right censored	S 0.2363 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.07461 [0.06234, 0.08541] S low 0.8585 [0.841, 0.8786] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[556.9, 556.9] resolved	[3.2e+04, ∞] right censored	S 0.1307 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.06234 [0.05486, 0.07004] S low 0.9009 [0.8872, 0.9156] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[767.4, 767.4] resolved	[3.2e+04, ∞] right censored	S 0.05192 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.02702 [0.02078, 0.03304] S low 0.9572 [0.9493, 0.9663] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1269, 1269] resolved	[3.2e+04, ∞] right censored	S 0.01491 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.004988 [0.002494, 0.007481] S low 0.9925 [0.9875, 0.9975] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_542_S0_H3_candidates_longest_cadence_x8_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x8 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[3.2e+04, ∞] right censored	S 0 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.0001562 [0, 0.0004685] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.0004156 [0, 0.001247] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9797 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.1966 [0.0349, 0.3847] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3906, 3906] resolved	[3.2e+04, ∞] right censored	S 0.9783 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.4823 [0.3896, 0.5738] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2015, 2015] resolved	[3.2e+04, ∞] right censored	S 0.9841 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.718 [0.6498, 0.7791] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[981.7, 981.7] resolved	[3.2e+04, ∞] right censored	S 0.9783 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.8603 [0.8356, 0.8832] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[392.1, 392.1] resolved	[3.2e+04, ∞] right censored	S 0.9591 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9175 [0.899, 0.936] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[104.9, 104.9] resolved	[3.2e+04, ∞] right censored	S 0.9648 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9302 [0.9252, 0.9352] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_543_S0_H3_candidates_longest_cadence_x8_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x8 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	inconclusive	inconclusive	[0, 80] unresolved below two cadences	[3.2e+04, ∞] right censored	S 0.9724 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.8354 [0.8204, 0.8504] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[3200, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[3200, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[800, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.0008313 [0, 0.00187] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	criterion met	inconclusive	[534.3, 534.3] resolved	[3200, ∞] right censored	S 0.9978 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.02328 [0.006234, 0.04489] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[146.2, 146.2] resolved	[3200, ∞] right censored	S 0.9913 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.1357 [0.1062, 0.1704] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[37.83, 37.83] resolved	[3200, ∞] right censored	S 0.984 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.3554 [0.3215, 0.3874] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.25, 18.25] resolved	[3200, ∞] right censored	S 0.9838 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.4985 [0.4701, 0.5299] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.83, 10.83] resolved	[3200, ∞] right censored	S 0.9799 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.6586 [0.6424, 0.6735] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[0, 8] unresolved below two cadences	[3200, ∞] right censored	S 0.9745 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.787 [0.7695, 0.802] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	inconclusive	inconclusive	[0, 8] unresolved below two cadences	[3200, ∞] right censored	S 0.9599 / W 1 H 3200 / 12800 95% bound check: ?	S multi 0.8628 [0.8454, 0.8803] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_544_S0_H3_candidates_longest_cadence_x8_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x8 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	inconclusive	inconclusive	[0, 8] unresolved below two cadences	[3200, ∞] right censored	S 0.9925 / W 1 H 3200 / 12800 95% bound check: ?	S multi 0.7793 [0.7456, 0.813] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_545_S0_H3_candidates_longest_cadence_x8_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x8 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1.001 [1, 1.002] N 12/12; valid 1	1.002 [1.002, 1.003] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.011 [1.002, 1.023] N 12/12; valid 1	1.005 [1.004, 1.007] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.868, low 0; forced mean no	1.212 [1.09, 1.34] N 12/12; valid 1	1.009 [1.007, 1.01] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.668 [1.558, 1.771] N 12/12; valid 1	1.009 [1.006, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_546_S0_H3_fingerprints_longest_cadence_x8_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x8 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3859, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9092, low 0; forced mean no	2.309 [2.248, 2.367] N 12/12; valid 1	1.024 [1.017, 1.031] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.482 [2.376, 2.593] N 2/2; valid 1	1.124 [1.11, 1.139] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.507 [2.432, 2.583] N 2/2; valid 1	1.25 [1.233, 1.268] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.008, 1.008] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.029 [1.029, 1.03] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.113, 1.117] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9581 [0.936, 0.9784] N 12/12; valid 1	1.484 [1.393, 1.597] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7913 [0.7772, 0.8073] N 12/12; valid 1	3.342 [3.279, 3.408] N 12/12; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_547_S0_H3_fingerprints_longest_cadence_x8_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x8 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.01559; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1407; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.8003 [0.7627, 0.8608] N 12/12; valid 1	2.556 [2.371, 2.688] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.03	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check single front; H 32000 multi 0.008105, low 0; forced mean yes	1.454 [1.386, 1.527] N 12/12; valid 1	0.4761 [0.4656, 0.4867] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.05	not declared	not declared	not declared	1.348 [1.277, 1.415] N 12/12; valid 1	0.4608 [0.45, 0.4716] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.1	not declared	not declared	not declared	1.206 [1.133, 1.282] N 12/12; valid 1	0.4371 [0.4272, 0.4482] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.3	not declared	not declared	not declared	0.654 [0.6049, 0.7068] N 2/2; valid 1	0.4247 [0.4159, 0.4336] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=1	not declared	not declared	not declared	0.07134 [0.05671, 0.08629] N 12/2; valid 1	0.4328 [0.4281, 0.4376] N 12/2; valid 1	count unavailable contrast unavailable
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_548_S0_H3_fingerprints_longest_cadence_x8_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x8 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.008] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.194 [1.022, 1.375] N 12/12; valid 1	1.013 [1.012, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.509 [1.389, 1.617] N 12/12; valid 1	1.029 [1.026, 1.031] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.01143, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8543, low 0; forced mean no	1.771 [1.657, 1.891] N 12/12; valid 1	1.049 [1.045, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.012 [1.942, 2.085] N 12/12; valid 1	1.076 [1.07, 1.083] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7014, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8103, low 0; forced mean no	2.11 [2.05, 2.176] N 12/12; valid 1	1.134 [1.123, 1.146] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_549_S0_H3_fingerprints_longest_cadence_x8_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x8 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.354 [2.32, 2.389] N 2/2; valid 1	1.305 [1.285, 1.325] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.215 [2.078, 2.359] N 2/2; valid 1	1.692 [1.633, 1.755] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.021 [1.004, 1.043] N 12/12; valid 1	1.007 [1.005, 1.009] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.121 [1.094, 1.153] N 12/12; valid 1	1.001 [0.9976, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.298 [1.265, 1.33] N 12/12; valid 1	1 [0.9938, 1.006] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_550_S0_H3_fingerprints_longest_cadence_x8_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x8 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1727, low 0; forced mean no	no eligible check multiple fronts; H 3200 multi 0.5106, low 0; forced mean no	1.434 [1.397, 1.473] N 12/12; valid 1	0.9973 [0.9902, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.567 [1.538, 1.592] N 12/12; valid 1	1.012 [1, 1.024] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5576, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7423, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6027, low 0; forced mean no	1.701 [1.68, 1.722] N 12/12; valid 1	1.033 [1.018, 1.05] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.762 [1.704, 1.821] N 2/2; valid 1	1.172 [1.093, 1.26] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=1	not declared	not declared	not declared	1.759 [1.752, 1.766] N 2/2; valid 1	1.25 [1.225, 1.276] N 2/2; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_551_S0_H3_fingerprints_longest_cadence_x8_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_552_S0_H3_pairs_longest_cadence_x8_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_553_S0_H3_pairs_longest_cadence_x8_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_554_S0_H3_pairs_longest_cadence_x8_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_555_S0_H3_pairs_longest_cadence_x8_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_556_S0_H3_pairs_longest_cadence_x8_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_557_S0_H3_pairs_longest_cadence_x8_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_558_S0_H3_pairs_longest_cadence_x8_primary_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_559_S0_H3_pairs_longest_cadence_x8_primary_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_560_S0_H3_pairs_longest_cadence_x8_primary_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x8 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.001093 [0.0002082, 0.002186] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.01808 [0.008105, 0.03096] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9888 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.2344 [0.1236, 0.3753] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2702, 2702] resolved	[3.2e+04, ∞] right censored	S 0.98 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.6394 [0.5521, 0.7211] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[504.9, 504.9] resolved	[3.2e+04, ∞] right censored	S 0.9444 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.953 [0.9426, 0.9634] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[130.7, 130.7] resolved	[3.2e+04, ∞] right censored	S 0.9131 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.99 [0.985, 0.995] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	criterion met	inconclusive	[82.42, 82.42] resolved	[3.2e+04, ∞] right censored	S 0.9286 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9663 [0.9626, 0.9701] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_561_S0_H3_candidates_longest_cadence_x8_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x8 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9577 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.00406 [0, 0.0114] S low 0.0368 [0, 0.1046] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	criterion met	[479.3, 479.3] resolved	[3.2e+04, ∞] right censored	S 0.2363 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.07461 [0.06234, 0.08541] S low 0.8585 [0.841, 0.8786] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[556.9, 556.9] resolved	[3.2e+04, ∞] right censored	S 0.1307 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.06234 [0.05486, 0.07004] S low 0.9009 [0.8872, 0.9156] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[767.4, 767.4] resolved	[3.2e+04, ∞] right censored	S 0.05192 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.02702 [0.02078, 0.03304] S low 0.9572 [0.9493, 0.9663] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1269, 1269] resolved	[3.2e+04, ∞] right censored	S 0.01491 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.004988 [0.002494, 0.007481] S low 0.9925 [0.9875, 0.9975] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_562_S0_H3_candidates_longest_cadence_x8_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x8 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[3.2e+04, ∞] right censored	S 0 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.0001562 [0, 0.0004685] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.0004156 [0, 0.001247] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9797 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.1966 [0.0349, 0.3847] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3906, 3906] resolved	[3.2e+04, ∞] right censored	S 0.9783 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.4823 [0.3896, 0.5738] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2015, 2015] resolved	[3.2e+04, ∞] right censored	S 0.9841 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.718 [0.6498, 0.7791] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[981.7, 981.7] resolved	[3.2e+04, ∞] right censored	S 0.9783 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.8603 [0.8356, 0.8832] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[392.1, 392.1] resolved	[3.2e+04, ∞] right censored	S 0.9591 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9175 [0.899, 0.936] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[104.9, 104.9] resolved	[3.2e+04, ∞] right censored	S 0.9648 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9302 [0.9252, 0.9352] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_563_S0_H3_candidates_longest_cadence_x8_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x8 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	criterion met	inconclusive	[0, 80] unresolved below two cadences	[3.2e+04, ∞] right censored	S 0.9724 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.8354 [0.8204, 0.8504] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[3200, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[3200, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[800, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.0008313 [0, 0.00187] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	criterion met	inconclusive	[534.3, 534.3] resolved	[3200, ∞] right censored	S 0.9978 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.02328 [0.006234, 0.04489] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[146.2, 146.2] resolved	[3200, ∞] right censored	S 0.9913 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.1357 [0.1062, 0.1704] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[37.83, 37.83] resolved	[3200, ∞] right censored	S 0.984 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.3554 [0.3215, 0.3874] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.25, 18.25] resolved	[3200, ∞] right censored	S 0.9838 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.4985 [0.4701, 0.5299] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.83, 10.83] resolved	[3200, ∞] right censored	S 0.9799 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.6586 [0.6424, 0.6735] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[0, 8] unresolved below two cadences	[3200, ∞] right censored	S 0.9745 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.787 [0.7695, 0.802] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	criterion met	inconclusive	[0, 8] unresolved below two cadences	[3200, ∞] right censored	S 0.9599 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.8628 [0.8454, 0.8803] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_564_S0_H3_candidates_longest_cadence_x8_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x8 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	criterion met	inconclusive	[0, 8] unresolved below two cadences	[3200, ∞] right censored	S 0.9925 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7793 [0.7456, 0.813] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_565_S0_H3_candidates_longest_cadence_x8_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x8 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1.001 [1, 1.002] N 12/12; valid 1	1.002 [1.002, 1.003] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.011 [1.002, 1.023] N 12/12; valid 1	1.005 [1.004, 1.007] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.868, low 0; forced mean no	1.212 [1.09, 1.34] N 12/12; valid 1	1.009 [1.007, 1.01] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.668 [1.558, 1.771] N 12/12; valid 1	1.009 [1.006, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_566_S0_H3_fingerprints_longest_cadence_x8_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x8 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3859, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9092, low 0; forced mean no	2.309 [2.248, 2.367] N 12/12; valid 1	1.024 [1.017, 1.031] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.482 [2.376, 2.593] N 2/2; valid 1	1.124 [1.11, 1.139] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.507 [2.432, 2.583] N 2/2; valid 1	1.25 [1.233, 1.268] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.008, 1.008] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.029 [1.029, 1.03] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.113, 1.117] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9581 [0.936, 0.9784] N 12/12; valid 1	1.484 [1.393, 1.597] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7913 [0.7772, 0.8073] N 12/12; valid 1	3.342 [3.279, 3.408] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_567_S0_H3_fingerprints_longest_cadence_x8_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x8 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.01559; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1407; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.8003 [0.7627, 0.8608] N 12/12; valid 1	2.556 [2.371, 2.688] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.03	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	single front; H 32000 qualified yes; same no multi 0.008105, low 0; forced mean yes	1.454 [1.386, 1.527] N 12/12; valid 1	0.4761 [0.4656, 0.4867] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.05	not declared	not declared	not declared	1.348 [1.277, 1.415] N 12/12; valid 1	0.4608 [0.45, 0.4716] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.1	not declared	not declared	not declared	1.206 [1.133, 1.282] N 12/12; valid 1	0.4371 [0.4272, 0.4482] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.3	not declared	not declared	not declared	0.654 [0.6049, 0.7068] N 2/2; valid 1	0.4247 [0.4159, 0.4336] N 2/2; valid 1	count yes contrast yes
Goryachev f=1	not declared	not declared	not declared	0.07134 [0.05671, 0.08629] N 12/2; valid 1	0.4328 [0.4281, 0.4376] N 12/2; valid 1	count yes contrast yes
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_568_S0_H3_fingerprints_longest_cadence_x8_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x8 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.008] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.194 [1.022, 1.375] N 12/12; valid 1	1.013 [1.012, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.509 [1.389, 1.617] N 12/12; valid 1	1.029 [1.026, 1.031] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.01143, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8543, low 0; forced mean no	1.771 [1.657, 1.891] N 12/12; valid 1	1.049 [1.045, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.012 [1.942, 2.085] N 12/12; valid 1	1.076 [1.07, 1.083] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7014, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8103, low 0; forced mean no	2.11 [2.05, 2.176] N 12/12; valid 1	1.134 [1.123, 1.146] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_569_S0_H3_fingerprints_longest_cadence_x8_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x8 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.354 [2.32, 2.389] N 2/2; valid 1	1.305 [1.285, 1.325] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.215 [2.078, 2.359] N 2/2; valid 1	1.692 [1.633, 1.755] N 2/2; valid 1	count yes contrast yes
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.021 [1.004, 1.043] N 12/12; valid 1	1.007 [1.005, 1.009] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.121 [1.094, 1.153] N 12/12; valid 1	1.001 [0.9976, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.298 [1.265, 1.33] N 12/12; valid 1	1 [0.9938, 1.006] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_570_S0_H3_fingerprints_longest_cadence_x8_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x8 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1727, low 0; forced mean no	no eligible check multiple fronts; H 3200 multi 0.5106, low 0; forced mean no	1.434 [1.397, 1.473] N 12/12; valid 1	0.9973 [0.9902, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.567 [1.538, 1.592] N 12/12; valid 1	1.012 [1, 1.024] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5576, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7423, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6027, low 0; forced mean no	1.701 [1.68, 1.722] N 12/12; valid 1	1.033 [1.018, 1.05] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.762 [1.704, 1.821] N 2/2; valid 1	1.172 [1.093, 1.26] N 2/2; valid 1	count yes contrast yes
Spring f=1	not declared	not declared	not declared	1.759 [1.752, 1.766] N 2/2; valid 1	1.25 [1.225, 1.276] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_571_S0_H3_fingerprints_longest_cadence_x8_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / criterion met	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_572_S0_H3_pairs_longest_cadence_x8_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_573_S0_H3_pairs_longest_cadence_x8_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_574_S0_H3_pairs_longest_cadence_x8_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_575_S0_H3_pairs_longest_cadence_x8_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_576_S0_H3_pairs_longest_cadence_x8_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_577_S0_H3_pairs_longest_cadence_x8_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_578_S0_H3_pairs_longest_cadence_x8_inclusive_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_579_S0_H3_pairs_longest_cadence_x8_inclusive_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_580_S0_H3_pairs_longest_cadence_x8_inclusive_009.csv

S0 · Separate geometric qualification · Per-model geometric qualification · common · operationally unforced conditions

Wave-pinning: heterogeneous across inputs
Included 11 / 12 designed inputs; finite qualified time intervals 3; finite qualified mobility intervals 11; conflicts 0.
Time: time CI includes geometry: 1; time below geometry: 2; unavailable: 9.
Mobility: Dx CI includes flat: 1; Dx decreases with size: 7; Dx increases with size: 3; unavailable: 1.
Input strengths by qualifier: compatible with flat Dx: 0.03; less than geometry: 0.05, 0.1, 0.3; more than geometry: 0.0001, 0.0003, 0.001, 0.003, 0.005, 0.01, 0.02; unavailable: 1.

Goryachev: more than geometry
Included 5 / 13 designed inputs; finite qualified time intervals 0; finite qualified mobility intervals 5; conflicts 0.
Time: unavailable: 13.
Mobility: Dx decreases with size: 5; unavailable: 8.
Input strengths by qualifier: more than geometry: 0.0001, 0.0003, 0.001, 0.003, 0.005; unavailable: 0.01, 0.015, 0.02, 0.03, 0.05, 0.1, 0.3, 1.

Separate geometric qualification. Main-view summaries are shown; source CSV retains all seven windows/cadences. Time excess uses direct persistence divided by actual C-ratio squared. Dx ratios convert short-lag global activity-heading fluctuations using actual C; they do not measure one tracked site's diffusion in multi-region states. Qualified single-region literal A3 heading/peak MSDs remain separate. A CI including flat Dx establishes compatibility only, never equivalence. Unbounded/undefined inference stays unavailable; increased Dx, angular persistence advantages, conflicts and heterogeneous inputs all remain visible. These labels do not change the 308 H1–H3 criteria. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S4_S5_final_geometry_20261020_001_S0_geometry_common_primary_1.csv

S0 · Separate geometric qualification · Per-model geometric qualification · common · operationally unforced conditions

Holmes 3: heterogeneous across inputs
Included 11 / 12 designed inputs; finite qualified time intervals 5; finite qualified mobility intervals 11; conflicts 0.
Time: time below geometry: 5; unavailable: 7.
Mobility: Dx CI includes flat: 3; Dx decreases with size: 2; Dx increases with size: 6; unavailable: 1.
Input strengths by qualifier: compatible with flat Dx: 0.0003, 0.003, 0.005; less than geometry: 0.01, 0.02, 0.03, 0.05, 0.1, 0.3; more than geometry: 0.0001, 0.001; unavailable: 1.

Spring: heterogeneous across inputs
Included 10 / 12 designed inputs; finite qualified time intervals 5; finite qualified mobility intervals 10; conflicts 0.
Time: time below geometry: 5; unavailable: 7.
Mobility: Dx CI includes flat: 2; Dx decreases with size: 1; Dx increases with size: 7; unavailable: 2.
Input strengths by qualifier: compatible with flat Dx: 0.001, 0.003; less than geometry: 0.0003, 0.005, 0.01, 0.02, 0.03, 0.05, 0.1; more than geometry: 0.0001; unavailable: 0.3, 1.

Separate geometric qualification. Main-view summaries are shown; source CSV retains all seven windows/cadences. Time excess uses direct persistence divided by actual C-ratio squared. Dx ratios convert short-lag global activity-heading fluctuations using actual C; they do not measure one tracked site's diffusion in multi-region states. Qualified single-region literal A3 heading/peak MSDs remain separate. A CI including flat Dx establishes compatibility only, never equivalence. Unbounded/undefined inference stays unavailable; increased Dx, angular persistence advantages, conflicts and heterogeneous inputs all remain visible. These labels do not change the 308 H1–H3 criteria. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S4_S5_final_geometry_20261020_002_S0_geometry_common_primary_2.csv

S0 · Separate geometric qualification · Per-model geometric qualification · common · all conditions including operationally forced

Wave-pinning: heterogeneous across inputs
Included 12 / 12 designed inputs; finite qualified time intervals 4; finite qualified mobility intervals 12; conflicts 0.
Time: time CI includes geometry: 1; time below geometry: 3; unavailable: 8.
Mobility: Dx CI includes flat: 1; Dx decreases with size: 7; Dx increases with size: 4.
Input strengths by qualifier: compatible with flat Dx: 0.03; less than geometry: 0.05, 0.1, 0.3, 1; more than geometry: 0.0001, 0.0003, 0.001, 0.003, 0.005, 0.01, 0.02.

Goryachev: heterogeneous across inputs
Included 12 / 13 designed inputs; finite qualified time intervals 0; finite qualified mobility intervals 12; conflicts 0.
Time: unavailable: 13.
Mobility: Dx decreases with size: 10; Dx increases with size: 2; unavailable: 1.
Input strengths by qualifier: less than geometry: 0.03, 0.05; more than geometry: 0.0001, 0.0003, 0.001, 0.003, 0.005, 0.01, 0.02, 0.1, 0.3, 1; unavailable: 0.015.

Separate geometric qualification. Main-view summaries are shown; source CSV retains all seven windows/cadences. Time excess uses direct persistence divided by actual C-ratio squared. Dx ratios convert short-lag global activity-heading fluctuations using actual C; they do not measure one tracked site's diffusion in multi-region states. Qualified single-region literal A3 heading/peak MSDs remain separate. A CI including flat Dx establishes compatibility only, never equivalence. Unbounded/undefined inference stays unavailable; increased Dx, angular persistence advantages, conflicts and heterogeneous inputs all remain visible. These labels do not change the 308 H1–H3 criteria. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S4_S5_final_geometry_20261020_003_S0_geometry_common_inclusive_1.csv

S0 · Separate geometric qualification · Per-model geometric qualification · common · all conditions including operationally forced

Holmes 3: heterogeneous across inputs
Included 12 / 12 designed inputs; finite qualified time intervals 6; finite qualified mobility intervals 12; conflicts 0.
Time: time below geometry: 6; unavailable: 6.
Mobility: Dx CI includes flat: 3; Dx decreases with size: 2; Dx increases with size: 7.
Input strengths by qualifier: compatible with flat Dx: 0.0003, 0.003, 0.005; less than geometry: 0.01, 0.02, 0.03, 0.05, 0.1, 0.3, 1; more than geometry: 0.0001, 0.001.

Spring: heterogeneous across inputs
Included 12 / 12 designed inputs; finite qualified time intervals 7; finite qualified mobility intervals 12; conflicts 0.
Time: time below geometry: 7; unavailable: 5.
Mobility: Dx CI includes flat: 2; Dx decreases with size: 1; Dx increases with size: 9.
Input strengths by qualifier: compatible with flat Dx: 0.001, 0.003; less than geometry: 0.0003, 0.005, 0.01, 0.02, 0.03, 0.05, 0.1, 0.3, 1; more than geometry: 0.0001.

Separate geometric qualification. Main-view summaries are shown; source CSV retains all seven windows/cadences. Time excess uses direct persistence divided by actual C-ratio squared. Dx ratios convert short-lag global activity-heading fluctuations using actual C; they do not measure one tracked site's diffusion in multi-region states. Qualified single-region literal A3 heading/peak MSDs remain separate. A CI including flat Dx establishes compatibility only, never equivalence. Unbounded/undefined inference stays unavailable; increased Dx, angular persistence advantages, conflicts and heterogeneous inputs all remain visible. These labels do not change the 308 H1–H3 criteria. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S4_S5_final_geometry_20261020_004_S0_geometry_common_inclusive_2.csv

S0 · Separate geometric qualification · Per-model geometric qualification · longest · operationally unforced conditions

Wave-pinning: heterogeneous across inputs
Included 11 / 12 designed inputs; finite qualified time intervals 3; finite qualified mobility intervals 11; conflicts 0.
Time: time CI includes geometry: 1; time below geometry: 2; unavailable: 9.
Mobility: Dx CI includes flat: 1; Dx decreases with size: 7; Dx increases with size: 3; unavailable: 1.
Input strengths by qualifier: compatible with flat Dx: 0.03; less than geometry: 0.05, 0.1, 0.3; more than geometry: 0.0001, 0.0003, 0.001, 0.003, 0.005, 0.01, 0.02; unavailable: 1.

Goryachev: more than geometry
Included 5 / 13 designed inputs; finite qualified time intervals 0; finite qualified mobility intervals 5; conflicts 0.
Time: unavailable: 13.
Mobility: Dx decreases with size: 5; unavailable: 8.
Input strengths by qualifier: more than geometry: 0.0001, 0.0003, 0.001, 0.003, 0.005; unavailable: 0.01, 0.015, 0.02, 0.03, 0.05, 0.1, 0.3, 1.

Separate geometric qualification. Main-view summaries are shown; source CSV retains all seven windows/cadences. Time excess uses direct persistence divided by actual C-ratio squared. Dx ratios convert short-lag global activity-heading fluctuations using actual C; they do not measure one tracked site's diffusion in multi-region states. Qualified single-region literal A3 heading/peak MSDs remain separate. A CI including flat Dx establishes compatibility only, never equivalence. Unbounded/undefined inference stays unavailable; increased Dx, angular persistence advantages, conflicts and heterogeneous inputs all remain visible. These labels do not change the 308 H1–H3 criteria. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S4_S5_final_geometry_20261020_005_S0_geometry_longest_primary_1.csv

S0 · Separate geometric qualification · Per-model geometric qualification · longest · operationally unforced conditions

Holmes 3: heterogeneous across inputs
Included 11 / 12 designed inputs; finite qualified time intervals 5; finite qualified mobility intervals 11; conflicts 0.
Time: time below geometry: 5; unavailable: 7.
Mobility: Dx CI includes flat: 3; Dx decreases with size: 2; Dx increases with size: 6; unavailable: 1.
Input strengths by qualifier: compatible with flat Dx: 0.0003, 0.003, 0.005; less than geometry: 0.01, 0.02, 0.03, 0.05, 0.1, 0.3; more than geometry: 0.0001, 0.001; unavailable: 1.

Spring: heterogeneous across inputs
Included 10 / 12 designed inputs; finite qualified time intervals 5; finite qualified mobility intervals 10; conflicts 0.
Time: time below geometry: 5; unavailable: 7.
Mobility: Dx CI includes flat: 3; Dx increases with size: 7; unavailable: 2.
Input strengths by qualifier: compatible with flat Dx: 0.0001, 0.001, 0.003; less than geometry: 0.0003, 0.005, 0.01, 0.02, 0.03, 0.05, 0.1; unavailable: 0.3, 1.

Separate geometric qualification. Main-view summaries are shown; source CSV retains all seven windows/cadences. Time excess uses direct persistence divided by actual C-ratio squared. Dx ratios convert short-lag global activity-heading fluctuations using actual C; they do not measure one tracked site's diffusion in multi-region states. Qualified single-region literal A3 heading/peak MSDs remain separate. A CI including flat Dx establishes compatibility only, never equivalence. Unbounded/undefined inference stays unavailable; increased Dx, angular persistence advantages, conflicts and heterogeneous inputs all remain visible. These labels do not change the 308 H1–H3 criteria. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S4_S5_final_geometry_20261020_006_S0_geometry_longest_primary_2.csv

S0 · Separate geometric qualification · Per-model geometric qualification · longest · all conditions including operationally forced

Wave-pinning: heterogeneous across inputs
Included 12 / 12 designed inputs; finite qualified time intervals 4; finite qualified mobility intervals 12; conflicts 0.
Time: time CI includes geometry: 1; time below geometry: 3; unavailable: 8.
Mobility: Dx CI includes flat: 1; Dx decreases with size: 7; Dx increases with size: 4.
Input strengths by qualifier: compatible with flat Dx: 0.03; less than geometry: 0.05, 0.1, 0.3, 1; more than geometry: 0.0001, 0.0003, 0.001, 0.003, 0.005, 0.01, 0.02.

Goryachev: heterogeneous across inputs
Included 12 / 13 designed inputs; finite qualified time intervals 0; finite qualified mobility intervals 12; conflicts 0.
Time: unavailable: 13.
Mobility: Dx decreases with size: 10; Dx increases with size: 2; unavailable: 1.
Input strengths by qualifier: less than geometry: 0.03, 0.05; more than geometry: 0.0001, 0.0003, 0.001, 0.003, 0.005, 0.01, 0.02, 0.1, 0.3, 1; unavailable: 0.015.

Separate geometric qualification. Main-view summaries are shown; source CSV retains all seven windows/cadences. Time excess uses direct persistence divided by actual C-ratio squared. Dx ratios convert short-lag global activity-heading fluctuations using actual C; they do not measure one tracked site's diffusion in multi-region states. Qualified single-region literal A3 heading/peak MSDs remain separate. A CI including flat Dx establishes compatibility only, never equivalence. Unbounded/undefined inference stays unavailable; increased Dx, angular persistence advantages, conflicts and heterogeneous inputs all remain visible. These labels do not change the 308 H1–H3 criteria. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S4_S5_final_geometry_20261020_007_S0_geometry_longest_inclusive_1.csv

S0 · Separate geometric qualification · Per-model geometric qualification · longest · all conditions including operationally forced

Holmes 3: heterogeneous across inputs
Included 12 / 12 designed inputs; finite qualified time intervals 6; finite qualified mobility intervals 12; conflicts 0.
Time: time below geometry: 6; unavailable: 6.
Mobility: Dx CI includes flat: 3; Dx decreases with size: 2; Dx increases with size: 7.
Input strengths by qualifier: compatible with flat Dx: 0.0003, 0.003, 0.005; less than geometry: 0.01, 0.02, 0.03, 0.05, 0.1, 0.3, 1; more than geometry: 0.0001, 0.001.

Spring: heterogeneous across inputs
Included 12 / 12 designed inputs; finite qualified time intervals 7; finite qualified mobility intervals 12; conflicts 0.
Time: time below geometry: 7; unavailable: 5.
Mobility: Dx CI includes flat: 3; Dx increases with size: 9.
Input strengths by qualifier: compatible with flat Dx: 0.0001, 0.001, 0.003; less than geometry: 0.0003, 0.005, 0.01, 0.02, 0.03, 0.05, 0.1, 0.3, 1.

Separate geometric qualification. Main-view summaries are shown; source CSV retains all seven windows/cadences. Time excess uses direct persistence divided by actual C-ratio squared. Dx ratios convert short-lag global activity-heading fluctuations using actual C; they do not measure one tracked site's diffusion in multi-region states. Qualified single-region literal A3 heading/peak MSDs remain separate. A CI including flat Dx establishes compatibility only, never equivalence. Unbounded/undefined inference stays unavailable; increased Dx, angular persistence advantages, conflicts and heterogeneous inputs all remain visible. These labels do not change the 308 H1–H3 criteria. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S4_S5_final_geometry_20261020_008_S0_geometry_longest_inclusive_2.csv

S1 · Final canonical design and forcing · Native forcing, reporter qualification and observation coordinates

$$I(q,t) = fI_{\mathrm{ref}} \exp\left[\sigma Z(q,t) - \frac{\sigma^2}{2}\right] \tag{S1}$$

$$R(t) = \frac{|M_1(t)|}{|M_{1,\,\mathrm{basal}}|} \tag{S2}$$

R uses the exact no-input native basal first-mode magnitude; it can exceed1.

Chemical Gaussian forcing: physical kernel1.5; native OU relaxation50.

Spring forcing is material attached: physical correlation length=0.15 C(t)/C_basal; relaxation5.

Fixed physical optical smoothing uses1.5 (Spring0.15), separately from the forcing coordinate.

Matched mean σ=0 and zero-input controls have no independent stochastic-seed bootstrap.

Native clocks, grids, source hashes, σ branches and retained-prefix dependencies remain in the census.

Common and longest observation arms select from these same sources; neither adds a duplicate cohort.

One-dimensional activity-heading persistence is distinct from empirical migration persistence.

Final canonical design and forcing. f uses each model’s native reference input; native kinetic/time units are not ranked across models. The original R convention is separate from Figure3 instantaneous |M1|/M0 coherence. Exact E4 prefix repairs and the unresolved historical Spring scalar-cadence sensitivity remain separate S12 qualifications. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_001_S1_forcing.csv

S1 · Final canonical design and forcing · All canonical paths and their dependency census

Context	Model	Noisy paths	Mean controls	Zero controls	Native paths	Gaussian cohorts	Seed lineages
common	Wave-pinning	432	46	5	483	432	432
common	Goryachev	444	47	5	496	444	444
common	Holmes 3	432	46	5	483	432	432
common	Spring	432	46	5	483	432	432
E1_extra	Wave-pinning	216	0	0	216	72	72
E1_extra	Goryachev	216	0	0	216	72	72
E1_extra	Holmes 3	216	0	0	216	72	72
E1_extra	Spring	216	0	0	216	72	72
E4_long	Wave-pinning	168	14	3	185	168	168
E4_long	Goryachev	216	18	3	237	216	216
E4_long	Holmes 3	144	12	3	159	144	144
E4_long	Spring	96	8	3	107	96	96

Final canonical design and forcing. Counts are native paths, not a pooled independent-seed count. Sigma branches share Gaussian cohorts; E4 prefixes retain lineage dependence. The accepted global census is3497 paths,3228 noisy paths,269 deterministic controls,2364 Gaussian cohorts and1990 seed lineages. E4 includes374 retained-prefix noisy paths and250 independent top-ups. Unaccepted attempted prefixes remain in the engineering archive, separately documented in S12. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_002_S1_population.csv

S1 · Final canonical design and forcing · Complete condition, seed and native observation inventory

Context	Model	C_0	f	σ	Kind	Paths / seed blocks	Native H	Scalar / profile Δt	Origin
common	Wave-pinning	20	0	0	zero	1 / 0	32000	5/40	E0
common	Wave-pinning	20	0.0001	0	mean	1 / 0	32000	5/40	E0
common	Wave-pinning	20	0.0001	1	noise	2 / 2	32000	5/40	E0
common	Wave-pinning	20	0.0003	0	mean	1 / 0	32000	5/40	E0
common	Wave-pinning	20	0.0003	1	noise	2 / 2	32000	5/40	E0
common	Wave-pinning	20	0.001	0	mean	1 / 0	32000	5/40	E0
common	Wave-pinning	20	0.001	1	noise	12 / 12	32000	5/40	E0
common	Wave-pinning	20	0.003	0	mean	1 / 0	32000	5/40	E0
common	Wave-pinning	20	0.003	1	noise	12 / 12	32000	5/40	E0
common	Wave-pinning	20	0.005	0	mean	1 / 0	32000	5/5	E2
common	Wave-pinning	20	0.005	1	noise	12 / 12	32000	5/5	E2
common	Wave-pinning	20	0.01	0	mean	1 / 0	32000	5/40	E0
common	Wave-pinning	20	0.01	1	noise	12 / 12	32000	5/40	E0
common	Wave-pinning	20	0.02	0	mean	1 / 0	32000	5/5	E2
common	Wave-pinning	20	0.02	1	noise	12 / 12	32000	5/5	E2
common	Wave-pinning	20	0.03	0	mean	1 / 0	32000	5/40	E0
common	Wave-pinning	20	0.03	1	noise	12 / 12	32000	5/40	E0
common	Wave-pinning	20	0.05	0	mean	1 / 0	32000	5/5	E2

Final canonical design and forcing. Seed blocks are counted within each condition only; they must not be summed across sigma branches, inputs or retained-prefix comparisons. Controls contribute0 independent stochastic seeds. Cadence lists are exact, including mixed native sources. Each page CSV retains every member’s complete canonical source record, archive hashes, cohort/lineage identity and prefix/top-up provenance. Native units differ across models. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_003_S1_conditions_01.csv

S1 · Final canonical design and forcing · Complete condition, seed and native observation inventory

Context	Model	C_0	f	σ	Kind	Paths / seed blocks	Native H	Scalar / profile Δt	Origin
common	Wave-pinning	20	0.05	1	noise	12 / 12	32000	5/5	E2
common	Wave-pinning	20	0.1	0	mean	1 / 0	32000	5/40	E0
common	Wave-pinning	20	0.1	1	noise	12 / 12	32000	5/40	E0
common	Wave-pinning	20	0.3	0	mean	1 / 0	32000	5/40	E0
common	Wave-pinning	20	0.3	1	noise	2 / 2	32000	5/40	E0
common	Wave-pinning	20	1	0	mean	1 / 0	32000	5/40	E0
common	Wave-pinning	20	1	1	noise	2 / 2	32000	5/40	E0
common	Wave-pinning	30	0	0	zero	1 / 0	32000	5/40	E3
common	Wave-pinning	30	0.001	0	mean	1 / 0	32000	5/40	E3
common	Wave-pinning	30	0.001	1	noise	12 / 12	32000	5/40	E3
common	Wave-pinning	30	0.003	0	mean	1 / 0	32000	5/40	E3
common	Wave-pinning	30	0.003	1	noise	12 / 12	32000	5/40	E3
common	Wave-pinning	30	0.01	0	mean	1 / 0	32000	5/40	E3
common	Wave-pinning	30	0.01	1	noise	12 / 12	32000	5/40	E3
common	Wave-pinning	30	0.03	0	mean	1 / 0	32000	5/40	E3
common	Wave-pinning	30	0.03	1	noise	12 / 12	32000	5/40	E3
common	Wave-pinning	30	0.1	0	mean	1 / 0	32000	5/40	E3
common	Wave-pinning	30	0.1	1	noise	12 / 12	32000	5/40	E3

Final canonical design and forcing. Seed blocks are counted within each condition only; they must not be summed across sigma branches, inputs or retained-prefix comparisons. Controls contribute0 independent stochastic seeds. Cadence lists are exact, including mixed native sources. Each page CSV retains every member's complete canonical source record, archive hashes, cohort/lineage identity and prefix/top-up provenance. Native units differ across models. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_004_S1_conditions_02.csv

S1 · Final canonical design and forcing · Complete condition, seed and native observation inventory

Context	Model	C_0	f	σ	Kind	Paths / seed blocks	Native H	Scalar / profile Δt	Origin
common	Wave-pinning	40	0	0	zero	1 / 0	32000	5/40	E0
common	Wave-pinning	40	0.0001	0	mean	1 / 0	32000	5/40	E0
common	Wave-pinning	40	0.0001	1	noise	2 / 2	32000	5/40	E0
common	Wave-pinning	40	0.0003	0	mean	1 / 0	32000	5/40	E0
common	Wave-pinning	40	0.0003	1	noise	2 / 2	32000	5/40	E0
common	Wave-pinning	40	0.001	0	mean	1 / 0	32000	5/40	E0
common	Wave-pinning	40	0.001	1	noise	12 / 12	32000	5/40	E0
common	Wave-pinning	40	0.003	0	mean	1 / 0	32000	5/40	E0
common	Wave-pinning	40	0.003	1	noise	12 / 12	32000	5/40	E0
common	Wave-pinning	40	0.005	0	mean	1 / 0	32000	5/5	E2
common	Wave-pinning	40	0.005	1	noise	12 / 12	32000	5/5	E2
common	Wave-pinning	40	0.01	0	mean	1 / 0	32000	5/40	E0
common	Wave-pinning	40	0.01	1	noise	12 / 12	32000	5/40	E0
common	Wave-pinning	40	0.02	0	mean	1 / 0	32000	5/5	E2
common	Wave-pinning	40	0.02	1	noise	12 / 12	32000	5/5	E2
common	Wave-pinning	40	0.03	0	mean	1 / 0	32000	5/40	E0
common	Wave-pinning	40	0.03	1	noise	12 / 12	32000	5/40	E0
common	Wave-pinning	40	0.05	0	mean	1 / 0	32000	5/5	E2

Final canonical design and forcing. Seed blocks are counted within each condition only; they must not be summed across sigma branches, inputs or retained-prefix comparisons. Controls contribute0 independent stochastic seeds. Cadence lists are exact, including mixed native sources. Each page CSV retains every member’s complete canonical source record, archive hashes, cohort/lineage identity and prefix/top-up provenance. Native units differ across models. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_005_S1_conditions_03.csv

S1 · Final canonical design and forcing · Complete condition, seed and native observation inventory

Context	Model	C_0	f	σ	Kind	Paths / seed blocks	Native H	Scalar / profile Δt	Origin
common	Wave-pinning	40	0.05	1	noise	12 / 12	32000	5/5	E2
common	Wave-pinning	40	0.1	0	mean	1 / 0	32000	5/40	E0
common	Wave-pinning	40	0.1	1	noise	12 / 12	32000	5/40	E0
common	Wave-pinning	40	0.3	0	mean	1 / 0	32000	5/40	E0
common	Wave-pinning	40	0.3	1	noise	2 / 2	32000	5/40	E0
common	Wave-pinning	40	1	0	mean	1 / 0	32000	5/40	E0
common	Wave-pinning	40	1	1	noise	2 / 2	32000	5/40	E0
common	Wave-pinning	50	0	0	zero	1 / 0	32000	5/40	E3
common	Wave-pinning	50	0.001	0	mean	1 / 0	32000	5/40	E3
common	Wave-pinning	50	0.001	1	noise	12 / 12	32000	5/40	E3
common	Wave-pinning	50	0.003	0	mean	1 / 0	32000	5/40	E3
common	Wave-pinning	50	0.003	1	noise	12 / 12	32000	5/40	E3
common	Wave-pinning	50	0.01	0	mean	1 / 0	32000	5/40	E3
common	Wave-pinning	50	0.01	1	noise	12 / 12	32000	5/40	E3
common	Wave-pinning	50	0.03	0	mean	1 / 0	32000	5/40	E3
common	Wave-pinning	50	0.03	1	noise	12 / 12	32000	5/40	E3
common	Wave-pinning	50	0.1	0	mean	1 / 0	32000	5/40	E3
common	Wave-pinning	50	0.1	1	noise	12 / 12	32000	5/40	E3

Final canonical design and forcing. Seed blocks are counted within each condition only; they must not be summed across sigma branches, inputs or retained-prefix comparisons. Controls contribute0 independent stochastic seeds. Cadence lists are exact, including mixed native sources. Each page CSV retains every member's complete canonical source record, archive hashes, cohort/lineage identity and prefix/top-up provenance. Native units differ across models. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_006_S1_conditions_04.csv

S1 · Final canonical design and forcing · Complete condition, seed and native observation inventory

Context	Model	C_0	f	σ	Kind	Paths / seed blocks	Native H	Scalar / profile Δt	Origin
common	Wave-pinning	60	0	0	zero	1 / 0	32000	5/40	E0
common	Wave-pinning	60	0.0001	0	mean	1 / 0	32000	5/40	E0
common	Wave-pinning	60	0.0001	1	noise	2 / 2	32000	5/40	E0
common	Wave-pinning	60	0.0003	0	mean	1 / 0	32000	5/40	E0
common	Wave-pinning	60	0.0003	1	noise	2 / 2	32000	5/40	E0
common	Wave-pinning	60	0.001	0	mean	1 / 0	32000	5/40	E0
common	Wave-pinning	60	0.001	1	noise	12 / 12	32000	5/40	E0
common	Wave-pinning	60	0.003	0	mean	1 / 0	32000	5/40	E0
common	Wave-pinning	60	0.003	1	noise	12 / 12	32000	5/40	E0
common	Wave-pinning	60	0.005	0	mean	1 / 0	32000	5/5	E2
common	Wave-pinning	60	0.005	1	noise	12 / 12	32000	5/5	E2
common	Wave-pinning	60	0.01	0	mean	1 / 0	32000	5/40	E0
common	Wave-pinning	60	0.01	1	noise	12 / 12	32000	5/40	E0
common	Wave-pinning	60	0.02	0	mean	1 / 0	32000	5/5	E2
common	Wave-pinning	60	0.02	1	noise	12 / 12	32000	5/5	E2
common	Wave-pinning	60	0.03	0	mean	1 / 0	32000	5/40	E0
common	Wave-pinning	60	0.03	1	noise	12 / 12	32000	5/40	E0
common	Wave-pinning	60	0.05	0	mean	1 / 0	32000	5/5	E2

Final canonical design and forcing. Seed blocks are counted within each condition only; they must not be summed across sigma branches, inputs or retained-prefix comparisons. Controls contribute0 independent stochastic seeds. Cadence lists are exact, including mixed native sources. Each page CSV retains every member’s complete canonical source record, archive hashes, cohort/lineage identity and prefix/top-up provenance. Native units differ across models. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_007_S1_conditions_05.csv

S1 · Final canonical design and forcing · Complete condition, seed and native observation inventory

Context	Model	C_0	f	σ	Kind	Paths / seed blocks	Native H	Scalar / profile Δt	Origin
common	Wave-pinning	60	0.05	1	noise	12 / 12	32000	5/5	E2
common	Wave-pinning	60	0.1	0	mean	1 / 0	32000	5/40	E0
common	Wave-pinning	60	0.1	1	noise	12 / 12	32000	5/40	E0
common	Wave-pinning	60	0.3	0	mean	1 / 0	32000	5/40	E0
common	Wave-pinning	60	0.3	1	noise	2 / 2	32000	5/40	E0
common	Wave-pinning	60	1	0	mean	1 / 0	32000	5/40	E0
common	Wave-pinning	60	1	1	noise	2 / 2	32000	5/40	E0
common	Goryachev	20	0	0	zero	1 / 0	32000	5/40	E0
common	Goryachev	20	0.0001	0	mean	1 / 0	32000	5/40	E0
common	Goryachev	20	0.0001	1	noise	2 / 2	32000	5/40	E0
common	Goryachev	20	0.0003	0	mean	1 / 0	32000	5/40	E0
common	Goryachev	20	0.0003	1	noise	2 / 2	32000	5/40	E0
common	Goryachev	20	0.001	0	mean	1 / 0	32000	5/40	E0
common	Goryachev	20	0.001	1	noise	12 / 12	32000	5/40	E0
common	Goryachev	20	0.003	0	mean	1 / 0	32000	5/40	E0
common	Goryachev	20	0.003	1	noise	12 / 12	32000	5/40	E0
common	Goryachev	20	0.005	0	mean	1 / 0	32000	5/5	E2
common	Goryachev	20	0.005	1	noise	12 / 12	32000	5/5	E2

Final canonical design and forcing. Seed blocks are counted within each condition only; they must not be summed across sigma branches, inputs or retained-prefix comparisons. Controls contribute0 independent stochastic seeds. Cadence lists are exact, including mixed native sources. Each page CSV retains every member’s complete canonical source record, archive hashes, cohort/lineage identity and prefix/top-up provenance. Native units differ across models. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_008_S1_conditions_06.csv

S1 · Final canonical design and forcing · Complete condition, seed and native observation inventory

Context	Model	C_0	f	σ	Kind	Paths / seed blocks	Native H	Scalar / profile Δt	Origin
common	Goryachev	20	0.01	0	mean	1 / 0	32000	5/40	E0
common	Goryachev	20	0.01	1	noise	12 / 12	32000	5/40	E0
common	Goryachev	20	0.02	0	mean	1 / 0	32000	5/5	E2
common	Goryachev	20	0.02	1	noise	12 / 12	32000	5/5	E2
common	Goryachev	20	0.03	0	mean	1 / 0	32000	5/40	E0
common	Goryachev	20	0.03	1	noise	12 / 12	32000	5/40	E0
common	Goryachev	20	0.05	0	mean	1 / 0	32000	5/5	E2
common	Goryachev	20	0.05	1	noise	12 / 12	32000	5/5	E2
common	Goryachev	20	0.1	0	mean	1 / 0	32000	5/40	E0
common	Goryachev	20	0.1	1	noise	12 / 12	32000	5/40	E0
common	Goryachev	20	0.3	0	mean	1 / 0	32000	5/40	E0
common	Goryachev	20	0.3	1	noise	2 / 2	32000	5/40	E0
common	Goryachev	20	1	0	mean	1 / 0	32000	5/40	E0
common	Goryachev	20	1	1	noise	2 / 2	32000	5/40	E0
common	Goryachev	30	0	0	zero	1 / 0	32000	5/40	E3
common	Goryachev	30	0.001	0	mean	1 / 0	32000	5/40	E3
common	Goryachev	30	0.001	1	noise	12 / 12	32000	5/40	E3
common	Goryachev	30	0.003	0	mean	1 / 0	32000	5/40	E3

Final canonical design and forcing. Seed blocks are counted within each condition only; they must not be summed across sigma branches, inputs or retained-prefix comparisons. Controls contribute0 independent stochastic seeds. Cadence lists are exact, including mixed native sources. Each page CSV retains every member’s complete canonical source record, archive hashes, cohort/lineage identity and prefix/top-up provenance. Native units differ across models. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_009_S1_conditions_07.csv

S1 · Final canonical design and forcing · Complete condition, seed and native observation inventory

Context	Model	C_0	f	σ	Kind	Paths / seed blocks	Native H	Scalar / profile Δt	Origin
common	Goryachev	30	0.003	1	noise	12 / 12	32000	5/40	E3
common	Goryachev	30	0.01	0	mean	1 / 0	32000	5/40	E3
common	Goryachev	30	0.01	1	noise	12 / 12	32000	5/40	E3
common	Goryachev	30	0.03	0	mean	1 / 0	32000	5/40	E3
common	Goryachev	30	0.03	1	noise	12 / 12	32000	5/40	E3
common	Goryachev	30	0.1	0	mean	1 / 0	32000	5/40	E3
common	Goryachev	30	0.1	1	noise	12 / 12	32000	5/40	E3
common	Goryachev	40	0	0	zero	1 / 0	32000	5/40	E0
common	Goryachev	40	0.0001	0	mean	1 / 0	32000	5/40	E0
common	Goryachev	40	0.0001	1	noise	2 / 2	32000	5/40	E0
common	Goryachev	40	0.0003	0	mean	1 / 0	32000	5/40	E0
common	Goryachev	40	0.0003	1	noise	2 / 2	32000	5/40	E0
common	Goryachev	40	0.001	0	mean	1 / 0	32000	5/40	E0
common	Goryachev	40	0.001	1	noise	12 / 12	32000	5/40	E0
common	Goryachev	40	0.003	0	mean	1 / 0	32000	5/40	E0
common	Goryachev	40	0.003	1	noise	12 / 12	32000	5/40	E0
common	Goryachev	40	0.005	0	mean	1 / 0	32000	5/5	E2
common	Goryachev	40	0.005	1	noise	12 / 12	32000	5/5	E2

Final canonical design and forcing. Seed blocks are counted within each condition only; they must not be summed across sigma branches, inputs or retained-prefix comparisons. Controls contribute0 independent stochastic seeds. Cadence lists are exact, including mixed native sources. Each page CSV retains every member’s complete canonical source record, archive hashes, cohort/lineage identity and prefix/top-up provenance. Native units differ across models. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_010_S1_conditions_08.csv

S1 · Final canonical design and forcing · Complete condition, seed and native observation inventory

Context	Model	C_0	f	σ	Kind	Paths / seed blocks	Native H	Scalar / profile Δt	Origin
common	Goryachev	40	0.01	0	mean	1 / 0	32000	5/40	E0
common	Goryachev	40	0.01	1	noise	12 / 12	32000	5/40	E0
common	Goryachev	40	0.02	0	mean	1 / 0	32000	5/5	E2
common	Goryachev	40	0.02	1	noise	12 / 12	32000	5/5	E2
common	Goryachev	40	0.03	0	mean	1 / 0	32000	5/40	E0
common	Goryachev	40	0.03	1	noise	12 / 12	32000	5/40	E0
common	Goryachev	40	0.05	0	mean	1 / 0	32000	5/5	E2
common	Goryachev	40	0.05	1	noise	12 / 12	32000	5/5	E2
common	Goryachev	40	0.1	0	mean	1 / 0	32000	5/40	E0
common	Goryachev	40	0.1	1	noise	12 / 12	32000	5/40	E0
common	Goryachev	40	0.3	0	mean	1 / 0	32000	5/40	E0
common	Goryachev	40	0.3	1	noise	2 / 2	32000	5/40	E0
common	Goryachev	40	1	0	mean	1 / 0	32000	5/40	E0
common	Goryachev	40	1	1	noise	2 / 2	32000	5/40	E0
common	Goryachev	50	0	0	zero	1 / 0	32000	5/40	E3
common	Goryachev	50	0.001	0	mean	1 / 0	32000	5/40	E3
common	Goryachev	50	0.001	1	noise	12 / 12	32000	5/40	E3
common	Goryachev	50	0.003	0	mean	1 / 0	32000	5/40	E3

Final canonical design and forcing. Seed blocks are counted within each condition only; they must not be summed across sigma branches, inputs or retained-prefix comparisons. Controls contribute0 independent stochastic seeds. Cadence lists are exact, including mixed native sources. Each page CSV retains every member's complete canonical source record, archive hashes, cohort/lineage identity and prefix/top-up provenance. Native units differ across models. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_011_S1_conditions_09.csv

S1 · Final canonical design and forcing · Complete condition, seed and native observation inventory

Context	Model	C_0	f	σ	Kind	Paths / seed blocks	Native H	Scalar / profile Δt	Origin
common	Goryachev	50	0.003	1	noise	12 / 12	32000	5/40	E3
common	Goryachev	50	0.01	0	mean	1 / 0	32000	5/40	E3
common	Goryachev	50	0.01	1	noise	12 / 12	32000	5/40	E3
common	Goryachev	50	0.03	0	mean	1 / 0	32000	5/40	E3
common	Goryachev	50	0.03	1	noise	12 / 12	32000	5/40	E3
common	Goryachev	50	0.1	0	mean	1 / 0	32000	5/40	E3
common	Goryachev	50	0.1	1	noise	12 / 12	32000	5/40	E3
common	Goryachev	60	0	0	zero	1 / 0	32000	5/40	E0
common	Goryachev	60	0.0001	0	mean	1 / 0	32000	5/40	E0
common	Goryachev	60	0.0001	1	noise	2 / 2	32000	5/40	E0
common	Goryachev	60	0.0003	0	mean	1 / 0	32000	5/40	E0
common	Goryachev	60	0.0003	1	noise	2 / 2	32000	5/40	E0
common	Goryachev	60	0.001	0	mean	1 / 0	32000	5/40	E0
common	Goryachev	60	0.001	1	noise	12 / 12	32000	5/40	E0
common	Goryachev	60	0.003	0	mean	1 / 0	32000	5/40	E0
common	Goryachev	60	0.003	1	noise	12 / 12	32000	5/40	E0
common	Goryachev	60	0.005	0	mean	1 / 0	32000	5/5	E2
common	Goryachev	60	0.005	1	noise	12 / 12	32000	5/5	E2

Final canonical design and forcing. Seed blocks are counted within each condition only; they must not be summed across sigma branches, inputs or retained-prefix comparisons. Controls contribute0 independent stochastic seeds. Cadence lists are exact, including mixed native sources. Each page CSV retains every member’s complete canonical source record, archive hashes, cohort/lineage identity and prefix/top-up provenance. Native units differ across models. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_012_S1_conditions_10.csv

S1 · Final canonical design and forcing · Complete condition, seed and native observation inventory

Context	Model	C_0	f	σ	Kind	Paths / seed blocks	Native H	Scalar / profile Δt	Origin
common	Goryachev	60	0.01	0	mean	1 / 0	32000	5/40	E0
common	Goryachev	60	0.01	1	noise	12 / 12	32000	5/40	E0
common	Goryachev	60	0.015	0	mean	1 / 0	32000	5/40	E0
common	Goryachev	60	0.015	1	noise	12 / 12	32000	5/40	E0
common	Goryachev	60	0.02	0	mean	1 / 0	32000	5/5	E2
common	Goryachev	60	0.02	1	noise	12 / 12	32000	5/5	E2
common	Goryachev	60	0.03	0	mean	1 / 0	32000	5/40	E0
common	Goryachev	60	0.03	1	noise	12 / 12	32000	5/40	E0
common	Goryachev	60	0.05	0	mean	1 / 0	32000	5/5	E2
common	Goryachev	60	0.05	1	noise	12 / 12	32000	5/5	E2
common	Goryachev	60	0.1	0	mean	1 / 0	32000	5/40	E0
common	Goryachev	60	0.1	1	noise	12 / 12	32000	5/40	E0
common	Goryachev	60	0.3	0	mean	1 / 0	32000	5/40	E0
common	Goryachev	60	0.3	1	noise	2 / 2	32000	5/40	E0
common	Goryachev	60	1	0	mean	1 / 0	32000	5/40	E0
common	Goryachev	60	1	1	noise	2 / 2	32000	5/40	E0
common	Holmes 3	20	0	0	zero	1 / 0	32000	5/40	E0
common	Holmes 3	20	0.0001	0	mean	1 / 0	32000	5/40	E0

Final canonical design and forcing. Seed blocks are counted within each condition only; they must not be summed across sigma branches, inputs or retained-prefix comparisons. Controls contribute0 independent stochastic seeds. Cadence lists are exact, including mixed native sources. Each page CSV retains every member’s complete canonical source record, archive hashes, cohort/lineage identity and prefix/top-up provenance. Native units differ across models. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_013_S1_conditions_11.csv

S1 · Final canonical design and forcing · Complete condition, seed and native observation inventory

Context	Model	C_0	f	σ	Kind	Paths / seed blocks	Native H	Scalar / profile Δt	Origin
common	Holmes 3	20	0.0001	1	noise	2 / 2	32000	5/40	E0
common	Holmes 3	20	0.0003	0	mean	1 / 0	32000	5/40	E0
common	Holmes 3	20	0.0003	1	noise	2 / 2	32000	5/40	E0
common	Holmes 3	20	0.001	0	mean	1 / 0	32000	5/40	E0
common	Holmes 3	20	0.001	1	noise	12 / 12	32000	5/40	E0
common	Holmes 3	20	0.003	0	mean	1 / 0	32000	5/40	E0
common	Holmes 3	20	0.003	1	noise	12 / 12	32000	5/40	E0
common	Holmes 3	20	0.005	0	mean	1 / 0	32000	5/5	E2
common	Holmes 3	20	0.005	1	noise	12 / 12	32000	5/5	E2
common	Holmes 3	20	0.01	0	mean	1 / 0	32000	5/40	E0
common	Holmes 3	20	0.01	1	noise	12 / 12	32000	5/40	E0
common	Holmes 3	20	0.02	0	mean	1 / 0	32000	5/5	E2
common	Holmes 3	20	0.02	1	noise	12 / 12	32000	5/5	E2
common	Holmes 3	20	0.03	0	mean	1 / 0	32000	5/40	E0
common	Holmes 3	20	0.03	1	noise	12 / 12	32000	5/40	E0
common	Holmes 3	20	0.05	0	mean	1 / 0	32000	5/5	E2
common	Holmes 3	20	0.05	1	noise	12 / 12	32000	5/5	E2
common	Holmes 3	20	0.1	0	mean	1 / 0	32000	5/40	E0

Final canonical design and forcing. Seed blocks are counted within each condition only; they must not be summed across sigma branches, inputs or retained-prefix comparisons. Controls contribute0 independent stochastic seeds. Cadence lists are exact, including mixed native sources. Each page CSV retains every member’s complete canonical source record, archive hashes, cohort/lineage identity and prefix/top-up provenance. Native units differ across models. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_014_S1_conditions_12.csv

S1 · Final canonical design and forcing · Complete condition, seed and native observation inventory

Context	Model	C_0	f	σ	Kind	Paths / seed blocks	Native H	Scalar / profile Δt	Origin
common	Holmes 3	20	0.1	1	noise	12 / 12	32000	5/40	E0
common	Holmes 3	20	0.3	0	mean	1 / 0	32000	5/40	E0
common	Holmes 3	20	0.3	1	noise	2 / 2	32000	5/40	E0
common	Holmes 3	20	1	0	mean	1 / 0	32000	5/40	E0
common	Holmes 3	20	1	1	noise	2 / 2	32000	5/40	E0
common	Holmes 3	30	0	0	zero	1 / 0	32000	5/40	E3
common	Holmes 3	30	0.001	0	mean	1 / 0	32000	5/40	E3
common	Holmes 3	30	0.001	1	noise	12 / 12	32000	5/40	E3
common	Holmes 3	30	0.003	0	mean	1 / 0	32000	5/40	E3
common	Holmes 3	30	0.003	1	noise	12 / 12	32000	5/40	E3
common	Holmes 3	30	0.01	0	mean	1 / 0	32000	5/40	E3
common	Holmes 3	30	0.01	1	noise	12 / 12	32000	5/40	E3
common	Holmes 3	30	0.03	0	mean	1 / 0	32000	5/40	E3
common	Holmes 3	30	0.03	1	noise	12 / 12	32000	5/40	E3
common	Holmes 3	30	0.1	0	mean	1 / 0	32000	5/40	E3
common	Holmes 3	30	0.1	1	noise	12 / 12	32000	5/40	E3
common	Holmes 3	40	0	0	zero	1 / 0	32000	5/40	E0
common	Holmes 3	40	0.0001	0	mean	1 / 0	32000	5/40	E0

Final canonical design and forcing. Seed blocks are counted within each condition only; they must not be summed across sigma branches, inputs or retained-prefix comparisons. Controls contribute0 independent stochastic seeds. Cadence lists are exact, including mixed native sources. Each page CSV retains every member’s complete canonical source record, archive hashes, cohort/lineage identity and prefix/top-up provenance. Native units differ across models. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_015_S1_conditions_13.csv

S1 · Final canonical design and forcing · Complete condition, seed and native observation inventory

Context	Model	C_0	f	σ	Kind	Paths / seed blocks	Native H	Scalar / profile Δt	Origin
common	Holmes 3	40	0.0001	1	noise	2 / 2	32000	5/40	E0
common	Holmes 3	40	0.0003	0	mean	1 / 0	32000	5/40	E0
common	Holmes 3	40	0.0003	1	noise	2 / 2	32000	5/40	E0
common	Holmes 3	40	0.001	0	mean	1 / 0	32000	5/40	E0
common	Holmes 3	40	0.001	1	noise	12 / 12	32000	5/40	E0
common	Holmes 3	40	0.003	0	mean	1 / 0	32000	5/40	E0
common	Holmes 3	40	0.003	1	noise	12 / 12	32000	5/40	E0
common	Holmes 3	40	0.005	0	mean	1 / 0	32000	5/5	E2
common	Holmes 3	40	0.005	1	noise	12 / 12	32000	5/5	E2
common	Holmes 3	40	0.01	0	mean	1 / 0	32000	5/40	E0
common	Holmes 3	40	0.01	1	noise	12 / 12	32000	5/40	E0
common	Holmes 3	40	0.02	0	mean	1 / 0	32000	5/5	E2
common	Holmes 3	40	0.02	1	noise	12 / 12	32000	5/5	E2
common	Holmes 3	40	0.03	0	mean	1 / 0	32000	5/40	E0
common	Holmes 3	40	0.03	1	noise	12 / 12	32000	5/40	E0
common	Holmes 3	40	0.05	0	mean	1 / 0	32000	5/5	E2
common	Holmes 3	40	0.05	1	noise	12 / 12	32000	5/5	E2
common	Holmes 3	40	0.1	0	mean	1 / 0	32000	5/40	E0

Final canonical design and forcing. Seed blocks are counted within each condition only; they must not be summed across sigma branches, inputs or retained-prefix comparisons. Controls contribute0 independent stochastic seeds. Cadence lists are exact, including mixed native sources. Each page CSV retains every member’s complete canonical source record, archive hashes, cohort/lineage identity and prefix/top-up provenance. Native units differ across models. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_016_S1_conditions_14.csv

S1 · Final canonical design and forcing · Complete condition, seed and native observation inventory

Context	Model	C_0	f	σ	Kind	Paths / seed blocks	Native H	Scalar / profile Δt	Origin
common	Holmes 3	40	0.1	1	noise	12 / 12	32000	5/40	E0
common	Holmes 3	40	0.3	0	mean	1 / 0	32000	5/40	E0
common	Holmes 3	40	0.3	1	noise	2 / 2	32000	5/40	E0
common	Holmes 3	40	1	0	mean	1 / 0	32000	5/40	E0
common	Holmes 3	40	1	1	noise	2 / 2	32000	5/40	E0
common	Holmes 3	50	0	0	zero	1 / 0	32000	5/40	E3
common	Holmes 3	50	0.001	0	mean	1 / 0	32000	5/40	E3
common	Holmes 3	50	0.001	1	noise	12 / 12	32000	5/40	E3
common	Holmes 3	50	0.003	0	mean	1 / 0	32000	5/40	E3
common	Holmes 3	50	0.003	1	noise	12 / 12	32000	5/40	E3
common	Holmes 3	50	0.01	0	mean	1 / 0	32000	5/40	E3
common	Holmes 3	50	0.01	1	noise	12 / 12	32000	5/40	E3
common	Holmes 3	50	0.03	0	mean	1 / 0	32000	5/40	E3
common	Holmes 3	50	0.03	1	noise	12 / 12	32000	5/40	E3
common	Holmes 3	50	0.1	0	mean	1 / 0	32000	5/40	E3
common	Holmes 3	50	0.1	1	noise	12 / 12	32000	5/40	E3
common	Holmes 3	60	0	0	zero	1 / 0	32000	5/40	E0
common	Holmes 3	60	0.0001	0	mean	1 / 0	32000	5/40	E0

Final canonical design and forcing. Seed blocks are counted within each condition only; they must not be summed across sigma branches, inputs or retained-prefix comparisons. Controls contribute0 independent stochastic seeds. Cadence lists are exact, including mixed native sources. Each page CSV retains every member’s complete canonical source record, archive hashes, cohort/lineage identity and prefix/top-up provenance. Native units differ across models. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_017_S1_conditions_15.csv

S1 · Final canonical design and forcing · Complete condition, seed and native observation inventory

Context	Model	C_0	f	σ	Kind	Paths / seed blocks	Native H	Scalar / profile Δt	Origin
common	Holmes 3	60	0.0001	1	noise	2 / 2	32000	5/40	E0
common	Holmes 3	60	0.0003	0	mean	1 / 0	32000	5/40	E0
common	Holmes 3	60	0.0003	1	noise	2 / 2	32000	5/40	E0
common	Holmes 3	60	0.001	0	mean	1 / 0	32000	5/40	E0
common	Holmes 3	60	0.001	1	noise	12 / 12	32000	5/40	E0
common	Holmes 3	60	0.003	0	mean	1 / 0	32000	5/40	E0
common	Holmes 3	60	0.003	1	noise	12 / 12	32000	5/40	E0
common	Holmes 3	60	0.005	0	mean	1 / 0	32000	5/5	E2
common	Holmes 3	60	0.005	1	noise	12 / 12	32000	5/5	E2
common	Holmes 3	60	0.01	0	mean	1 / 0	32000	5/40	E0
common	Holmes 3	60	0.01	1	noise	12 / 12	32000	5/40	E0
common	Holmes 3	60	0.02	0	mean	1 / 0	32000	5/5	E2
common	Holmes 3	60	0.02	1	noise	12 / 12	32000	5/5	E2
common	Holmes 3	60	0.03	0	mean	1 / 0	32000	5/40	E0
common	Holmes 3	60	0.03	1	noise	12 / 12	32000	5/40	E0
common	Holmes 3	60	0.05	0	mean	1 / 0	32000	5/5	E2
common	Holmes 3	60	0.05	1	noise	12 / 12	32000	5/5	E2
common	Holmes 3	60	0.1	0	mean	1 / 0	32000	5/40	E0

Final canonical design and forcing. Seed blocks are counted within each condition only; they must not be summed across sigma branches, inputs or retained-prefix comparisons. Controls contribute0 independent stochastic seeds. Cadence lists are exact, including mixed native sources. Each page CSV retains every member’s complete canonical source record, archive hashes, cohort/lineage identity and prefix/top-up provenance. Native units differ across models. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_018_S1_conditions_16.csv

S1 · Final canonical design and forcing · Complete condition, seed and native observation inventory

Context	Model	C_0	f	σ	Kind	Paths / seed blocks	Native H	Scalar / profile Δt	Origin
common	Holmes 3	60	0.1	1	noise	12 / 12	32000	5/40	E0
common	Holmes 3	60	0.3	0	mean	1 / 0	32000	5/40	E0
common	Holmes 3	60	0.3	1	noise	2 / 2	32000	5/40	E0
common	Holmes 3	60	1	0	mean	1 / 0	32000	5/40	E0
common	Holmes 3	60	1	1	noise	2 / 2	32000	5/40	E0
common	Spring	1	0	0	zero	1 / 0	3200	0.5/4	E0
common	Spring	1	0.0001	0	mean	1 / 0	3200	0.5/4	E0
common	Spring	1	0.0001	1	noise	2 / 2	3200	0.5/4	E0
common	Spring	1	0.0003	0	mean	1 / 0	3200	0.5/4	E0
common	Spring	1	0.0003	1	noise	2 / 2	3200	0.5/4	E0
common	Spring	1	0.001	0	mean	1 / 0	3200	0.5/4	E0
common	Spring	1	0.001	1	noise	12 / 12	3200	0.5/4	E0
common	Spring	1	0.003	0	mean	1 / 0	3200	0.5/4	E0
common	Spring	1	0.003	1	noise	12 / 12	3200	0.5/4	E0
common	Spring	1	0.005	0	mean	1 / 0	3200	0.5/0.5	E2
common	Spring	1	0.005	1	noise	12 / 12	3200	0.5/0.5	E2
common	Spring	1	0.01	0	mean	1 / 0	3200	0.5/4	E0
common	Spring	1	0.01	1	noise	12 / 12	3200	0.5/4	E0

Final canonical design and forcing. Seed blocks are counted within each condition only; they must not be summed across sigma branches, inputs or retained-prefix comparisons. Controls contribute0 independent stochastic seeds. Cadence lists are exact, including mixed native sources. Each page CSV retains every member's complete canonical source record, archive hashes, cohort/lineage identity and prefix/top-up provenance. Native units differ across models. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_019_S1_conditions_17.csv

S1 · Final canonical design and forcing · Complete condition, seed and native observation inventory

Context	Model	C_0	f	σ	Kind	Paths / seed blocks	Native H	Scalar / profile Δt	Origin
common	Spring	1	0.02	0	mean	1 / 0	3200	0.5/0.5	E2
common	Spring	1	0.02	1	noise	12 / 12	3200	0.5/0.5	E2
common	Spring	1	0.03	0	mean	1 / 0	3200	0.5/4	E0
common	Spring	1	0.03	1	noise	12 / 12	3200	0.5/4	E0
common	Spring	1	0.05	0	mean	1 / 0	3200	0.5/0.5	E2
common	Spring	1	0.05	1	noise	12 / 12	3200	0.5/0.5	E2
common	Spring	1	0.1	0	mean	1 / 0	3200	0.5/4	E0
common	Spring	1	0.1	1	noise	12 / 12	3200	0.5/4	E0
common	Spring	1	0.3	0	mean	1 / 0	3200	0.5/4	E0
common	Spring	1	0.3	1	noise	2 / 2	3200	0.5/4	E0
common	Spring	1	1	0	mean	1 / 0	3200	0.5/4	E0
common	Spring	1	1	1	noise	2 / 2	3200	0.5/4	E0
common	Spring	1.5	0	0	zero	1 / 0	3200	0.5/4	E3
common	Spring	1.5	0.001	0	mean	1 / 0	3200	0.5/4	E3
common	Spring	1.5	0.001	1	noise	12 / 12	3200	0.5/4	E3
common	Spring	1.5	0.003	0	mean	1 / 0	3200	0.5/4	E3
common	Spring	1.5	0.003	1	noise	12 / 12	3200	0.5/4	E3
common	Spring	1.5	0.01	0	mean	1 / 0	3200	0.5/4	E3

Final canonical design and forcing. Seed blocks are counted within each condition only; they must not be summed across sigma branches, inputs or retained-prefix comparisons. Controls contribute0 independent stochastic seeds. Cadence lists are exact, including mixed native sources. Each page CSV retains every member's complete canonical source record, archive hashes, cohort/lineage identity and prefix/top-up provenance. Native units differ across models. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_020_S1_conditions_18.csv

S1 · Final canonical design and forcing · Complete condition, seed and native observation inventory

Context	Model	C_0	f	σ	Kind	Paths / seed blocks	Native H	Scalar / profile Δt	Origin
common	Spring	1.5	0.01	1	noise	12 / 12	3200	0.5/4	E3
common	Spring	1.5	0.03	0	mean	1 / 0	3200	0.5/4	E3
common	Spring	1.5	0.03	1	noise	12 / 12	3200	0.5/4	E3
common	Spring	1.5	0.1	0	mean	1 / 0	3200	0.5/4	E3
common	Spring	1.5	0.1	1	noise	12 / 12	3200	0.5/4	E3
common	Spring	2	0	0	zero	1 / 0	3200	0.5/4	E0
common	Spring	2	0.0001	0	mean	1 / 0	3200	0.5/4	E0
common	Spring	2	0.0001	1	noise	2 / 2	3200	0.5/4	E0
common	Spring	2	0.0003	0	mean	1 / 0	3200	0.5/4	E0
common	Spring	2	0.0003	1	noise	2 / 2	3200	0.5/4	E0
common	Spring	2	0.001	0	mean	1 / 0	3200	0.5/4	E0
common	Spring	2	0.001	1	noise	12 / 12	3200	0.5/4	E0
common	Spring	2	0.003	0	mean	1 / 0	3200	0.5/4	E0
common	Spring	2	0.003	1	noise	12 / 12	3200	0.5/4	E0
common	Spring	2	0.005	0	mean	1 / 0	3200	0.5/0.5	E2
common	Spring	2	0.005	1	noise	12 / 12	3200	0.5/0.5	E2
common	Spring	2	0.01	0	mean	1 / 0	3200	0.5/4	E0
common	Spring	2	0.01	1	noise	12 / 12	3200	0.5/4	E0

Final canonical design and forcing. Seed blocks are counted within each condition only; they must not be summed across sigma branches, inputs or retained-prefix comparisons. Controls contribute0 independent stochastic seeds. Cadence lists are exact, including mixed native sources. Each page CSV retains every member’s complete canonical source record, archive hashes, cohort/lineage identity and prefix/top-up provenance. Native units differ across models. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_021_S1_conditions_19.csv

S1 · Final canonical design and forcing · Complete condition, seed and native observation inventory

Context	Model	C_0	f	σ	Kind	Paths / seed blocks	Native H	Scalar / profile Δt	Origin
common	Spring	2	0.02	0	mean	1 / 0	3200	0.5/0.5	E2
common	Spring	2	0.02	1	noise	12 / 12	3200	0.5/0.5	E2
common	Spring	2	0.03	0	mean	1 / 0	3200	0.5/4	E0
common	Spring	2	0.03	1	noise	12 / 12	3200	0.5/4	E0
common	Spring	2	0.05	0	mean	1 / 0	3200	0.5/0.5	E2
common	Spring	2	0.05	1	noise	12 / 12	3200	0.5/0.5	E2
common	Spring	2	0.1	0	mean	1 / 0	3200	0.5/4	E0
common	Spring	2	0.1	1	noise	12 / 12	3200	0.5/4	E0
common	Spring	2	0.3	0	mean	1 / 0	3200	0.5/4	E0
common	Spring	2	0.3	1	noise	2 / 2	3200	0.5/4	E0
common	Spring	2	1	0	mean	1 / 0	3200	0.5/4	E0
common	Spring	2	1	1	noise	2 / 2	3200	0.5/4	E0
common	Spring	2.5	0	0	zero	1 / 0	3200	0.5/4	E3
common	Spring	2.5	0.001	0	mean	1 / 0	3200	0.5/4	E3
common	Spring	2.5	0.001	1	noise	12 / 12	3200	0.5/4	E3
common	Spring	2.5	0.003	0	mean	1 / 0	3200	0.5/4	E3
common	Spring	2.5	0.003	1	noise	12 / 12	3200	0.5/4	E3
common	Spring	2.5	0.01	0	mean	1 / 0	3200	0.5/4	E3

Final canonical design and forcing. Seed blocks are counted within each condition only; they must not be summed across sigma branches, inputs or retained-prefix comparisons. Controls contribute0 independent stochastic seeds. Cadence lists are exact, including mixed native sources. Each page CSV retains every member’s complete canonical source record, archive hashes, cohort/lineage identity and prefix/top-up provenance. Native units differ across models. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_022_S1_conditions_20.csv

S1 · Final canonical design and forcing · Complete condition, seed and native observation inventory

Context	Model	C_0	f	σ	Kind	Paths / seed blocks	Native H	Scalar / profile Δt	Origin
common	Spring	2.5	0.01	1	noise	12 / 12	3200	0.5/4	E3
common	Spring	2.5	0.03	0	mean	1 / 0	3200	0.5/4	E3
common	Spring	2.5	0.03	1	noise	12 / 12	3200	0.5/4	E3
common	Spring	2.5	0.1	0	mean	1 / 0	3200	0.5/4	E3
common	Spring	2.5	0.1	1	noise	12 / 12	3200	0.5/4	E3
common	Spring	3	0	0	zero	1 / 0	3200	0.5/4	E0
common	Spring	3	0.0001	0	mean	1 / 0	3200	0.5/4	E0
common	Spring	3	0.0001	1	noise	2 / 2	3200	0.5/4	E0
common	Spring	3	0.0003	0	mean	1 / 0	3200	0.5/4	E0
common	Spring	3	0.0003	1	noise	2 / 2	3200	0.5/4	E0
common	Spring	3	0.001	0	mean	1 / 0	3200	0.5/4	E0
common	Spring	3	0.001	1	noise	12 / 12	3200	0.5/4	E0
common	Spring	3	0.003	0	mean	1 / 0	3200	0.5/4	E0
common	Spring	3	0.003	1	noise	12 / 12	3200	0.5/4	E0
common	Spring	3	0.005	0	mean	1 / 0	3200	0.5/0.5	E2
common	Spring	3	0.005	1	noise	12 / 12	3200	0.5/0.5	E2
common	Spring	3	0.01	0	mean	1 / 0	3200	0.5/4	E0
common	Spring	3	0.01	1	noise	12 / 12	3200	0.5/4	E0

Final canonical design and forcing. Seed blocks are counted within each condition only; they must not be summed across sigma branches, inputs or retained-prefix comparisons. Controls contribute0 independent stochastic seeds. Cadence lists are exact, including mixed native sources. Each page CSV retains every member's complete canonical source record, archive hashes, cohort/lineage identity and prefix/top-up provenance. Native units differ across models. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_023_S1_conditions_21.csv

S1 · Final canonical design and forcing · Complete condition, seed and native observation inventory

Context	Model	C_0	f	σ	Kind	Paths / seed blocks	Native H	Scalar / profile Δt	Origin
common	Spring	3	0.02	0	mean	1 / 0	3200	0.5/0.5	E2
common	Spring	3	0.02	1	noise	12 / 12	3200	0.5/0.5	E2
common	Spring	3	0.03	0	mean	1 / 0	3200	0.5/4	E0
common	Spring	3	0.03	1	noise	12 / 12	3200	0.5/4	E0
common	Spring	3	0.05	0	mean	1 / 0	3200	0.5/0.5	E2
common	Spring	3	0.05	1	noise	12 / 12	3200	0.5/0.5	E2
common	Spring	3	0.1	0	mean	1 / 0	3200	0.5/4	E0
common	Spring	3	0.1	1	noise	12 / 12	3200	0.5/4	E0
common	Spring	3	0.3	0	mean	1 / 0	3200	0.5/4	E0
common	Spring	3	0.3	1	noise	2 / 2	3200	0.5/4	E0
common	Spring	3	1	0	mean	1 / 0	3200	0.5/4	E0
common	Spring	3	1	1	noise	2 / 2	3200	0.5/4	E0
E1_extra	Wave-pinning	20	0.03	0.25	noise	12 / 12	32000	5/40	E1
E1_extra	Wave-pinning	20	0.03	0.5	noise	12 / 12	32000	5/40	E1
E1_extra	Wave-pinning	20	0.03	2	noise	12 / 12	32000	5/40	E1
E1_extra	Wave-pinning	20	0.1	0.25	noise	12 / 12	32000	5/40	E1
E1_extra	Wave-pinning	20	0.1	0.5	noise	12 / 12	32000	5/40	E1
E1_extra	Wave-pinning	20	0.1	2	noise	12 / 12	32000	5/40	E1

Final canonical design and forcing. Seed blocks are counted within each condition only; they must not be summed across sigma branches, inputs or retained-prefix comparisons. Controls contribute0 independent stochastic seeds. Cadence lists are exact, including mixed native sources. Each page CSV retains every member’s complete canonical source record, archive hashes, cohort/lineage identity and prefix/top-up provenance. Native units differ across models. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_024_S1_conditions_22.csv

S1 · Final canonical design and forcing · Complete condition, seed and native observation inventory

Context	Model	C_0	f	σ	Kind	Paths / seed blocks	Native H	Scalar / profile Δt	Origin
E1_extra	Wave-pinning	40	0.03	0.25	noise	12 / 12	32000	5/40	E1
E1_extra	Wave-pinning	40	0.03	0.5	noise	12 / 12	32000	5/40	E1
E1_extra	Wave-pinning	40	0.03	2	noise	12 / 12	32000	5/40	E1
E1_extra	Wave-pinning	40	0.1	0.25	noise	12 / 12	32000	5/40	E1
E1_extra	Wave-pinning	40	0.1	0.5	noise	12 / 12	32000	5/40	E1
E1_extra	Wave-pinning	40	0.1	2	noise	12 / 12	32000	5/40	E1
E1_extra	Wave-pinning	60	0.03	0.25	noise	12 / 12	32000	5/40	E1
E1_extra	Wave-pinning	60	0.03	0.5	noise	12 / 12	32000	5/40	E1
E1_extra	Wave-pinning	60	0.03	2	noise	12 / 12	32000	5/40	E1
E1_extra	Wave-pinning	60	0.1	0.25	noise	12 / 12	32000	5/40	E1
E1_extra	Wave-pinning	60	0.1	0.5	noise	12 / 12	32000	5/40	E1
E1_extra	Wave-pinning	60	0.1	2	noise	12 / 12	32000	5/40	E1
E1_extra	Goryachev	20	0.02	0.25	noise	12 / 12	32000	5/40	E1
E1_extra	Goryachev	20	0.02	0.5	noise	12 / 12	32000	5/40	E1
E1_extra	Goryachev	20	0.02	2	noise	12 / 12	32000	5/40	E1
E1_extra	Goryachev	20	0.03	0.25	noise	12 / 12	32000	5/40	E1
E1_extra	Goryachev	20	0.03	0.5	noise	12 / 12	32000	5/40	E1
E1_extra	Goryachev	20	0.03	2	noise	12 / 12	32000	5/40	E1

Final canonical design and forcing. Seed blocks are counted within each condition only; they must not be summed across sigma branches, inputs or retained-prefix comparisons. Controls contribute0 independent stochastic seeds. Cadence lists are exact, including mixed native sources. Each page CSV retains every member's complete canonical source record, archive hashes, cohort/lineage identity and prefix/top-up provenance. Native units differ across models. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_025_S1_conditions_23.csv

S1 · Final canonical design and forcing · Complete condition, seed and native observation inventory

Context	Model	C_0	f	σ	Kind	Paths / seed blocks	Native H	Scalar / profile Δt	Origin
E1_extra	Goryachev	40	0.02	0.25	noise	12 / 12	32000	5/40	E1
E1_extra	Goryachev	40	0.02	0.5	noise	12 / 12	32000	5/40	E1
E1_extra	Goryachev	40	0.02	2	noise	12 / 12	32000	5/40	E1
E1_extra	Goryachev	40	0.03	0.25	noise	12 / 12	32000	5/40	E1
E1_extra	Goryachev	40	0.03	0.5	noise	12 / 12	32000	5/40	E1
E1_extra	Goryachev	40	0.03	2	noise	12 / 12	32000	5/40	E1
E1_extra	Goryachev	60	0.02	0.25	noise	12 / 12	32000	5/40	E1
E1_extra	Goryachev	60	0.02	0.5	noise	12 / 12	32000	5/40	E1
E1_extra	Goryachev	60	0.02	2	noise	12 / 12	32000	5/40	E1
E1_extra	Goryachev	60	0.03	0.25	noise	12 / 12	32000	5/40	E1
E1_extra	Goryachev	60	0.03	0.5	noise	12 / 12	32000	5/40	E1
E1_extra	Goryachev	60	0.03	2	noise	12 / 12	32000	5/40	E1
E1_extra	Holmes 3	20	0.03	0.25	noise	12 / 12	32000	5/40	E1
E1_extra	Holmes 3	20	0.03	0.5	noise	12 / 12	32000	5/40	E1
E1_extra	Holmes 3	20	0.03	2	noise	12 / 12	32000	5/40	E1
E1_extra	Holmes 3	20	0.1	0.25	noise	12 / 12	32000	5/40	E1
E1_extra	Holmes 3	20	0.1	0.5	noise	12 / 12	32000	5/40	E1
E1_extra	Holmes 3	20	0.1	2	noise	12 / 12	32000	5/40	E1

Final canonical design and forcing. Seed blocks are counted within each condition only; they must not be summed across sigma branches, inputs or retained-prefix comparisons. Controls contribute0 independent stochastic seeds. Cadence lists are exact, including mixed native sources. Each page CSV retains every member’s complete canonical source record, archive hashes, cohort/lineage identity and prefix/top-up provenance. Native units differ across models. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_026_S1_conditions_24.csv

S1 · Final canonical design and forcing · Complete condition, seed and native observation inventory

Context	Model	C_0	f	σ	Kind	Paths / seed blocks	Native H	Scalar / profile Δt	Origin
E1_extra	Holmes 3	40	0.03	0.25	noise	12 / 12	32000	5/40	E1
E1_extra	Holmes 3	40	0.03	0.5	noise	12 / 12	32000	5/40	E1
E1_extra	Holmes 3	40	0.03	2	noise	12 / 12	32000	5/40	E1
E1_extra	Holmes 3	40	0.1	0.25	noise	12 / 12	32000	5/40	E1
E1_extra	Holmes 3	40	0.1	0.5	noise	12 / 12	32000	5/40	E1
E1_extra	Holmes 3	40	0.1	2	noise	12 / 12	32000	5/40	E1
E1_extra	Holmes 3	60	0.03	0.25	noise	12 / 12	32000	5/40	E1
E1_extra	Holmes 3	60	0.03	0.5	noise	12 / 12	32000	5/40	E1
E1_extra	Holmes 3	60	0.03	2	noise	12 / 12	32000	5/40	E1
E1_extra	Holmes 3	60	0.1	0.25	noise	12 / 12	32000	5/40	E1
E1_extra	Holmes 3	60	0.1	0.5	noise	12 / 12	32000	5/40	E1
E1_extra	Holmes 3	60	0.1	2	noise	12 / 12	32000	5/40	E1
E1_extra	Spring	1	0.03	0.25	noise	12 / 12	3200	0.5/4	E1
E1_extra	Spring	1	0.03	0.5	noise	12 / 12	3200	0.5/4	E1
E1_extra	Spring	1	0.03	2	noise	12 / 12	3200	0.5/4	E1
E1_extra	Spring	1	0.1	0.25	noise	12 / 12	3200	0.5/4	E1
E1_extra	Spring	1	0.1	0.5	noise	12 / 12	3200	0.5/4	E1
E1_extra	Spring	1	0.1	2	noise	12 / 12	3200	0.5/4	E1

Final canonical design and forcing. Seed blocks are counted within each condition only; they must not be summed across sigma branches, inputs or retained-prefix comparisons. Controls contribute0 independent stochastic seeds. Cadence lists are exact, including mixed native sources. Each page CSV retains every member’s complete canonical source record, archive hashes, cohort/lineage identity and prefix/top-up provenance. Native units differ across models. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_027_S1_conditions_25.csv

S1 · Final canonical design and forcing · Complete condition, seed and native observation inventory

Context	Model	C_0	f	σ	Kind	Paths / seed blocks	Native H	Scalar / profile Δt	Origin
E1_extra	Spring	2	0.03	0.25	noise	12 / 12	3200	0.5/4	E1
E1_extra	Spring	2	0.03	0.5	noise	12 / 12	3200	0.5/4	E1
E1_extra	Spring	2	0.03	2	noise	12 / 12	3200	0.5/4	E1
E1_extra	Spring	2	0.1	0.25	noise	12 / 12	3200	0.5/4	E1
E1_extra	Spring	2	0.1	0.5	noise	12 / 12	3200	0.5/4	E1
E1_extra	Spring	2	0.1	2	noise	12 / 12	3200	0.5/4	E1
E1_extra	Spring	3	0.03	0.25	noise	12 / 12	3200	0.5/4	E1
E1_extra	Spring	3	0.03	0.5	noise	12 / 12	3200	0.5/4	E1
E1_extra	Spring	3	0.03	2	noise	12 / 12	3200	0.5/4	E1
E1_extra	Spring	3	0.1	0.25	noise	12 / 12	3200	0.5/4	E1
E1_extra	Spring	3	0.1	0.5	noise	12 / 12	3200	0.5/4	E1
E1_extra	Spring	3	0.1	2	noise	12 / 12	3200	0.5/4	E1
E4_long	Wave-pinning	20	0	0	zero	1 / 0	128000	5/5	E4
E4_long	Wave-pinning	20	0.0001	0	mean	1 / 0	128000	5/5	E4
E4_long	Wave-pinning	20	0.0001	1	noise	12 / 12	128000	5/5	E4
E4_long	Wave-pinning	20	0.0003	0	mean	1 / 0	128000	5/5	E4
E4_long	Wave-pinning	20	0.0003	1	noise	12 / 12	128000	5/5	E4
E4_long	Wave-pinning	20	0.001	0	mean	1 / 0	128000	5/5	E4

Final canonical design and forcing. Seed blocks are counted within each condition only; they must not be summed across sigma branches, inputs or retained-prefix comparisons. Controls contribute0 independent stochastic seeds. Cadence lists are exact, including mixed native sources. Each page CSV retains every member's complete canonical source record, archive hashes, cohort/lineage identity and prefix/top-up provenance. Native units differ across models. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_028_S1_conditions_26.csv

S1 · Final canonical design and forcing · Complete condition, seed and native observation inventory

Context	Model	C_0	f	σ	Kind	Paths / seed blocks	Native H	Scalar / profile Δt	Origin
E4_long	Wave-pinning	20	0.001	1	noise	12 / 12	128000	5/5	E4
E4_long	Wave-pinning	20	0.003	0	mean	1 / 0	128000	5/5	E4
E4_long	Wave-pinning	20	0.003	1	noise	12 / 12	128000	5/5	E4
E4_long	Wave-pinning	40	0	0	zero	1 / 0	128000	5/5	E4
E4_long	Wave-pinning	40	0.0001	0	mean	1 / 0	128000	5/5	E4
E4_long	Wave-pinning	40	0.0001	1	noise	12 / 12	128000	5/5	E4
E4_long	Wave-pinning	40	0.0003	0	mean	1 / 0	128000	5/5	E4
E4_long	Wave-pinning	40	0.0003	1	noise	12 / 12	128000	5/5	E4
E4_long	Wave-pinning	40	0.001	0	mean	1 / 0	128000	5/5	E4
E4_long	Wave-pinning	40	0.001	1	noise	12 / 12	128000	5/5	E4
E4_long	Wave-pinning	40	0.003	0	mean	1 / 0	128000	5/5	E4
E4_long	Wave-pinning	40	0.003	1	noise	12 / 12	128000	5/5	E4
E4_long	Wave-pinning	40	0.01	0	mean	1 / 0	128000	5/5	E4
E4_long	Wave-pinning	40	0.01	1	noise	12 / 12	128000	5/5	E4
E4_long	Wave-pinning	60	0	0	zero	1 / 0	128000	5/5	E4
E4_long	Wave-pinning	60	0.0001	0	mean	1 / 0	128000	5/5	E4
E4_long	Wave-pinning	60	0.0001	1	noise	12 / 12	128000	5/5	E4
E4_long	Wave-pinning	60	0.0003	0	mean	1 / 0	128000	5/5	E4

Final canonical design and forcing. Seed blocks are counted within each condition only; they must not be summed across sigma branches, inputs or retained-prefix comparisons. Controls contribute0 independent stochastic seeds. Cadence lists are exact, including mixed native sources. Each page CSV retains every member’s complete canonical source record, archive hashes, cohort/lineage identity and prefix/top-up provenance. Native units differ across models. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_029_S1_conditions_27.csv

S1 · Final canonical design and forcing · Complete condition, seed and native observation inventory

Context	Model	C_0	f	σ	Kind	Paths / seed blocks	Native H	Scalar / profile Δt	Origin
E4_long	Wave-pinning	60	0.0003	1	noise	12 / 12	128000	5/5	E4
E4_long	Wave-pinning	60	0.001	0	mean	1 / 0	128000	5/5	E4
E4_long	Wave-pinning	60	0.001	1	noise	12 / 12	128000	5/5	E4
E4_long	Wave-pinning	60	0.003	0	mean	1 / 0	128000	5/5	E4
E4_long	Wave-pinning	60	0.003	1	noise	12 / 12	128000	5/5	E4
E4_long	Wave-pinning	60	0.01	0	mean	1 / 0	128000	5/5	E4
E4_long	Wave-pinning	60	0.01	1	noise	12 / 12	128000	5/5	E4
E4_long	Goryachev	20	0	0	zero	1 / 0	128000	5/5	E4
E4_long	Goryachev	20	0.0001	0	mean	1 / 0	128000	5/5	E4
E4_long	Goryachev	20	0.0001	1	noise	12 / 12	128000	5/5	E4
E4_long	Goryachev	20	0.0003	0	mean	1 / 0	128000	5/5	E4
E4_long	Goryachev	20	0.0003	1	noise	12 / 12	128000	5/5	E4
E4_long	Goryachev	20	0.001	0	mean	1 / 0	128000	5/5	E4
E4_long	Goryachev	20	0.001	1	noise	12 / 12	128000	5/5	E4
E4_long	Goryachev	20	0.003	0	mean	1 / 0	128000	5/5	E4
E4_long	Goryachev	20	0.003	1	noise	12 / 12	128000	5/5	E4
E4_long	Goryachev	20	1	0	mean	1 / 0	128000	5/5	E4
E4_long	Goryachev	20	1	1	noise	12 / 12	128000	5/5	E4

Final canonical design and forcing. Seed blocks are counted within each condition only; they must not be summed across sigma branches, inputs or retained-prefix comparisons. Controls contribute0 independent stochastic seeds. Cadence lists are exact, including mixed native sources. Each page CSV retains every member’s complete canonical source record, archive hashes, cohort/lineage identity and prefix/top-up provenance. Native units differ across models. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_030_S1_conditions_28.csv

S1 · Final canonical design and forcing · Complete condition, seed and native observation inventory

Context	Model	C_0	f	σ	Kind	Paths / seed blocks	Native H	Scalar / profile Δt	Origin
E4_long	Goryachev	40	0	0	zero	1 / 0	128000	5/5	E4
E4_long	Goryachev	40	0.0001	0	mean	1 / 0	128000	5/5	E4
E4_long	Goryachev	40	0.0001	1	noise	12 / 12	128000	5/5	E4
E4_long	Goryachev	40	0.0003	0	mean	1 / 0	128000	5/5	E4
E4_long	Goryachev	40	0.0003	1	noise	12 / 12	128000	5/5	E4
E4_long	Goryachev	40	0.001	0	mean	1 / 0	128000	5/5	E4
E4_long	Goryachev	40	0.001	1	noise	12 / 12	128000	5/5	E4
E4_long	Goryachev	40	0.003	0	mean	1 / 0	128000	5/5	E4
E4_long	Goryachev	40	0.003	1	noise	12 / 12	128000	5/5	E4
E4_long	Goryachev	40	0.01	0	mean	1 / 0	128000	5/5	E4
E4_long	Goryachev	40	0.01	1	noise	12 / 12	128000	5/5	E4
E4_long	Goryachev	40	0.1	0	mean	1 / 0	128000	5/5	E4
E4_long	Goryachev	40	0.1	1	noise	12 / 12	128000	5/5	E4
E4_long	Goryachev	60	0	0	zero	1 / 0	128000	5/5	E4
E4_long	Goryachev	60	0.0001	0	mean	1 / 0	128000	5/5	E4
E4_long	Goryachev	60	0.0001	1	noise	12 / 12	128000	5/5	E4
E4_long	Goryachev	60	0.0003	0	mean	1 / 0	128000	5/5	E4
E4_long	Goryachev	60	0.0003	1	noise	12 / 12	128000	5/5	E4

Final canonical design and forcing. Seed blocks are counted within each condition only; they must not be summed across sigma branches, inputs or retained-prefix comparisons. Controls contribute0 independent stochastic seeds. Cadence lists are exact, including mixed native sources. Each page CSV retains every member’s complete canonical source record, archive hashes, cohort/lineage identity and prefix/top-up provenance. Native units differ across models. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_031_S1_conditions_29.csv

S1 · Final canonical design and forcing · Complete condition, seed and native observation inventory

Context	Model	C_0	f	σ	Kind	Paths / seed blocks	Native H	Scalar / profile Δt	Origin
E4_long	Goryachev	60	0.001	0	mean	1 / 0	128000	5/5	E4
E4_long	Goryachev	60	0.001	1	noise	12 / 12	128000	5/5	E4
E4_long	Goryachev	60	0.003	0	mean	1 / 0	128000	5/5	E4
E4_long	Goryachev	60	0.003	1	noise	12 / 12	128000	5/5	E4
E4_long	Goryachev	60	0.01	0	mean	1 / 0	128000	5/5	E4
E4_long	Goryachev	60	0.01	1	noise	12 / 12	128000	5/5	E4
E4_long	Goryachev	60	0.015	0	mean	1 / 0	128000	5/5	E4
E4_long	Goryachev	60	0.015	1	noise	12 / 12	128000	5/5	E4
E4_long	Goryachev	60	0.02	0	mean	1 / 0	128000	5/5	E4
E4_long	Goryachev	60	0.02	1	noise	12 / 12	128000	5/5	E4
E4_long	Holmes 3	20	0	0	zero	1 / 0	128000	5/5	E4
E4_long	Holmes 3	20	0.0001	0	mean	1 / 0	128000	5/5	E4
E4_long	Holmes 3	20	0.0001	1	noise	12 / 12	128000	5/5	E4
E4_long	Holmes 3	20	0.0003	0	mean	1 / 0	128000	5/5	E4
E4_long	Holmes 3	20	0.0003	1	noise	12 / 12	128000	5/5	E4
E4_long	Holmes 3	20	0.001	0	mean	1 / 0	128000	5/5	E4
E4_long	Holmes 3	20	0.001	1	noise	12 / 12	128000	5/5	E4
E4_long	Holmes 3	20	0.003	0	mean	1 / 0	128000	5/5	E4

Final canonical design and forcing. Seed blocks are counted within each condition only; they must not be summed across sigma branches, inputs or retained-prefix comparisons. Controls contribute0 independent stochastic seeds. Cadence lists are exact, including mixed native sources. Each page CSV retains every member's complete canonical source record, archive hashes, cohort/lineage identity and prefix/top-up provenance. Native units differ across models. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_032_S1_conditions_30.csv

S1 · Final canonical design and forcing · Complete condition, seed and native observation inventory

Context	Model	C_0	f	σ	Kind	Paths / seed blocks	Native H	Scalar / profile Δt	Origin
E4_long	Holmes 3	20	0.003	1	noise	12 / 12	128000	5/5	E4
E4_long	Holmes 3	40	0	0	zero	1 / 0	128000	5/5	E4
E4_long	Holmes 3	40	0.0001	0	mean	1 / 0	128000	5/5	E4
E4_long	Holmes 3	40	0.0001	1	noise	12 / 12	128000	5/5	E4
E4_long	Holmes 3	40	0.0003	0	mean	1 / 0	128000	5/5	E4
E4_long	Holmes 3	40	0.0003	1	noise	12 / 12	128000	5/5	E4
E4_long	Holmes 3	40	0.001	0	mean	1 / 0	128000	5/5	E4
E4_long	Holmes 3	40	0.001	1	noise	12 / 12	128000	5/5	E4
E4_long	Holmes 3	40	0.003	0	mean	1 / 0	128000	5/5	E4
E4_long	Holmes 3	40	0.003	1	noise	12 / 12	128000	5/5	E4
E4_long	Holmes 3	60	0	0	zero	1 / 0	128000	5/5	E4
E4_long	Holmes 3	60	0.0001	0	mean	1 / 0	128000	5/5	E4
E4_long	Holmes 3	60	0.0001	1	noise	12 / 12	128000	5/5	E4
E4_long	Holmes 3	60	0.0003	0	mean	1 / 0	128000	5/5	E4
E4_long	Holmes 3	60	0.0003	1	noise	12 / 12	128000	5/5	E4
E4_long	Holmes 3	60	0.001	0	mean	1 / 0	128000	5/5	E4
E4_long	Holmes 3	60	0.001	1	noise	12 / 12	128000	5/5	E4
E4_long	Holmes 3	60	0.003	0	mean	1 / 0	128000	5/5	E4

Final canonical design and forcing. Seed blocks are counted within each condition only; they must not be summed across sigma branches, inputs or retained-prefix comparisons. Controls contribute0 independent stochastic seeds. Cadence lists are exact, including mixed native sources. Each page CSV retains every member's complete canonical source record, archive hashes, cohort/lineage identity and prefix/top-up provenance. Native units differ across models. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_033_S1_conditions_31.csv

S1 · Final canonical design and forcing · Complete condition, seed and native observation inventory

Context	Model	C_0	f	σ	Kind	Paths / seed blocks	Native H	Scalar / profile Δt	Origin
E4_long	Holmes 3	60	0.003	1	noise	12 / 12	128000	5/5	E4
E4_long	Spring	1	0	0	zero	1 / 0	12800	0.5/0.5	E4
E4_long	Spring	1	0.0001	0	mean	1 / 0	12800	0.5/0.5	E4
E4_long	Spring	1	0.0001	1	noise	12 / 12	12800	0.5/0.5	E4
E4_long	Spring	1	0.0003	0	mean	1 / 0	12800	0.5/0.5	E4
E4_long	Spring	1	0.0003	1	noise	12 / 12	12800	0.5/0.5	E4
E4_long	Spring	2	0	0	zero	1 / 0	12800	0.5/0.5	E4
E4_long	Spring	2	0.0001	0	mean	1 / 0	12800	0.5/0.5	E4
E4_long	Spring	2	0.0001	1	noise	12 / 12	12800	0.5/0.5	E4
E4_long	Spring	2	0.0003	0	mean	1 / 0	12800	0.5/0.5	E4
E4_long	Spring	2	0.0003	1	noise	12 / 12	12800	0.5/0.5	E4
E4_long	Spring	2	0.001	0	mean	1 / 0	12800	0.5/0.5	E4
E4_long	Spring	2	0.001	1	noise	12 / 12	12800	0.5/0.5	E4
E4_long	Spring	3	0	0	zero	1 / 0	12800	0.5/0.5	E4
E4_long	Spring	3	0.0001	0	mean	1 / 0	12800	0.5/0.5	E4
E4_long	Spring	3	0.0001	1	noise	12 / 12	12800	0.5/0.5	E4
E4_long	Spring	3	0.0003	0	mean	1 / 0	12800	0.5/0.5	E4
E4_long	Spring	3	0.0003	1	noise	12 / 12	12800	0.5/0.5	E4

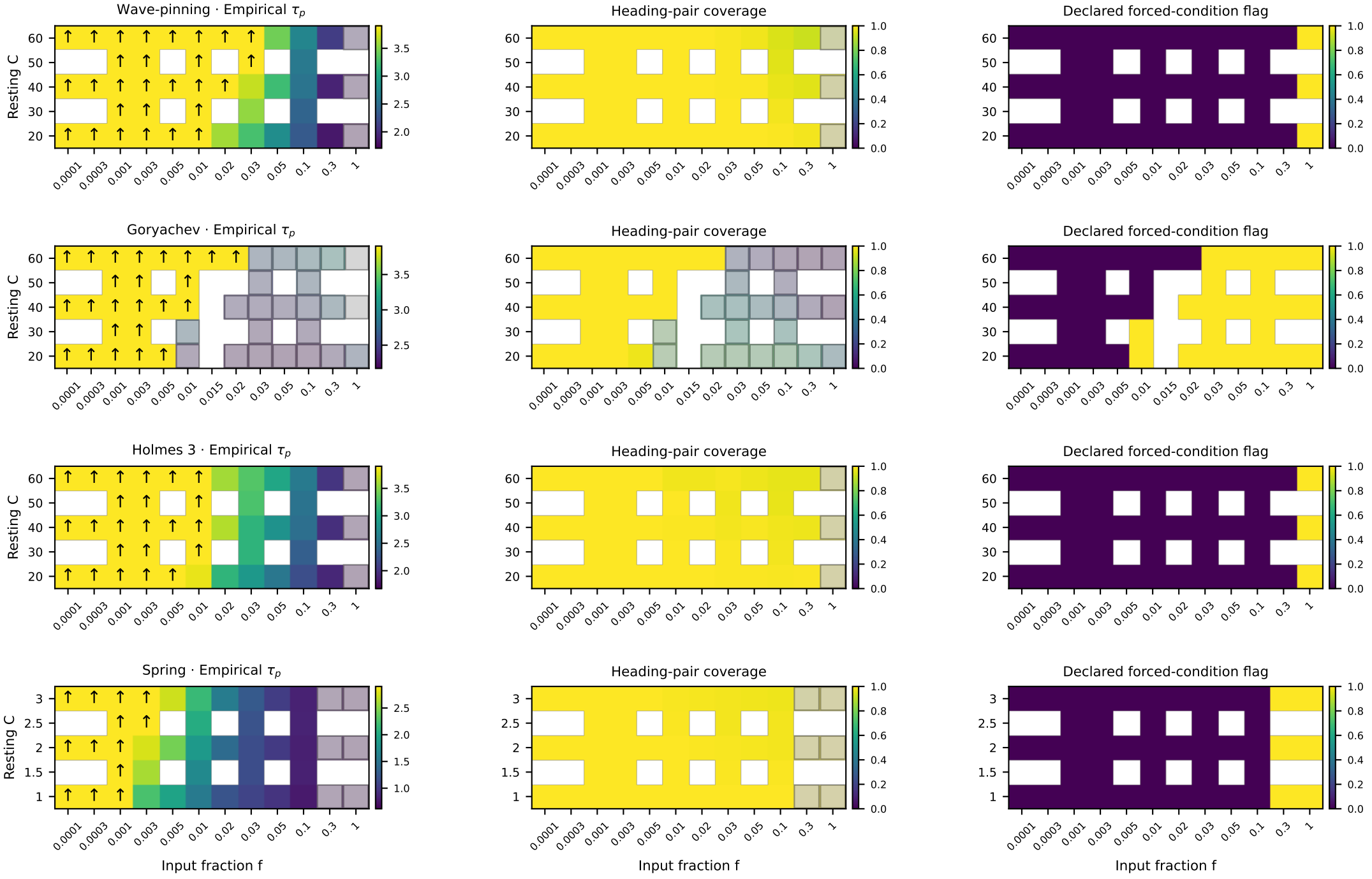
Final canonical design and forcing. Seed blocks are counted within each condition only; they must not be summed across sigma branches, inputs or retained-prefix comparisons. Controls contribute0 independent stochastic seeds. Cadence lists are exact, including mixed native sources. Each page CSV retains every member's complete canonical source record, archive hashes, cohort/lineage identity and prefix/top-up provenance. Native units differ across models. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_034_S1_conditions_32.csv

S1 · Final canonical design and forcing · Complete condition, seed and native observation inventory

Context	Model	C_0	f	σ	Kind	Paths / seed blocks	Native H	Scalar / profile Δt	Origin
E4_long	Spring	3	0.001	0	mean	1 / 0	12800	0.5/0.5	E4
E4_long	Spring	3	0.001	1	noise	12 / 12	12800	0.5/0.5	E4

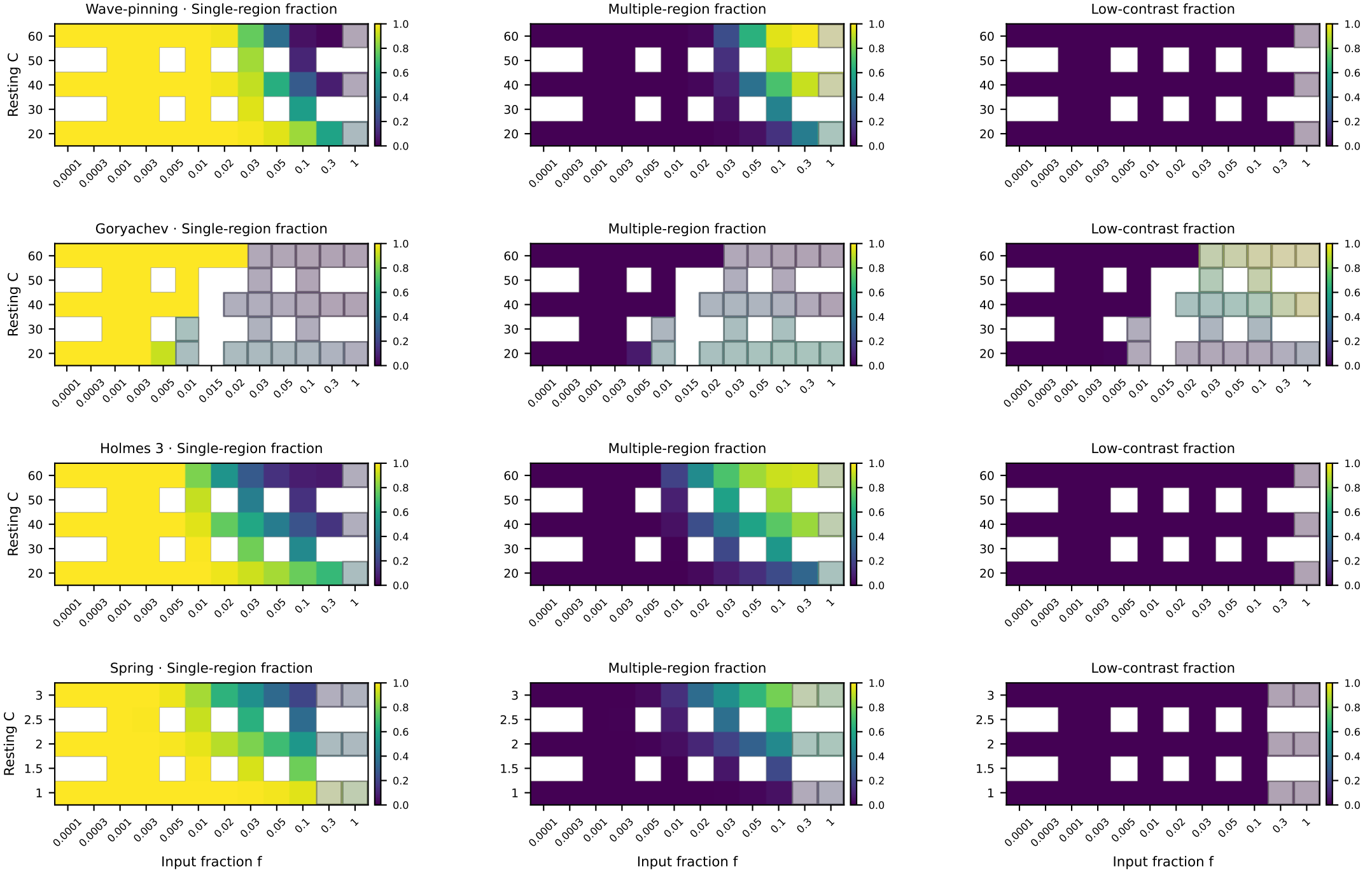
Final canonical design and forcing. Seed blocks are counted within each condition only; they must not be summed across sigma branches, inputs or retained-prefix comparisons. Controls contribute0 independent stochastic seeds. Cadence lists are exact, including mixed native sources. Each page CSV retains every member’s complete canonical source record, archive hashes, cohort/lineage identity and prefix/top-up provenance. Native units differ across models. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S1_final_design_20261020_035_S1_conditions_33.csv

S2 · Common observations · Condition overview · log₁₀ native persistence

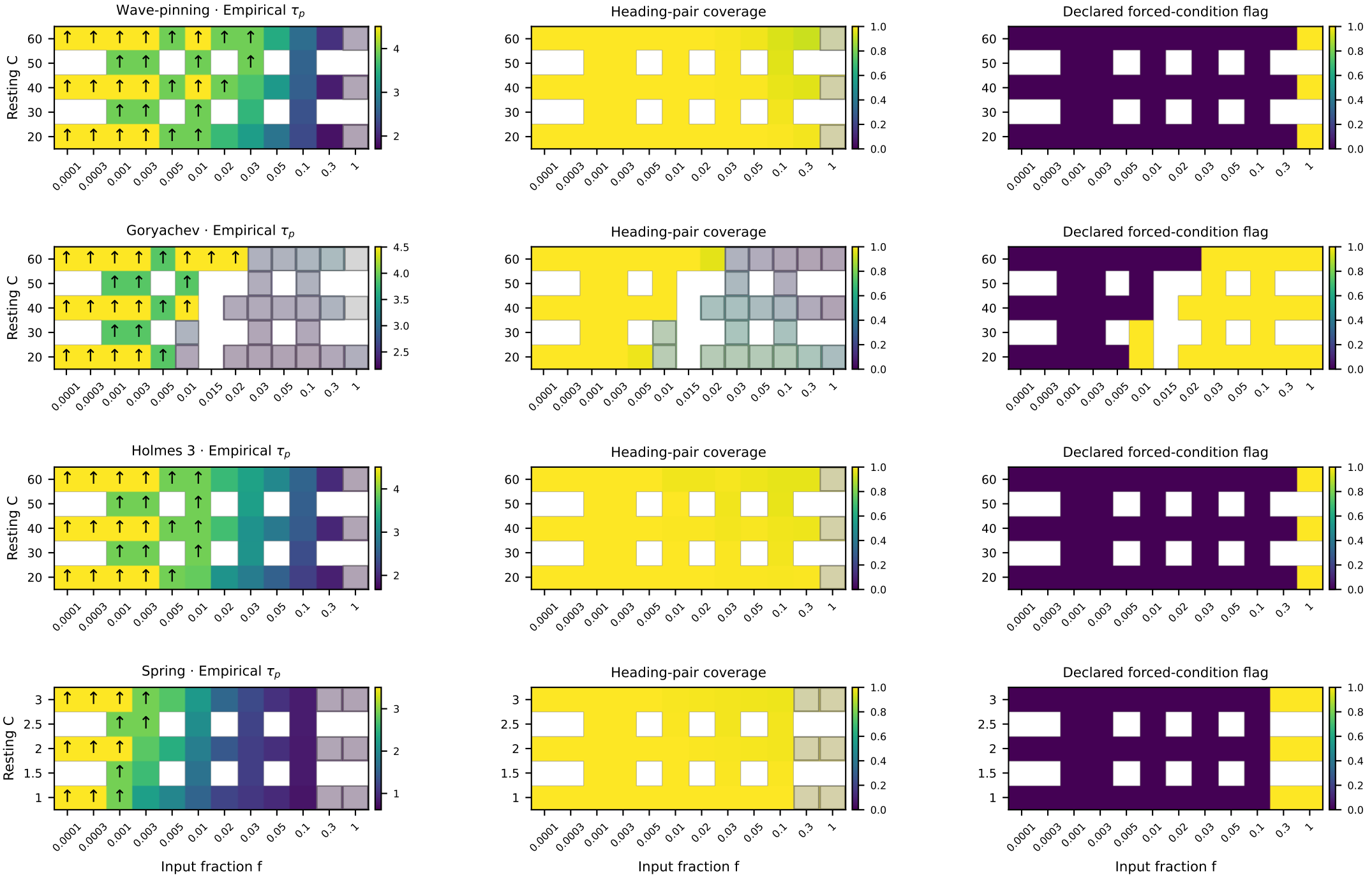


Common observations. All declared input strengths in this observation arm are shown. Grey overlays retain forced conditions; the exact display flag and its components remain in CSV. Arrows preserve censoring and cadence limits. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_canonical_measurements_20261020_001_S2_1.csv

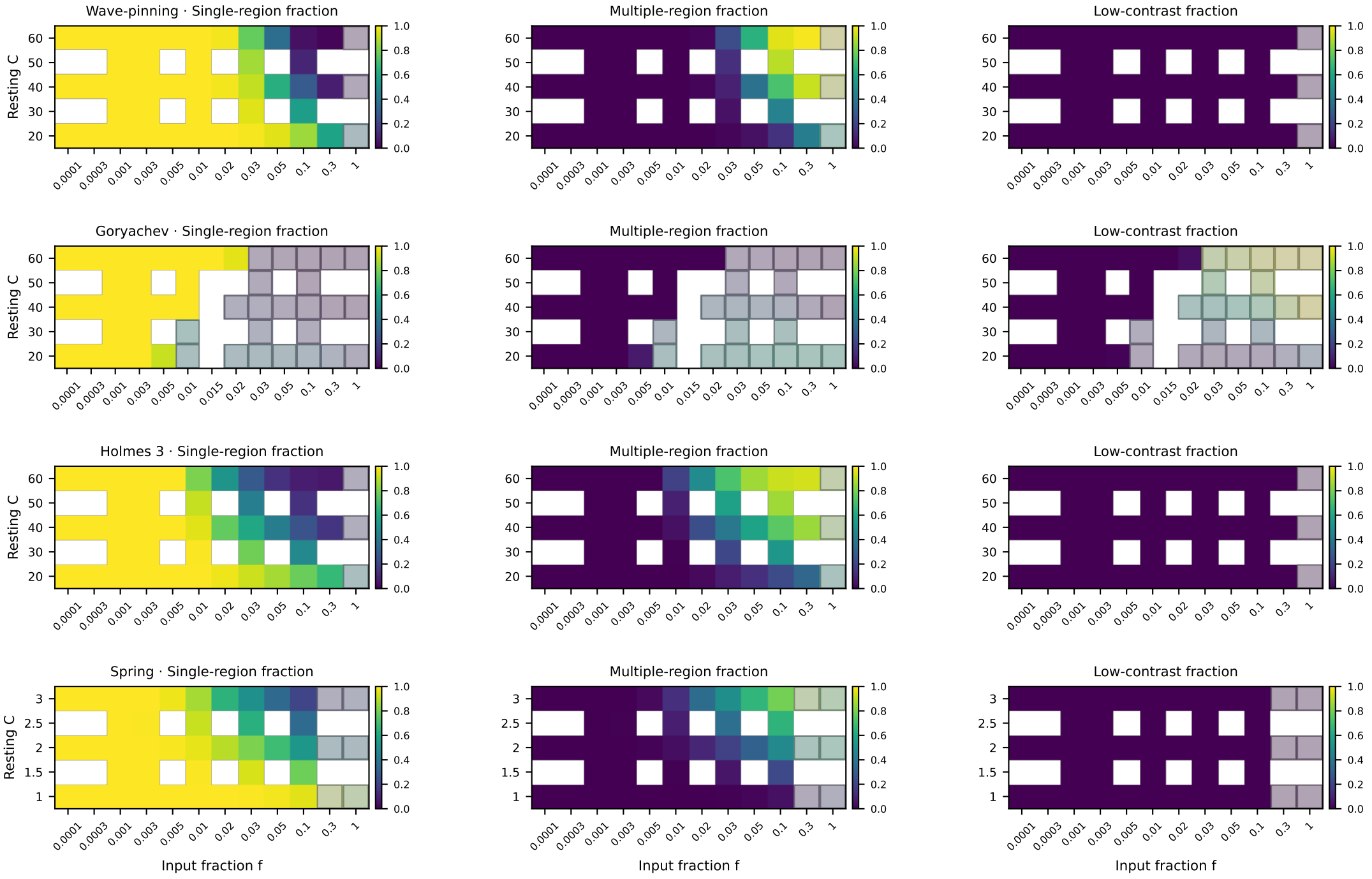
S2 · Common observations · Condition overview · state fractions



Common observations. All declared input strengths in this observation arm are shown. Grey overlays retain forced conditions; the exact display flag and its components remain in CSV. Arrows preserve censoring and cadence limits. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_canonical_measurements_20261020_002_S2_2.csv

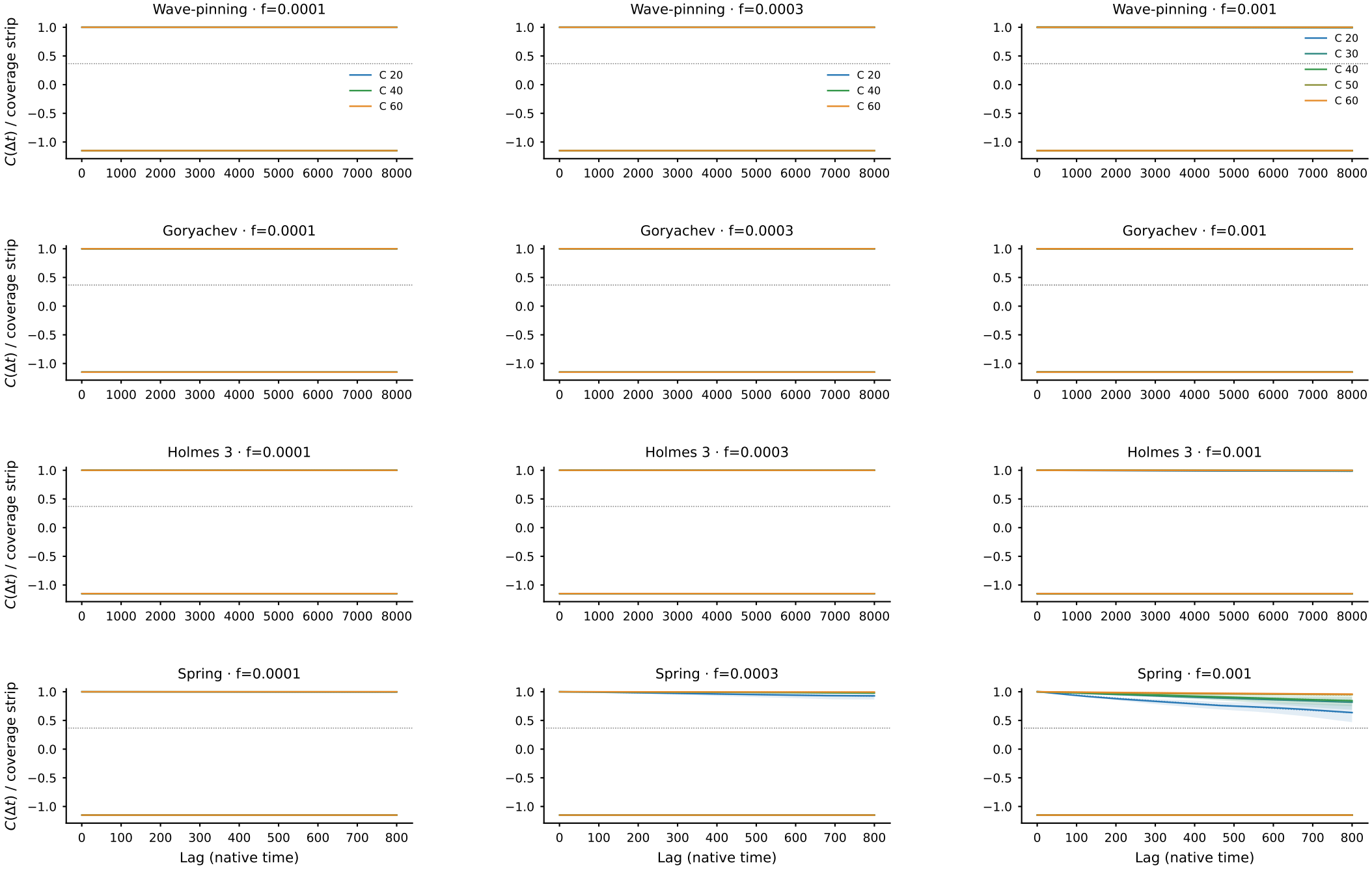


Longest qualified observations. All declared input strengths in this observation arm are shown. Grey overlays retain forced conditions; the exact display flag and its components remain in CSV. Arrows preserve censoring and cadence limits. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_canonical_measurements_20261020_001_S2_1.csv



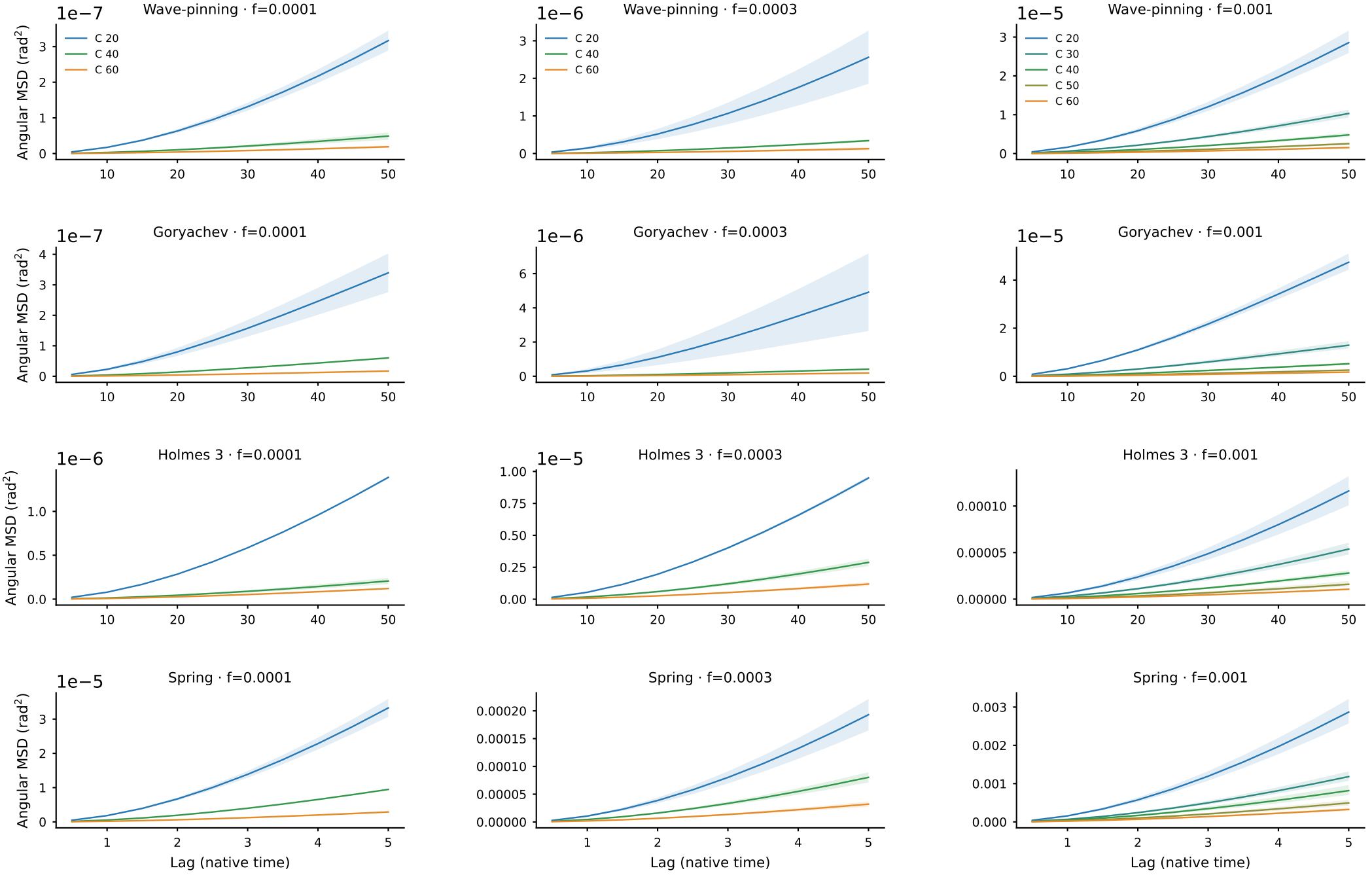
Longest qualified observations. All declared input strengths in this observation arm are shown. Grey overlays retain forced conditions; the exact display flag and its components remain in CSV. Arrows preserve censoring and cadence limits. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_canonical_measurements_20261020_002_S2_2.csv

S3 · Common observations · All heading correlations



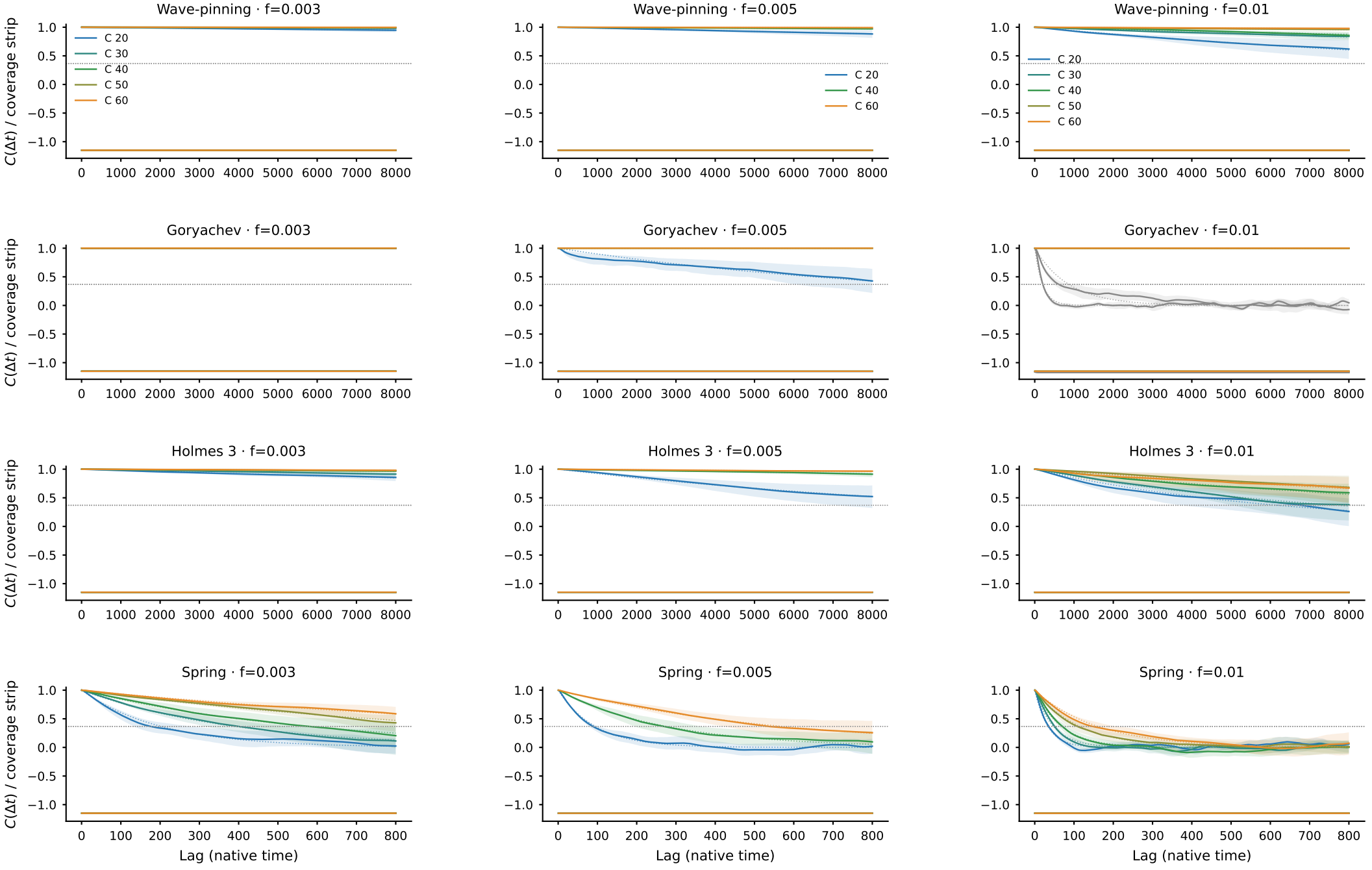
Common observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_canonical_measurements_20261020_003_S3_1_cosine.csv.gz

S3 · Common observations · Short-lag angular MSD



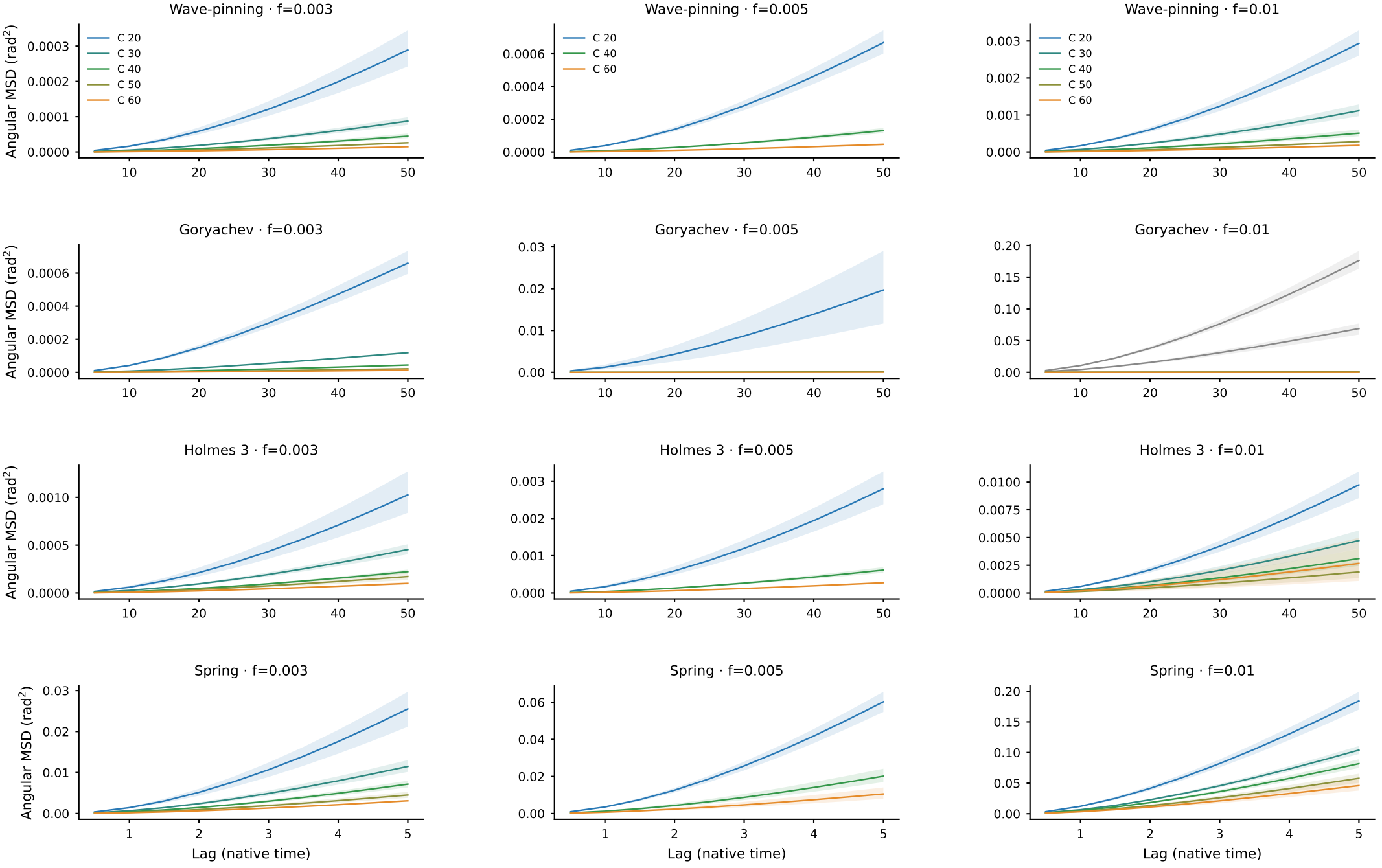
Common observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_canonical_measurements_20261020_004_S3_1_angular_msd.csv.gz

S3 · Common observations · All heading correlations



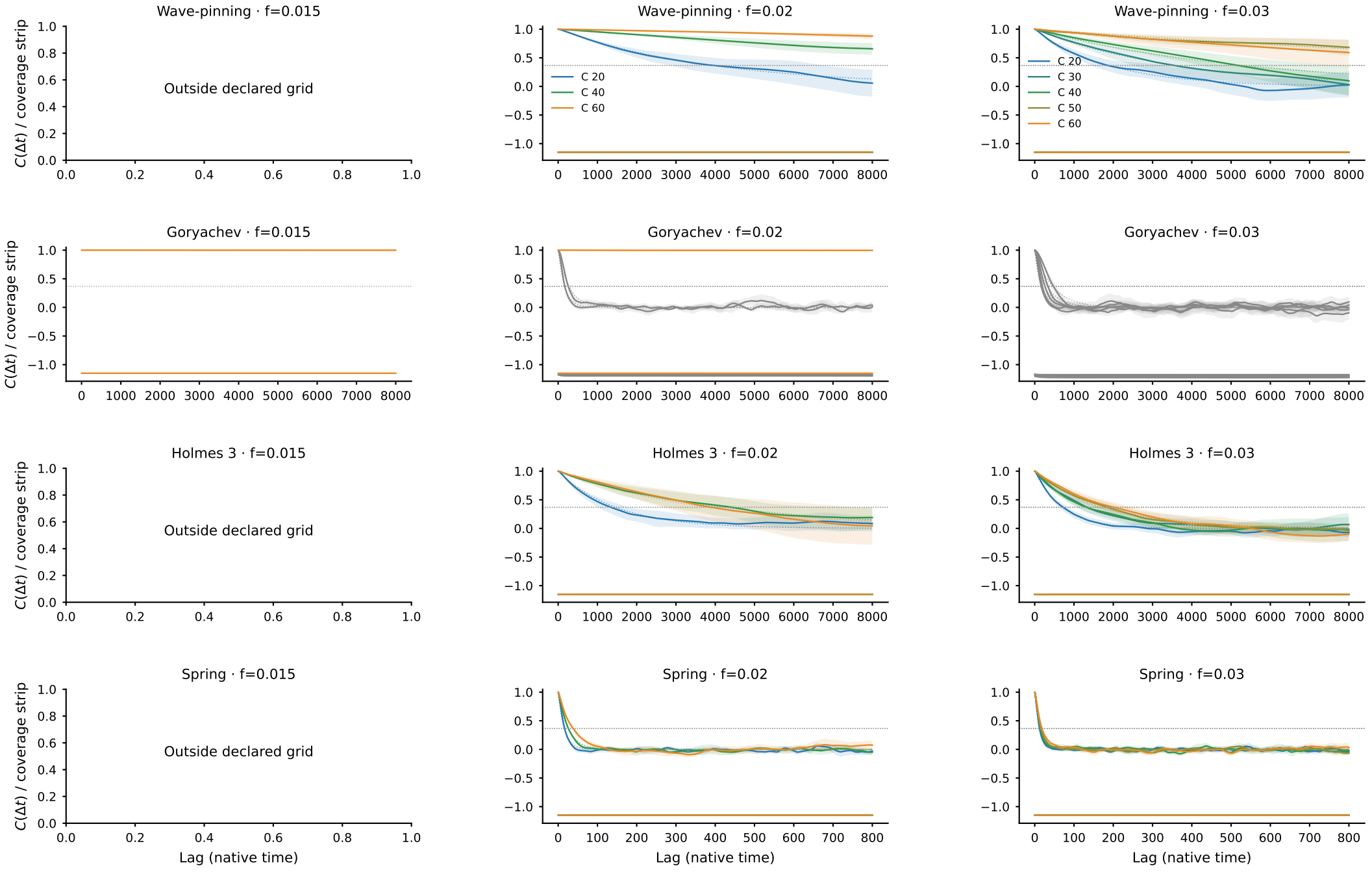
Common observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_canonical_measurements_20261020_005_S3_2_cosine.csv.gz

S3 · Common observations · Short-lag angular MSD



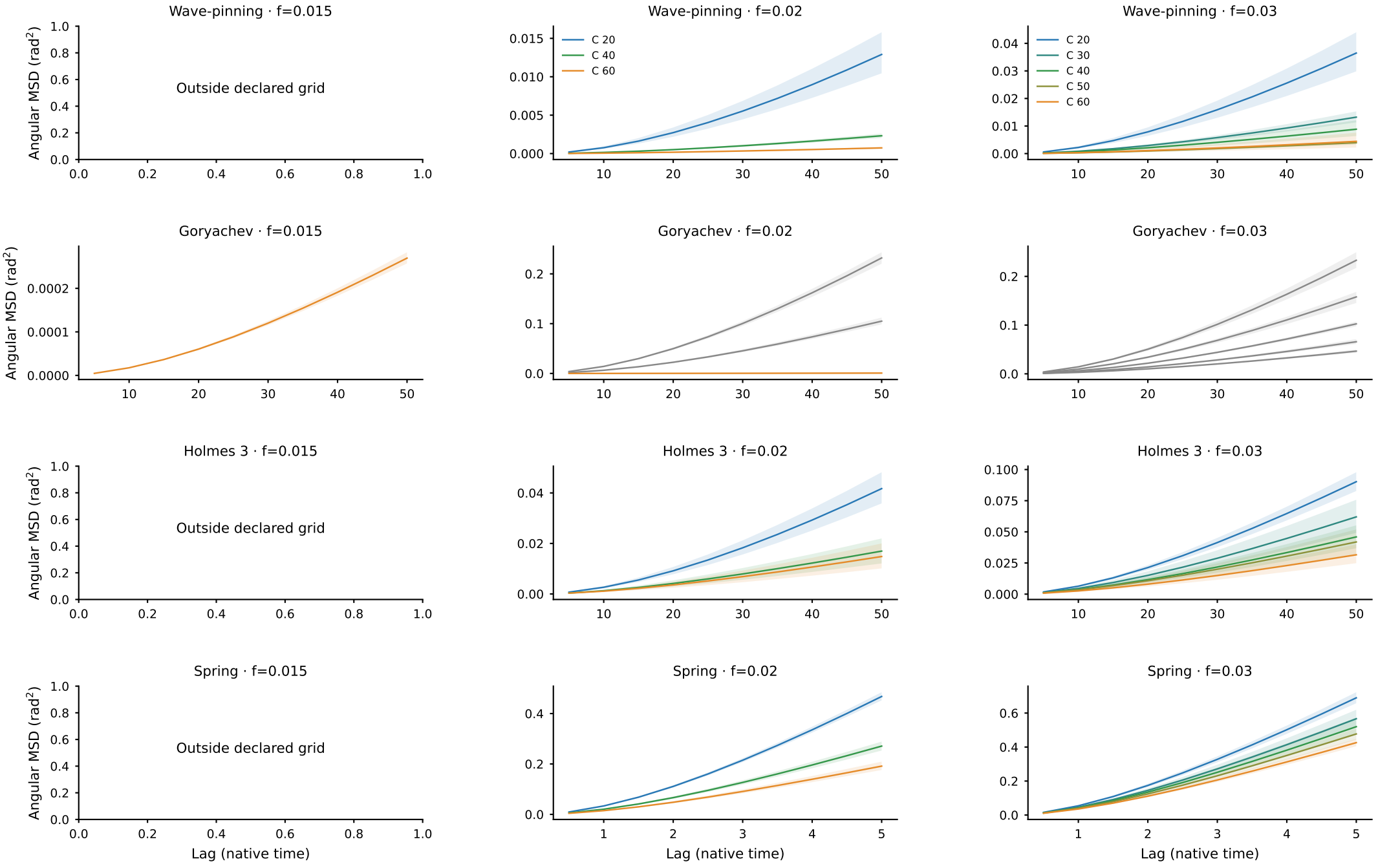
Common observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_canonical_measurements_20261020_006_S3_2_angular_msd.csv.gz

S3 · Common observations · All heading correlations



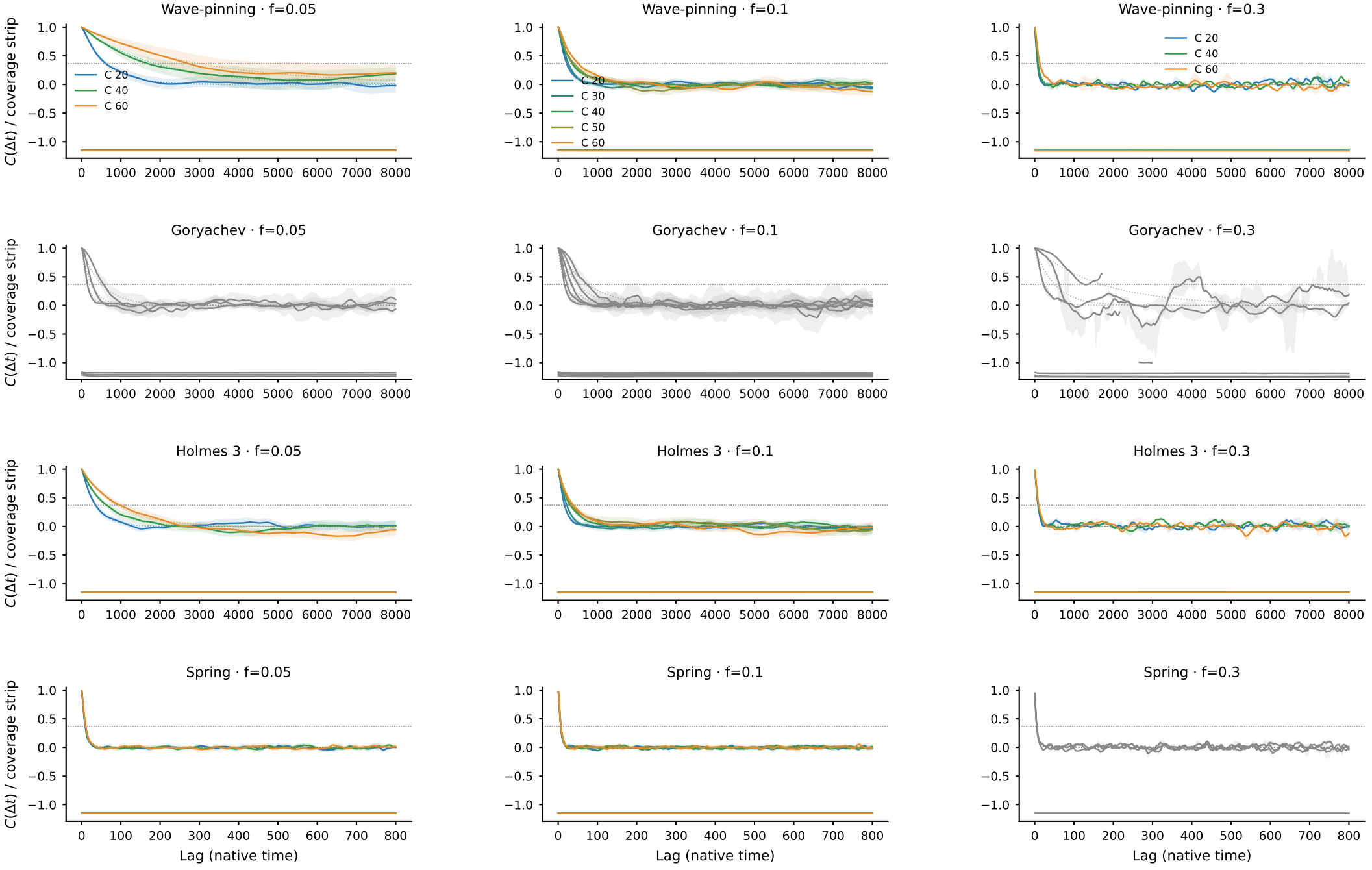
Common observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_canonical_measurements_20261020_007_S3_3_cosine.csv.gz

S3 · Common observations · Short-lag angular MSD



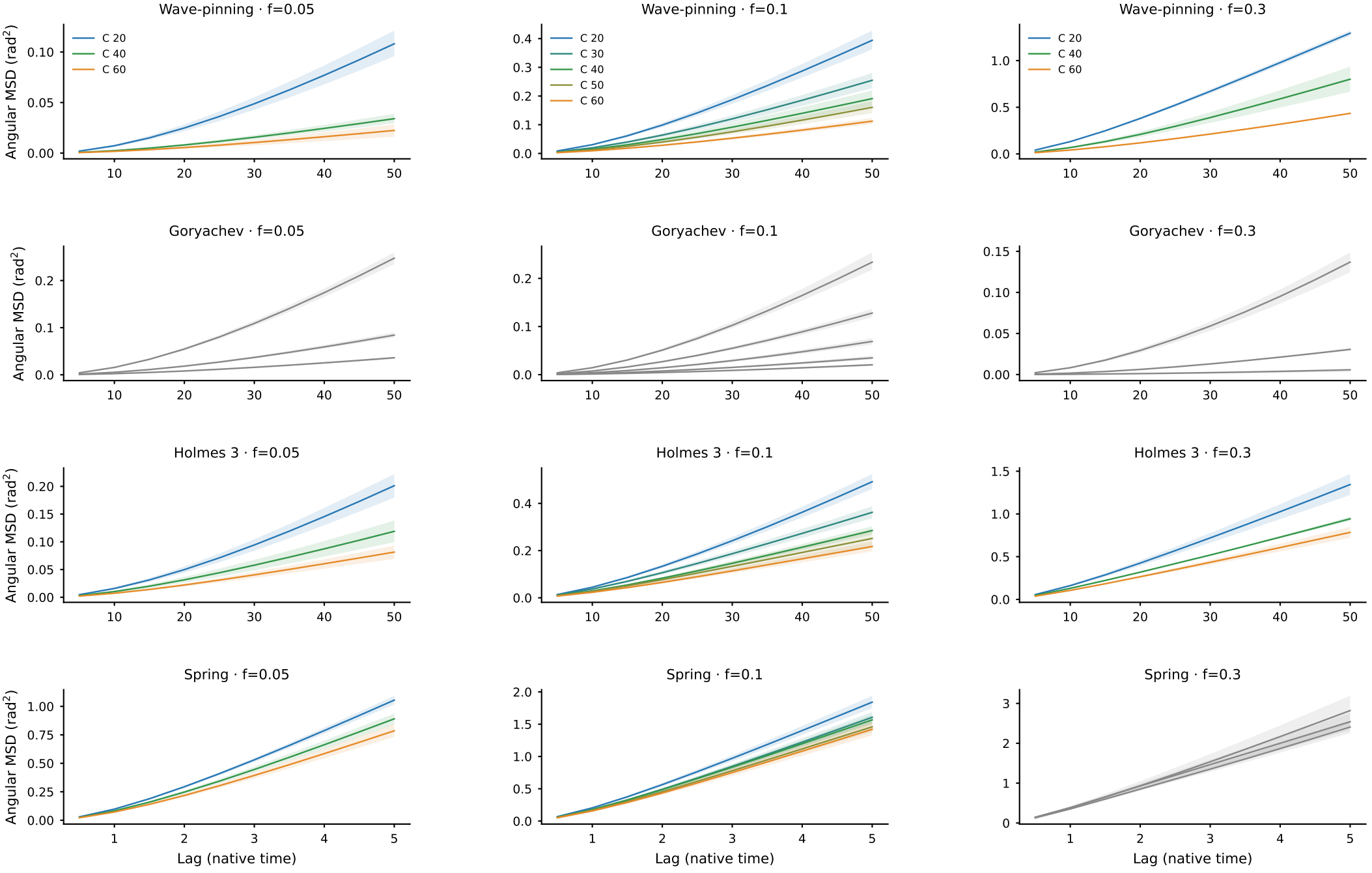
Common observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_canonical_measurements_20261020_008_S3_3_angular_msd.csv.gz

S3 · Common observations · All heading correlations



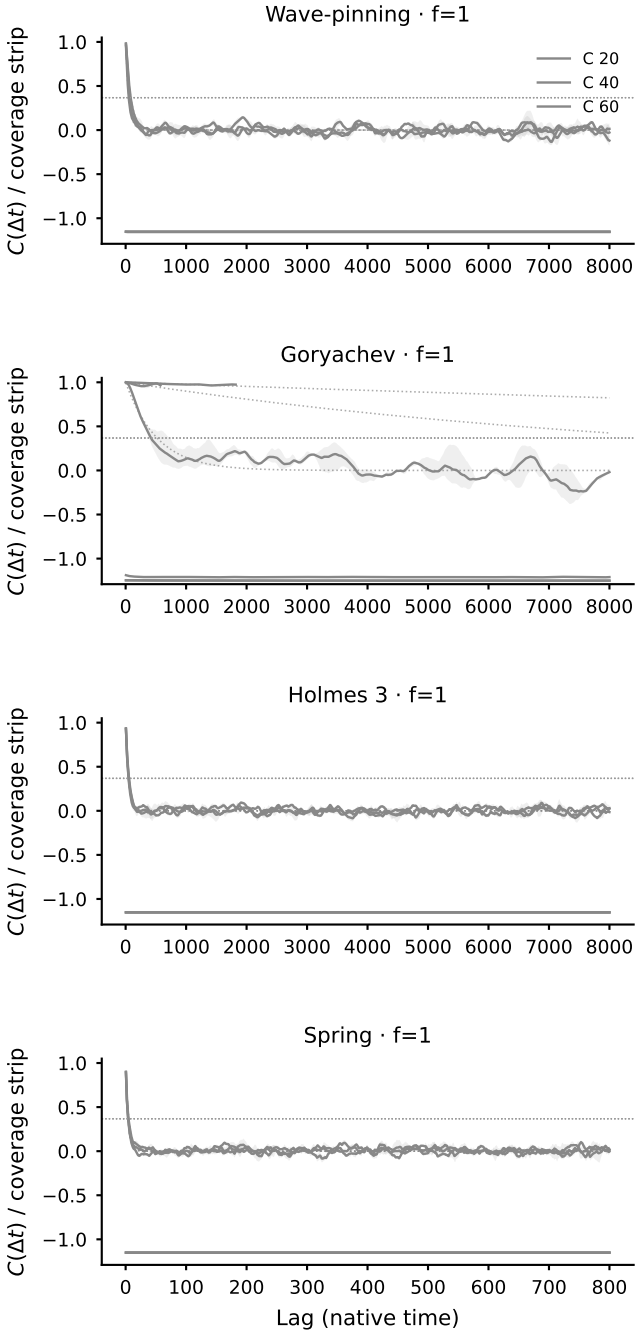
Common observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_canonical_measurements_20261020_009_S3_4_cosine.csv.gz

S3 · Common observations · Short-lag angular MSD



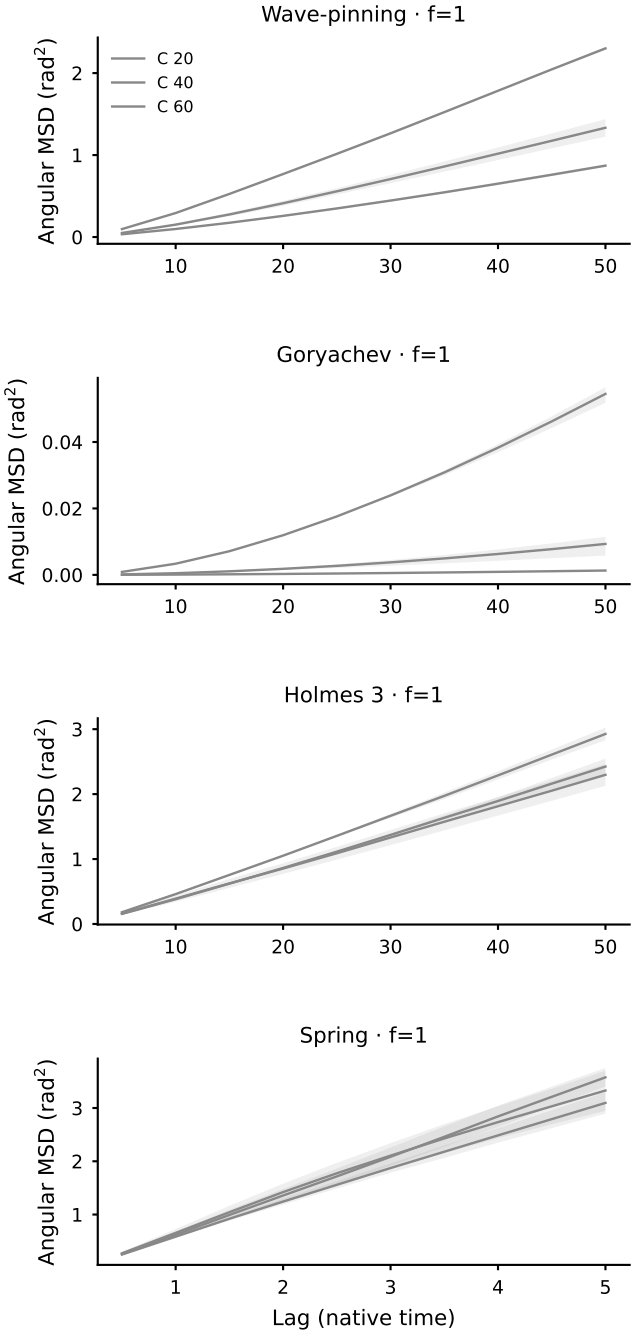
Common observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_canonical_measurements_20261020_010_S3_4_angular_msd.csv.gz

S3 · Common observations · All heading correlations



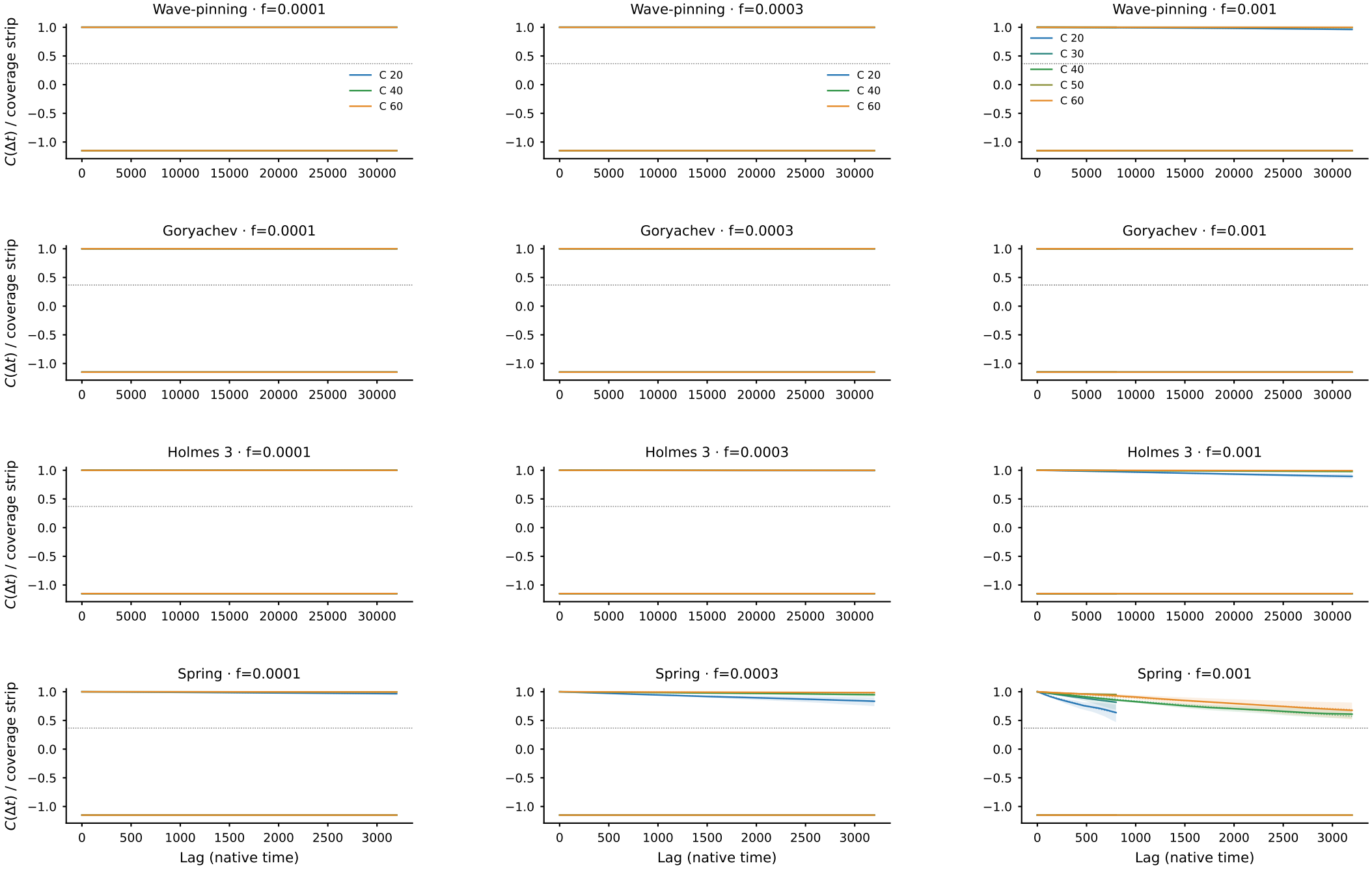
Common observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_canonical_measurements_20261020_011_S3_5_cosine.csv

S3 · Common observations · Short-lag angular MSD



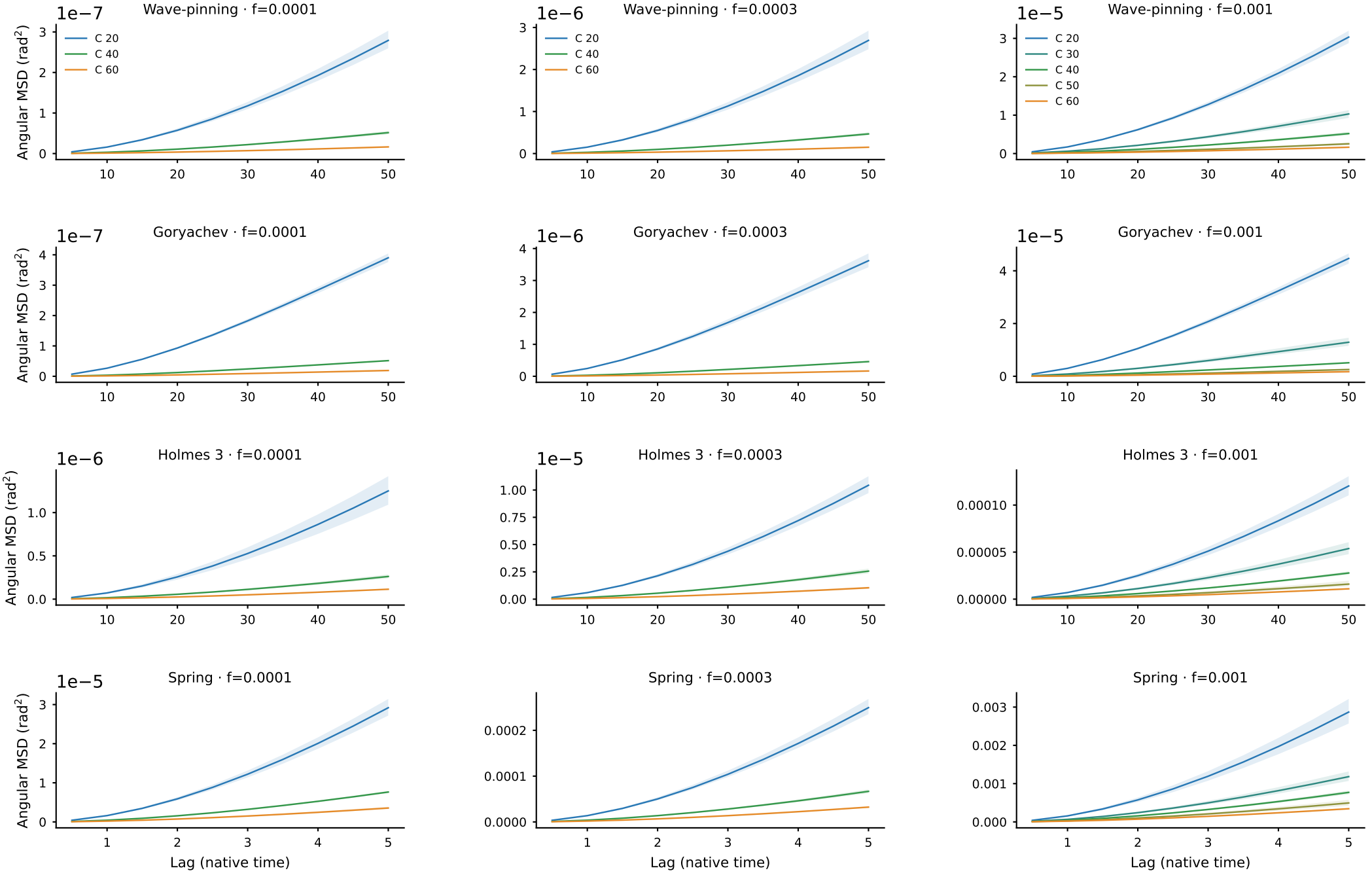
Common observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_canonical_measurements_20261020_012_S3_5_angular_msd.csv

S3 · Longest qualified observations · All heading correlations



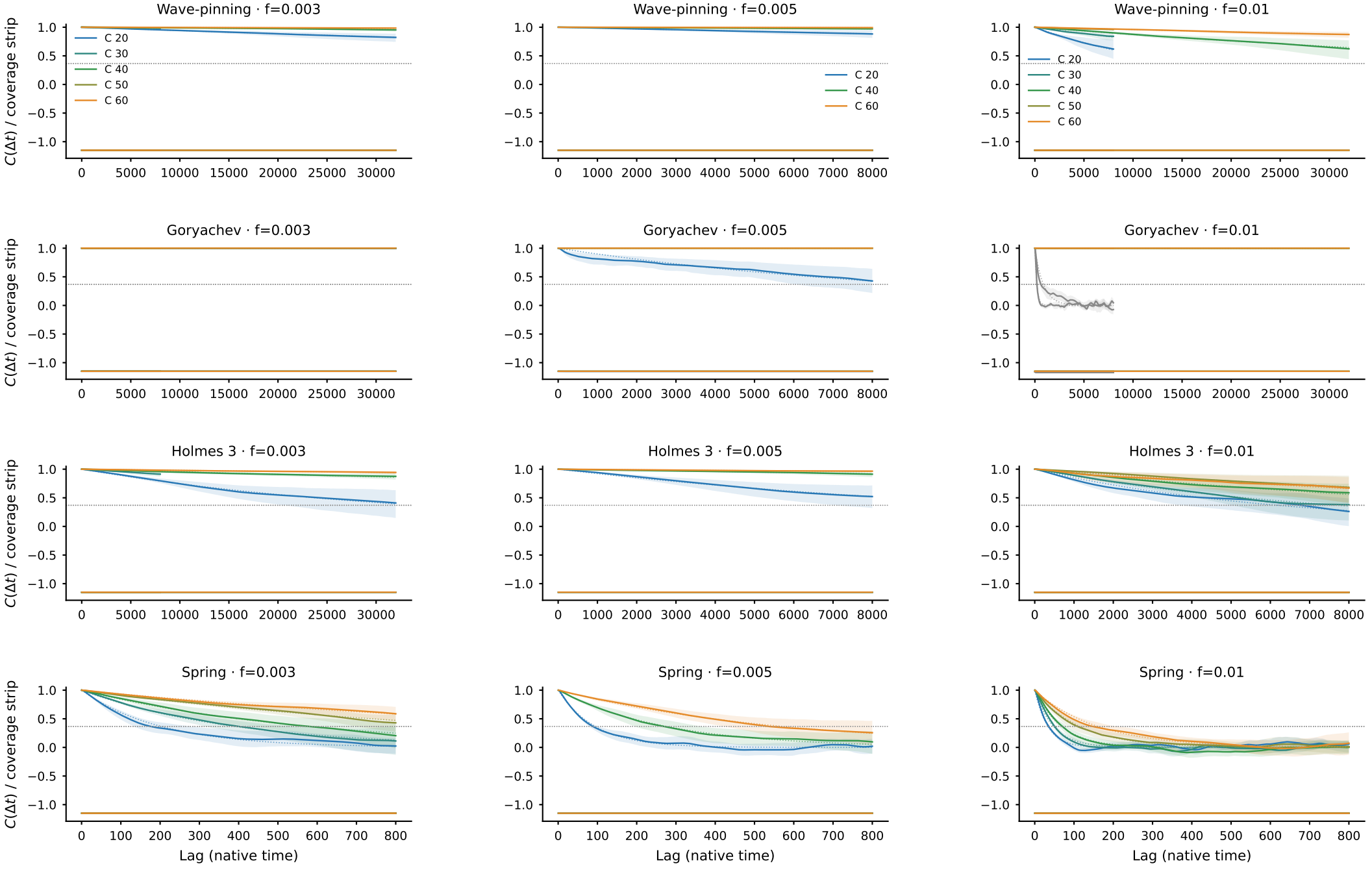
Longest qualified observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_canonical_measurements_20261020_003_S3_1_cosine.csv.gz

S3 · Longest qualified observations · Short-lag angular MSD



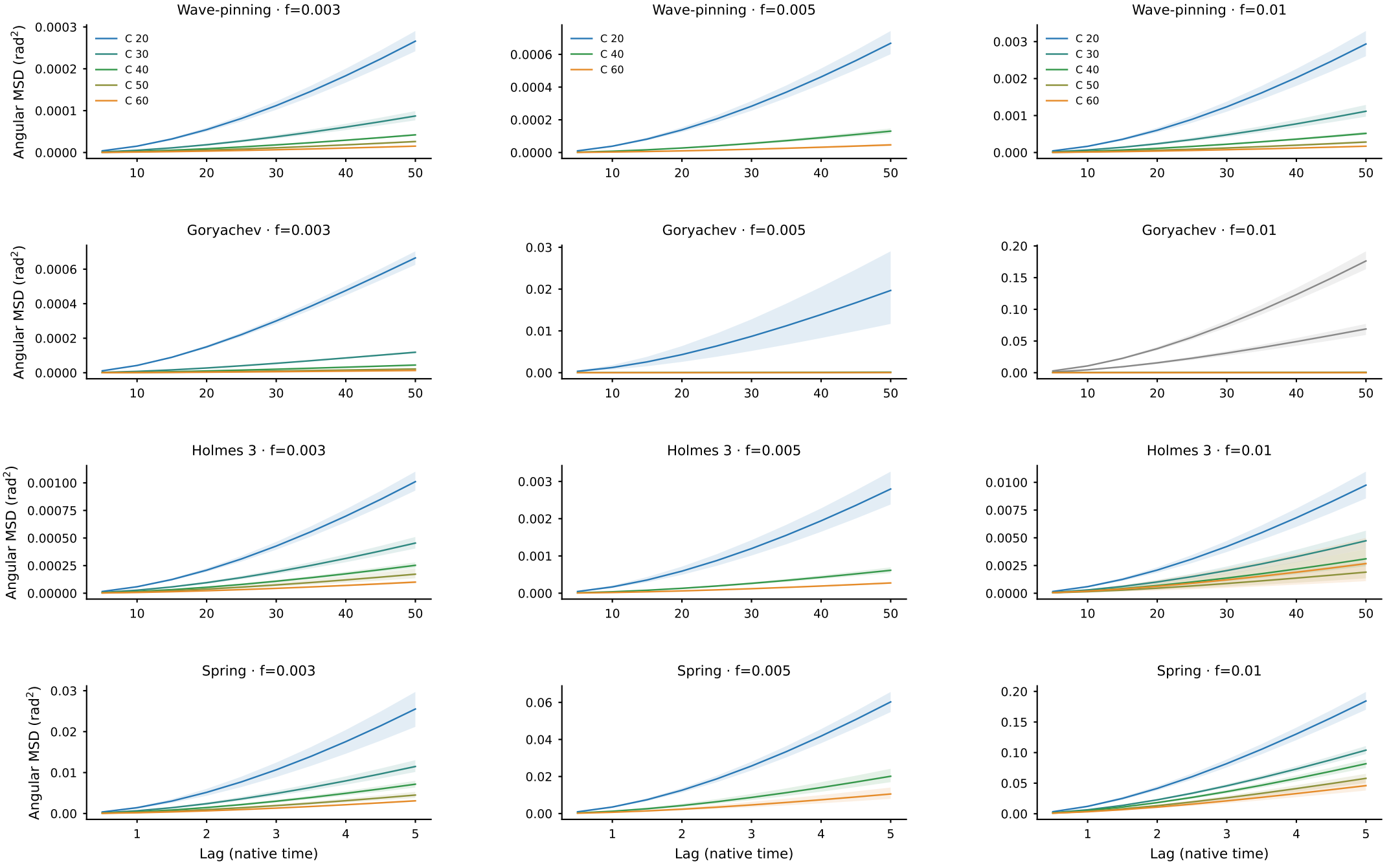
Longest qualified observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_canonical_measurements_20261020_004_S3_1_angular_msd.csv.gz

S3 · Longest qualified observations · All heading correlations



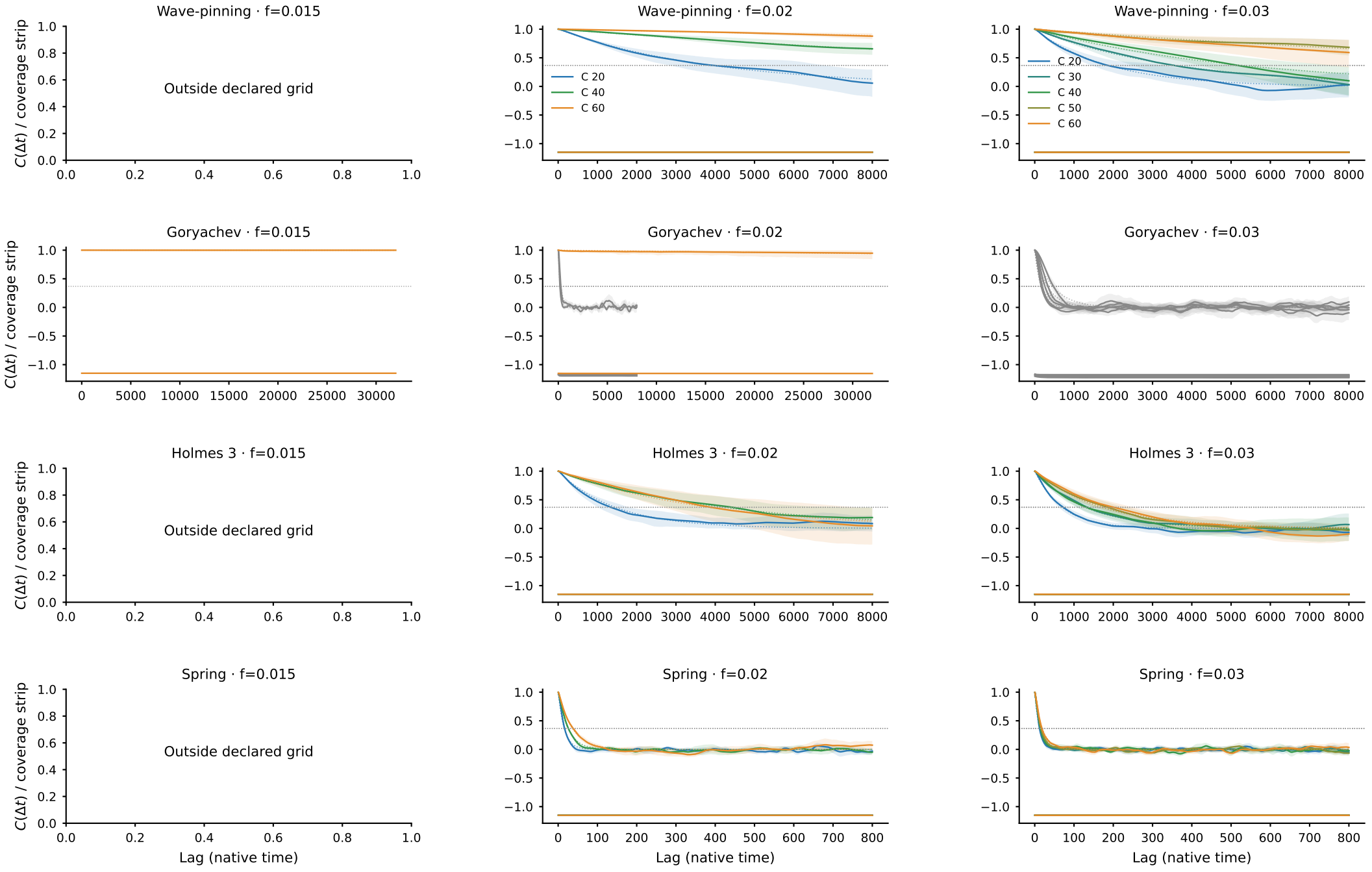
Longest qualified observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_canonical_measurements_20261020_005_S3_2_cosine.csv.gz

S3 · Longest qualified observations · Short-lag angular MSD

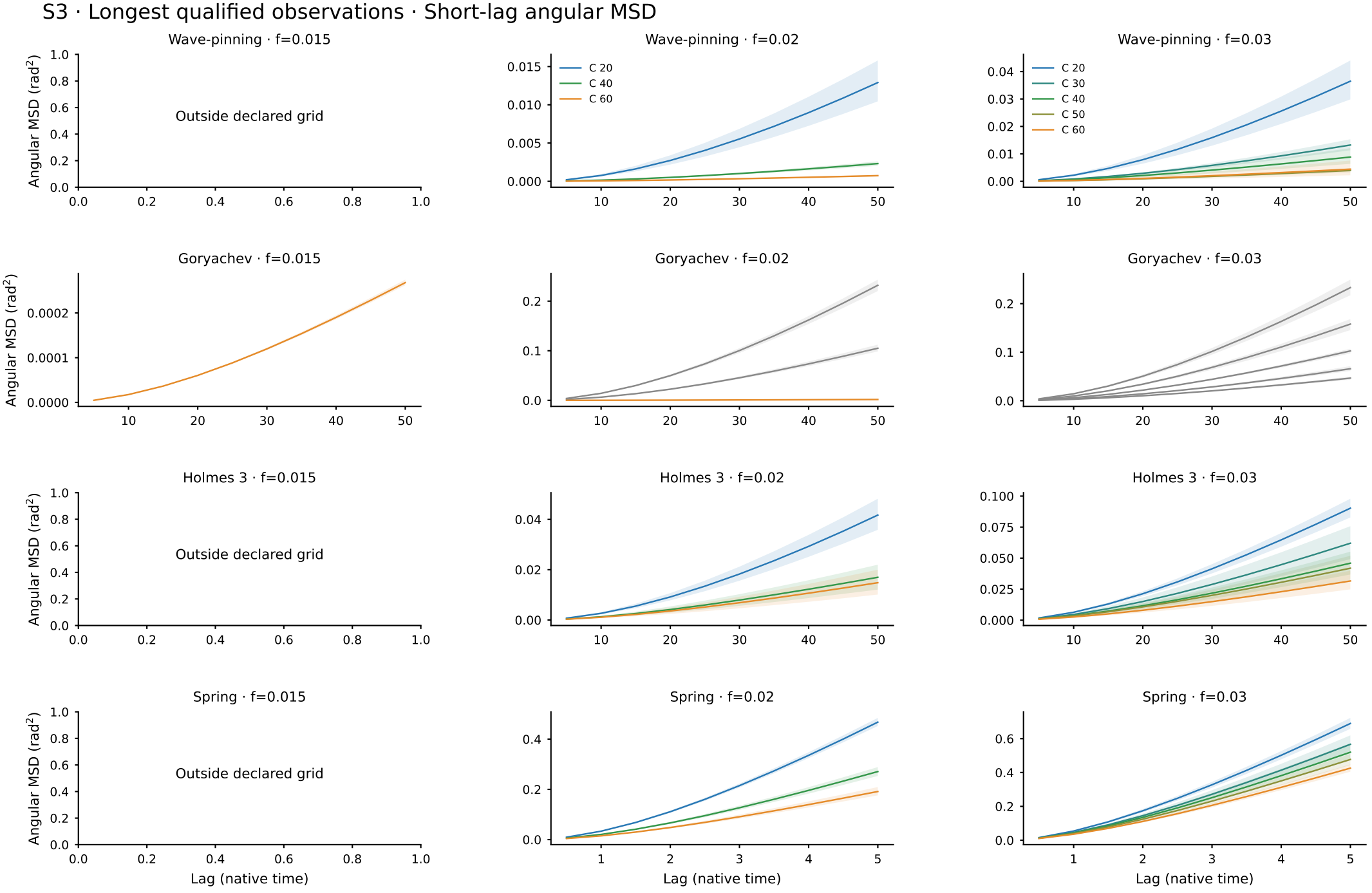


Longest qualified observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_canonical_measurements_20261020_006_S3_2_angular_msd.csv.gz

S3 · Longest qualified observations · All heading correlations

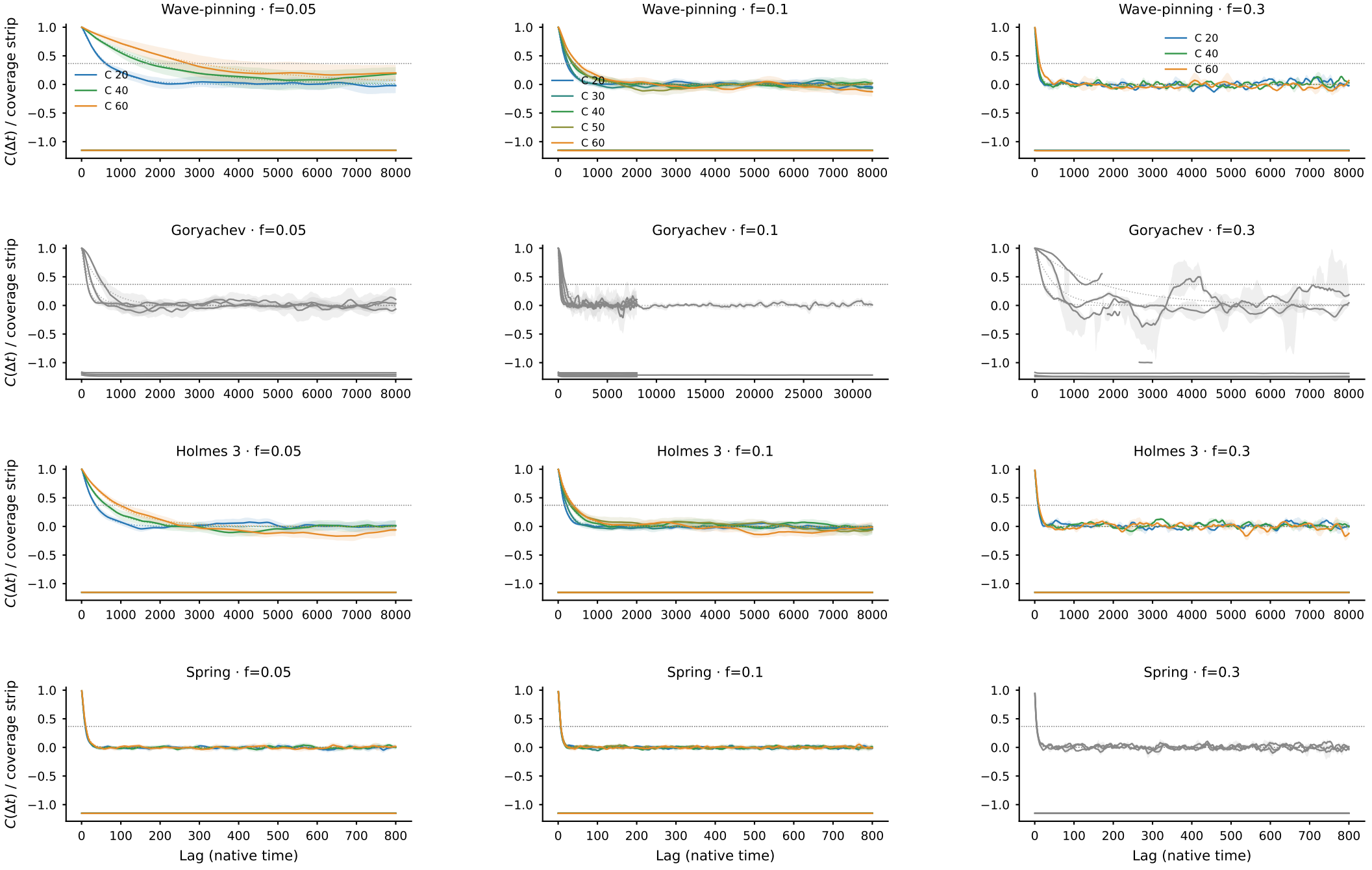


Longest qualified observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_canonical_measurements_20261020_007_S3_3_cosine.csv.gz



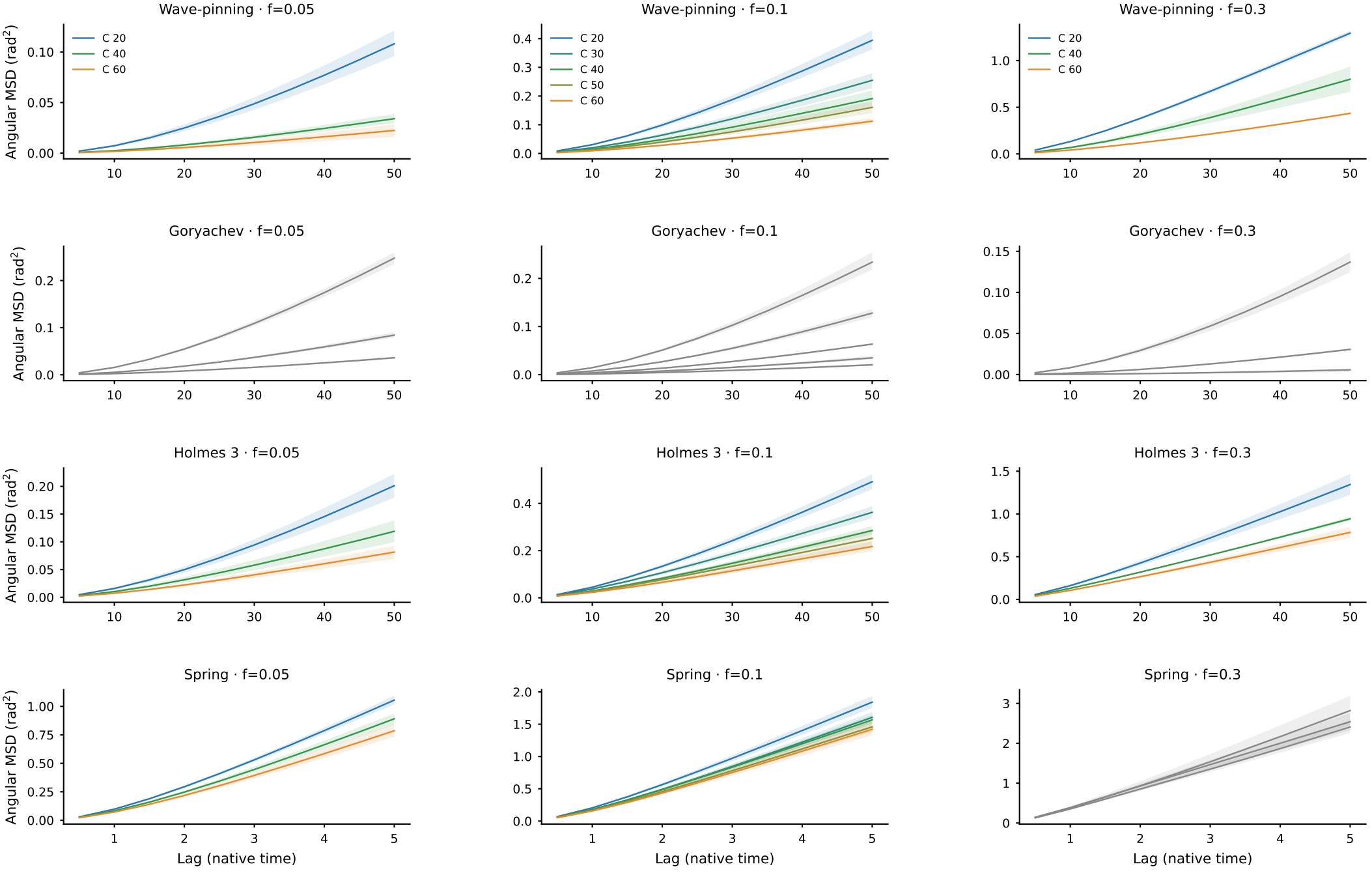
Longest qualified observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_canonical_measurements_20261020_008_S3_3_angular_msd.csv.gz

S3 · Longest qualified observations · All heading correlations



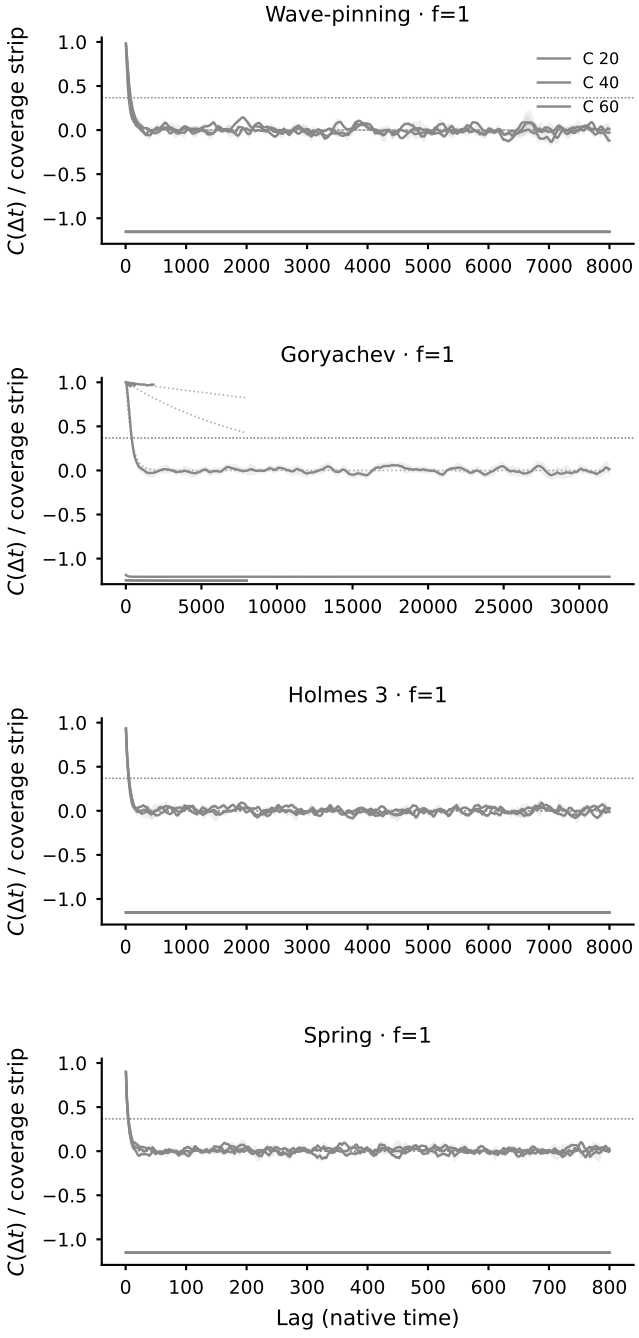
Longest qualified observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_canonical_measurements_20261020_009_S3_4_cosine.csv.gz

S3 · Longest qualified observations · Short-lag angular MSD



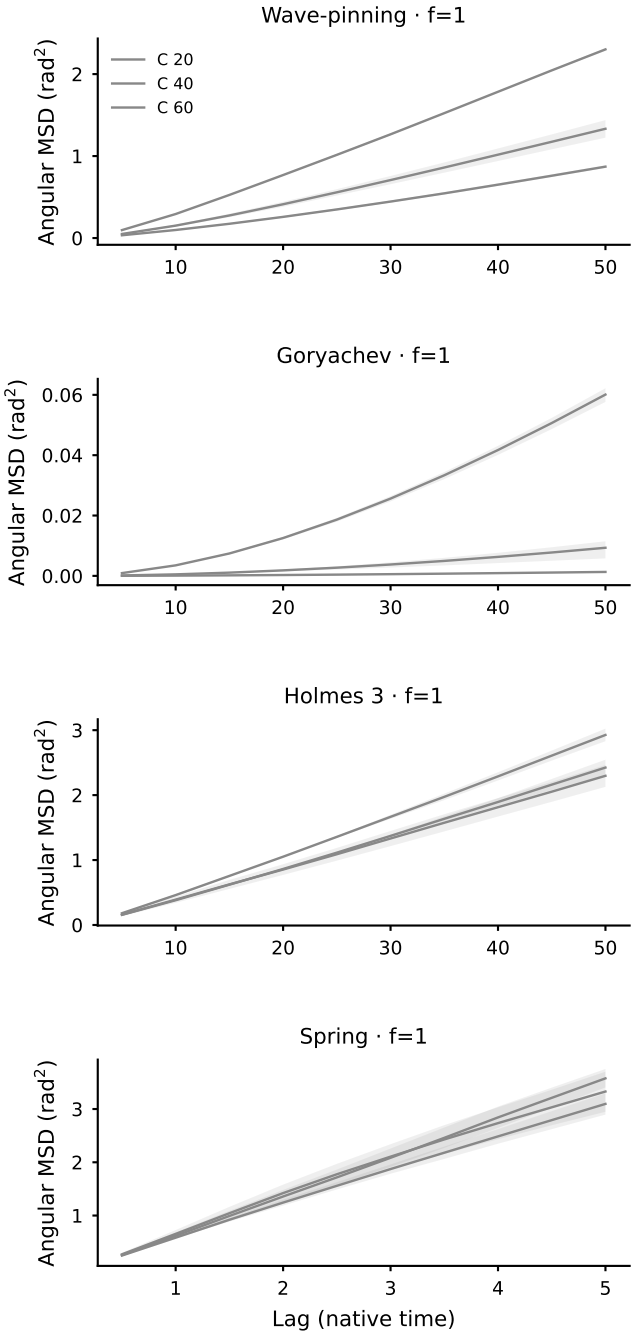
Longest qualified observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_canonical_measurements_20261020_010_S3_4_angular_msd.csv.gz

S3 · Longest qualified observations · All heading correlations



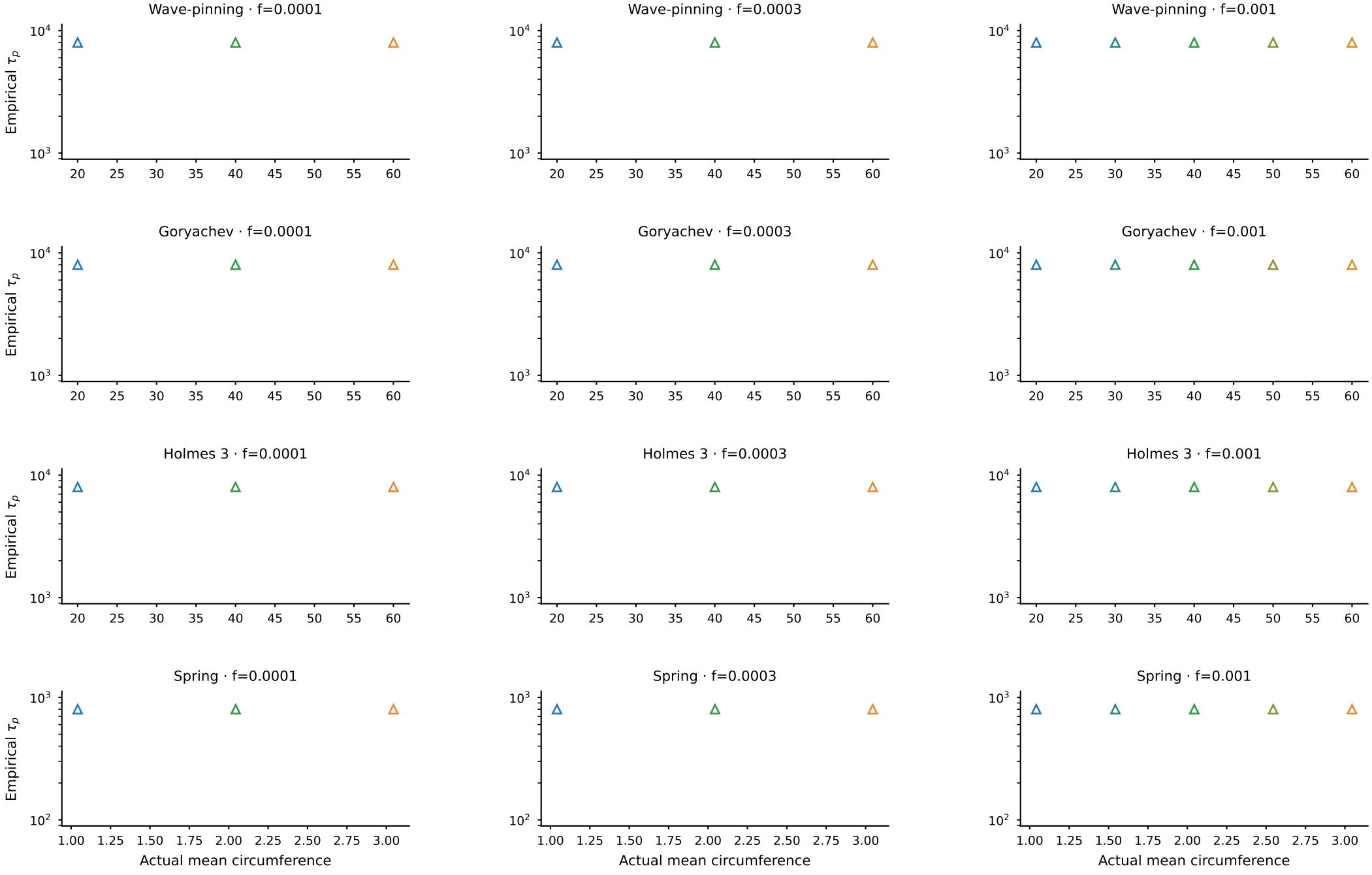
Longest qualified observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_canonical_measurements_20261020_011_S3_5_cosine.csv

S3 · Longest qualified observations · Short-lag angular MSD



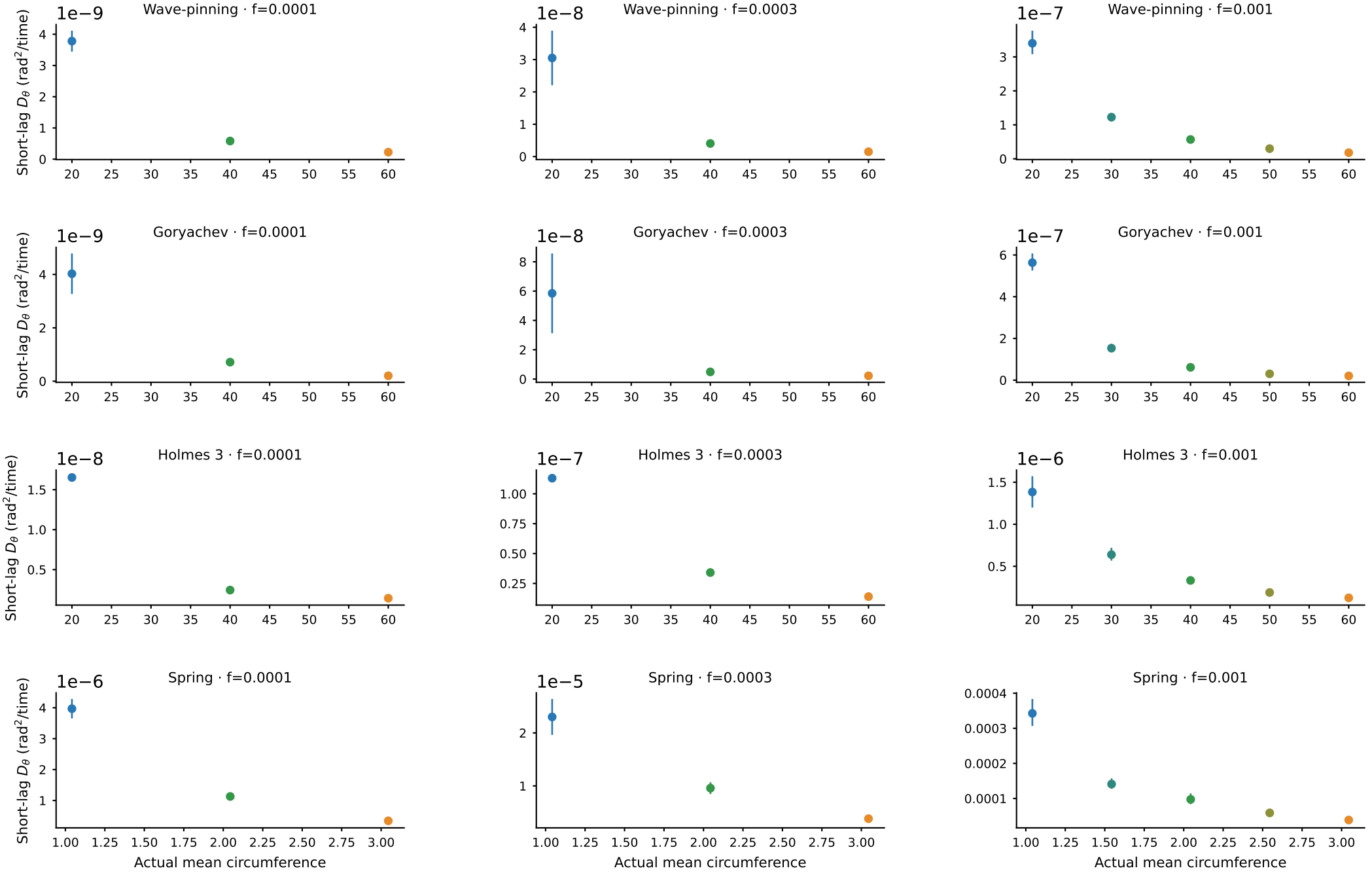
Longest qualified observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_canonical_measurements_20261020_012_S3_5_angular_msd.csv

S4 · Common observations · Persistence and angular mobility across size



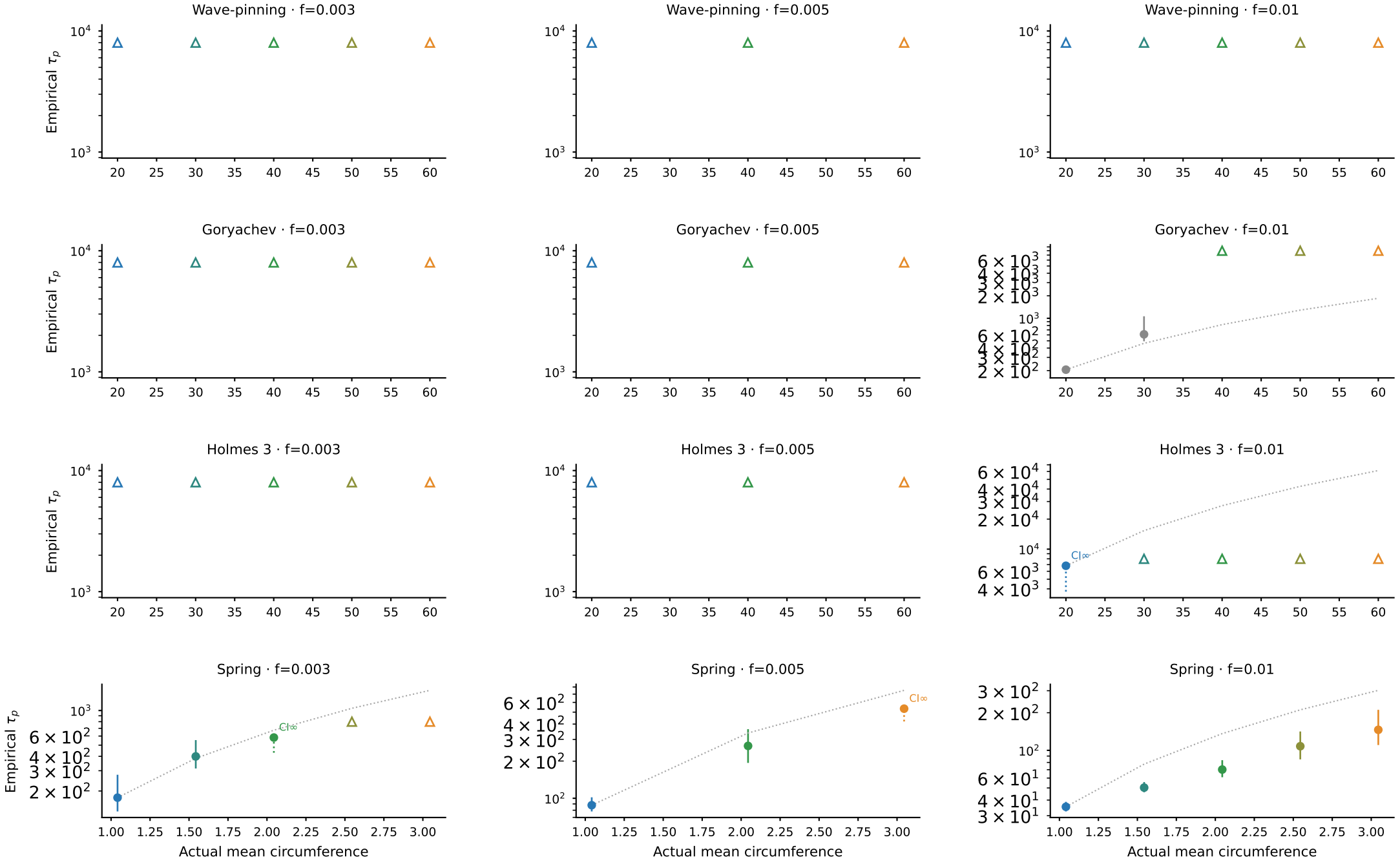
Common observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_canonical_measurements_20261020_013_S4_1_tau_p.csv

S4 · Common observations · Persistence and angular mobility across size



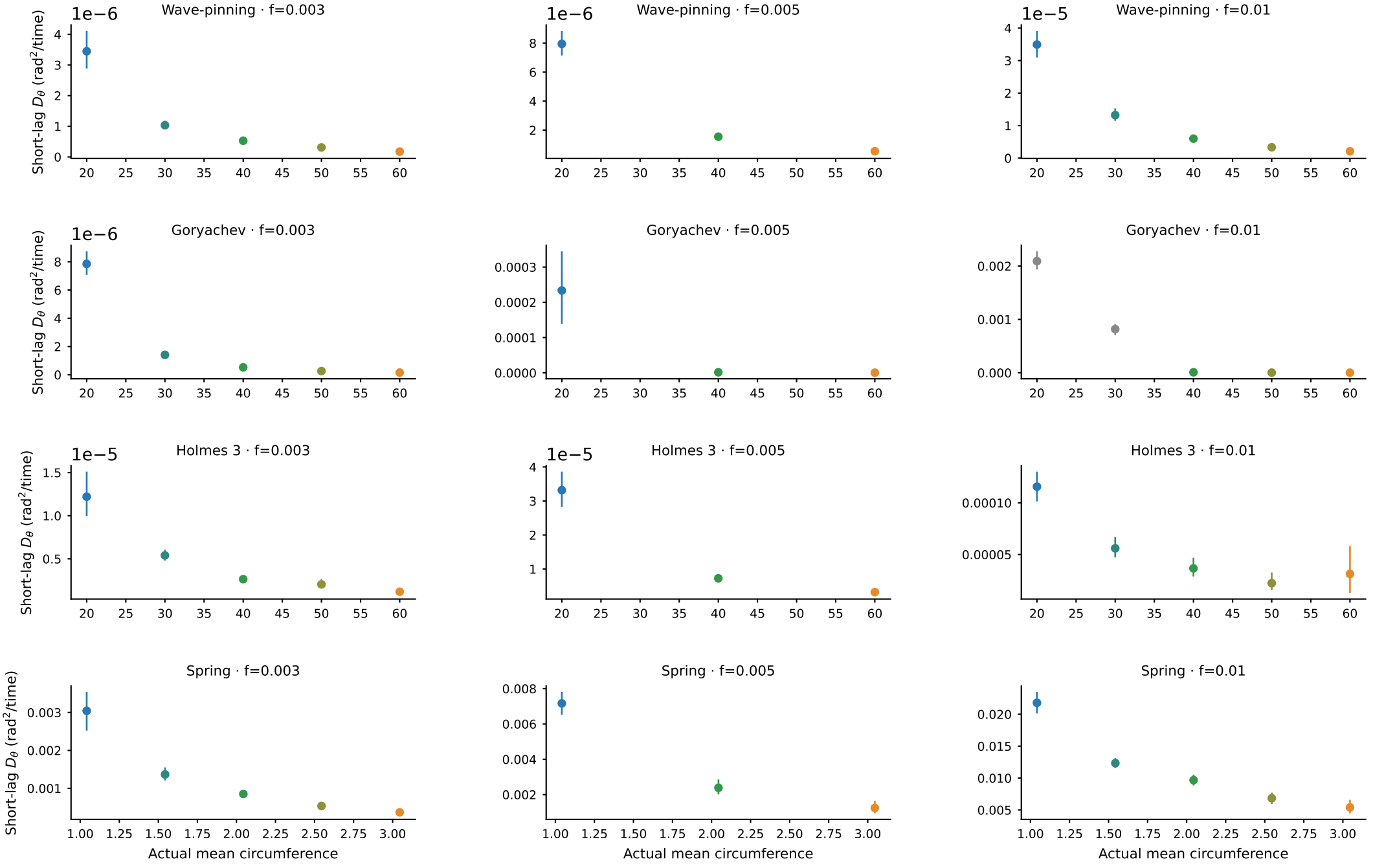
Common observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_canonical_measurements_20261020_014_S4_1_Dtheta.csv

S4 · Common observations · Persistence and angular mobility across size



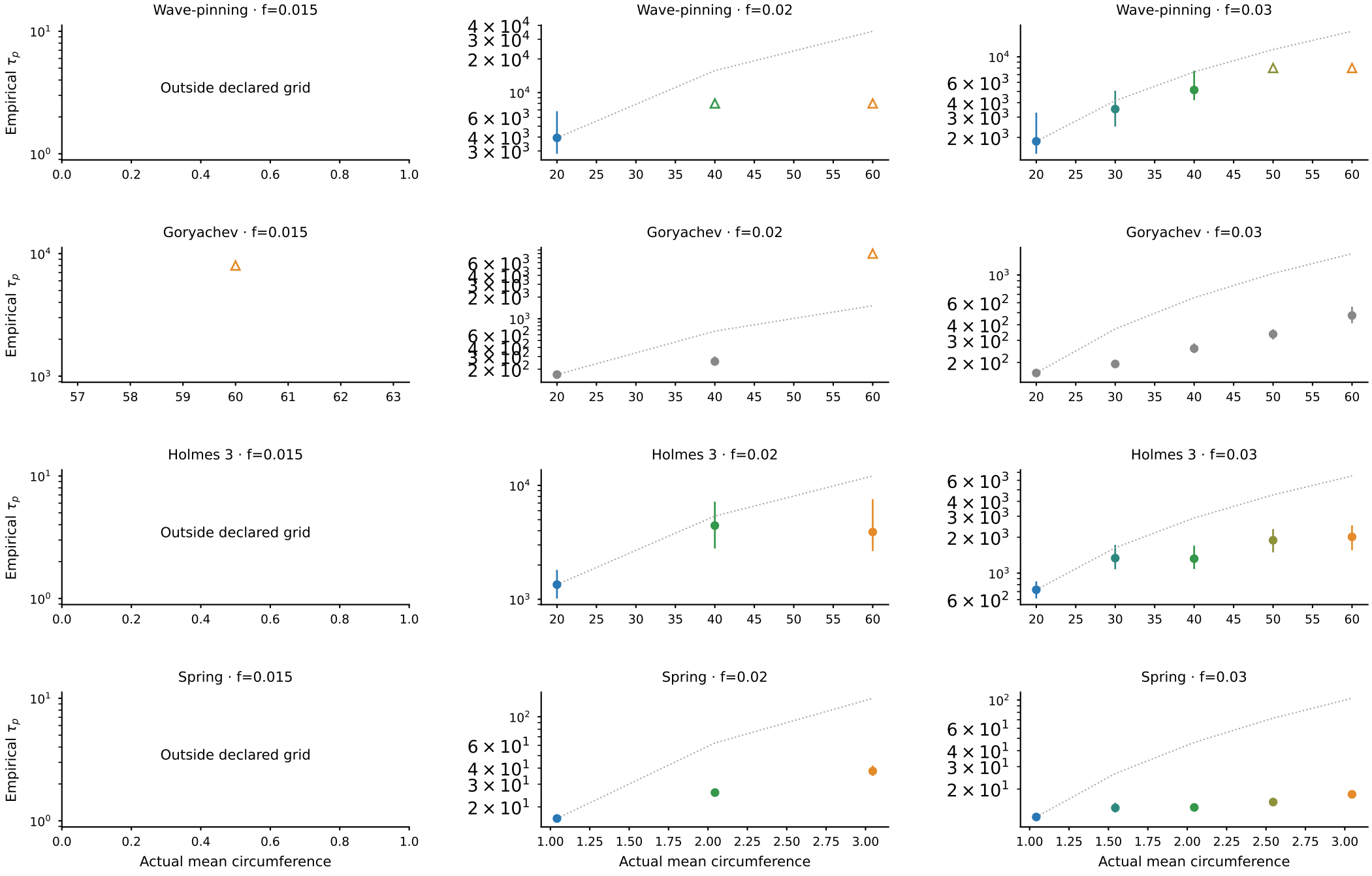
Common observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV. CI∞ next to a finite point marks an upper-unbounded bootstrap CI; finite lower endpoints are retained. CI[0,∞) spans the full nonnegative range.
CSV: figure4_common_canonical_measurements_20261020_015_S4_2_tau_p.csv

S4 · Common observations · Persistence and angular mobility across size



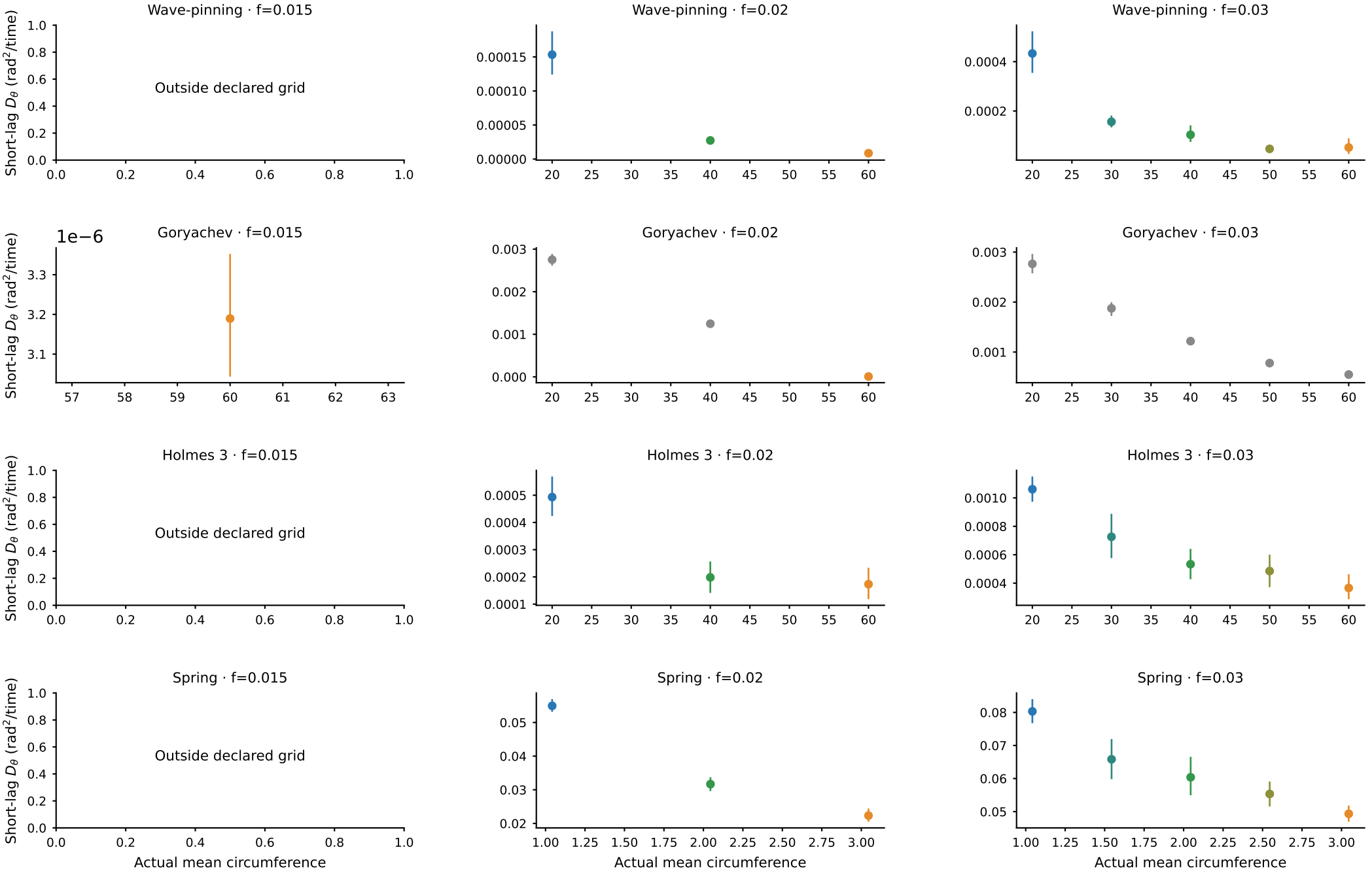
Common observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV. CI∞ next to a finite point marks an upper-unbounded bootstrap CI; finite lower endpoints are retained. CI[0,∞) spans the full nonnegative range.
CSV: figure4_common_canonical_measurements_20261020_016_S4_2_Dtheta.csv

S4 · Common observations · Persistence and angular mobility across size



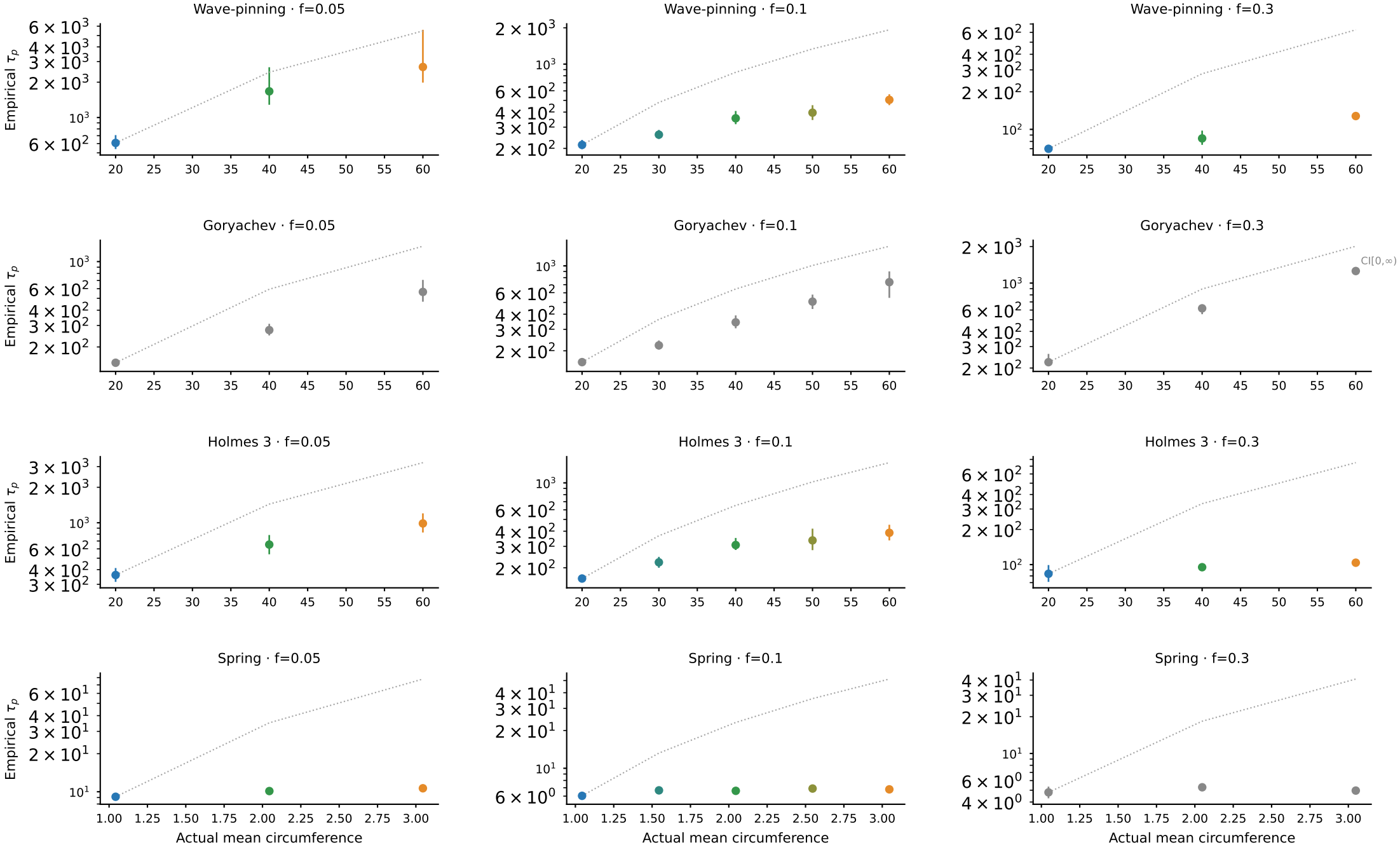
Common observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_canonical_measurements_20261020_017_S4_3_tau_p.csv

S4 · Common observations · Persistence and angular mobility across size



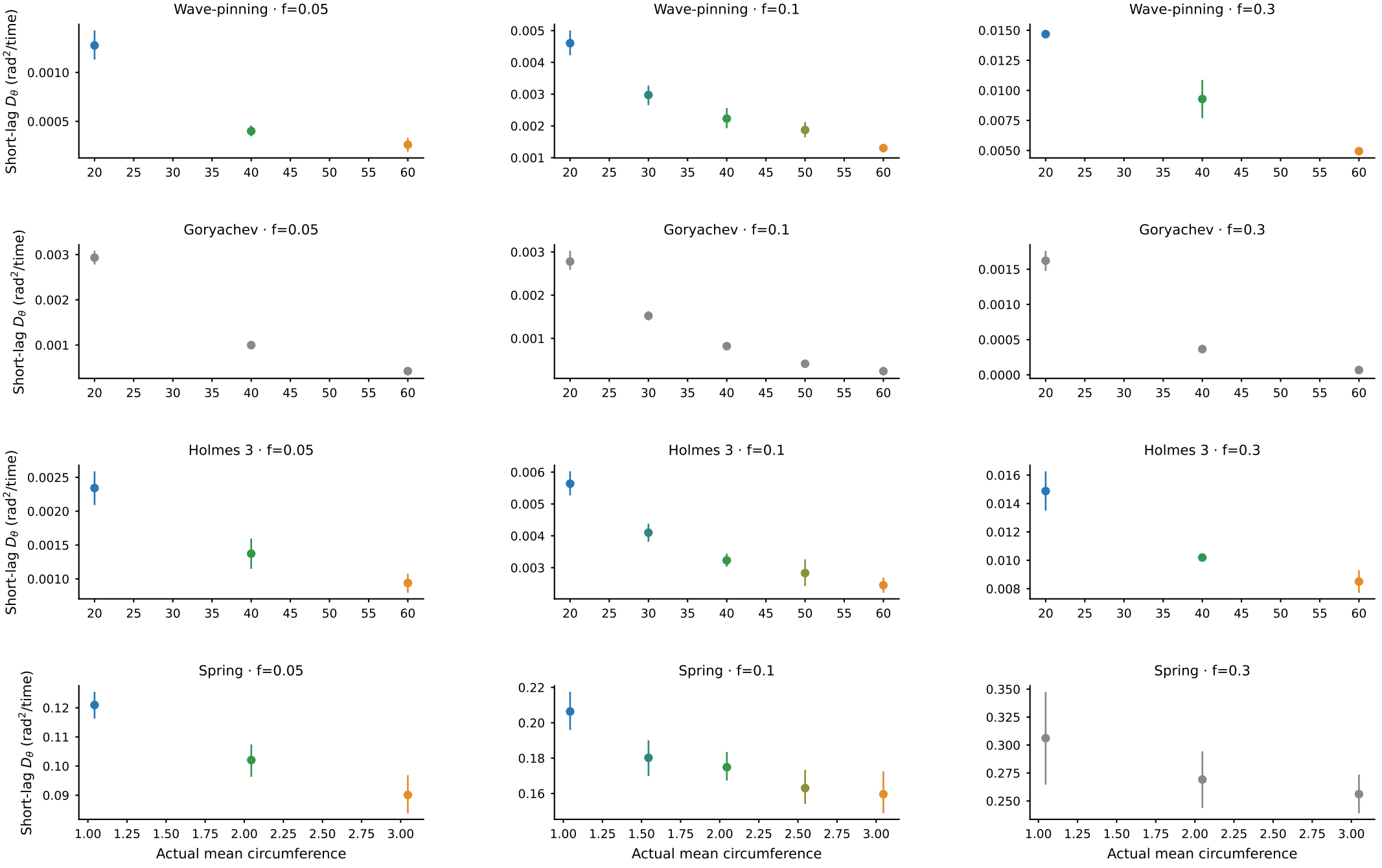
Common observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_canonical_measurements_20261020_018_S4_3_Dtheta.csv

S4 · Common observations · Persistence and angular mobility across size



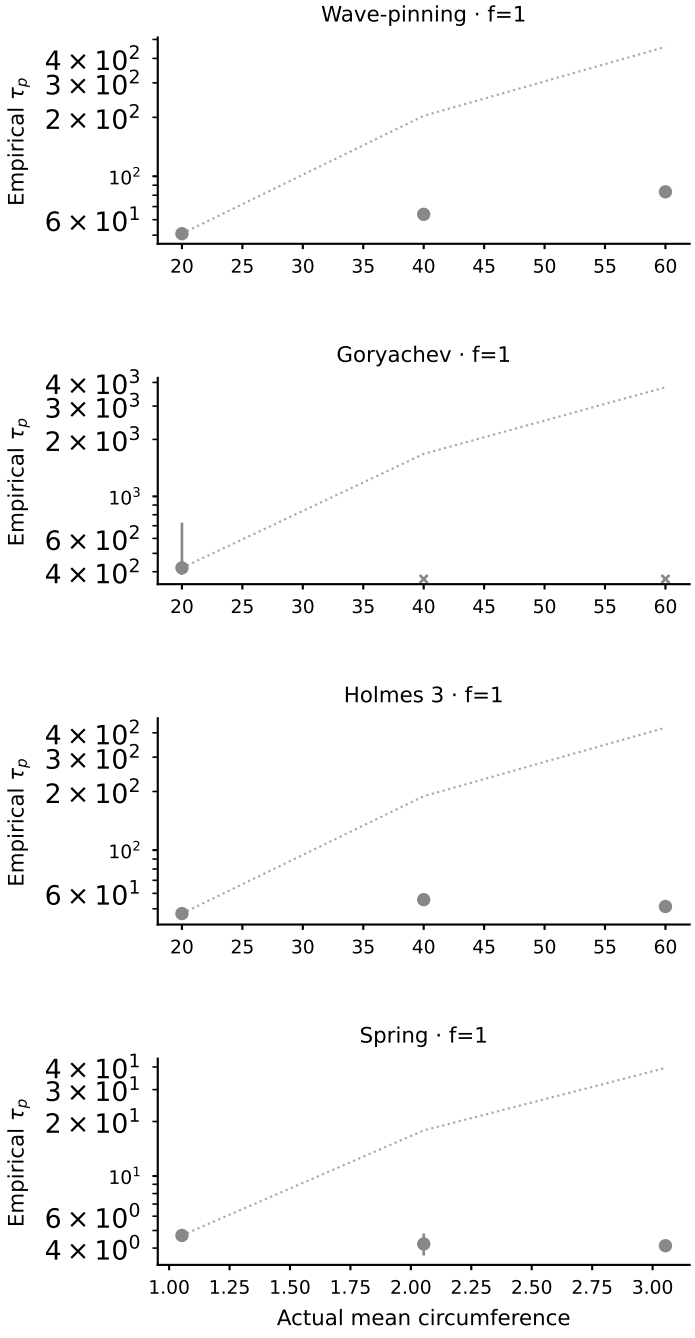
Common observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV. CI∞ next to a finite point marks an upper-unbounded bootstrap CI; finite lower endpoints are retained. CI[0,∞) spans the full nonnegative range.
CSV: figure4_common_canonical_measurements_20261020_019_S4_4_tau_p.csv

S4 · Common observations · Persistence and angular mobility across size



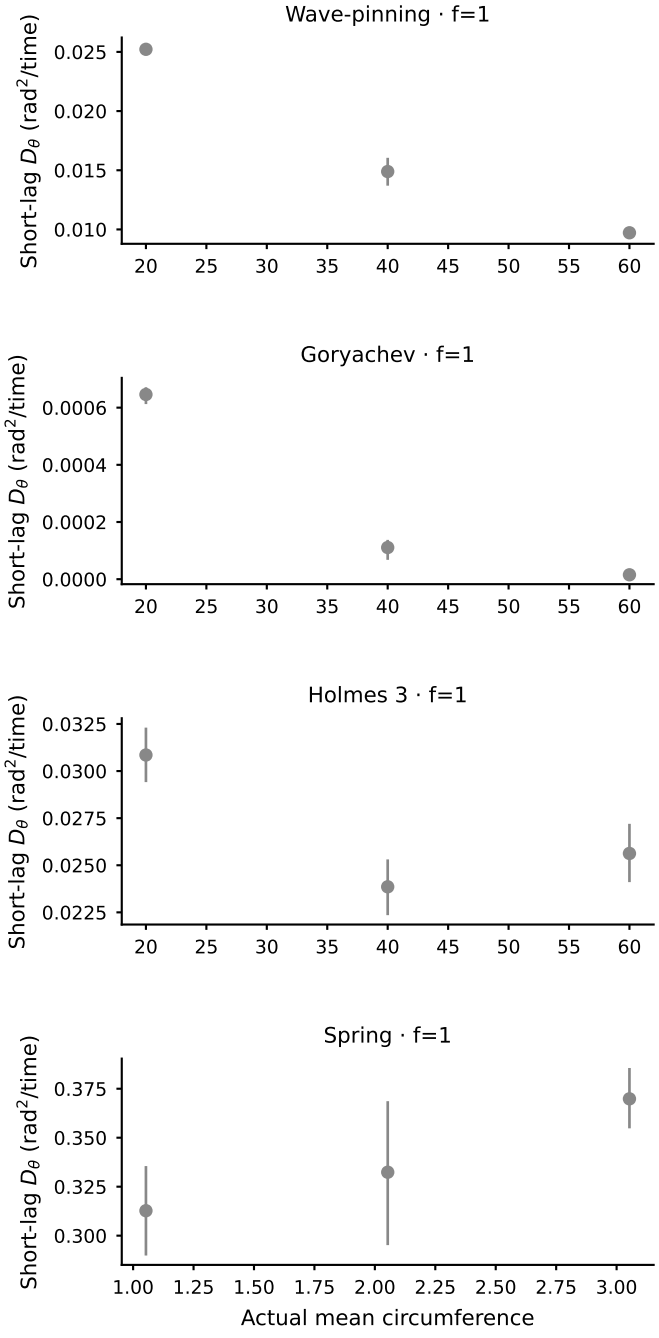
Common observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV. CI∞ next to a finite point marks an upper-unbounded bootstrap CI; finite lower endpoints are retained. CI[0,∞) spans the full nonnegative range.
CSV: figure4_common_canonical_measurements_20261020_020_S4_4_Dtheta.csv

S4 · Common observations · Persistence and angular mobility across size



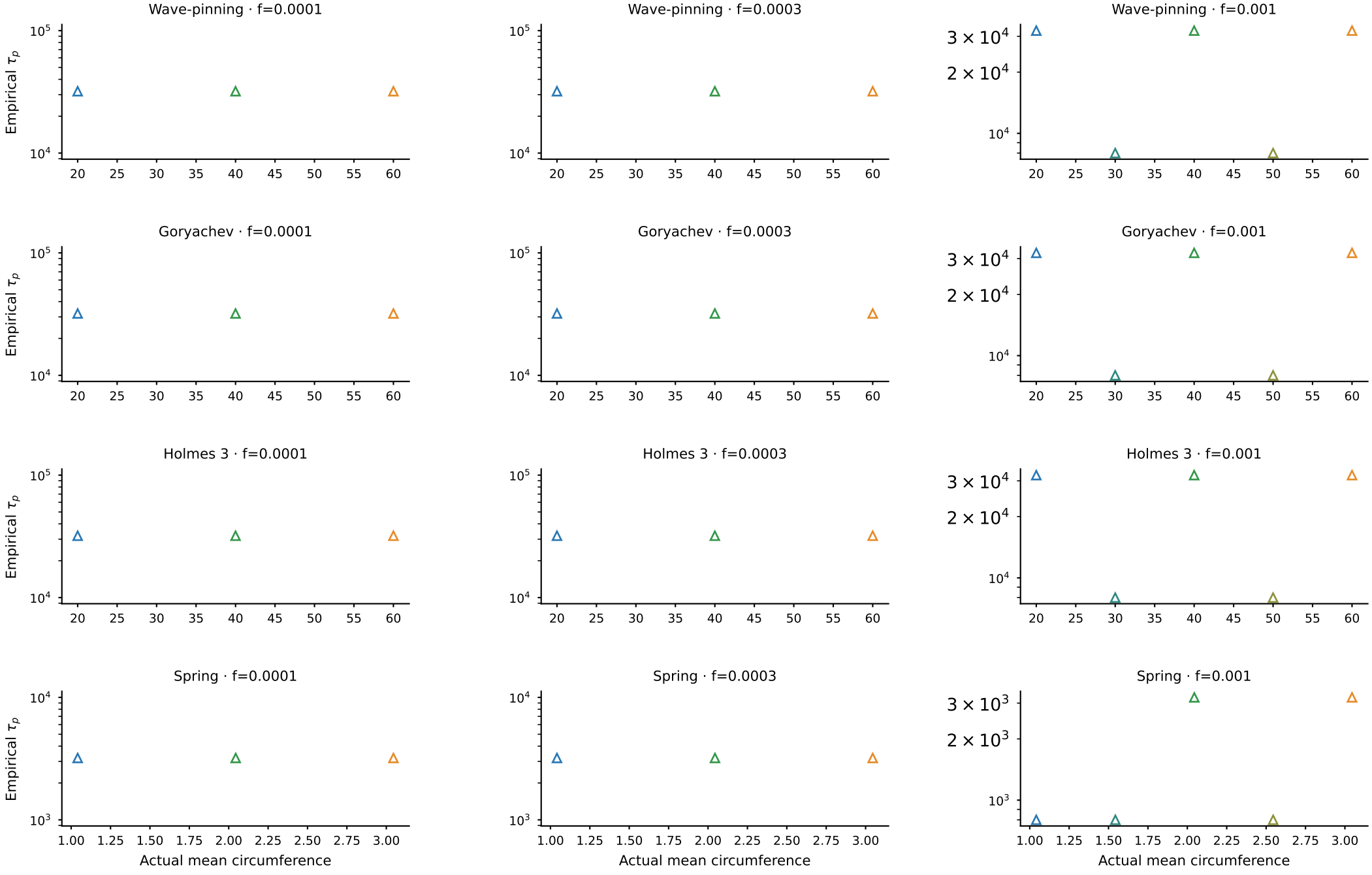
Common observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_canonical_measurements_20261020_021_S4_5_tau_p.csv

S4 · Common observations · Persistence and angular mobility across size



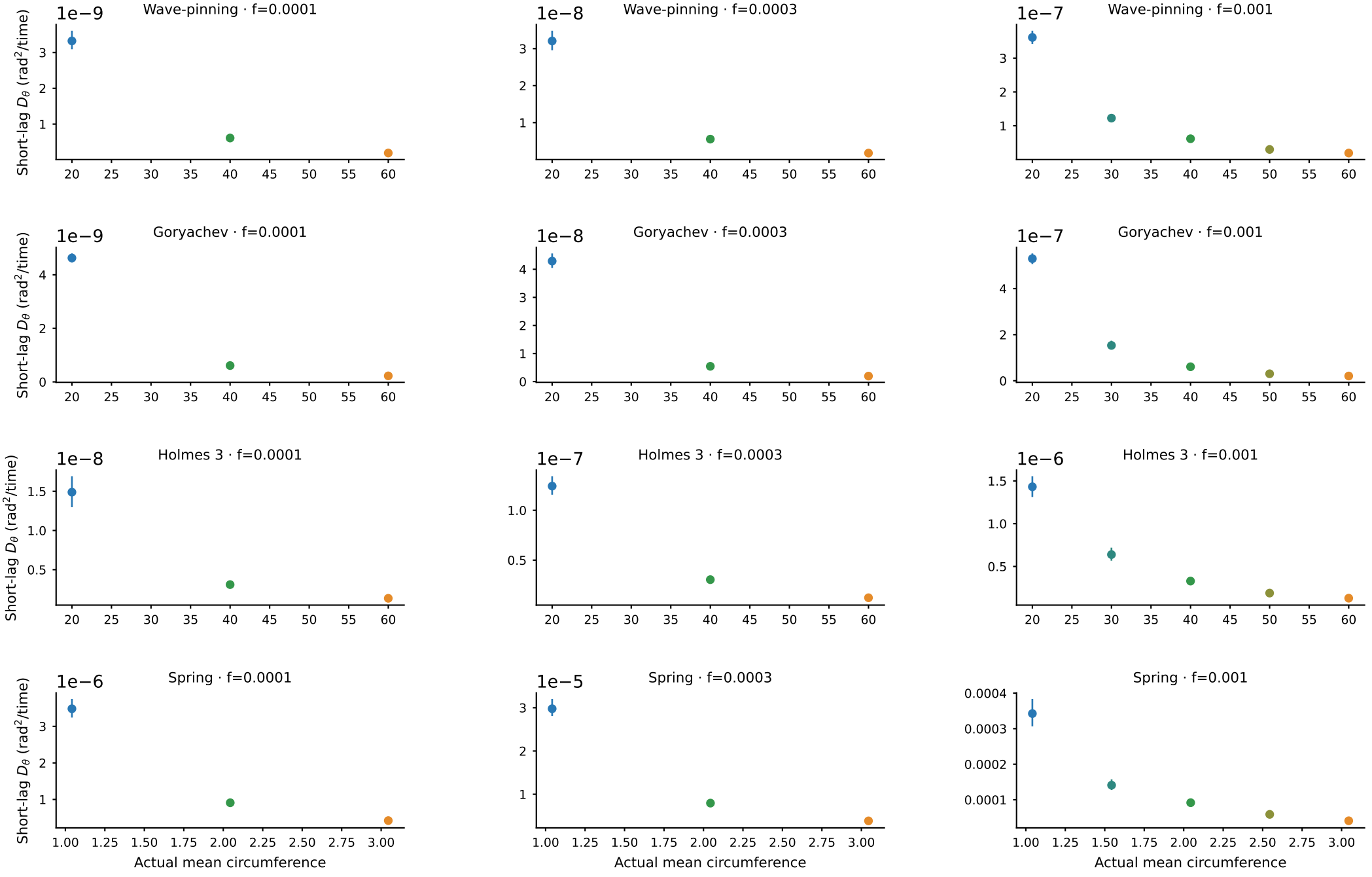
Common observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_canonical_measurements_20261020_022_S4_5_Dtheta.csv

S4 · Longest qualified observations · Persistence and angular mobility across size



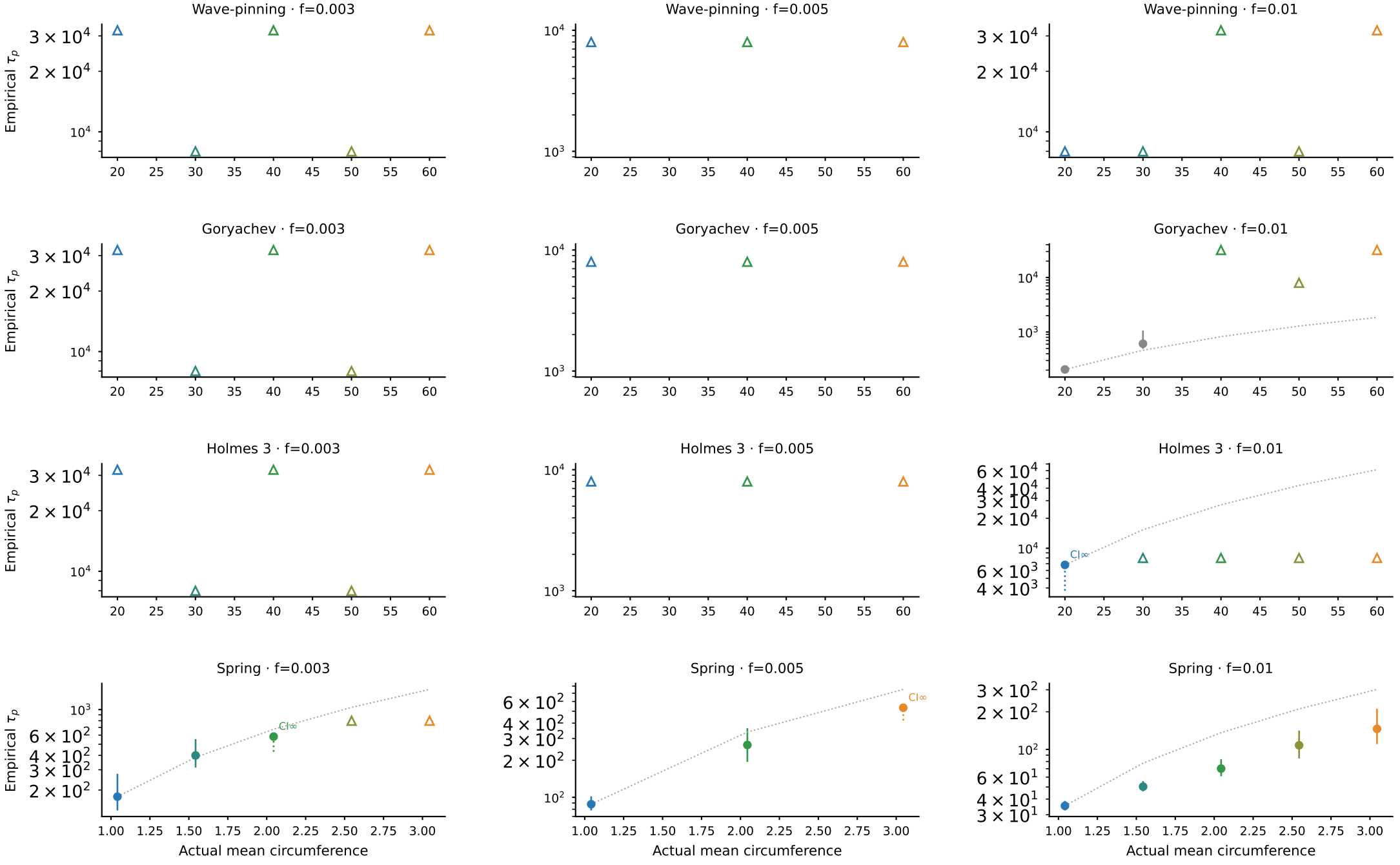
Longest qualified observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_canonical_measurements_20261020_013_S4_1_tau_p.csv

S4 · Longest qualified observations · Persistence and angular mobility across size



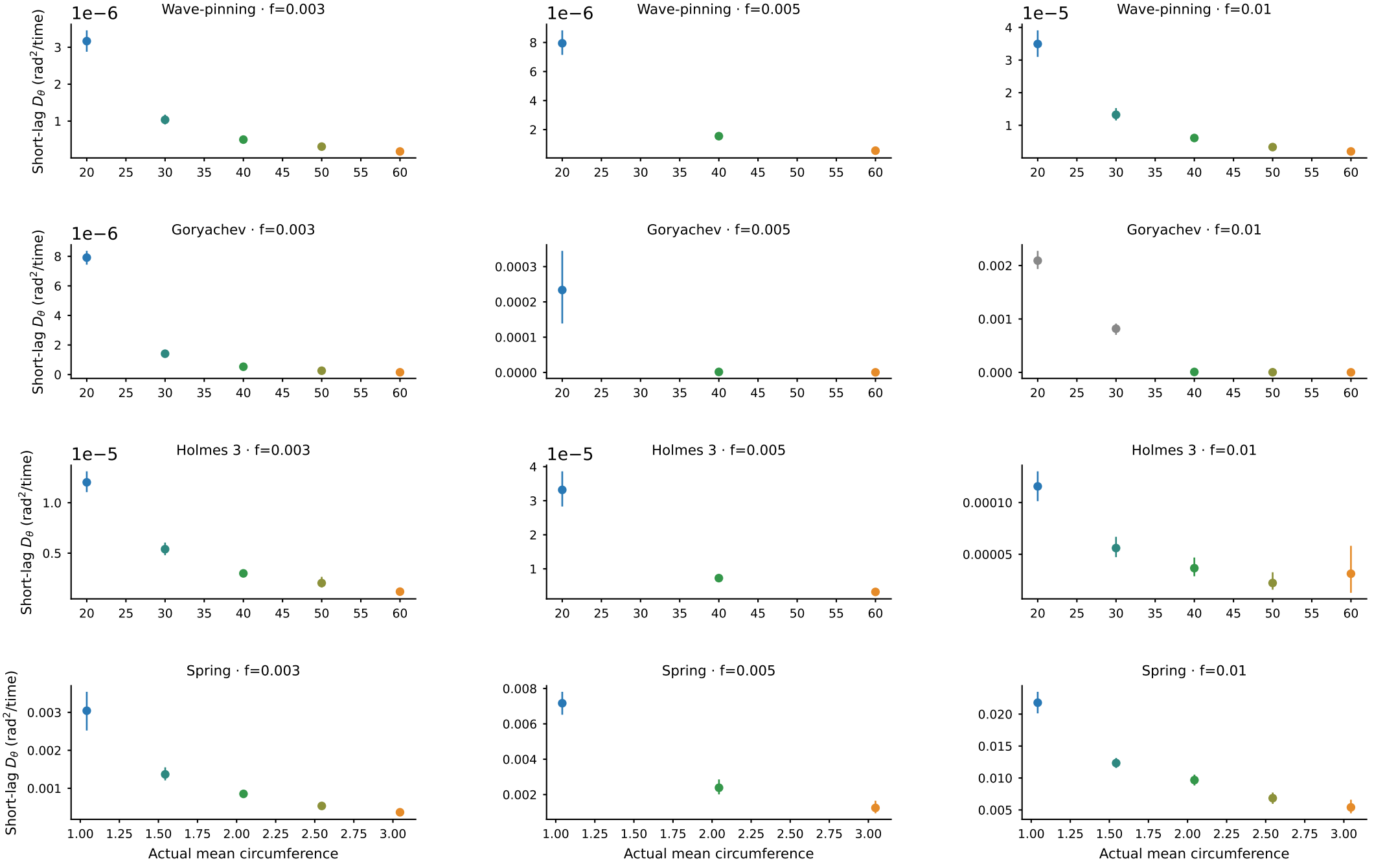
Longest qualified observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_canonical_measurements_20261020_014_S4_1_Dtheta.csv

S4 · Longest qualified observations · Persistence and angular mobility across size



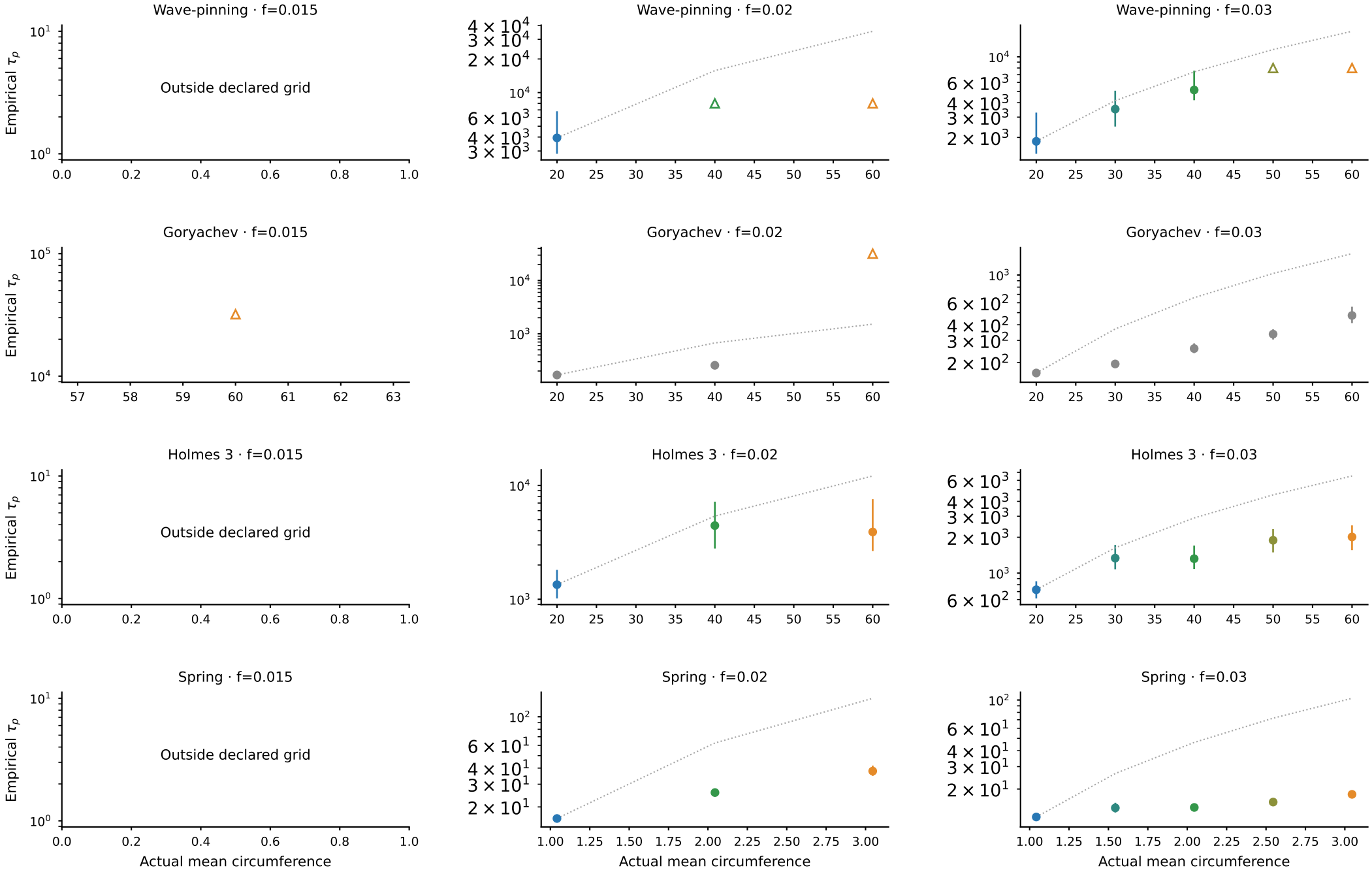
Longest qualified observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C^2 reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV. CI∞ next to a finite point marks an upper-unbounded bootstrap CI; finite lower endpoints are retained. CI[0,∞) spans the full nonnegative range.
CSV: figure4_longest_canonical_measurements_20261020_015_S4_2_tau_p.csv

S4 · Longest qualified observations · Persistence and angular mobility across size



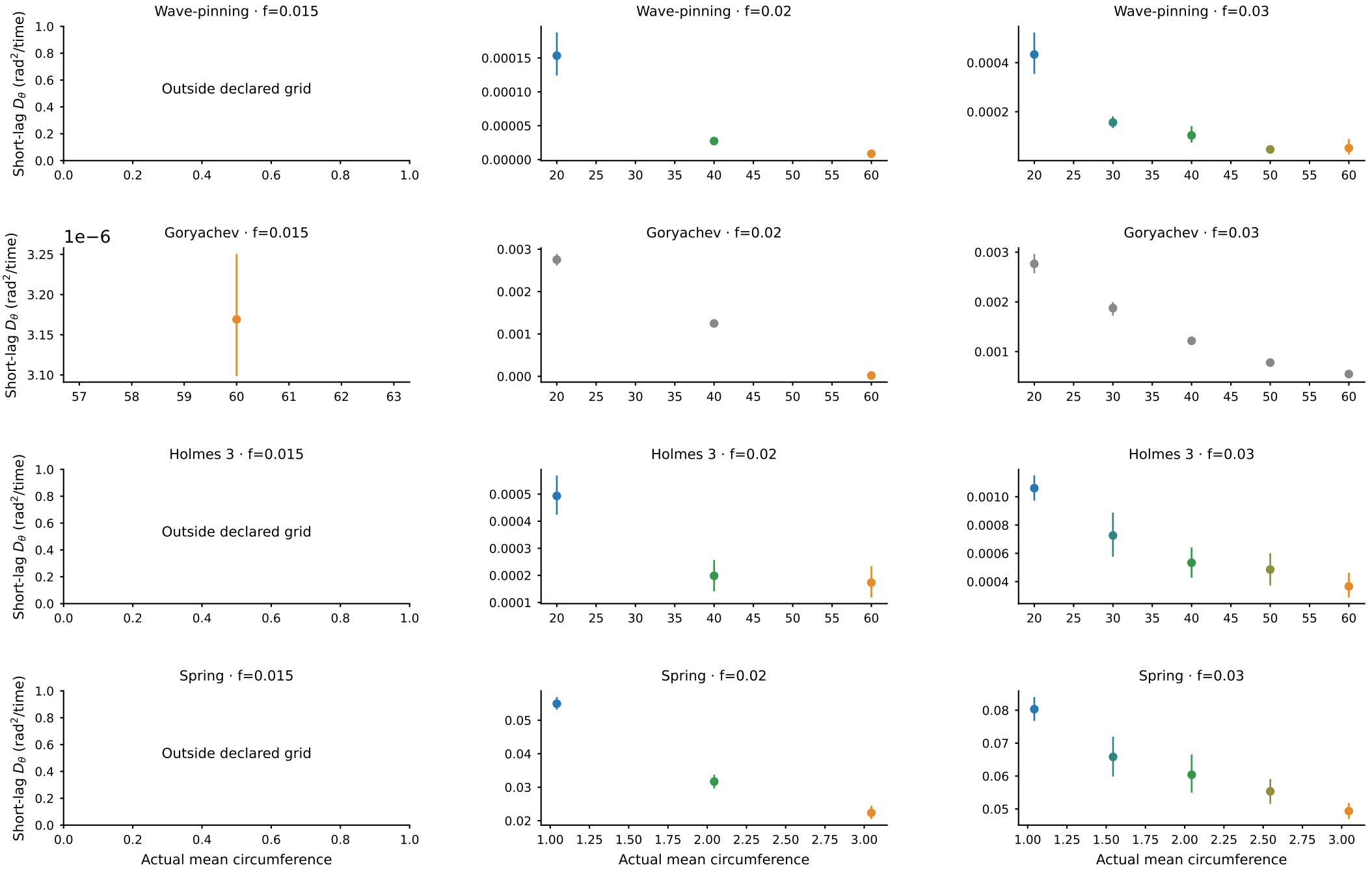
Longest qualified observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV. CI∞ next to a finite point marks an upper-unbounded bootstrap CI; finite lower endpoints are retained. CI[0,∞) spans the full nonnegative range.
CSV: figure4_longest_canonical_measurements_20261020_016_S4_2_Dtheta.csv

S4 · Longest qualified observations · Persistence and angular mobility across size

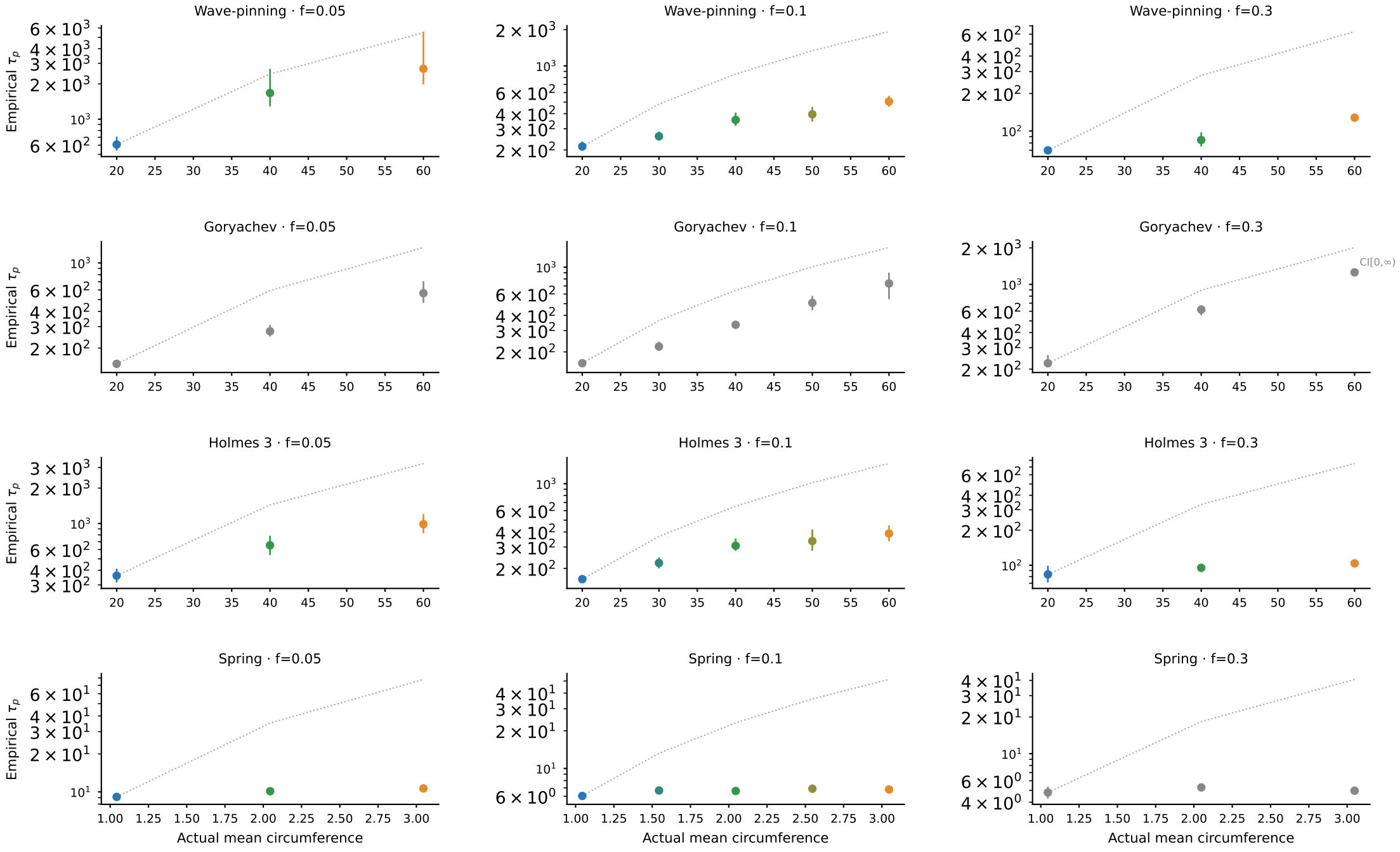


Longest qualified observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_canonical_measurements_20261020_017_S4_3_tau_p.csv

S4 · Longest qualified observations · Persistence and angular mobility across size

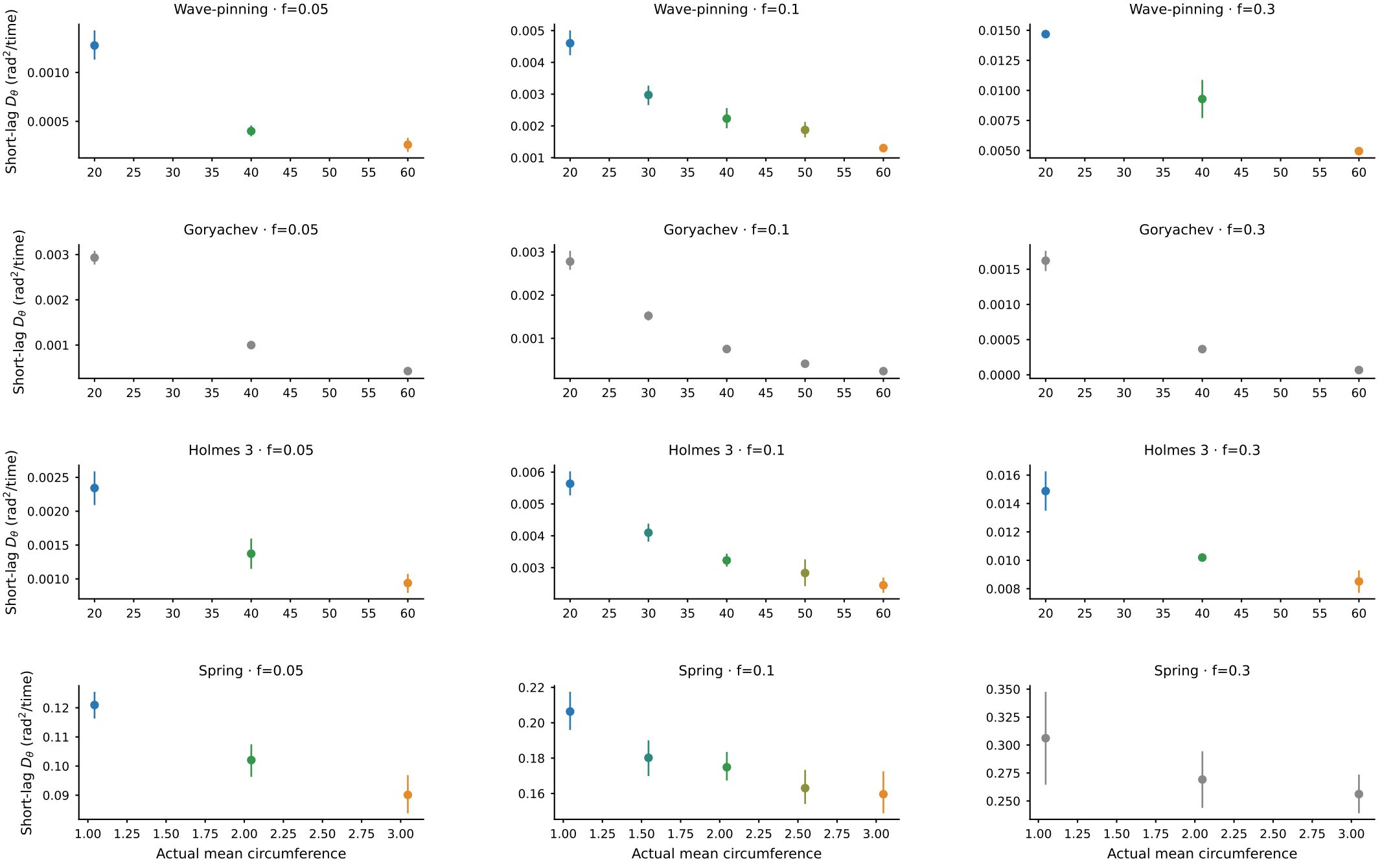


Longest qualified observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_canonical_measurements_20261020_018_S4_3_Dtheta.csv



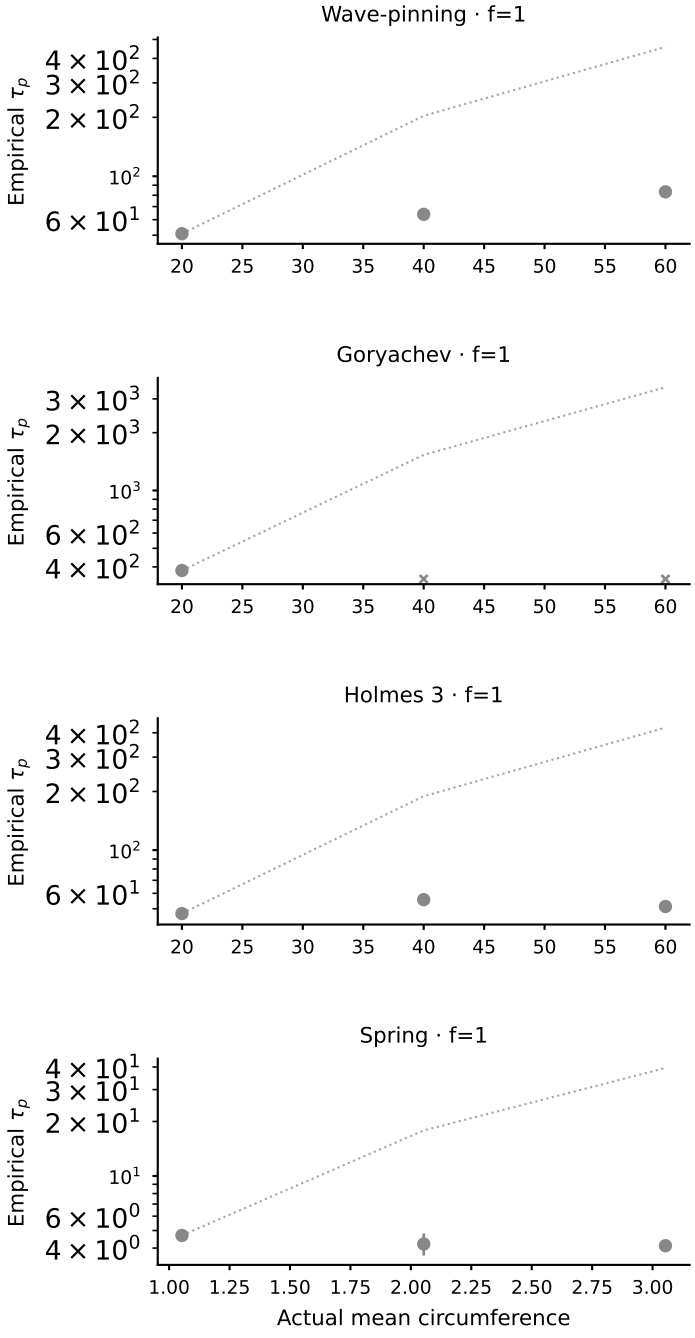
Longest qualified observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C^2 reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV. CI∞ next to a finite point marks an upper-unbounded bootstrap CI; finite lower endpoints are retained. CI[0,∞) spans the full nonnegative range.
CSV: figure4_longest_canonical_measurements_20261020_019_S4_4_tau_p.csv

S4 · Longest qualified observations · Persistence and angular mobility across size



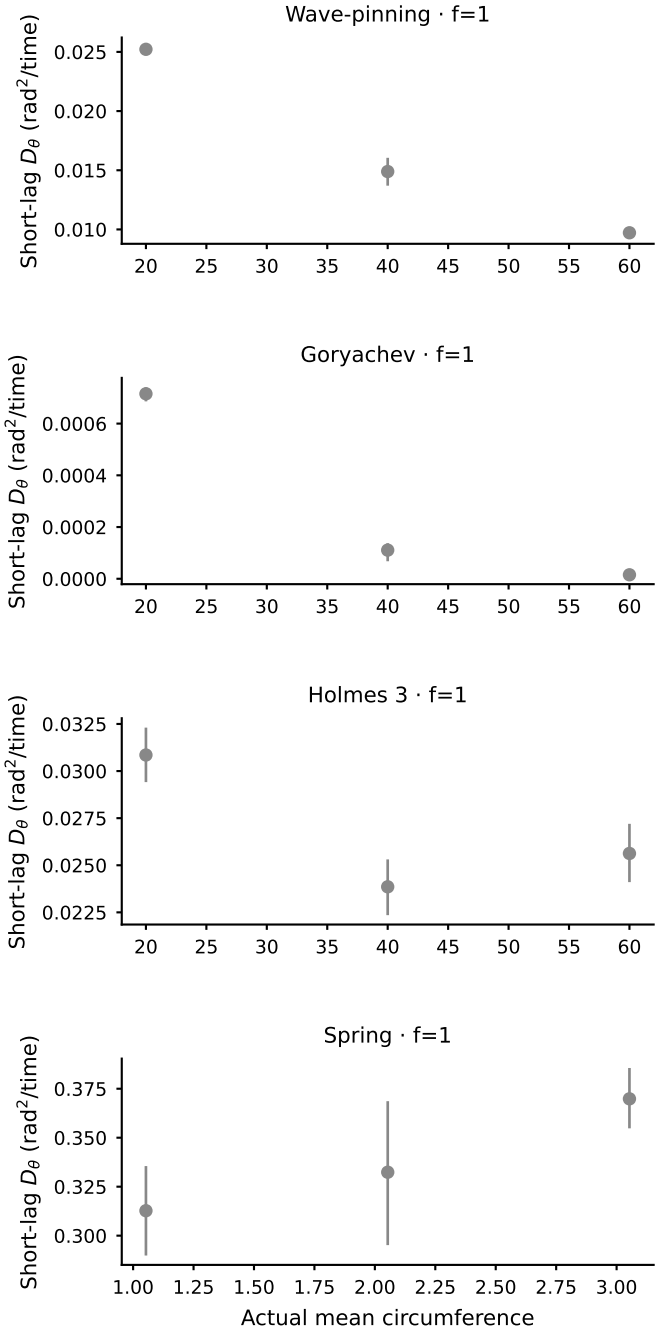
Longest qualified observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV. CI∞ next to a finite point marks an upper-unbounded bootstrap CI; finite lower endpoints are retained. CI[0,∞) spans the full nonnegative range.
CSV: figure4_longest_canonical_measurements_20261020_020_S4_4_Dtheta.csv

S4 · Longest qualified observations · Persistence and angular mobility across size



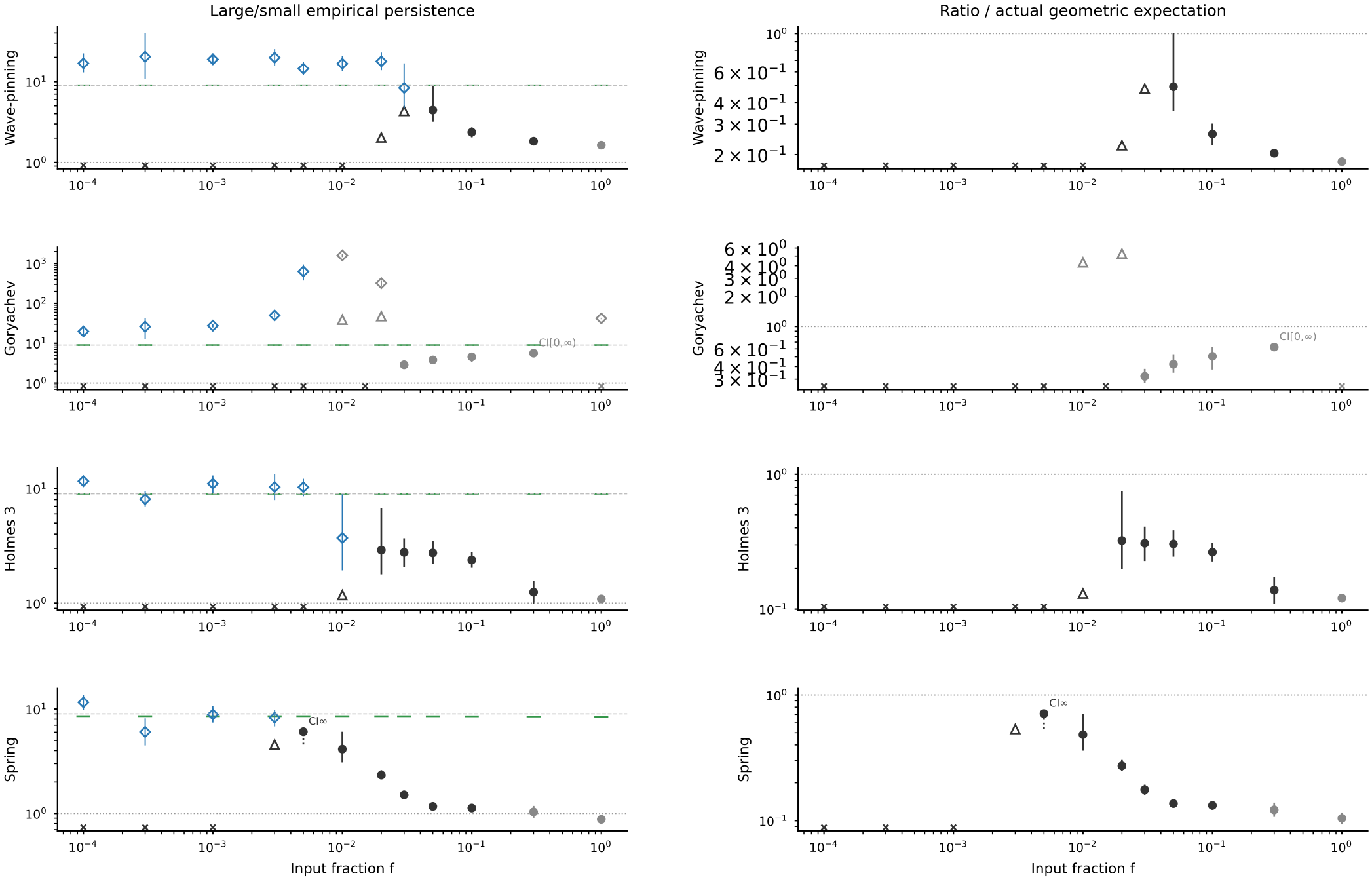
Longest qualified observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_canonical_measurements_20261020_021_S4_5_tau_p.csv

S4 · Longest qualified observations · Persistence and angular mobility across size



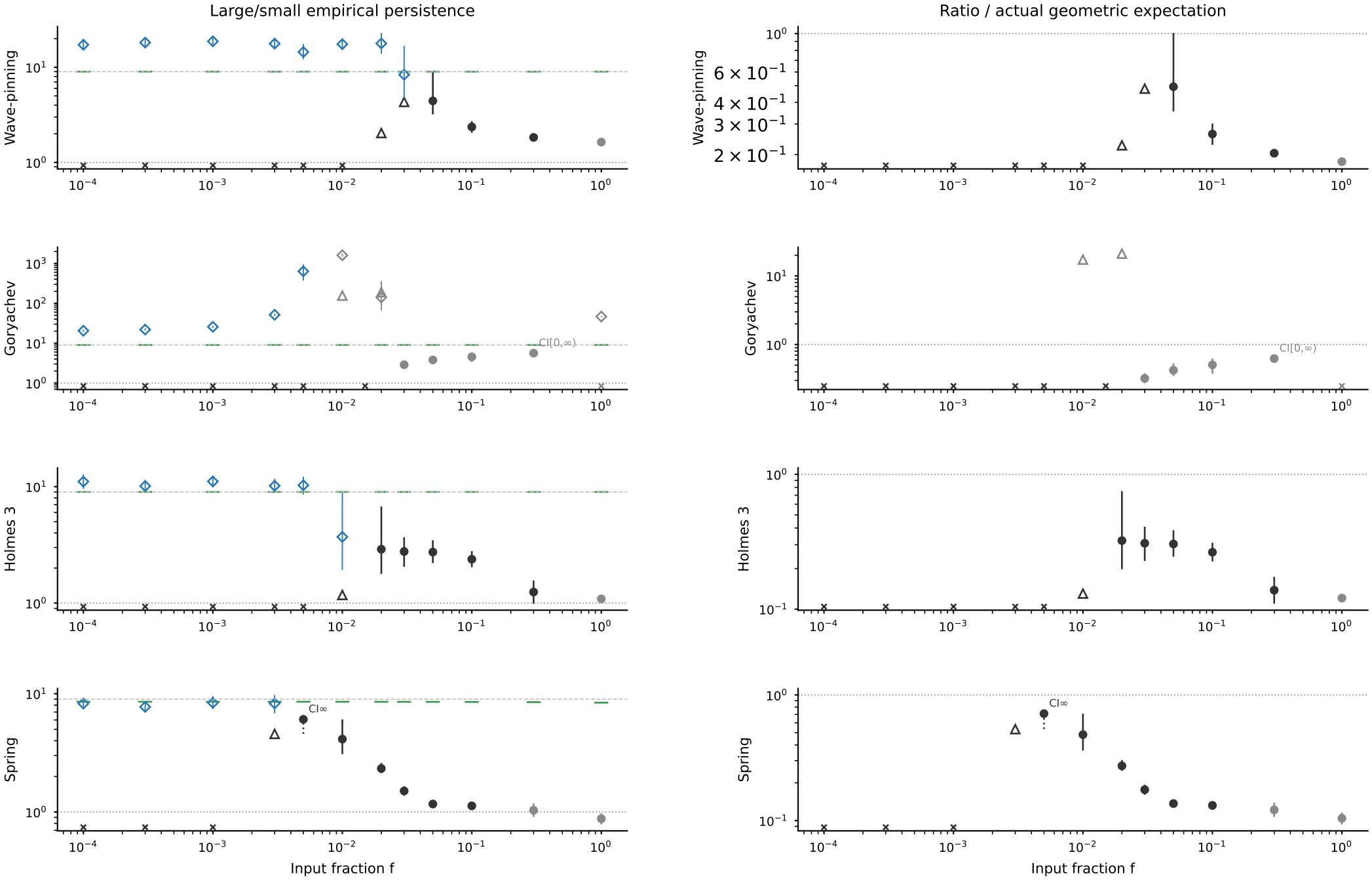
Longest qualified observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_canonical_measurements_20261020_022_S4_5_Dtheta.csv

S4 · Common qualified observations · Size advantage and geometric reference



Common qualified observations. Triangles preserve finite-window or resolution bounds. Two censored times do not yield a finite ratio. Open diamonds show the short-lag global activity-heading small/large Dθ proxy: blue for unforced conditions and grey for forced conditions. This proxy does not resolve a direct persistence ratio or measure one tracked site's diffusion in multi-region states. Green marks: measured circumference-ratio squared; dashed 9: resting-size reference. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV. CI∞ next to a finite point marks an upper-unbounded bootstrap CI; finite lower endpoints are retained. CI[0,∞) spans the full nonnegative range.
CSV: figure4_ratio_scope_replacements_20261020_001_S4_ratios.csv

S4 · Longest qualified observations · Size advantage and geometric reference



Longest qualified observations. Triangles preserve finite-window or resolution bounds. Two censored times do not yield a finite ratio. Open diamonds show the short-lag global activity-heading small/large Dθ proxy: blue for unforced conditions and grey for forced conditions. This proxy does not resolve a direct persistence ratio or measure one tracked site's diffusion in multi-region states. Green marks: measured circumference-ratio squared; dashed 9: resting-size reference. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV. CI∞ next to a finite point marks an upper-unbounded bootstrap CI; finite lower endpoints are retained. CI[0,∞) spans the full nonnegative range.
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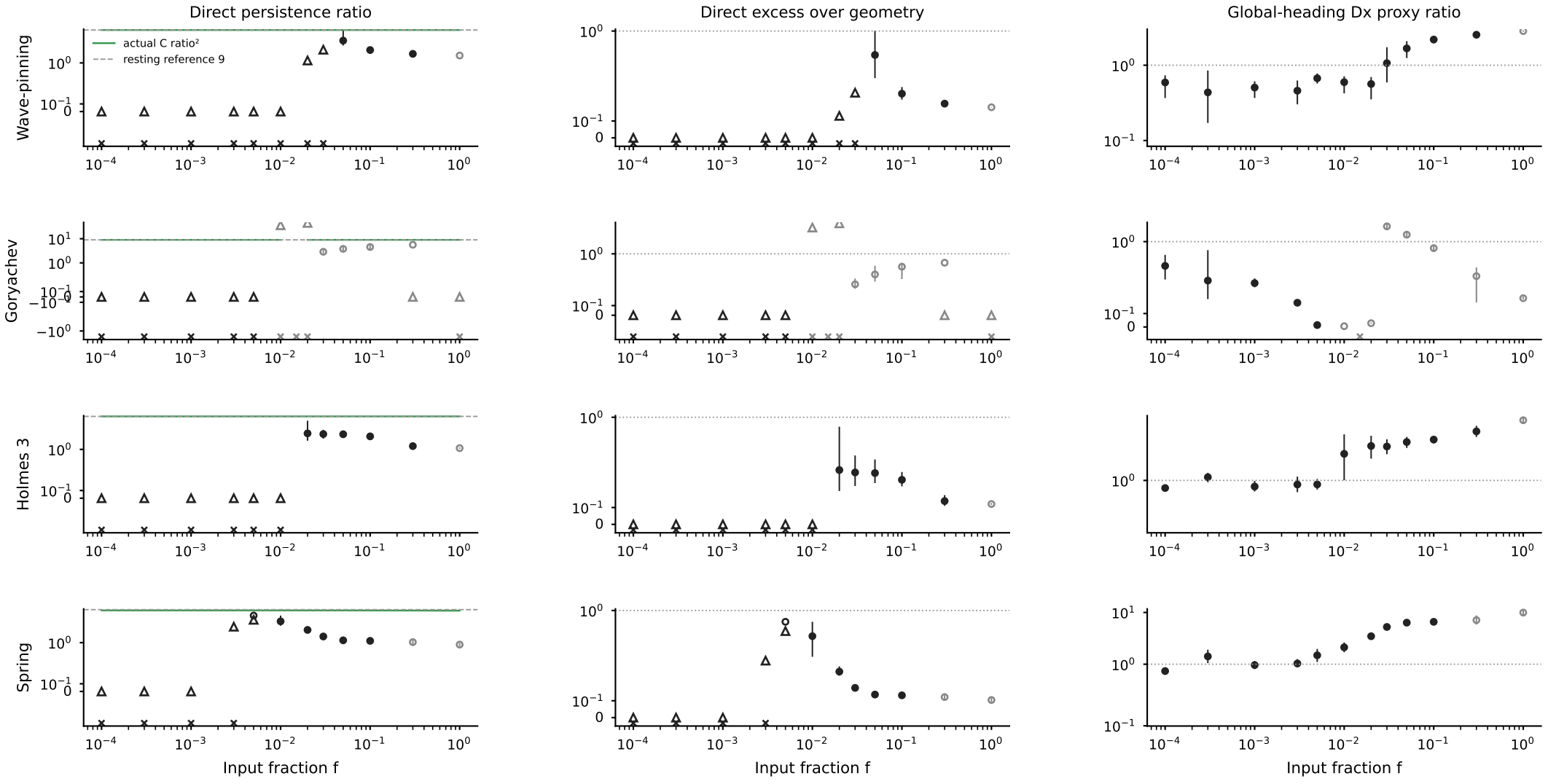
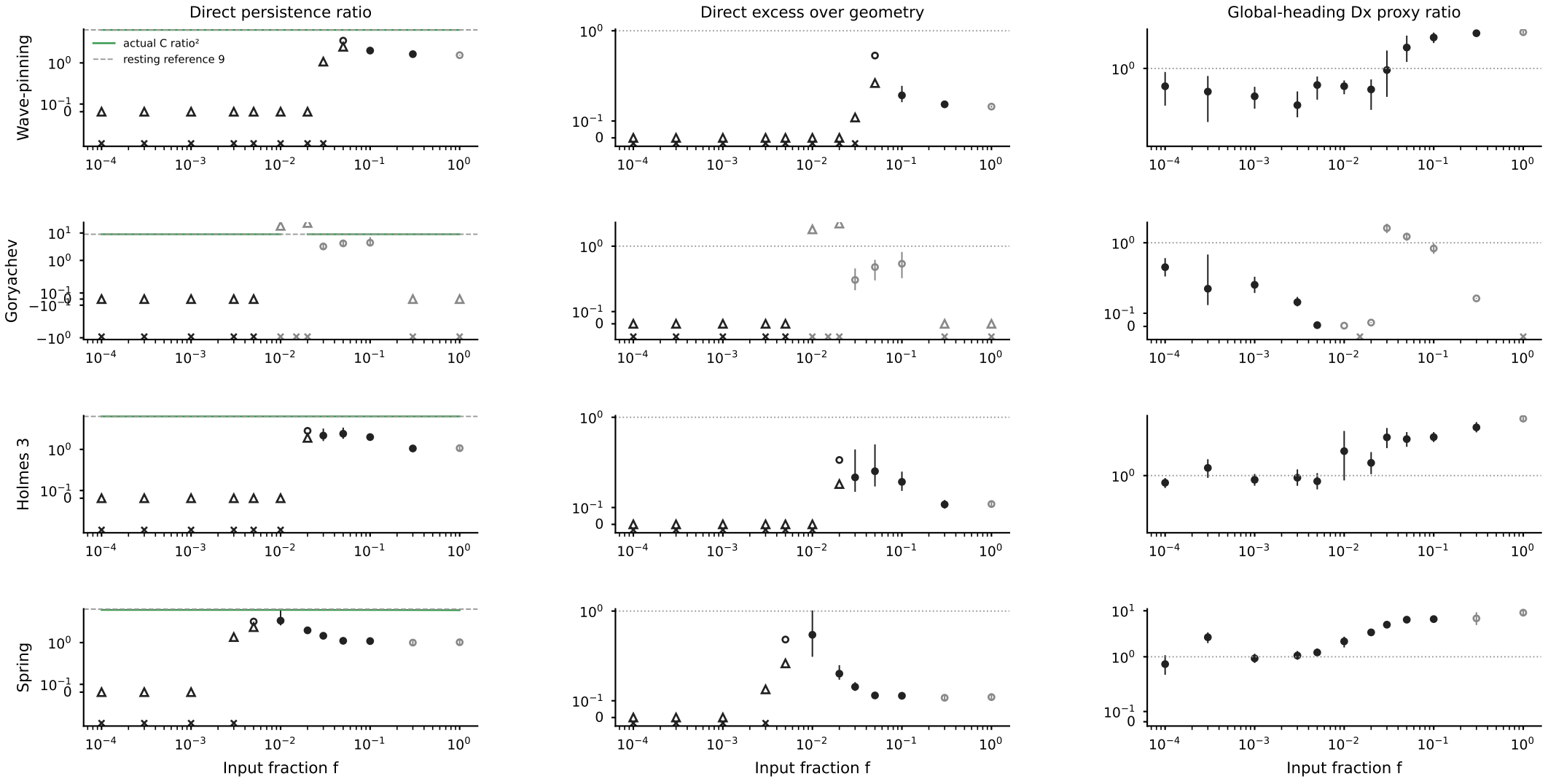
S4 · Separate geometric qualification · Direct time and short-lag mobility · common · main · operationally unforced conditions



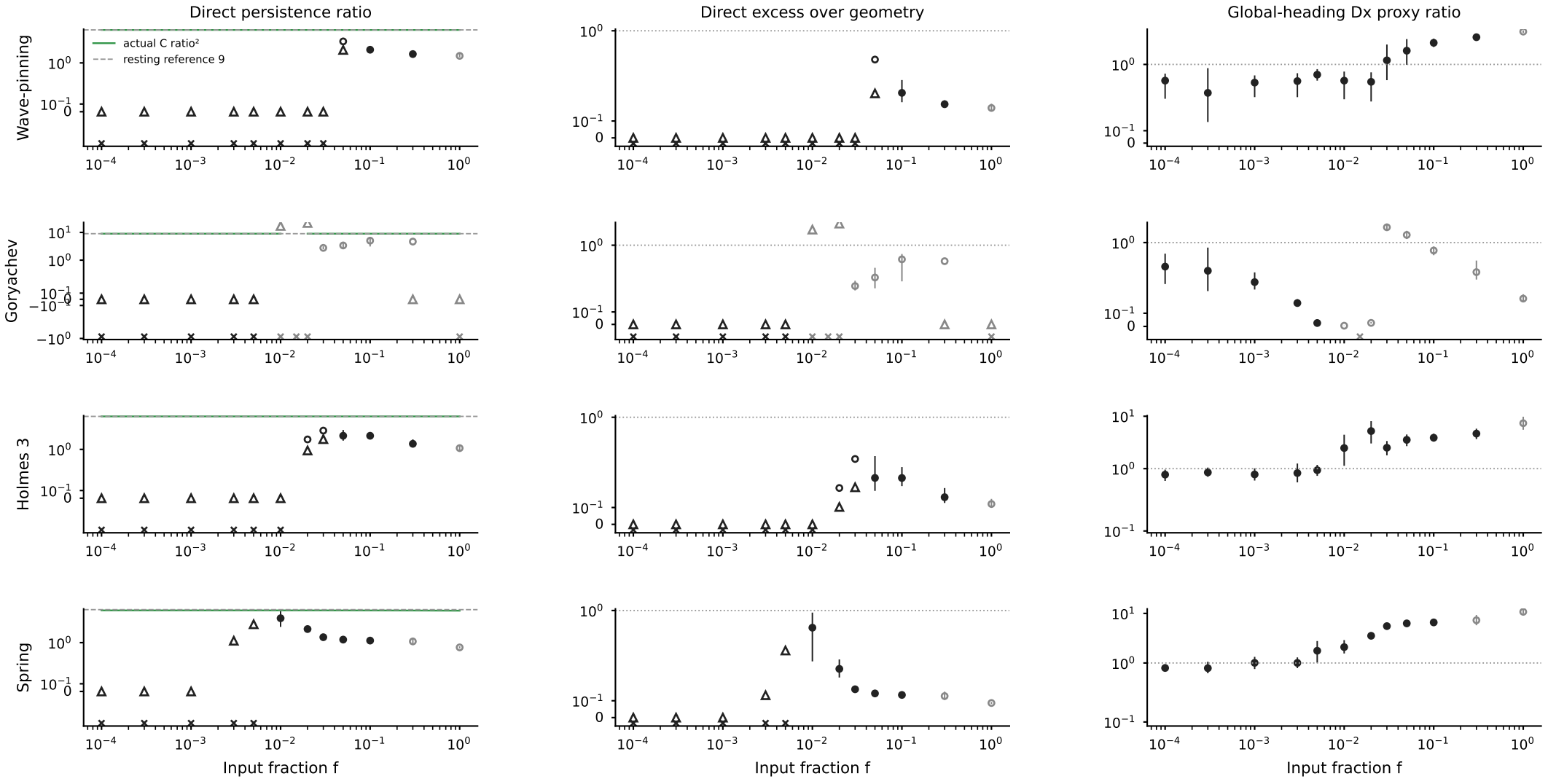
Separate geometric qualification. Columns keep direct time, direct excess-persistence CI and short-lag global-heading Dx-proxy CI separate. Dx converts global activity-heading fluctuations using actual C; in multi-region states it is not one tracked site's diffusion. Qualified single-region literal A3 heading/peak MSDs remain separate. Green is actual C-ratio²; grey dashed9 is the resting reference. Filled points have finite qualified inference; open circles retain raw points without it. Open triangles retain a finite confidence endpoint when the other is unbounded; × is unidentified, not zero. Symlog preserves signed raw estimates. CI including1 means compatibility, not equivalence. Increased Dx, persistence advantages and conflicts remain visible; this descriptor does not alter H1–H3. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S4_S5_final_geometry_20261020_009_S4_S5_geometry_common_primary_main.csv

S4 · Separate geometric qualification · Direct time and short-lag mobility · common · analysis first half · operationally unforced conditions



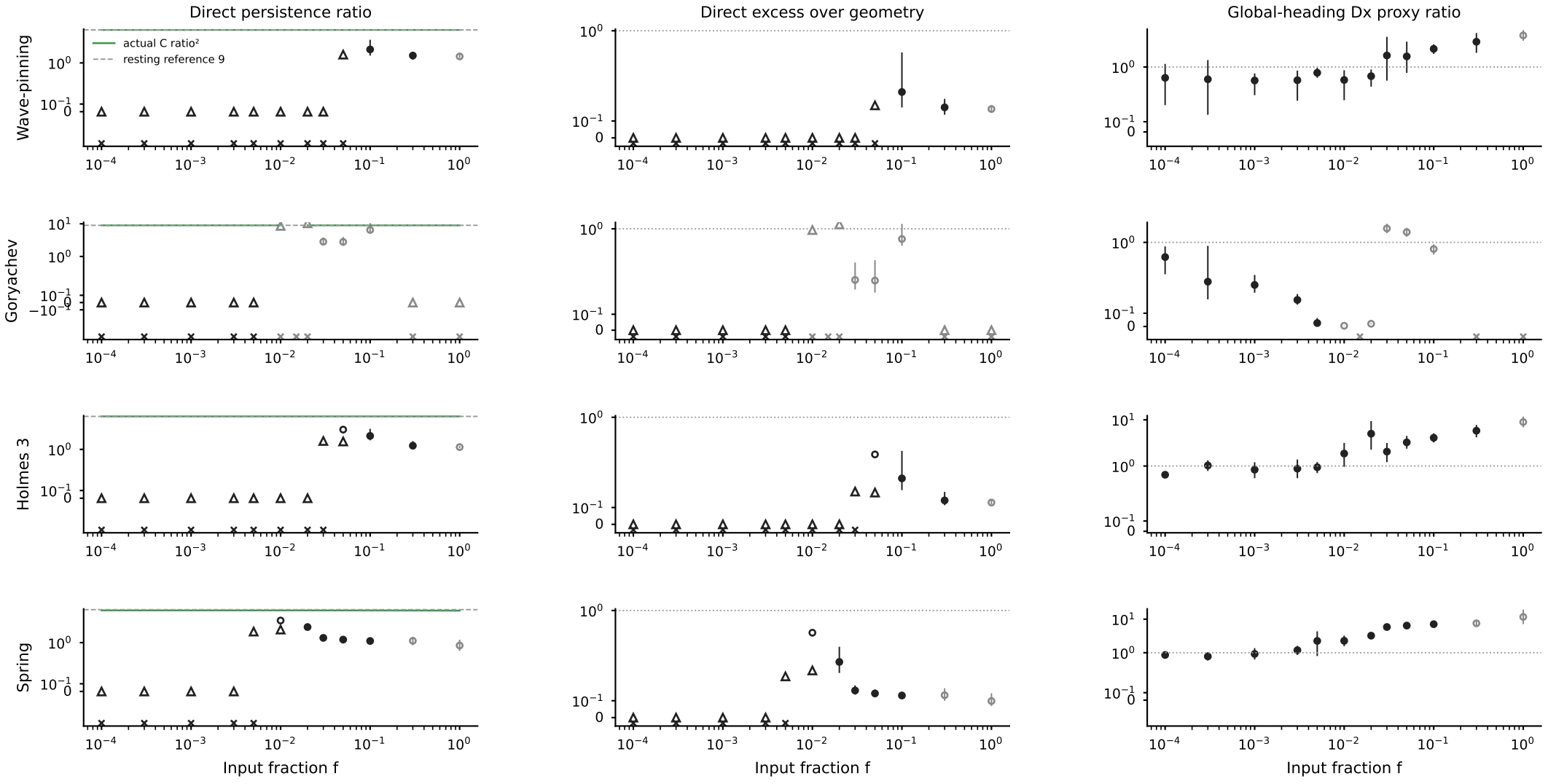
Separate geometric qualification. Columns keep direct time, direct excess-persistence CI and short-lag global-heading Dx-proxy CI separate. Dx converts global activity-heading fluctuations using actual C; in multi-region states it is not one tracked site's diffusion. Qualified single-region literal A3 heading/peak MSDs remain separate. Green is actual C-ratio²; grey dashed9 is the resting reference. Filled points have finite qualified inference; open circles retain raw points without it. Open triangles retain a finite confidence endpoint when the other is unbounded; × is unidentified, not zero. Symlog preserves signed raw estimates. CI including1 means compatibility, not equivalence. Increased Dx, persistence advantages and conflicts remain visible; this descriptor does not alter H1–H3. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S4_S5_final_geometry_20261020_010_S4_S5_geometry_common_primary_analysis_first_half.csv

S4 · Separate geometric qualification · Direct time and short-lag mobility · common · analysis second half · operationally unforced conditions



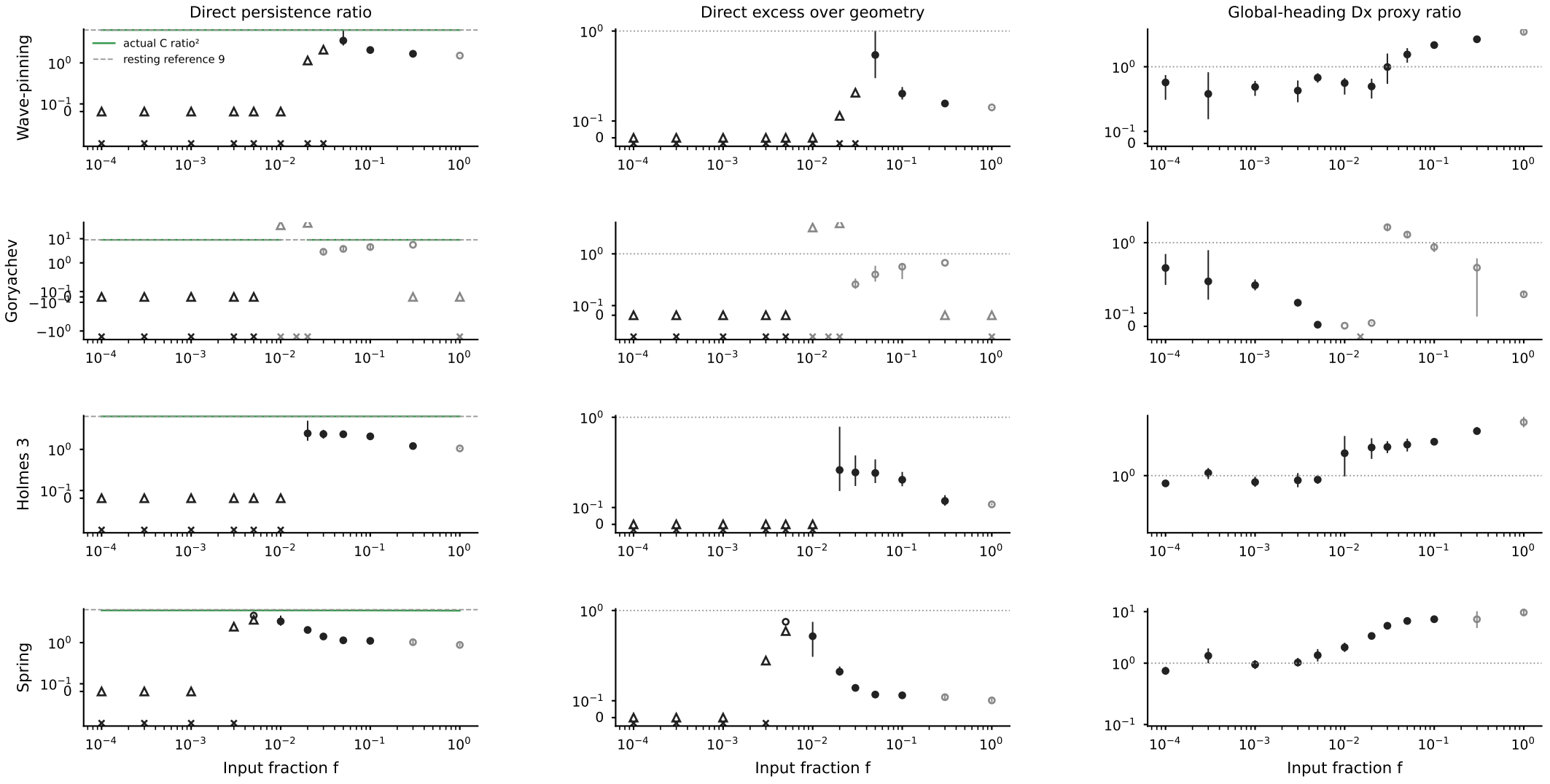
Separate geometric qualification. Columns keep direct time, direct excess-persistence CI and short-lag global-heading Dx-proxy CI separate. Dx converts global activity-heading fluctuations using actual C; in multi-region states it is not one tracked site's diffusion. Qualified single-region literal A3 heading/peak MSDs remain separate. Green is actual C-ratio²; grey dashed9 is the resting reference. Filled points have finite qualified inference; open circles retain raw points without it. Open triangles retain a finite confidence endpoint when the other is unbounded; × is unidentified, not zero. Symlog preserves signed raw estimates. CI including1 means compatibility, not equivalence. Increased Dx, persistence advantages and conflicts remain visible; this descriptor does not alter H1–H3. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S4_S5_final_geometry_20261020_011_S4_S5_geometry_common_primary_analysis_second_half.csv

S4 · Separate geometric qualification · Direct time and short-lag mobility · common · late quarter · operationally unforced conditions



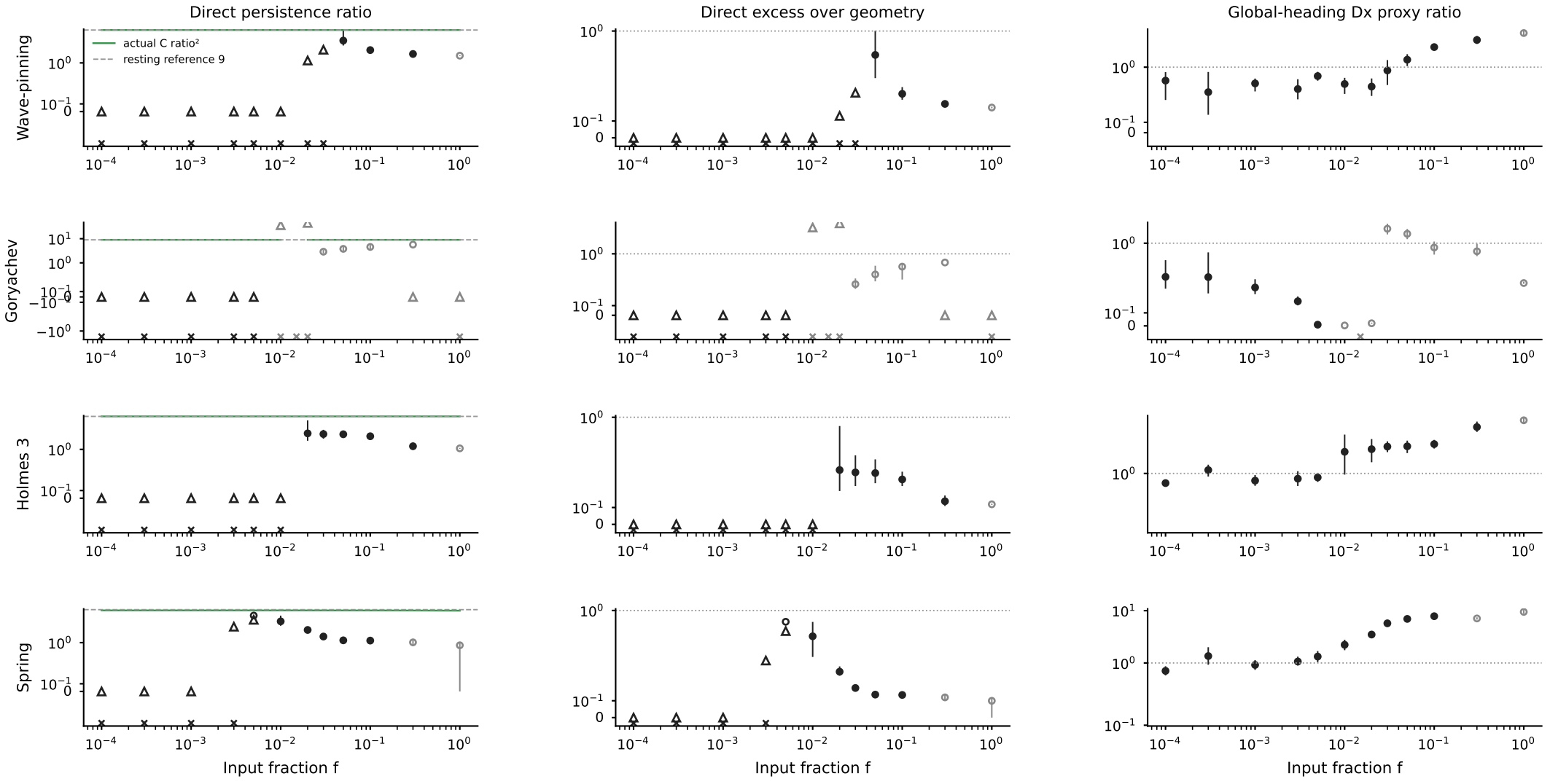
Separate geometric qualification. Columns keep direct time, direct excess-persistence CI and short-lag global-heading Dx-proxy CI separate. Dx converts global activity-heading fluctuations using actual C; in multi-region states it is not one tracked site's diffusion. Qualified single-region literal A3 heading/peak MSDs remain separate. Green is actual C-ratio²; grey dashed9 is the resting reference. Filled points have finite qualified inference; open circles retain raw points without it. Open triangles retain a finite confidence endpoint when the other is unbounded; × is unidentified, not zero. Symlog preserves signed raw estimates. CI including1 means compatibility, not equivalence. Increased Dx, persistence advantages and conflicts remain visible; this descriptor does not alter H1–H3. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S4_S5_final_geometry_20261020_012_S4_S5_geometry_common_primary_late_quarter.csv

S4 · Separate geometric qualification · Direct time and short-lag mobility · common · cadence x2 · operationally unforced conditions



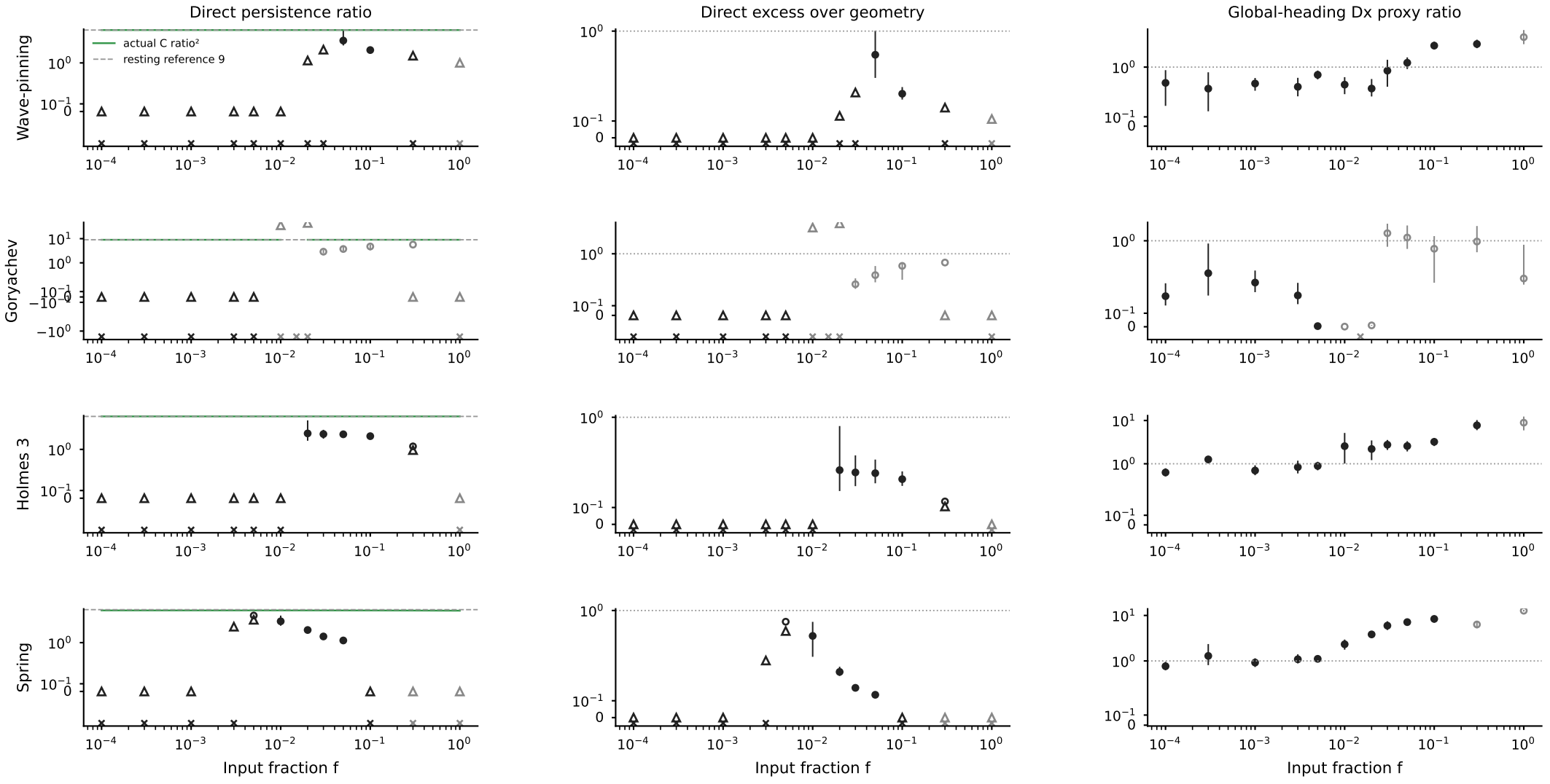
Separate geometric qualification. Columns keep direct time, direct excess-persistence CI and short-lag global-heading Dx-proxy CI separate. Dx converts global activity-heading fluctuations using actual C; in multi-region states it is not one tracked site's diffusion. Qualified single-region literal A3 heading/peak MSDs remain separate. Green is actual C-ratio²; grey dashed9 is the resting reference. Filled points have finite qualified inference; open circles retain raw points without it. Open triangles retain a finite confidence endpoint when the other is unbounded; × is unidentified, not zero. Symlog preserves signed raw estimates. CI including1 means compatibility, not equivalence. Increased Dx, persistence advantages and conflicts remain visible; this descriptor does not alter H1–H3. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S4_S5_final_geometry_20261020_013_S4_S5_geometry_common_primary_cadence_x2.csv

S4 · Separate geometric qualification · Direct time and short-lag mobility · common · cadence x4 · operationally unforced conditions



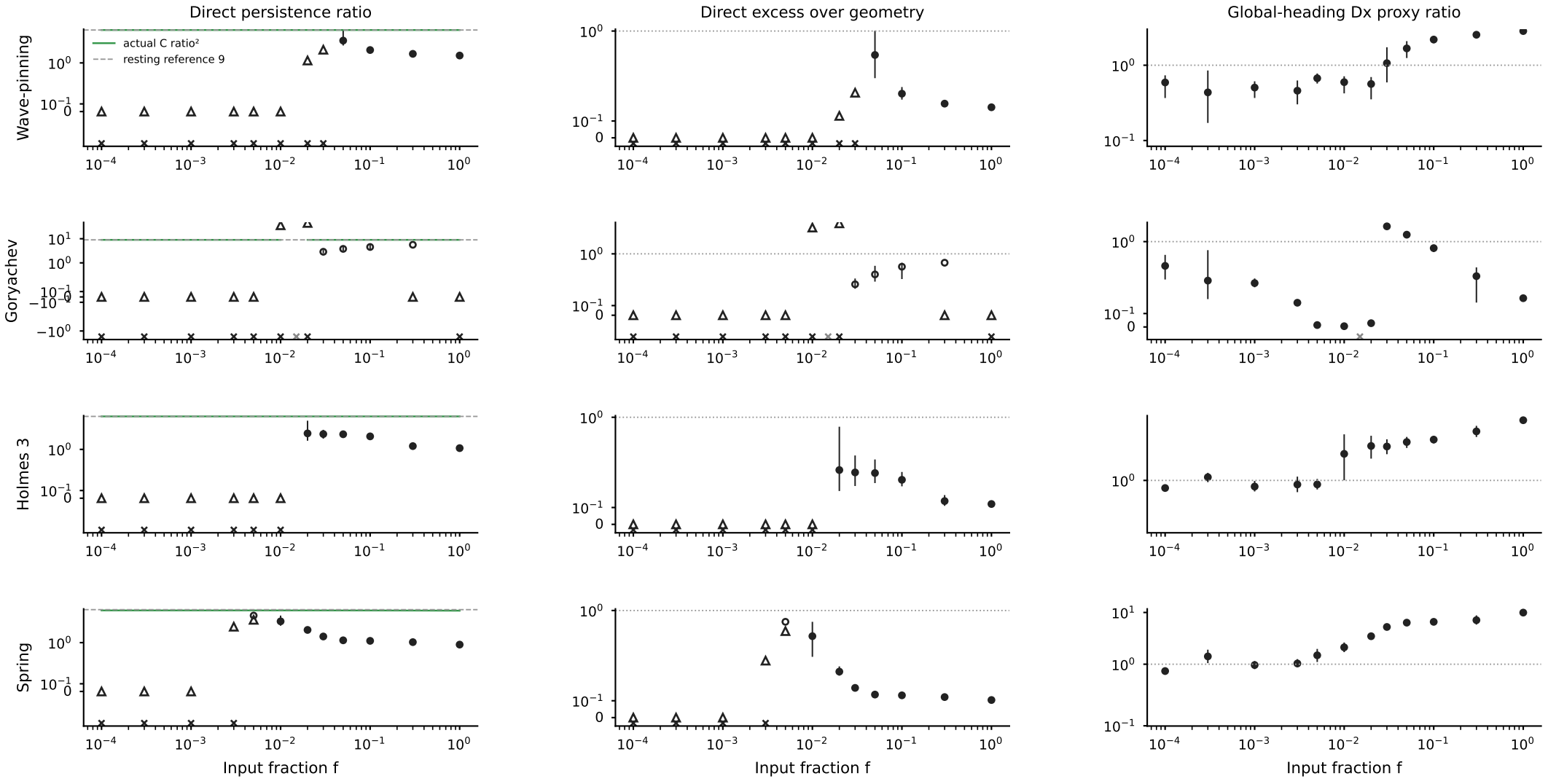
Separate geometric qualification. Columns keep direct time, direct excess-persistence CI and short-lag global-heading Dx-proxy CI separate. Dx converts global activity-heading fluctuations using actual C; in multi-region states it is not one tracked site's diffusion. Qualified single-region literal A3 heading/peak MSDs remain separate. Green is actual C-ratio²; grey dashed9 is the resting reference. Filled points have finite qualified inference; open circles retain raw points without it. Open triangles retain a finite confidence endpoint when the other is unbounded; × is unidentified, not zero. Symlog preserves signed raw estimates. CI including1 means compatibility, not equivalence. Increased Dx, persistence advantages and conflicts remain visible; this descriptor does not alter H1–H3. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S4_S5_final_geometry_20261020_014_S4_S5_geometry_common_primary_cadence_x4.csv

S4 · Separate geometric qualification · Direct time and short-lag mobility · common · cadence x8 · operationally unforced conditions



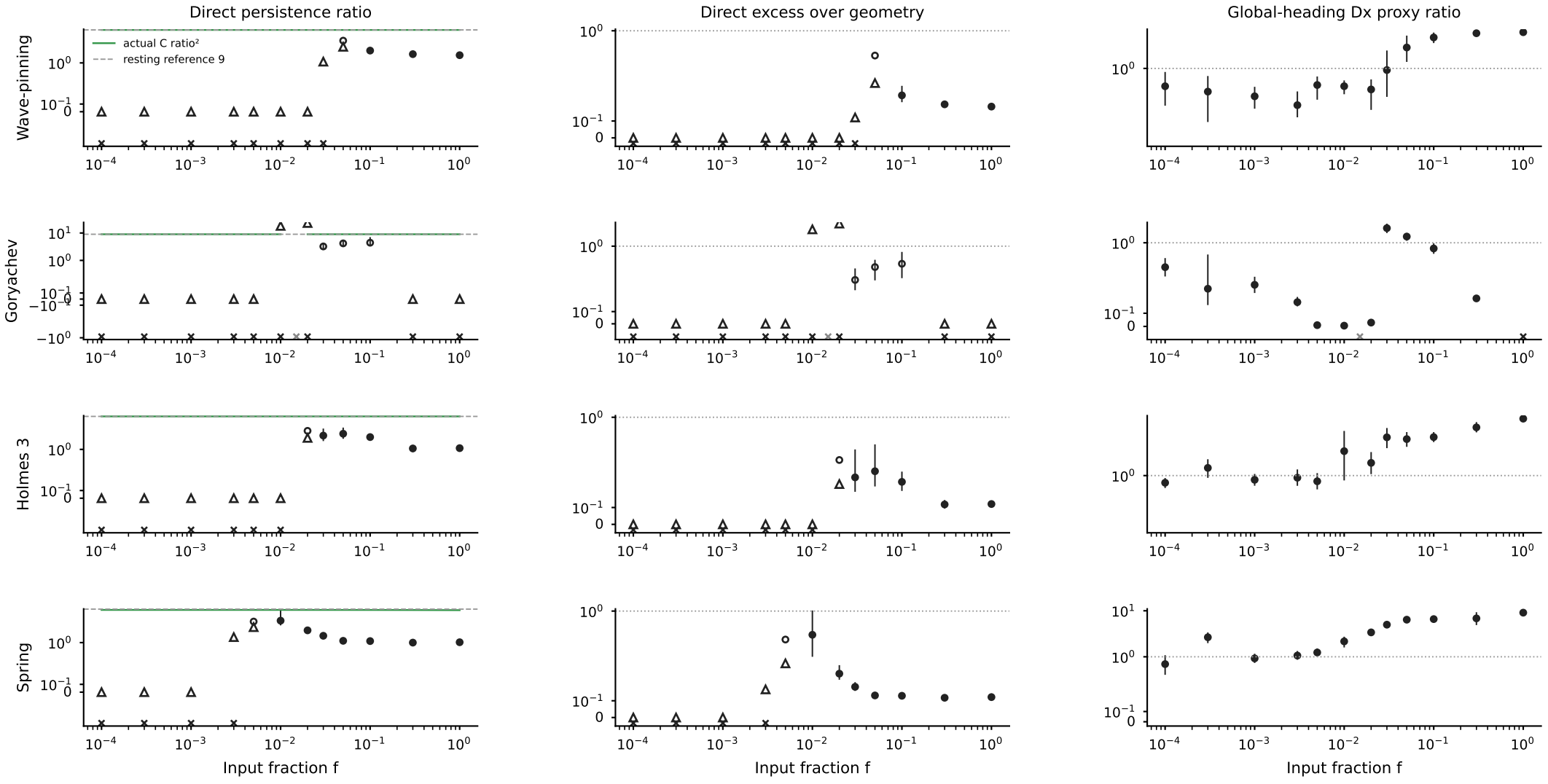
Separate geometric qualification. Columns keep direct time, direct excess-persistence CI and short-lag global-heading Dx-proxy CI separate. Dx converts global activity-heading fluctuations using actual C; in multi-region states it is not one tracked site's diffusion. Qualified single-region literal A3 heading/peak MSDs remain separate. Green is actual C-ratio²; grey dashed9 is the resting reference. Filled points have finite qualified inference; open circles retain raw points without it. Open triangles retain a finite confidence endpoint when the other is unbounded; × is unidentified, not zero. Symlog preserves signed raw estimates. CI including1 means compatibility, not equivalence. Increased Dx, persistence advantages and conflicts remain visible; this descriptor does not alter H1–H3. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S4_S5_final_geometry_20261020_015_S4_S5_geometry_common_primary_cadence_x8.csv

S4 · Separate geometric qualification · Direct time and short-lag mobility · common · main · all conditions including operationally forced



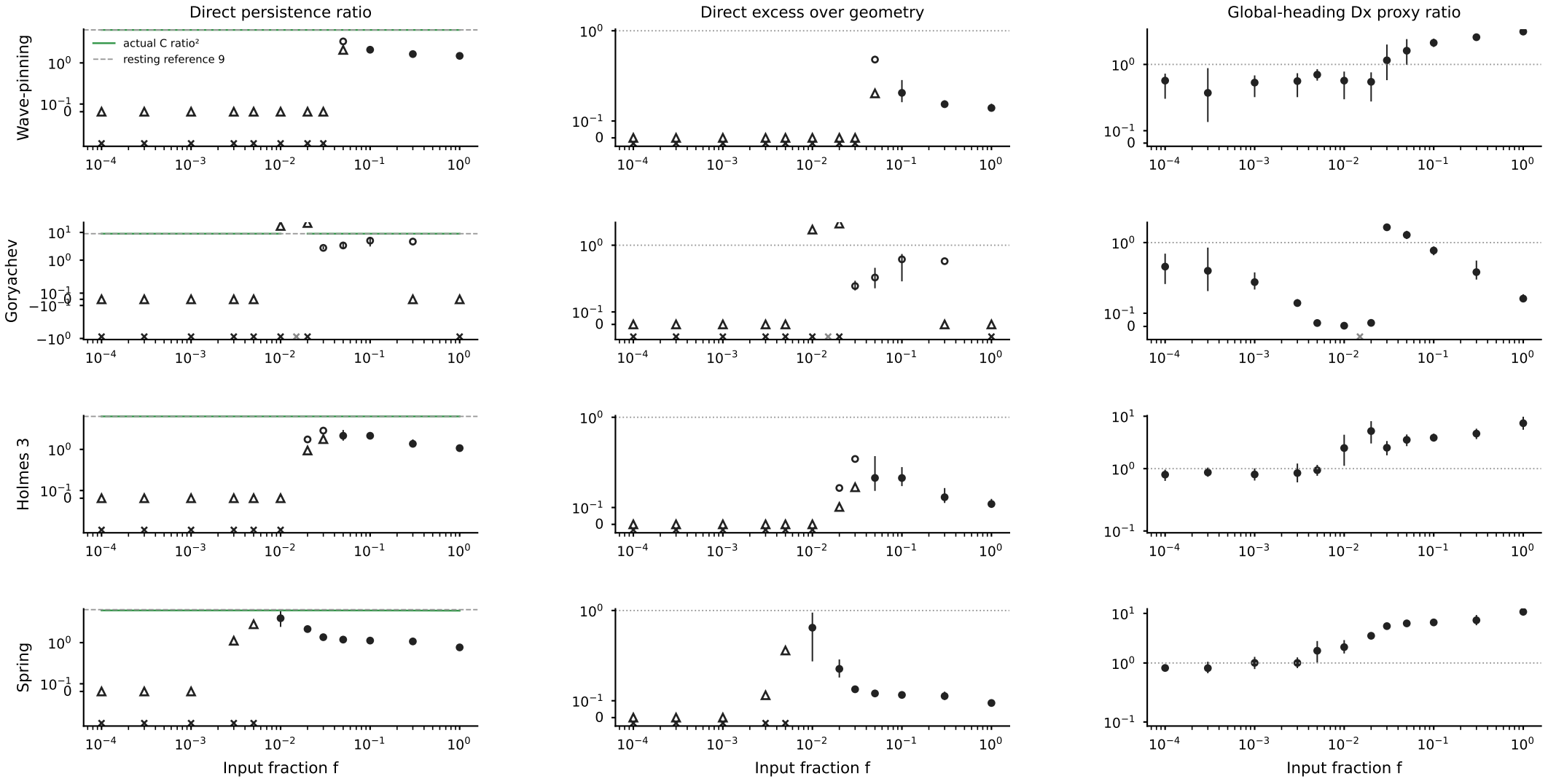
Separate geometric qualification. Columns keep direct time, direct excess-persistence CI and short-lag global-heading Dx-proxy CI separate. Dx converts global activity-heading fluctuations using actual C; in multi-region states it is not one tracked site's diffusion. Qualified single-region literal A3 heading/peak MSDs remain separate. Green is actual C-ratio²; grey dashed9 is the resting reference. Filled points have finite qualified inference; open circles retain raw points without it. Open triangles retain a finite confidence endpoint when the other is unbounded; × is unidentified, not zero. Symlog preserves signed raw estimates. CI including1 means compatibility, not equivalence. Increased Dx, persistence advantages and conflicts remain visible; this descriptor does not alter H1–H3. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S4_S5_final_geometry_20261020_016_S4_S5_geometry_common_inclusive_main.csv

S4 · Separate geometric qualification · Direct time and short-lag mobility · common · analysis first half
· all conditions including operationally forced



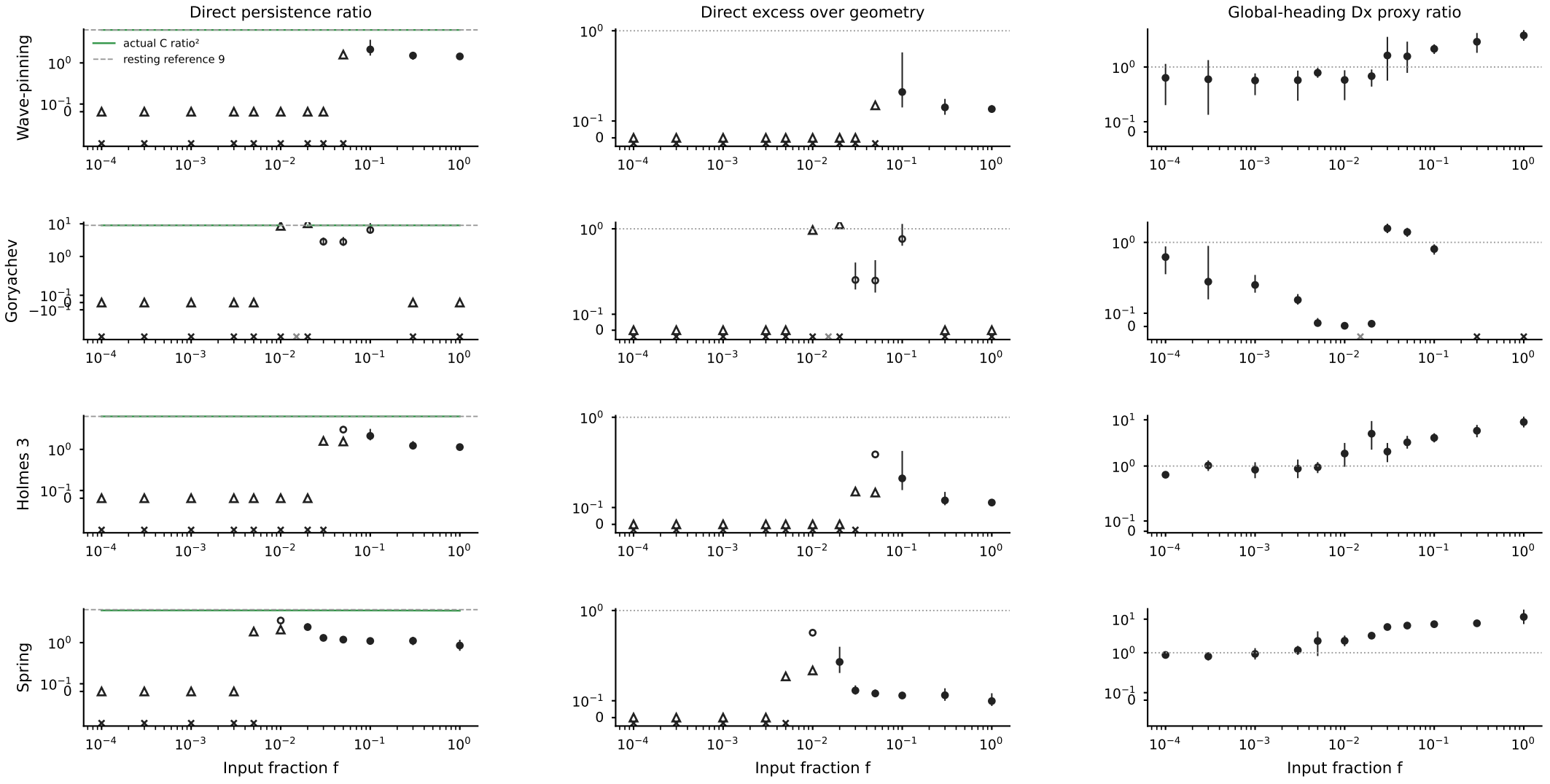
Separate geometric qualification. Columns keep direct time, direct excess-persistence CI and short-lag global-heading Dx-proxy CI separate. Dx converts global activity-heading fluctuations using actual C; in multi-region states it is not one tracked site's diffusion. Qualified single-region literal A3 heading/peak MSDs remain separate. Green is actual C-ratio²; grey dashed9 is the resting reference. Filled points have finite qualified inference; open circles retain raw points without it. Open triangles retain a finite confidence endpoint when the other is unbounded; × is unidentified, not zero. Symlog preserves signed raw estimates. CI including1 means compatibility, not equivalence. Increased Dx, persistence advantages and conflicts remain visible; this descriptor does not alter H1–H3. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S4_S5_final_geometry_20261020_017_S4_S5_geometry_common_inclusive_analysis_first_half.csv

S4 · Separate geometric qualification · Direct time and short-lag mobility · common · analysis second half · all conditions including operationally forced



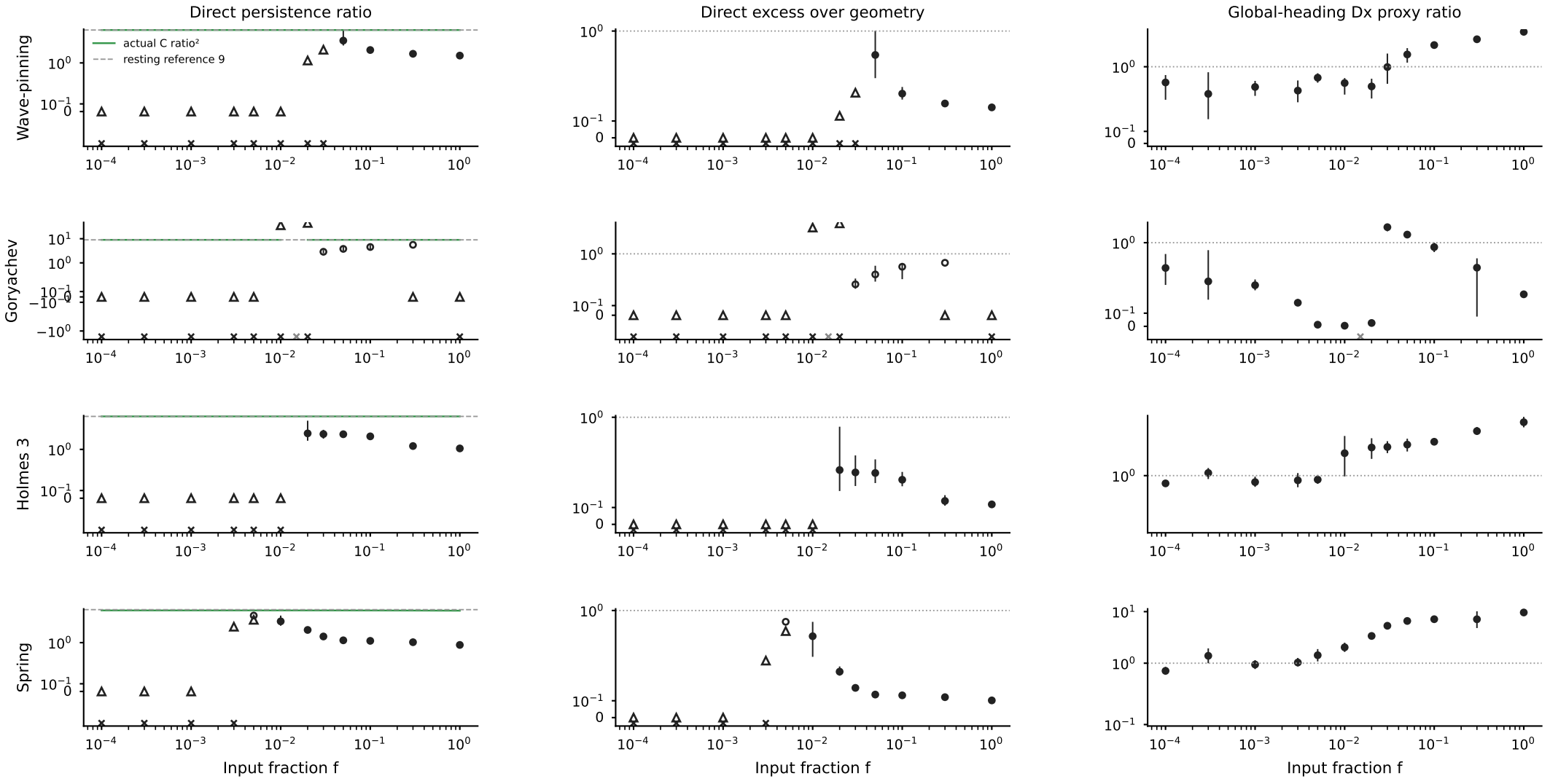
Separate geometric qualification. Columns keep direct time, direct excess-persistence CI and short-lag global-heading Dx-proxy CI separate. Dx converts global activity-heading fluctuations using actual C; in multi-region states it is not one tracked site's diffusion. Qualified single-region literal A3 heading/peak MSDs remain separate. Green is actual C-ratio²; grey dashed9 is the resting reference. Filled points have finite qualified inference; open circles retain raw points without it. Open triangles retain a finite confidence endpoint when the other is unbounded; × is unidentified, not zero. Symlog preserves signed raw estimates. CI including1 means compatibility, not equivalence. Increased Dx, persistence advantages and conflicts remain visible; this descriptor does not alter H1–H3. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S4_S5_final_geometry_20261020_018_S4_S5_geometry_common_inclusive_analysis_second_half.csv

S4 · Separate geometric qualification · Direct time and short-lag mobility · common · late quarter · all conditions including operationally forced



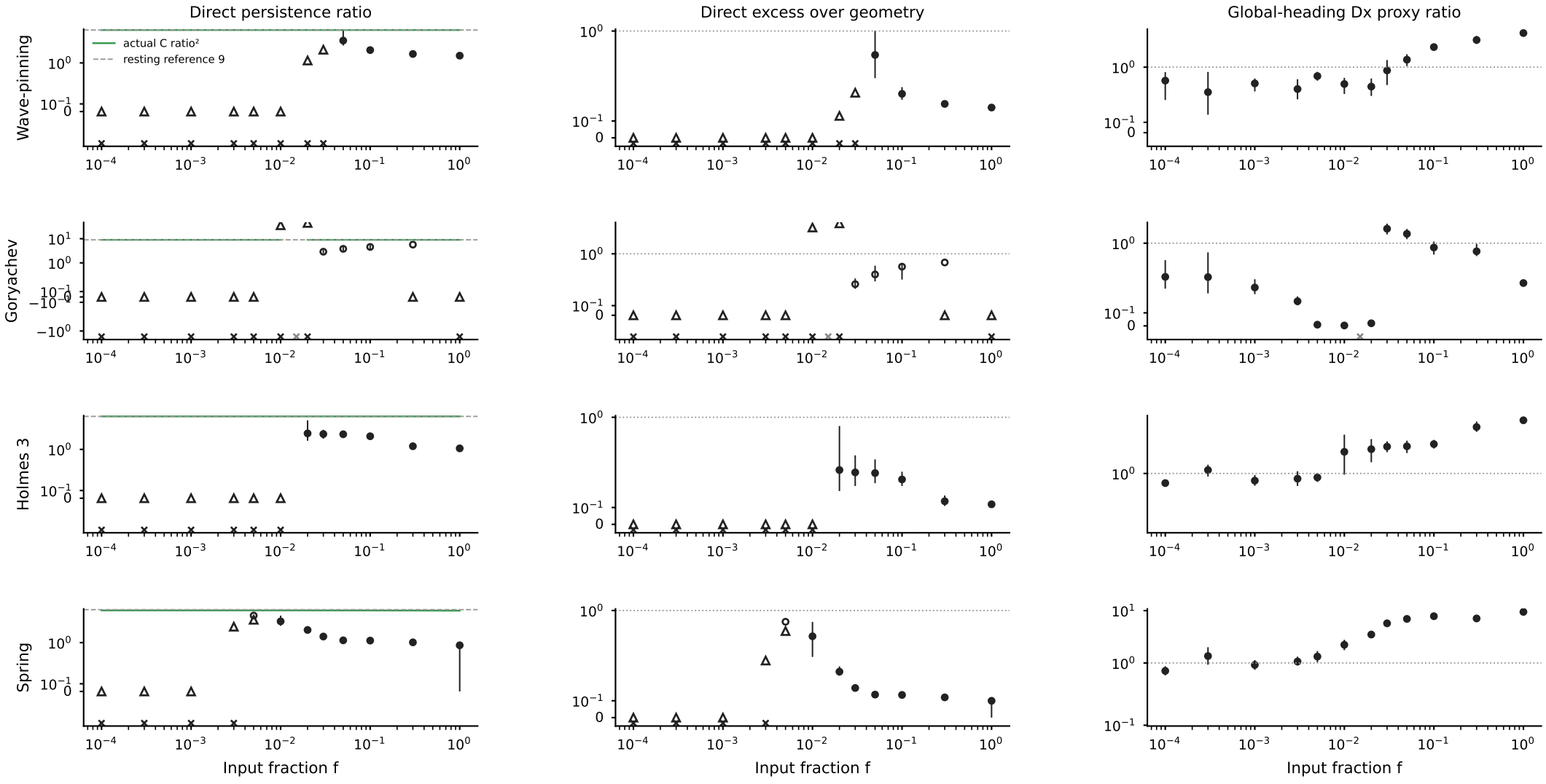
Separate geometric qualification. Columns keep direct time, direct excess-persistence CI and short-lag global-heading Dx-proxy CI separate. Dx converts global activity-heading fluctuations using actual C; in multi-region states it is not one tracked site's diffusion. Qualified single-region literal A3 heading/peak MSDs remain separate. Green is actual C-ratio²; grey dashed9 is the resting reference. Filled points have finite qualified inference; open circles retain raw points without it. Open triangles retain a finite confidence endpoint when the other is unbounded; × is unidentified, not zero. Symlog preserves signed raw estimates. CI including1 means compatibility, not equivalence. Increased Dx, persistence advantages and conflicts remain visible; this descriptor does not alter H1–H3. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S4_S5_final_geometry_20261020_019_S4_S5_geometry_common_inclusive_late_quarter.csv

S4 · Separate geometric qualification · Direct time and short-lag mobility · common · cadence x2 · all conditions including operationally forced



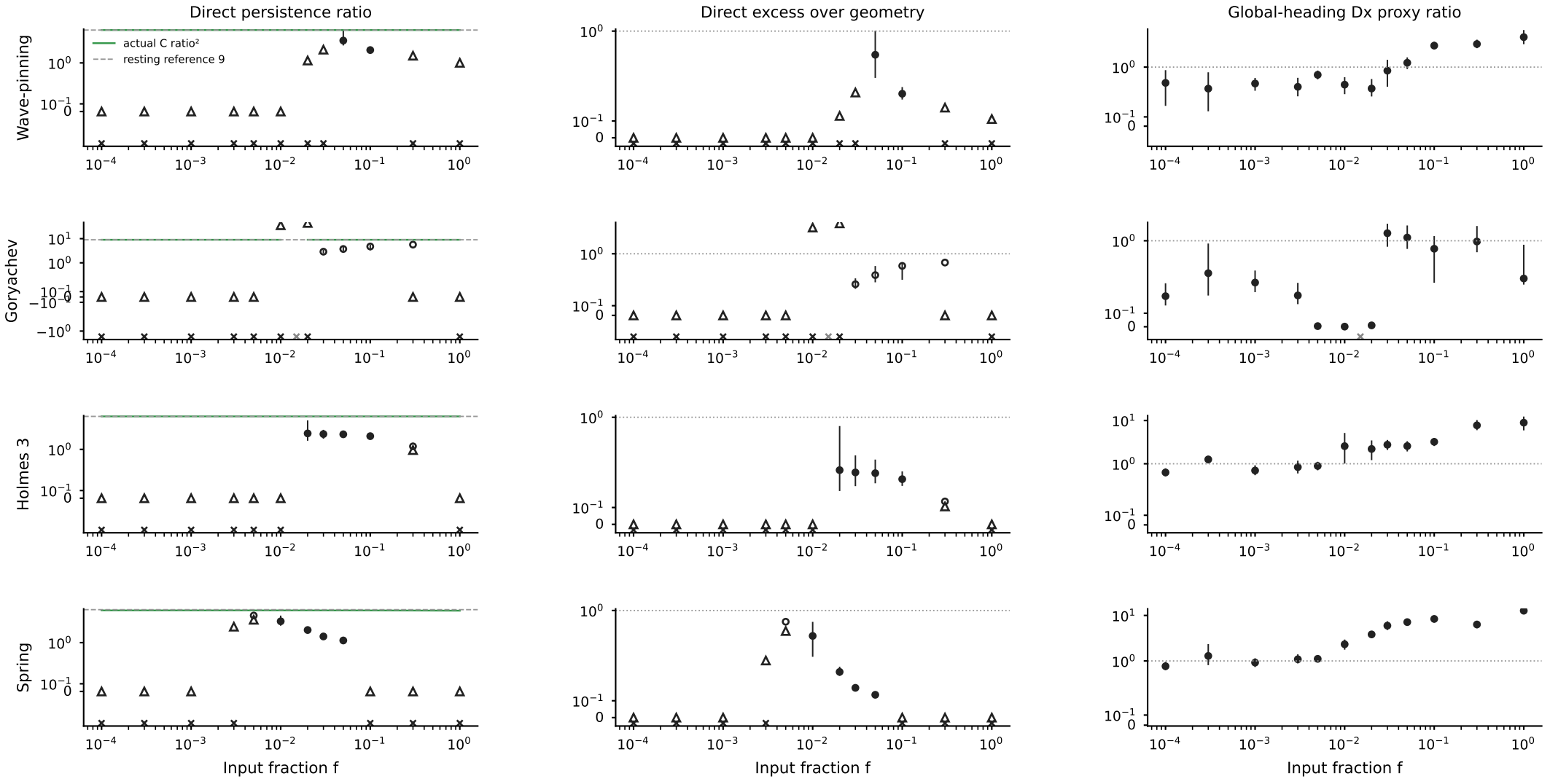
Separate geometric qualification. Columns keep direct time, direct excess-persistence CI and short-lag global-heading Dx-proxy CI separate. Dx converts global activity-heading fluctuations using actual C; in multi-region states it is not one tracked site's diffusion. Qualified single-region literal A3 heading/peak MSDs remain separate. Green is actual C-ratio²; grey dashed9 is the resting reference. Filled points have finite qualified inference; open circles retain raw points without it. Open triangles retain a finite confidence endpoint when the other is unbounded; × is unidentified, not zero. Symlog preserves signed raw estimates. CI including1 means compatibility, not equivalence. Increased Dx, persistence advantages and conflicts remain visible; this descriptor does not alter H1–H3. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S4_S5_final_geometry_20261020_020_S4_S5_geometry_common_inclusive_cadence_x2.csv

S4 · Separate geometric qualification · Direct time and short-lag mobility · common · cadence x4 · all conditions including operationally forced



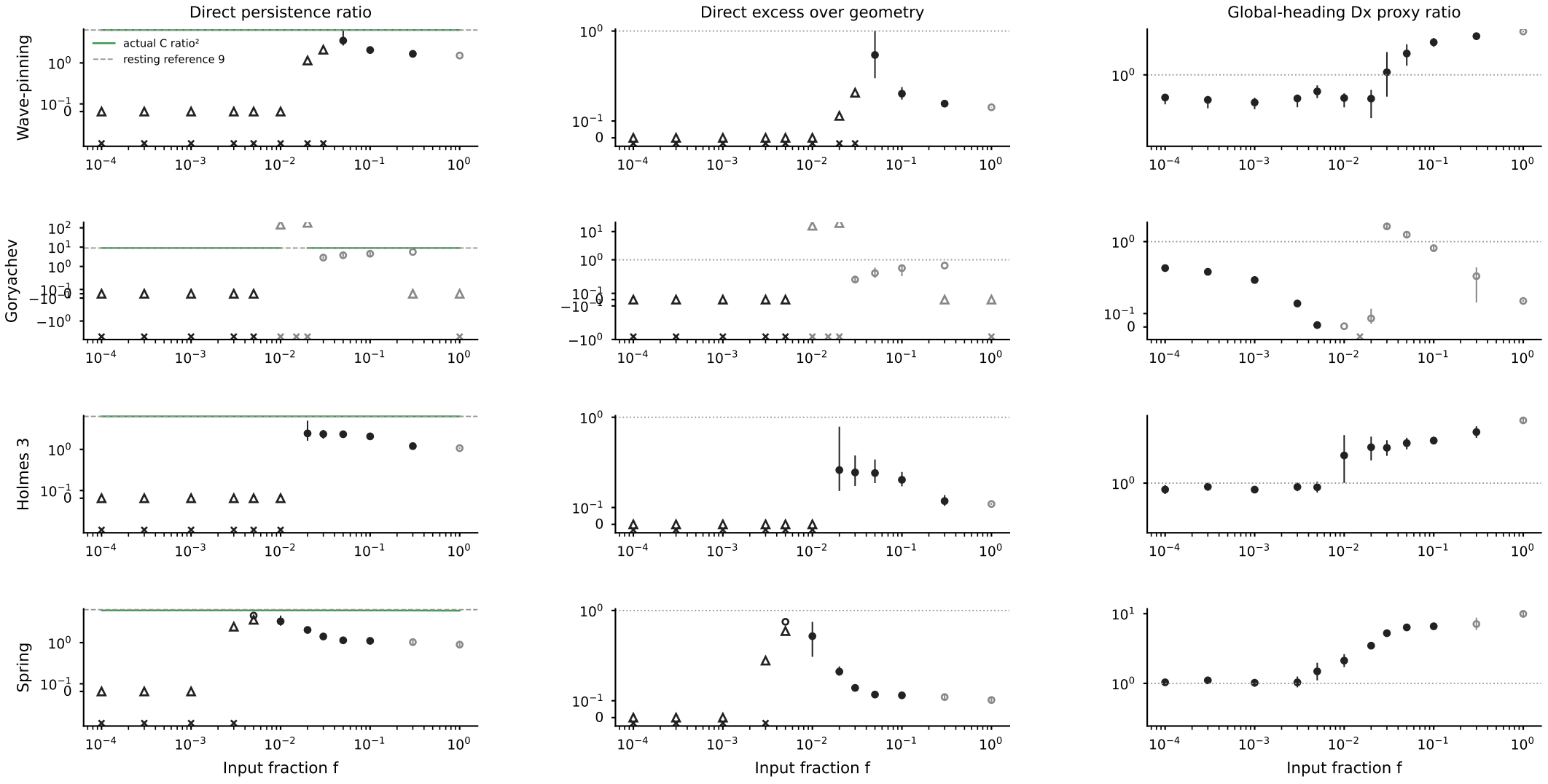
Separate geometric qualification. Columns keep direct time, direct excess-persistence CI and short-lag global-heading Dx-proxy CI separate. Dx converts global activity-heading fluctuations using actual C; in multi-region states it is not one tracked site's diffusion. Qualified single-region literal A3 heading/peak MSDs remain separate. Green is actual C-ratio²; grey dashed9 is the resting reference. Filled points have finite qualified inference; open circles retain raw points without it. Open triangles retain a finite confidence endpoint when the other is unbounded; × is unidentified, not zero. Symlog preserves signed raw estimates. CI including1 means compatibility, not equivalence. Increased Dx, persistence advantages and conflicts remain visible; this descriptor does not alter H1–H3. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S4_S5_final_geometry_20261020_021_S4_S5_geometry_common_inclusive_cadence_x4.csv

S4 · Separate geometric qualification · Direct time and short-lag mobility · common · cadence x8 · all conditions including operationally forced



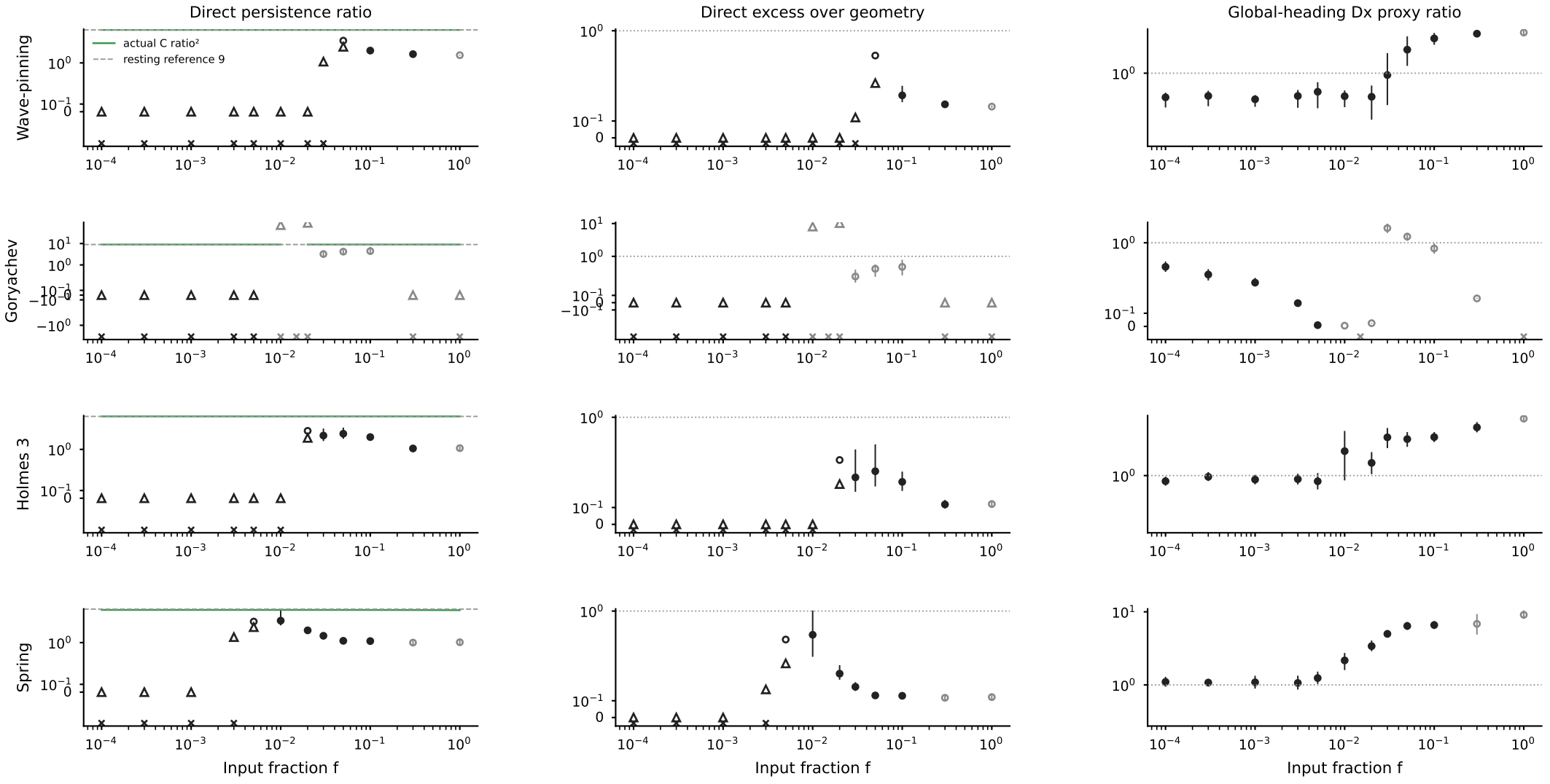
Separate geometric qualification. Columns keep direct time, direct excess-persistence CI and short-lag global-heading Dx-proxy CI separate. Dx converts global activity-heading fluctuations using actual C; in multi-region states it is not one tracked site's diffusion. Qualified single-region literal A3 heading/peak MSDs remain separate. Green is actual C-ratio²; grey dashed9 is the resting reference. Filled points have finite qualified inference; open circles retain raw points without it. Open triangles retain a finite confidence endpoint when the other is unbounded; × is unidentified, not zero. Symlog preserves signed raw estimates. CI including1 means compatibility, not equivalence. Increased Dx, persistence advantages and conflicts remain visible; this descriptor does not alter H1–H3. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S4_S5_final_geometry_20261020_022_S4_S5_geometry_common_inclusive_cadence_x8.csv

S4 · Separate geometric qualification · Direct time and short-lag mobility · longest · main · operationally unforced conditions



Separate geometric qualification. Columns keep direct time, direct excess-persistence CI and short-lag global-heading Dx-proxy CI separate. Dx converts global activity-heading fluctuations using actual C; in multi-region states it is not one tracked site's diffusion. Qualified single-region literal A3 heading/peak MSDs remain separate. Green is actual C-ratio²; grey dashed9 is the resting reference. Filled points have finite qualified inference; open circles retain raw points without it. Open triangles retain a finite confidence endpoint when the other is unbounded; × is unidentified, not zero. Symlog preserves signed raw estimates. CI including1 means compatibility, not equivalence. Increased Dx, persistence advantages and conflicts remain visible; this descriptor does not alter H1–H3. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S4_S5_final_geometry_20261020_023_S4_S5_geometry_longest_primary_main.csv

S4 · Separate geometric qualification · Direct time and short-lag mobility · longest · analysis first half · operationally unforced conditions



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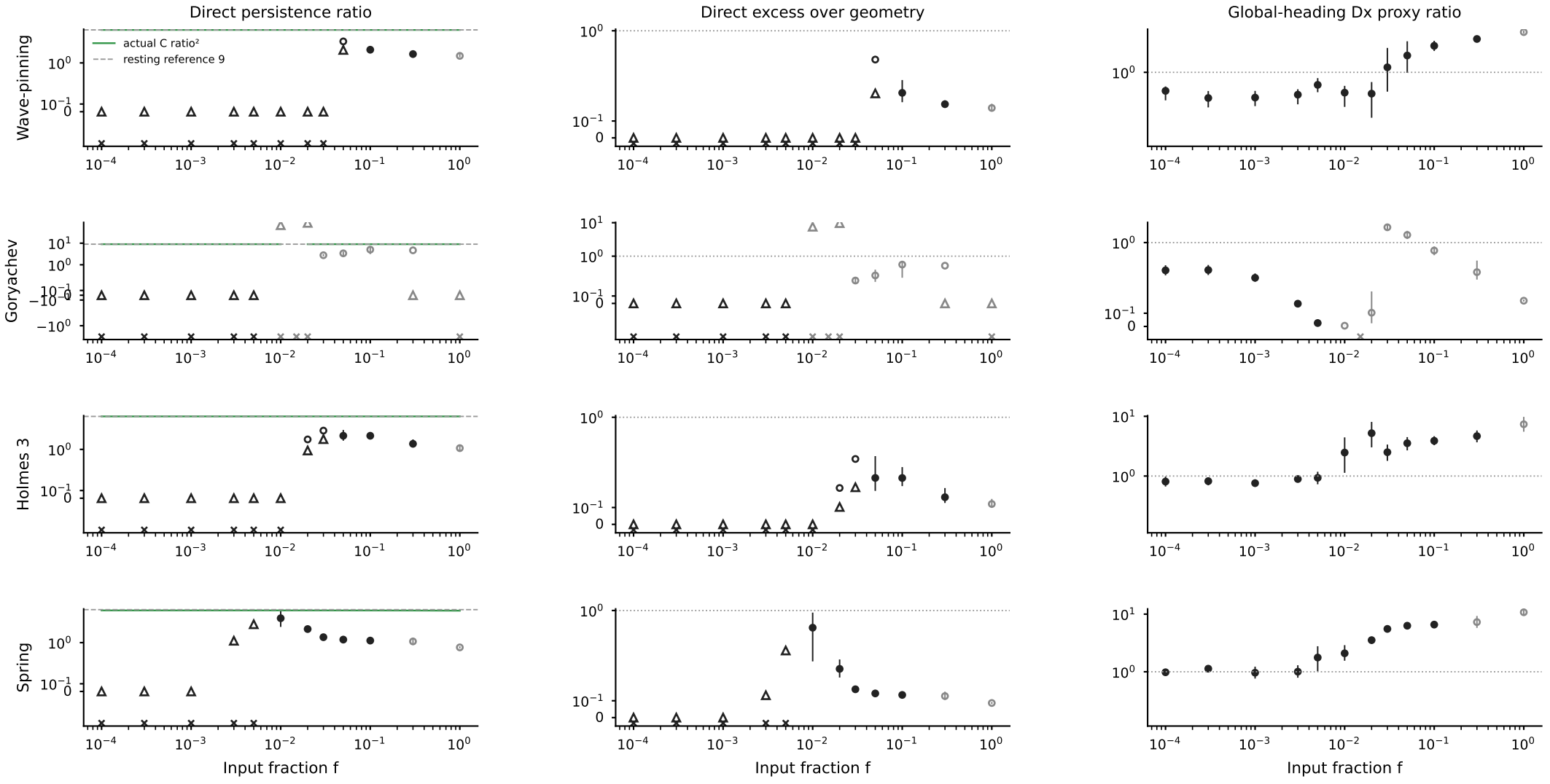
CSV: figure4_S0_S4_S5_final_geometry_20261020_024_S4_S5_geometry_longest_primary_analysis_first_half.csv

S4 · Separate geometric qualification · Direct time and short-lag mobility · longest · analysis second half · operationally unforced conditions



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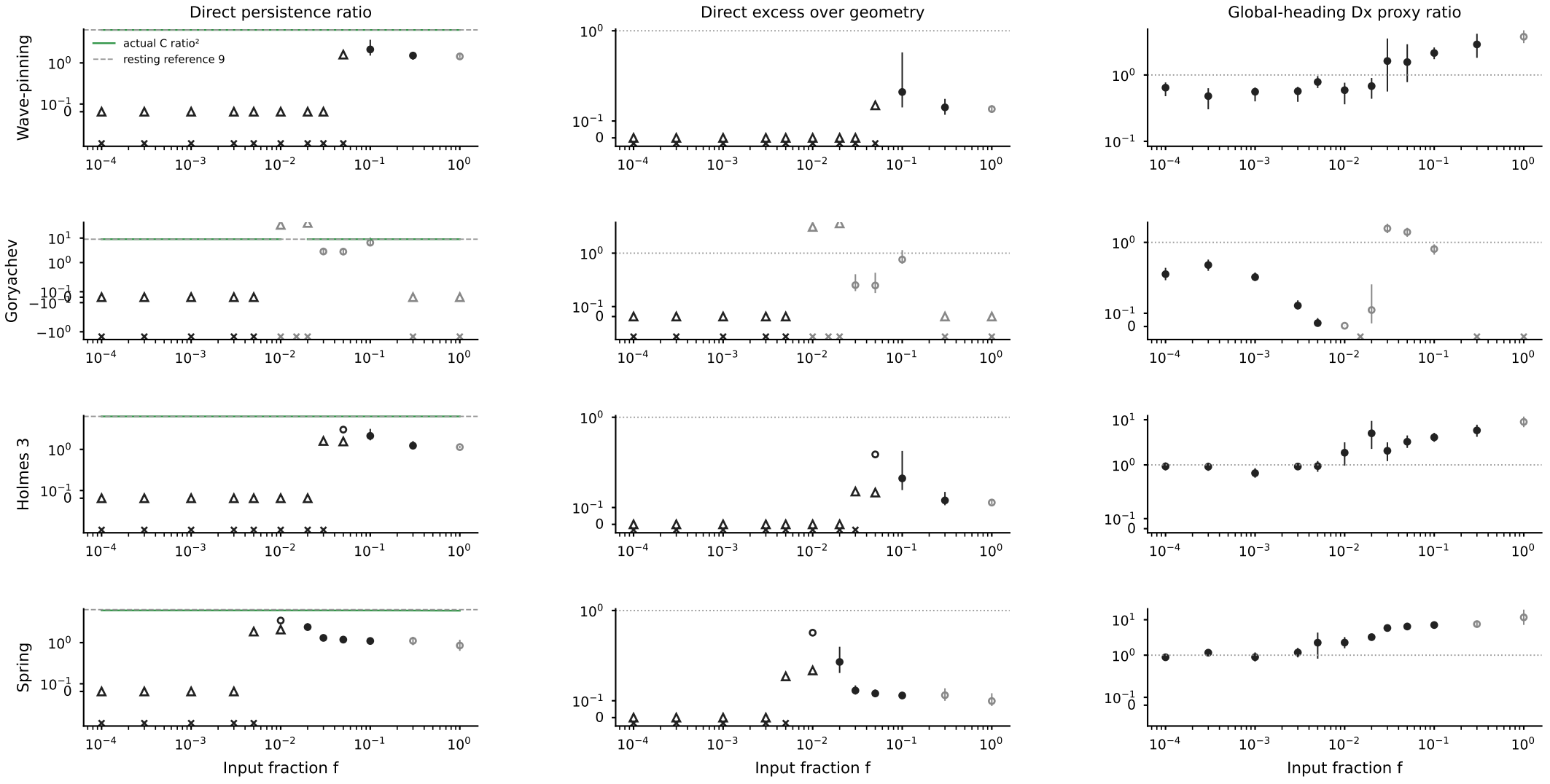
CSV: figure4_S0_S4_S5_final_geometry_20261020_025_S4_S5_geometry_longest_primary_analysis_second_half.csv

S4 · Separate geometric qualification · Direct time and short-lag mobility · longest · late quarter · operationally unforced conditions



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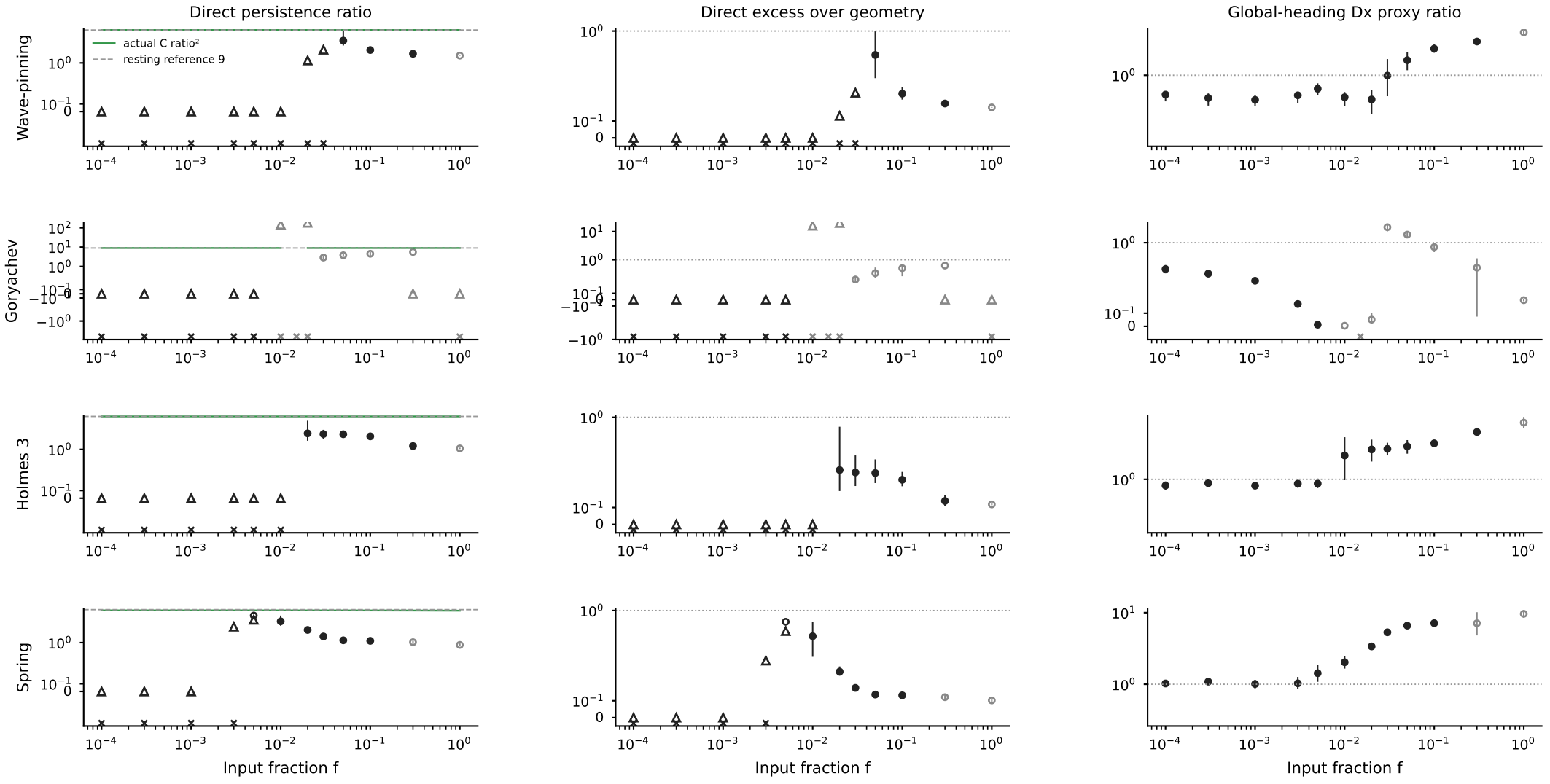
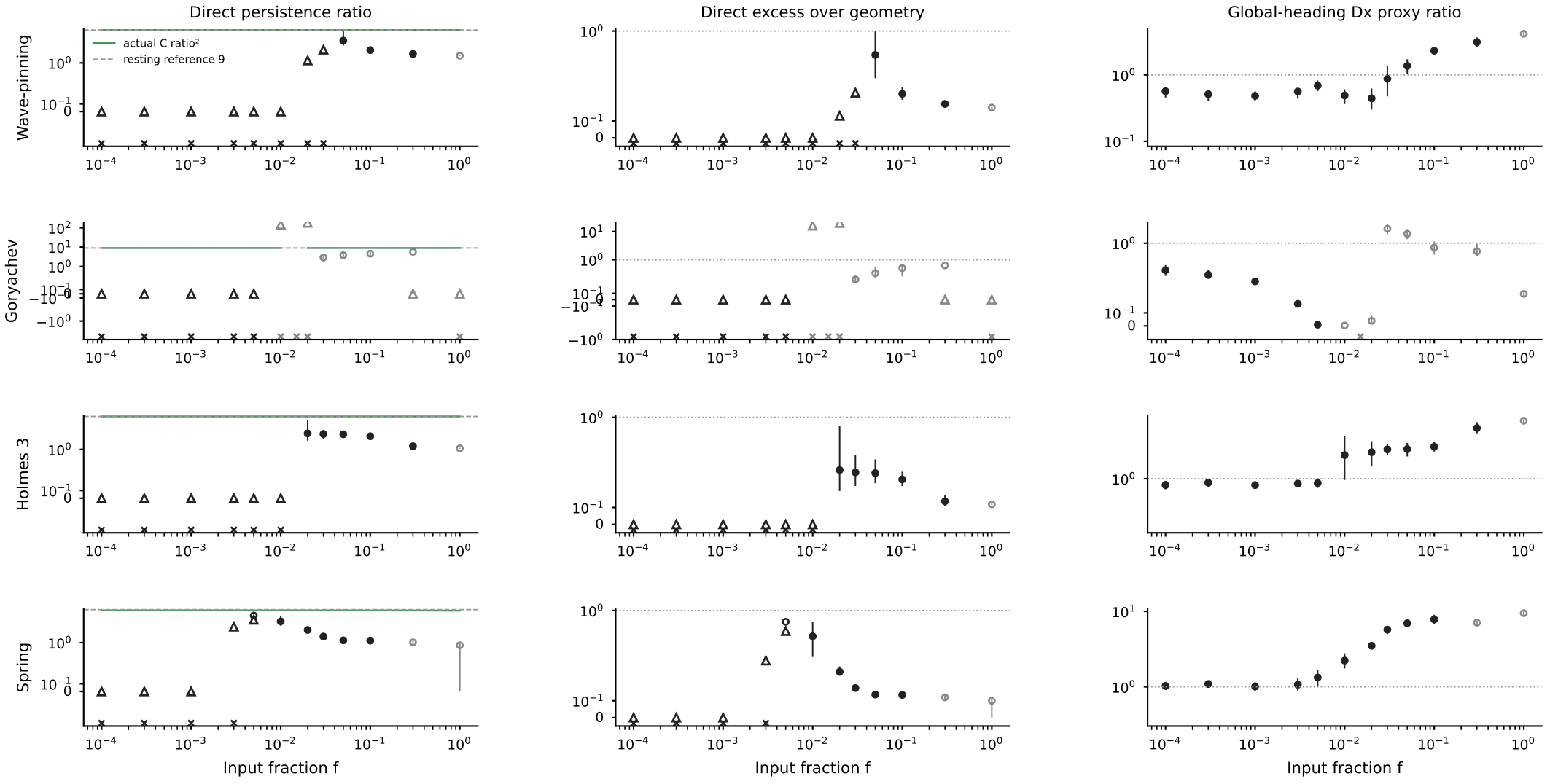
CSV: figure4_S0_S4_S5_final_geometry_20261020_026_S4_S5_geometry_longest_primary_late_quarter.csv

S4 · Separate geometric qualification · Direct time and short-lag mobility · longest · cadence x2 · operationally unforced conditions



Separate geometric qualification. Columns keep direct time, direct excess-persistence CI and short-lag global-heading Dx-proxy CI separate. Dx converts global activity-heading fluctuations using actual C; in multi-region states it is not one tracked site's diffusion. Qualified single-region literal A3 heading/peak MSDs remain separate. Green is actual C-ratio²; grey dashed9 is the resting reference. Filled points have finite qualified inference; open circles retain raw points without it. Open triangles retain a finite confidence endpoint when the other is unbounded; × is unidentified, not zero. Symlog preserves signed raw estimates. CI including1 means compatibility, not equivalence. Increased Dx, persistence advantages and conflicts remain visible; this descriptor does not alter H1–H3. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S4_S5_final_geometry_20261020_027_S4_S5_geometry_longest_primary_cadence_x2.csv

S4 · Separate geometric qualification · Direct time and short-lag mobility · longest · cadence x4 · operationally unforced conditions



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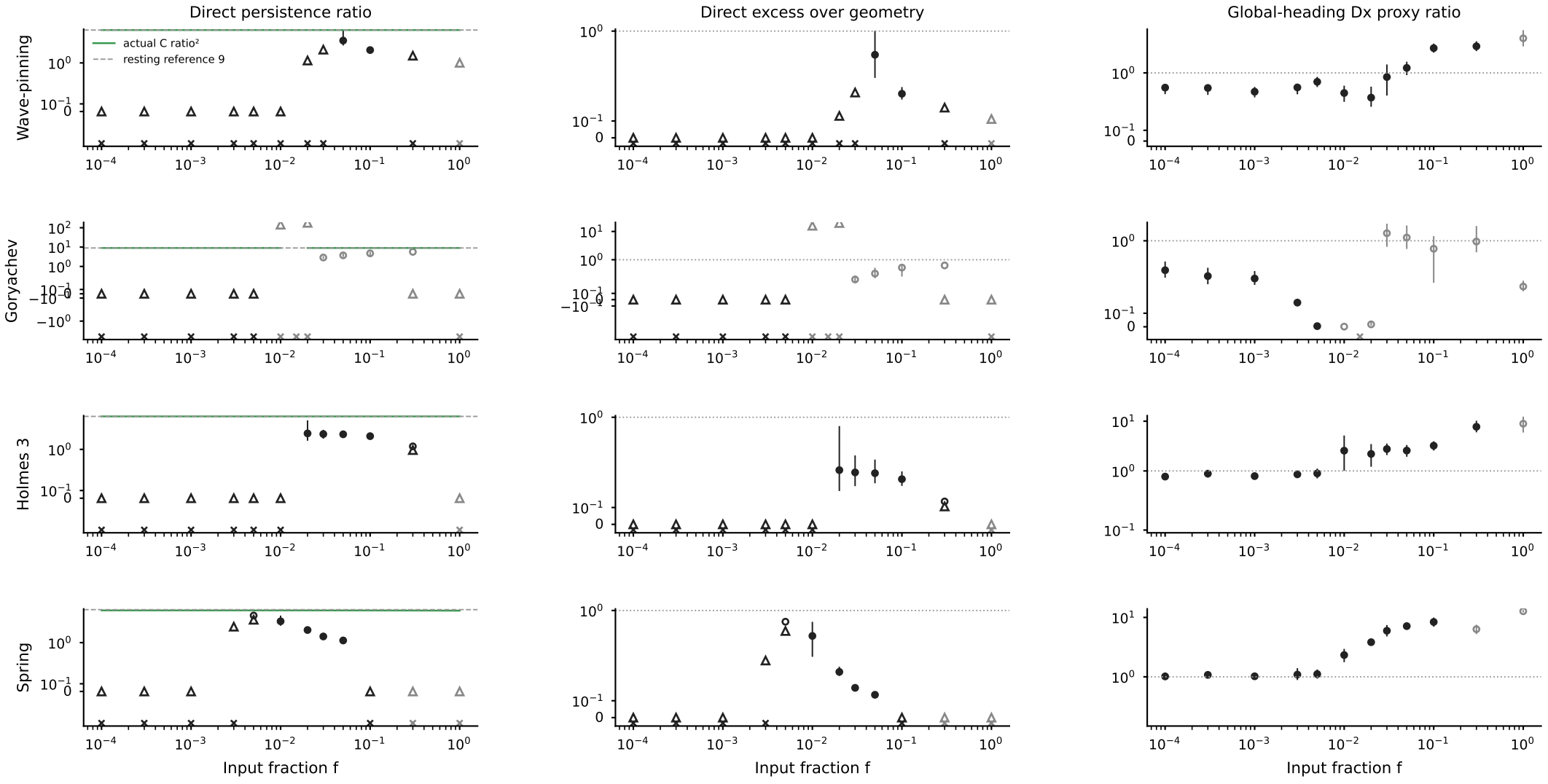
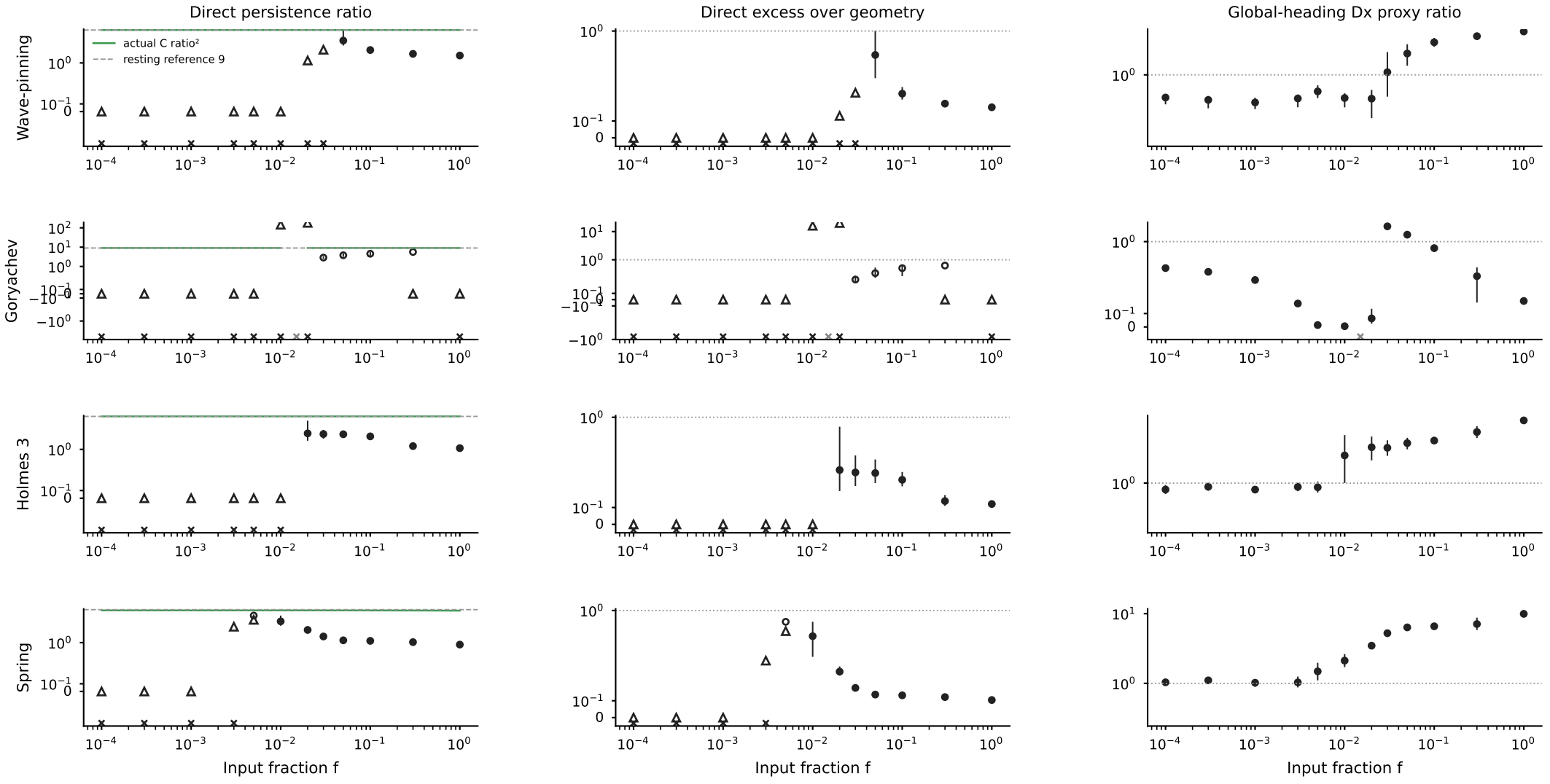
CSV: figure4_S0_S4_S5_final_geometry_20261020_028_S4_S5_geometry_longest_primary_cadence_x4.csv

S4 · Separate geometric qualification · Direct time and short-lag mobility · longest · cadence x8 · operationally unforced conditions



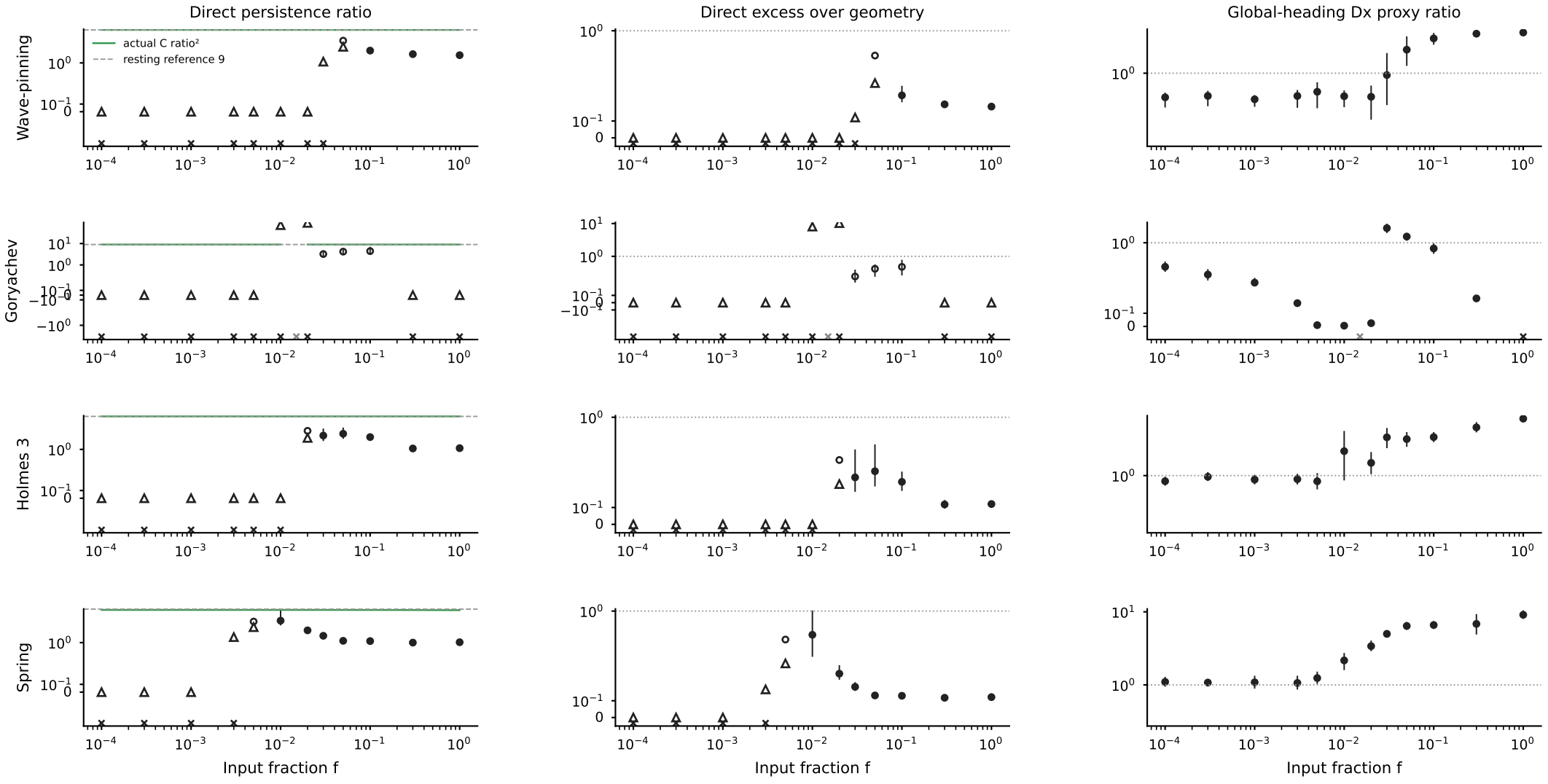
Separate geometric qualification. Columns keep direct time, direct excess-persistence CI and short-lag global-heading Dx-proxy CI separate. Dx converts global activity-heading fluctuations using actual C; in multi-region states it is not one tracked site's diffusion. Qualified single-region literal A3 heading/peak MSDs remain separate. Green is actual C-ratio²; grey dashed9 is the resting reference. Filled points have finite qualified inference; open circles retain raw points without it. Open triangles retain a finite confidence endpoint when the other is unbounded; × is unidentified, not zero. Symlog preserves signed raw estimates. CI including1 means compatibility, not equivalence. Increased Dx, persistence advantages and conflicts remain visible; this descriptor does not alter H1–H3. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S4_S5_final_geometry_20261020_029_S4_S5_geometry_longest_primary_cadence_x8.csv

S4 · Separate geometric qualification · Direct time and short-lag mobility · longest · main · all conditions including operationally forced



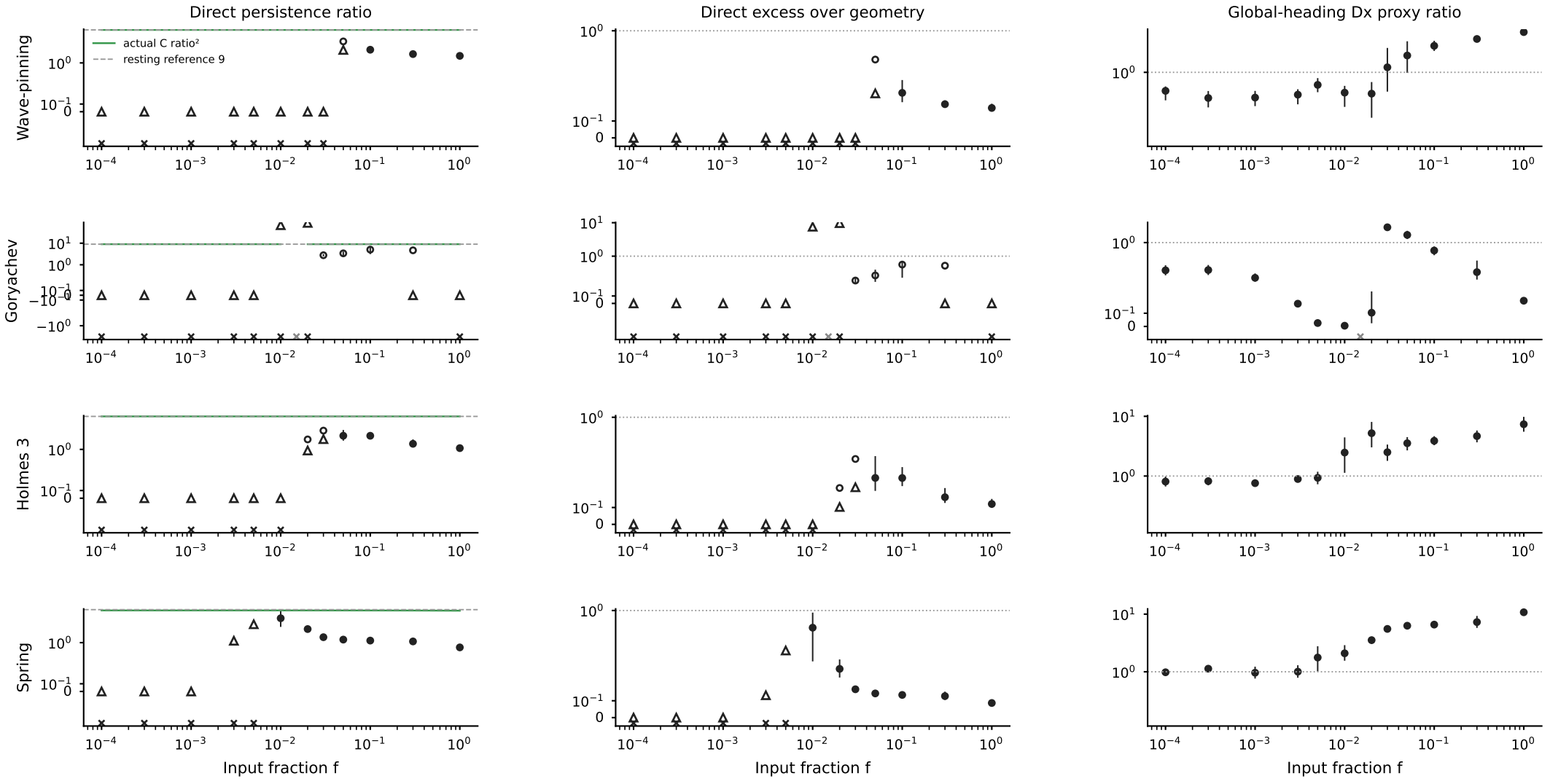
Separate geometric qualification. Columns keep direct time, direct excess-persistence CI and short-lag global-heading Dx-proxy CI separate. Dx converts global activity-heading fluctuations using actual C; in multi-region states it is not one tracked site's diffusion. Qualified single-region literal A3 heading/peak MSDs remain separate. Green is actual C-ratio²; grey dashed9 is the resting reference. Filled points have finite qualified inference; open circles retain raw points without it. Open triangles retain a finite confidence endpoint when the other is unbounded; × is unidentified, not zero. Symlog preserves signed raw estimates. CI including1 means compatibility, not equivalence. Increased Dx, persistence advantages and conflicts remain visible; this descriptor does not alter H1–H3. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S4_S5_final_geometry_20261020_030_S4_S5_geometry_longest_inclusive_main.csv

S4 · Separate geometric qualification · Direct time and short-lag mobility · longest · analysis first half · all conditions including operationally forced



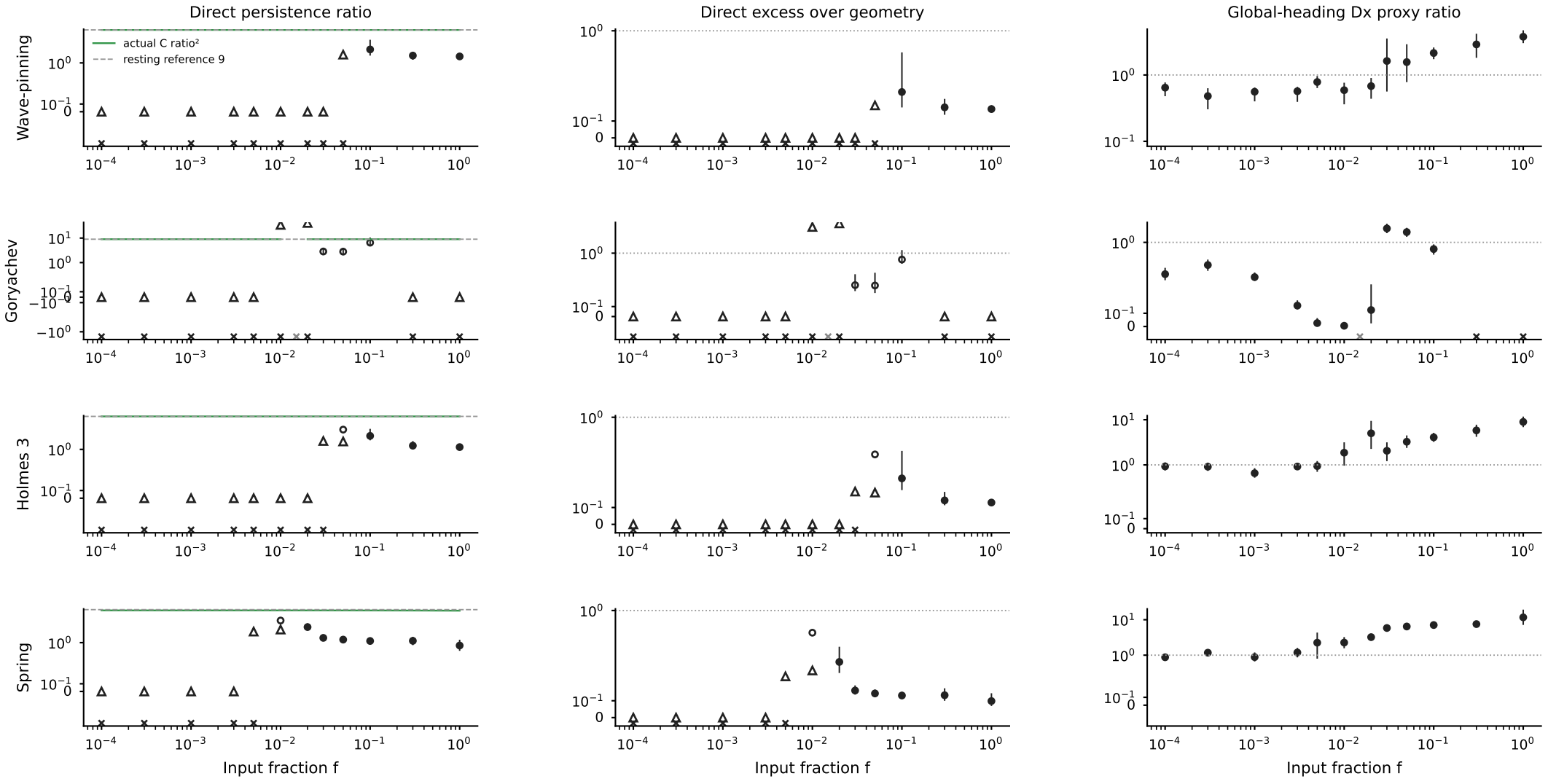
Separate geometric qualification. Columns keep direct time, direct excess-persistence CI and short-lag global-heading Dx-proxy CI separate. Dx converts global activity-heading fluctuations using actual C; in multi-region states it is not one tracked site's diffusion. Qualified single-region literal A3 heading/peak MSDs remain separate. Green is actual C-ratio²; grey dashed9 is the resting reference. Filled points have finite qualified inference; open circles retain raw points without it. Open triangles retain a finite confidence endpoint when the other is unbounded; × is unidentified, not zero. Symlog preserves signed raw estimates. CI including1 means compatibility, not equivalence. Increased Dx, persistence advantages and conflicts remain visible; this descriptor does not alter H1–H3. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S4_S5_final_geometry_20261020_031_S4_S5_geometry_longest_inclusive_analysis_first_half.csv

S4 · Separate geometric qualification · Direct time and short-lag mobility · longest · analysis second half · all conditions including operationally forced



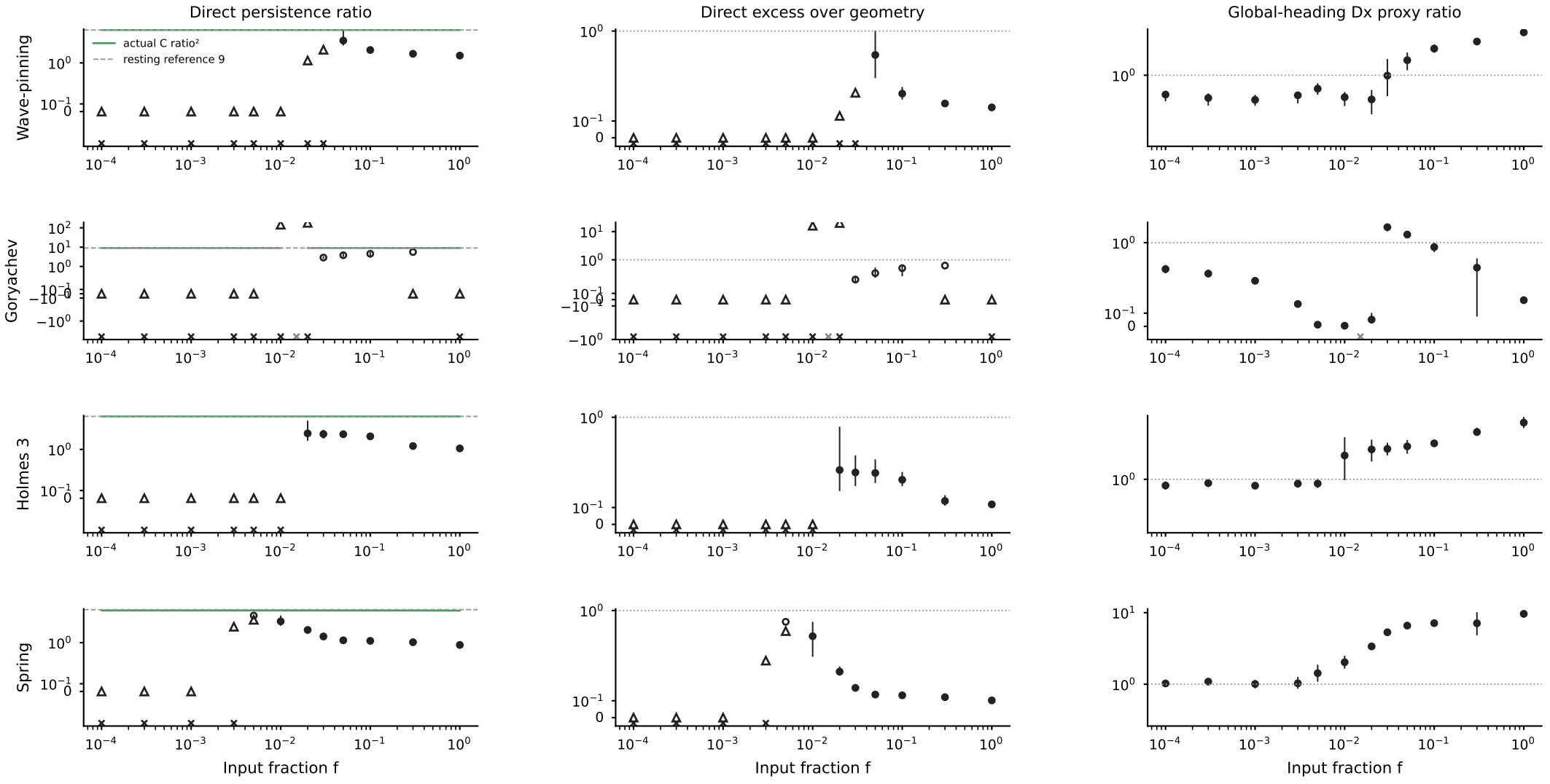
Separate geometric qualification. Columns keep direct time, direct excess-persistence CI and short-lag global-heading Dx-proxy CI separate. Dx converts global activity-heading fluctuations using actual C; in multi-region states it is not one tracked site's diffusion. Qualified single-region literal A3 heading/peak MSDs remain separate. Green is actual C-ratio²; grey dashed9 is the resting reference. Filled points have finite qualified inference; open circles retain raw points without it. Open triangles retain a finite confidence endpoint when the other is unbounded; × is unidentified, not zero. Symlog preserves signed raw estimates. CI including1 means compatibility, not equivalence. Increased Dx, persistence advantages and conflicts remain visible; this descriptor does not alter H1–H3. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S4_S5_final_geometry_20261020_032_S4_S5_geometry_longest_inclusive_analysis_second_half.csv

S4 · Separate geometric qualification · Direct time and short-lag mobility · longest · late quarter · all conditions including operationally forced



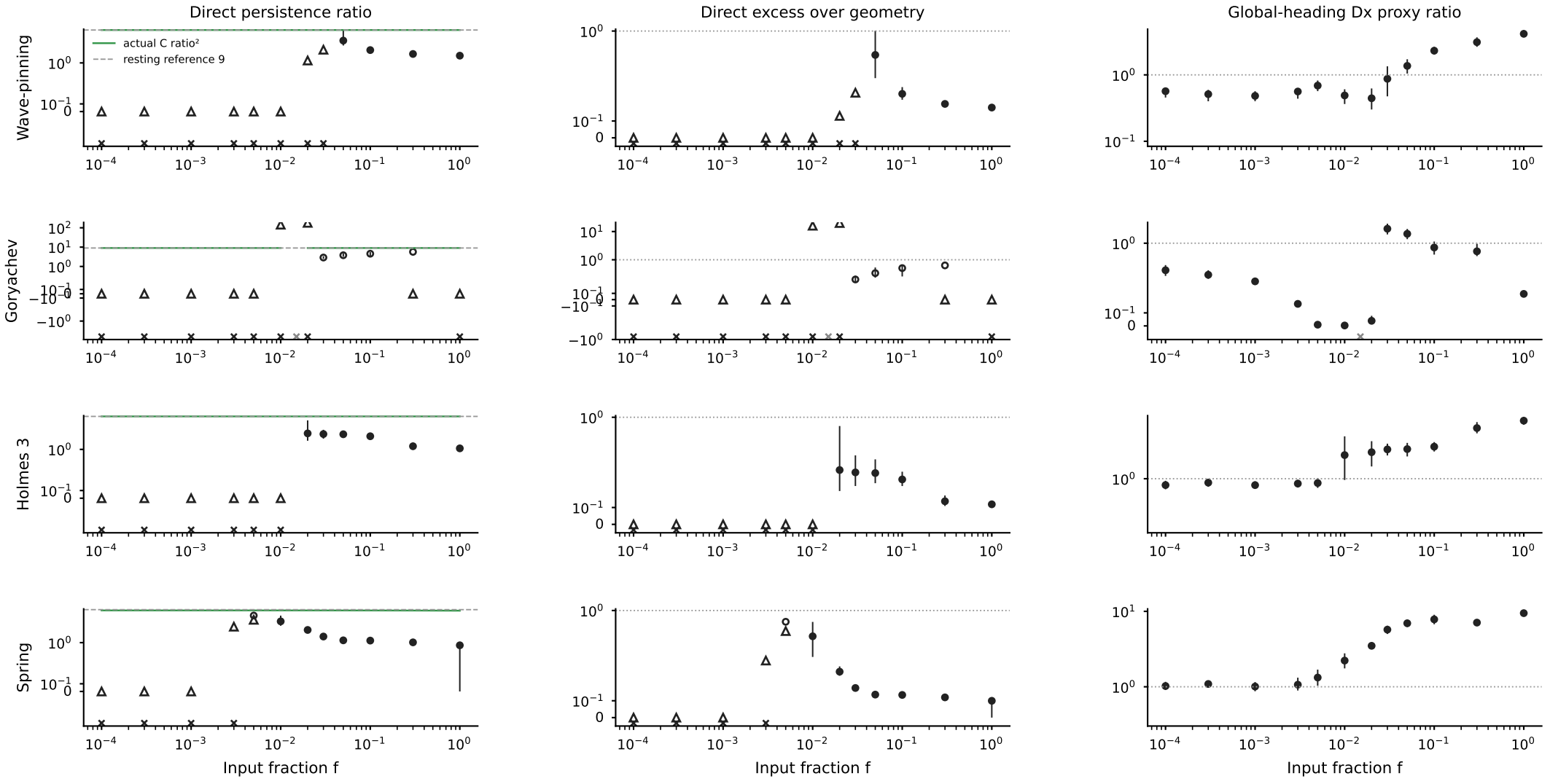
Separate geometric qualification. Columns keep direct time, direct excess-persistence CI and short-lag global-heading Dx-proxy CI separate. Dx converts global activity-heading fluctuations using actual C; in multi-region states it is not one tracked site's diffusion. Qualified single-region literal A3 heading/peak MSDs remain separate. Green is actual C-ratio²; grey dashed9 is the resting reference. Filled points have finite qualified inference; open circles retain raw points without it. Open triangles retain a finite confidence endpoint when the other is unbounded; × is unidentified, not zero. Symlog preserves signed raw estimates. CI including1 means compatibility, not equivalence. Increased Dx, persistence advantages and conflicts remain visible; this descriptor does not alter H1–H3. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S4_S5_final_geometry_20261020_033_S4_S5_geometry_longest_inclusive_late_quarter.csv

S4 · Separate geometric qualification · Direct time and short-lag mobility · longest · cadence x2 · all conditions including operationally forced



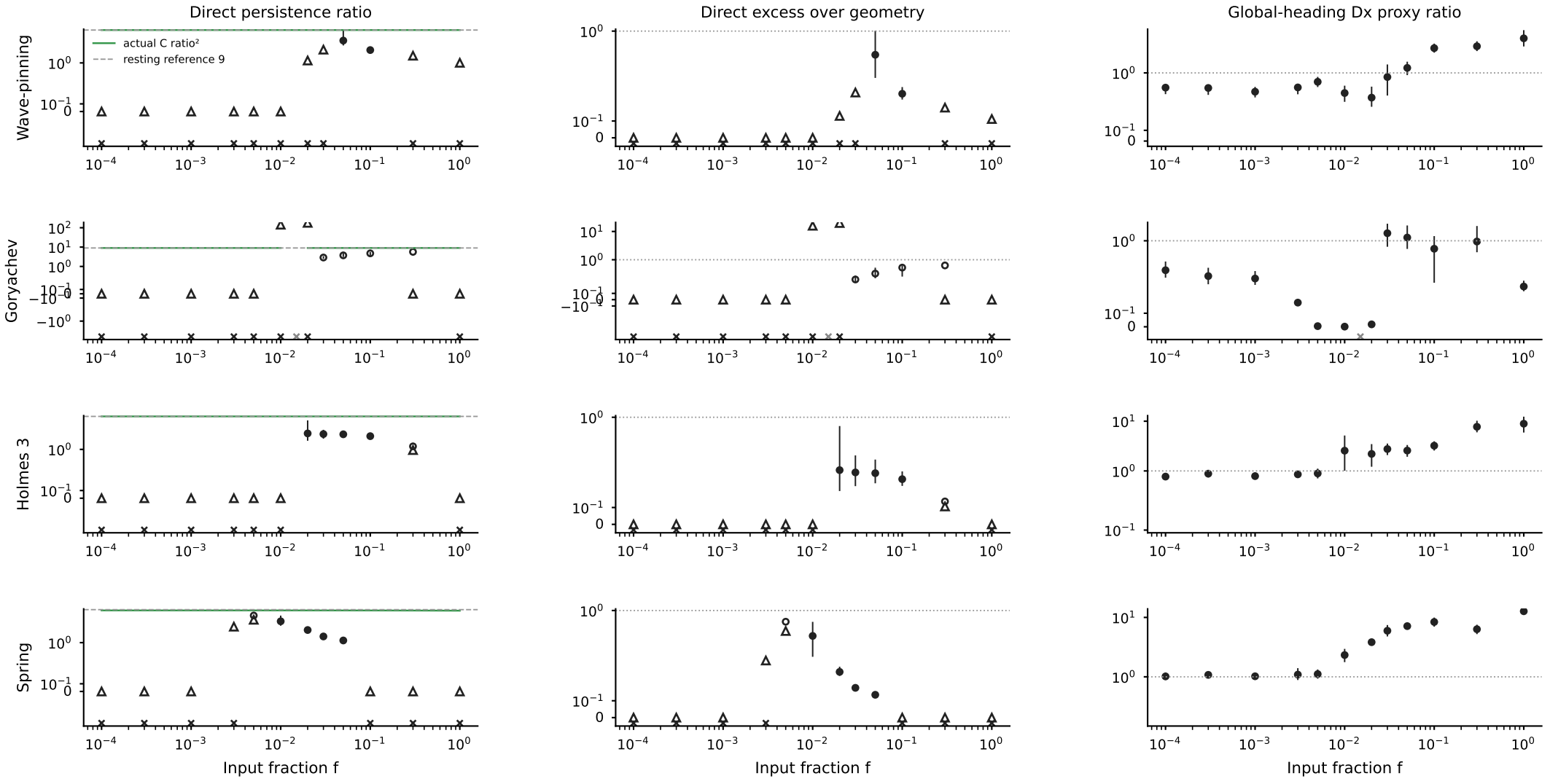
Separate geometric qualification. Columns keep direct time, direct excess-persistence CI and short-lag global-heading Dx-proxy CI separate. Dx converts global activity-heading fluctuations using actual C; in multi-region states it is not one tracked site's diffusion. Qualified single-region literal A3 heading/peak MSDs remain separate. Green is actual C-ratio²; grey dashed9 is the resting reference. Filled points have finite qualified inference; open circles retain raw points without it. Open triangles retain a finite confidence endpoint when the other is unbounded; × is unidentified, not zero. Symlog preserves signed raw estimates. CI including1 means compatibility, not equivalence. Increased Dx, persistence advantages and conflicts remain visible; this descriptor does not alter H1–H3. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S4_S5_final_geometry_20261020_034_S4_S5_geometry_longest_inclusive_cadence_x2.csv

S4 · Separate geometric qualification · Direct time and short-lag mobility · longest · cadence x4 · all conditions including operationally forced



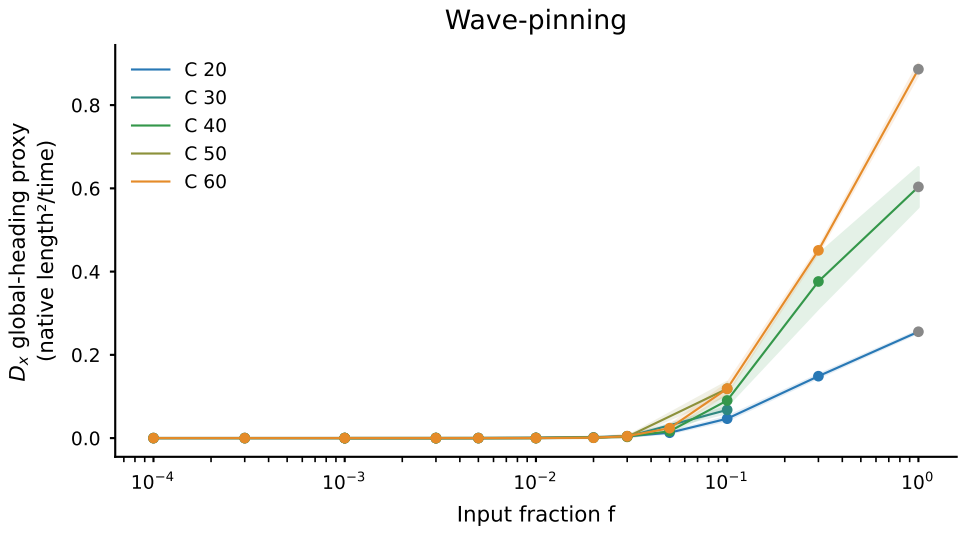
Separate geometric qualification. Columns keep direct time, direct excess-persistence CI and short-lag global-heading Dx-proxy CI separate. Dx converts global activity-heading fluctuations using actual C; in multi-region states it is not one tracked site's diffusion. Qualified single-region literal A3 heading/peak MSDs remain separate. Green is actual C-ratio²; grey dashed9 is the resting reference. Filled points have finite qualified inference; open circles retain raw points without it. Open triangles retain a finite confidence endpoint when the other is unbounded; × is unidentified, not zero. Symlog preserves signed raw estimates. CI including1 means compatibility, not equivalence. Increased Dx, persistence advantages and conflicts remain visible; this descriptor does not alter H1–H3. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S4_S5_final_geometry_20261020_035_S4_S5_geometry_longest_inclusive_cadence_x4.csv

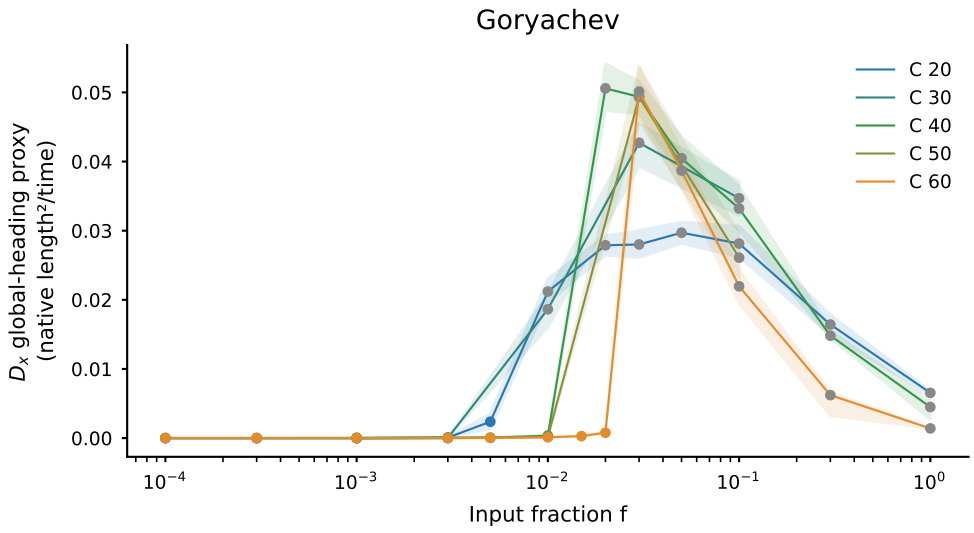
S4 · Separate geometric qualification · Direct time and short-lag mobility · longest · cadence x8 · all conditions including operationally forced

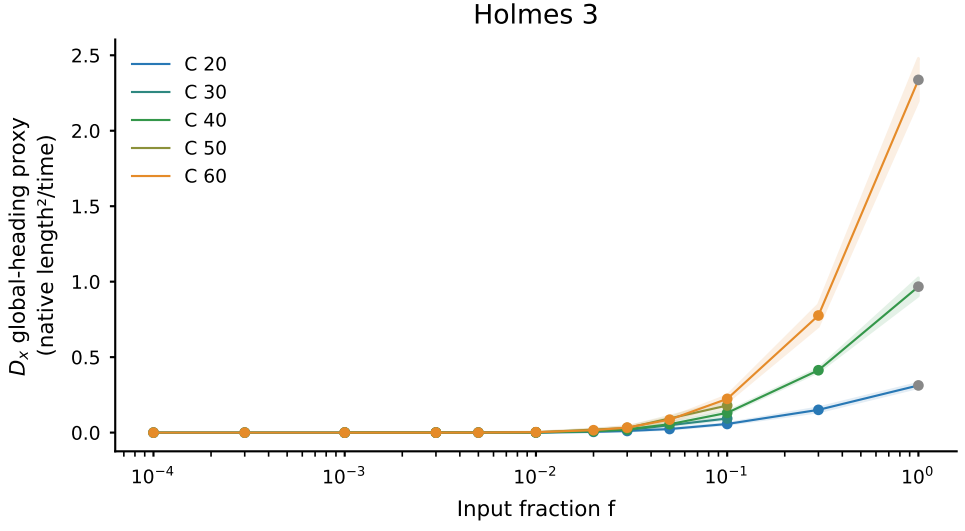


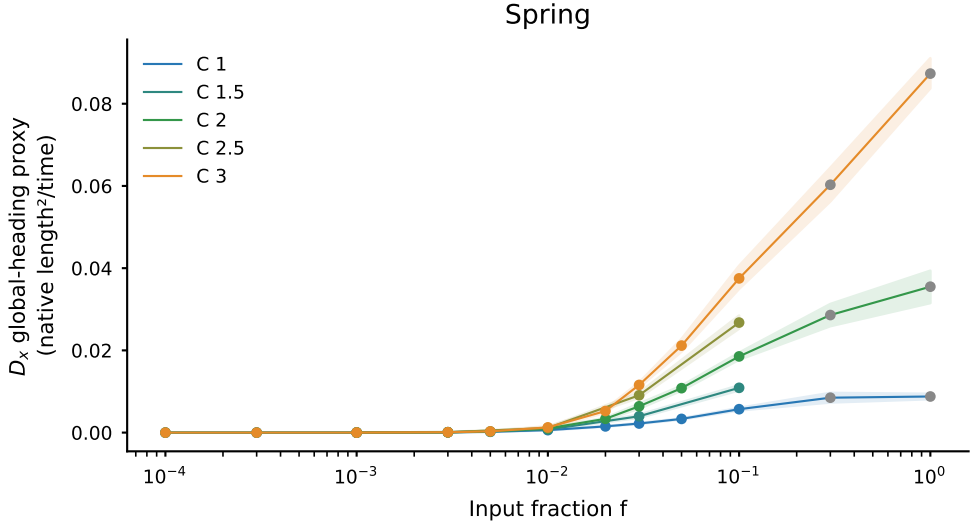
Separate geometric qualification. Columns keep direct time, direct excess-persistence CI and short-lag global-heading Dx-proxy CI separate. Dx converts global activity-heading fluctuations using actual C; in multi-region states it is not one tracked site's diffusion. Qualified single-region literal A3 heading/peak MSDs remain separate. Green is actual C-ratio²; grey dashed9 is the resting reference. Filled points have finite qualified inference; open circles retain raw points without it. Open triangles retain a finite confidence endpoint when the other is unbounded; × is unidentified, not zero. Symlog preserves signed raw estimates. CI including1 means compatibility, not equivalence. Increased Dx, persistence advantages and conflicts remain visible; this descriptor does not alter H1–H3. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S4_S5_final_geometry_20261020_036_S4_S5_geometry_longest_inclusive_cadence_x8.csv

S5 · Common observations · Global-heading mobility proxy



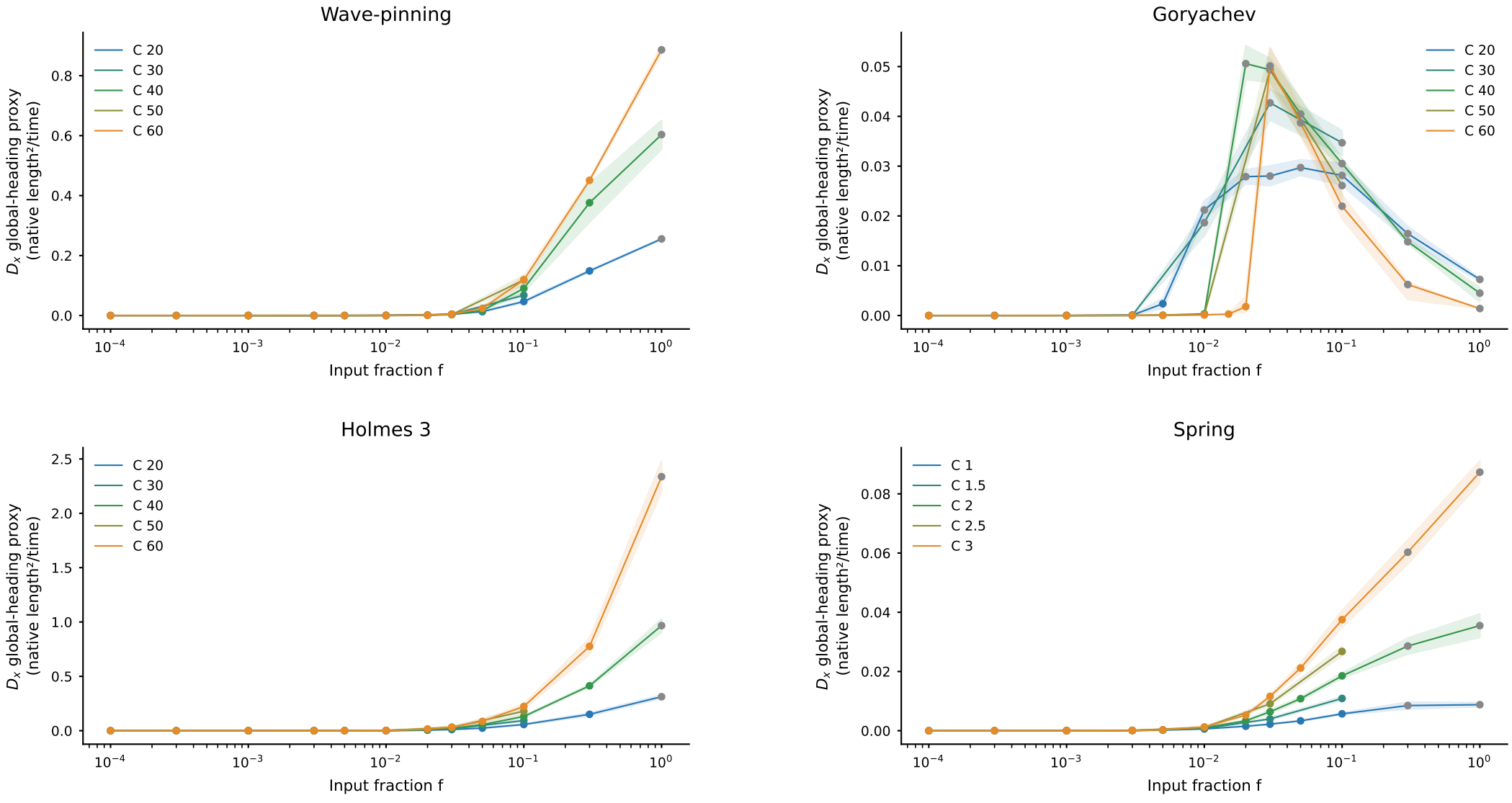






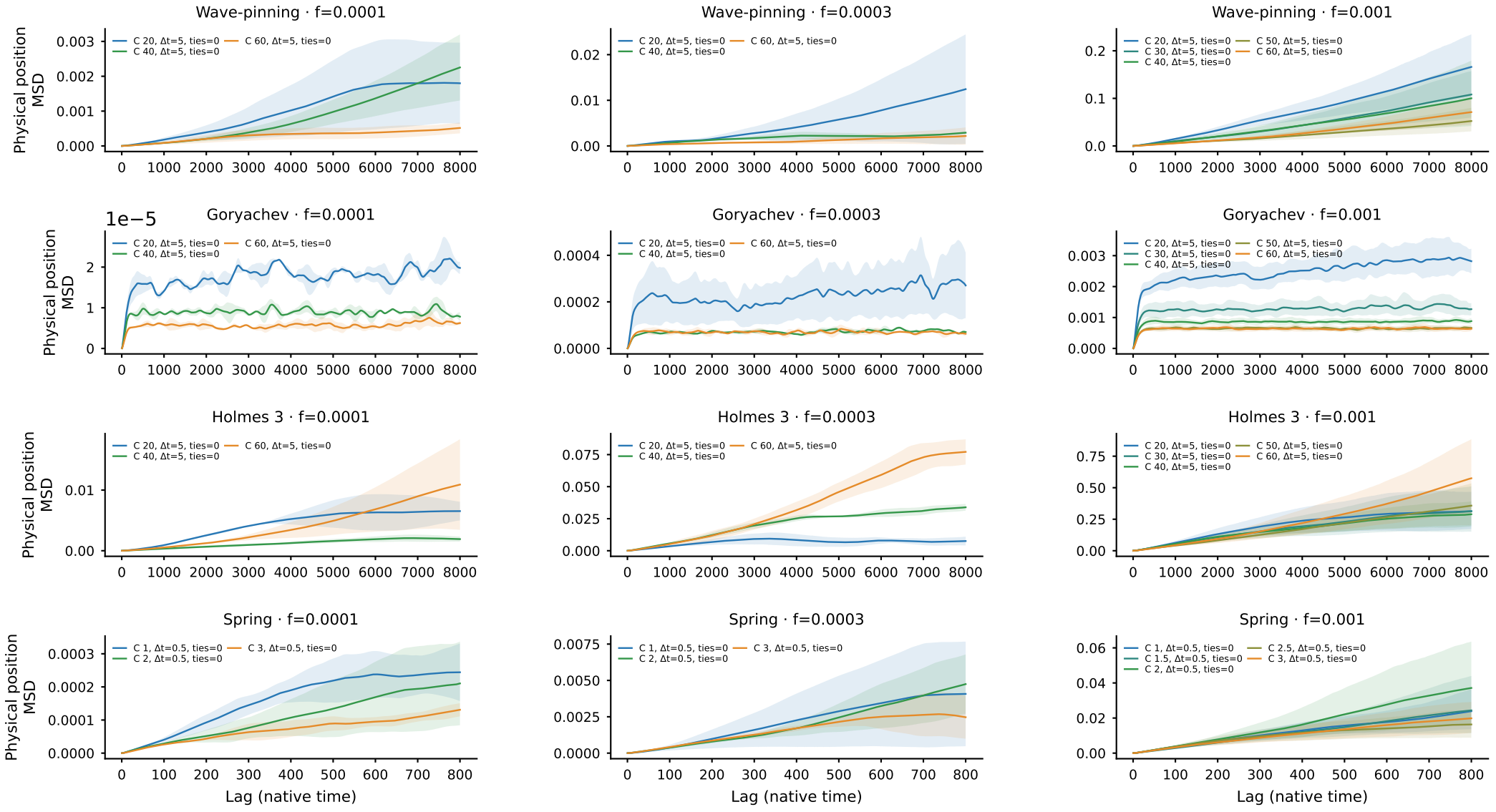
Common observations. Dx = Dθ × (actual mean C / 2π)² converts the signed short-lag global activity-heading estimate using each window's actual mean circumference. In multi-region states this proxy is not diffusion of one tracked site. It is distinct from qualified single-region literal heading/peak coordinate MSD and is not a slope fit to θC(t)/(2π). Increased Dx and angular persistence advantages are both retained when they coexist. Grey retains forced conditions. Native length/time units differ across models. No free-diffusion or migration velocity law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV. CI∞ next to a finite point marks an upper-unbounded bootstrap CI; finite lower endpoints are retained. CI[0,∞) spans the full nonnegative range.
CSV: figure4_common_canonical_measurements_20261020_024_S5_Dx.csv

S5 · Longest qualified observations · Global-heading mobility proxy



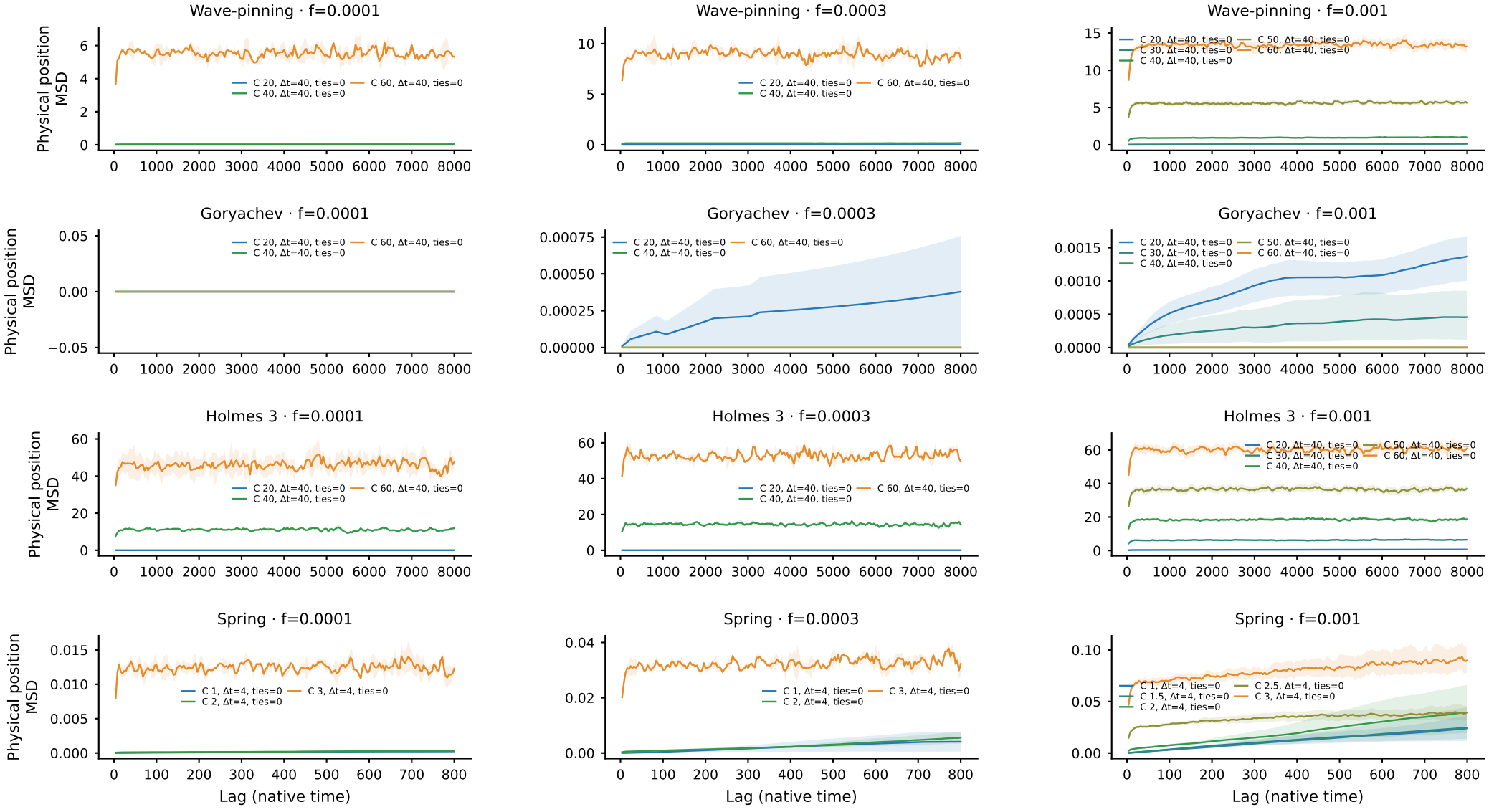
Longest qualified observations. Dx = Dθ × (actual mean C / 2π)² converts the signed short-lag global activity-heading estimate using each window's actual mean circumference. In multi-region states this proxy is not diffusion of one tracked site. It is distinct from qualified single-region literal heading/peak coordinate MSD and is not a slope fit to θC(t)/(2π). Increased Dx and angular persistence advantages are both retained when they coexist. Grey retains forced conditions. Native length/time units differ across models. No free-diffusion or migration velocity law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV. CI∞ next to a finite point marks an upper-unbounded bootstrap CI; finite lower endpoints are retained. CI[0,∞) spans the full nonnegative range.
CSV: figure4_longest_canonical_measurements_20261020_024_S5_Dx.csv

S5 · Common selected observations literal material-origin physical-site measurements · Literal heading position · A3 primary split · σ=1



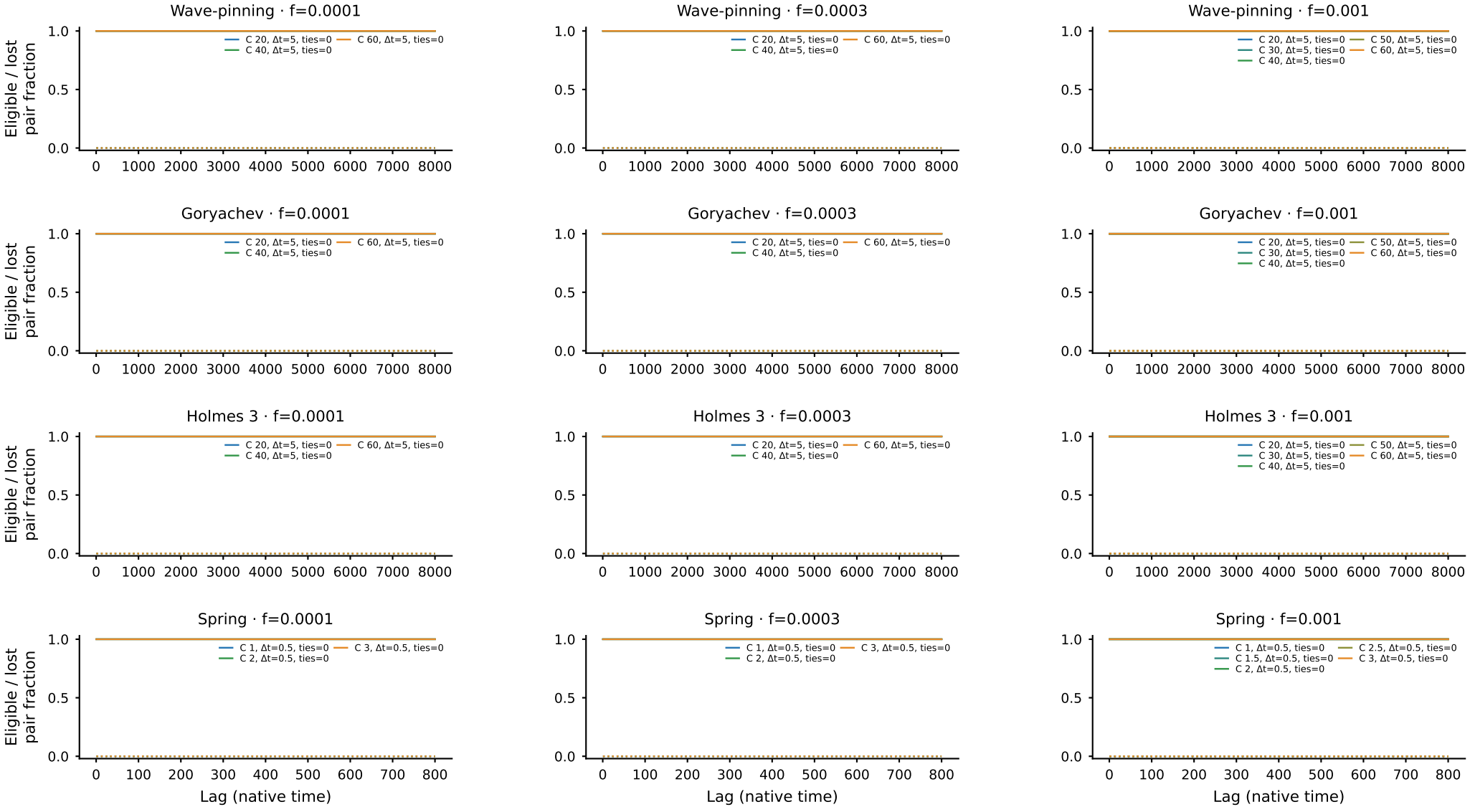
Common selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M₁|/|M₁,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_001_S5_literal_sigma1_heading_literal_actualC_MSD_1.csv.gz

S5 · Common selected observations literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · σ=1



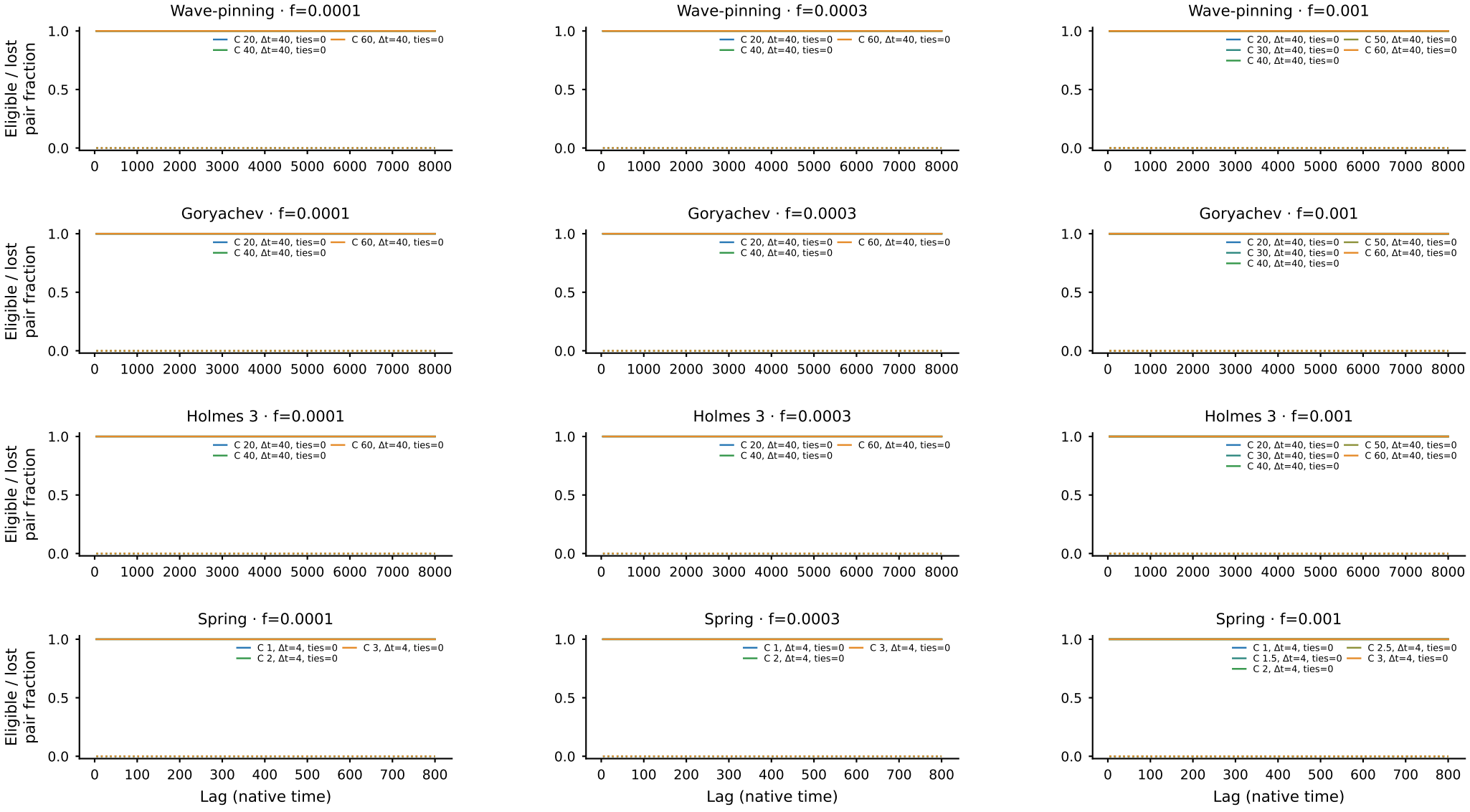
Common selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M_1|/|M_1,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_002_S5_literal_sigma1_profile_peak_literal_actualC_MSD_1.csv

S5 · Common selected observations literal material-origin physical-site measurements · Literal heading position · A3 primary split · pair coverage and ambiguity loss · σ=1



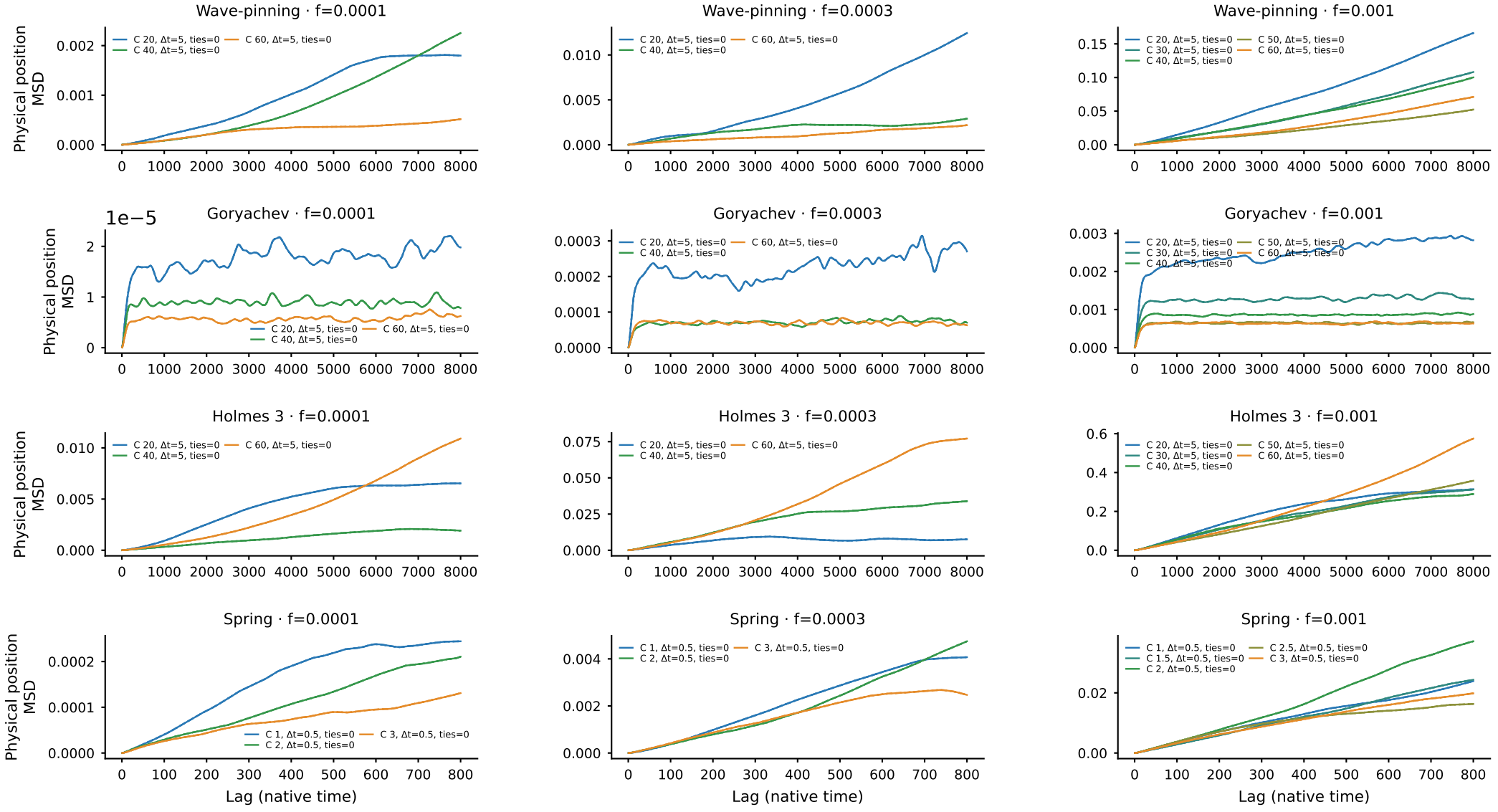
Common selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M₁|/|M₁,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_003_S5_literal_sigma1_heading_literal_actualC_coverage_1.csv.gz

S5 · Common selected observations literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · pair coverage and ambiguity loss · σ=1



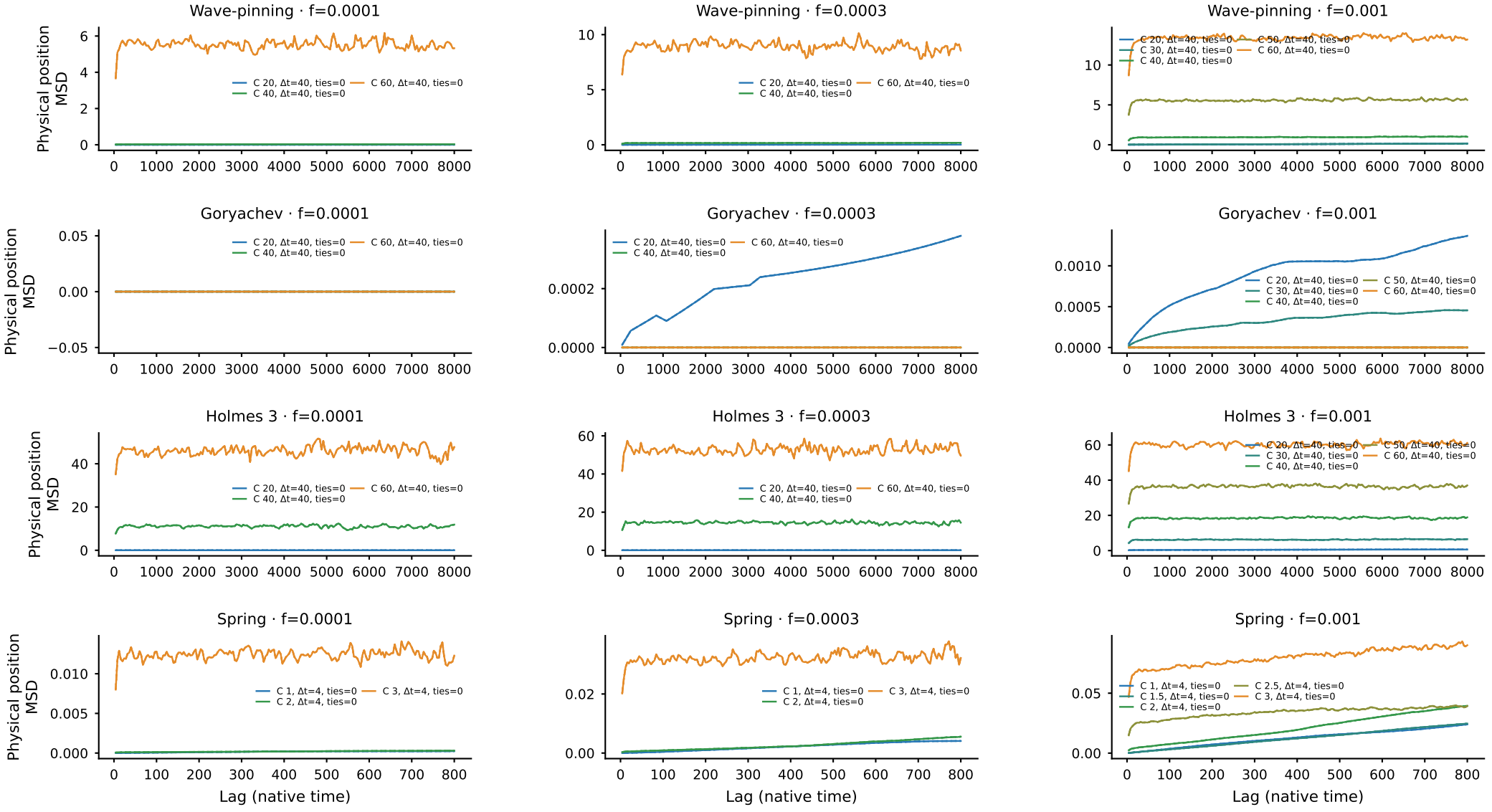
Common selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_004_S5_literal_sigma1_profile_peak_literal_actualC_coverage_1.csv

S5 · Common selected observations literal material-origin physical-site measurements · Literal heading position · paired route sensitivities · split / +π / −π · σ=1



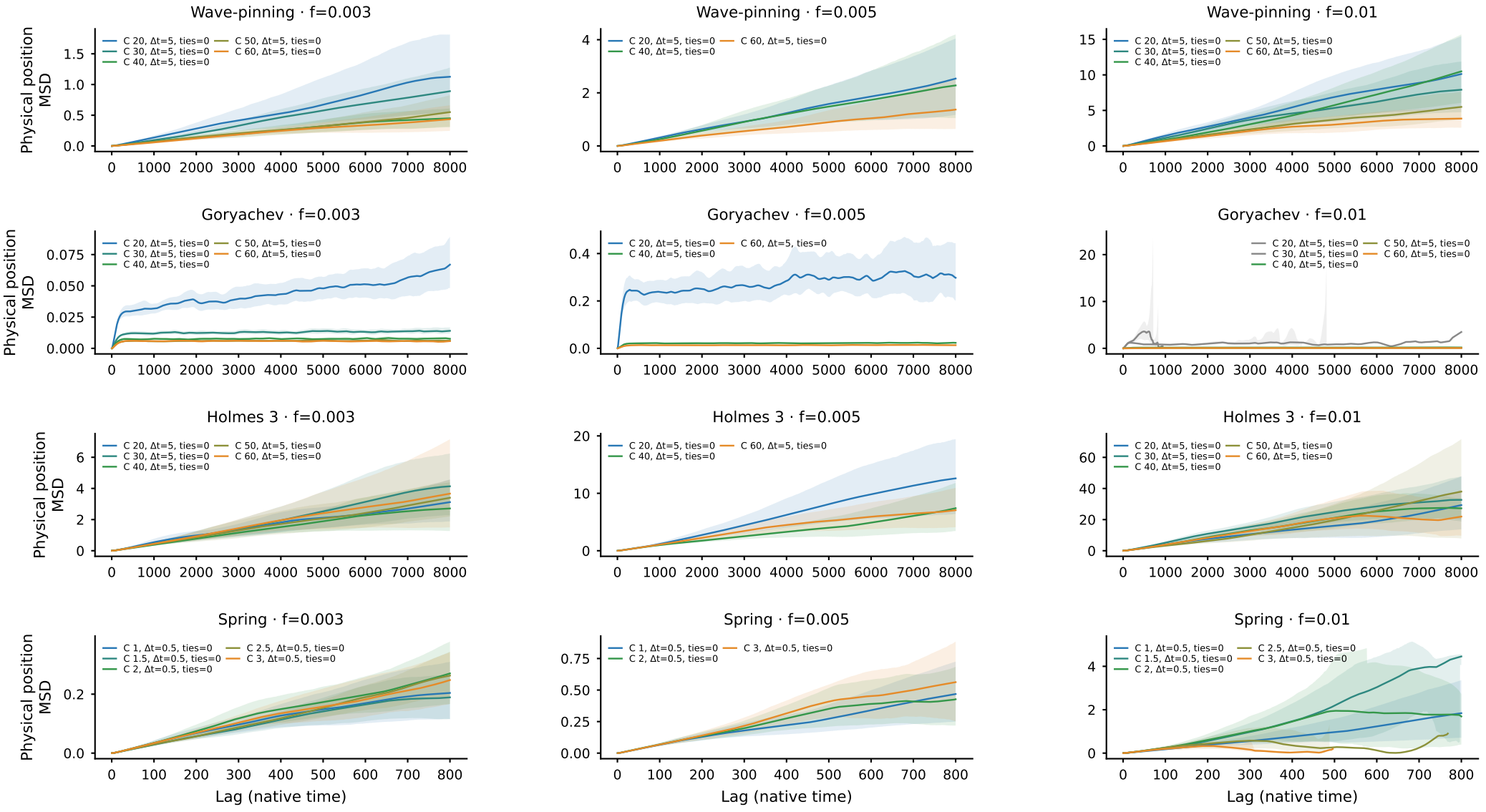
Common selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M₁|/|M₁,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_005_S5_literal_sigma1_heading_literal_actualC_routes_1.csv.gz

S5 · Common selected observations literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · split / +π / −π · σ=1



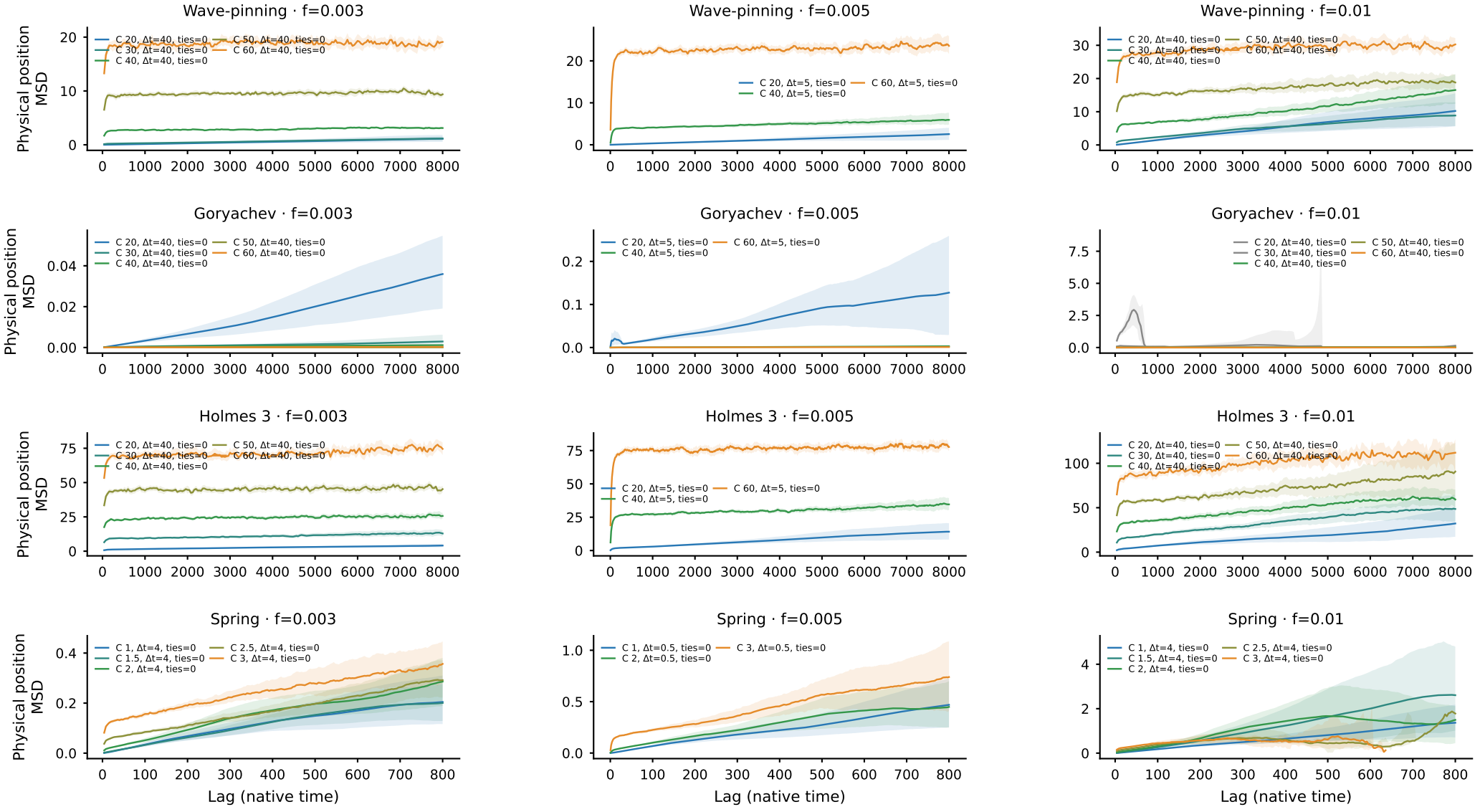
Common selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M_1|/|M_1,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_006_S5_literal_sigma1_profile_peak_literal_actualC_routes_1.csv

S5 · Common selected observations literal material-origin physical-site measurements · Literal heading position · A3 primary split · σ=1



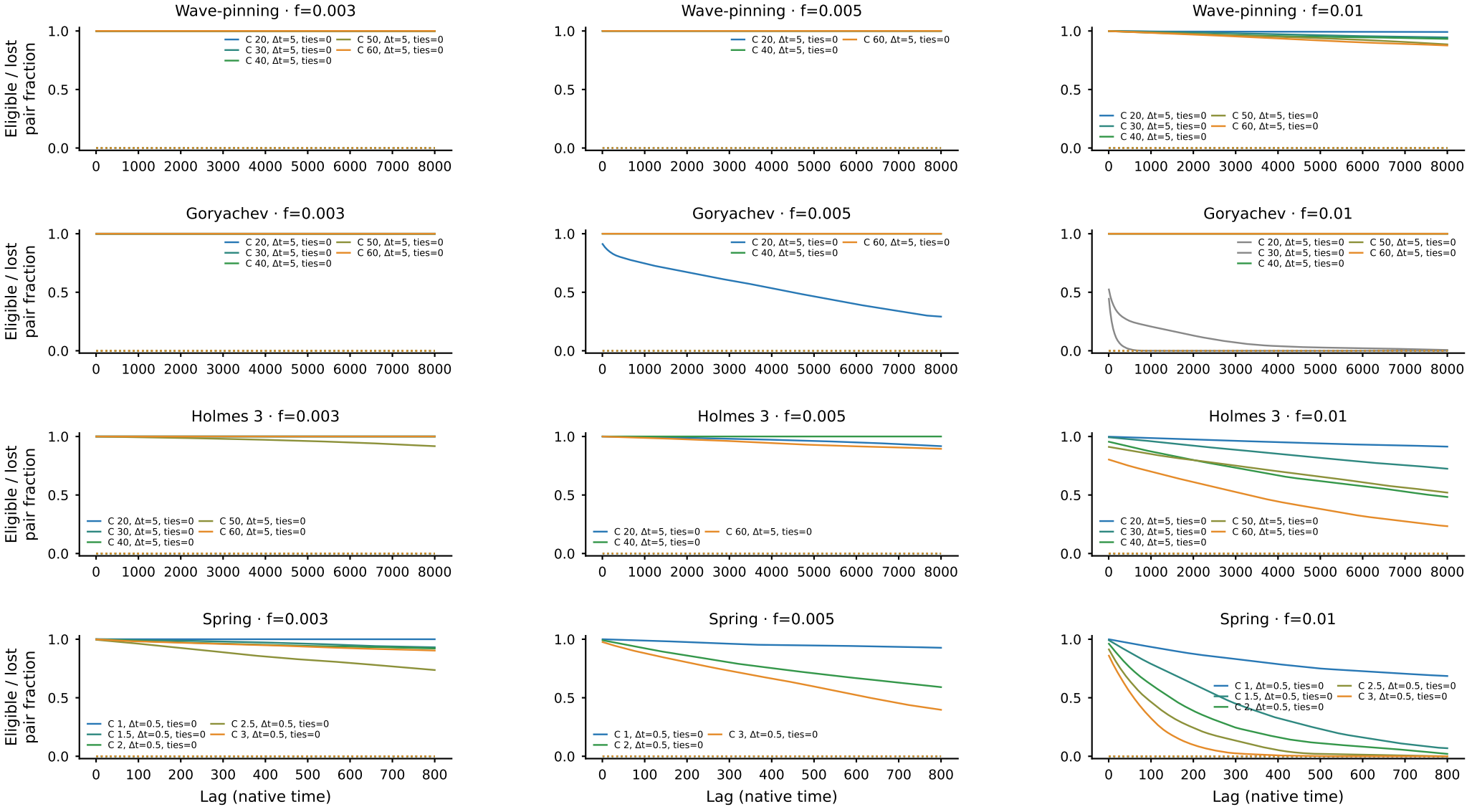
Common selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M_1|/|M_1,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_007_S5_literal_sigma1_heading_literal_actualC_MSD_2.csv.gz

S5 · Common selected observations literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · σ=1



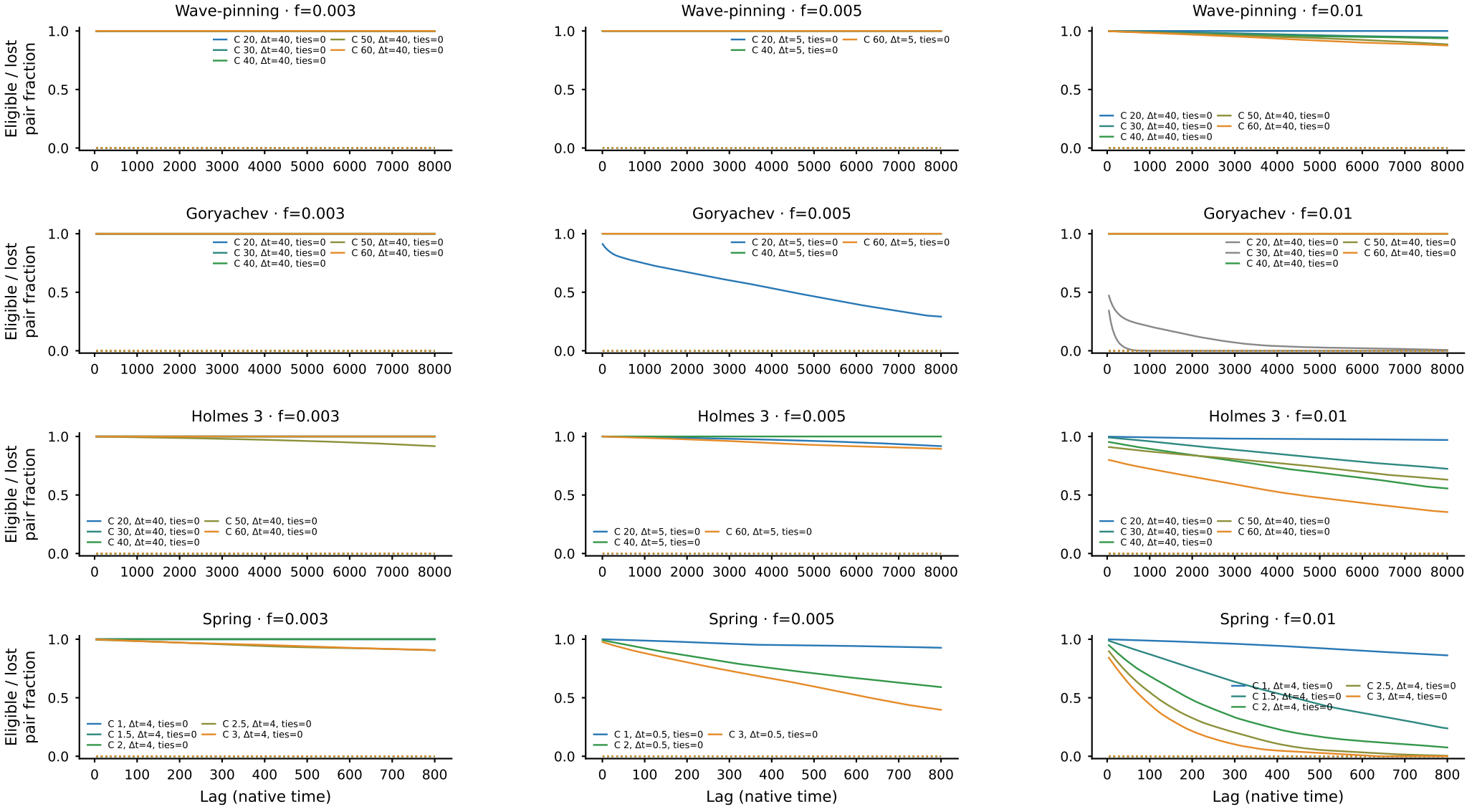
Common selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M_1|/|M_1,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_008_S5_literal_sigma1_profile_peak_literal_actualC_MSD_2.csv

S5 · Common selected observations literal material-origin physical-site measurements · Literal heading position · A3 primary split · pair coverage and ambiguity loss · σ=1



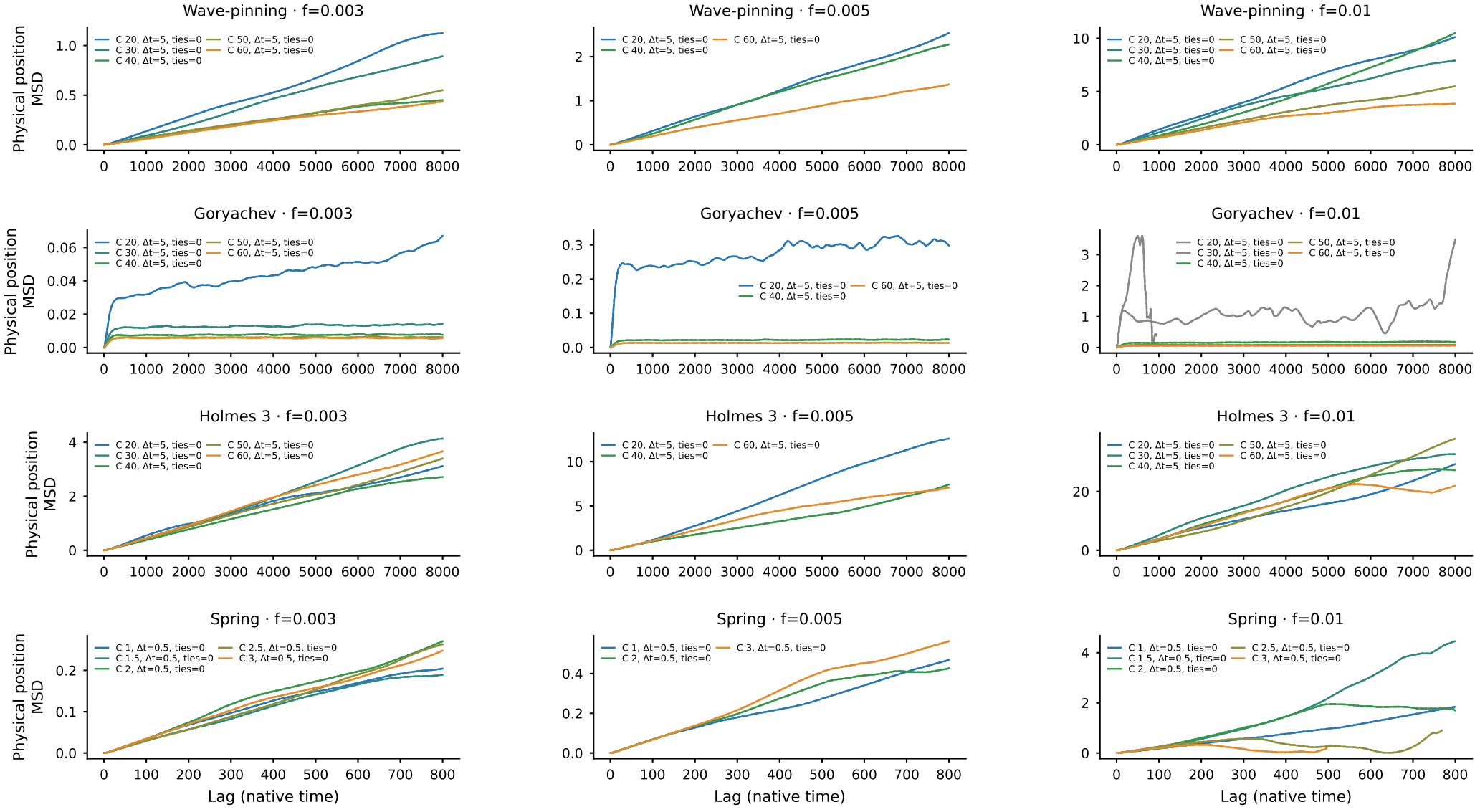
Common selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M₁|/|M₁,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_009_S5_literal_sigma1_heading_literal_actualC_coverage_2.csv.gz

S5 · Common selected observations literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · pair coverage and ambiguity loss · σ=1



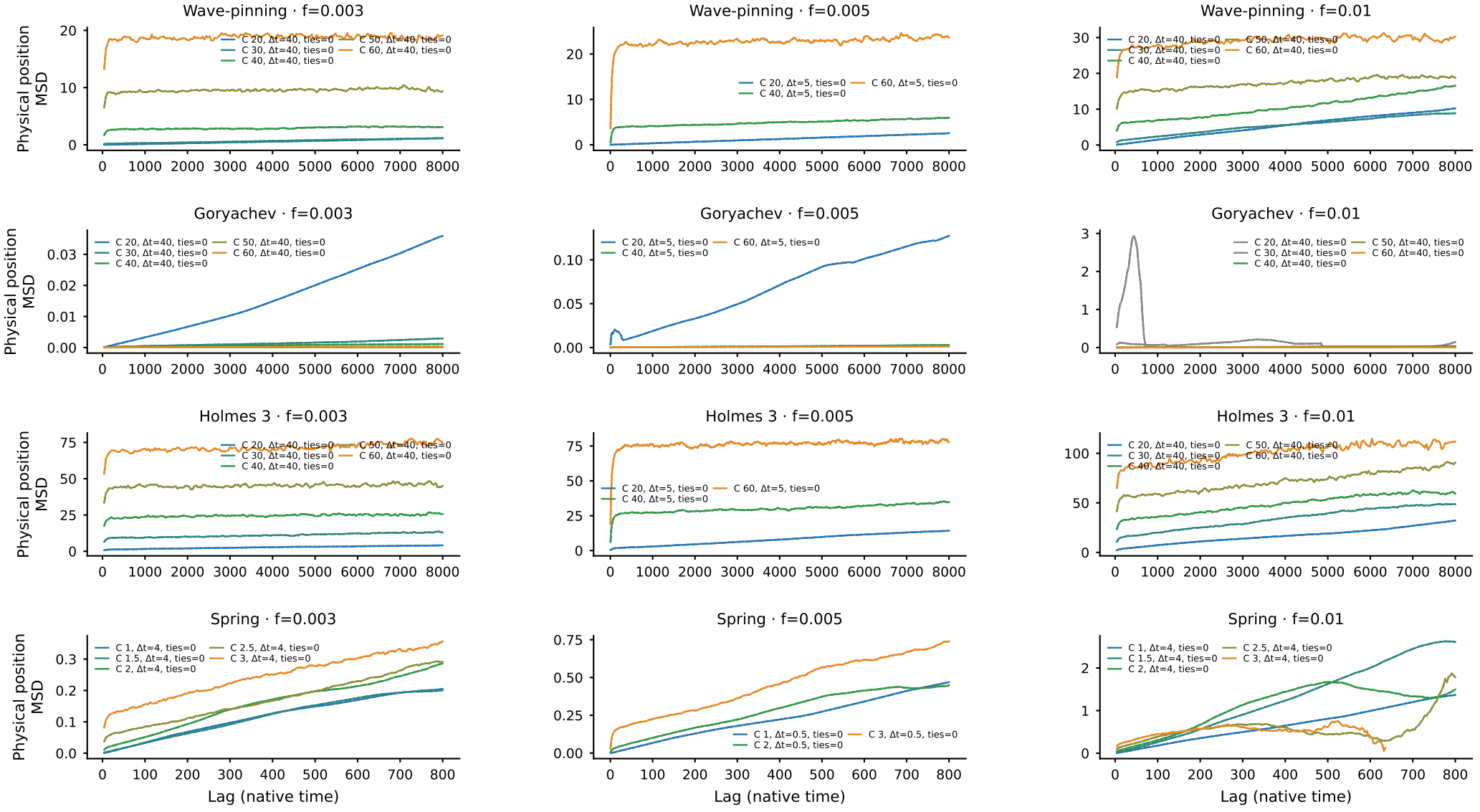
Common selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M_1|/|M_1,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_010_S5_literal_sigma1_profile_peak_literal_actualC_coverage_2.csv

S5 · Common selected observations literal material-origin physical-site measurements · Literal heading position · paired route sensitivities · split / +π / −π · σ=1



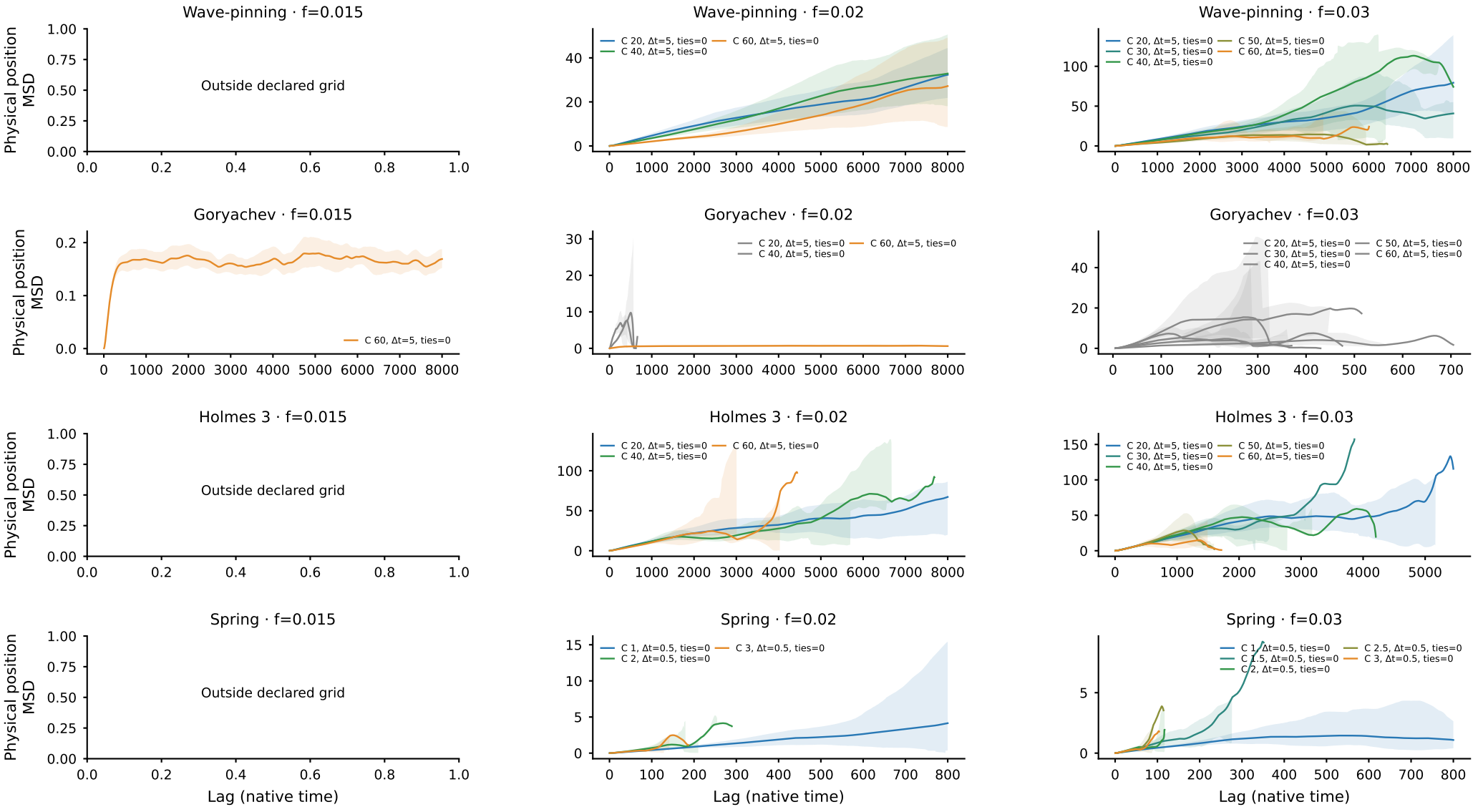
Common selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M_1|/|M_1,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_011_S5_literal_sigma1_heading_literal_actualC_routes_2.csv.gz

S5 · Common selected observations literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · split / +π / −π · σ=1



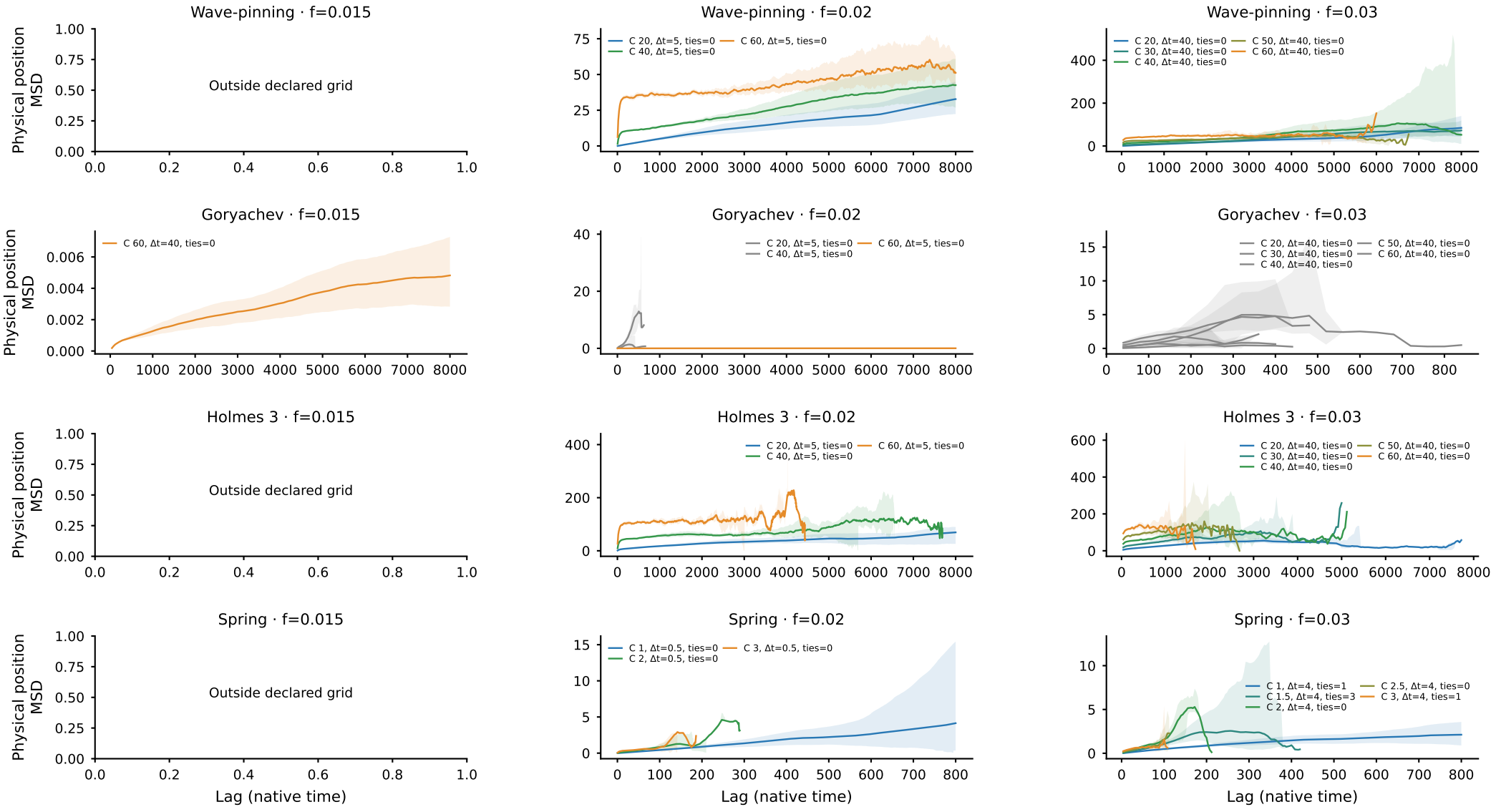
Common selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_012_S5_literal_sigma1_profile_peak_literal_actualC_routes_2.csv.gz

S5 · Common selected observations literal material-origin physical-site measurements · Literal heading position · A3 primary split · σ=1



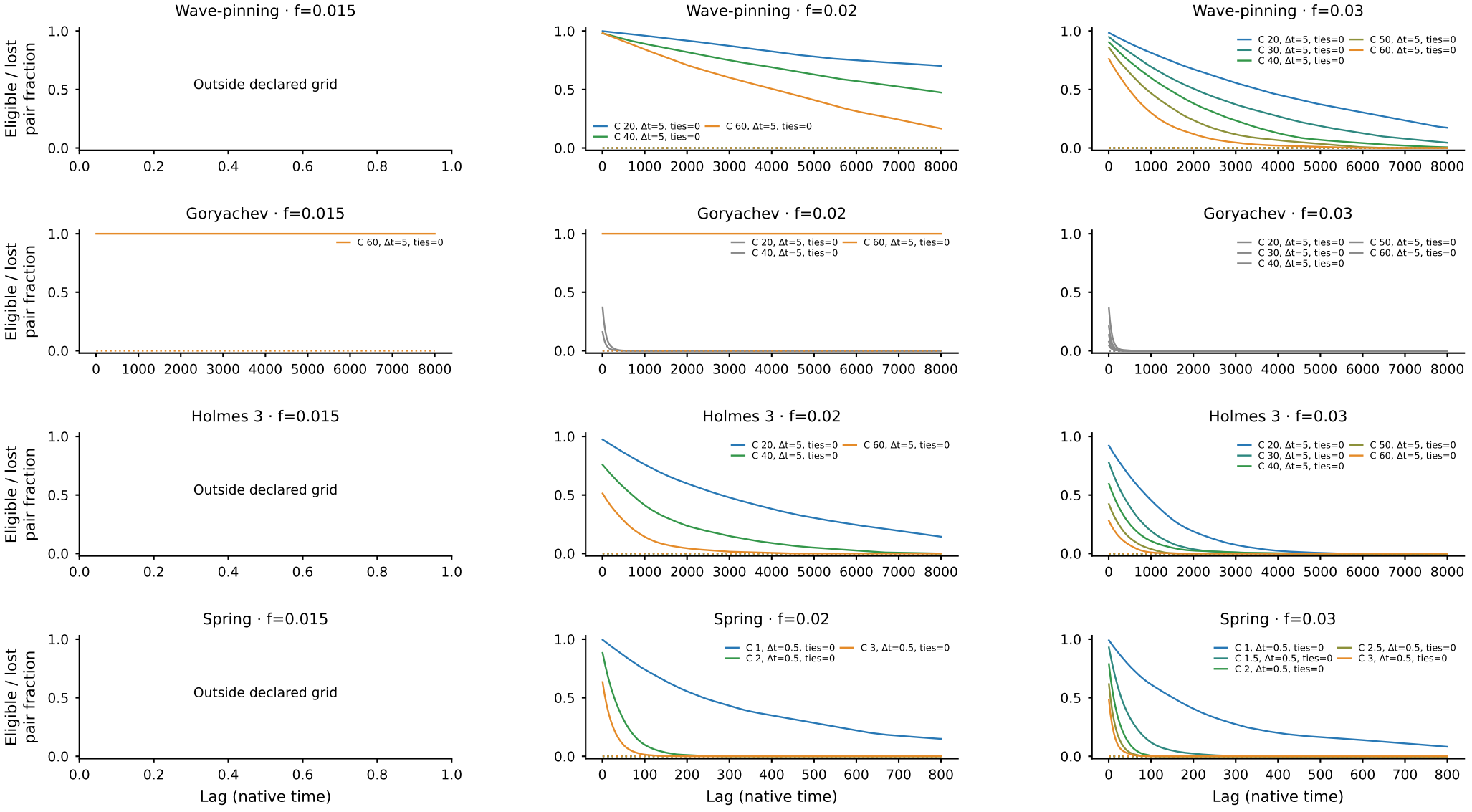
Common selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_1,\text{basal}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_013_S5_literal_sigma1_heading_literal_actualC_MSD_3.csv.gz

S5 · Common selected observations literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · σ=1



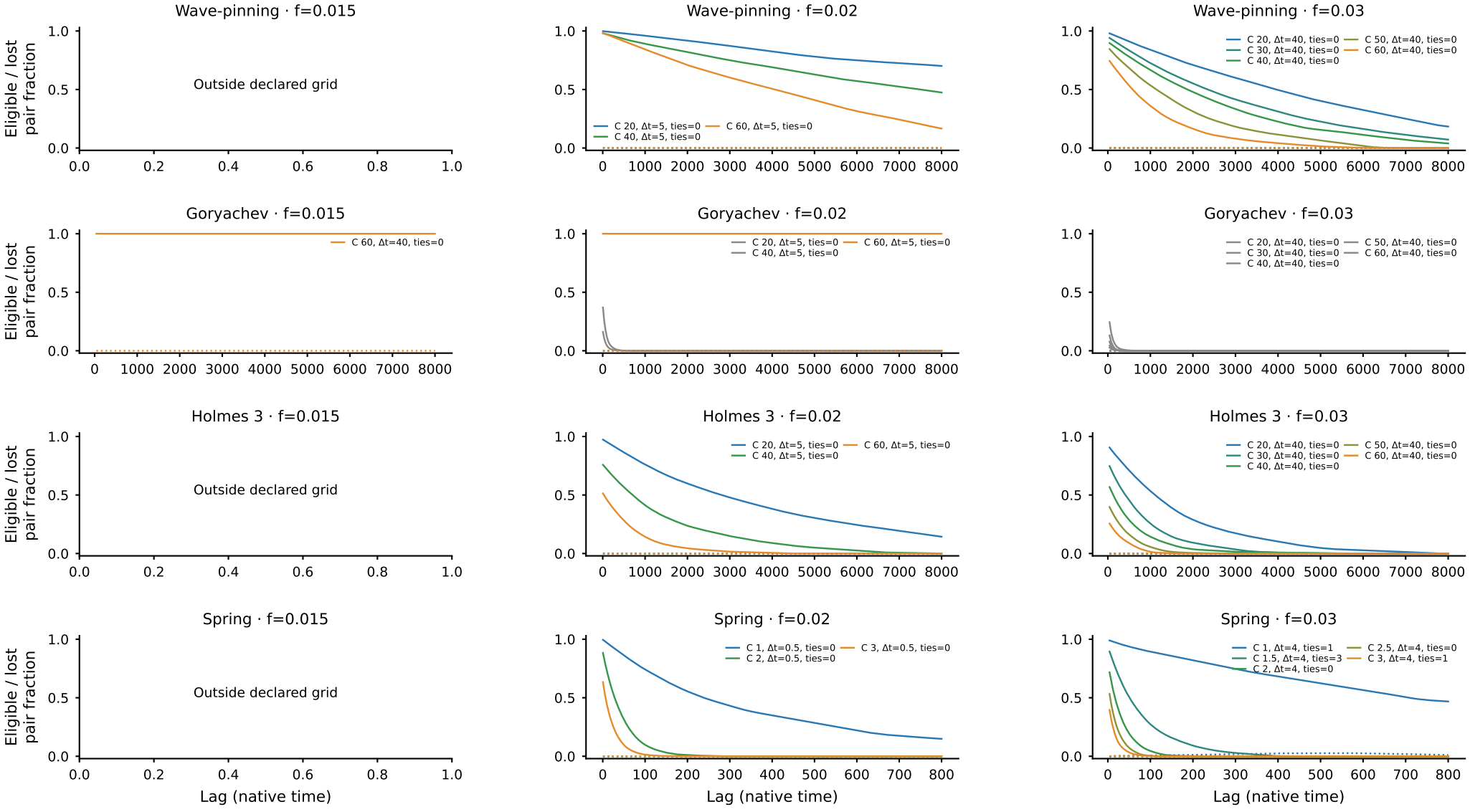
Common selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_1,\text{basal}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_014_S5_literal_sigma1_profile_peak_literal_actualC_MSD_3.csv

S5 · Common selected observations literal material-origin physical-site measurements · Literal heading position · A3 primary split · pair coverage and ambiguity loss · σ=1



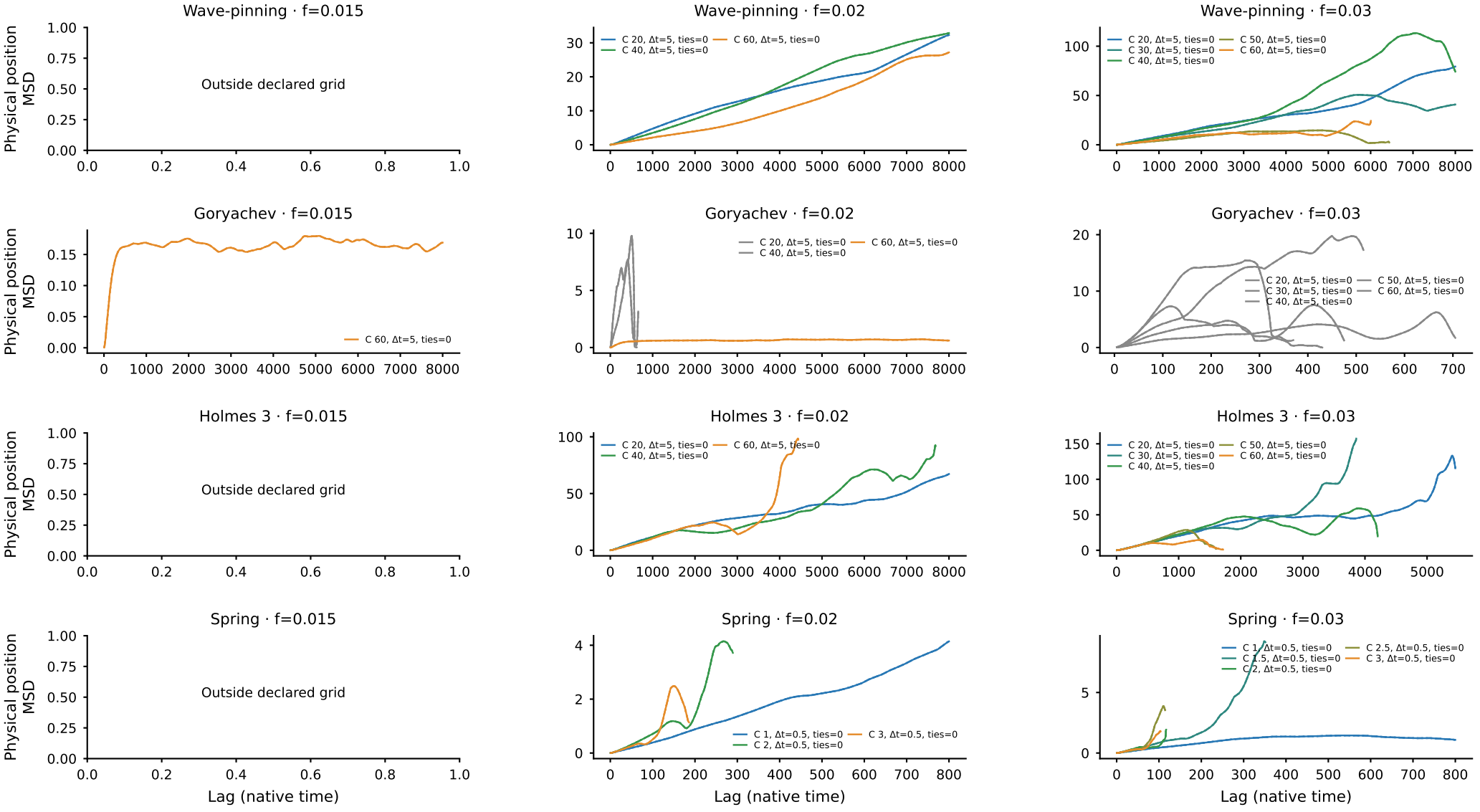
Common selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M_1|/|M_1,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_015_S5_literal_sigma1_heading_literal_actualC_coverage_3.csv.gz

S5 · Common selected observations literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · pair coverage and ambiguity loss · σ=1



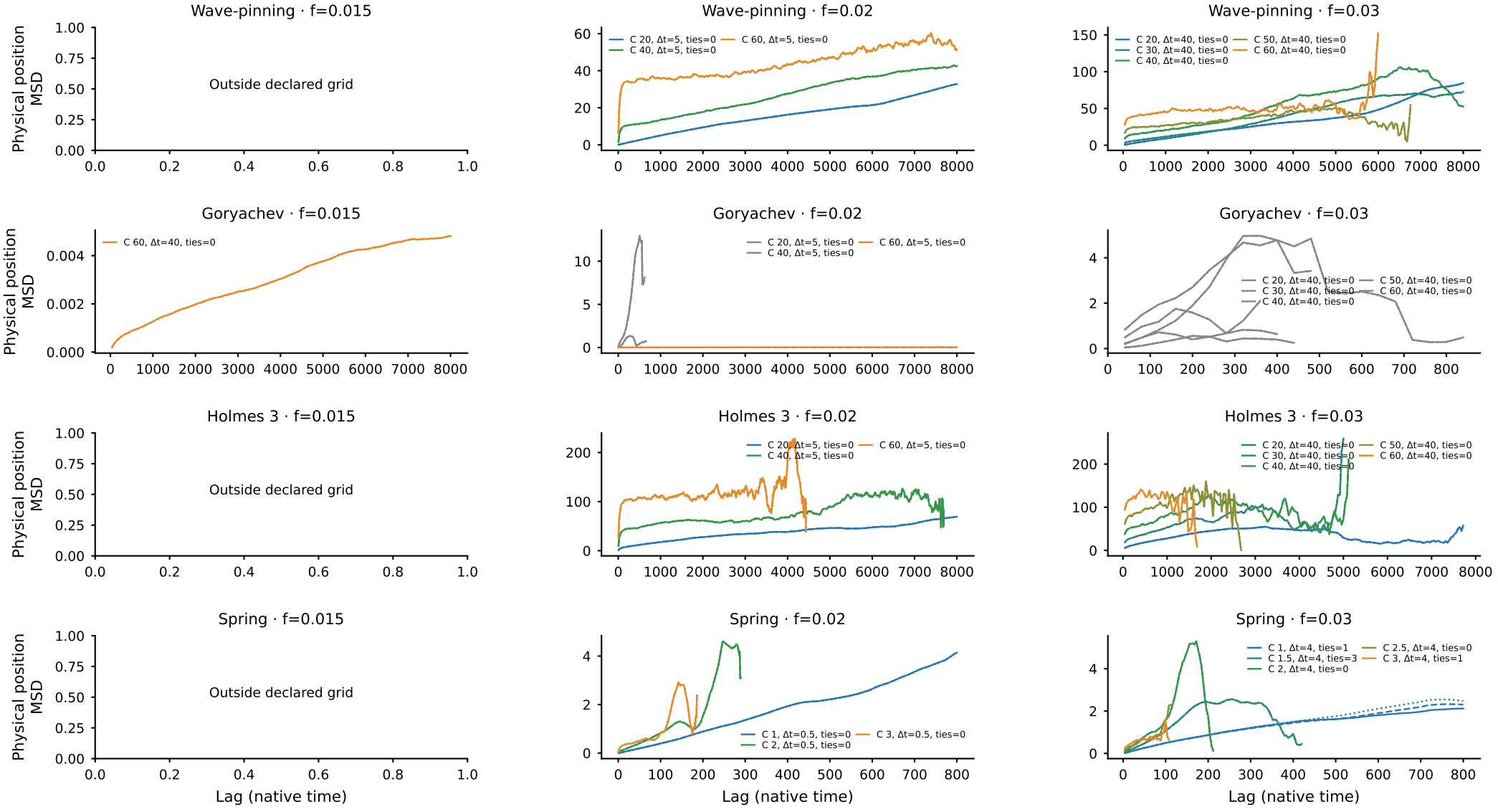
Common selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_1,\text{basal}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_016_S5_literal_sigma1_profile_peak_literal_actualC_coverage_3.csv

S5 · Common selected observations literal material-origin physical-site measurements · Literal heading position · paired route sensitivities · split / +π / −π · σ=1



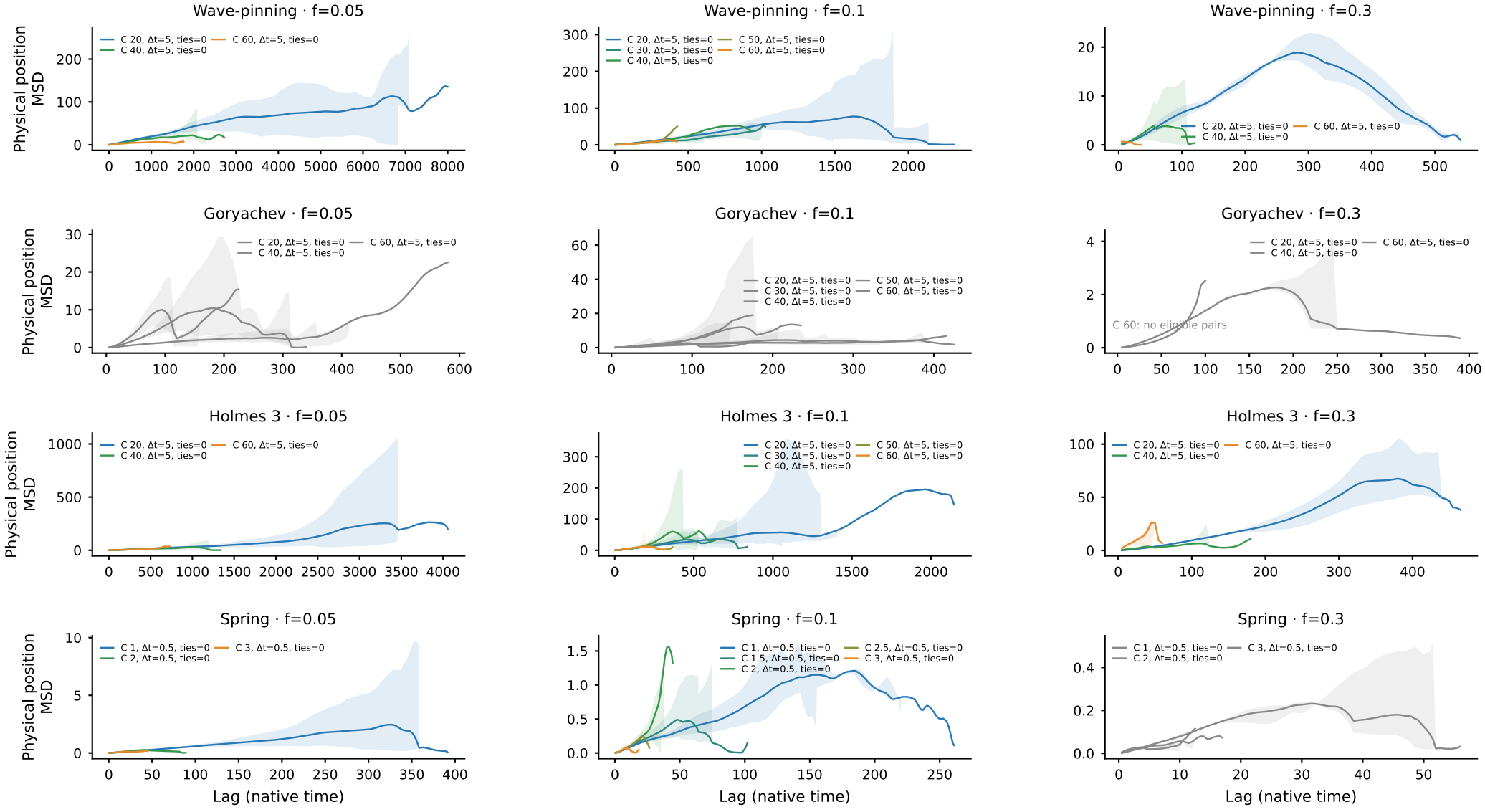
Common selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_017_S5_literal_sigma1_heading_literal_actualC_routes_3.csv.gz

S5 · Common selected observations literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · split / +π / −π · σ=1



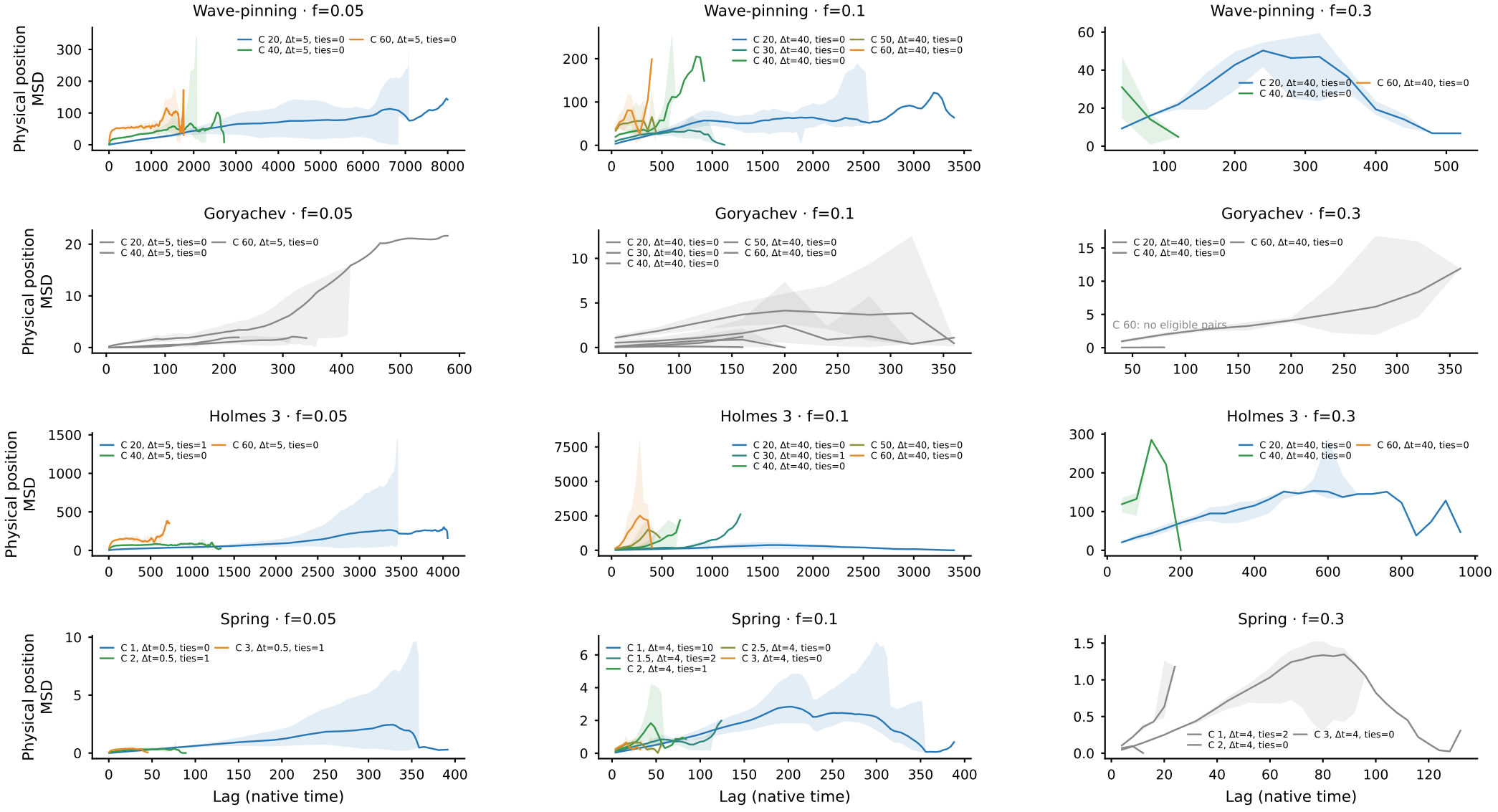
Common selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_1,\text{basal}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_018_S5_literal_sigma1_profile_peak_literal_actualC_routes_3.csv.gz

S5 · Common selected observations literal material-origin physical-site measurements · Literal heading position · A3 primary split · σ=1



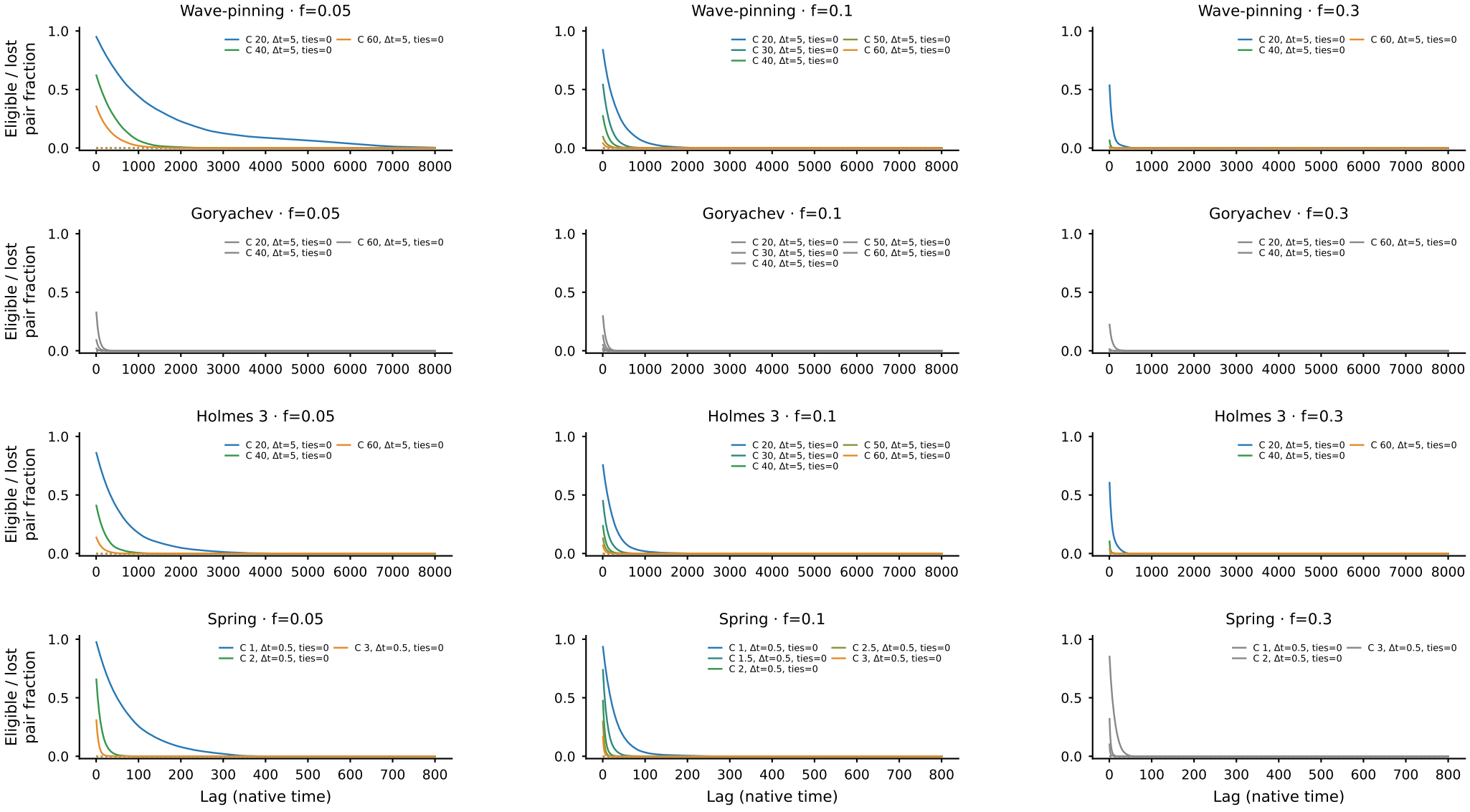
Common selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_1,\text{basal}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_019_S5_literal_sigma1_heading_literal_actualC_MSD_4.csv.gz

S5 · Common selected observations literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · σ=1



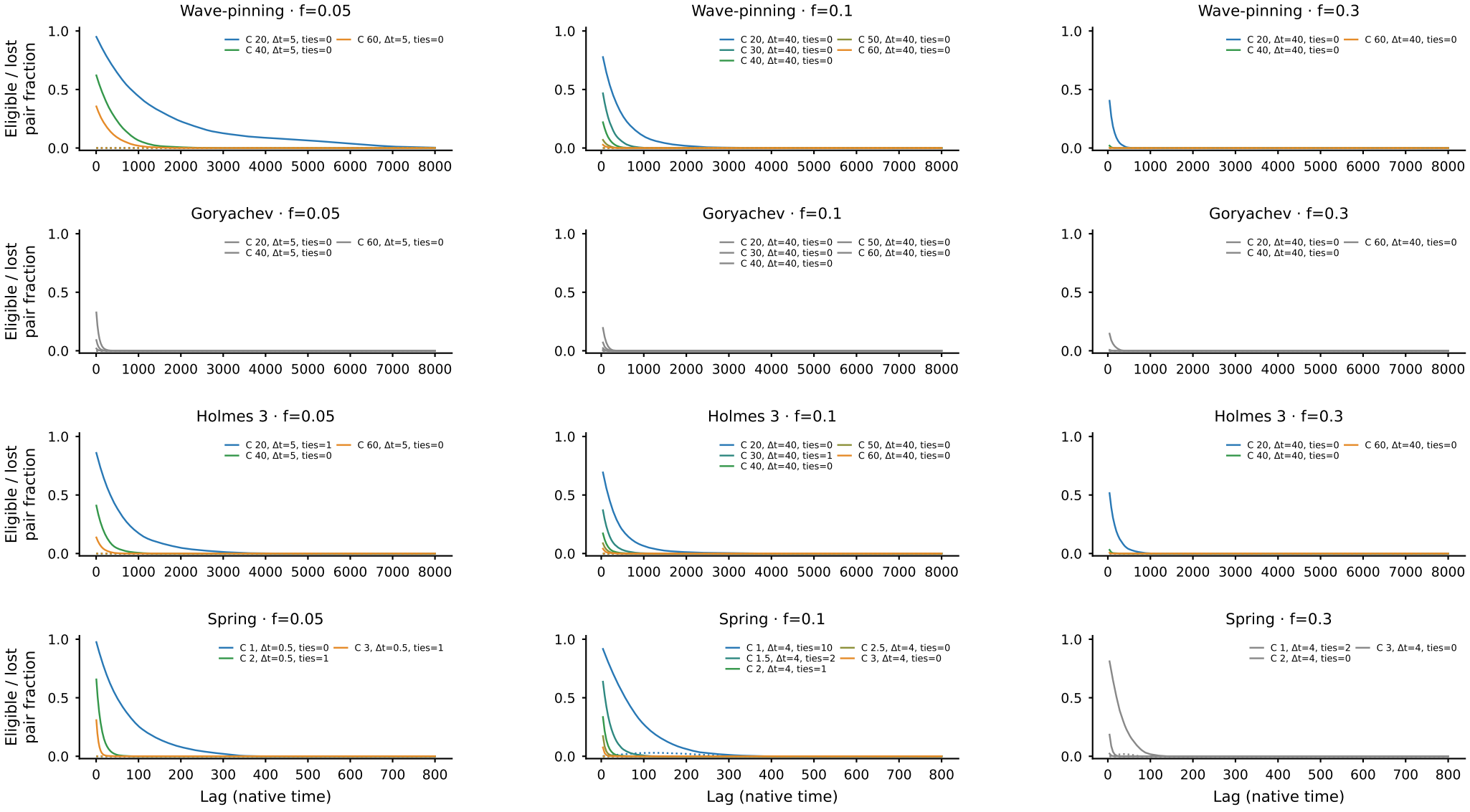
Common selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_020_S5_literal_sigma1_profile_peak_literal_actualC_MSD_4.csv

S5 · Common selected observations literal material-origin physical-site measurements · Literal heading position · A3 primary split · pair coverage and ambiguity loss · σ=1



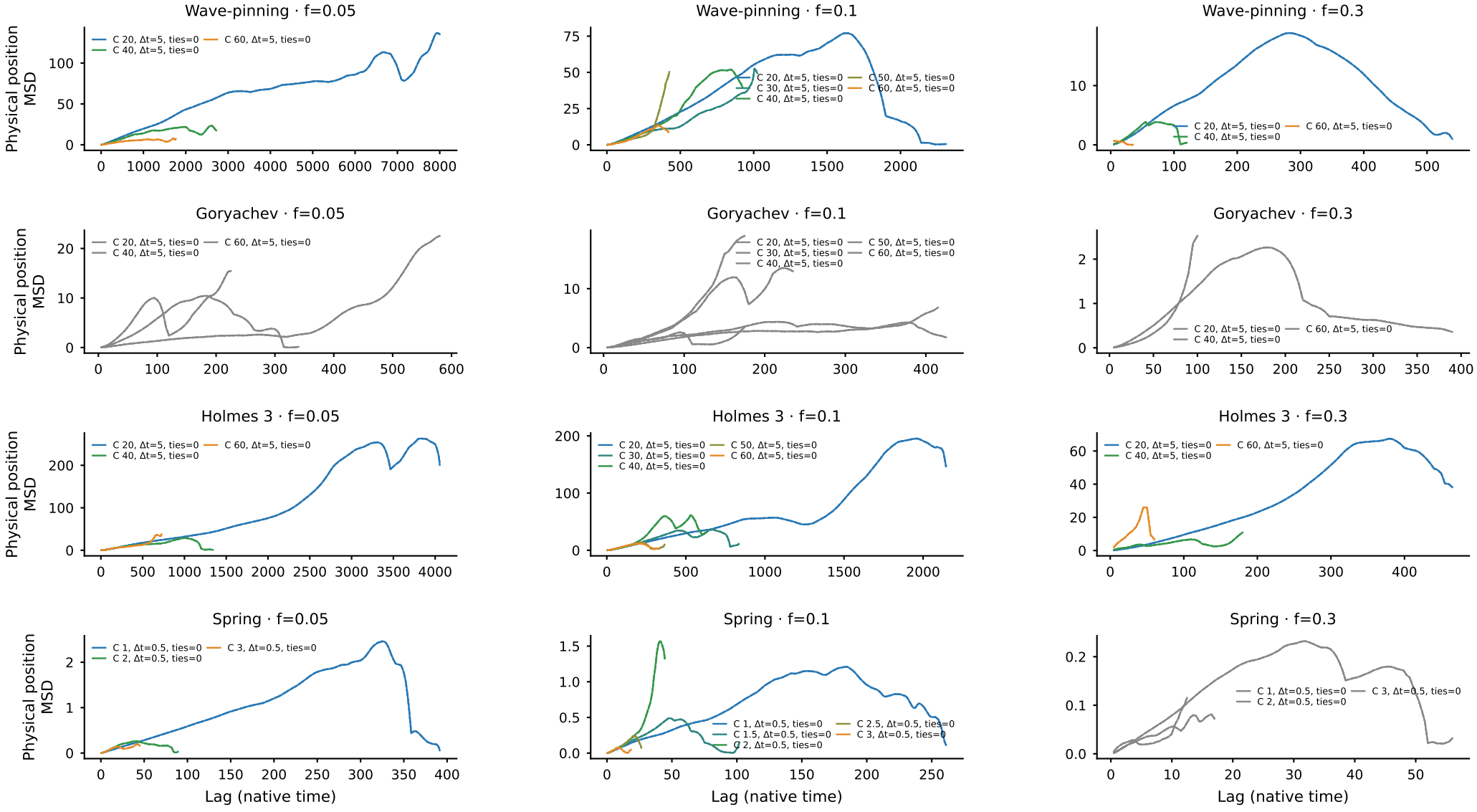
Common selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_021_S5_literal_sigma1_heading_literal_actualC_coverage_4.csv.gz

S5 · Common selected observations literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · pair coverage and ambiguity loss · σ=1



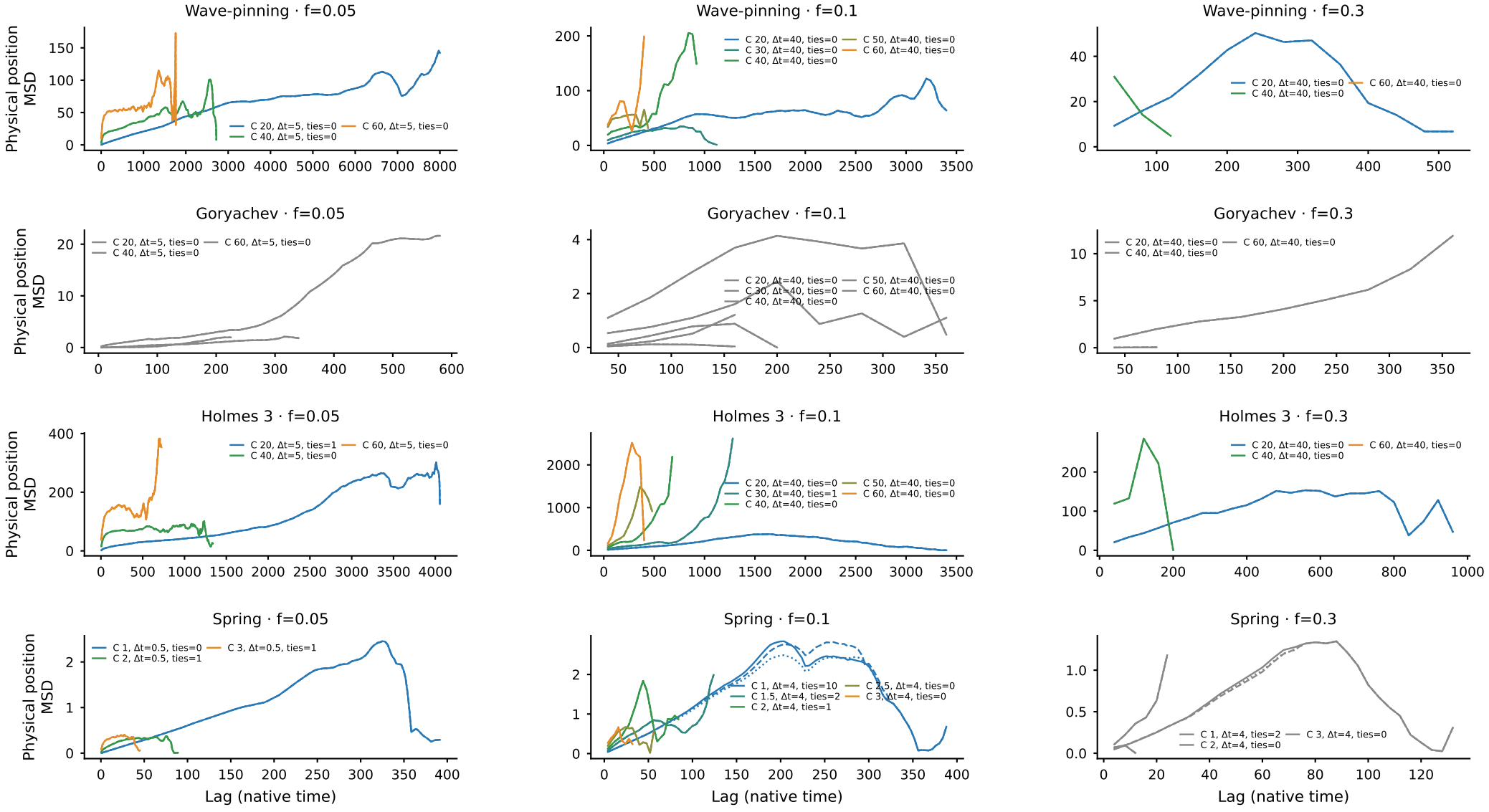
Common selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M₁|/|M₁,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_022_S5_literal_sigma1_profile_peak_literal_actualC_coverage_4.csv

S5 · Common selected observations literal material-origin physical-site measurements · Literal heading position · paired route sensitivities · split / +π / −π · σ=1



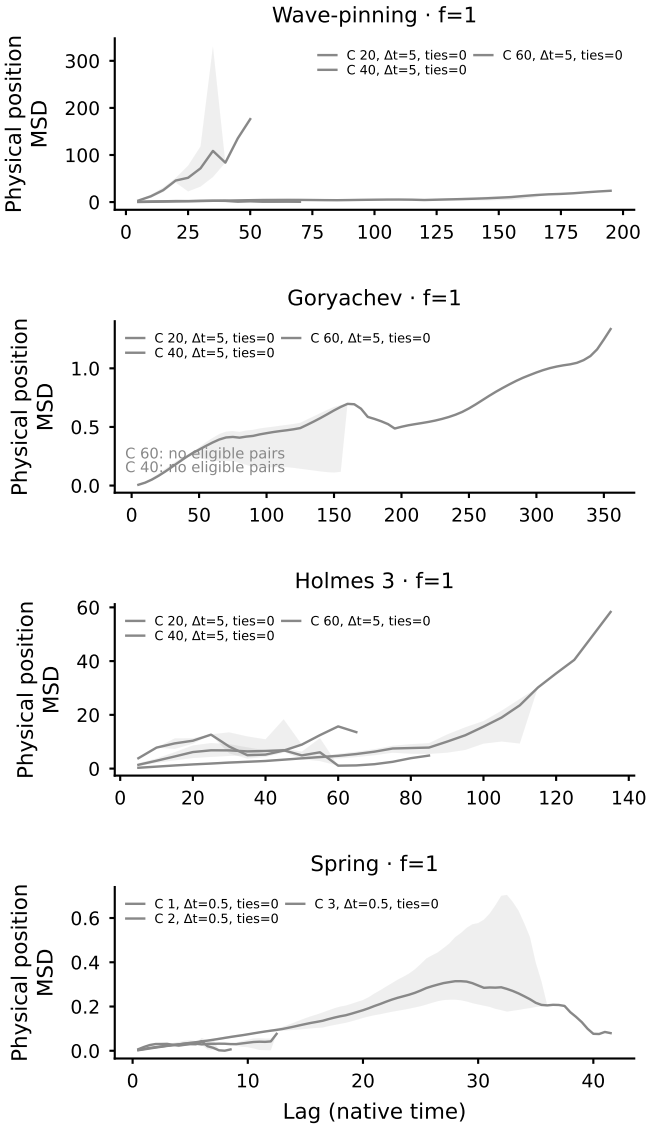
Common selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_1,\text{basal}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_023_S5_literal_sigma1_heading_literal_actualC_routes_4.csv.gz

S5 · Common selected observations literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · split / +π / −π · σ=1



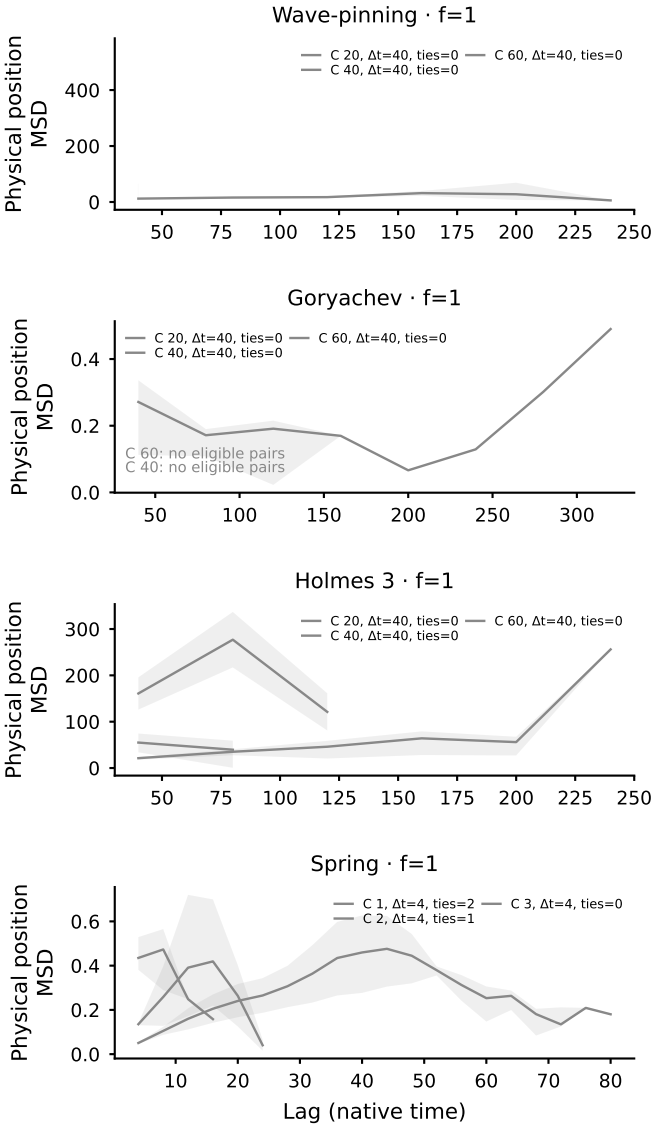
Common selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_024_S5_literal_sigma1_profile_peak_literal_actualC_routes_4.csv.gz

S5 · Common selected observations literal material-origin physical-site measurements · Literal heading position · A3 primary split · σ=1



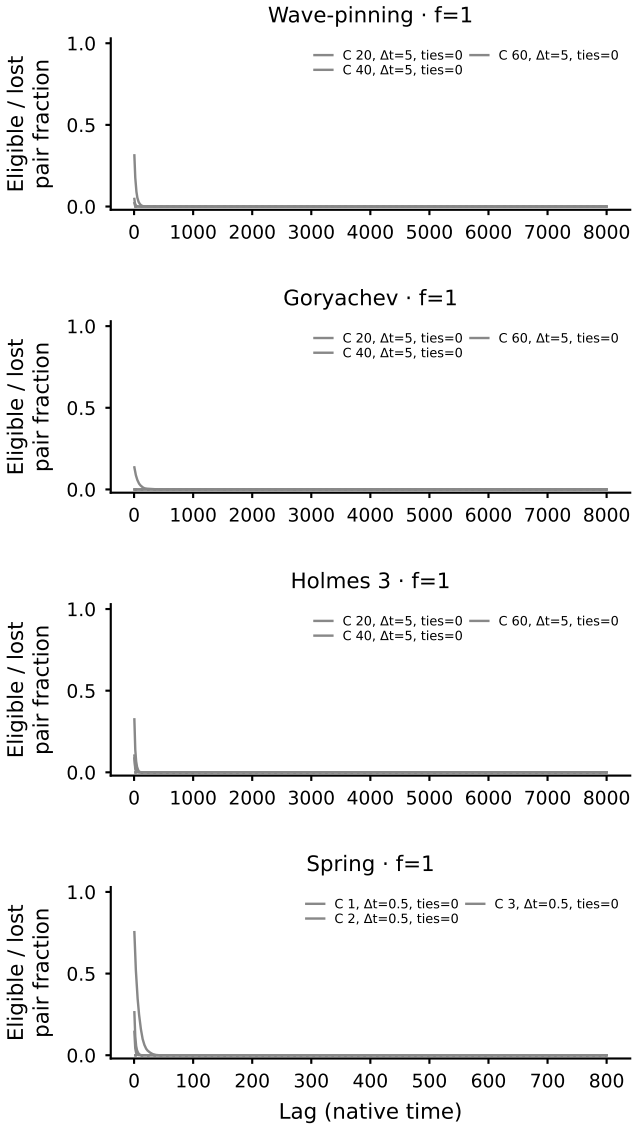
Common selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M₁|/|M₁,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_025_S5_literal_sigma1_heading_literal_actualC_MSD_5.csv

S5 · Common selected observations literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · σ=1



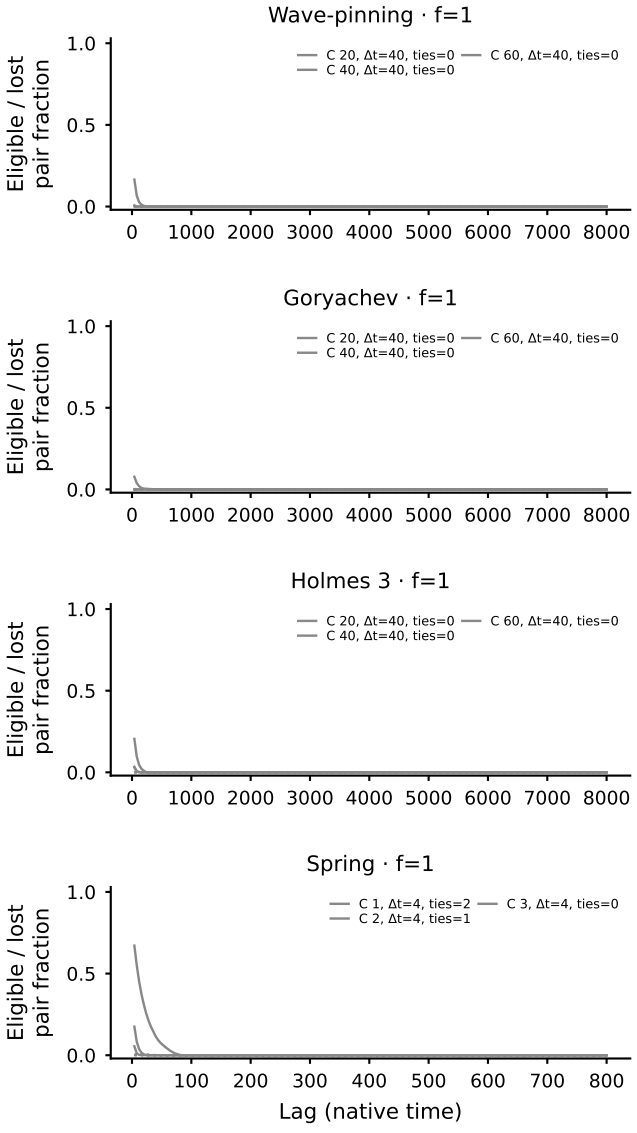
Common selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M_1|/|M_1,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_026_S5_literal_sigma1_profile_peak_literal_actualC_MSD_5.csv

S5 · Common selected observations literal material-origin physical-site measurements · Literal heading position · A3 primary split · pair coverage and ambiguity loss · σ=1



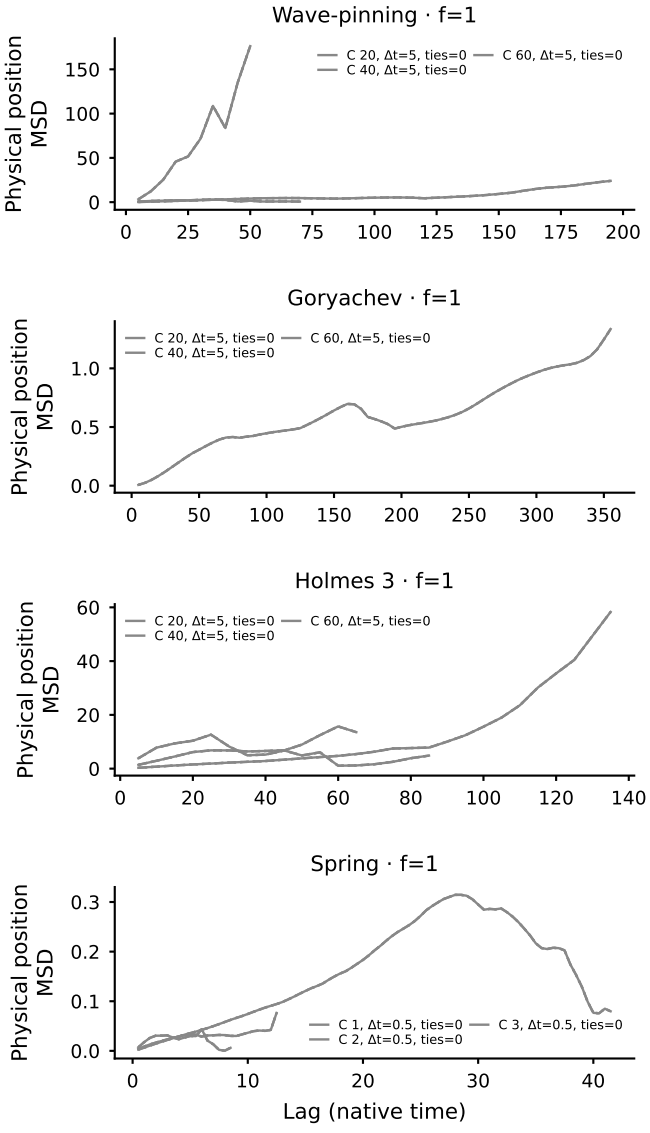
Common selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M_1|/|M_1,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_027_S5_literal_sigma1_heading_literal_actualC_coverage_5.csv

S5 · Common selected observations literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · pair coverage and ambiguity loss · σ=1



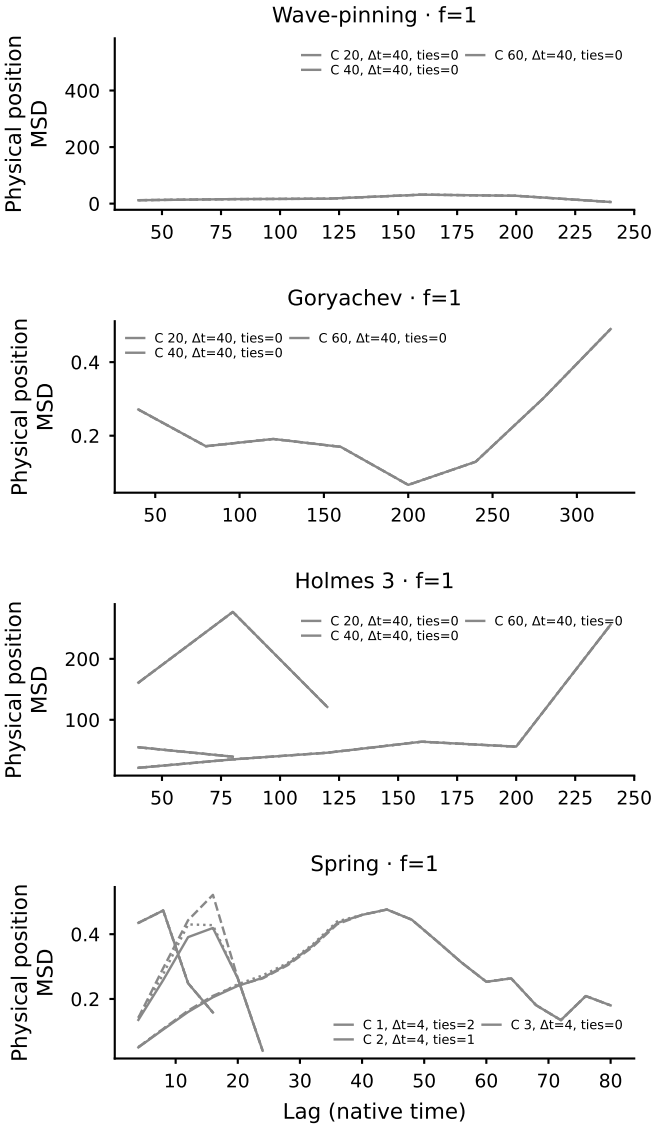
Common selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M₁|/|M₁,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_028_S5_literal_sigma1_profile_peak_literal_actualC_coverage_5.csv

S5 · Common selected observations literal material-origin physical-site measurements · Literal heading position · paired route sensitivities · split / +π / −π · σ=1



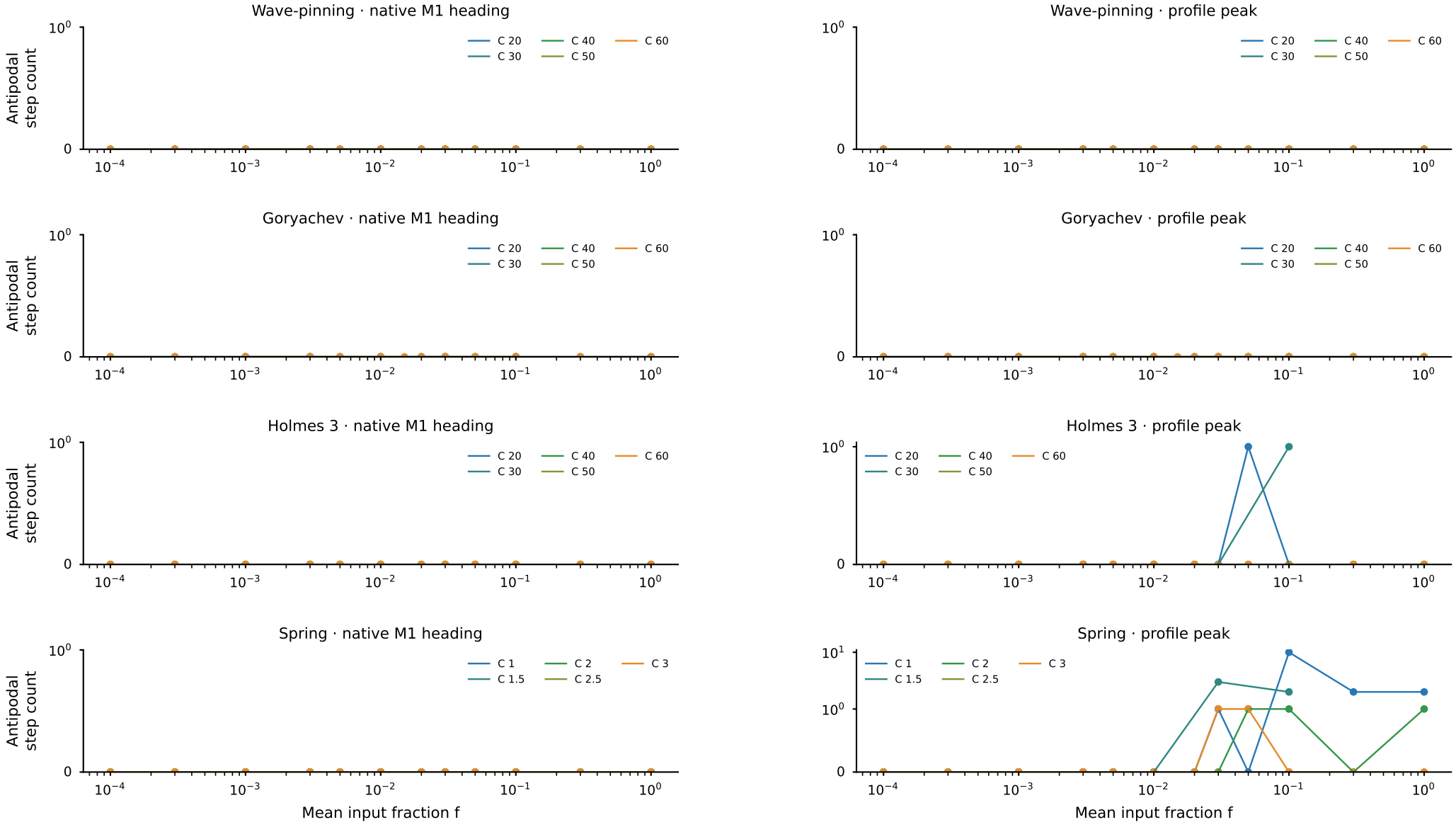
Common selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M₁|/|M₁,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_029_S5_literal_sigma1_heading_literal_actualC_routes_5.csv.gz

S5 · Common selected observations literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · split / +π / −π · σ=1



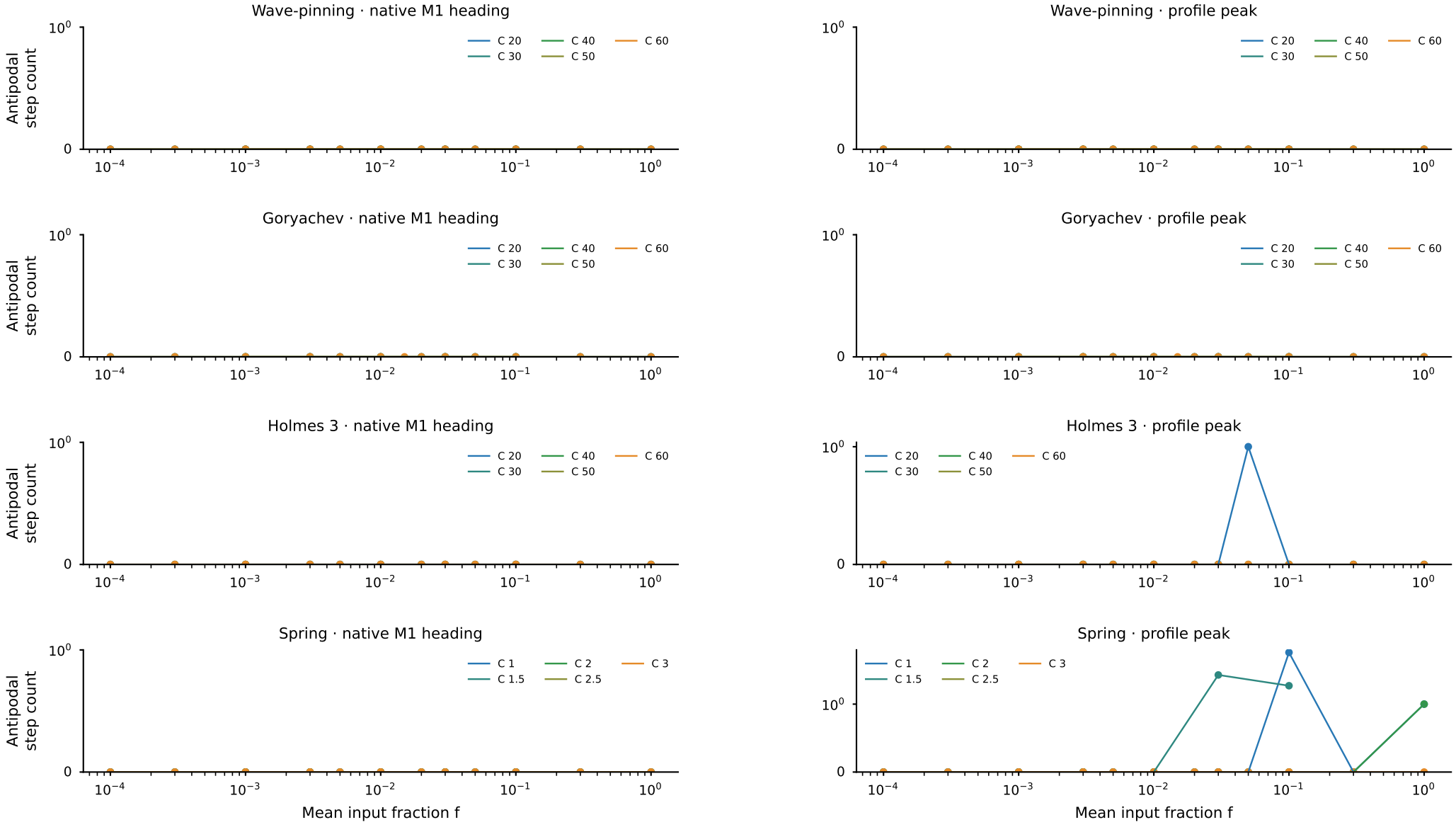
Common selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M_1|/|M_1,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_030_S5_literal_sigma1_profile_peak_literal_actualC_routes_5.csv

S5 · Common selected observations literal material-origin physical-site measurements · Native antipodal ambiguity · main · σ=1



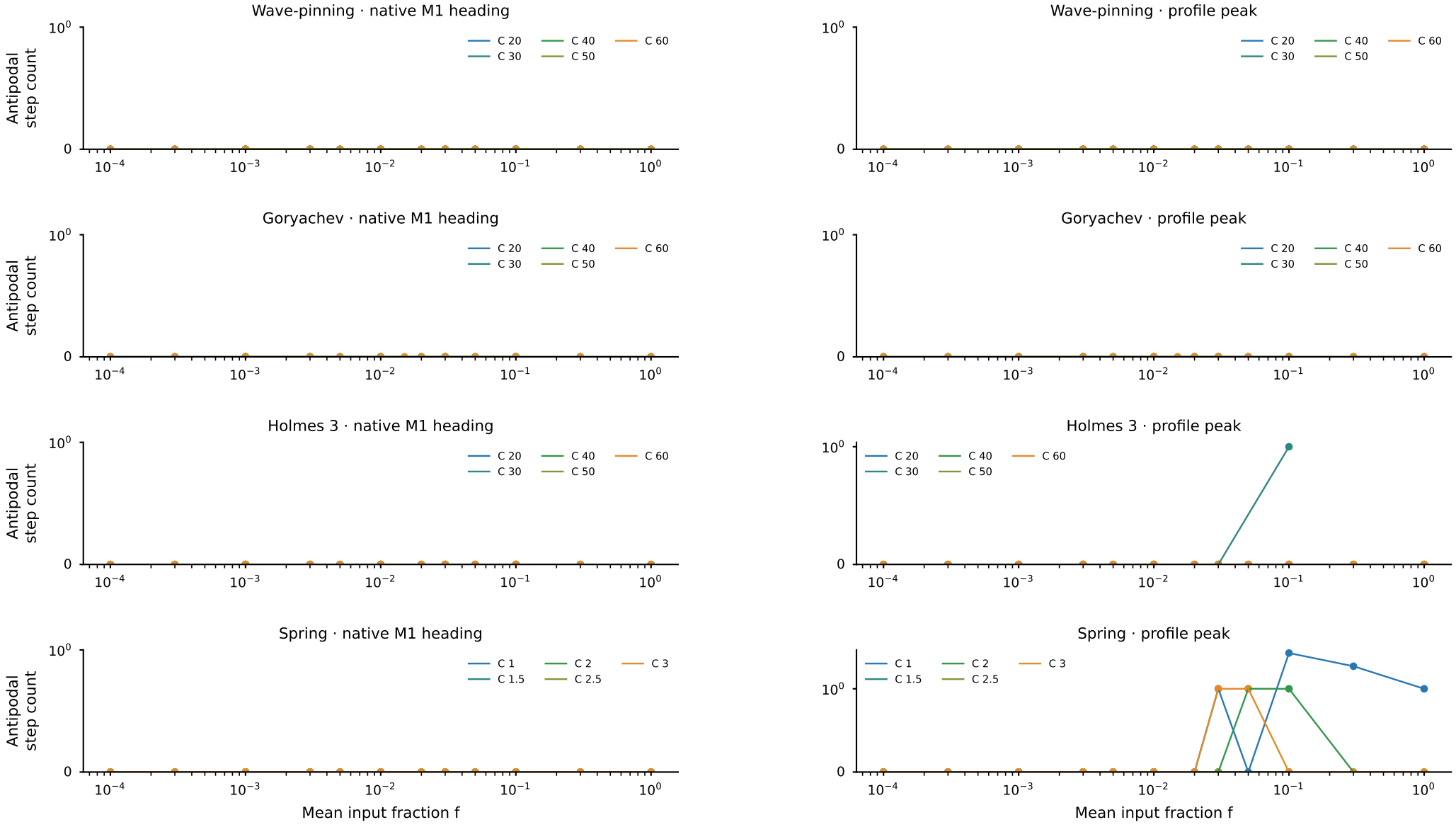
Common selected observations literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_031_S5_ambiguity_sigma1_main.csv

S5 · Common selected observations literal material-origin physical-site measurements · Native antipodal ambiguity · analysis first half · σ=1



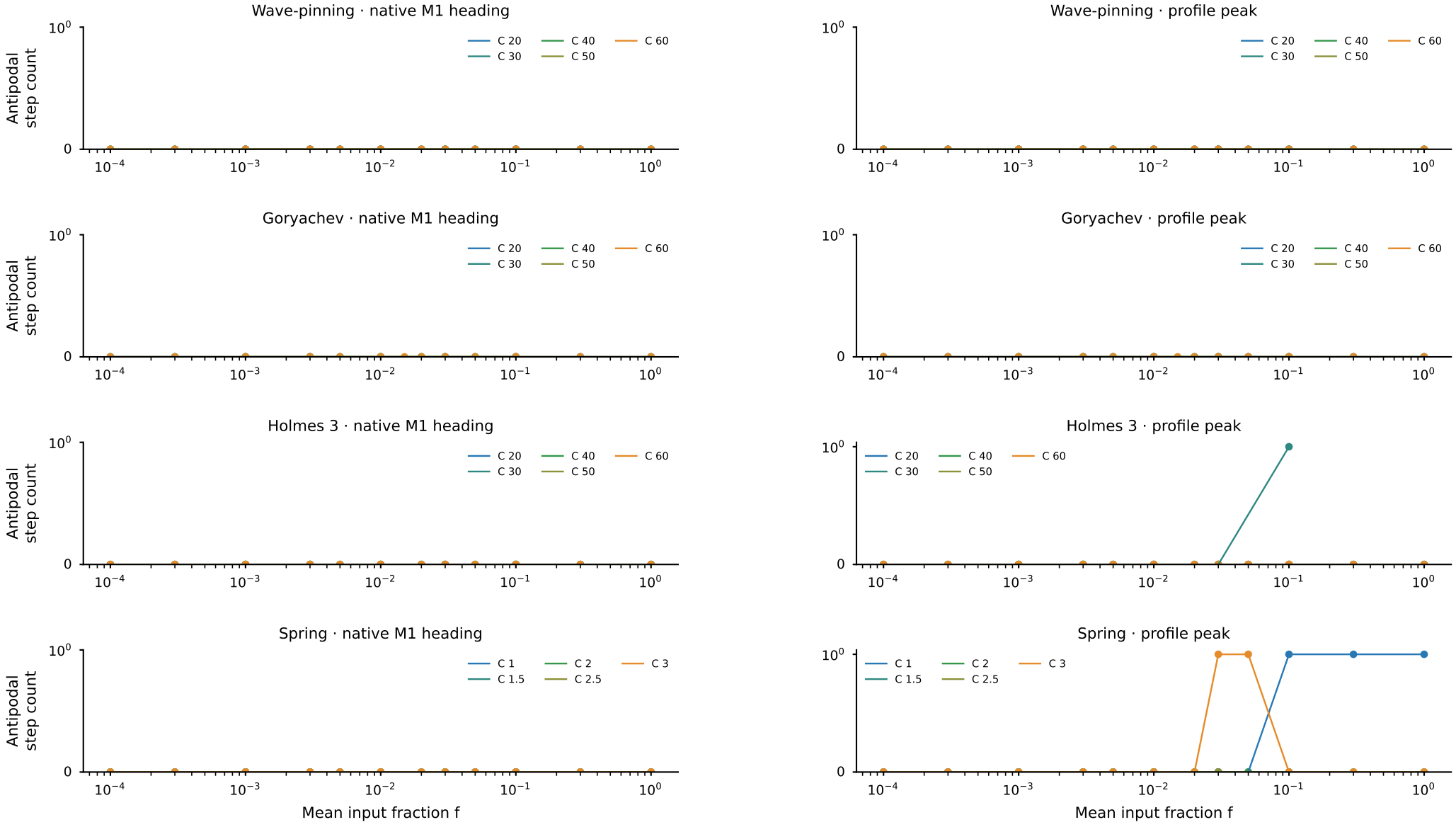
Common selected observations literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_032_S5_ambiguity_sigma1_analysis_first_half.csv

S5 · Common selected observations literal material-origin physical-site measurements · Native antipodal ambiguity · analysis second half · σ=1



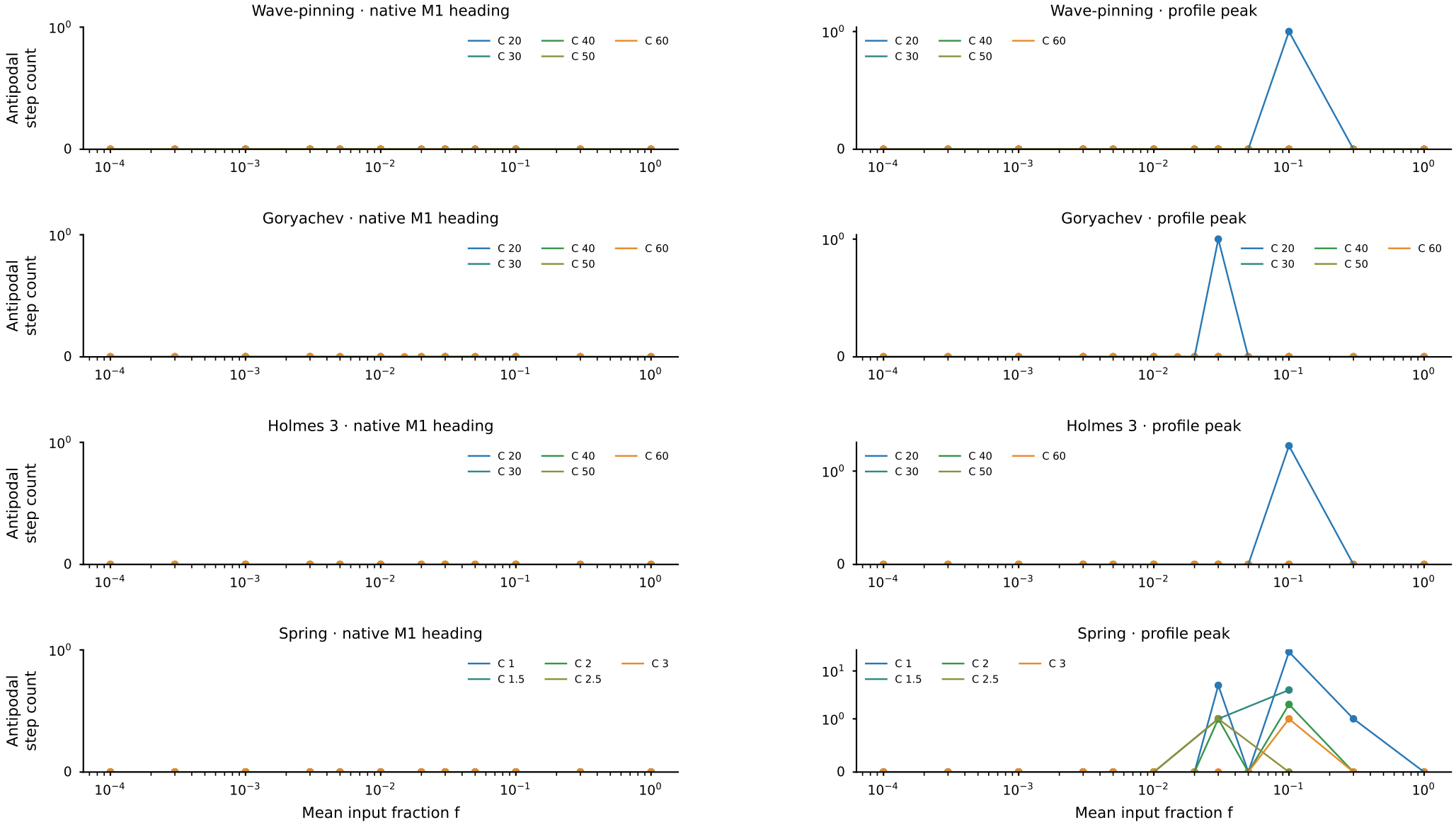
Common selected observations literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_033_S5_ambiguity_sigma1_analysis_second_half.csv

S5 · Common selected observations literal material-origin physical-site measurements · Native antipodal ambiguity · late quarter · σ=1



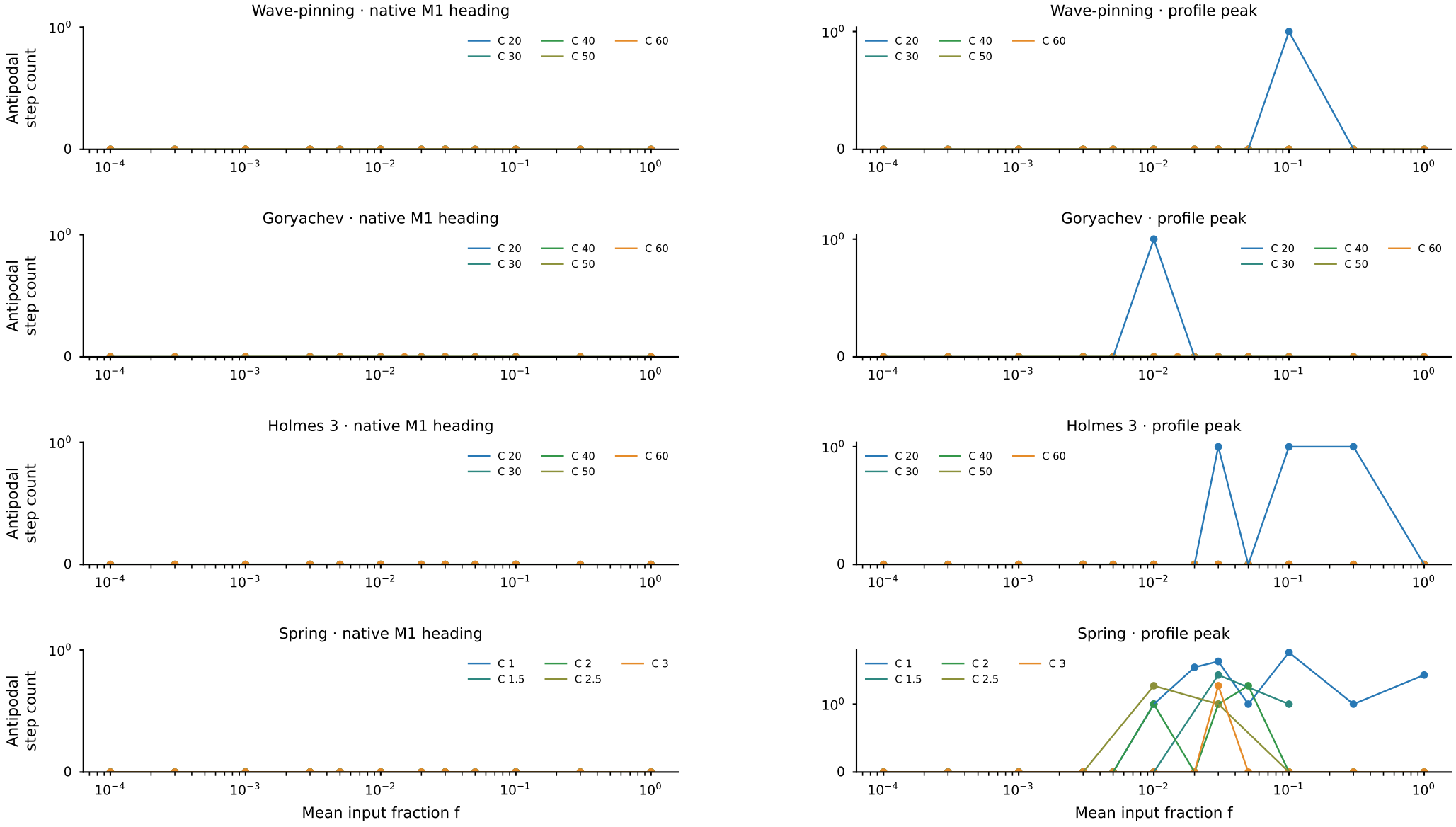
Common selected observations literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_034_S5_ambiguity_sigma1_late_quarter.csv

S5 · Common selected observations literal material-origin physical-site measurements · Native antipodal ambiguity · cadence x2 · σ=1



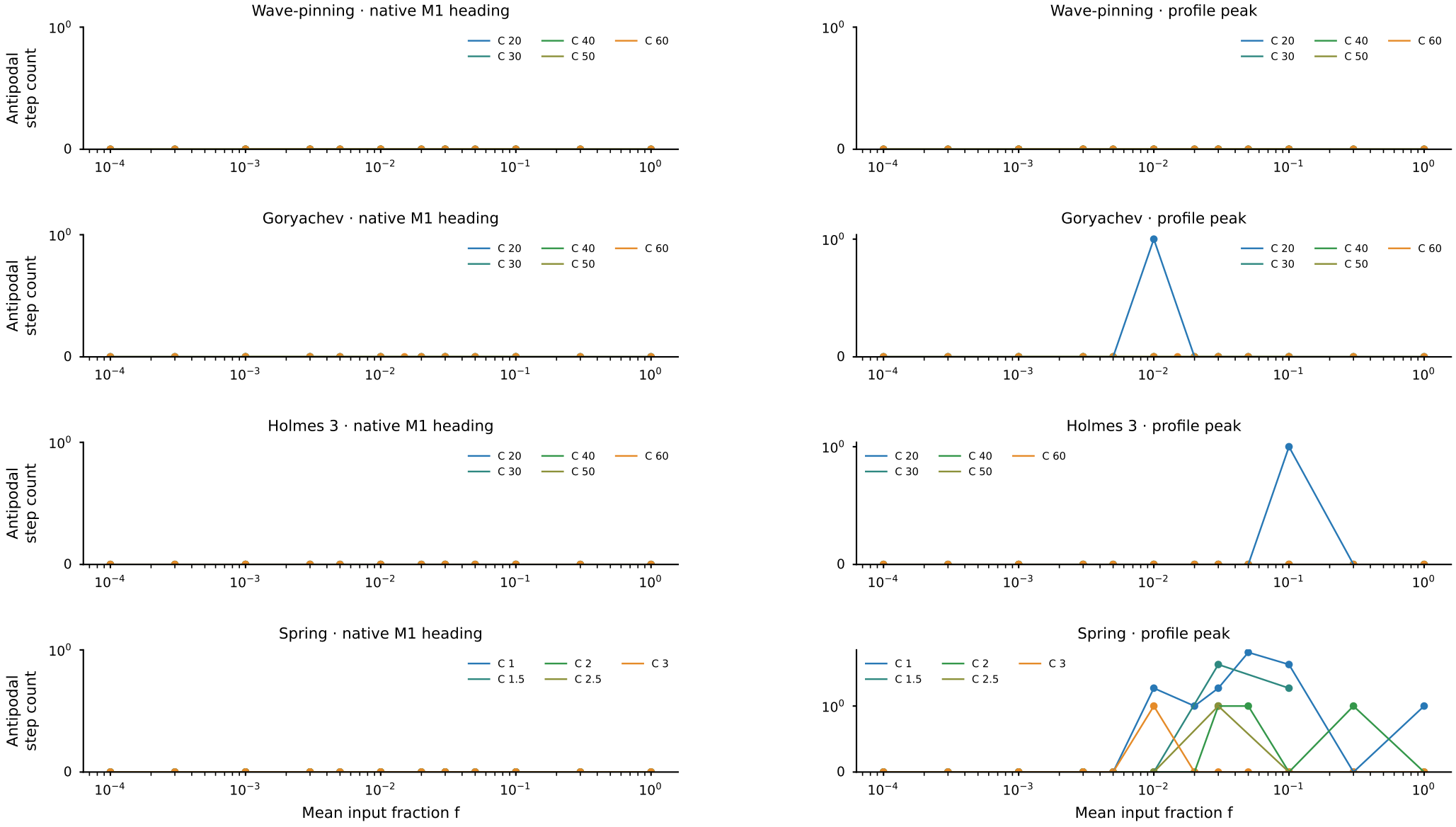
Common selected observations literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_035_S5_ambiguity_sigma1_cadence_x2.csv

S5 · Common selected observations literal material-origin physical-site measurements · Native antipodal ambiguity · cadence x4 · σ=1



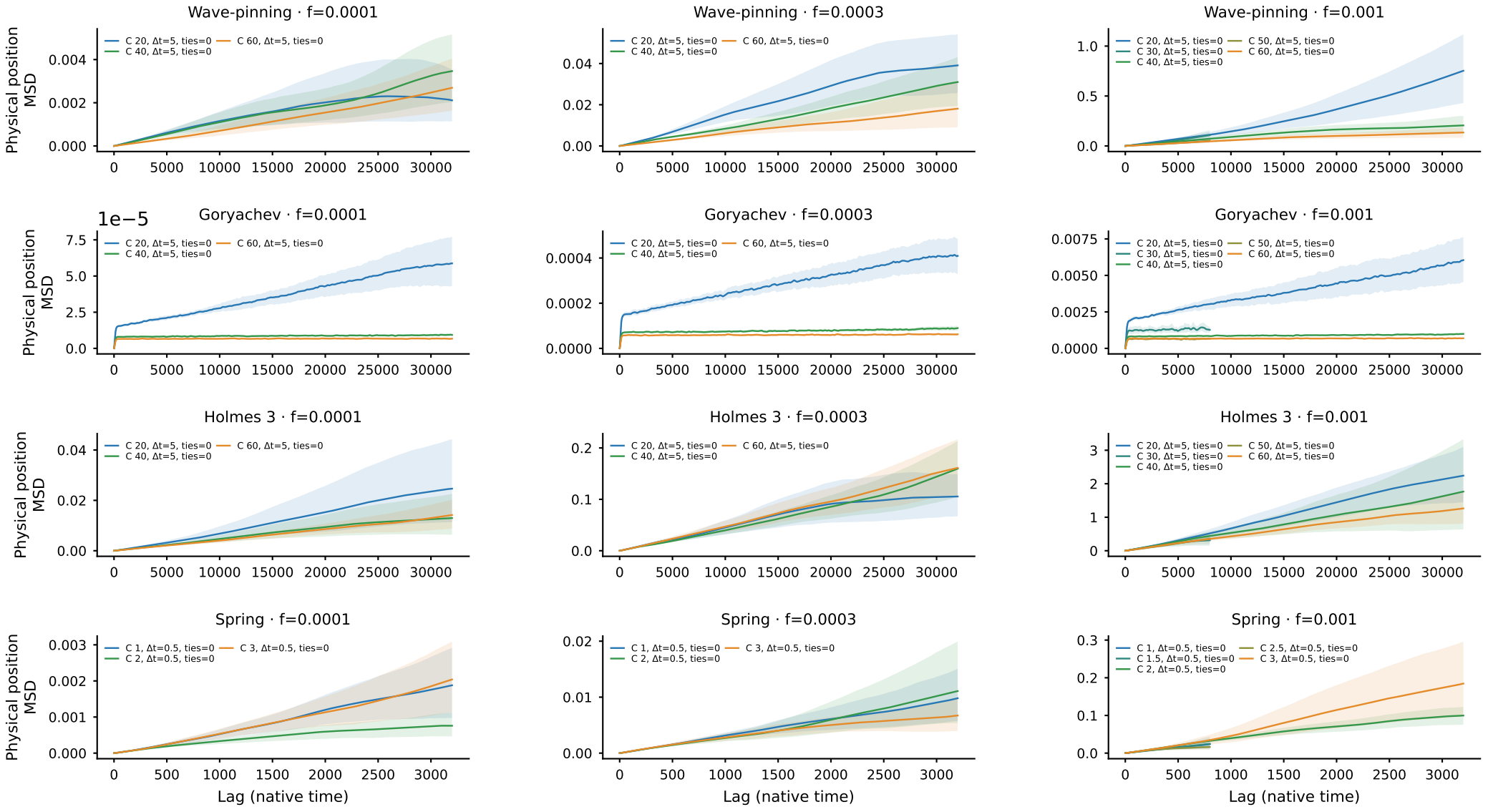
Common selected observations literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_036_S5_ambiguity_sigma1_cadence_x4.csv

S5 · Common selected observations literal material-origin physical-site measurements · Native antipodal ambiguity · cadence x8 · σ=1



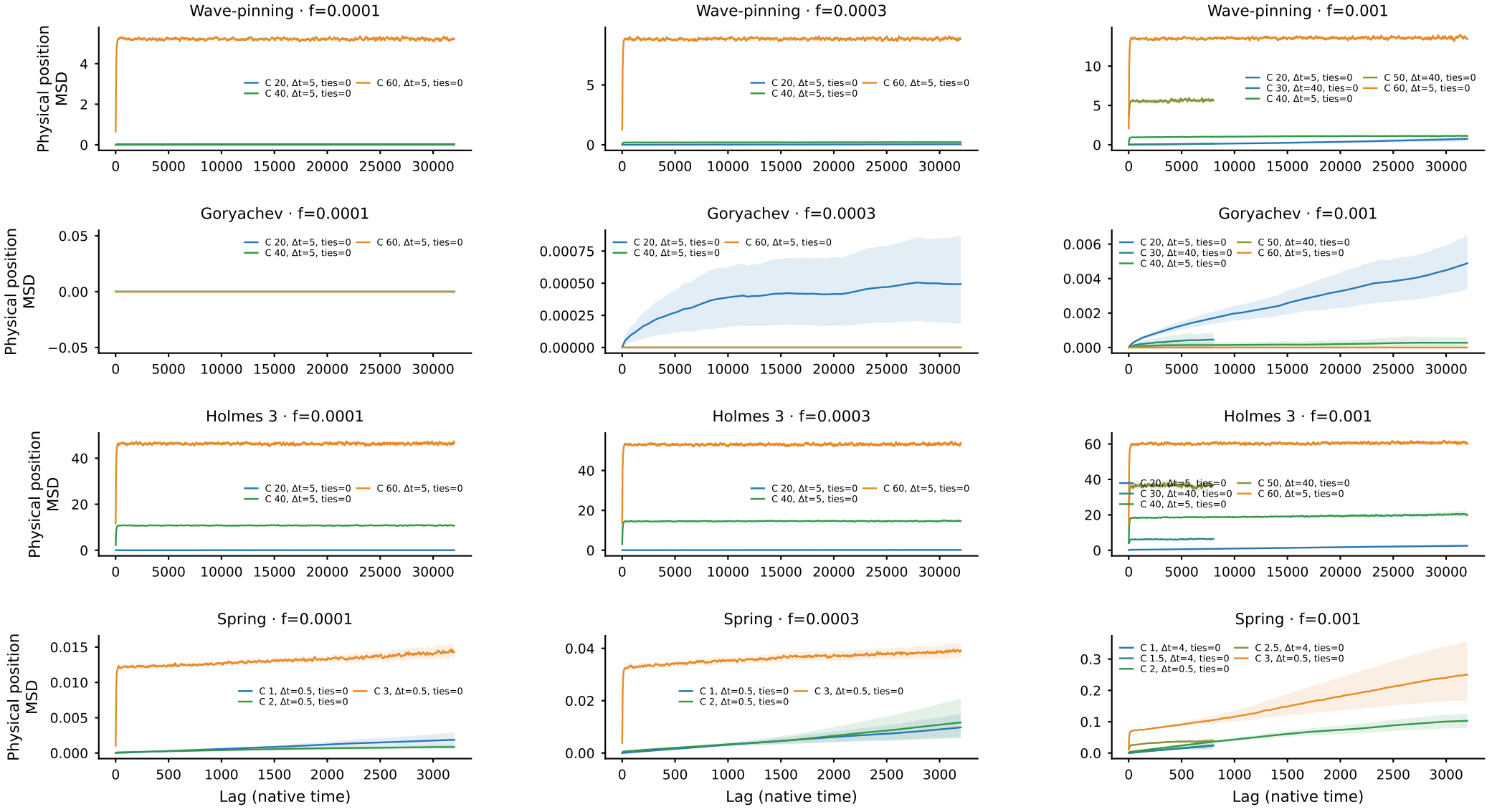
Common selected observations literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_literal_canonical_20261020_037_S5_ambiguity_sigma1_cadence_x8.csv

S5 · Longest selected observations literal material-origin physical-site measurements · Literal heading position · A3 primary split · σ=1



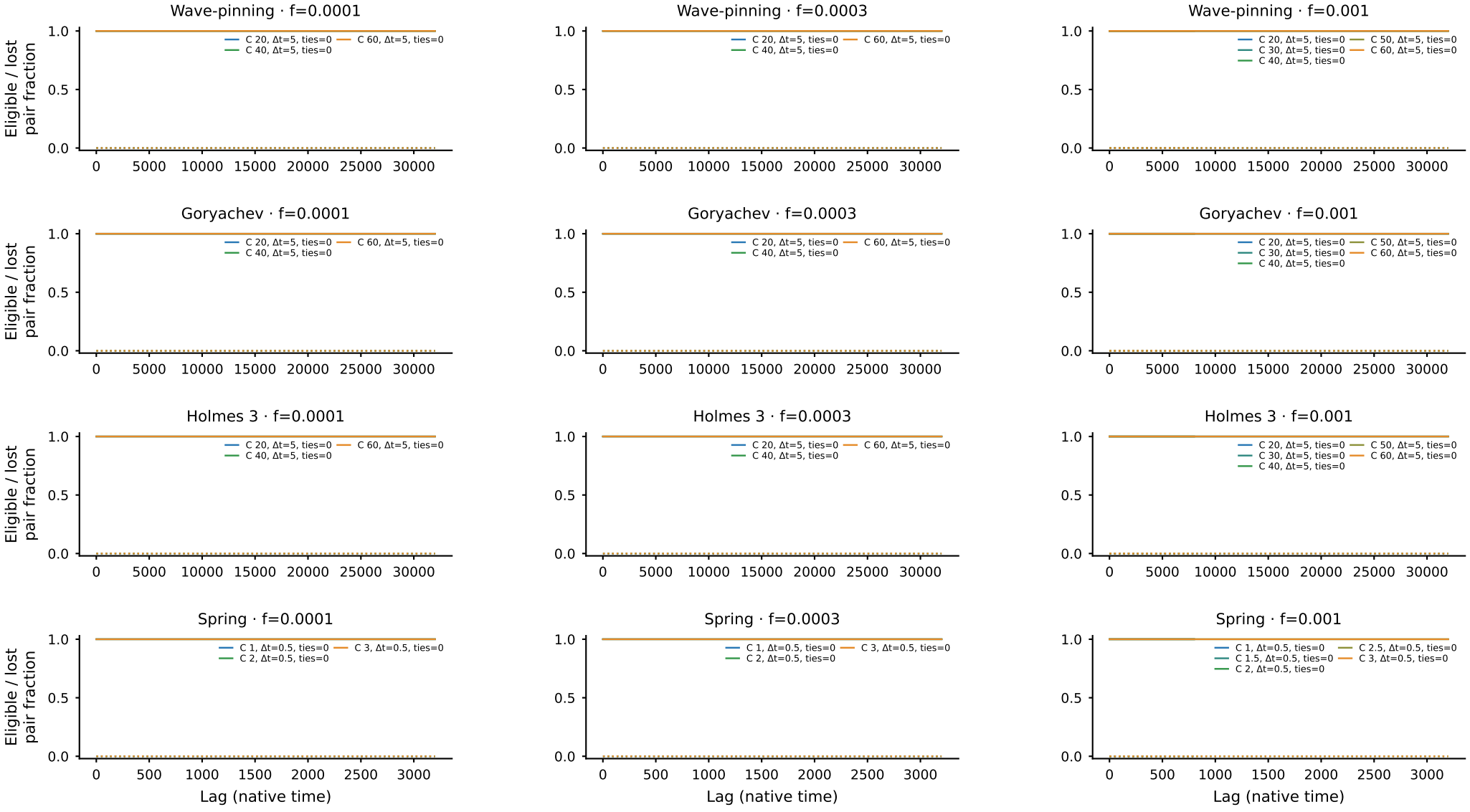
Longest selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_1,\text{basal}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_001_S5_literal_sigma1_heading_literal_actualC_MSD_1.csv.gz

S5 · Longest selected observations literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · σ=1



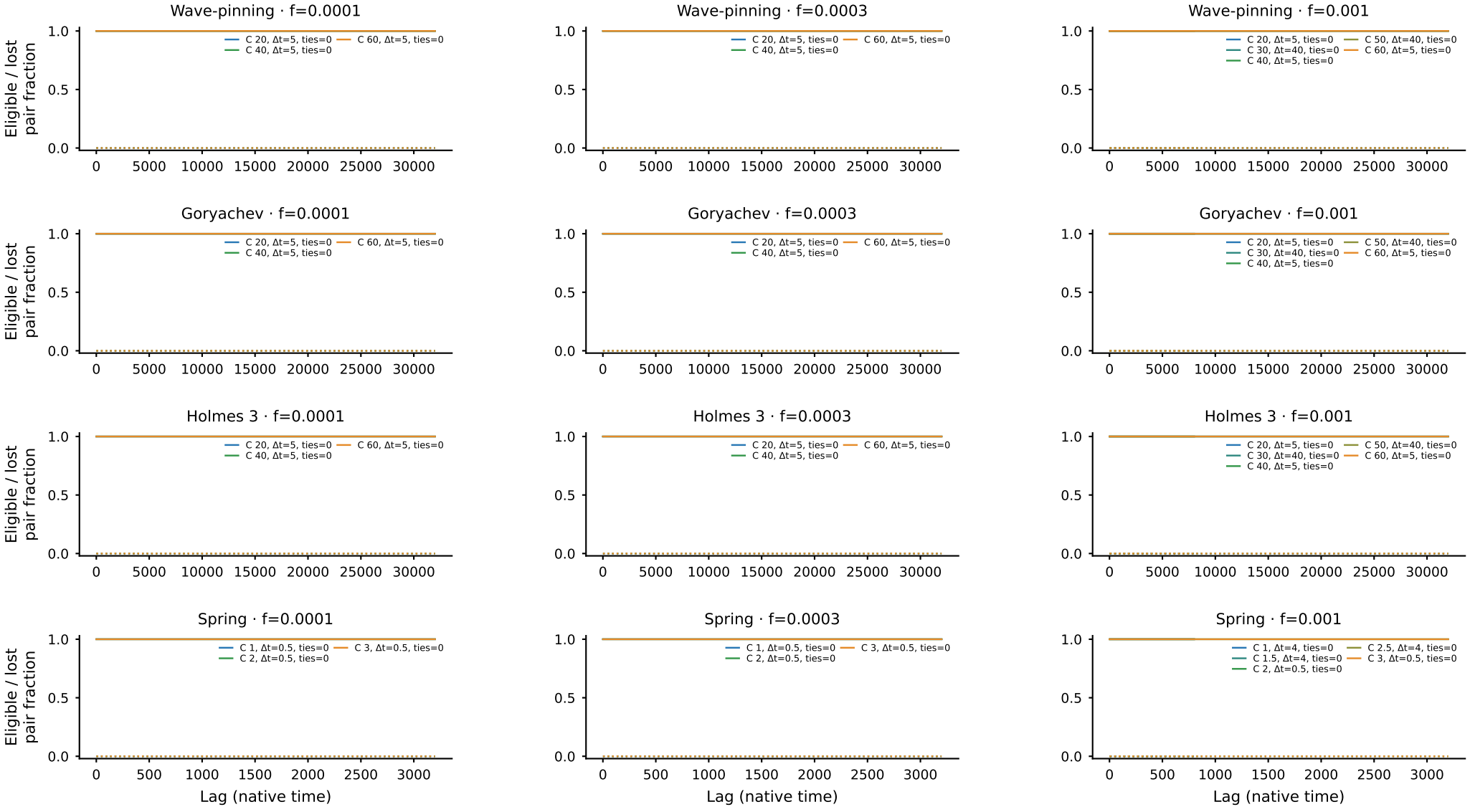
Longest selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_002_S5_literal_sigma1_profile_peak_literal_actualC_MSD_1.csv.gz

S5 · Longest selected observations literal material-origin physical-site measurements · Literal heading position · A3 primary split · pair coverage and ambiguity loss · σ=1



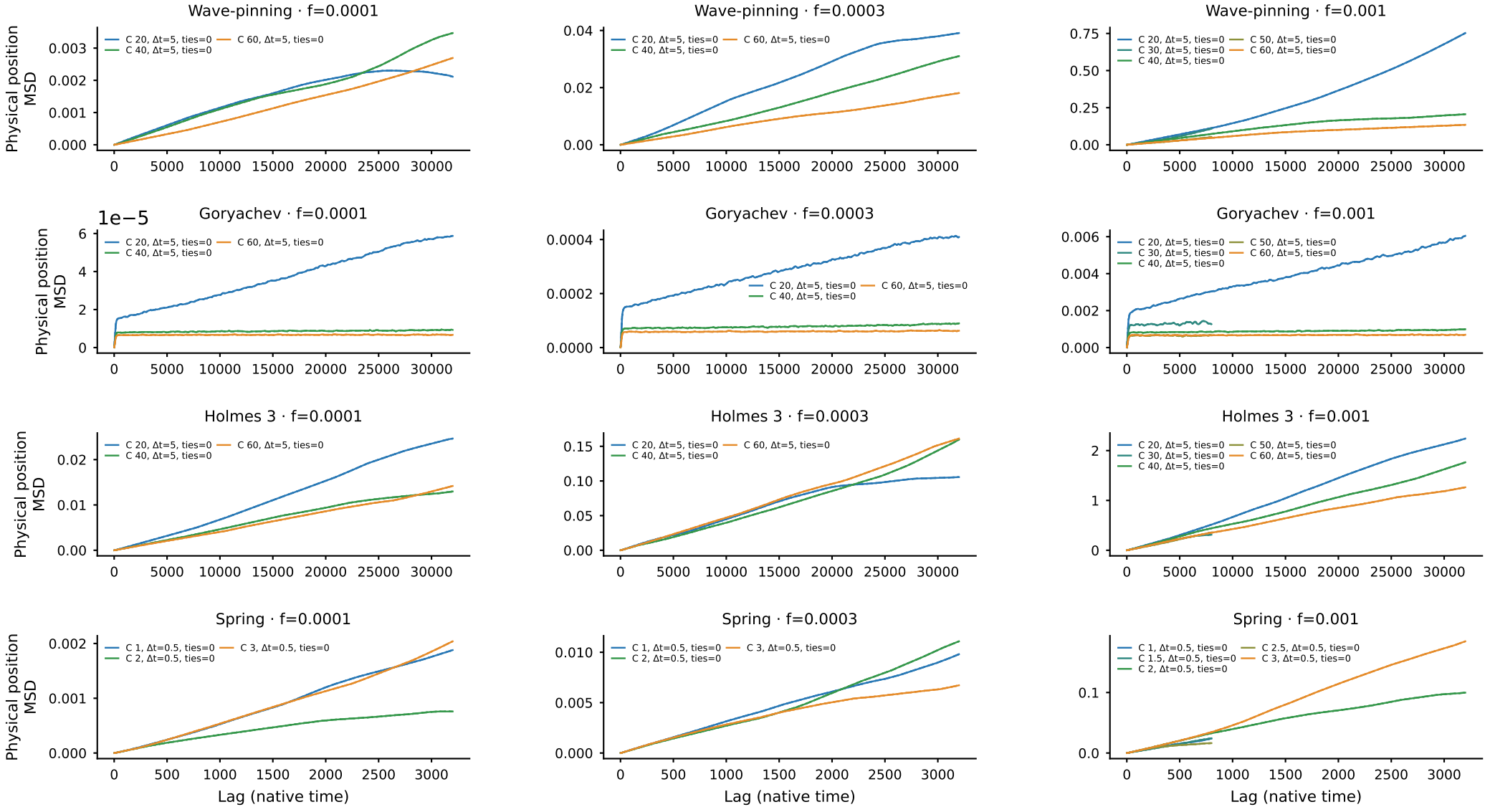
Longest selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M_1|/|M_1,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_003_S5_literal_sigma1_heading_literal_actualC_coverage_1.csv.gz

S5 · Longest selected observations literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · pair coverage and ambiguity loss · σ=1



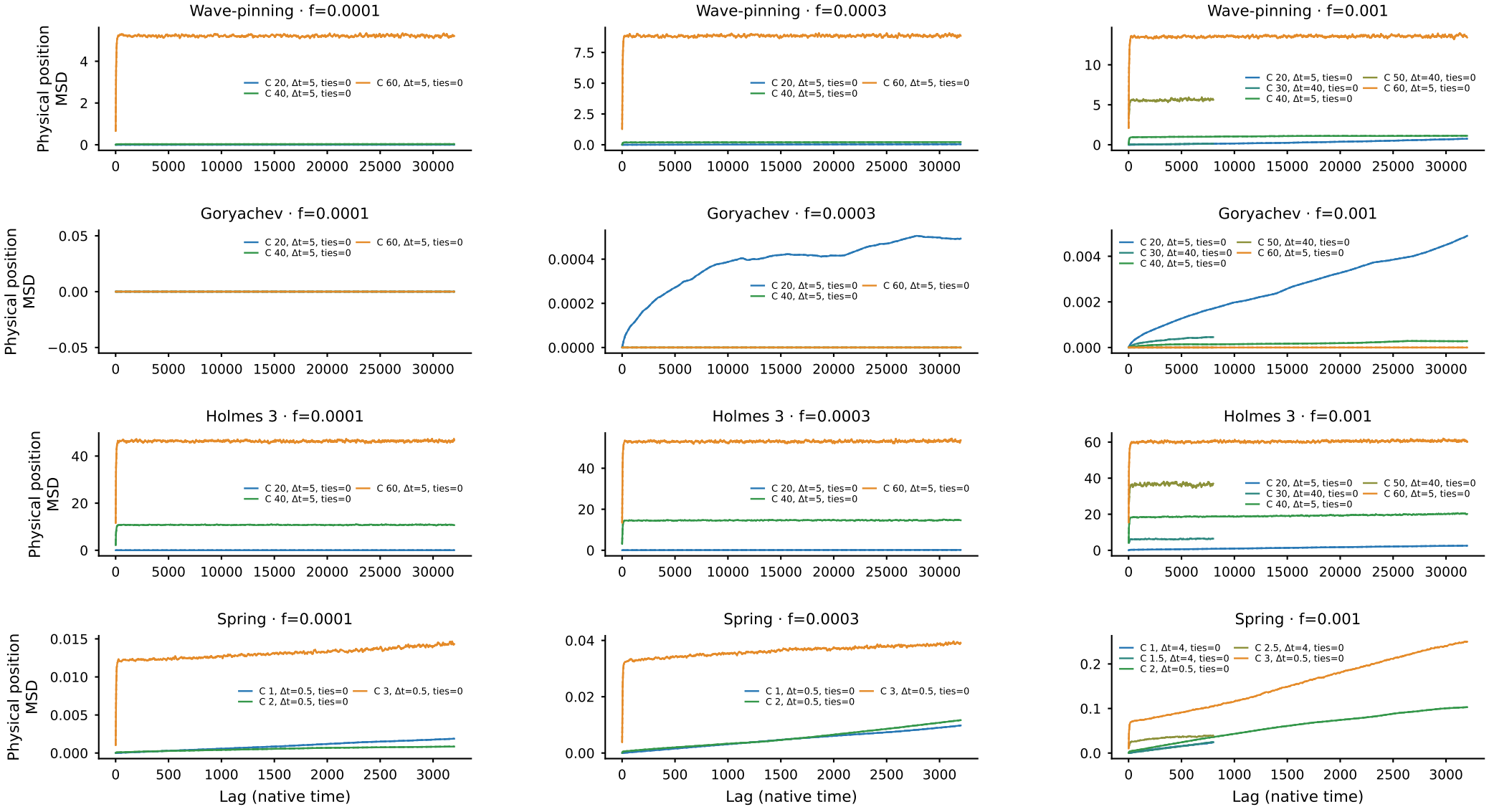
Longest selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_004_S5_literal_sigma1_profile_peak_literal_actualC_coverage_1.csv.gz

S5 · Longest selected observations literal material-origin physical-site measurements · Literal heading position · paired route sensitivities · split / +π / −π · σ=1



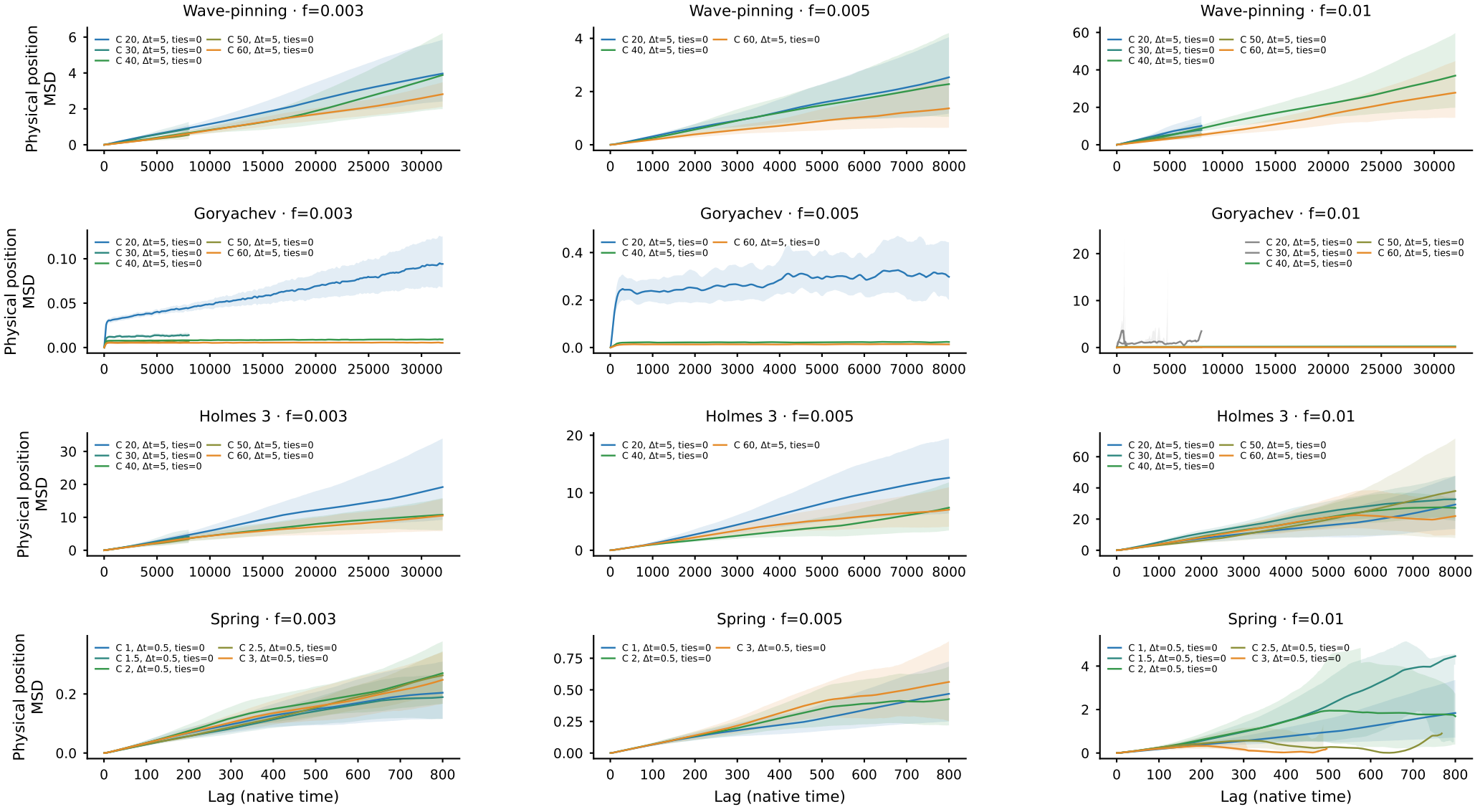
Longest selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_005_S5_literal_sigma1_heading_literal_actualC_routes_1.csv.gz

S5 · Longest selected observations literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · split / +π / −π · σ=1



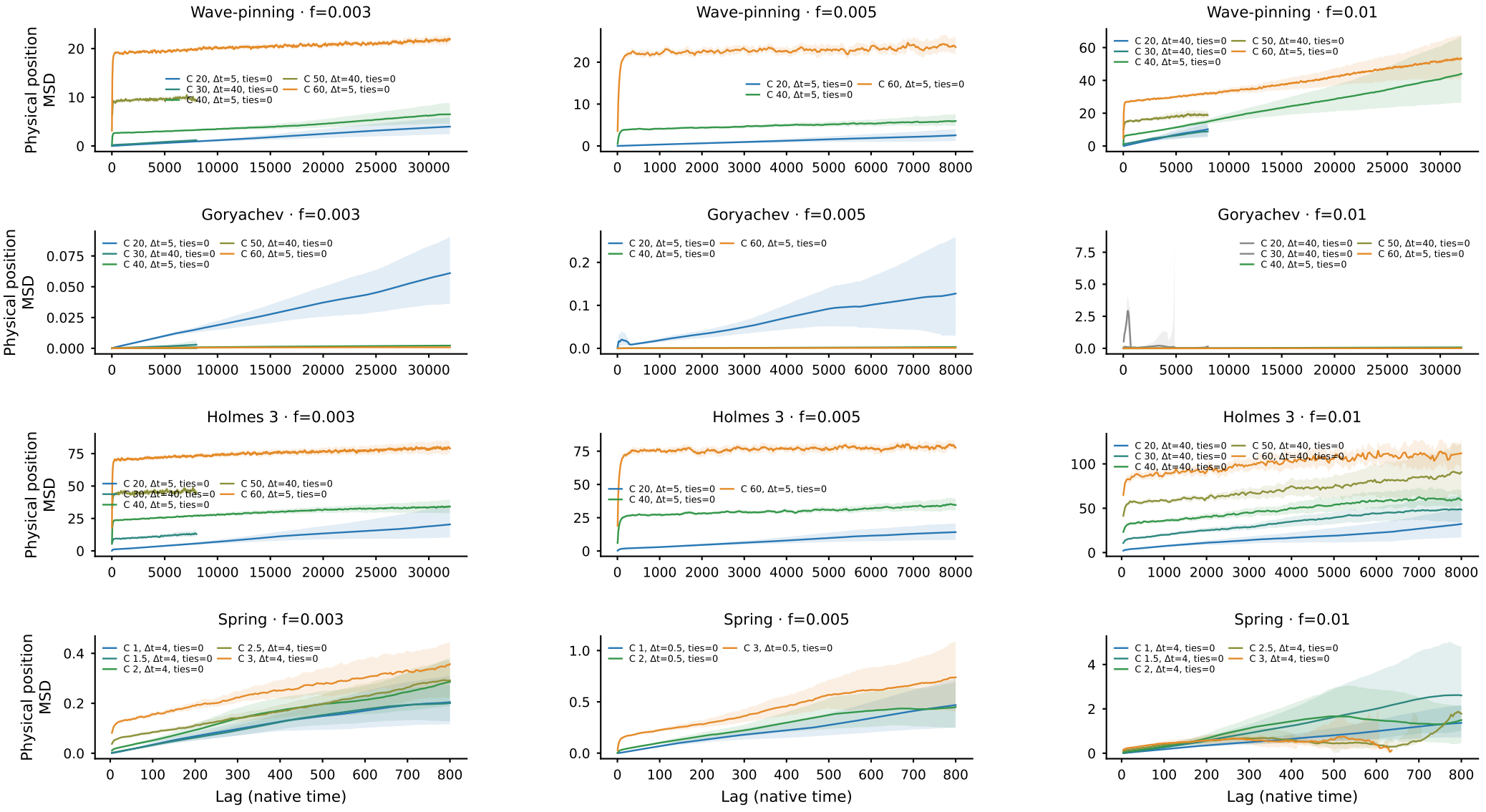
Longest selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_006_S5_literal_sigma1_profile_peak_literal_actualC_routes_1.csv.gz

S5 · Longest selected observations literal material-origin physical-site measurements · Literal heading position · A3 primary split · σ=1



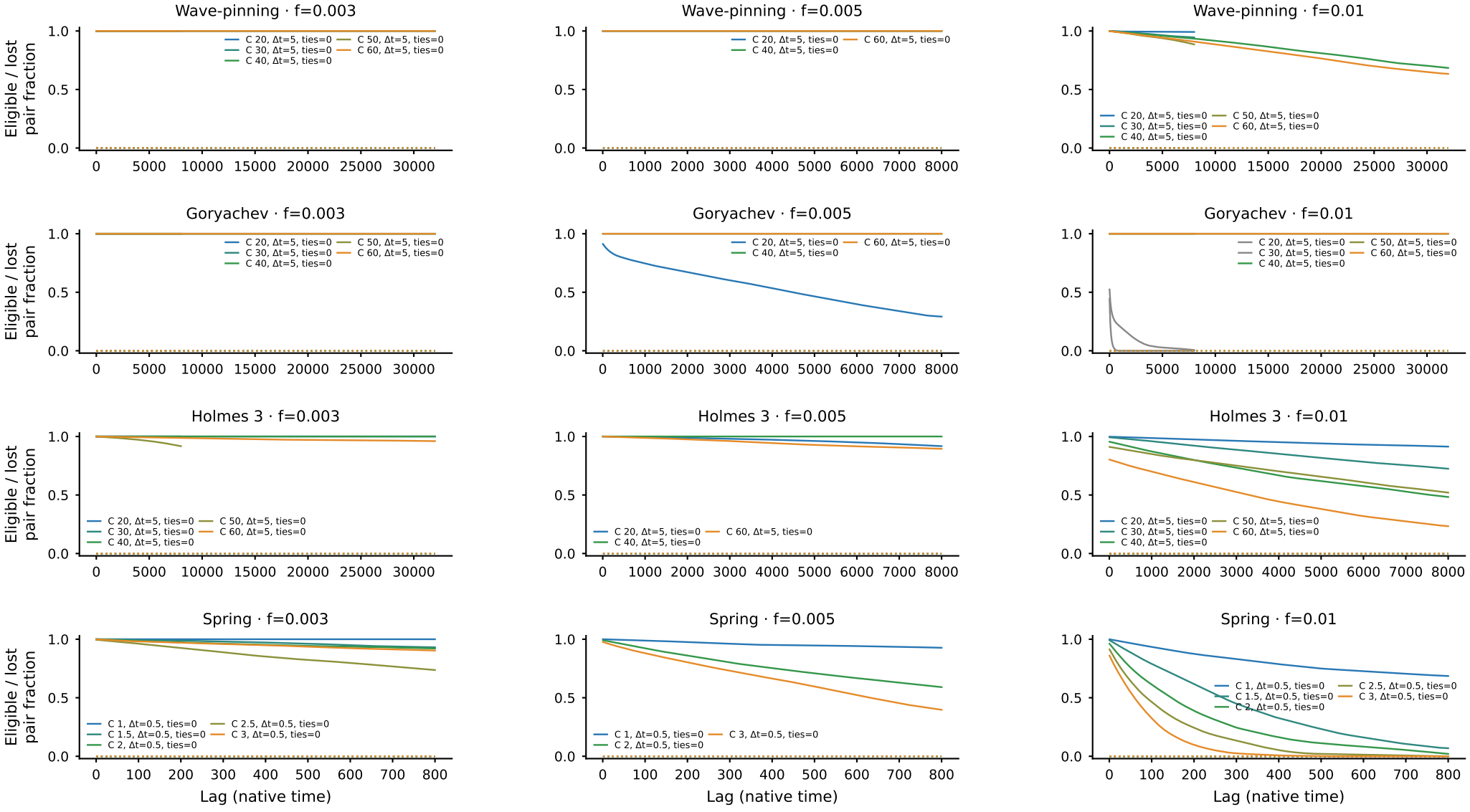
Longest selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_007_S5_literal_sigma1_heading_literal_actualC_MSD_2.csv.gz

S5 · Longest selected observations literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · σ=1



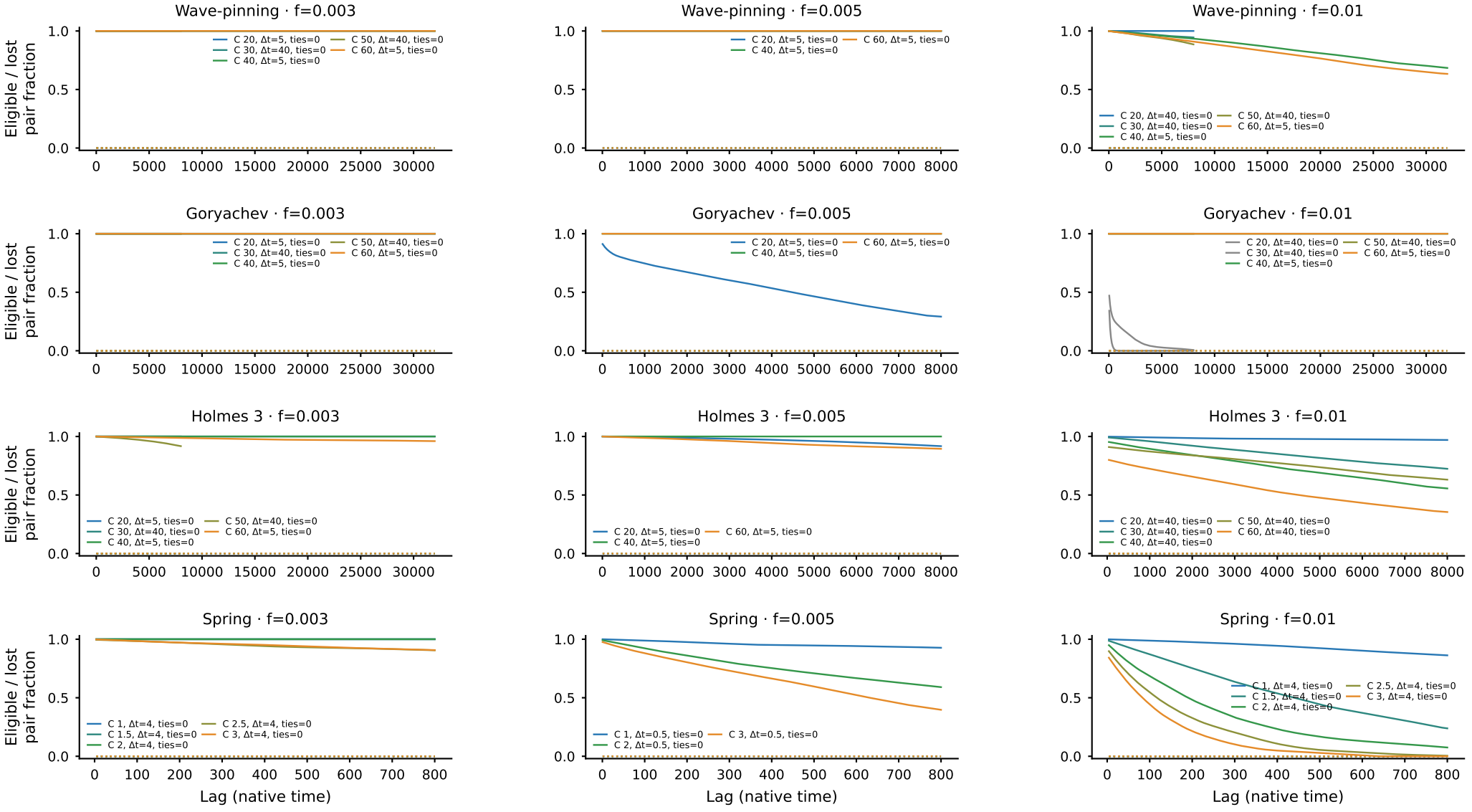
Longest selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\text{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_008_S5_literal_sigma1_profile_peak_literal_actualC_MSD_2.csv.gz

S5 · Longest selected observations literal material-origin physical-site measurements · Literal heading position · A3 primary split · pair coverage and ambiguity loss · σ=1



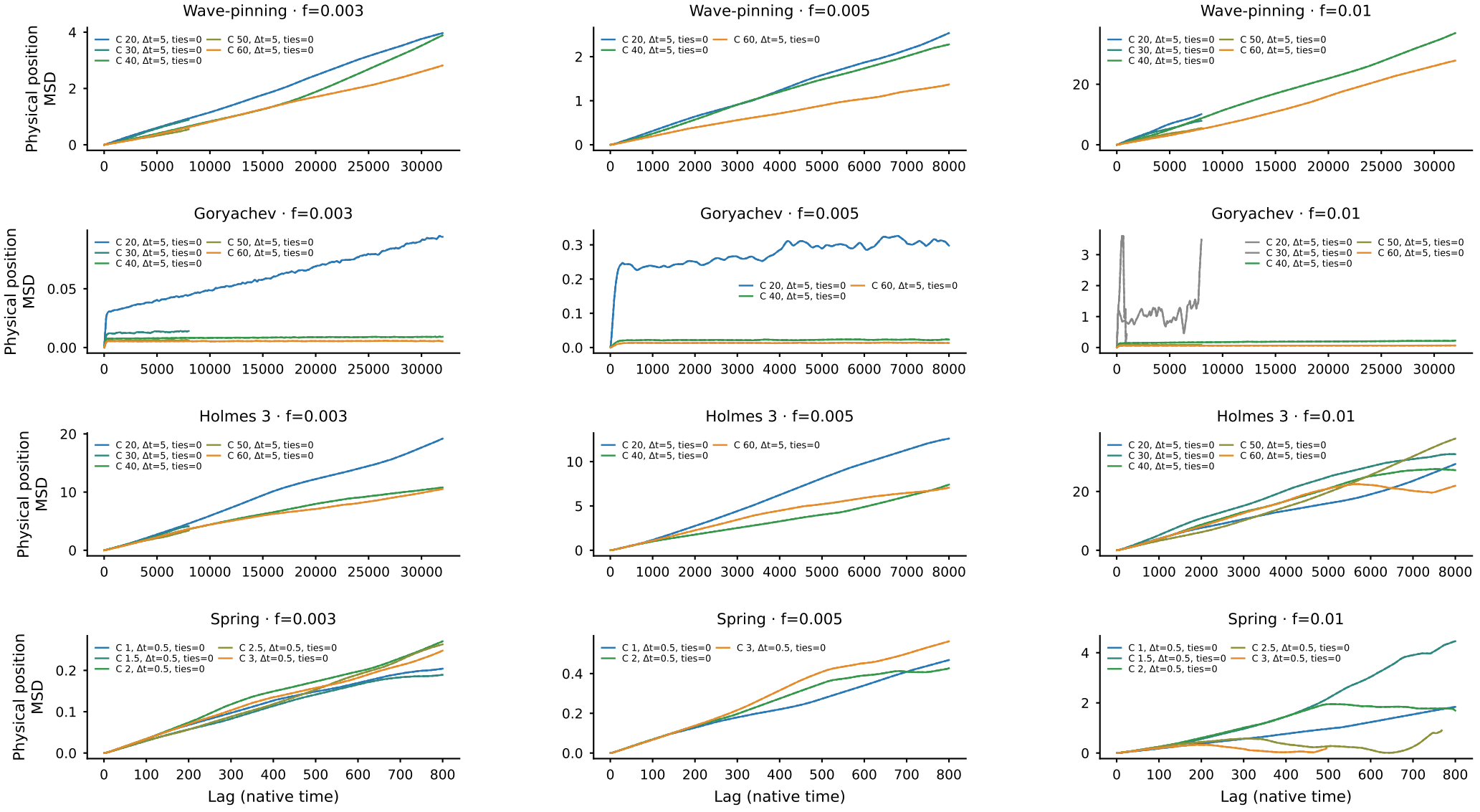
Longest selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M_1|/|M_1,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_009_S5_literal_sigma1_heading_literal_actualC_coverage_2.csv.gz

S5 · Longest selected observations literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · pair coverage and ambiguity loss · σ=1



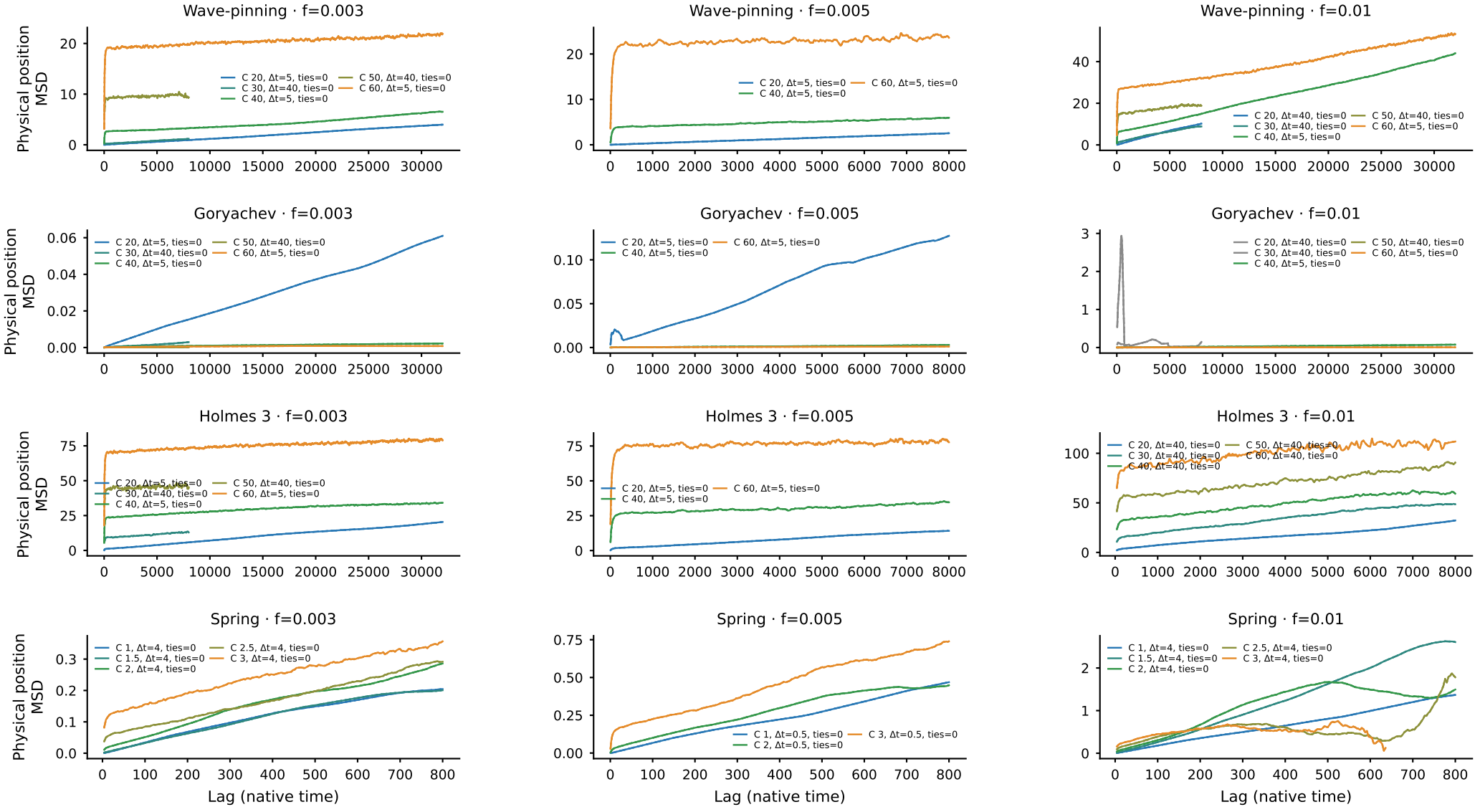
Longest selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M_1|/|M_1,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_010_S5_literal_sigma1_profile_peak_literal_actualC_coverage_2.csv.gz

S5 · Longest selected observations literal material-origin physical-site measurements · Literal heading position · paired route sensitivities · split / +π / −π · σ=1



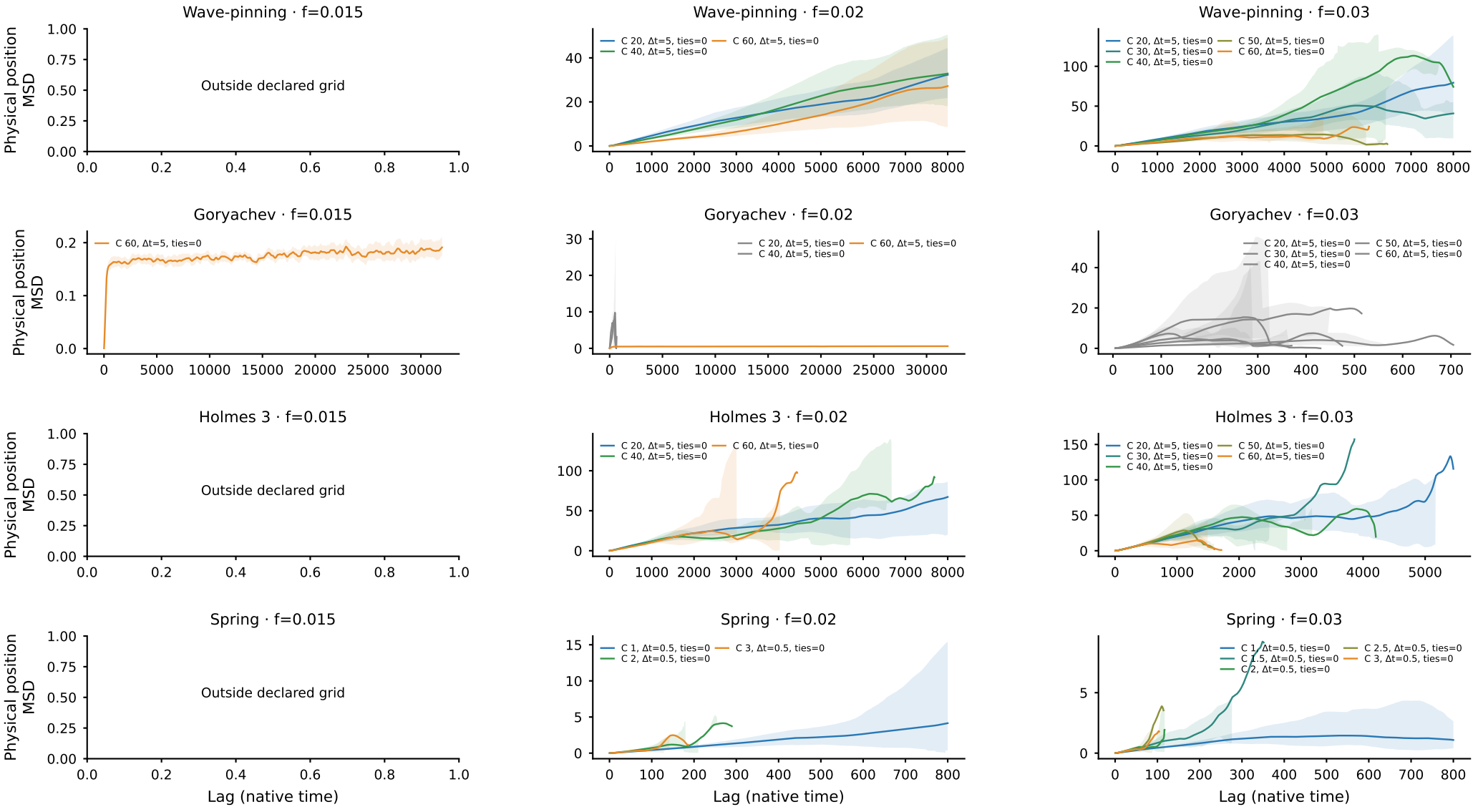
Longest selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_011_S5_literal_sigma1_heading_literal_actualC_routes_2.csv.gz

S5 · Longest selected observations literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · split / +π / −π · σ=1



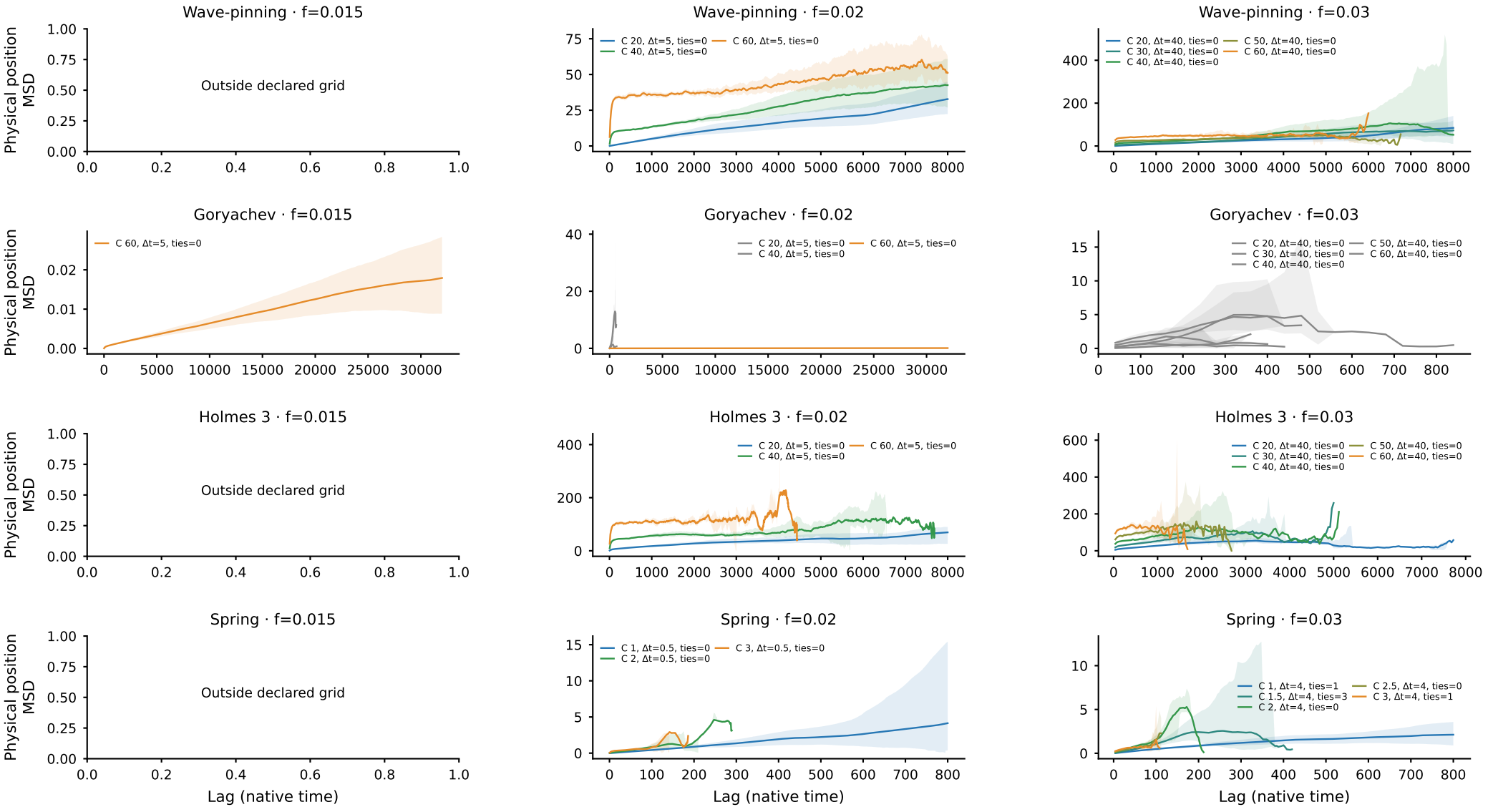
Longest selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_012_S5_literal_sigma1_profile_peak_literal_actualC_routes_2.csv.gz

S5 · Longest selected observations literal material-origin physical-site measurements · Literal heading position · A3 primary split · σ=1



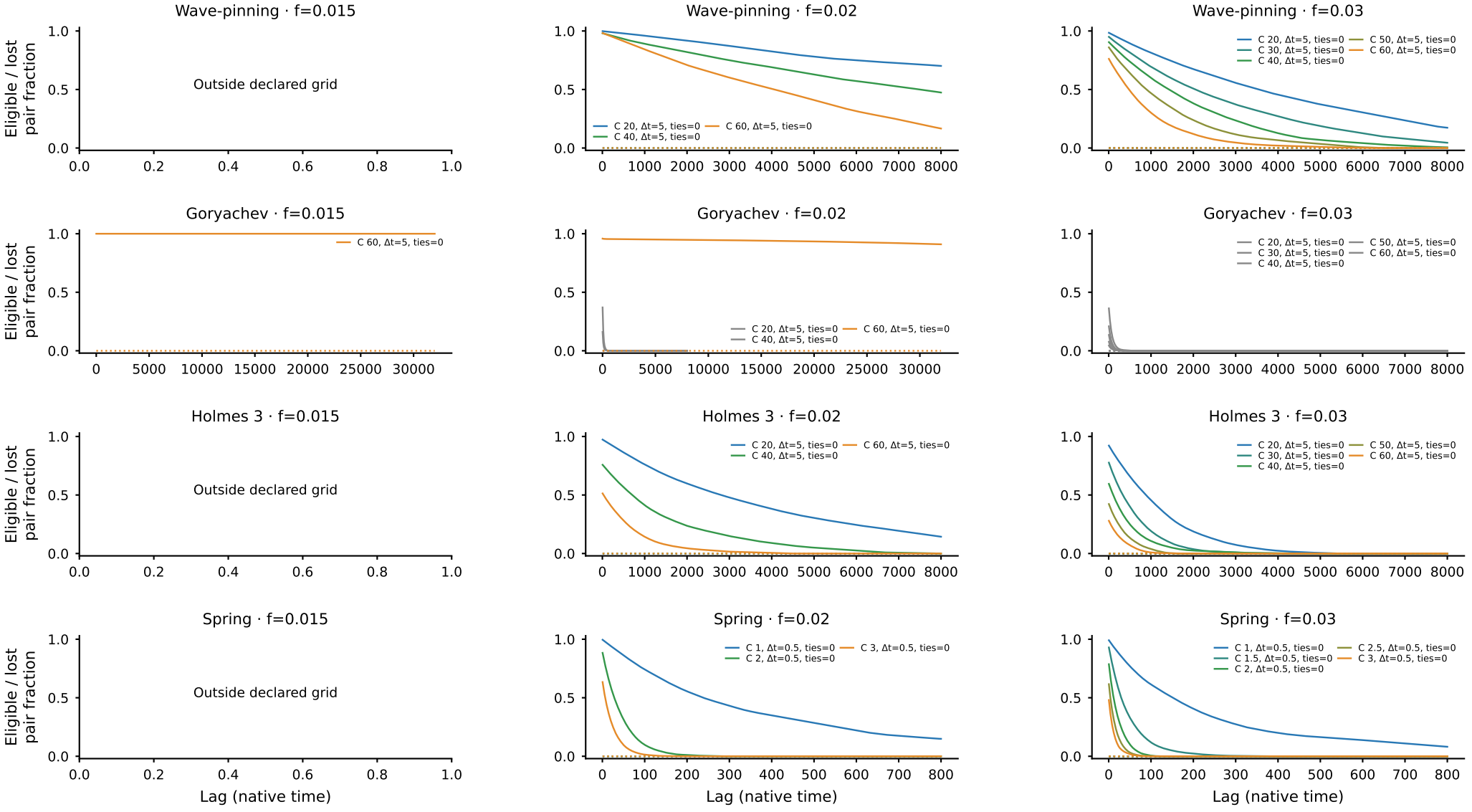
Longest selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_013_S5_literal_sigma1_heading_literal_actualC_MSD_3.csv.gz

S5 · Longest selected observations literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · σ=1



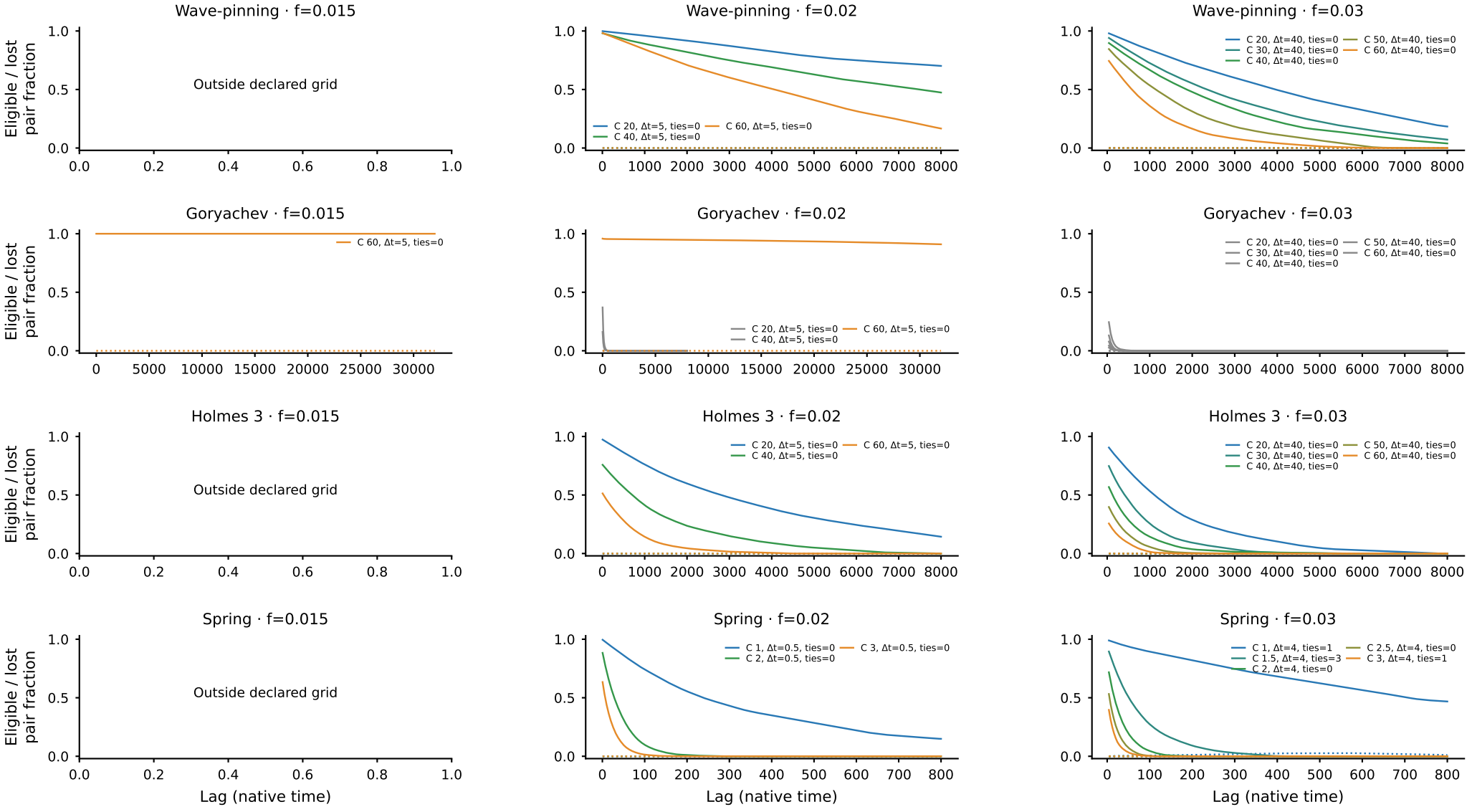
Longest selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M_1|/|M_1,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_014_S5_literal_sigma1_profile_peak_literal_actualC_MSD_3.csv

S5 · Longest selected observations literal material-origin physical-site measurements · Literal heading position · A3 primary split · pair coverage and ambiguity loss · σ=1



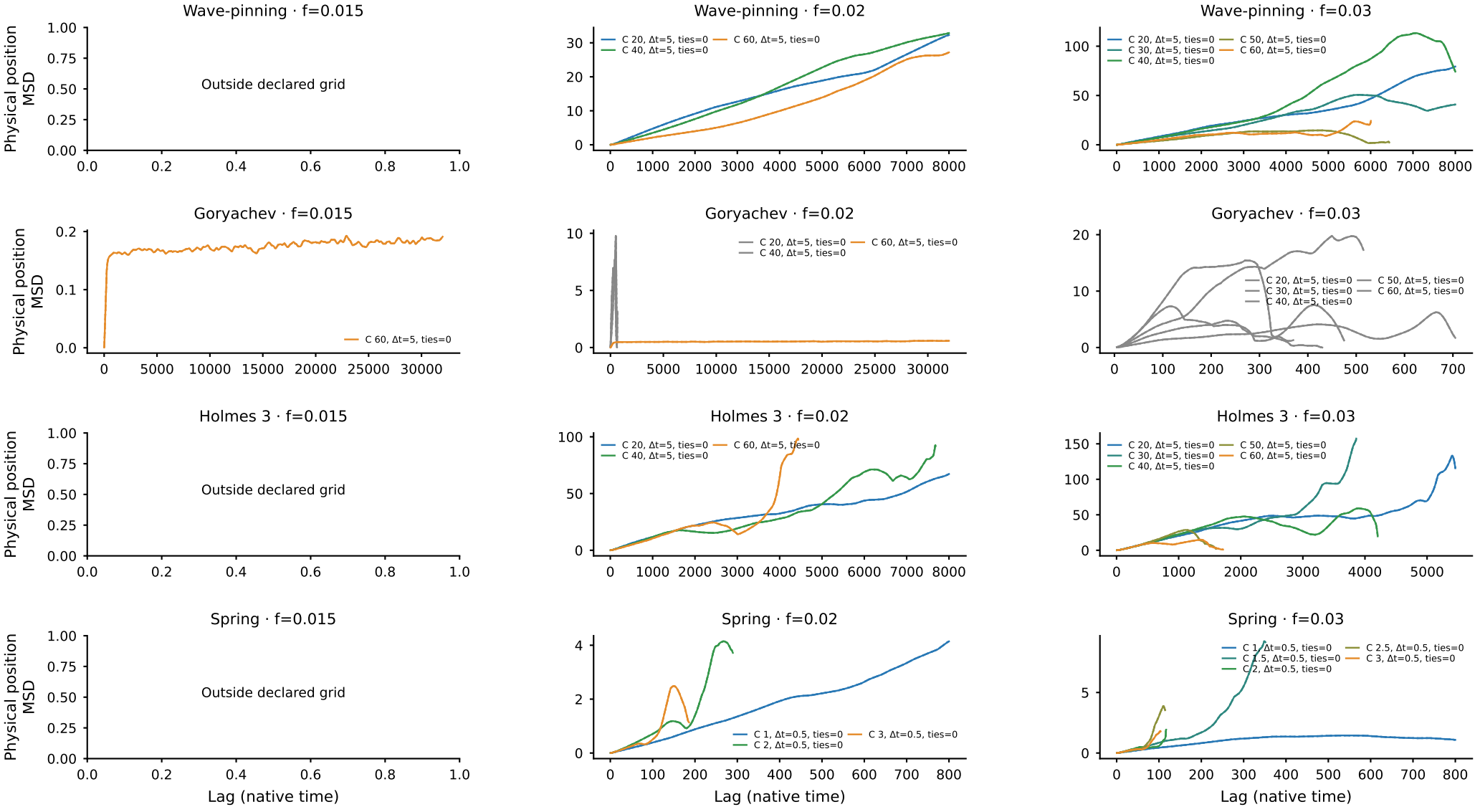
Longest selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_1,\text{basal}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_015_S5_literal_sigma1_heading_literal_actualC_coverage_3.csv.gz

S5 · Longest selected observations literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · pair coverage and ambiguity loss · σ=1



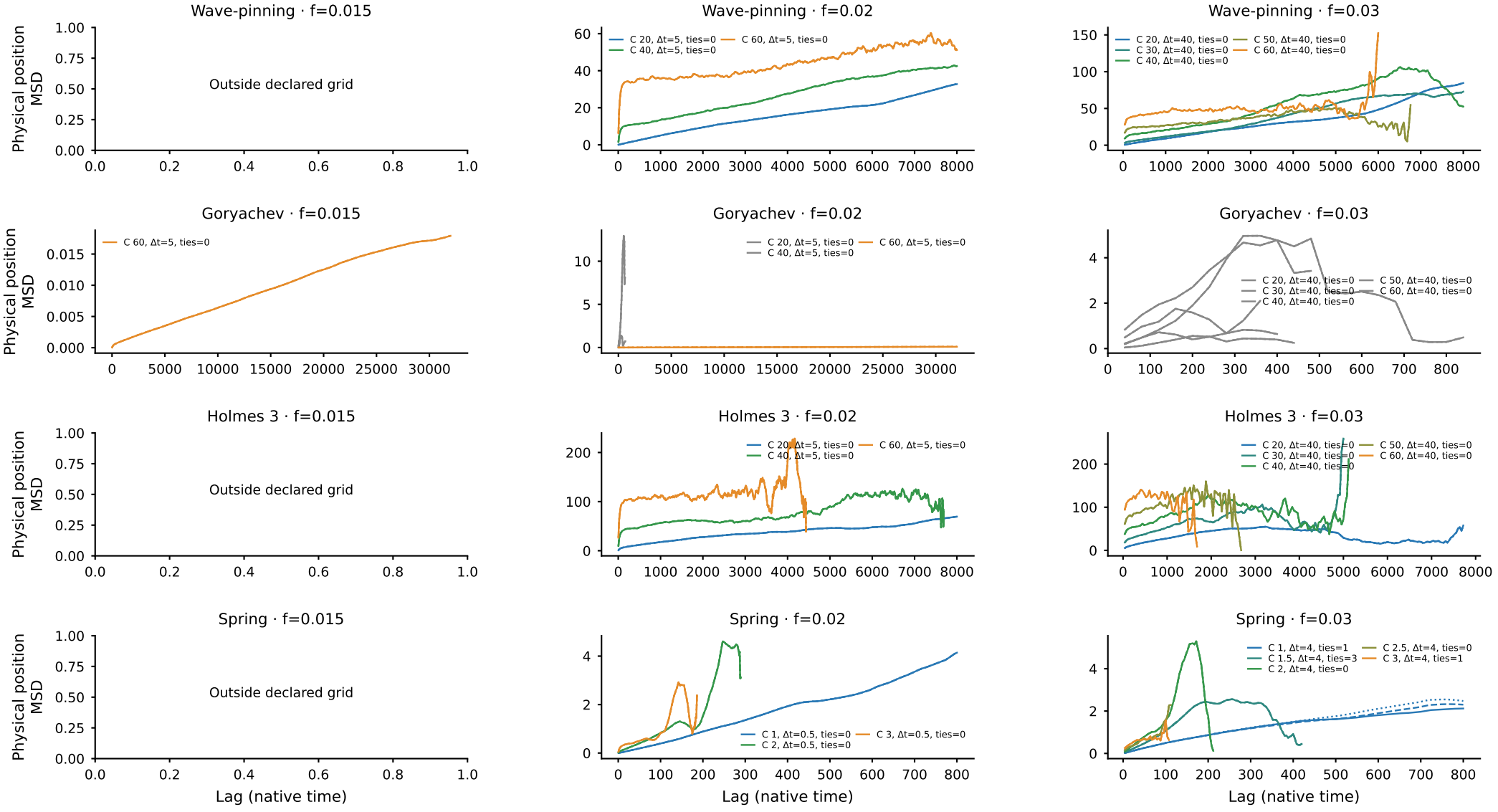
Longest selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M_1|/|M_1,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_016_S5_literal_sigma1_profile_peak_literal_actualC_coverage_3.csv

S5 · Longest selected observations literal material-origin physical-site measurements · Literal heading position · paired route sensitivities · split / +π / −π · σ=1



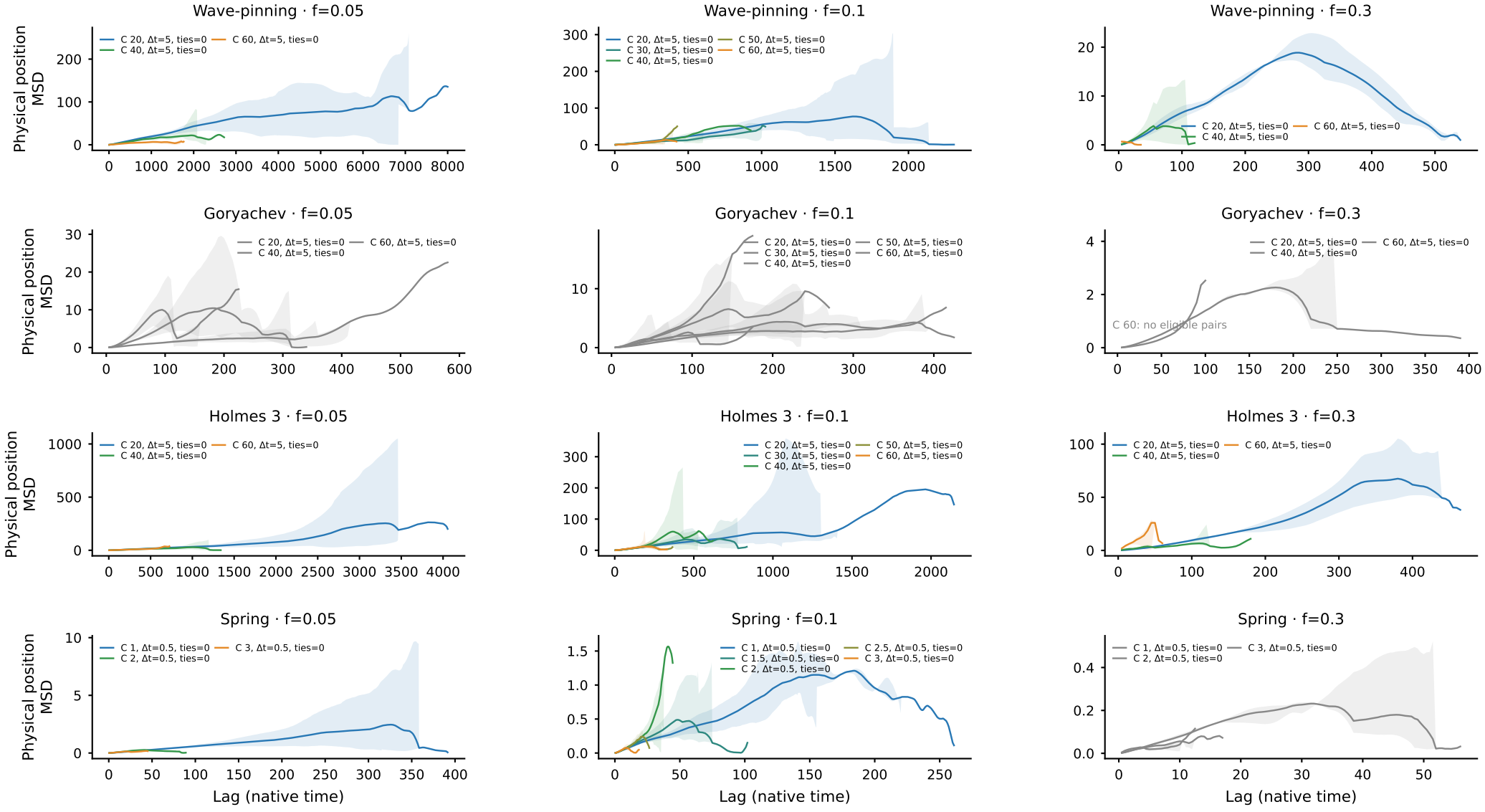
Longest selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M₁|/|M₁,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_017_S5_literal_sigma1_heading_literal_actualC_routes_3.csv.gz

S5 · Longest selected observations literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · split / +π / −π · σ=1



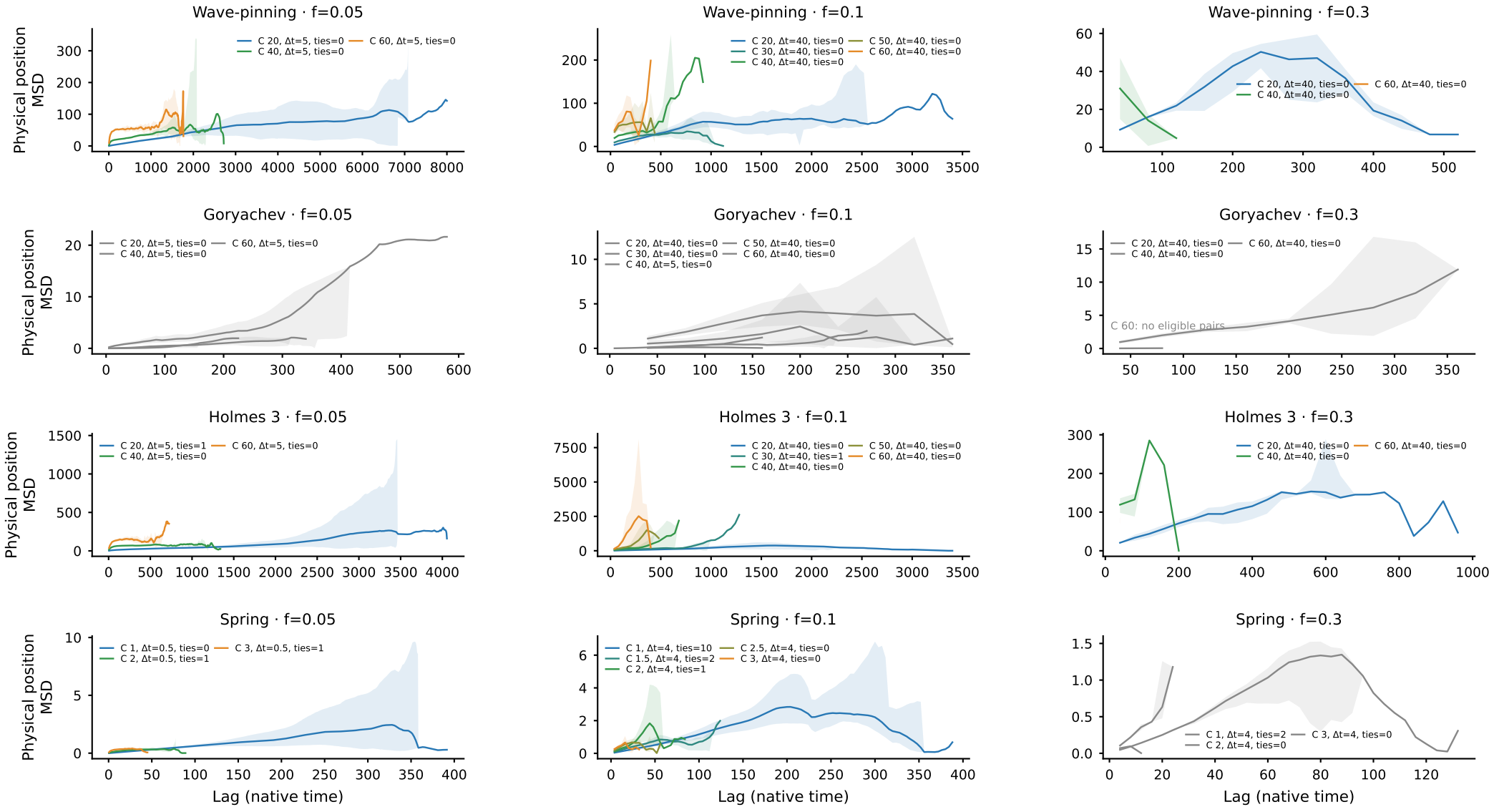
Longest selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M_1|/|M_1,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_018_S5_literal_sigma1_profile_peak_literal_actualC_routes_3.csv.gz

S5 · Longest selected observations literal material-origin physical-site measurements · Literal heading position · A3 primary split · σ=1



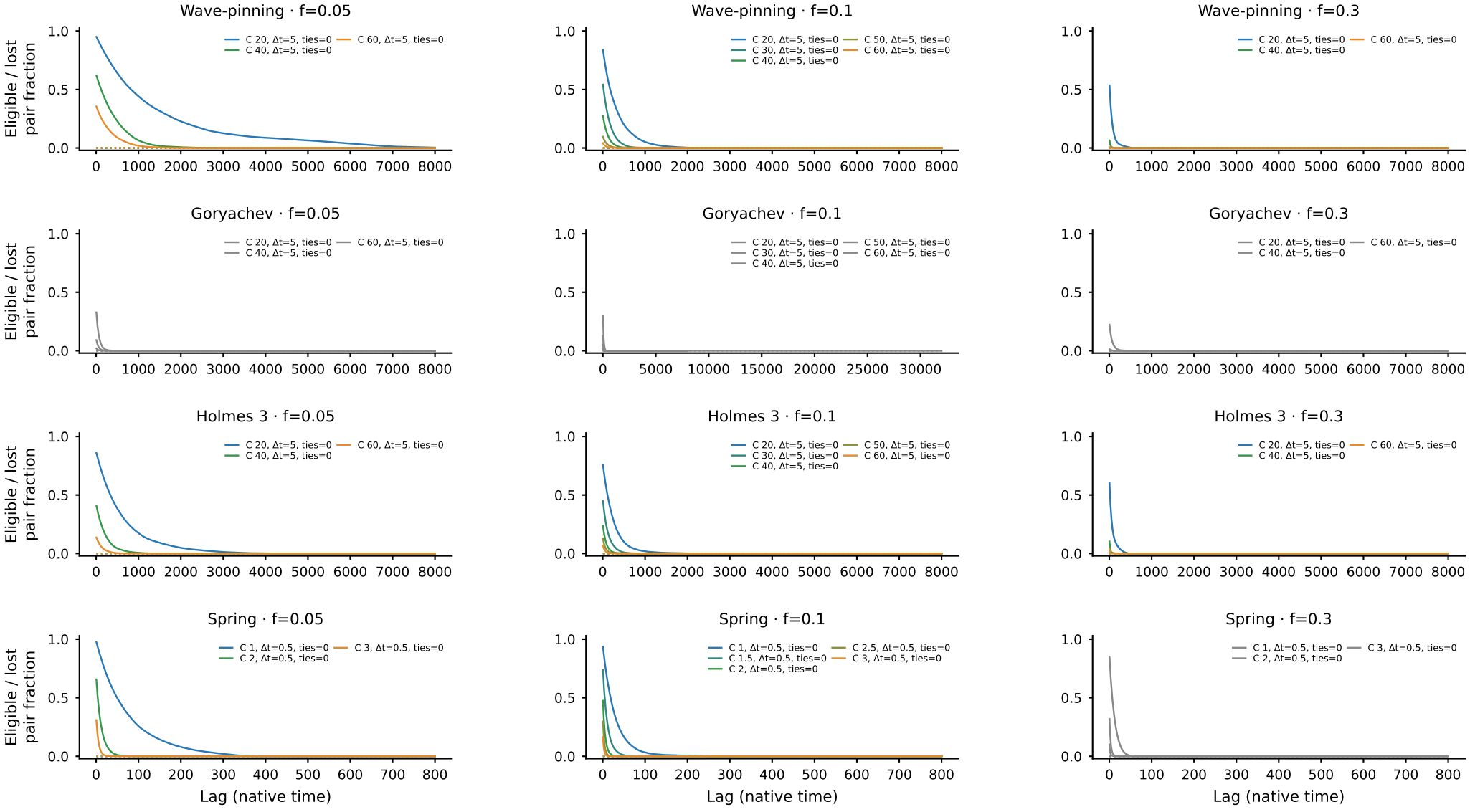
Longest selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_019_S5_literal_sigma1_heading_literal_actualC_MSD_4.csv.gz

S5 · Longest selected observations literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · σ=1



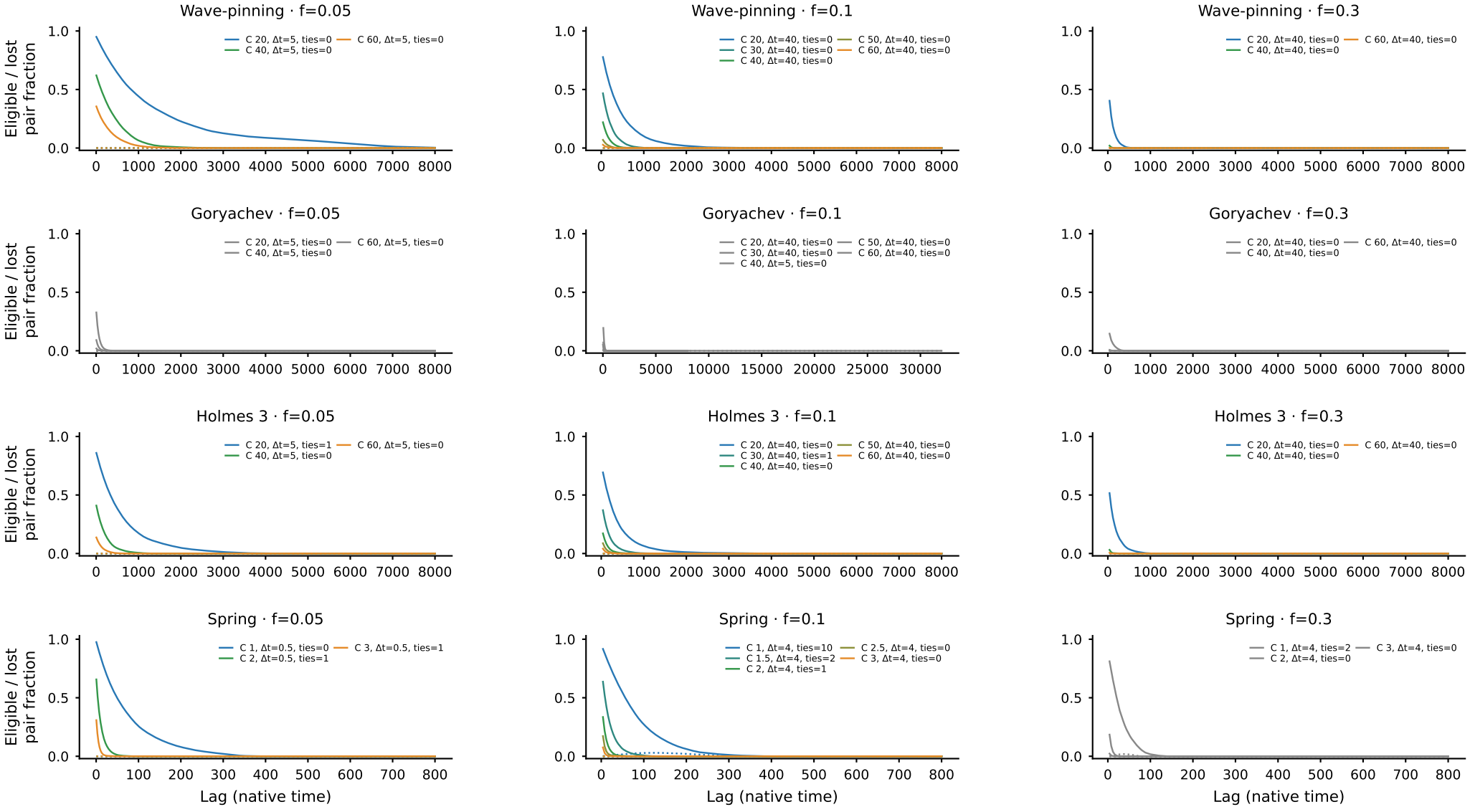
Longest selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_020_S5_literal_sigma1_profile_peak_literal_actualC_MSD_4.csv

S5 · Longest selected observations literal material-origin physical-site measurements · Literal heading position · A3 primary split · pair coverage and ambiguity loss · σ=1



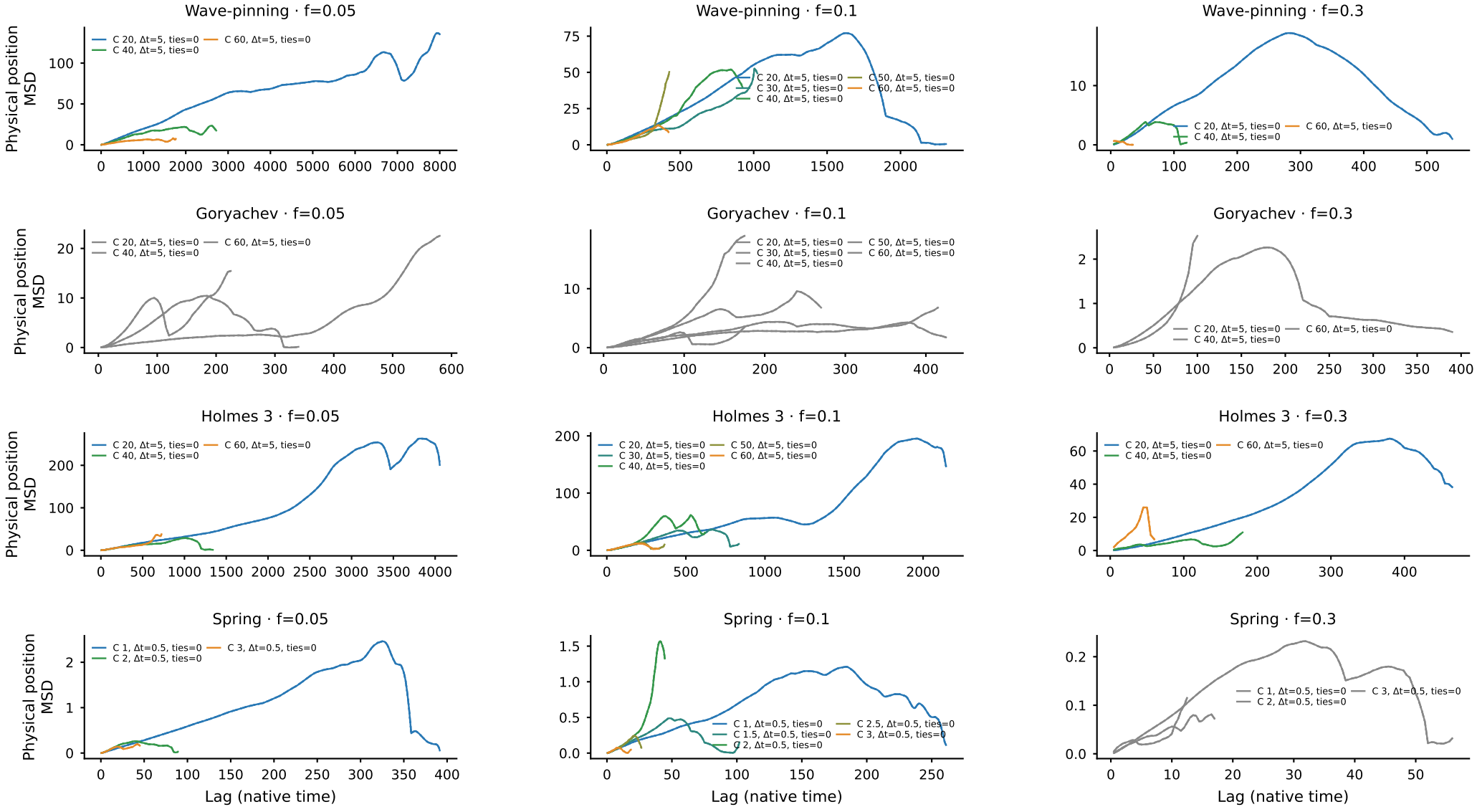
Longest selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_021_S5_literal_sigma1_heading_literal_actualC_coverage_4.csv.gz

S5 · Longest selected observations literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · pair coverage and ambiguity loss · σ=1



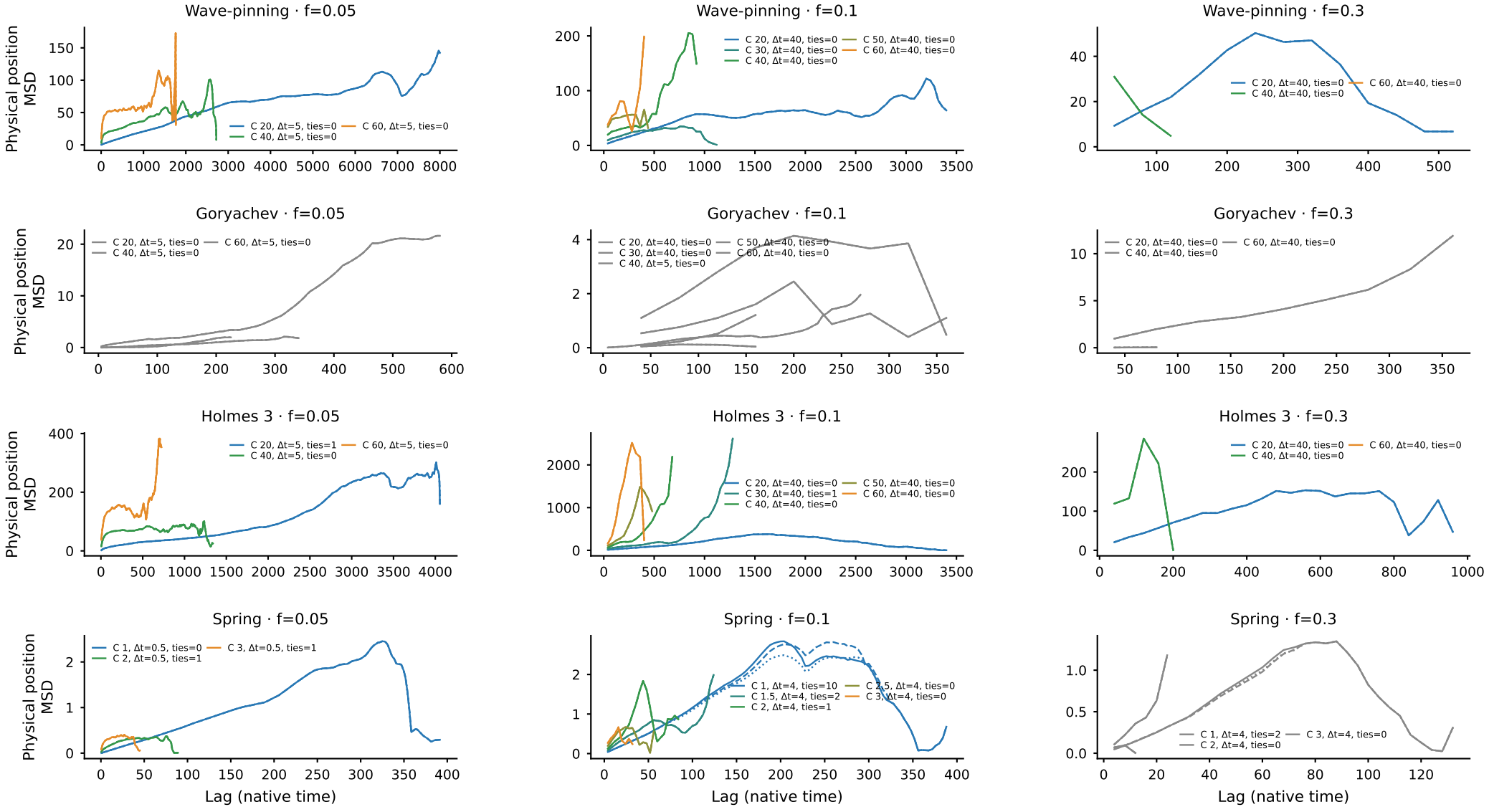
Longest selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M₁|/|M₁,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_022_S5_literal_sigma1_profile_peak_literal_actualC_coverage_4.csv

S5 · Longest selected observations literal material-origin physical-site measurements · Literal heading position · paired route sensitivities · split / +π / −π · σ=1



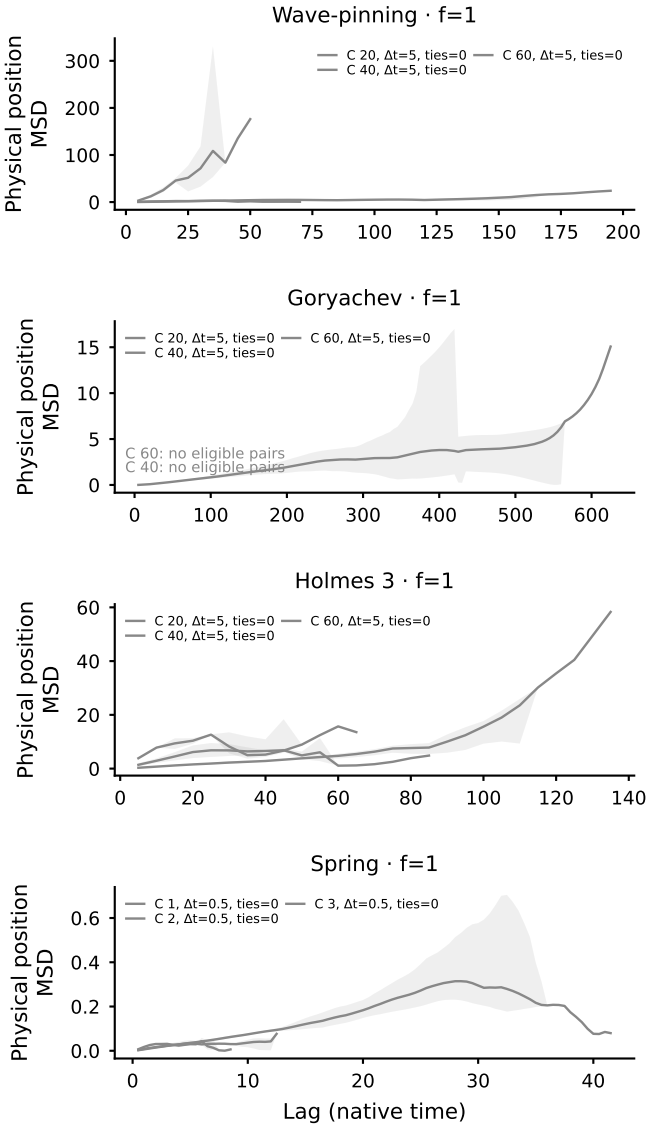
Longest selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_023_S5_literal_sigma1_heading_literal_actualC_routes_4.csv.gz

S5 · Longest selected observations literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · split / +π / −π · σ=1



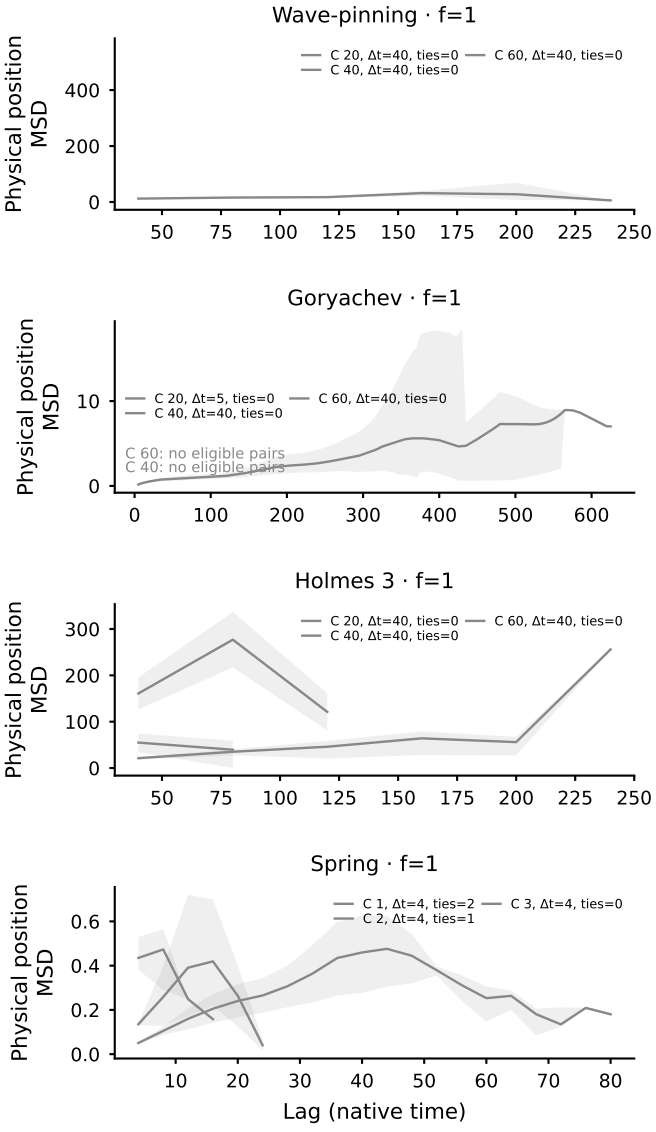
Longest selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M_1|/|M_1,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_024_S5_literal_sigma1_profile_peak_literal_actualC_routes_4.csv.gz

S5 · Longest selected observations literal material-origin physical-site measurements · Literal heading position · A3 primary split · σ=1



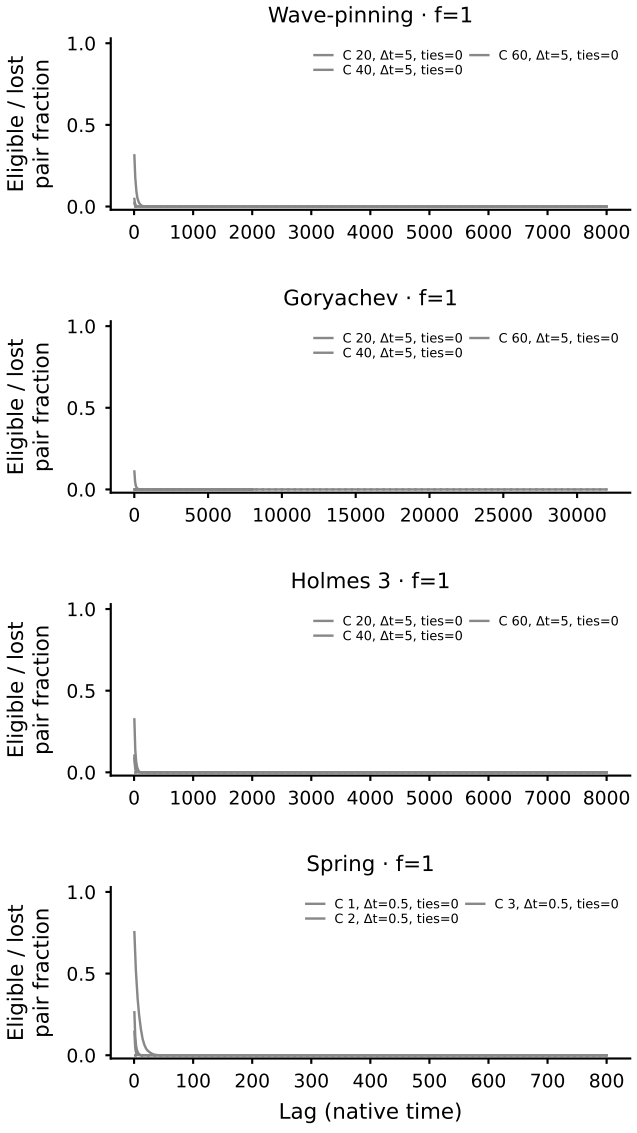
Longest selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M_1|/|M_1,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_025_S5_literal_sigma1_heading_literal_actualC_MSD_5.csv

S5 · Longest selected observations literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · σ=1



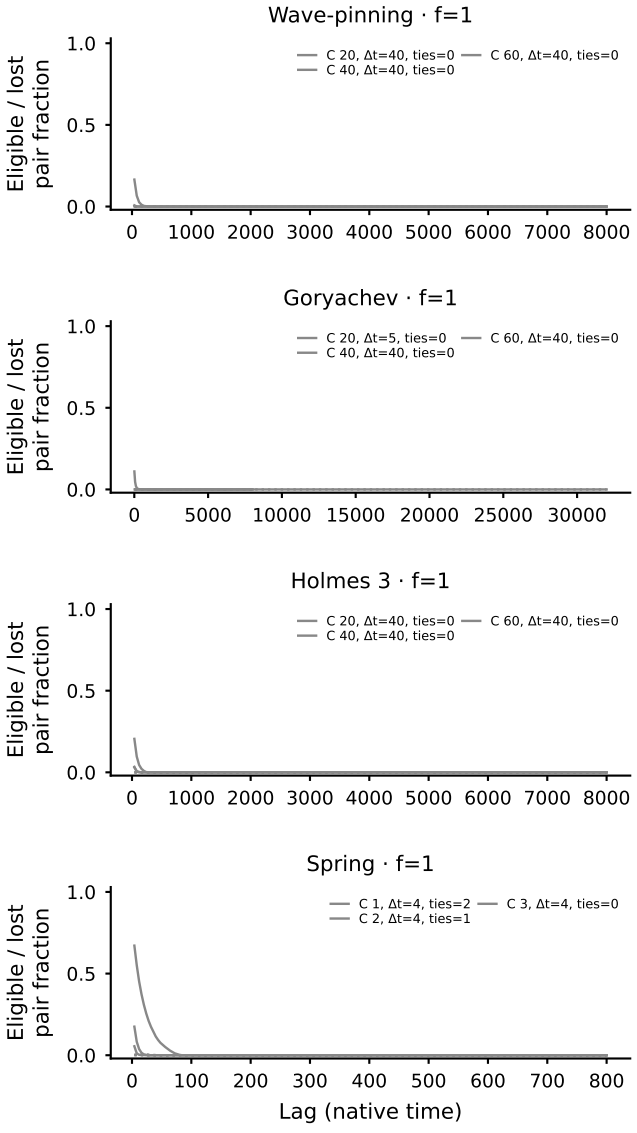
Longest selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M₁|/|M₁,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_026_S5_literal_sigma1_profile_peak_literal_actualC_MSD_5.csv

S5 · Longest selected observations literal material-origin physical-site measurements · Literal heading position · A3 primary split · pair coverage and ambiguity loss · σ=1



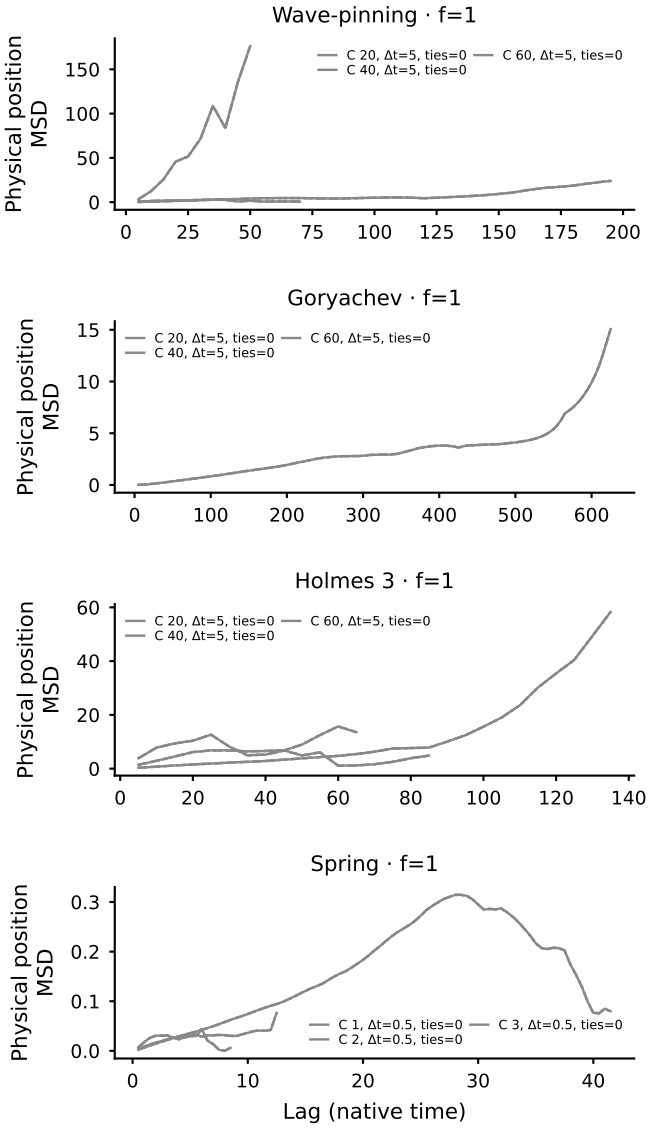
Longest selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M₁|/|M₁,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_027_S5_literal_sigma1_heading_literal_actualC_coverage_5.csv

S5 · Longest selected observations literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · pair coverage and ambiguity loss · σ=1



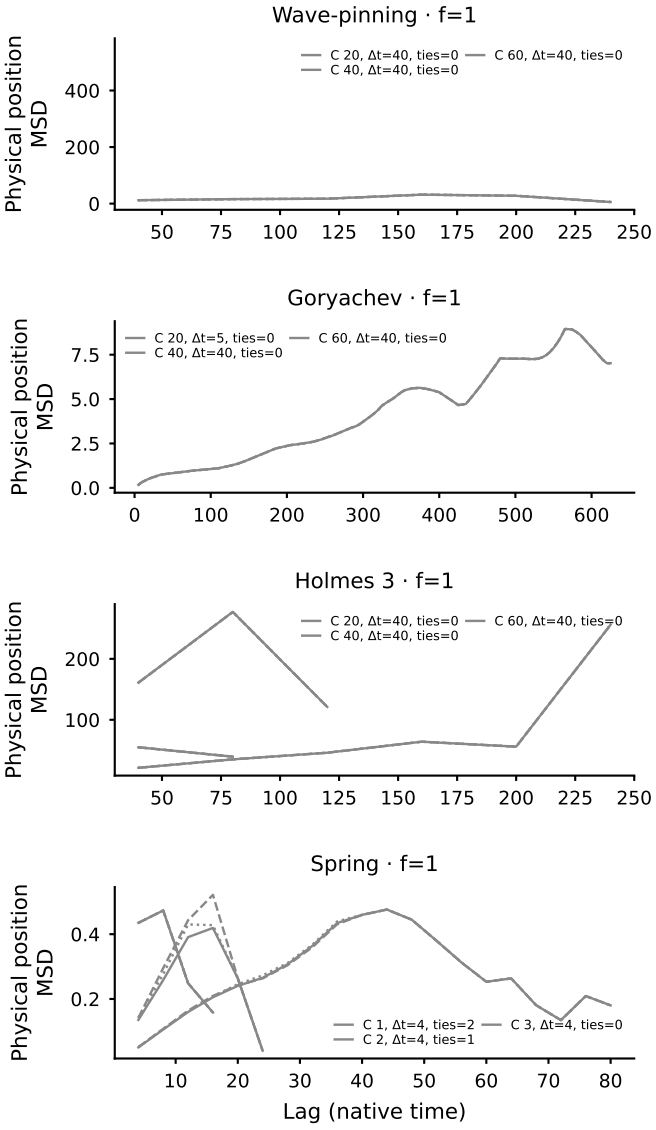
Longest selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M₁|/|M₁,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_028_S5_literal_sigma1_profile_peak_literal_actualC_coverage_5.csv

S5 · Longest selected observations literal material-origin physical-site measurements · Literal heading position · paired route sensitivities · split / +π / −π · σ=1



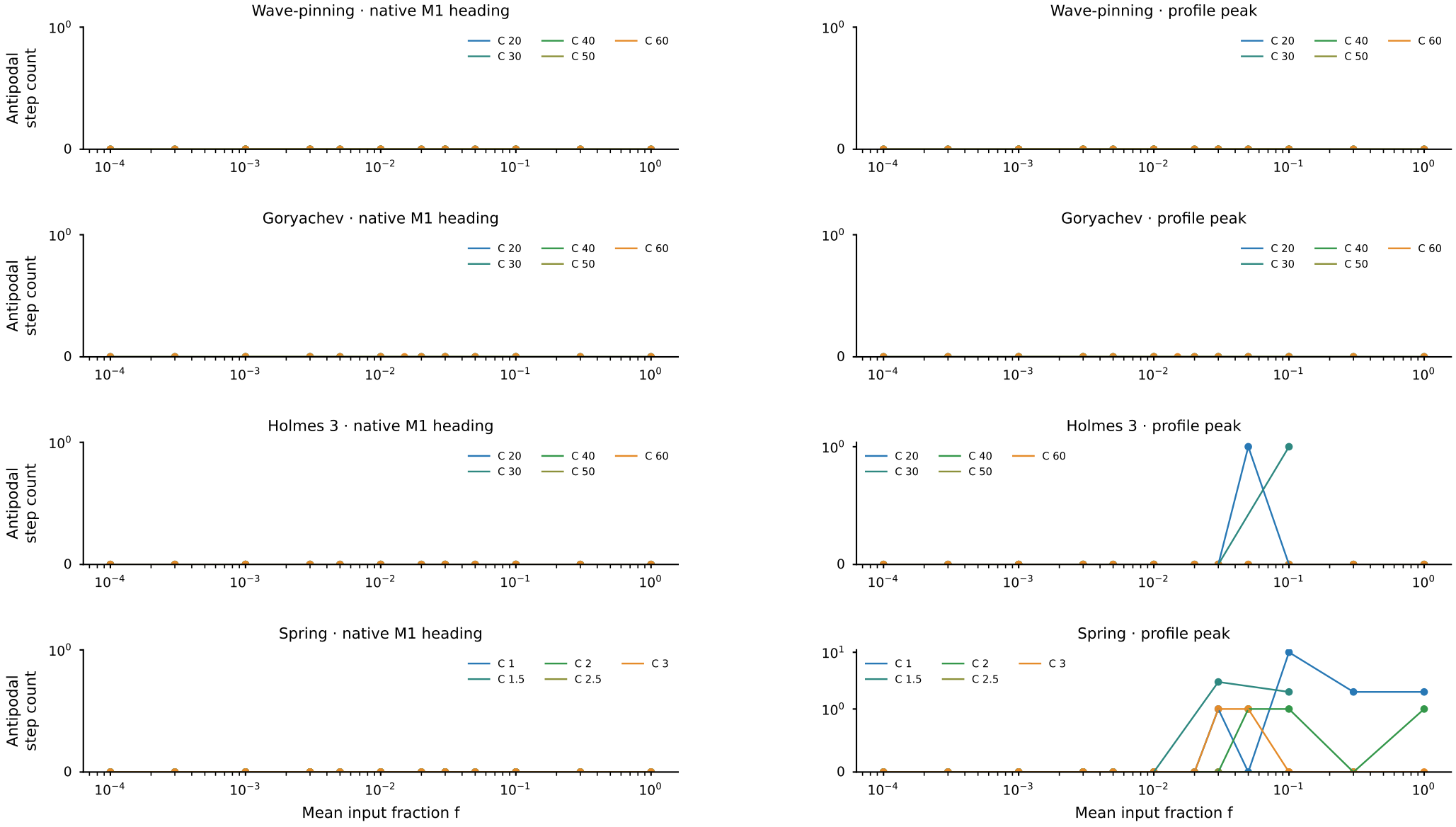
Longest selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M₁|/|M₁,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_029_S5_literal_sigma1_heading_literal_actualC_routes_5.csv.gz

S5 · Longest selected observations literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · split / +π / −π · σ=1



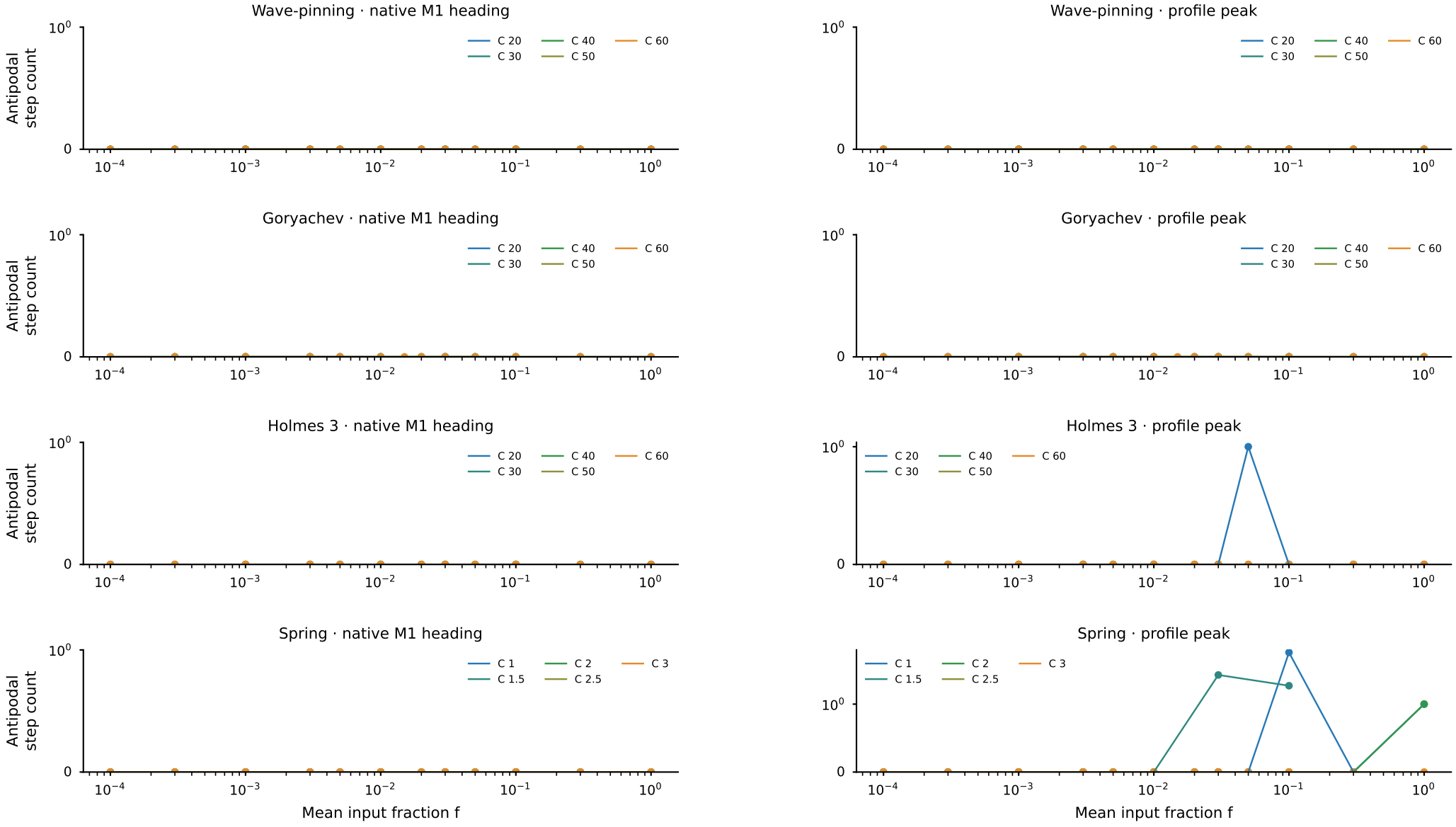
Longest selected observations literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M_1|/|M_1,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_030_S5_literal_sigma1_profile_peak_literal_actualC_routes_5.csv

S5 · Longest selected observations literal material-origin physical-site measurements · Native antipodal ambiguity · main · σ=1



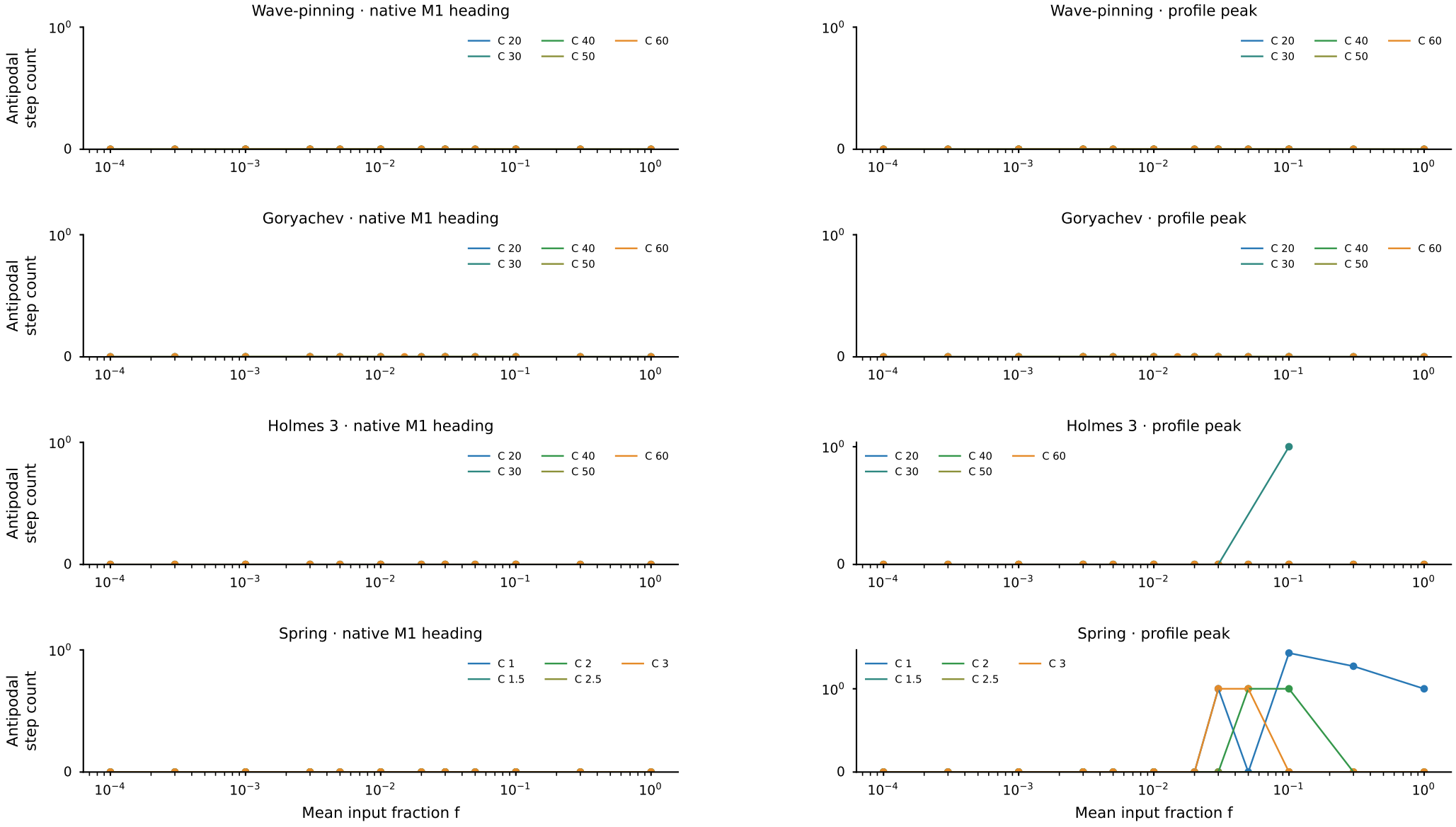
Longest selected observations literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_031_S5_ambiguity_sigma1_main.csv

S5 · Longest selected observations literal material-origin physical-site measurements · Native antipodal ambiguity · analysis first half · σ=1



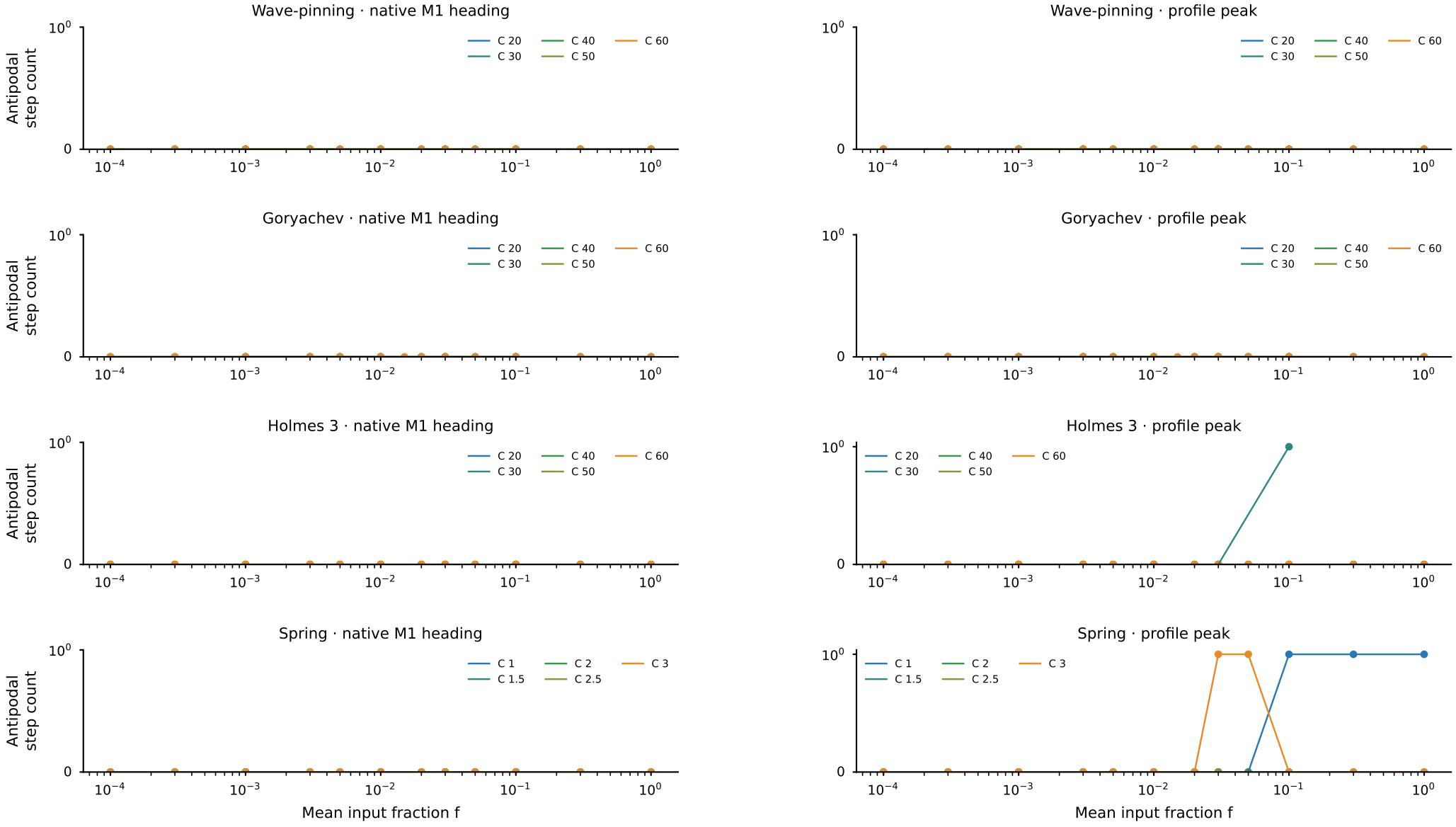
Longest selected observations literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_032_S5_ambiguity_sigma1_analysis_first_half.csv

S5 · Longest selected observations literal material-origin physical-site measurements · Native antipodal ambiguity · analysis second half · σ=1



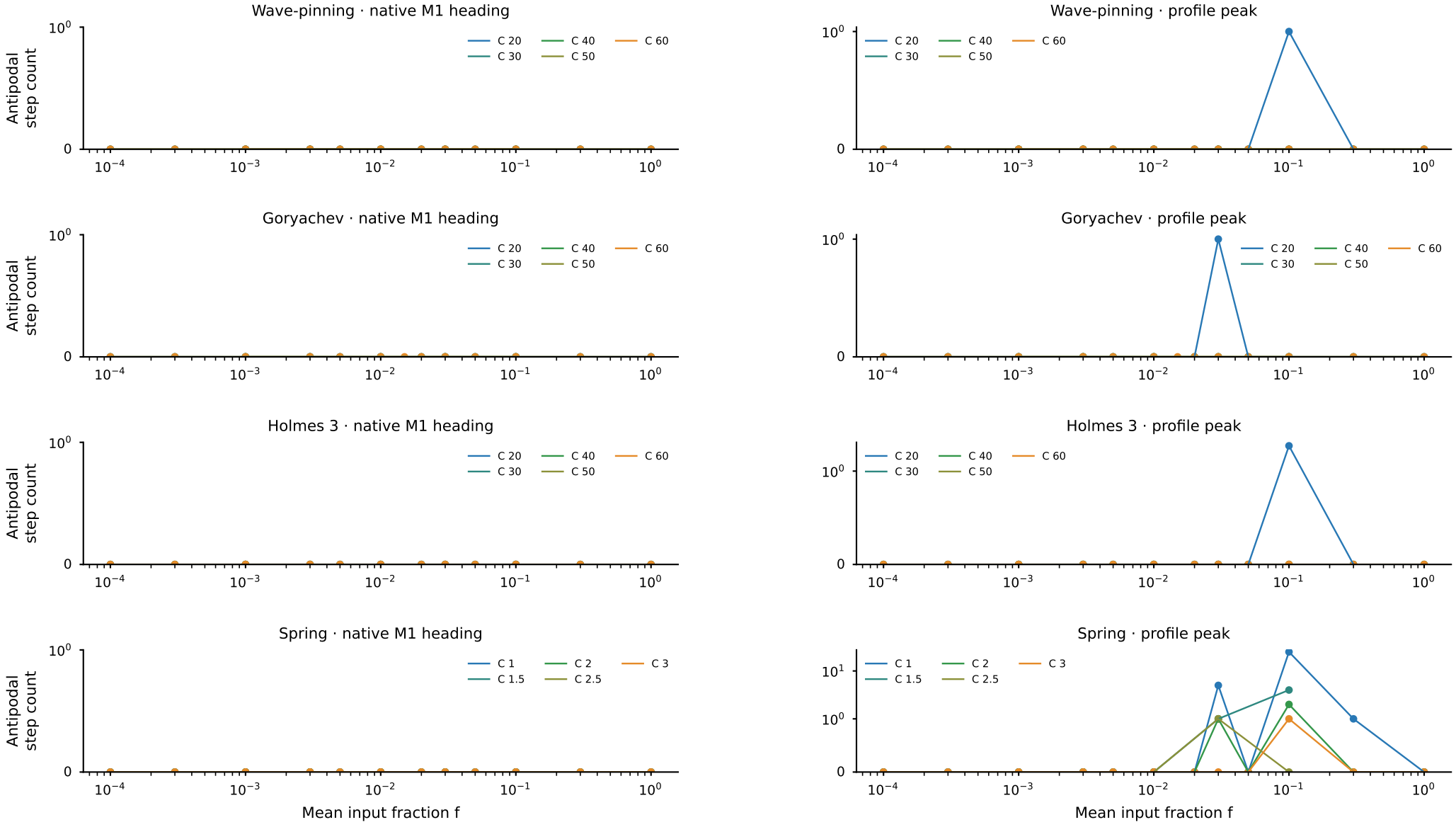
Longest selected observations literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_033_S5_ambiguity_sigma1_analysis_second_half.csv

S5 · Longest selected observations literal material-origin physical-site measurements · Native antipodal ambiguity · late quarter · σ=1



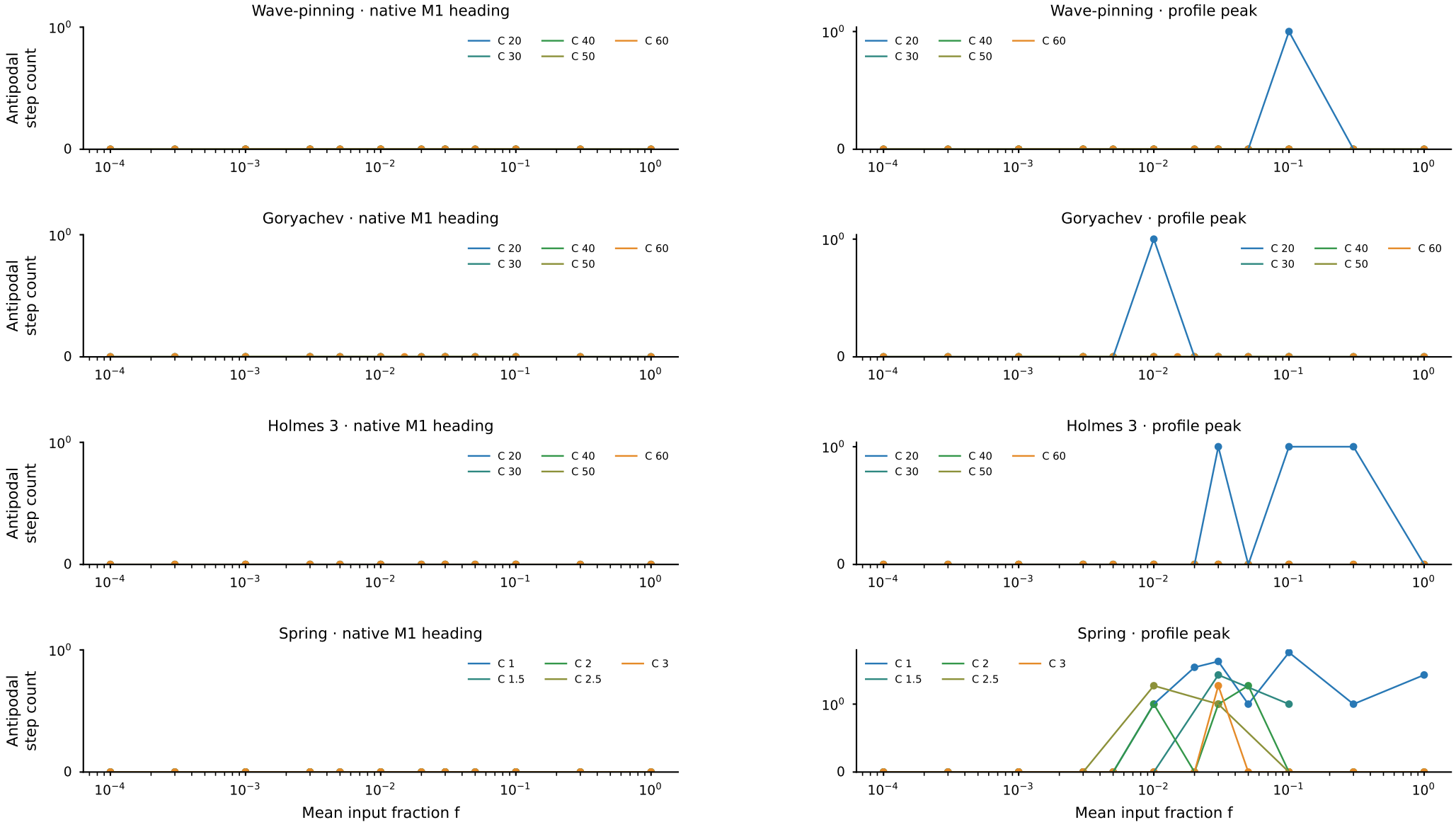
Longest selected observations literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_034_S5_ambiguity_sigma1_late_quarter.csv

S5 · Longest selected observations literal material-origin physical-site measurements · Native antipodal ambiguity · cadence x2 · σ=1



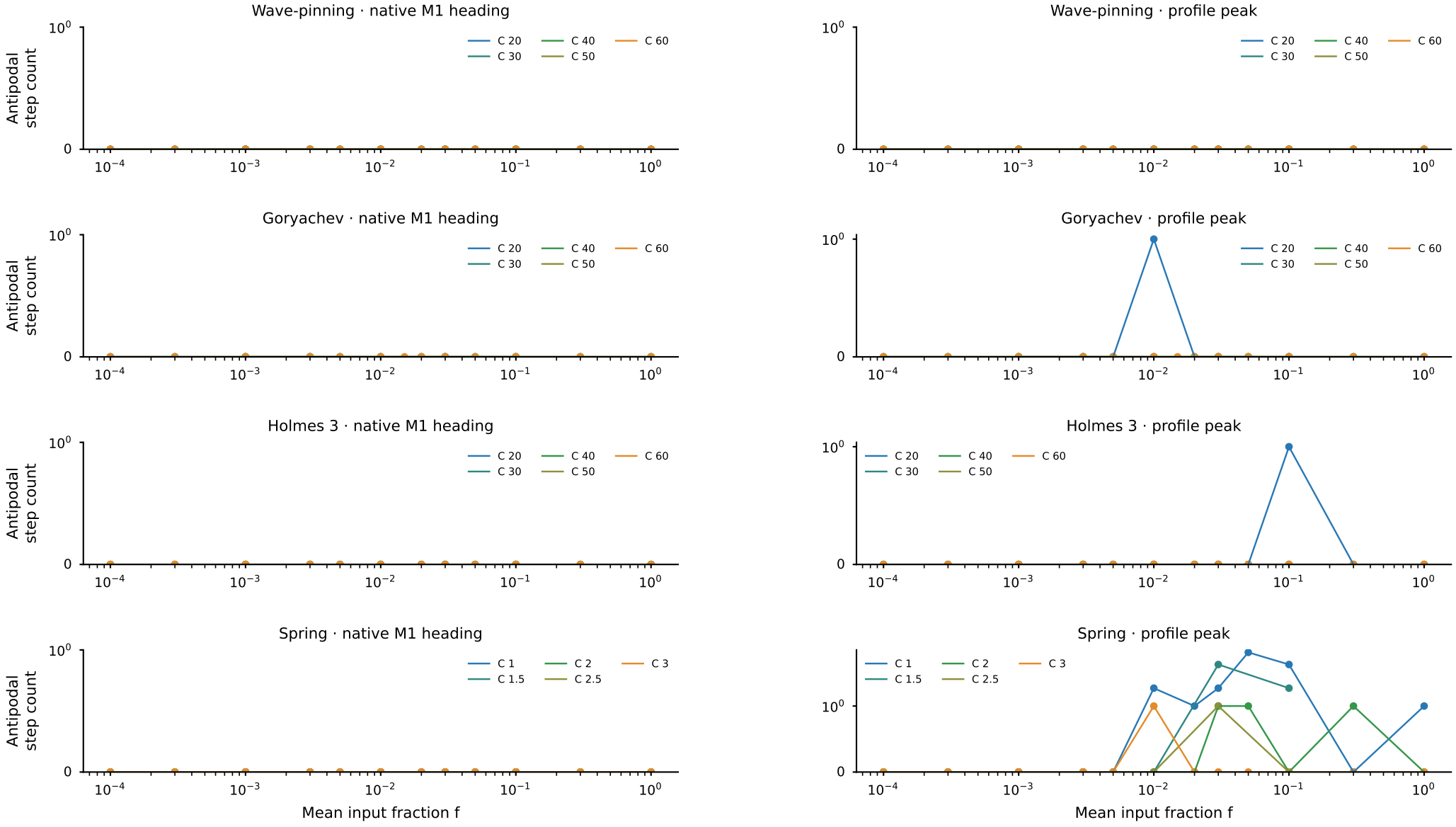
Longest selected observations literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_035_S5_ambiguity_sigma1_cadence_x2.csv

S5 · Longest selected observations literal material-origin physical-site measurements · Native antipodal ambiguity · cadence x4 · σ=1



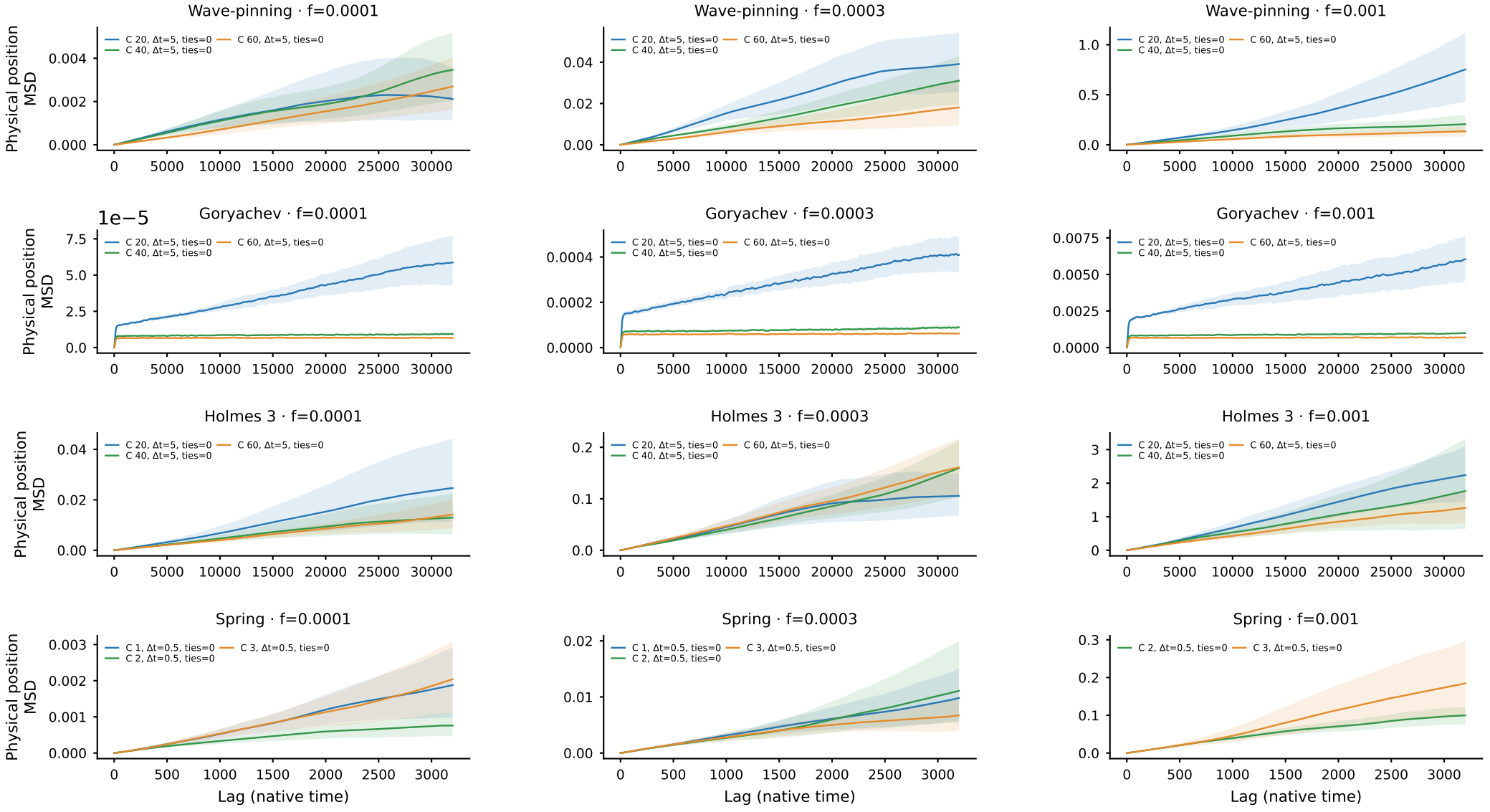
Longest selected observations literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_036_S5_ambiguity_sigma1_cadence_x4.csv

S5 · Longest selected observations literal material-origin physical-site measurements · Native antipodal ambiguity · cadence x8 · σ=1



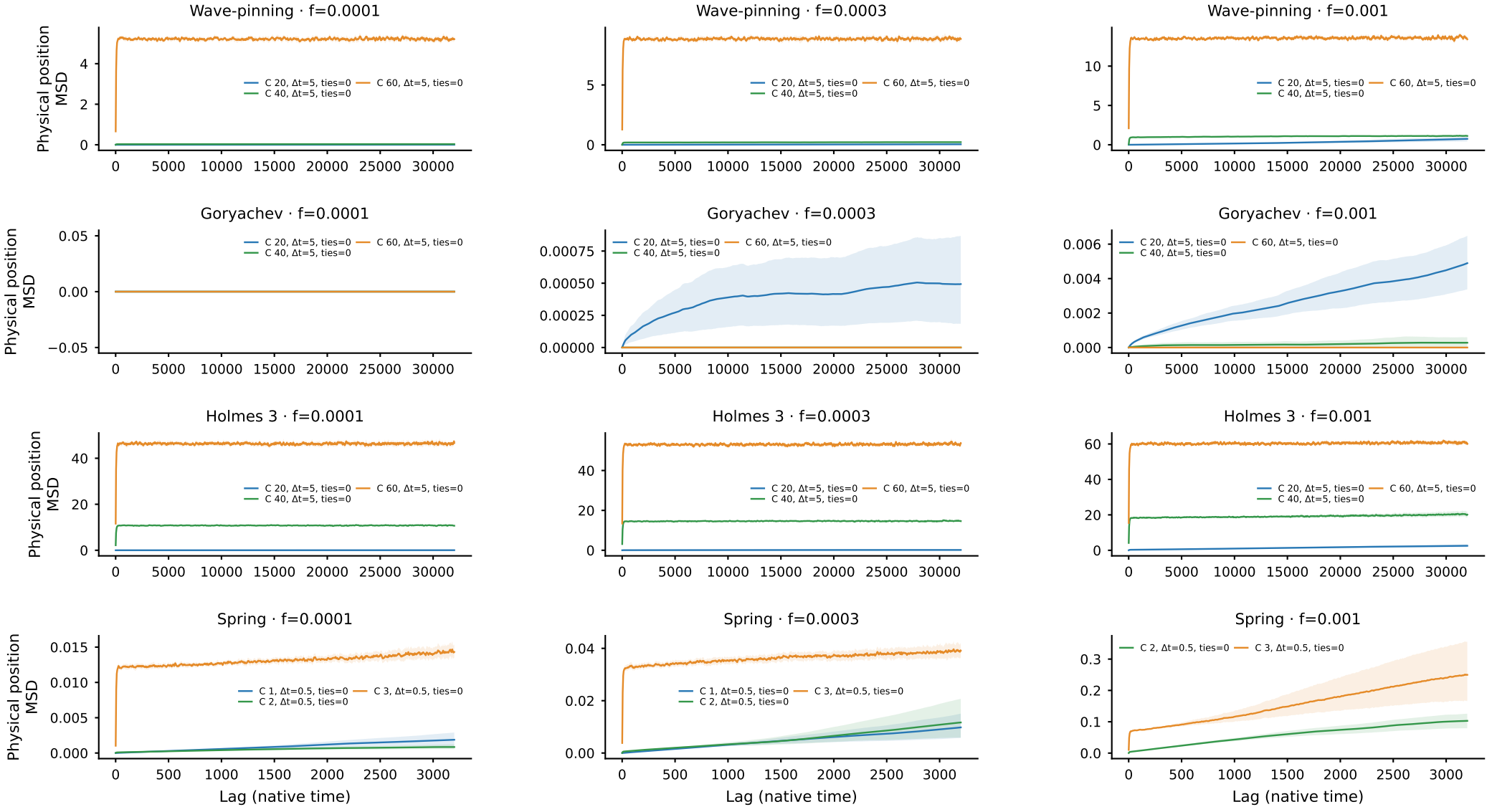
Longest selected observations literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_literal_canonical_20261020_037_S5_ambiguity_sigma1_cadence_x8.csv

S5 · E4 literal material-origin physical-site measurements · Literal heading position · A3 primary split · σ=1



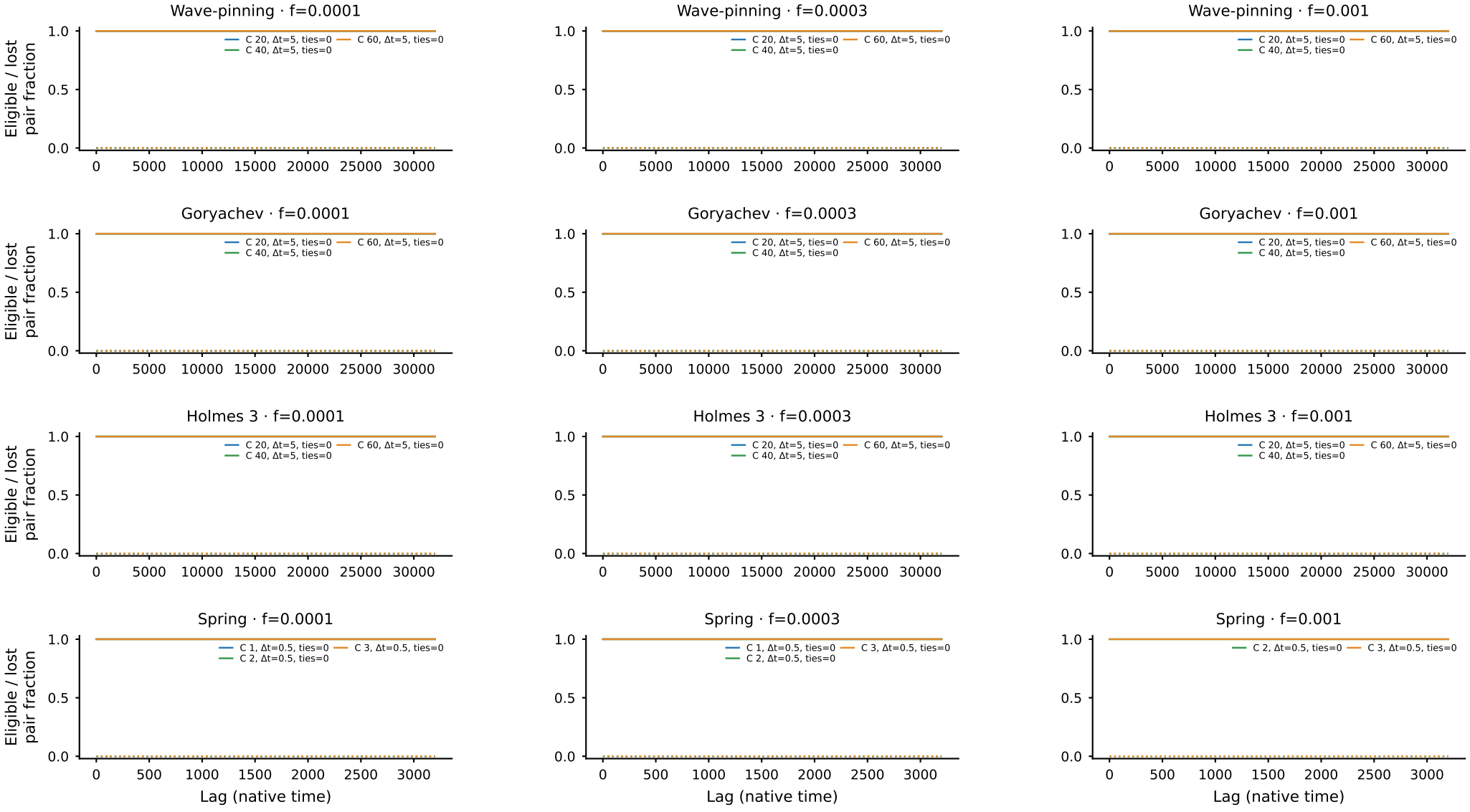
E4 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M₁|/|M₁,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_E4_literal_native_20261020_001_S5_literal_sigma1_heading_literal_actualC_MSD_1.csv.gz

S5 · E4 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · σ=1



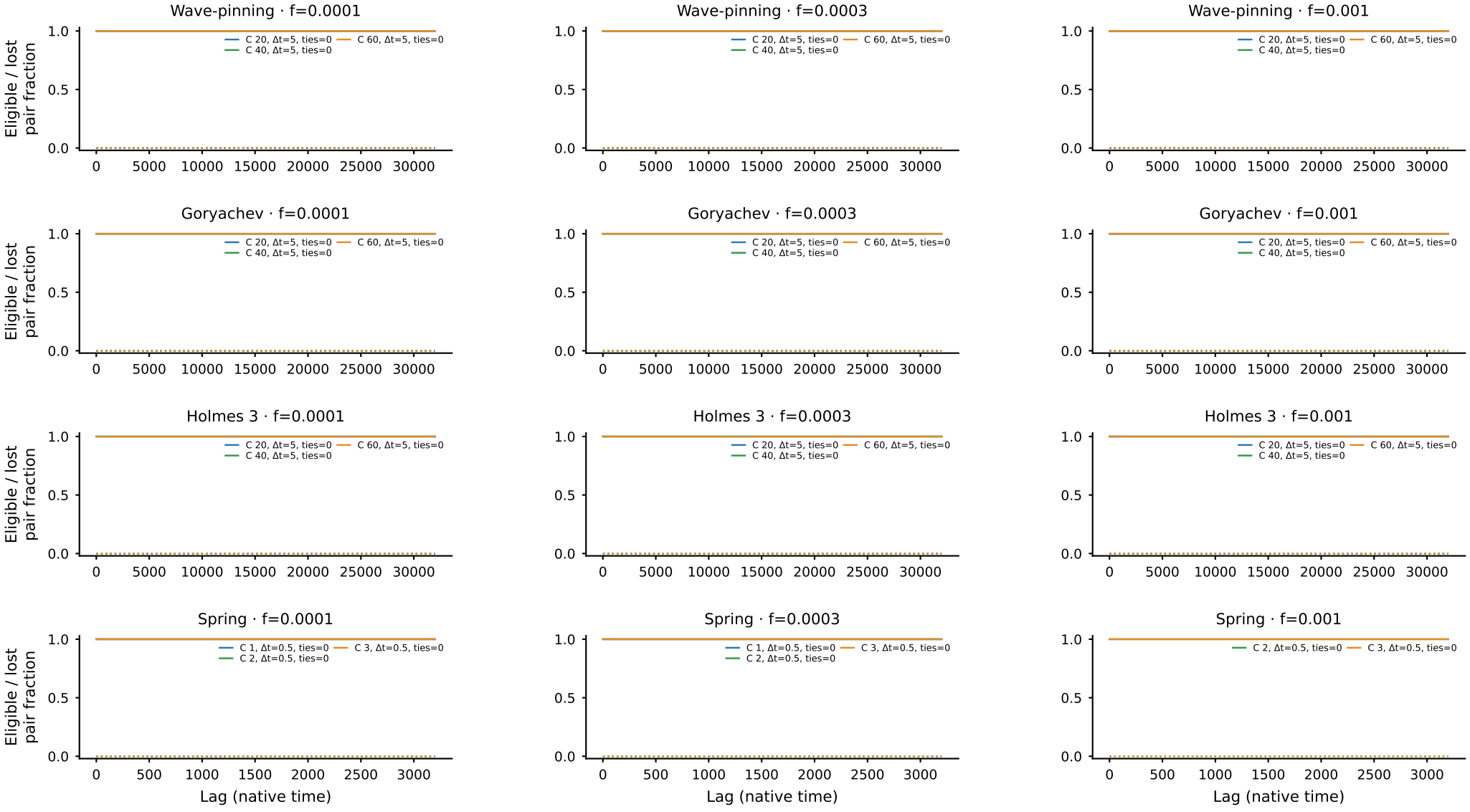
E4 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_E4_literal_native_20261020_002_S5_literal_sigma1_profile_peak_literal_actualC_MSD_1.csv.gz

S5 · E4 literal material-origin physical-site measurements · Literal heading position · A3 primary split · pair coverage and ambiguity loss · σ=1



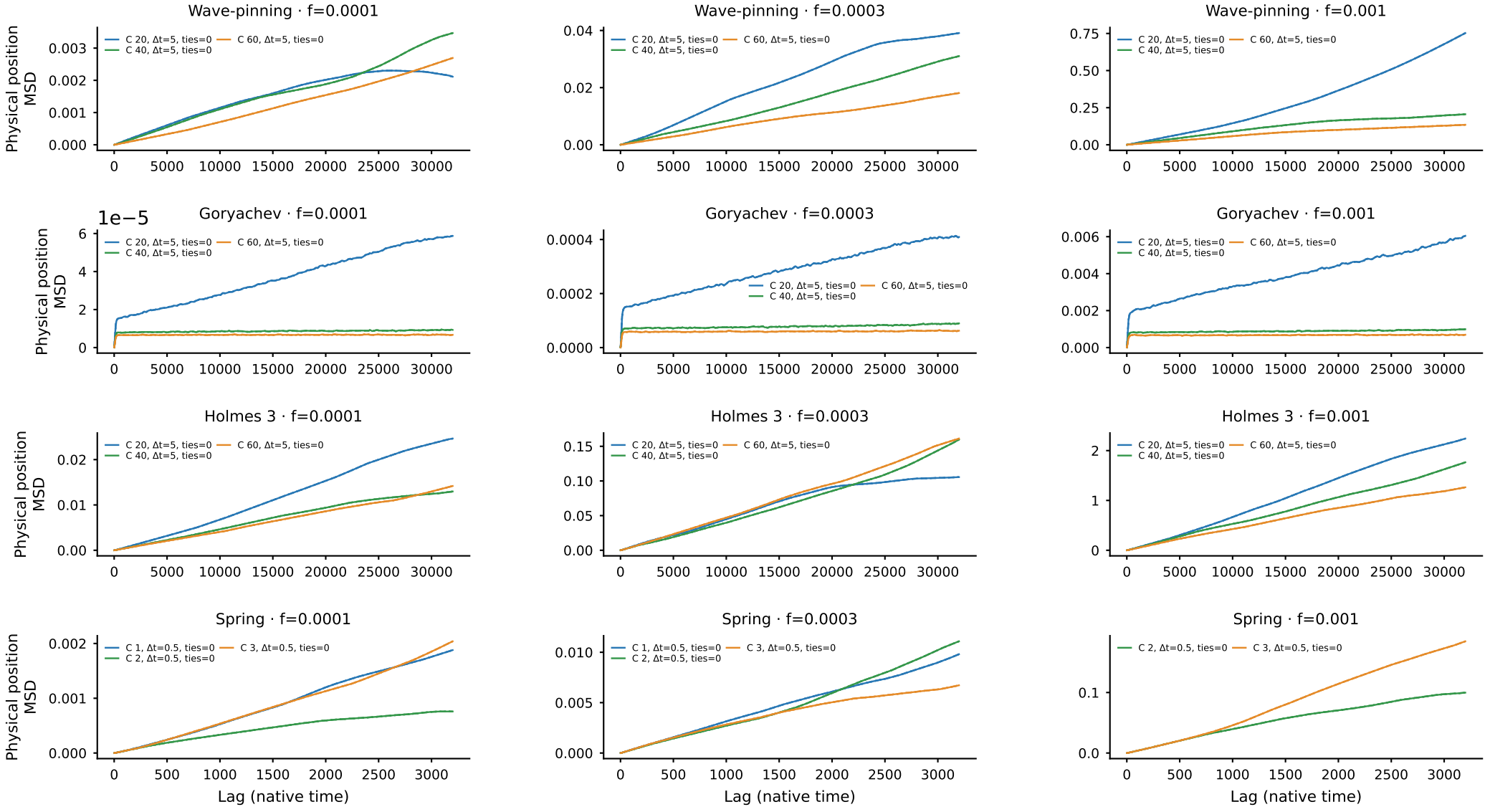
E4 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_E4_literal_native_20261020_003_S5_literal_sigma1_heading_literal_actualC_coverage_1.csv.gz

S5 · E4 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · pair coverage and ambiguity loss · σ=1



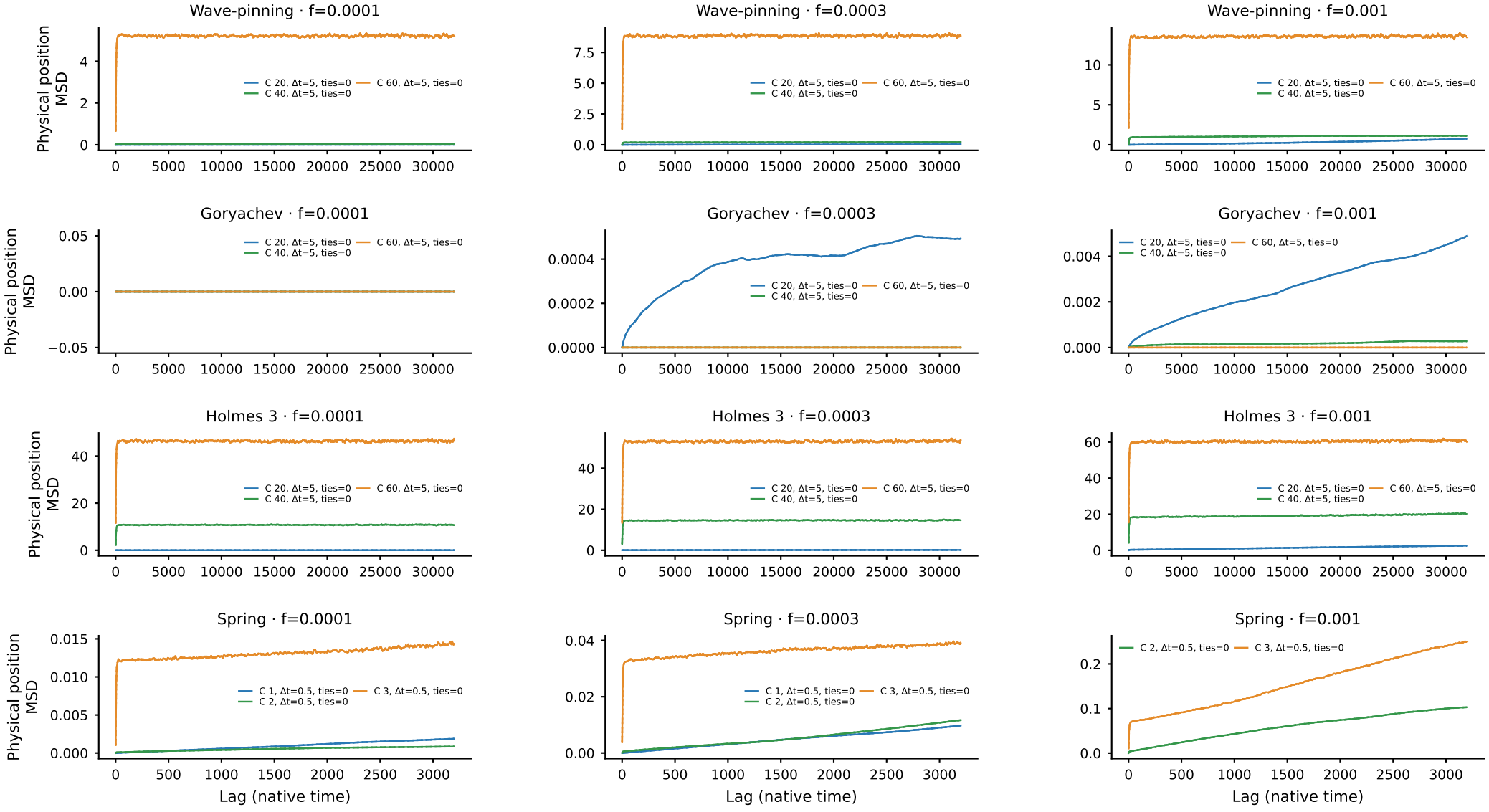
E4 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_E4_literal_native_20261020_004_S5_literal_sigma1_profile_peak_literal_actualC_coverage_1.csv.gz

S5 · E4 literal material-origin physical-site measurements · Literal heading position · paired route sensitivities · split / +π / −π · σ=1



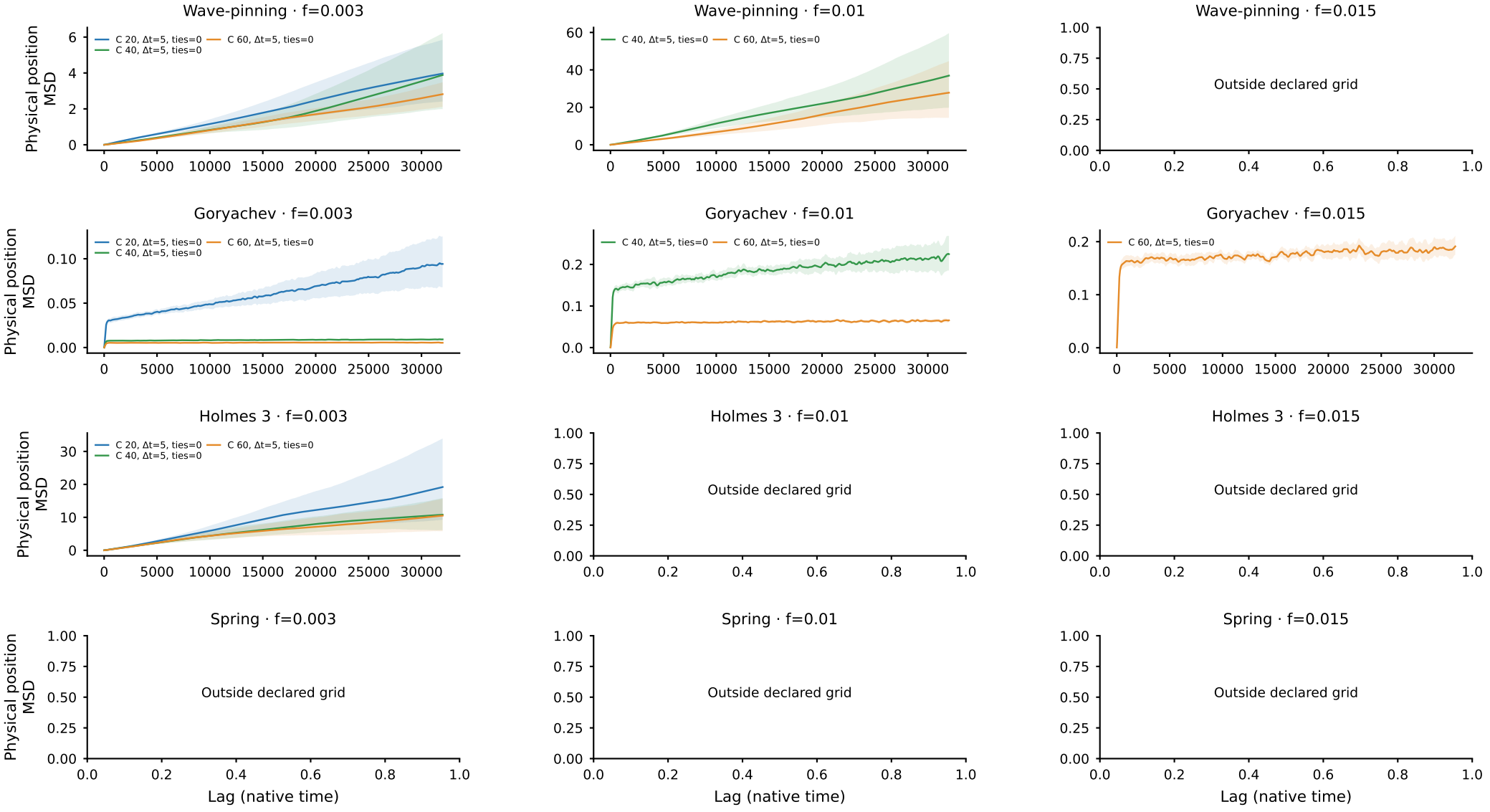
E4 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_E4_literal_native_20261020_005_S5_literal_sigma1_heading_literal_actualC_routes_1.csv.gz

S5 · E4 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · split / +π / −π · σ=1



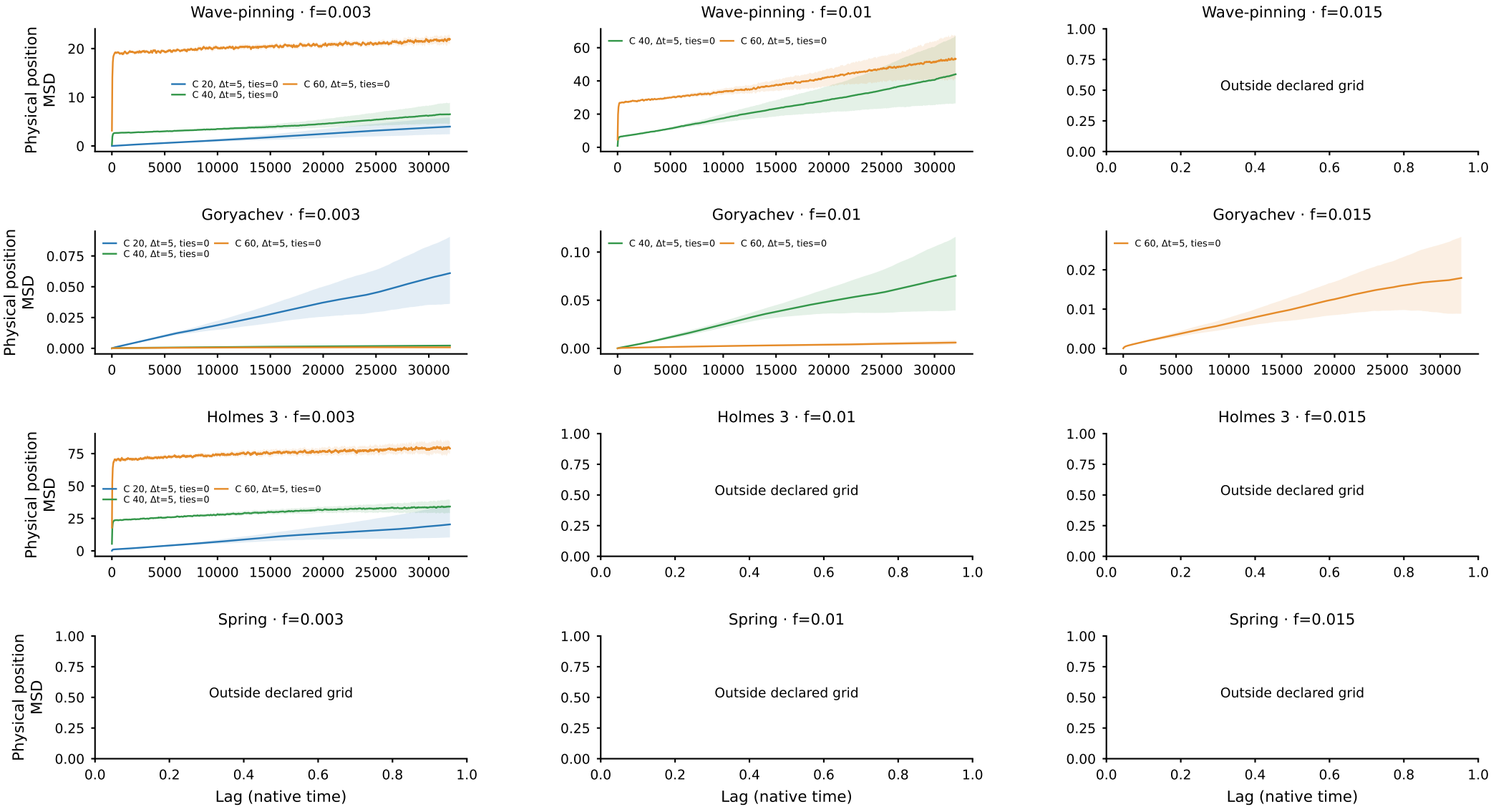
E4 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M₁|/|M₁,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_E4_literal_native_20261020_006_S5_literal_sigma1_profile_peak_literal_actualC_routes_1.csv.gz

S5 · E4 literal material-origin physical-site measurements · Literal heading position · A3 primary split · σ=1



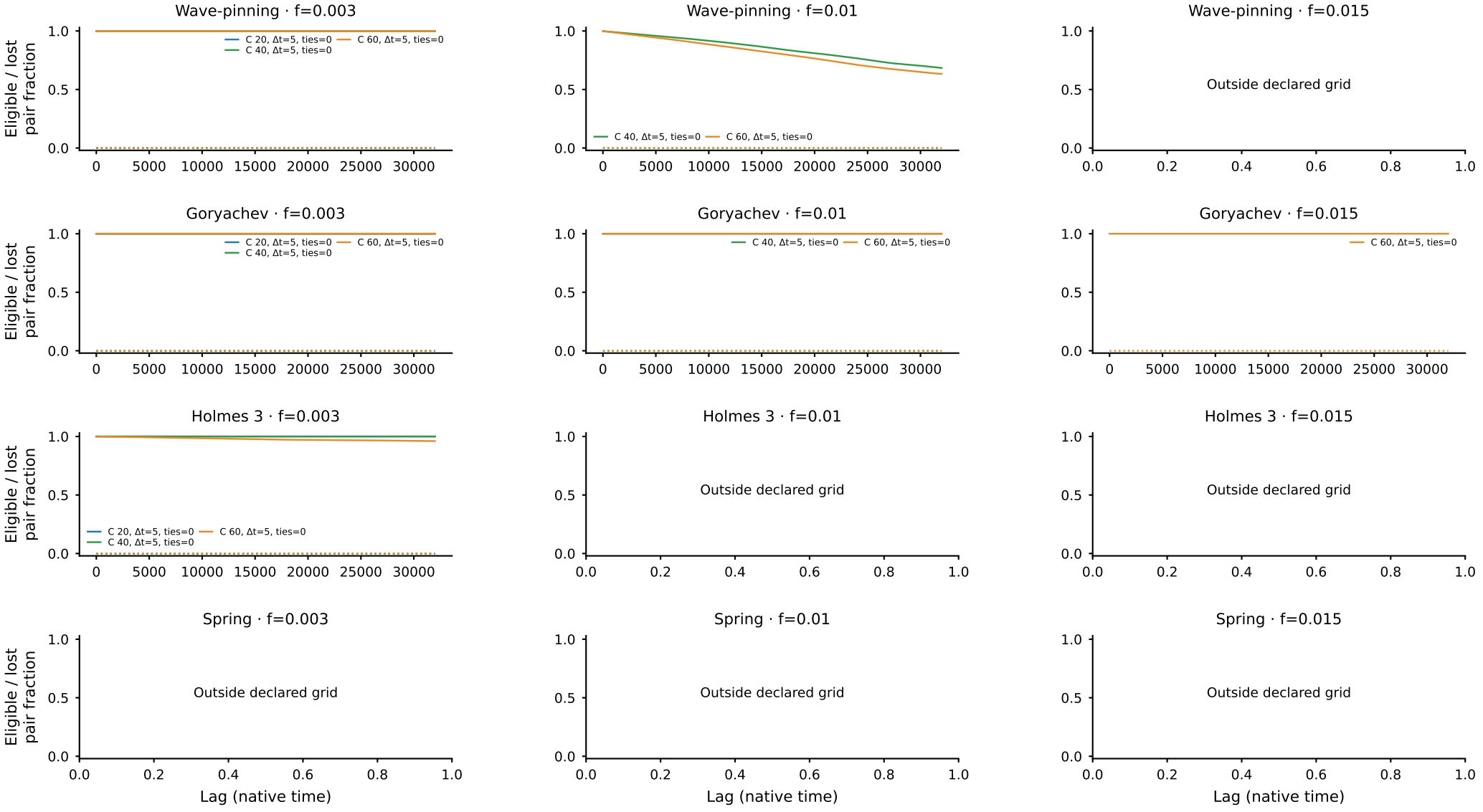
E4 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M₁|/|M₁,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_E4_literal_native_20261020_007_S5_literal_sigma1_heading_literal_actualC_MSD_2.csv.gz

S5 · E4 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · σ=1



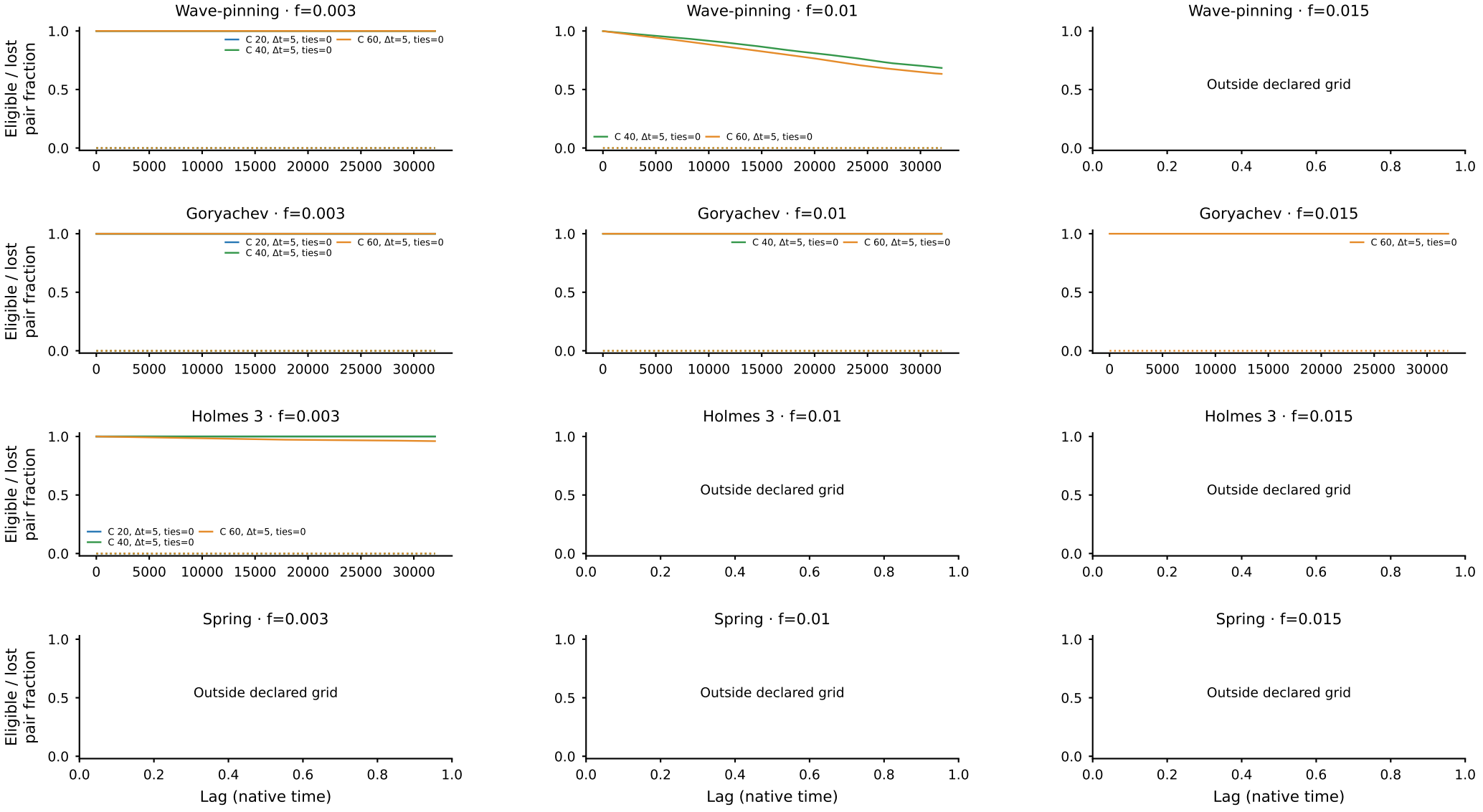
E4 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M₁|/|M₁,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_E4_literal_native_20261020_008_S5_literal_sigma1_profile_peak_literal_actualC_MSD_2.csv.gz

S5 · E4 literal material-origin physical-site measurements · Literal heading position · A3 primary split · pair coverage and ambiguity loss · σ=1



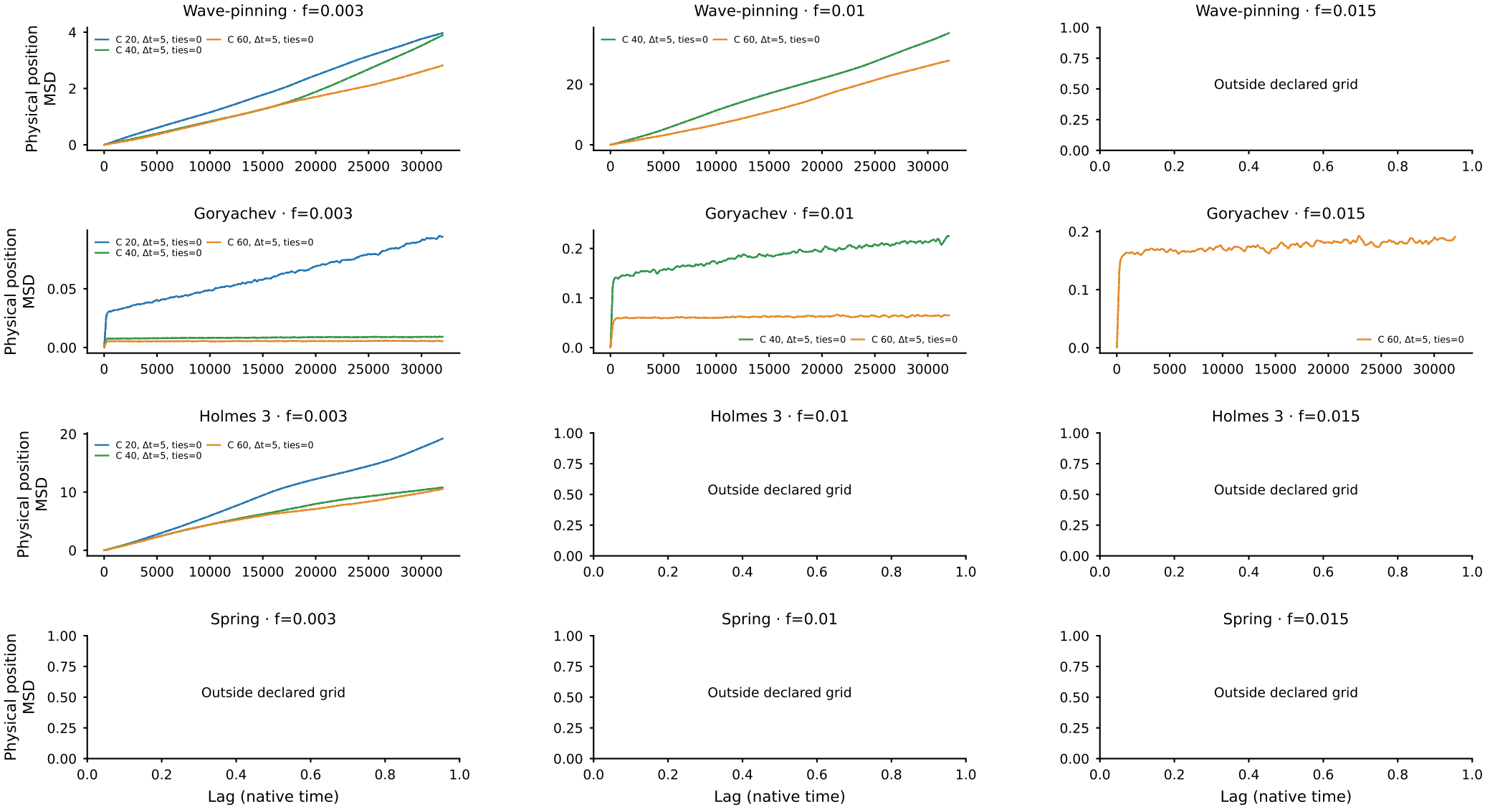
E4 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,basal}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_E4_literal_native_20261020_009_S5_literal_sigma1_heading_literal_actualC_coverage_2.csv.gz

S5 · E4 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · pair coverage and ambiguity loss · σ=1



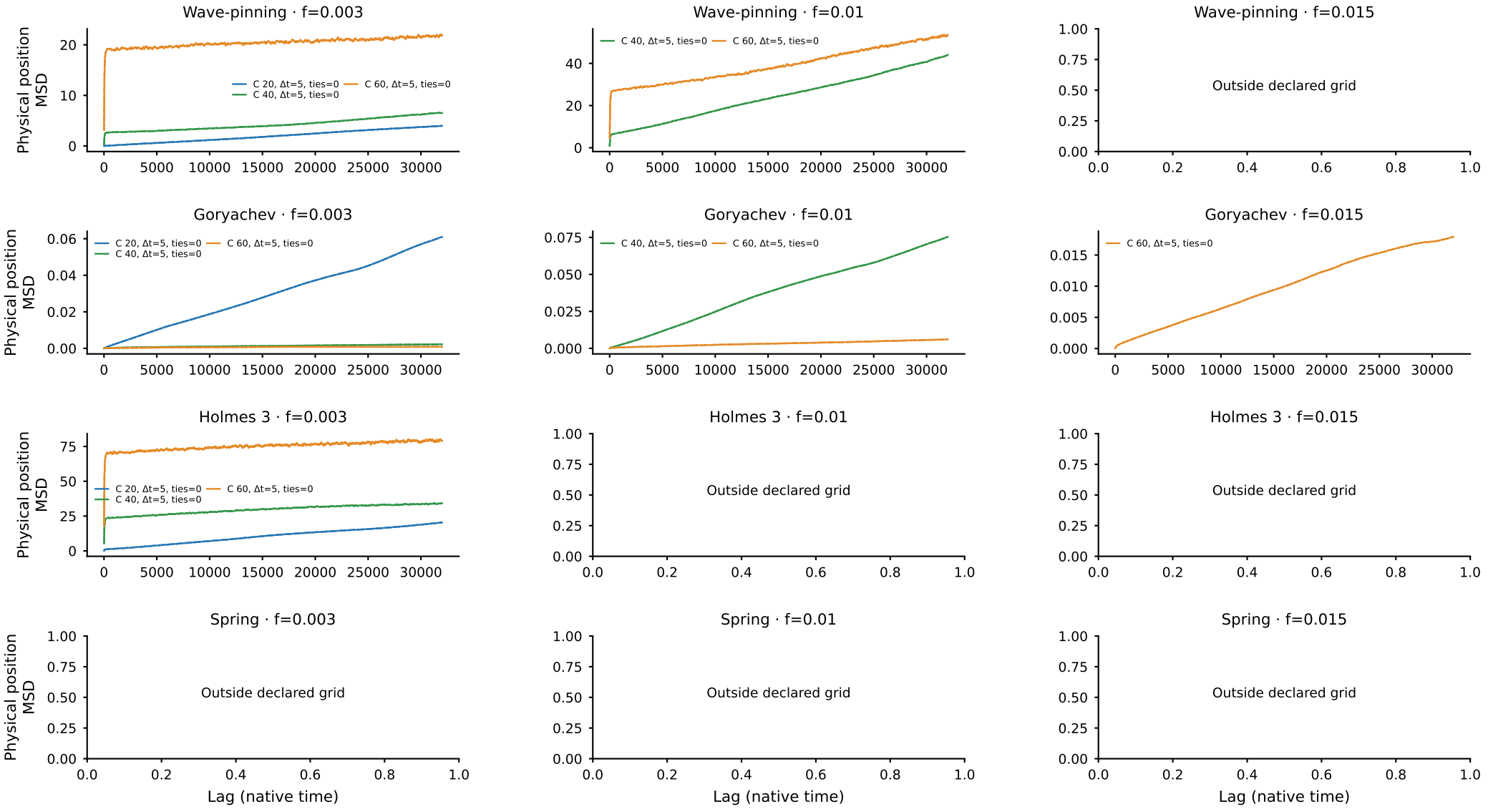
E4 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_E4_literal_native_20261020_010_S5_literal_sigma1_profile_peak_literal_actualC_coverage_2.csv.gz

S5 · E4 literal material-origin physical-site measurements · Literal heading position · paired route sensitivities · split / +π / −π · σ=1

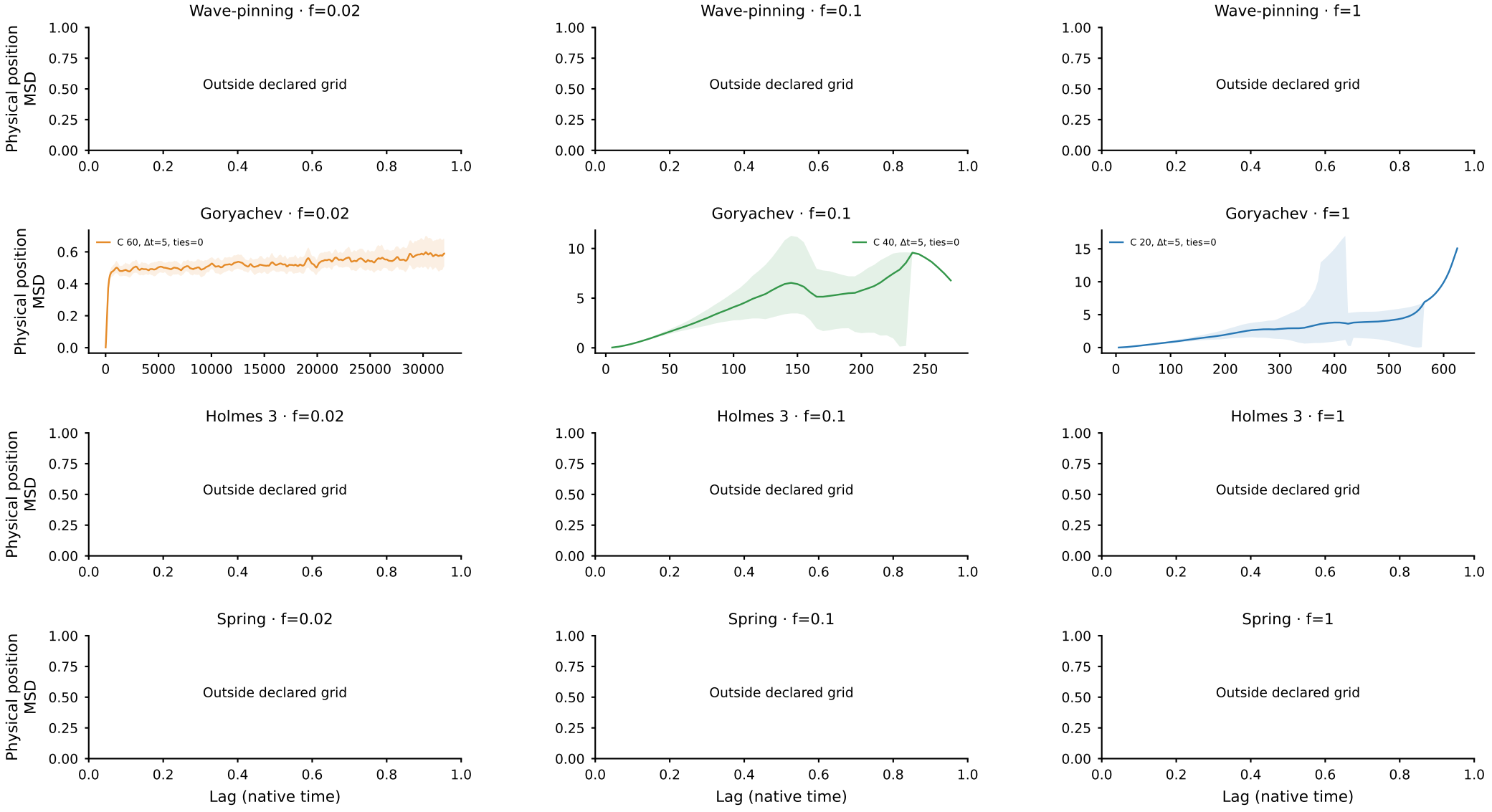


E4 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M₁|/|M₁,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_E4_literal_native_20261020_011_S5_literal_sigma1_heading_literal_actualC_routes_2.csv.gz

S5 · E4 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · split / +π / −π · σ=1

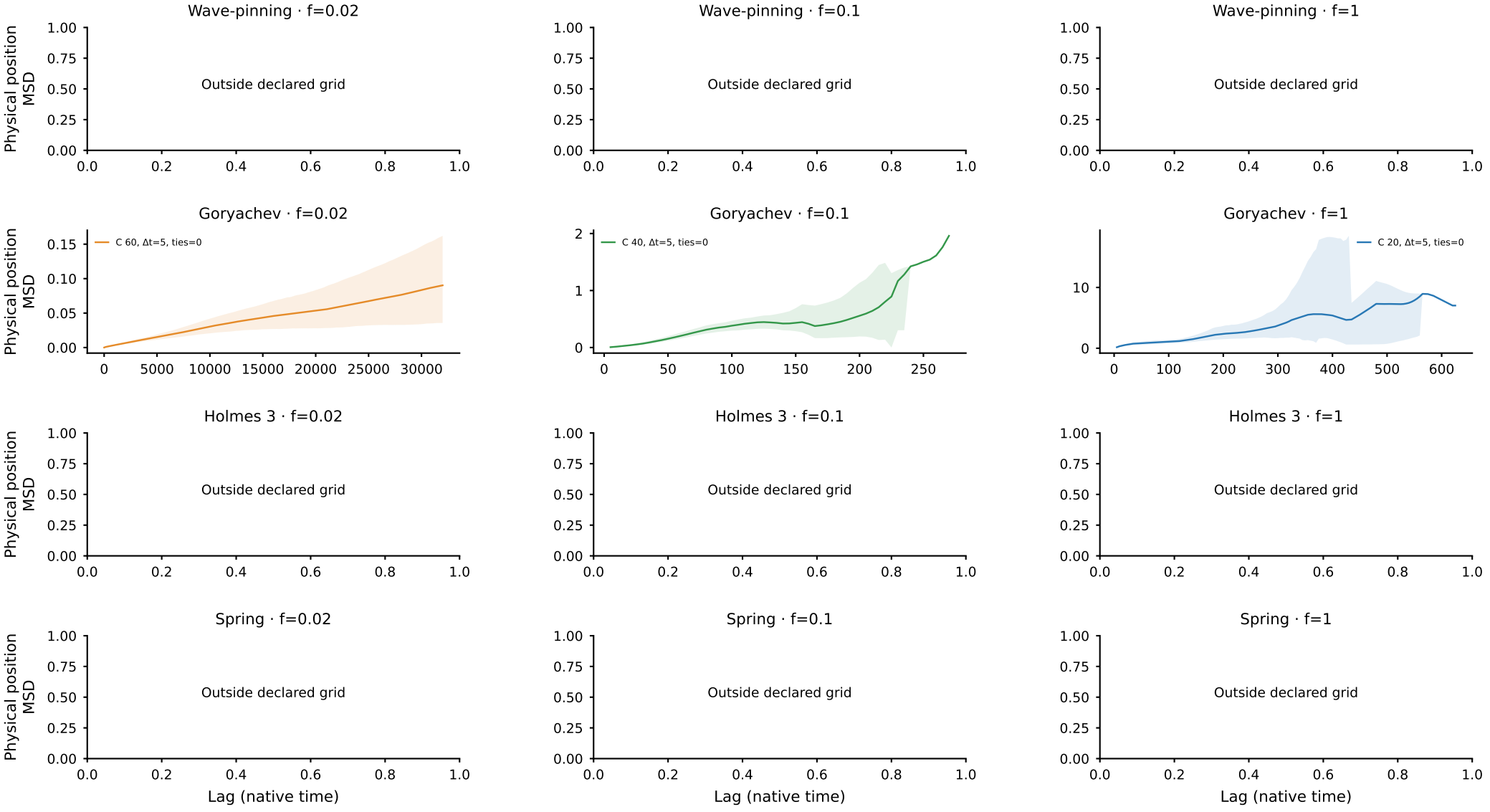


E4 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M₁|/|M₁,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_E4_literal_native_20261020_012_S5_literal_sigma1_profile_peak_literal_actualC_routes_2.csv.gz



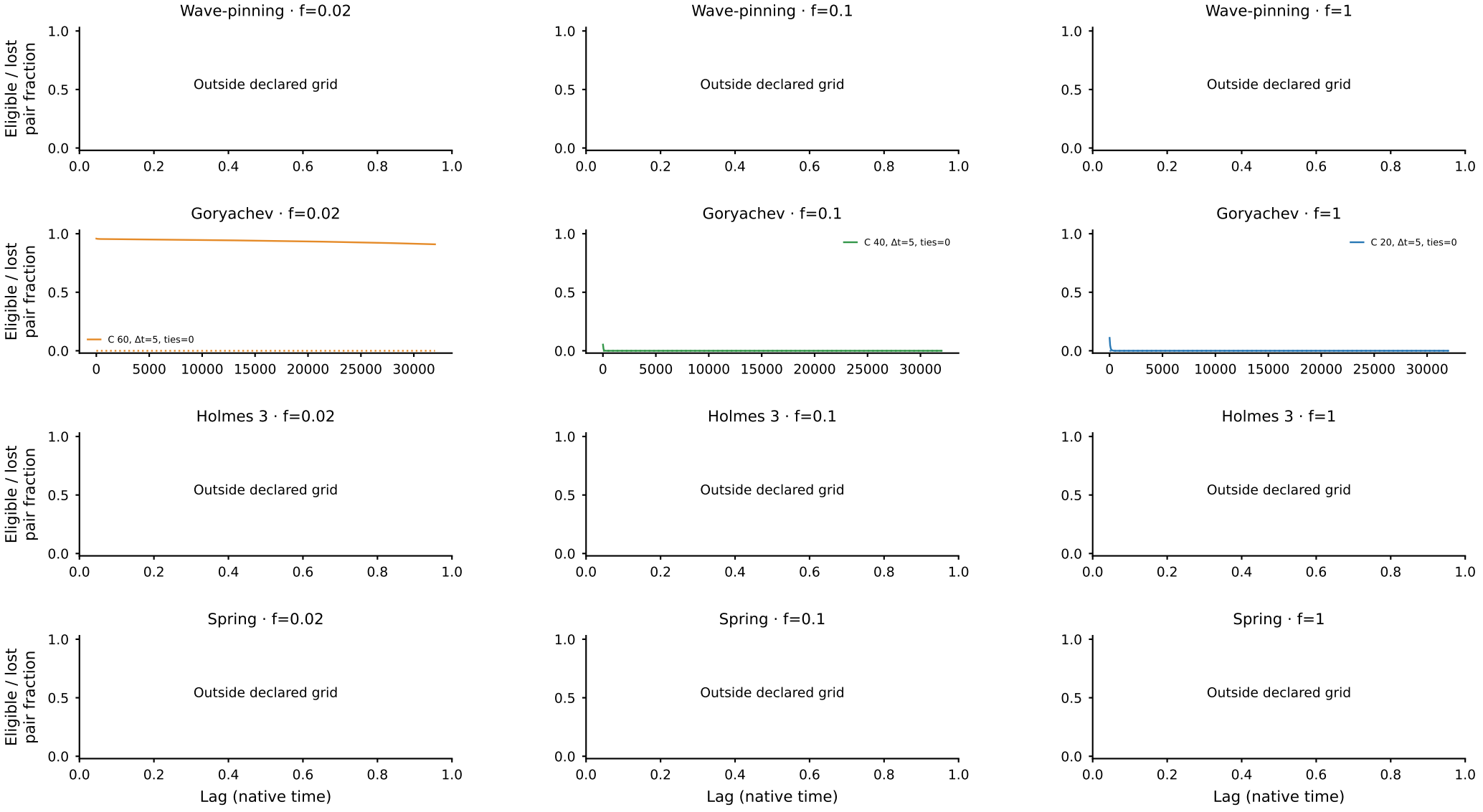
E4 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M_1|/|M_1,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_E4_literal_native_20261020_013_S5_literal_sigma1_heading_literal_actualC_MSD_3.csv

S5 · E4 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · σ=1



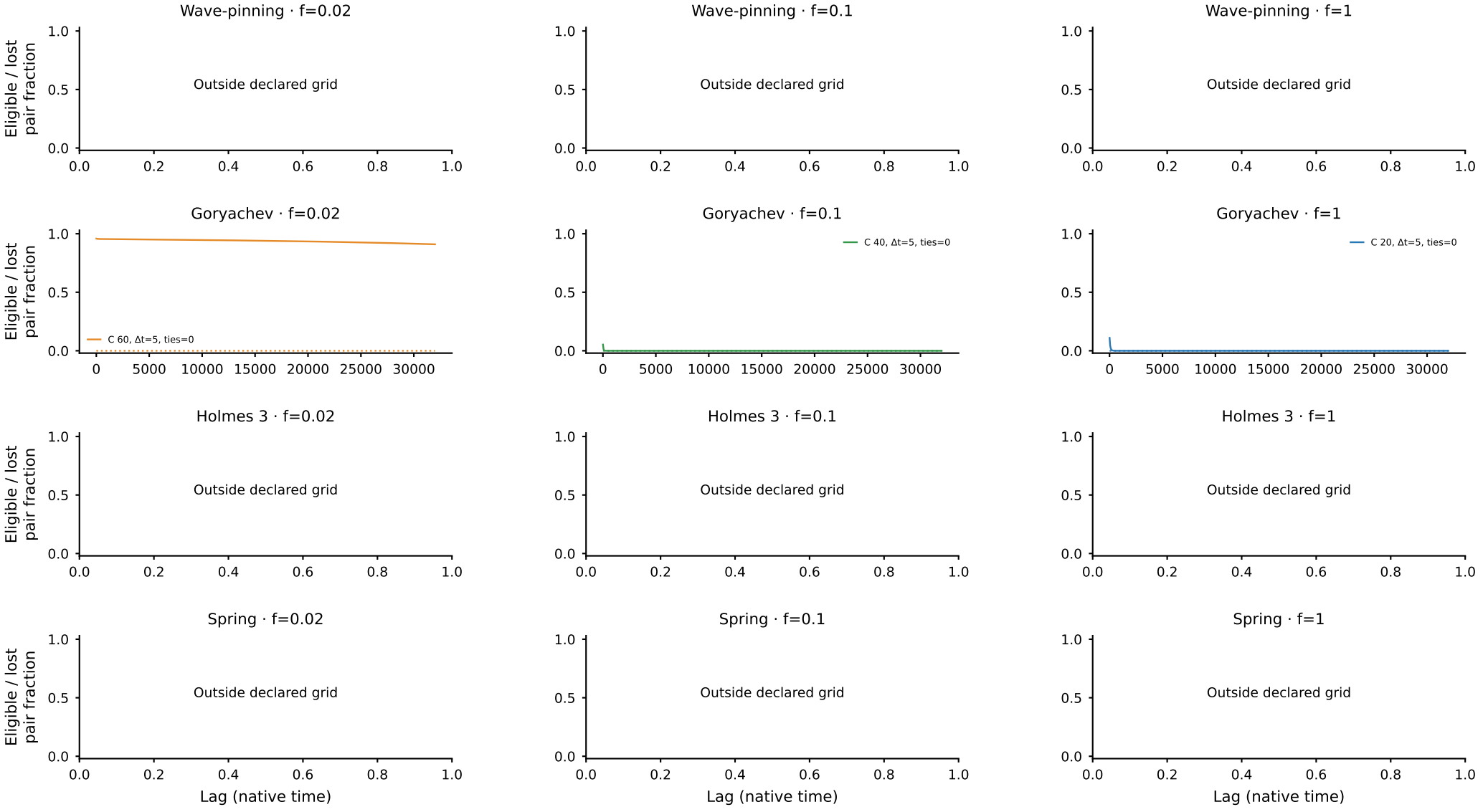
E4 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M_1|/|M_1,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_E4_literal_native_20261020_014_S5_literal_sigma1_profile_peak_literal_actualC_MSD_3.csv

S5 · E4 literal material-origin physical-site measurements · Literal heading position · A3 primary split · pair coverage and ambiguity loss · σ=1



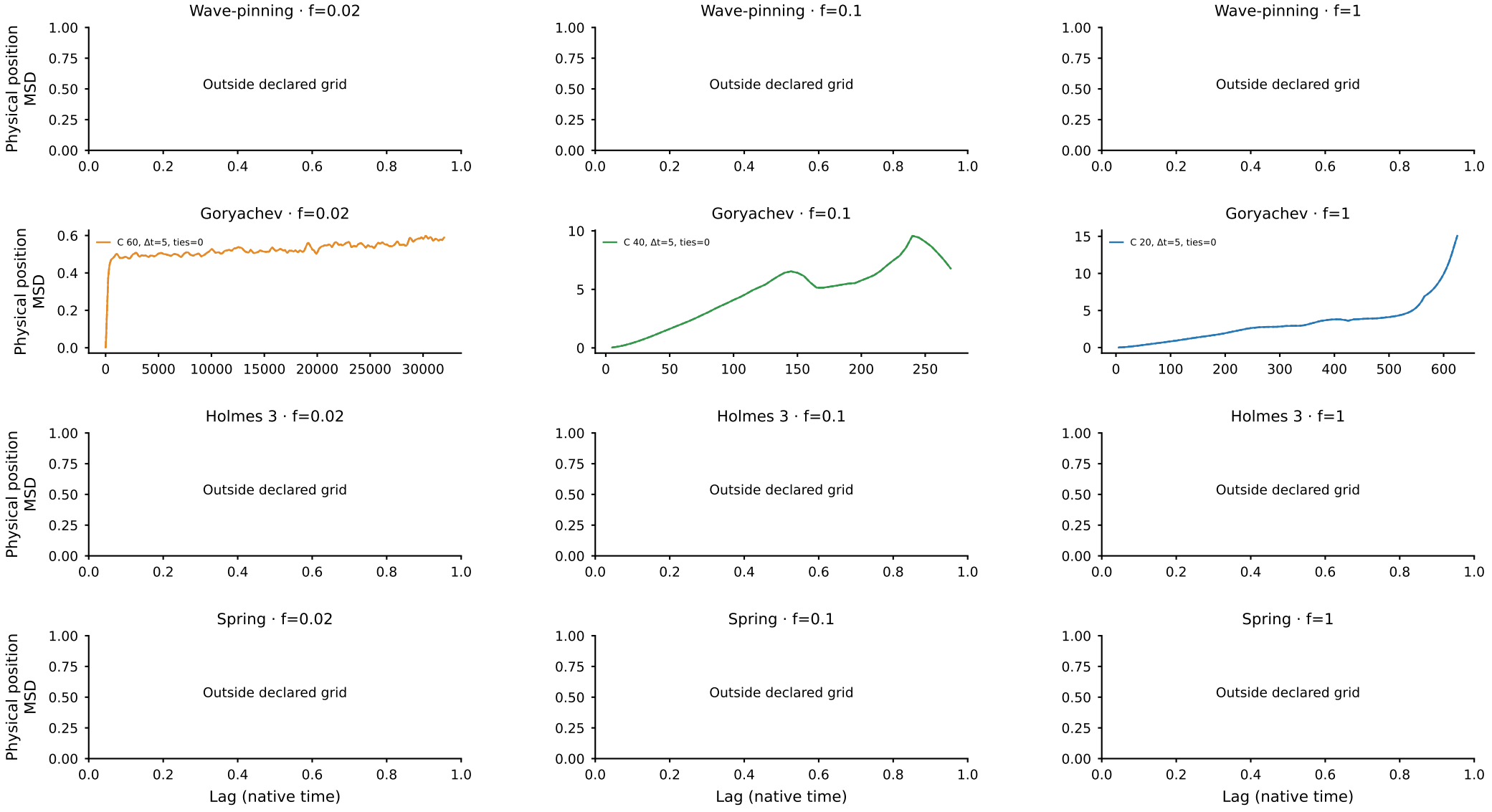
E4 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M₁|/|M₁,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_E4_literal_native_20261020_015_S5_literal_sigma1_heading_literal_actualC_coverage_3.csv

S5 · E4 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · pair coverage and ambiguity loss · σ=1



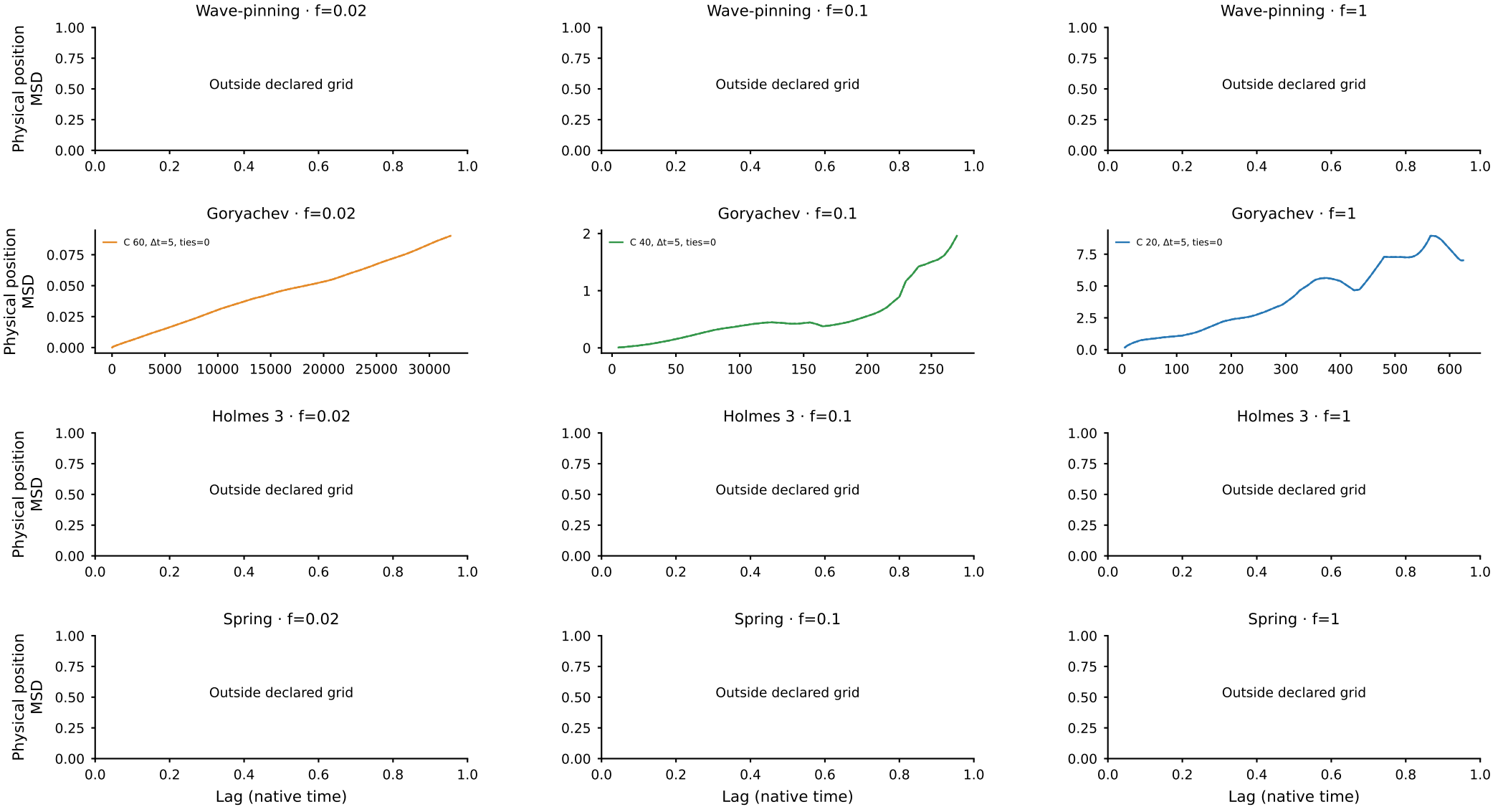
E4 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M_1|/|M_1,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_E4_literal_native_20261020_016_S5_literal_sigma1_profile_peak_literal_actualC_coverage_3.csv

S5 · E4 literal material-origin physical-site measurements · Literal heading position · paired route sensitivities · split / +π / −π · σ=1



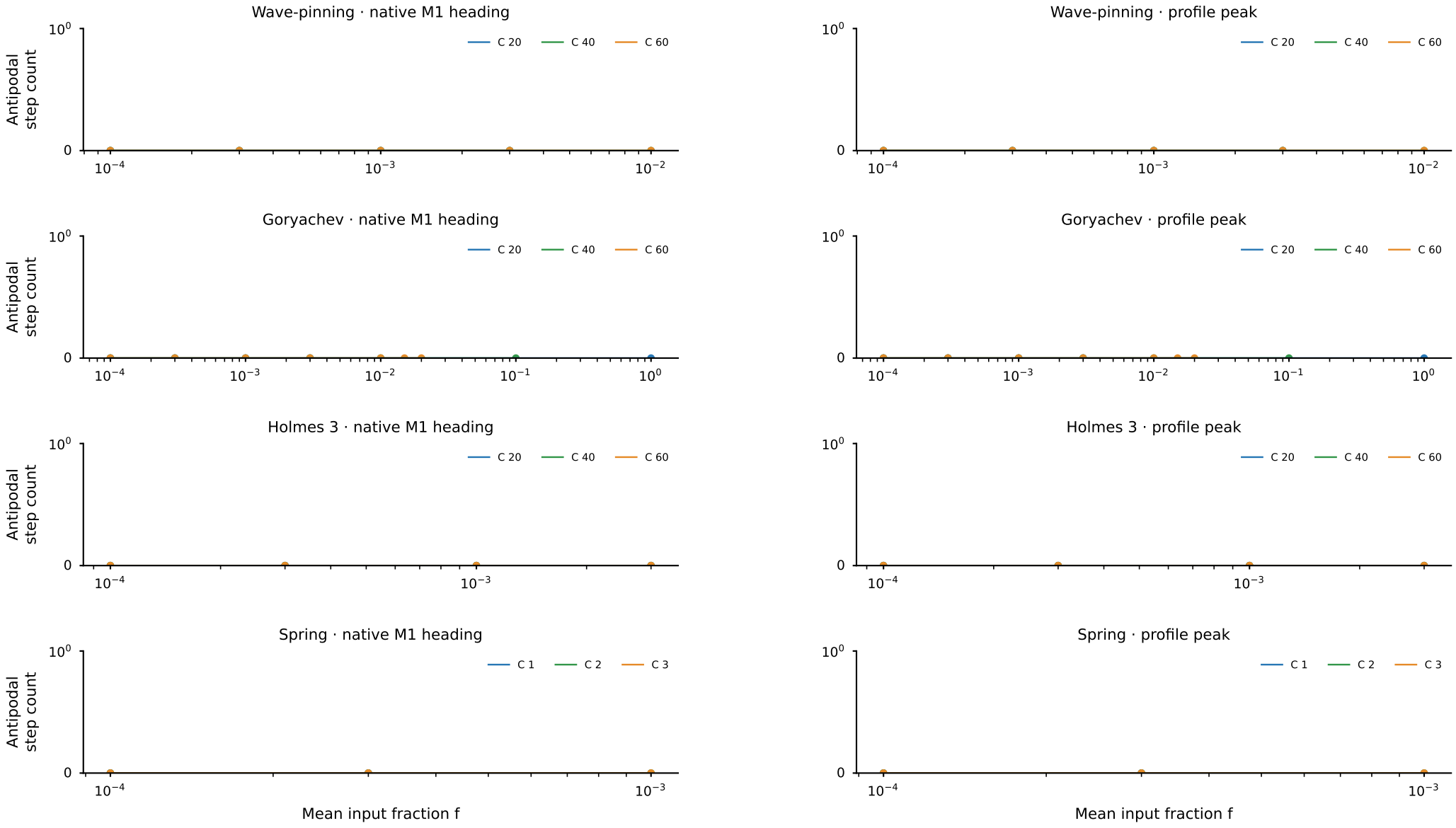
E4 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_E4_literal_native_20261020_017_S5_literal_sigma1_heading_literal_actualC_routes_3.csv.gz

S5 · E4 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · split / +π / −π · σ=1



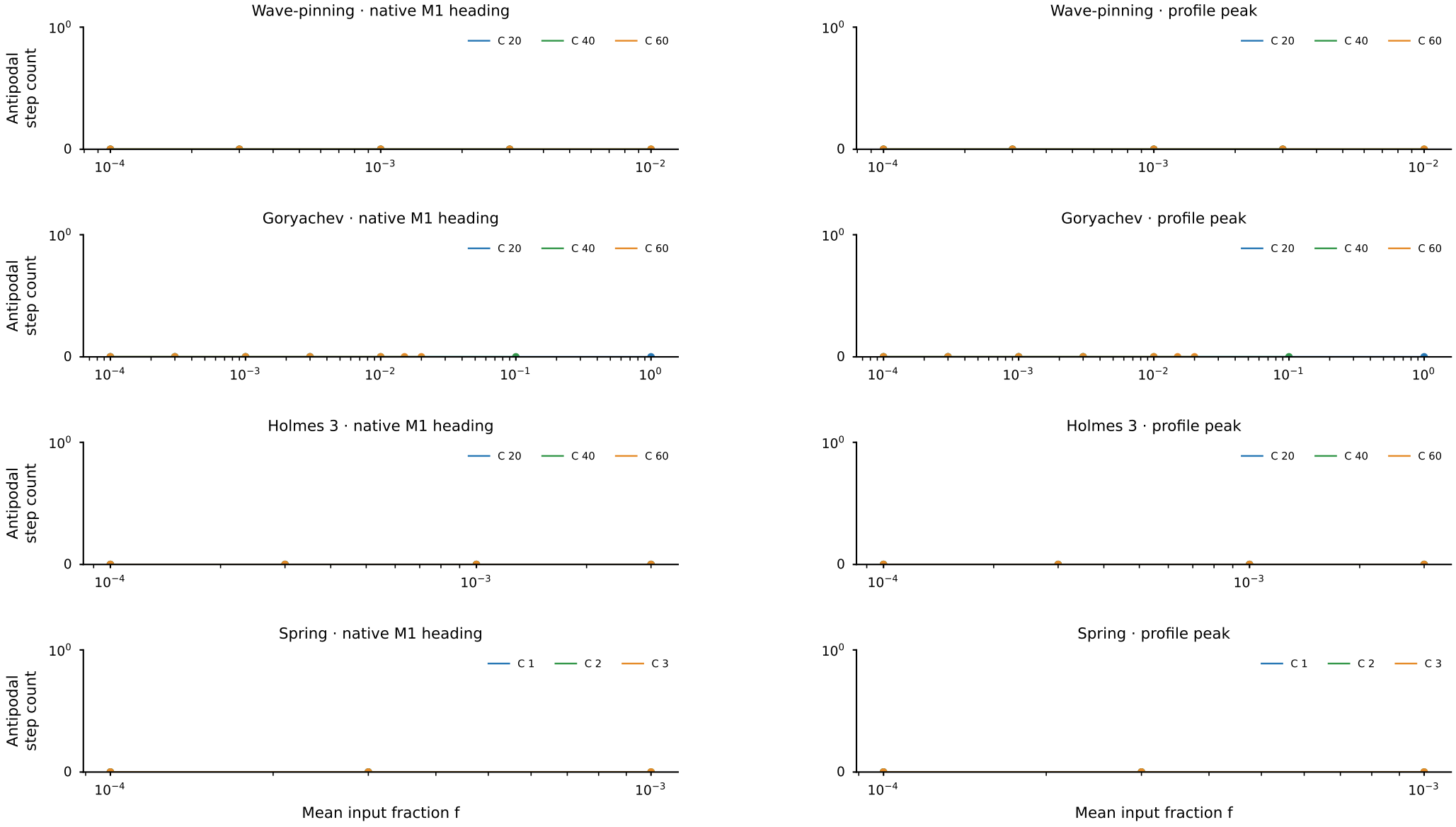
E4 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_E4_literal_native_20261020_018_S5_literal_sigma1_profile_peak_literal_actualC_routes_3.csv.gz

S5 · E4 literal material-origin physical-site measurements · Native antipodal ambiguity · main · σ=1



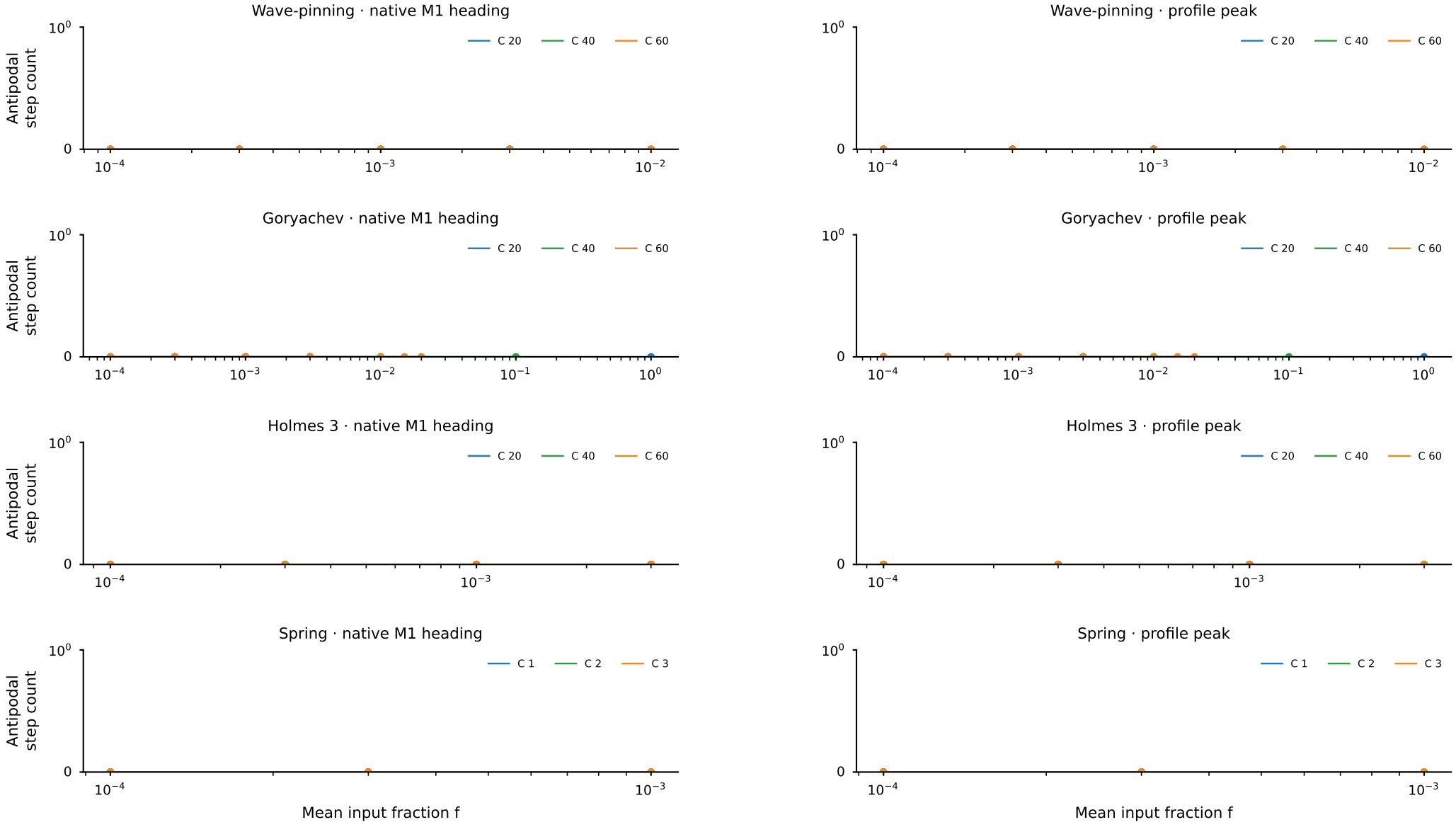
E4 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_E4_literal_native_20261020_019_S5_ambiguity_sigma1_main.csv

S5 · E4 literal material-origin physical-site measurements · Native antipodal ambiguity · analysis first half · σ=1



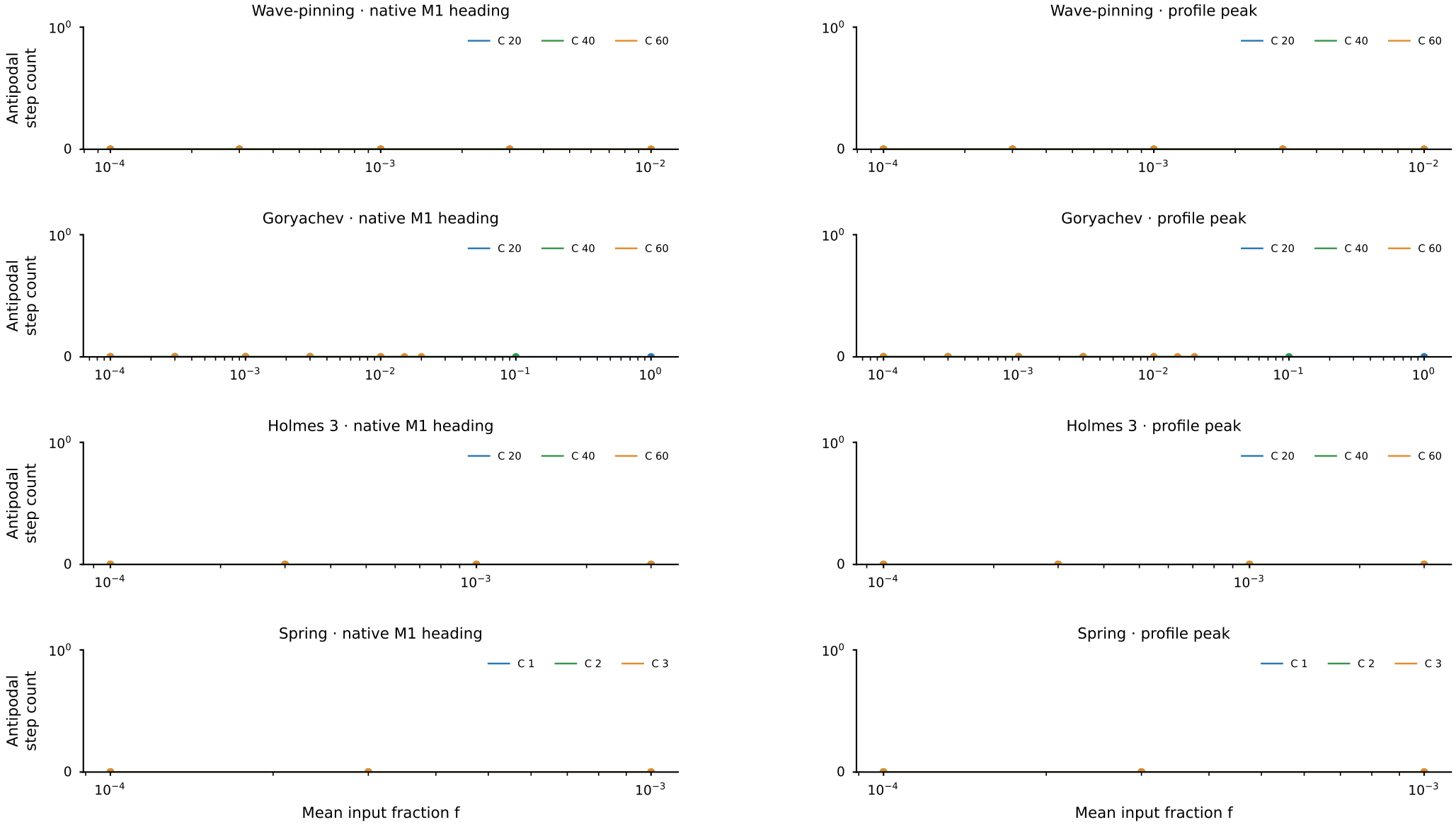
E4 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_E4_literal_native_20261020_020_S5_ambiguity_sigma1_analysis_first_half.csv

S5 · E4 literal material-origin physical-site measurements · Native antipodal ambiguity · analysis second half · σ=1



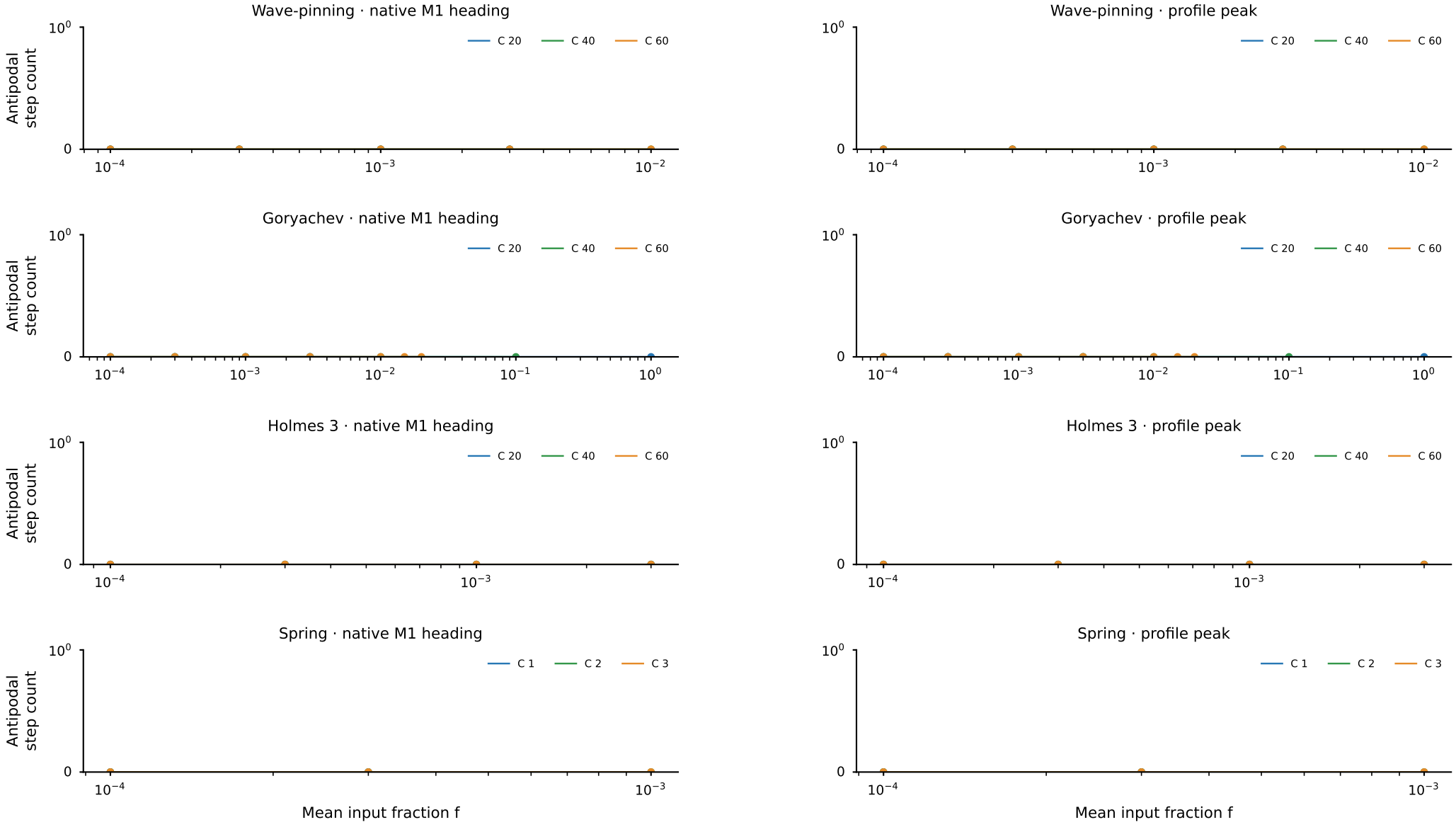
E4 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_E4_literal_native_20261020_021_S5_ambiguity_sigma1_analysis_second_half.csv

S5 · E4 literal material-origin physical-site measurements · Native antipodal ambiguity · late quarter · σ=1



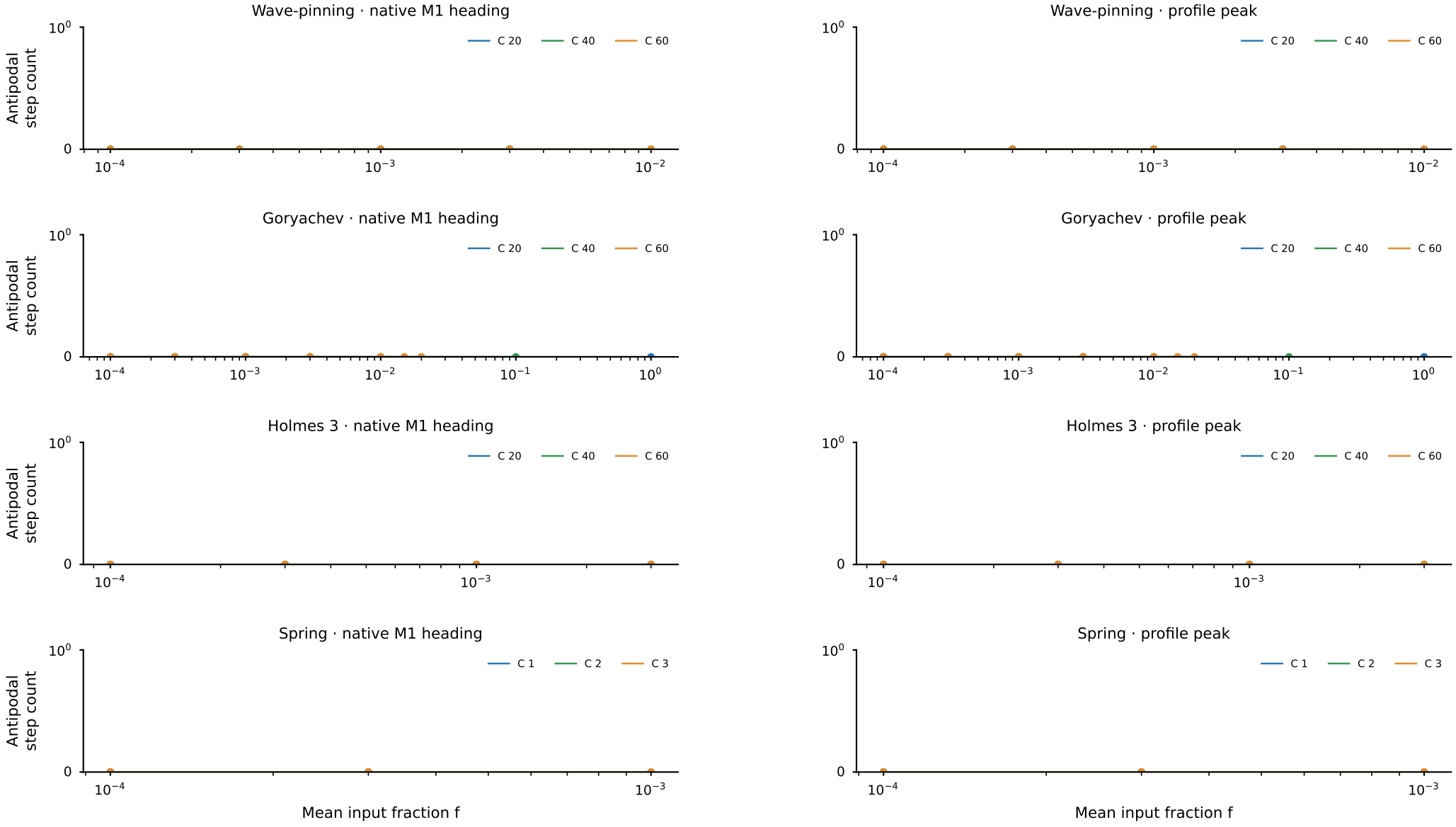
E4 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_E4_literal_native_20261020_022_S5_ambiguity_sigma1_late_quarter.csv

S5 · E4 literal material-origin physical-site measurements · Native antipodal ambiguity · cadence x2 · σ=1



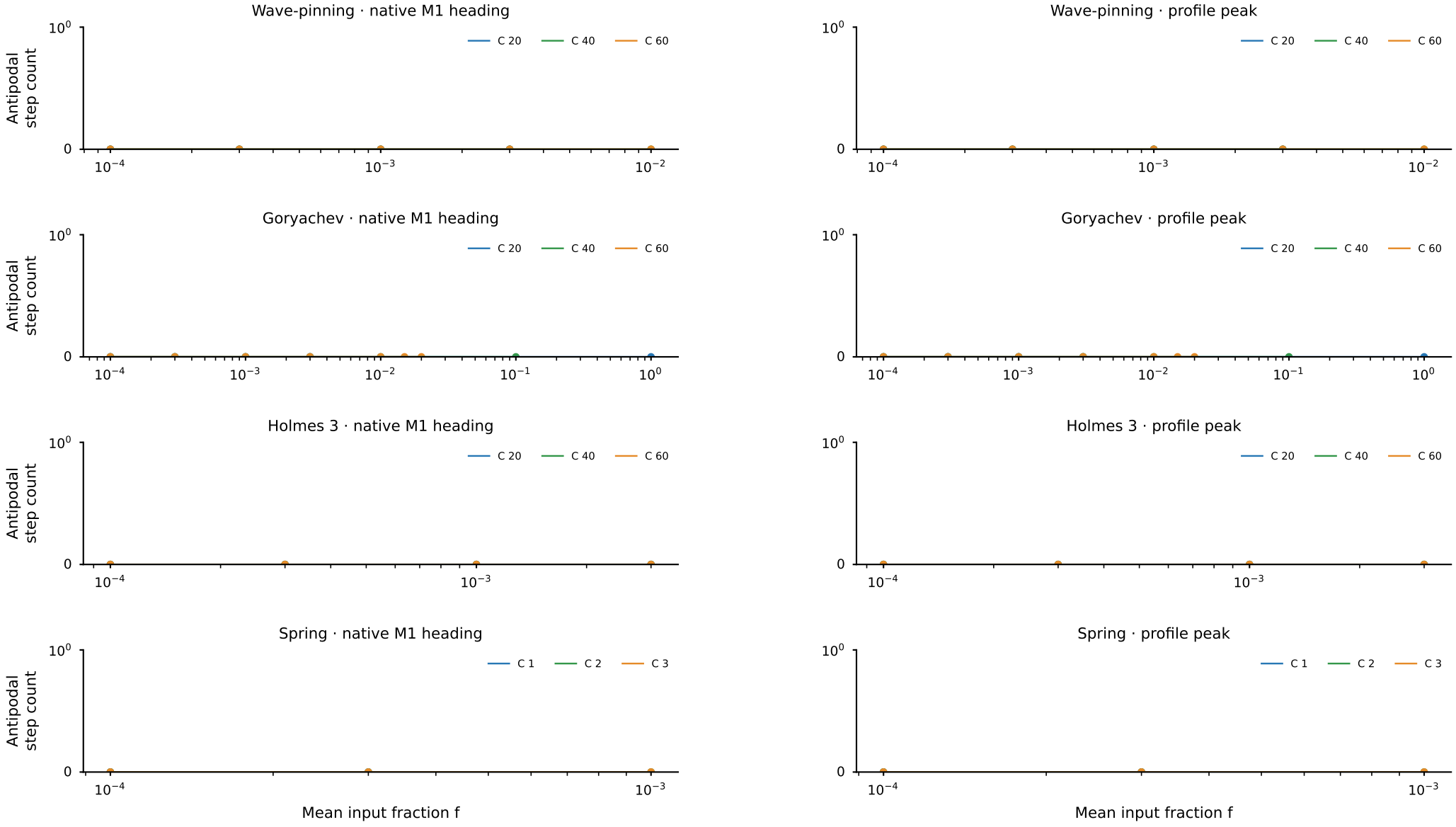
E4 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_E4_literal_native_20261020_023_S5_ambiguity_sigma1_cadence_x2.csv

S5 · E4 literal material-origin physical-site measurements · Native antipodal ambiguity · cadence x4 · σ=1



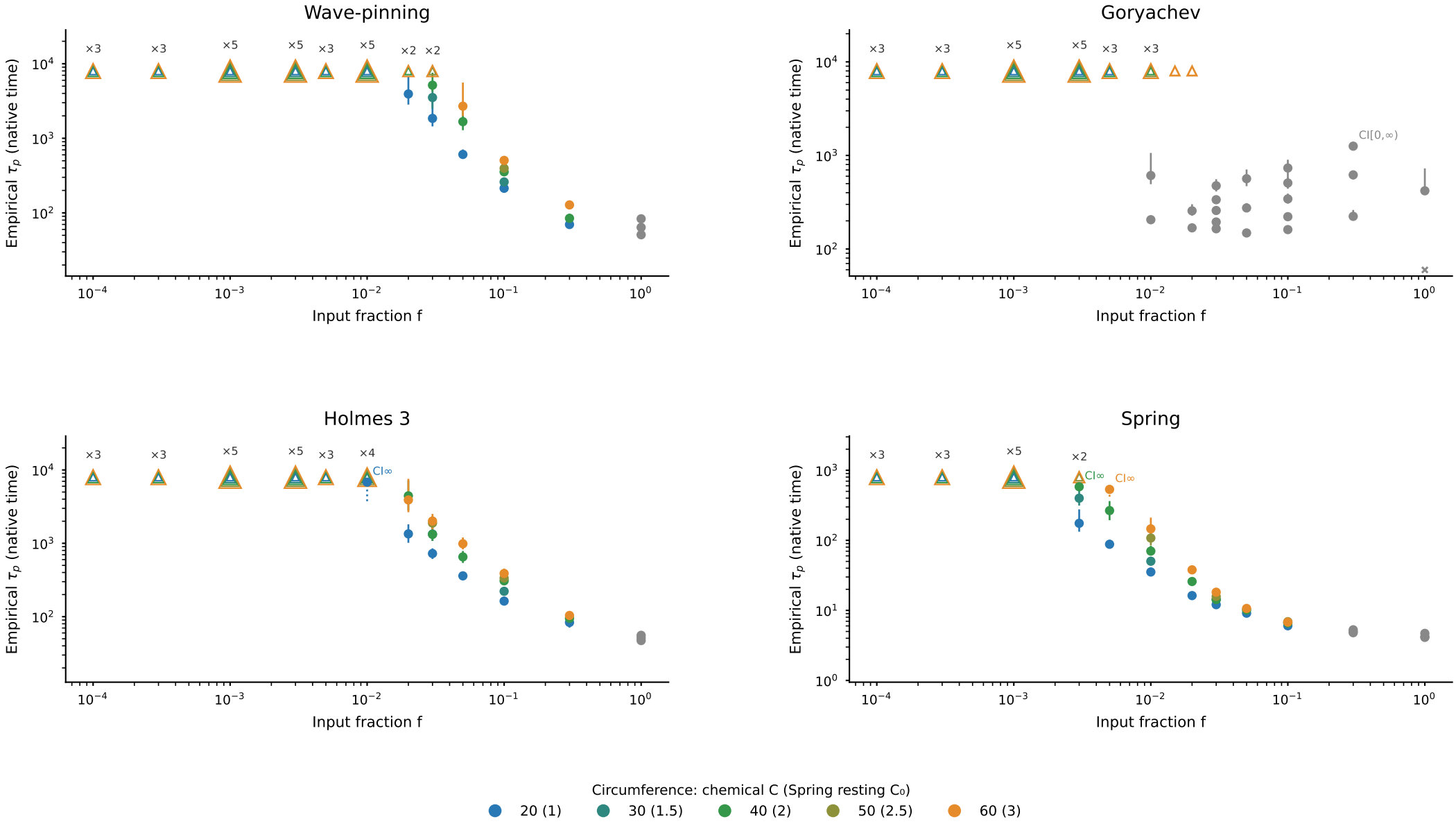
E4 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_E4_literal_native_20261020_024_S5_ambiguity_sigma1_cadence_x4.csv

S5 · E4 literal material-origin physical-site measurements · Native antipodal ambiguity · cadence x8 · σ=1



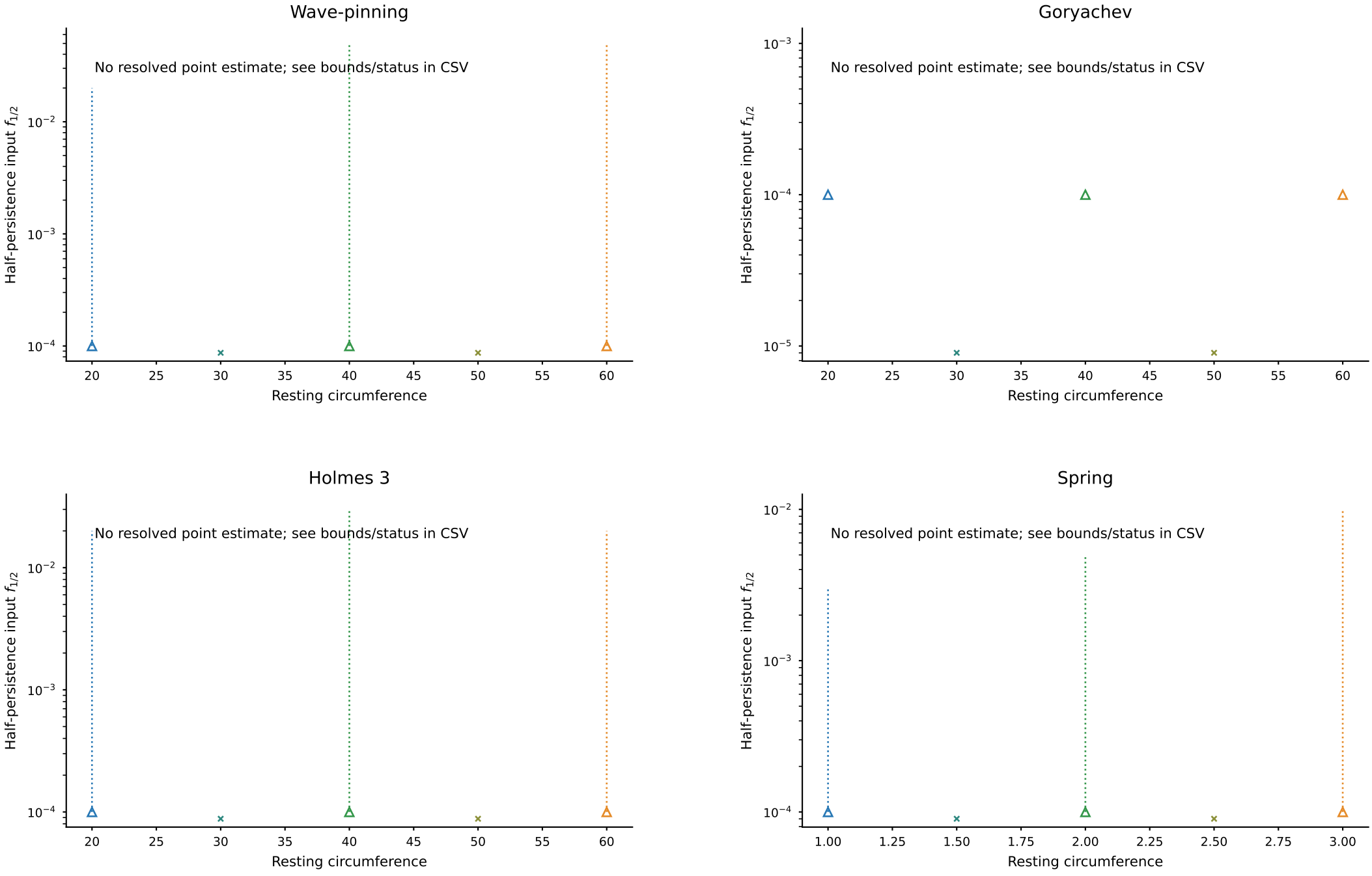
E4 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_E4_literal_native_20261020_025_S5_ambiguity_sigma1_cadence_x8.csv

S6 · Common observations · Empirical persistence against input strength



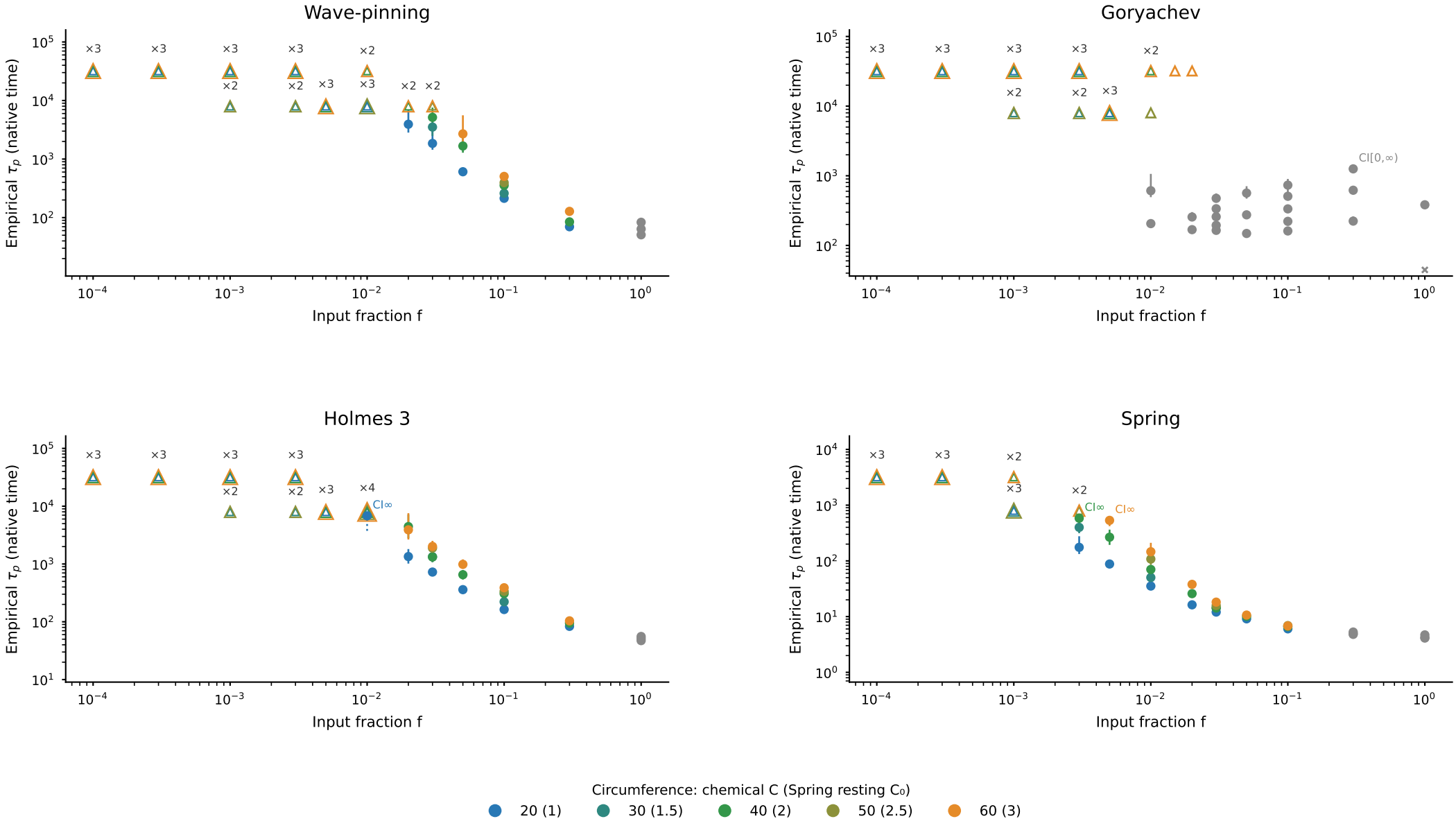
Common observations. Upward triangles are empirical lower bounds; downward triangles are unresolved below two cadences. Nested triangles at one unchanged coordinate retain coincident sizes; ×n gives their count. Their exact sizes, glyph sizes and coordinates remain in CSV. The shared legend maps chemical C to Spring resting C₀; grey retains forced observations. Bootstrap CI markers are separate from empirical bounds. Exponential fits never replace empirical measurements. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV. CI∞ next to a finite point marks an upper-unbounded bootstrap CI; finite lower endpoints are retained. CI[0,∞) spans the full nonnegative range.
CSV: figure4_common_canonical_measurements_20261020_025_S6_tau.csv

S6 · Common observations · Noise tolerance with finite-window uncertainty



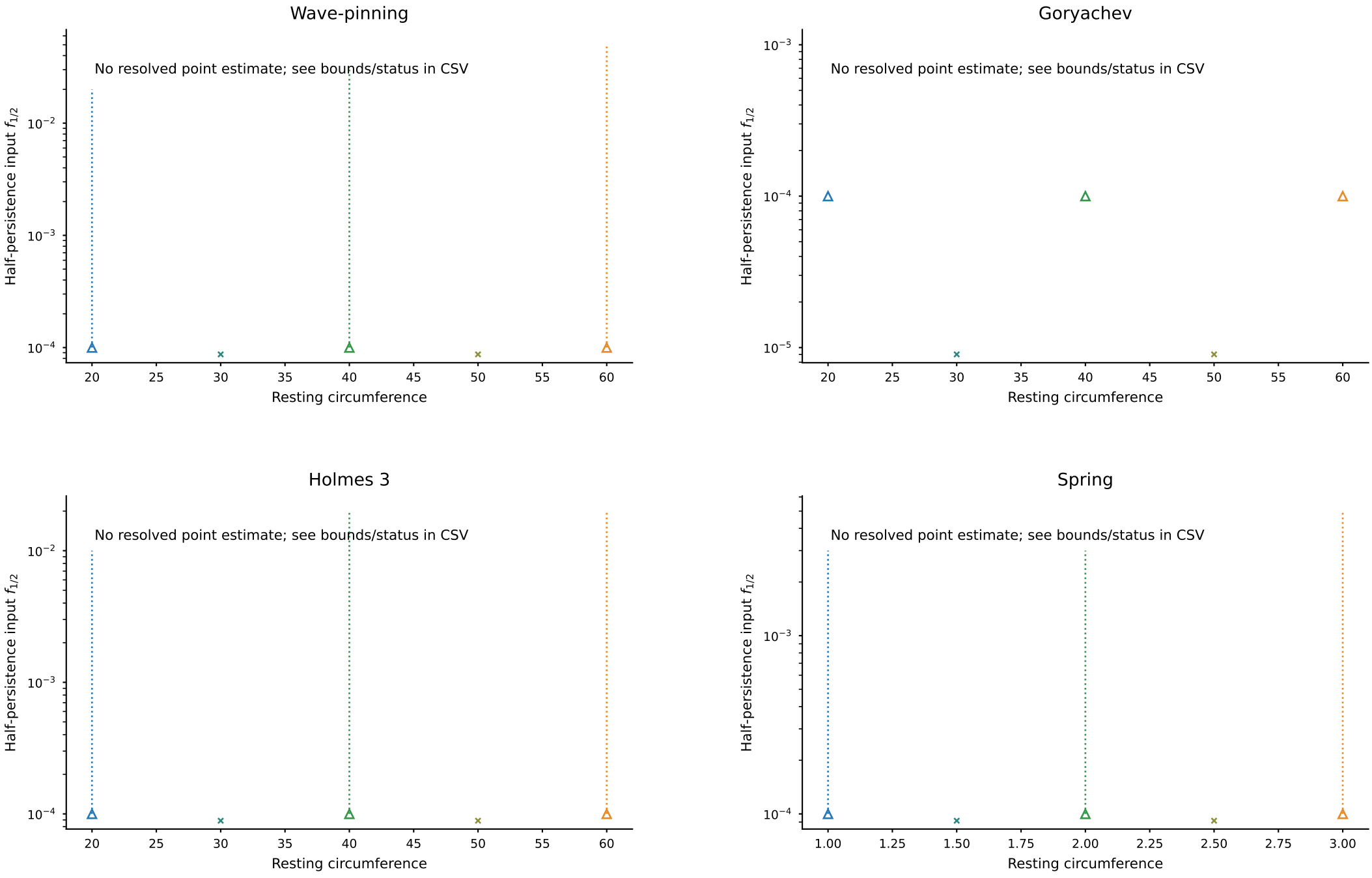
Common observations. First crossing of half the actual f=0.0001 persistence, using log-f interpolation only when identified. Censored references remain bounded or inconclusive. Intermediate-size grids lacking the declared f=0.0001 baseline retain unavailable markers/reasons; f=0.001 is never substituted and no tolerance curve bridges them. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_common_canonical_measurements_20261020_026_S6_fhalf.csv

S6 · Longest qualified observations · Empirical persistence against input strength



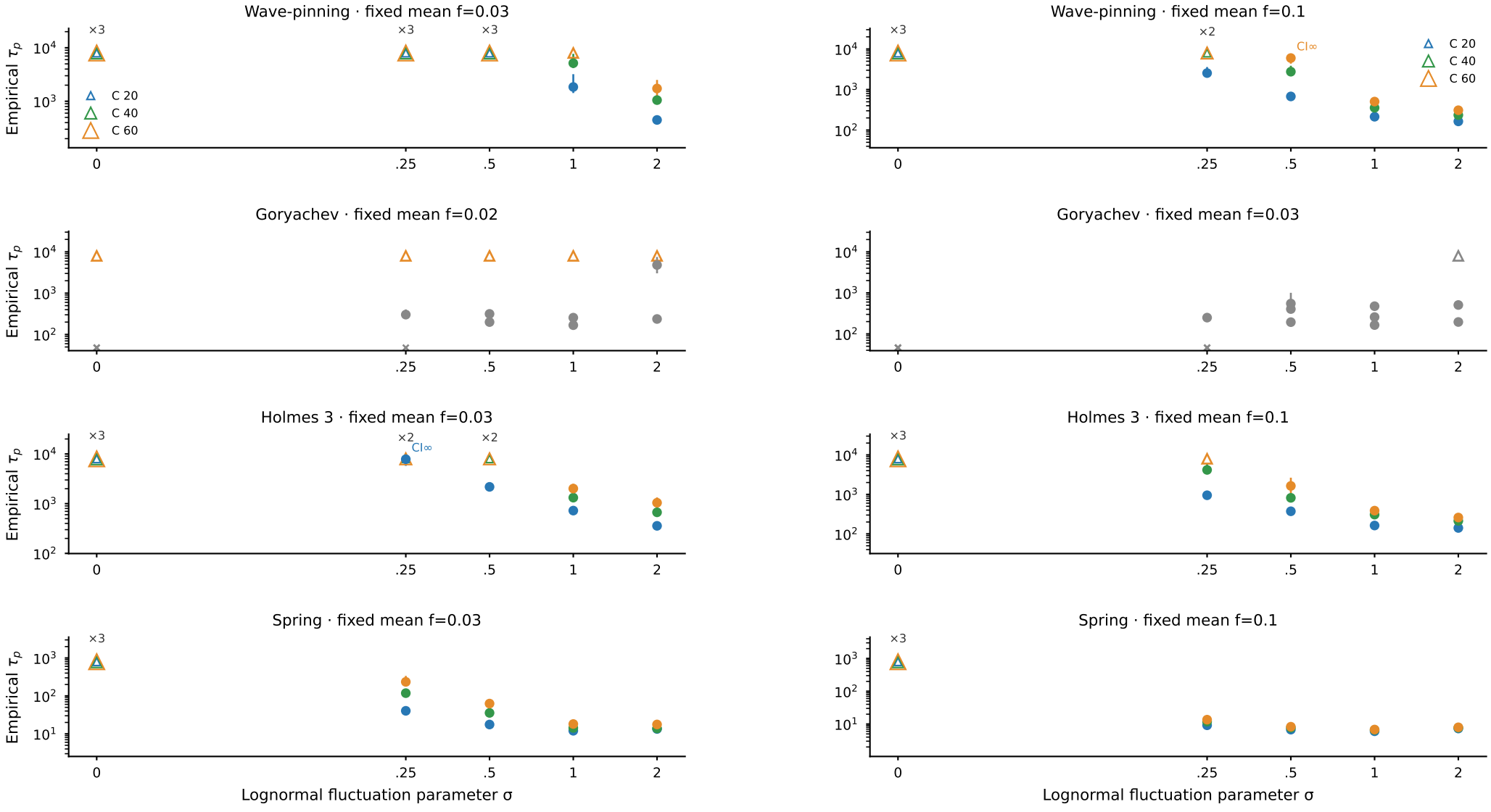
Longest qualified observations. Upward triangles are empirical lower bounds; downward triangles are unresolved below two cadences. Nested triangles at one unchanged coordinate retain coincident sizes; ×n gives their count. Their exact sizes, glyph sizes and coordinates remain in CSV. The shared legend maps chemical C to Spring resting C₀; grey retains forced observations. Bootstrap CI markers are separate from empirical bounds. Exponential fits never replace empirical measurements. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV. CI∞ next to a finite point marks an upper-unbounded bootstrap CI; finite lower endpoints are retained. CI[0,∞) spans the full nonnegative range.
CSV: figure4_longest_canonical_measurements_20261020_025_S6_tau.csv

S6 · Longest qualified observations · Noise tolerance with finite-window uncertainty



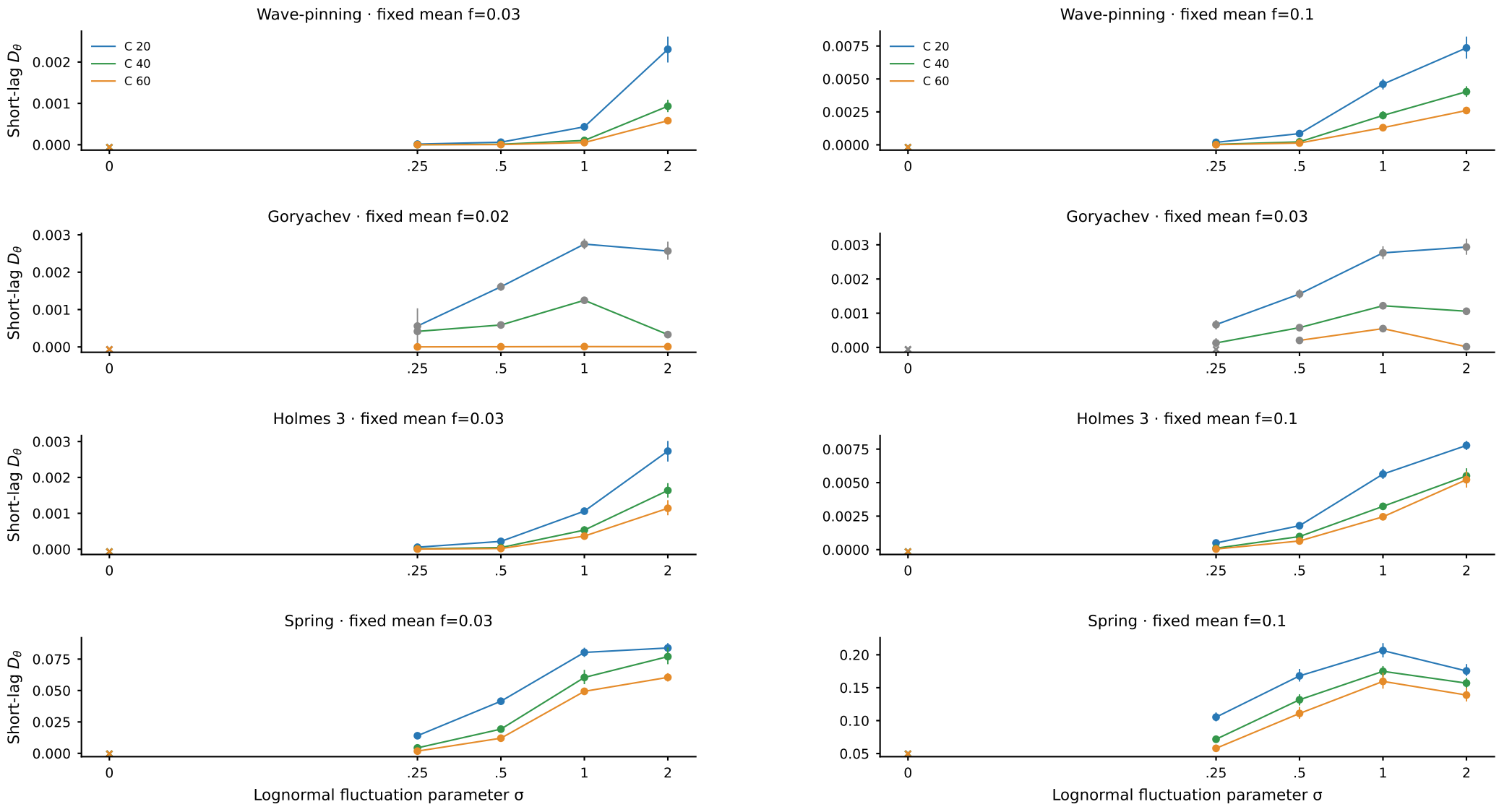
Longest qualified observations. First crossing of half the actual f=0.0001 persistence, using log-f interpolation only when identified. Censored references remain bounded or inconclusive. Intermediate-size grids lacking the declared f=0.0001 baseline retain unavailable markers/reasons; f=0.001 is never substituted and no tolerance curve bridges them. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_longest_canonical_measurements_20261020_026_S6_fhalf.csv

S7 · E1 fixed-mean σ sweep · Fixed-mean fluctuation sweep



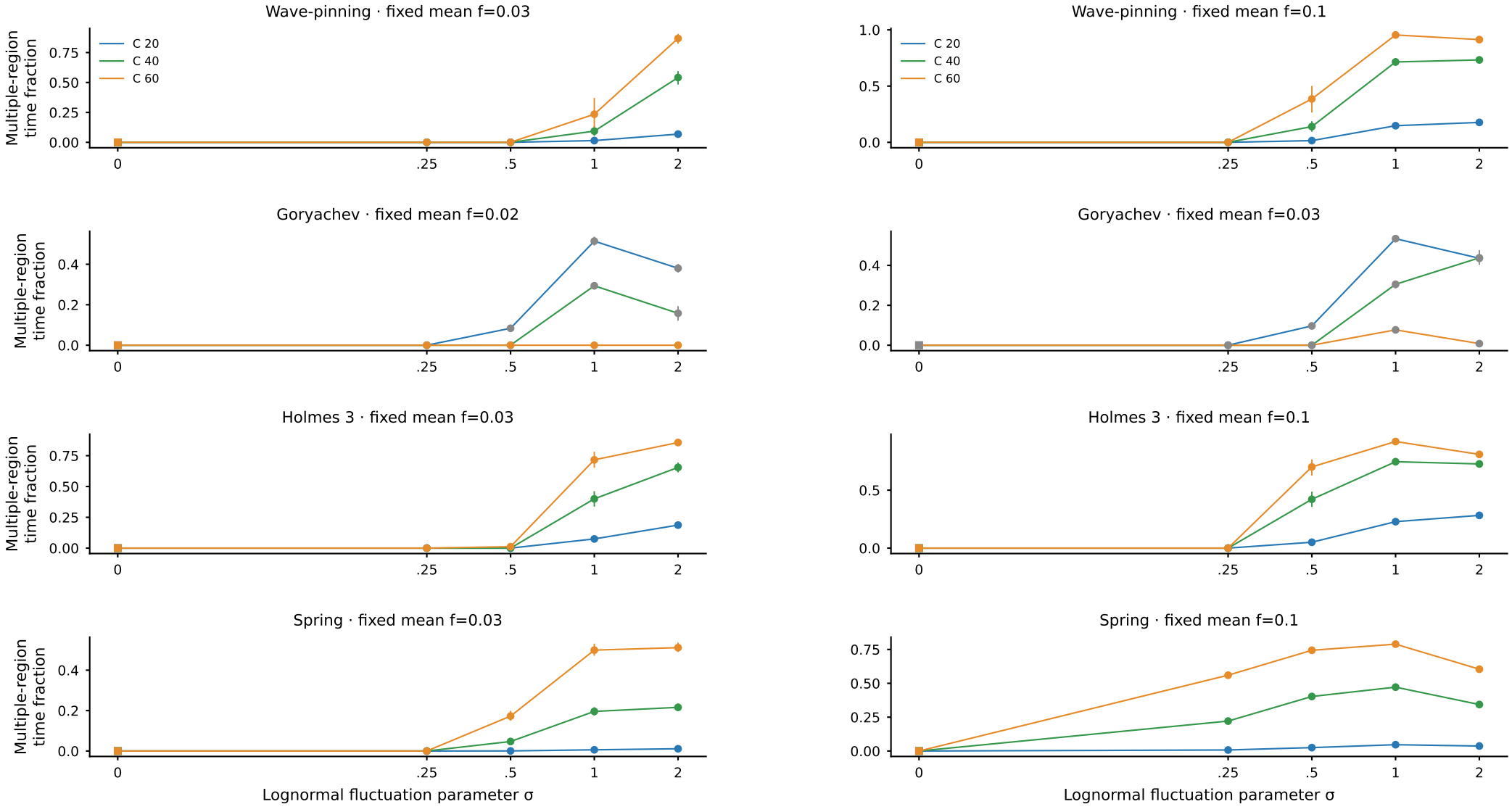
E1 fixed-mean σ sweep. At σ=0, resolved deterministic means use squares; triangles retain bounds and × marks unidentified values. Nested triangles and ×n retain coincident sizes at unchanged coordinates. Chemical C20/40/60 correspond to Spring resting $C_0$1/2/3. Controls have no ensemble CI. Other arms use twelve common latent Gaussian seed blocks, verified by knot hashes, and whole-seed paired bootstrap. Grey uses combined mean/E1 qualification; σ is not mean input. Native time units differ; bounds remain bounds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV. CI∞ next to a finite point marks an upper-unbounded bootstrap CI; finite lower endpoints are retained. CI[0,∞) spans the full nonnegative range.
CSV: figure4_S7_fixed_mean_completed_v3_20261020_001_S7_tau_p.csv

S7 · E1 fixed-mean σ sweep · Fixed-mean fluctuation sweep



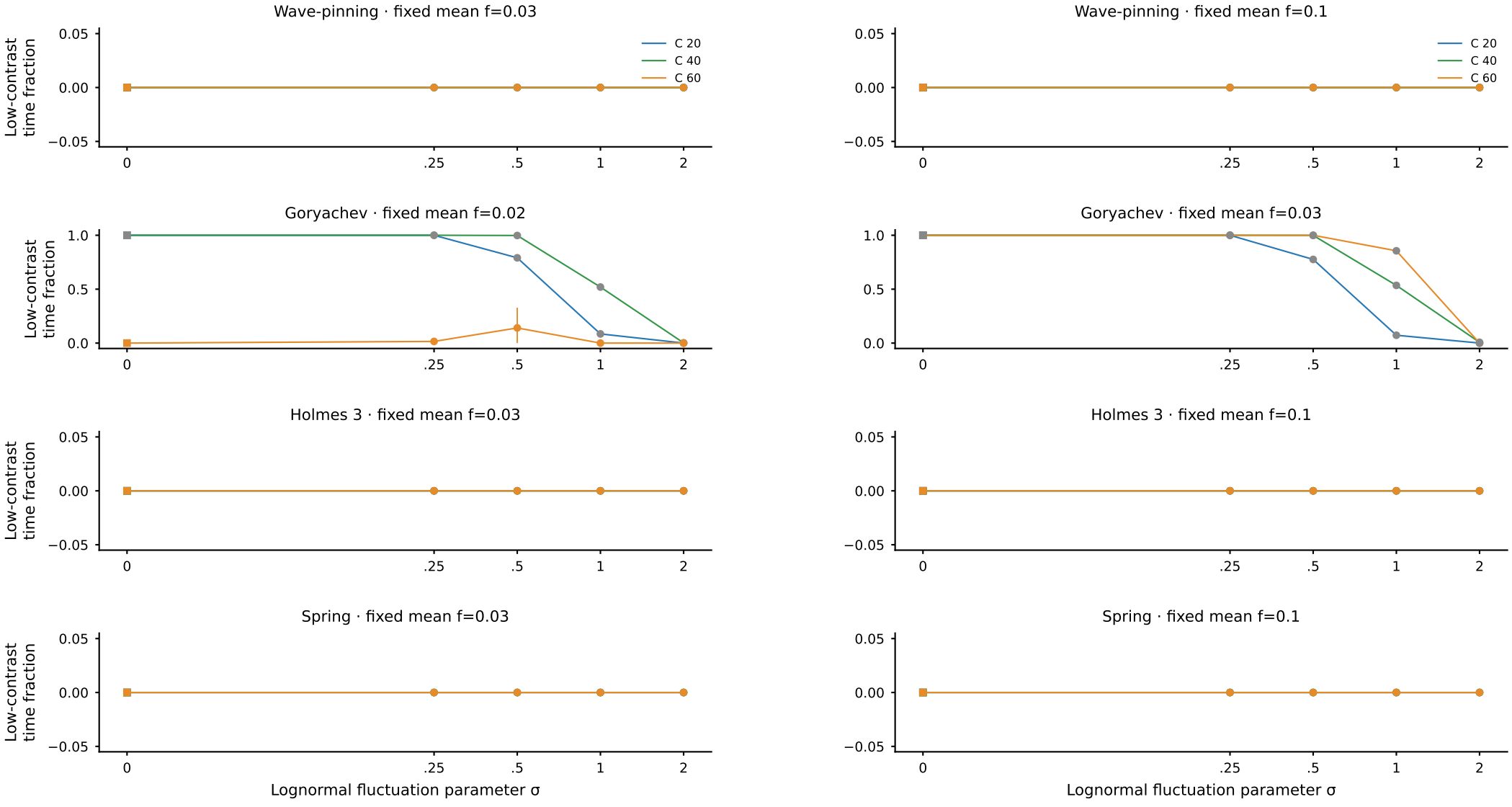
E1 fixed-mean σ sweep. At σ=0, resolved deterministic means use squares; triangles retain bounds and × marks unidentified values. Nested triangles and ×n retain coincident sizes at unchanged coordinates. Chemical C20/40/60 correspond to Spring resting C₀1/2/3. Controls have no ensemble CI. Other arms use twelve common latent Gaussian seed blocks, verified by knot hashes, and whole-seed paired bootstrap. Grey uses combined mean/E1 qualification; σ is not mean input. Native time units differ; bounds remain bounds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S7_fixed_mean_completed_v3_20261020_002_S7_Dtheta.csv

S7 · E1 fixed-mean σ sweep · Fixed-mean fluctuation sweep



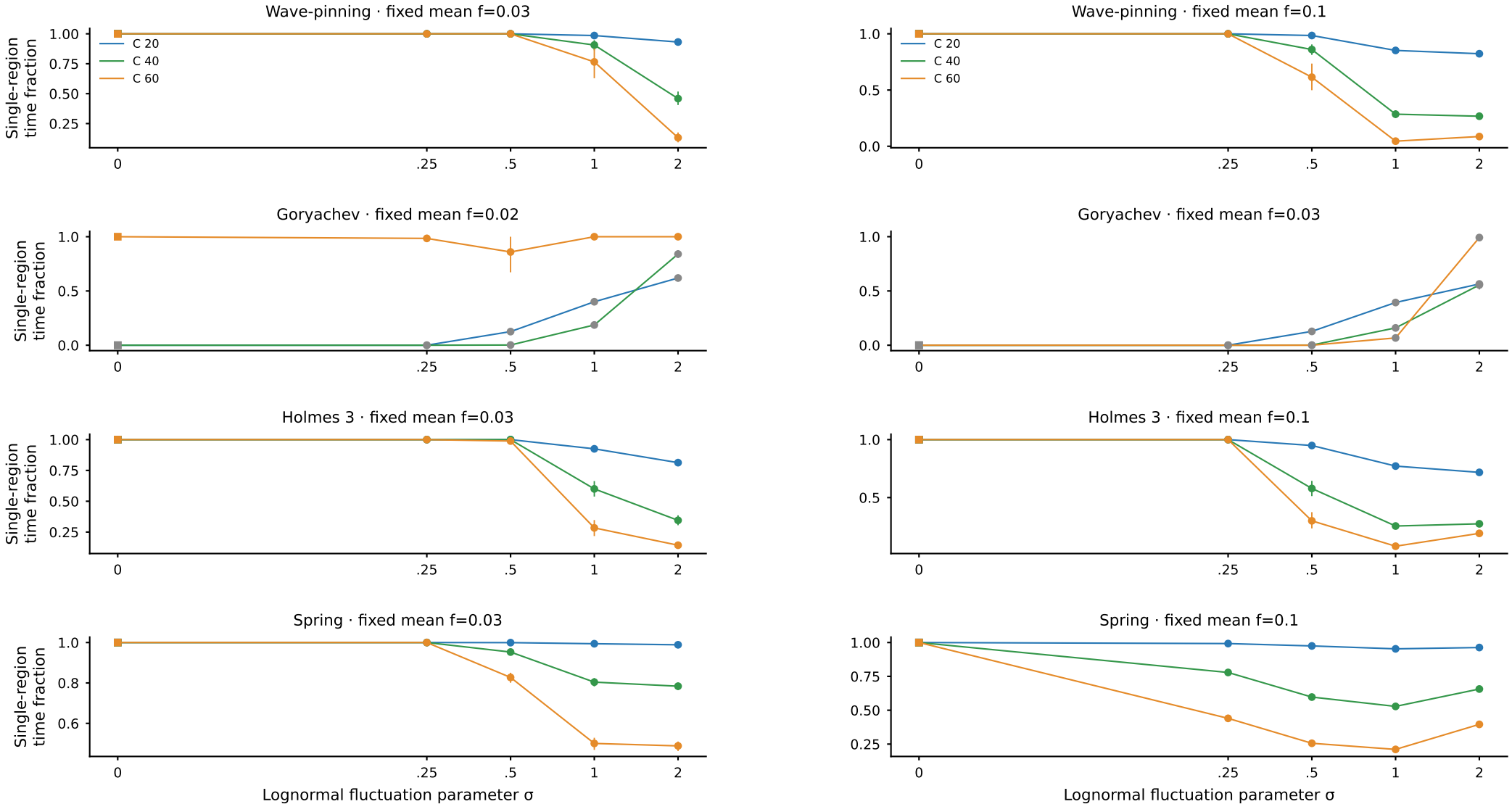
E1 fixed-mean σ sweep. At σ=0, resolved deterministic means use squares; triangles retain bounds and × marks unidentified values. Nested triangles and ×n retain coincident sizes at unchanged coordinates. Chemical C20/40/60 correspond to Spring resting C₀1/2/3. Controls have no ensemble CI. Other arms use twelve common latent Gaussian seed blocks, verified by knot hashes, and whole-seed paired bootstrap. Grey uses combined mean/E1 qualification; σ is not mean input. Native time units differ; bounds remain bounds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S7_fixed_mean_completed_v3_20261020_003_S7_multiple_fraction.csv

S7 · E1 fixed-mean σ sweep · Fixed-mean fluctuation sweep



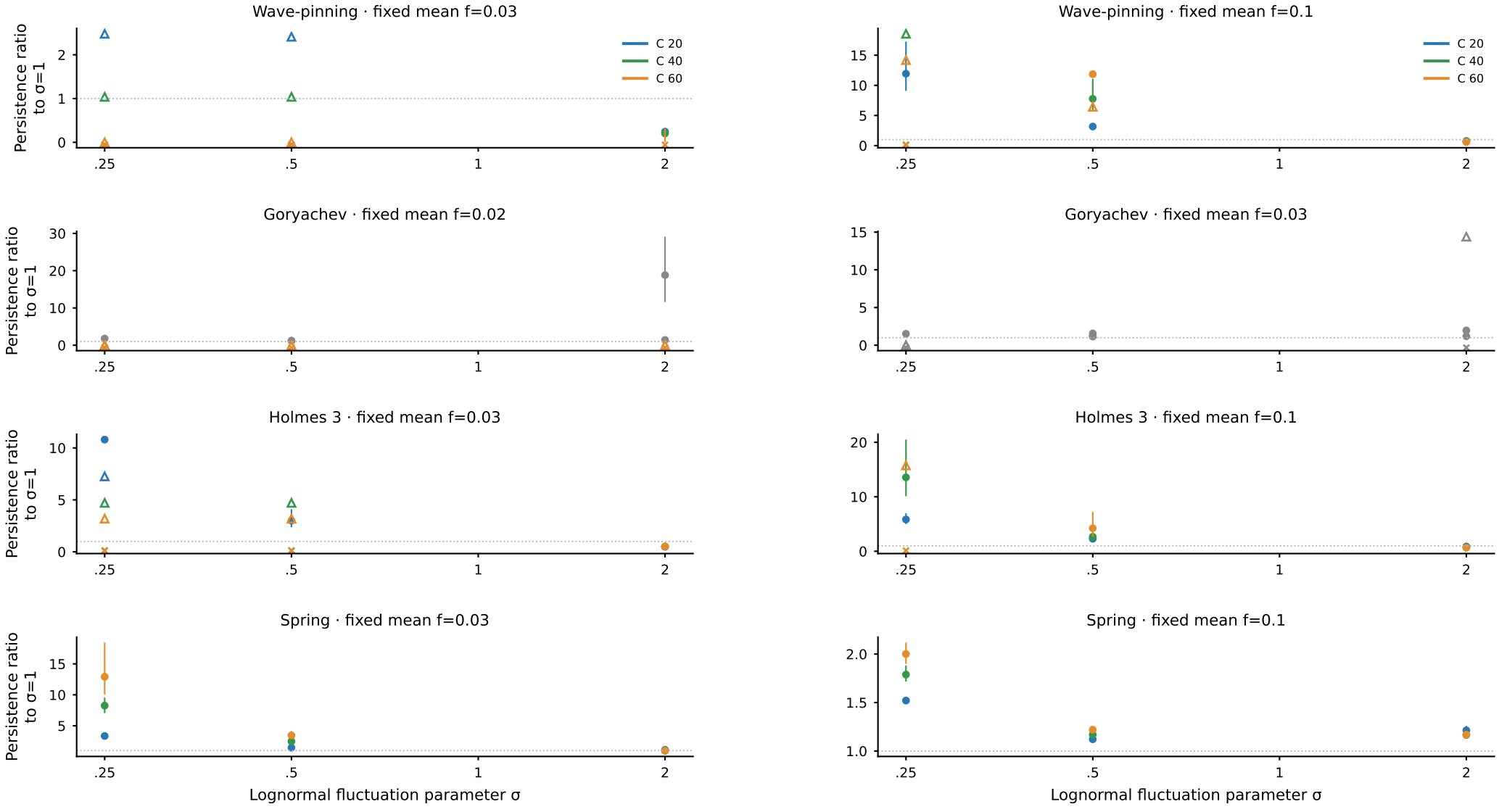
E1 fixed-mean σ sweep. At σ=0, resolved deterministic means use squares; triangles retain bounds and × marks unidentified values. Nested triangles and ×n retain coincident sizes at unchanged coordinates. Chemical C20/40/60 correspond to Spring resting $C_0$1/2/3. Controls have no ensemble CI. Other arms use twelve common latent Gaussian seed blocks, verified by knot hashes, and whole-seed paired bootstrap. Grey uses combined mean/E1 qualification; σ is not mean input. Native time units differ; bounds remain bounds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S7_fixed_mean_completed_v3_20261020_004_S7_low_contrast_fraction.csv

S7 · E1 fixed-mean σ sweep · Fixed-mean fluctuation sweep



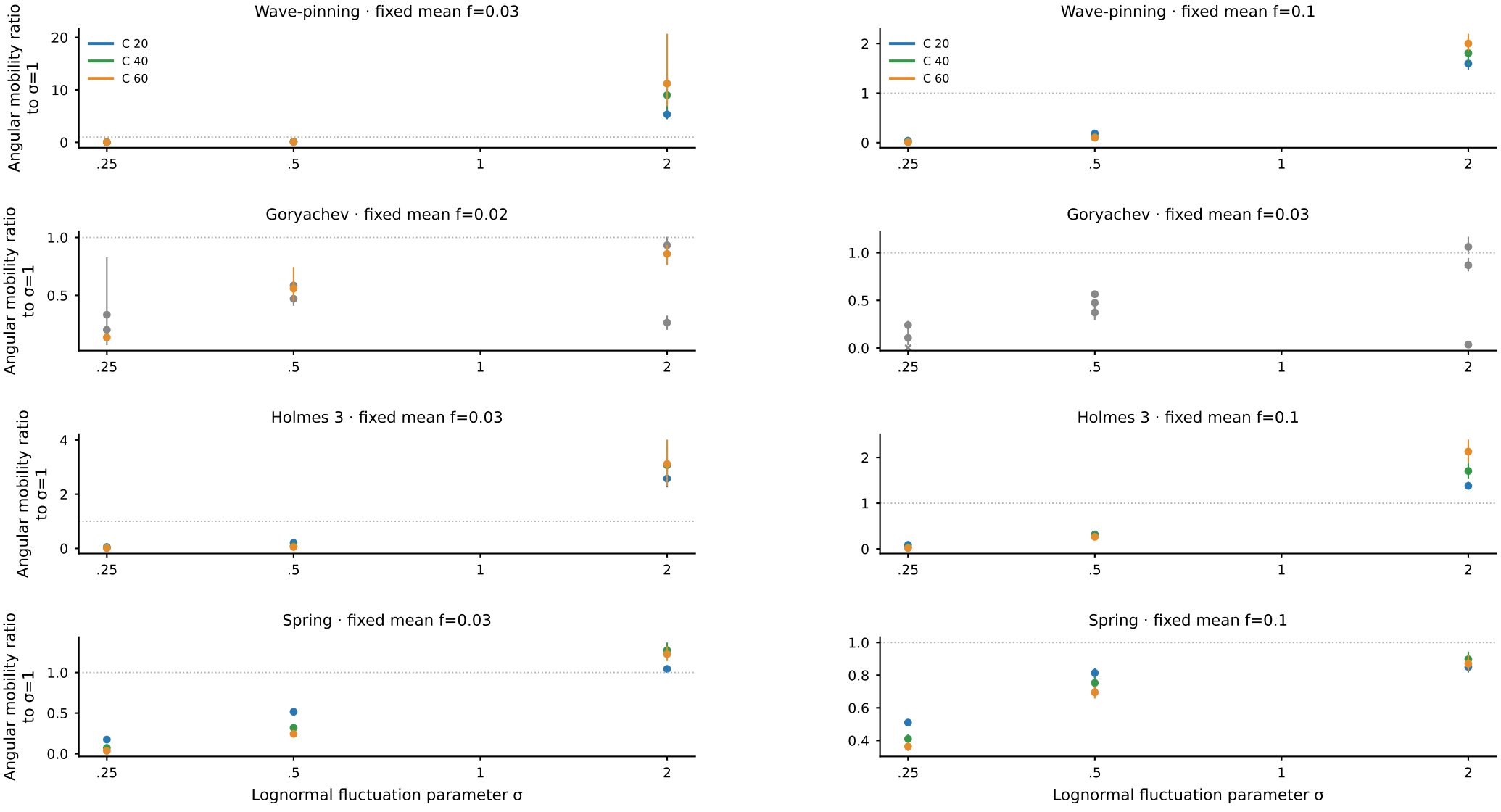
E1 fixed-mean σ sweep. At σ=0, resolved deterministic means use squares; triangles retain bounds and × marks unidentified values. Nested triangles and ×n retain coincident sizes at unchanged coordinates. Chemical C20/40/60 correspond to Spring resting $C_0$1/2/3. Controls have no ensemble CI. Other arms use twelve common latent Gaussian seed blocks, verified by knot hashes, and whole-seed paired bootstrap. Grey uses combined mean/E1 qualification; σ is not mean input. Native time units differ; bounds remain bounds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S7_fixed_mean_completed_v3_20261020_005_S7_single_fraction.csv

S7 · E1 fixed-mean σ sweep · Persistence ratio to σ=1



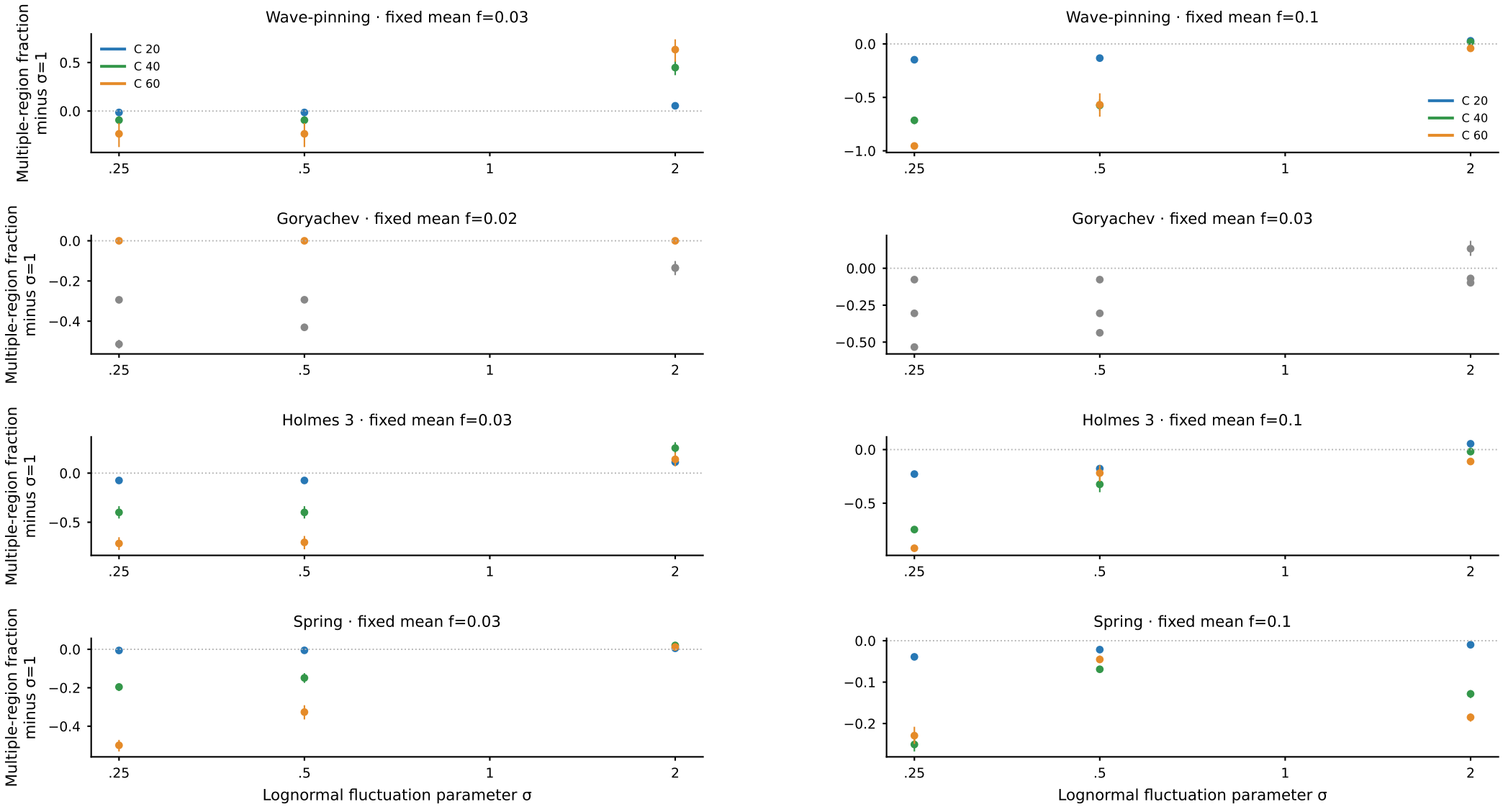
E1 fixed-mean σ sweep. Direct paired comparisons use twelve identical latent Gaussian seed blocks across σ. Grey marks retain combined-qualified forced conditions. Vertical ranges are bootstrap confidence intervals; open upward triangles mark finite lower confidence endpoints when upper endpoints are unbounded. × marks an unidentified point; no midpoint is invented. The σ=1 reference line is an identity, not an additional ensemble. Every unavailable value and valid-bootstrap fraction is retained in CSV. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S7_fixed_mean_completed_v3_20261020_006_S7_paired_tau_ratio.csv

S7 · E1 fixed-mean σ sweep · Angular mobility ratio to σ=1



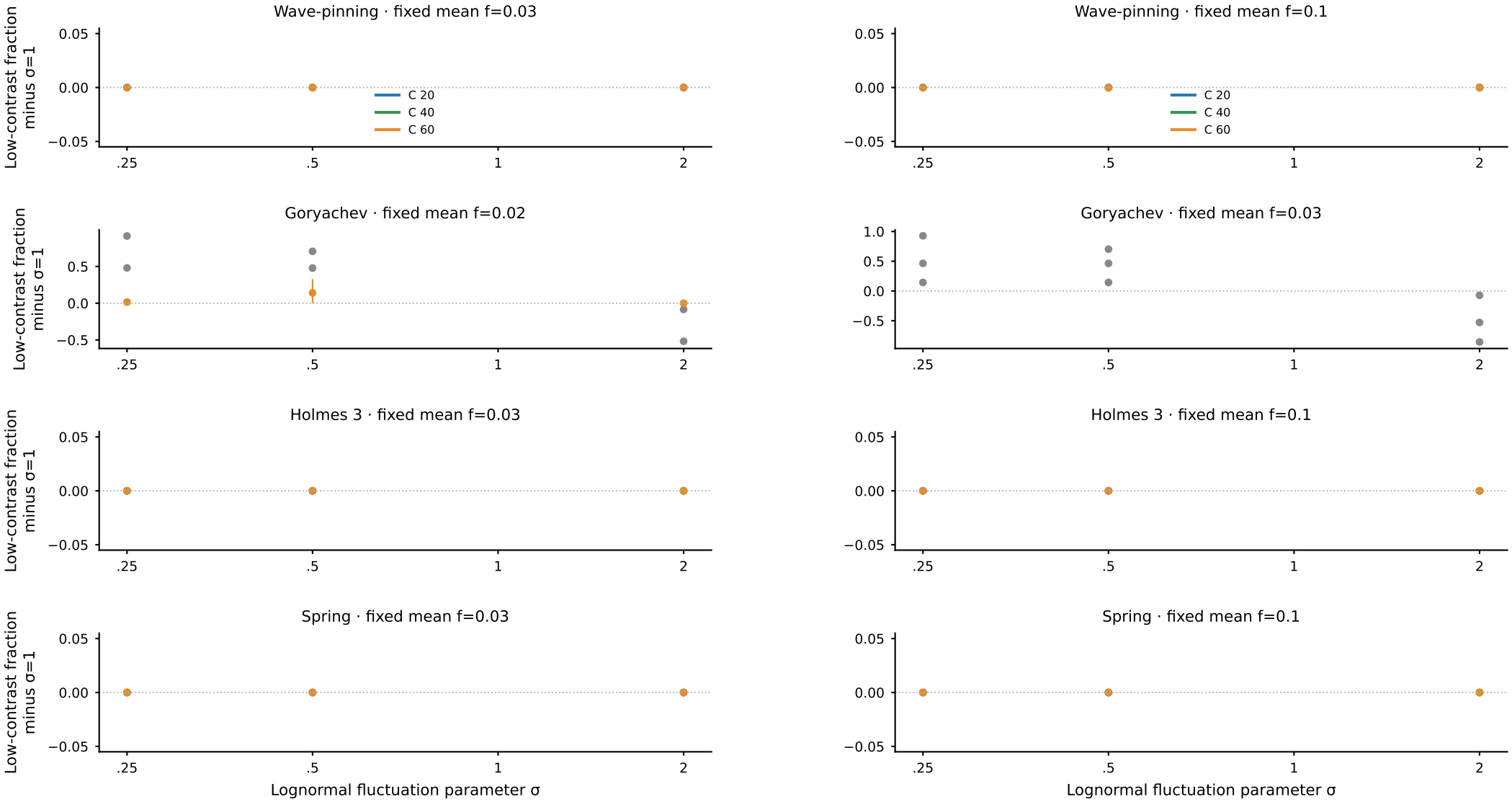
E1 fixed-mean σ sweep. Direct paired comparisons use twelve identical latent Gaussian seed blocks across σ. Grey marks retain combined-qualified forced conditions. Vertical ranges are bootstrap confidence intervals; open upward triangles mark finite lower confidence endpoints when upper endpoints are unbounded. × marks an unidentified point; no midpoint is invented. The σ=1 reference line is an identity, not an additional ensemble. Every unavailable value and valid-bootstrap fraction is retained in CSV. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S7_fixed_mean_completed_v3_20261020_007_S7_paired_Dtheta_ratio.csv

S7 · E1 fixed-mean σ sweep · Multiple-region fraction minus σ=1



E1 fixed-mean σ sweep. Direct paired comparisons use twelve identical latent Gaussian seed blocks across σ. Grey marks retain combined-qualified forced conditions. Vertical ranges are bootstrap confidence intervals; open upward triangles mark finite lower confidence endpoints when upper endpoints are unbounded. × marks an unidentified point; no midpoint is invented. The σ=1 reference line is an identity, not an additional ensemble. Every unavailable value and valid-bootstrap fraction is retained in CSV. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S7_fixed_mean_completed_v3_20261020_008_S7_paired_multiple_fronts_difference.csv

S7 · E1 fixed-mean σ sweep · Low-contrast fraction minus σ=1



E1 fixed-mean σ sweep. Direct paired comparisons use twelve identical latent Gaussian seed blocks across σ. Grey marks retain combined-qualified forced conditions. Vertical ranges are bootstrap confidence intervals; open upward triangles mark finite lower confidence endpoints when upper endpoints are unbounded. × marks an unidentified point; no midpoint is invented. The σ=1 reference line is an identity, not an additional ensemble. Every unavailable value and valid-bootstrap fraction is retained in CSV. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S7_fixed_mean_completed_v3_20261020_009_S7_paired_low_contrast_difference.csv

S7 · E1 fixed-mean σ sweep · Observed low-σ and matched-mean qualification

Wave-pinning

f=0.03, C=20: E1 qualification no; lower-σ abnormal regime persists no; matched mean agrees no.
f=0.03, C=40: E1 qualification no; lower-σ abnormal regime persists no; matched mean agrees no.
f=0.03, C=60: E1 qualification no; lower-σ abnormal regime persists no; matched mean agrees no.
f=0.1, C=20: E1 qualification no; lower-σ abnormal regime persists no; matched mean agrees no.
f=0.1, C=40: E1 qualification no; lower-σ abnormal regime persists no; matched mean agrees no.
f=0.1, C=60: E1 qualification no; lower-σ abnormal regime persists no; matched mean agrees no.

Goryachev

f=0.02, C=20: E1 qualification no; lower-σ abnormal regime persists no; matched mean agrees no.
f=0.02, C=40: E1 qualification yes; lower-σ abnormal regime persists yes; matched mean agrees yes.
f=0.02, C=60: E1 qualification no; lower-σ abnormal regime persists no; matched mean agrees no.
f=0.03, C=20: E1 qualification no; lower-σ abnormal regime persists no; matched mean agrees no.
f=0.03, C=40: E1 qualification yes; lower-σ abnormal regime persists yes; matched mean agrees yes.
f=0.03, C=60: E1 qualification yes; lower-σ abnormal regime persists yes; matched mean agrees yes.

Holmes 3

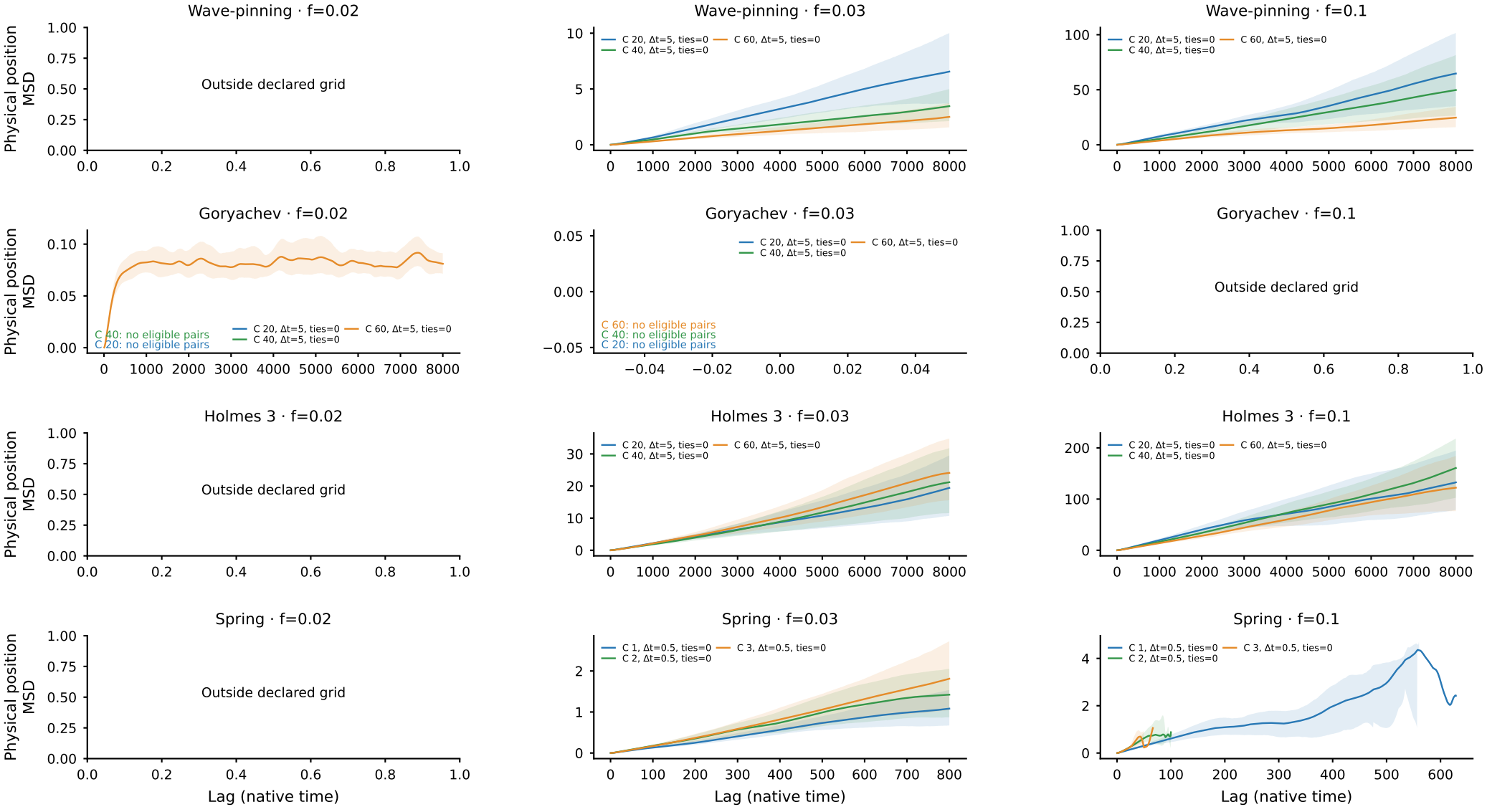
f=0.03, C=20: E1 qualification no; lower-σ abnormal regime persists no; matched mean agrees no.
f=0.03, C=40: E1 qualification no; lower-σ abnormal regime persists no; matched mean agrees no.
f=0.03, C=60: E1 qualification no; lower-σ abnormal regime persists no; matched mean agrees no.
f=0.1, C=20: E1 qualification no; lower-σ abnormal regime persists no; matched mean agrees no.
f=0.1, C=40: E1 qualification no; lower-σ abnormal regime persists no; matched mean agrees no.
f=0.1, C=60: E1 qualification no; lower-σ abnormal regime persists no; matched mean agrees no.

Spring

f=0.03, C=1: E1 qualification no; lower-σ abnormal regime persists no; matched mean agrees no.
f=0.03, C=2: E1 qualification no; lower-σ abnormal regime persists no; matched mean agrees no.
f=0.03, C=3: E1 qualification no; lower-σ abnormal regime persists no; matched mean agrees no.
f=0.1, C=1: E1 qualification no; lower-σ abnormal regime persists no; matched mean agrees no.
f=0.1, C=2: E1 qualification no; lower-σ abnormal regime persists no; matched mean agrees no.
f=0.1, C=3: E1 qualification no; lower-σ abnormal regime persists yes; matched mean agrees no.

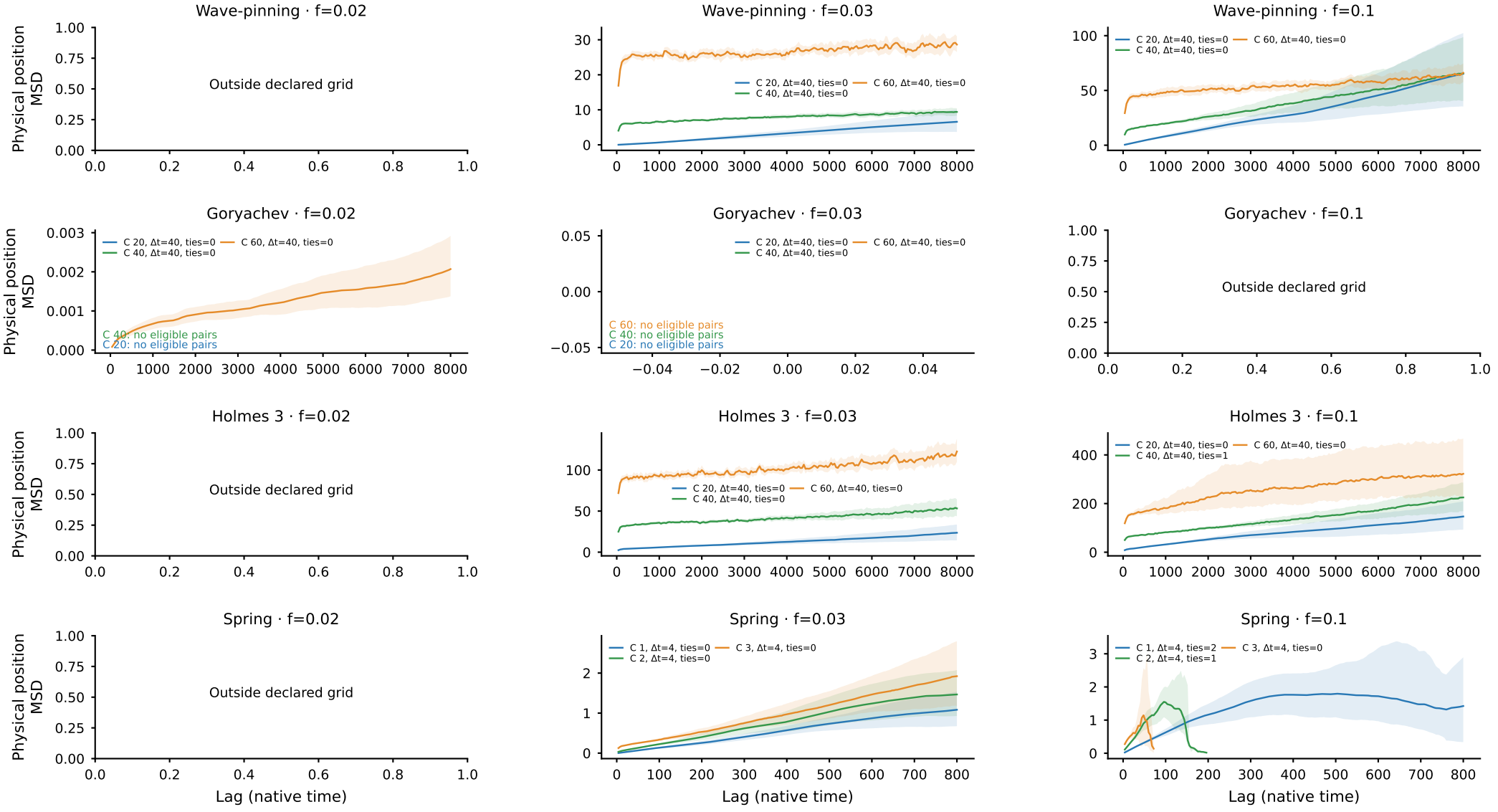
E1 fixed-mean σ sweep. The approved operational rule uses the same abnormal majority at σ=1,0.5,0.25 and the observed matched σ=0 mean. This is finite-control evidence, not an analytic σ→0 limit. The original matched-mean forced flag remains a separate field. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S7_fixed_mean_completed_v3_20261020_010_S7_forced_flags.csv

S5 · E1 literal material-origin physical-site measurements · Literal heading position · A3 primary split · σ=0.25



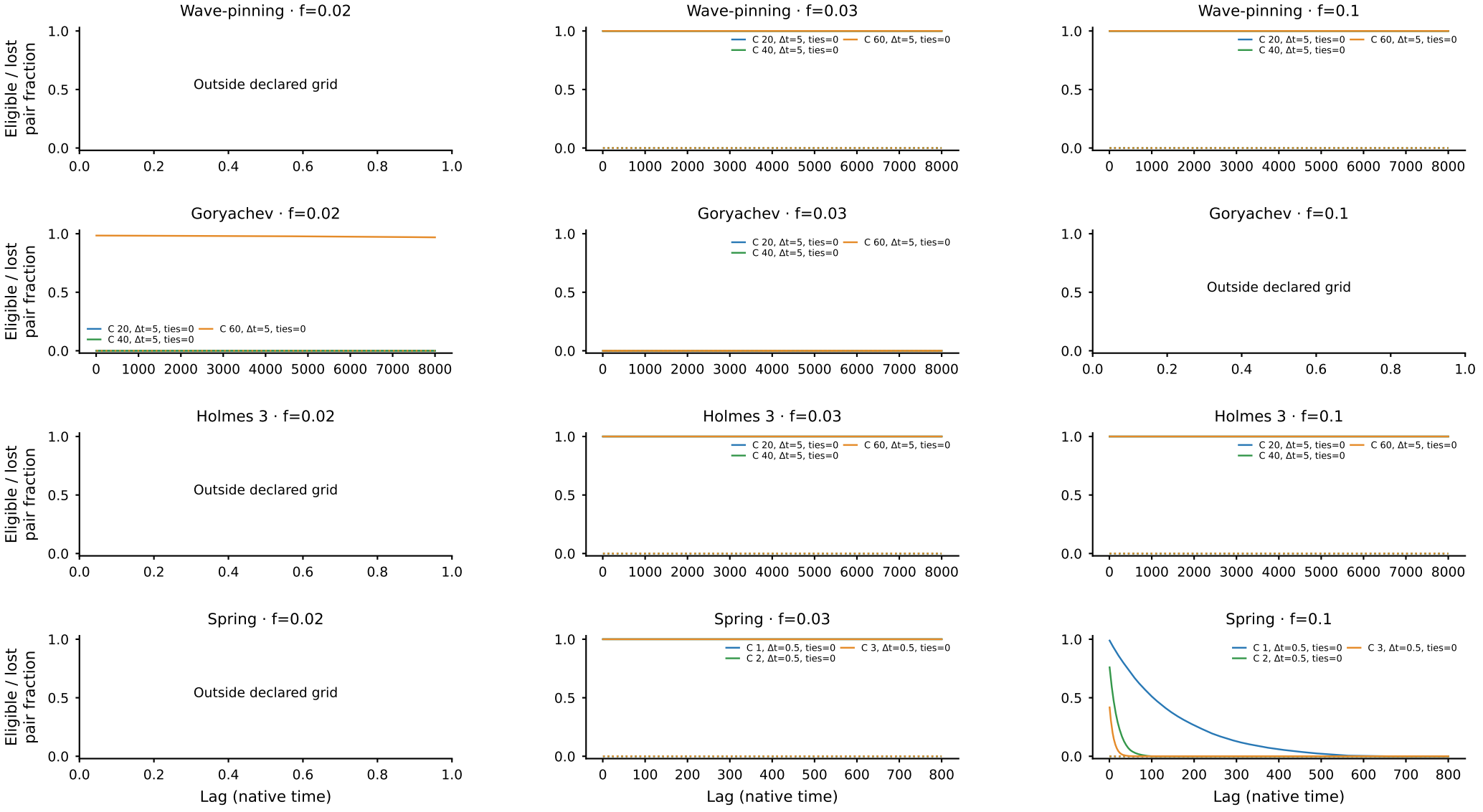
E1 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_001_S5_literal_sigma0.25_heading_literal_actualC_MSD_1.csv

S5 · E1 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · σ=0.25



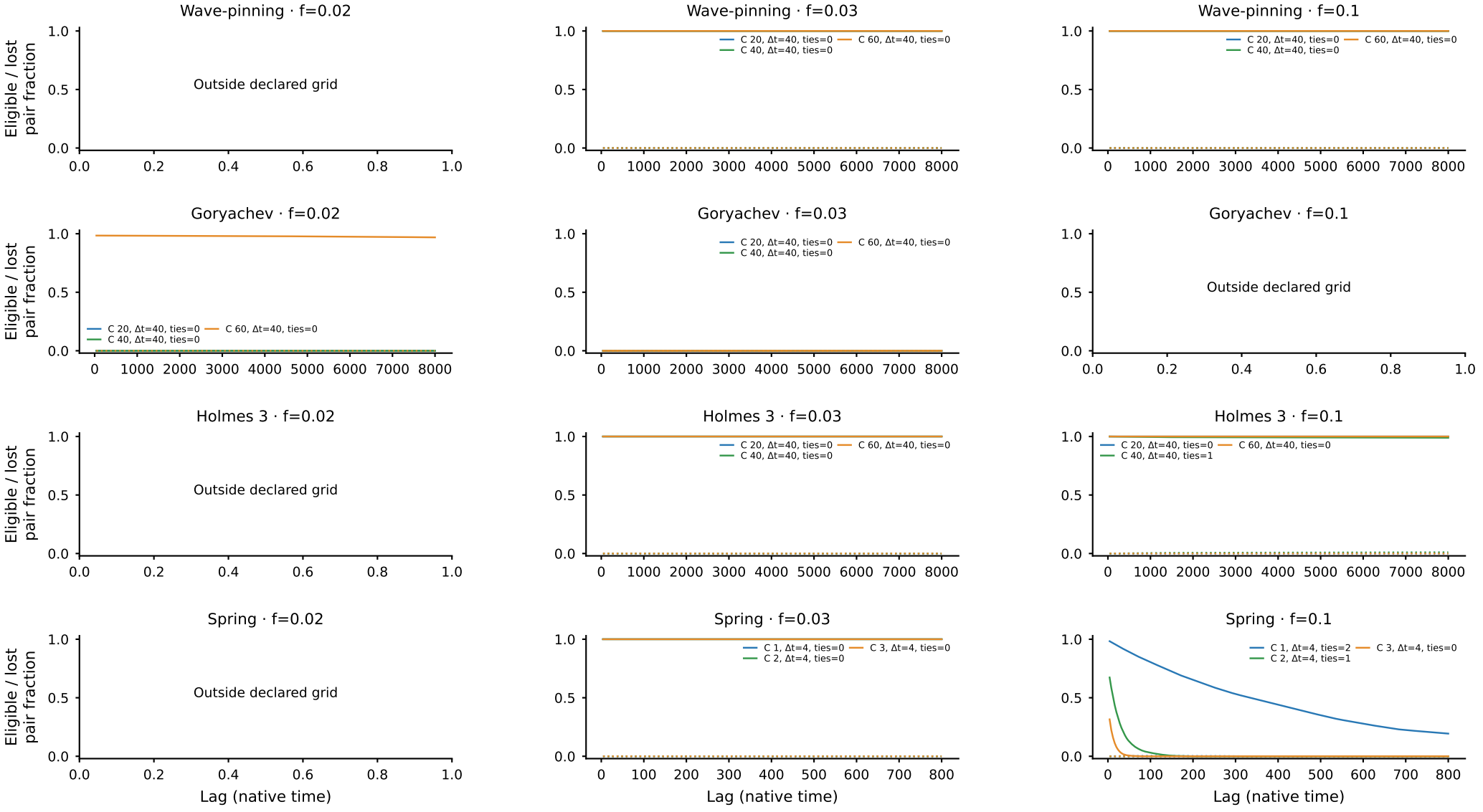
E1 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M₁|/|M₁,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_002_S5_literal_sigma0.25_profile_peak_literal_actualC_MSD_1.csv

S5 · E1 literal material-origin physical-site measurements · Literal heading position · A3 primary split · pair coverage and ambiguity loss · σ=0.25



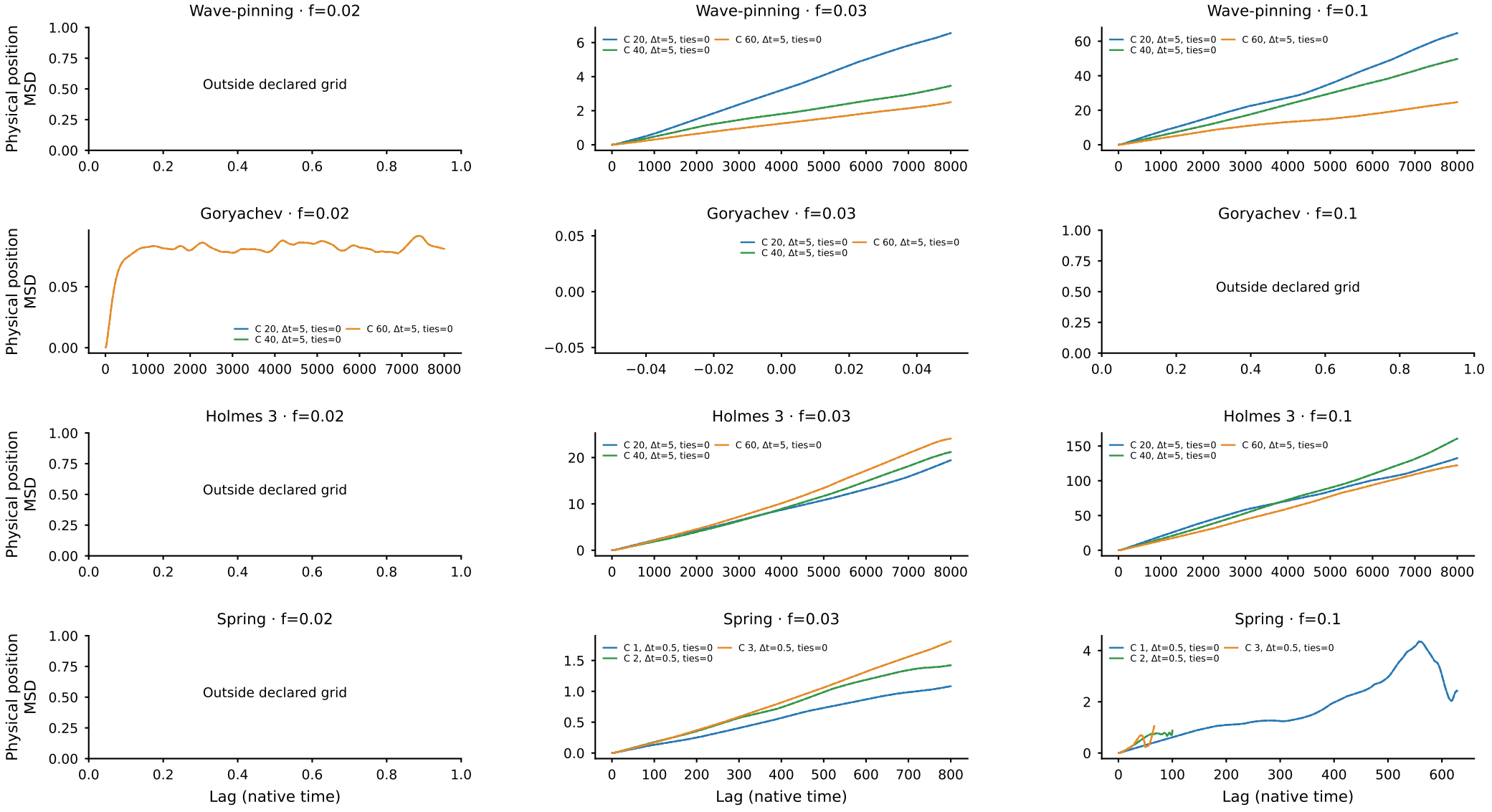
E1 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,basal}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_003_S5_literal_sigma0.25_heading_literal_actualC_coverage_1.csv

S5 · E1 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · pair coverage and ambiguity loss · σ=0.25



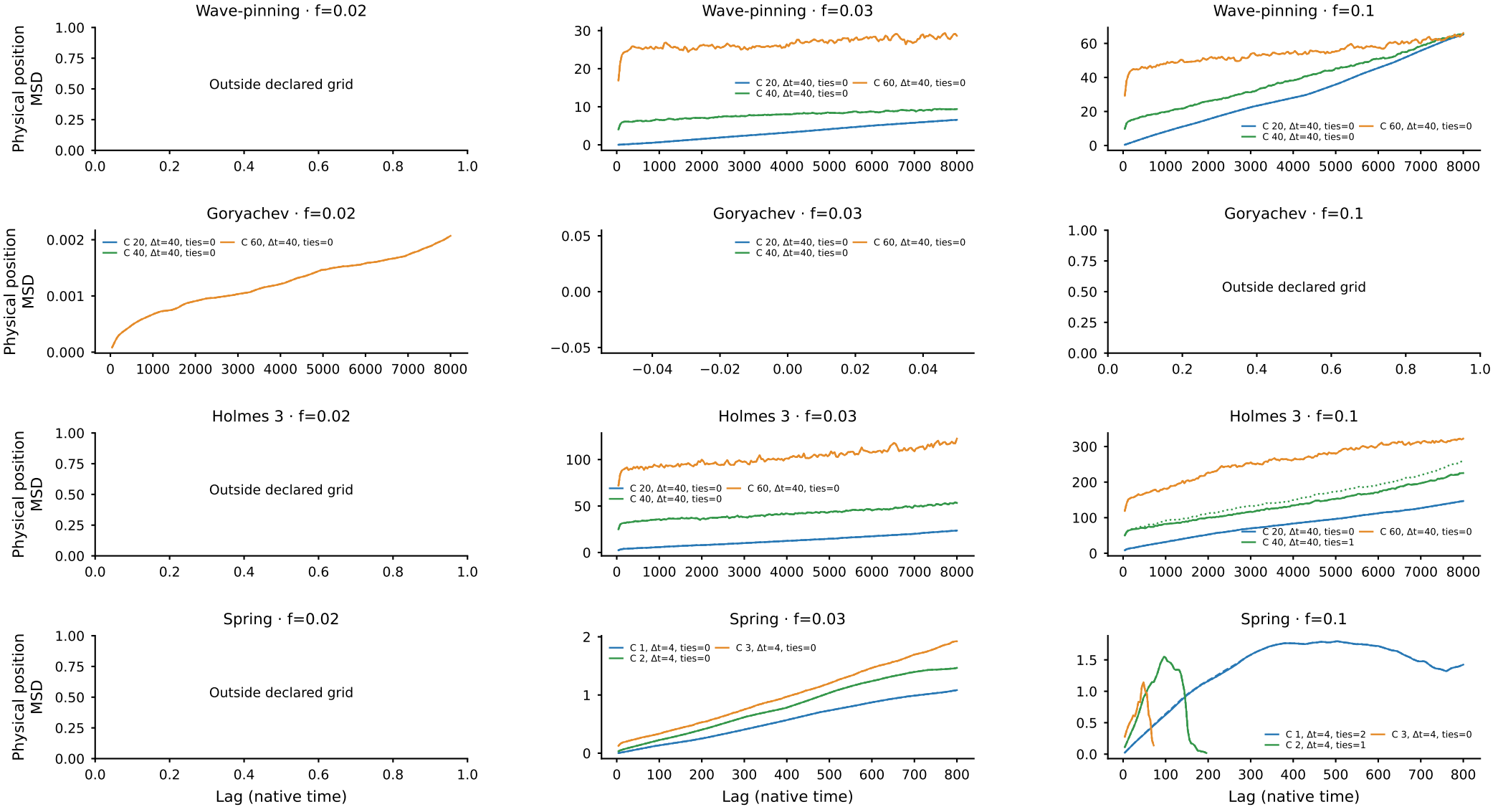
E1 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_004_S5_literal_sigma0.25_profile_peak_literal_actualC_coverage_1.csv

S5 · E1 literal material-origin physical-site measurements · Literal heading position · paired route sensitivities · split / +π / −π · σ=0.25



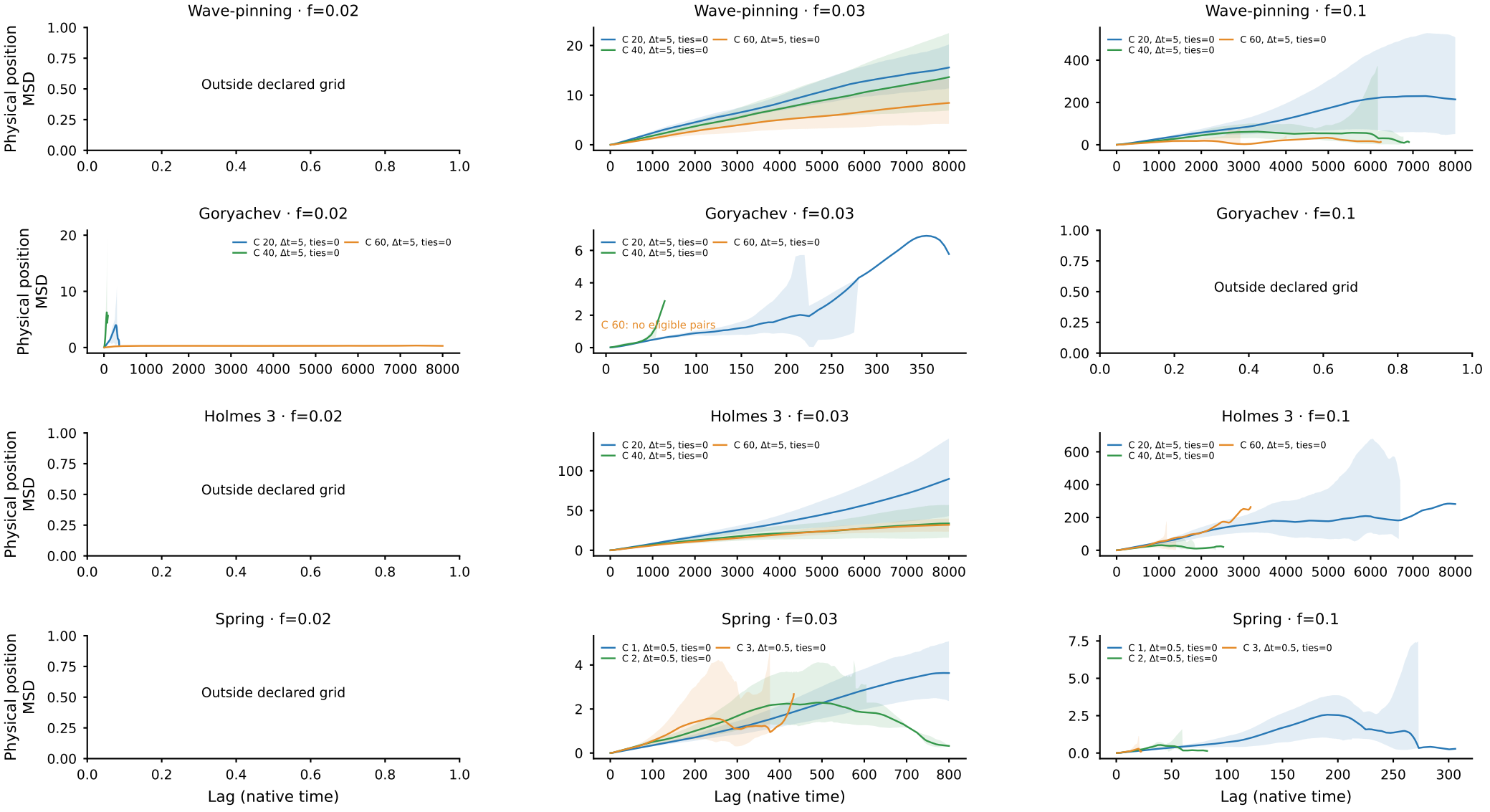
E1 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_005_S5_literal_sigma0.25_heading_literal_actualC_routes_1.csv.gz

S5 · E1 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · split / +π / −π · σ=0.25



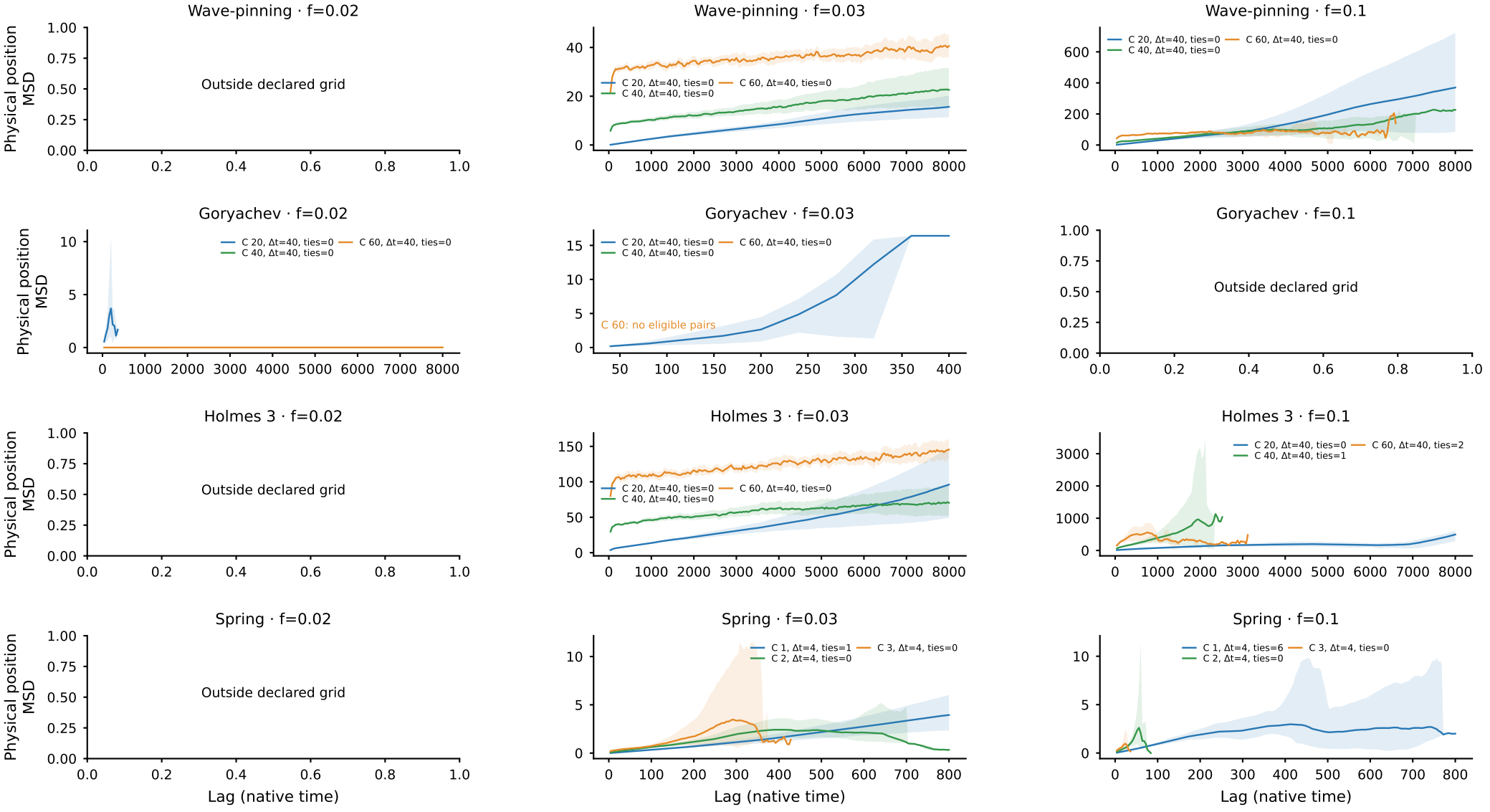
E1 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M_1|/|M_1,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_006_S5_literal_sigma0.25_profile_peak_literal_actualC_routes_1.csv

S5 · E1 literal material-origin physical-site measurements · Literal heading position · A3 primary split · σ=0.5



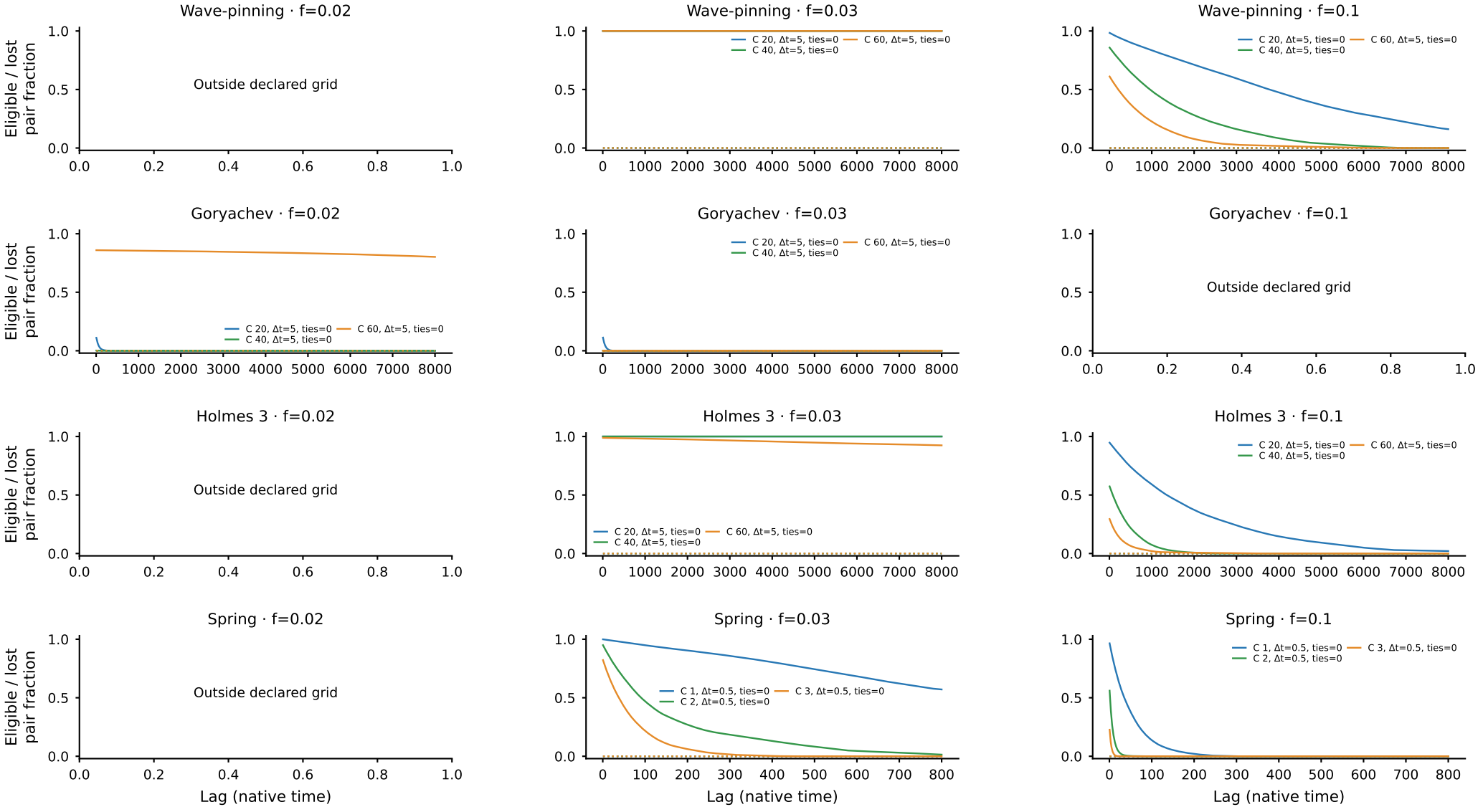
E1 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_007_S5_literal_sigma0.5_heading_literal_actualC_MSD_1.csv

S5 · E1 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · σ=0.5



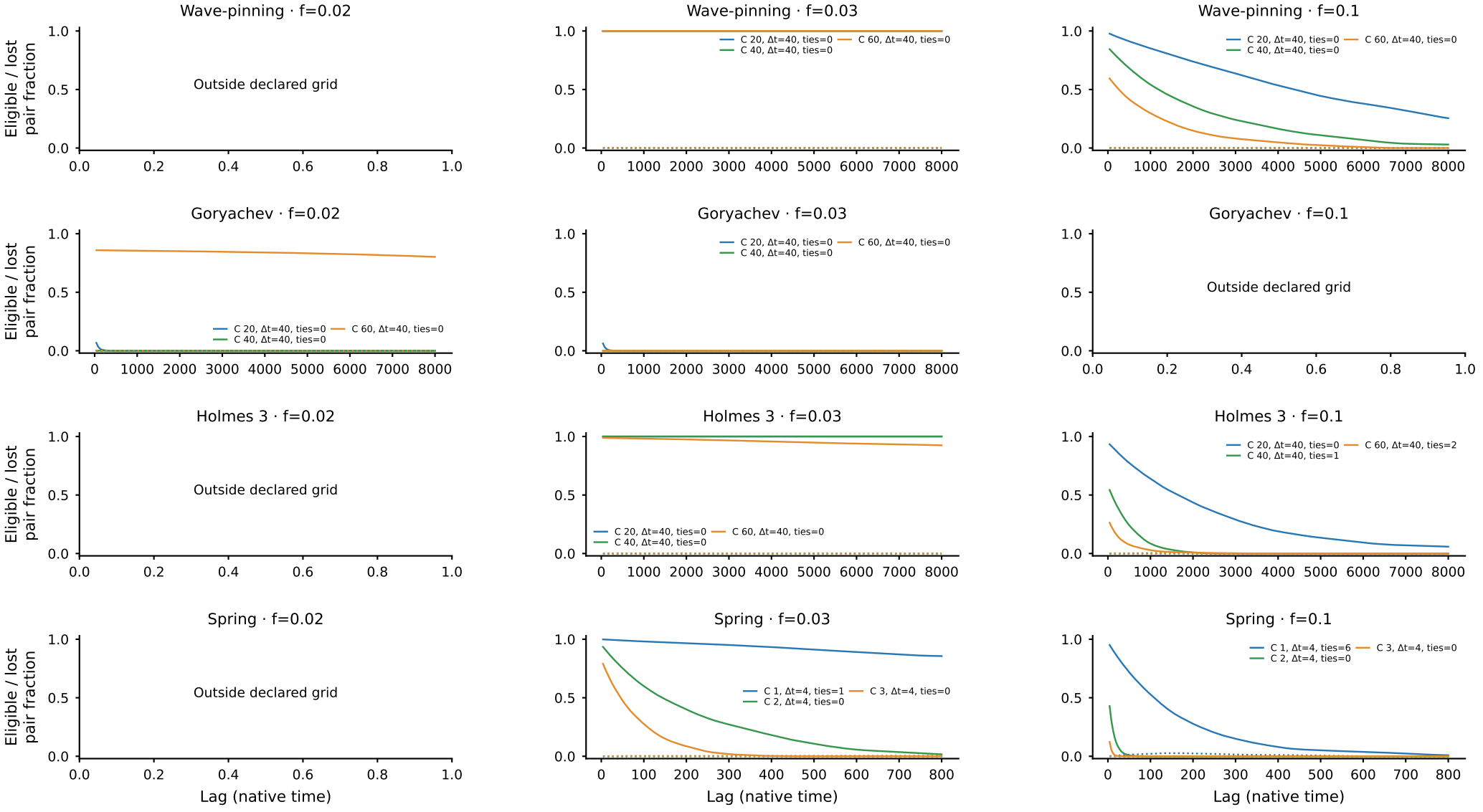
E1 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_008_S5_literal_sigma0.5_profile_peak_literal_actualC_MSD_1.csv

S5 · E1 literal material-origin physical-site measurements · Literal heading position · A3 primary split · pair coverage and ambiguity loss · σ=0.5



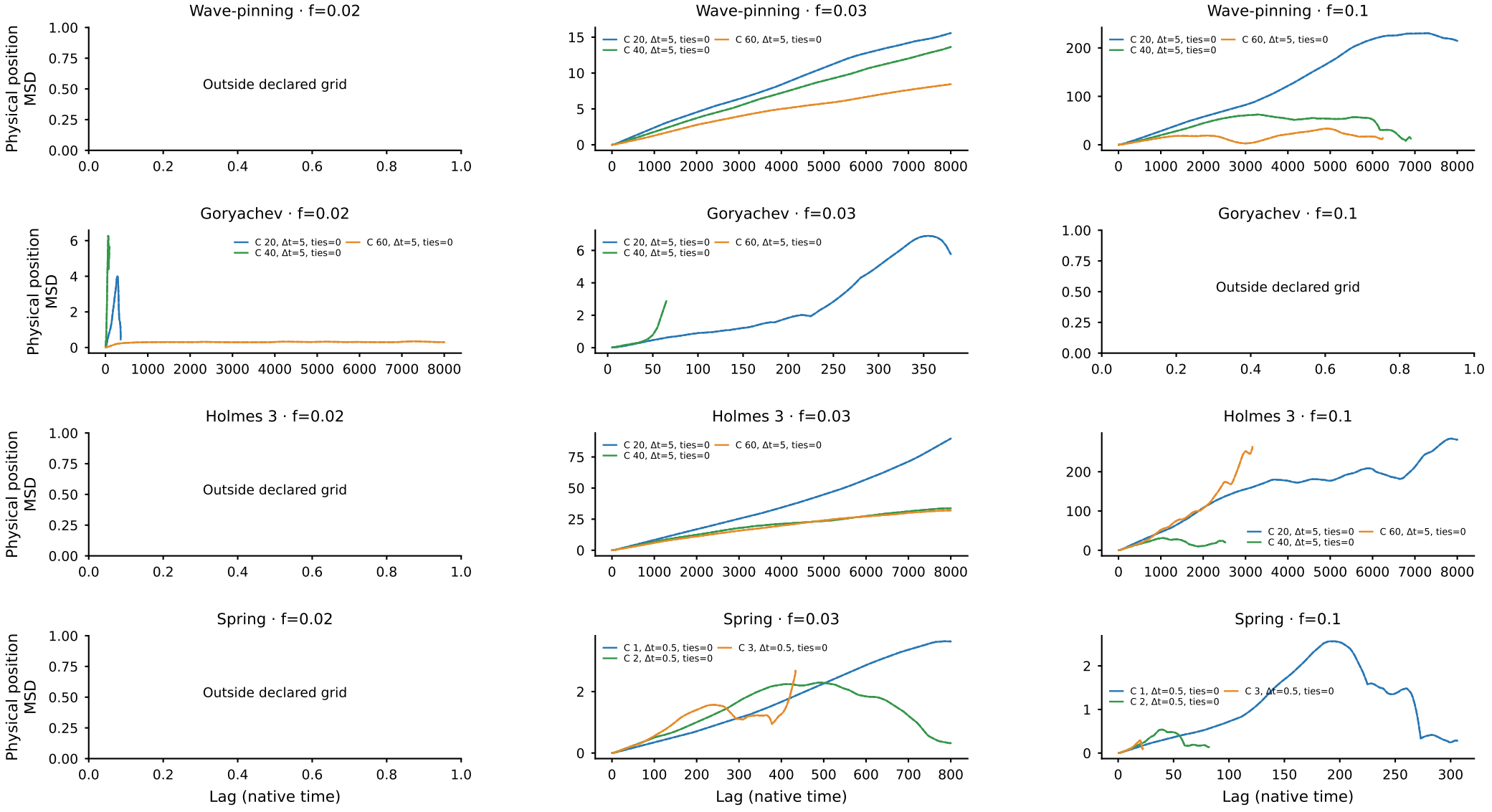
E1 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,basal}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_009_S5_literal_sigma0.5_heading_literal_actualC_coverage_1.csv

S5 · E1 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · pair coverage and ambiguity loss · σ=0.5



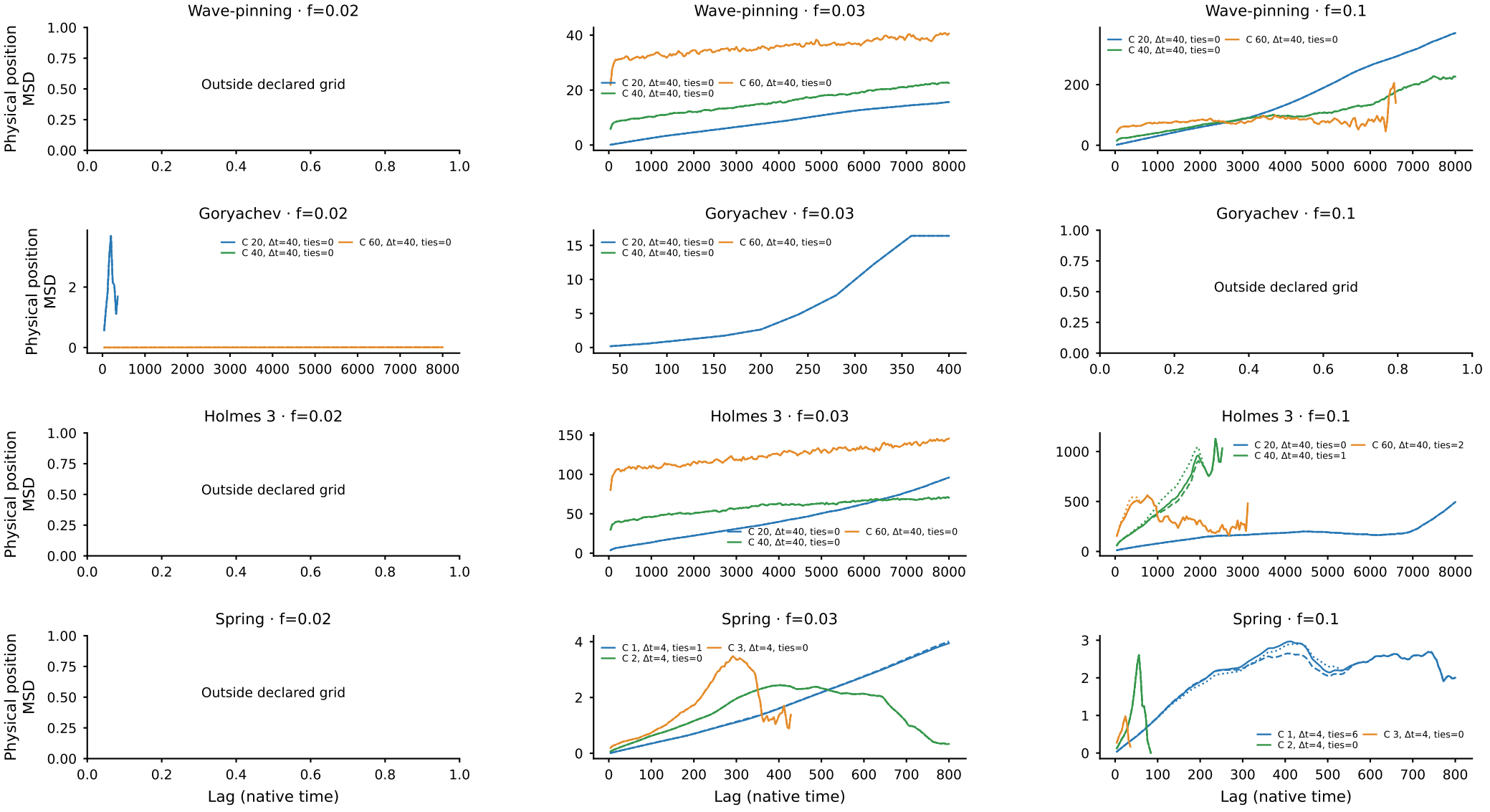
E1 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M₁|/|M₁,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_010_S5_literal_sigma0.5_profile_peak_literal_actualC_coverage_1.csv

S5 · E1 literal material-origin physical-site measurements · Literal heading position · paired route sensitivities · split / +π / −π · σ=0.5



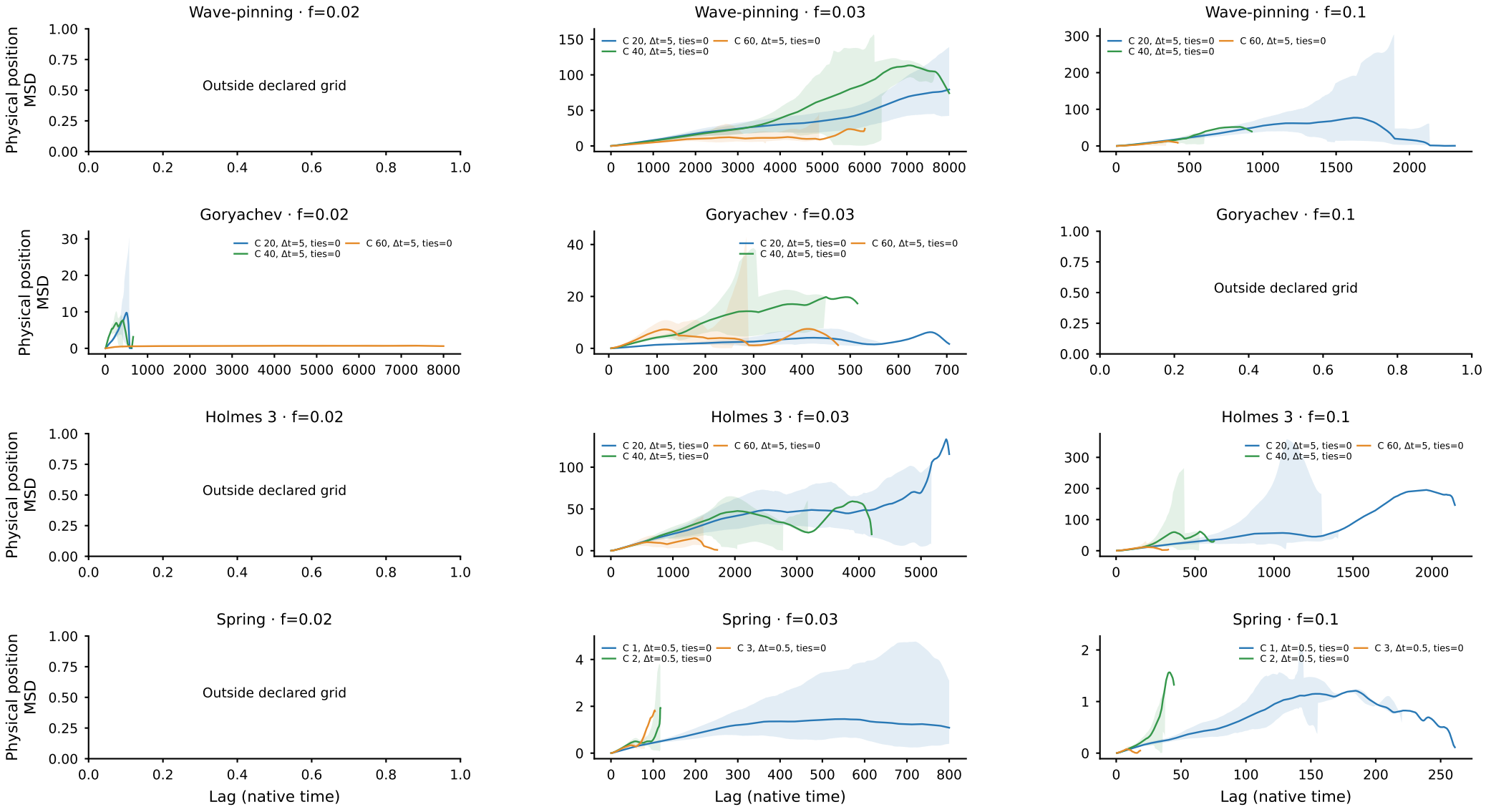
E1 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_011_S5_literal_sigma0.5_heading_literal_actualC_routes_1.csv.gz

S5 · E1 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · split / +π / −π · σ=0.5



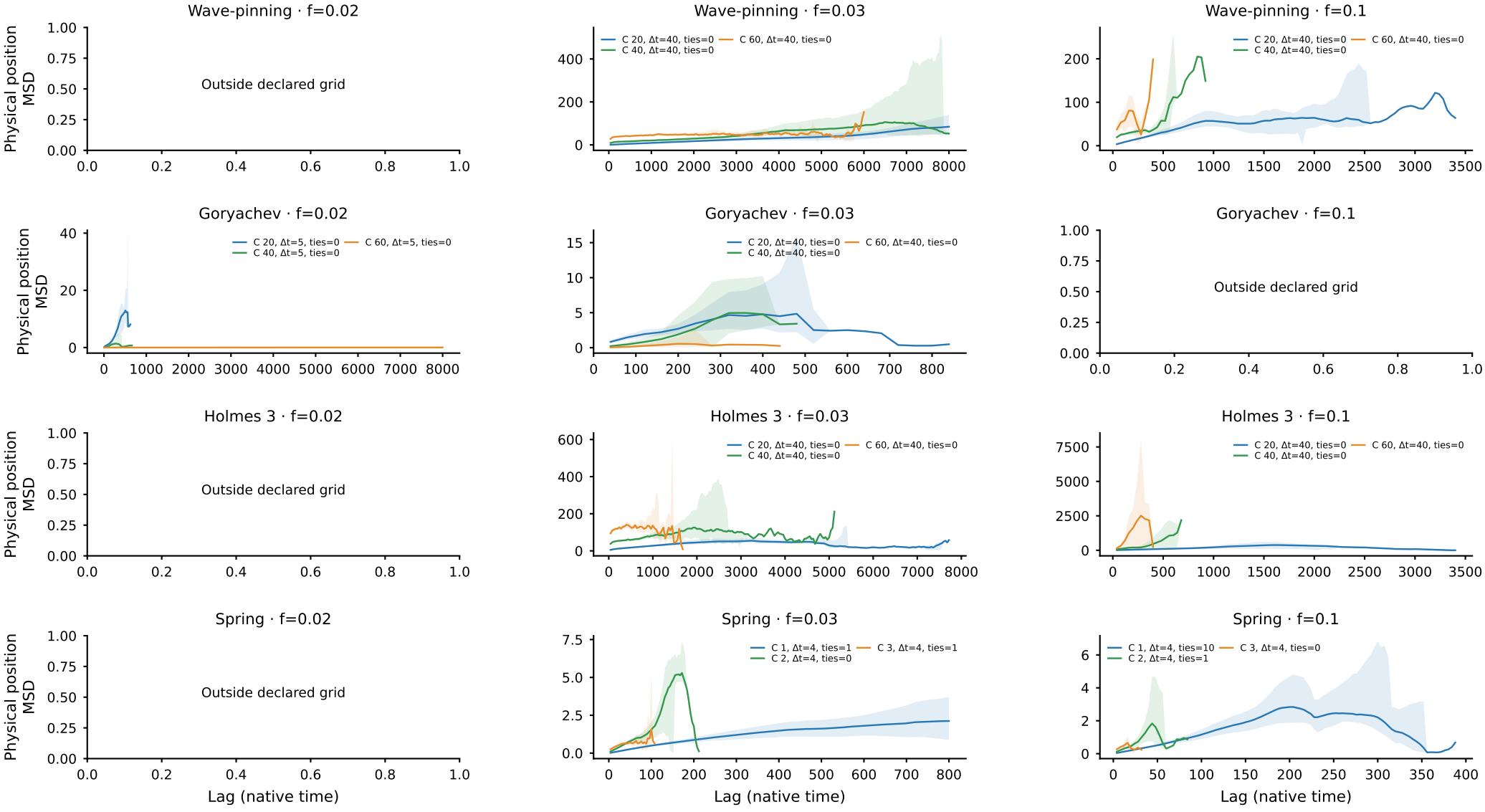
E1 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_012_S5_literal_sigma0.5_profile_peak_literal_actualC_routes_1.csv

S5 · E1 literal material-origin physical-site measurements · Literal heading position · A3 primary split · σ=1



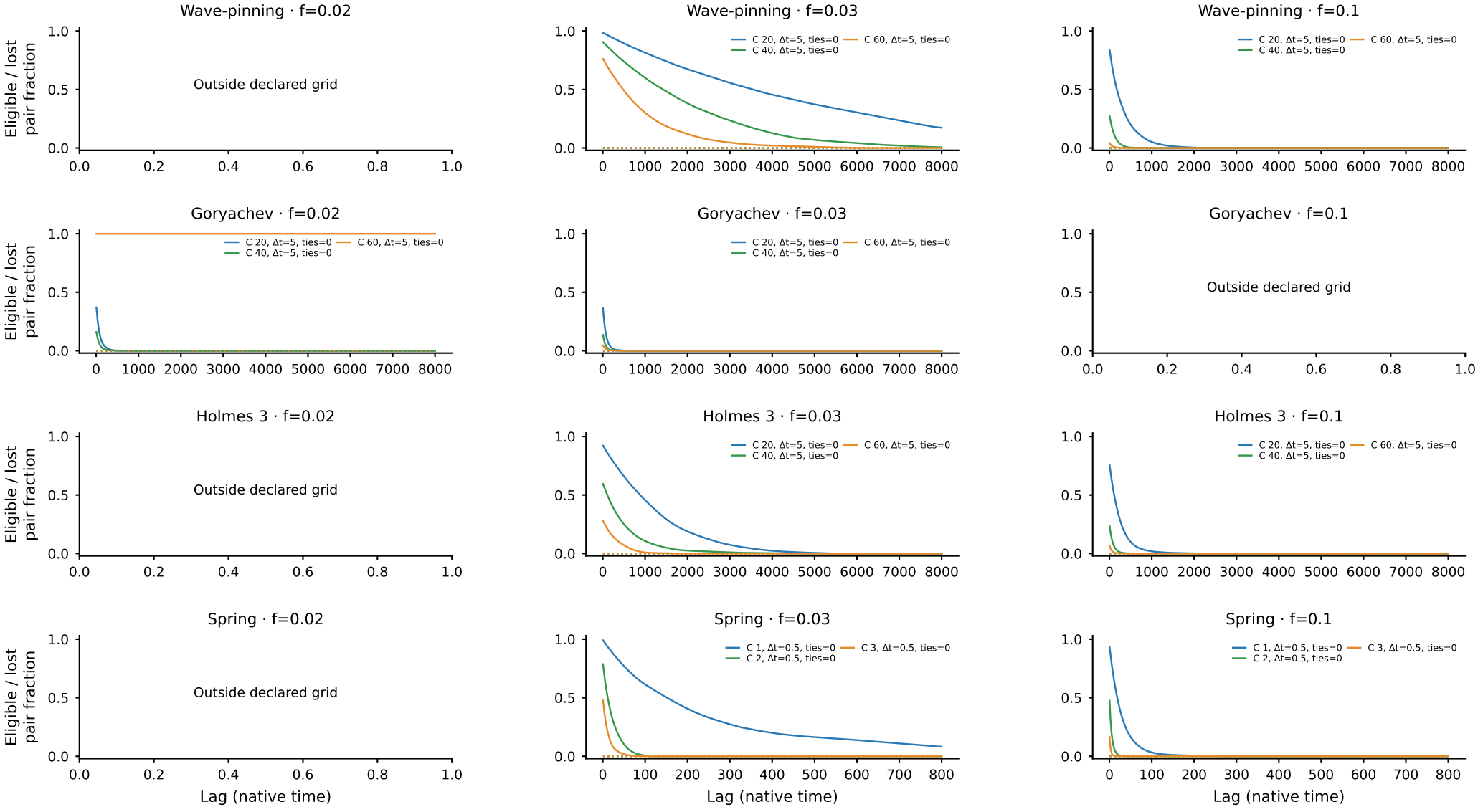
E1 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,basal}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_013_S5_literal_sigma1_heading_literal_actualC_MSD_1.csv

S5 · E1 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · σ=1



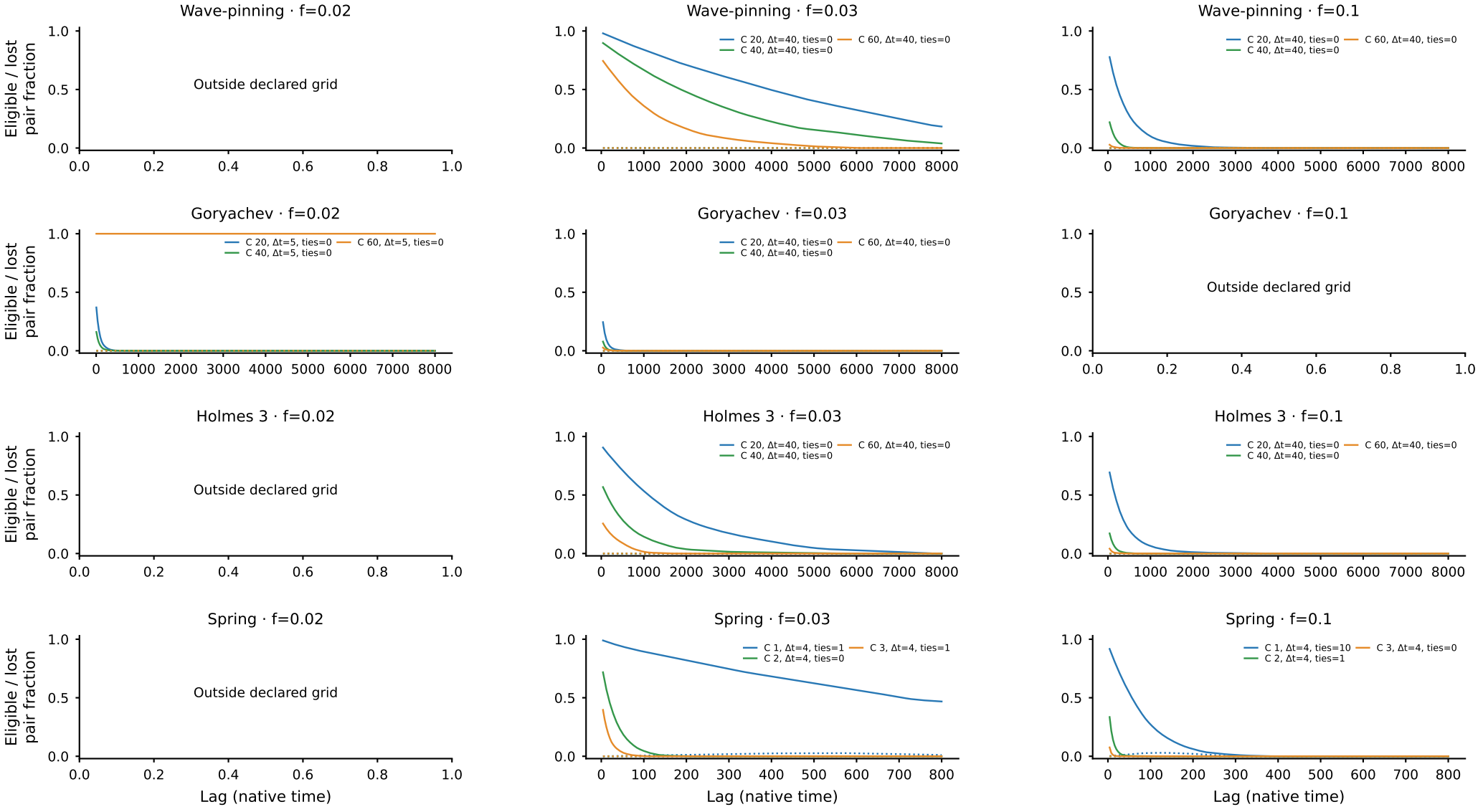
E1 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_014_S5_literal_sigma1_profile_peak_literal_actualC_MSD_1.csv

S5 · E1 literal material-origin physical-site measurements · Literal heading position · A3 primary split · pair coverage and ambiguity loss · σ=1



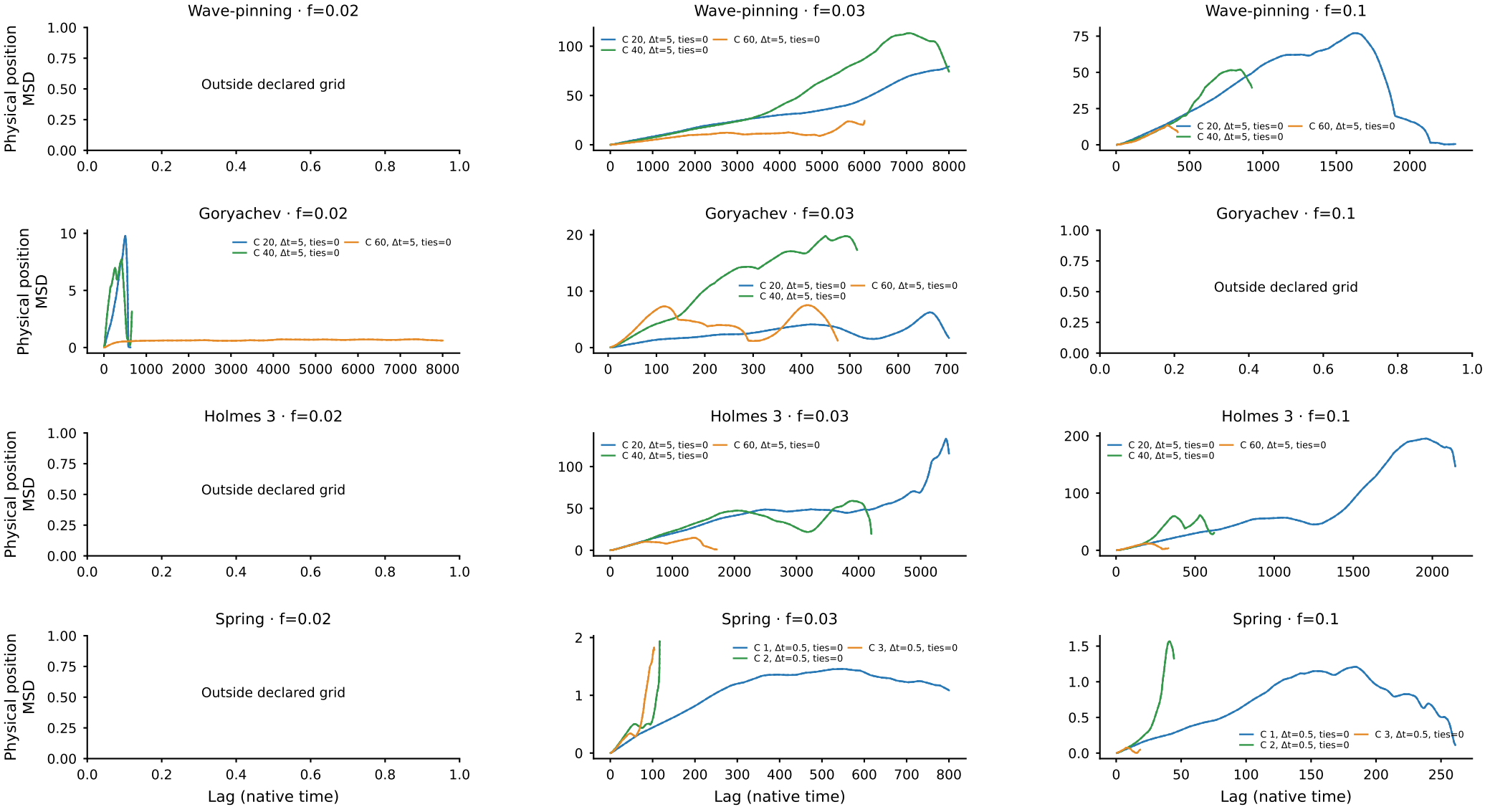
E1 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,basal}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_015_S5_literal_sigma1_heading_literal_actualC_coverage_1.csv

S5 · E1 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · pair coverage and ambiguity loss · σ=1



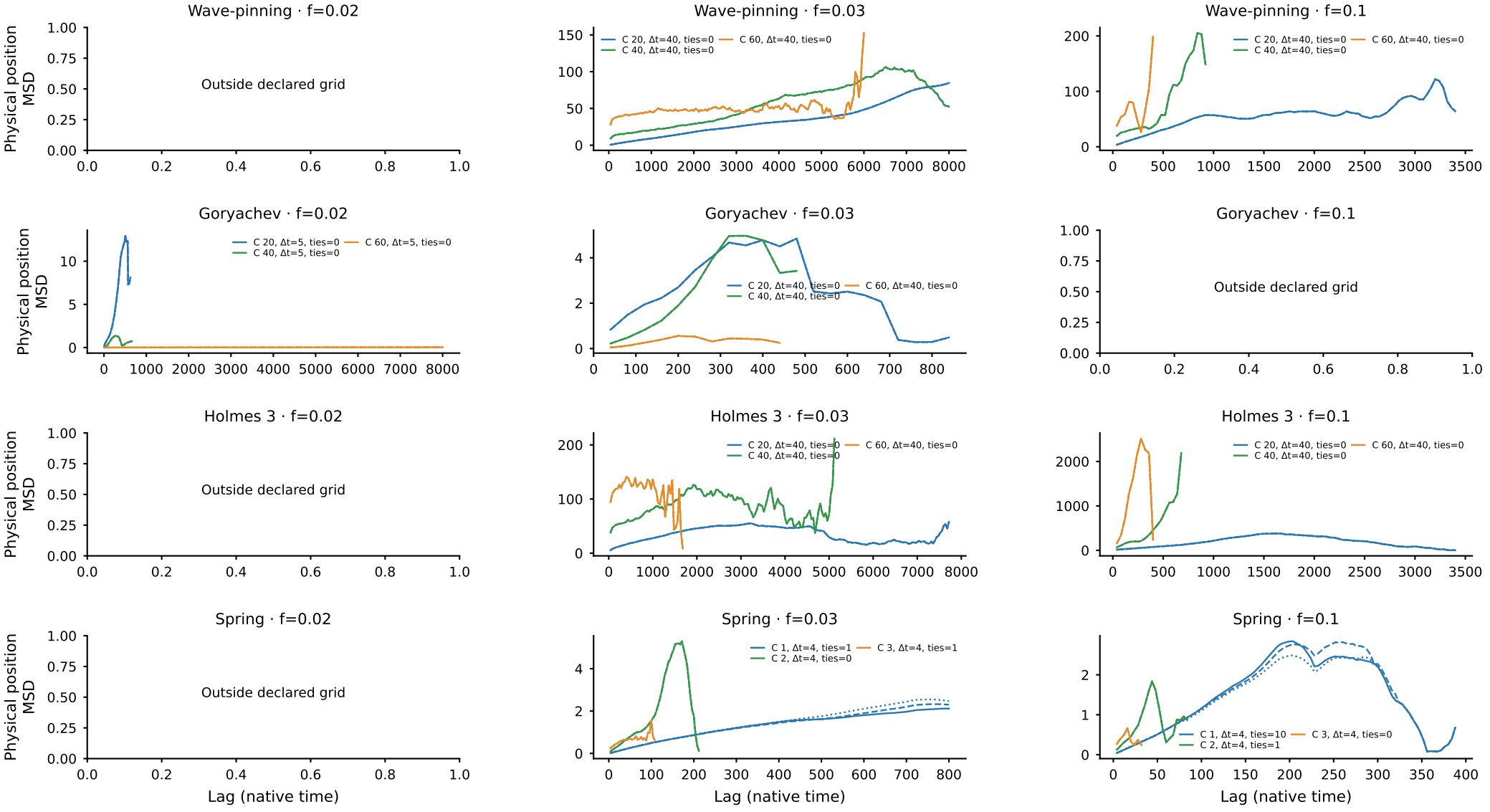
E1 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,basal}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_016_S5_literal_sigma1_profile_peak_literal_actualC_coverage_1.csv

S5 · E1 literal material-origin physical-site measurements · Literal heading position · paired route sensitivities · split / +π / −π · σ=1



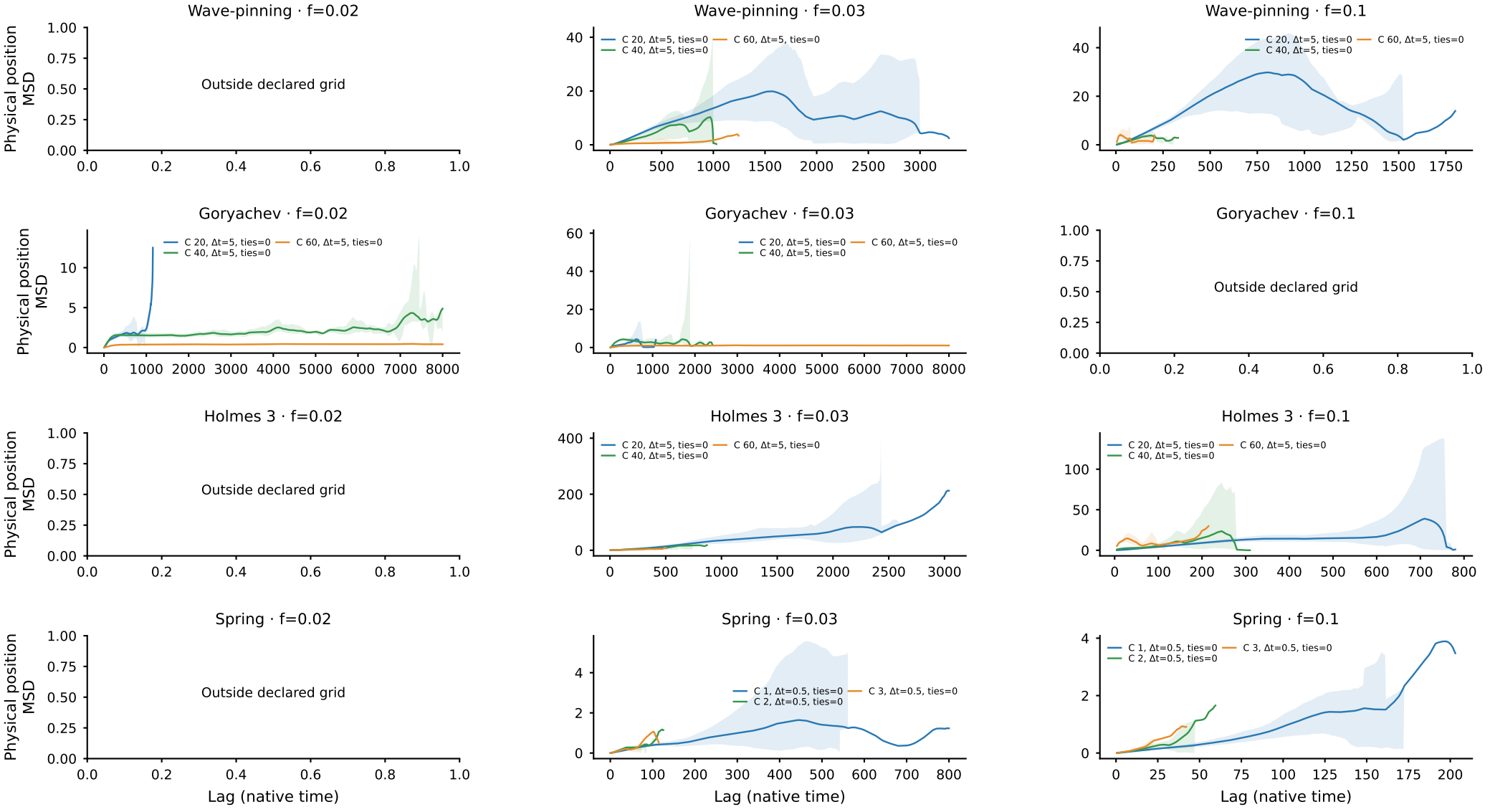
E1 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_017_S5_literal_sigma1_heading_literal_actualC_routes_1.csv.gz

S5 · E1 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · split / +π / −π · σ=1



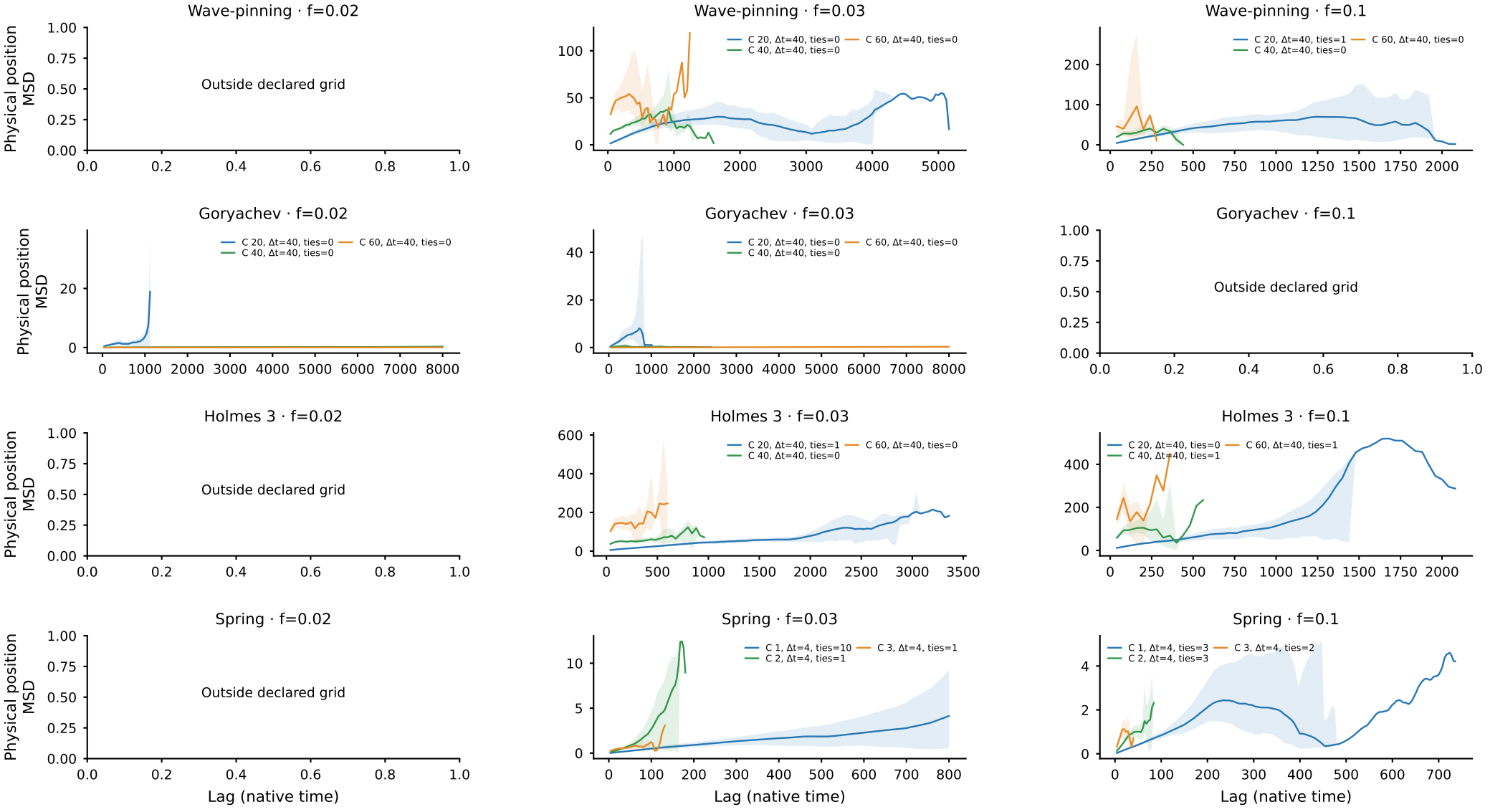
E1 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_018_S5_literal_sigma1_profile_peak_literal_actualC_routes_1.csv

S5 · E1 literal material-origin physical-site measurements · Literal heading position · A3 primary split · σ=2



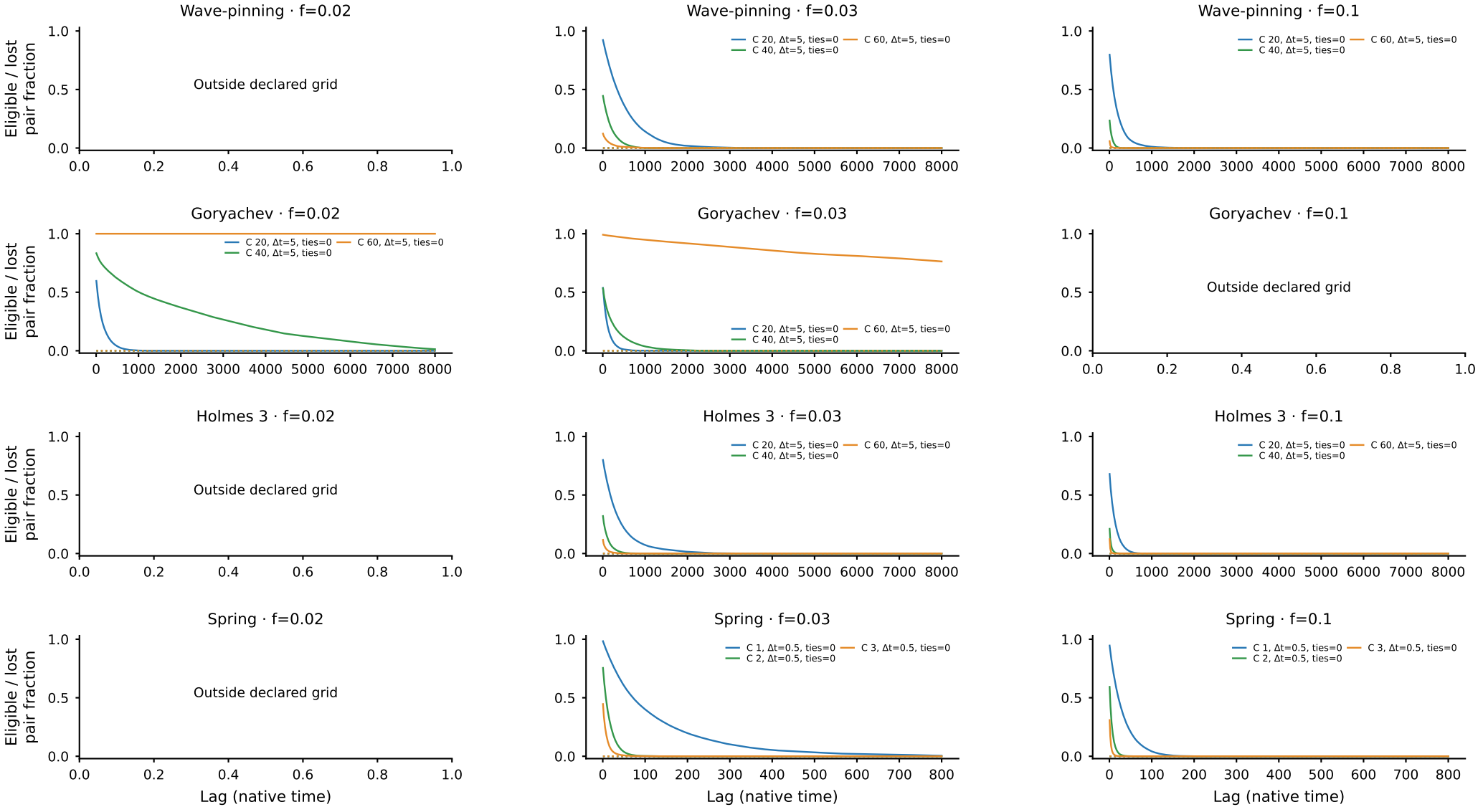
E1 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_019_S5_literal_sigma2_heading_literal_actualC_MSD_1.csv

S5 · E1 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · σ=2



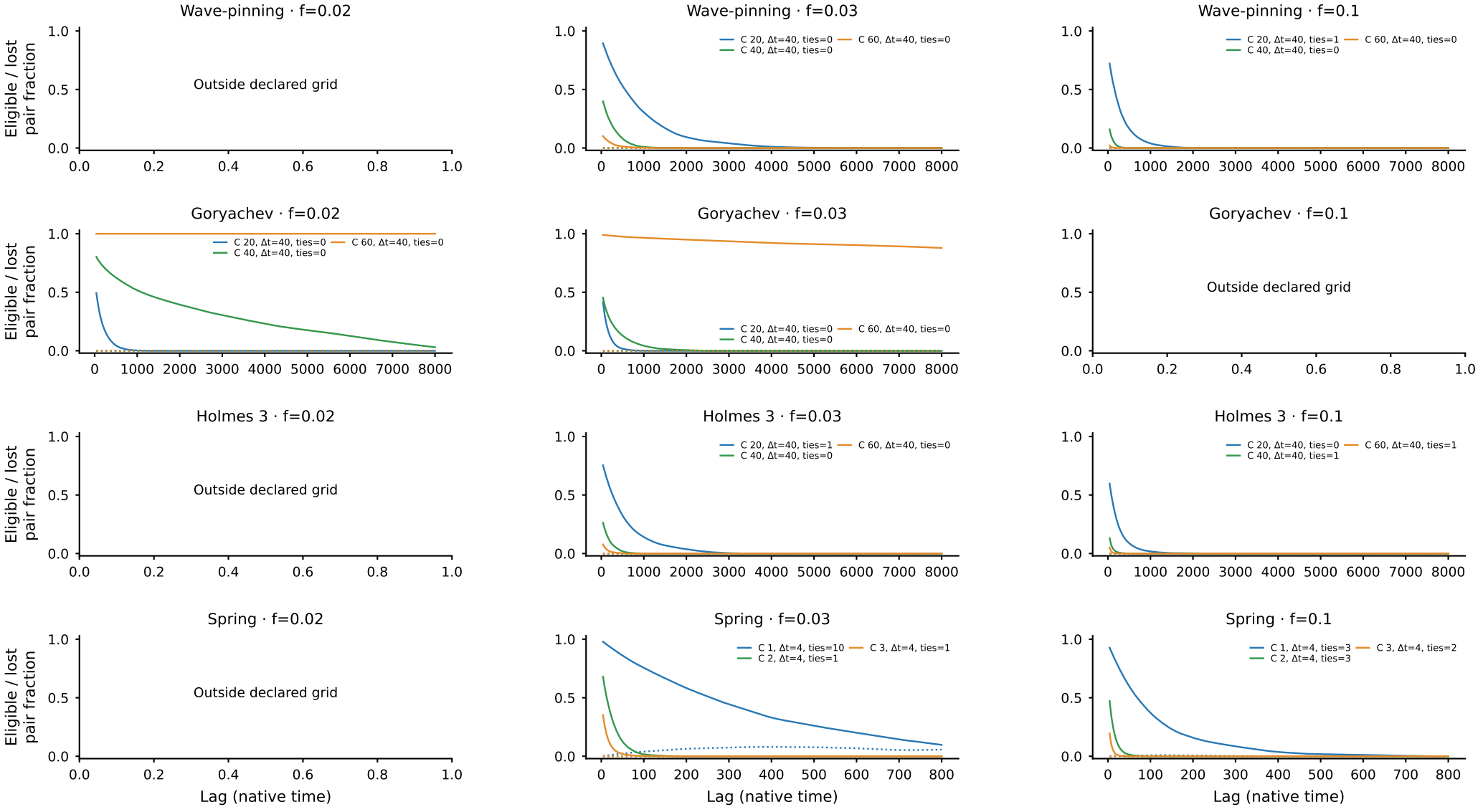
E1 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M_1|/|M_1,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_020_S5_literal_sigma2_profile_peak_literal_actualC_MSD_1.csv

S5 · E1 literal material-origin physical-site measurements · Literal heading position · A3 primary split · pair coverage and ambiguity loss · σ=2



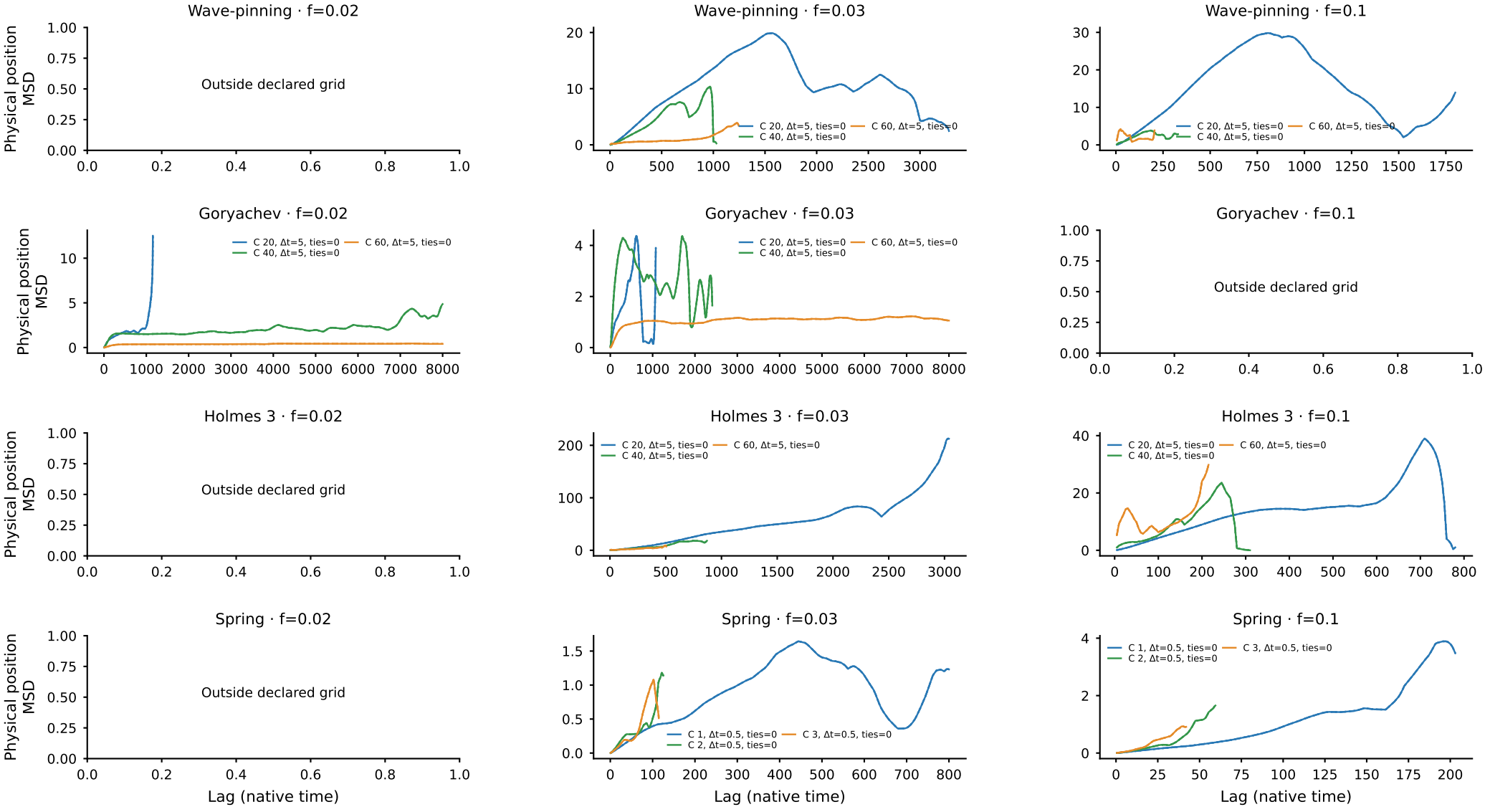
E1 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M₁|/|M₁,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_021_S5_literal_sigma2_heading_literal_actualC_coverage_1.csv

S5 · E1 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · pair coverage and ambiguity loss · σ=2



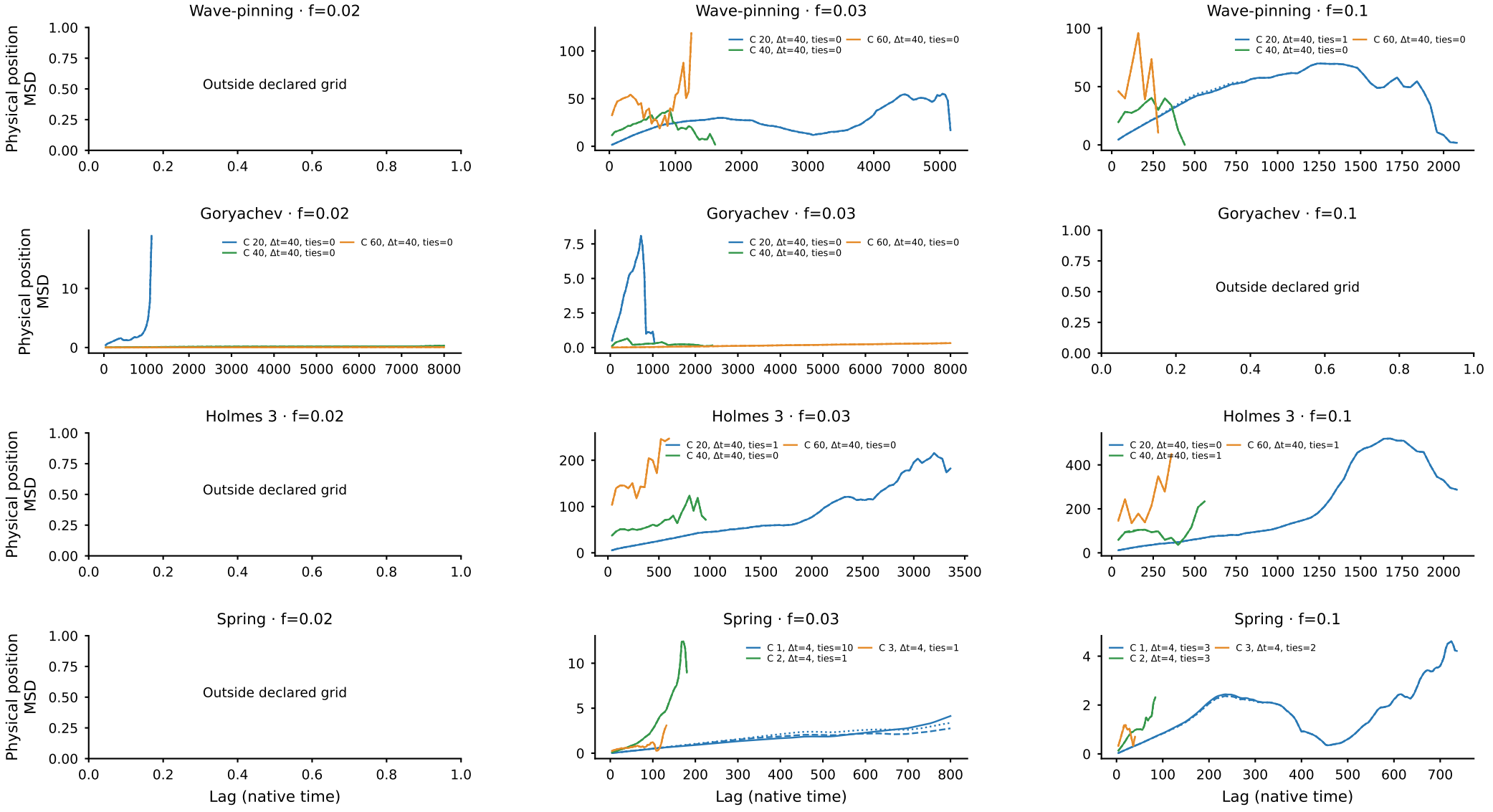
E1 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,basal}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_022_S5_literal_sigma2_profile_peak_literal_actualC_coverage_1.csv

S5 · E1 literal material-origin physical-site measurements · Literal heading position · paired route sensitivities · split / +π / −π · σ=2



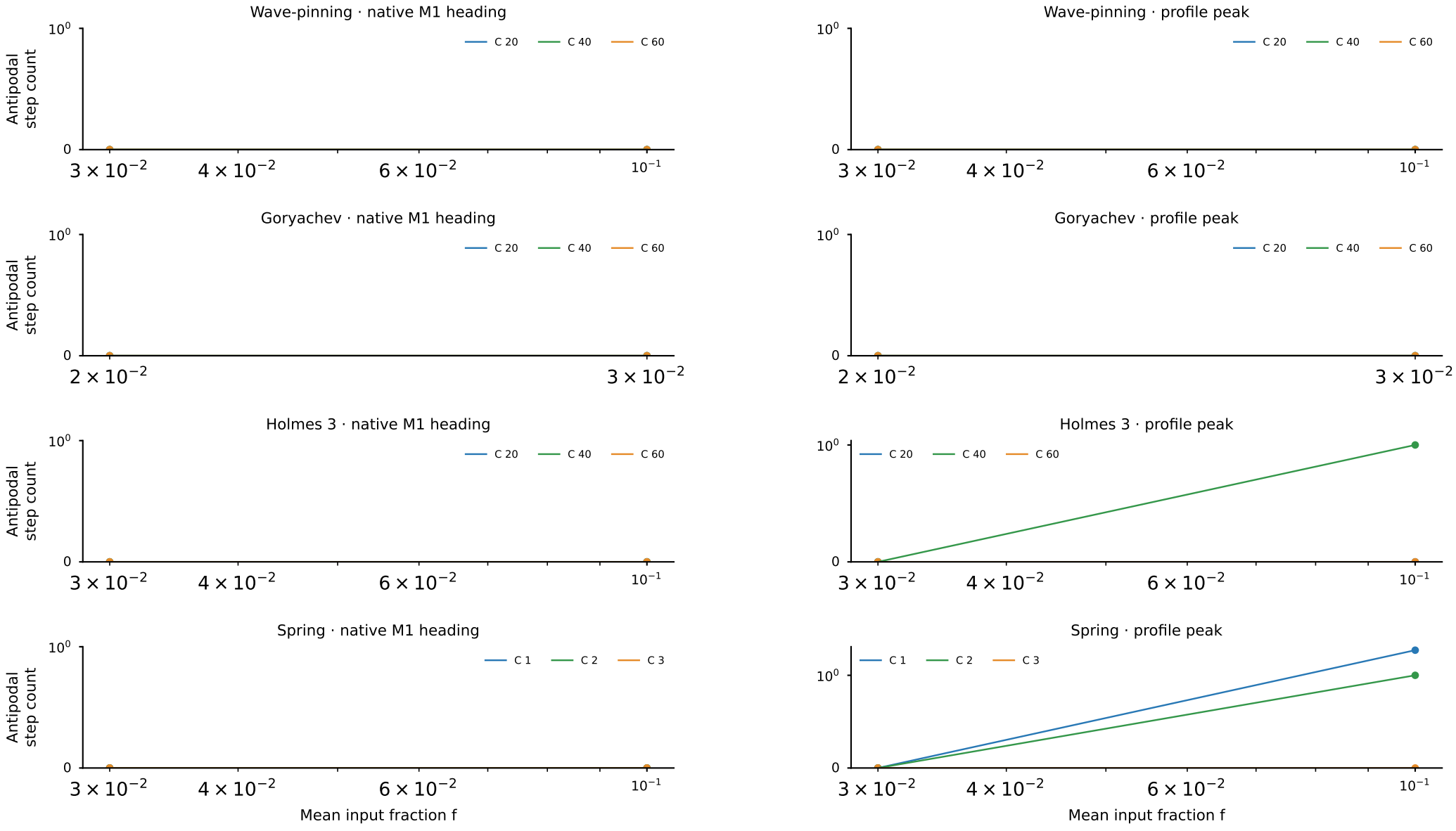
E1 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_023_S5_literal_sigma2_heading_literal_actualC_routes_1.csv.gz

S5 · E1 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · split / +π / −π · σ=2



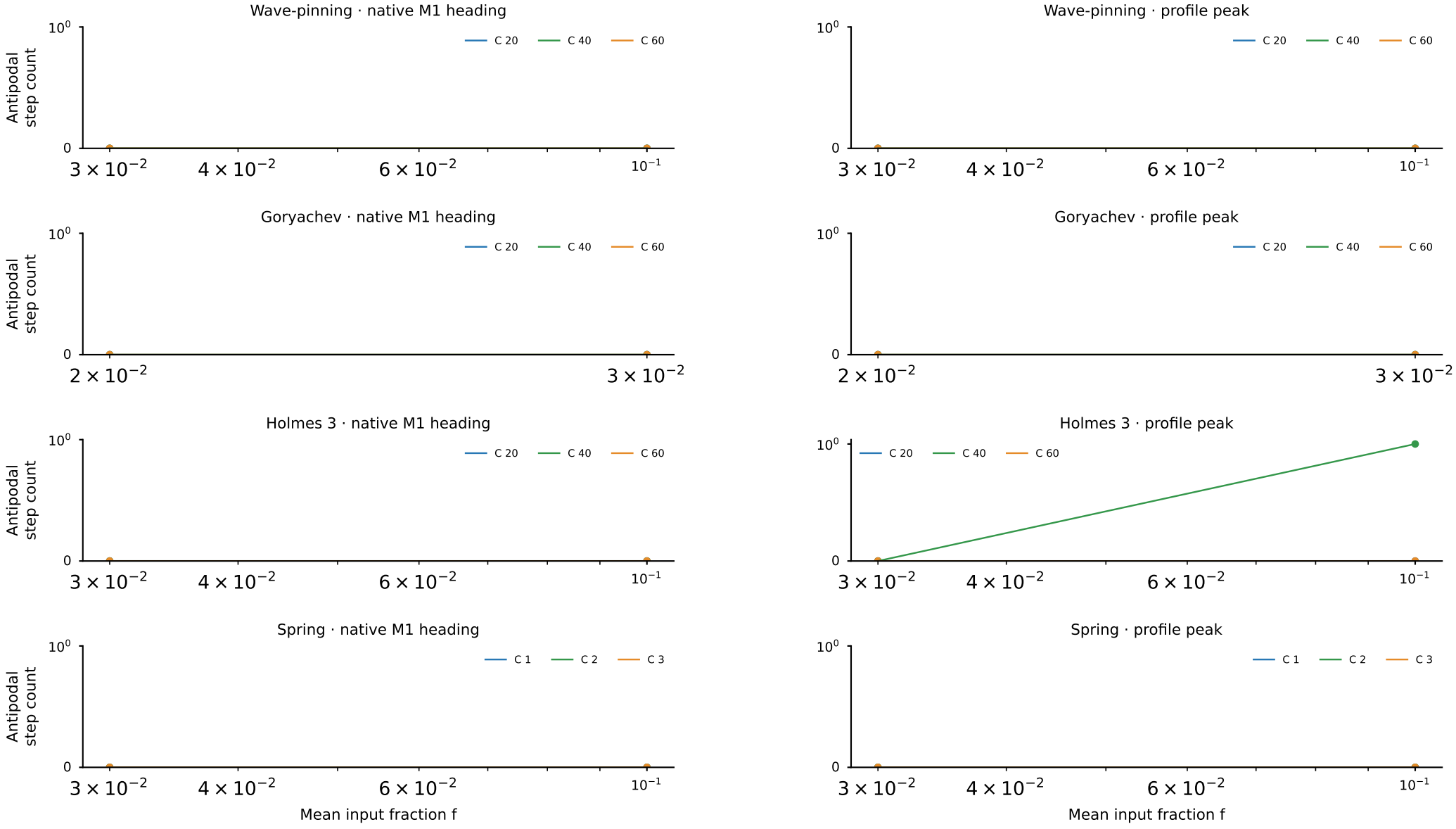
E1 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M_1|/|M_1,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_024_S5_literal_sigma2_profile_peak_literal_actualC_routes_1.csv

S5 · E1 literal material-origin physical-site measurements · Native antipodal ambiguity · main · σ=0.25



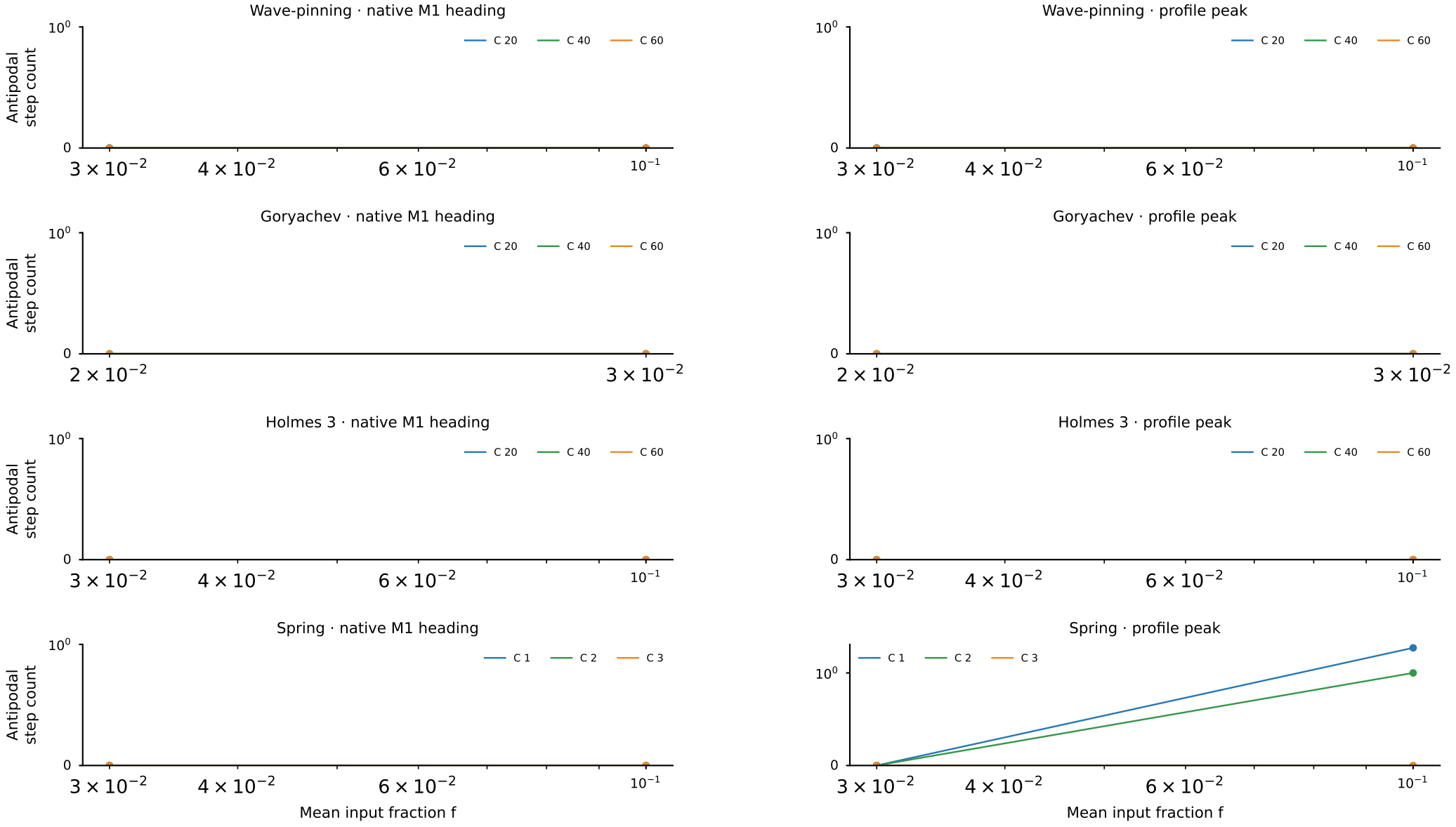
E1 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_025_S5_ambiguity_sigma0.25_main.csv

S5 · E1 literal material-origin physical-site measurements · Native antipodal ambiguity · analysis first half · σ=0.25



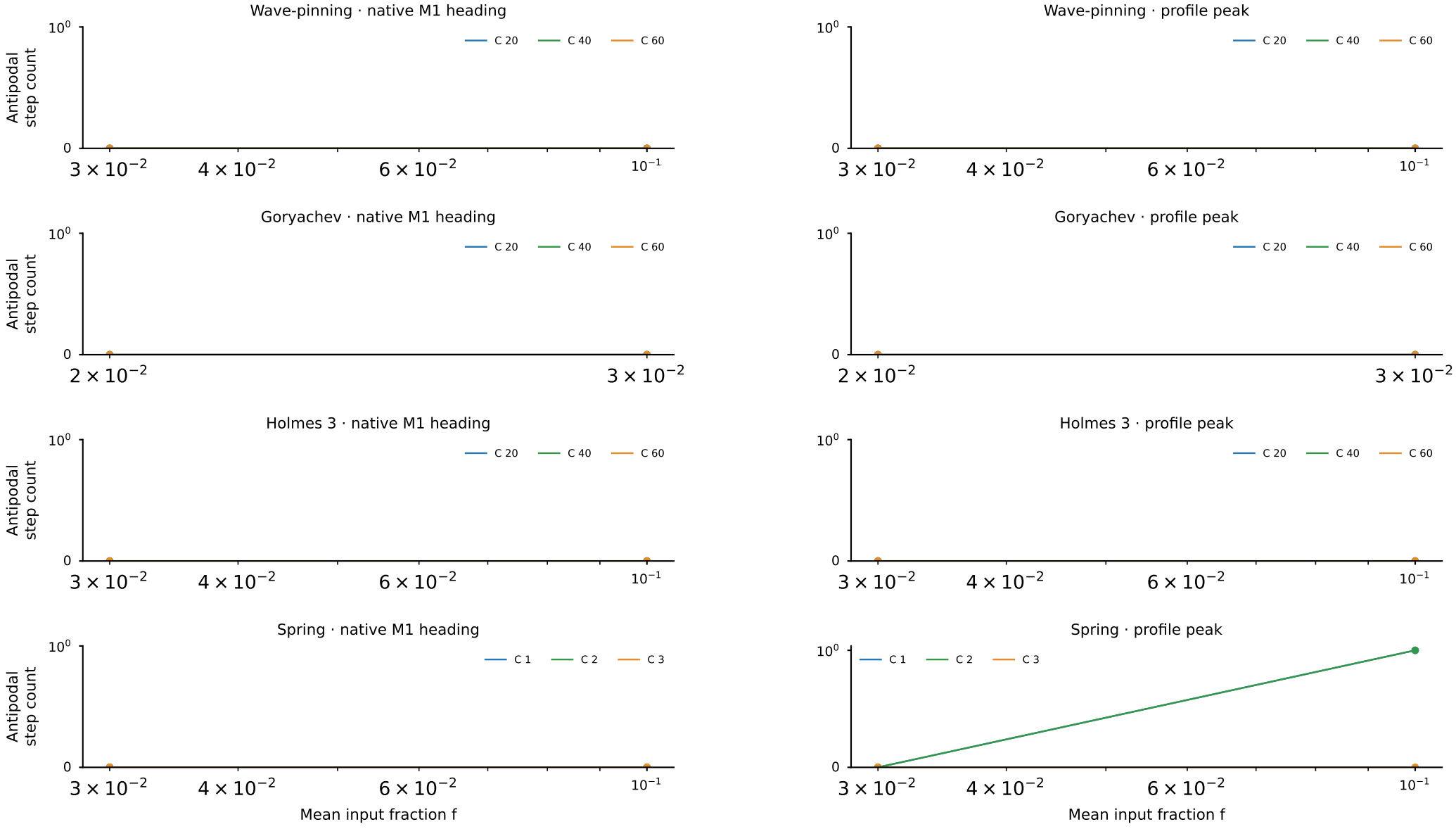
E1 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_026_S5_ambiguity_sigma0.25_analysis_first_half.csv

S5 · E1 literal material-origin physical-site measurements · Native antipodal ambiguity · analysis second half · σ=0.25



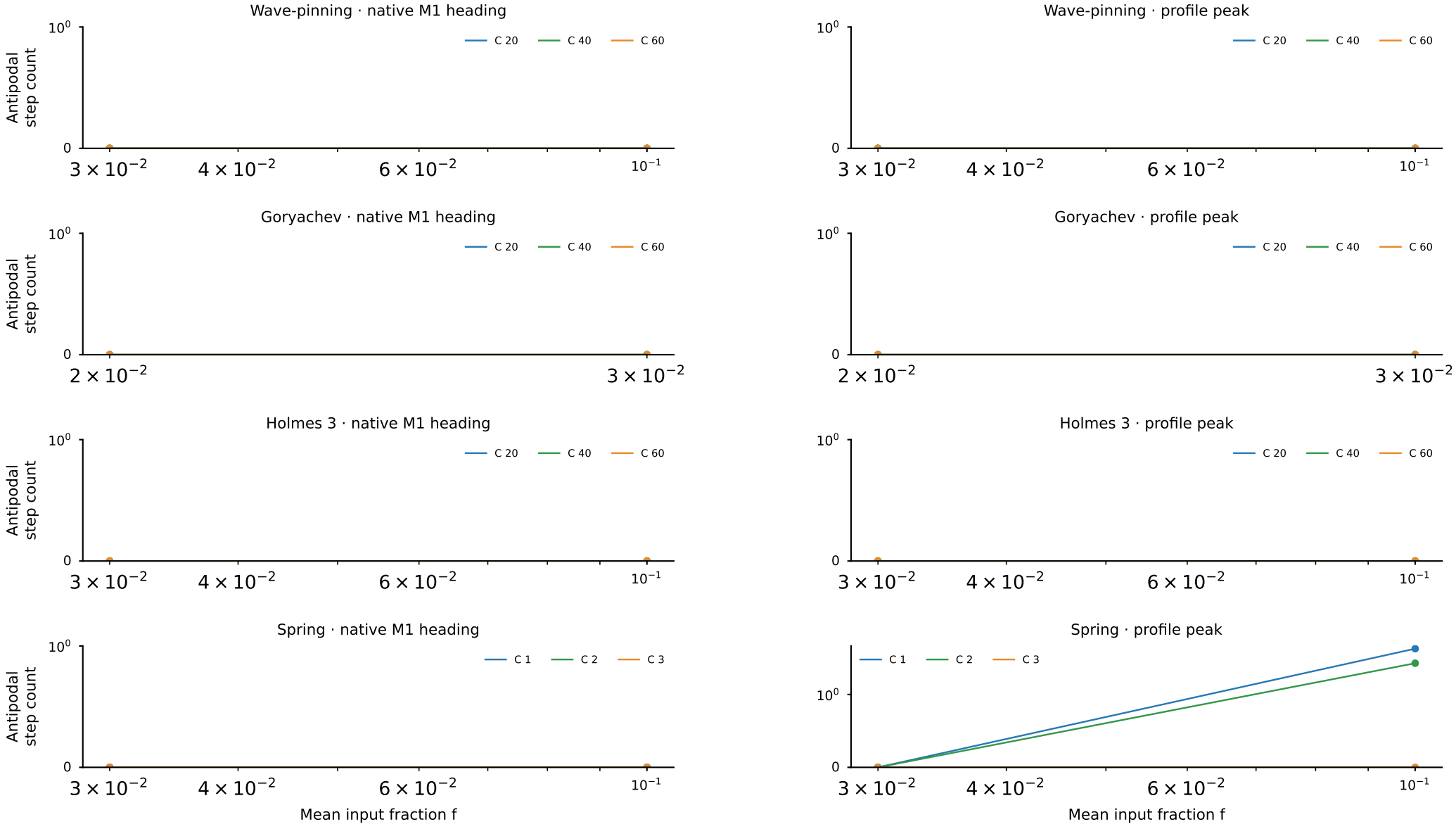
E1 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_027_S5_ambiguity_sigma0.25_analysis_second_half.csv

S5 · E1 literal material-origin physical-site measurements · Native antipodal ambiguity · late quarter · σ=0.25



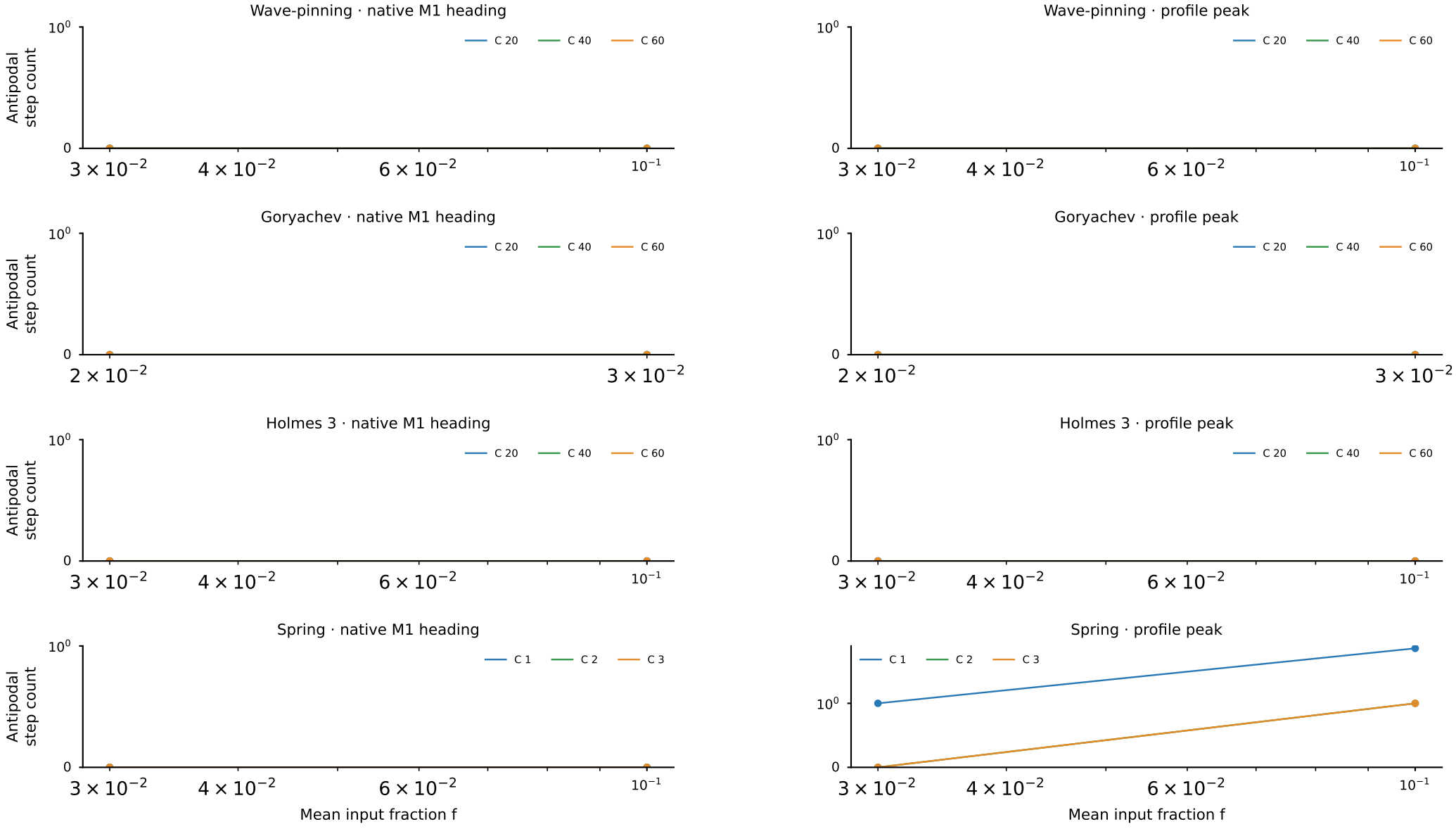
E1 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_028_S5_ambiguity_sigma0.25_late_quarter.csv

S5 · E1 literal material-origin physical-site measurements · Native antipodal ambiguity · cadence x2 · σ=0.25



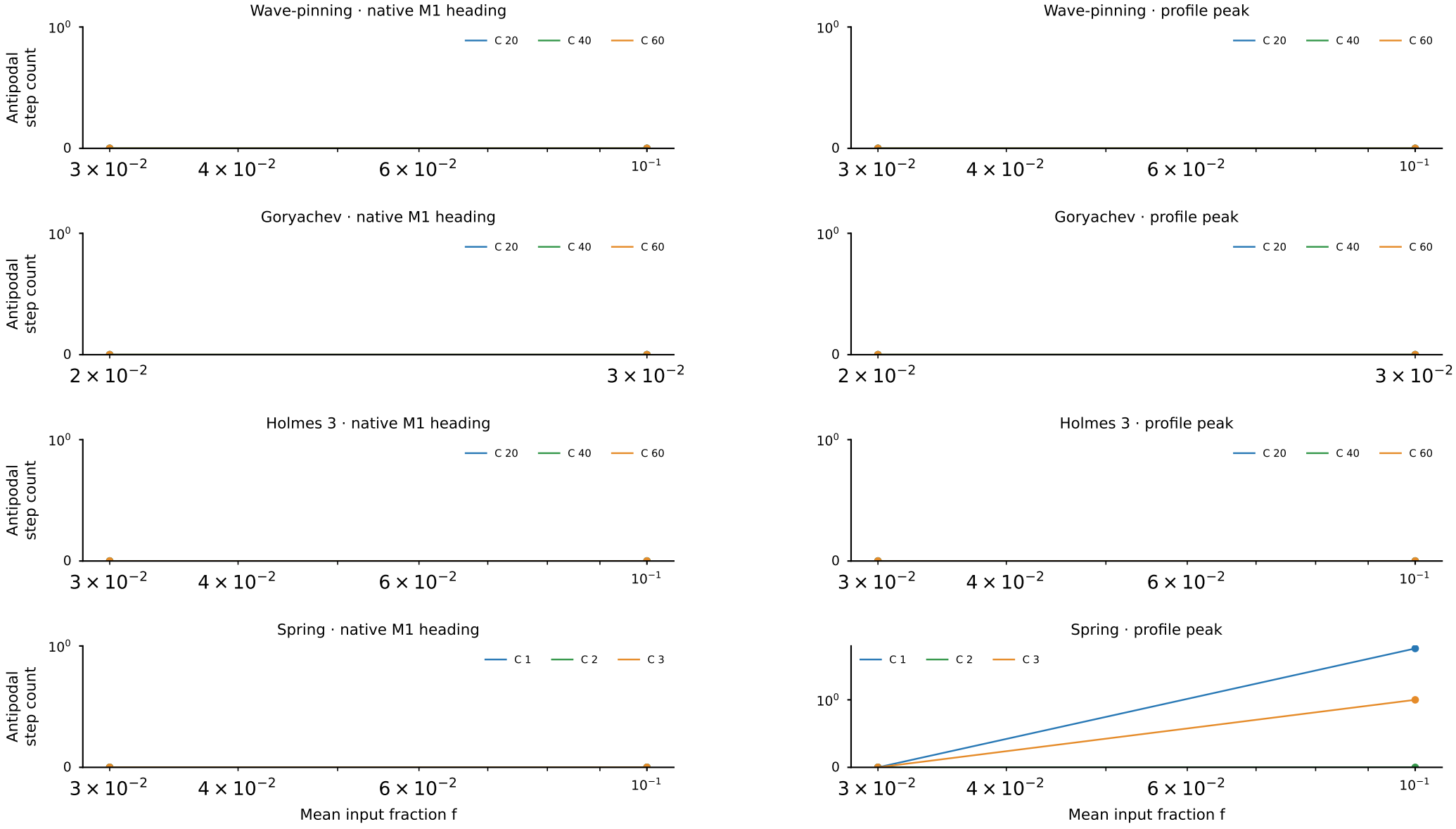
E1 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_029_S5_ambiguity_sigma0.25_cadence_x2.csv

S5 · E1 literal material-origin physical-site measurements · Native antipodal ambiguity · cadence x4 · σ=0.25



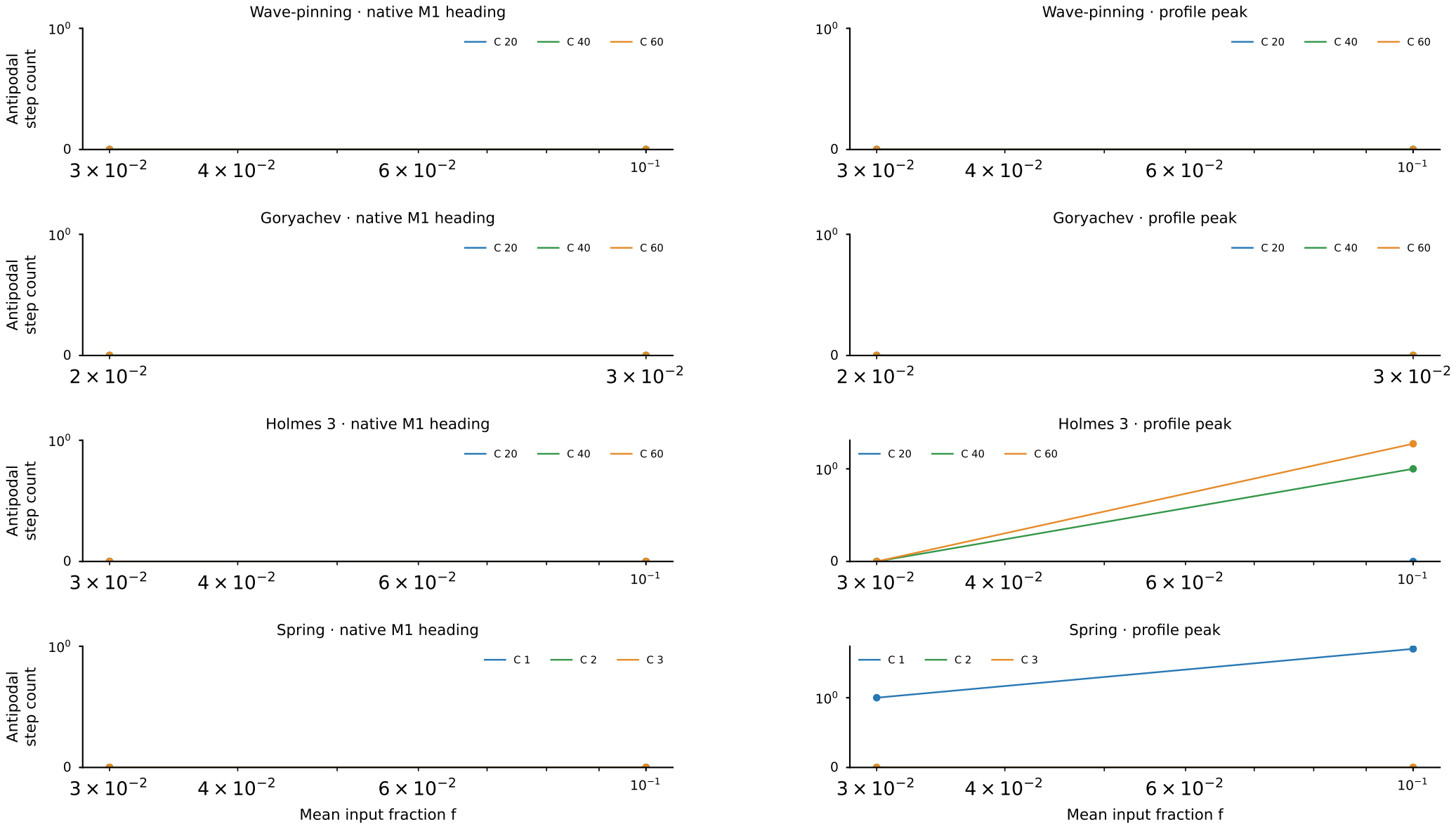
E1 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_030_S5_ambiguity_sigma0.25_cadence_x4.csv

S5 · E1 literal material-origin physical-site measurements · Native antipodal ambiguity · cadence x8 · σ=0.25



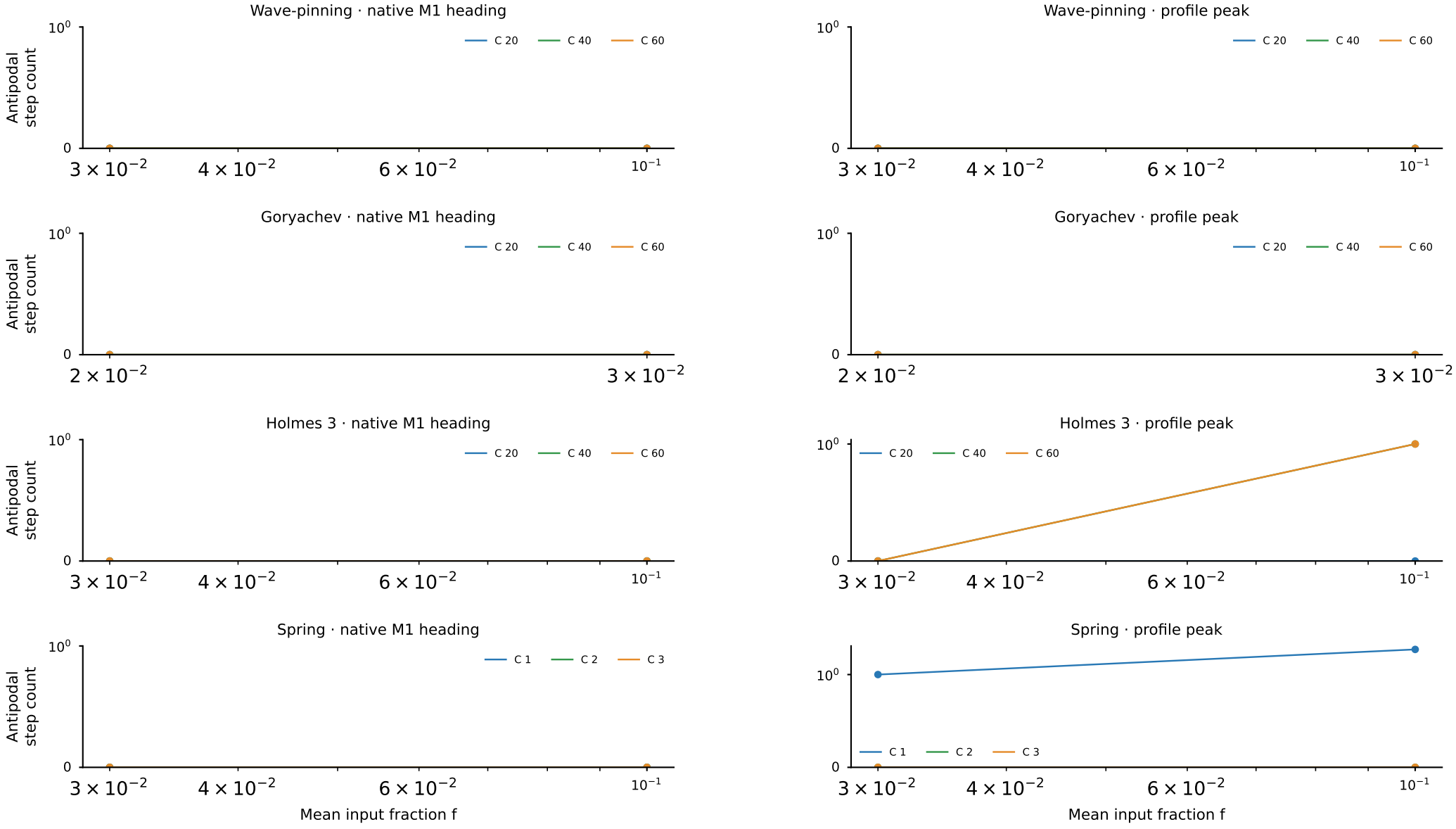
E1 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_031_S5_ambiguity_sigma0.25_cadence_x8.csv

S5 · E1 literal material-origin physical-site measurements · Native antipodal ambiguity · main · σ=0.5



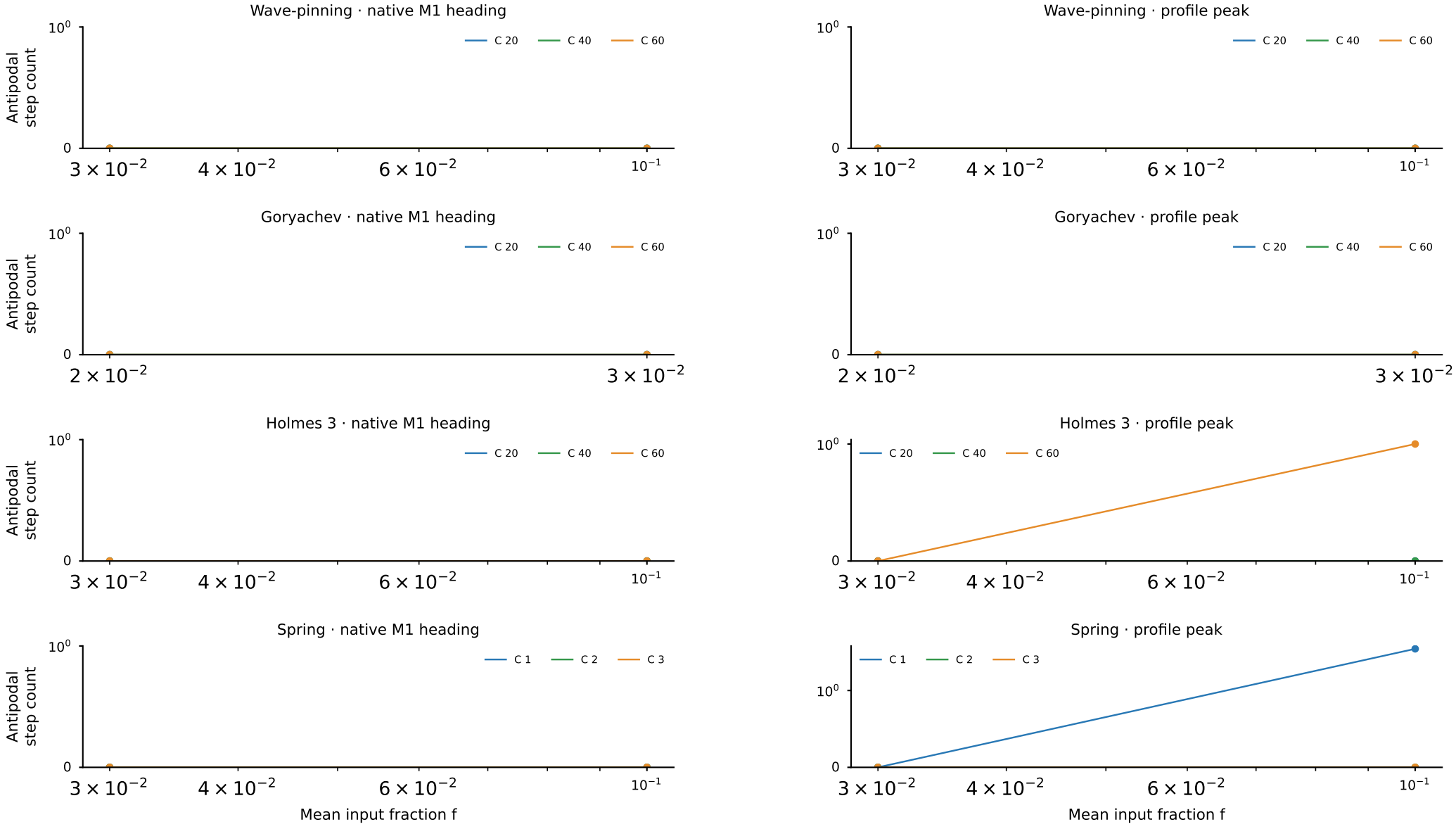
E1 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_032_S5_ambiguity_sigma0.5_main.csv

S5 · E1 literal material-origin physical-site measurements · Native antipodal ambiguity · analysis first half · σ=0.5



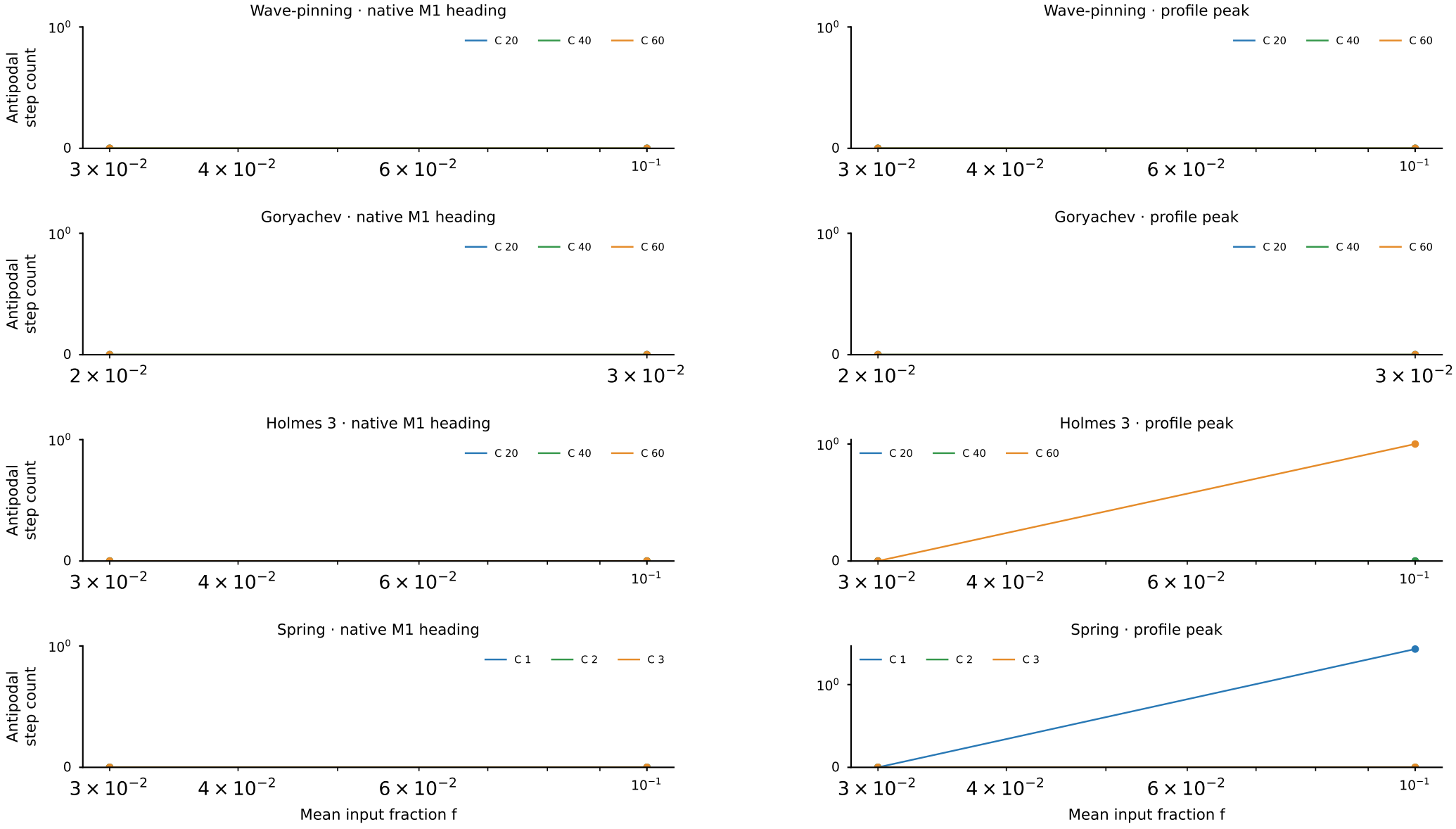
E1 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_033_S5_ambiguity_sigma0.5_analysis_first_half.csv

S5 · E1 literal material-origin physical-site measurements · Native antipodal ambiguity · analysis second half · σ=0.5



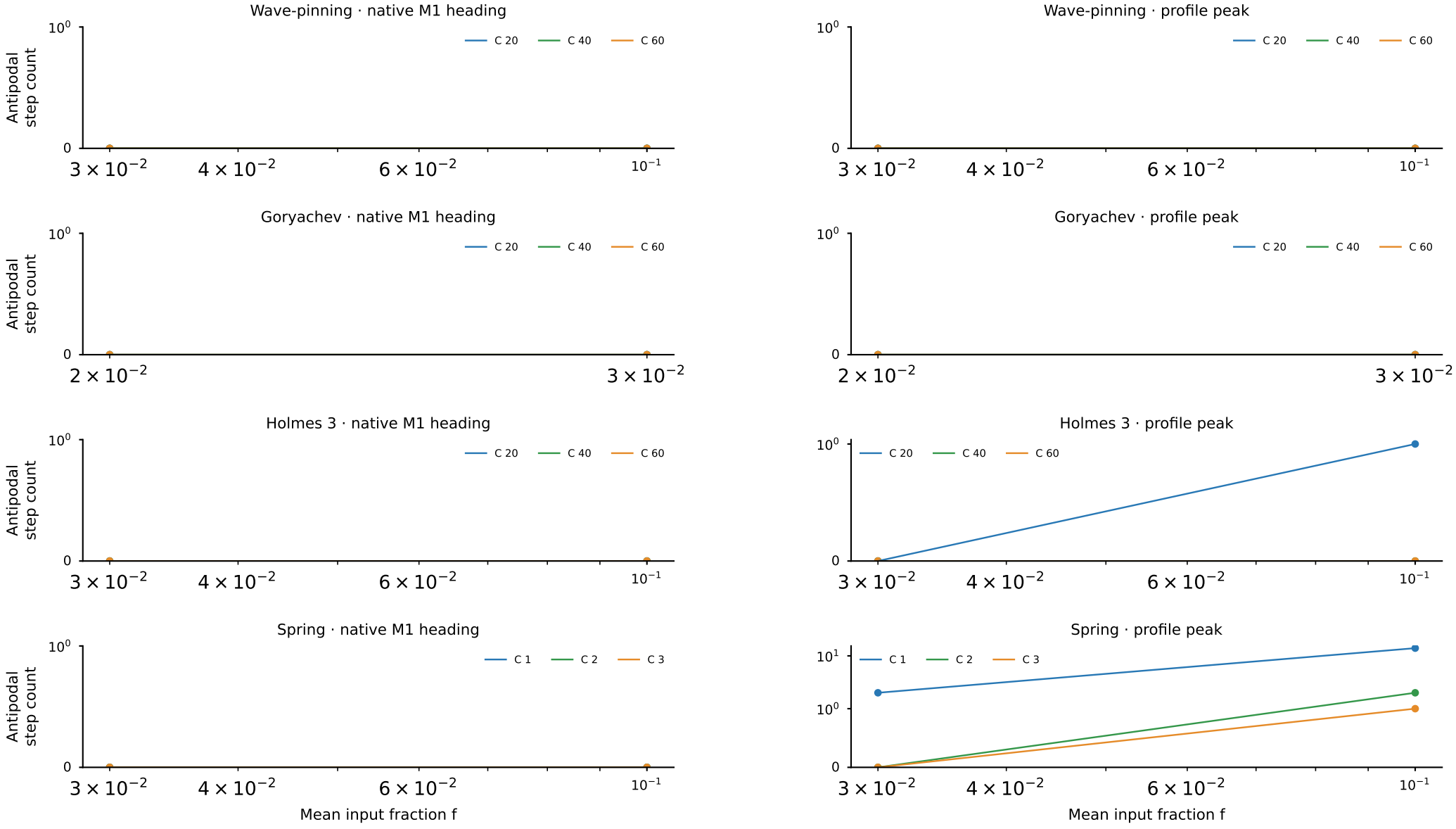
E1 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_034_S5_ambiguity_sigma0.5_analysis_second_half.csv

S5 · E1 literal material-origin physical-site measurements · Native antipodal ambiguity · late quarter · σ=0.5



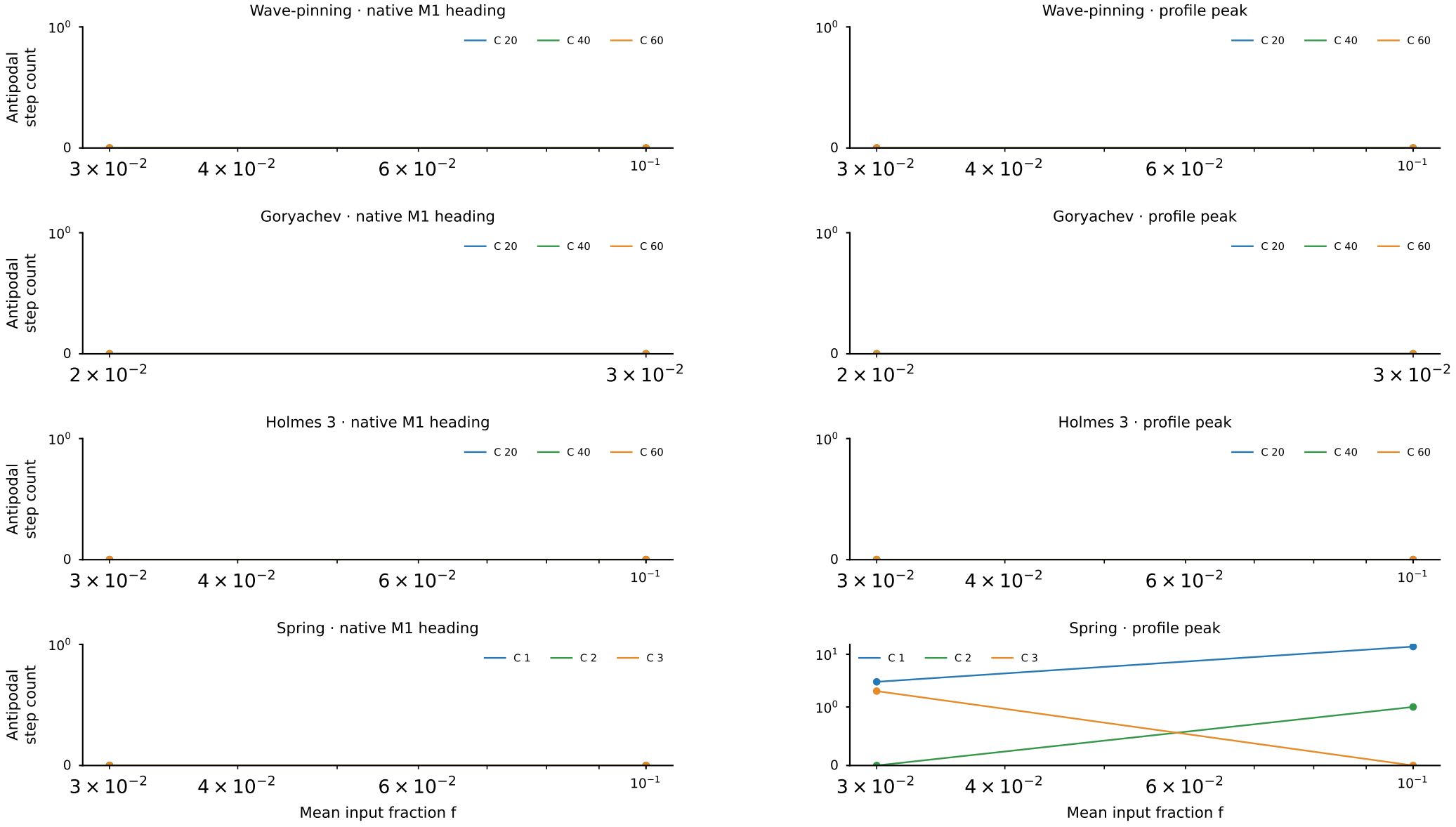
E1 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_035_S5_ambiguity_sigma0.5_late_quarter.csv

S5 · E1 literal material-origin physical-site measurements · Native antipodal ambiguity · cadence x2 · σ=0.5



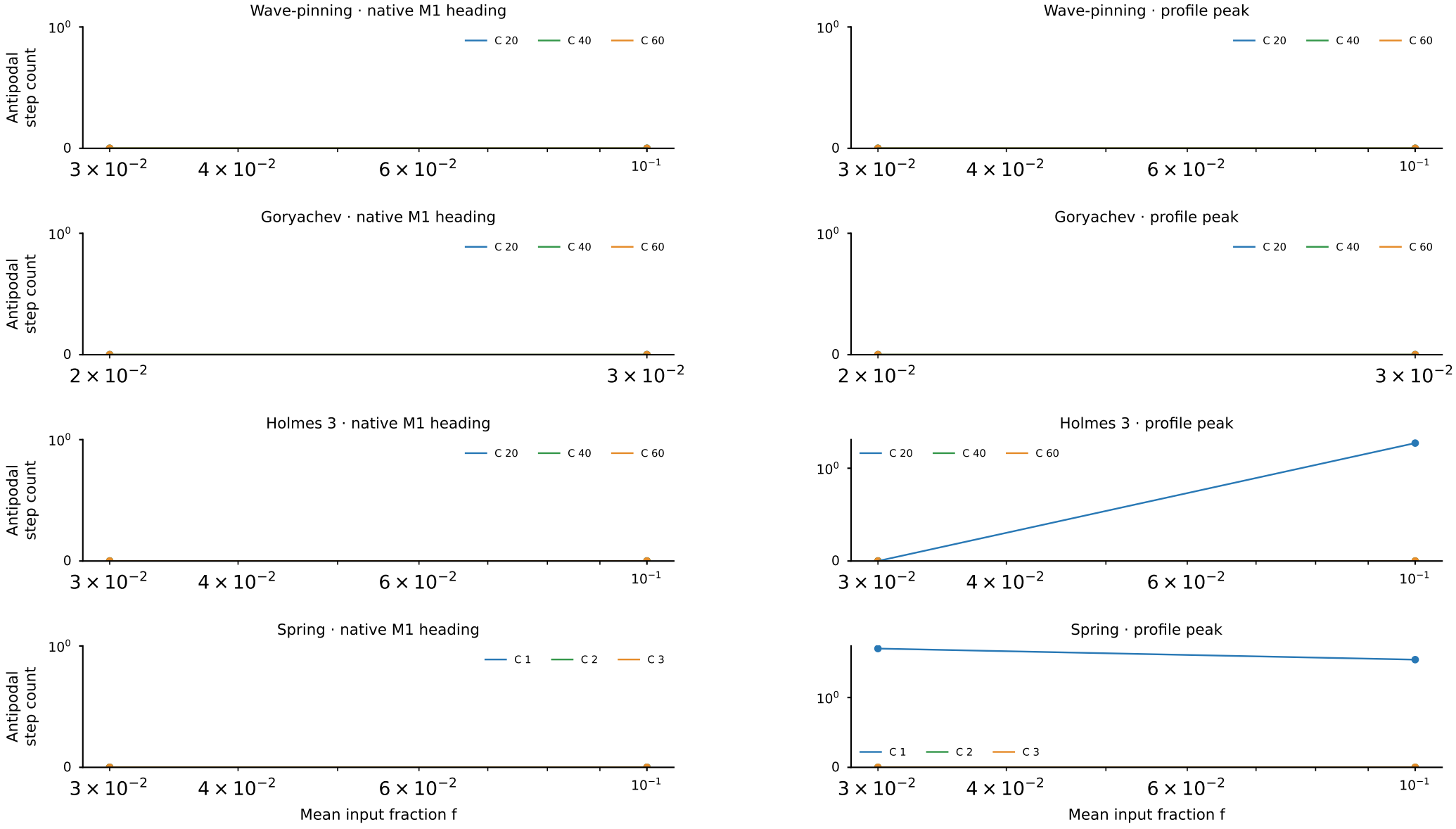
E1 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_036_S5_ambiguity_sigma0.5_cadence_x2.csv

S5 · E1 literal material-origin physical-site measurements · Native antipodal ambiguity · cadence x4 · σ=0.5



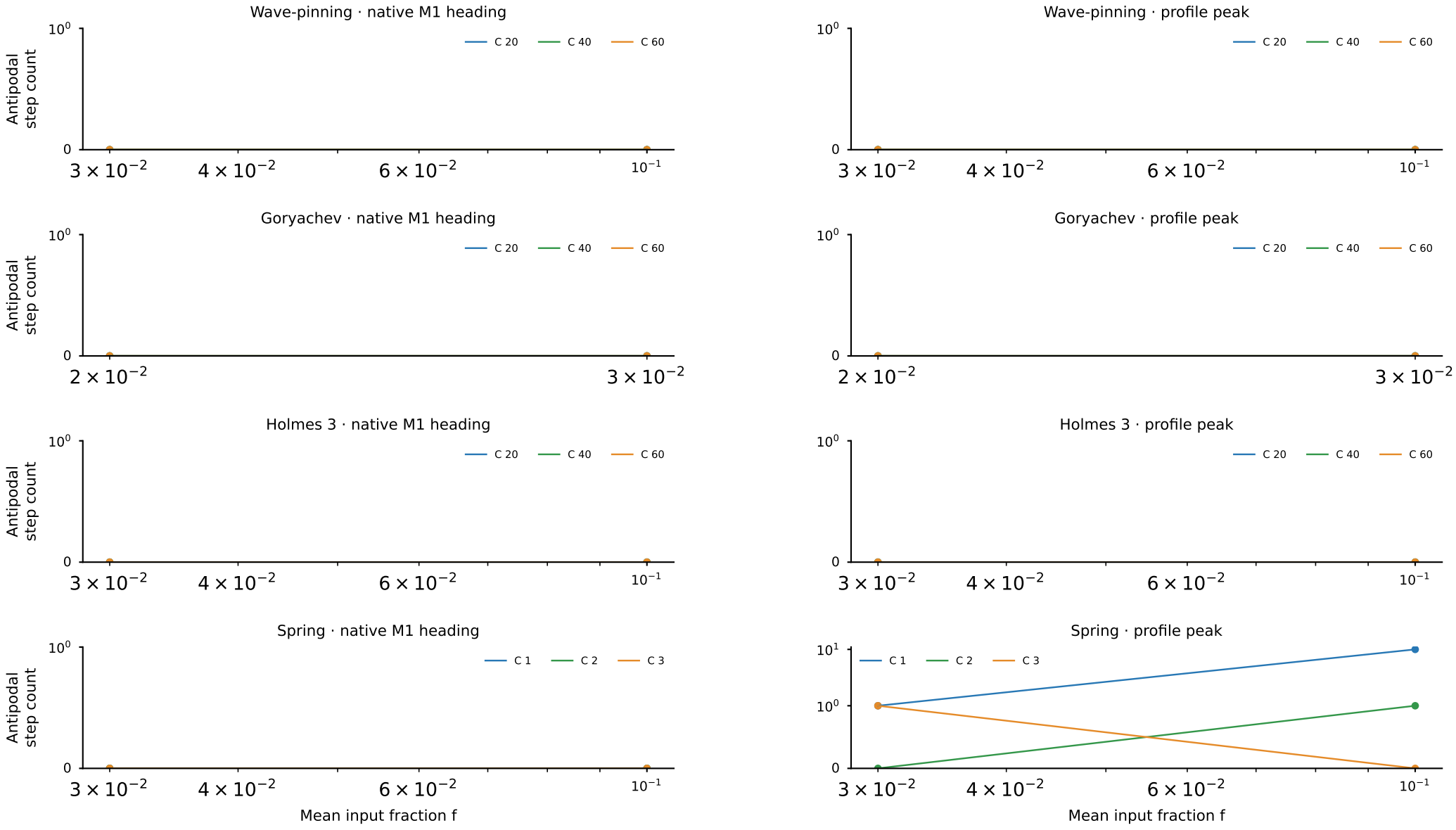
E1 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_037_S5_ambiguity_sigma0.5_cadence_x4.csv

S5 · E1 literal material-origin physical-site measurements · Native antipodal ambiguity · cadence x8 · σ=0.5



E1 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_038_S5_ambiguity_sigma0.5_cadence_x8.csv

S5 · E1 literal material-origin physical-site measurements · Native antipodal ambiguity · main · σ=1



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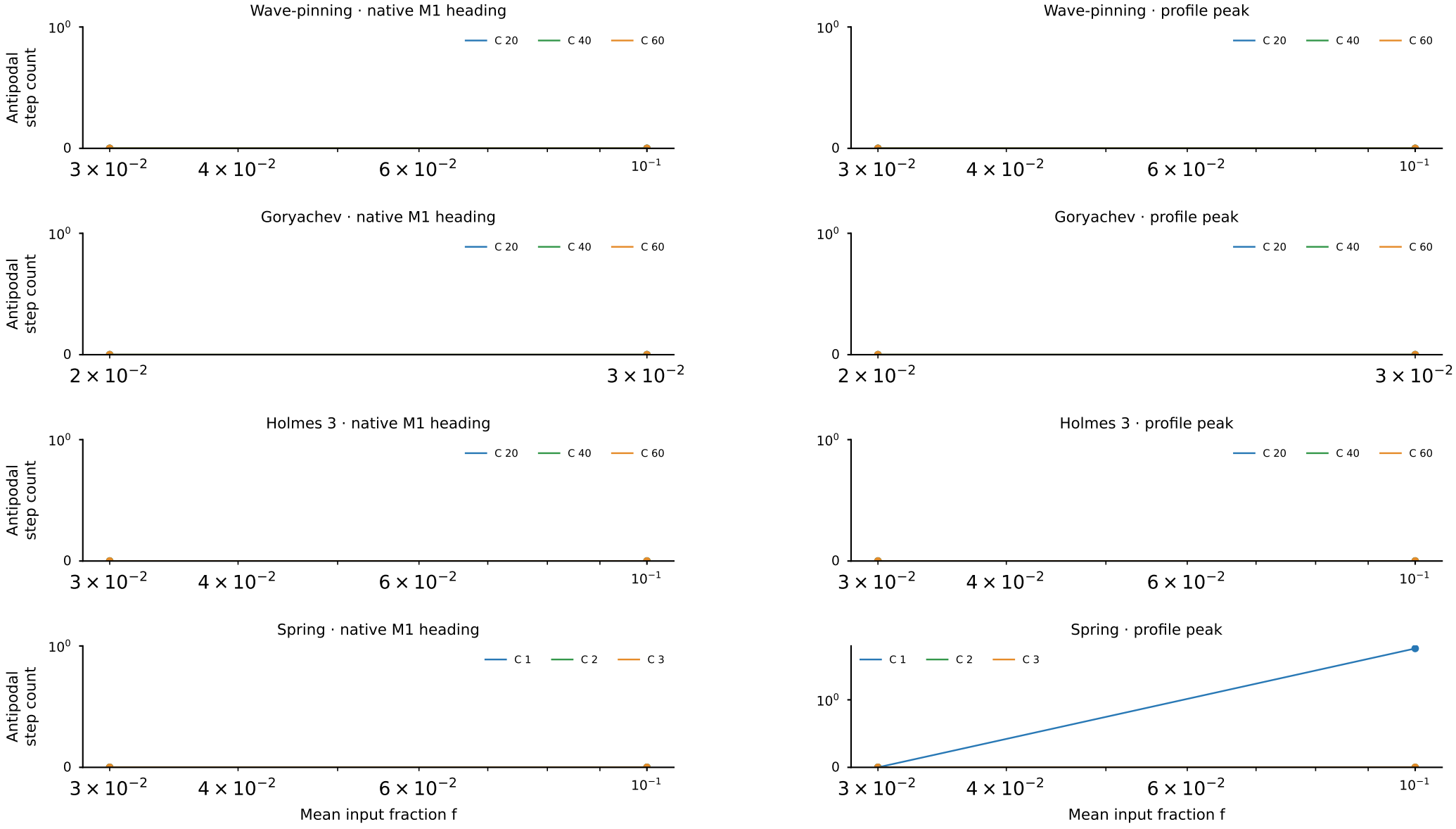
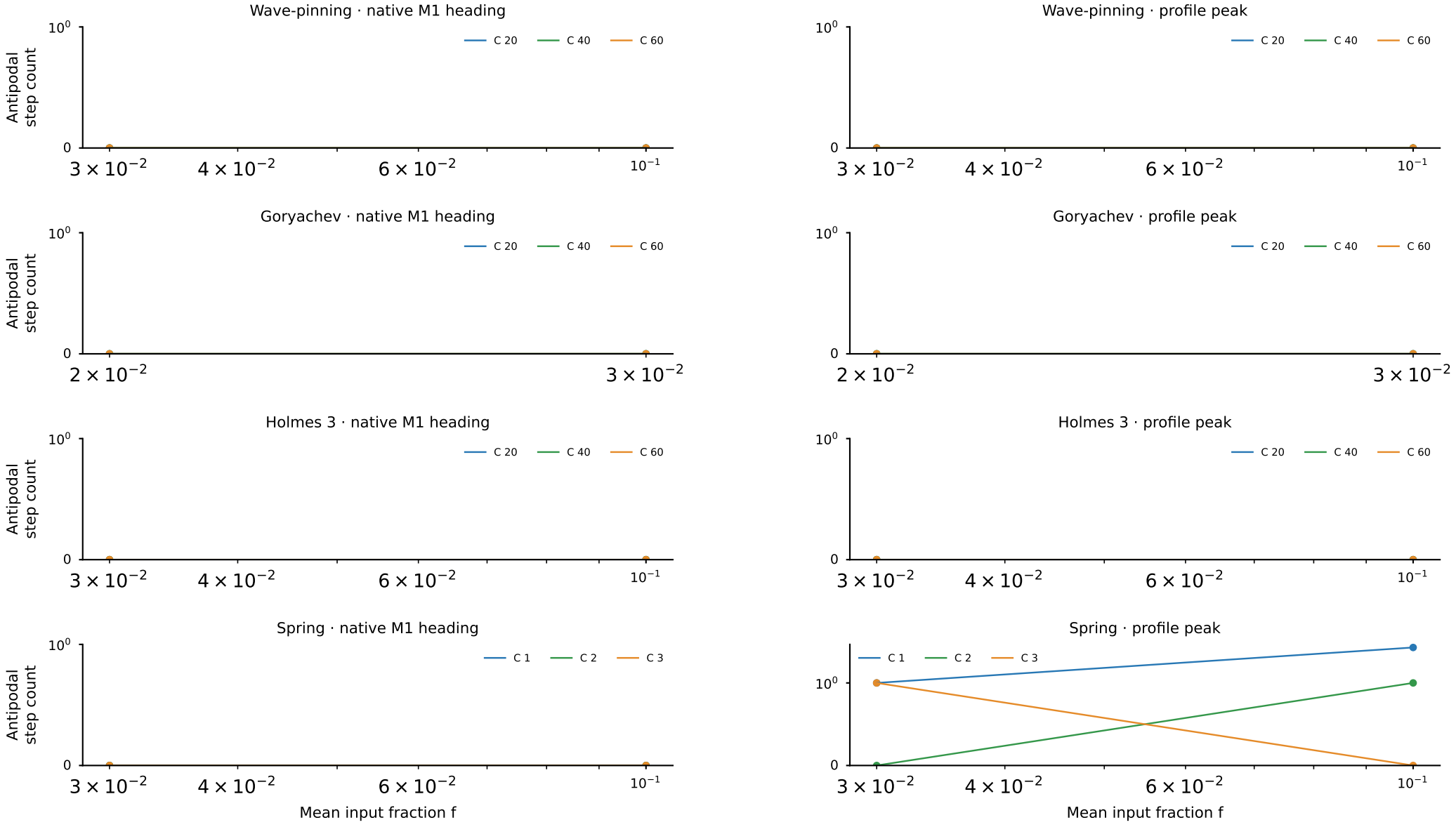
CSV: figure4_S5_literal_E1_final_layout_20261020_039_S5_ambiguity_sigma1_main.csv

S5 · E1 literal material-origin physical-site measurements · Native antipodal ambiguity · analysis first half · σ=1



E1 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_040_S5_ambiguity_sigma1_analysis_first_half.csv

S5 · E1 literal material-origin physical-site measurements · Native antipodal ambiguity · analysis second half · σ=1



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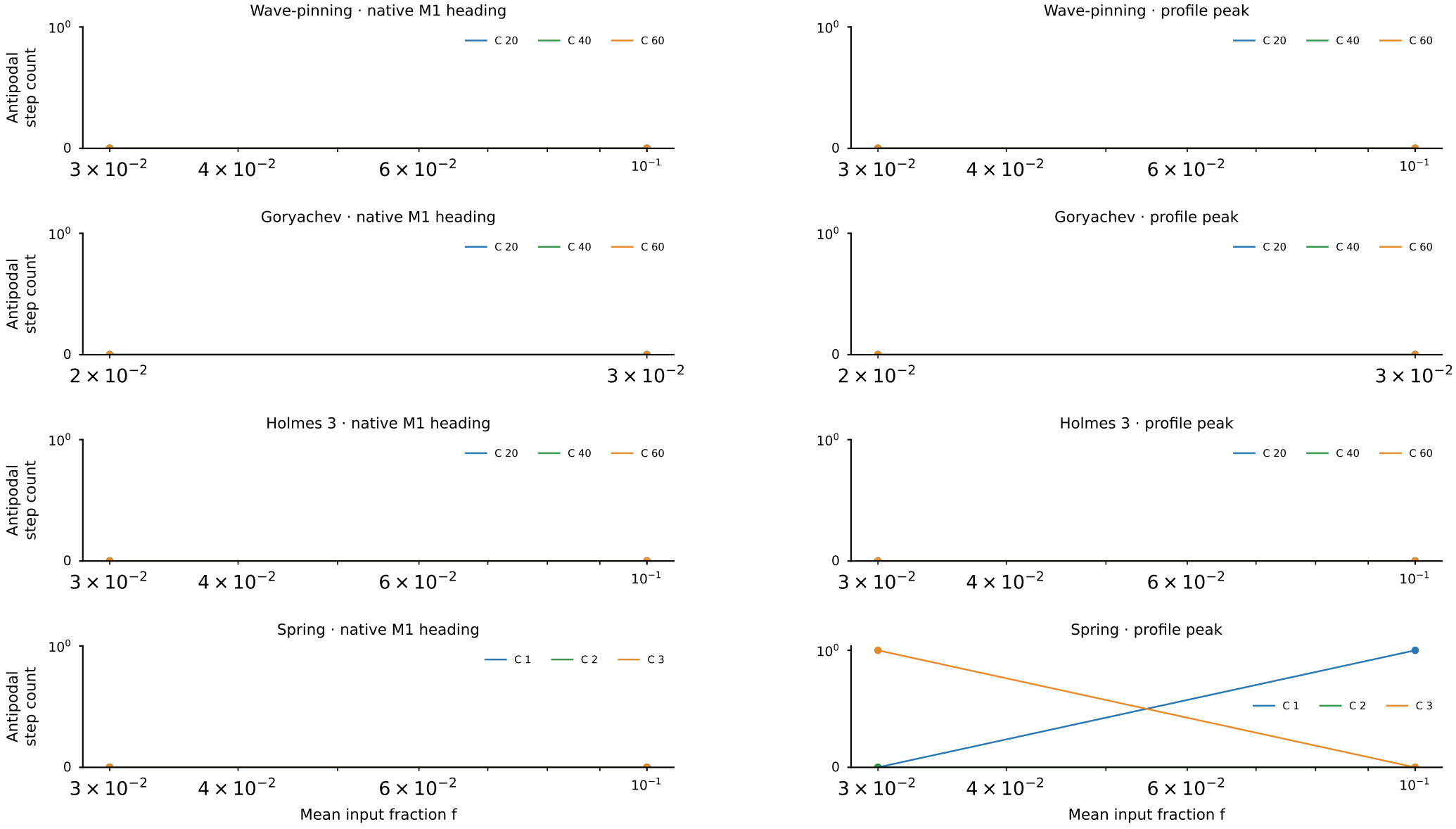
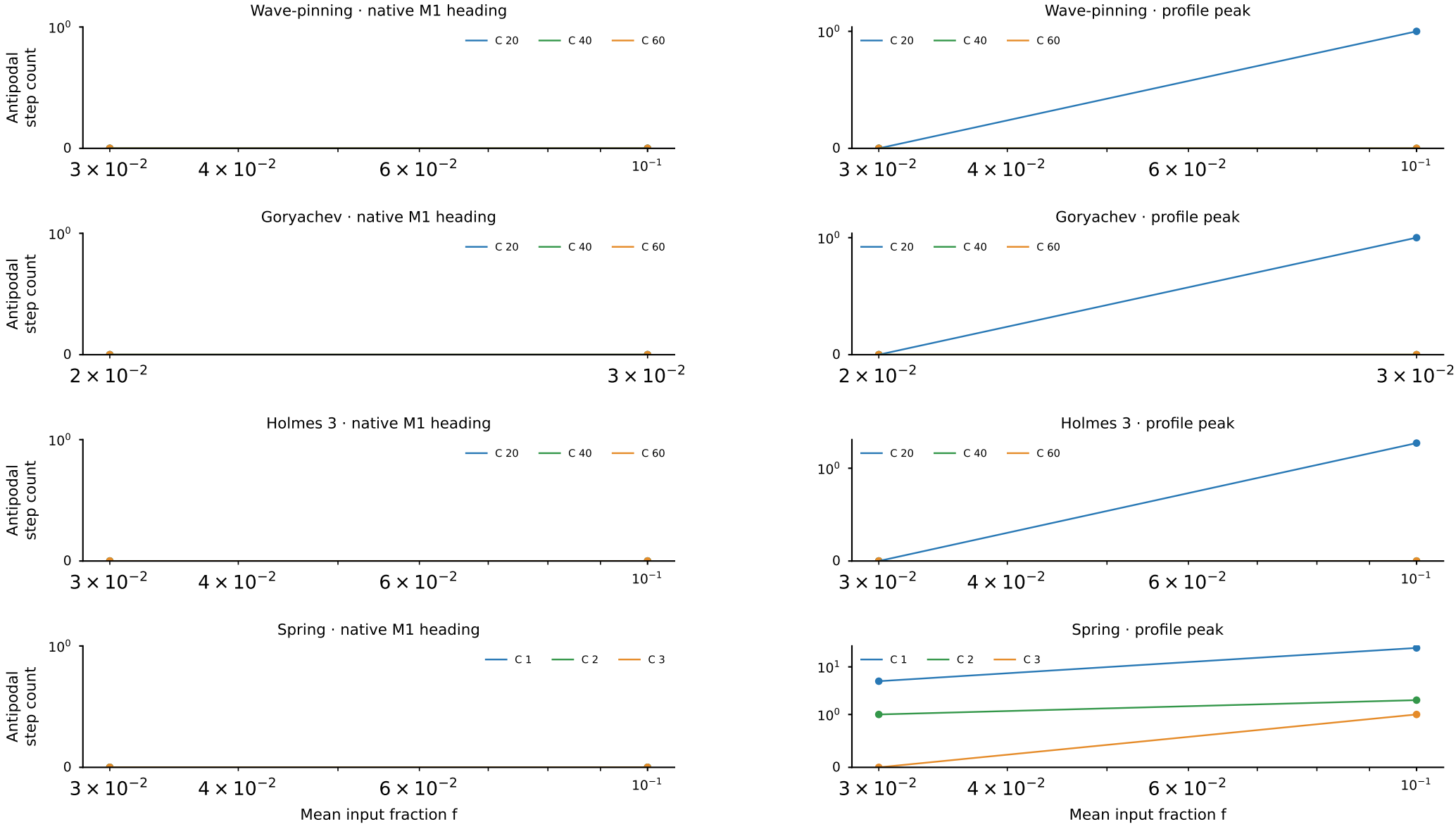
CSV: figure4_S5_literal_E1_final_layout_20261020_041_S5_ambiguity_sigma1_analysis_second_half.csv

S5 · E1 literal material-origin physical-site measurements · Native antipodal ambiguity · late quarter · σ=1



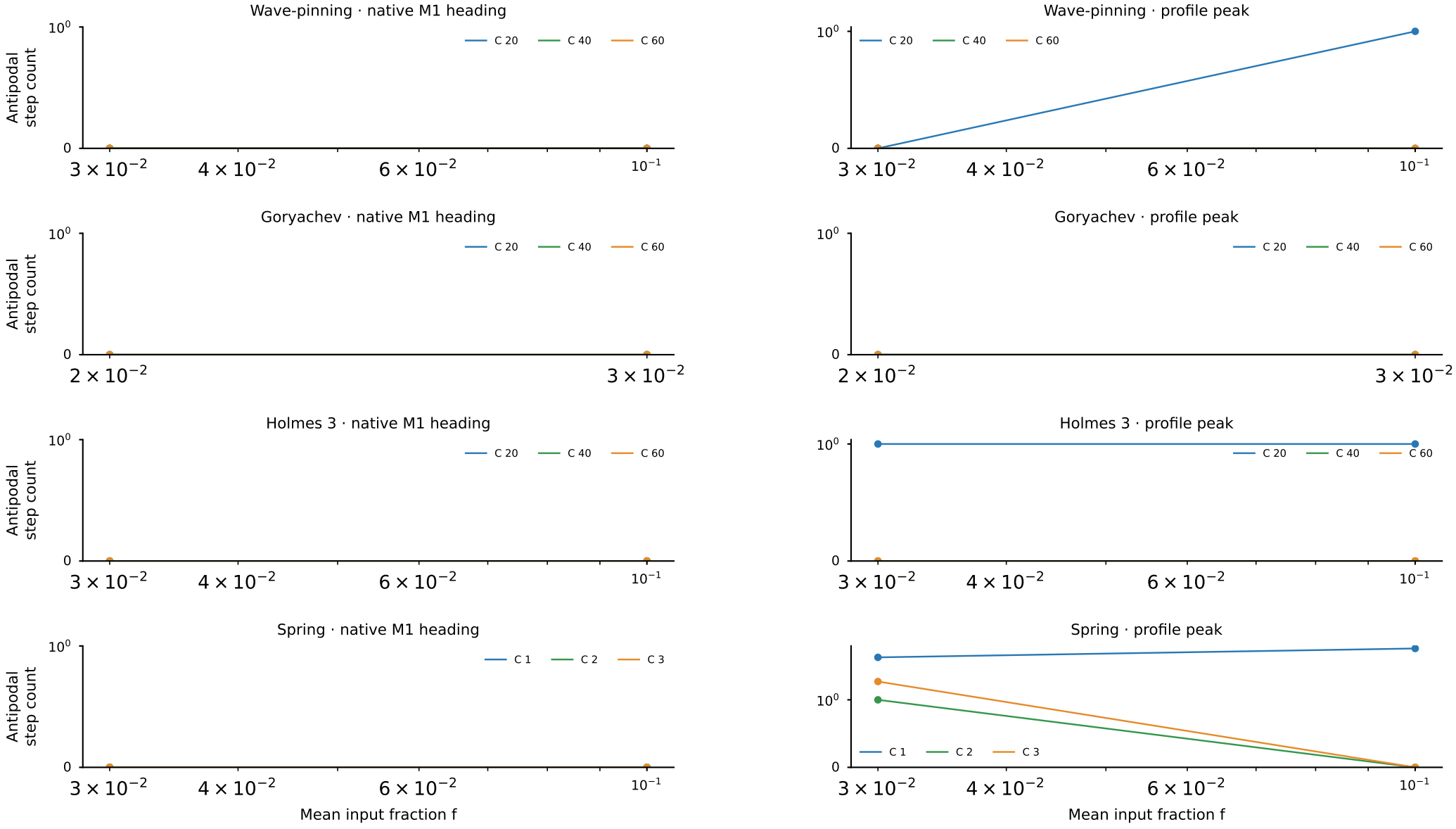
E1 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_042_S5_ambiguity_sigma1_late_quarter.csv

S5 · E1 literal material-origin physical-site measurements · Native antipodal ambiguity · cadence x2 · σ=1



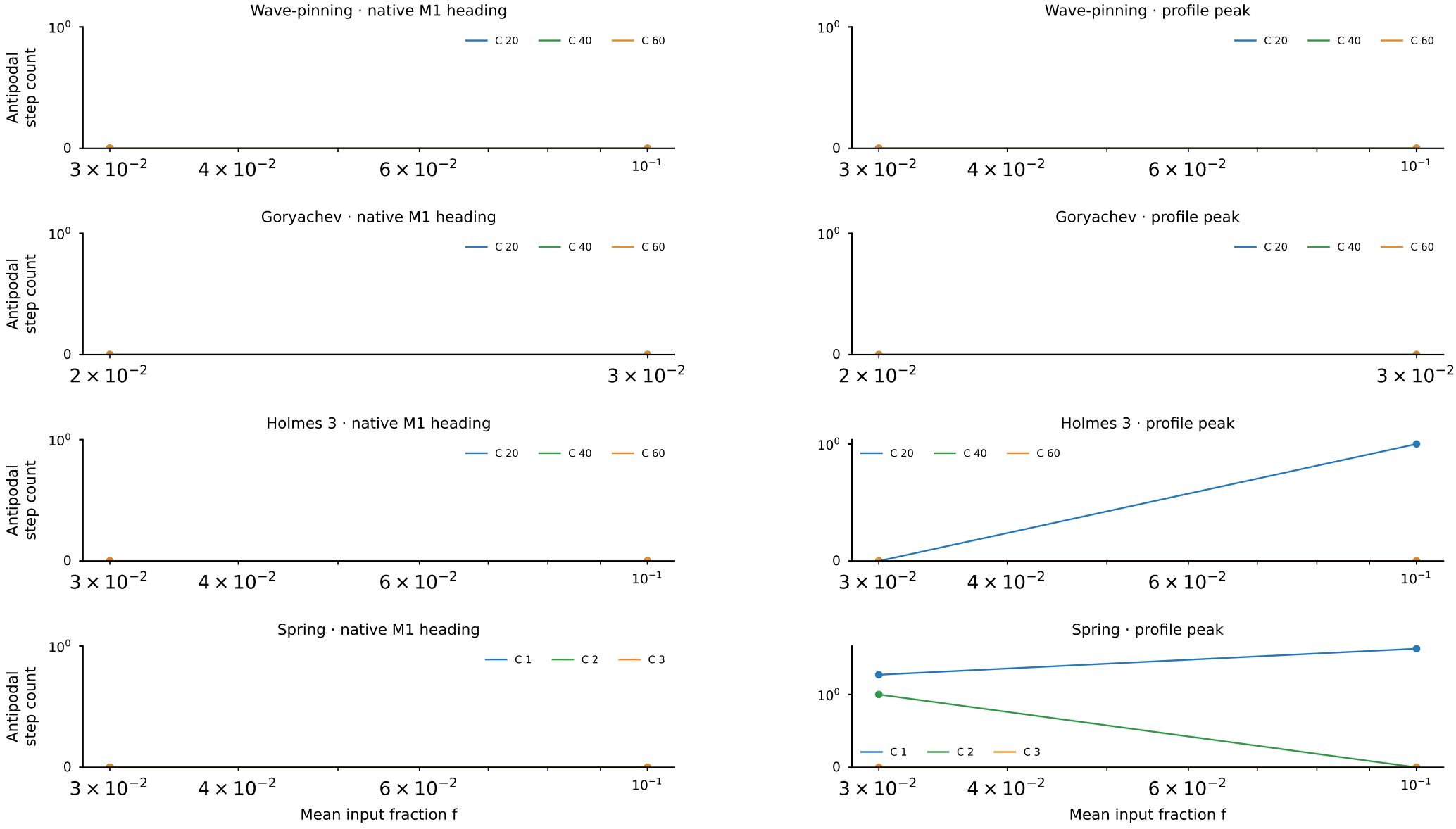
E1 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_043_S5_ambiguity_sigma1_cadence_x2.csv

S5 · E1 literal material-origin physical-site measurements · Native antipodal ambiguity · cadence x4 · σ=1



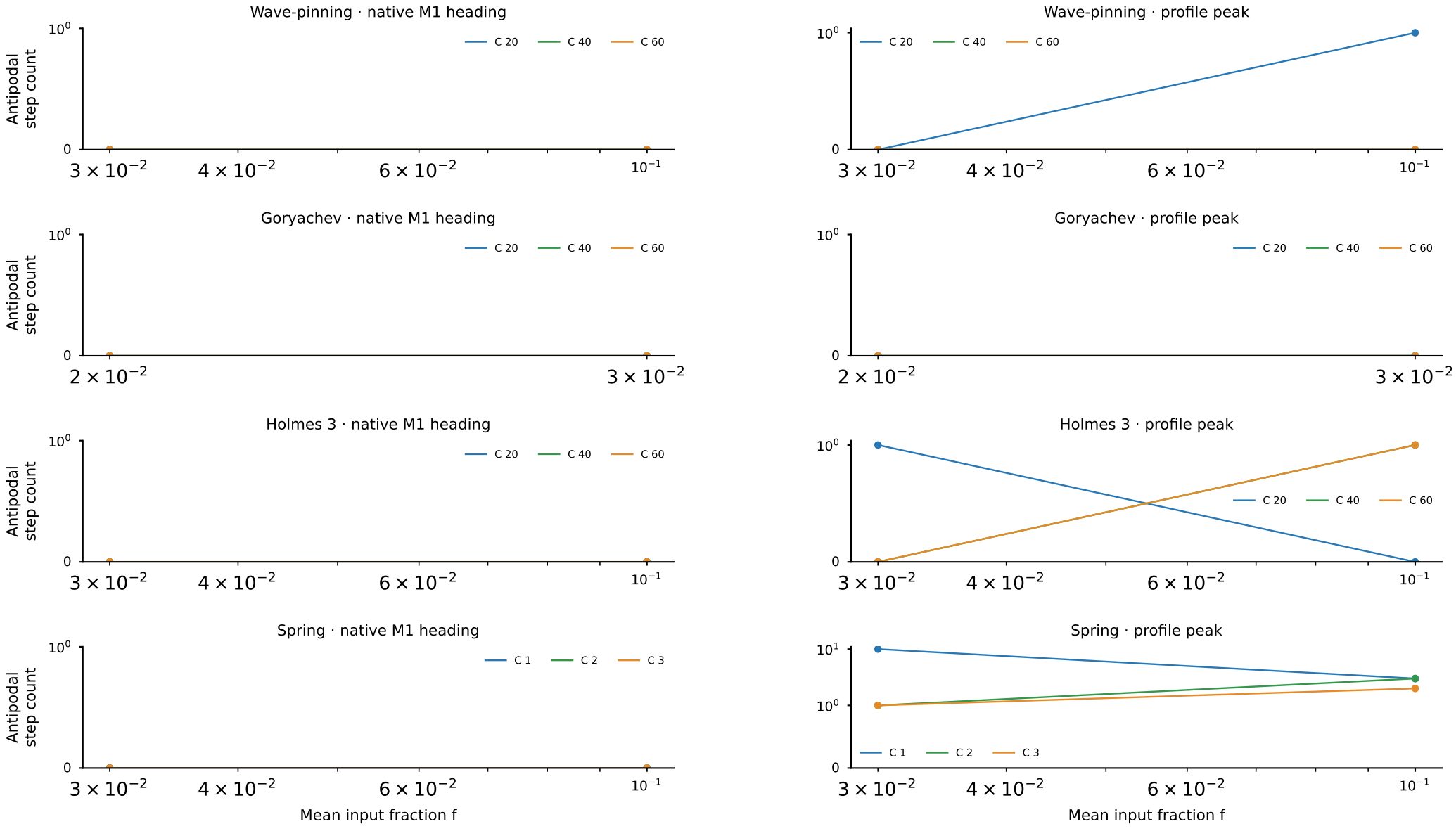
E1 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_044_S5_ambiguity_sigma1_cadence_x4.csv

S5 · E1 literal material-origin physical-site measurements · Native antipodal ambiguity · cadence x8 · σ=1



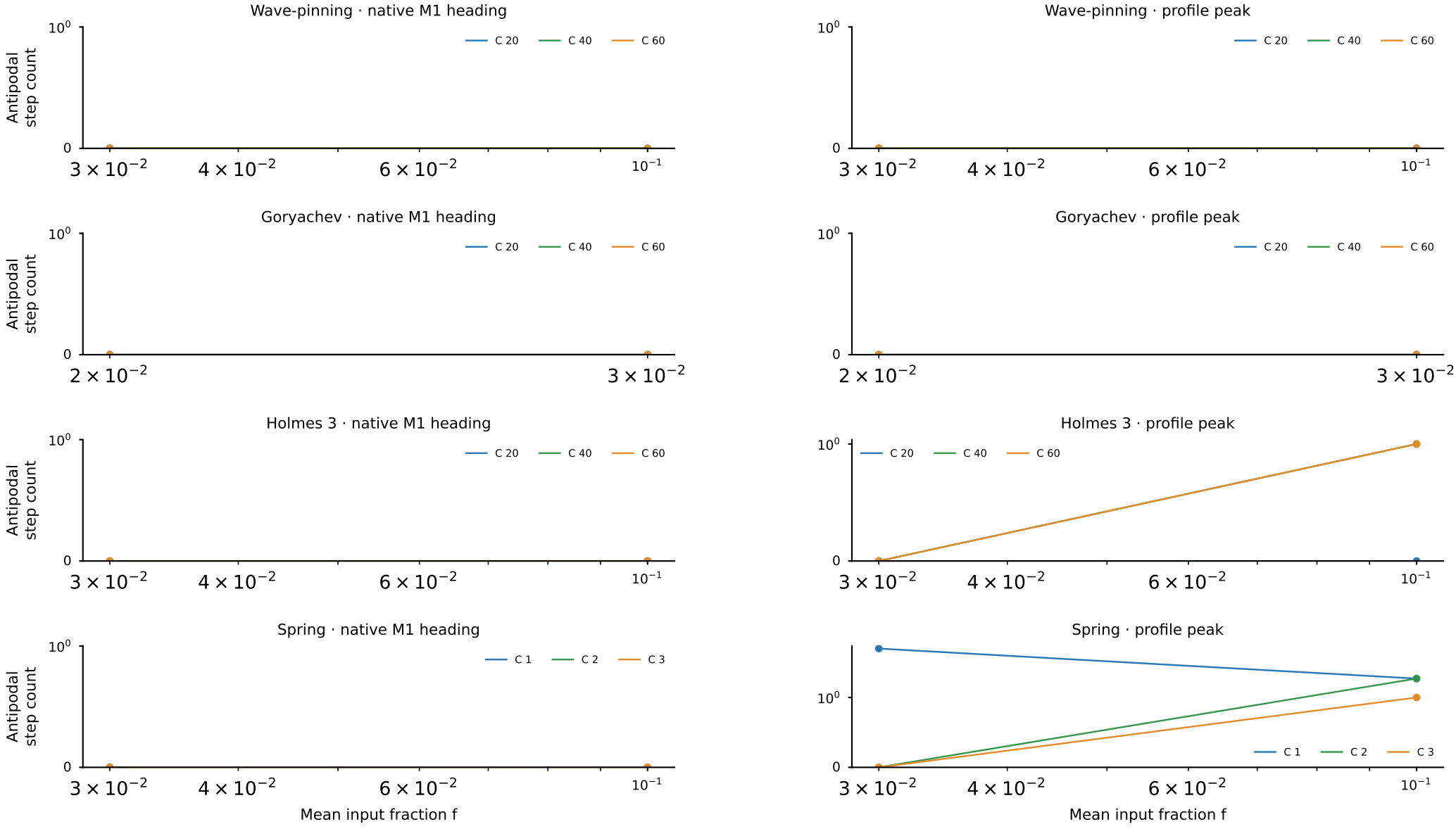
E1 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_045_S5_ambiguity_sigma1_cadence_x8.csv

S5 · E1 literal material-origin physical-site measurements · Native antipodal ambiguity · main · σ=2



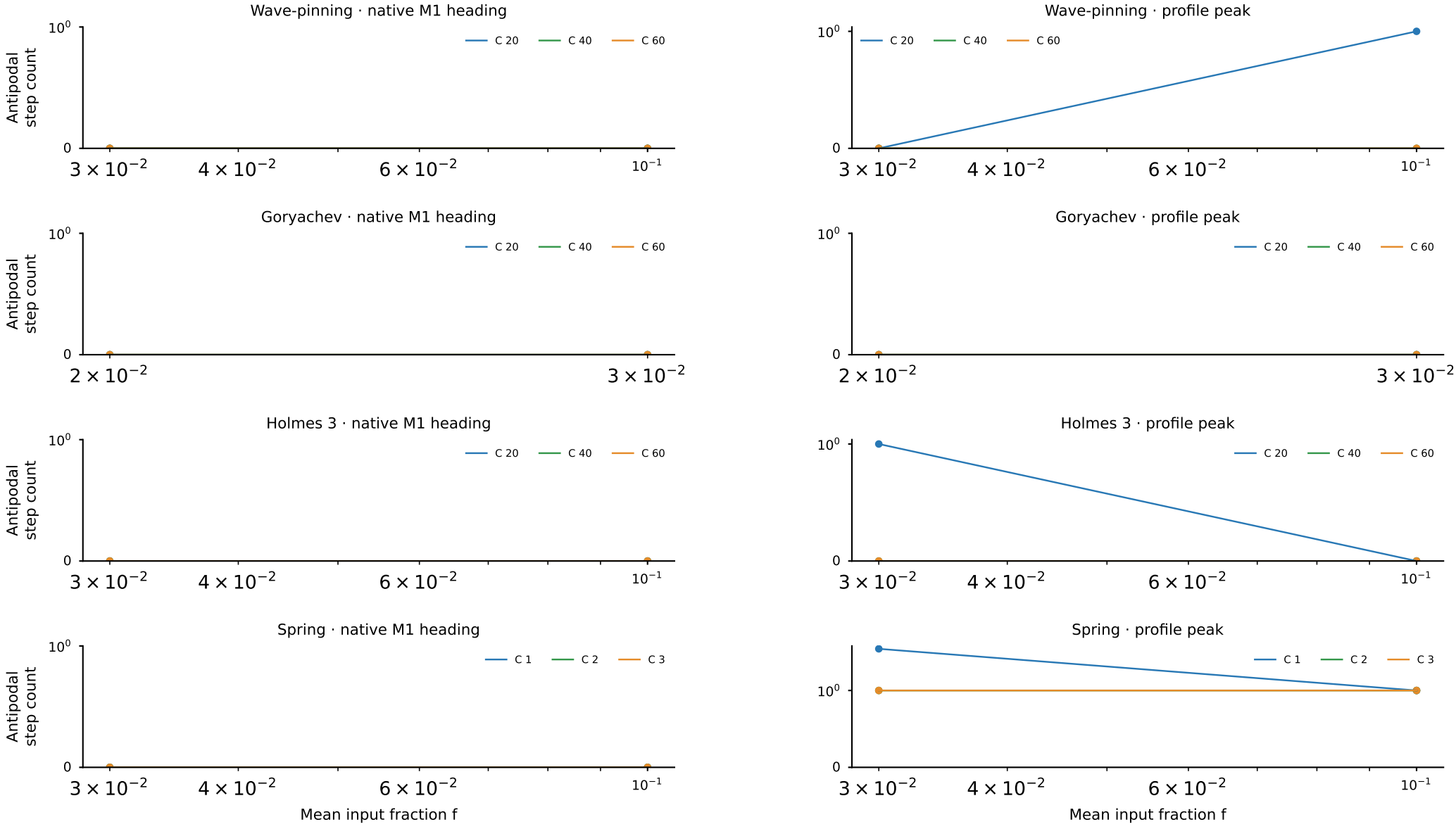
E1 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_046_S5_ambiguity_sigma2_main.csv

S5 · E1 literal material-origin physical-site measurements · Native antipodal ambiguity · analysis first half · σ=2



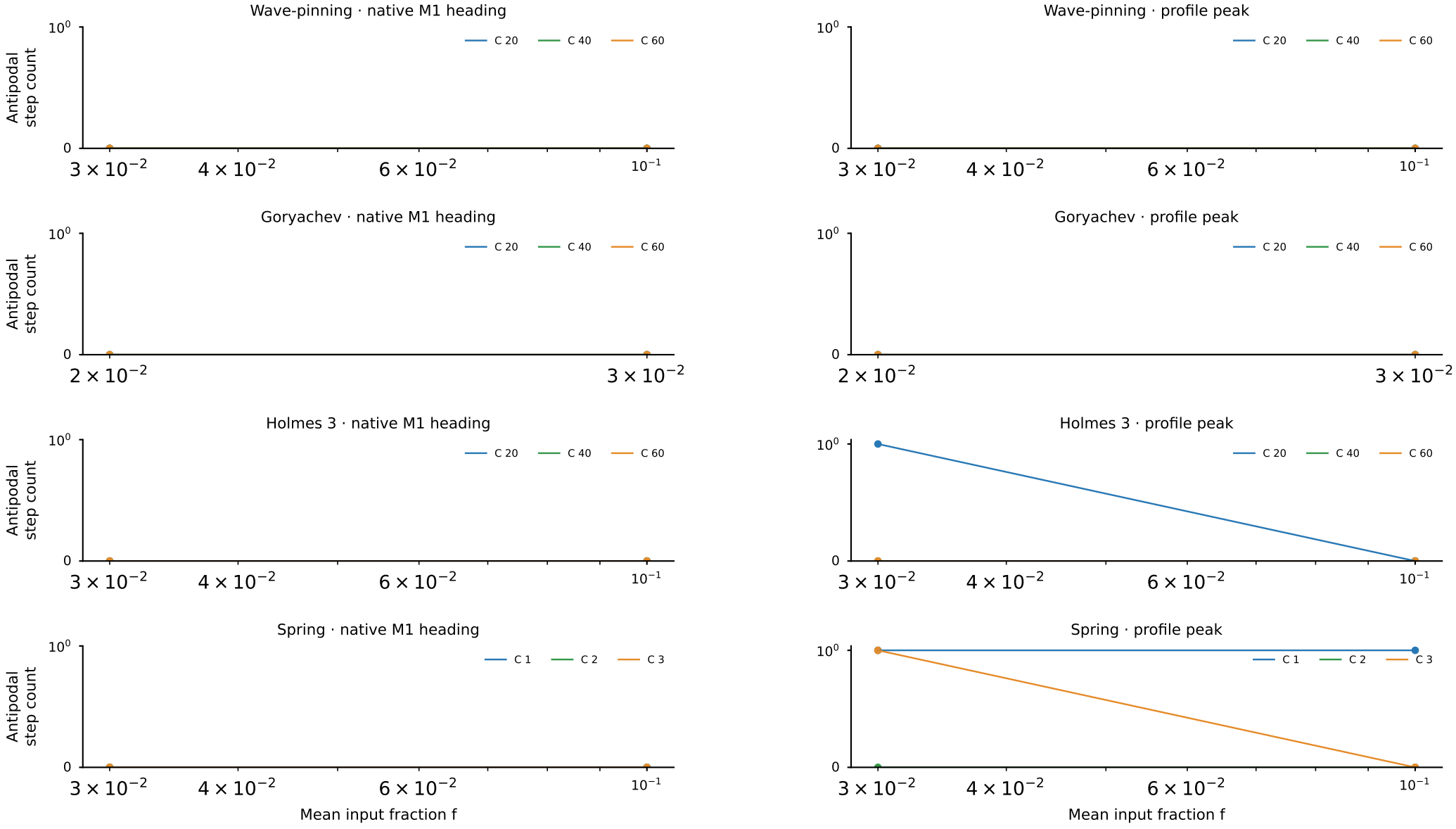
E1 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_047_S5_ambiguity_sigma2_analysis_first_half.csv

S5 · E1 literal material-origin physical-site measurements · Native antipodal ambiguity · analysis second half · σ=2



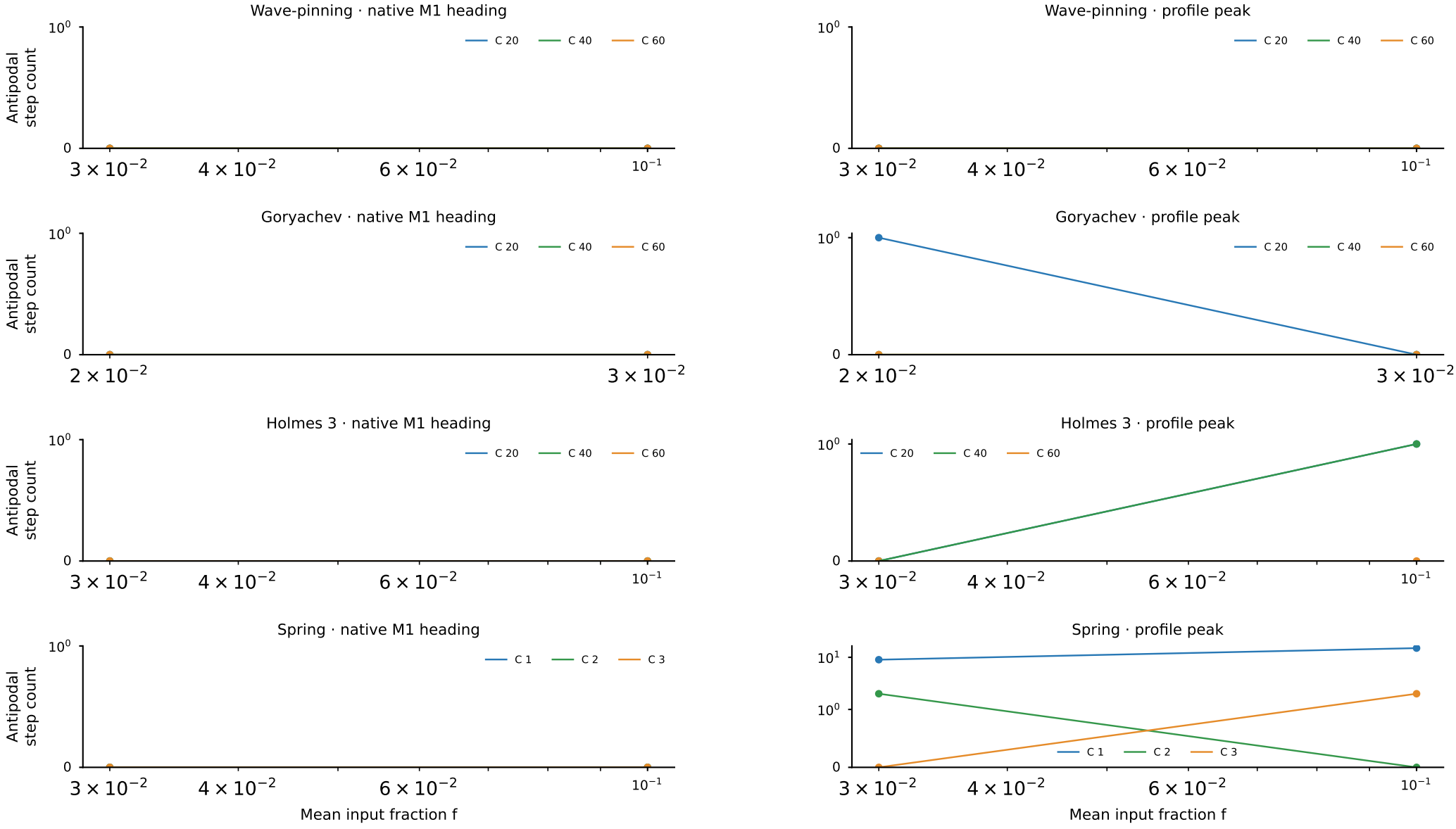
E1 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_048_S5_ambiguity_sigma2_analysis_second_half.csv

S5 · E1 literal material-origin physical-site measurements · Native antipodal ambiguity · late quarter · σ=2



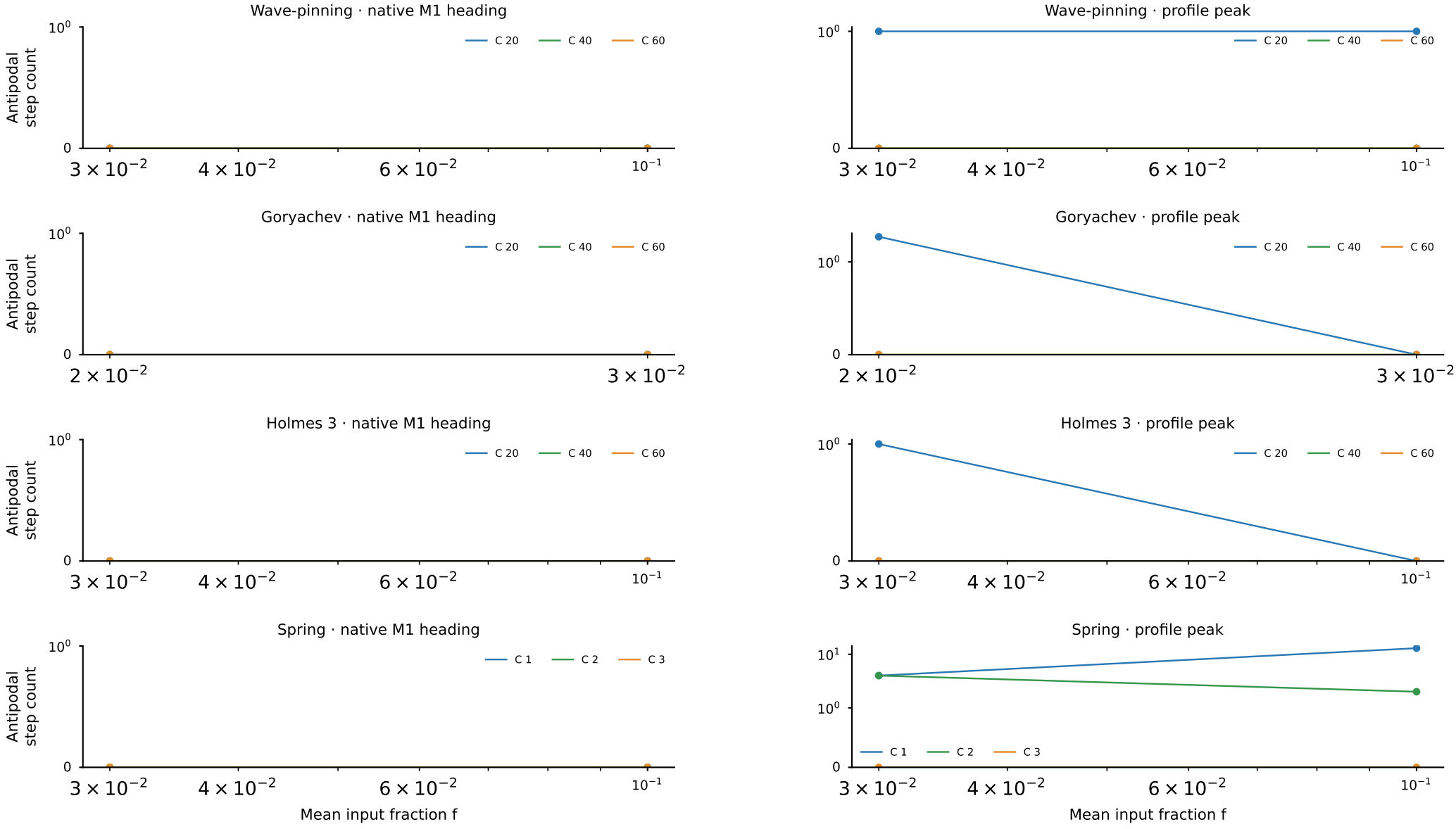
E1 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_049_S5_ambiguity_sigma2_late_quarter.csv

S5 · E1 literal material-origin physical-site measurements · Native antipodal ambiguity · cadence x2 · σ=2



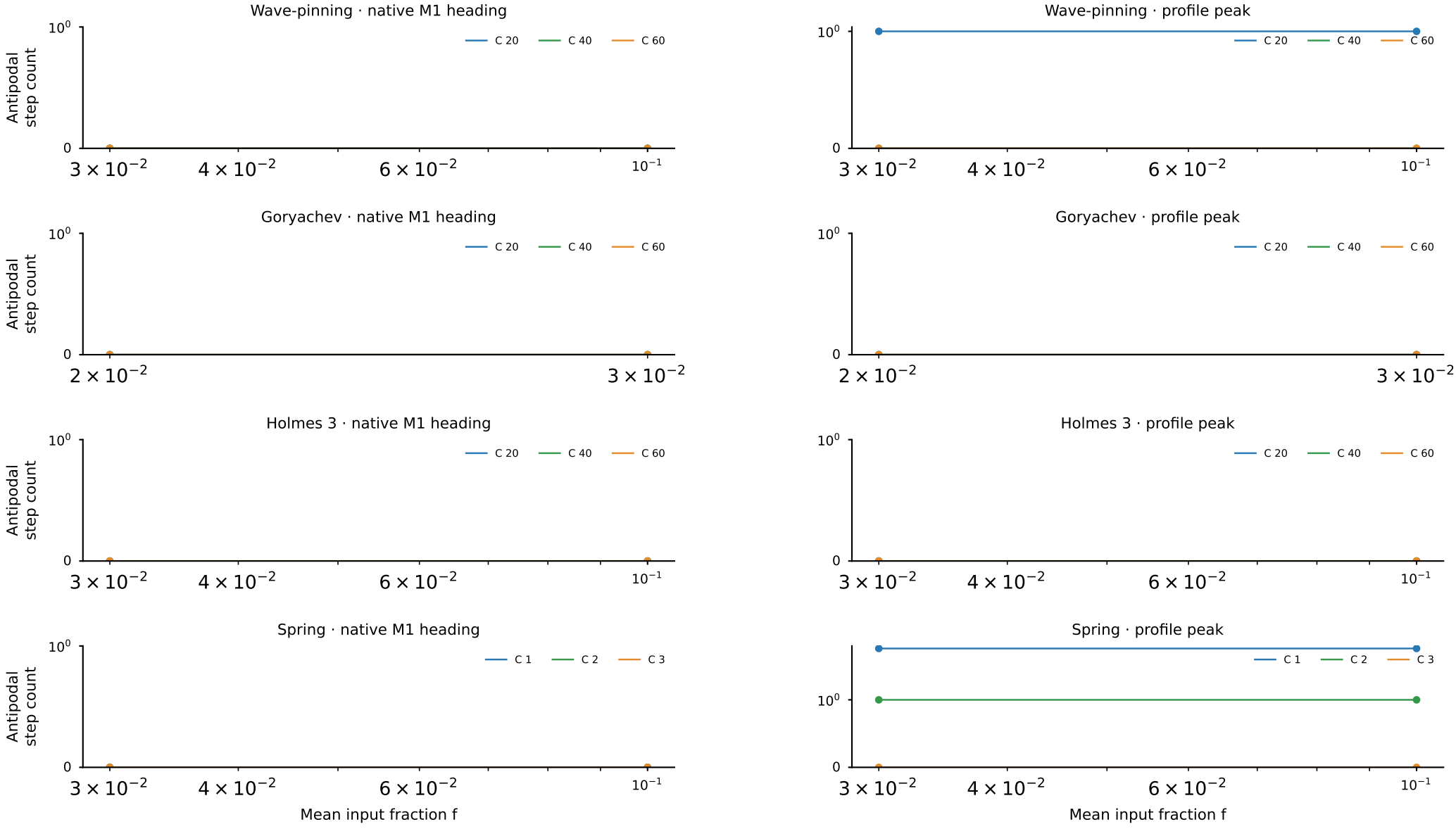
E1 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_050_S5_ambiguity_sigma2_cadence_x2.csv

S5 · E1 literal material-origin physical-site measurements · Native antipodal ambiguity · cadence x4 · σ=2

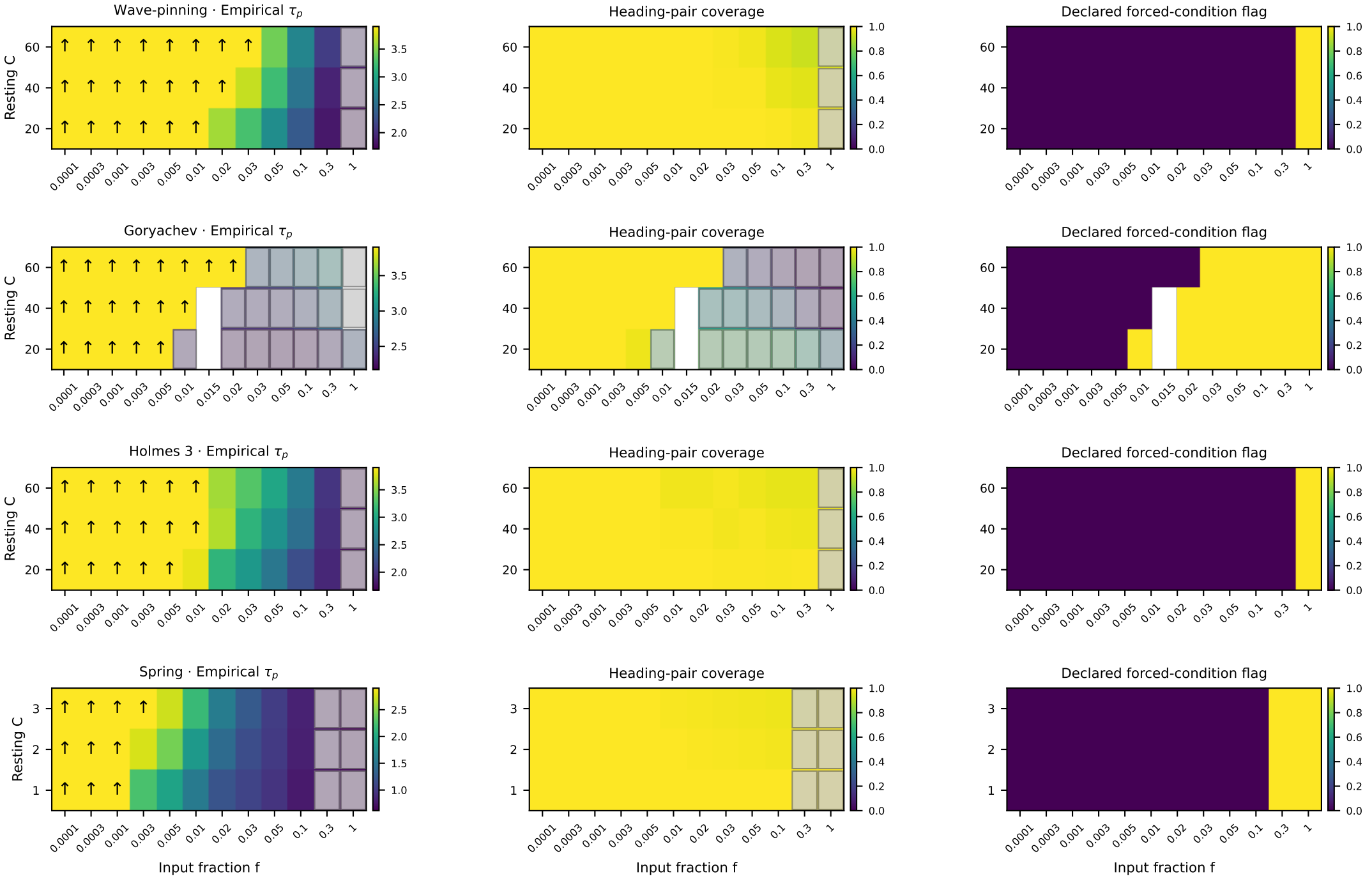


E1 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_051_S5_ambiguity_sigma2_cadence_x4.csv

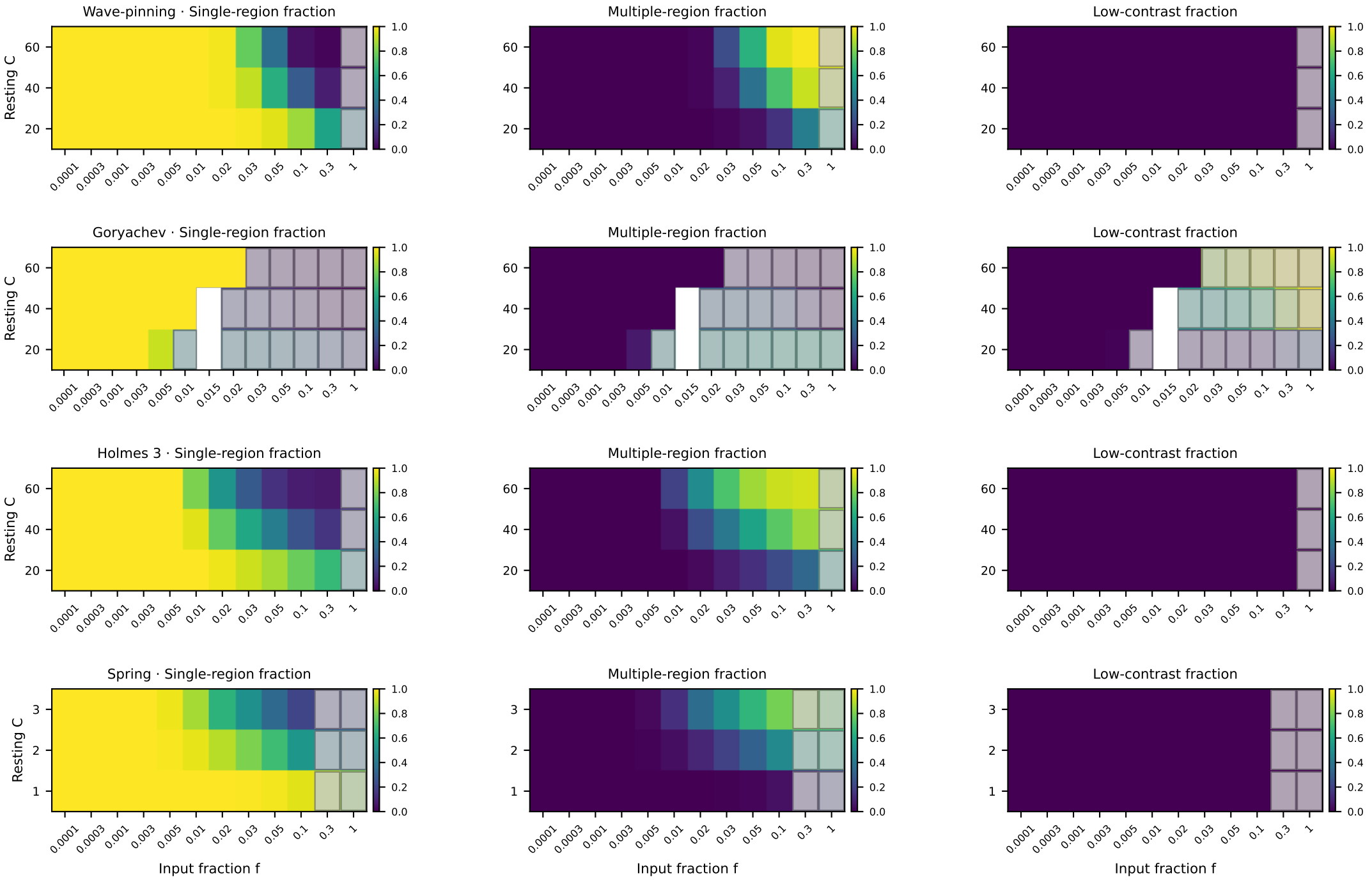
S5 · E1 literal material-origin physical-site measurements · Native antipodal ambiguity · cadence x8 · σ=2



E1 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E1_final_layout_20261020_052_S5_ambiguity_sigma2_cadence_x8.csv

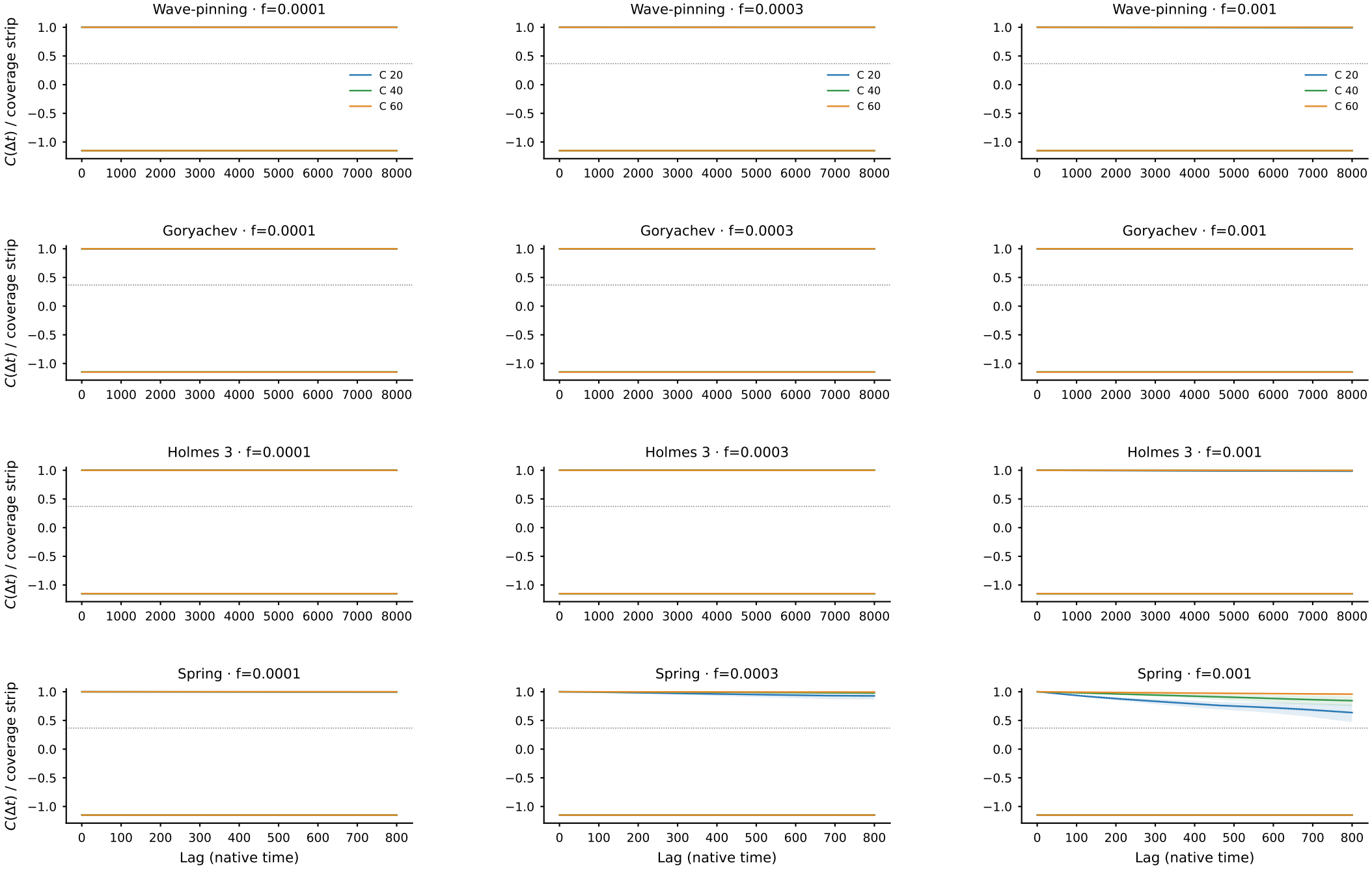


E2 completed native observations. All declared input strengths in this observation arm are shown. Grey overlays retain forced conditions; the exact display flag and its components remain in CSV. Arrows preserve censoring and cadence limits. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S8_dense_native_completed_20261020_001_S2_1.csv

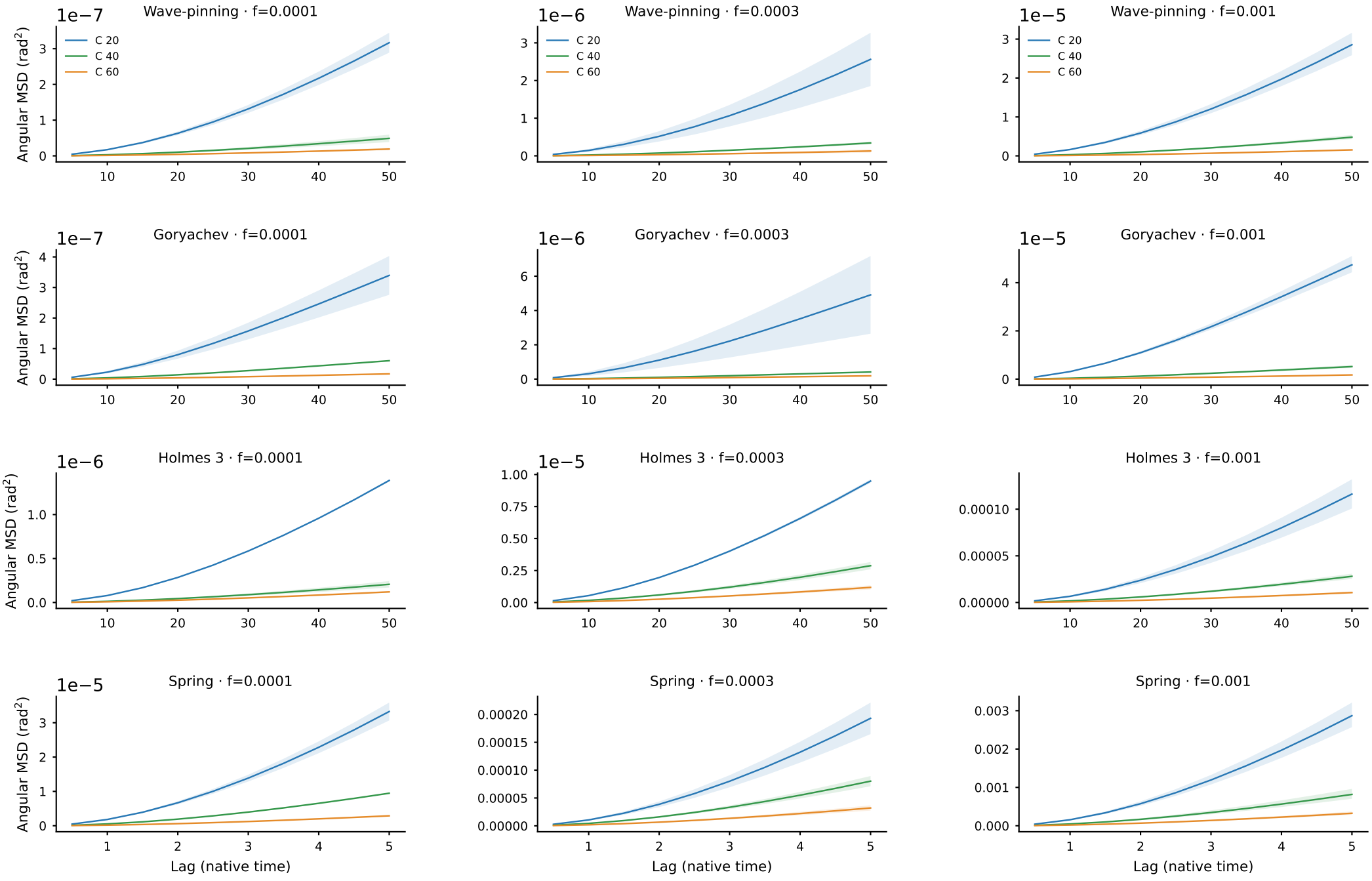


E2 completed native observations. All declared input strengths in this observation arm are shown. Grey overlays retain forced conditions; the exact display flag and its components remain in CSV. Arrows preserve censoring and cadence limits. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S8_dense_native_completed_20261020_002_S2_2.csv

S8 · E2 completed native observations · All heading correlations

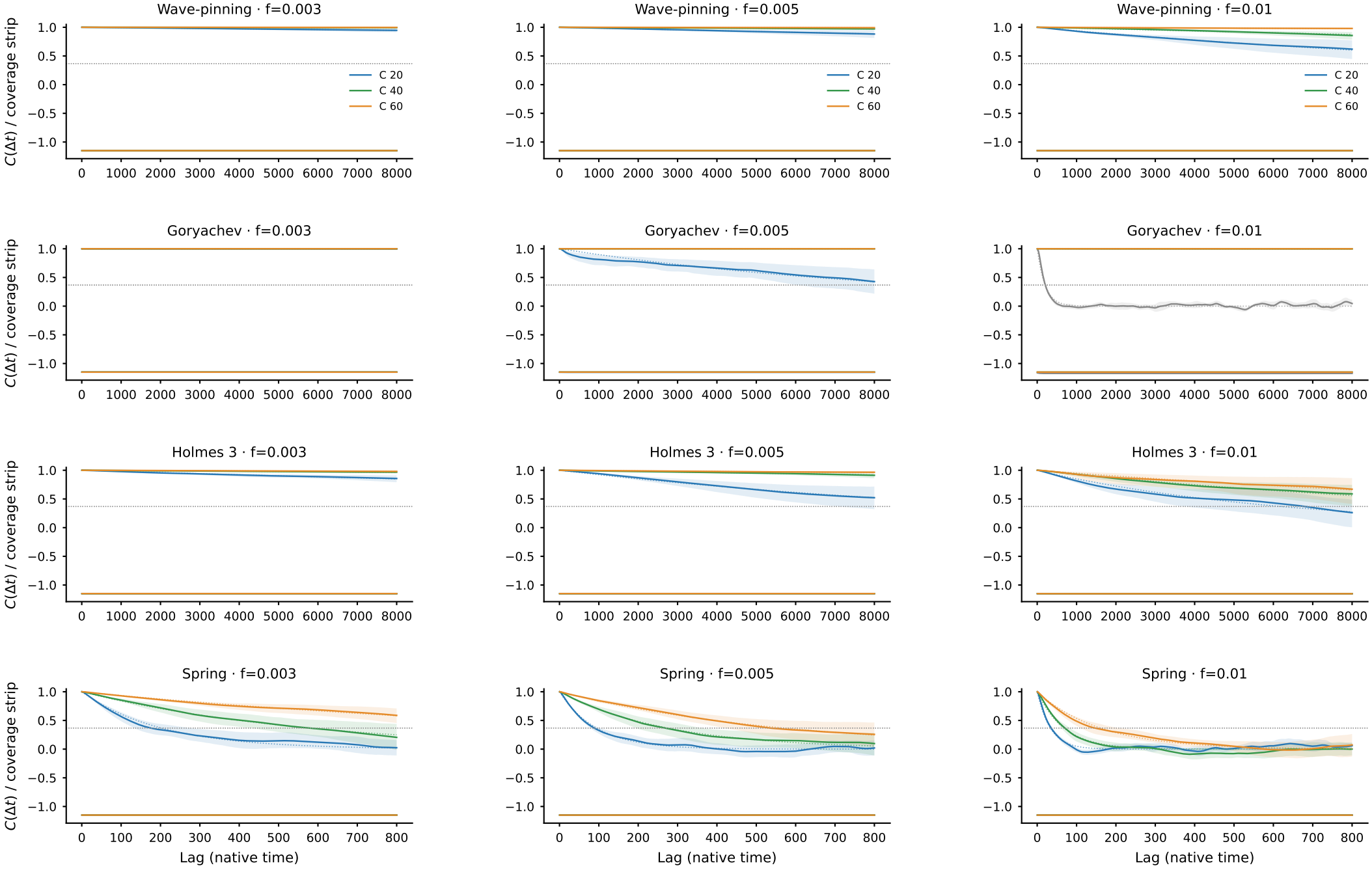


E2 completed native observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S8_dense_native_completed_20261020_003_S3_1_cosine.csv.gz



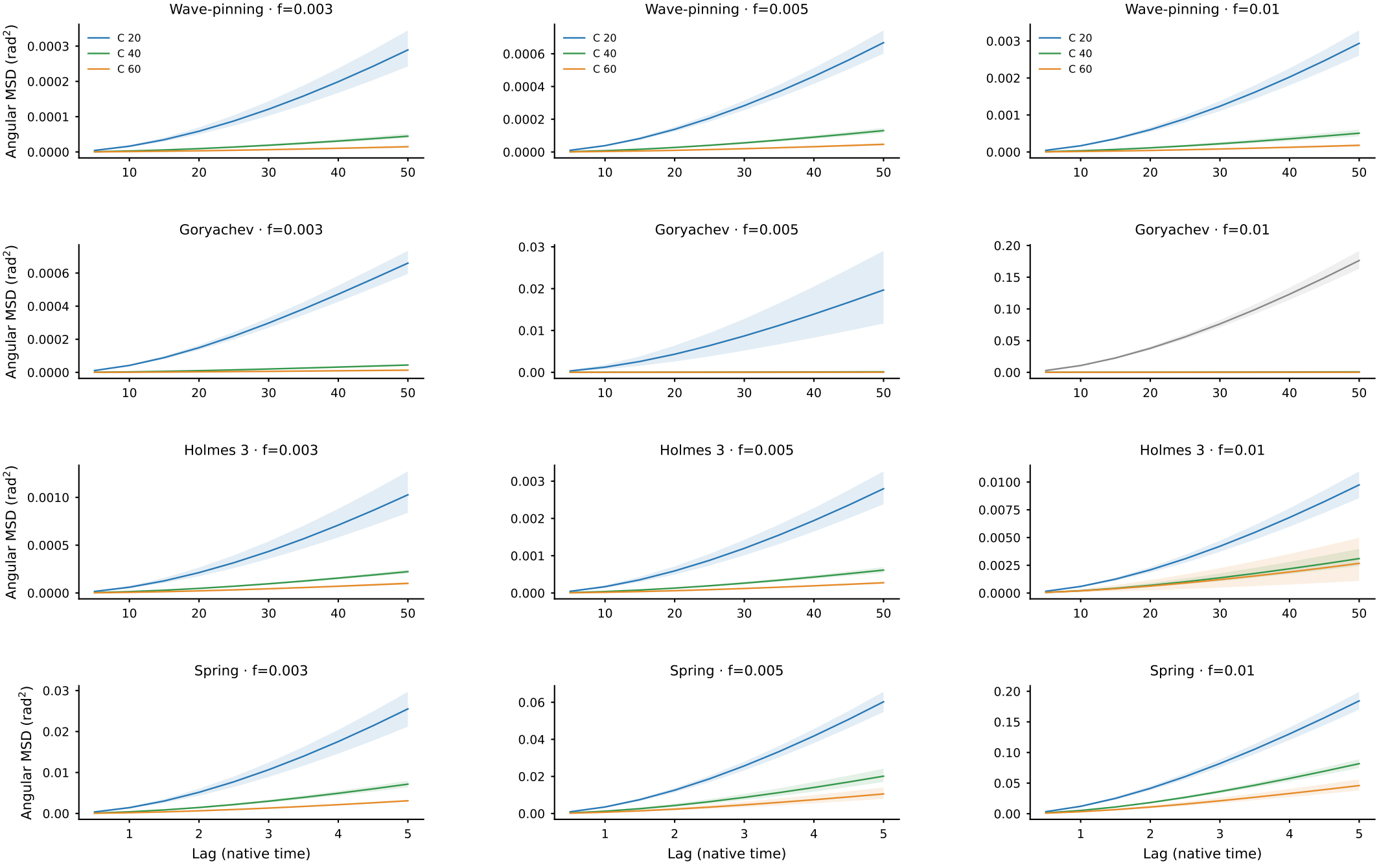
E2 completed native observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S8_dense_native_completed_20261020_004_S3_1_angular_msd.csv.gz

S8 · E2 completed native observations · All heading correlations



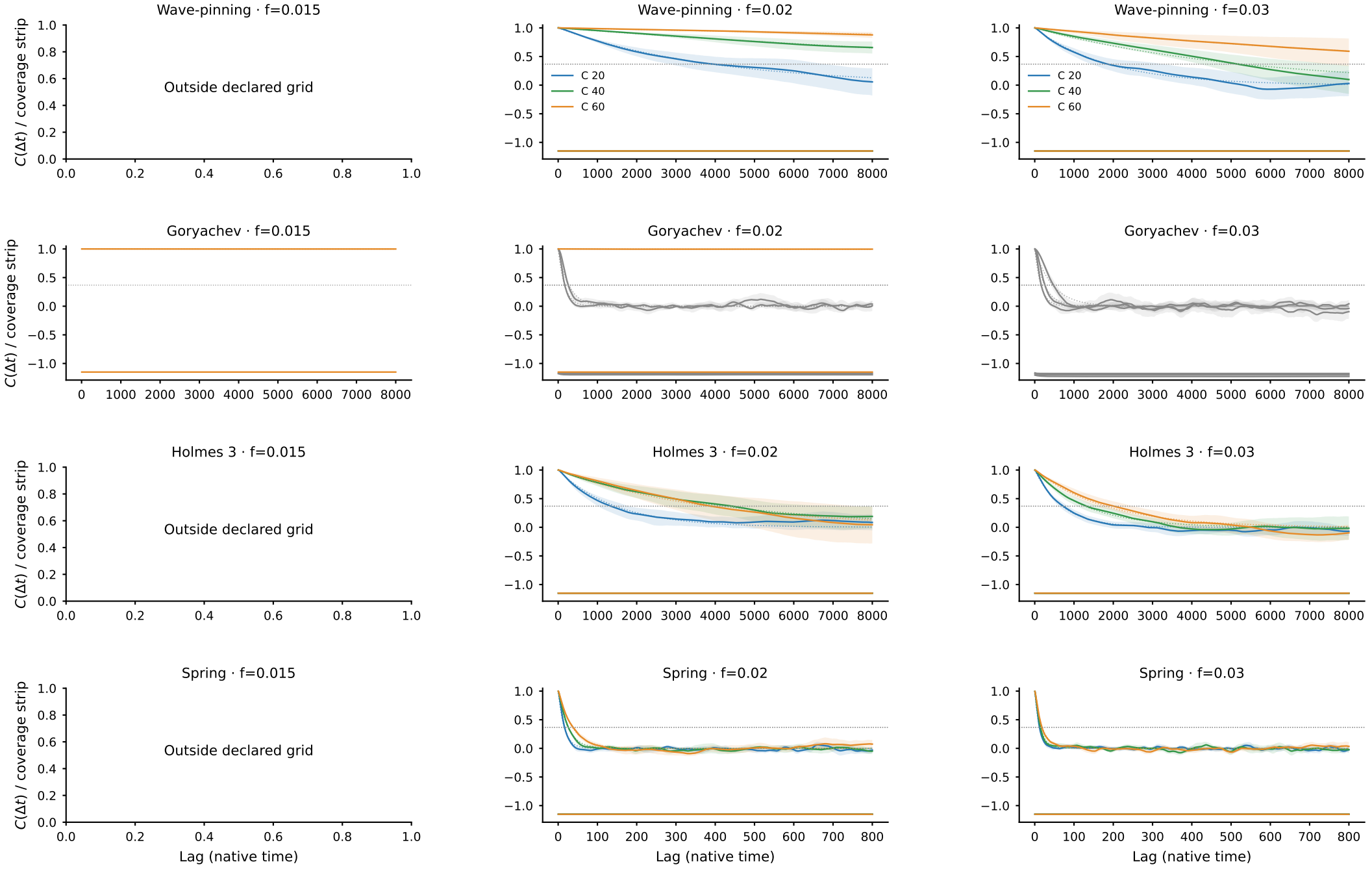
E2 completed native observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S8_dense_native_completed_20261020_005_S3_2_cosine.csv.gz

S8 · E2 completed native observations · Short-lag angular MSD

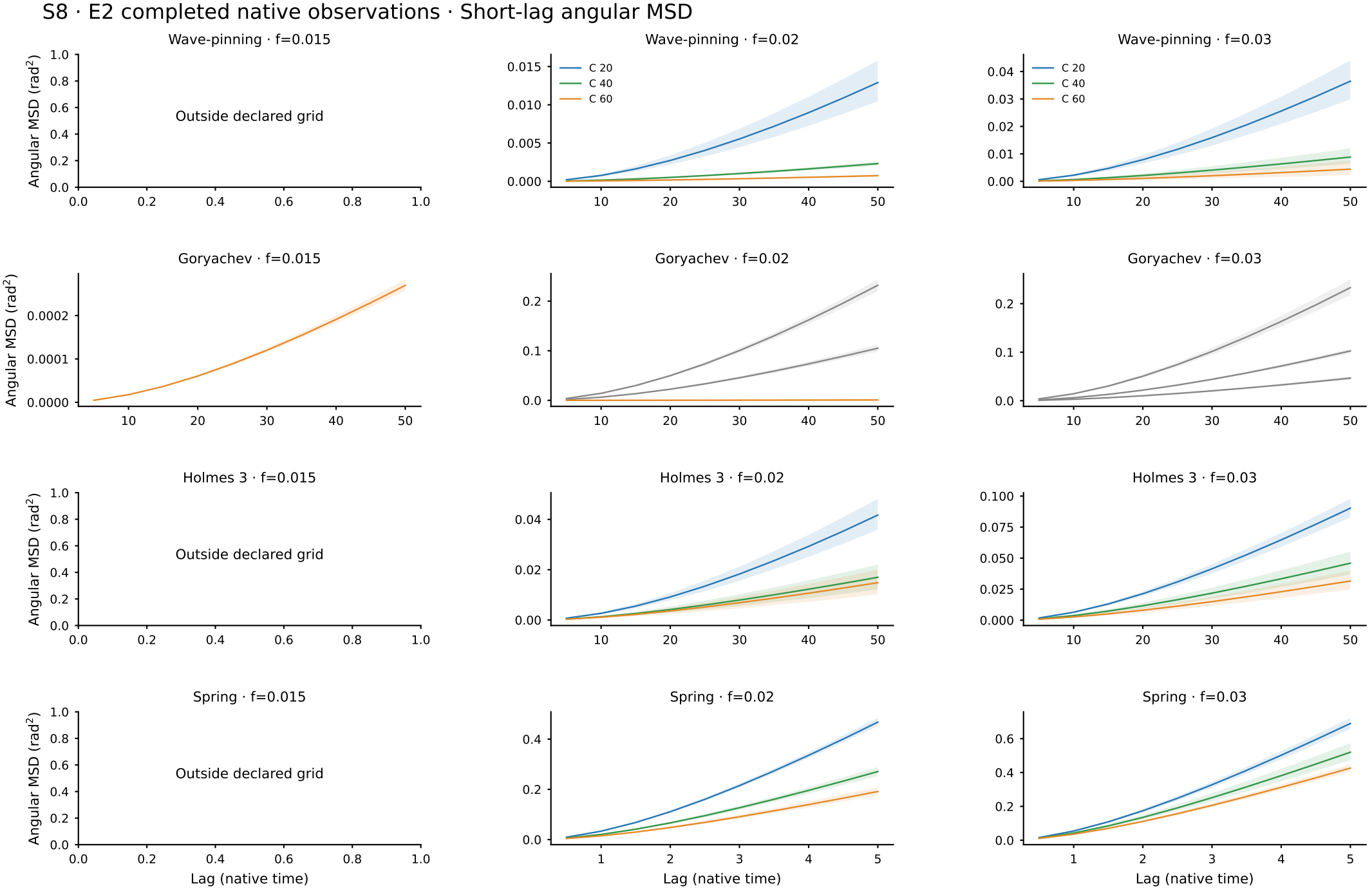


E2 completed native observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S8_dense_native_completed_20261020_006_S3_2_angular_msd.csv.gz

S8 · E2 completed native observations · All heading correlations

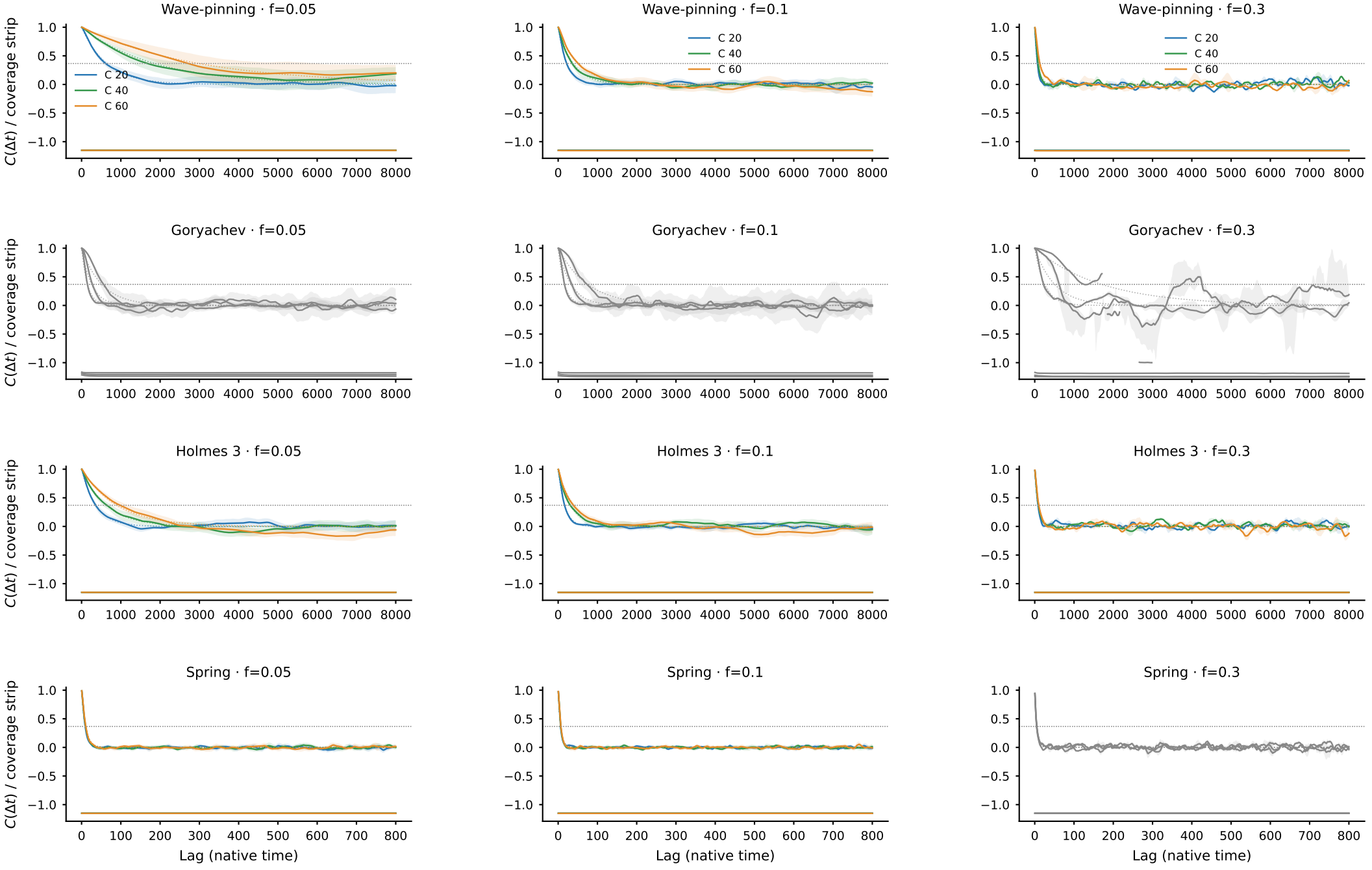


E2 completed native observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S8_dense_native_completed_20261020_007_S3_3_cosine.csv



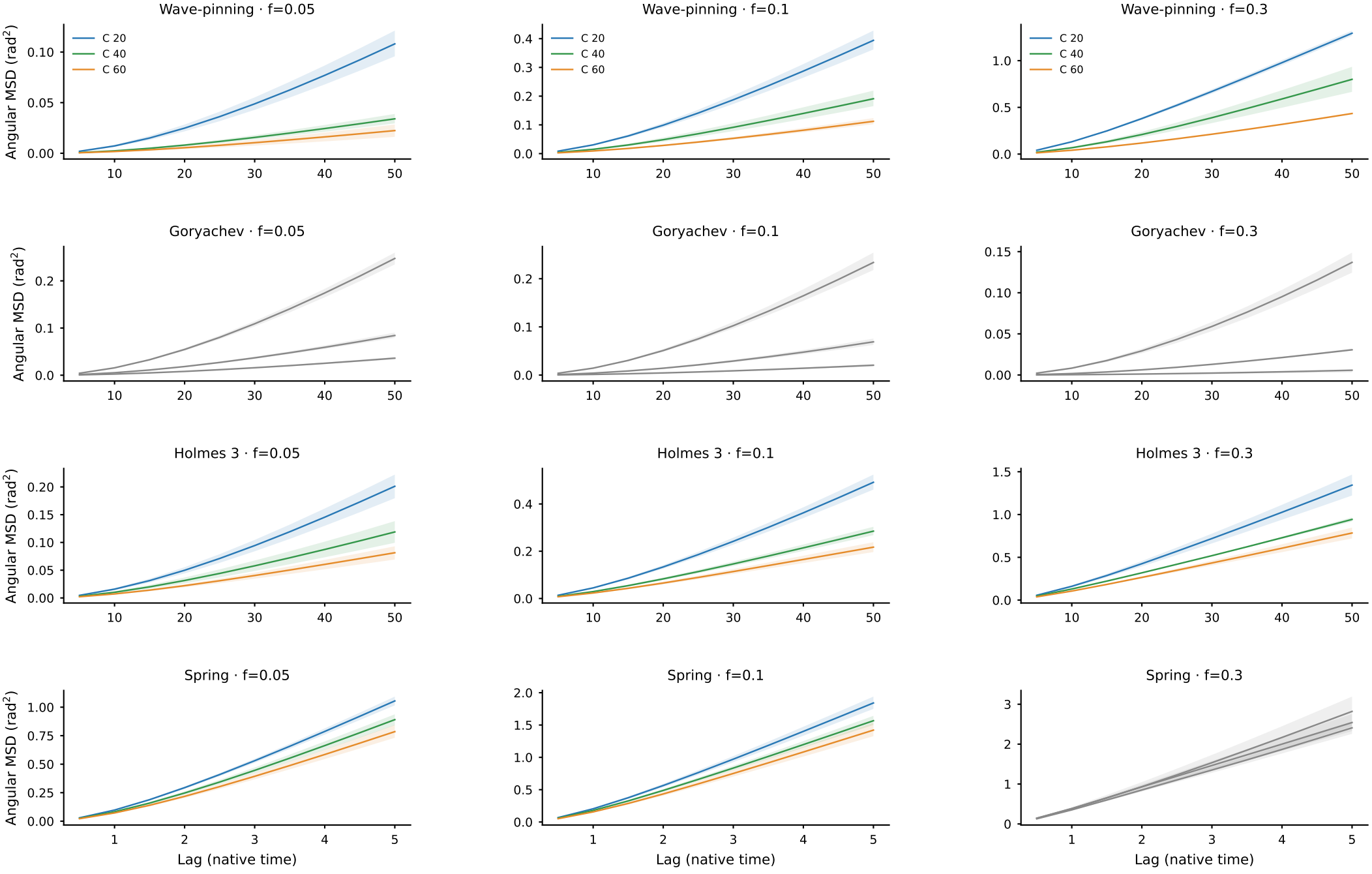
E2 completed native observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S8_dense_native_completed_20261020_008_S3_3_angular_msd.csv

S8 · E2 completed native observations · All heading correlations



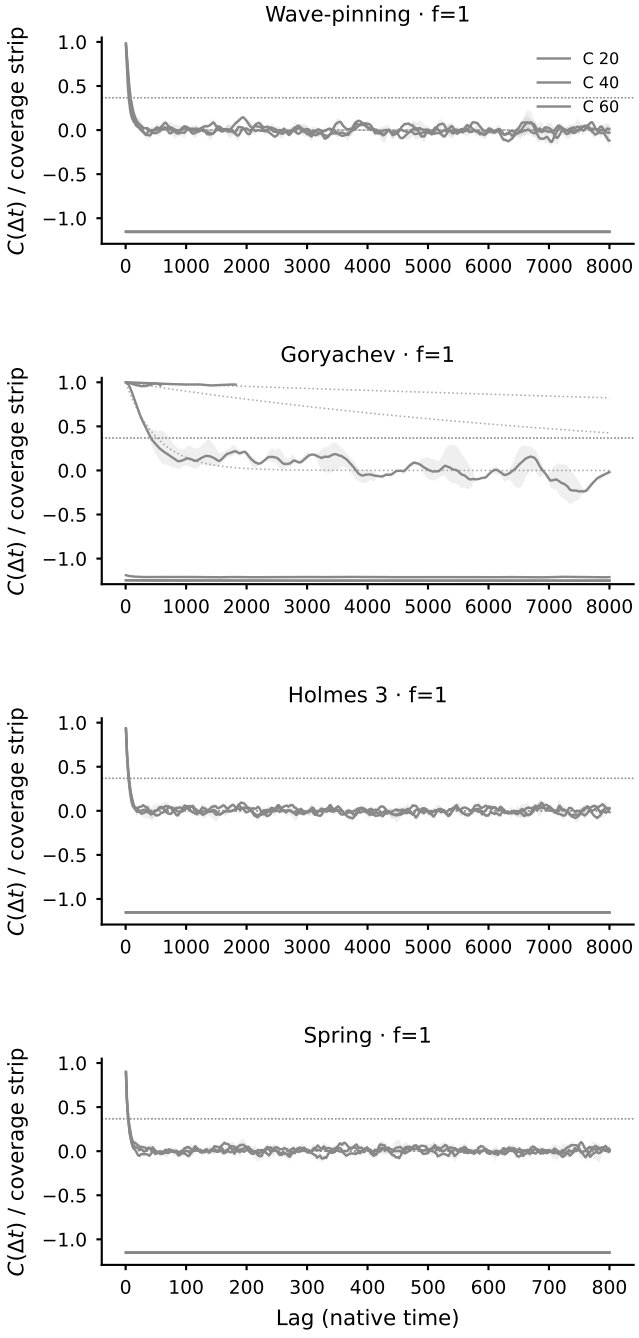
E2 completed native observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S8_dense_native_completed_20261020_009_S3_4_cosine.csv.gz

S8 · E2 completed native observations · Short-lag angular MSD



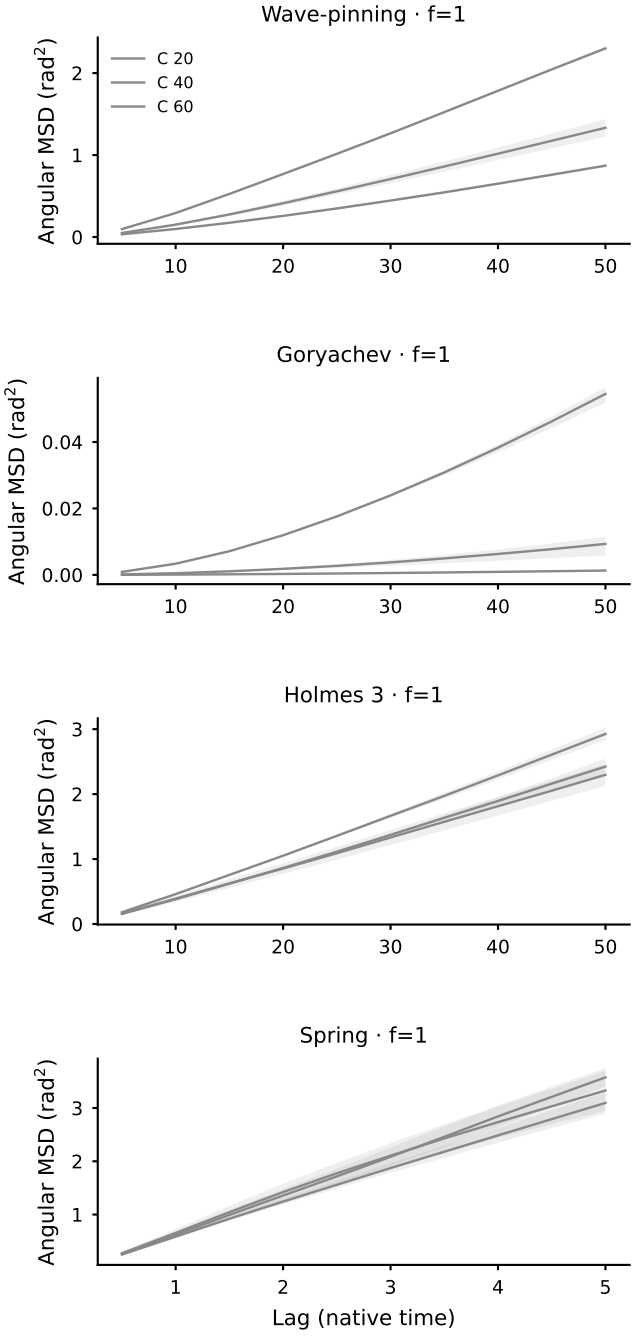
E2 completed native observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S8_dense_native_completed_20261020_010_S3_4_angular_msd.csv.gz

S8 · E2 completed native observations · All heading correlations

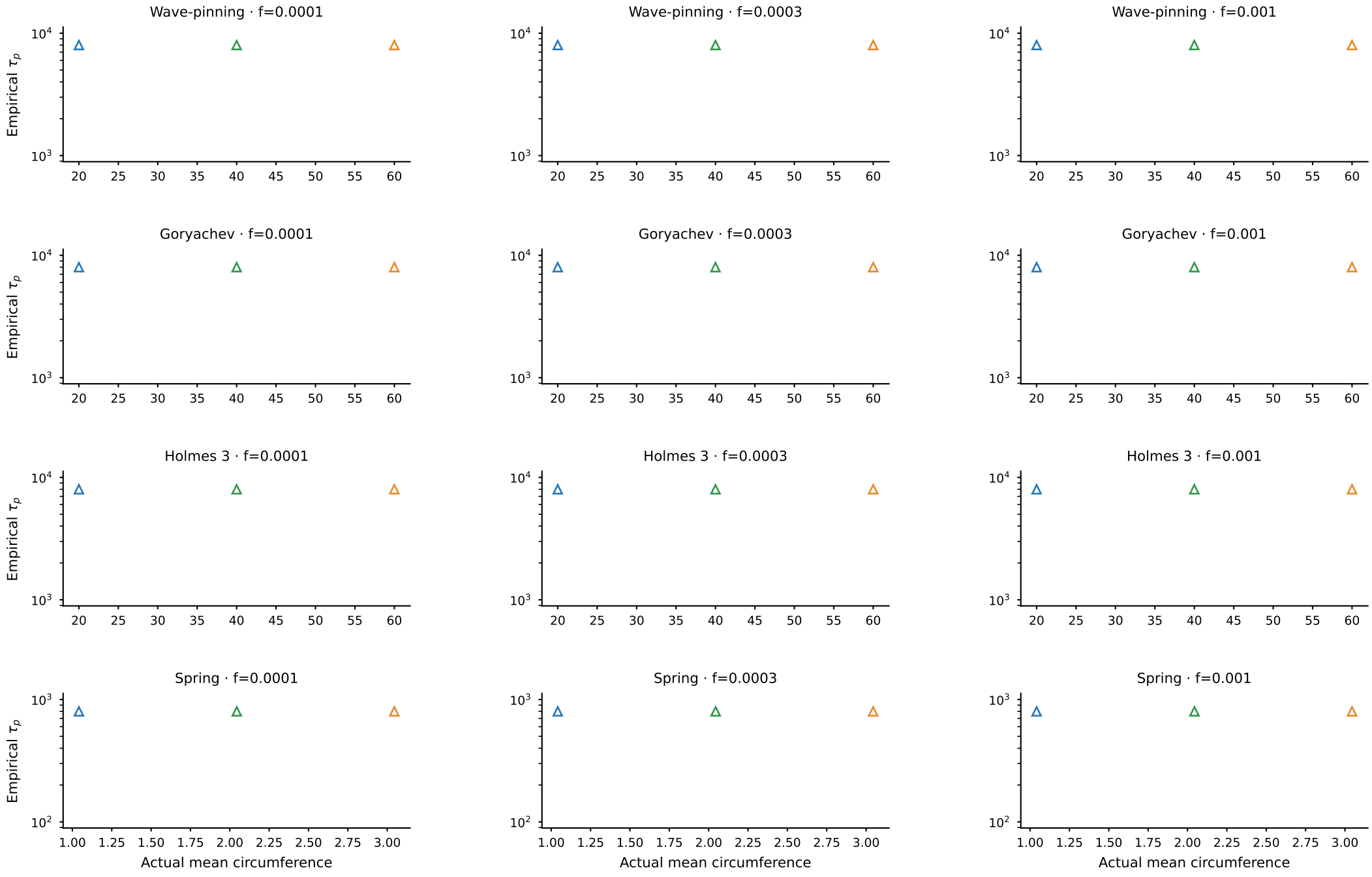


E2 completed native observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S8_dense_native_completed_20261020_011_S3_5_cosine.csv

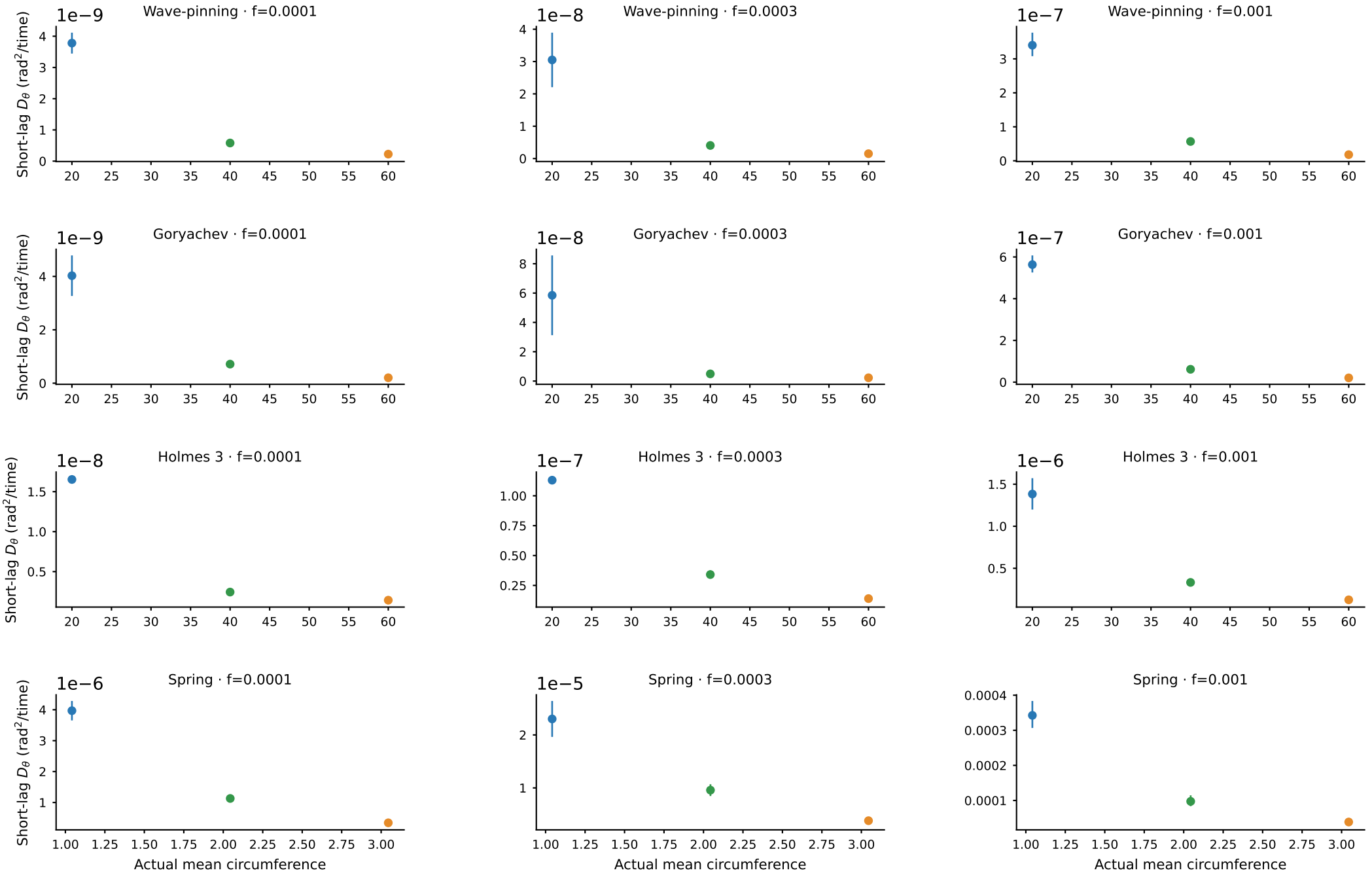
S8 · E2 completed native observations · Short-lag angular MSD



E2 completed native observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S8_dense_native_completed_20261020_012_S3_5_angular_msd.csv

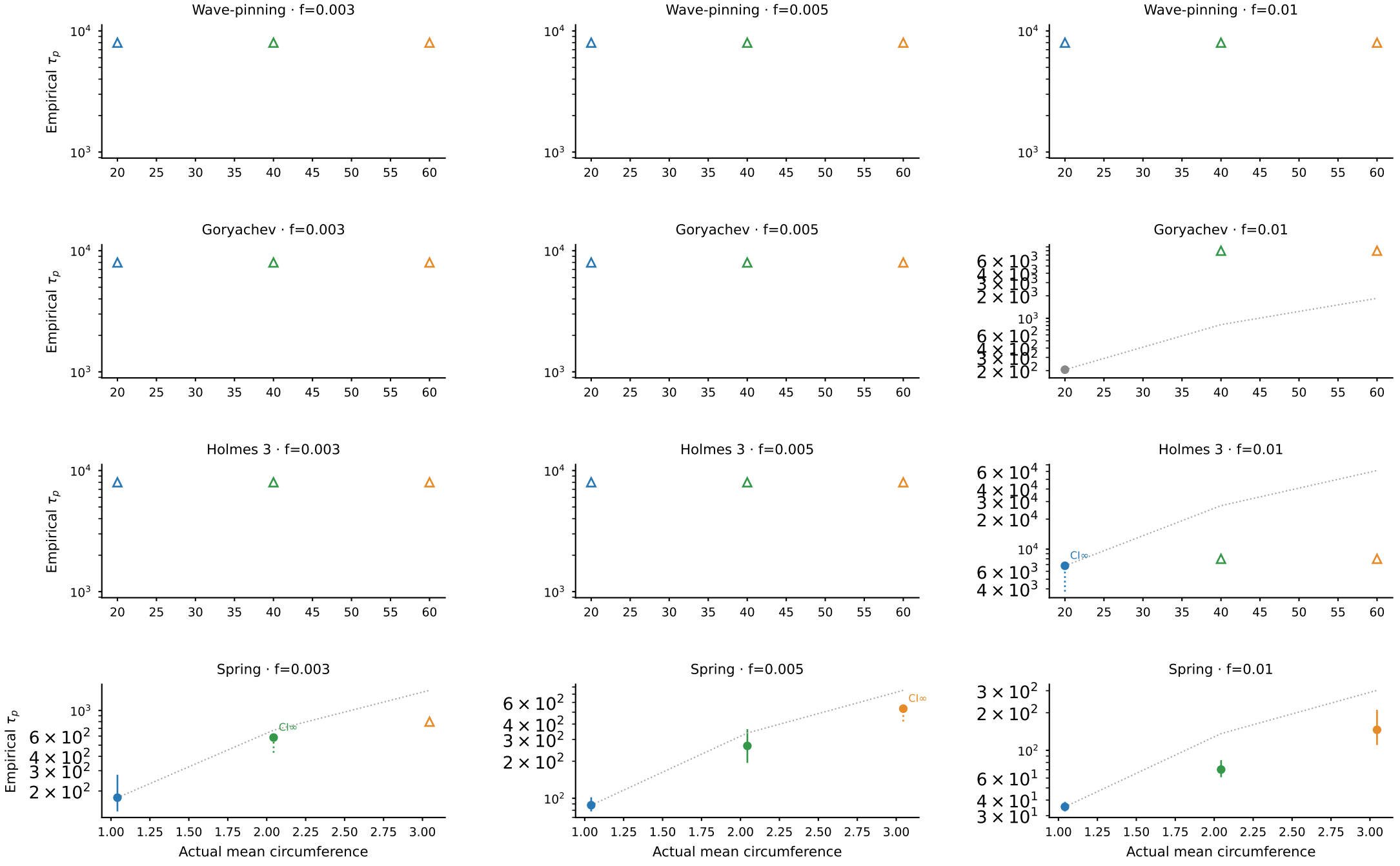


E2 completed native observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S8_dense_native_completed_20261020_013_S4_1_tau_p.csv



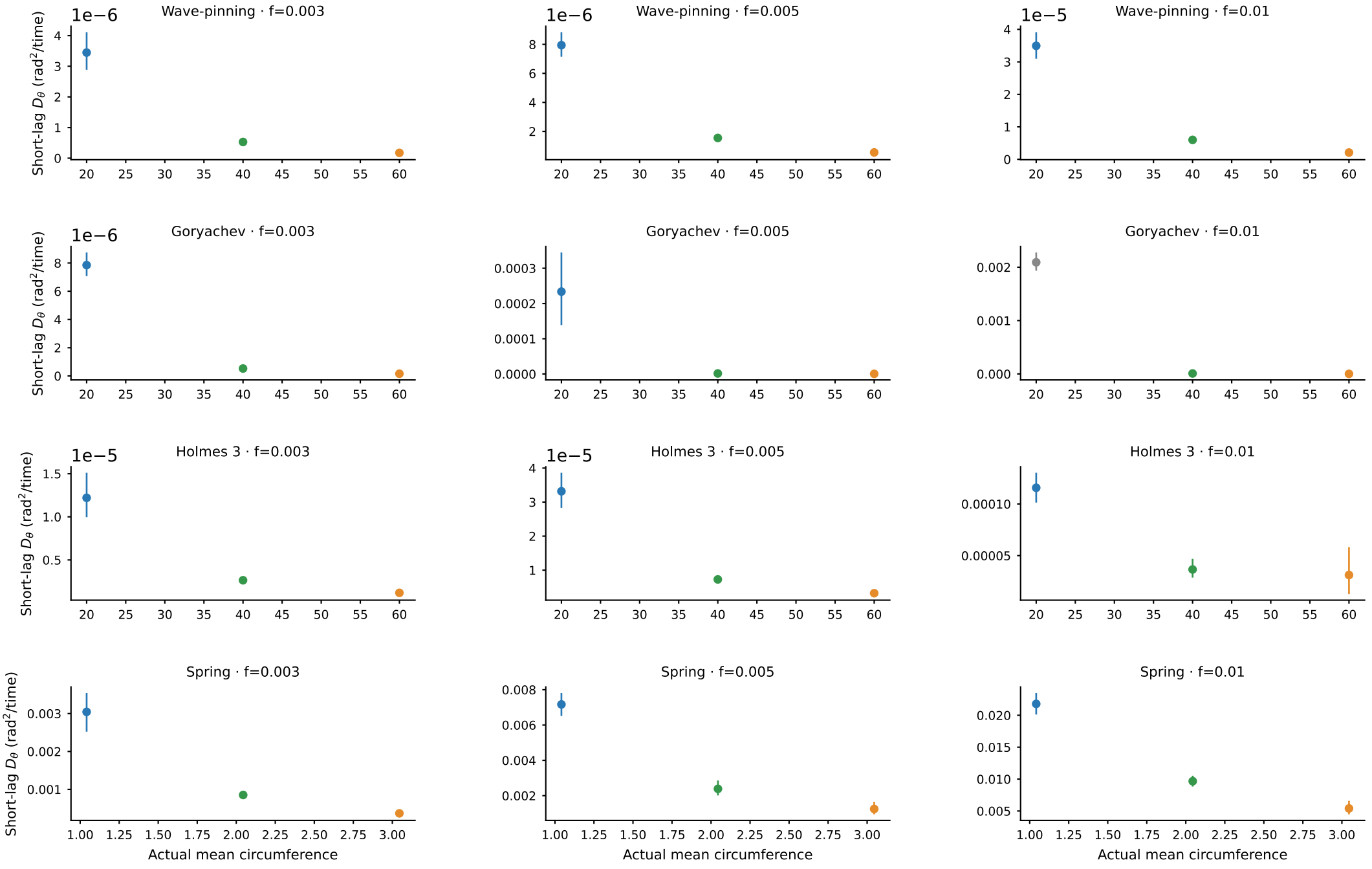
E2 completed native observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S8_dense_native_completed_20261020_014_S4_1_Dtheta.csv

S8 · E2 completed native observations · Persistence and angular mobility across size



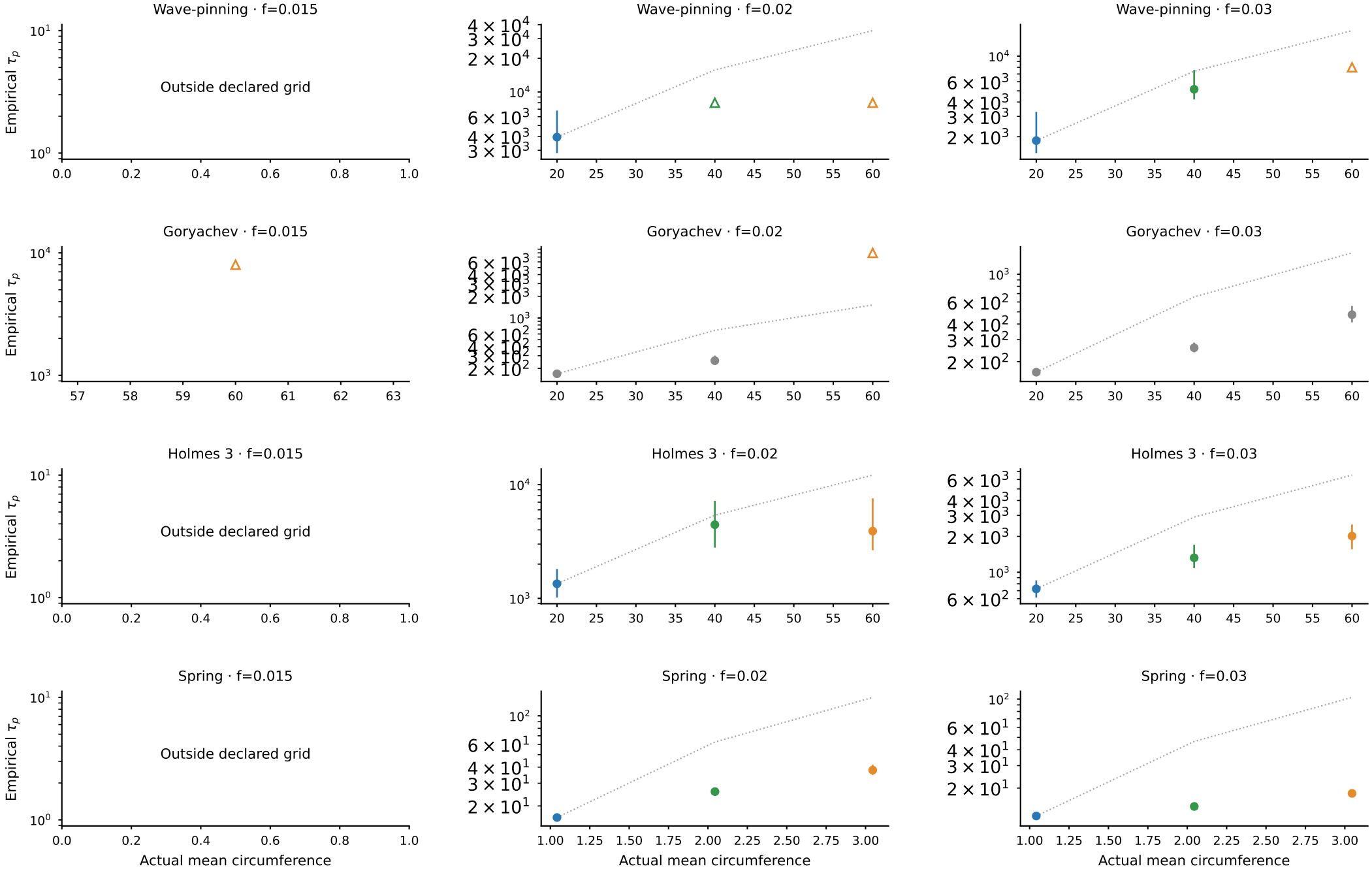
E2 completed native observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV. CI∞ next to a finite point marks an upper-unbounded bootstrap CI; finite lower endpoints are retained. CI[0,∞) spans the full nonnegative range.
CSV: figure4_S8_dense_native_completed_20261020_015_S4_2_tau_p.csv

S8 · E2 completed native observations · Persistence and angular mobility across size



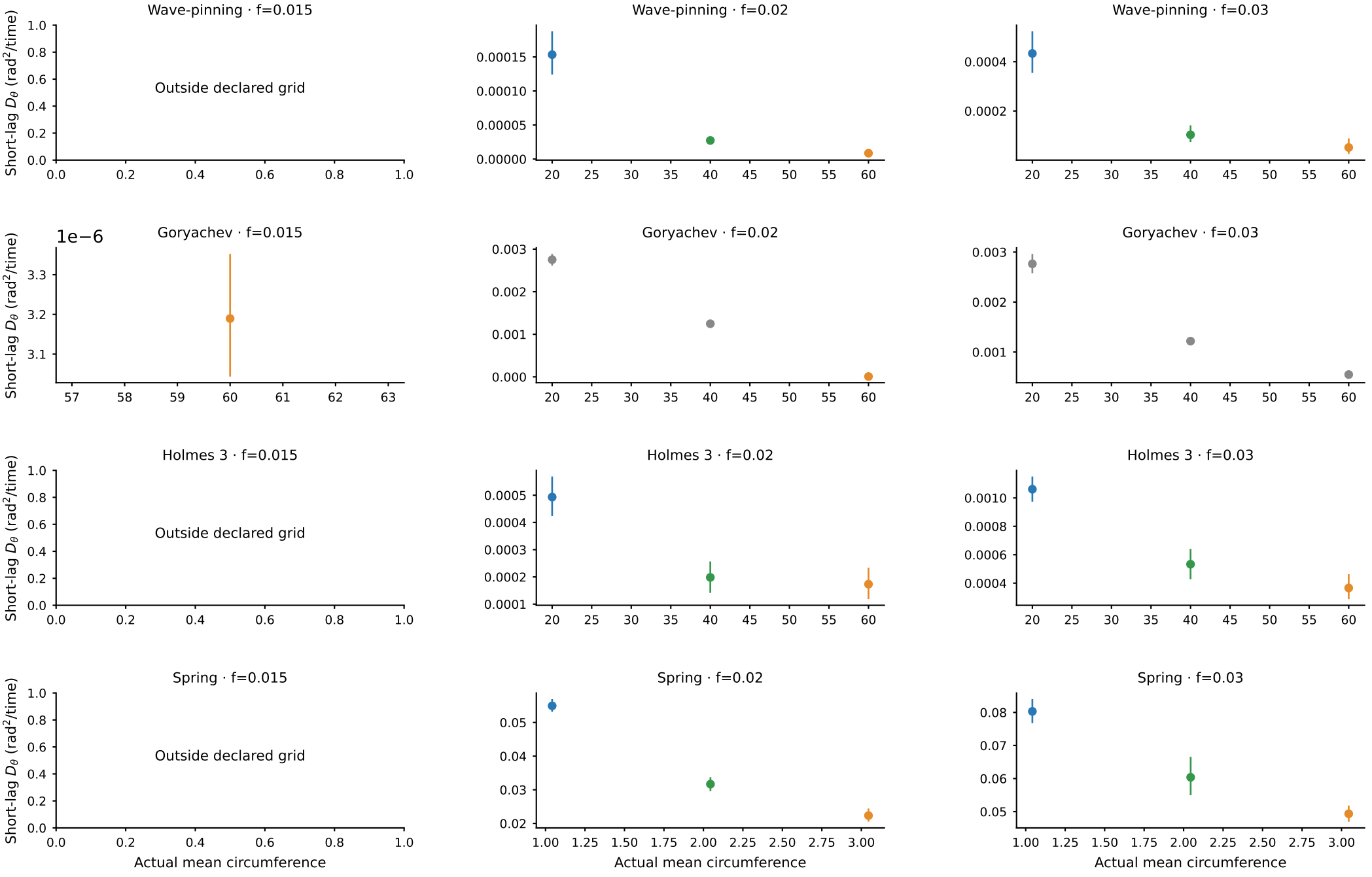
E2 completed native observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV. CI∞ next to a finite point marks an upper-unbounded bootstrap CI; finite lower endpoints are retained. CI[0,∞) spans the full nonnegative range.
CSV: figure4_S8_dense_native_completed_20261020_016_S4_2_Dtheta.csv

S8 · E2 completed native observations · Persistence and angular mobility across size



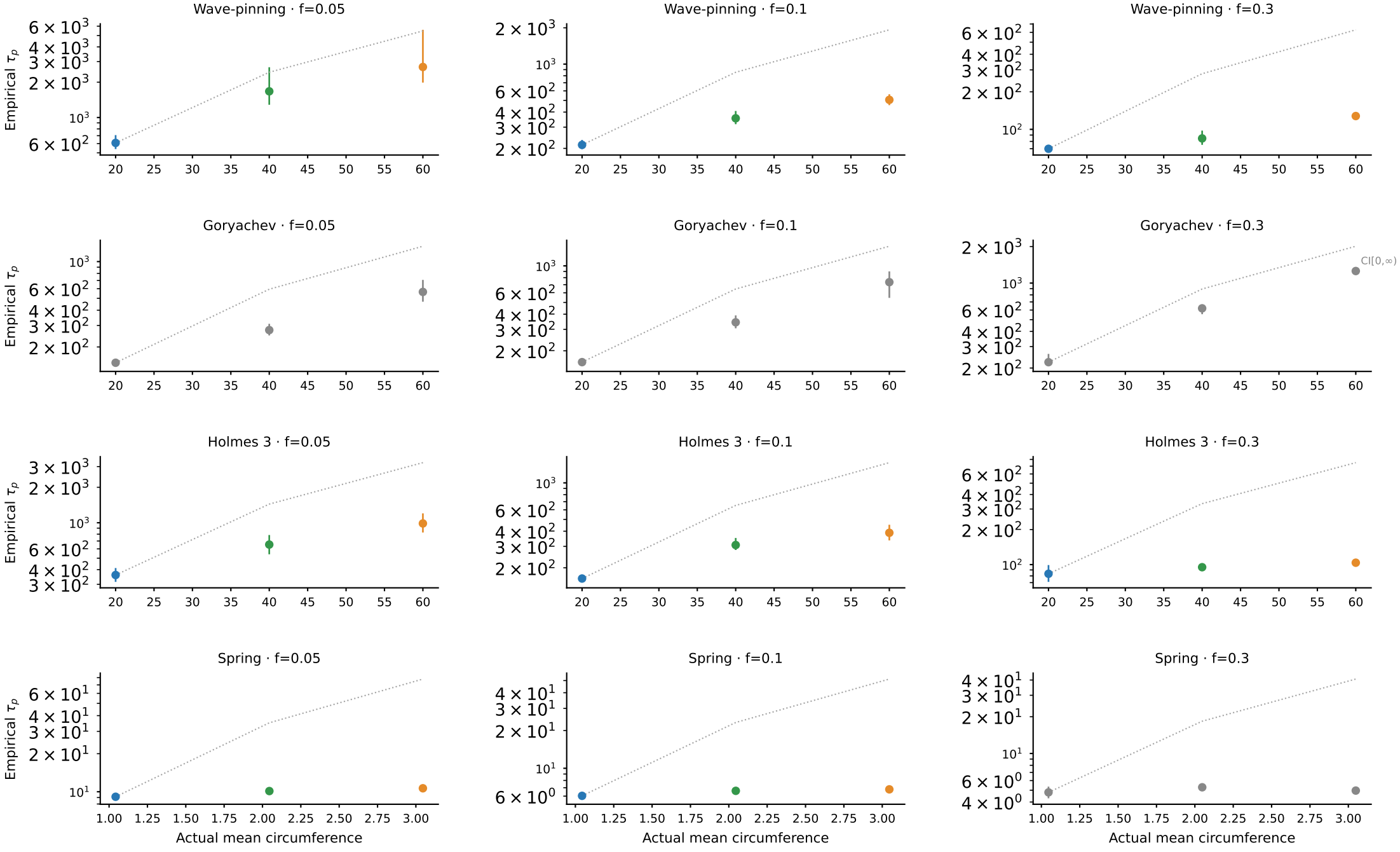
E2 completed native observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S8_dense_native_completed_20261020_017_S4_3_tau_p.csv

S8 · E2 completed native observations · Persistence and angular mobility across size



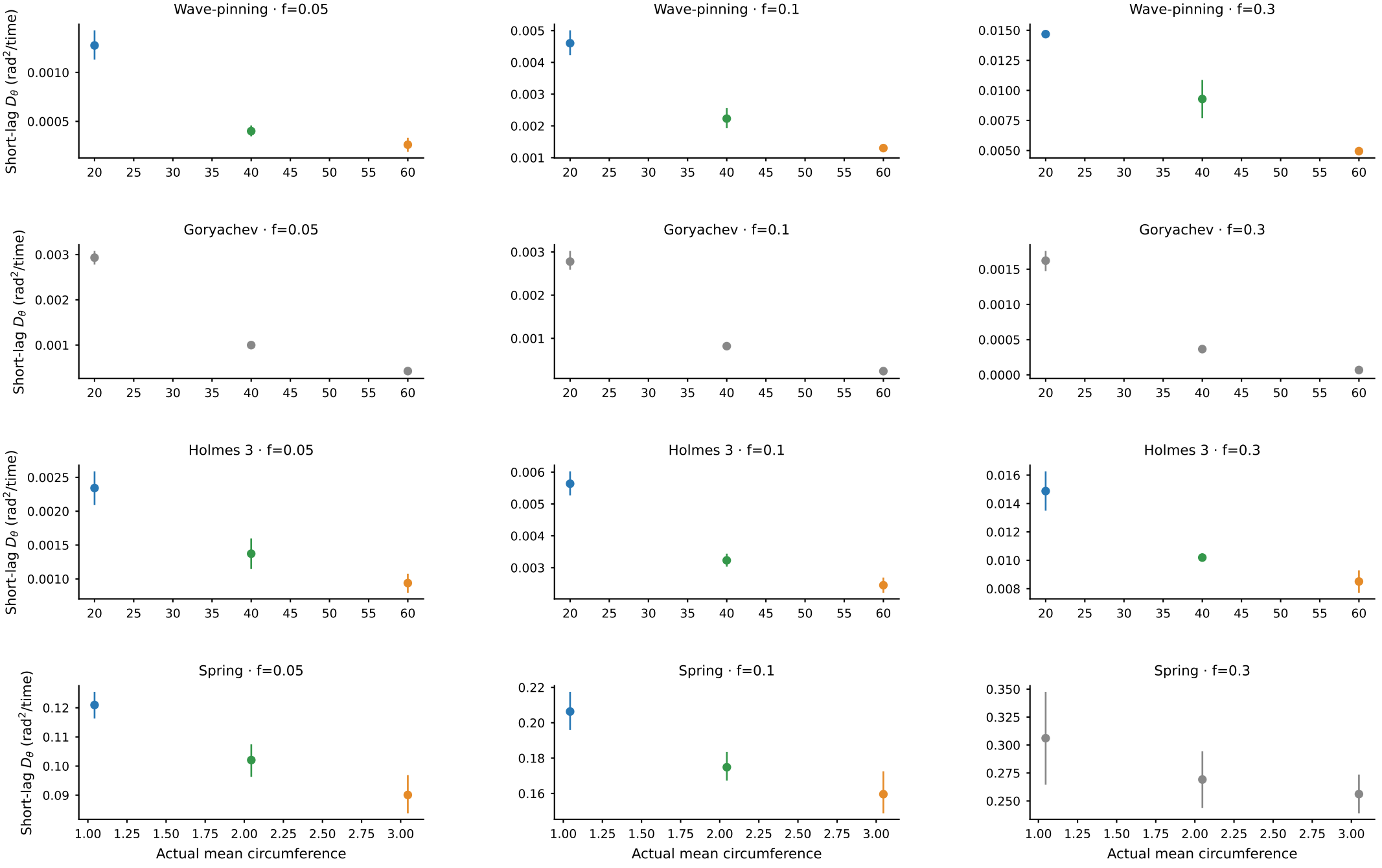
E2 completed native observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S8_dense_native_completed_20261020_018_S4_3_Dtheta.csv

S8 · E2 completed native observations · Persistence and angular mobility across size



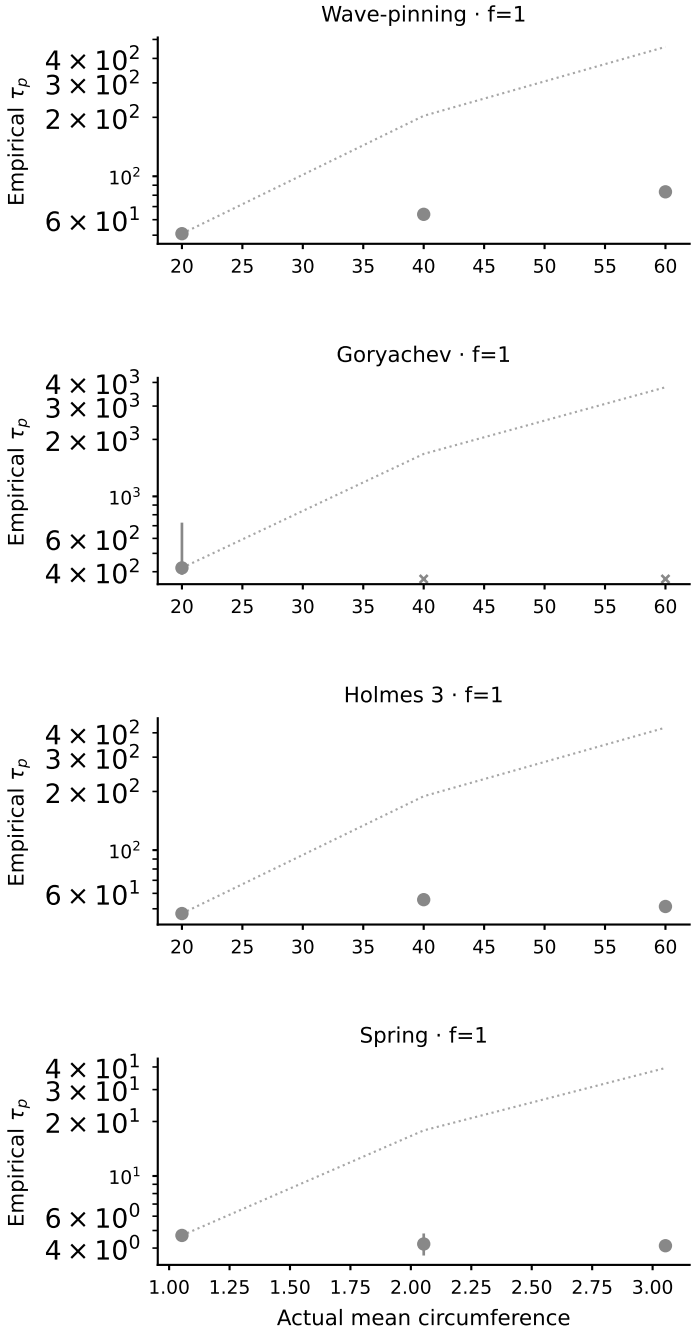
E2 completed native observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C^2 reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV. CI∞ next to a finite point marks an upper-unbounded bootstrap CI; finite lower endpoints are retained. CI[0,∞) spans the full nonnegative range.
CSV: figure4_S8_dense_native_completed_20261020_019_S4_4_tau_p.csv

S8 · E2 completed native observations · Persistence and angular mobility across size



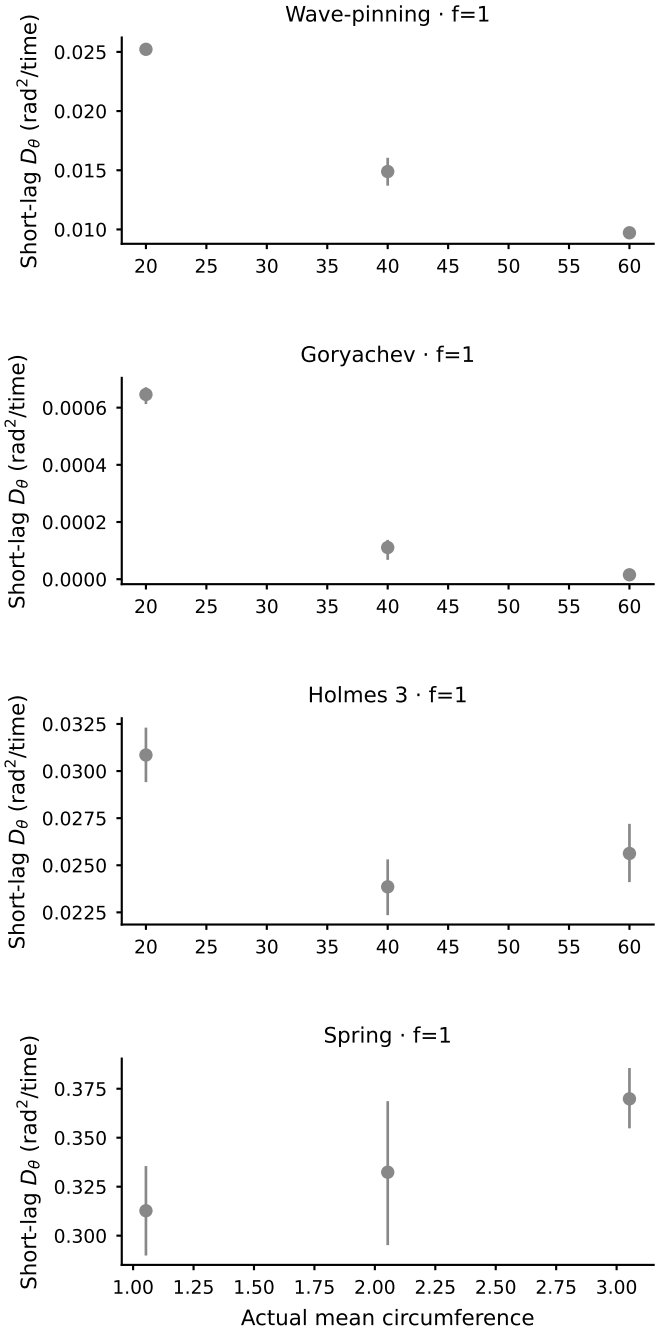
E2 completed native observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV. CI∞ next to a finite point marks an upper-unbounded bootstrap CI; finite lower endpoints are retained. CI[0,∞) spans the full nonnegative range.
CSV: figure4_S8_dense_native_completed_20261020_020_S4_4_Dtheta.csv

S8 · E2 completed native observations · Persistence and angular mobility across size



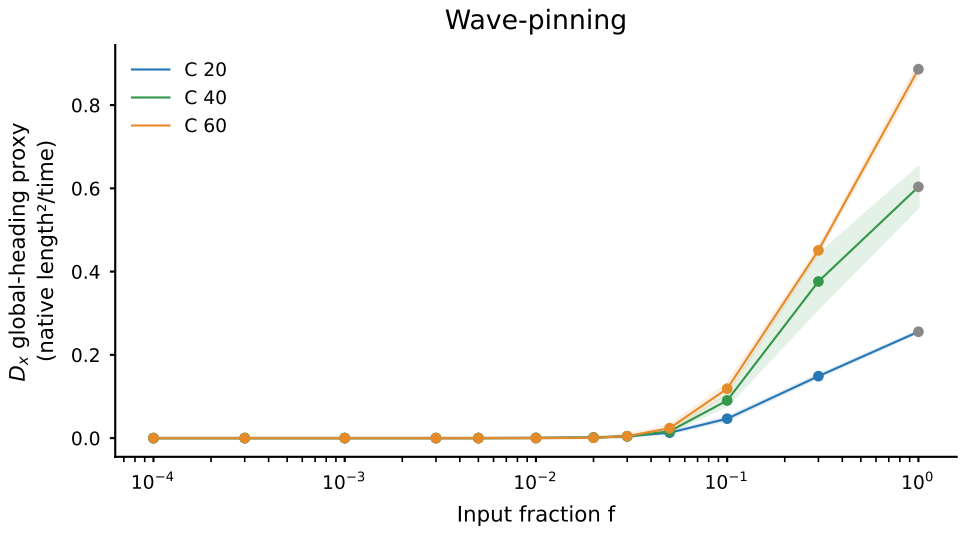
E2 completed native observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S8_dense_native_completed_20261020_021_S4_5_tau_p.csv

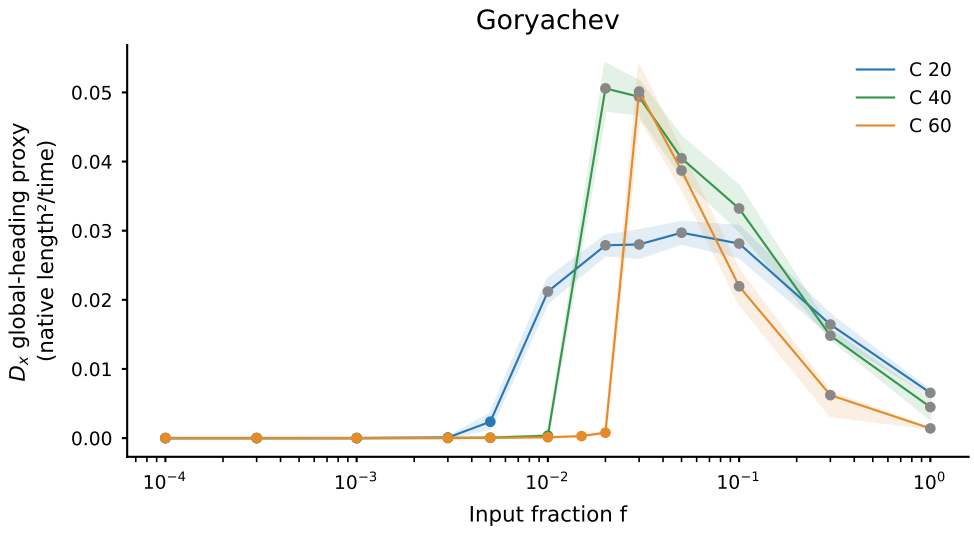
S8 · E2 completed native observations · Persistence and angular mobility across size

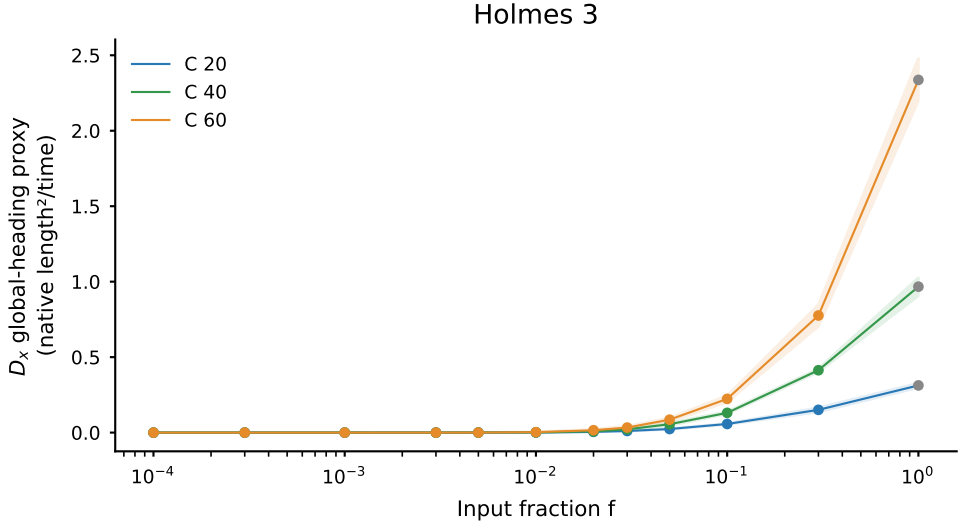


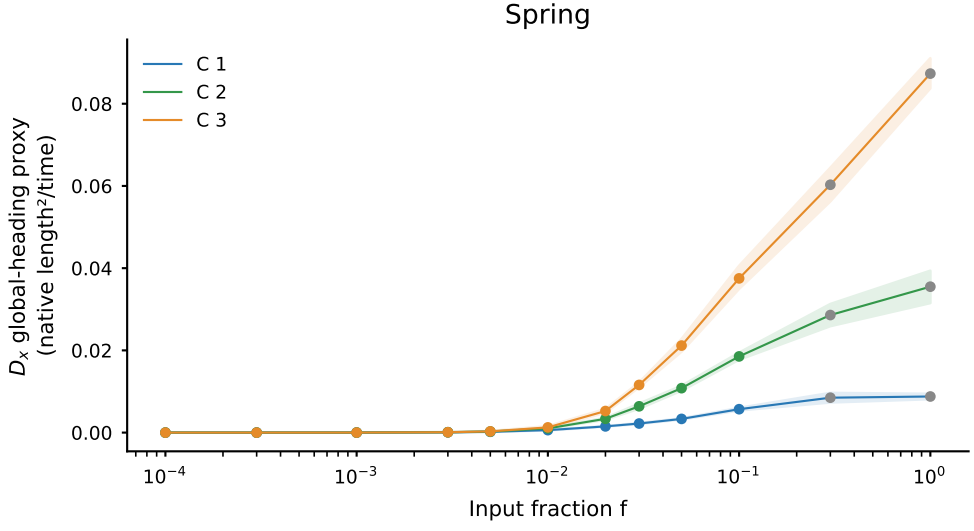
E2 completed native observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S8_dense_native_completed_20261020_022_S4_5_Dtheta.csv

S8 · E2 completed native observations · Global-heading mobility proxy



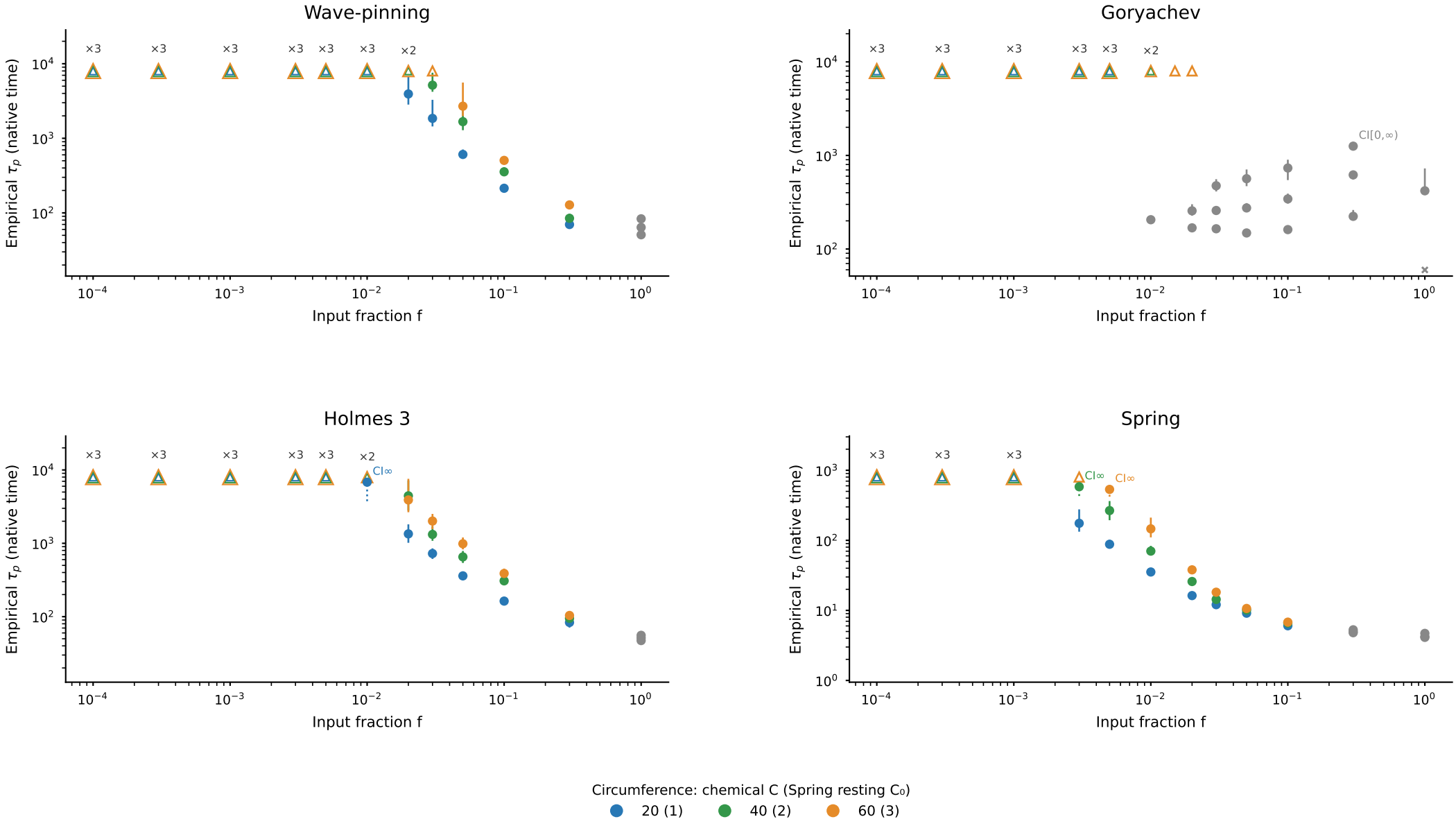






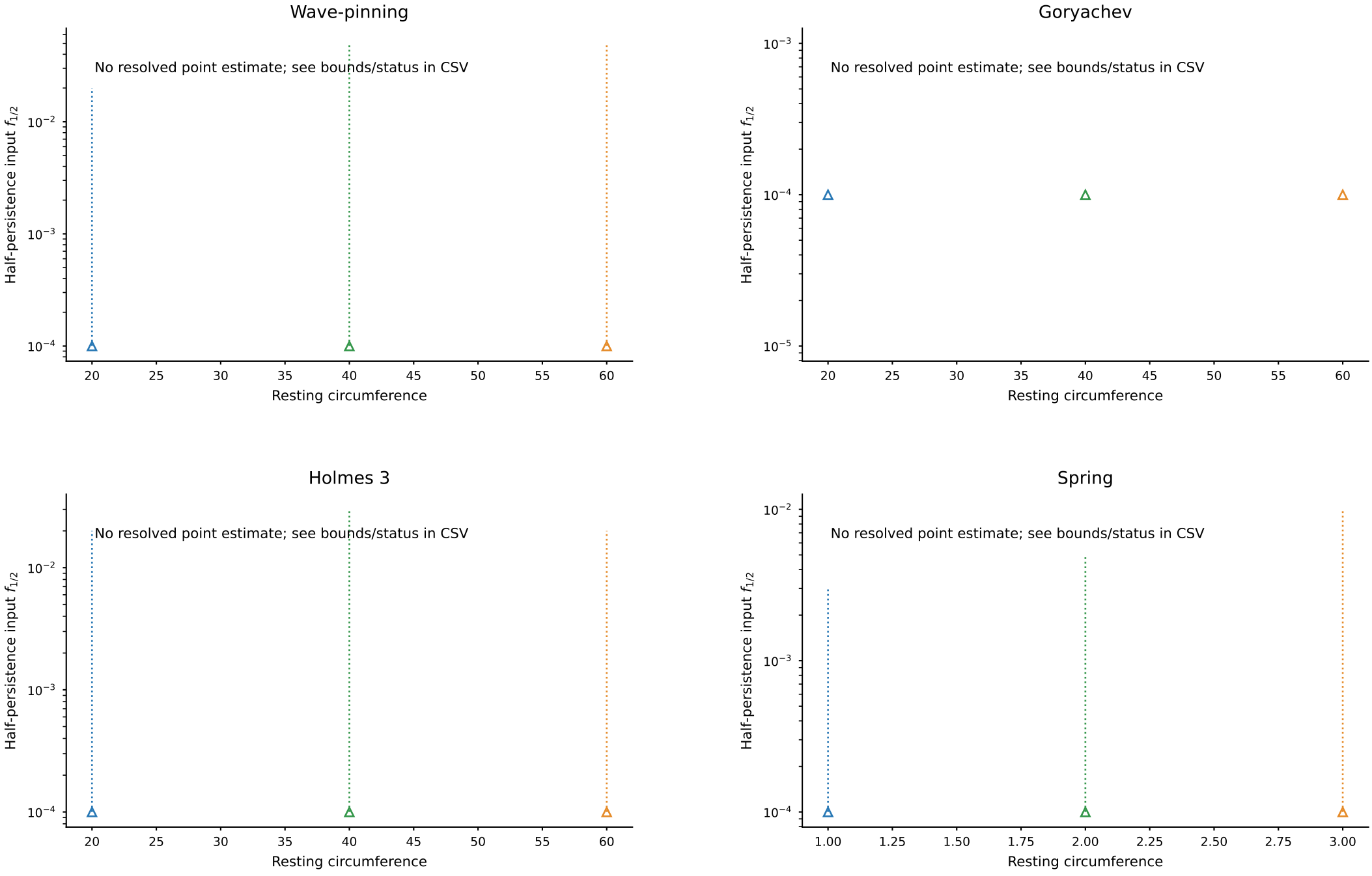
E2 completed native observations. Dx = Dθ × (actual mean C / 2π)² converts the signed short-lag global activity-heading estimate using each window's actual mean circumference. In multi-region states this proxy is not diffusion of one tracked site. It is distinct from qualified single-region literal heading/peak coordinate MSD and is not a slope fit to θC(t)/(2π). Increased Dx and angular persistence advantages are both retained when they coexist. Grey retains forced conditions. Native length/time units differ across models. No free-diffusion or migration velocity law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV. CI∞ next to a finite point marks an upper-unbounded bootstrap CI; finite lower endpoints are retained. CI[0,∞) spans the full nonnegative range.
CSV: figure4_S8_dense_native_completed_20261020_024_S5_Dx.csv

S8 · E2 completed native observations · Empirical persistence against input strength



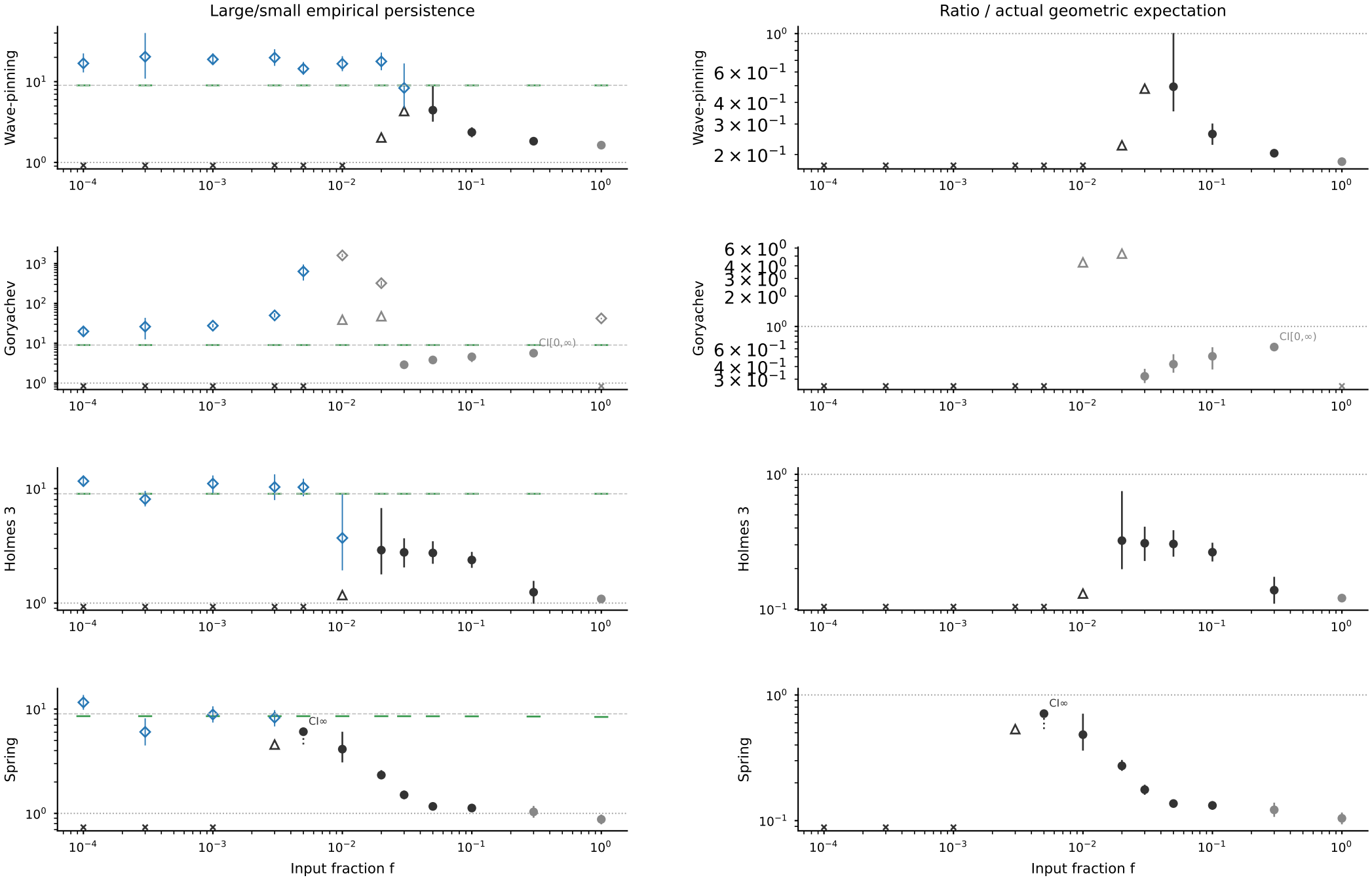
E2 completed native observations. Upward triangles are empirical lower bounds; downward triangles are unresolved below two cadences. Nested triangles at one unchanged coordinate retain coincident sizes; ×n gives their count. Their exact sizes, glyph sizes and coordinates remain in CSV. The shared legend maps chemical C to Spring resting C₀; grey retains forced observations. Bootstrap CI markers are separate from empirical bounds. Exponential fits never replace empirical measurements. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV. CI∞ next to a finite point marks an upper-unbounded bootstrap CI; finite lower endpoints are retained. CI[0,∞) spans the full nonnegative range.
CSV: figure4_S8_dense_native_completed_20261020_025_S6_tau.csv

S8 · E2 completed native observations · Noise tolerance with finite-window uncertainty



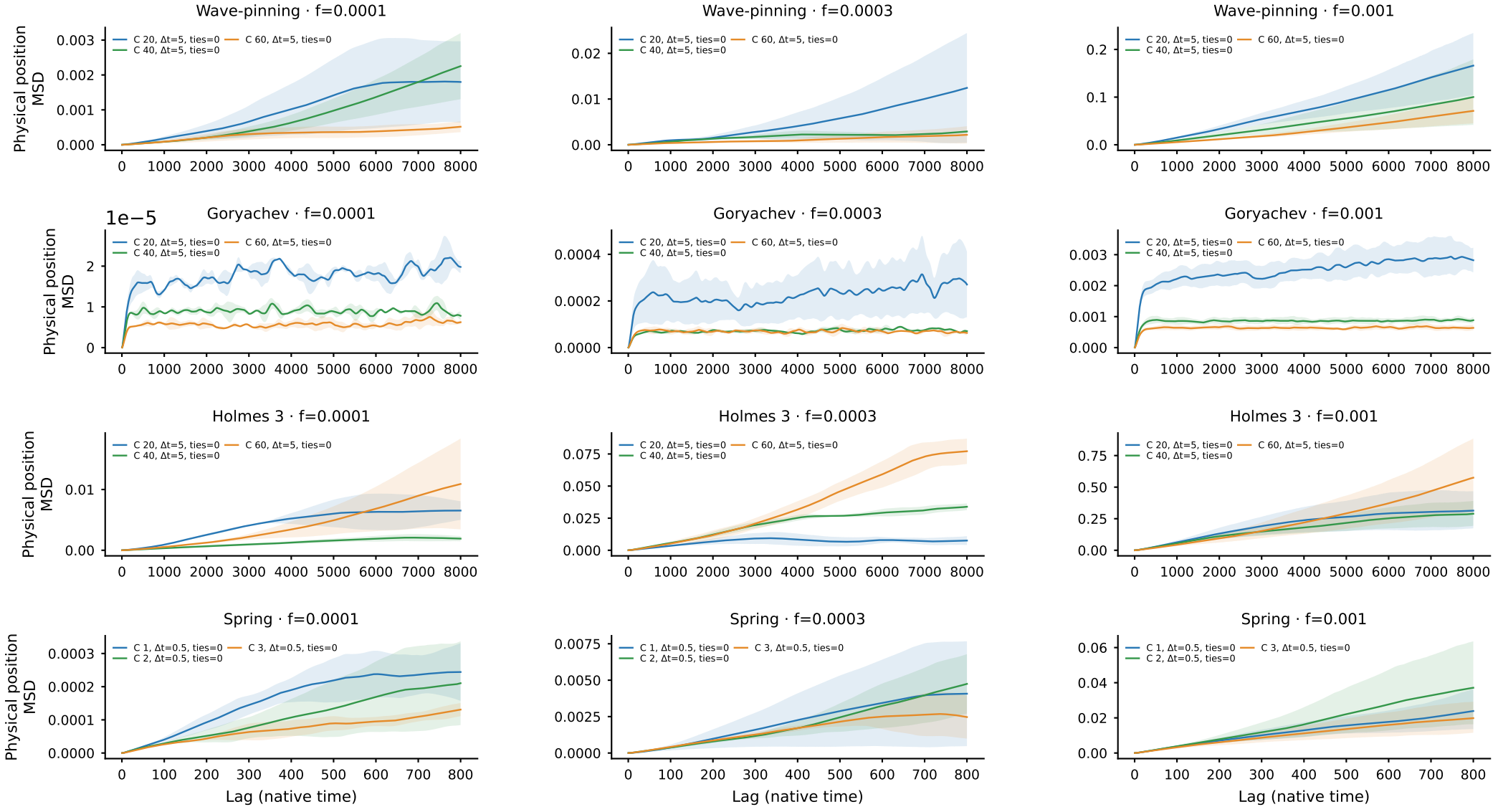
E2 completed native observations. First crossing of half the actual f=0.0001 persistence, using log-f interpolation only when identified. Censored references remain bounded or inconclusive. Intermediate-size grids lacking the declared f=0.0001 baseline retain unavailable markers/reasons; f=0.001 is never substituted and no tolerance curve bridges them. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S8_dense_native_completed_20261020_026_S6_fhalf.csv

S4 · E2 dense native observations · Size advantage and geometric reference



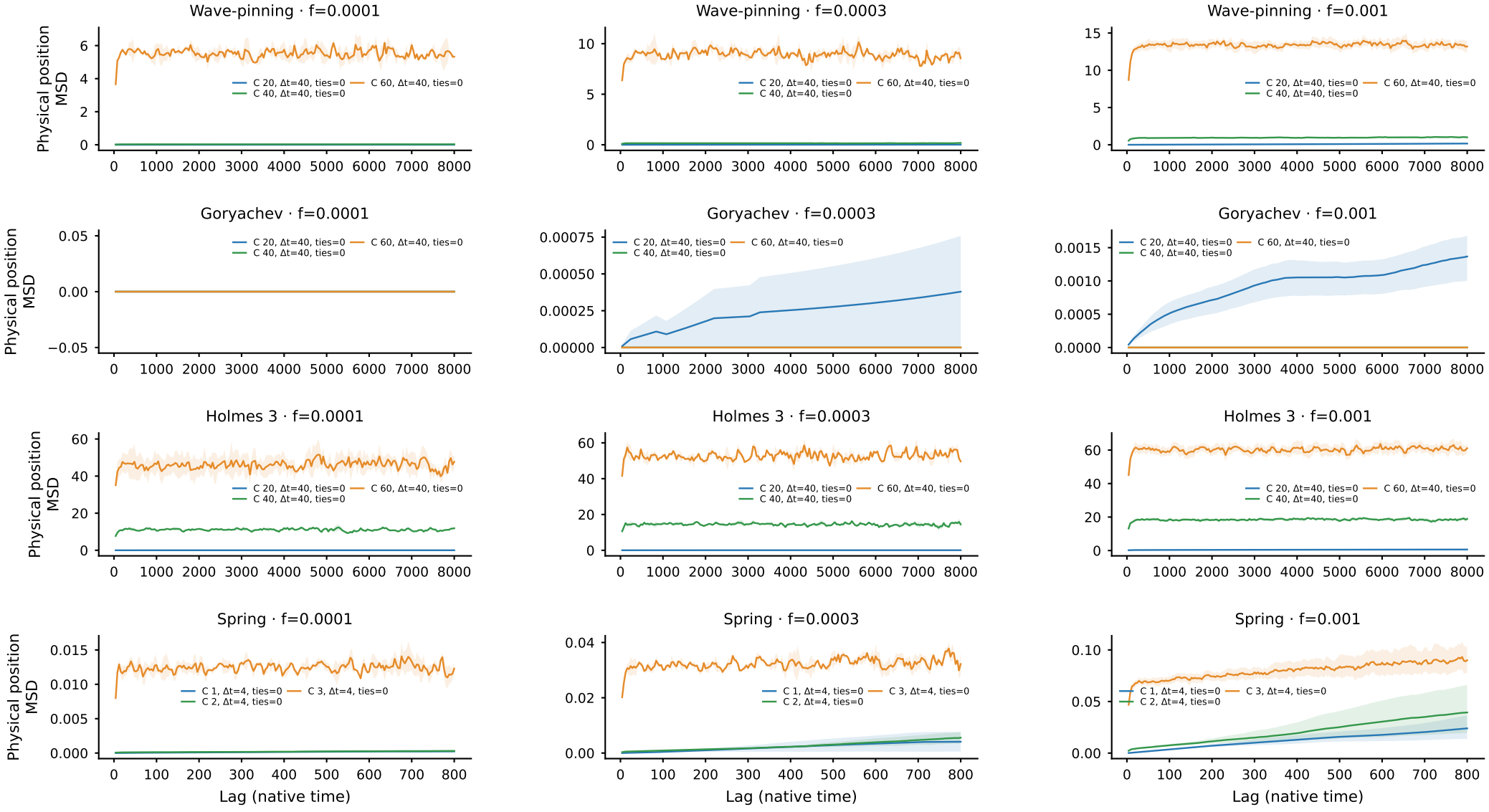
E2 dense native observations. Triangles preserve finite-window or resolution bounds. Two censored times do not yield a finite ratio. Open diamonds show the short-lag global activity-heading small/large Dθ proxy: blue for unforced conditions and grey for forced conditions. This proxy does not resolve a direct persistence ratio or measure one tracked site's diffusion in multi-region states. Green marks: measured circumference-ratio squared; dashed 9: resting-size reference. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV. CI∞ next to a finite point marks an upper-unbounded bootstrap CI; finite lower endpoints are retained. CI[0,∞) spans the full nonnegative range.
CSV: figure4_ratio_scope_replacements_20261020_003_S4_ratios.csv

S5 · E2 literal material-origin physical-site measurements · Literal heading position · A3 primary split · σ=1



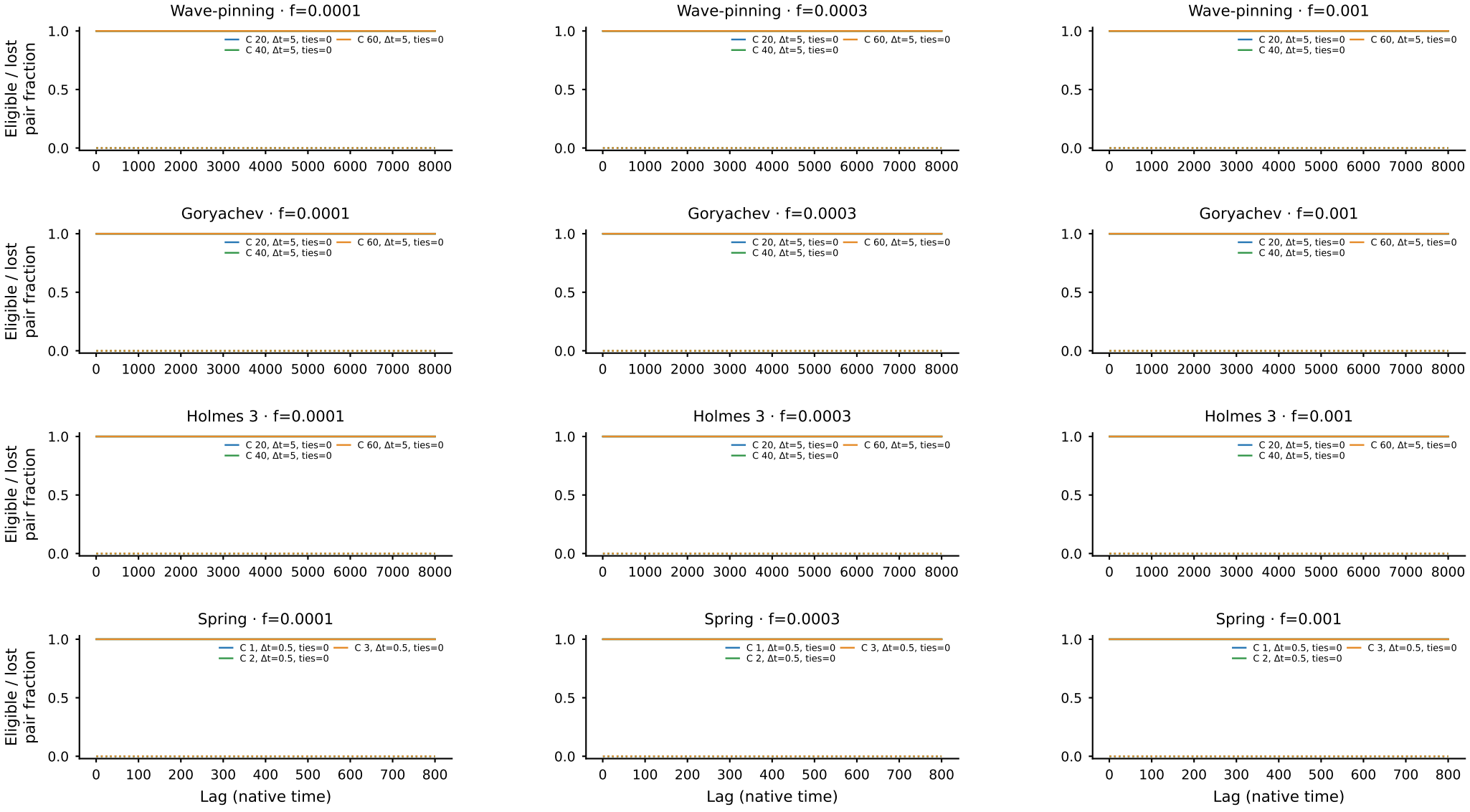
E2 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M₁|/|M₁,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_001_S5_literal_sigma1_heading_literal_actualC_MSD_1.csv.gz

S5 · E2 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · σ=1



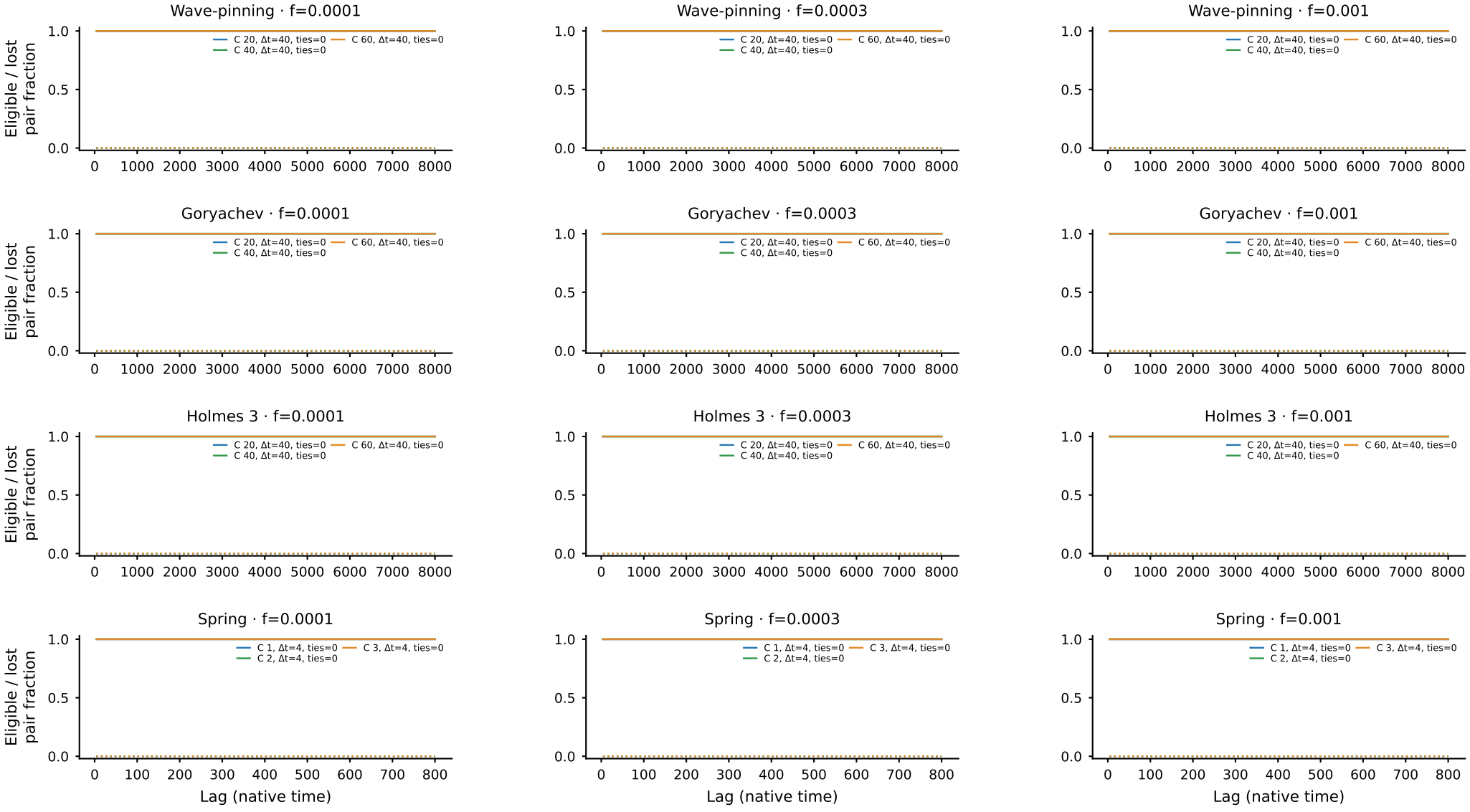
E2 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M₁|/|M₁,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_002_S5_literal_sigma1_profile_peak_literal_actualC_MSD_1.csv

S5 · E2 literal material-origin physical-site measurements · Literal heading position · A3 primary split · pair coverage and ambiguity loss · σ=1



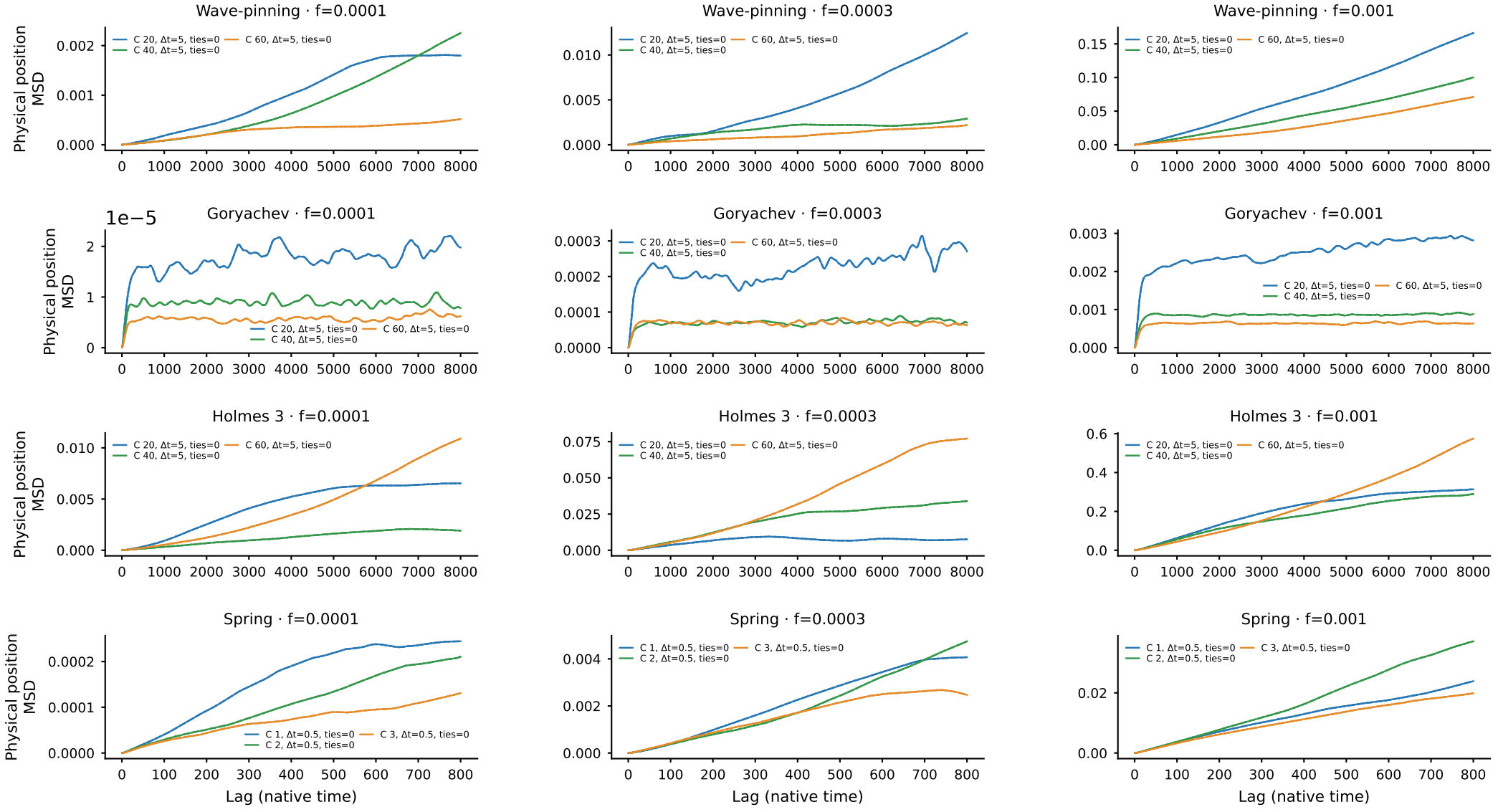
E2 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_003_S5_literal_sigma1_heading_literal_actualC_coverage_1.csv.gz

S5 · E2 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · pair coverage and ambiguity loss · σ=1



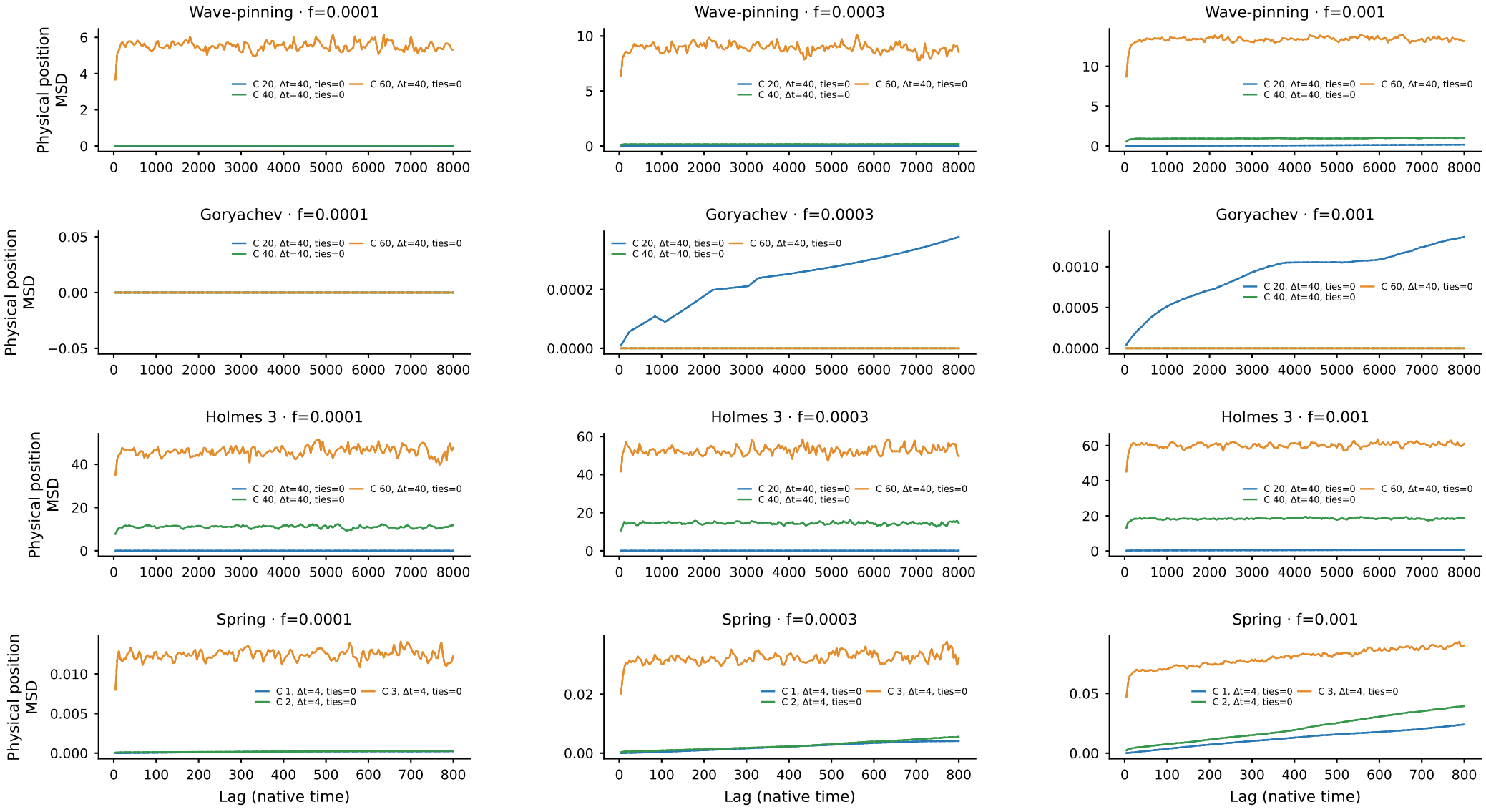
E2 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M₁|/|M₁,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_004_S5_literal_sigma1_profile_peak_literal_actualC_coverage_1.csv

S5 · E2 literal material-origin physical-site measurements · Literal heading position · paired route sensitivities · split / +π / −π · σ=1



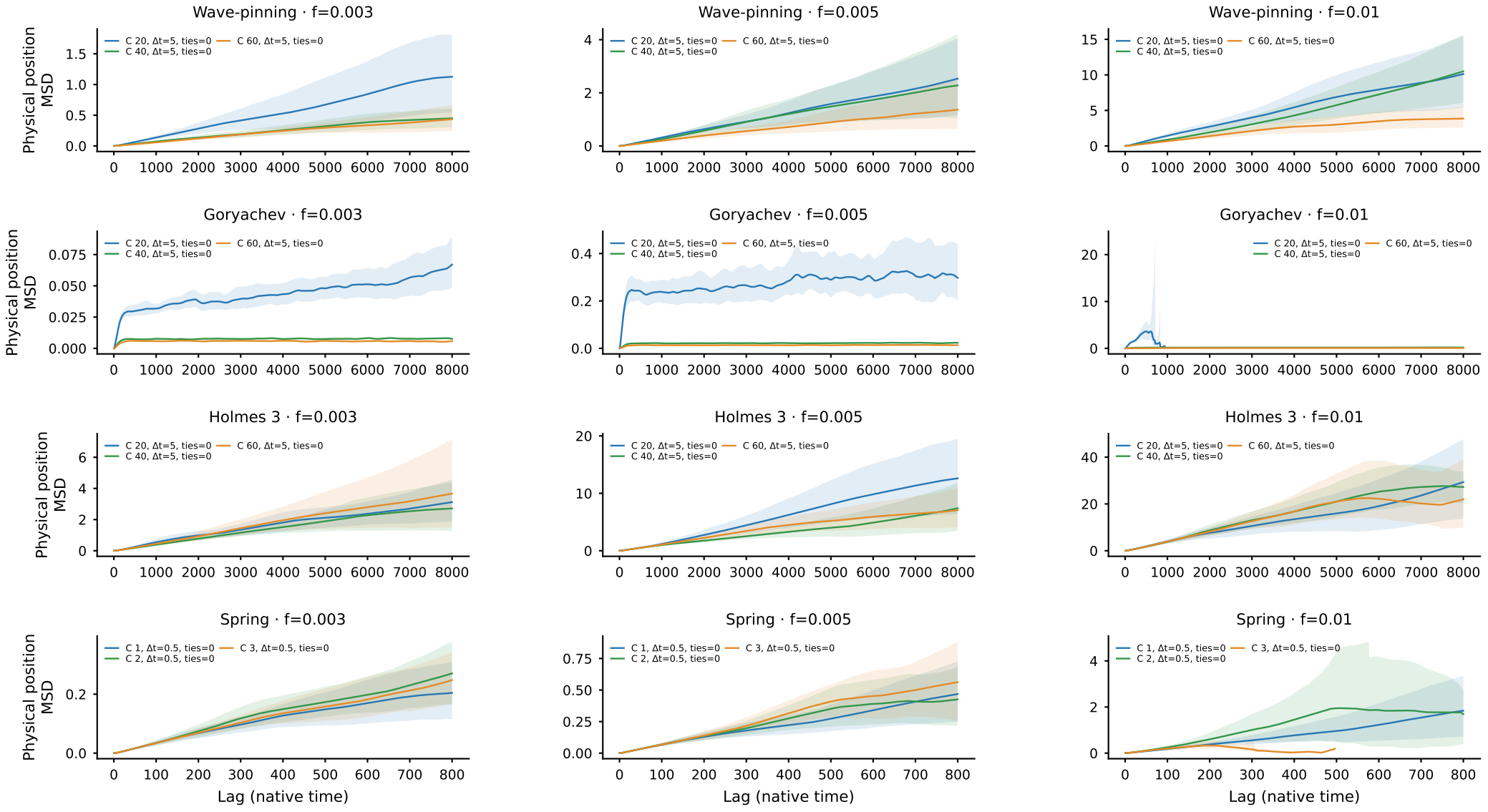
E2 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_005_S5_literal_sigma1_heading_literal_actualC_routes_1.csv.gz

S5 · E2 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · split / +π / −π · σ=1



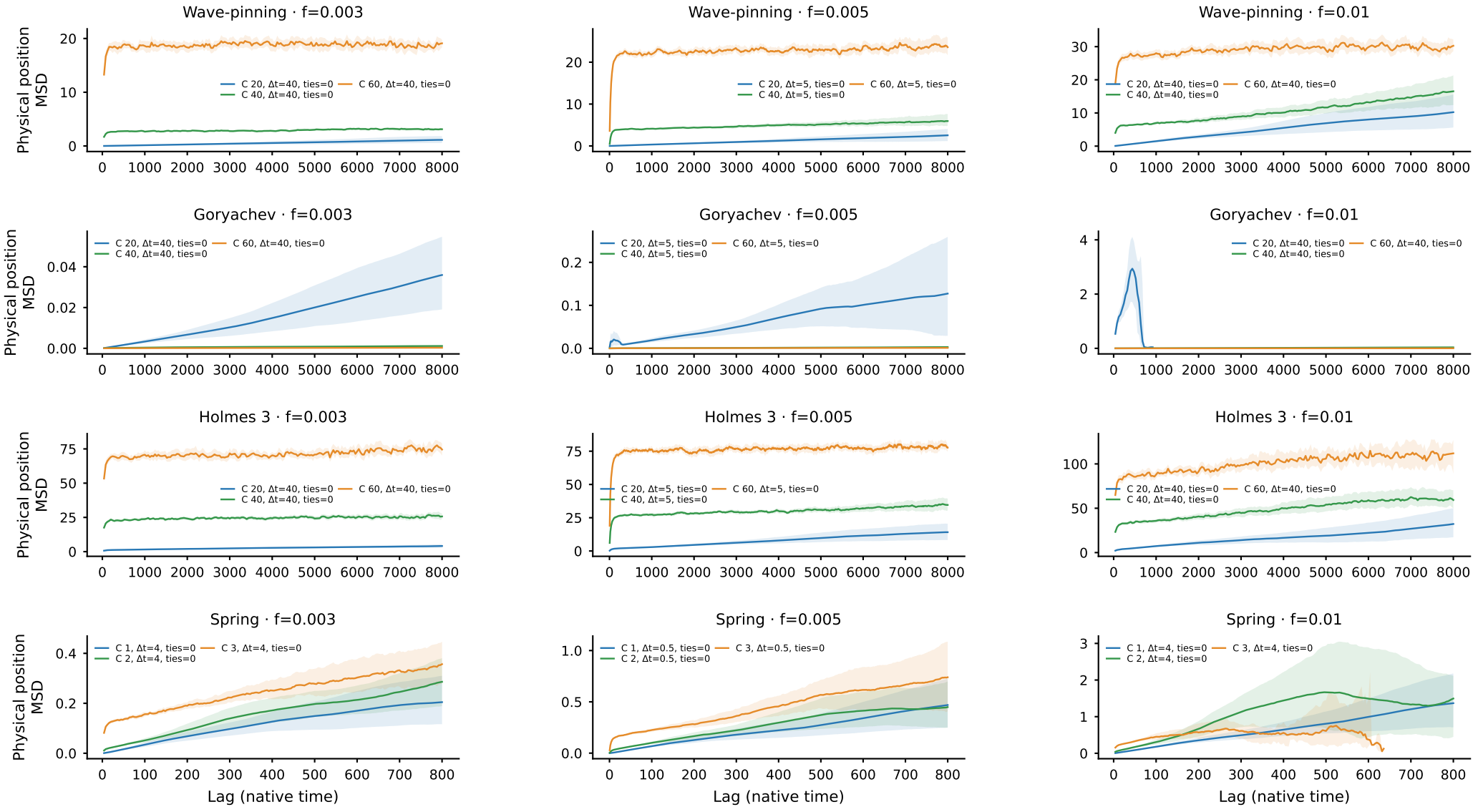
E2 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M_1|/|M_1,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_006_S5_literal_sigma1_profile_peak_literal_actualC_routes_1.csv

S5 · E2 literal material-origin physical-site measurements · Literal heading position · A3 primary split · σ=1



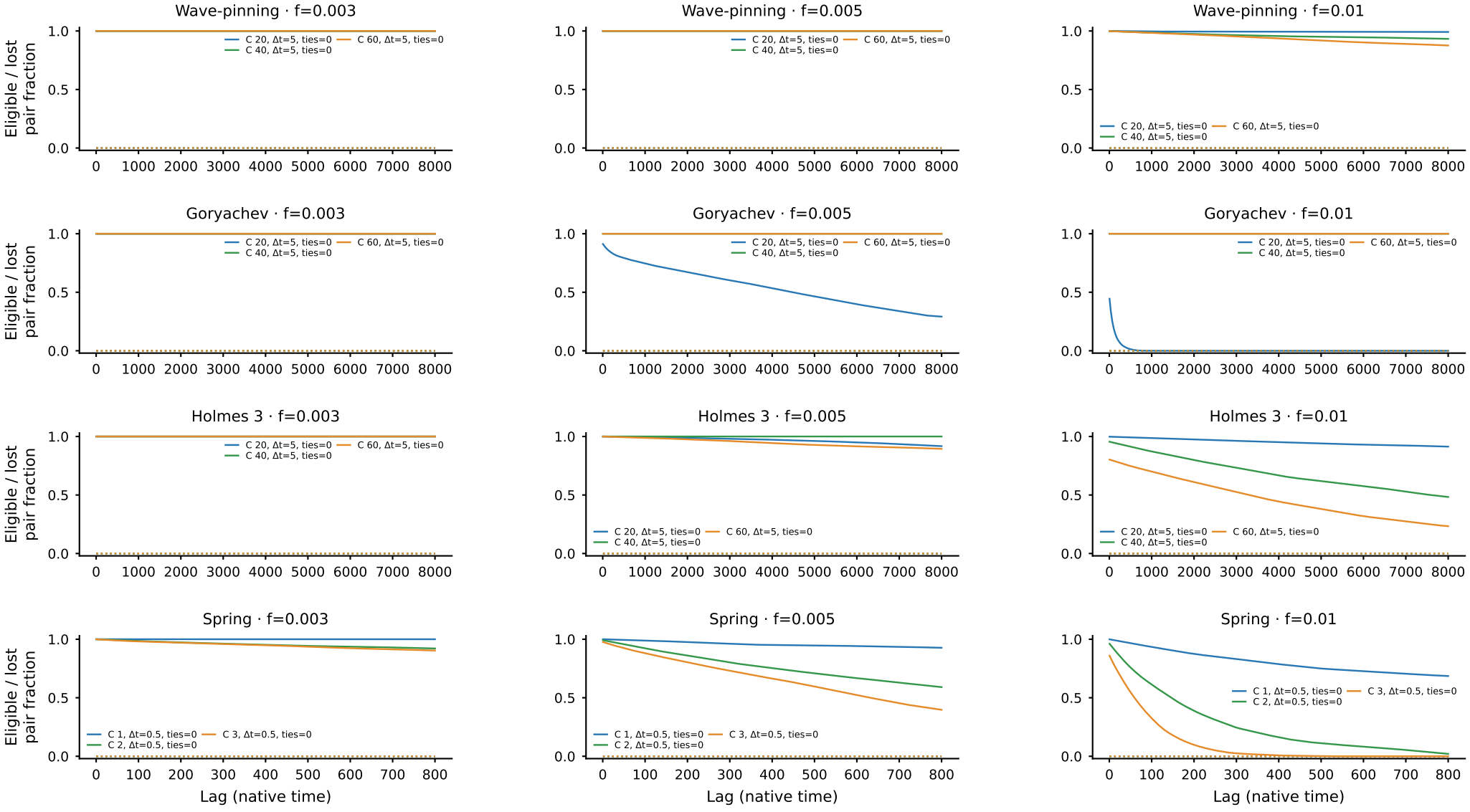
E2 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_007_S5_literal_sigma1_heading_literal_actualC_MSD_2.csv.gz

S5 · E2 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · σ=1



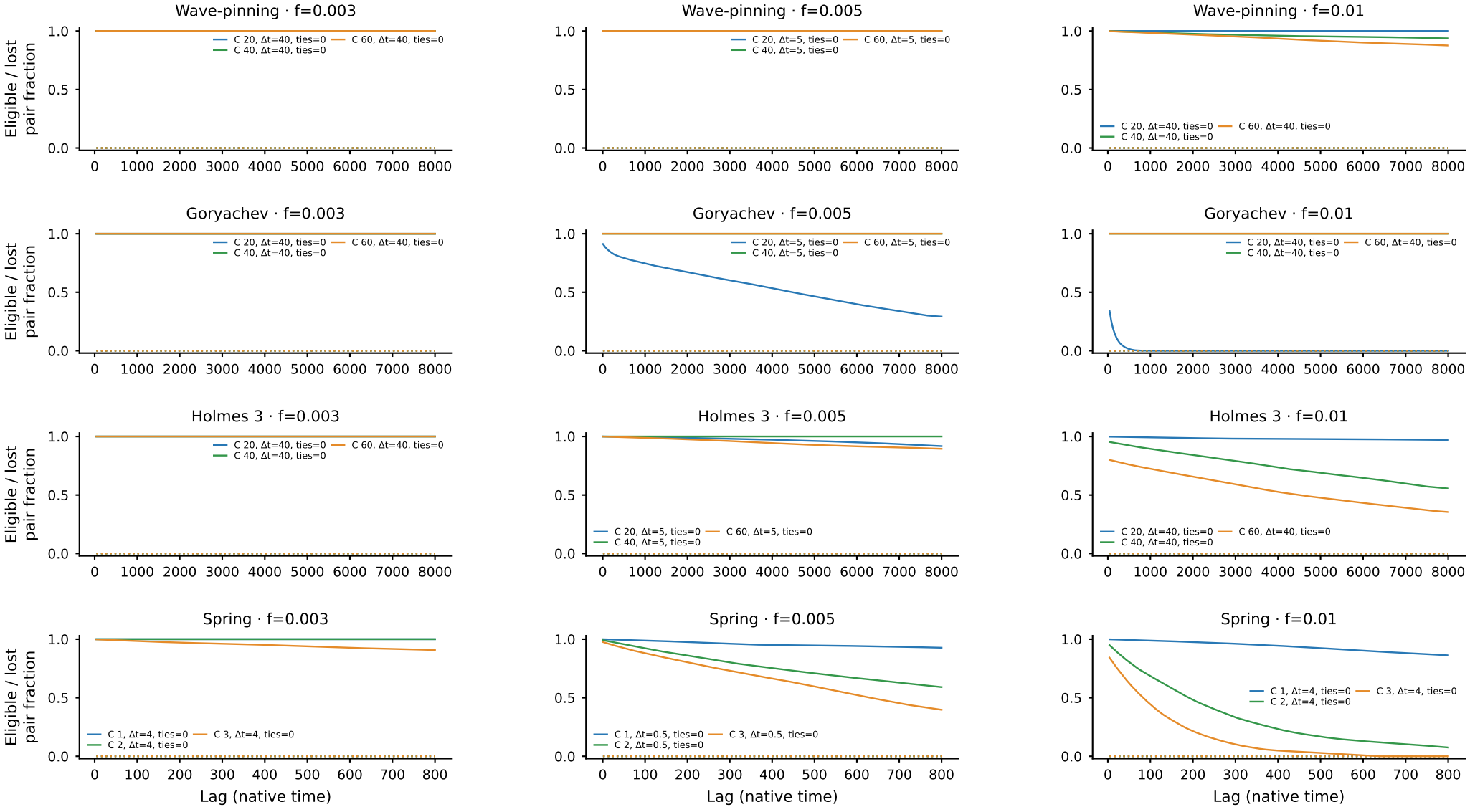
E2 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_008_S5_literal_sigma1_profile_peak_literal_actualC_MSD_2.csv

S5 · E2 literal material-origin physical-site measurements · Literal heading position · A3 primary split · pair coverage and ambiguity loss · σ=1



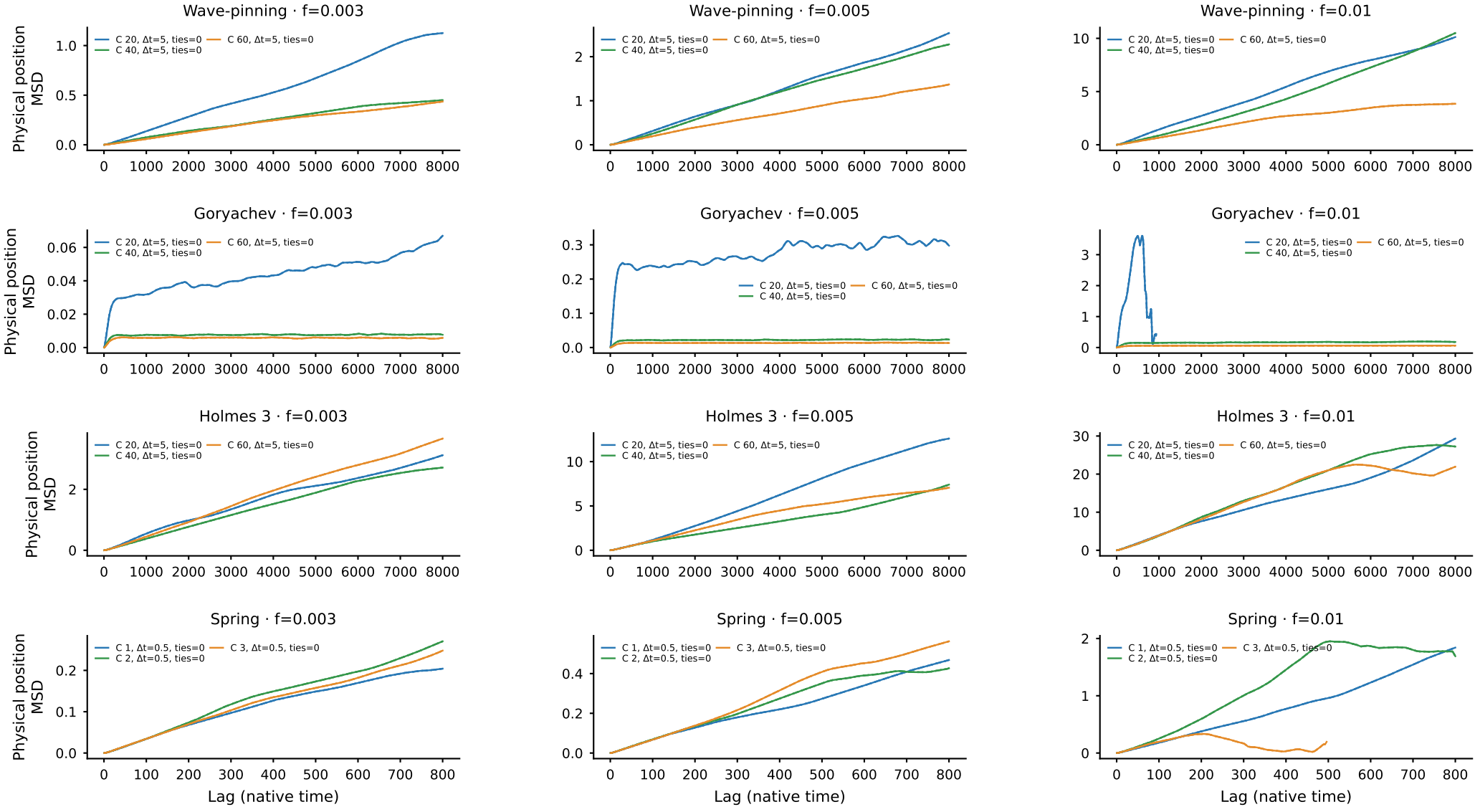
E2 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_009_S5_literal_sigma1_heading_literal_actualC_coverage_2.csv.gz

S5 · E2 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · pair coverage and ambiguity loss · σ=1



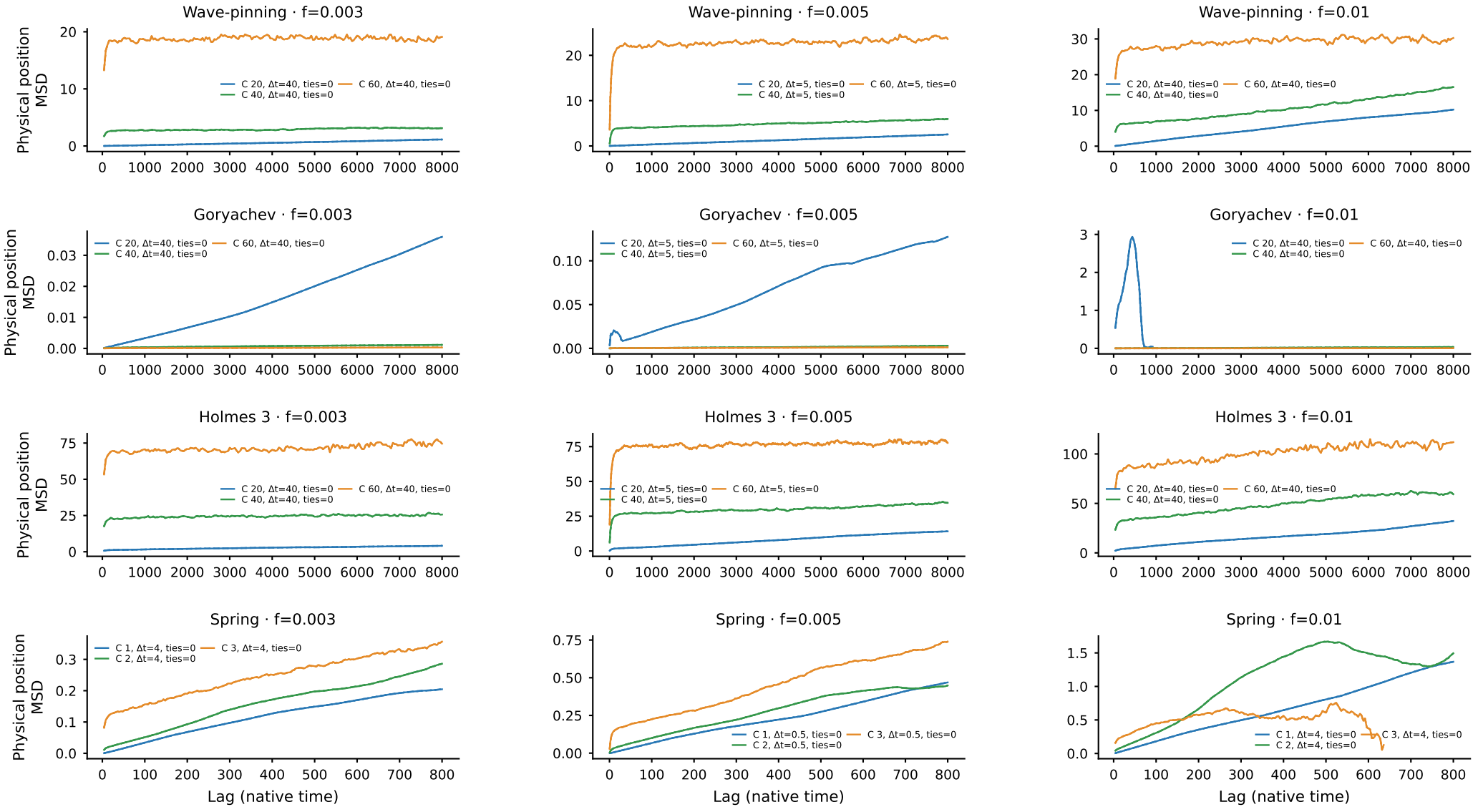
E2 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_010_S5_literal_sigma1_profile_peak_literal_actualC_coverage_2.csv

S5 · E2 literal material-origin physical-site measurements · Literal heading position · paired route sensitivities · split / +π / −π · σ=1



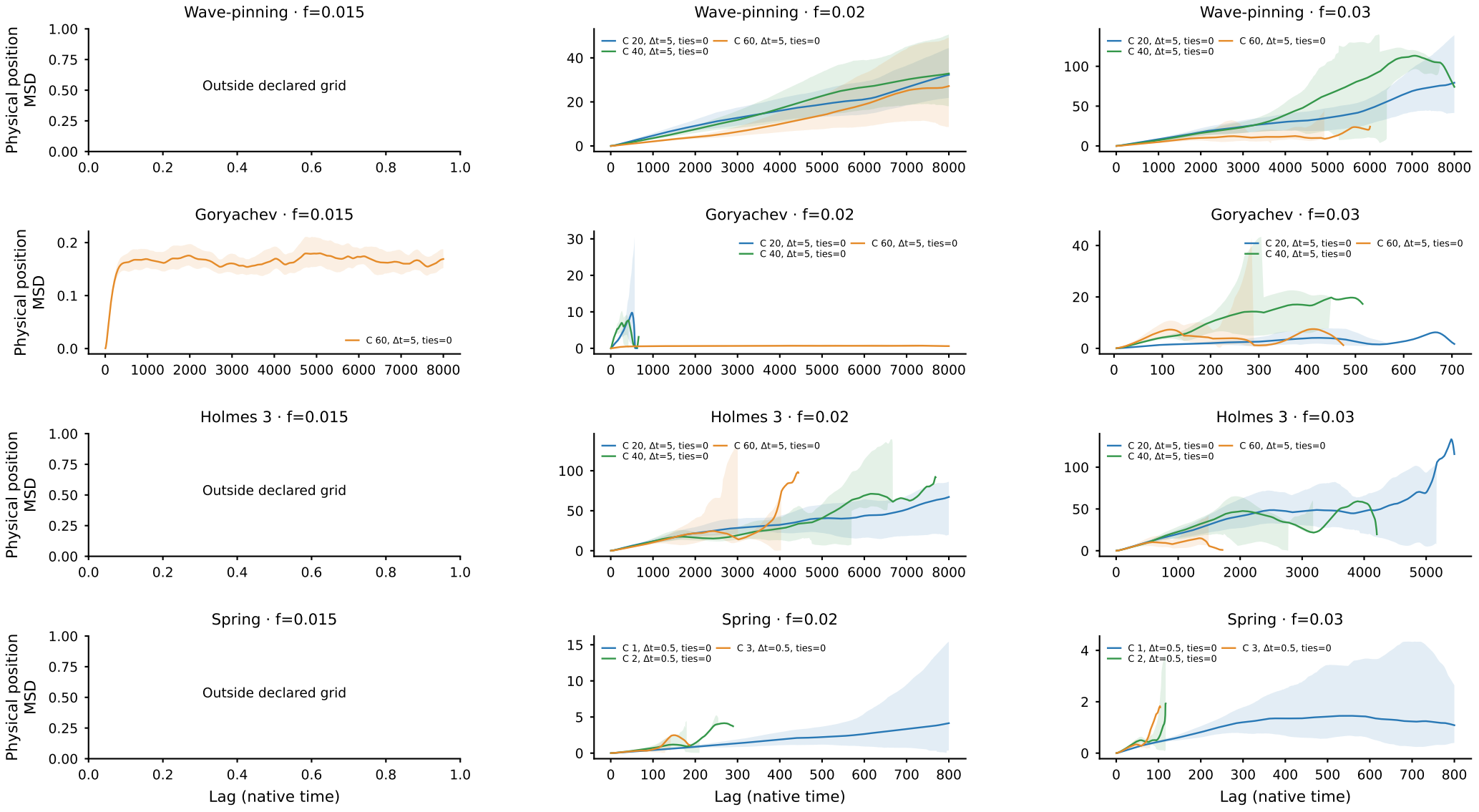
E2 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_011_S5_literal_sigma1_heading_literal_actualC_routes_2.csv.gz

S5 · E2 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · split / +π / −π · σ=1



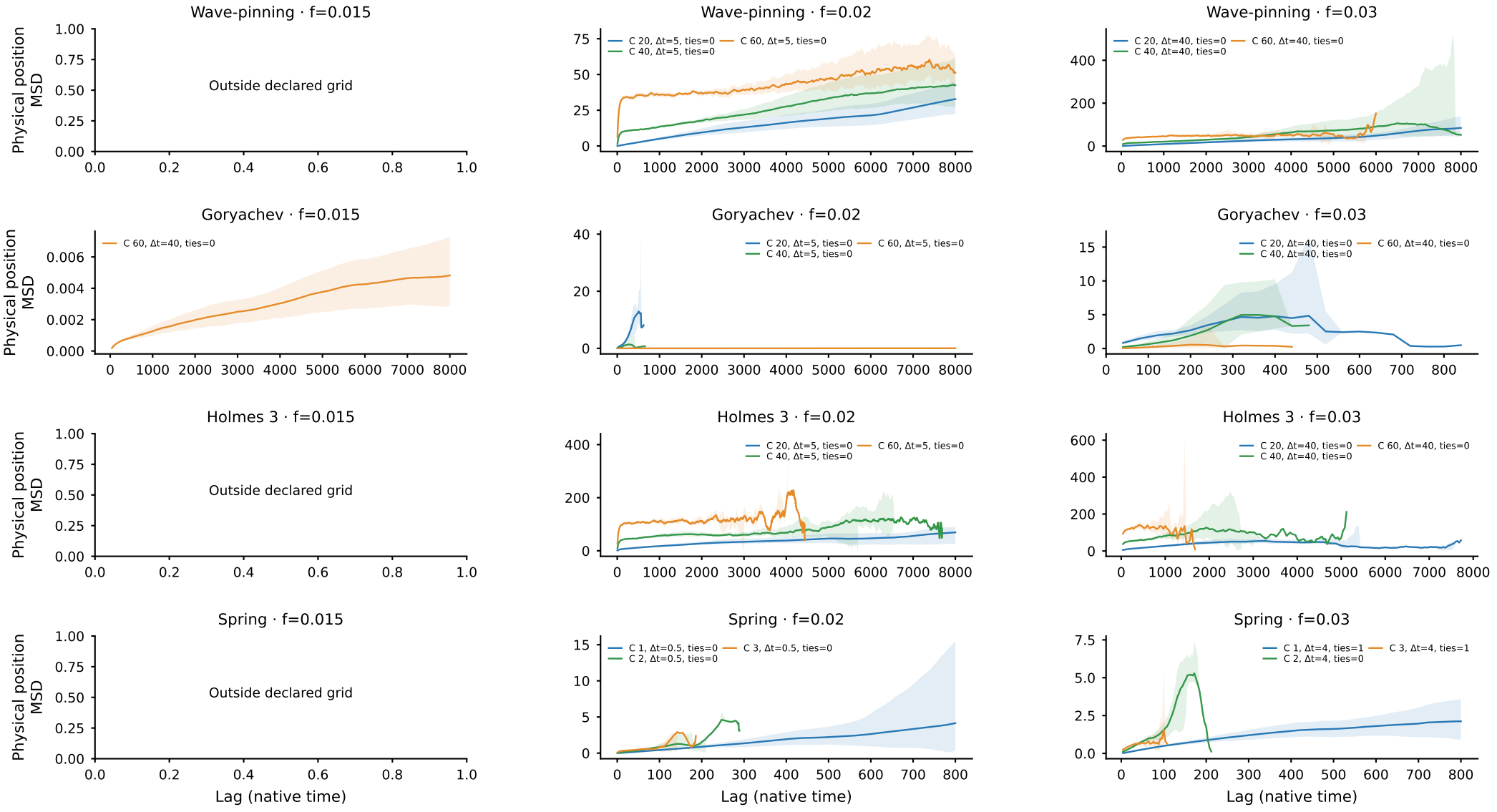
E2 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_012_S5_literal_sigma1_profile_peak_literal_actualC_routes_2.csv.gz

S5 · E2 literal material-origin physical-site measurements · Literal heading position · A3 primary split · σ=1



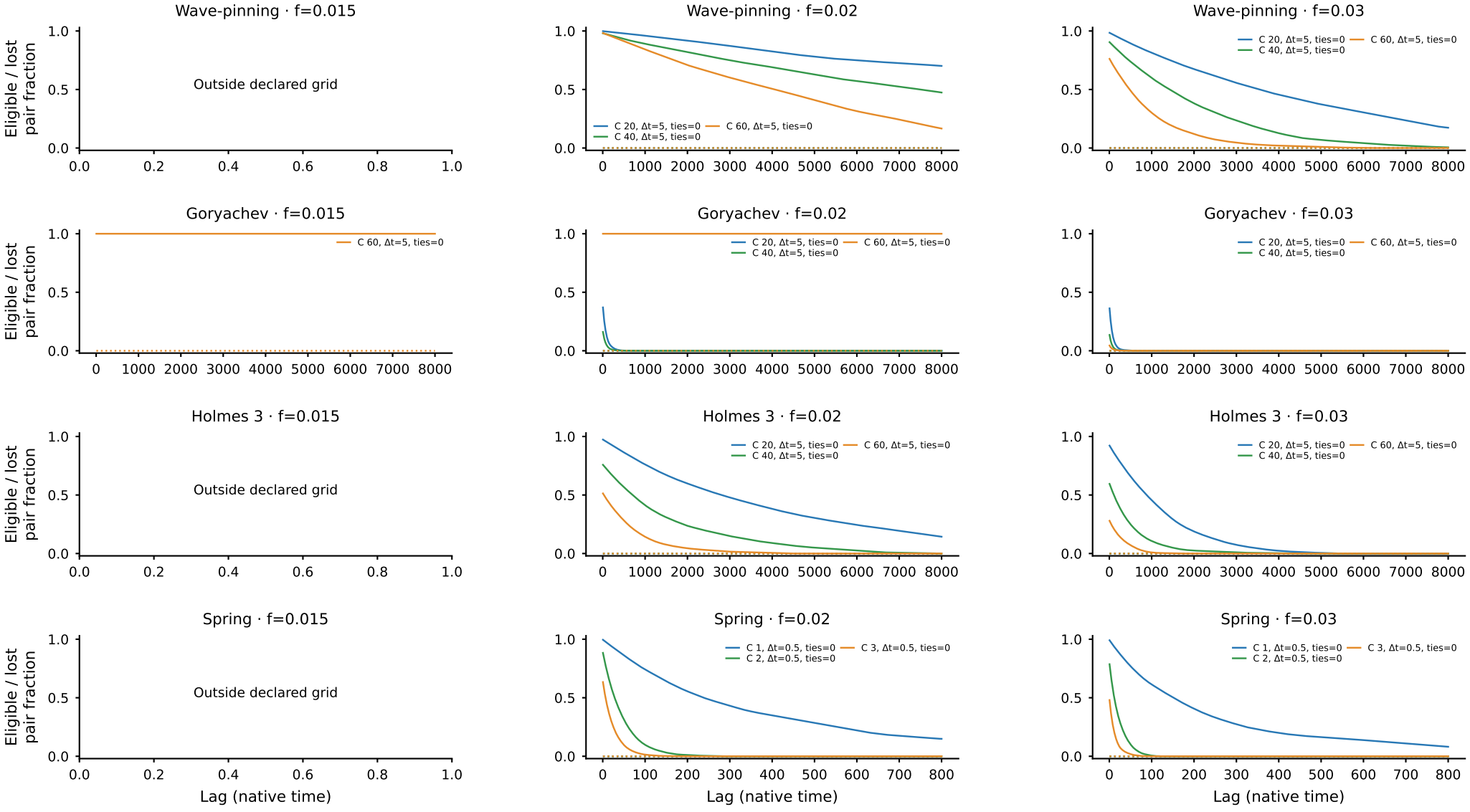
E2 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_013_S5_literal_sigma1_heading_literal_actualC_MSD_3.csv

S5 · E2 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · σ=1



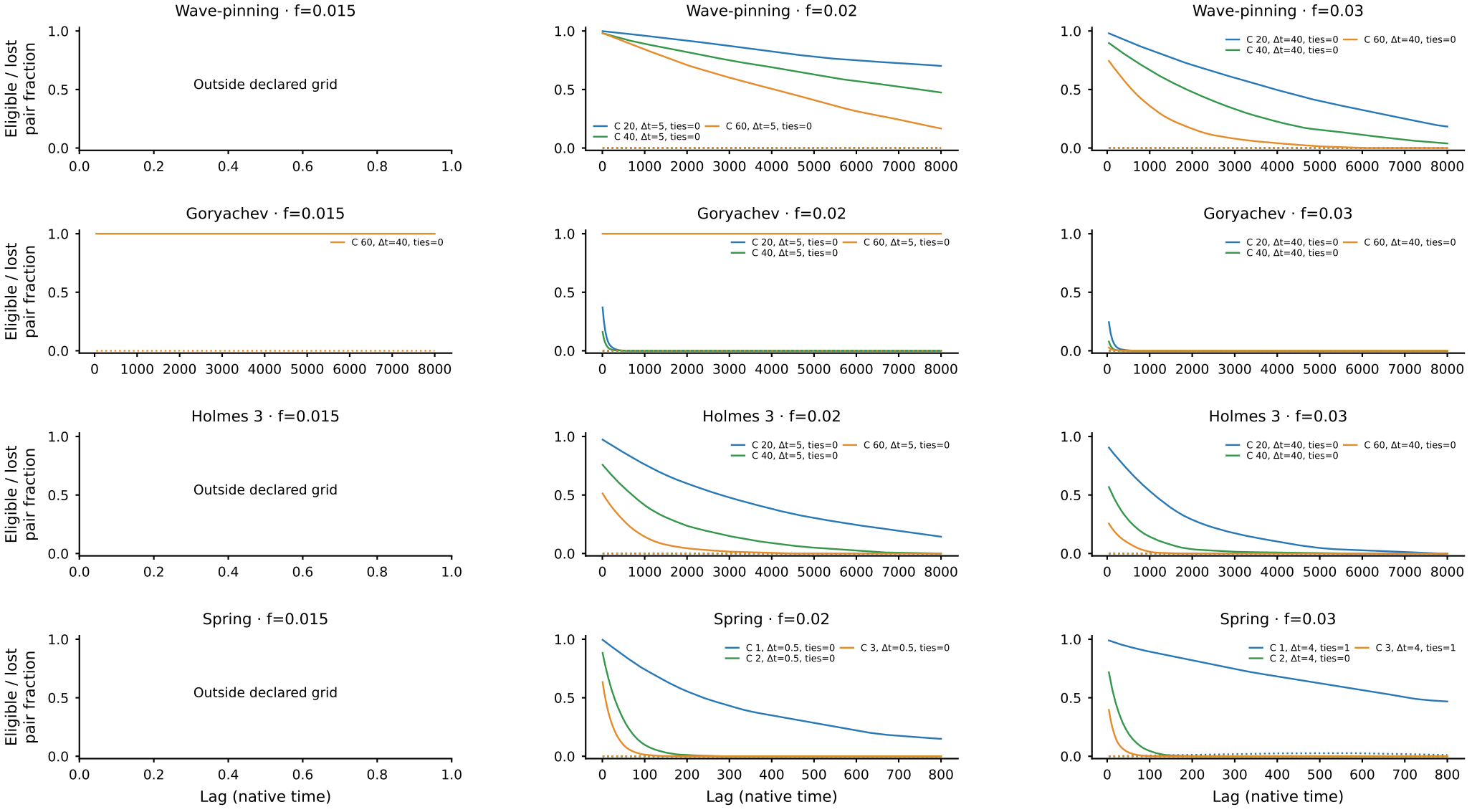
E2 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M₁|/|M₁,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_014_S5_literal_sigma1_profile_peak_literal_actualC_MSD_3.csv

S5 · E2 literal material-origin physical-site measurements · Literal heading position · A3 primary split · pair coverage and ambiguity loss · σ=1



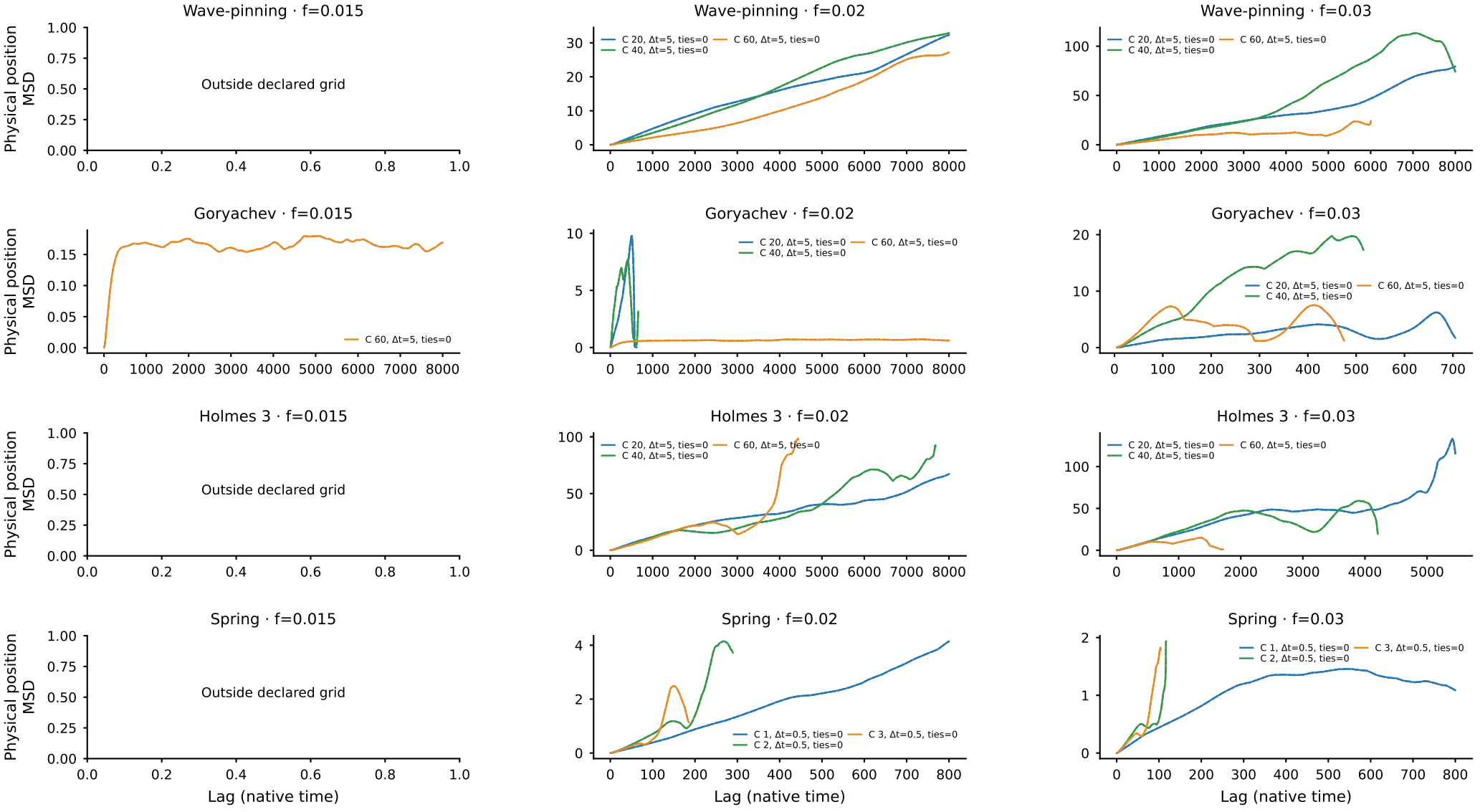
E2 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M₁|/|M₁,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_015_S5_literal_sigma1_heading_literal_actualC_coverage_3.csv

S5 · E2 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · pair coverage and ambiguity loss · σ=1



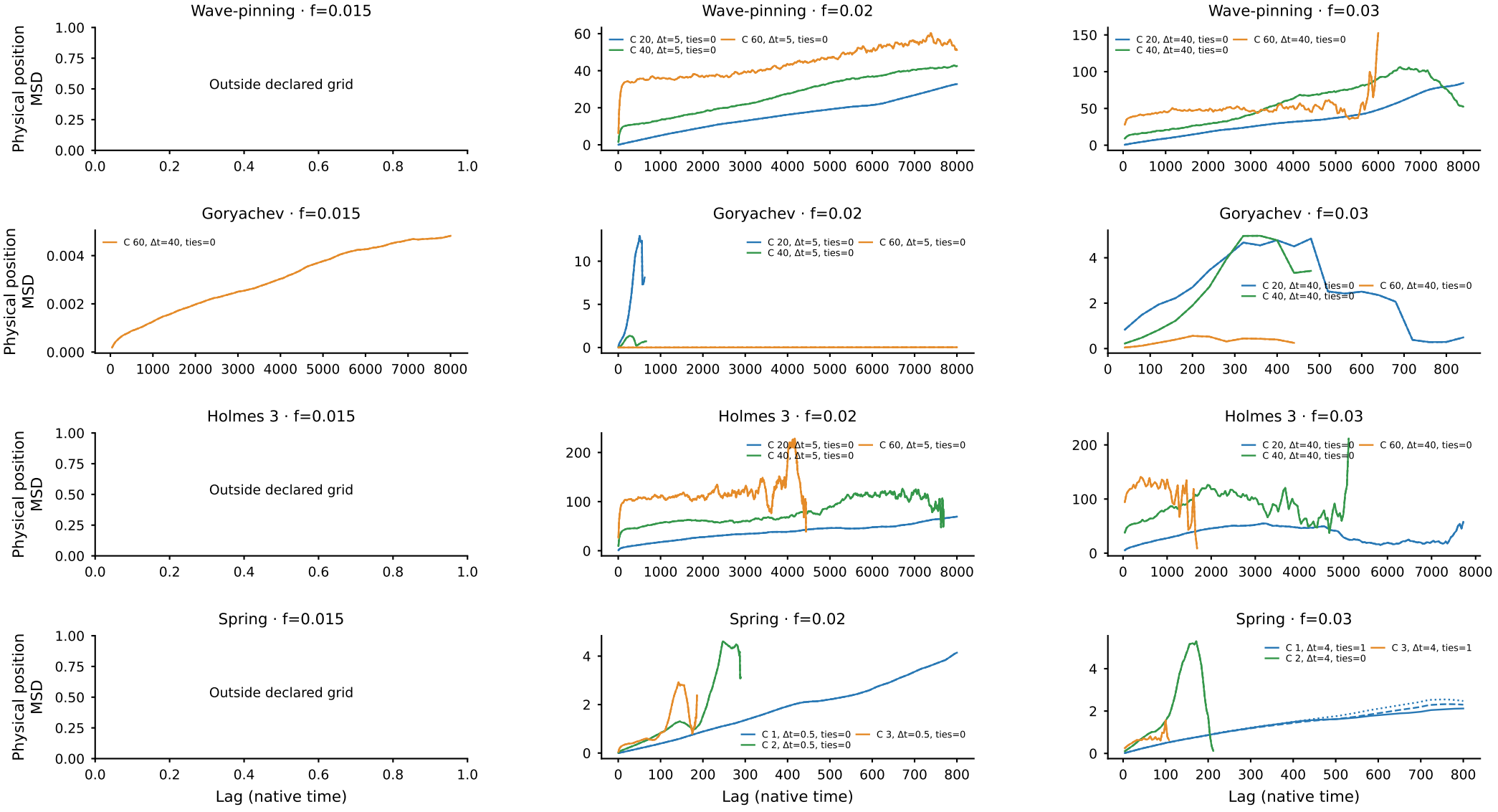
E2 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M₁|/|M₁,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_016_S5_literal_sigma1_profile_peak_literal_actualC_coverage_3.csv

S5 · E2 literal material-origin physical-site measurements · Literal heading position · paired route sensitivities · split / +π / −π · σ=1



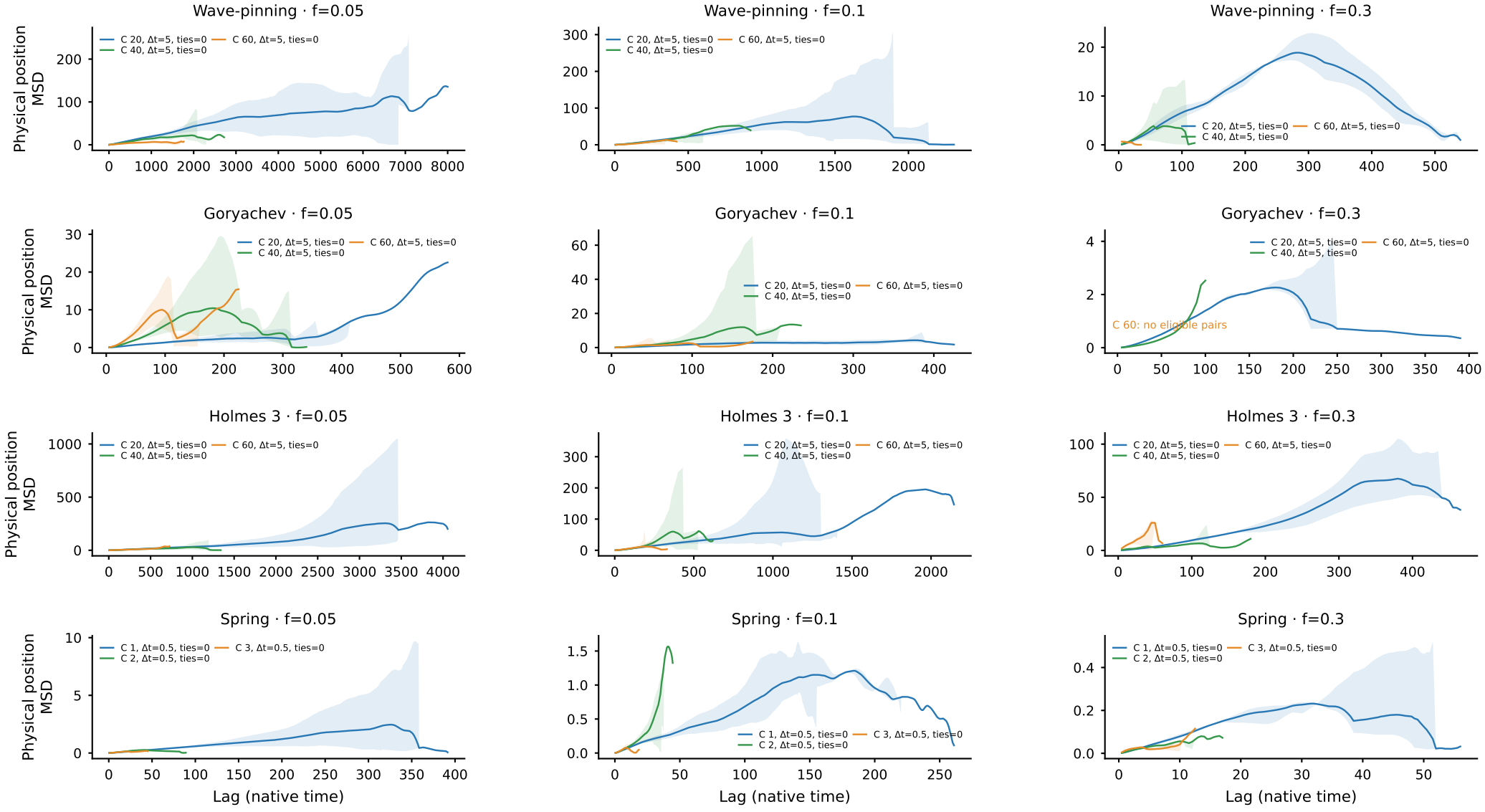
E2 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_017_S5_literal_sigma1_heading_literal_actualC_routes_3.csv.gz

S5 · E2 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · split / +π / −π · σ=1



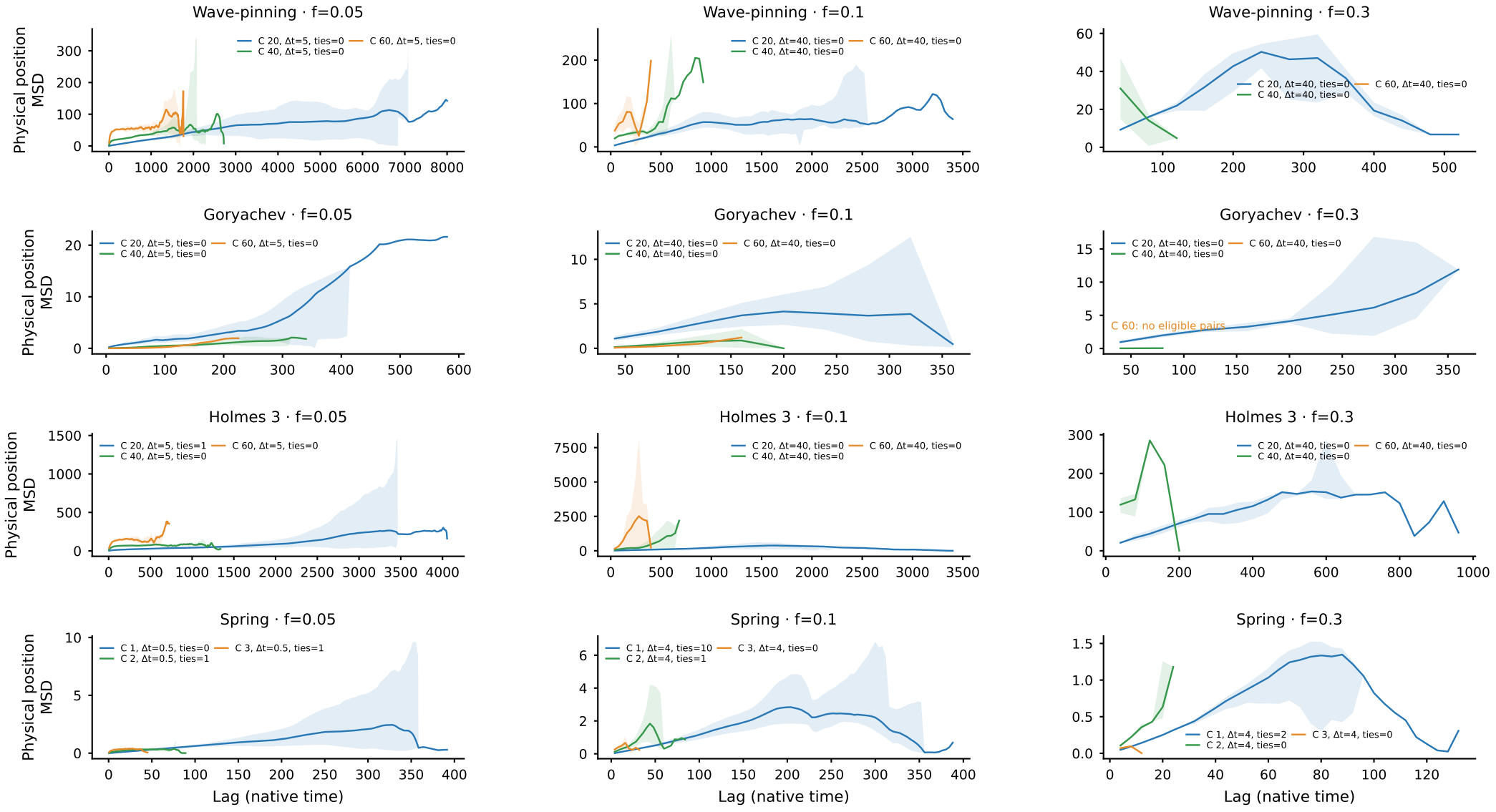
E2 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_018_S5_literal_sigma1_profile_peak_literal_actualC_routes_3.csv.gz

S5 · E2 literal material-origin physical-site measurements · Literal heading position · A3 primary split · σ=1



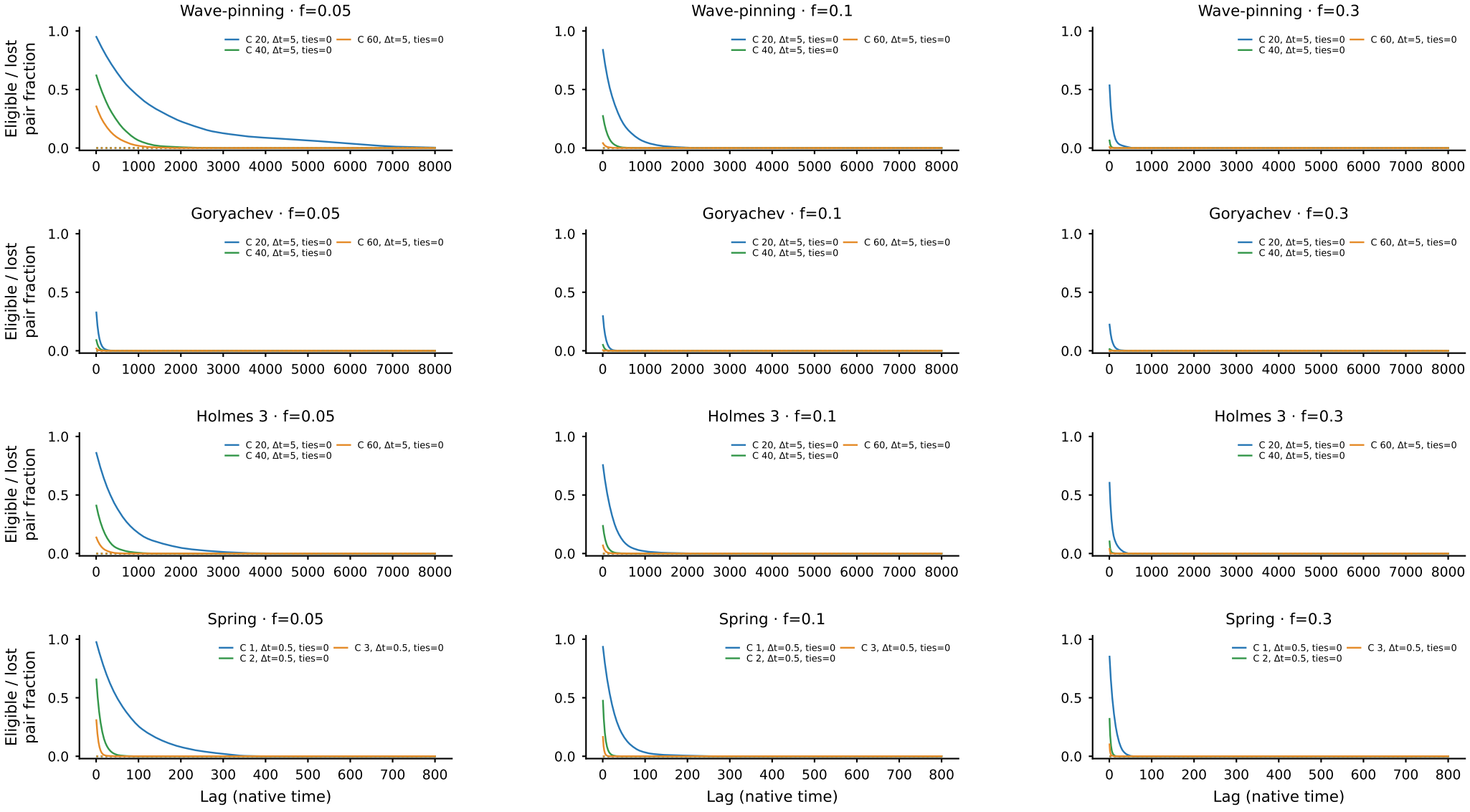
E2 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_019_S5_literal_sigma1_heading_literal_actualC_MSD_4.csv.gz

S5 · E2 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · σ=1



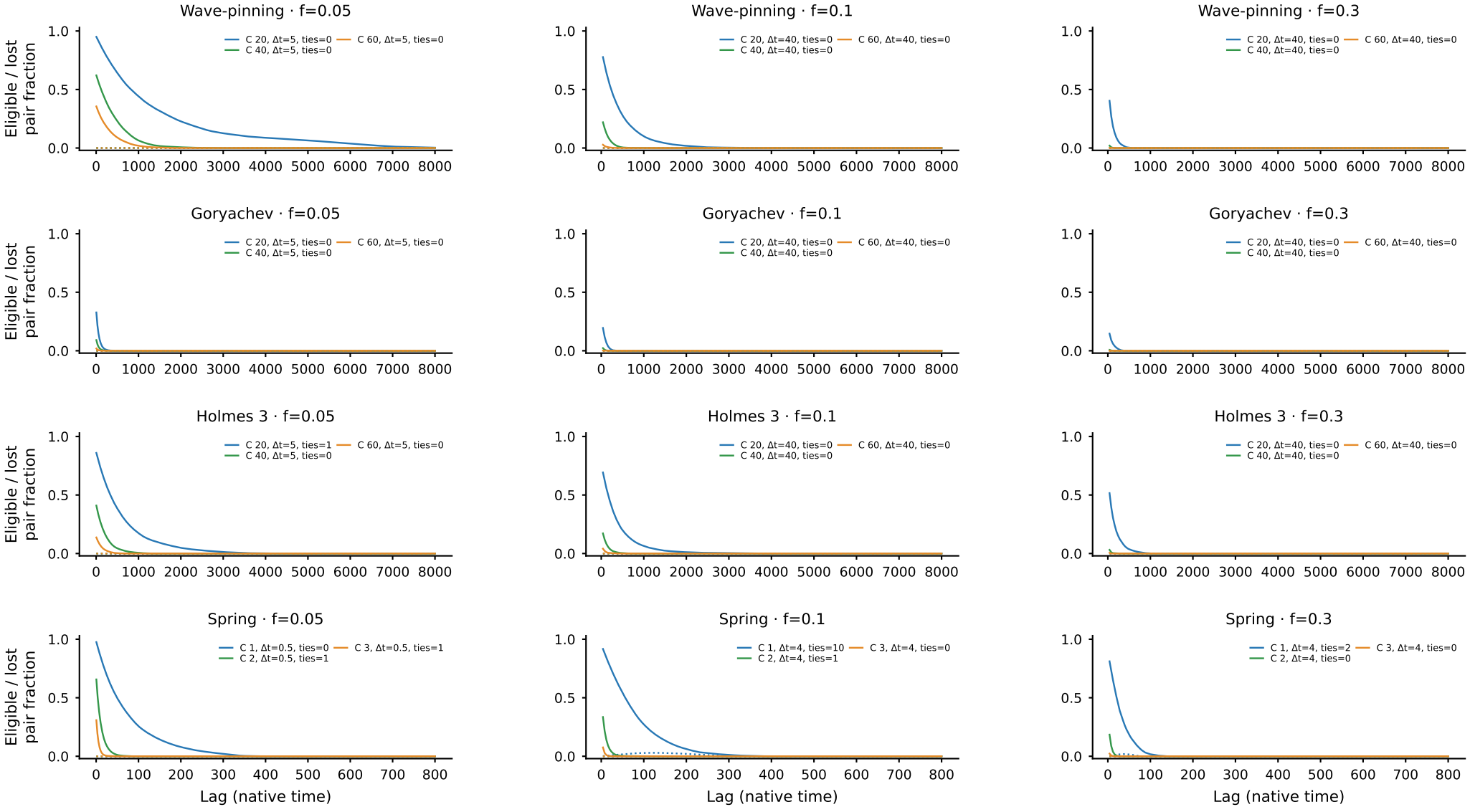
E2 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_020_S5_literal_sigma1_profile_peak_literal_actualC_MSD_4.csv

S5 · E2 literal material-origin physical-site measurements · Literal heading position · A3 primary split · pair coverage and ambiguity loss · σ=1



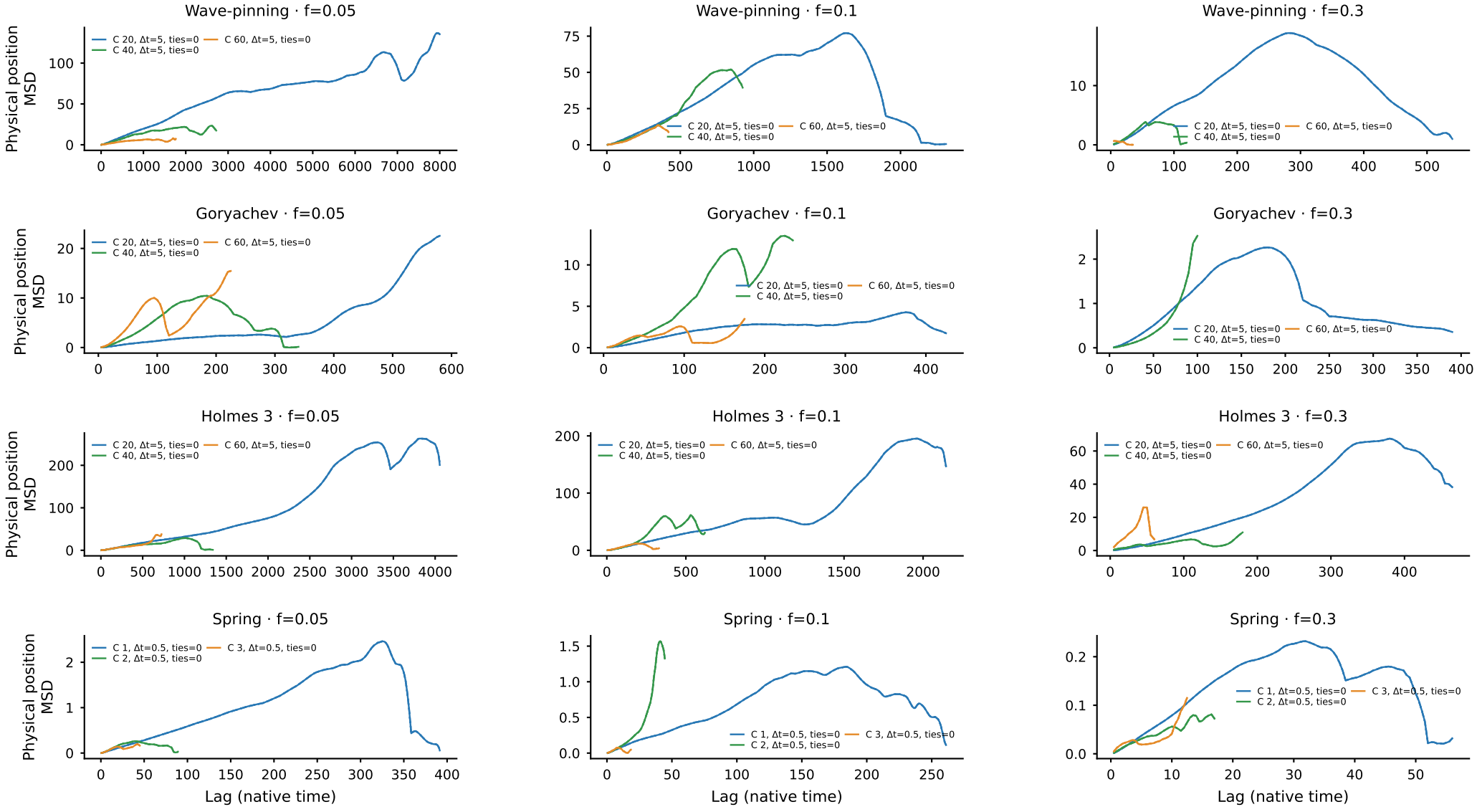
E2 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_021_S5_literal_sigma1_heading_literal_actualC_coverage_4.csv.gz

S5 · E2 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · pair coverage and ambiguity loss · σ=1



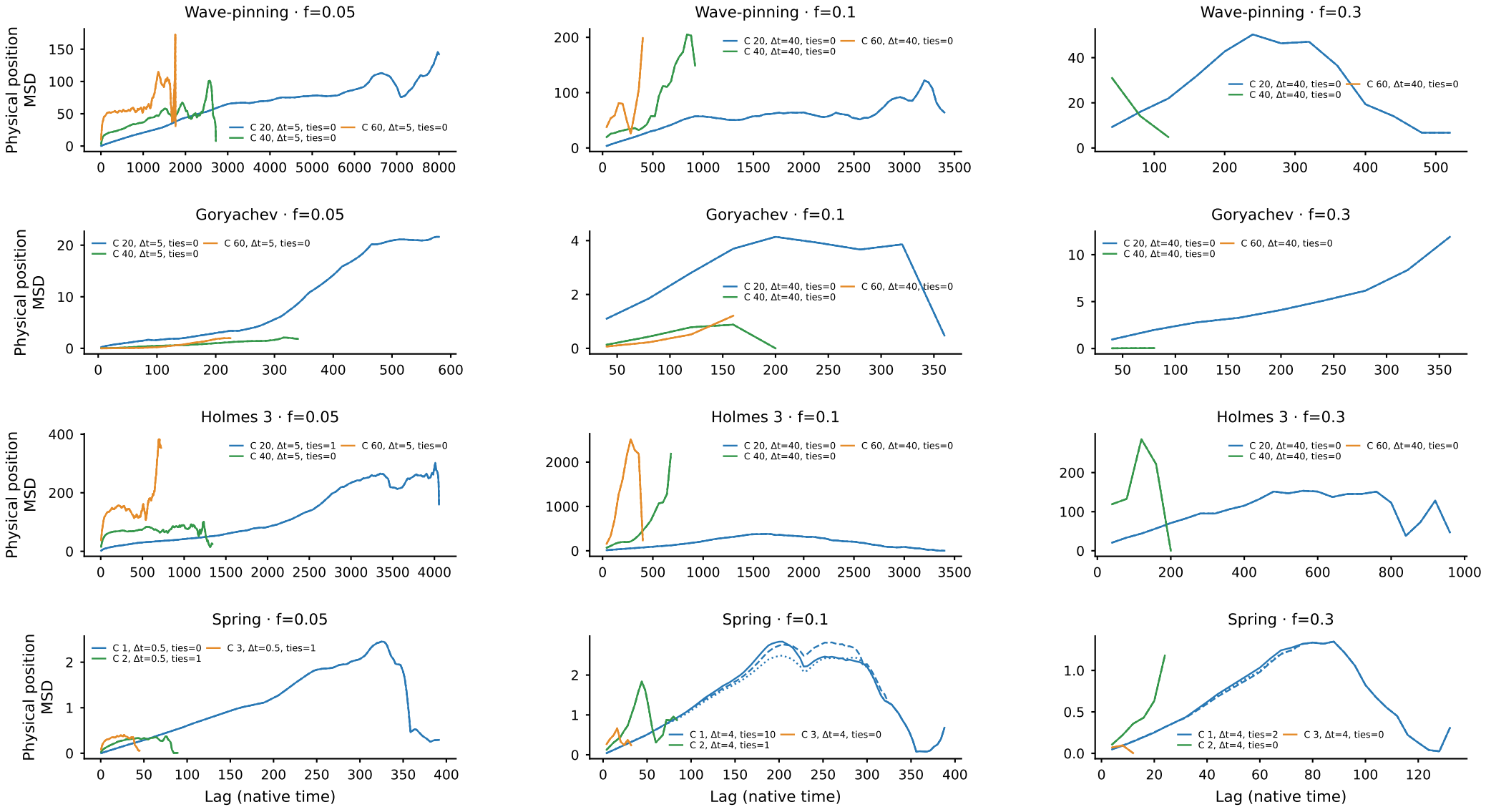
E2 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M₁|/|M₁,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_022_S5_literal_sigma1_profile_peak_literal_actualC_coverage_4.csv

S5 · E2 literal material-origin physical-site measurements · Literal heading position · paired route sensitivities · split / +π / −π · σ=1



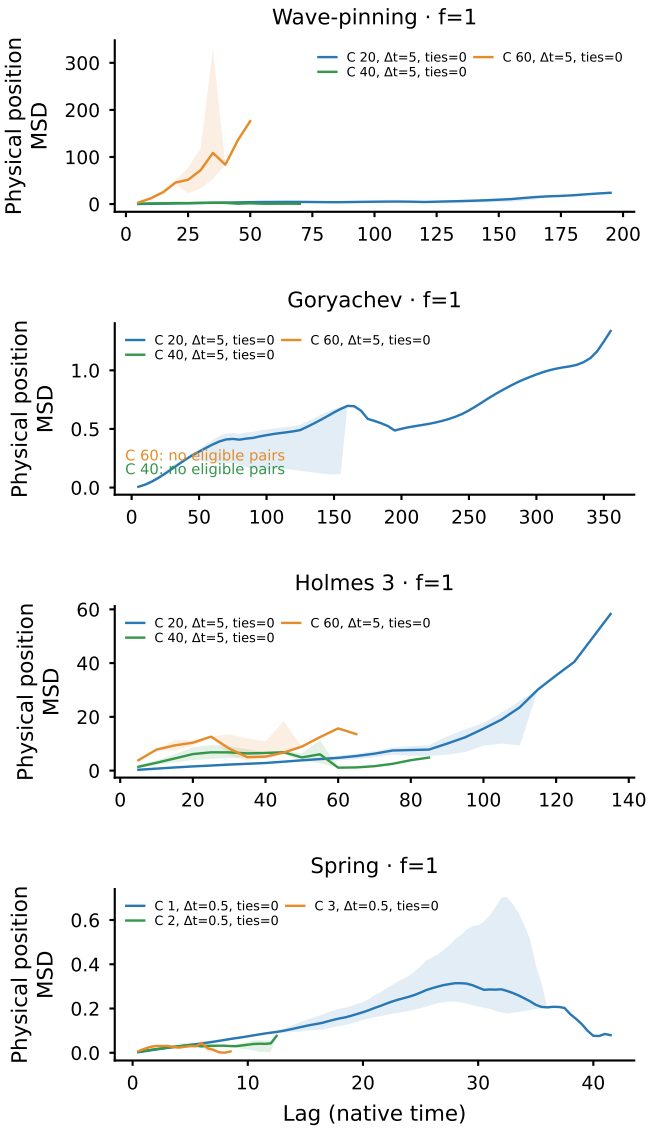
E2 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_023_S5_literal_sigma1_heading_literal_actualC_routes_4.csv.gz

S5 · E2 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · split / +π / −π · σ=1



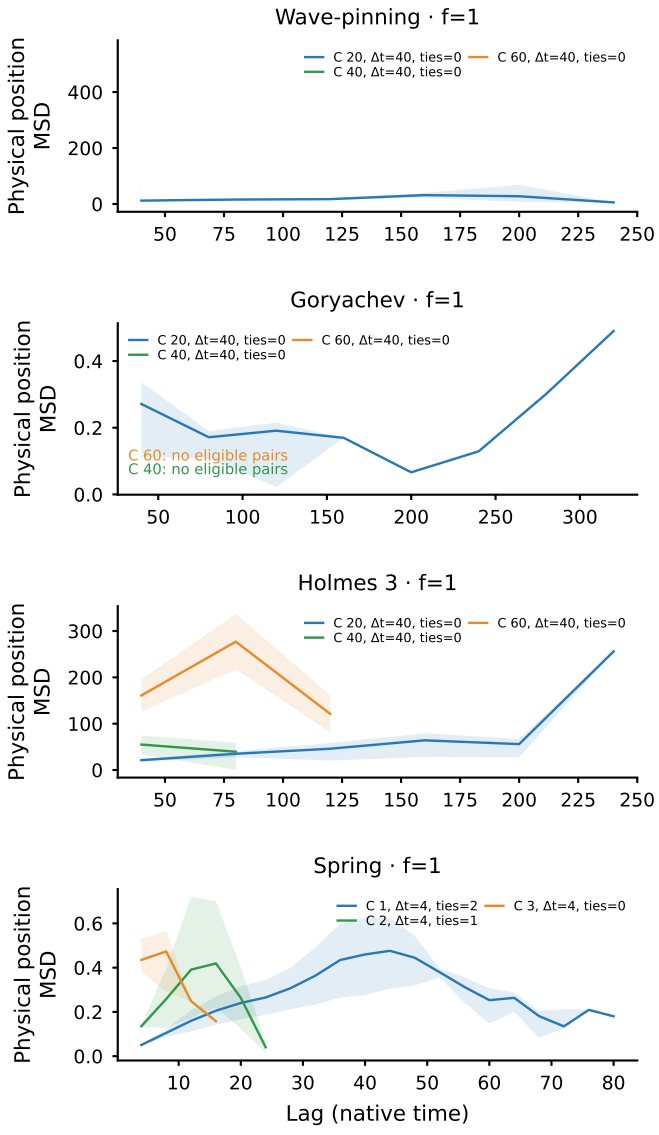
E2 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_024_S5_literal_sigma1_profile_peak_literal_actualC_routes_4.csv.gz

S5 · E2 literal material-origin physical-site measurements · Literal heading position · A3 primary split · σ=1



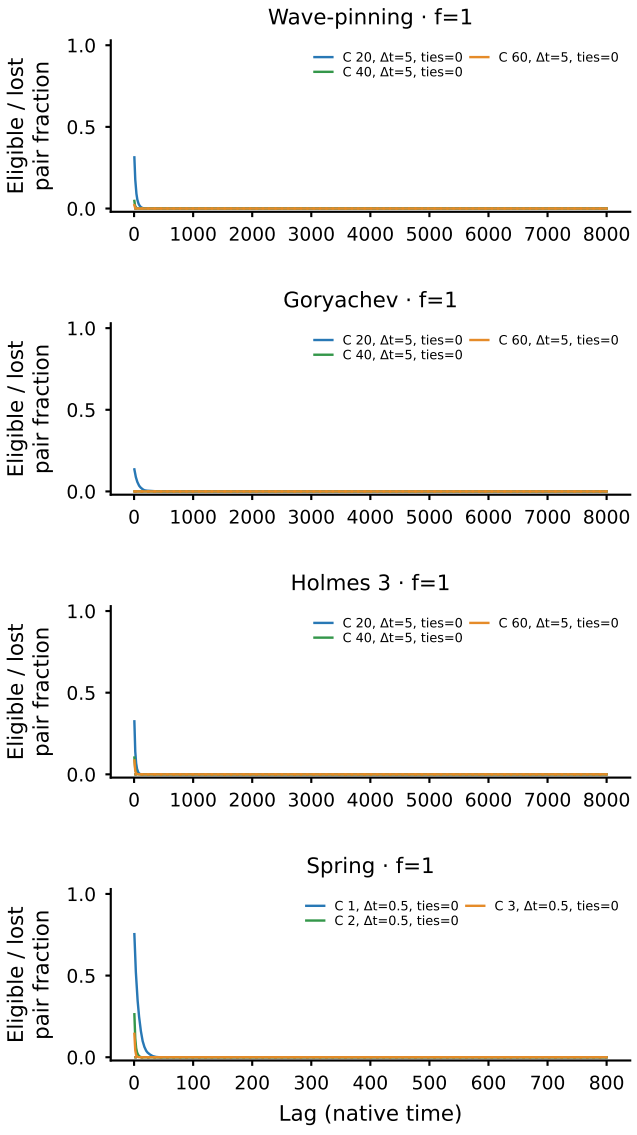
E2 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,basal}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_025_S5_literal_sigma1_heading_literal_actualC_MSD_5.csv

S5 · E2 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · σ=1



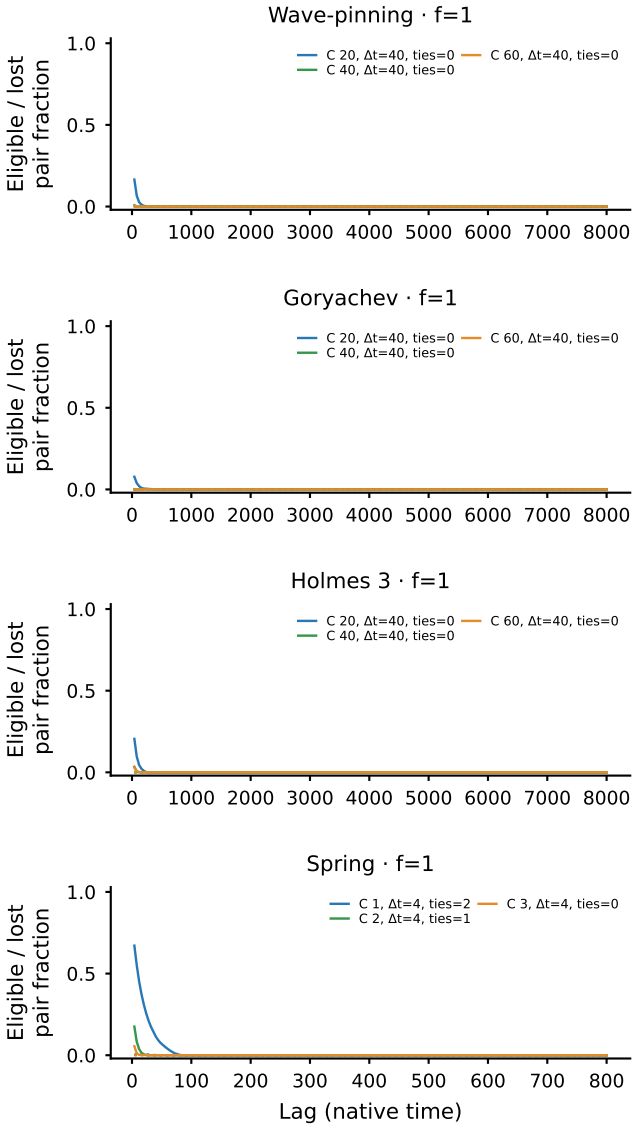
E2 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M₁|/|M₁,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_026_S5_literal_sigma1_profile_peak_literal_actualC_MSD_5.csv

S5 · E2 literal material-origin physical-site measurements · Literal heading position · A3 primary split · pair coverage and ambiguity loss · σ=1



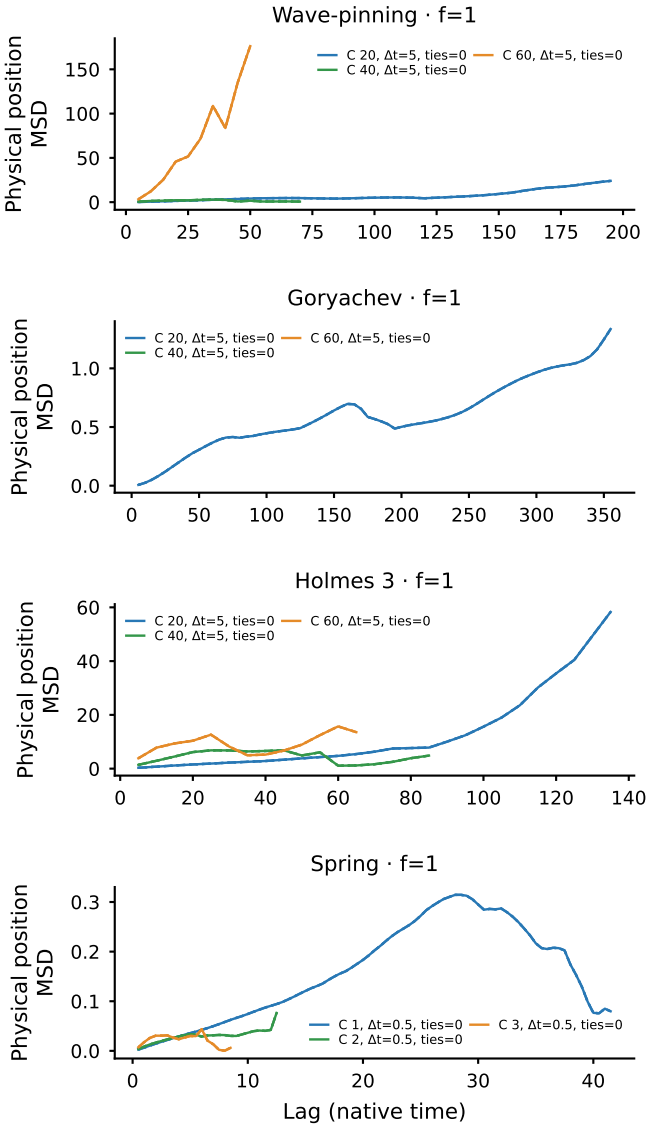
E2 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_027_S5_literal_sigma1_heading_literal_actualC_coverage_5.csv

S5 · E2 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · pair coverage and ambiguity loss · σ=1



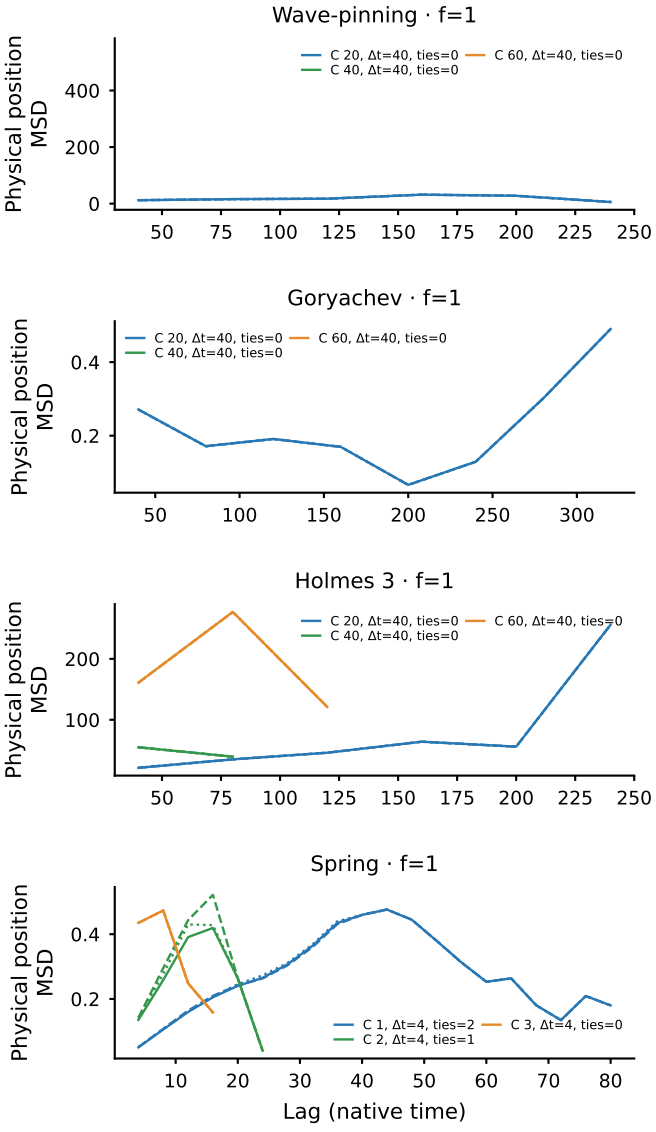
E2 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses R=|M_1|/|M_1,basal|, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_028_S5_literal_sigma1_profile_peak_literal_actualC_coverage_5.csv

S5 · E2 literal material-origin physical-site measurements · Literal heading position · paired route sensitivities · split / +π / −π · σ=1



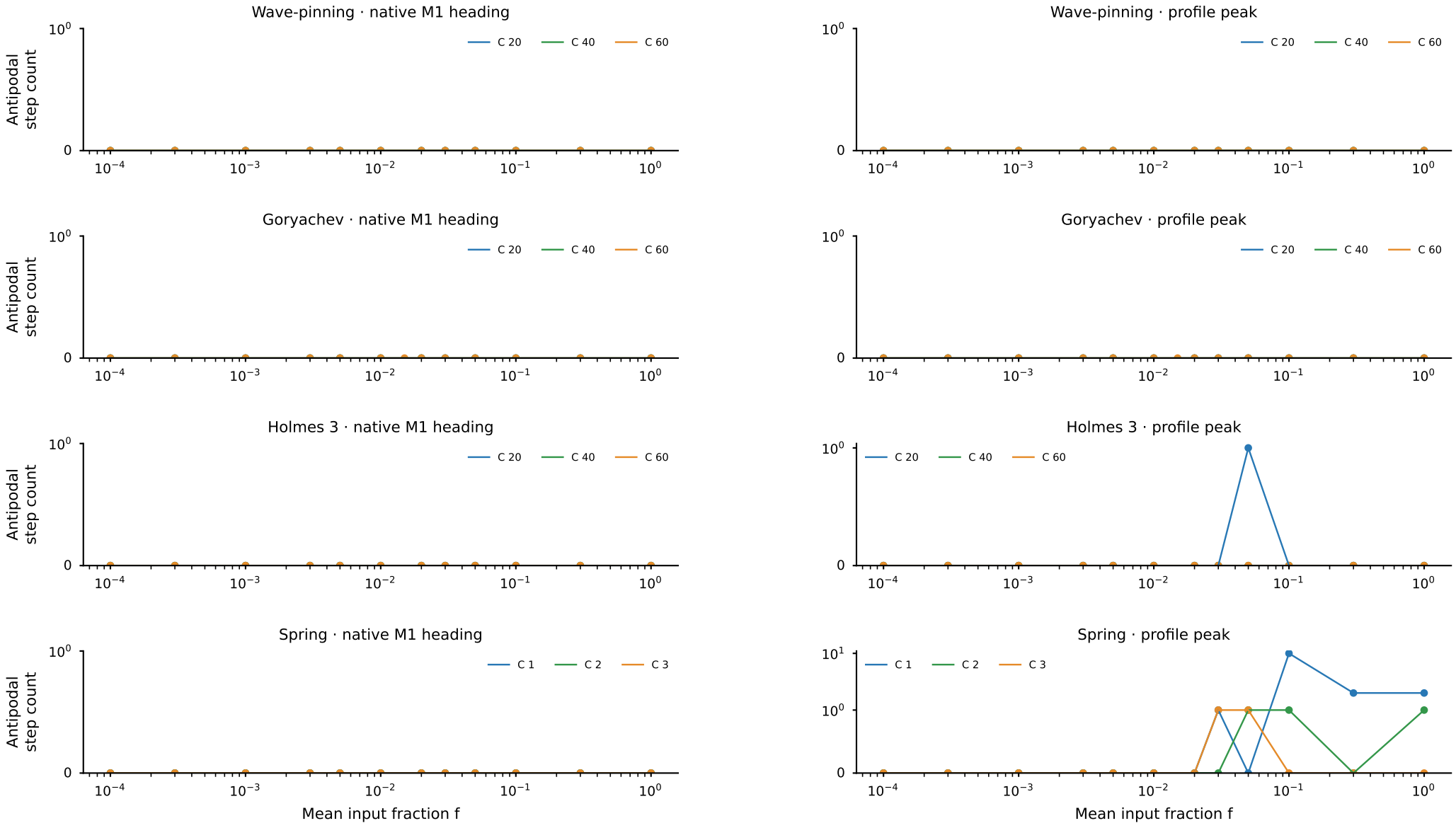
E2 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,\mathrm{basal}}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_029_S5_literal_sigma1_heading_literal_actualC_routes_5.csv.gz

S5 · E2 literal material-origin physical-site measurements · Literal profile-maximum position · sensitivity · split / +π / −π · σ=1



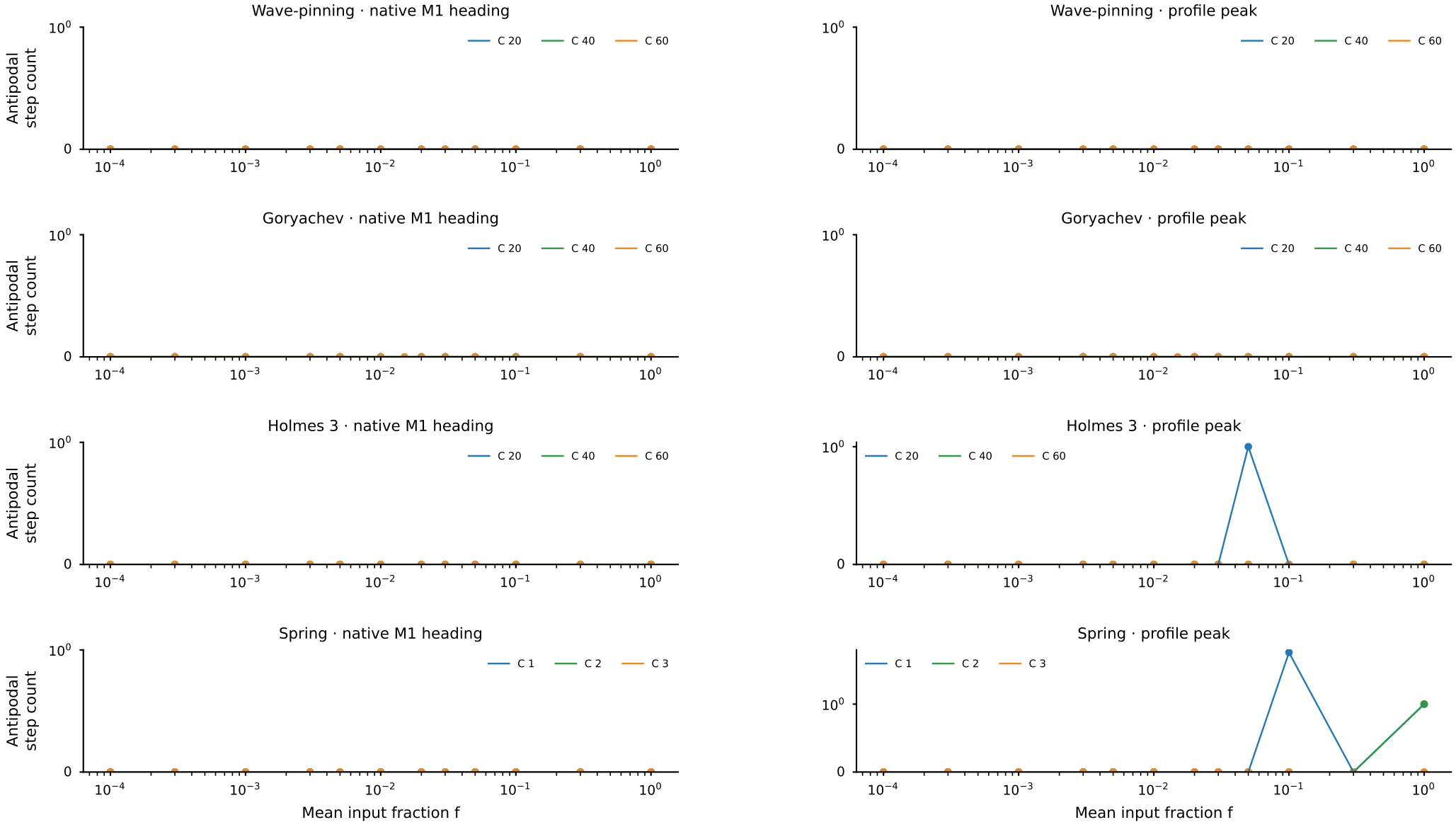
E2 literal material-origin physical-site measurements. Literal x=θC(t)/(2π) uses native material origin q=0 and section initial angle in [0,2π). Primary tracks split at unresolved antipodal steps (tolerance 10^{-12} rad); both native frames remain. Native M_1 phase uses $R=|M_1|/|M_{1,basal}|$, contrast and single-region qualification. Solid lines split tracks; dotted pair coverage shows ambiguity loss. Route panels compare solid split, dashed +π and dotted −π: paired deterministic sensitivities, not measured directions, independent seeds or uncertainty bounds. Legends give actual cadence Δt and tie count; peak sampling may equal scalar cadence. Shading is a 2000-whole-seed interval; unavailable lags remain gaps. Grey marks qualified forced conditions when canonical flags are supplied; checkpoints remain descriptive. These qualified single-region site MSDs differ from the global-heading-derived Dθ→Dx proxy, which is not one tracked site's diffusion in multi-region states. No migration law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_030_S5_literal_sigma1_profile_peak_literal_actualC_routes_5.csv

S5 · E2 literal material-origin physical-site measurements · Native antipodal ambiguity · main · σ=1



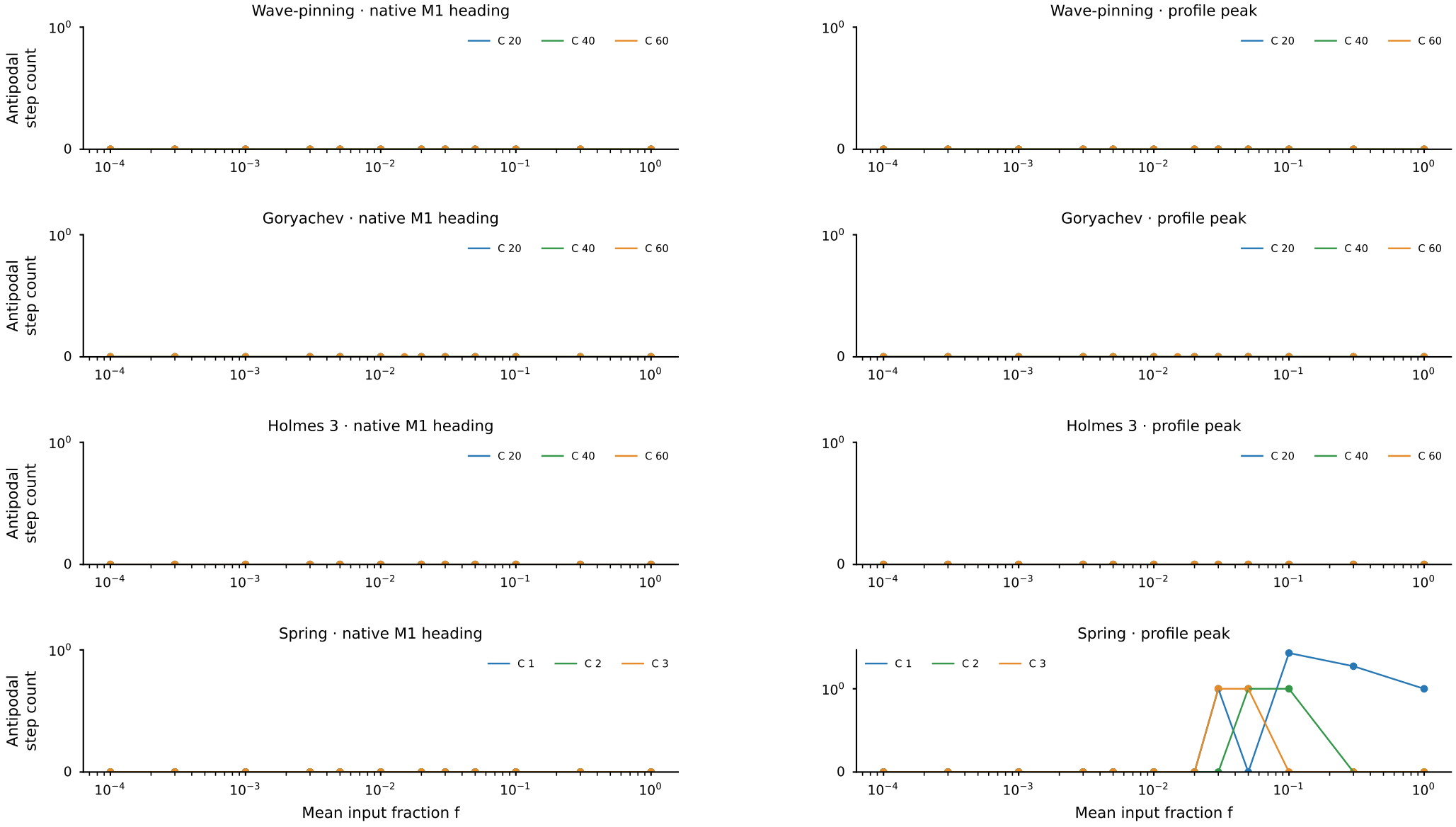
E2 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_031_S5_ambiguity_sigma1_main.csv

S5 · E2 literal material-origin physical-site measurements · Native antipodal ambiguity · analysis first half · σ=1



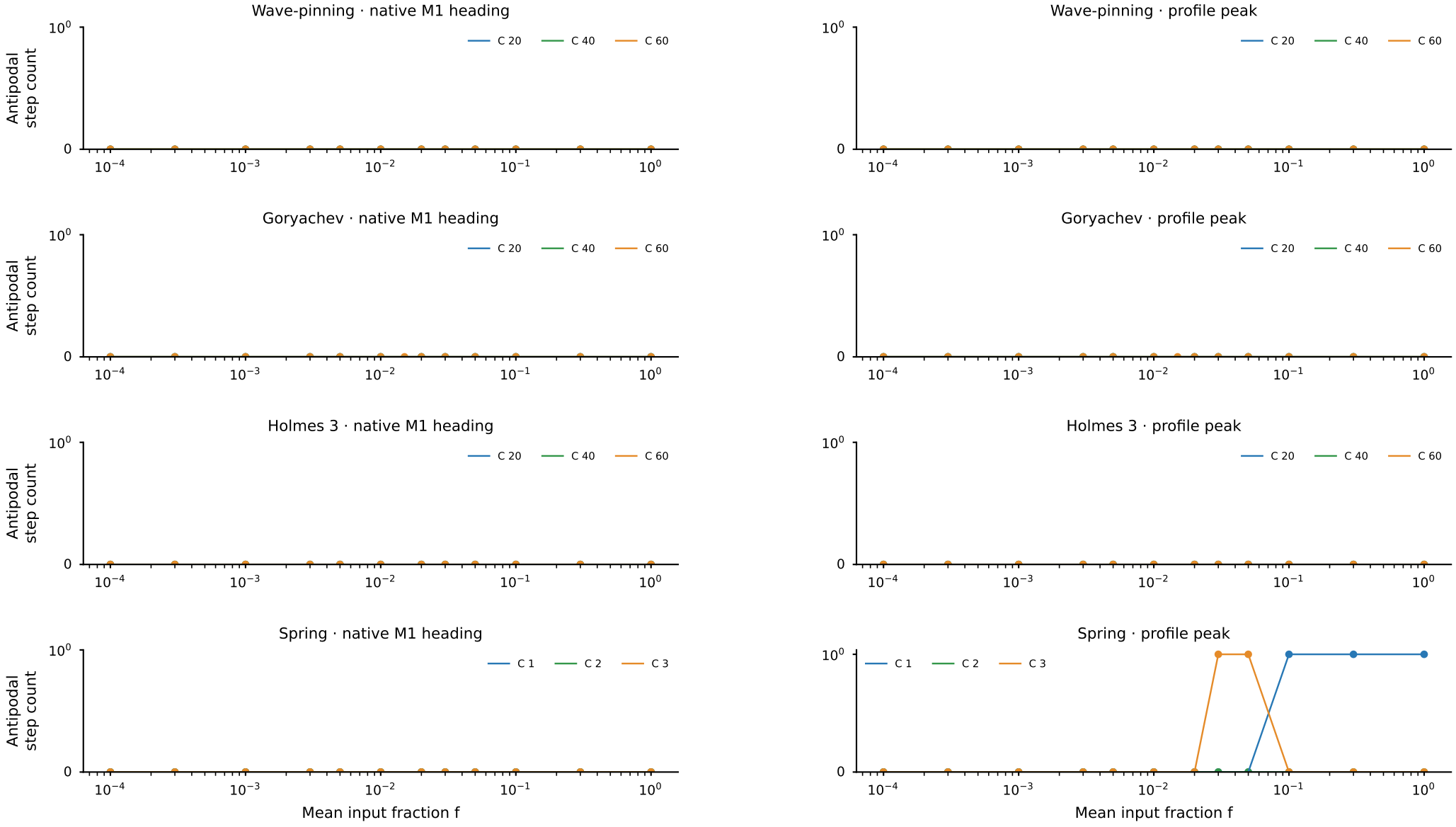
E2 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_032_S5_ambiguity_sigma1_analysis_first_half.csv

S5 · E2 literal material-origin physical-site measurements · Native antipodal ambiguity · analysis second half · σ=1



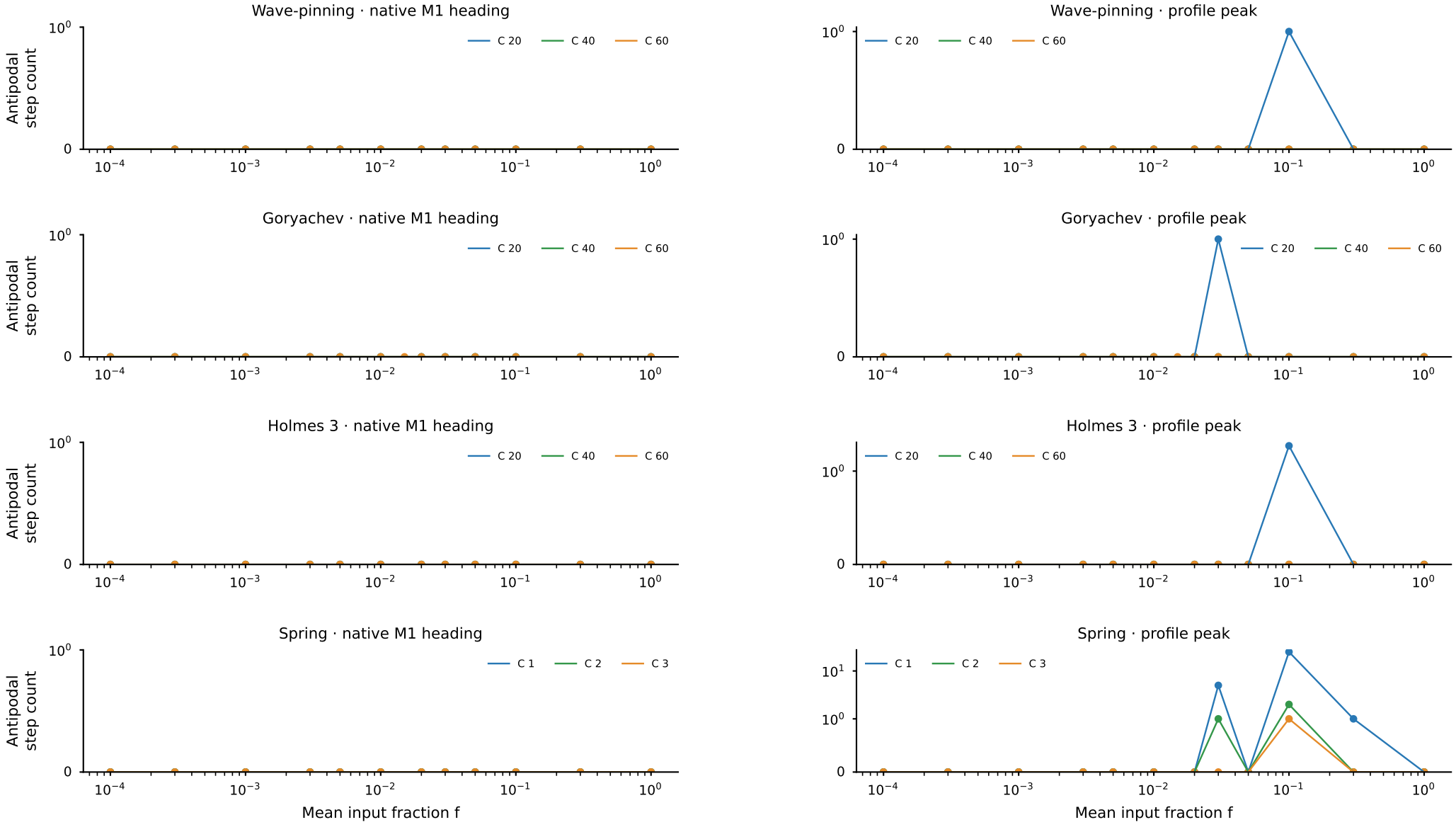
E2 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_033_S5_ambiguity_sigma1_analysis_second_half.csv

S5 · E2 literal material-origin physical-site measurements · Native antipodal ambiguity · late quarter · σ=1



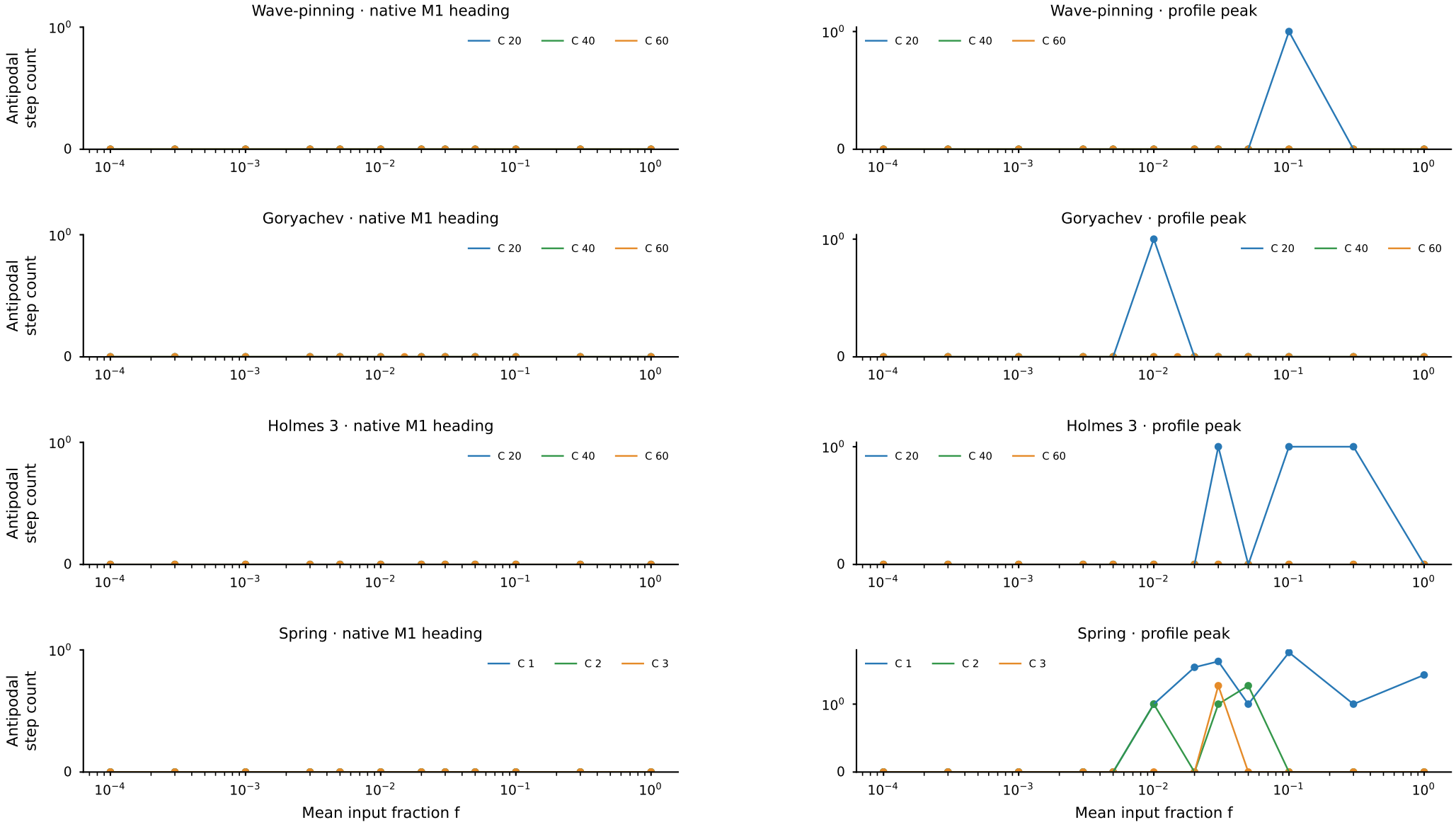
E2 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_034_S5_ambiguity_sigma1_late_quarter.csv

S5 · E2 literal material-origin physical-site measurements · Native antipodal ambiguity · cadence x2 · σ=1



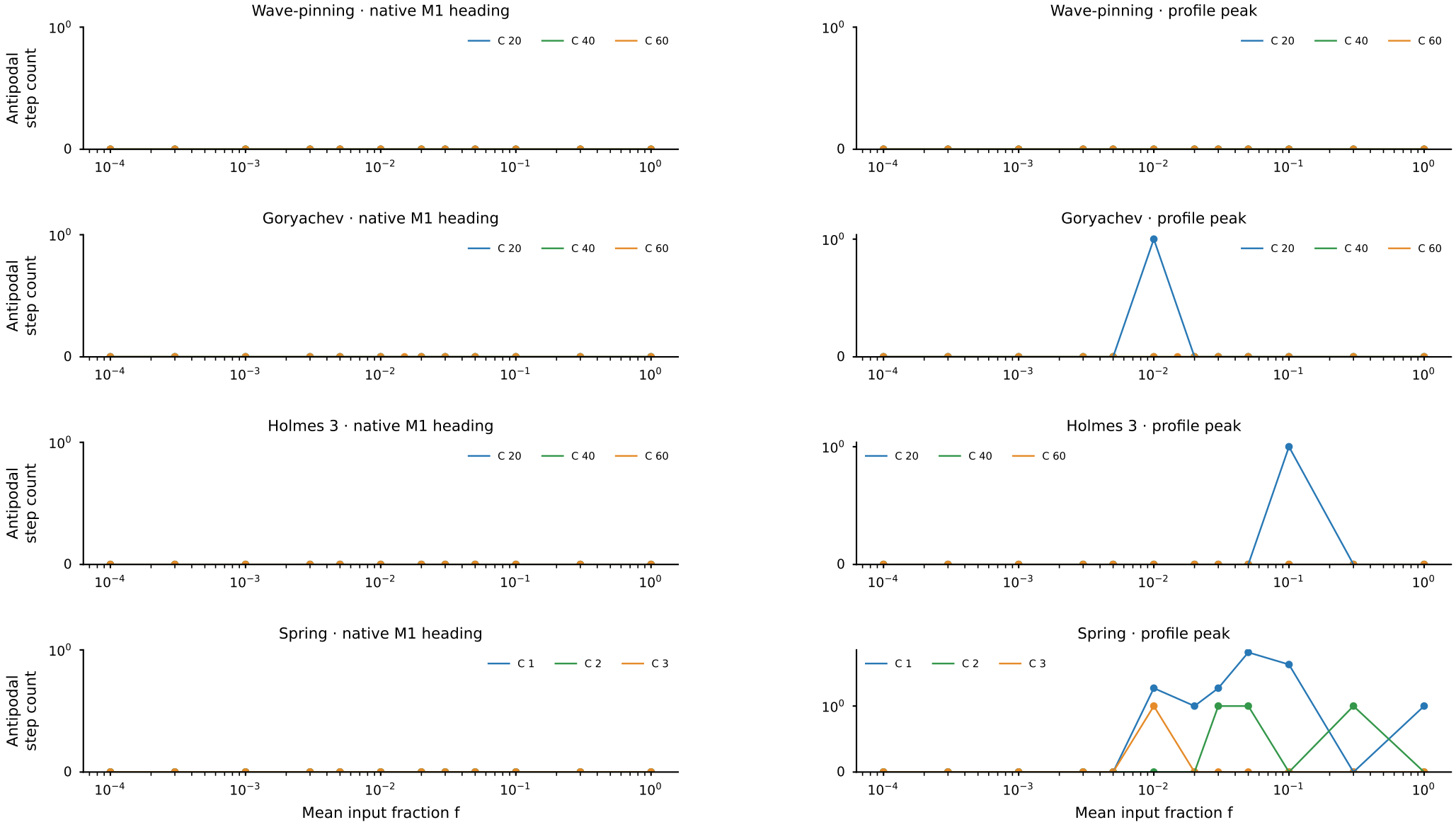
E2 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_035_S5_ambiguity_sigma1_cadence_x2.csv

S5 · E2 literal material-origin physical-site measurements · Native antipodal ambiguity · cadence x4 · σ=1

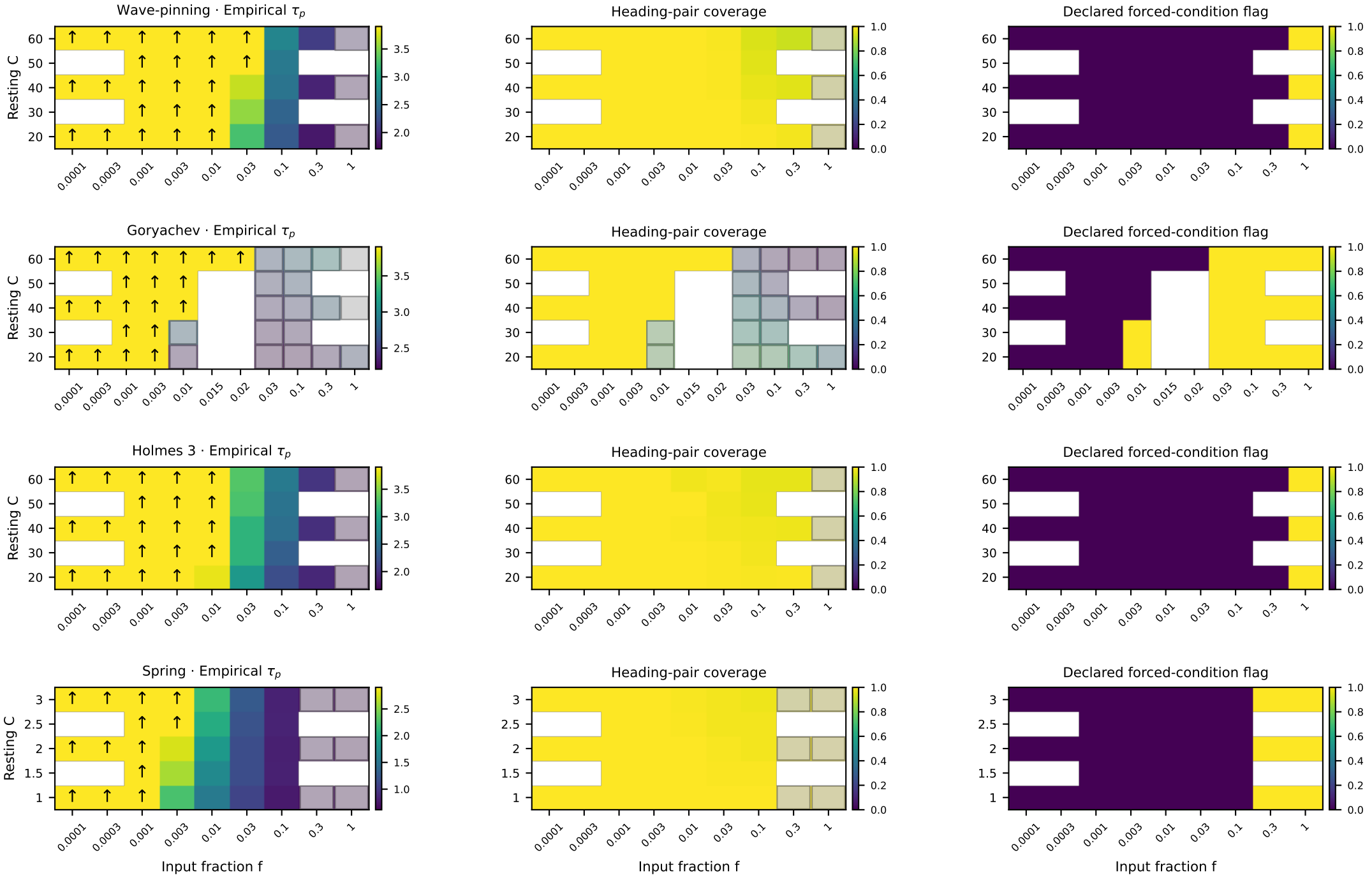


E2 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_036_S5_ambiguity_sigma1_cadence_x4.csv

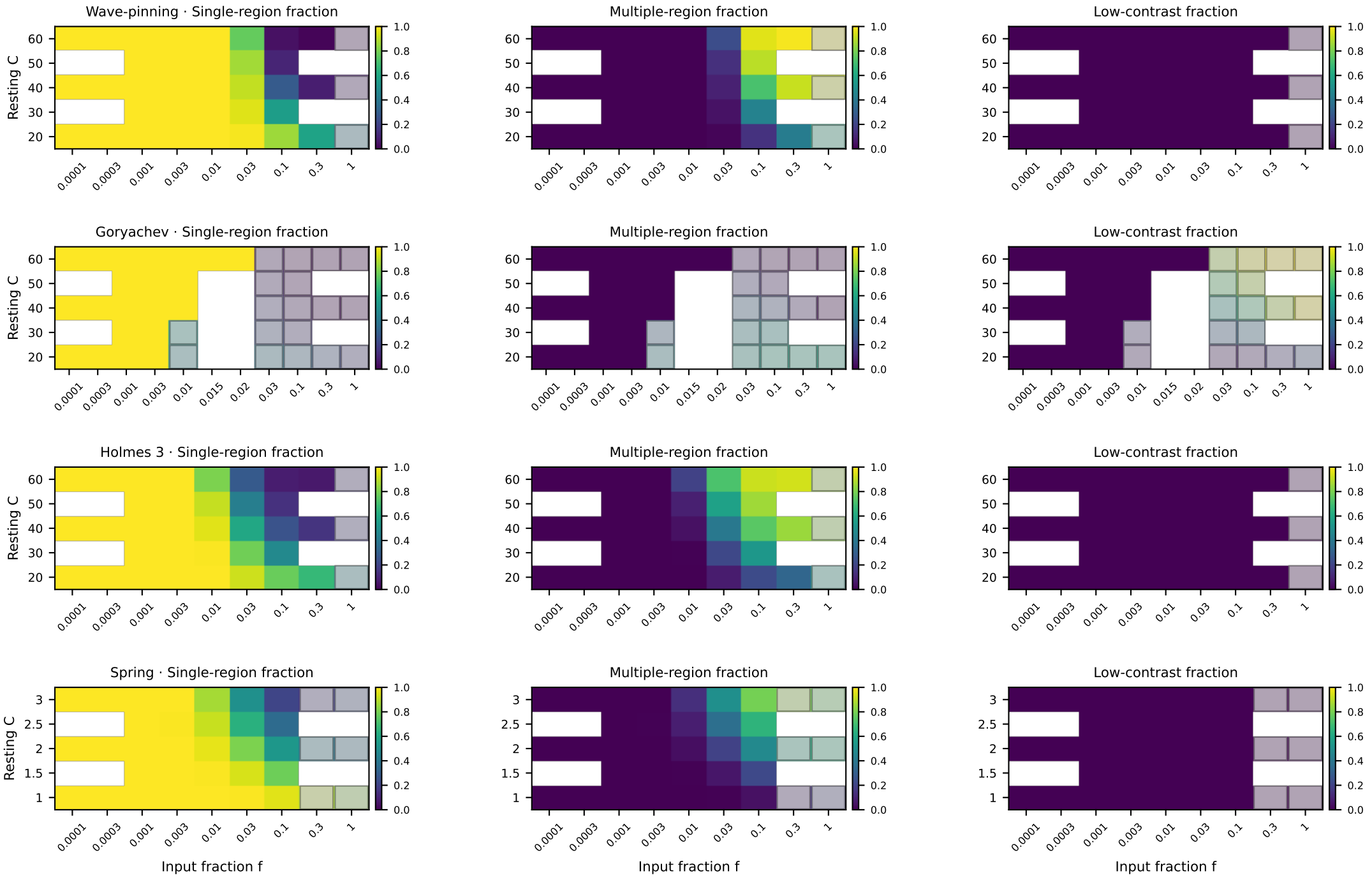
S5 · E2 literal material-origin physical-site measurements · Native antipodal ambiguity · cadence x8 · σ=1



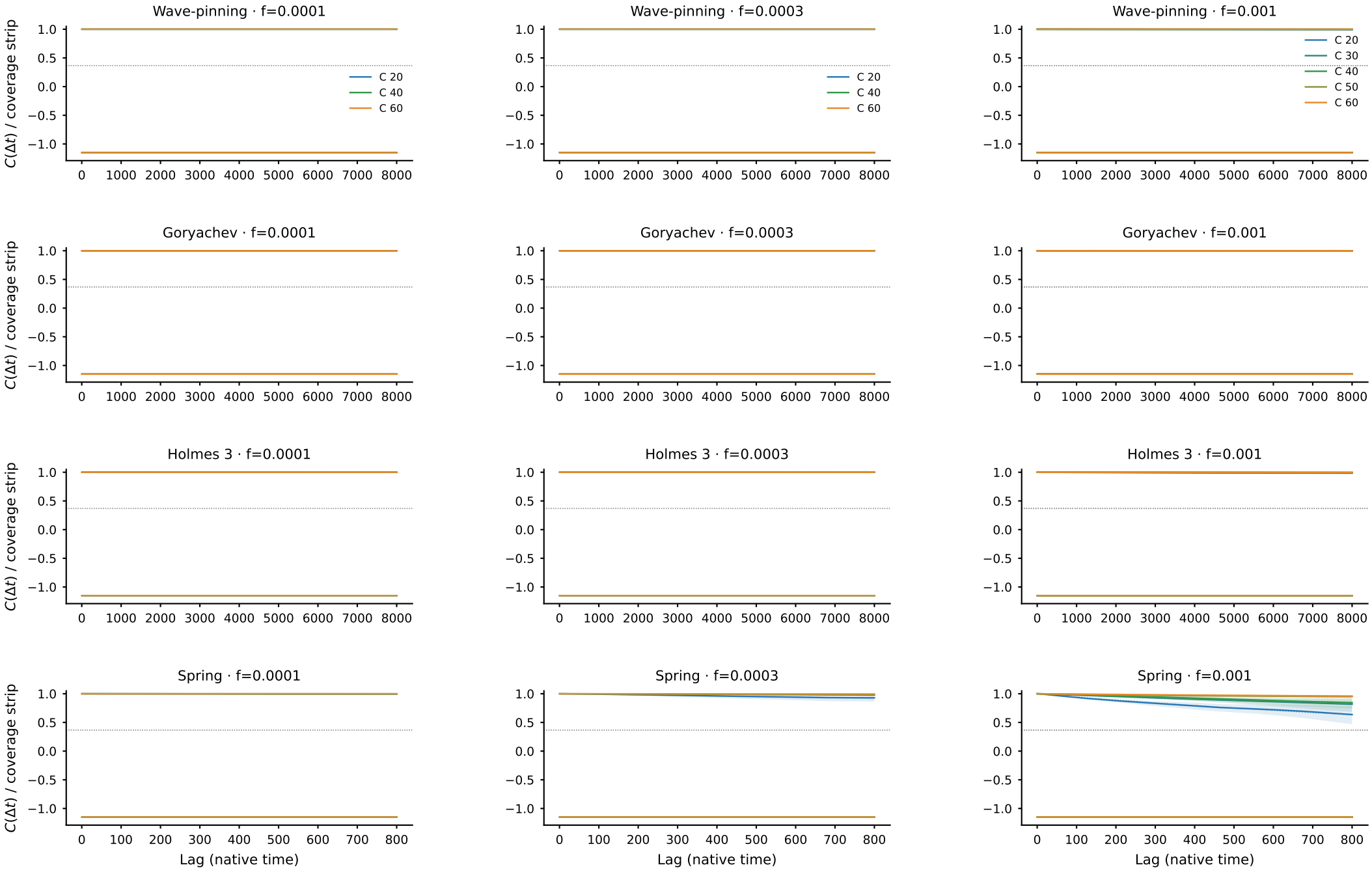
E2 literal material-origin physical-site measurements. Counts sum unresolved sampled antipodal edges across the existing whole-seed ensemble; stage contexts can overlap and do not add independent seeds. All seven declared views are retained. Primary literal coordinates split track labels at each tie, preserving both endpoint frames; eligible pair loss is shown in the paired coverage pages and retained at every lag in the source tables. Plus/minus-π routes are paired deterministic sensitivities. Exact native cadence, qualified/total frames, adjacent steps, and track counts before/after splitting remain in each case's page CSV. Zero means no observed tied step at this sampling, not continuous-path identifiability. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S5_literal_E2_final_layout_20261020_037_S5_ambiguity_sigma1_cadence_x8.csv



E3 completed native observations. All declared input strengths in this observation arm are shown. Grey overlays retain forced conditions; the exact display flag and its components remain in CSV. Arrows preserve censoring and cadence limits. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_E3_sizes_final_layout_20261020_001_S2_1.csv

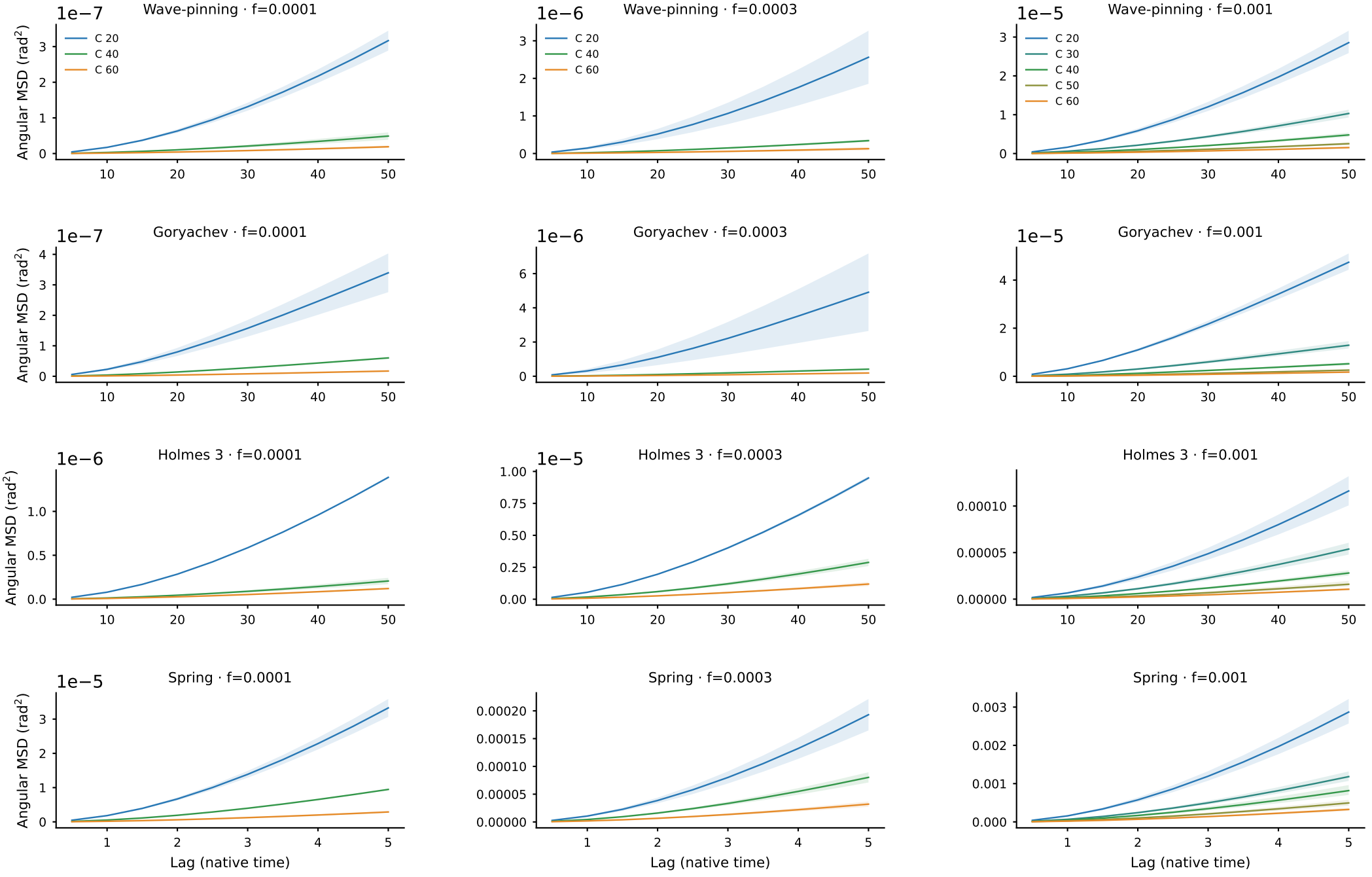


E3 completed native observations. All declared input strengths in this observation arm are shown. Grey overlays retain forced conditions; the exact display flag and its components remain in CSV. Arrows preserve censoring and cadence limits. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_E3_sizes_final_layout_20261020_002_S2_2.csv

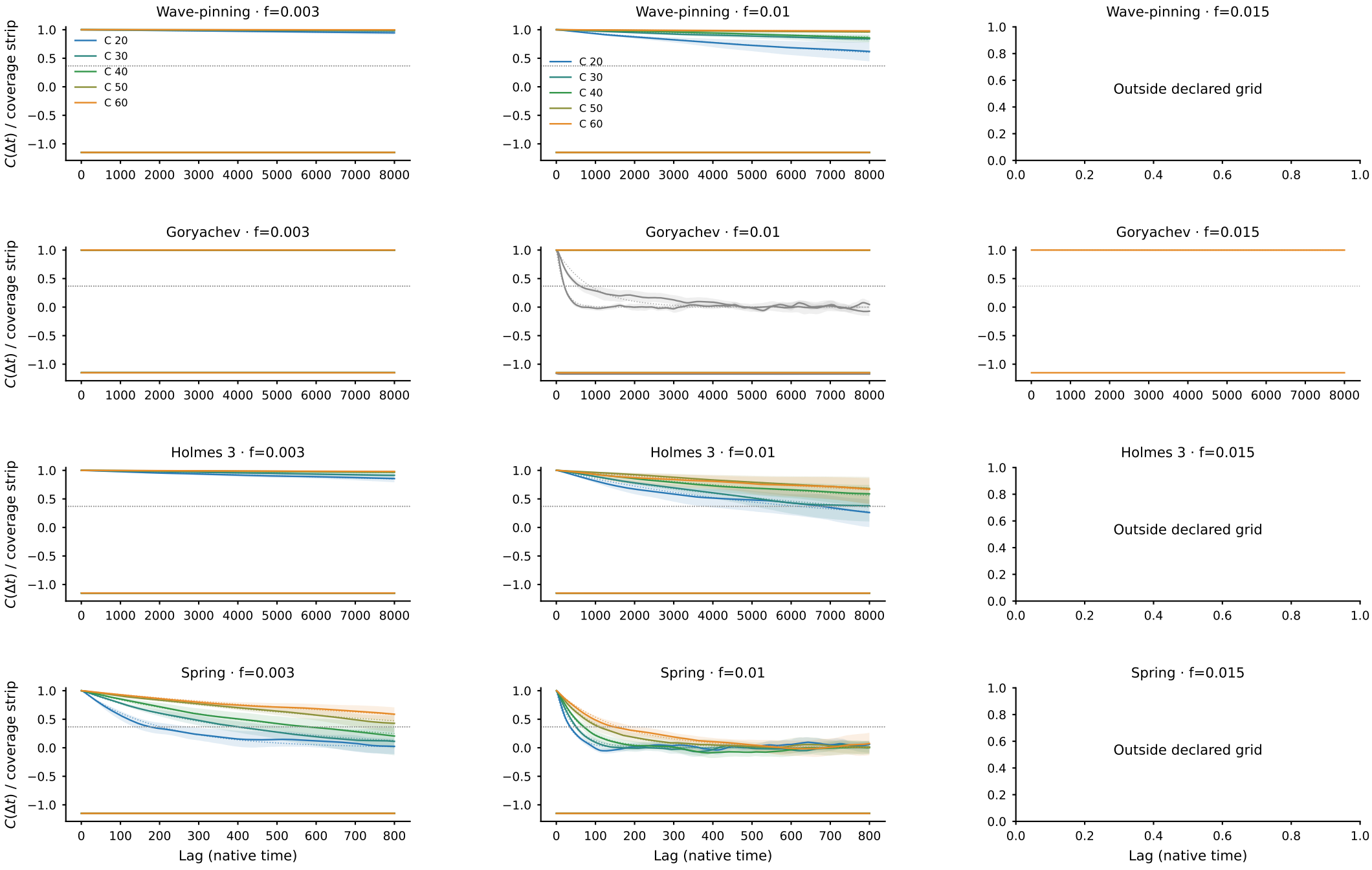


E3 completed native observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_E3_sizes_final_layout_20261020_003_S3_1_cosine.csv.gz

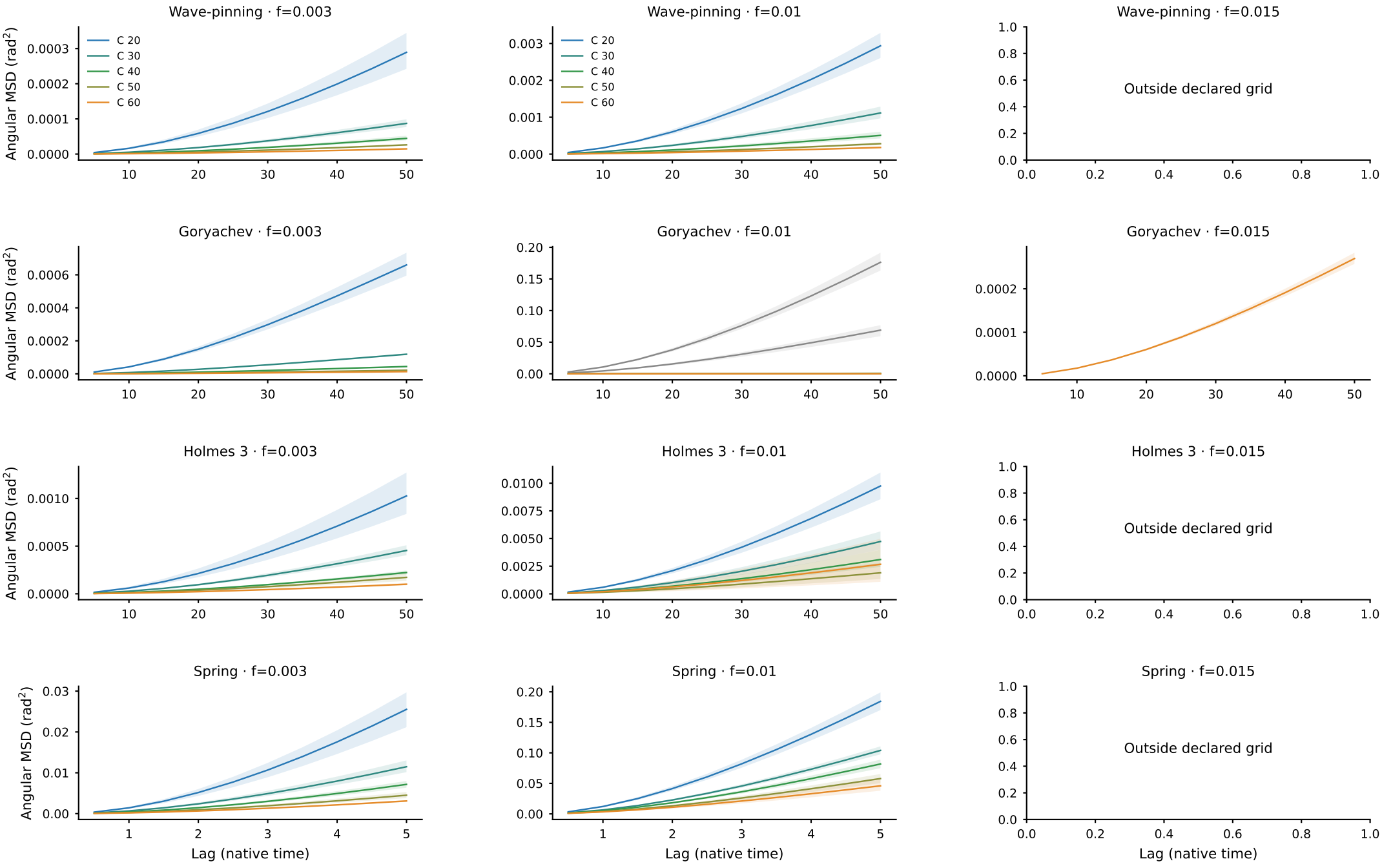
S3 · E3 completed native observations · Short-lag angular MSD



E3 completed native observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_E3_sizes_final_layout_20261020_004_S3_1_angular_msd.csv.gz

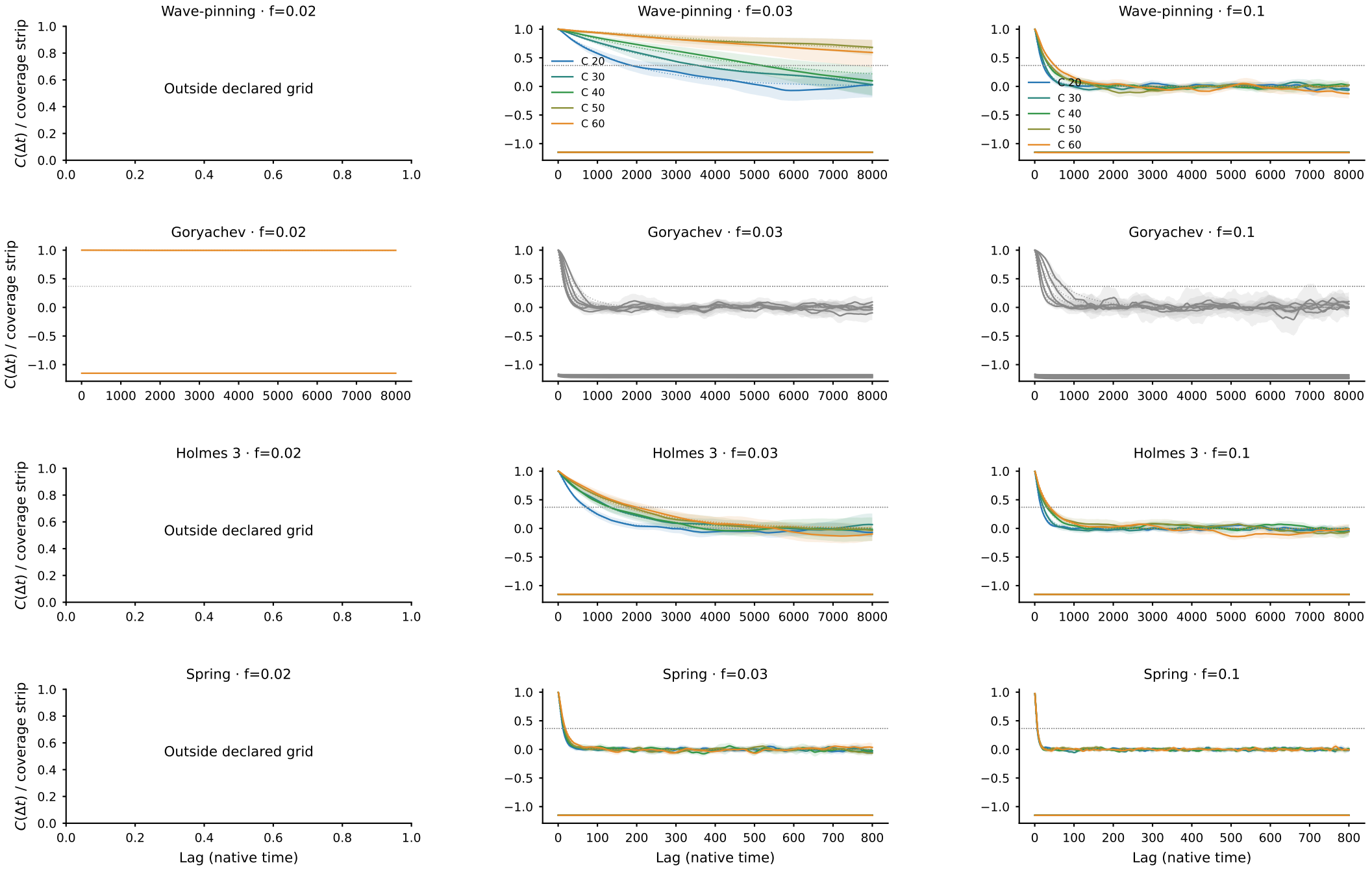


E3 completed native observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_E3_sizes_final_layout_20261020_005_S3_2_cosine.csv.gz

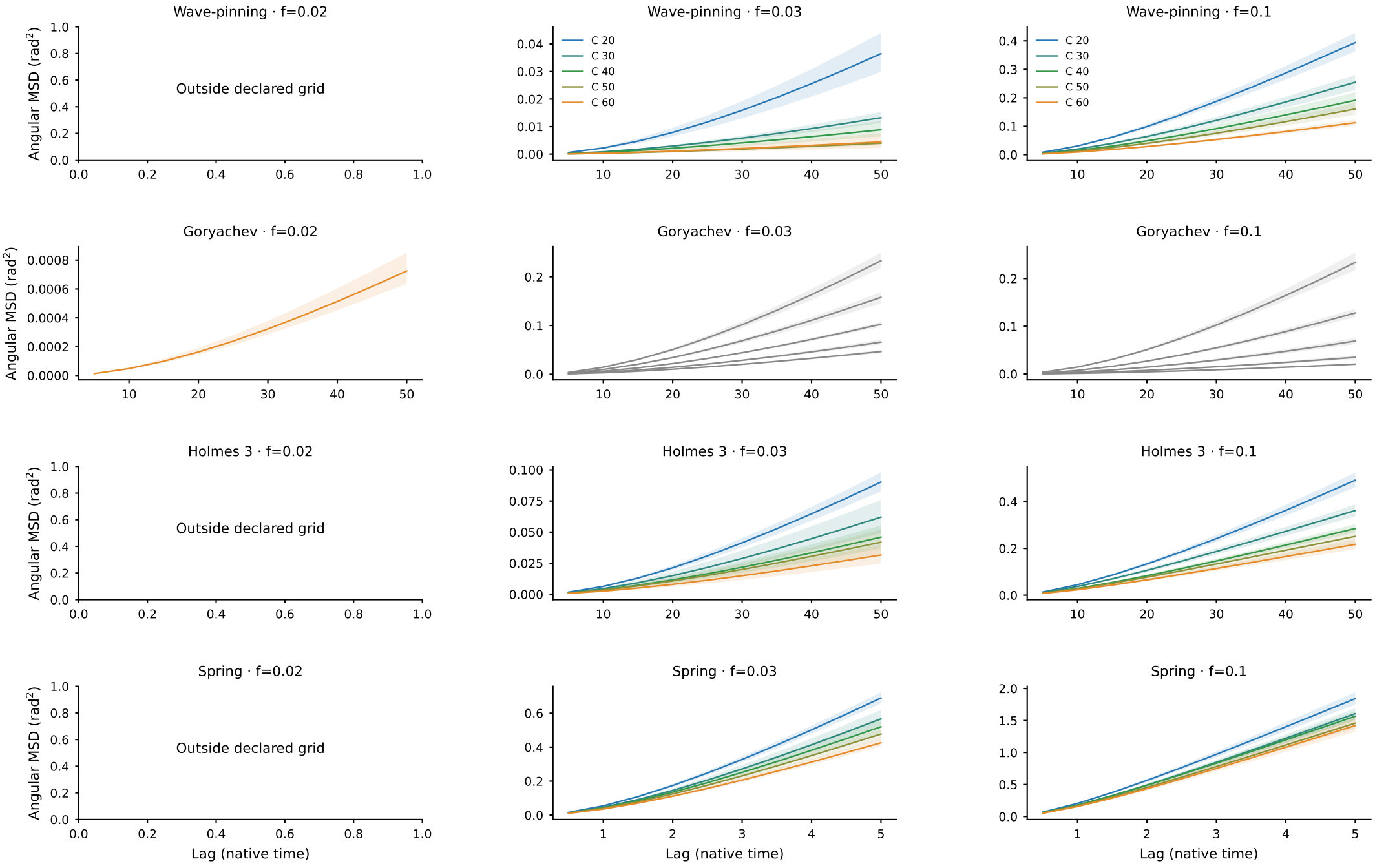


E3 completed native observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_E3_sizes_final_layout_20261020_006_S3_2_angular_msd.csv.gz

S3 · E3 completed native observations · All heading correlations

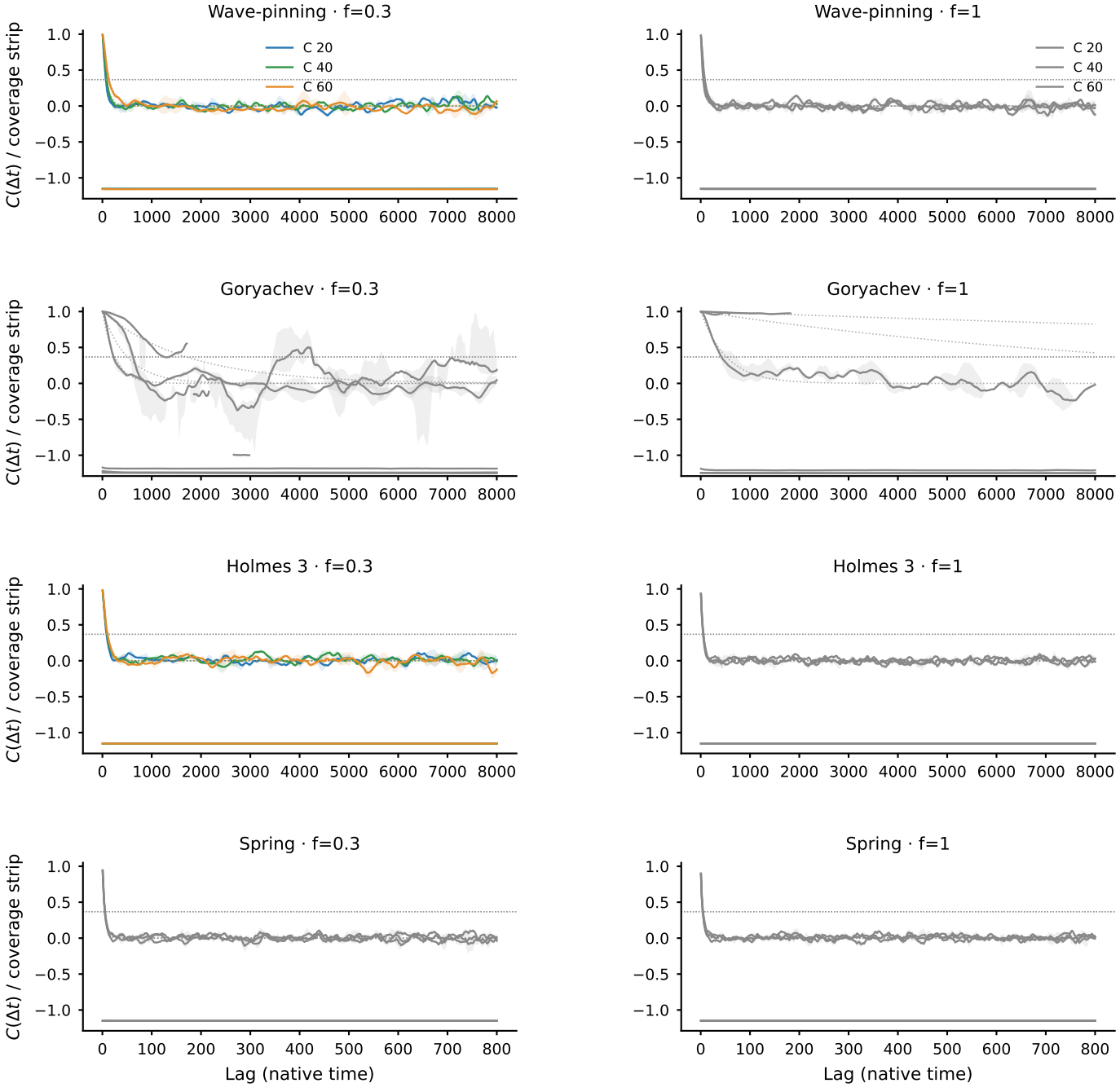


E3 completed native observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_E3_sizes_final_layout_20261020_007_S3_3_cosine.csv.gz



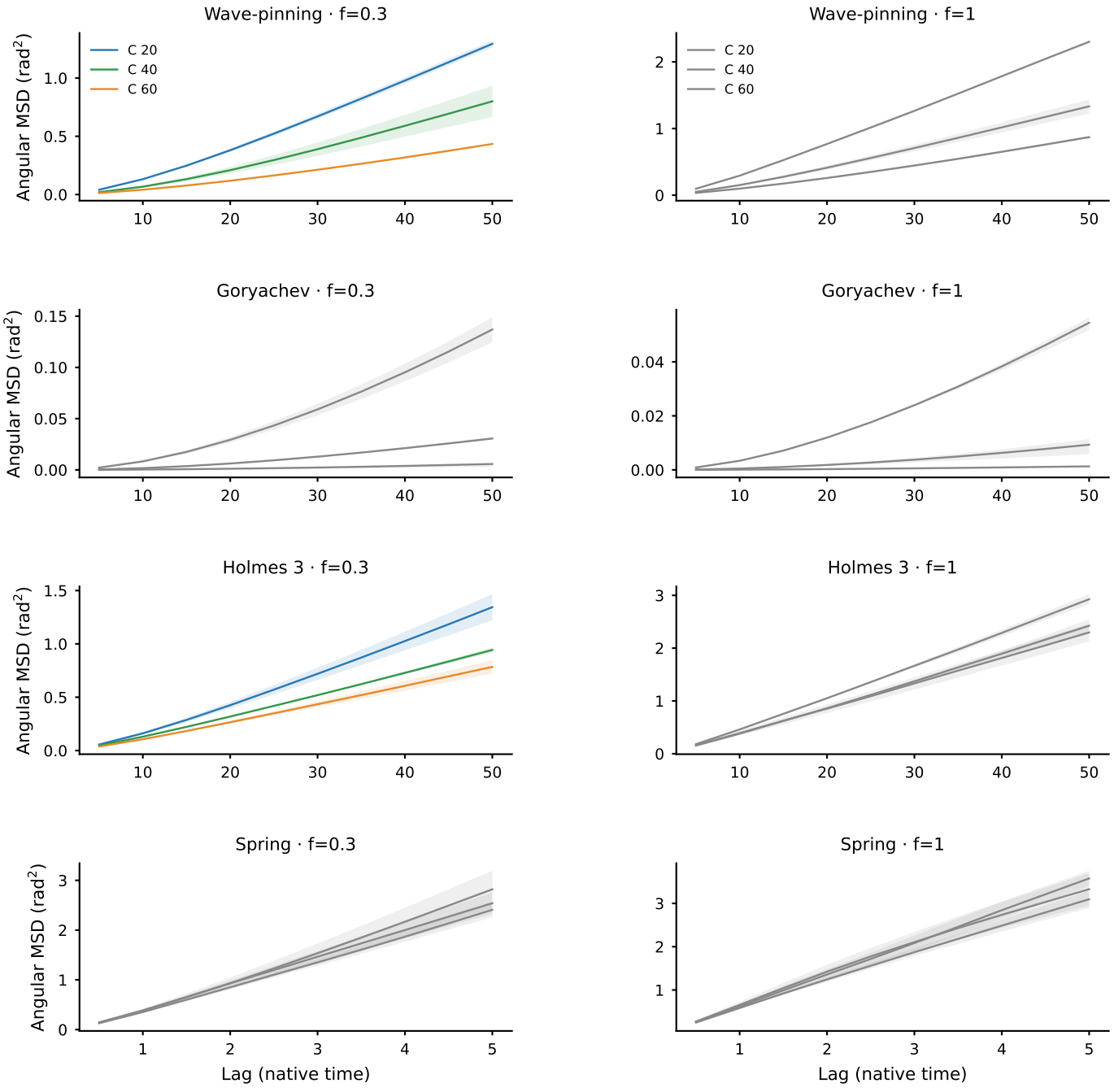
E3 completed native observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_E3_sizes_final_layout_20261020_008_S3_3_angular_msd.csv.gz

S3 · E3 completed native observations · All heading correlations



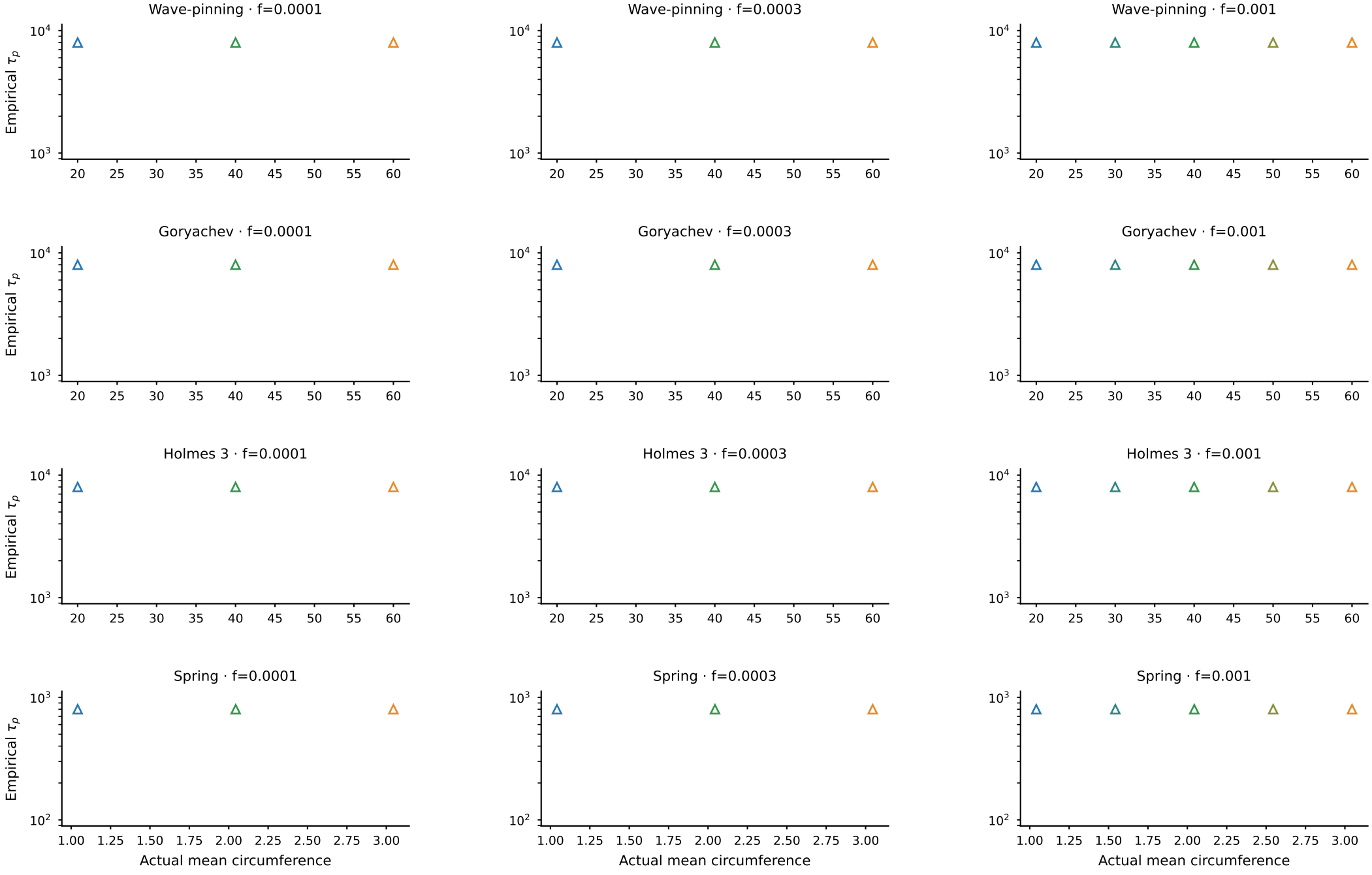
E3 completed native observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_E3_sizes_final_layout_20261020_009_S3_4_cosine.csv

S3 · E3 completed native observations · Short-lag angular MSD



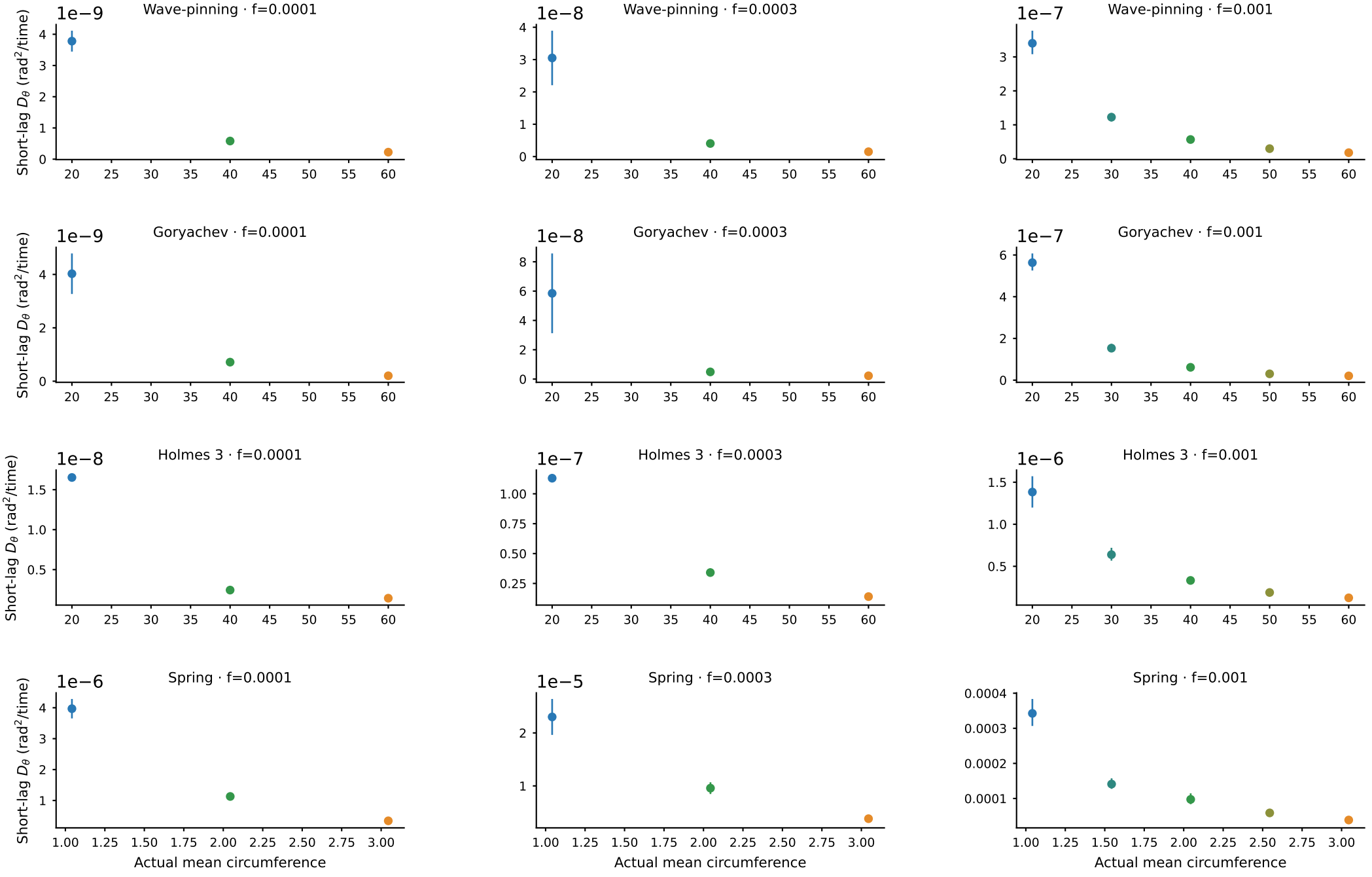
E3 completed native observations. Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. The full cosine range [−1,1] remains visible; coverage strips span −1.25 to −1.15 for coverage 0 to 1. MSD tracks never bridge invalid headings. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_E3_sizes_final_layout_20261020_010_S3_4_angular_msd.csv

S4 · E3 completed native observations · Persistence and angular mobility across size

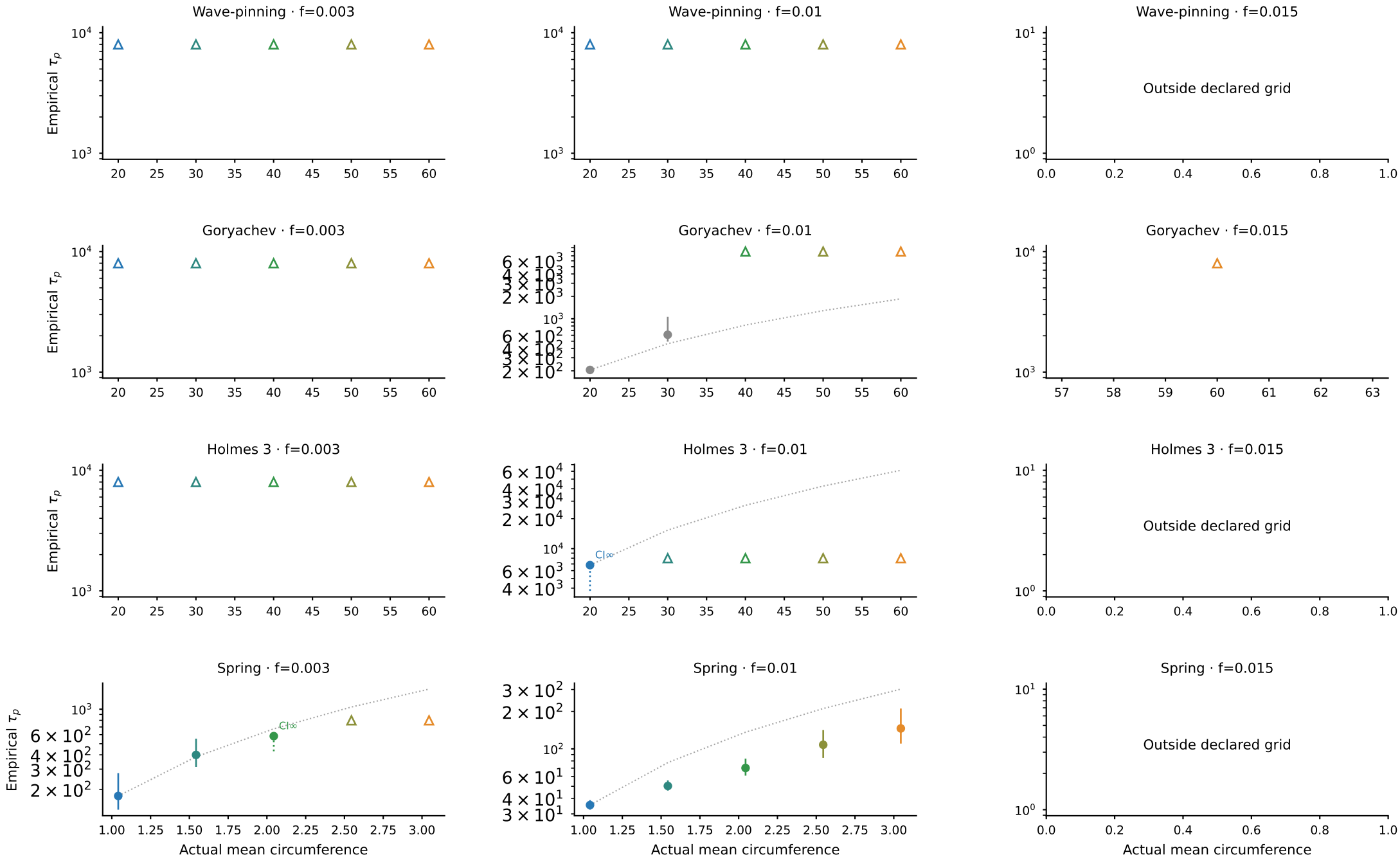


E3 completed native observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_E3_sizes_final_layout_20261020_011_S4_1_tau_p.csv

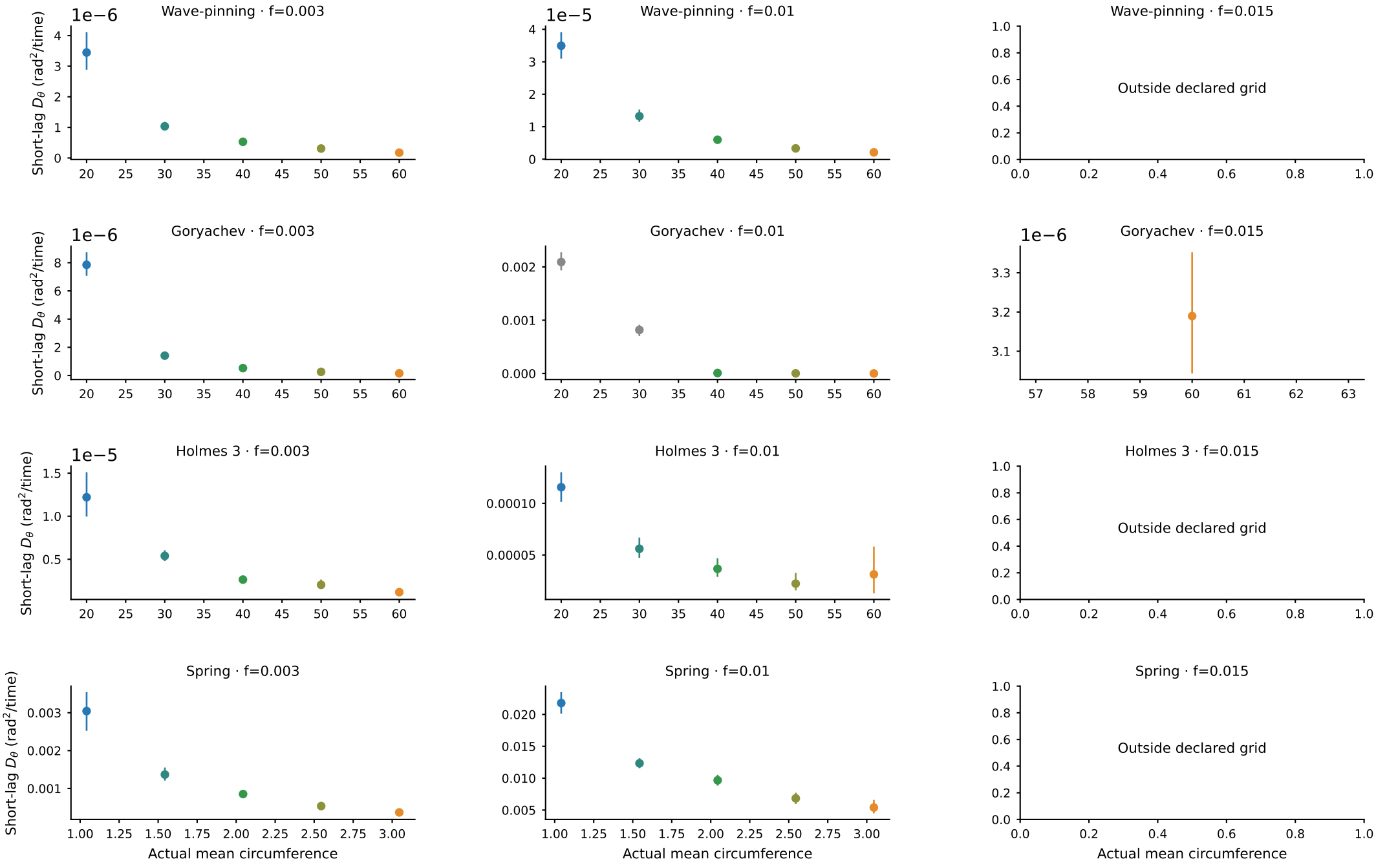
S4 · E3 completed native observations · Persistence and angular mobility across size



E3 completed native observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_E3_sizes_final_layout_20261020_012_S4_1_Dtheta.csv

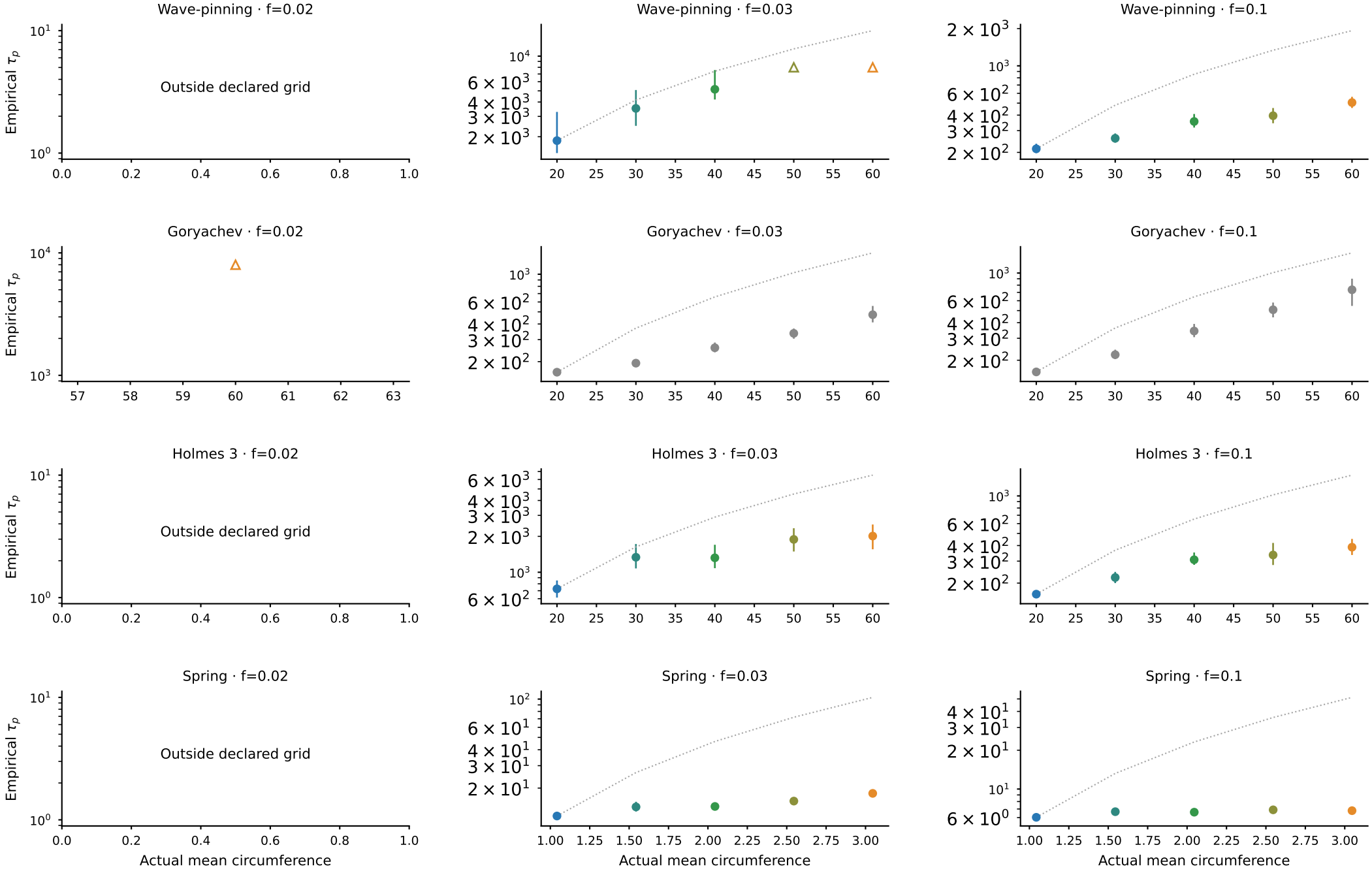


E3 completed native observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C^2 reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV. CI∞ next to a finite point marks an upper-unbounded bootstrap CI; finite lower endpoints are retained. CI[0,∞) spans the full nonnegative range.
CSV: figure4_E3_sizes_final_layout_20261020_013_S4_2_tau_p.csv

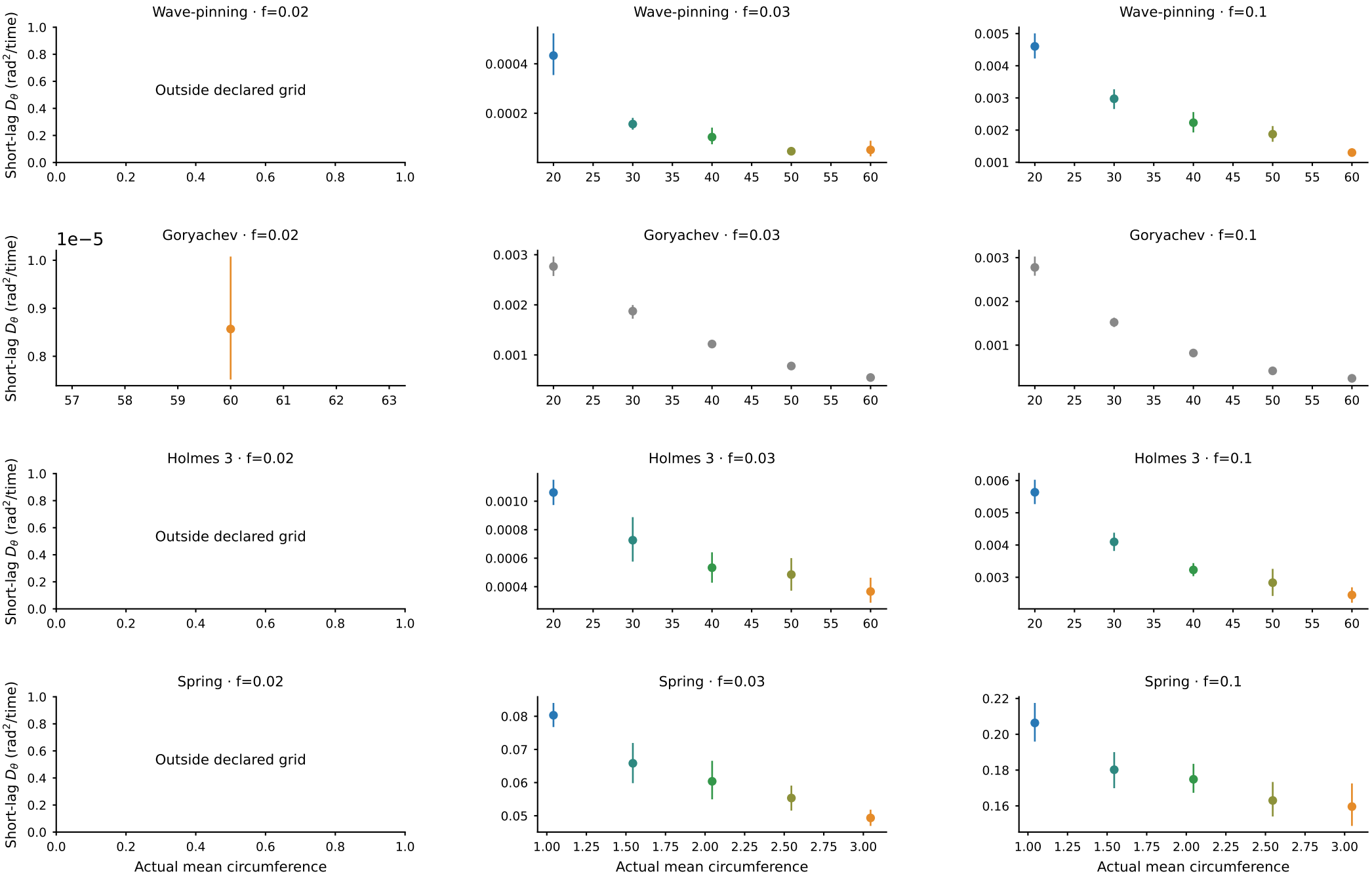


E3 completed native observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV. CI∞ next to a finite point marks an upper-unbounded bootstrap CI; finite lower endpoints are retained. CI[0,∞) spans the full nonnegative range.
CSV: figure4_E3_sizes_final_layout_20261020_014_S4_2_Dtheta.csv

S4 · E3 completed native observations · Persistence and angular mobility across size

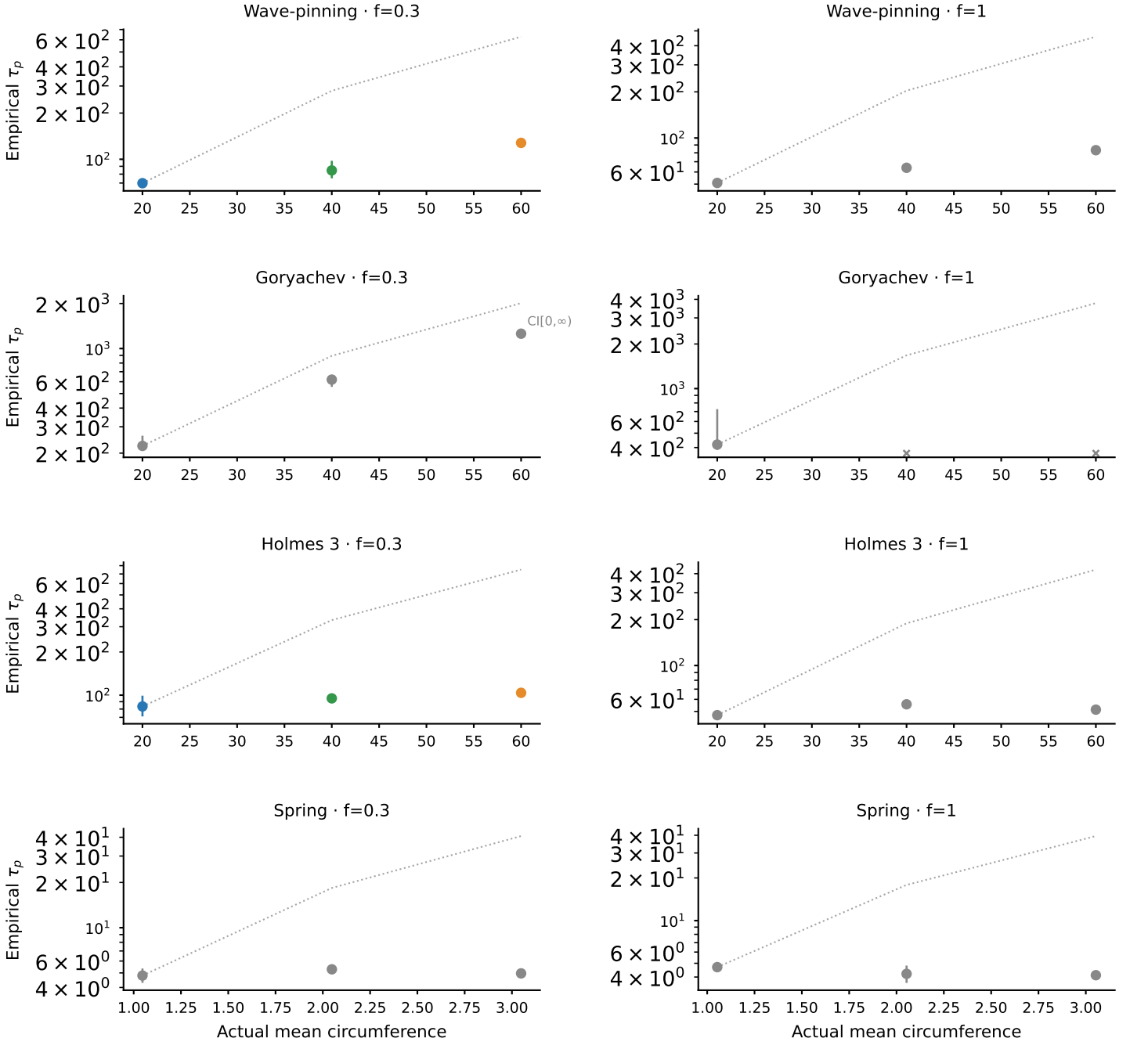


E3 completed native observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C^2 reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_E3_sizes_final_layout_20261020_015_S4_3_tau_p.csv



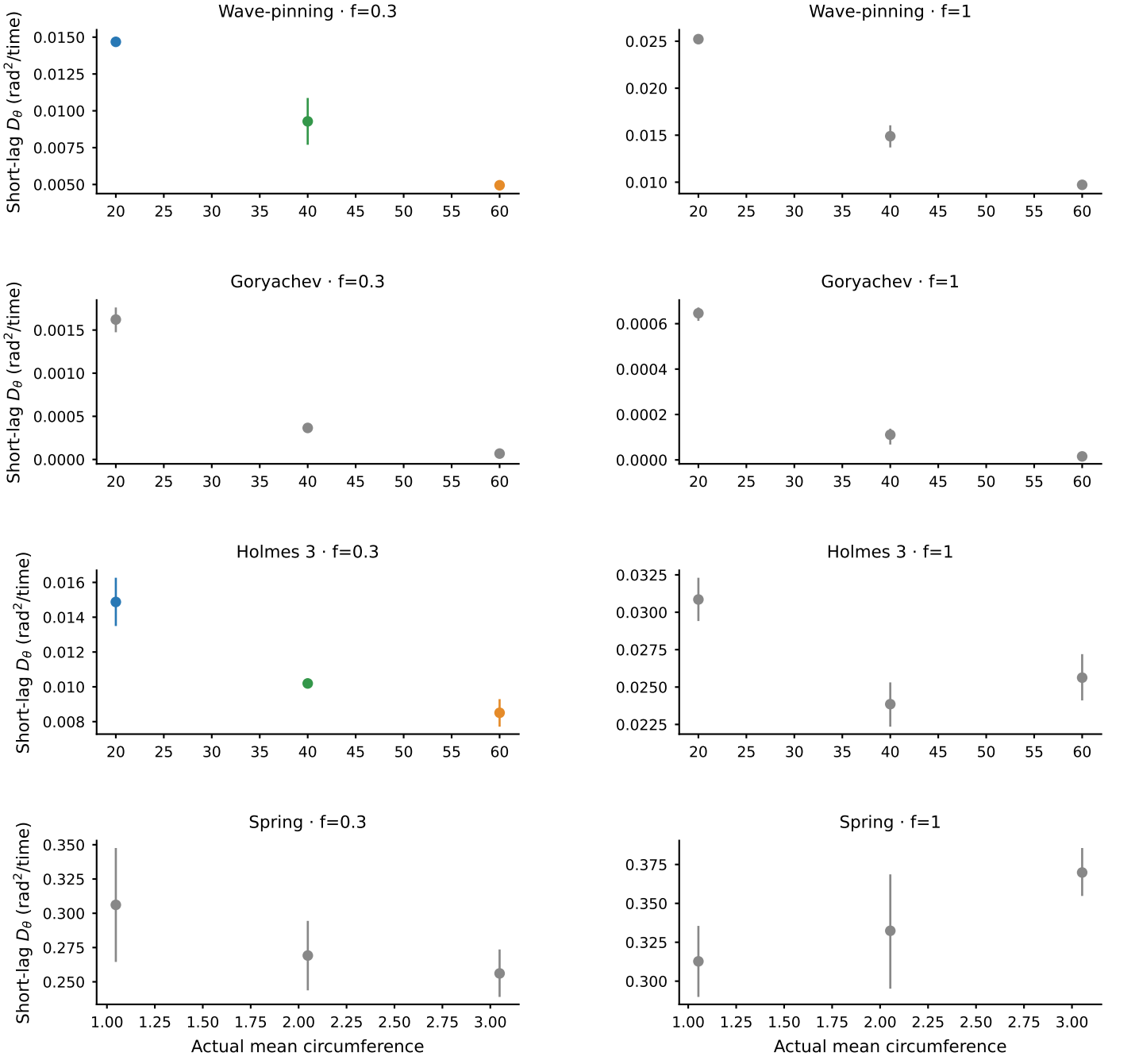
E3 completed native observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_E3_sizes_final_layout_20261020_016_S4_3_Dtheta.csv

S4 · E3 completed native observations · Persistence and angular mobility across size



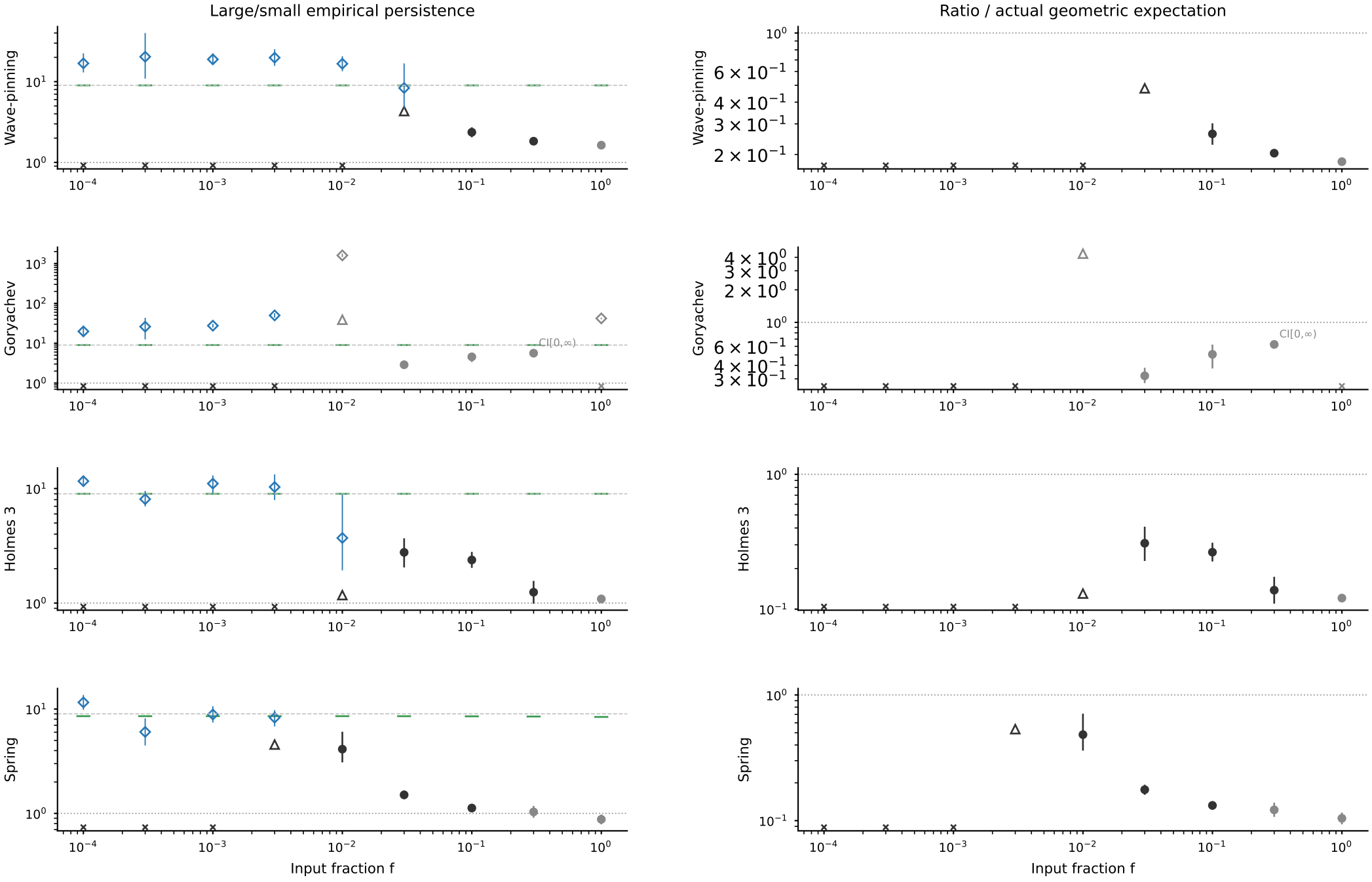
E3 completed native observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV. CI∞ next to a finite point marks an upper-unbounded bootstrap CI; finite lower endpoints are retained. CI[0,∞) spans the full nonnegative range.
CSV: figure4_E3_sizes_final_layout_20261020_017_S4_4_tau_p.csv

S4 · E3 completed native observations · Persistence and angular mobility across size



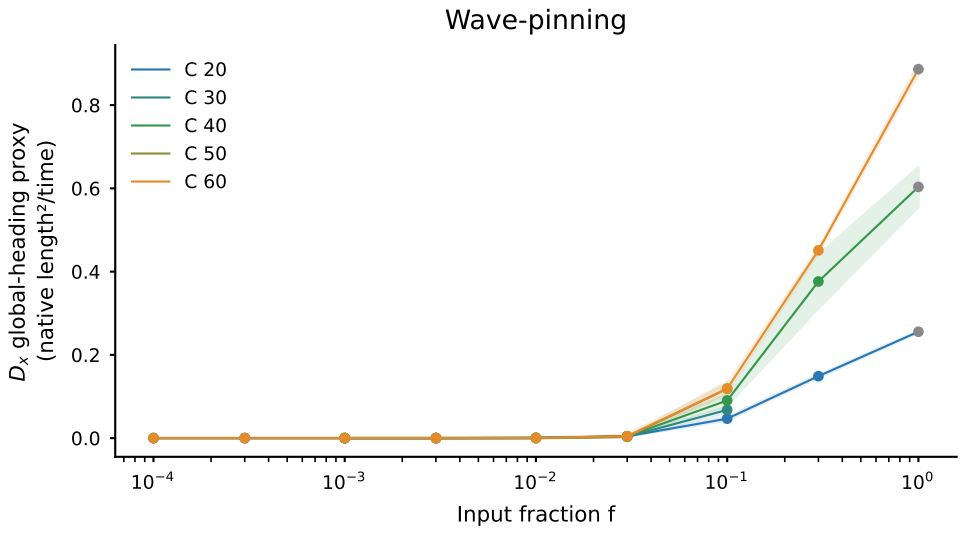
E3 completed native observations. Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV. CI∞ next to a finite point marks an upper-unbounded bootstrap CI; finite lower endpoints are retained. CI[0,∞) spans the full nonnegative range.
CSV: figure4_E3_sizes_final_layout_20261020_018_S4_4_Dtheta.csv

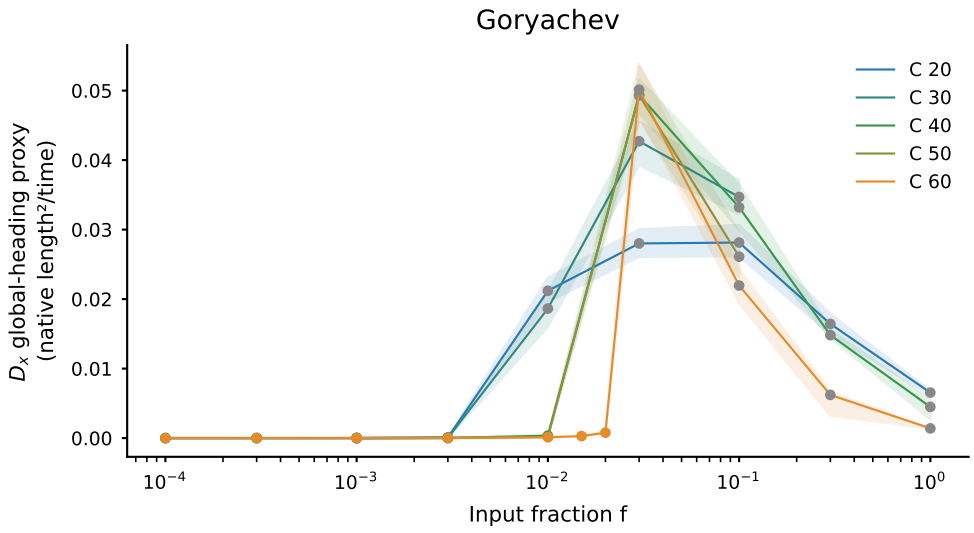
S4 · E3 completed native observations · Size advantage and geometric reference

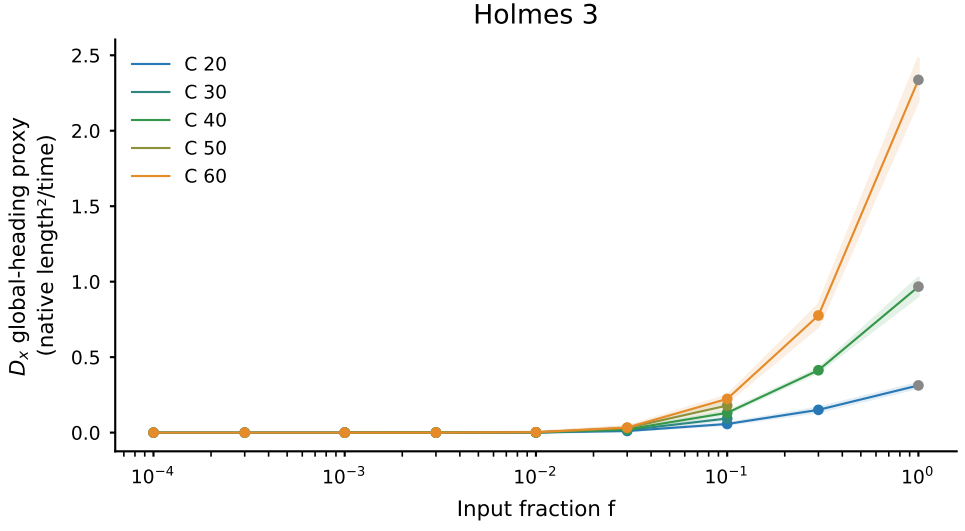


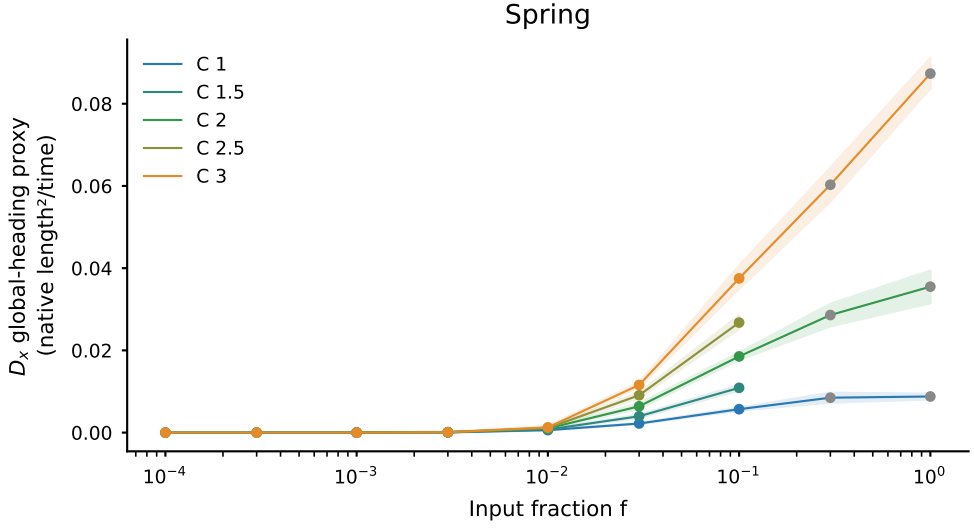
E3 completed native observations. Triangles preserve finite-window or resolution bounds. Two censored times do not yield a finite ratio. Open diamonds show the short-lag global activity-heading small/large Dθ proxy: blue for unforced conditions and grey for forced conditions. This proxy does not resolve a direct persistence ratio or measure one tracked site's diffusion in multi-region states. Green marks: measured circumference-ratio squared; dashed 9: resting-size reference. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV. CI∞ next to a finite point marks an upper-unbounded bootstrap CI; finite lower endpoints are retained. CI[0,∞) spans the full nonnegative range.
CSV: figure4_E3_sizes_final_layout_20261020_019_S4_ratios.csv

S5 · E3 completed native observations · Global-heading mobility proxy



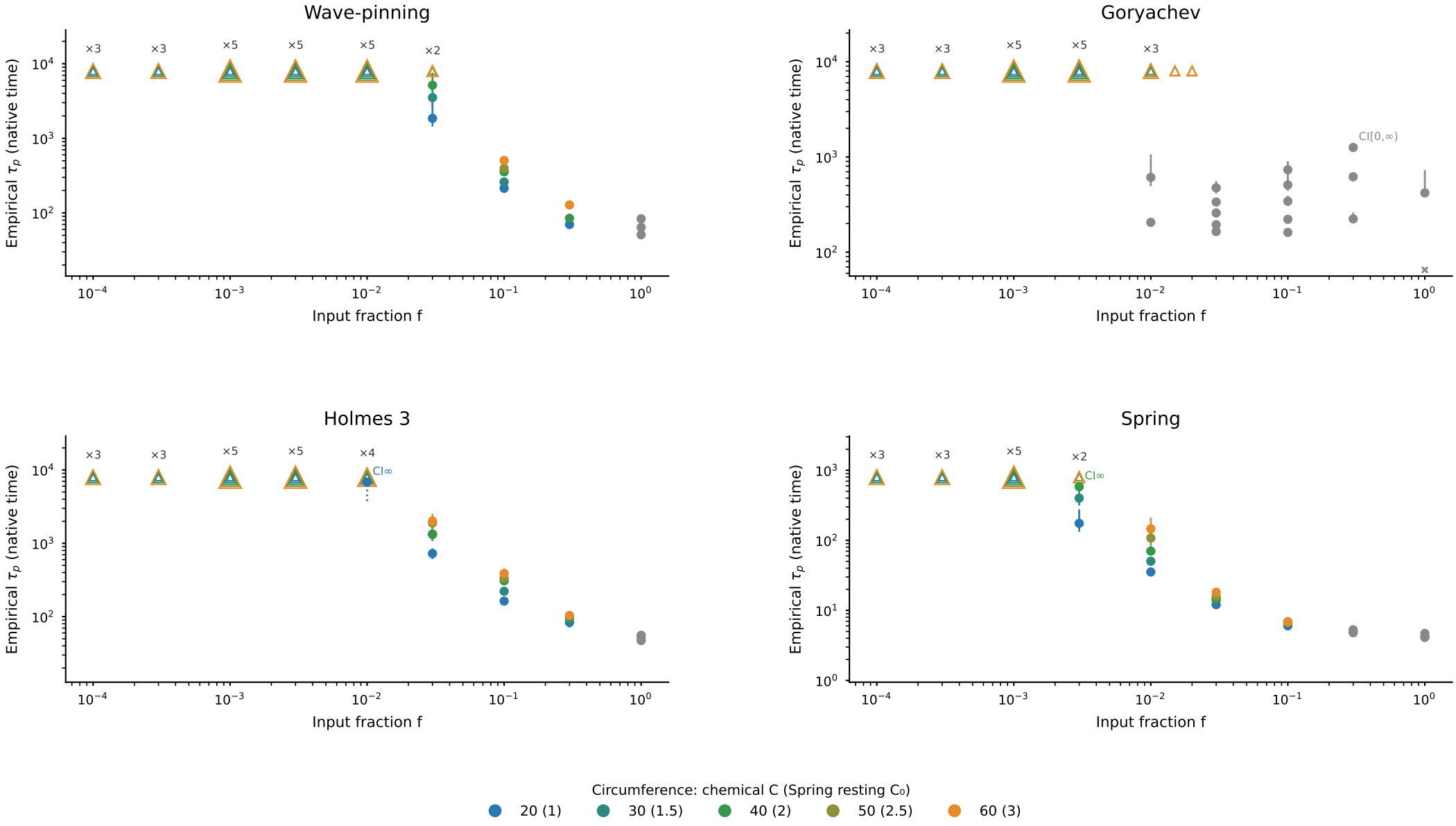






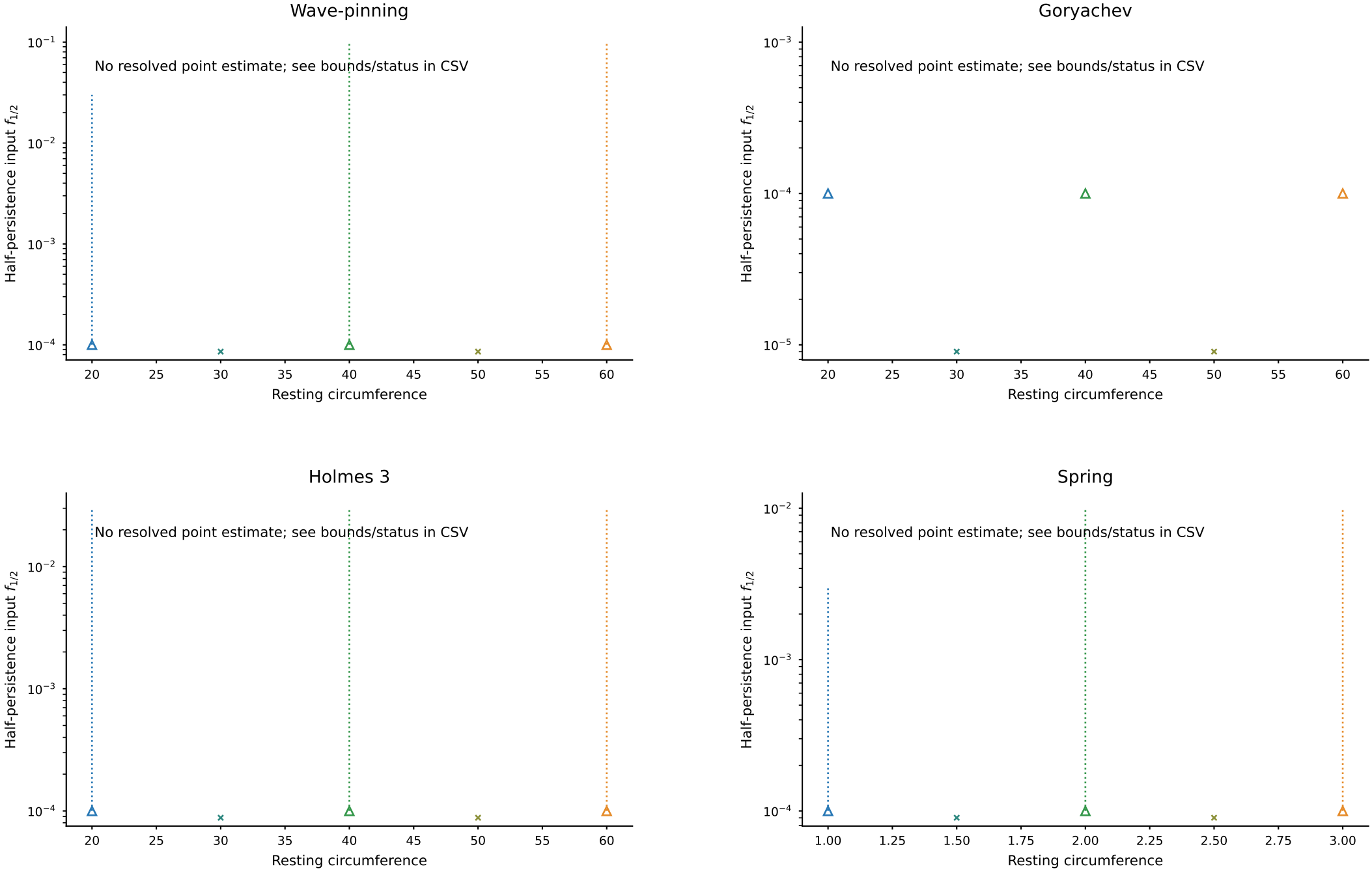
E3 completed native observations. Dx = Dθ × (actual mean C / 2π)² converts the signed short-lag global activity-heading estimate using each window's actual mean circumference. In multi-region states this proxy is not diffusion of one tracked site. It is distinct from qualified single-region literal heading/peak coordinate MSD and is not a slope fit to θC(t)/(2π). Increased Dx and angular persistence advantages are both retained when they coexist. Grey retains forced conditions. Native length/time units differ across models. No free-diffusion or migration velocity law is assumed. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV. CI∞ next to a finite point marks an upper-unbounded bootstrap CI; finite lower endpoints are retained. CI[0,∞) spans the full nonnegative range.
CSV: figure4_E3_sizes_final_layout_20261020_020_S5_Dx.csv

S6 · E3 completed native observations · Empirical persistence against input strength



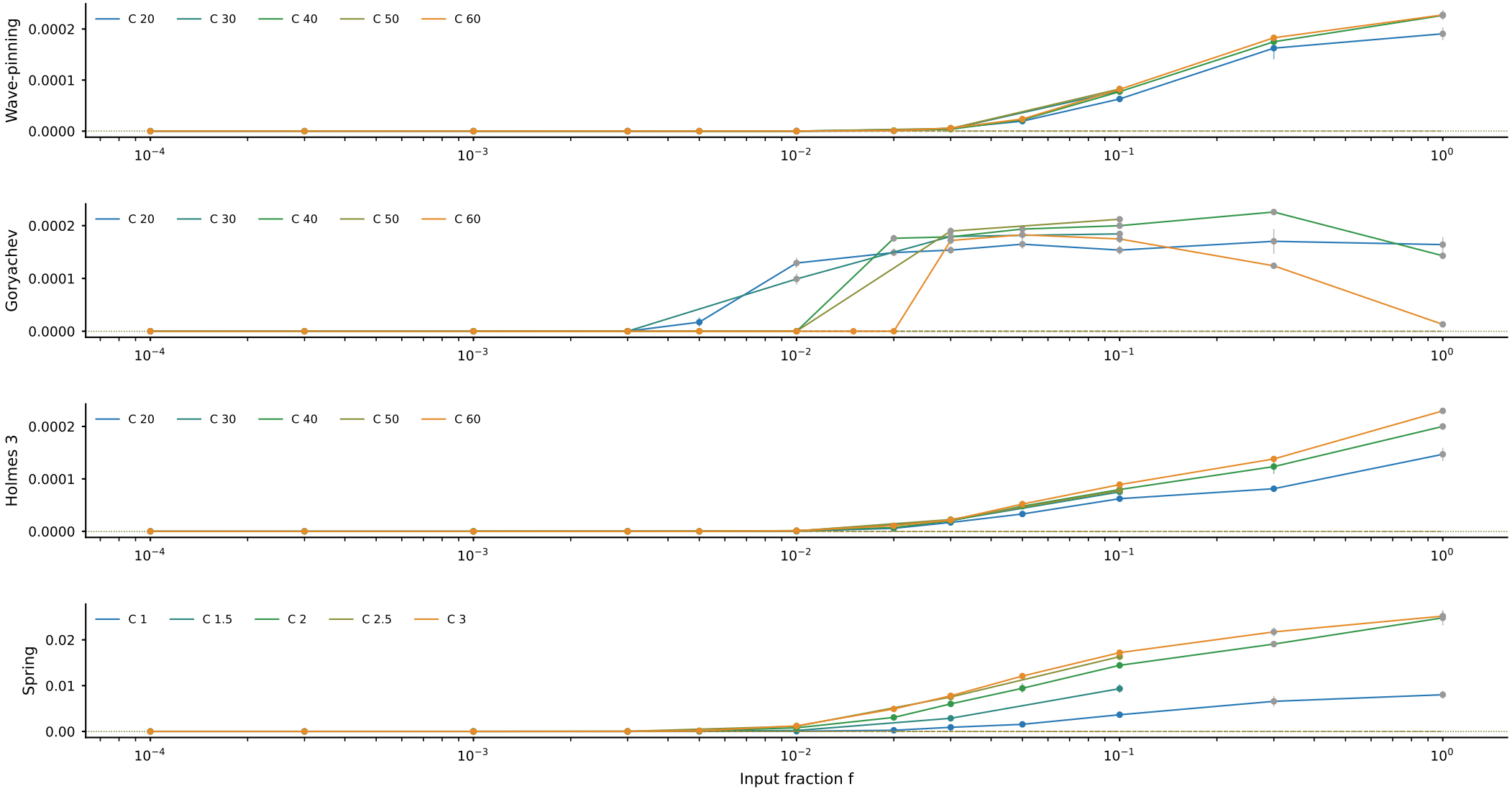
E3 completed native observations. Upward triangles are empirical lower bounds; downward triangles are unresolved below two cadences. Nested triangles at one unchanged coordinate retain coincident sizes; ×n gives their count. Their exact sizes, glyph sizes and coordinates remain in CSV. The shared legend maps chemical C to Spring resting C₀; grey retains forced observations. Bootstrap CI markers are separate from empirical bounds. Exponential fits never replace empirical measurements. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV. CI∞ next to a finite point marks an upper-unbounded bootstrap CI; finite lower endpoints are retained. CI[0,∞) spans the full nonnegative range.
CSV: figure4_E3_sizes_final_layout_20261020_021_S6_tau.csv

S6 · E3 completed native observations · Noise tolerance with finite-window uncertainty



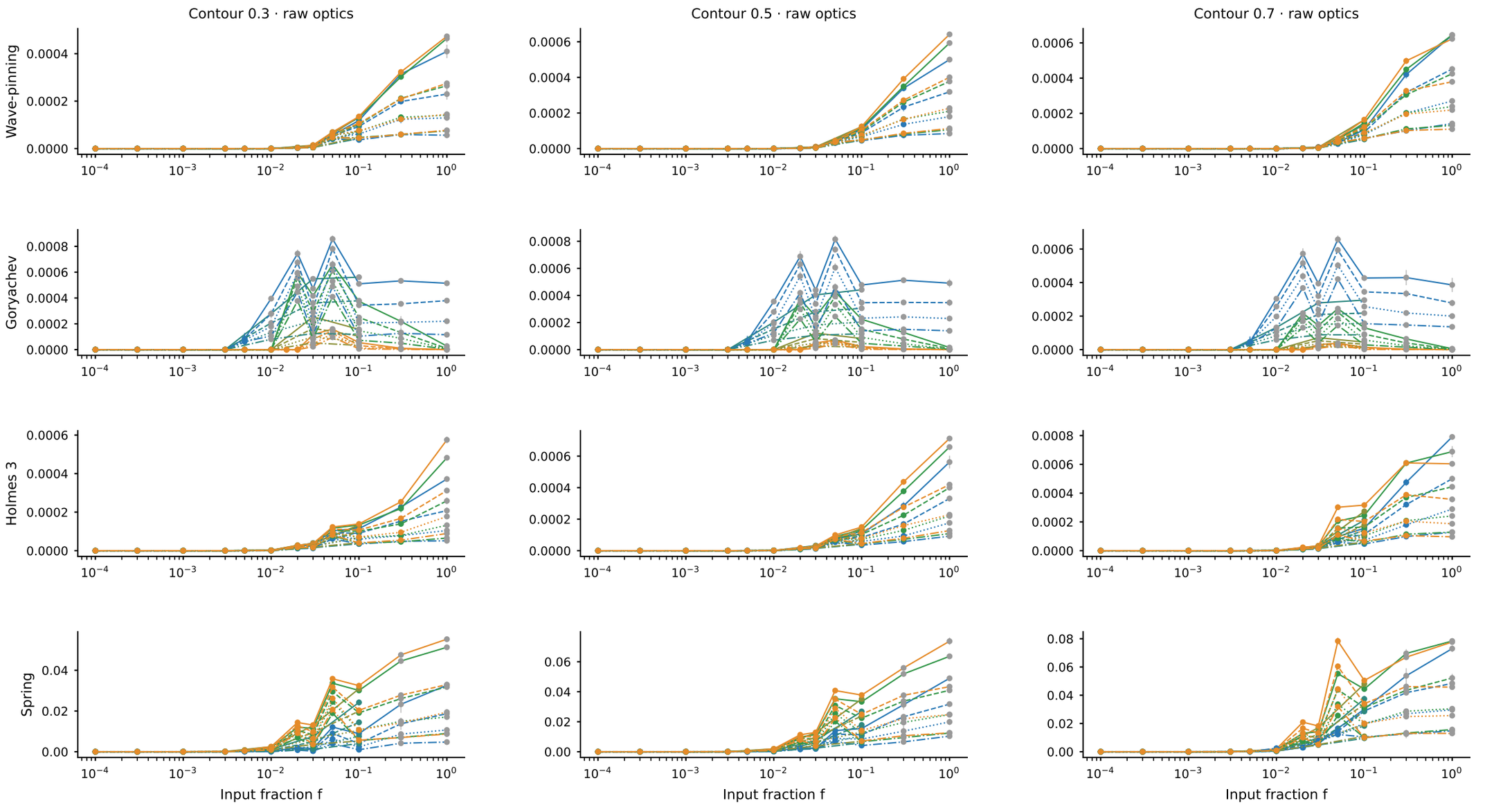
E3 completed native observations. First crossing of half the actual f=0.0001 persistence, using log-f interpolation only when identified. Censored references remain bounded or inconclusive. Intermediate-size grids lacking the declared f=0.0001 baseline retain unavailable markers/reasons; f=0.001 is never substituted and no tolerance curve bridges them. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_E3_sizes_final_layout_20261020_022_S6_fhalf.csv

S9 · Common canonical A5 observations · Region births per native time per physical perimeter



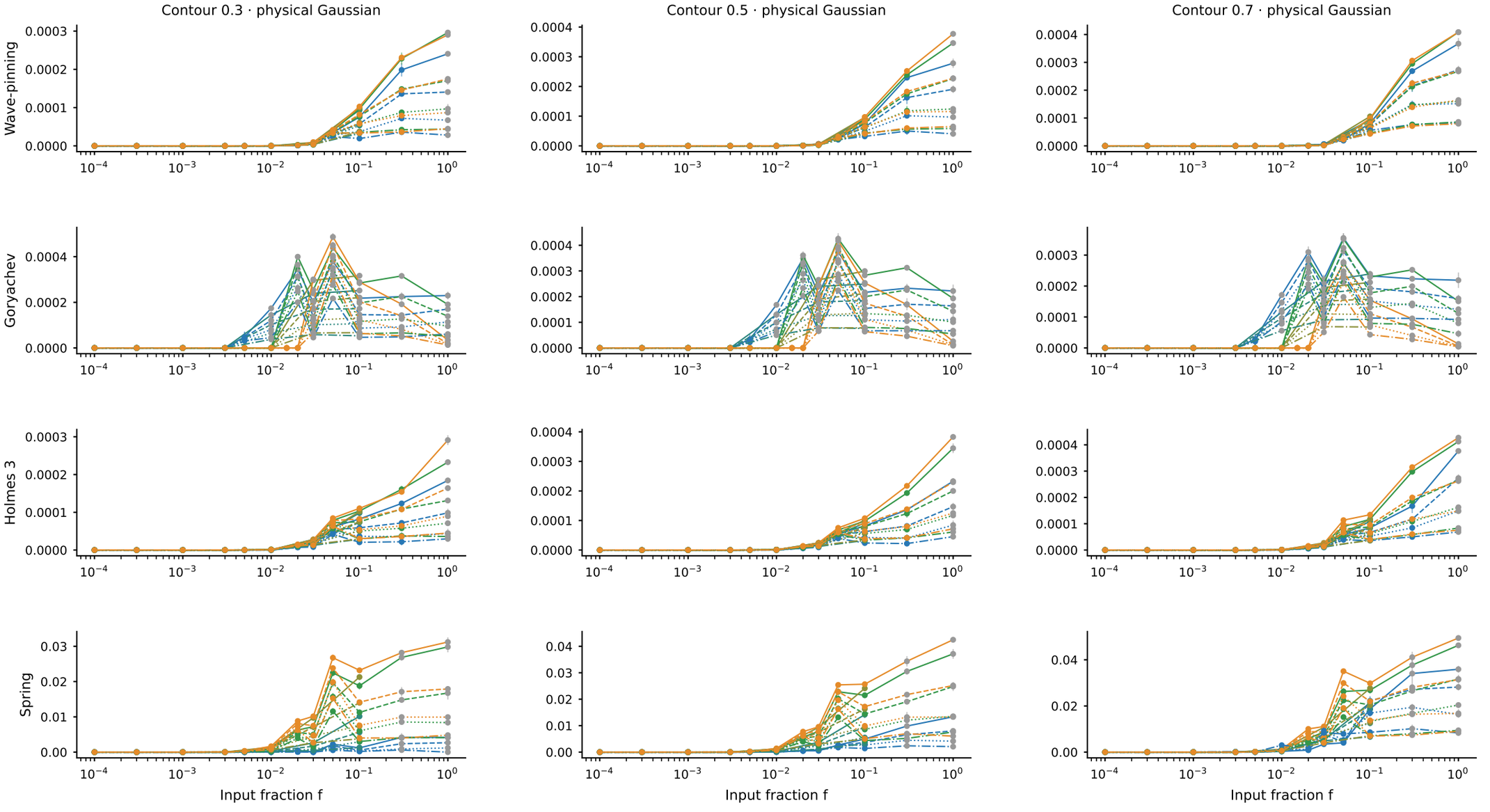
Common canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_001_S9_births_per_time_per_perimeter.csv

S9 · Common canonical A5 observations · Region births per native time per physical perimeter · full native-factor sensitivity



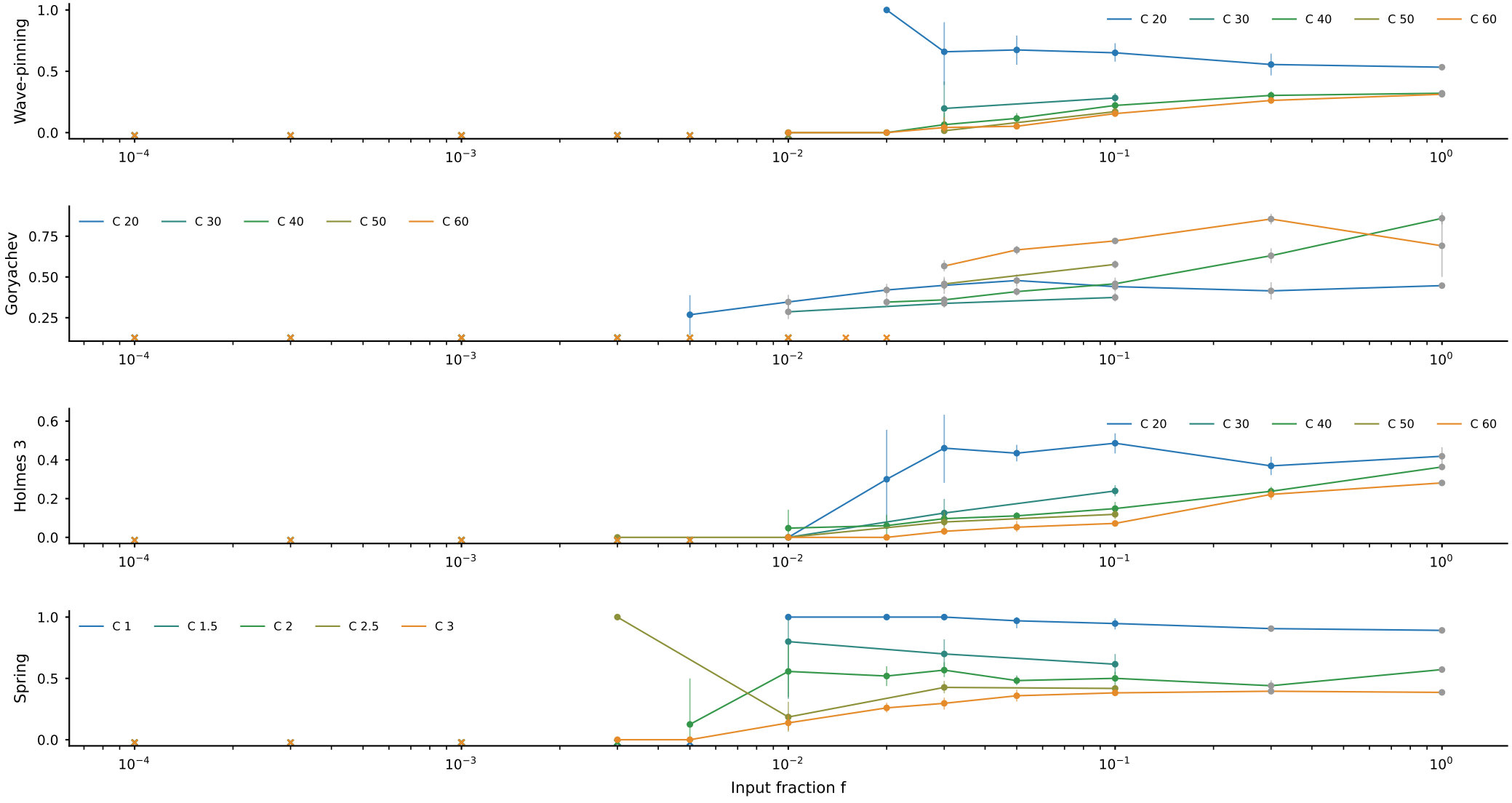
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_002_S9_births_per_time_per_perimeter_raw_factorial.csv

S9 · Common canonical A5 observations · Region births per native time per physical perimeter · full native-factor sensitivity



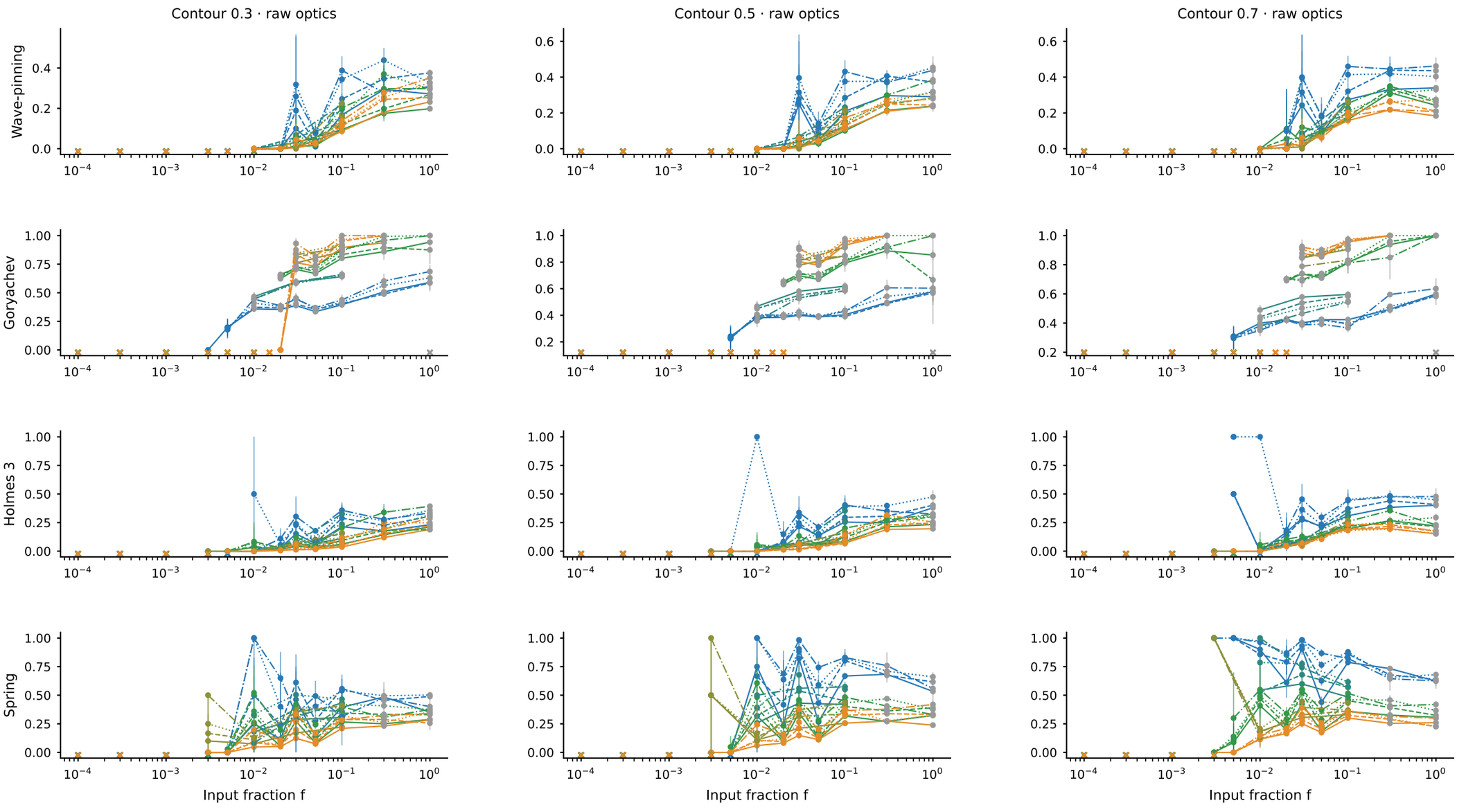
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_003_S9_births_per_time_per_perimeter_smoothed_factorial.csv

S9 · Common canonical A5 observations · Fraction of trajectories with an unscorable takeover track



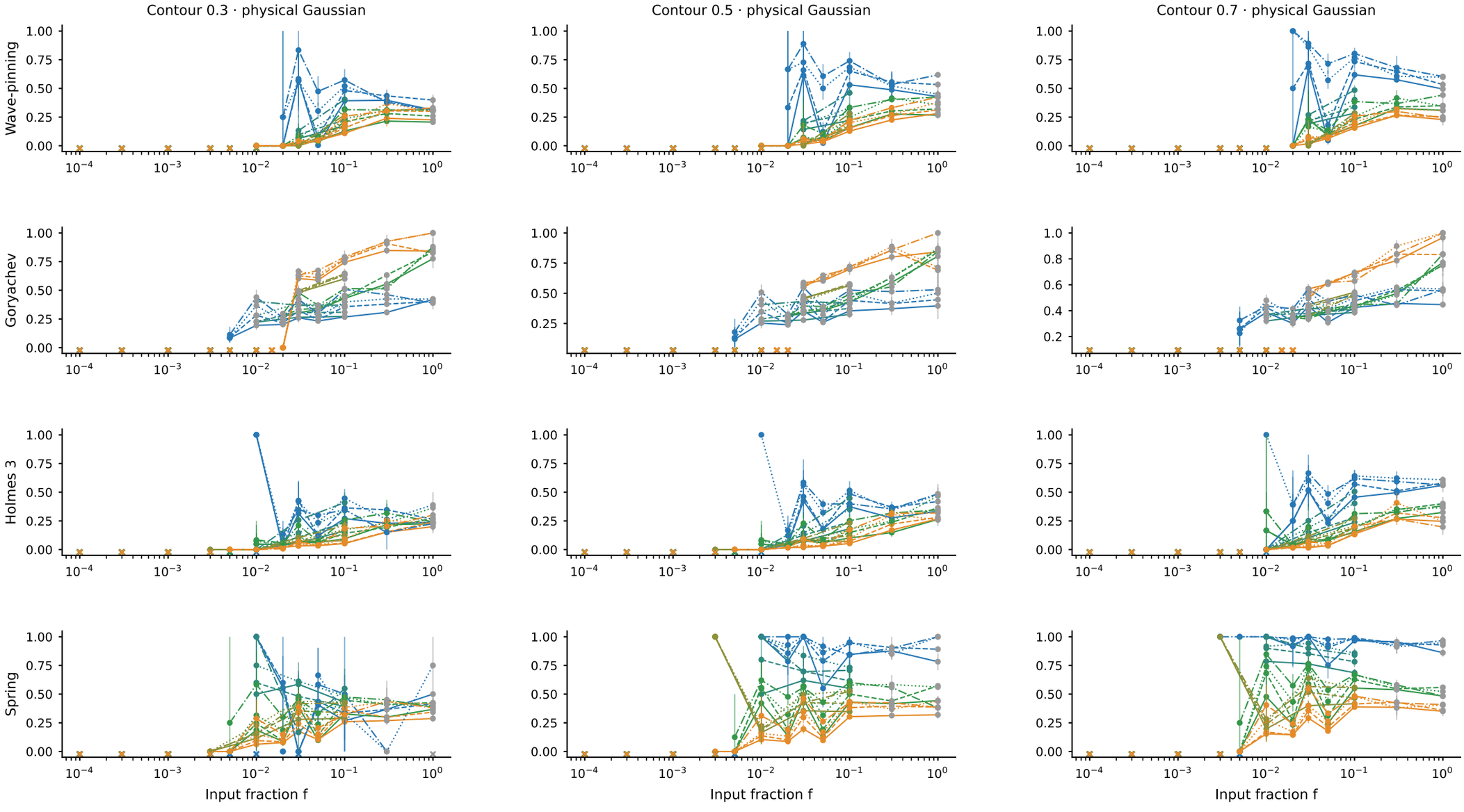
Common canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_004_S9_ever_unscorable_takeover_fraction.csv

S9 · Common canonical A5 observations · Fraction of trajectories with an unscorable takeover track · full native-factor sensitivity



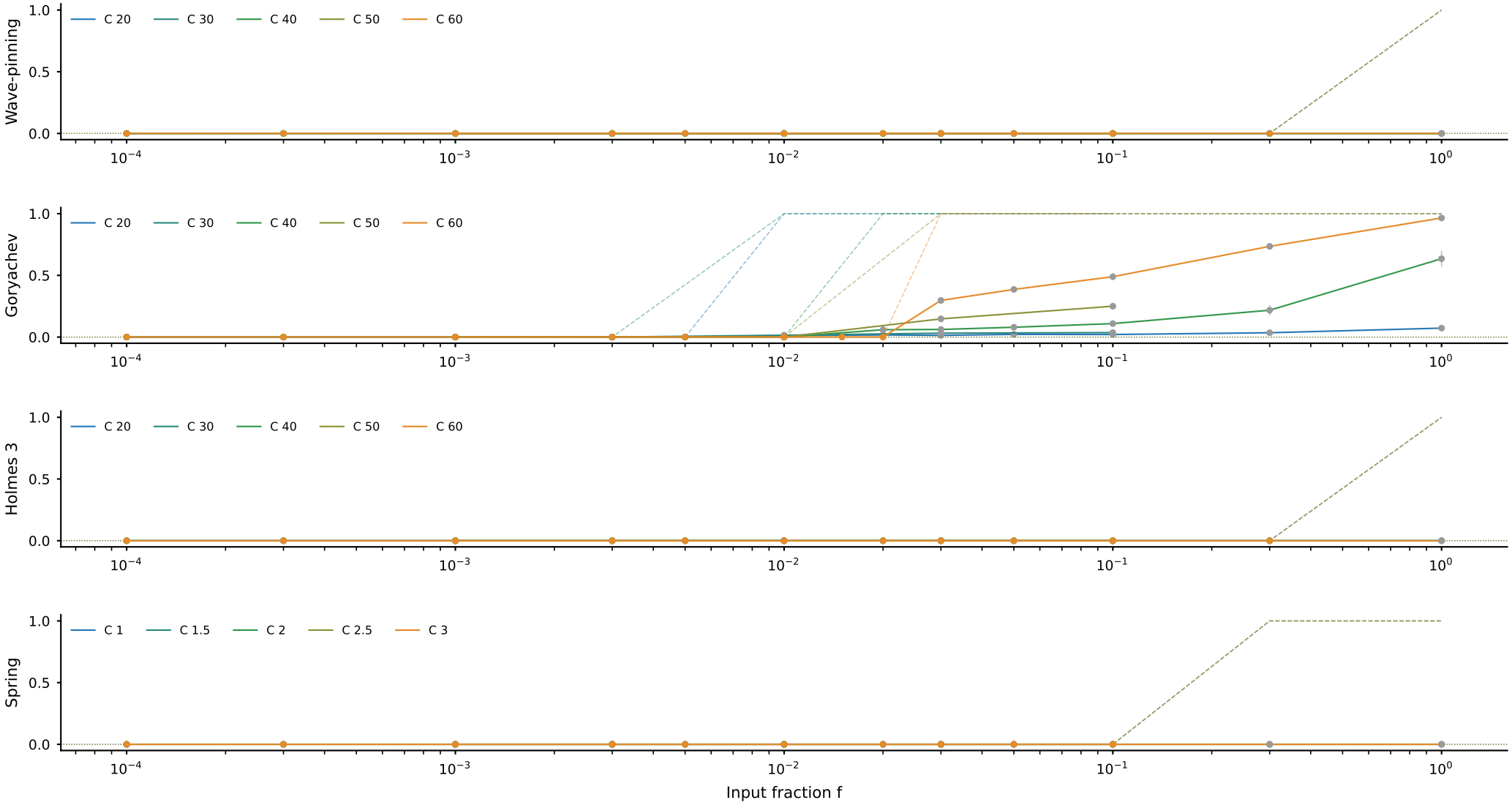
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_005_S9_ever_unscorable_takeover_fraction_raw_factorial.csv

S9 · Common canonical A5 observations · Fraction of trajectories with an unscorable takeover track · full native-factor sensitivity



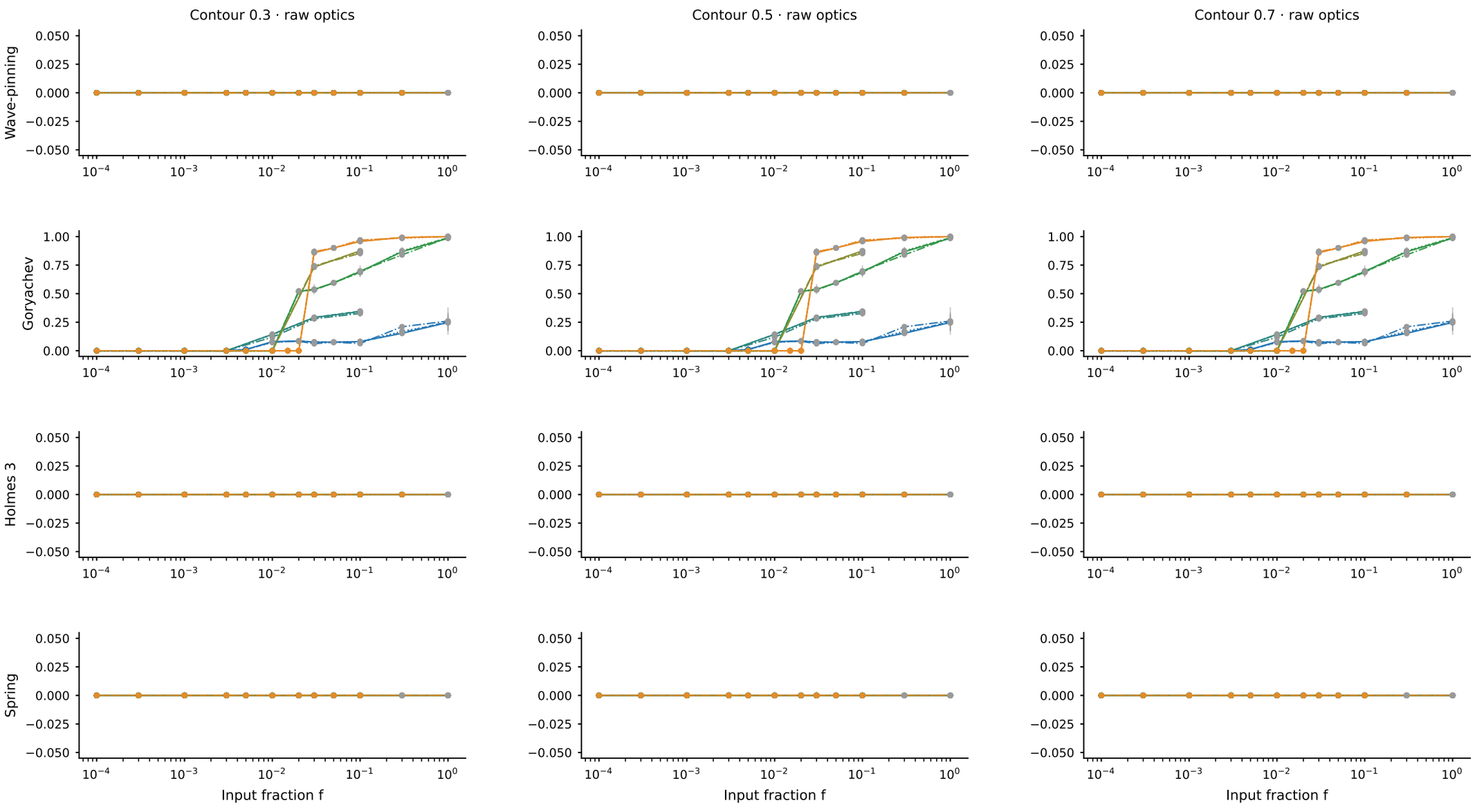
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_006_S9_ever_unscorable_takeover_fraction_smoothed_factorial.csv

S9 · Common canonical A5 observations · Time fraction below the contrast gate



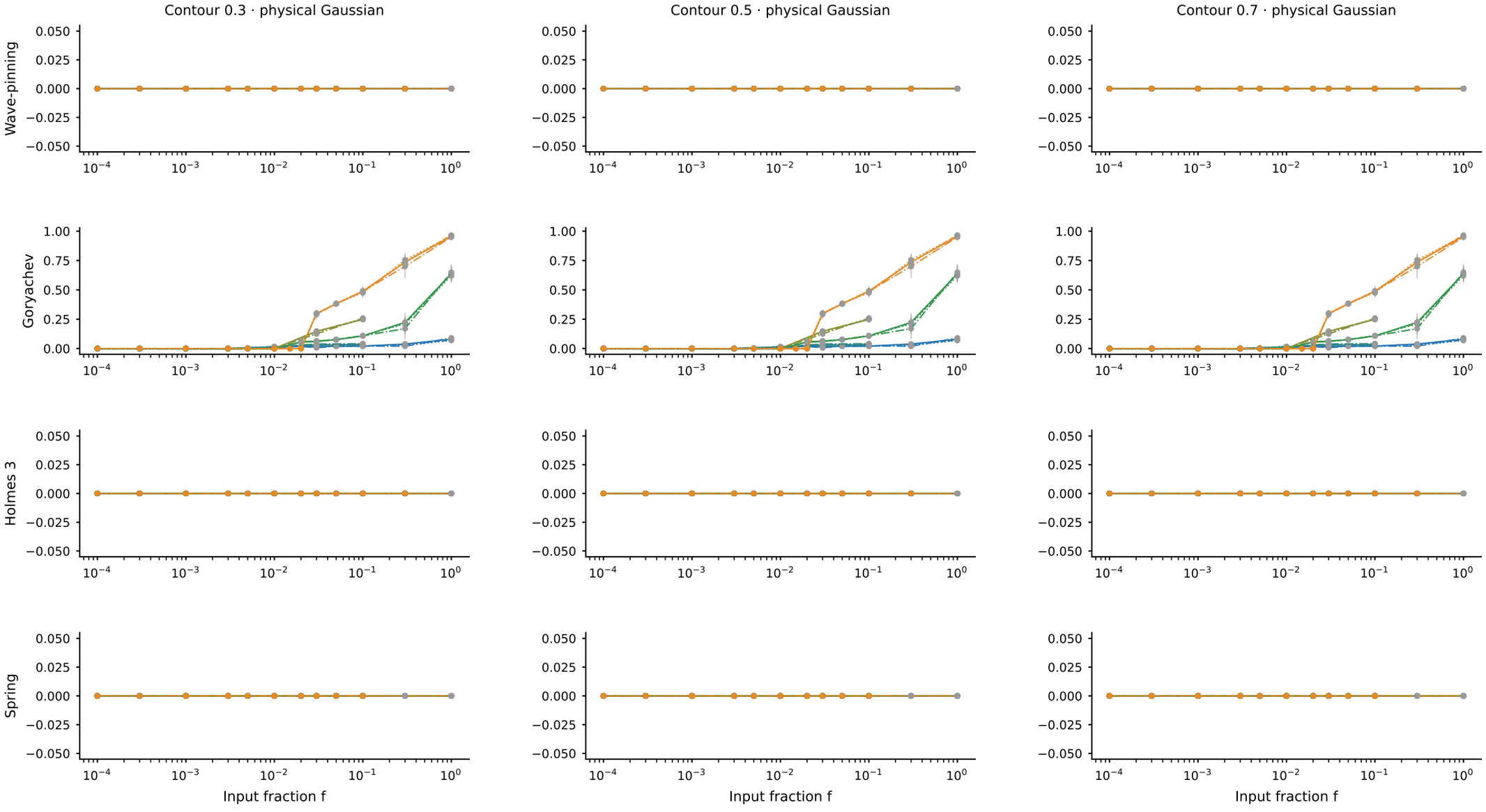
Common canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_007_S9_low_contrast_time_fraction.csv

S9 · Common canonical A5 observations · Time fraction below the contrast gate · full native-factor sensitivity



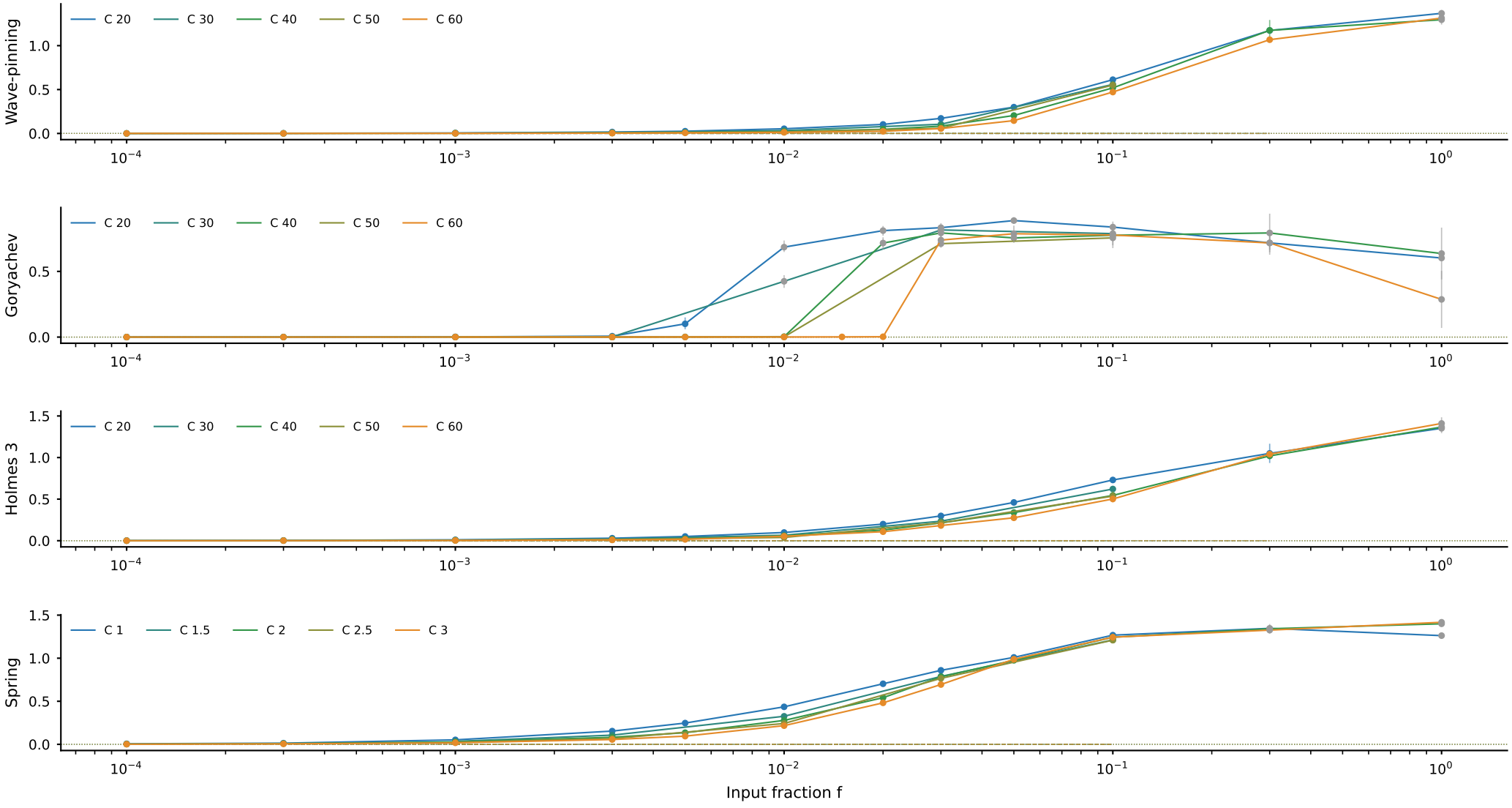
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_008_S9_low_contrast_time_fraction_raw_factorial.csv

S9 · Common canonical A5 observations · Time fraction below the contrast gate · full native-factor sensitivity



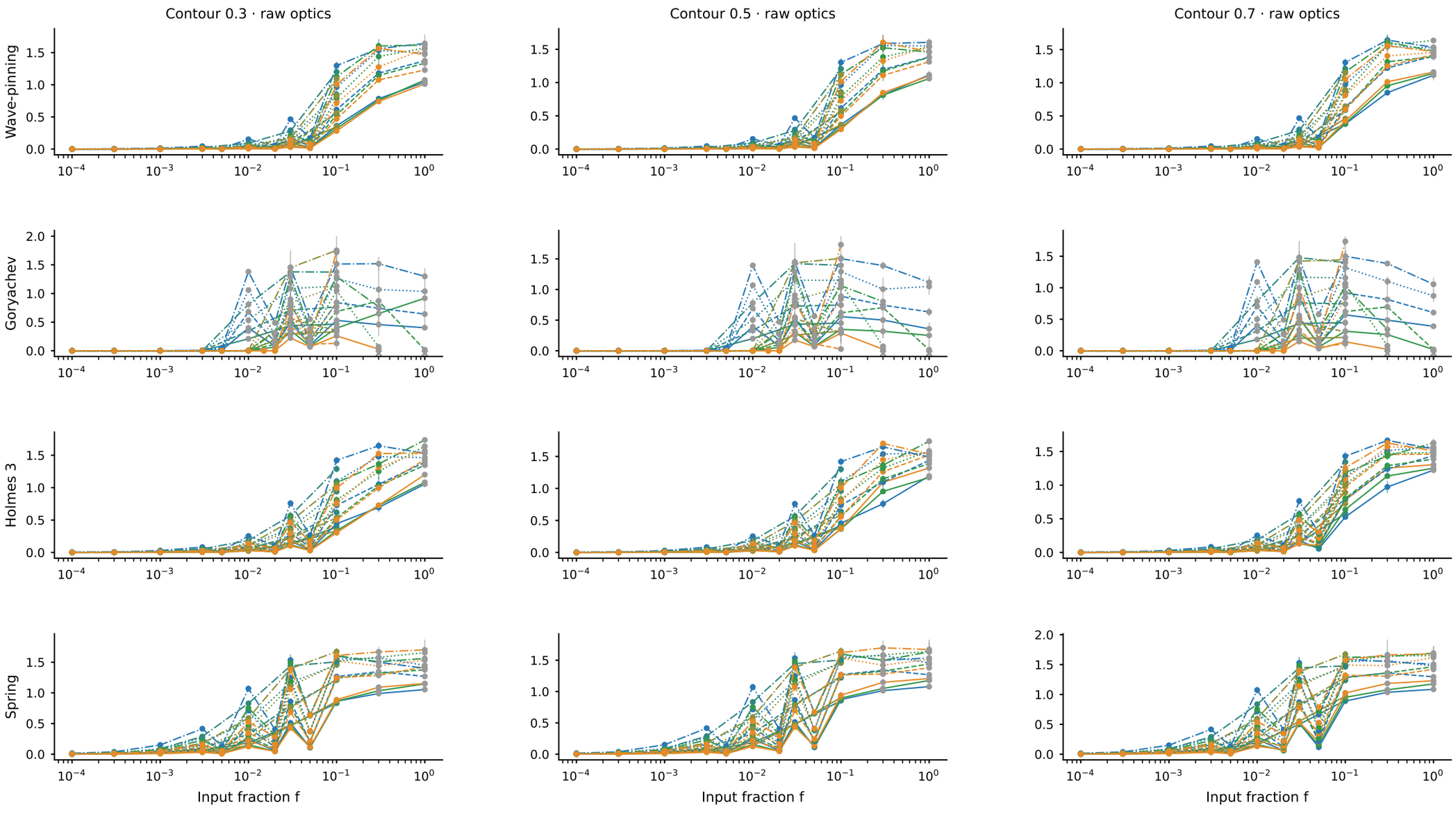
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_009_S9_low_contrast_time_fraction_smoothed_factorial.csv

S9 · Common canonical A5 observations · Mean absolute dominant-heading turn over 1 observation frames



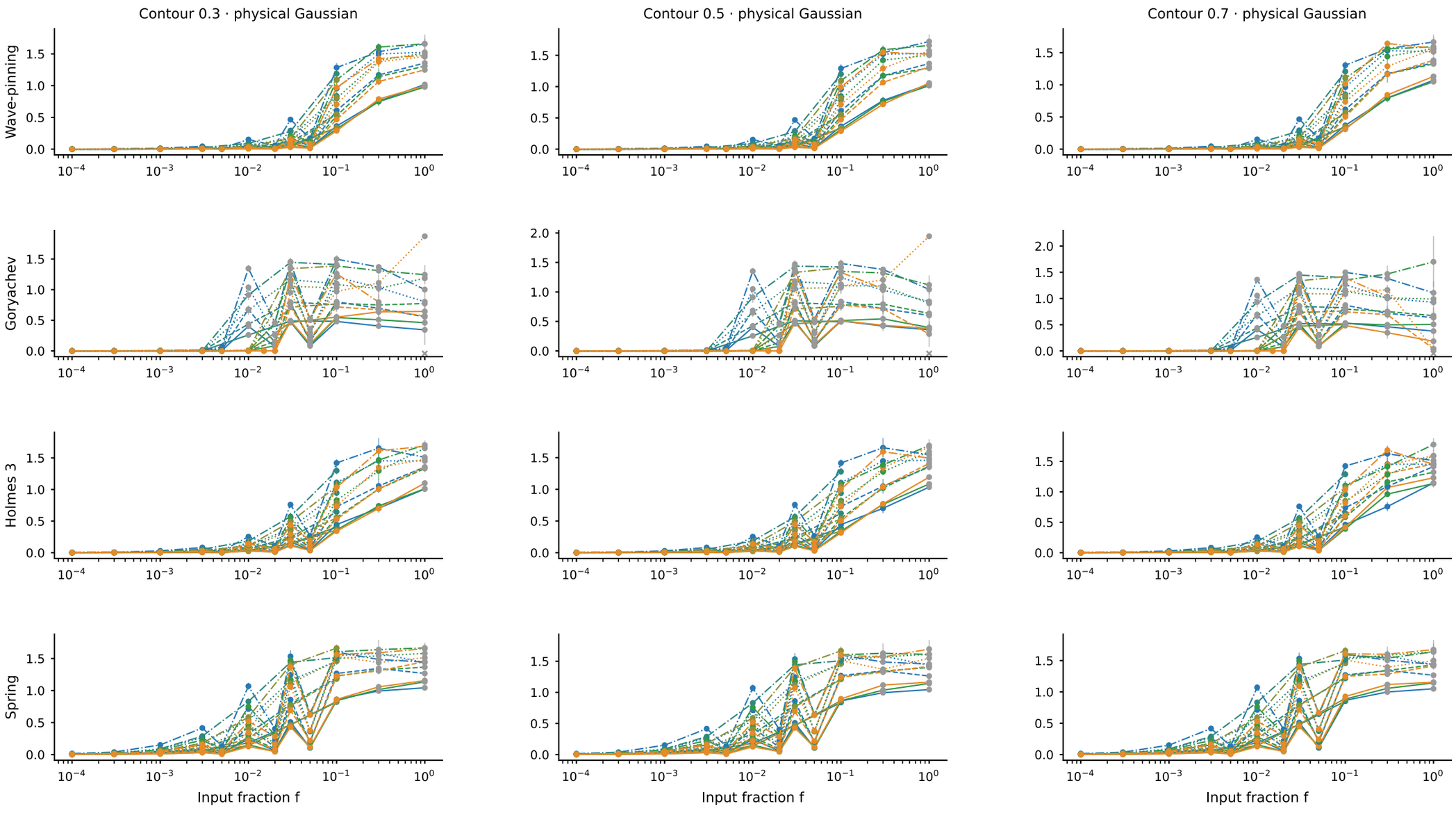
Common canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_010_S9_mean_absolute_turn_1_frames.csv

S9 · Common canonical A5 observations · Mean absolute dominant-heading turn over 1 observation frames · full native-factor sensitivity



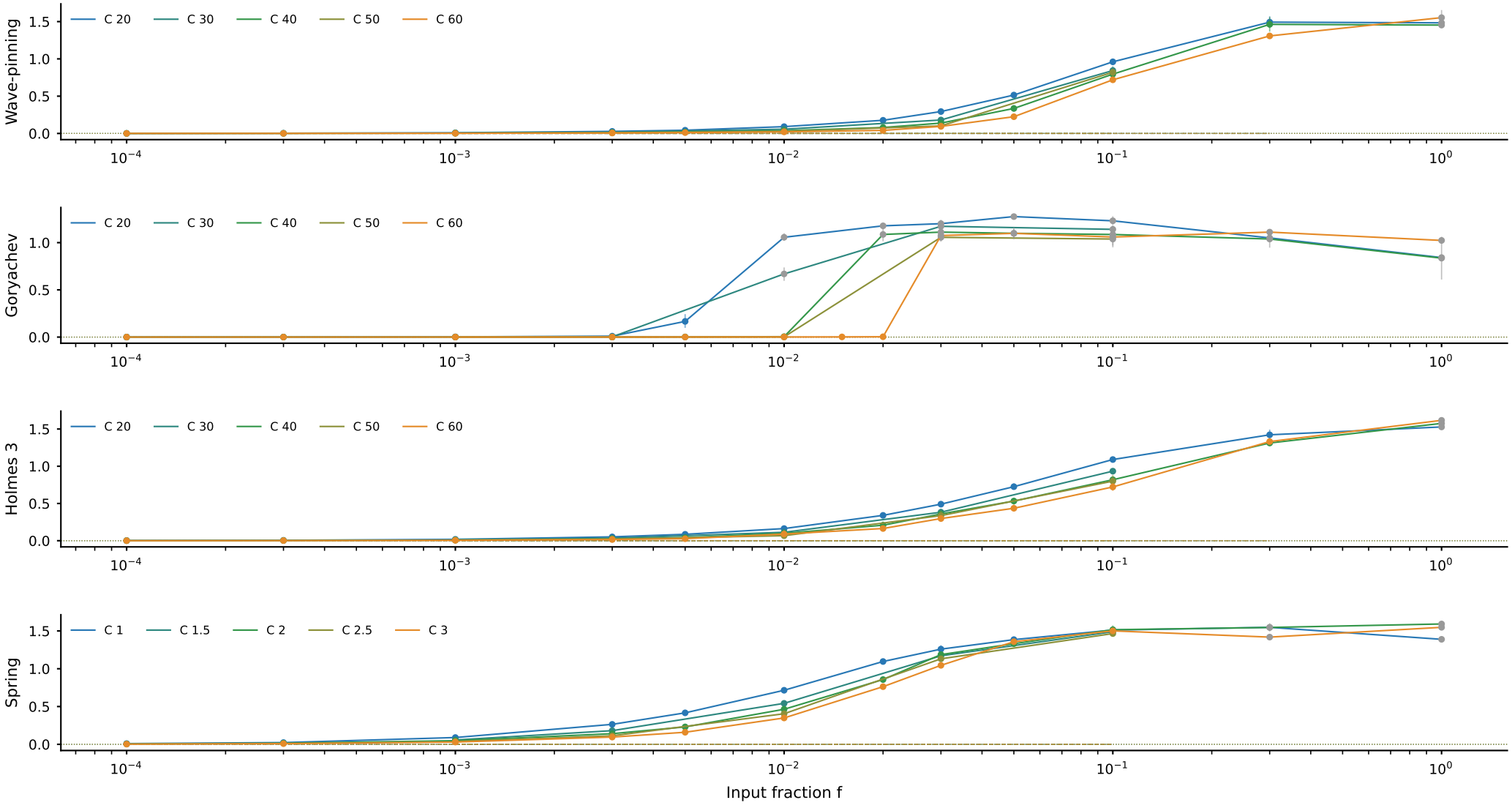
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_011_S9_mean_absolute_turn_1_frames_raw_factorial.csv

S9 · Common canonical A5 observations · Mean absolute dominant-heading turn over 1 observation frames · full native-factor sensitivity



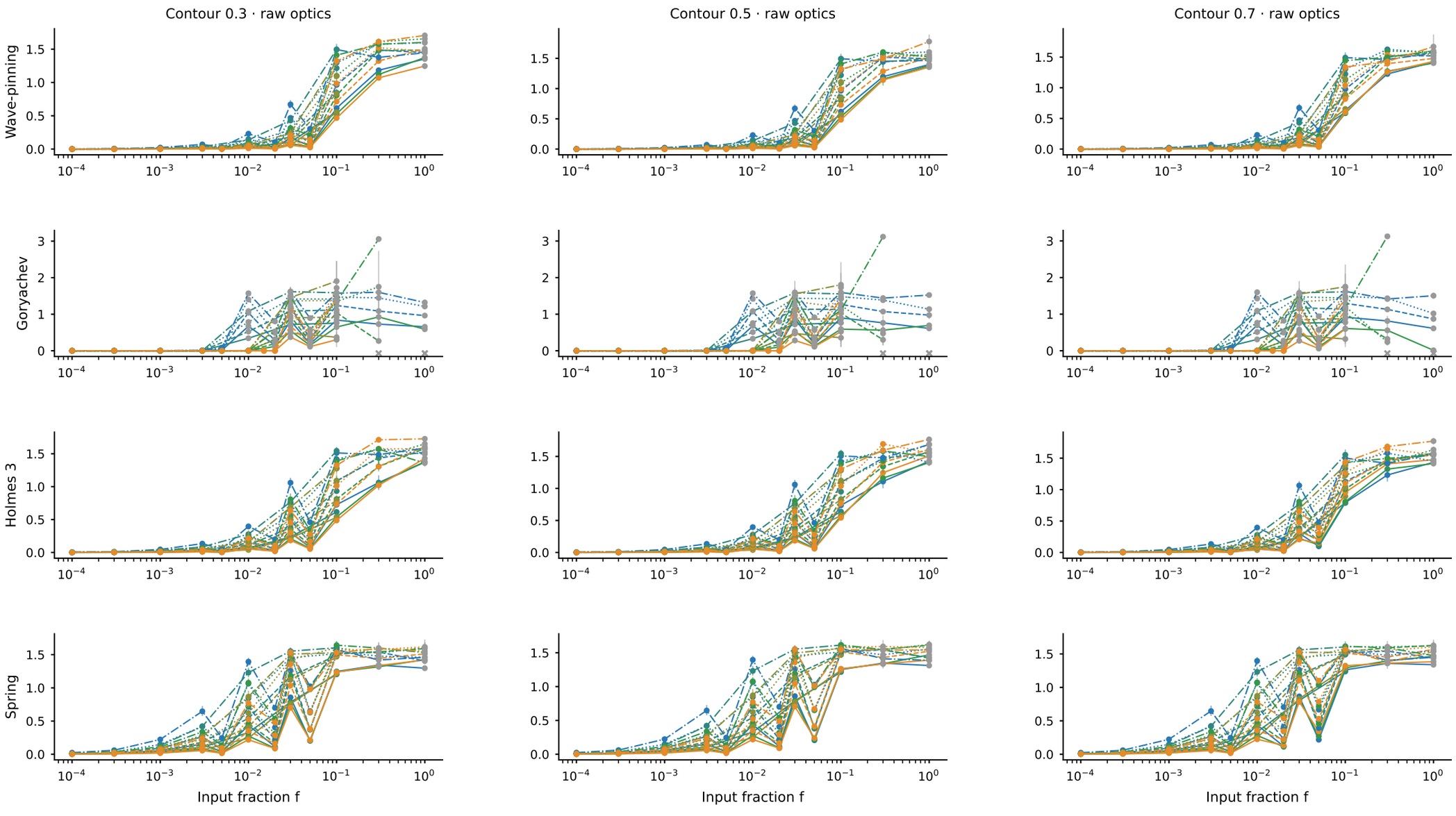
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_012_S9_mean_absolute_turn_1_frames_smoothed_factorial.csv

S9 · Common canonical A5 observations · Mean absolute dominant-heading turn over 2 observation frames



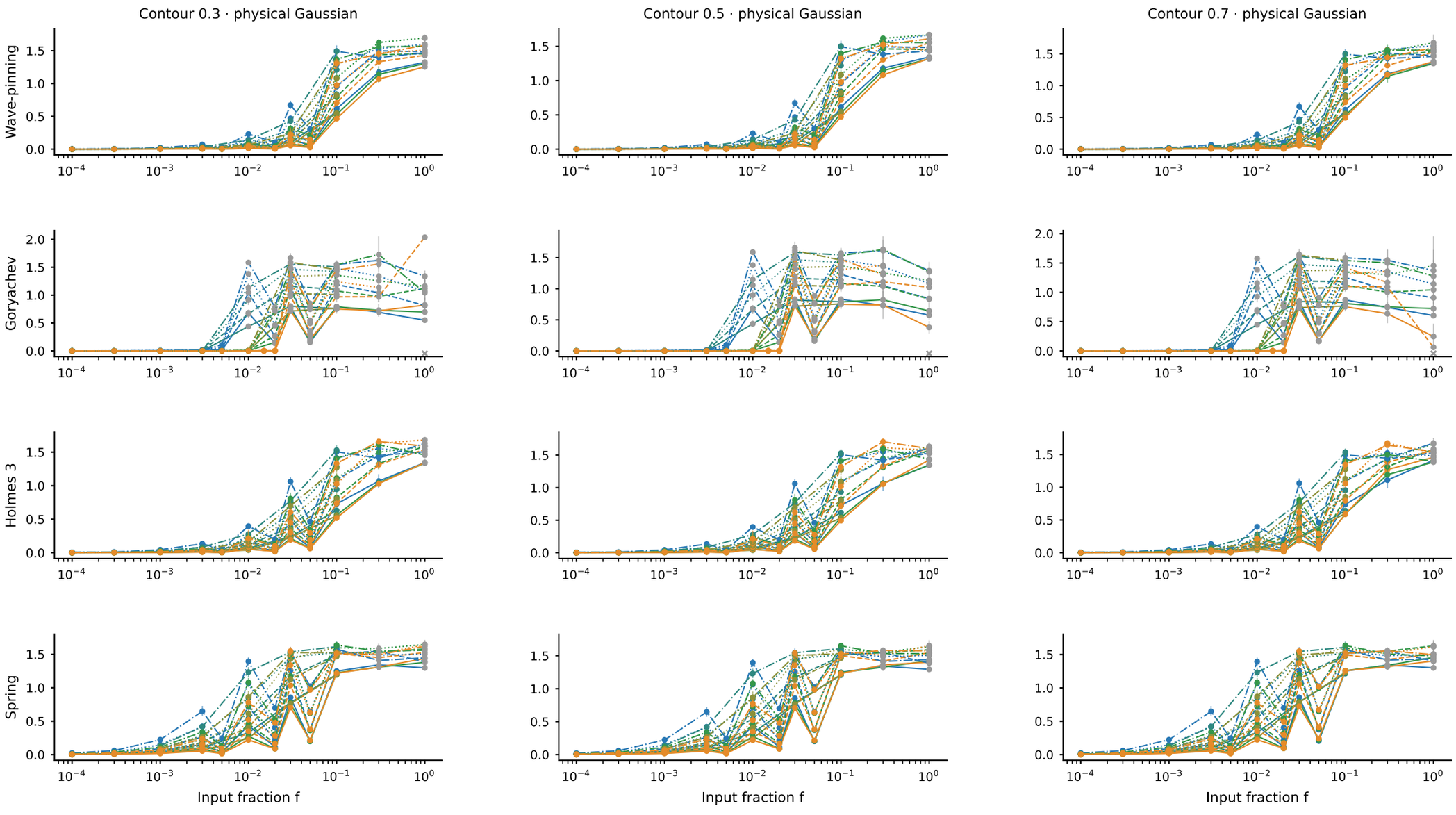
Common canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_013_S9_mean_absolute_turn_2_frames.csv

S9 · Common canonical A5 observations · Mean absolute dominant-heading turn over 2 observation frames · full native-factor sensitivity



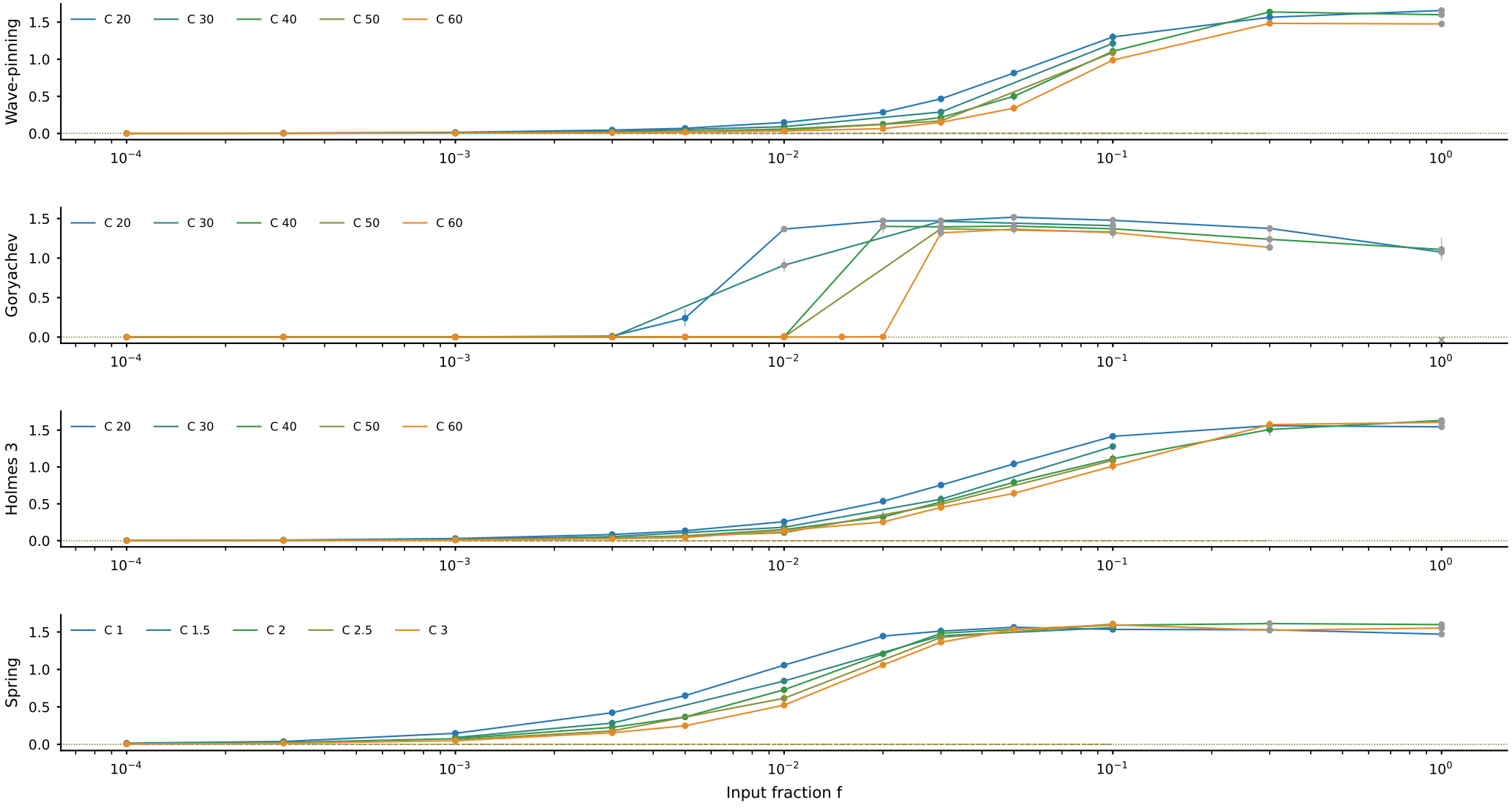
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_014_S9_mean_absolute_turn_2_frames_raw_factorial.csv

S9 · Common canonical A5 observations · Mean absolute dominant-heading turn over 2 observation frames · full native-factor sensitivity



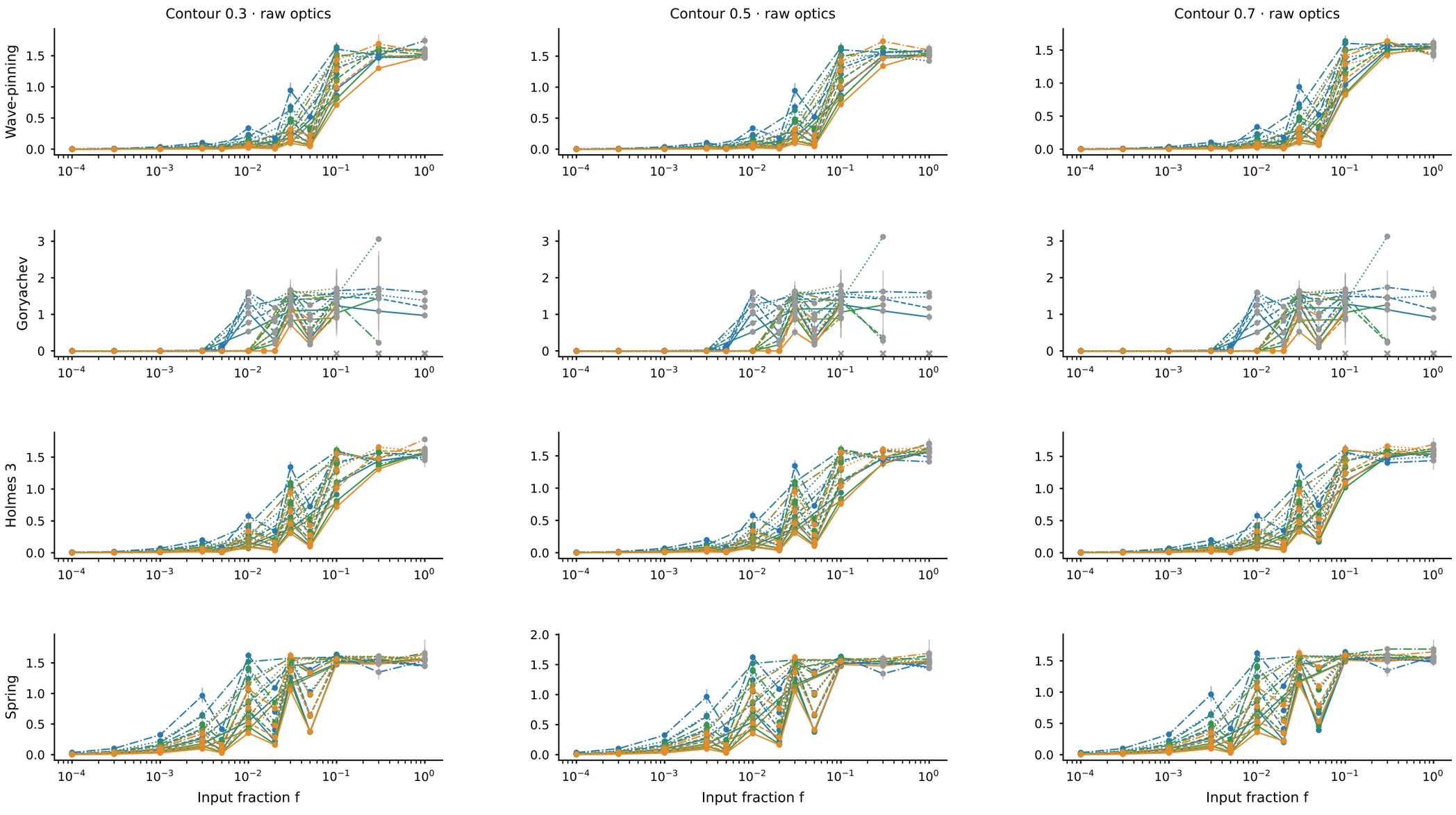
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_015_S9_mean_absolute_turn_2_frames_smoothed_factorial.csv

S9 · Common canonical A5 observations · Mean absolute dominant-heading turn over 4 observation frames



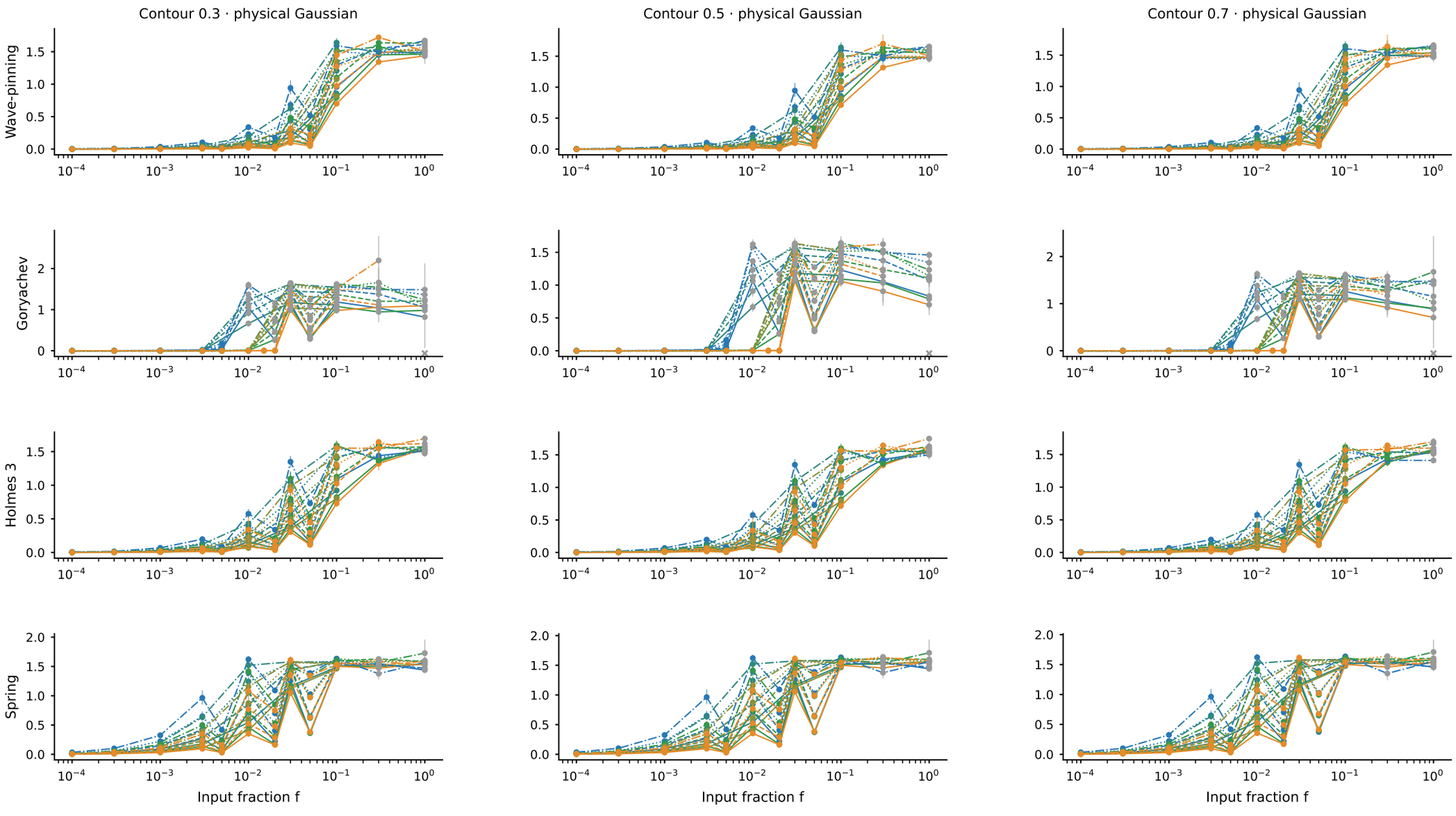
Common canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_016_S9_mean_absolute_turn_4_frames.csv

S9 · Common canonical A5 observations · Mean absolute dominant-heading turn over 4 observation frames · full native-factor sensitivity



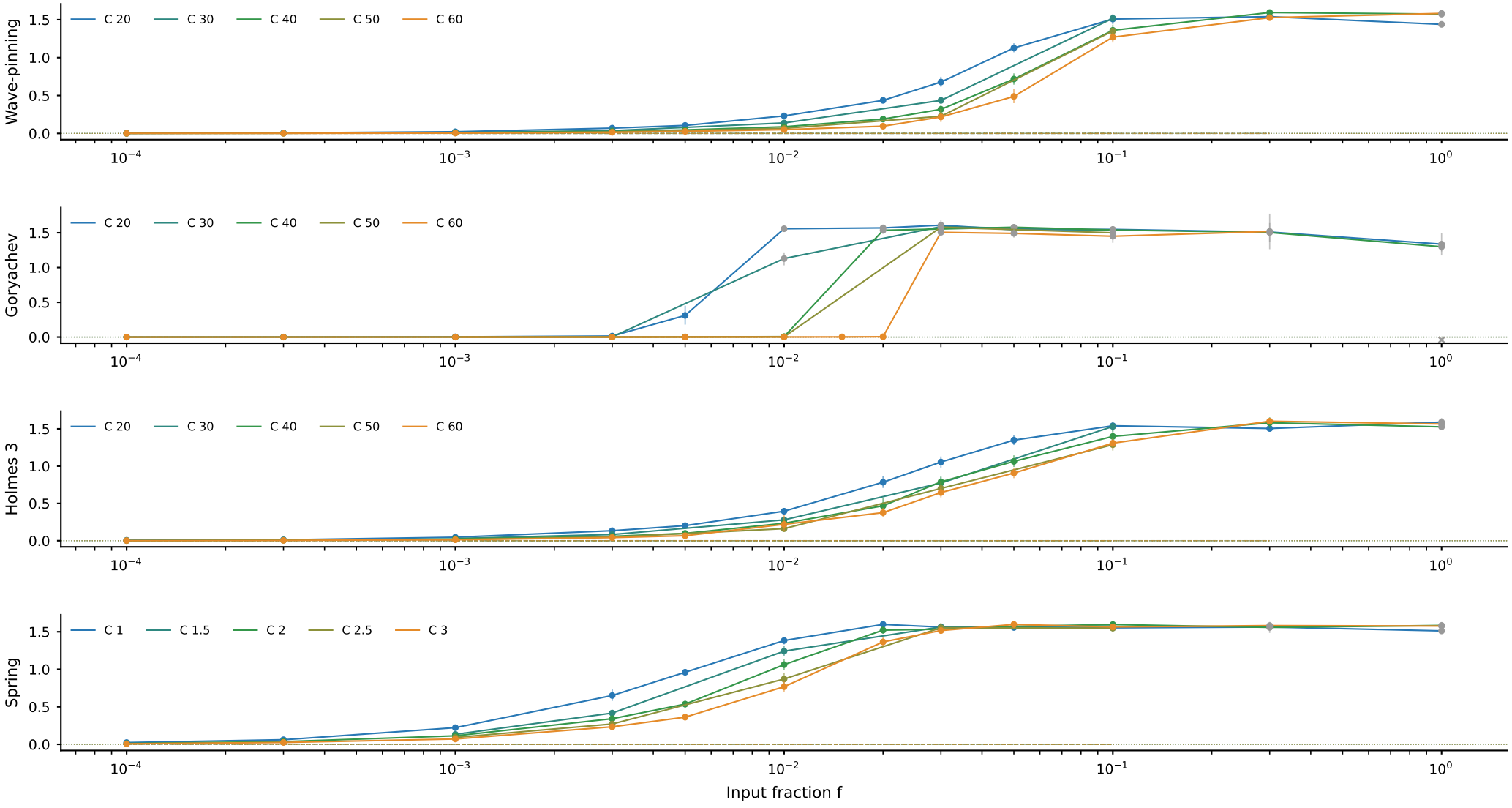
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_017_S9_mean_absolute_turn_4_frames_raw_factorial.csv

S9 · Common canonical A5 observations · Mean absolute dominant-heading turn over 4 observation frames · full native-factor sensitivity



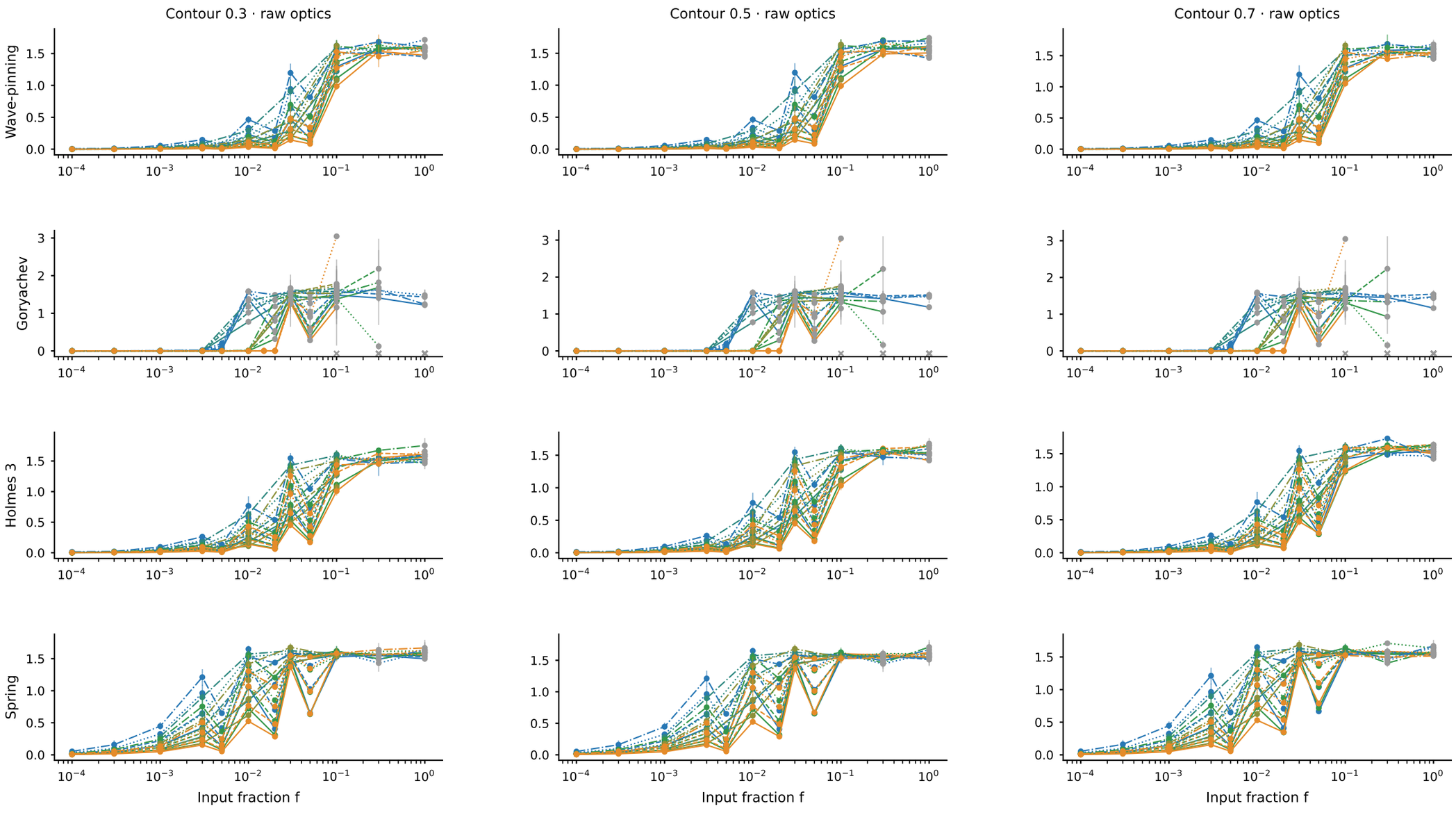
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_018_S9_mean_absolute_turn_4_frames_smoothed_factorial.csv

S9 · Common canonical A5 observations · Mean absolute dominant-heading turn over 8 observation frames



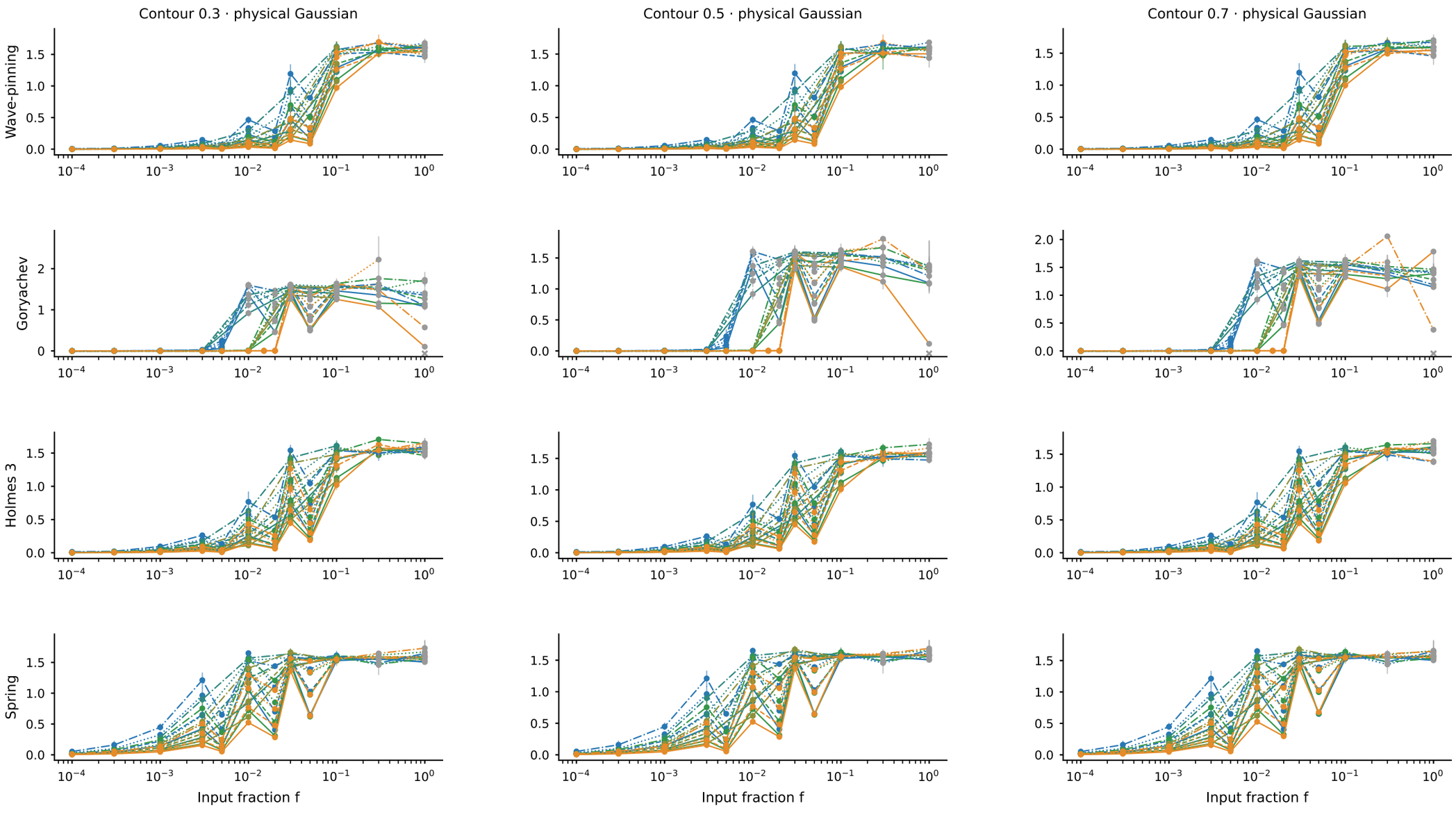
Common canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_019_S9_mean_absolute_turn_8_frames.csv

S9 · Common canonical A5 observations · Mean absolute dominant-heading turn over 8 observation frames · full native-factor sensitivity



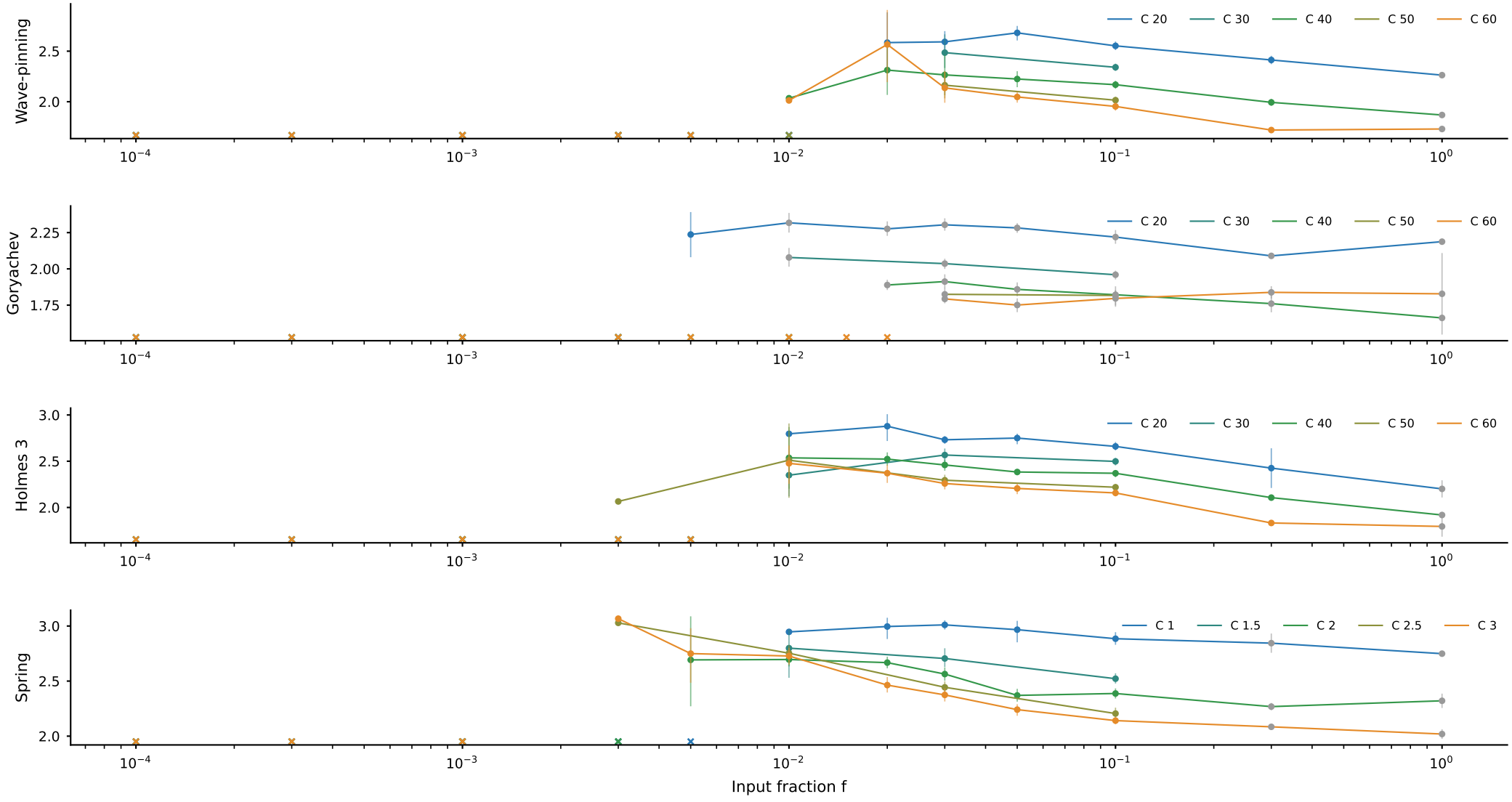
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_020_S9_mean_absolute_turn_8_frames_raw_factorial.csv

S9 · Common canonical A5 observations · Mean absolute dominant-heading turn over 8 observation frames · full native-factor sensitivity



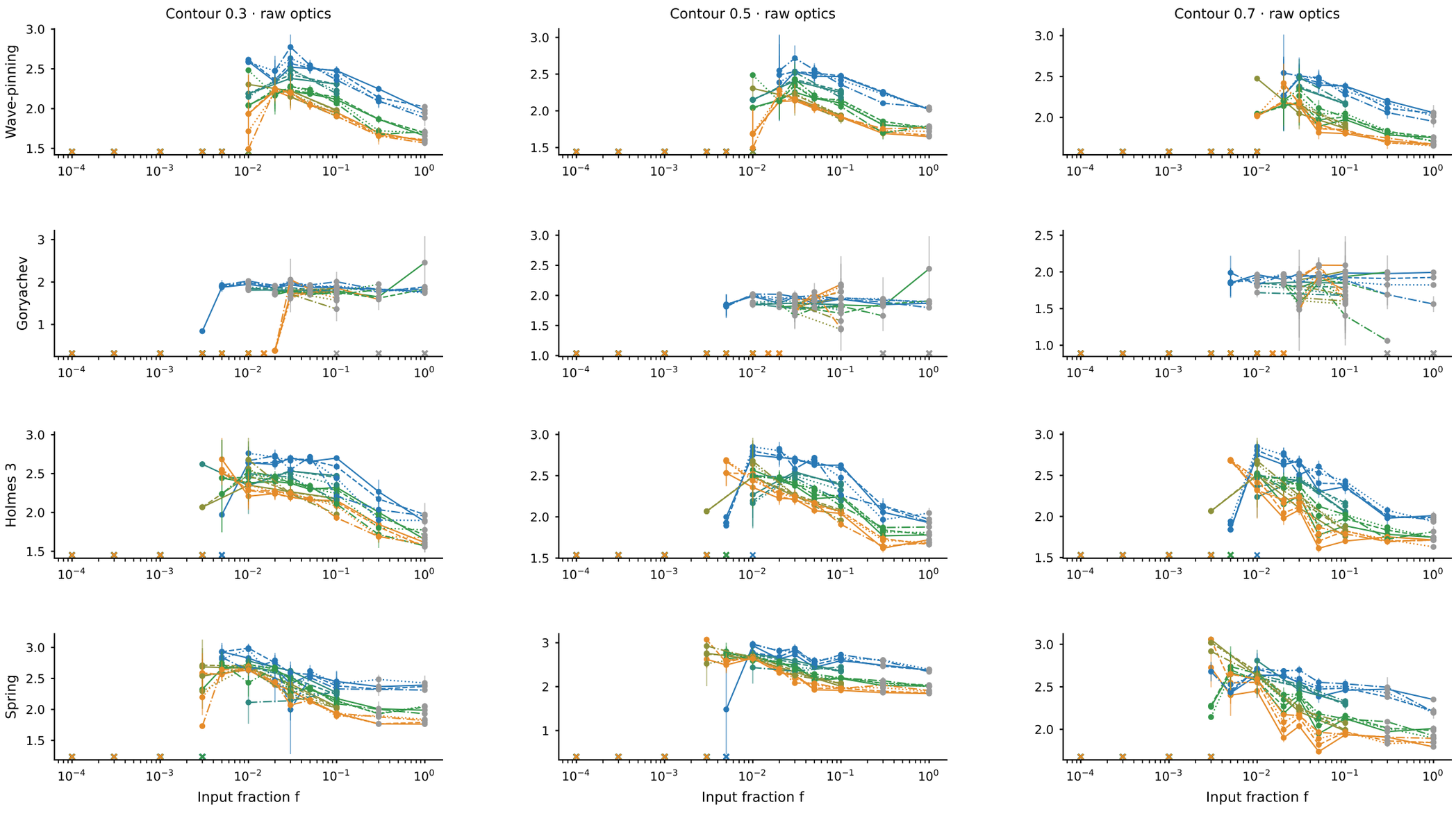
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_021_S9_mean_absolute_turn_8_frames_smoothed_factorial.csv

S9 · Common canonical A5 observations · Mean newborn angle from the previous dominant region



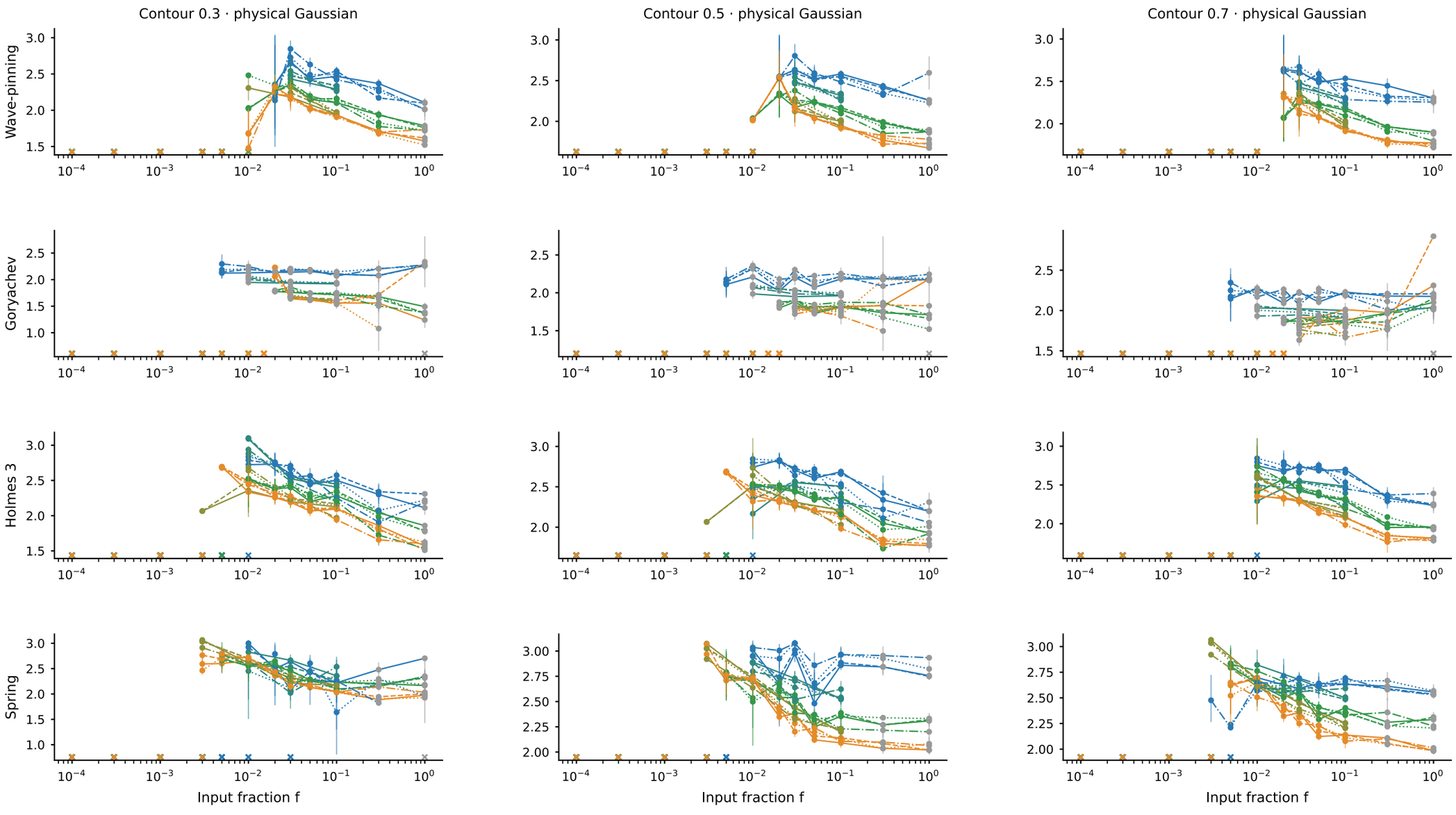
Common canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_022_S9_mean_birth_angle.csv

S9 · Common canonical A5 observations · Mean newborn angle from the previous dominant region · full native-factor sensitivity



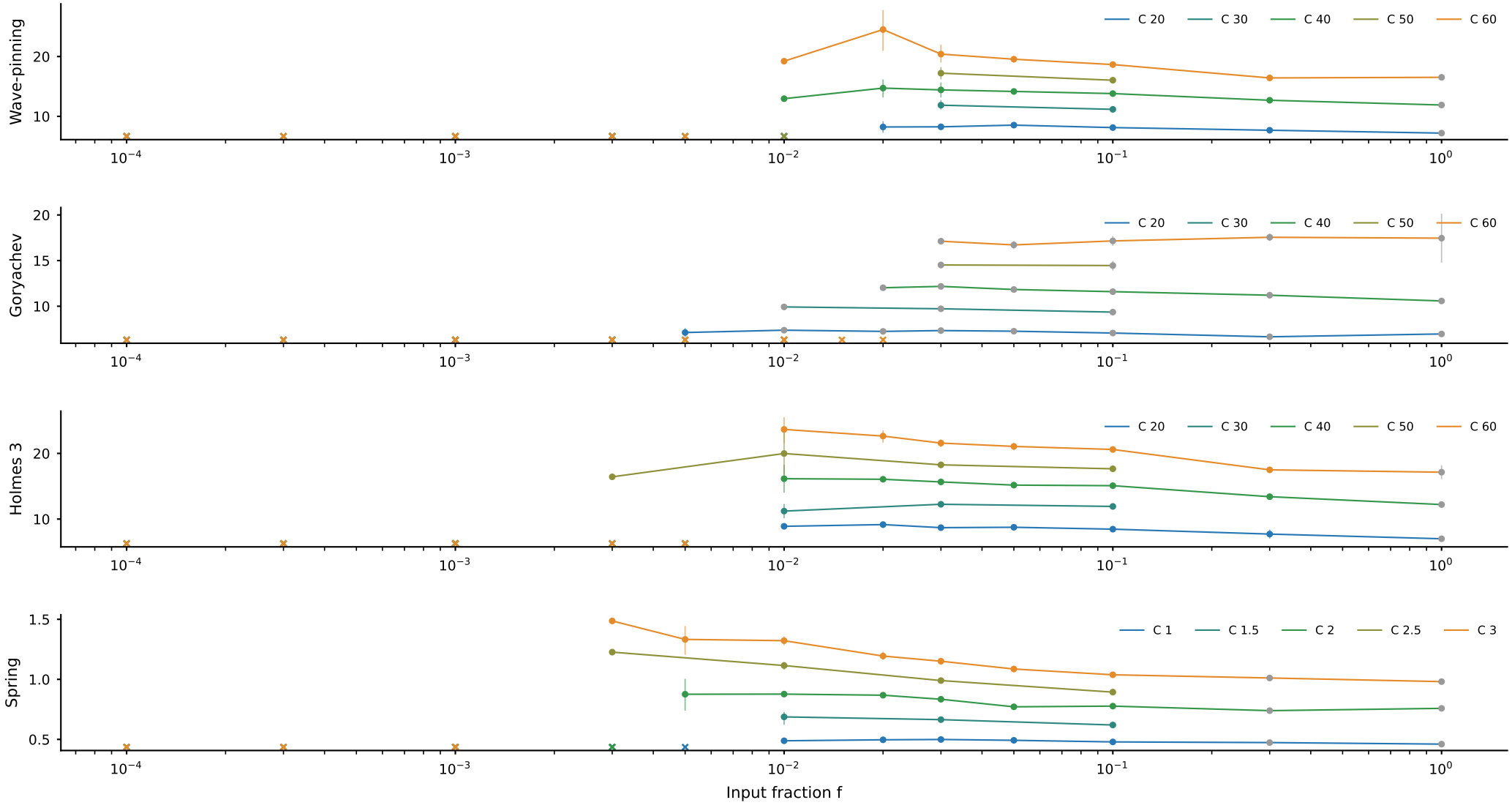
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_023_S9_mean_birth_angle_raw_factorial.csv

S9 · Common canonical A5 observations · Mean newborn angle from the previous dominant region · full native-factor sensitivity



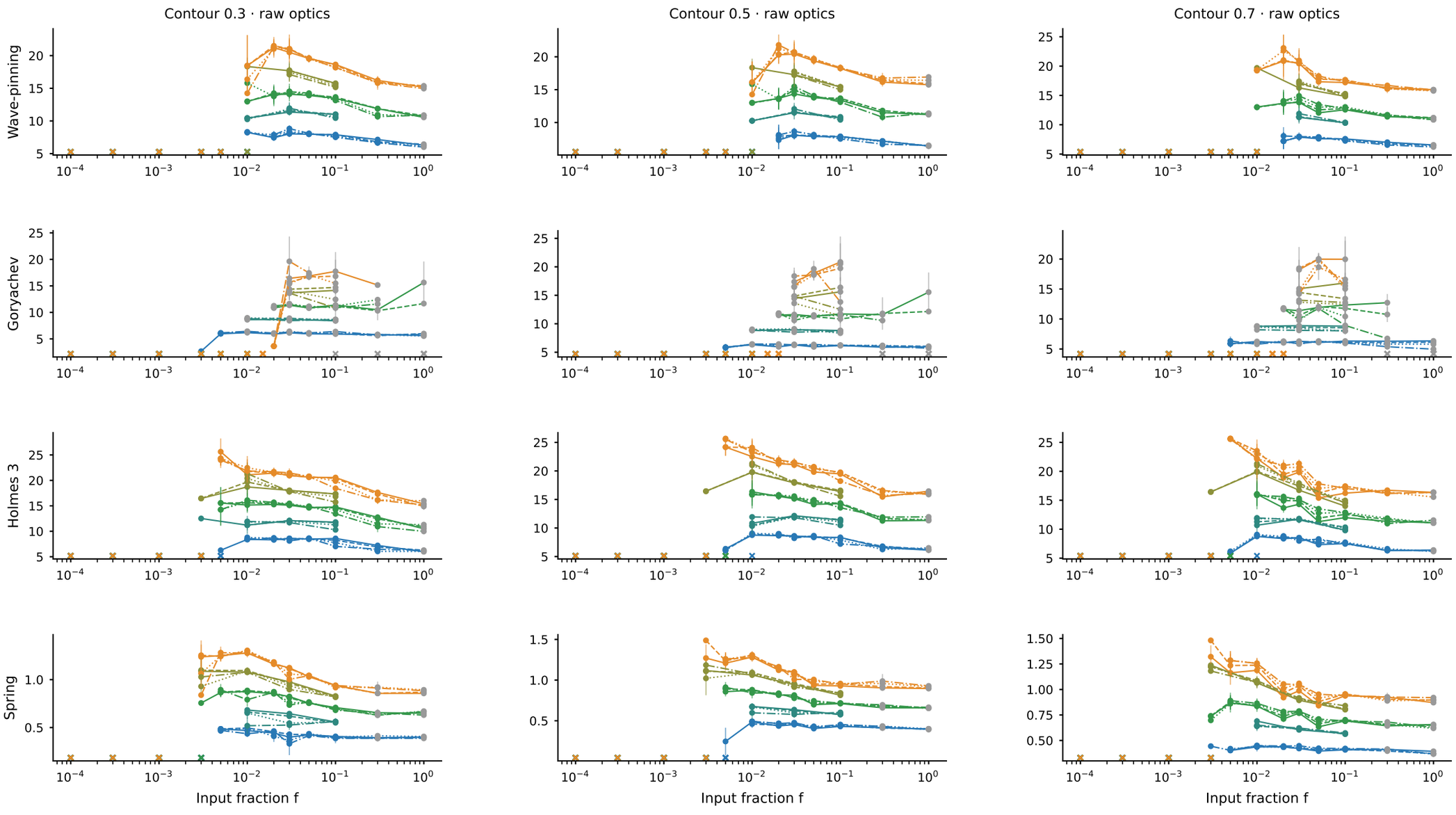
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_024_S9_mean_birth_angle_smoothed_factorial.csv

S9 · Common canonical A5 observations · Mean newborn physical arc distance



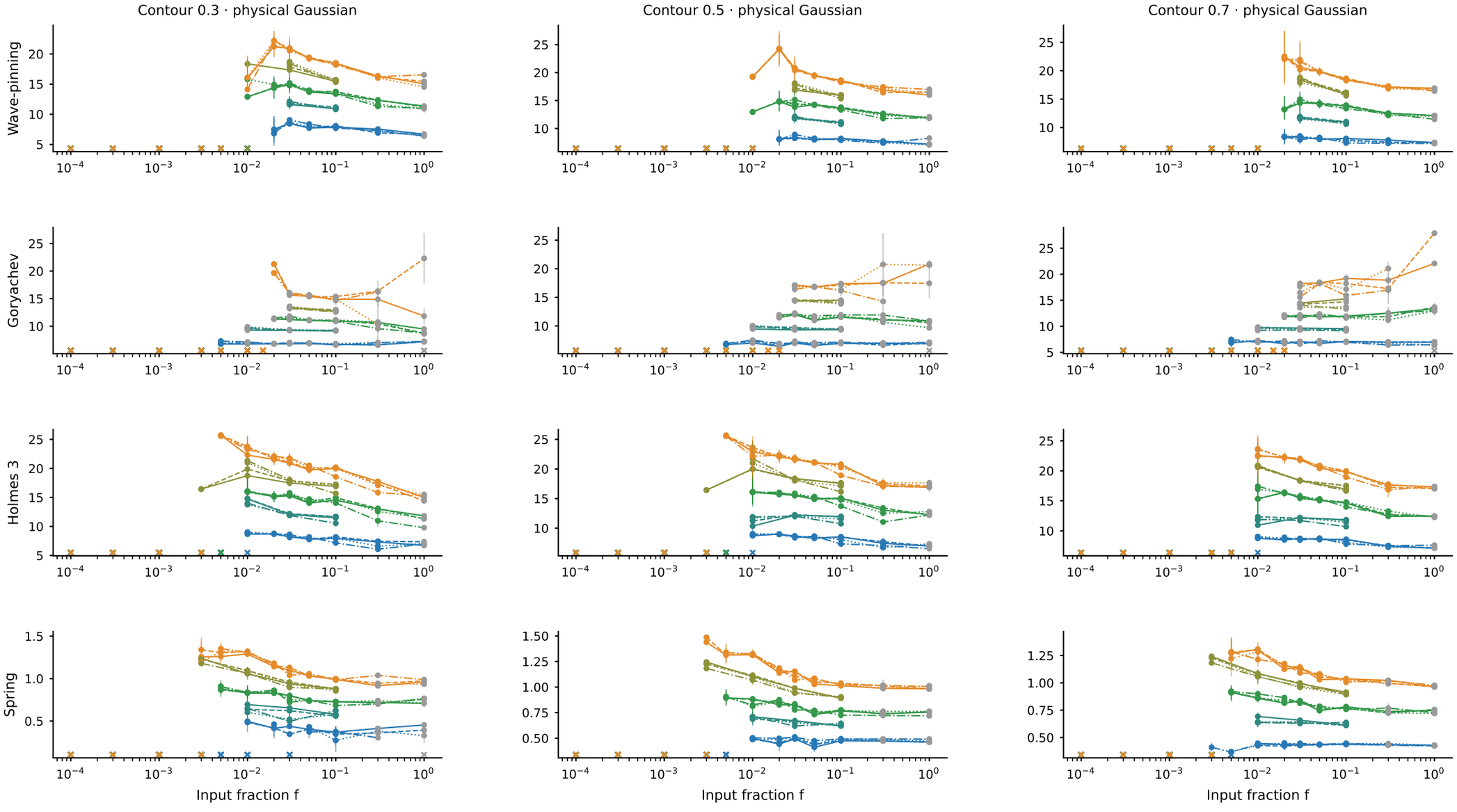
Common canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_025_S9_mean_birth_physical_distance.csv

S9 · Common canonical A5 observations · Mean newborn physical arc distance · full native-factor sensitivity



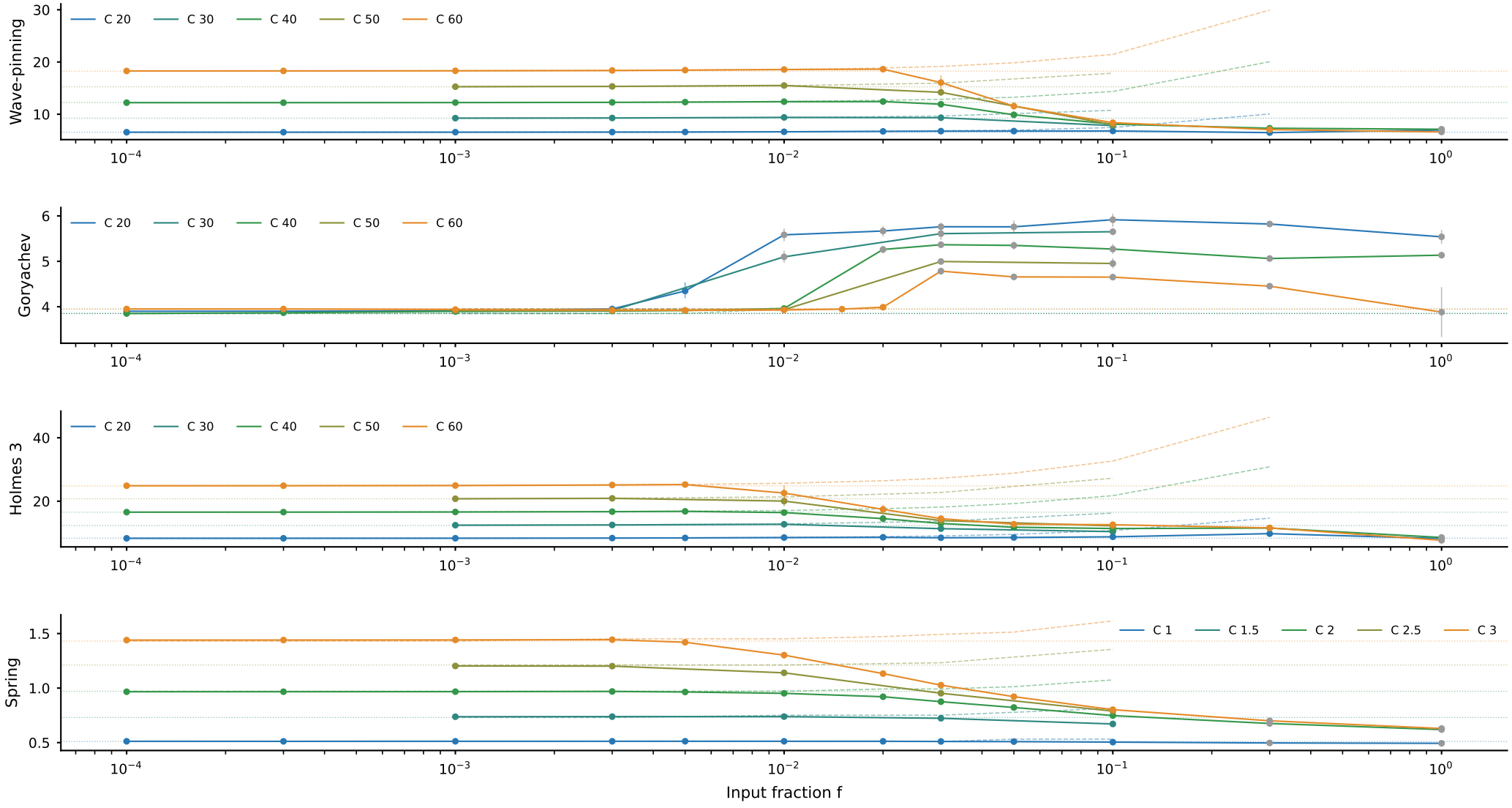
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_026_S9_mean_birth_physical_distance_raw_factorial.csv

S9 · Common canonical A5 observations · Mean newborn physical arc distance · full native-factor sensitivity



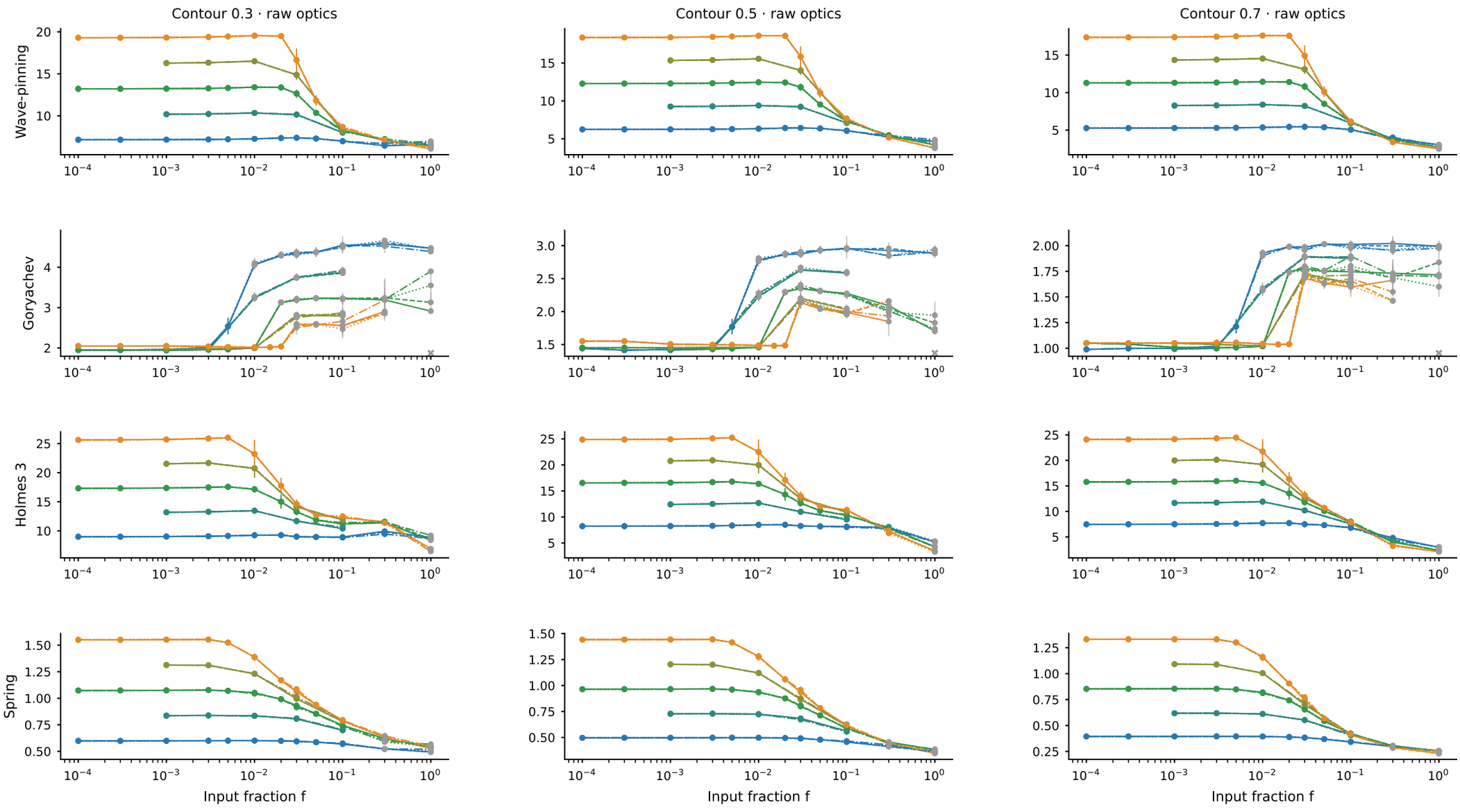
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_027_S9_mean_birth_physical_distance_smoothed_factorial.csv

S9 · Common canonical A5 observations · Mean physical region width



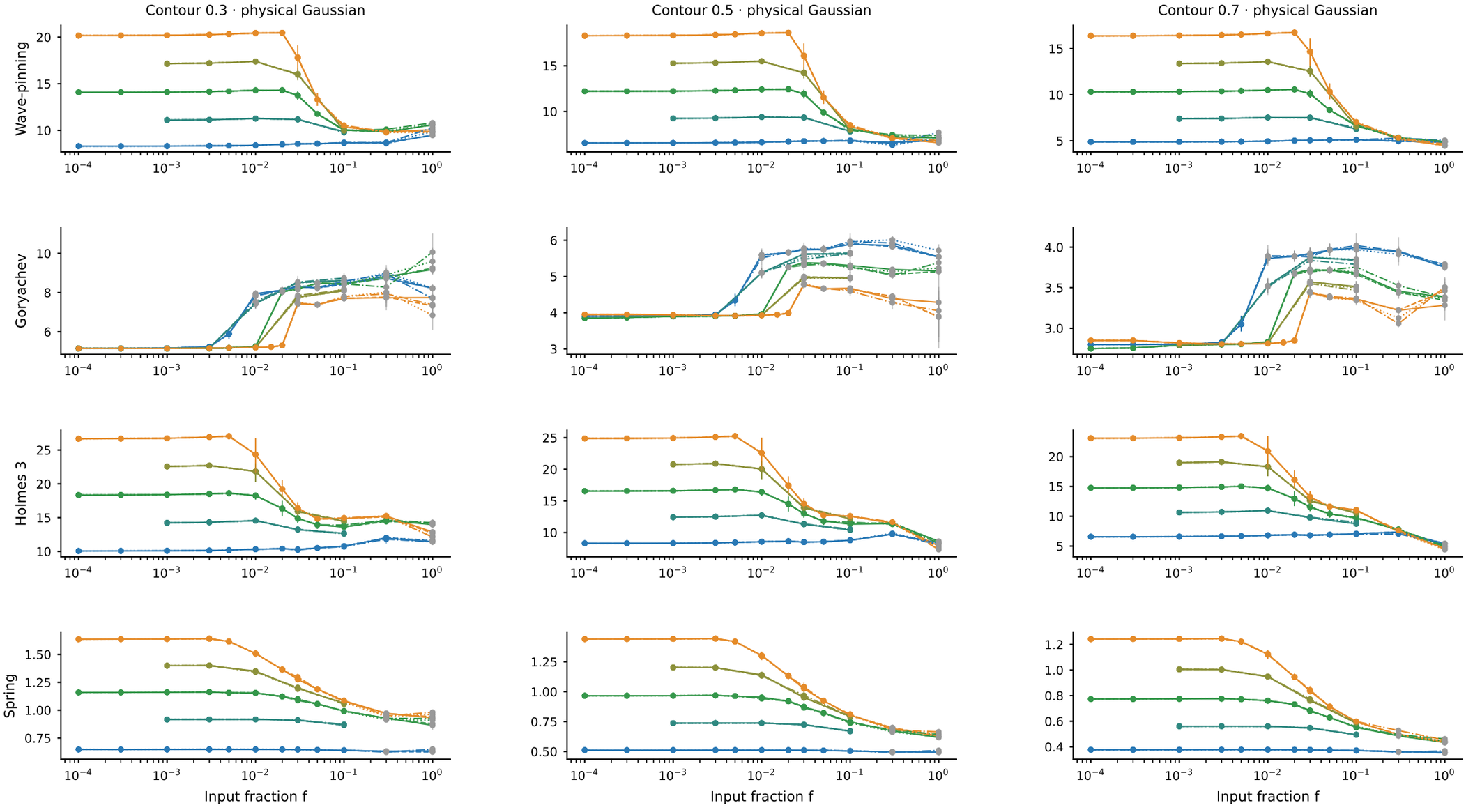
Common canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_028_S9_mean_region_width.csv

S9 · Common canonical A5 observations · Mean physical region width · full native-factor sensitivity



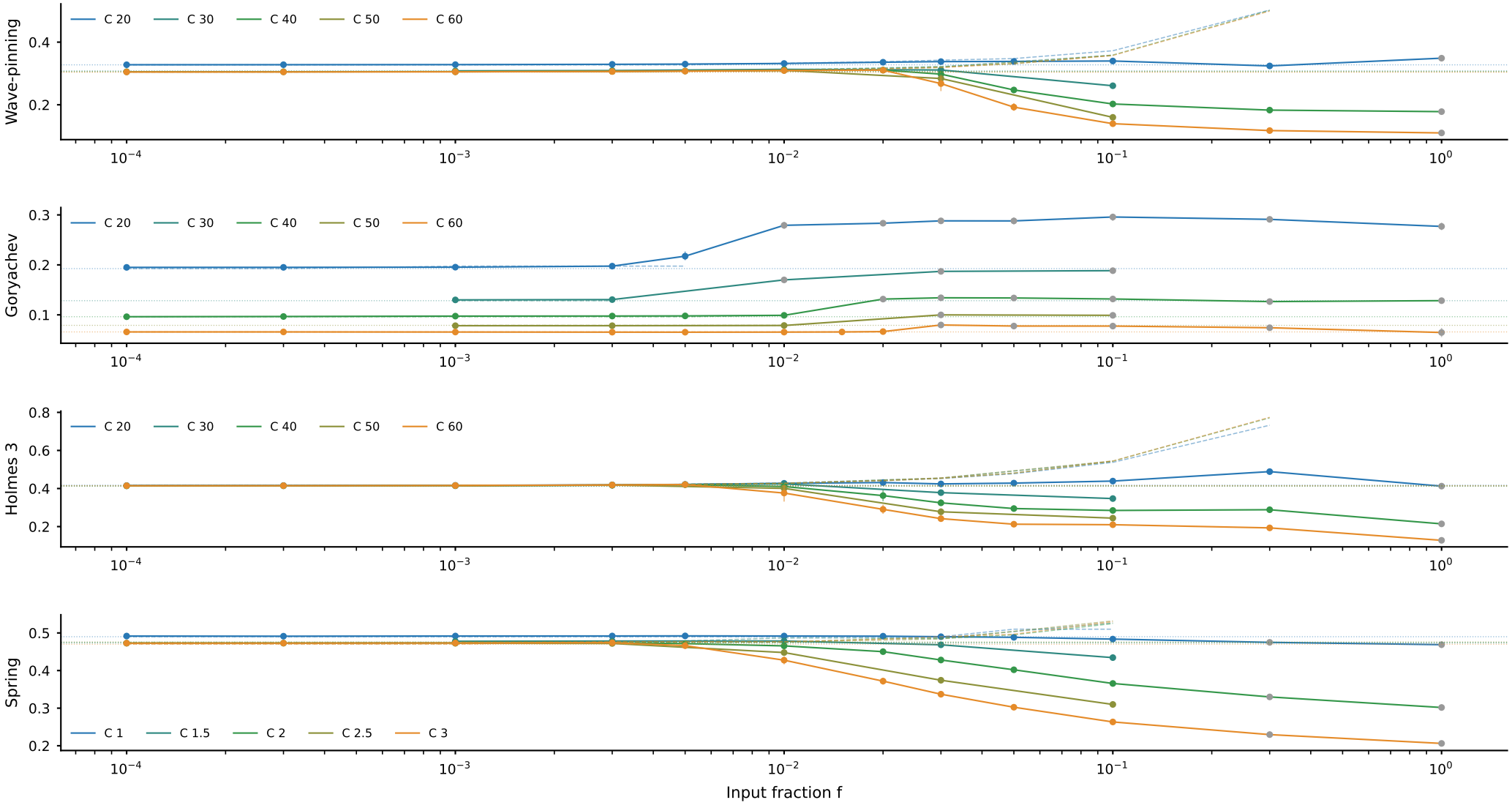
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_029_S9_mean_region_width_raw_factorial.csv

S9 · Common canonical A5 observations · Mean physical region width · full native-factor sensitivity



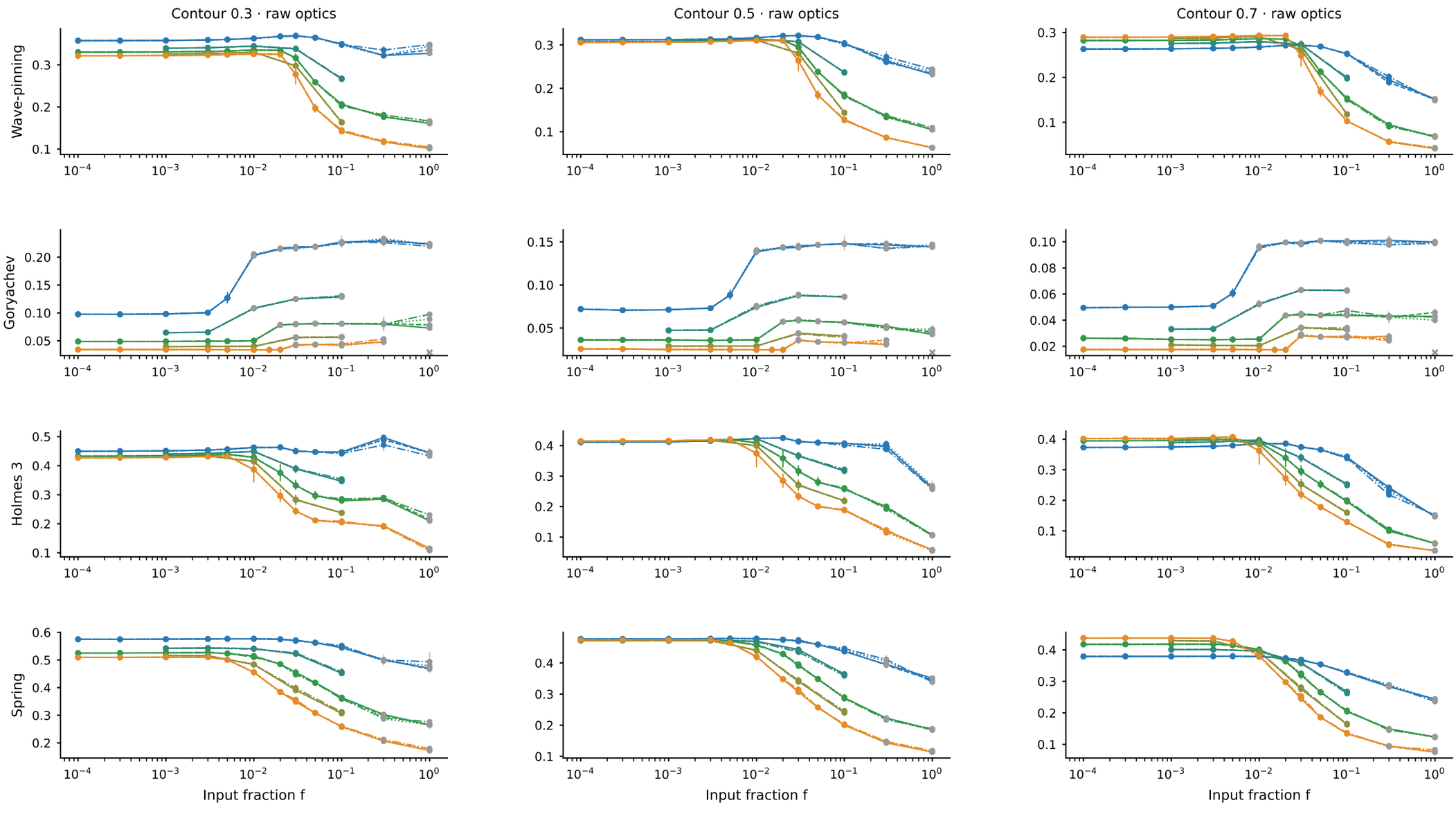
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_030_S9_mean_region_width_smoothed_factorial.csv

S9 · Common canonical A5 observations · Mean region width / actual circumference



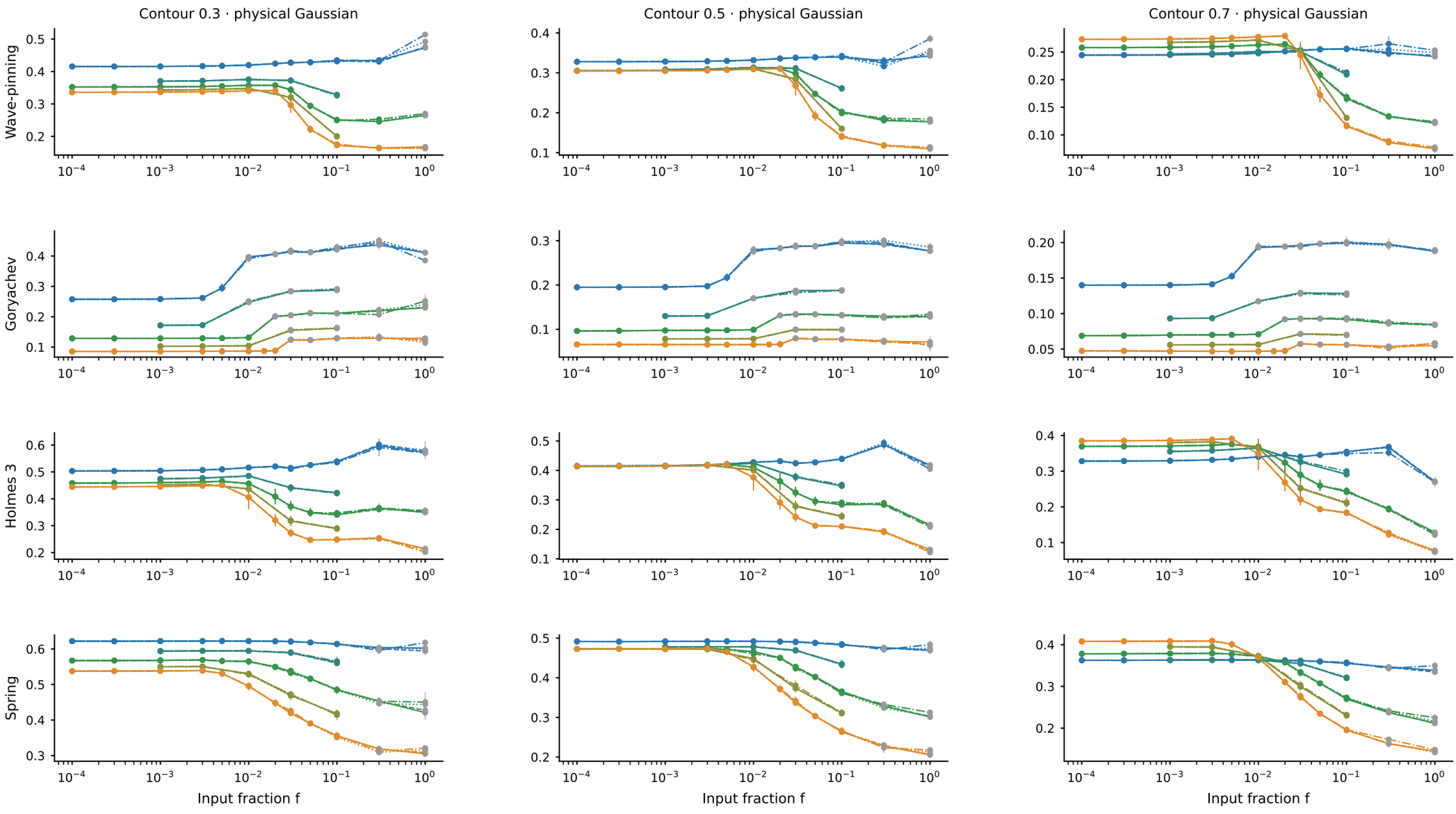
Common canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_031_S9_mean_region_width_fraction.csv

S9 · Common canonical A5 observations · Mean region width / actual circumference · full native-factor sensitivity



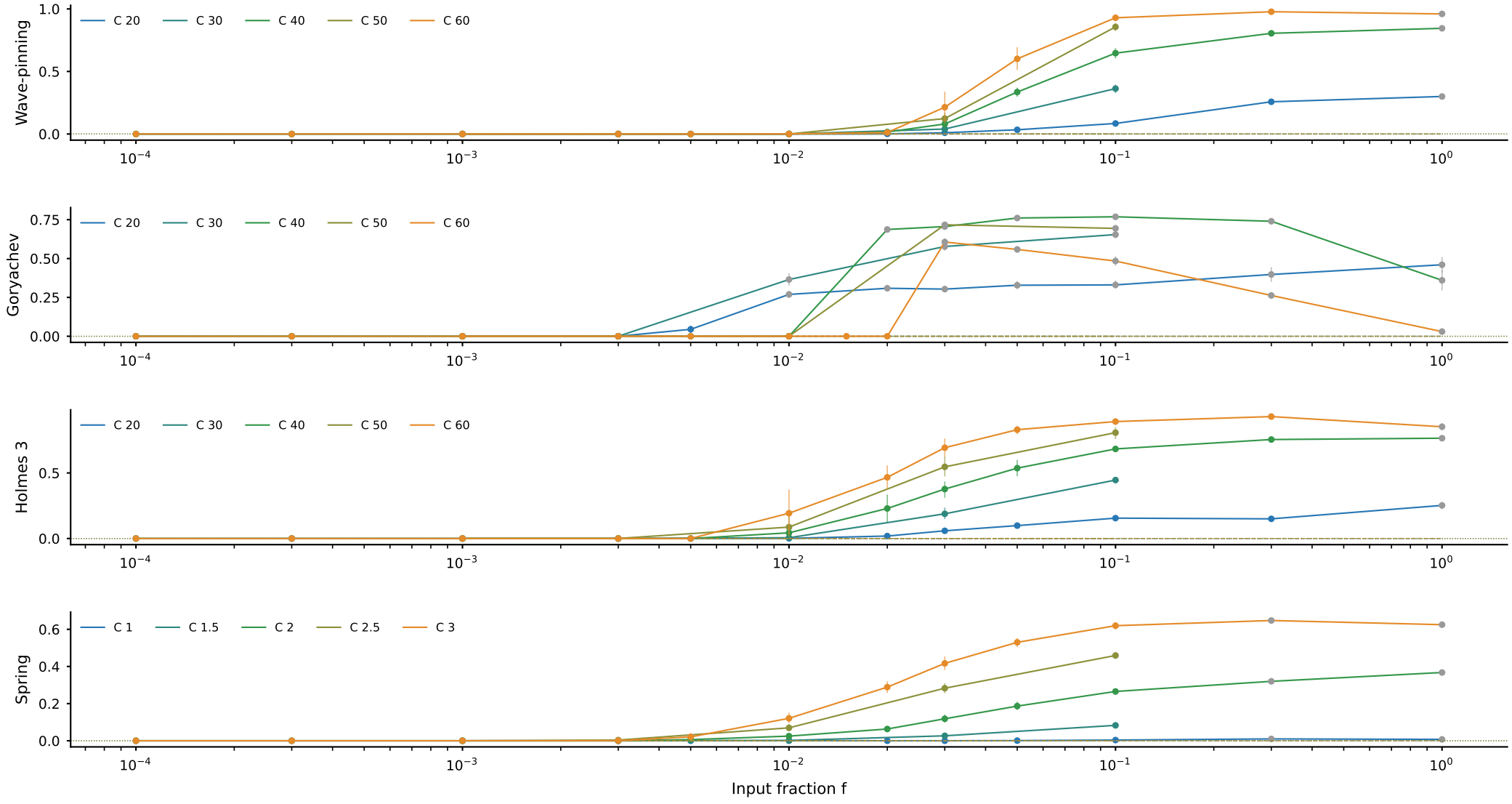
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_032_S9_mean_region_width_fraction_raw_factorial.csv

S9 · Common canonical A5 observations · Mean region width / actual circumference · full native-factor sensitivity



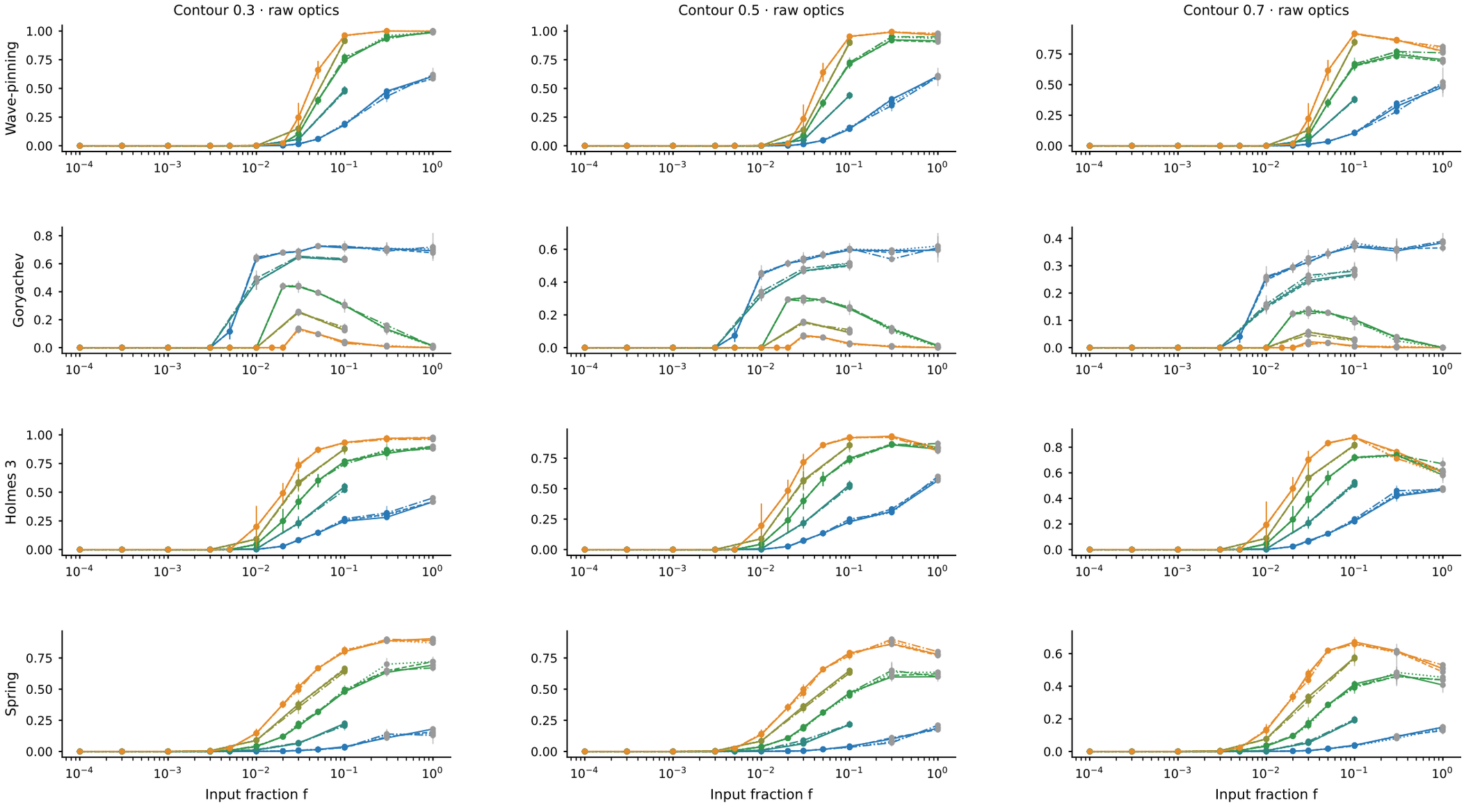
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_033_S9_mean_region_width_fraction_smoothed_factorial.csv

S9 · Common canonical A5 observations · Time fraction with multiple qualified regions



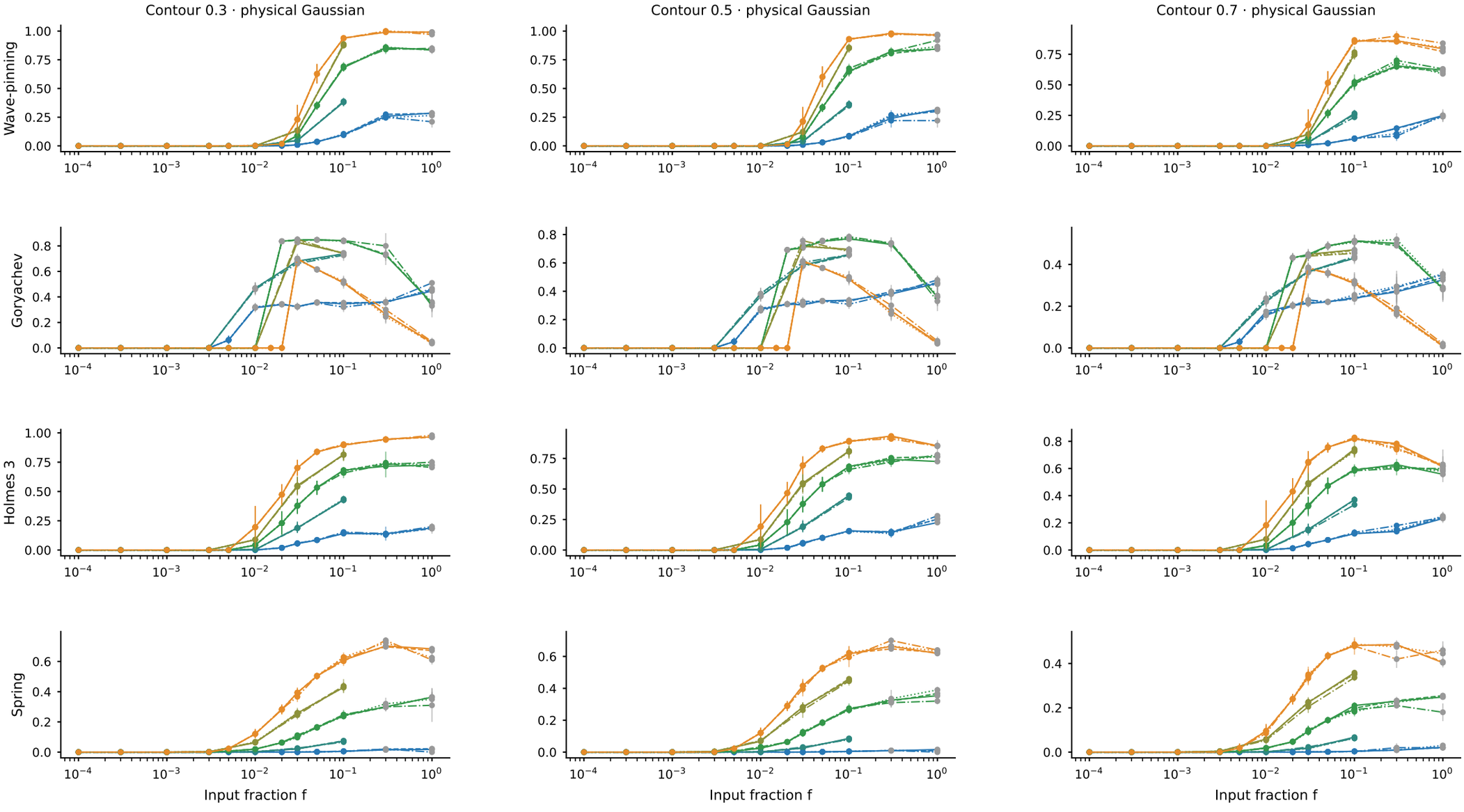
Common canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_034_S9_multiple_front_fraction.csv

S9 · Common canonical A5 observations · Time fraction with multiple qualified regions · full native-factor sensitivity



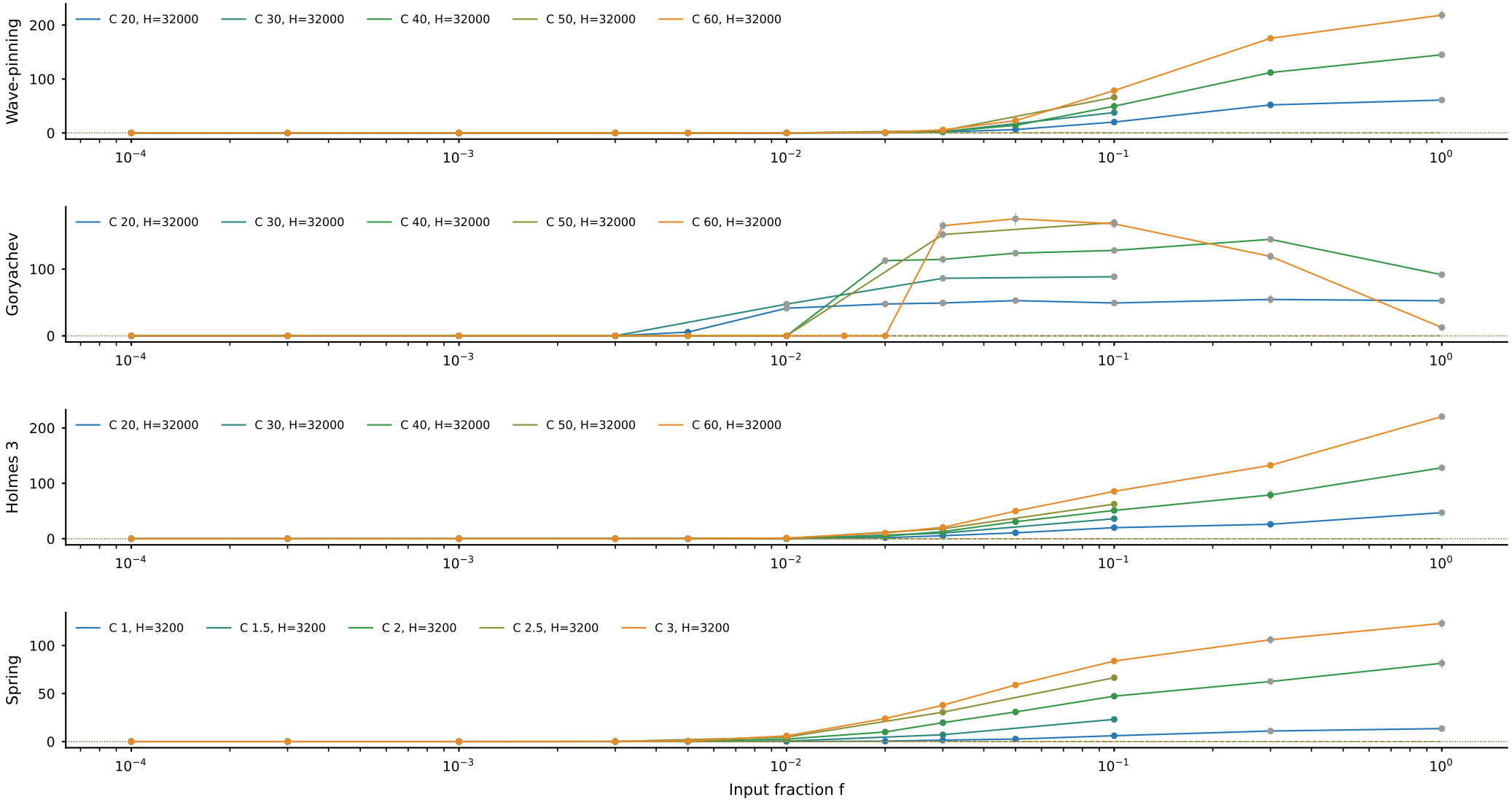
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_035_S9_multiple_front_fraction_raw_factorial.csv

S9 · Common canonical A5 observations · Time fraction with multiple qualified regions · full native-factor sensitivity



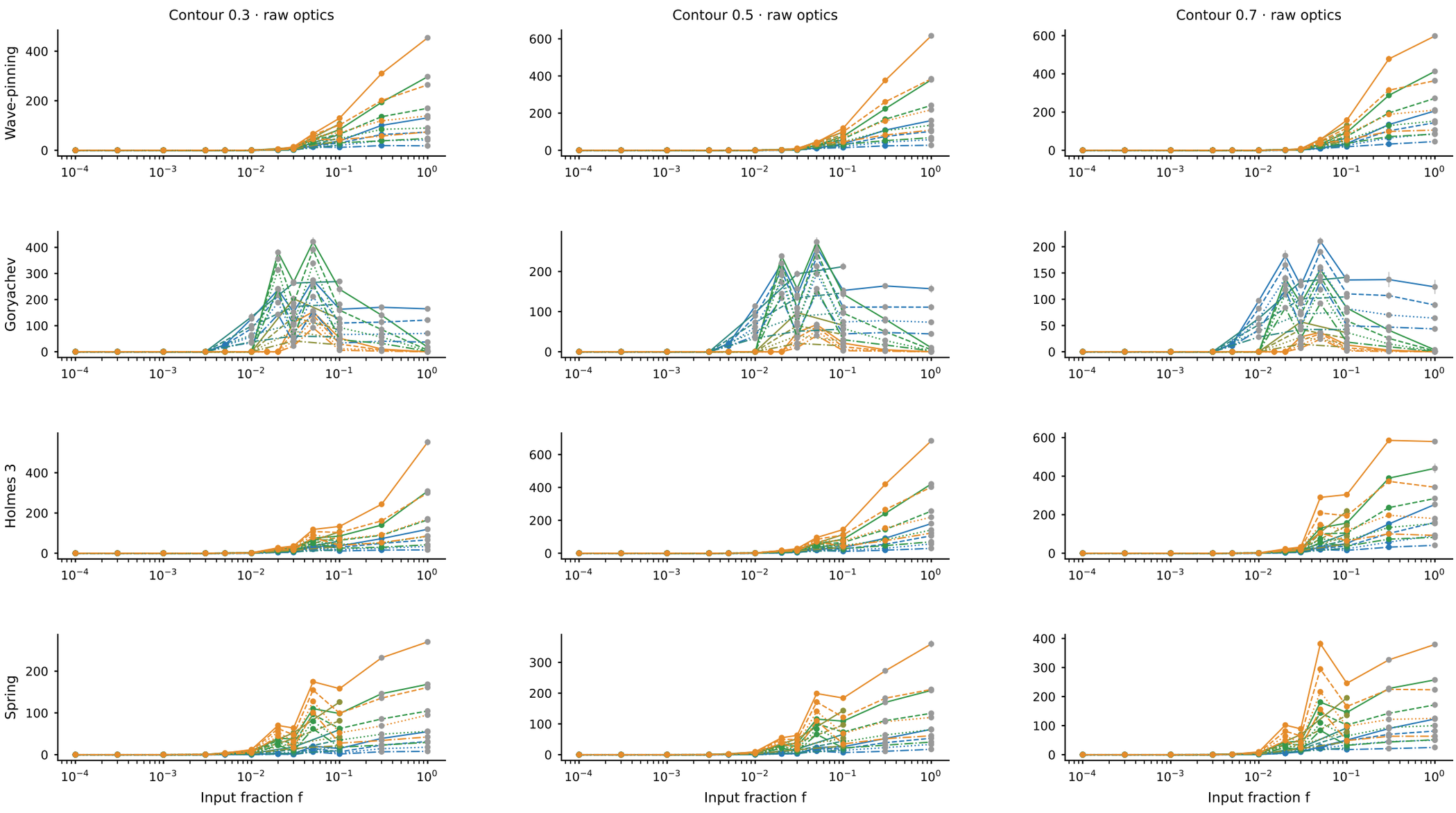
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_036_S9_multiple_front_fraction_smoothed_factorial.csv

S9 · Common canonical A5 observations · Observed region births per trajectory



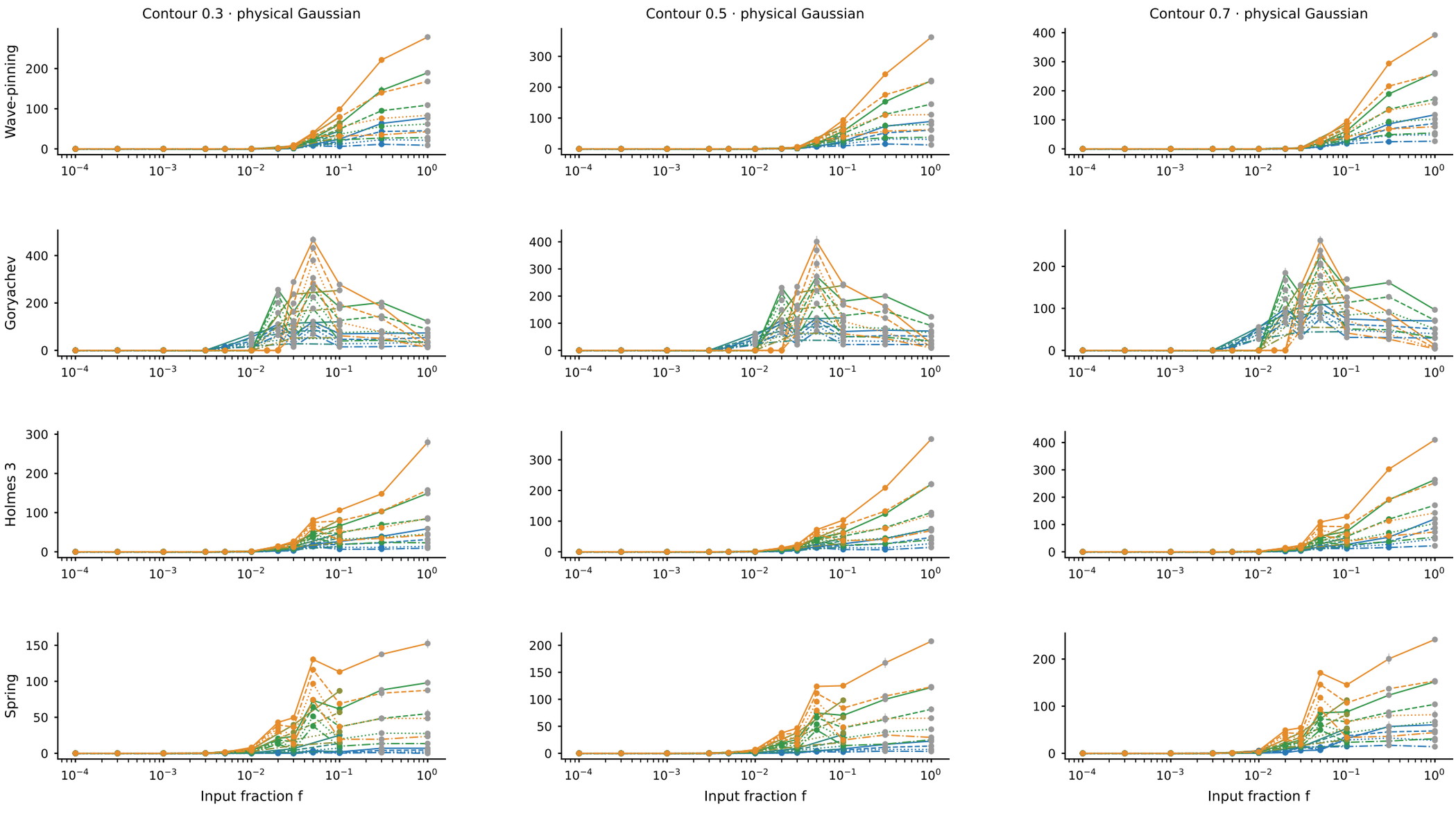
Common canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_037_S9_observed_births.csv

S9 · Common canonical A5 observations · Observed region births per trajectory · full native-factor sensitivity



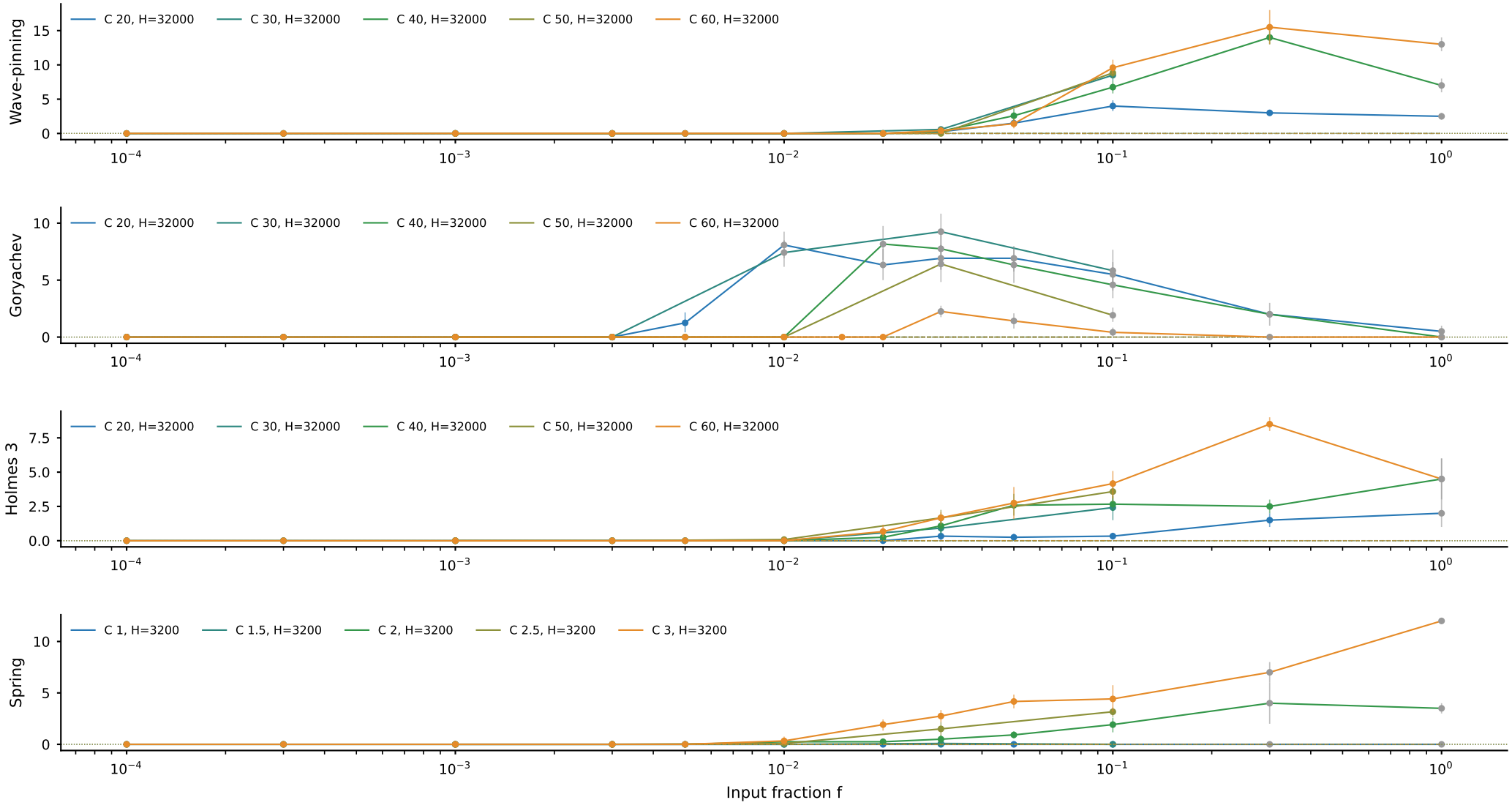
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_038_S9_observed_births_raw_factorial.csv

S9 · Common canonical A5 observations · Observed region births per trajectory · full native-factor sensitivity



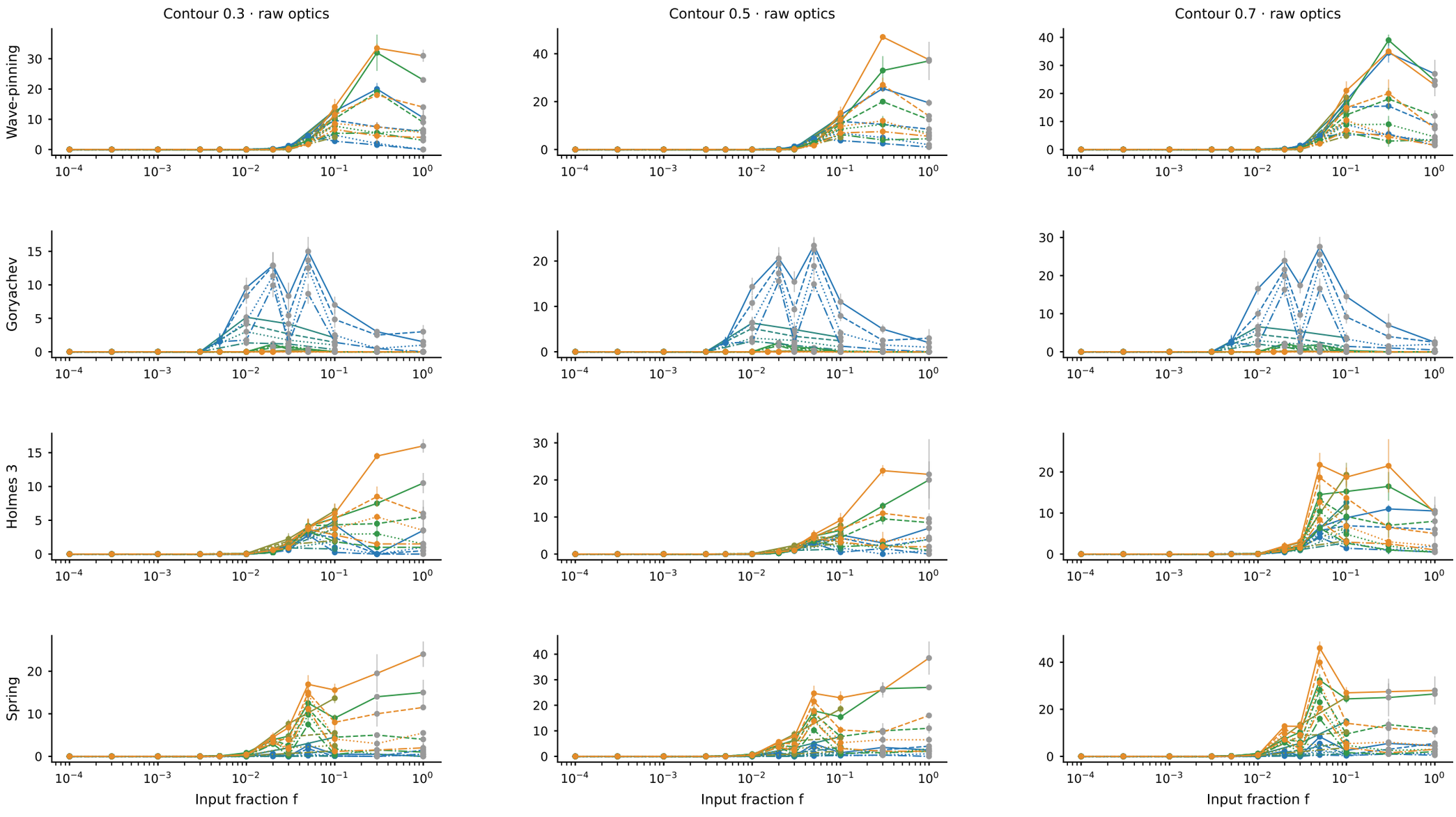
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_039_S9_observed_births_smoothed_factorial.csv

S9 · Common canonical A5 observations · Literal pairwise takeover sensitivity per trajectory



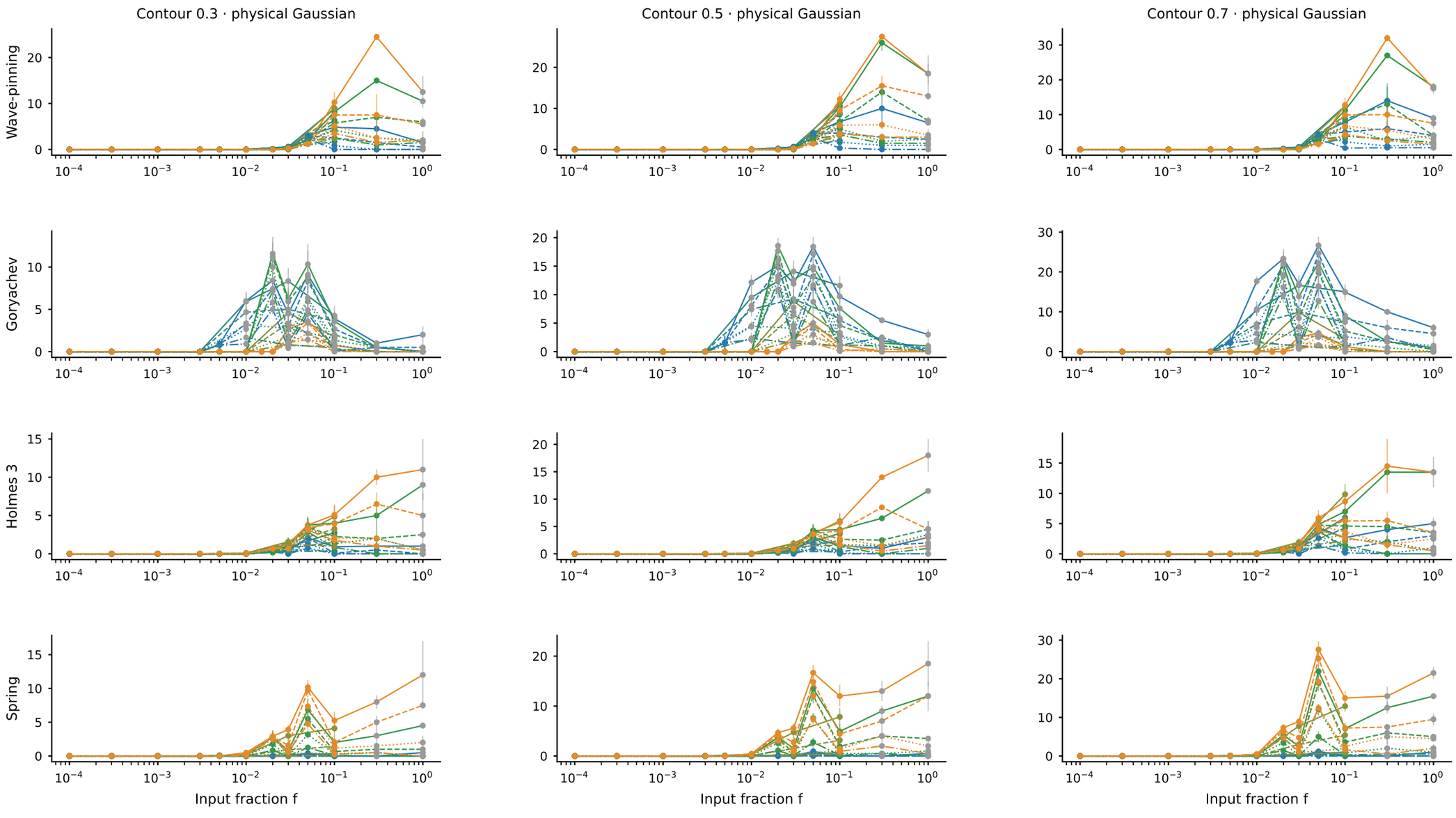
Common canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_040_S9_observed_pairwise_takeovers.csv

S9 · Common canonical A5 observations · Literal pairwise takeover sensitivity per trajectory · full native-factor sensitivity



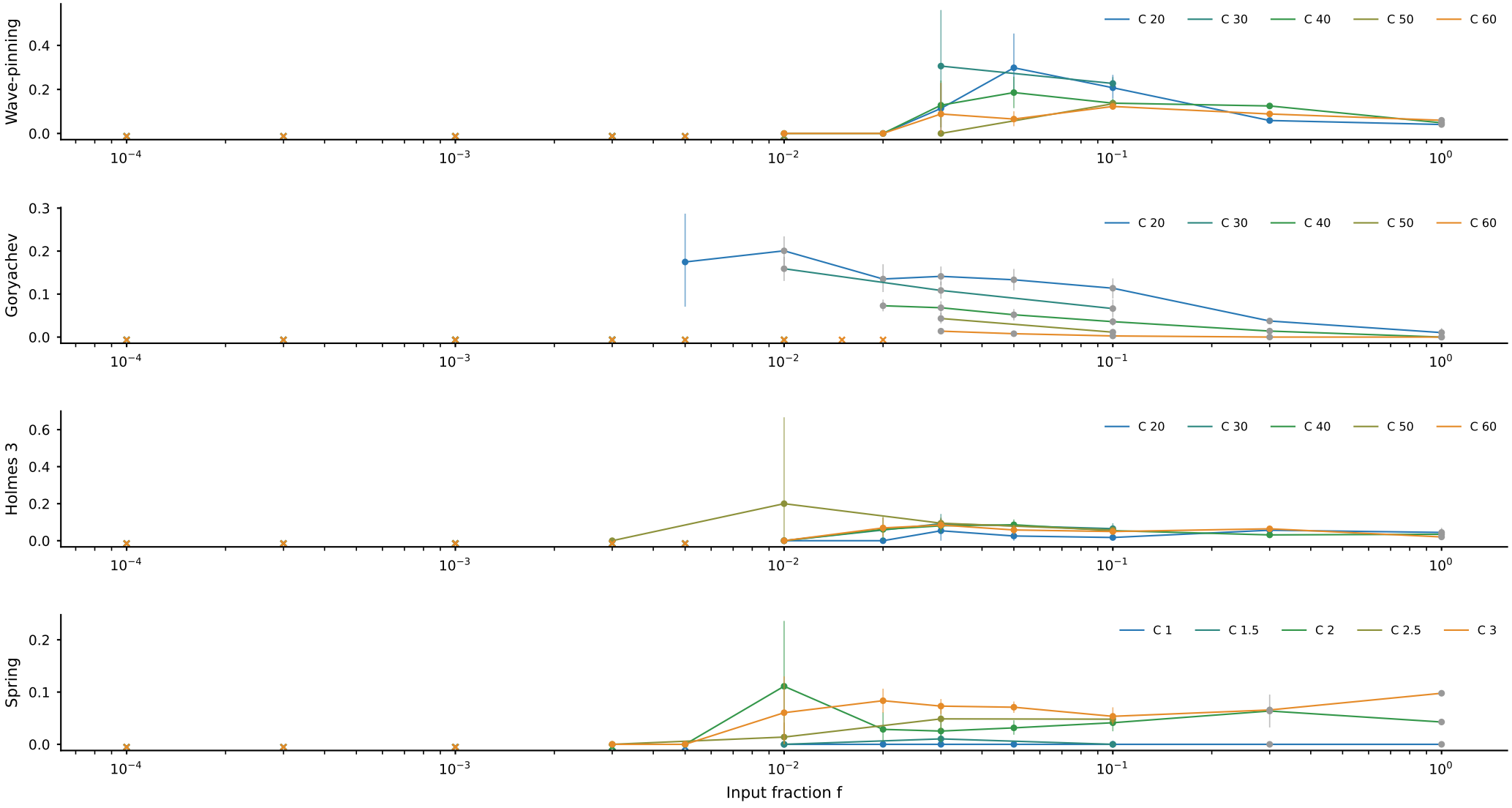
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_041_S9_observed_pairwise_takeovers_raw_factorial.csv

S9 · Common canonical A5 observations · Literal pairwise takeover sensitivity per trajectory · full native-factor sensitivity



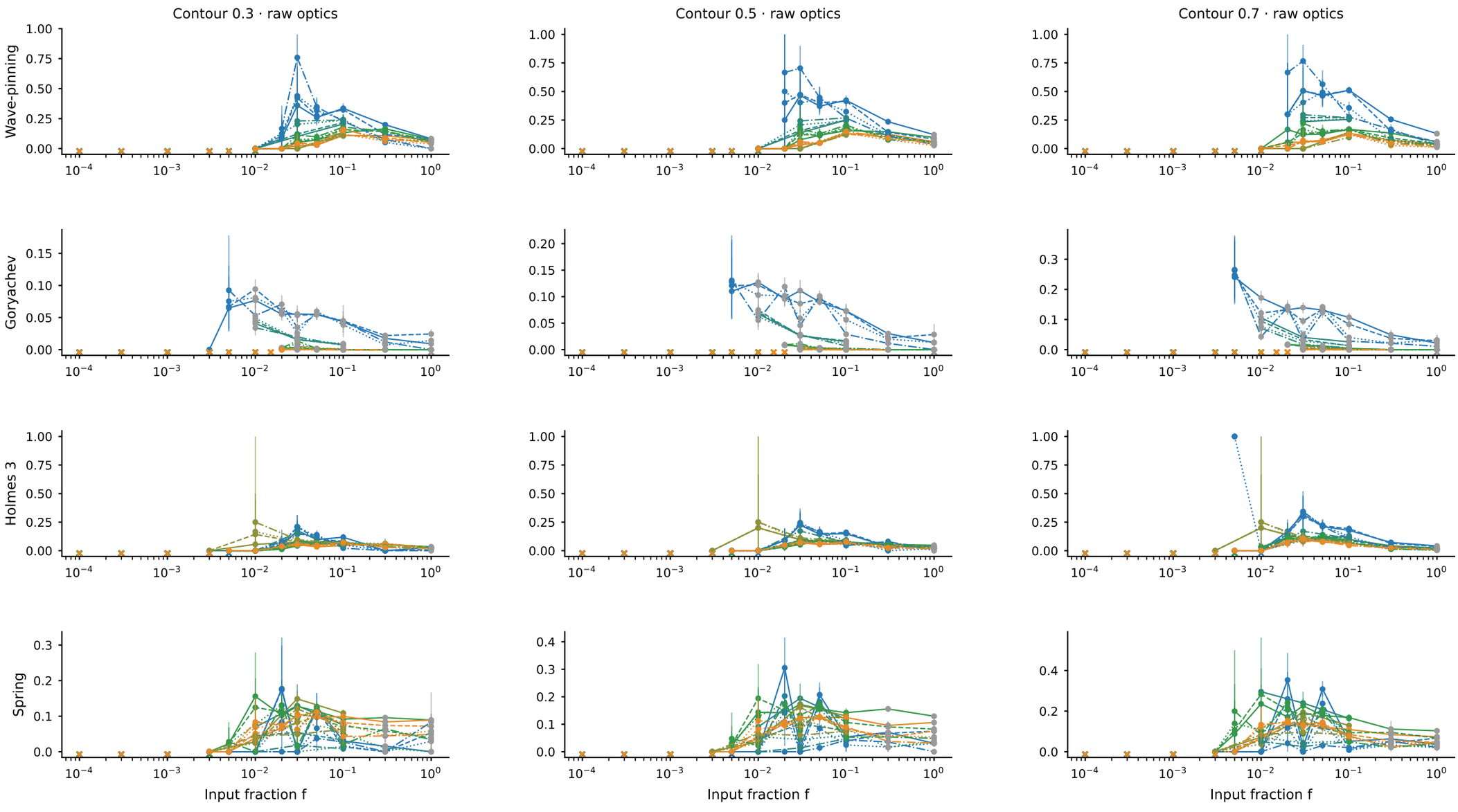
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_042_S9_observed_pairwise_takeovers_smoothed_factorial.csv

S9 · Common canonical A5 observations · Literal pairwise takeover sensitivity per birth



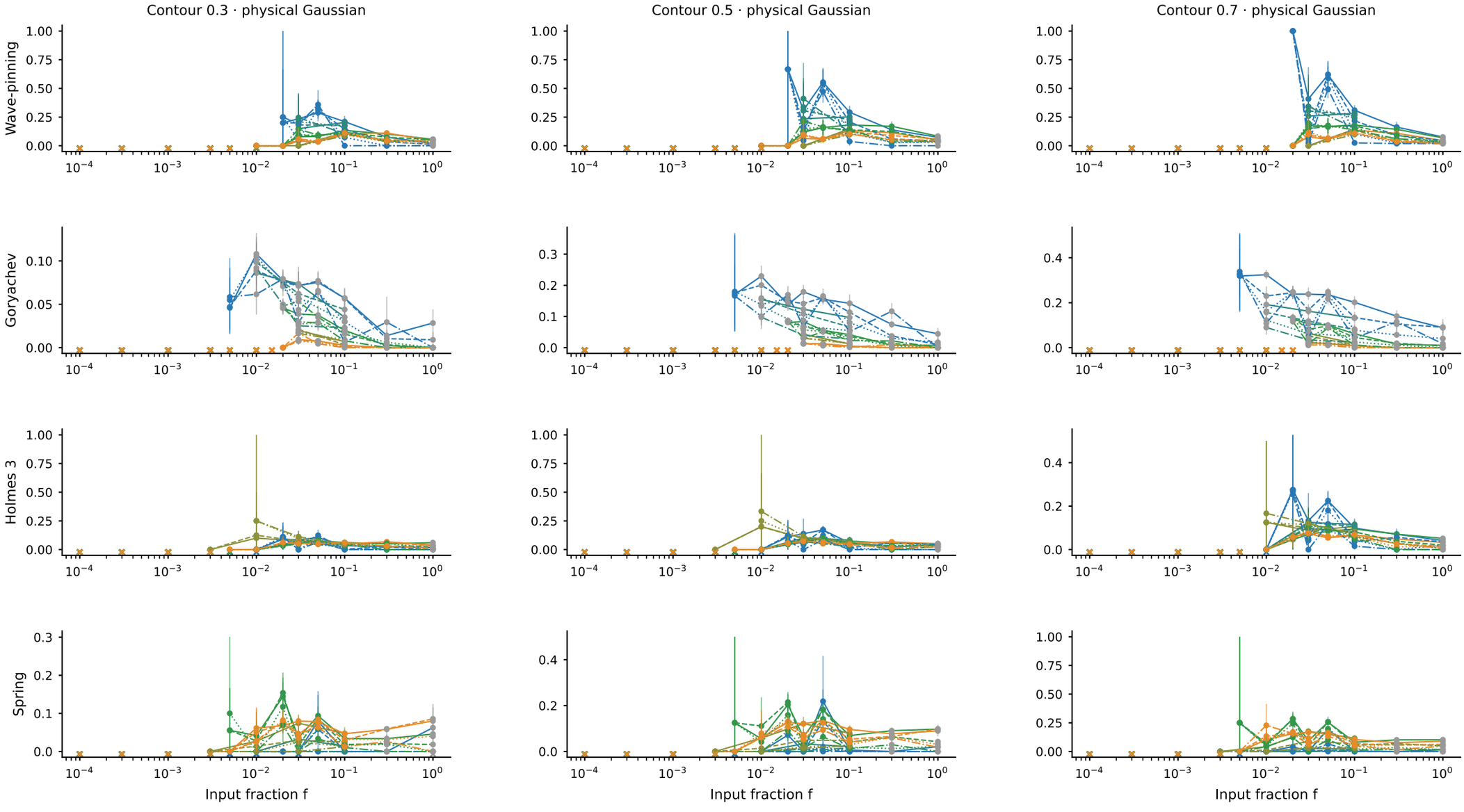
Common canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_043_S9_observed_pairwise_takeovers_per_birth.csv

S9 · Common canonical A5 observations · Literal pairwise takeover sensitivity per birth · full native-factor sensitivity



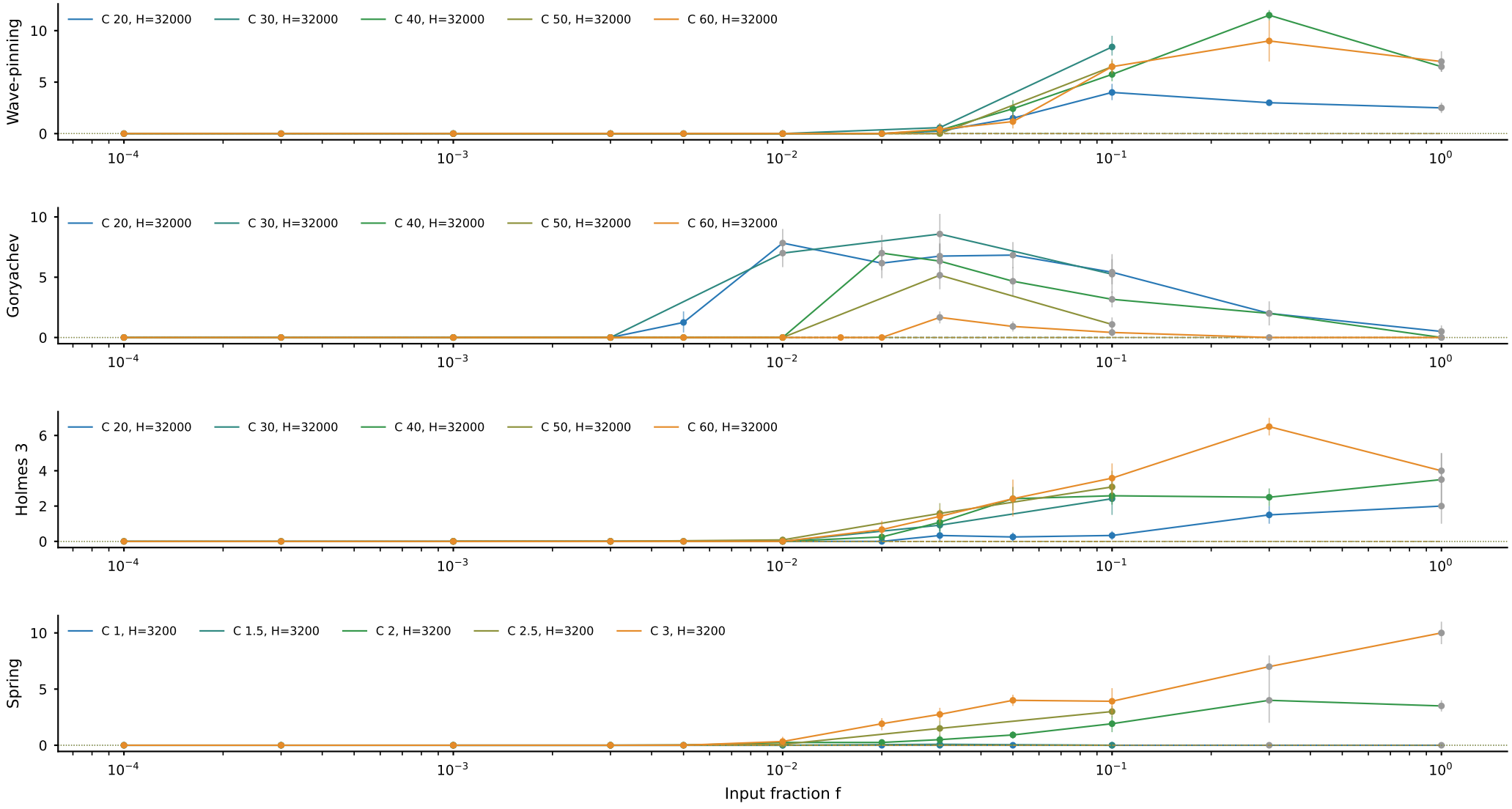
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_044_S9_observed_pairwise_takeovers_per_birth_raw_factorial.csv

S9 · Common canonical A5 observations · Literal pairwise takeover sensitivity per birth · full native-factor sensitivity



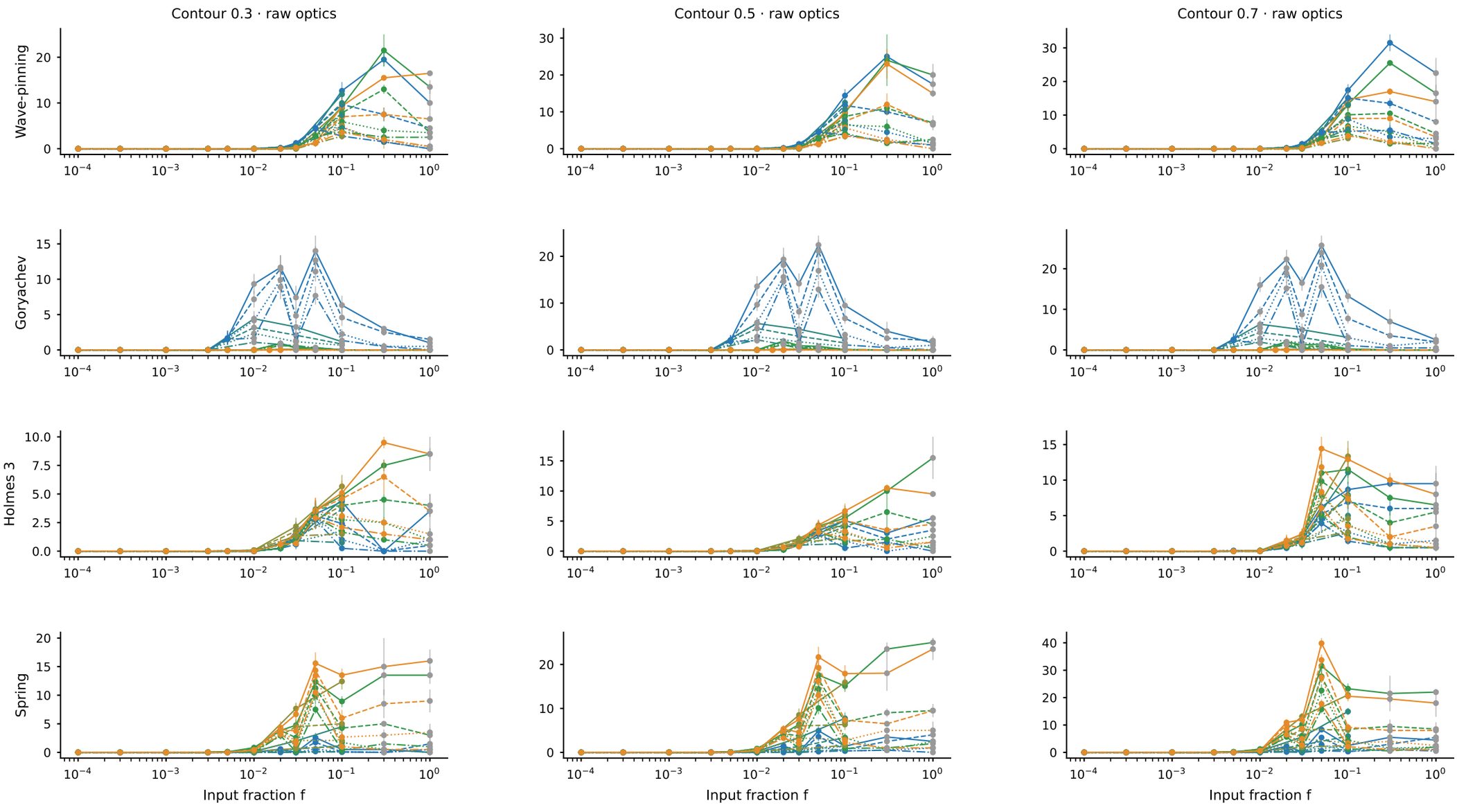
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_045_S9_observed_pairwise_takeovers_per_birth_smoothed_factorial.csv

S9 · Common canonical A5 observations · Observed globally dominant newborn takeovers per trajectory



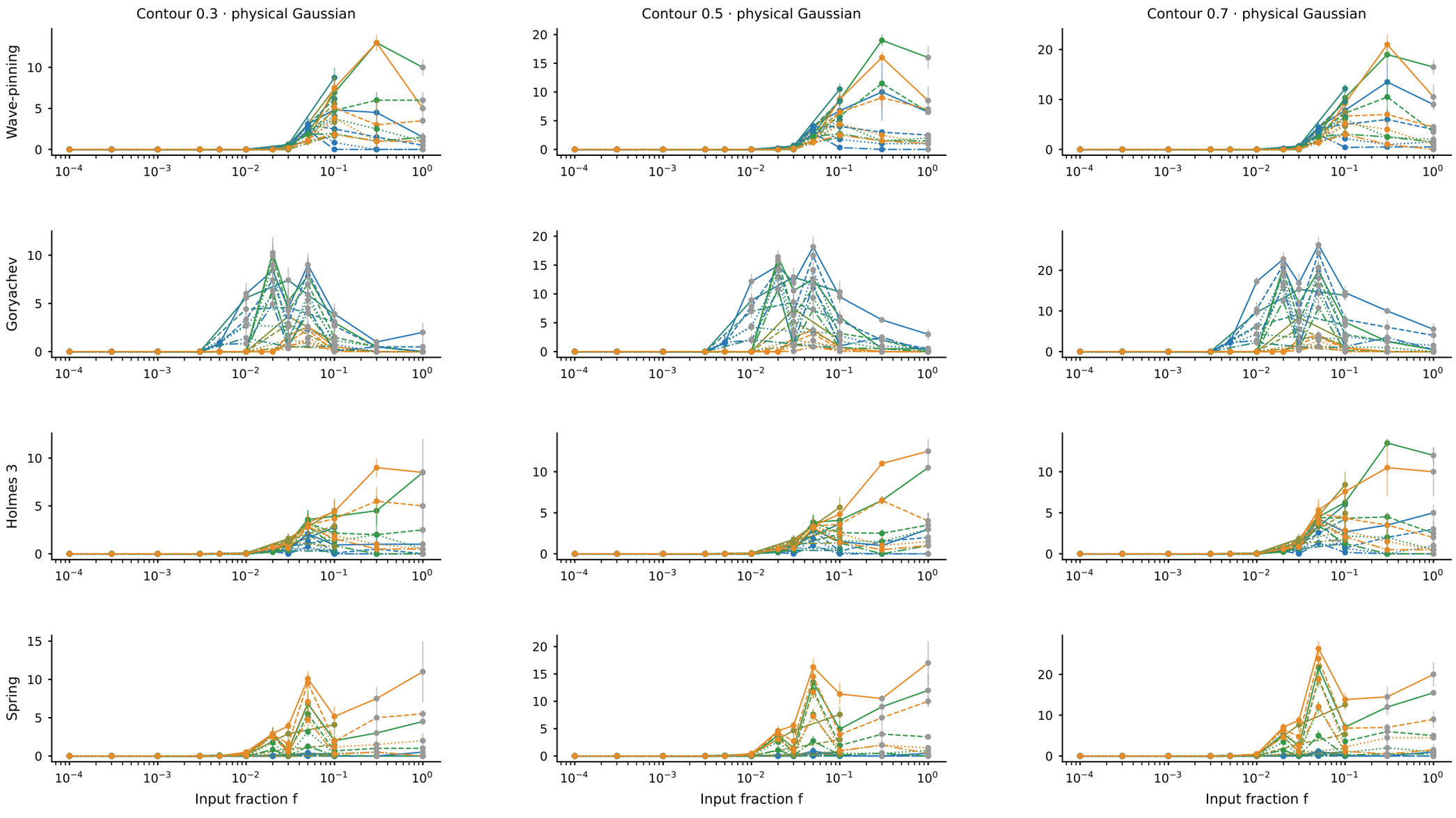
Common canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_046_S9_observed_takeovers.csv

S9 · Common canonical A5 observations · Observed globally dominant newborn takeovers per trajectory · full native-factor sensitivity



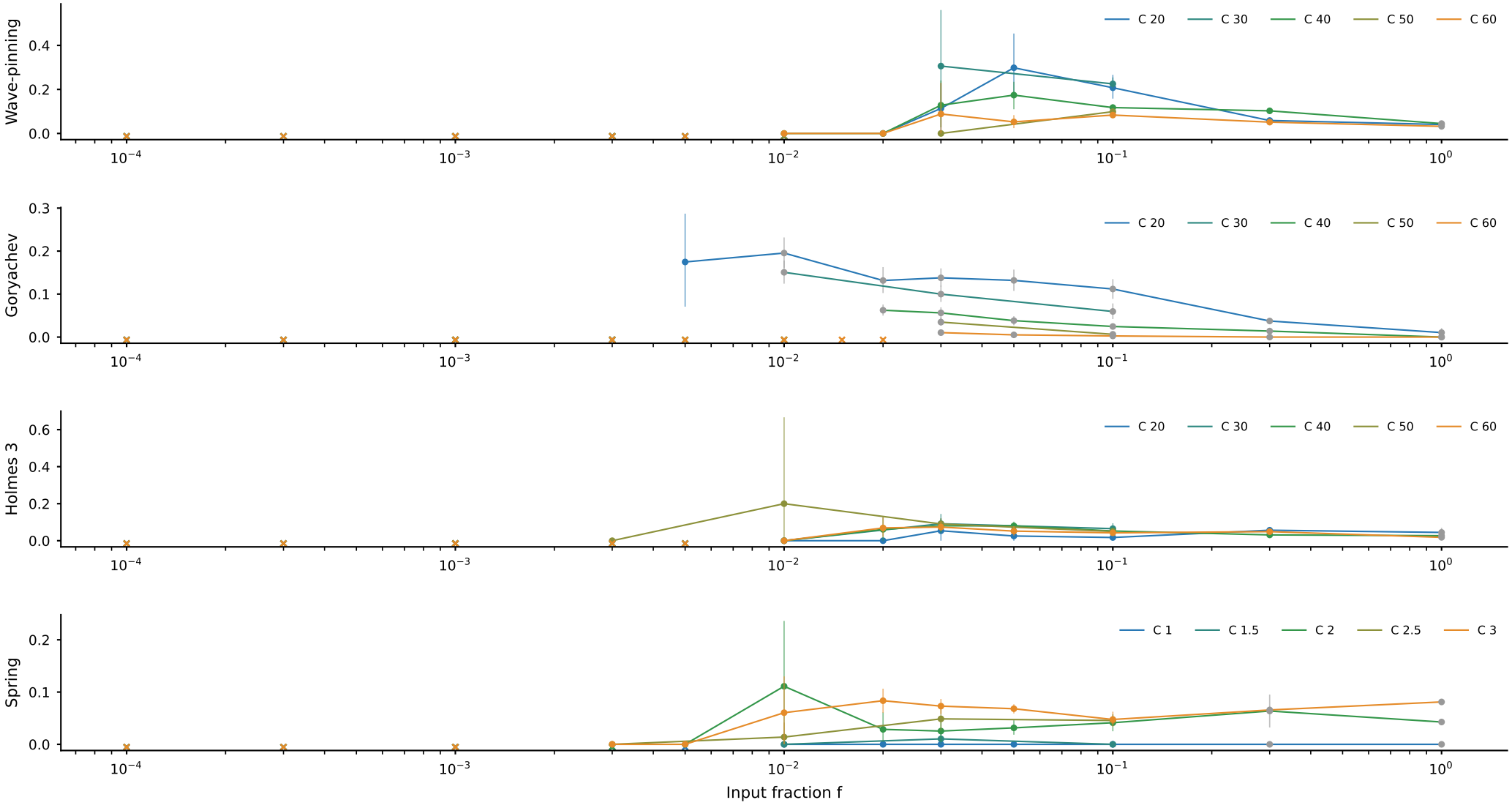
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_047_S9_observed_takeovers_raw_factorial.csv

S9 · Common canonical A5 observations · Observed globally dominant newborn takeovers per trajectory · full native-factor sensitivity



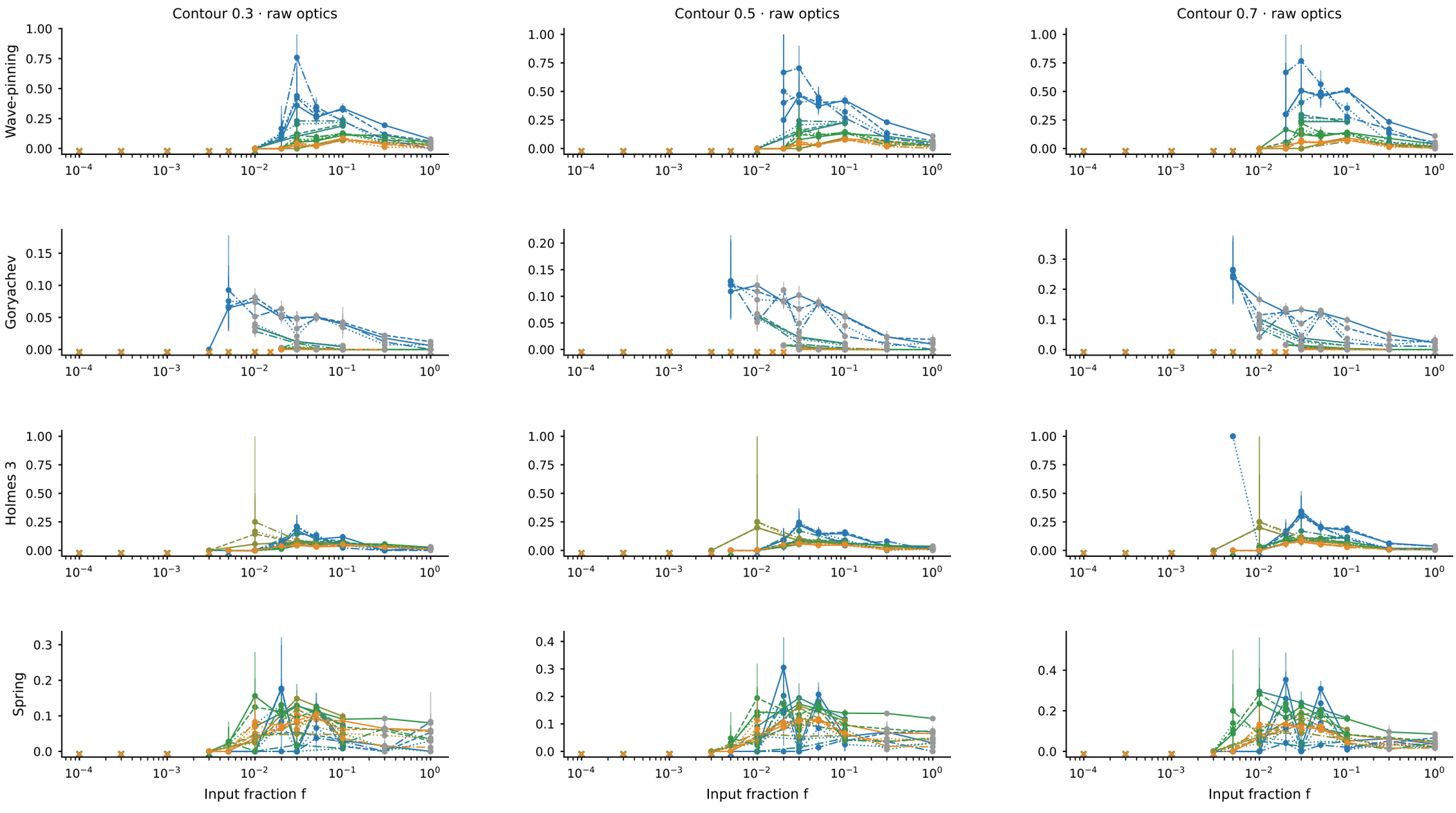
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_048_S9_observed_takeovers_smoothed_factorial.csv

S9 · Common canonical A5 observations · Globally dominant newborn takeovers per birth



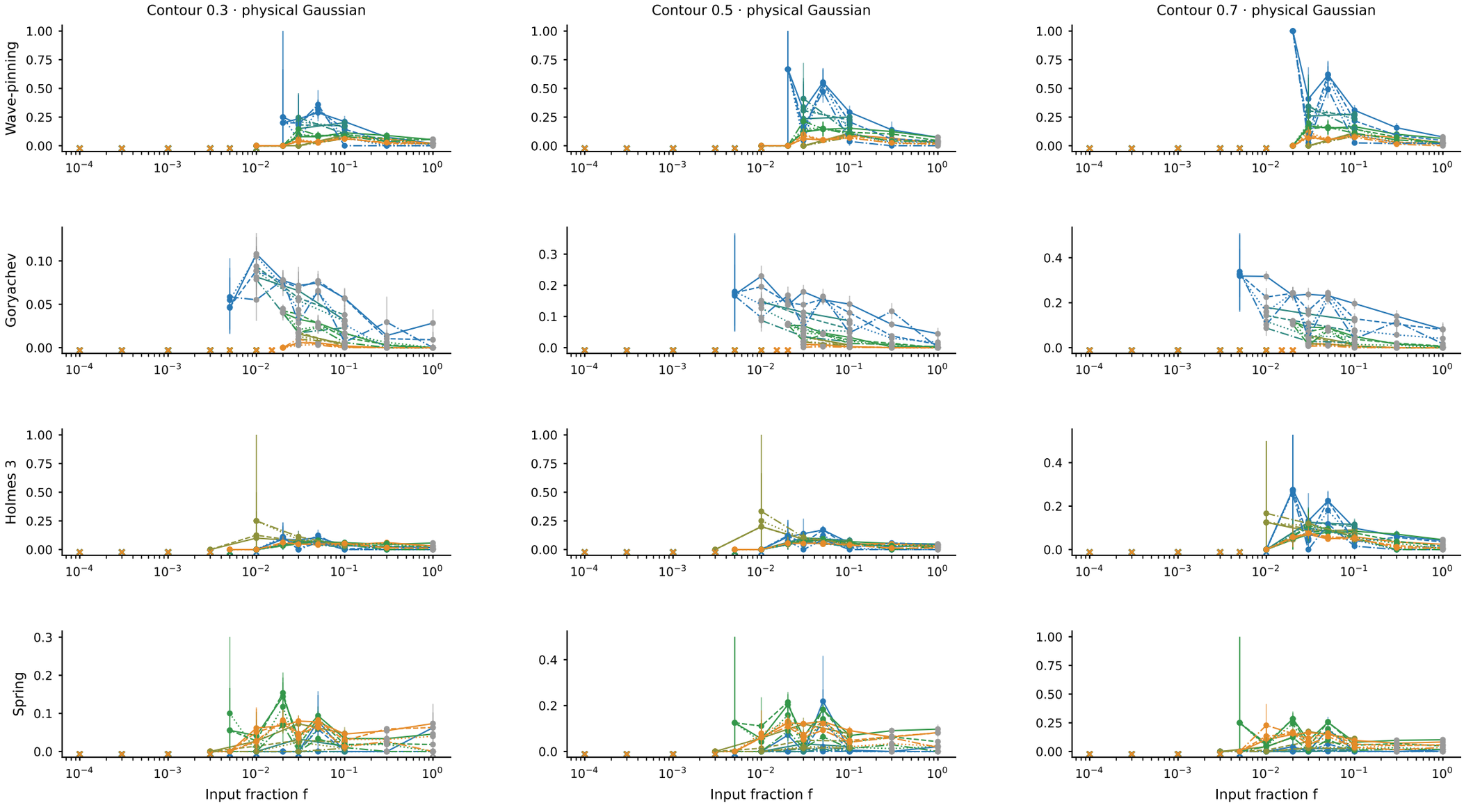
Common canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_049_S9_observed_takeovers_per_birth.csv

S9 · Common canonical A5 observations · Globally dominant newborn takeovers per birth · full native-factor sensitivity



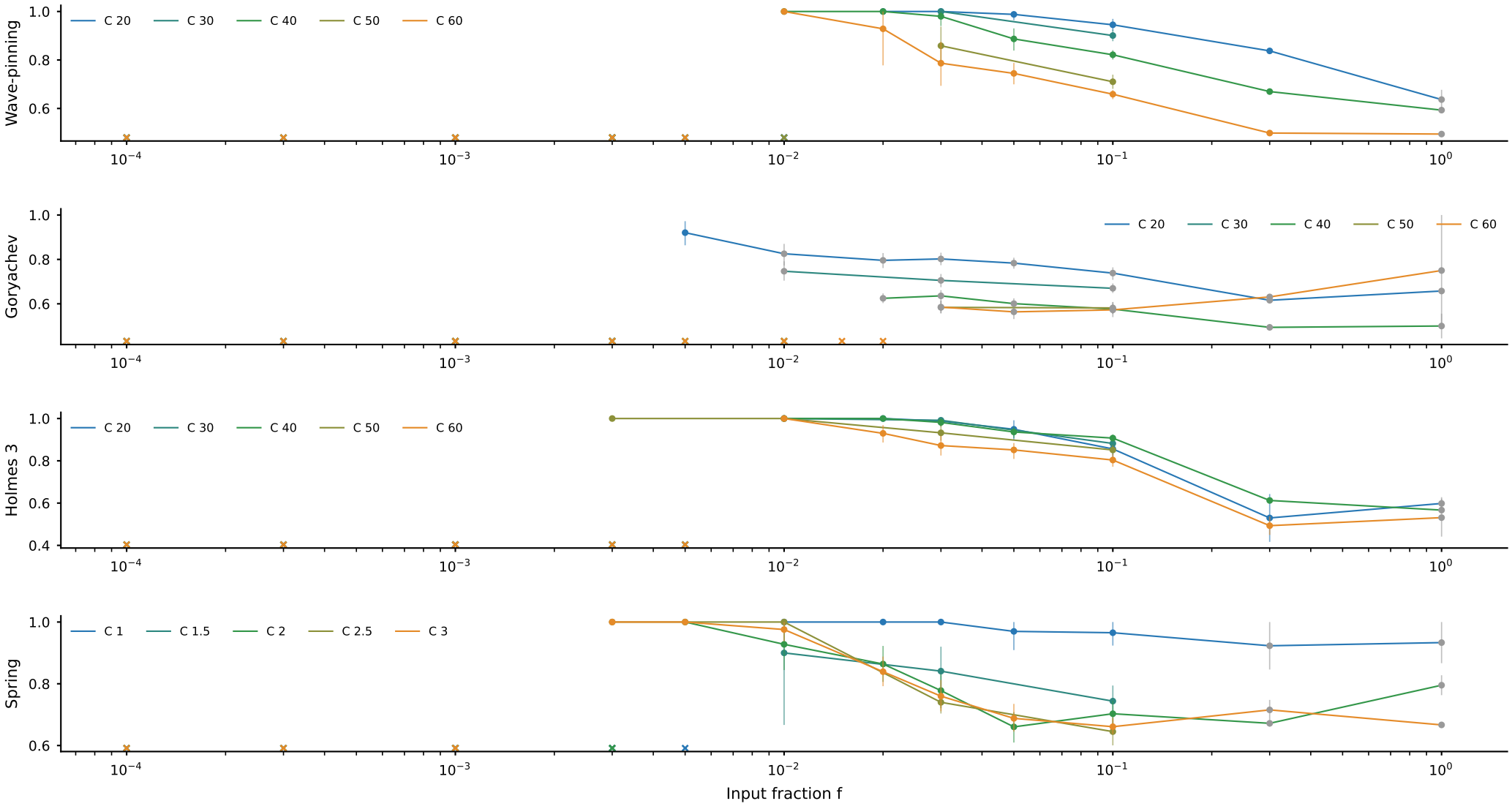
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_050_S9_observed_takeovers_per_birth_raw_factorial.csv

S9 · Common canonical A5 observations · Globally dominant newborn takeovers per birth · full native-factor sensitivity



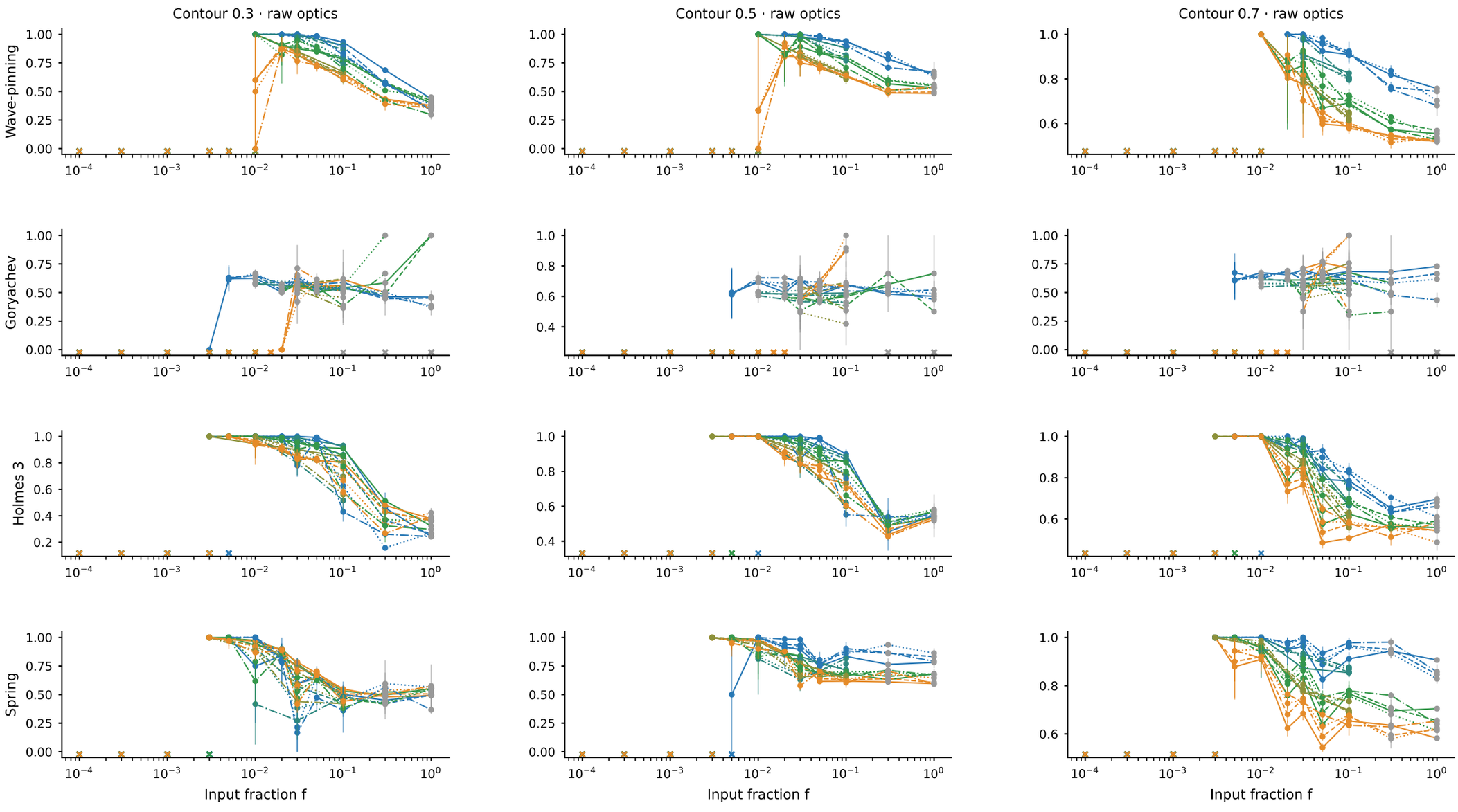
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_051_S9_observed_takeovers_per_birth_smoothed_factorial.csv

S9 · Common canonical A5 observations · Fraction of births opposite the previous dominant region



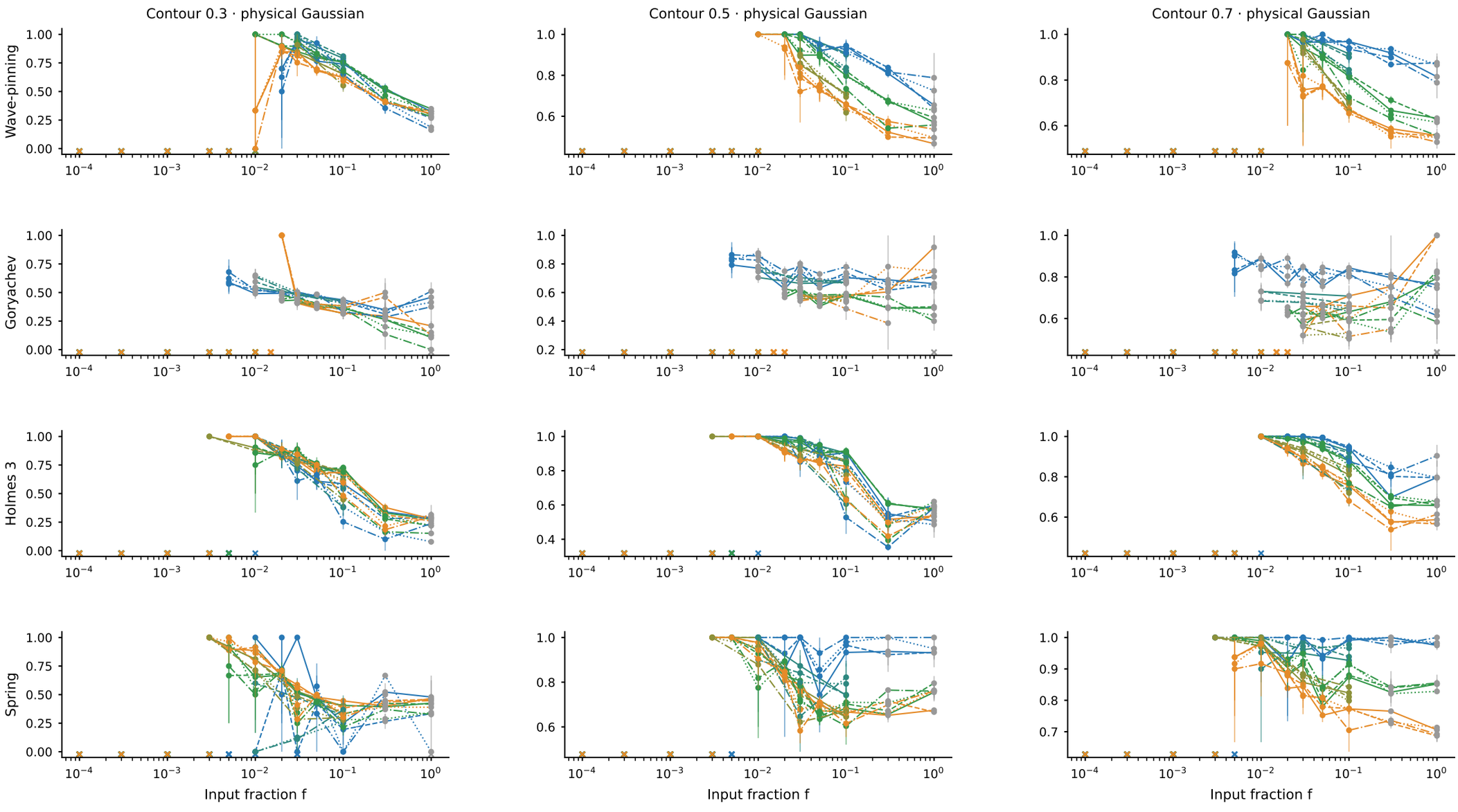
Common canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_052_S9_opposite_birth_fraction.csv

S9 · Common canonical A5 observations · Fraction of births opposite the previous dominant region · full native-factor sensitivity



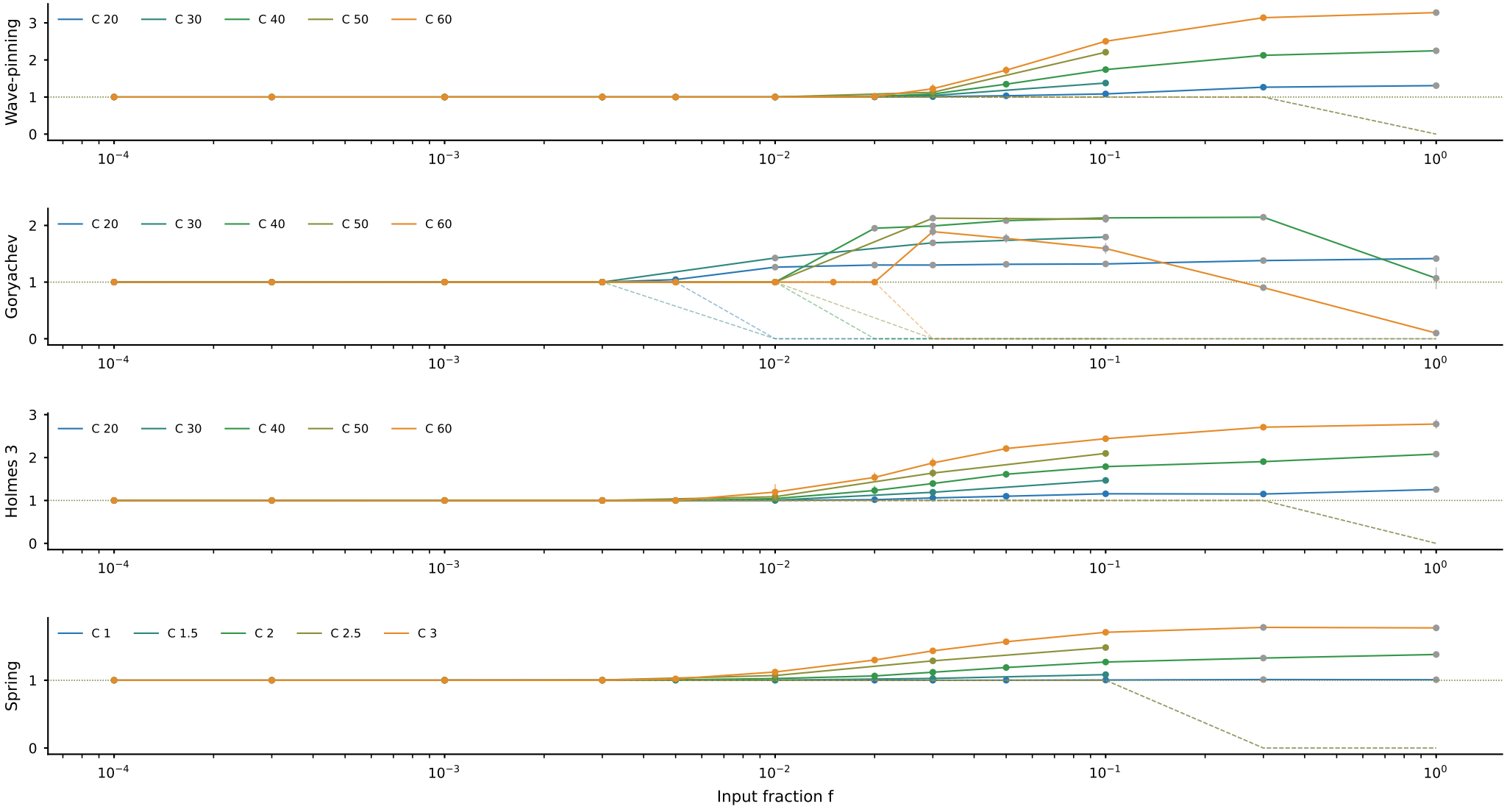
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_053_S9_opposite_birth_fraction_raw_factorial.csv

S9 · Common canonical A5 observations · Fraction of births opposite the previous dominant region · full native-factor sensitivity



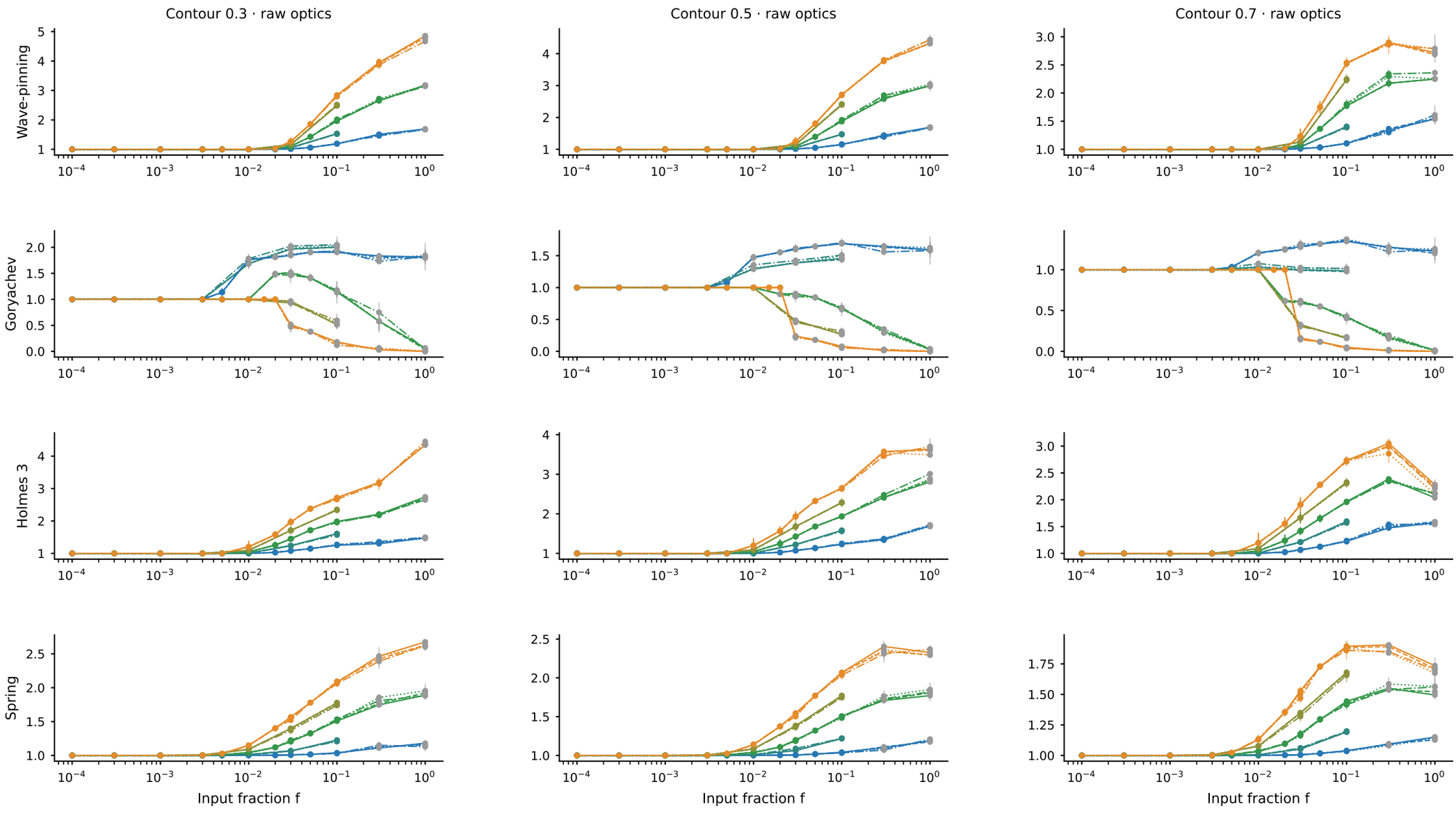
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_054_S9_opposite_birth_fraction_smoothed_factorial.csv

S9 · Common canonical A5 observations · Mean simultaneous qualified-region count



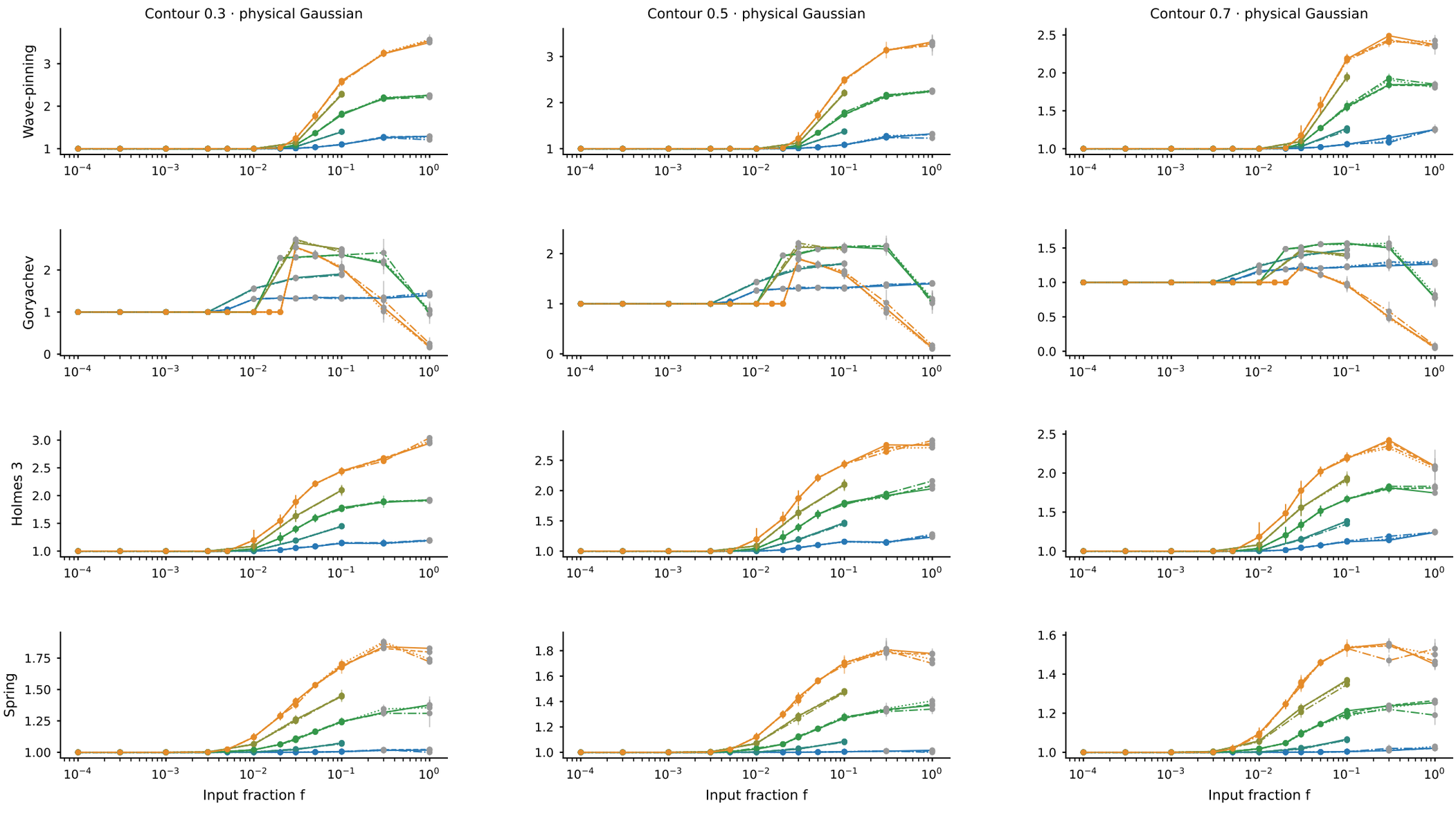
Common canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_055_S9_region_count_mean.csv

S9 · Common canonical A5 observations · Mean simultaneous qualified-region count · full native-factor sensitivity



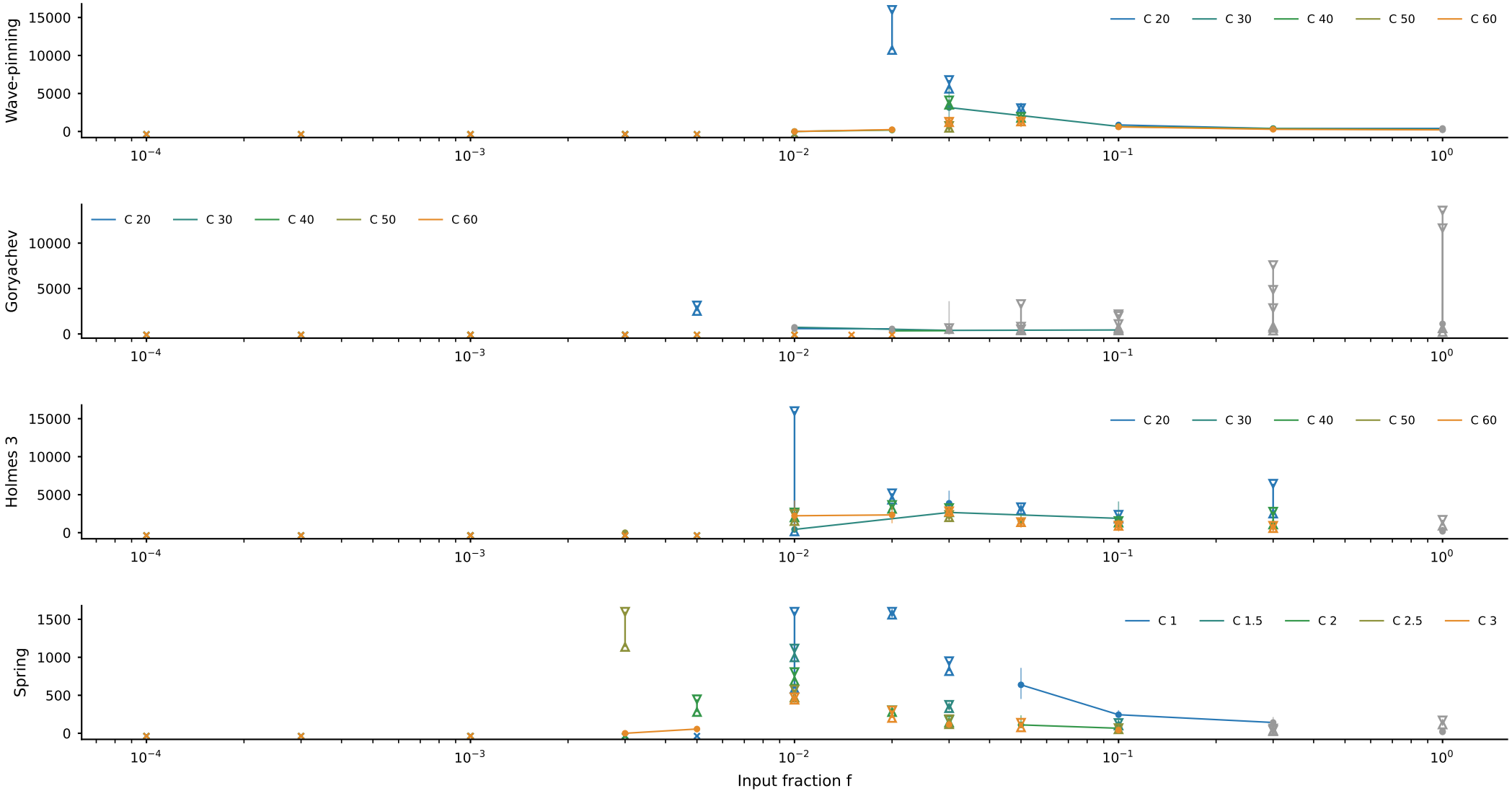
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_056_S9_region_count_mean_raw_factorial.csv

S9 · Common canonical A5 observations · Mean simultaneous qualified-region count · full native-factor sensitivity



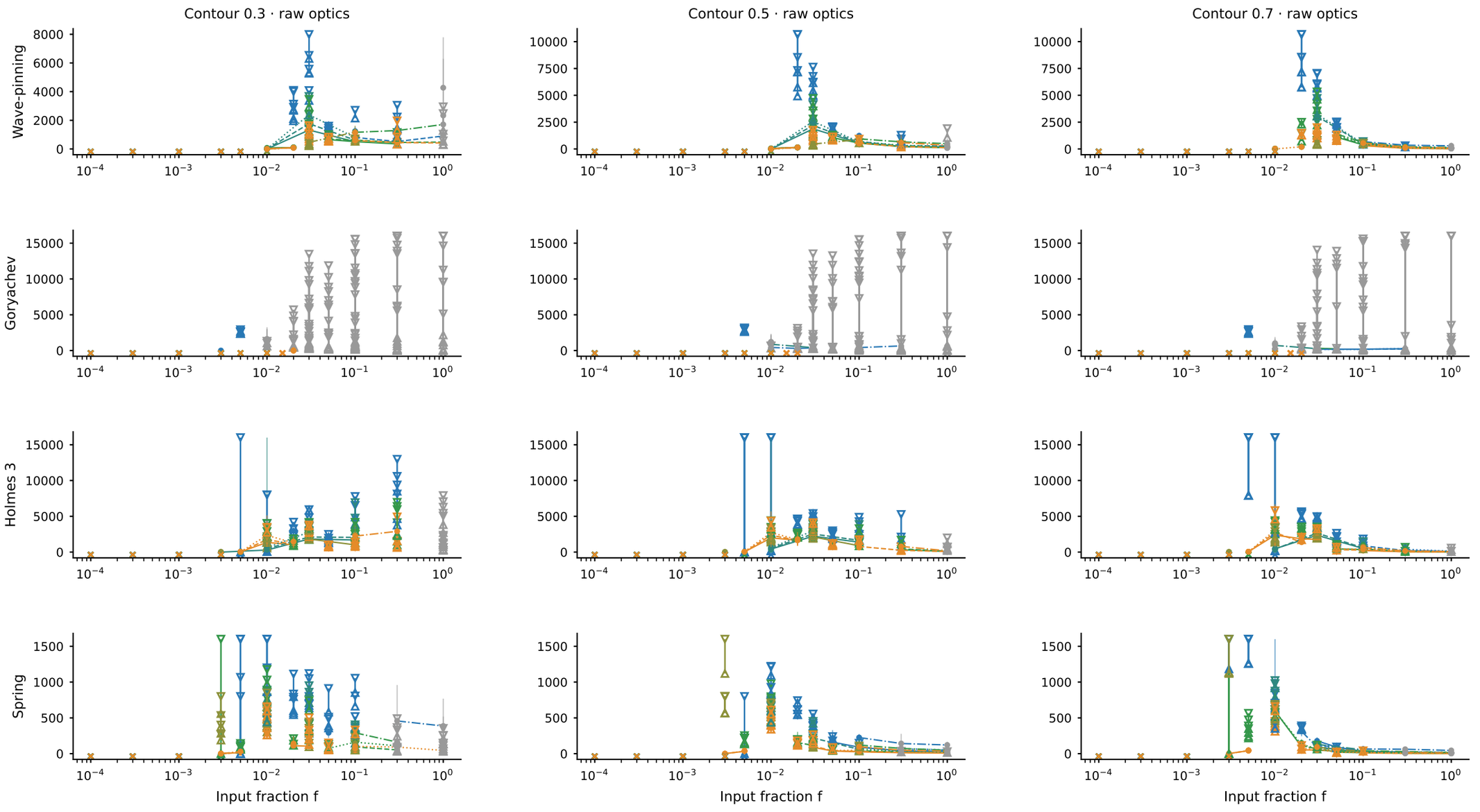
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_057_S9_region_count_mean_smoothed_factorial.csv

S9 · Common canonical A5 observations · Lower-censoring restricted mean lifetime



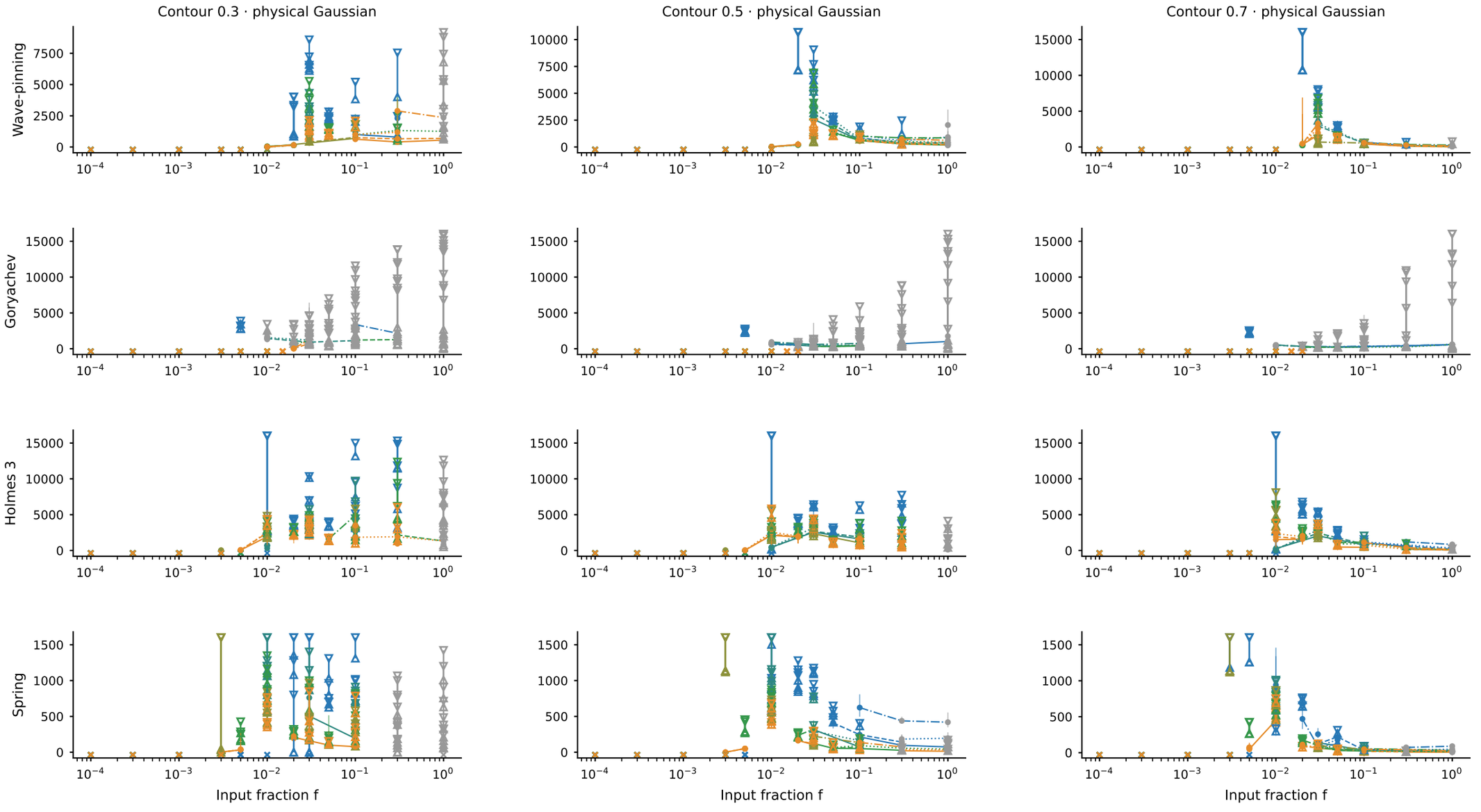
Common canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_058_S9_restricted_mean_lifetime_lower.csv

S9 · Common canonical A5 observations · Lower-censoring restricted mean lifetime · full native-factor sensitivity



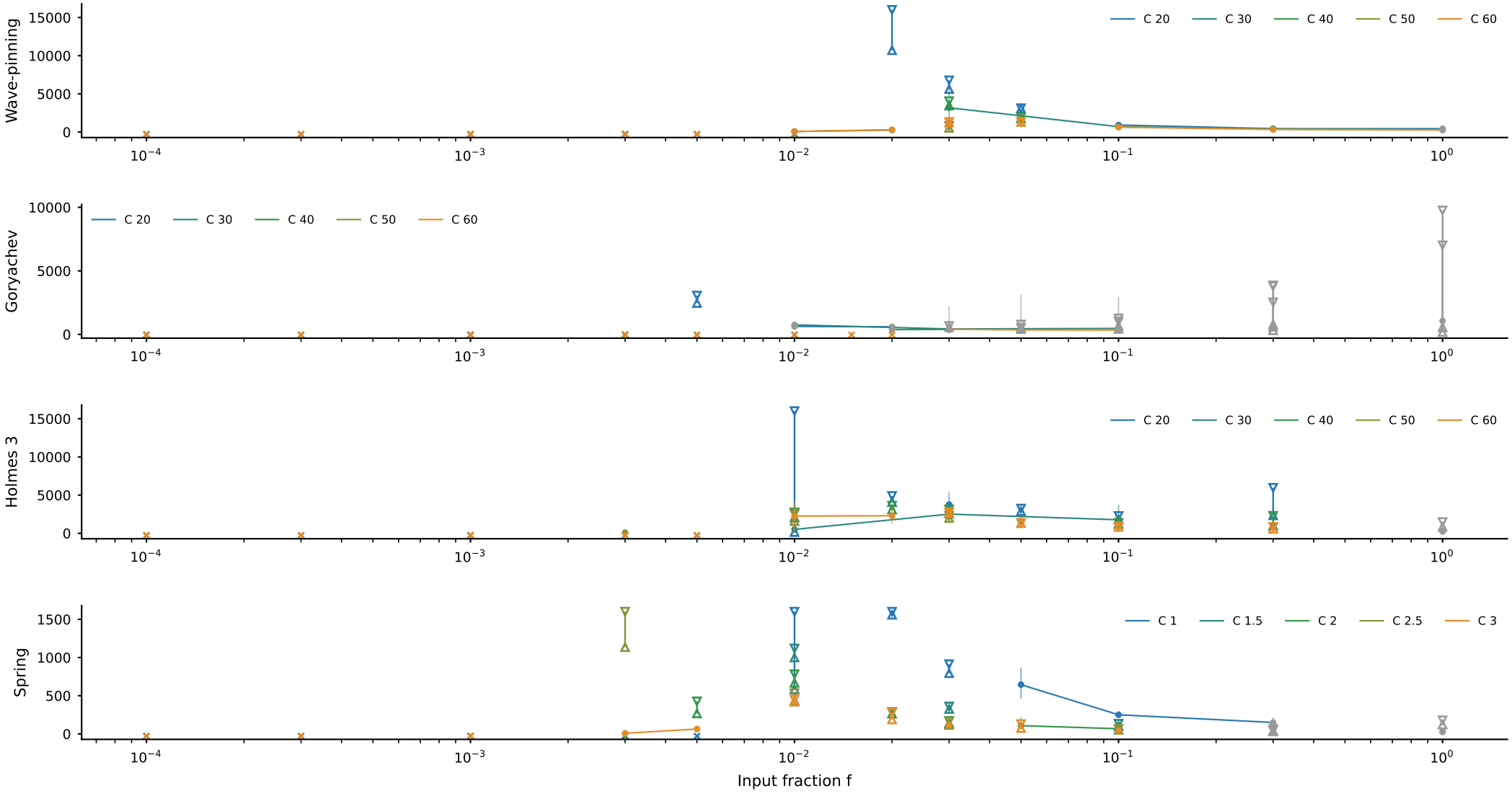
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_059_S9_restricted_mean_lifetime_lower_raw_factorial.csv

S9 · Common canonical A5 observations · Lower-censoring restricted mean lifetime · full native-factor sensitivity



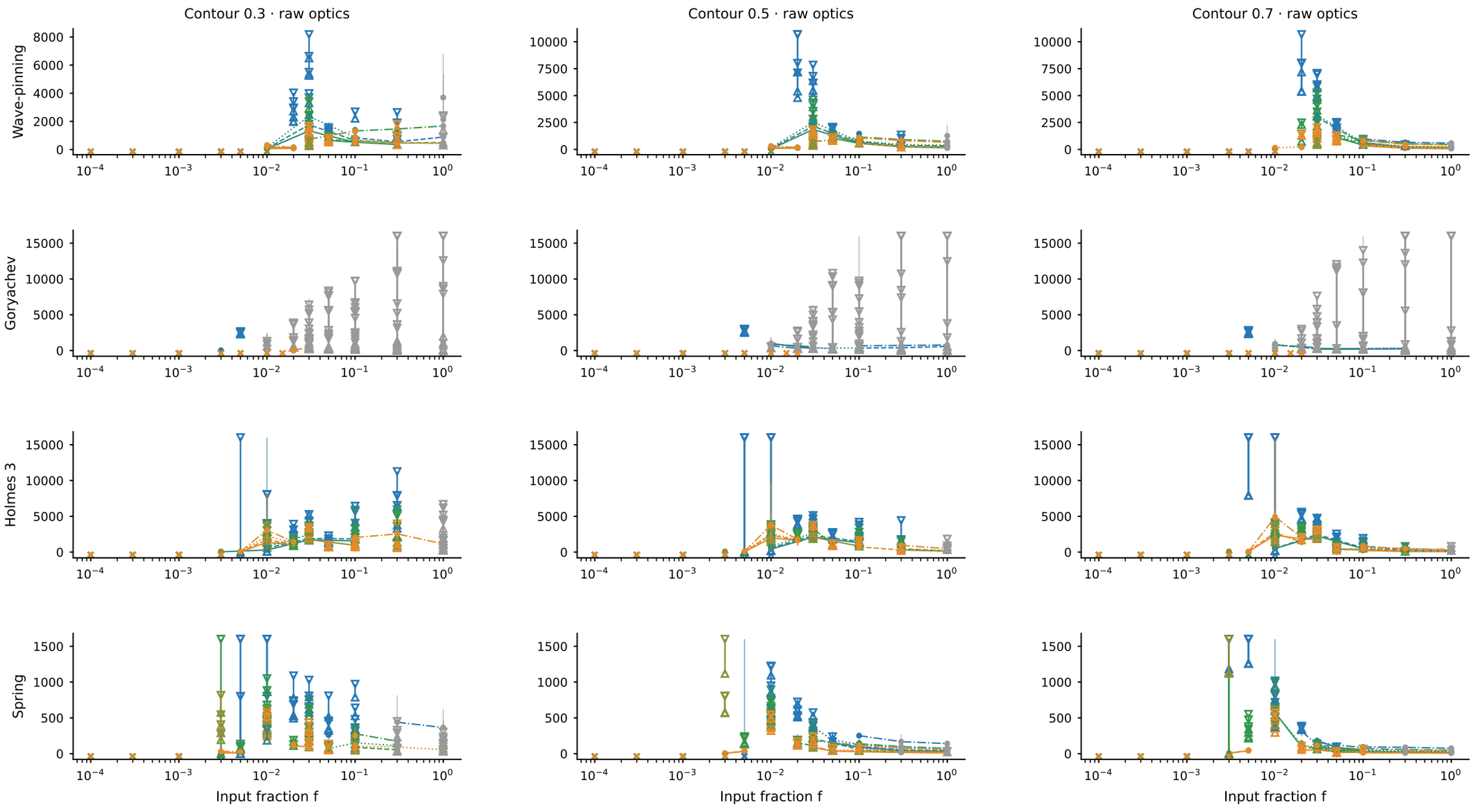
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_060_S9_restricted_mean_lifetime_lower_smoothed_factorial.csv

S9 · Common canonical A5 observations · Observed-frame restricted mean lifetime



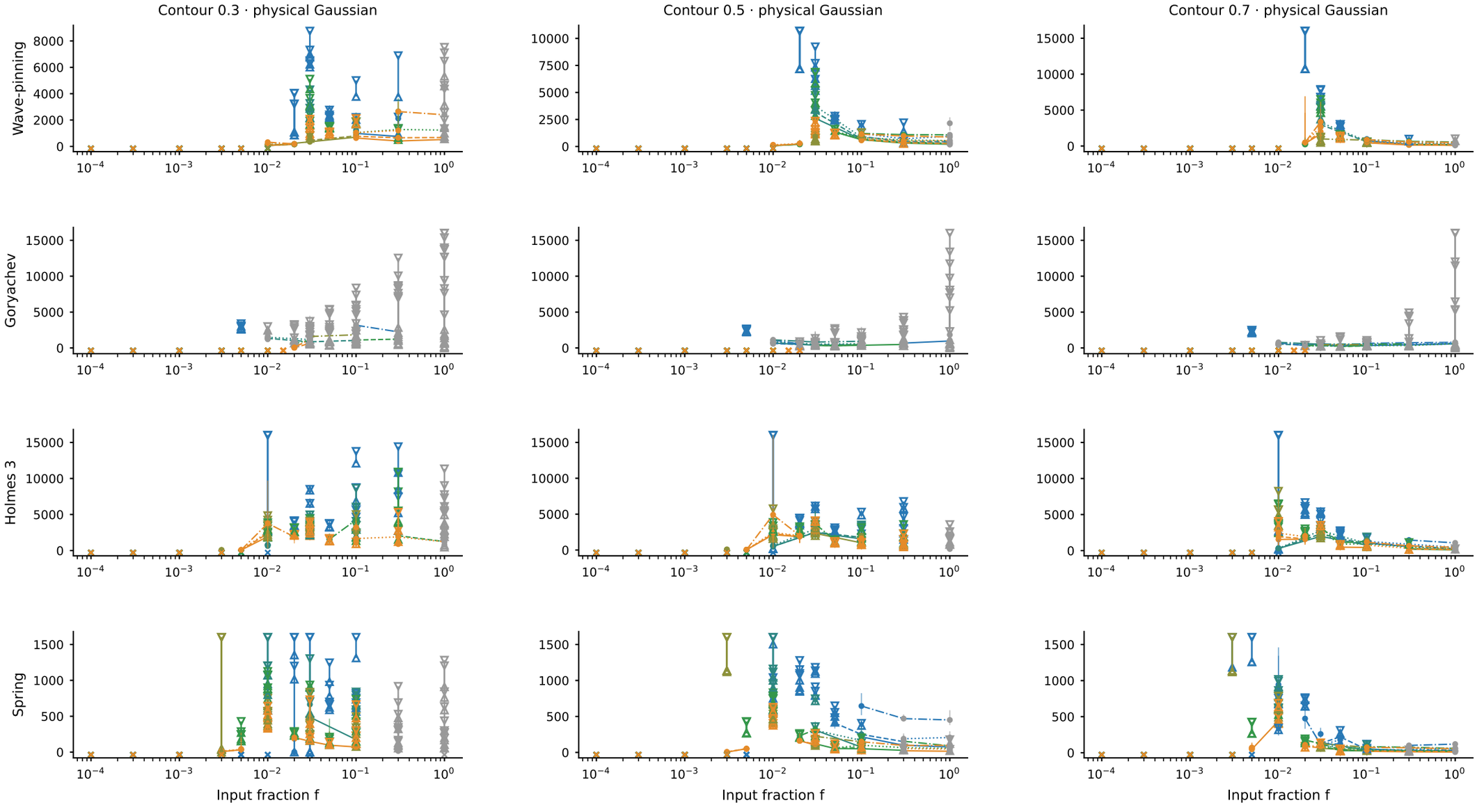
Common canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_061_S9_restricted_mean_lifetime_observed.csv

S9 · Common canonical A5 observations · Observed-frame restricted mean lifetime · full native-factor sensitivity



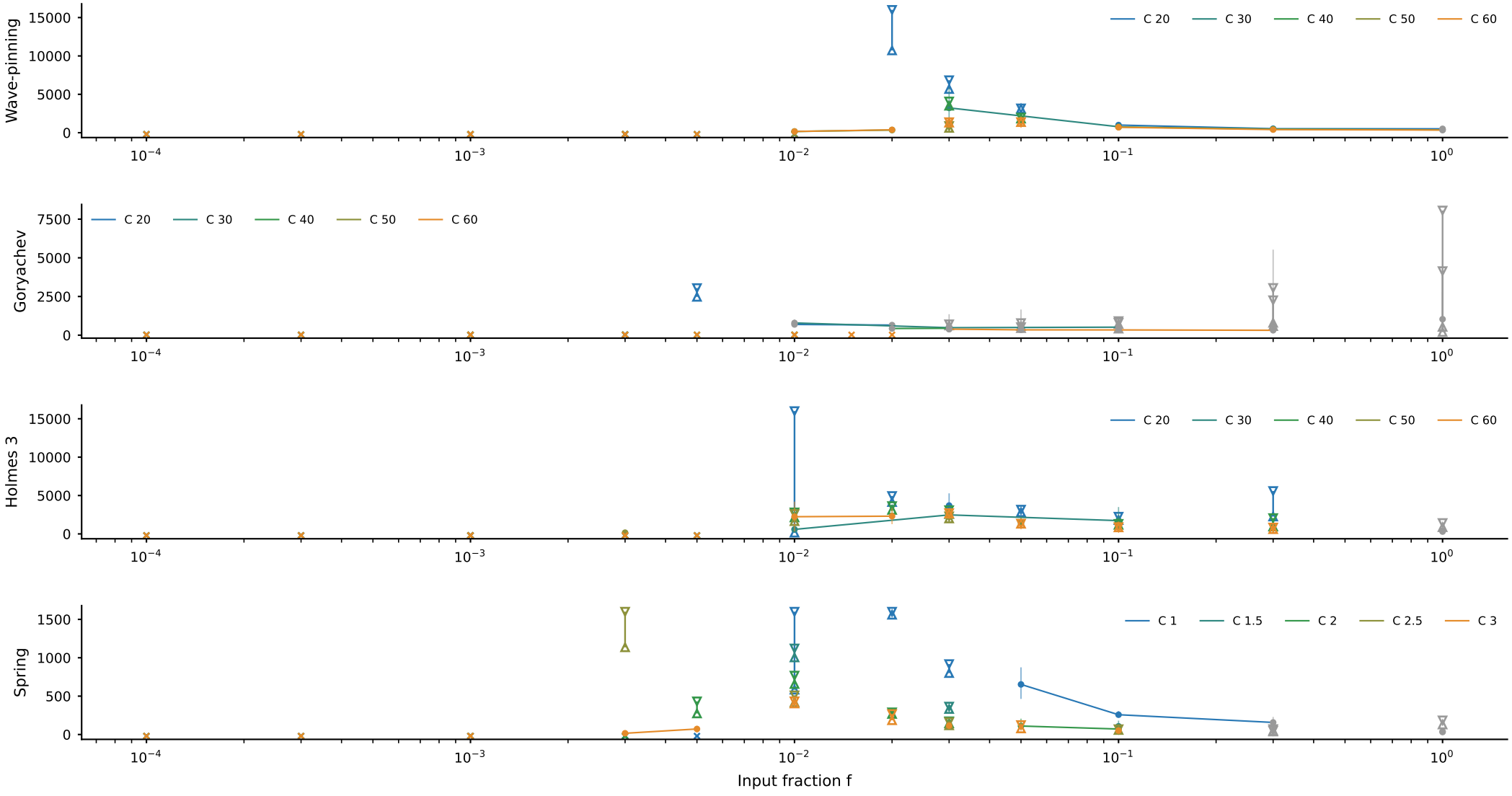
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_062_S9_restricted_mean_lifetime_observed_raw_factorial.csv

S9 · Common canonical A5 observations · Observed-frame restricted mean lifetime · full native-factor sensitivity



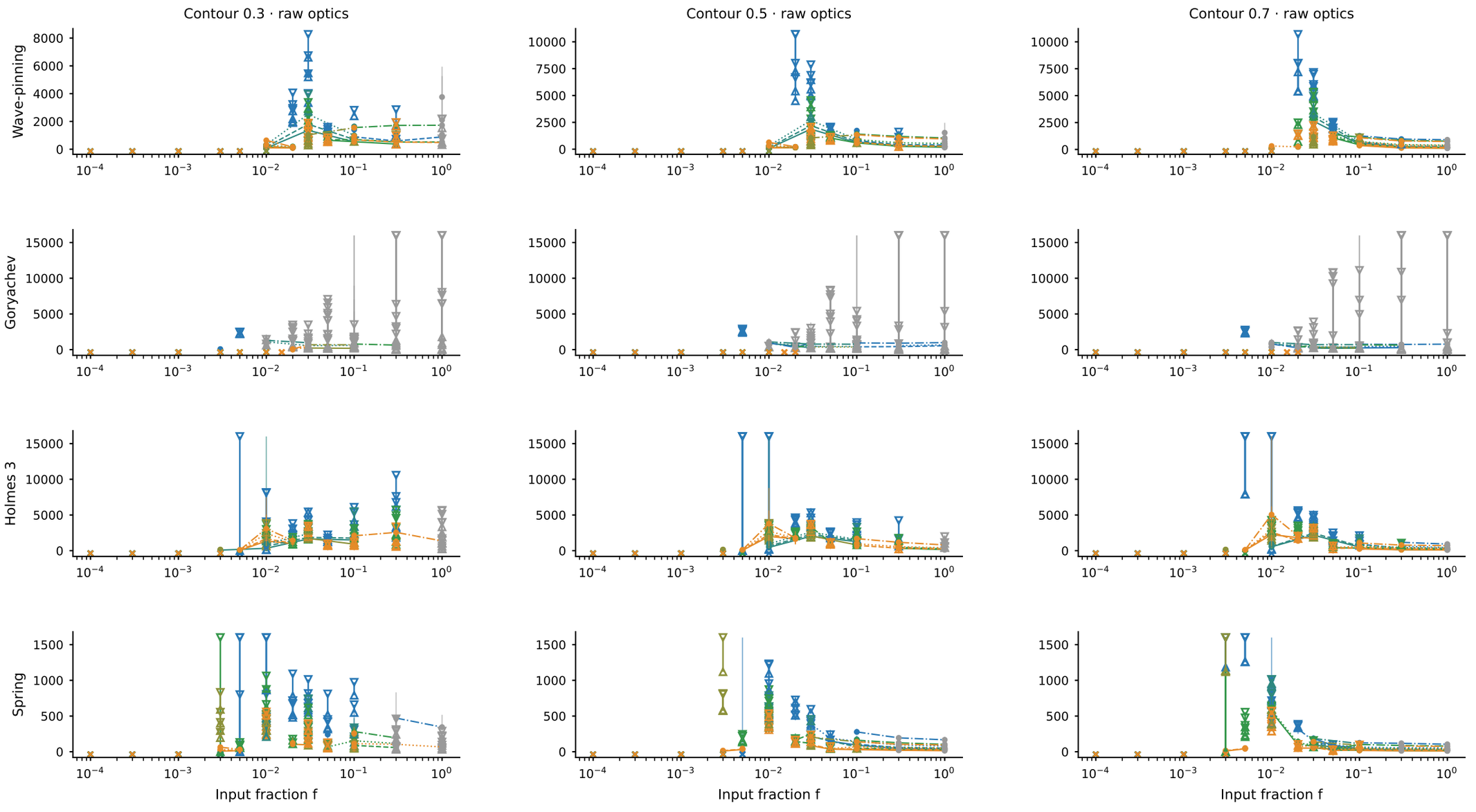
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_063_S9_restricted_mean_lifetime_observed_smoothed_factorial.csv

S9 · Common canonical A5 observations · Upper-censoring restricted mean lifetime



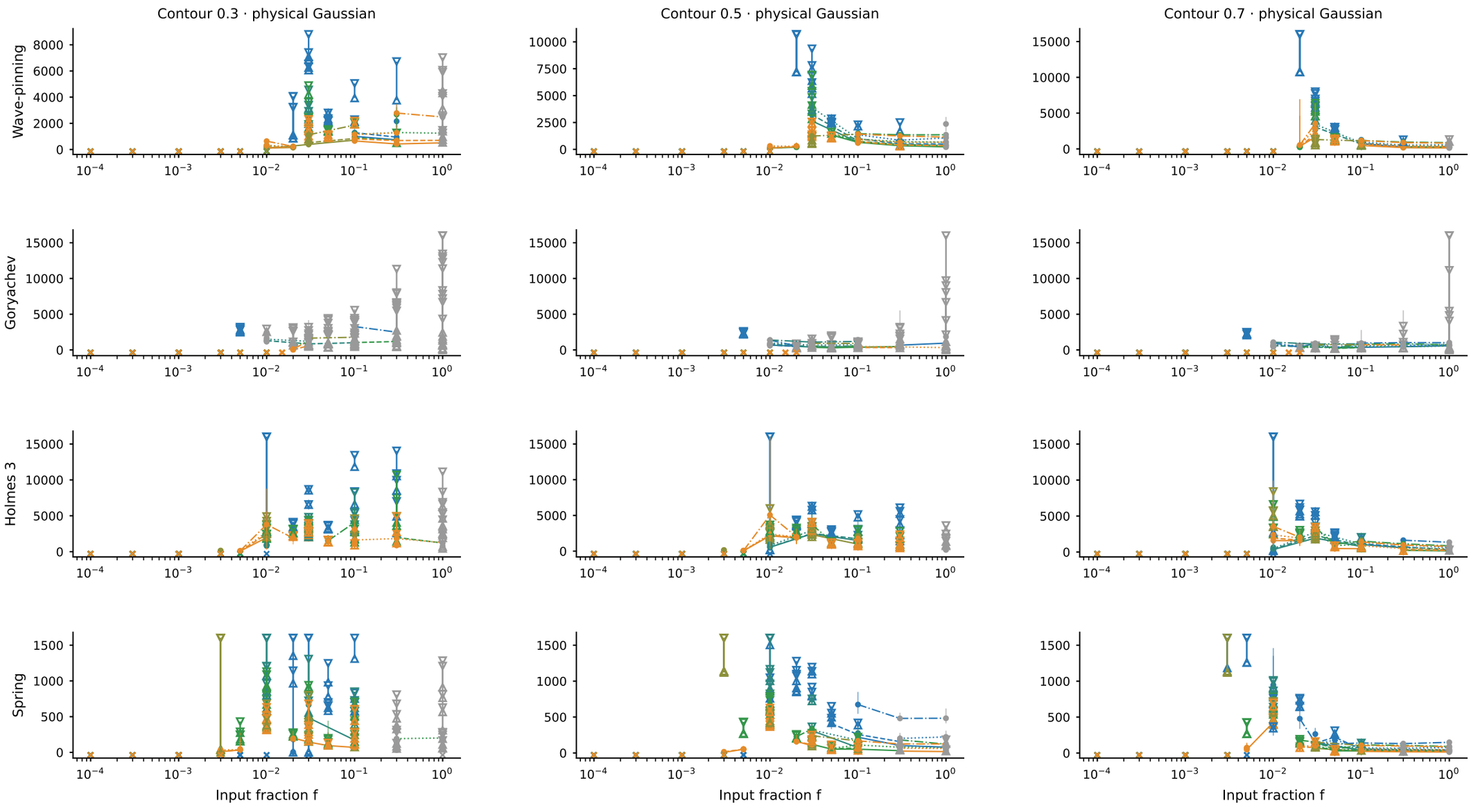
Common canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_064_S9_restricted_mean_lifetime_upper.csv

S9 · Common canonical A5 observations · Upper-censoring restricted mean lifetime · full native-factor sensitivity



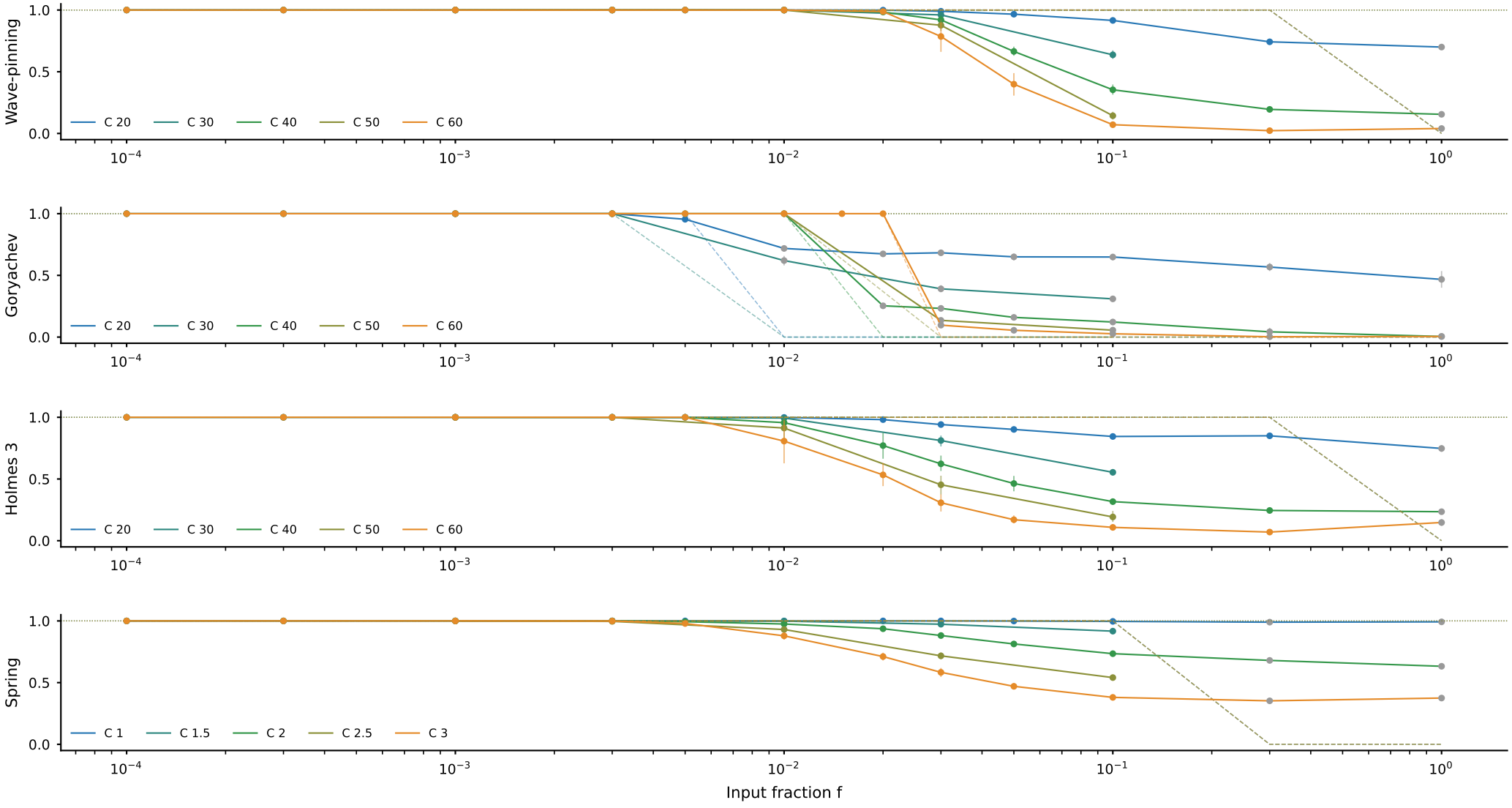
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_065_S9_restricted_mean_lifetime_upper_raw_factorial.csv

S9 · Common canonical A5 observations · Upper-censoring restricted mean lifetime · full native-factor sensitivity



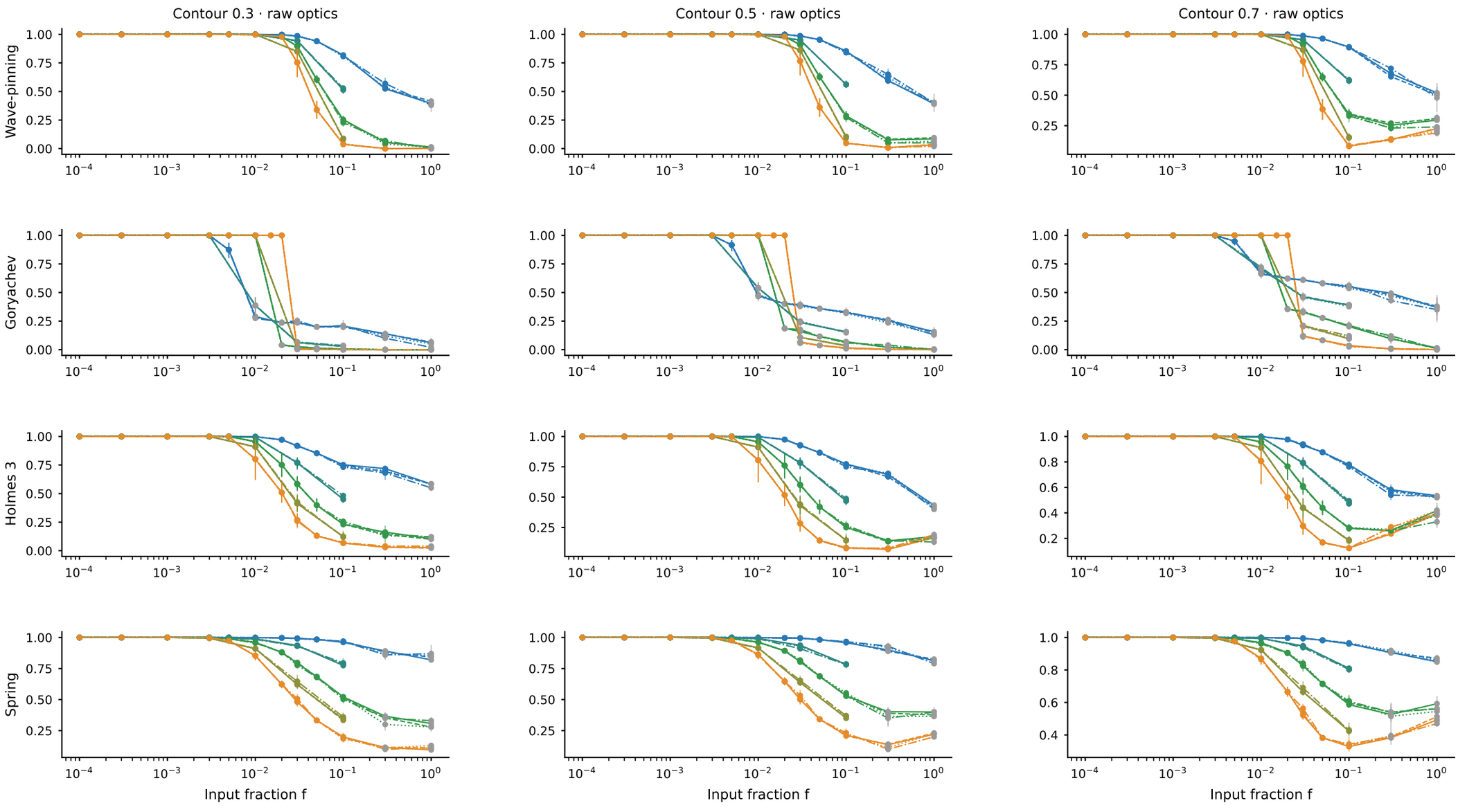
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_066_S9_restricted_mean_lifetime_upper_smoothed_factorial.csv

S9 · Common canonical A5 observations · Time fraction with one qualified region



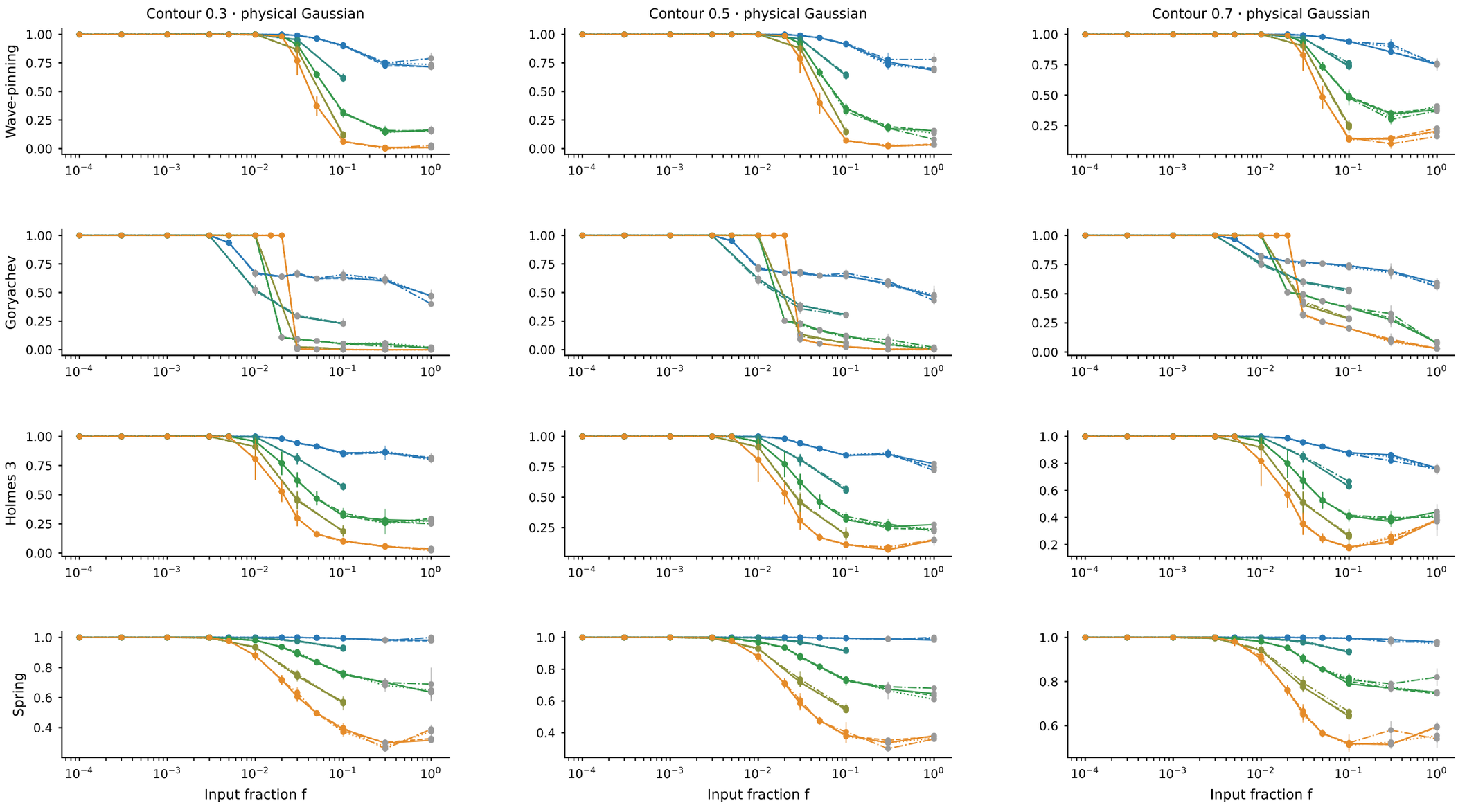
Common canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_067_S9_single_front_fraction.csv

S9 · Common canonical A5 observations · Time fraction with one qualified region · full native-factor sensitivity



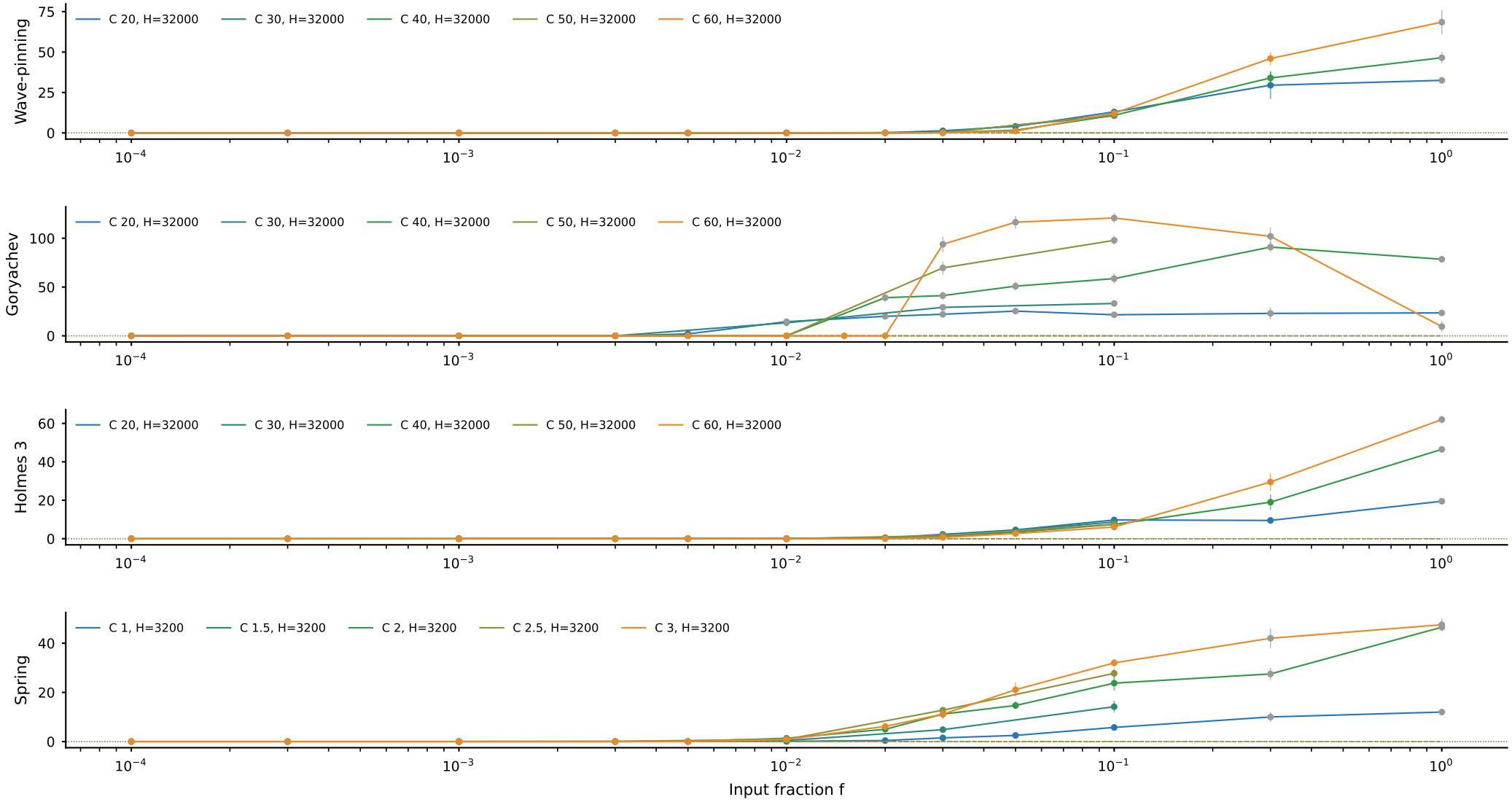
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_068_S9_single_front_fraction_raw_factorial.csv

S9 · Common canonical A5 observations · Time fraction with one qualified region · full native-factor sensitivity



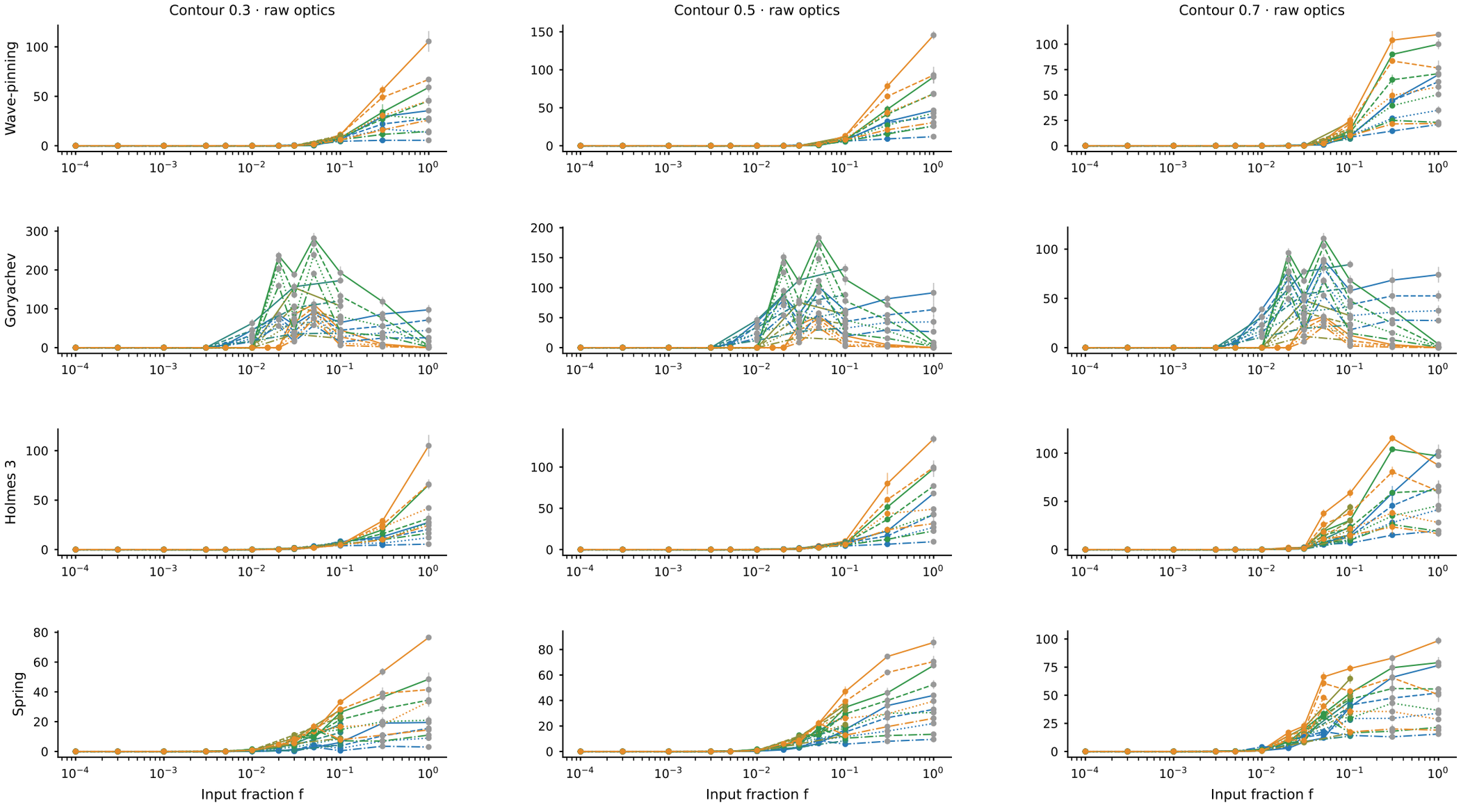
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_069_S9_single_front_fraction_smoothed_factorial.csv

S9 · Common canonical A5 observations · Tracks whose takeover cannot be scored



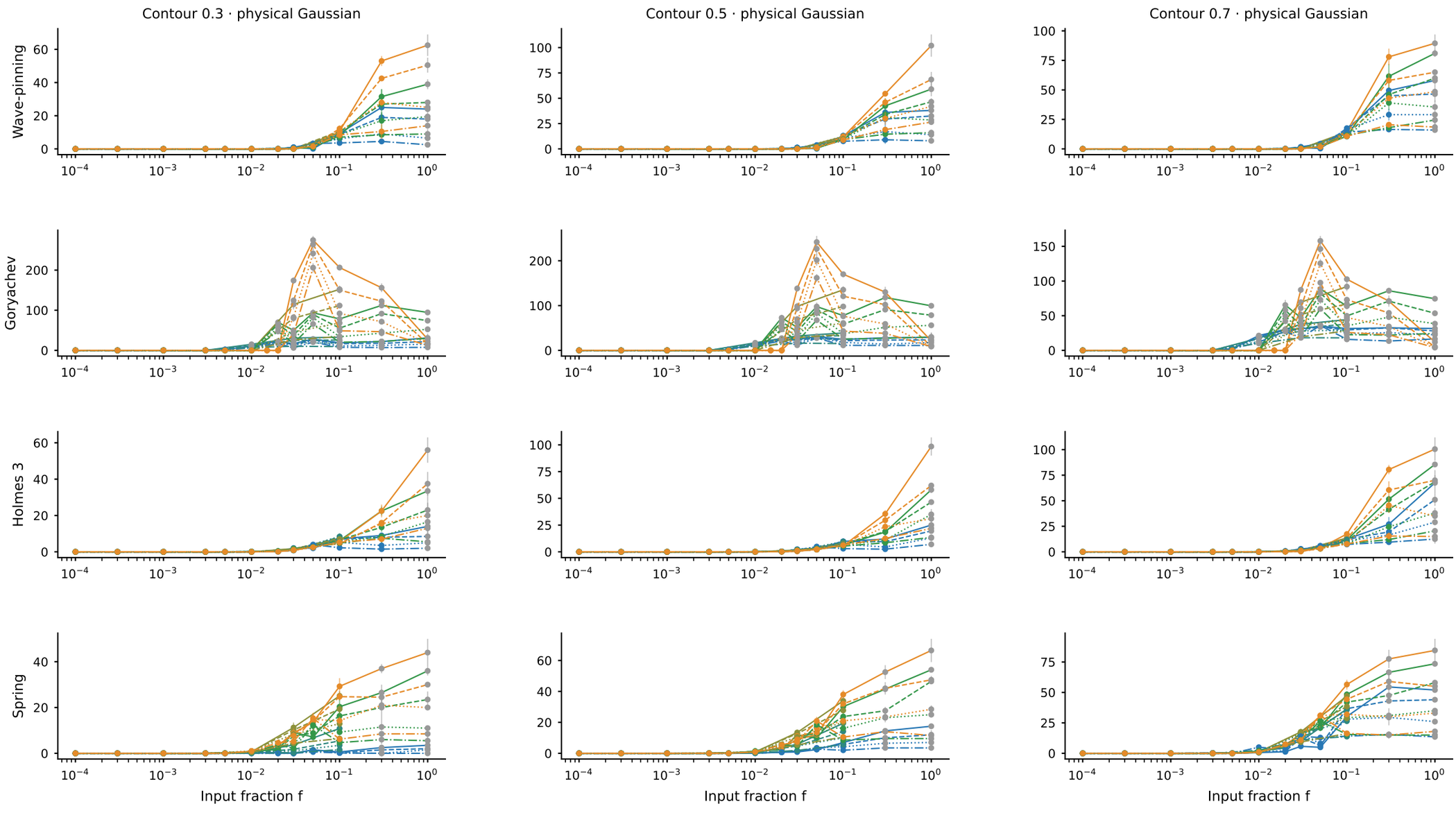
Common canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_070_S9_unscorable_takeover_tracks.csv

S9 · Common canonical A5 observations · Tracks whose takeover cannot be scored · full native-factor sensitivity



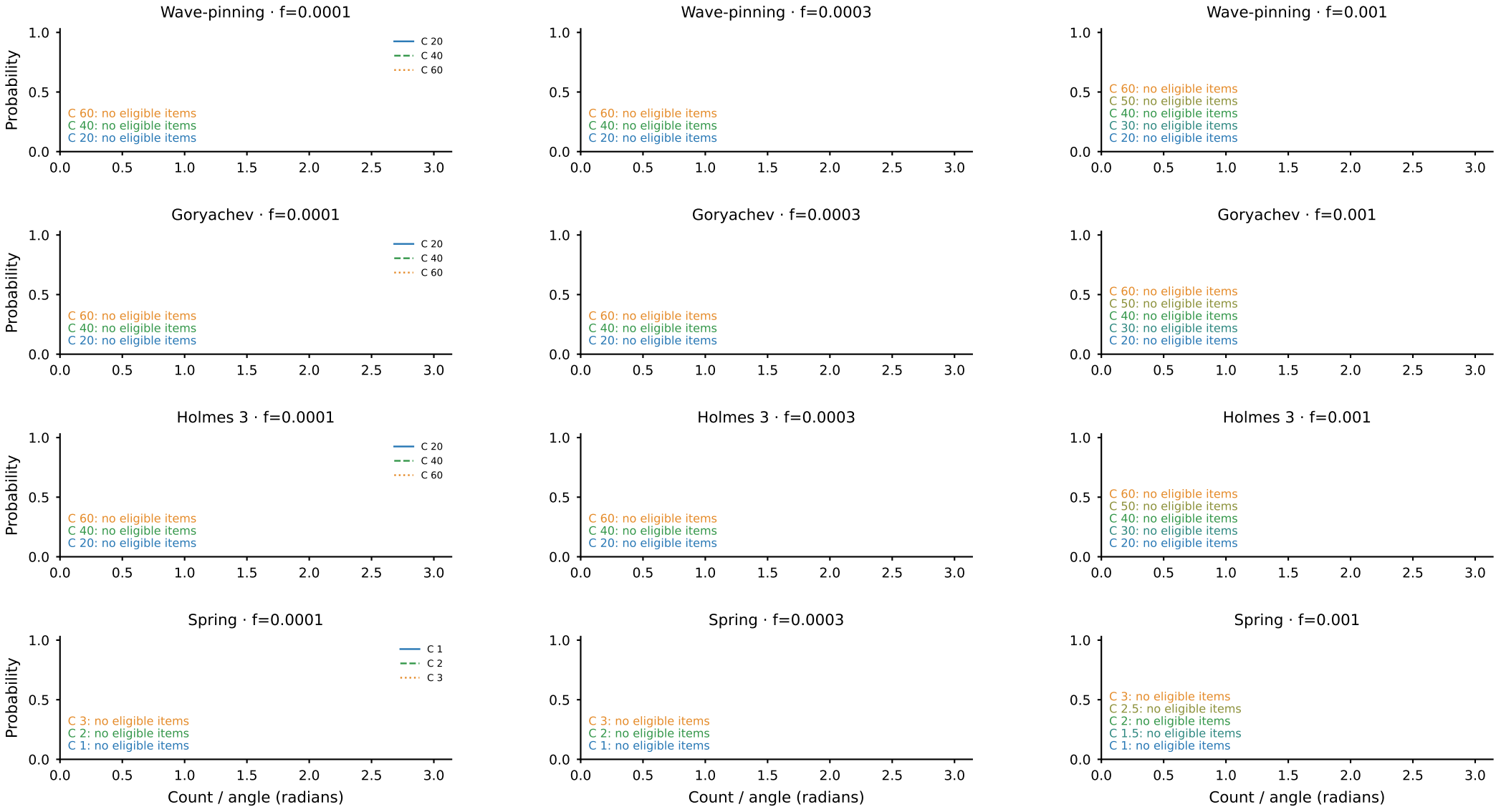
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_071_S9_unscorable_takeover_tracks_raw_factorial.csv

S9 · Common canonical A5 observations · Tracks whose takeover cannot be scored · full native-factor sensitivity



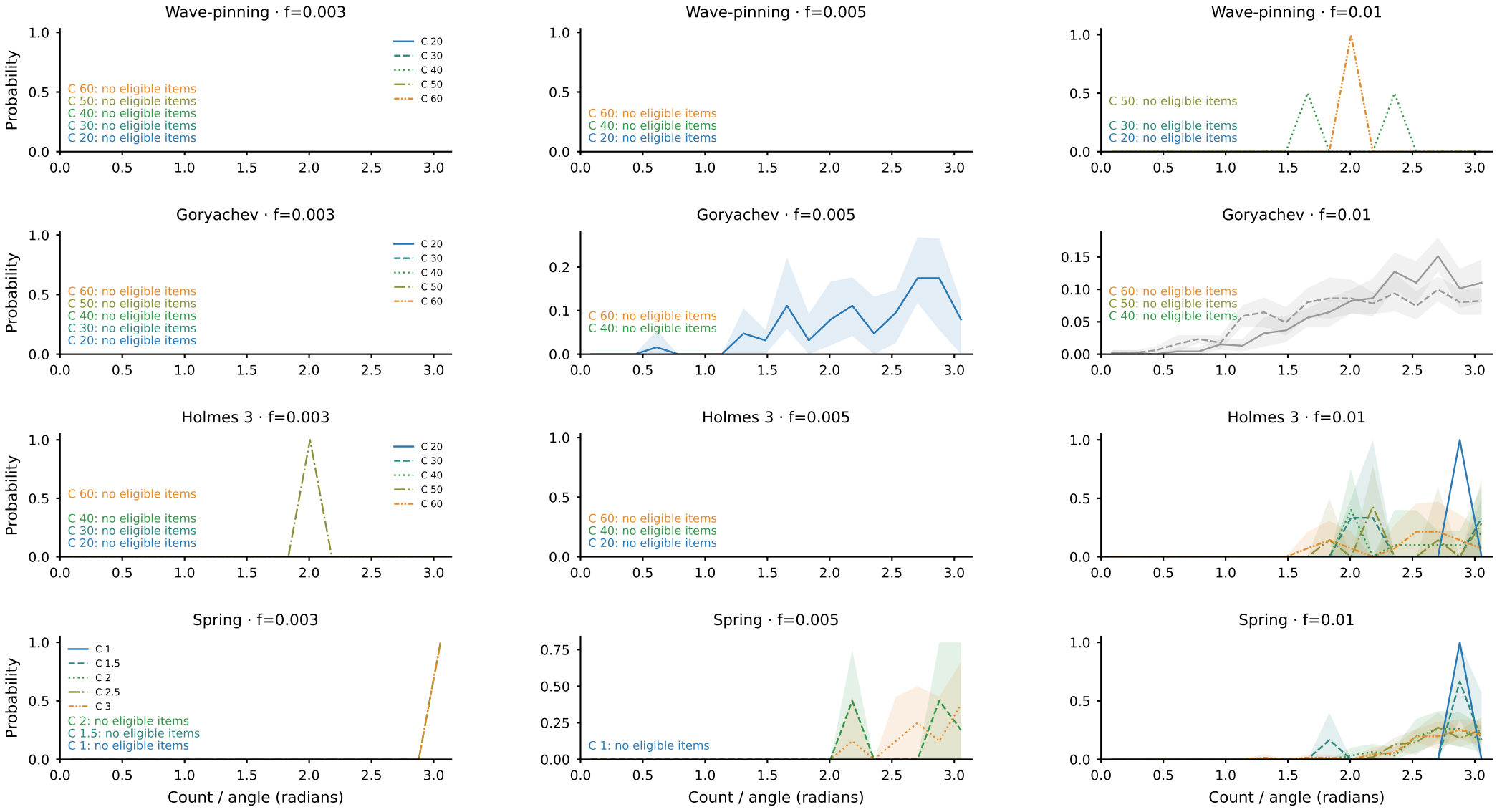
Common canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_072_S9_unscorable_takeover_tracks_smoothed_factorial.csv

S9 · Common canonical A5 observations · birth angle from previous dominant



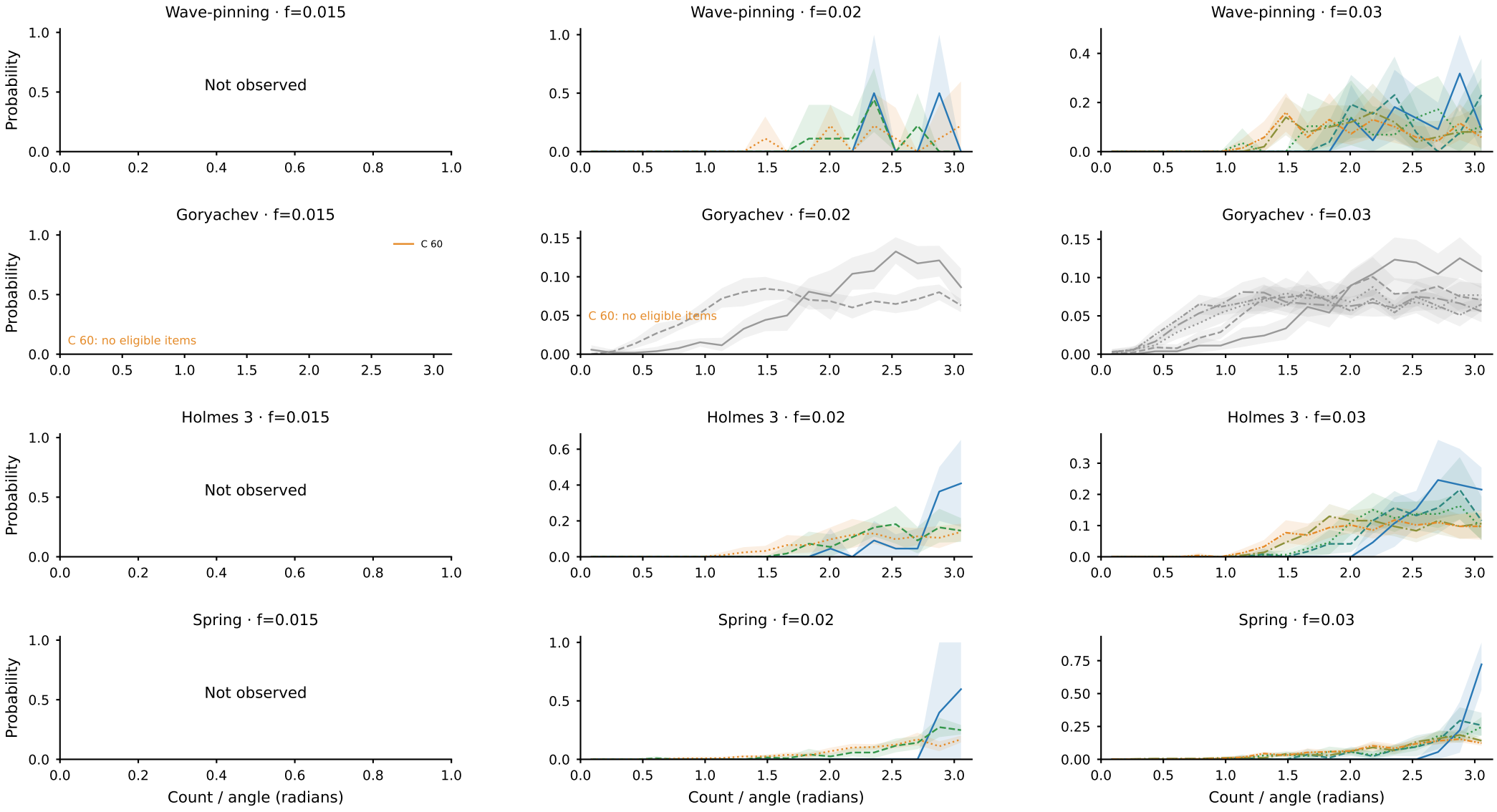
Common canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_073_S9_birth_angle_from_previous_dominant_None_0.csv

S9 · Common canonical A5 observations · birth angle from previous dominant



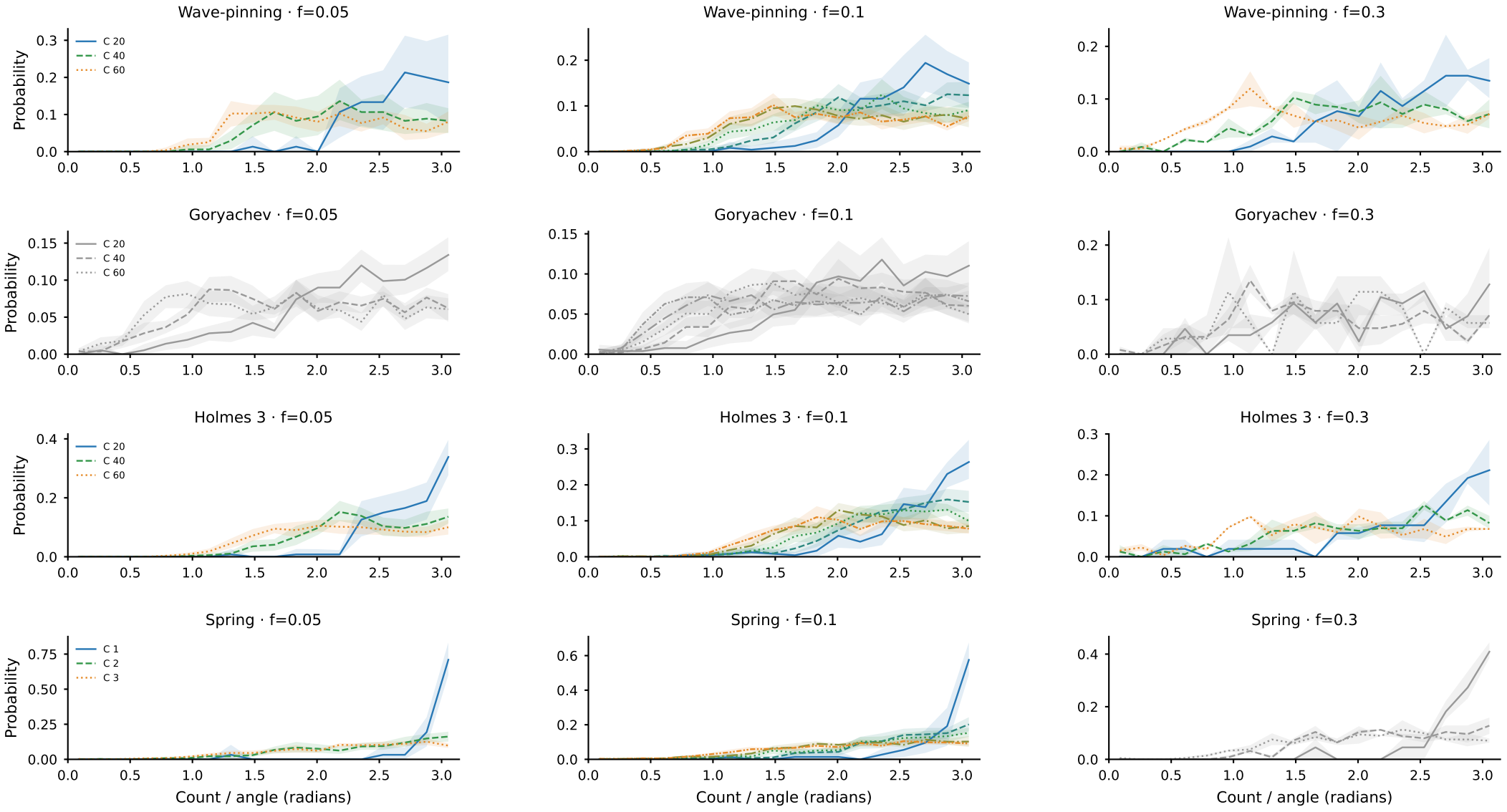
Common canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_074_S9_birth_angle_from_previous_dominant_None_3.csv

S9 · Common canonical A5 observations · birth angle from previous dominant



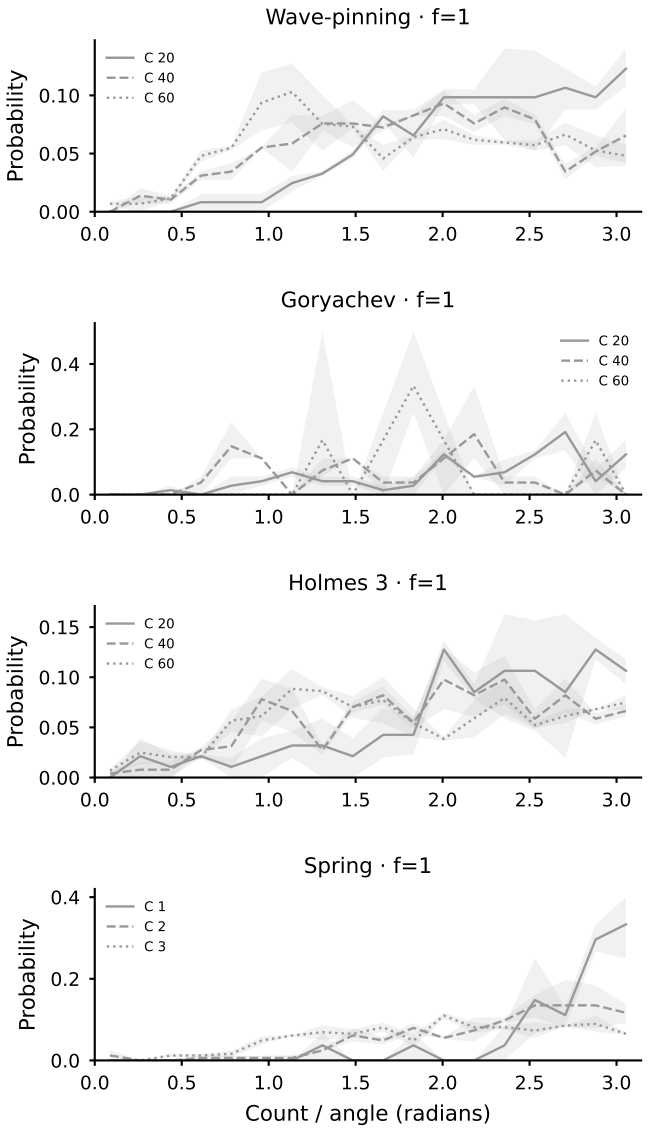
Common canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_075_S9_birth_angle_from_previous_dominant_None_6.csv

S9 · Common canonical A5 observations · birth angle from previous dominant



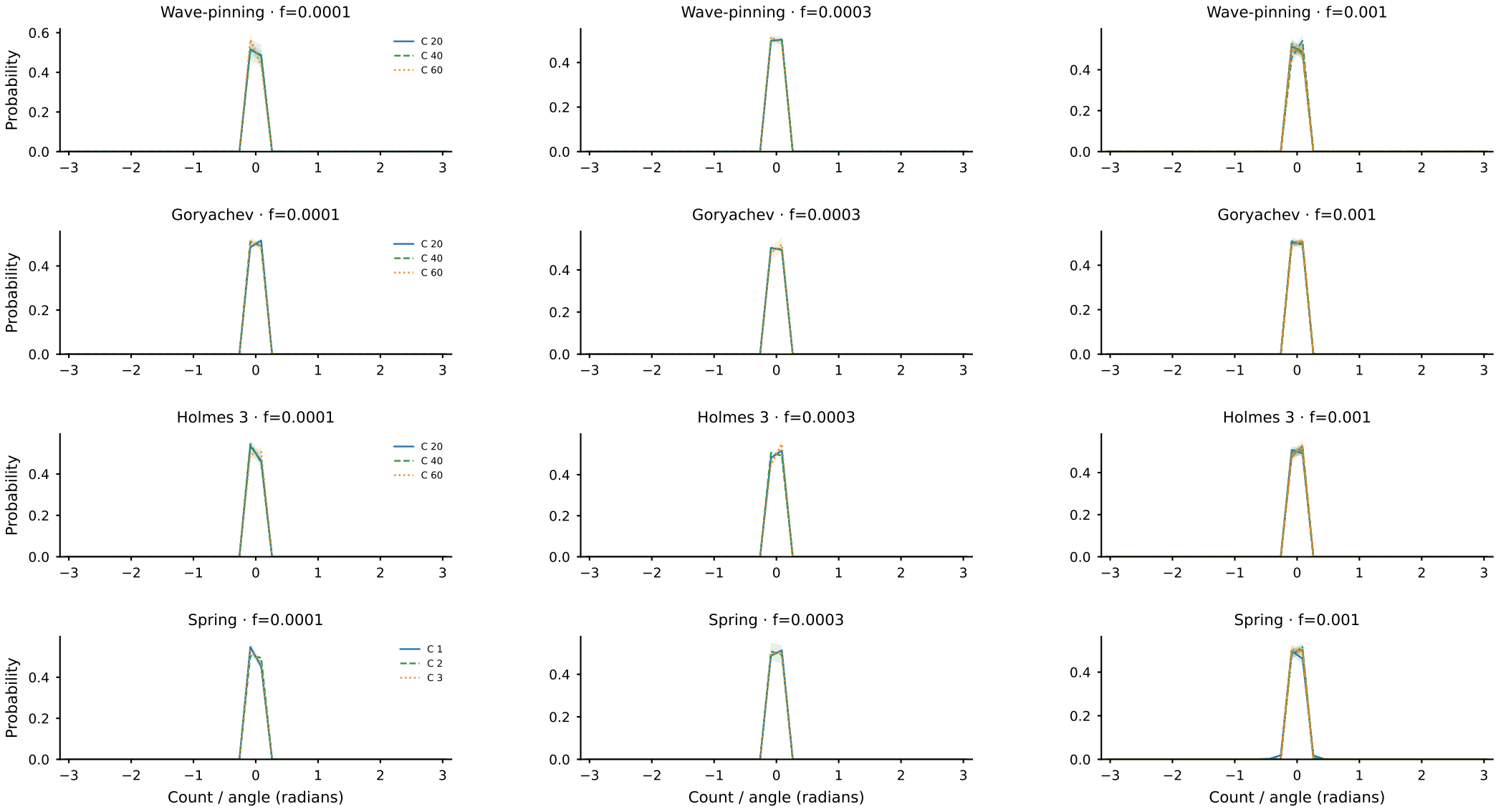
Common canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_076_S9_birth_angle_from_previous_dominant_None_9.csv

S9 · Common canonical A5 observations · birth angle from previous dominant



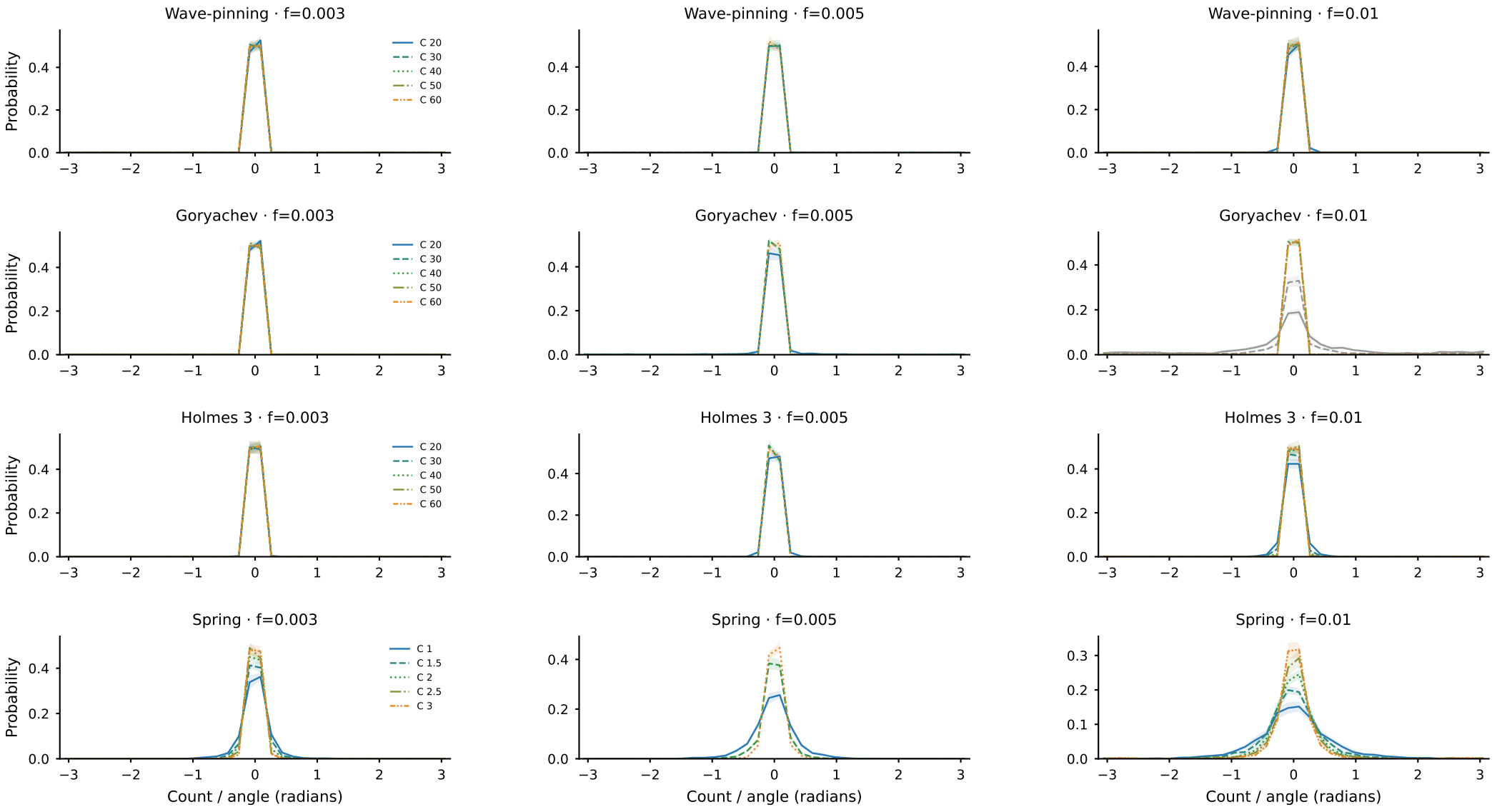
Common canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_077_S9_birth_angle_from_previous_dominant_None_12.csv

S9 · Common canonical A5 observations · dominant heading turn · 1 observation frames



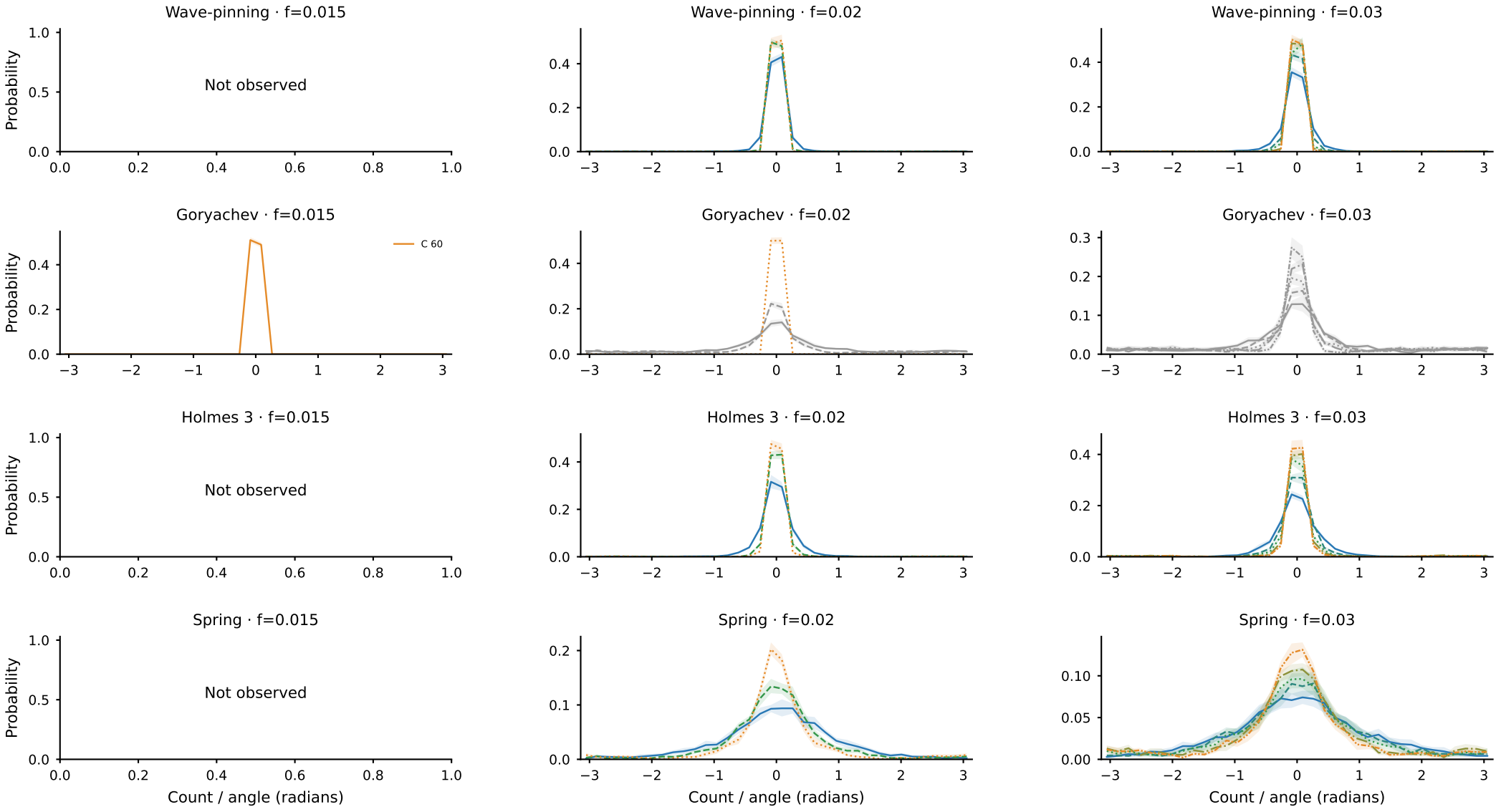
Common canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_078_S9_dominant_heading_turn_1.0_0.csv

S9 · Common canonical A5 observations · dominant heading turn · 1 observation frames



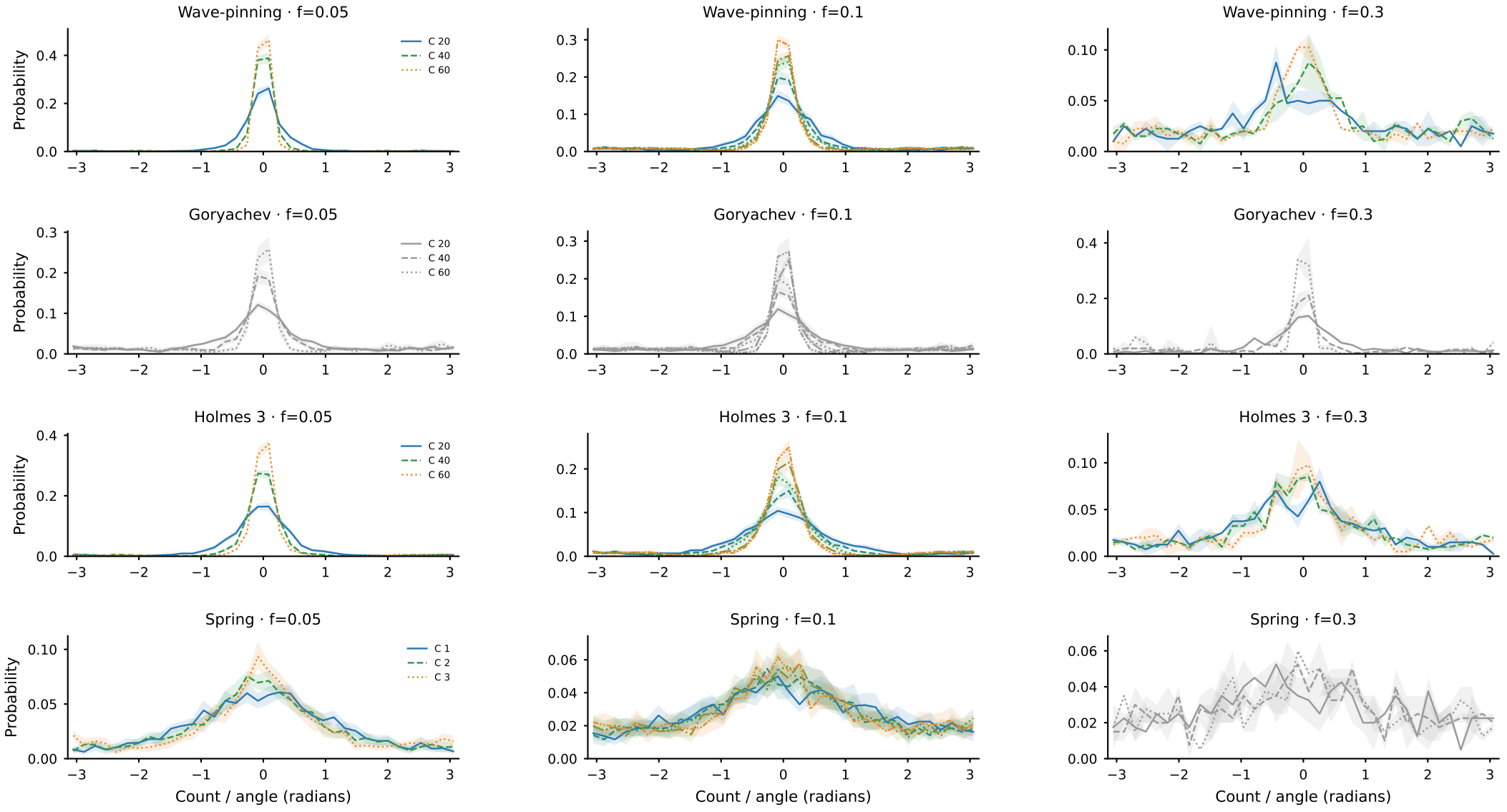
Common canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_079_S9_dominant_heading_turn_1.0_3.csv

S9 · Common canonical A5 observations · dominant heading turn · 1 observation frames



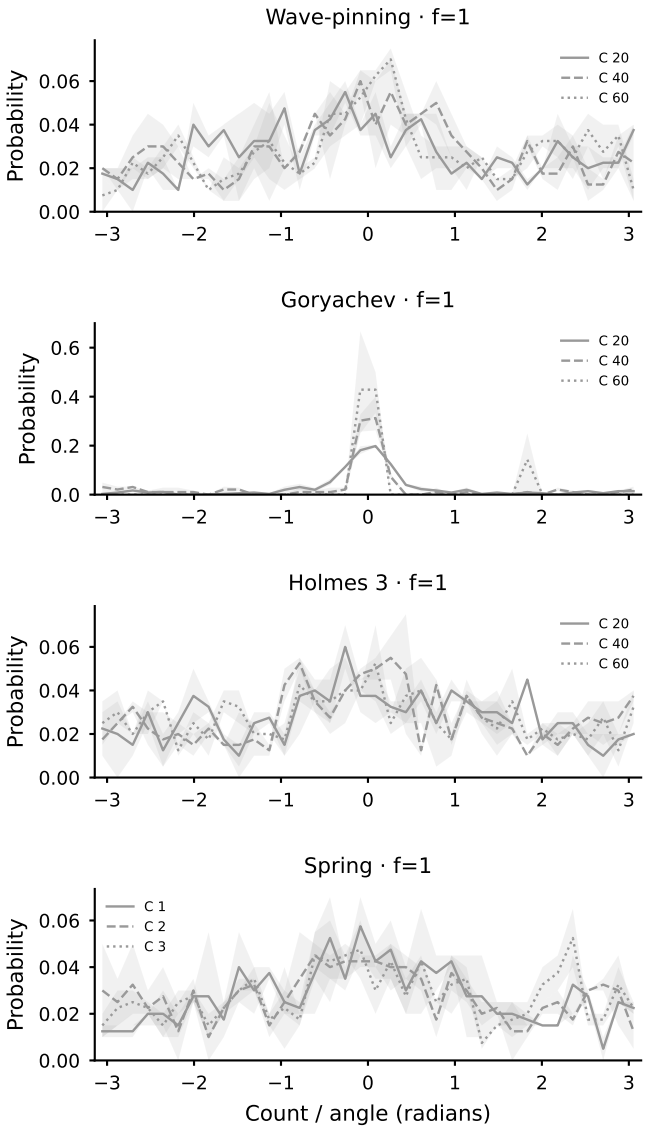
Common canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_080_S9_dominant_heading_turn_1.0_6.csv

S9 · Common canonical A5 observations · dominant heading turn · 1 observation frames



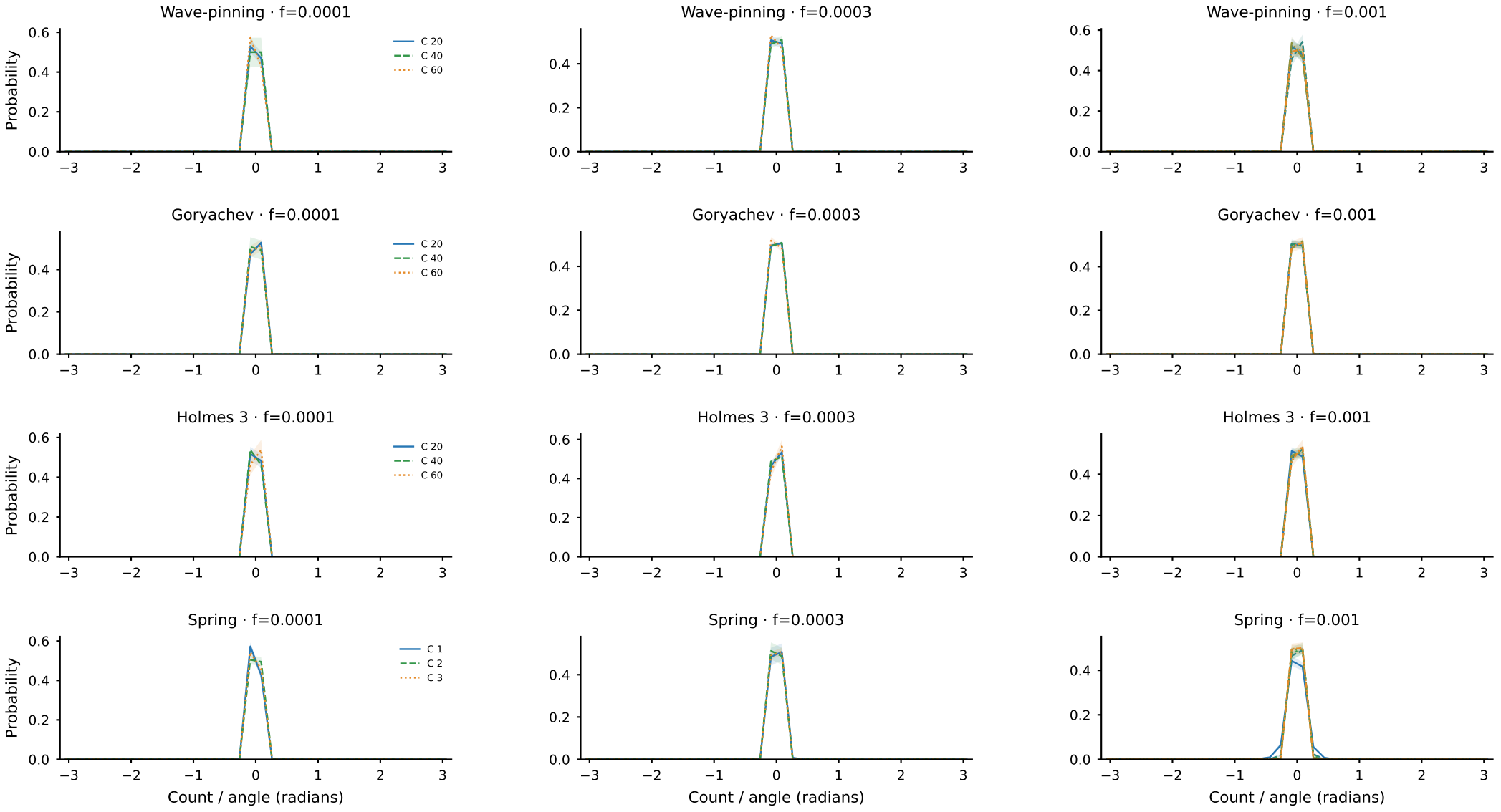
Common canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_081_S9_dominant_heading_turn_1.0_9.csv

S9 · Common canonical A5 observations · dominant heading turn · 1 observation frames



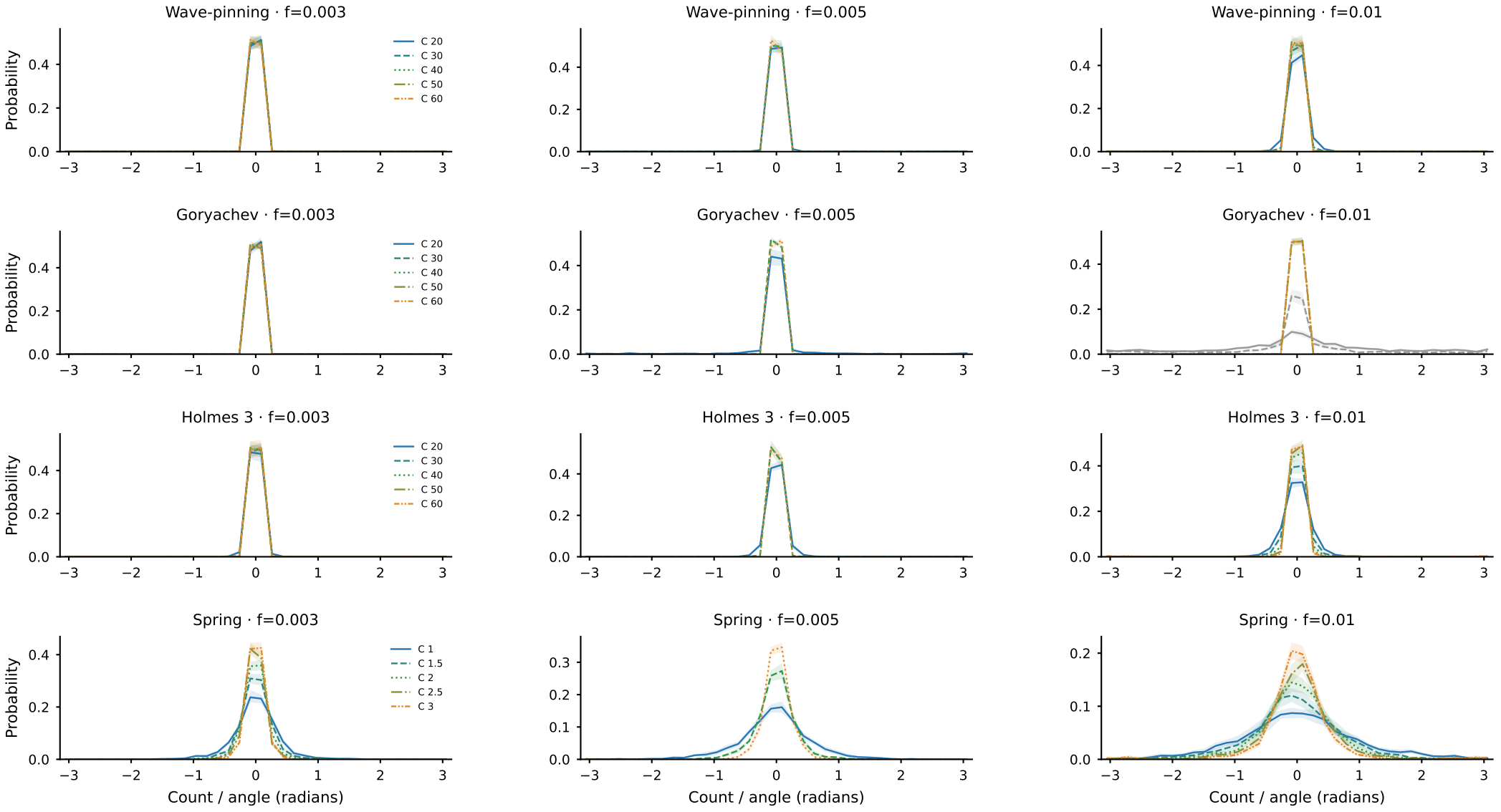
Common canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_082_S9_dominant_heading_turn_1.0_12.csv

S9 · Common canonical A5 observations · dominant heading turn · 2 observation frames



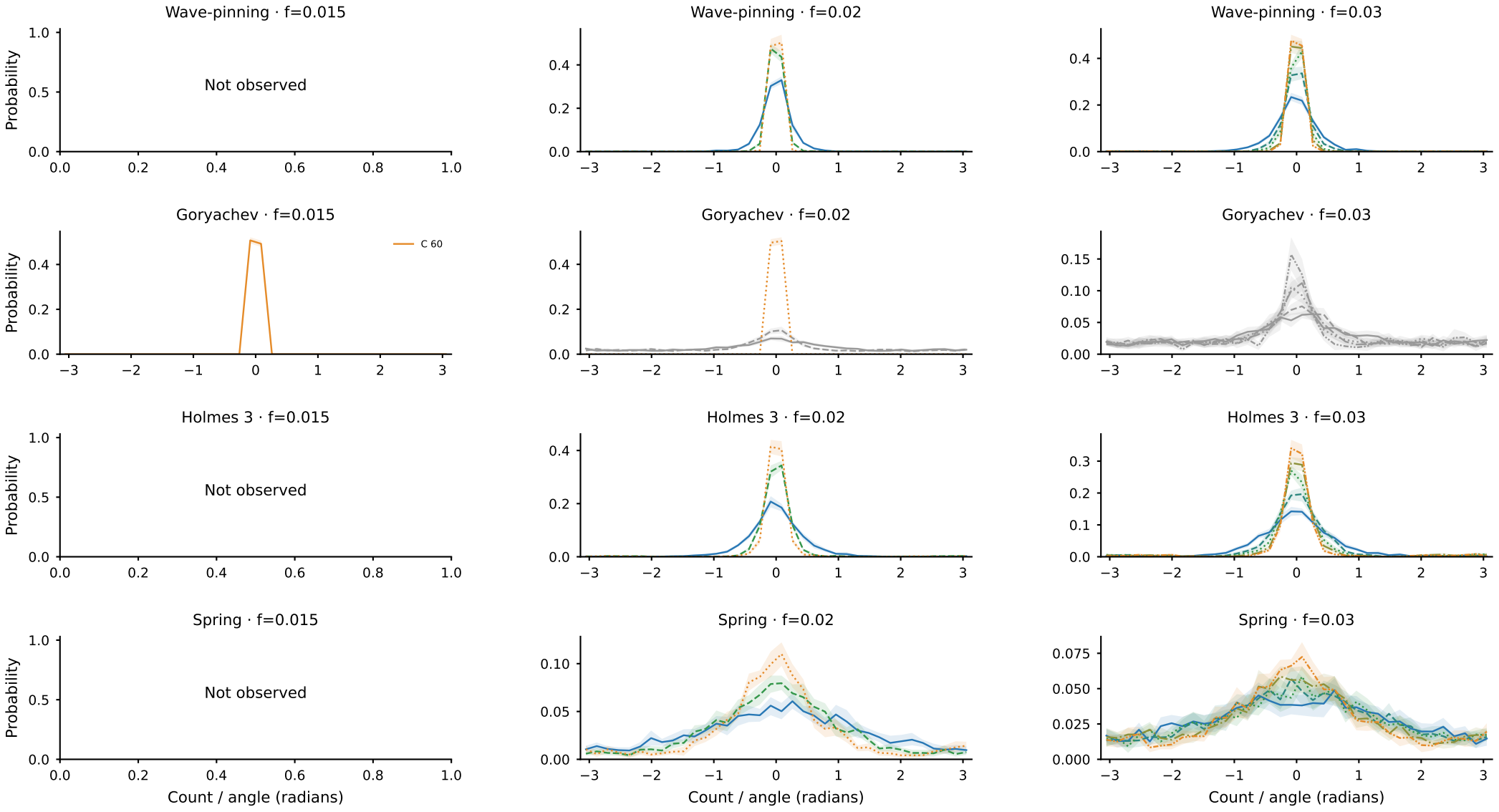
Common canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_083_S9_dominant_heading_turn_2.0_0.csv

S9 · Common canonical A5 observations · dominant heading turn · 2 observation frames



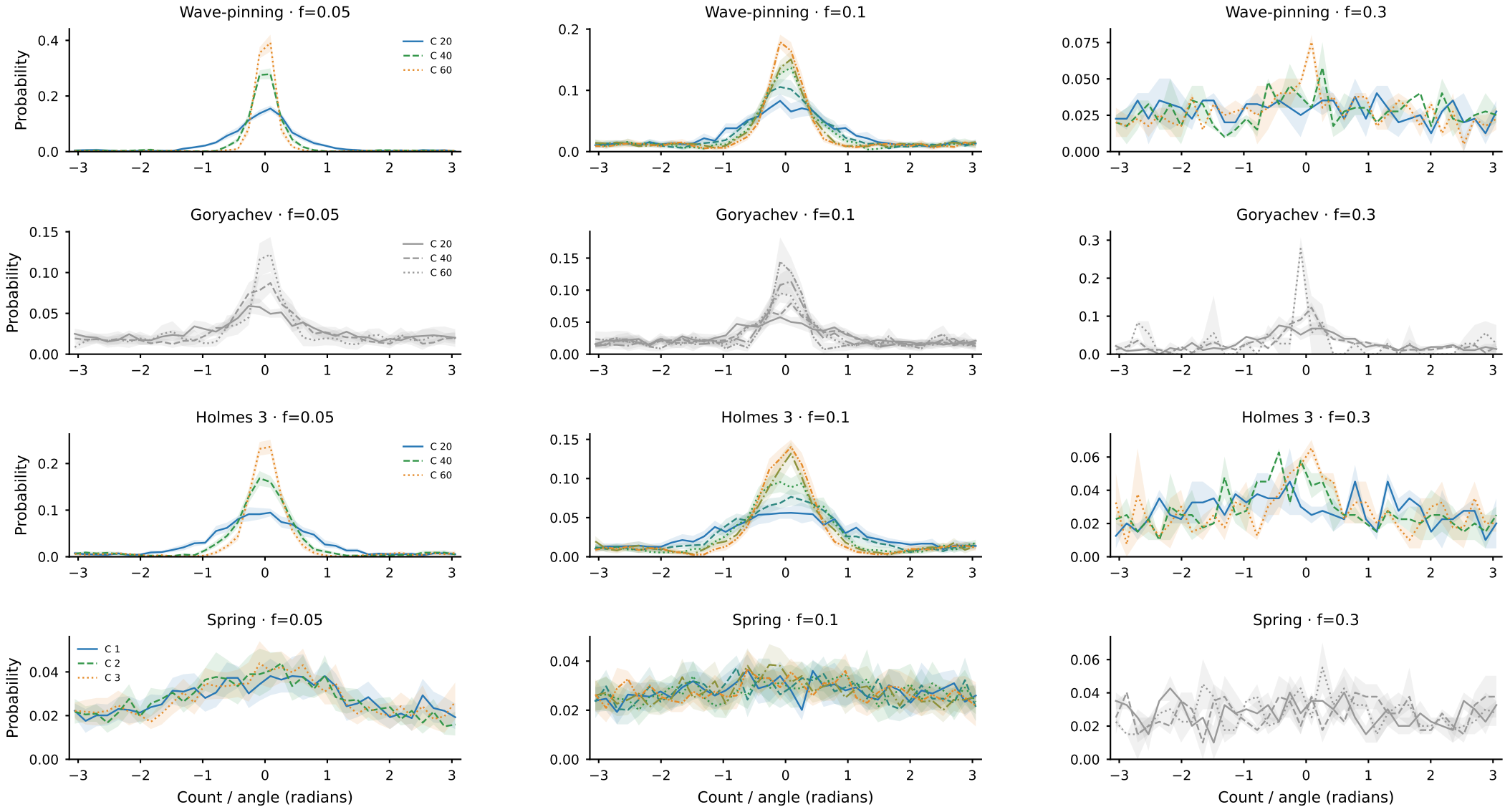
Common canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_084_S9_dominant_heading_turn_2.0_3.csv

S9 · Common canonical A5 observations · dominant heading turn · 2 observation frames



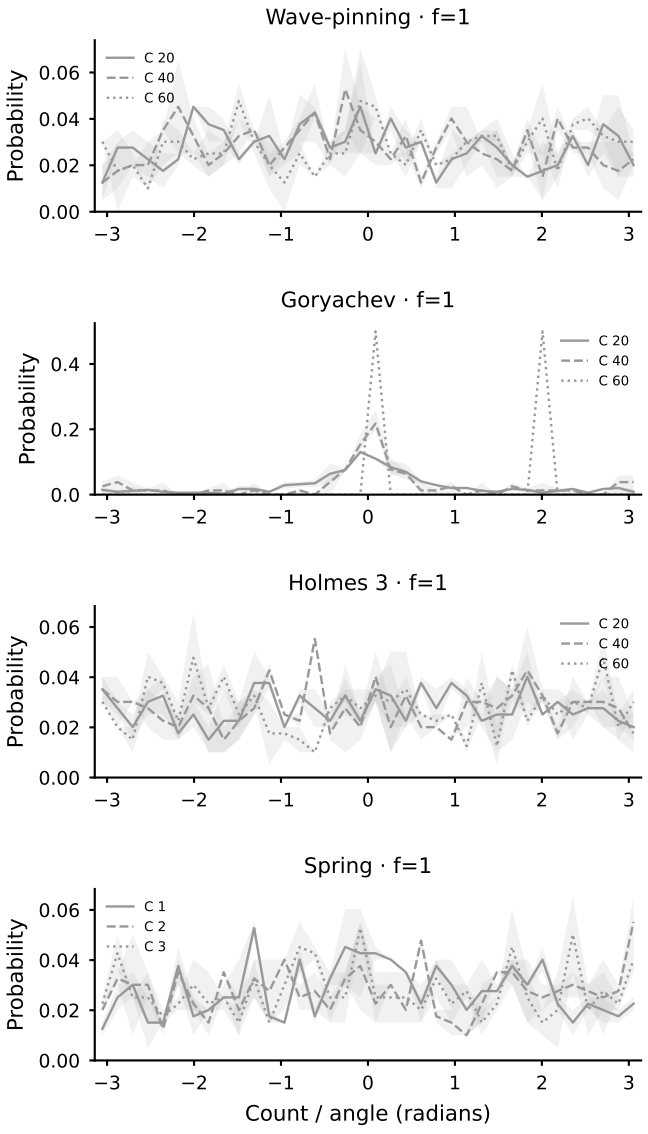
Common canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_085_S9_dominant_heading_turn_2.0_6.csv

S9 · Common canonical A5 observations · dominant heading turn · 2 observation frames



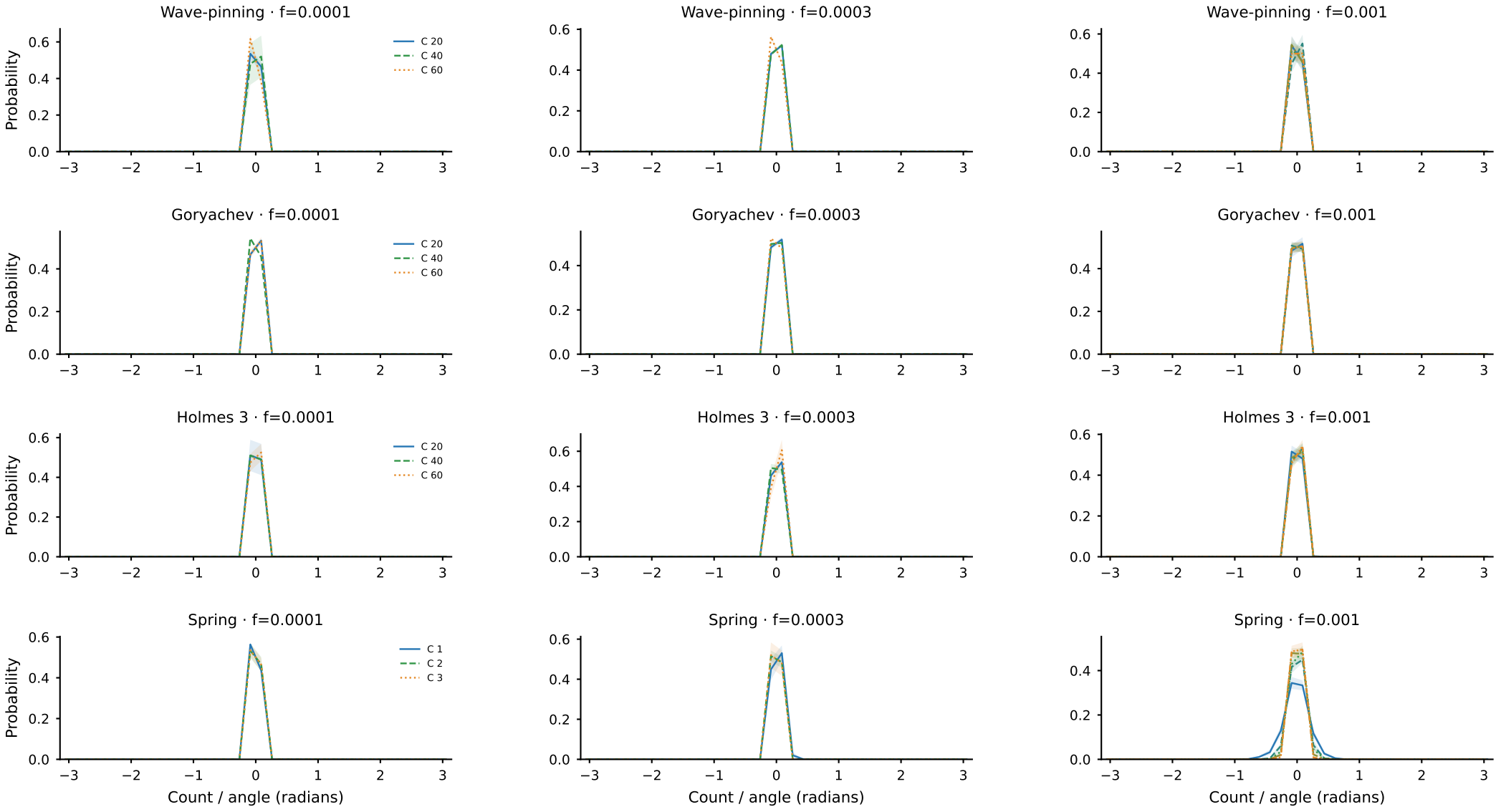
Common canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_086_S9_dominant_heading_turn_2.0_9.csv

S9 · Common canonical A5 observations · dominant heading turn · 2 observation frames



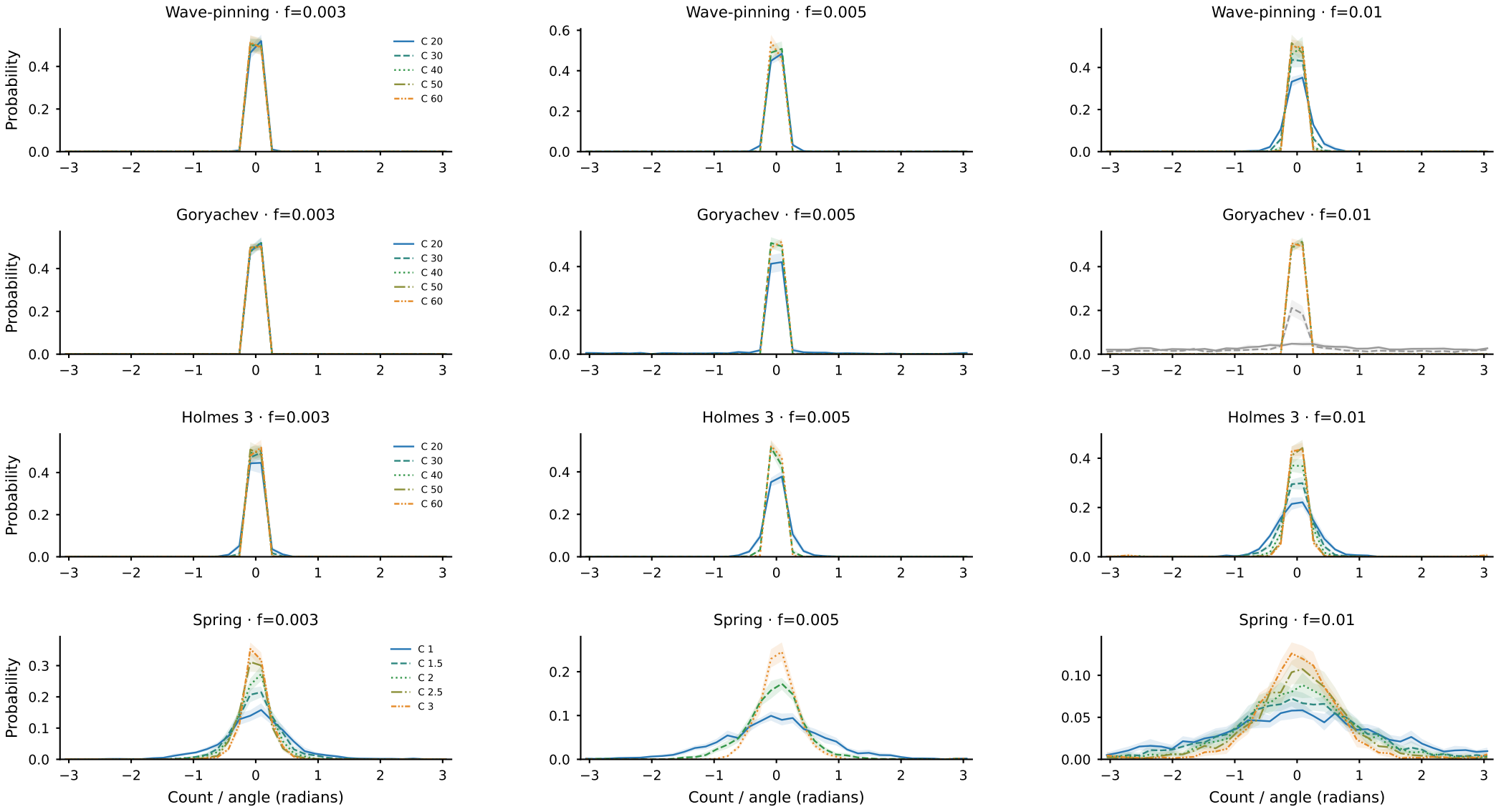
Common canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_087_S9_dominant_heading_turn_2.0_12.csv

S9 · Common canonical A5 observations · dominant heading turn · 4 observation frames



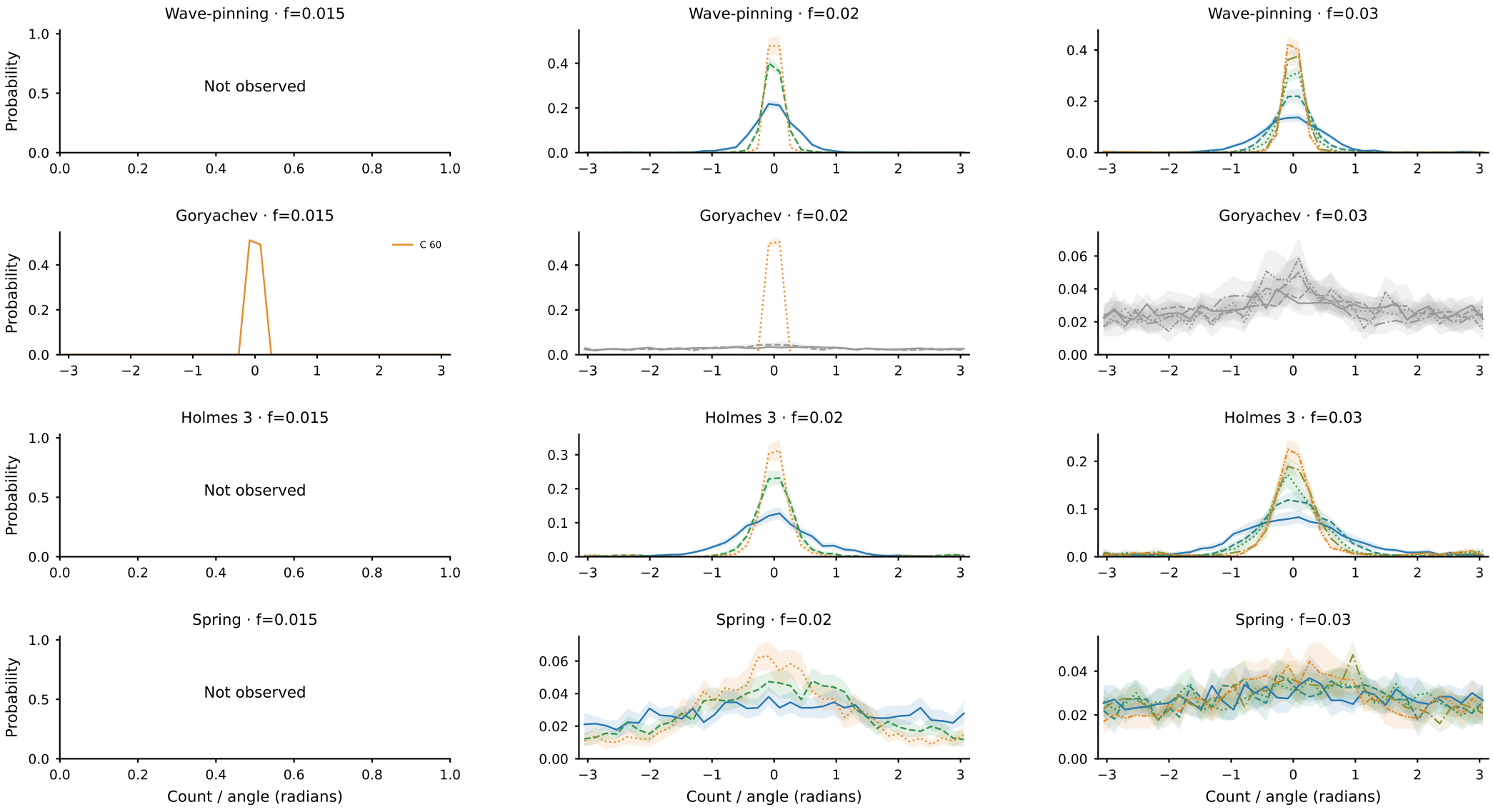
Common canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_088_S9_dominant_heading_turn_4.0_0.csv

S9 · Common canonical A5 observations · dominant heading turn · 4 observation frames



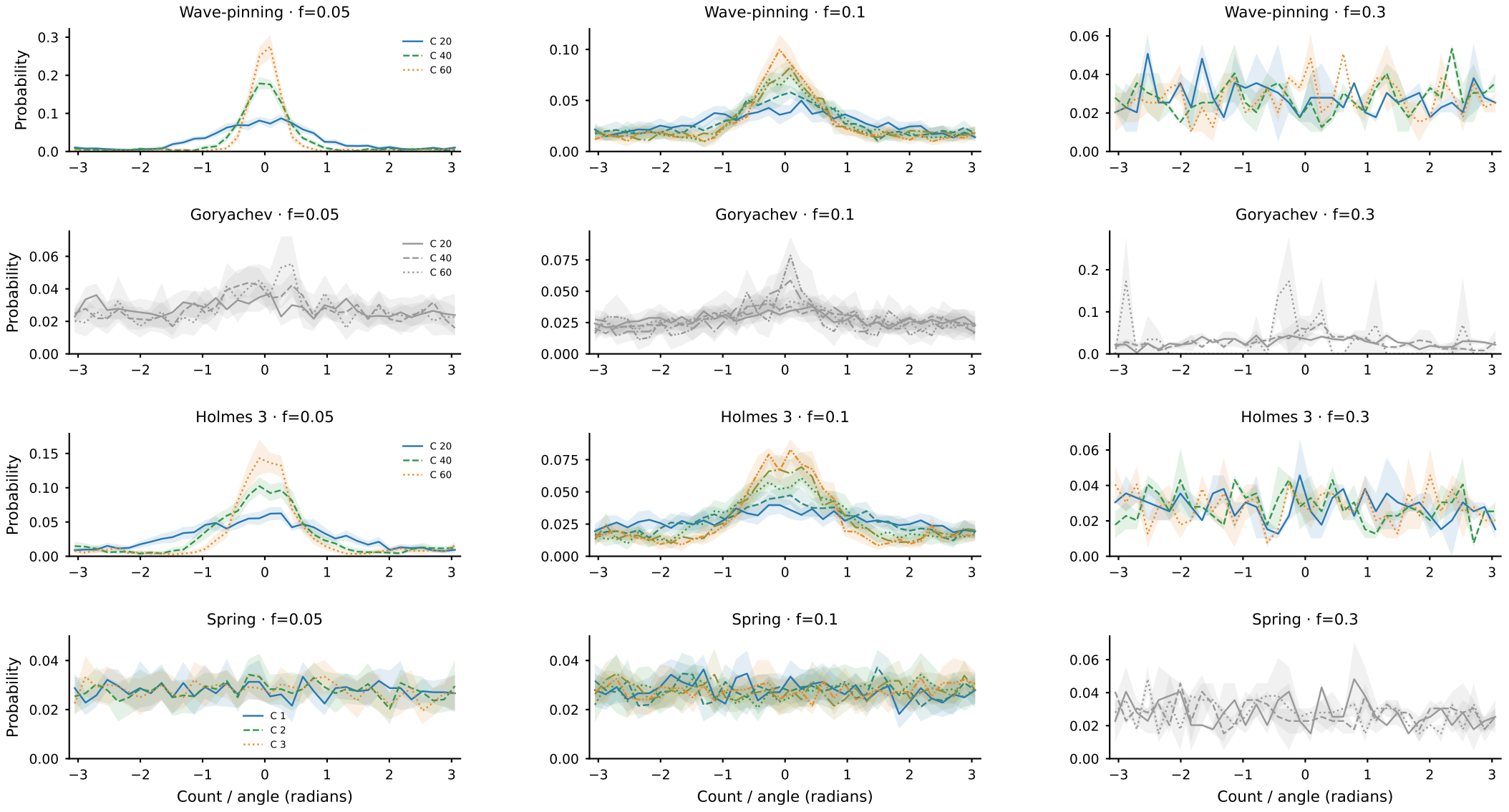
Common canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_089_S9_dominant_heading_turn_4.0_3.csv

S9 · Common canonical A5 observations · dominant heading turn · 4 observation frames



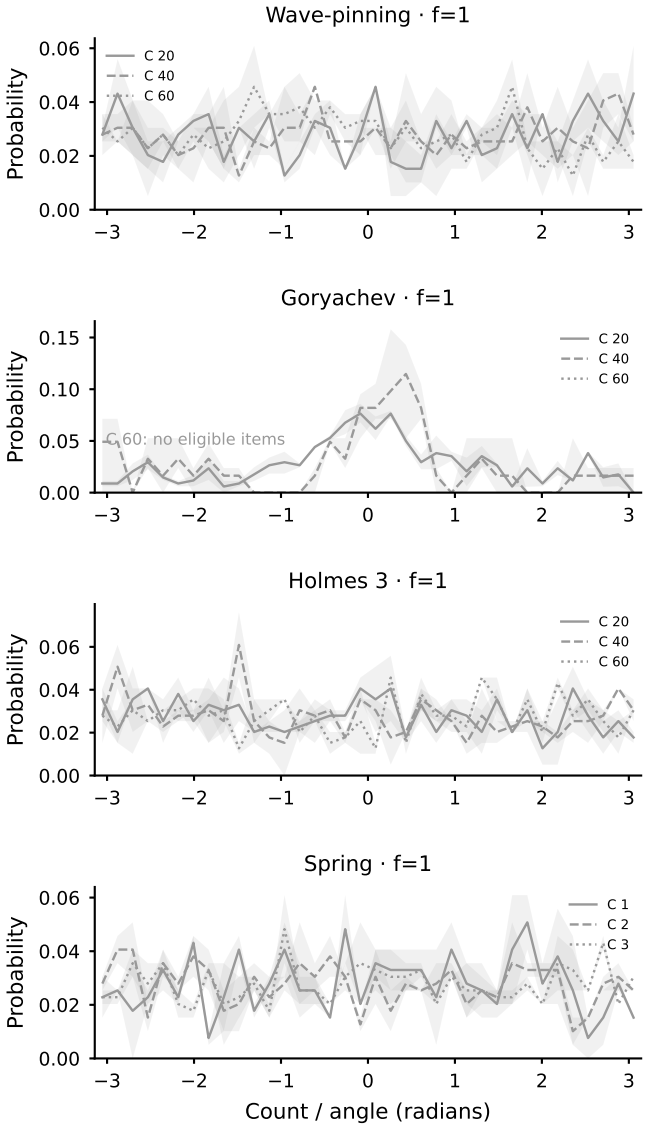
Common canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_090_S9_dominant_heading_turn_4.0_6.csv

S9 · Common canonical A5 observations · dominant heading turn · 4 observation frames



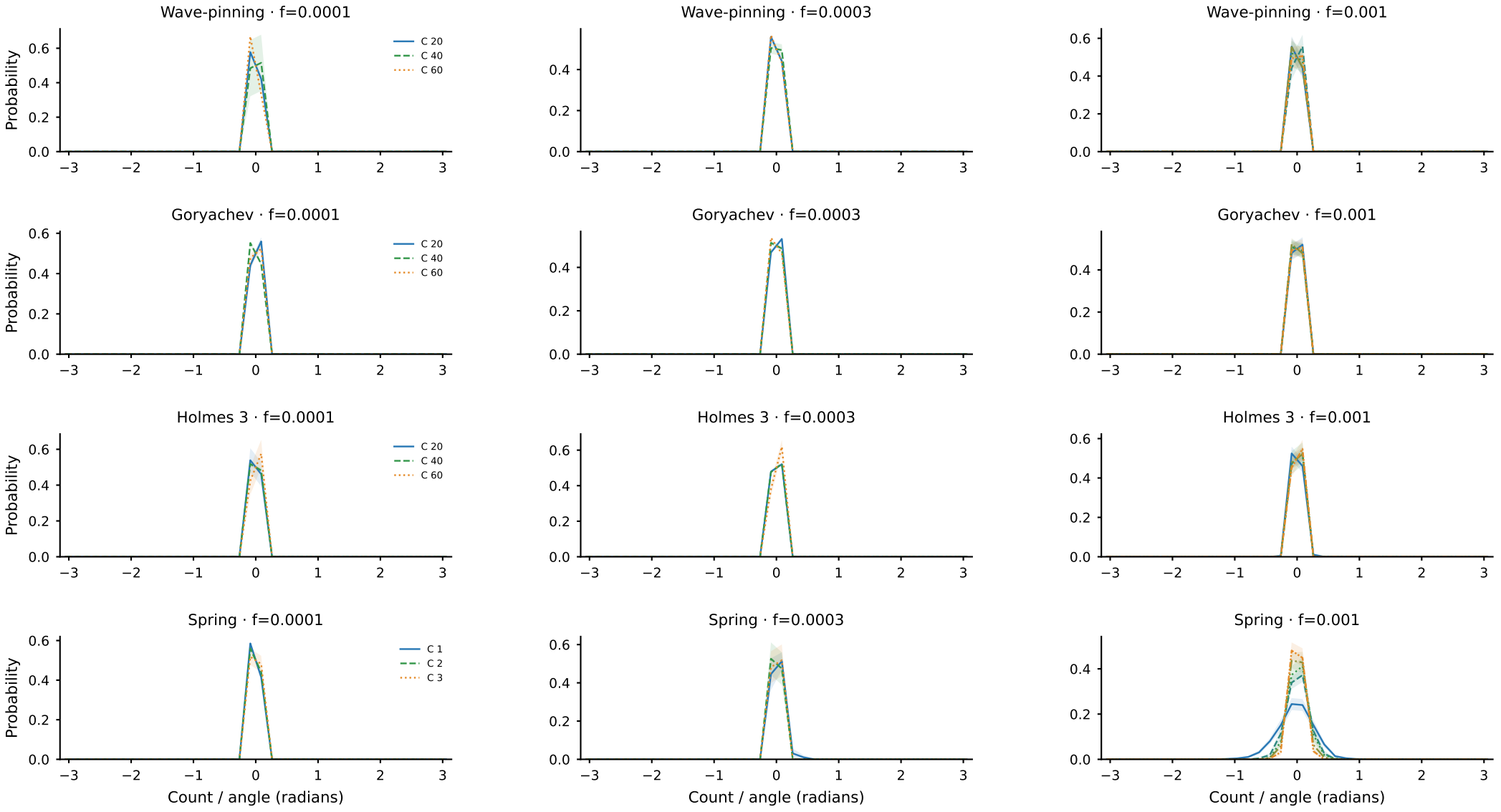
Common canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_091_S9_dominant_heading_turn_4.0_9.csv

S9 · Common canonical A5 observations · dominant heading turn · 4 observation frames



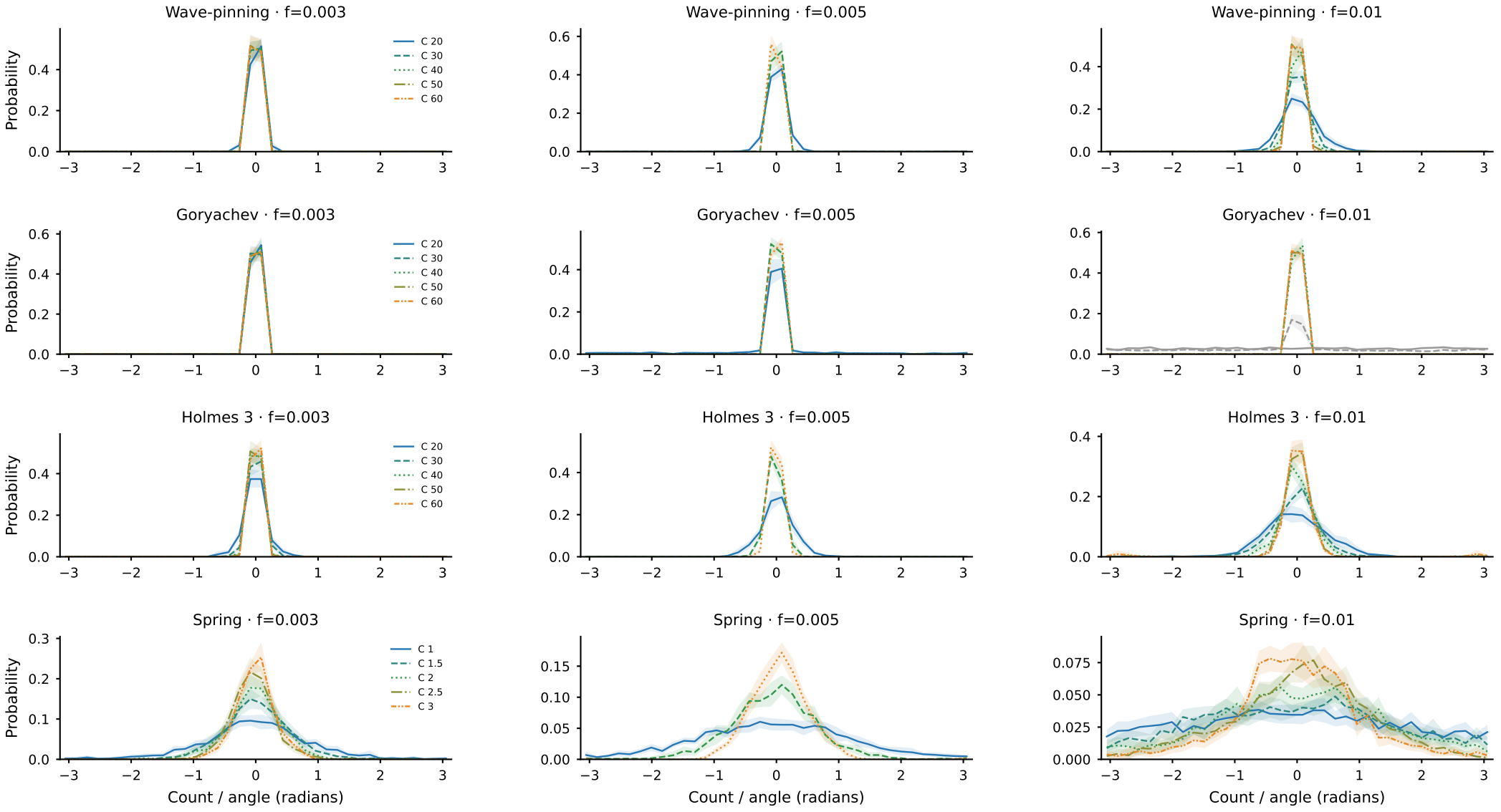
Common canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_092_S9_dominant_heading_turn_4.0_12.csv

S9 · Common canonical A5 observations · dominant heading turn · 8 observation frames



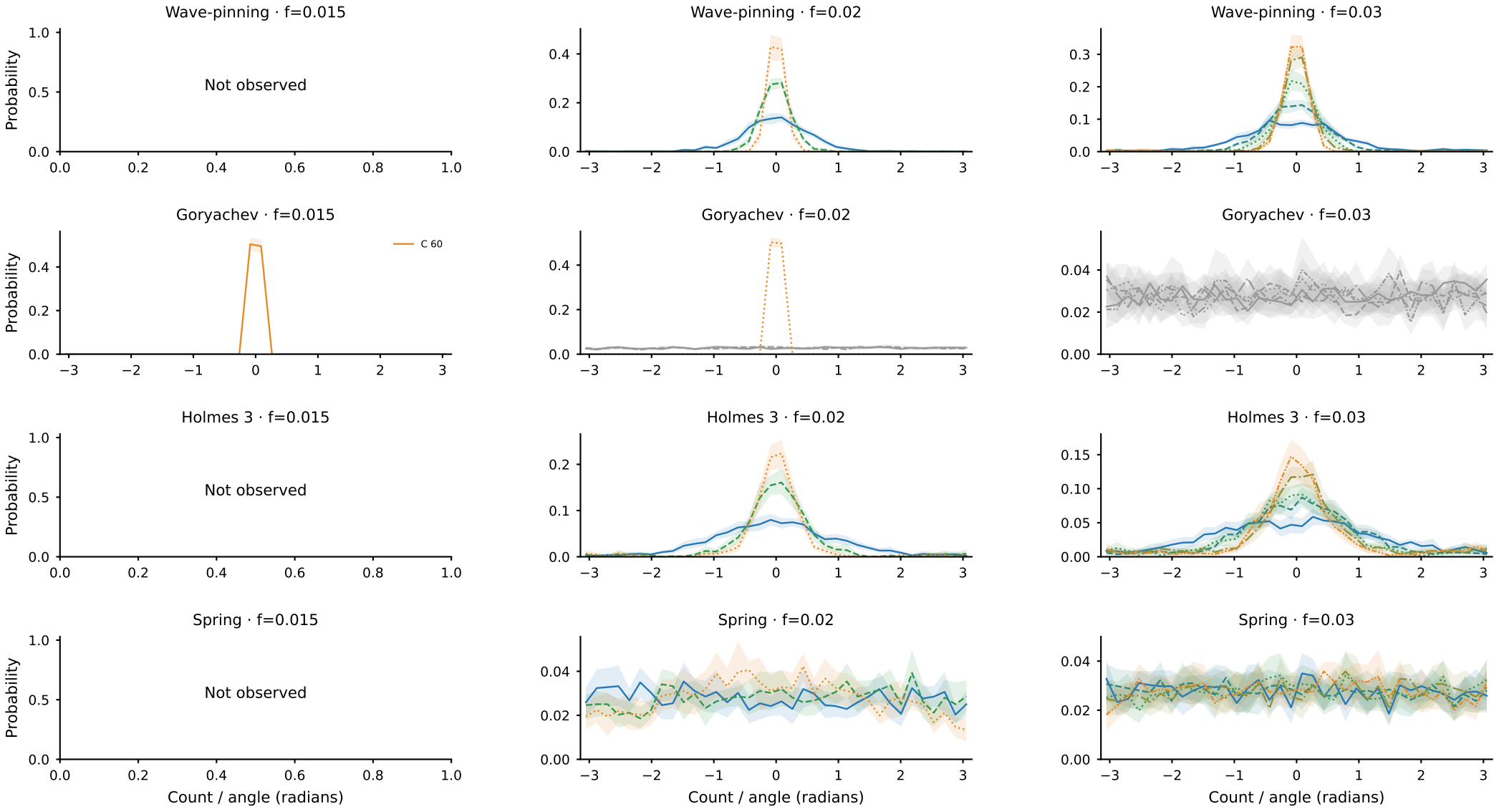
Common canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_093_S9_dominant_heading_turn_8.0_0.csv

S9 · Common canonical A5 observations · dominant heading turn · 8 observation frames



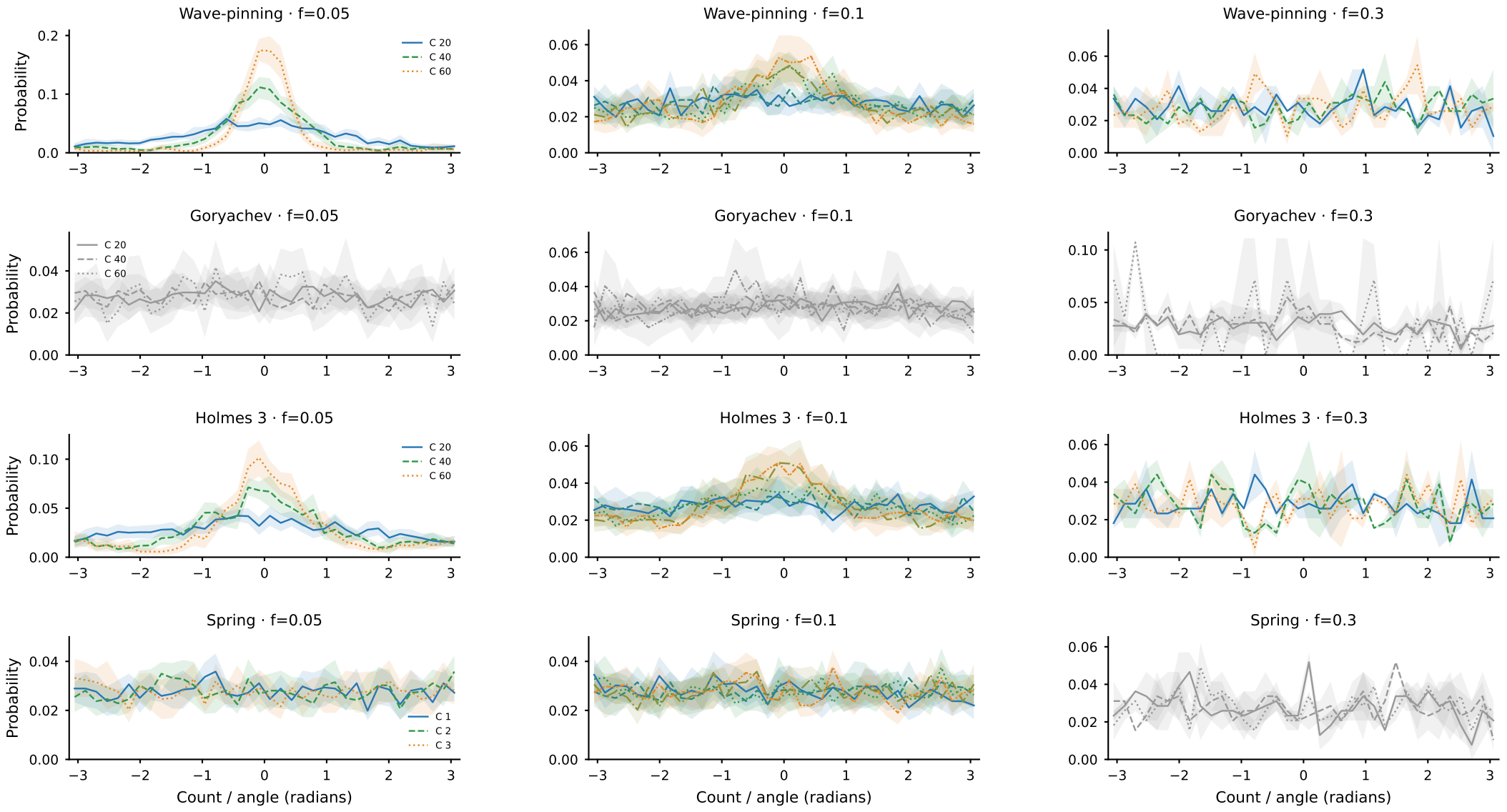
Common canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_094_S9_dominant_heading_turn_8.0_3.csv

S9 · Common canonical A5 observations · dominant heading turn · 8 observation frames



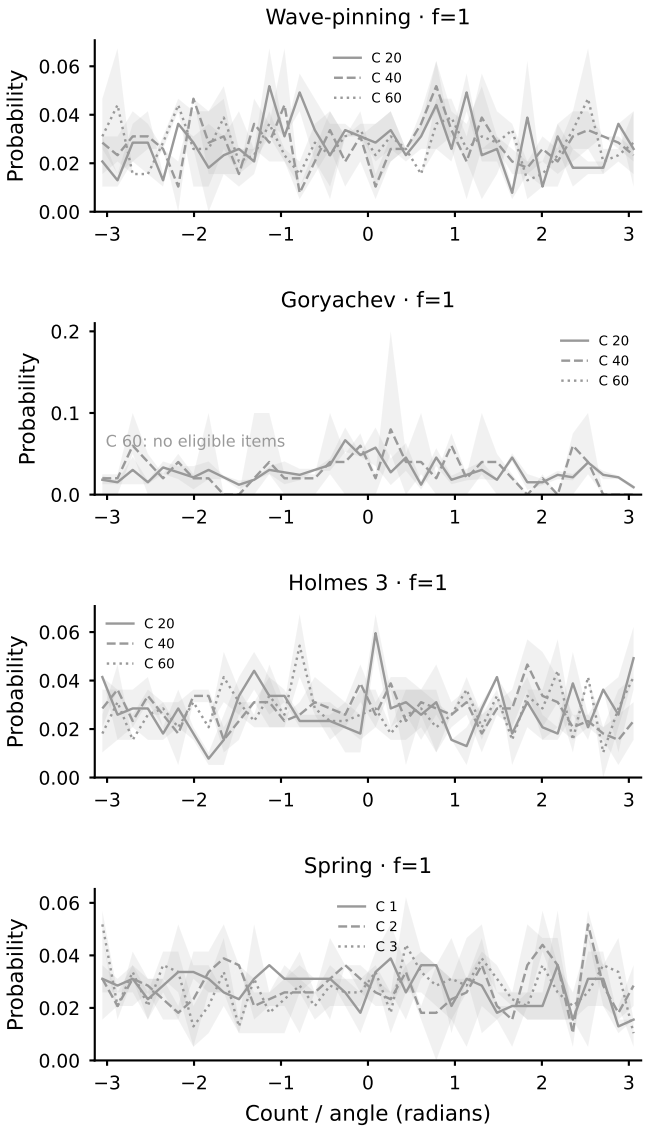
Common canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_095_S9_dominant_heading_turn_8.0_6.csv

S9 · Common canonical A5 observations · dominant heading turn · 8 observation frames



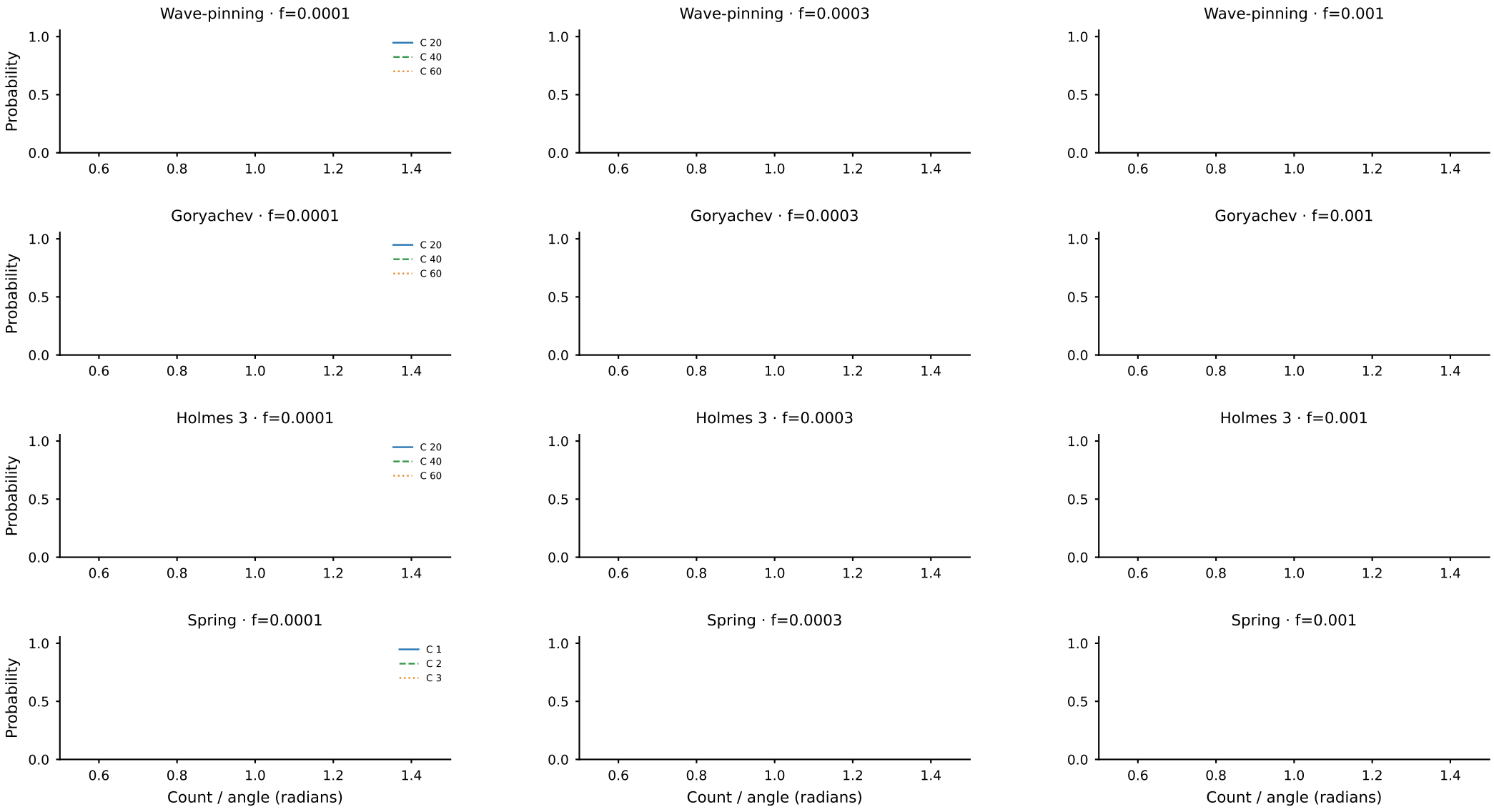
Common canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_096_S9_dominant_heading_turn_8.0_9.csv

S9 · Common canonical A5 observations · dominant heading turn · 8 observation frames



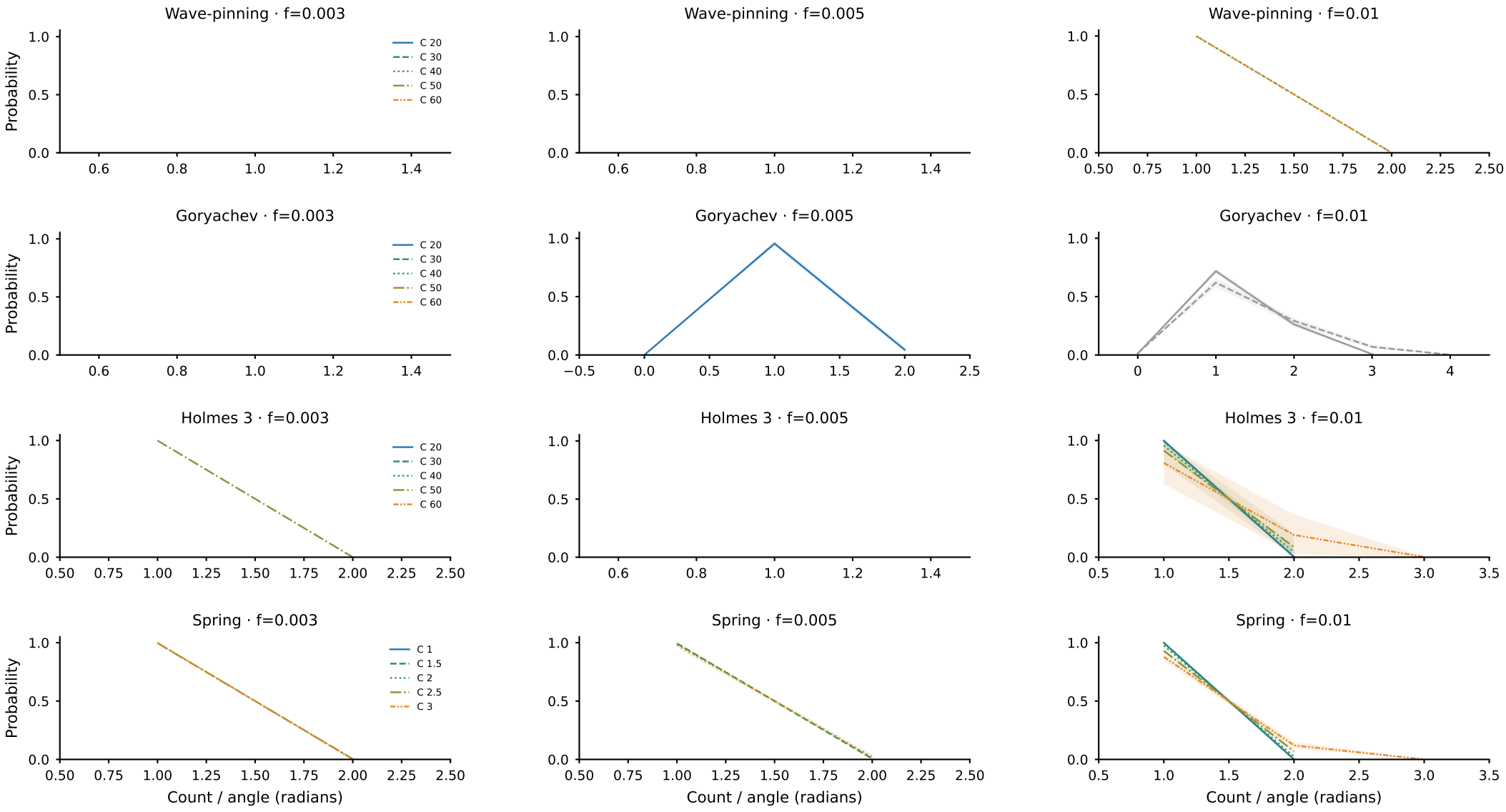
Common canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_097_S9_dominant_heading_turn_8.0_12.csv

S9 · Common canonical A5 observations · simultaneous region count



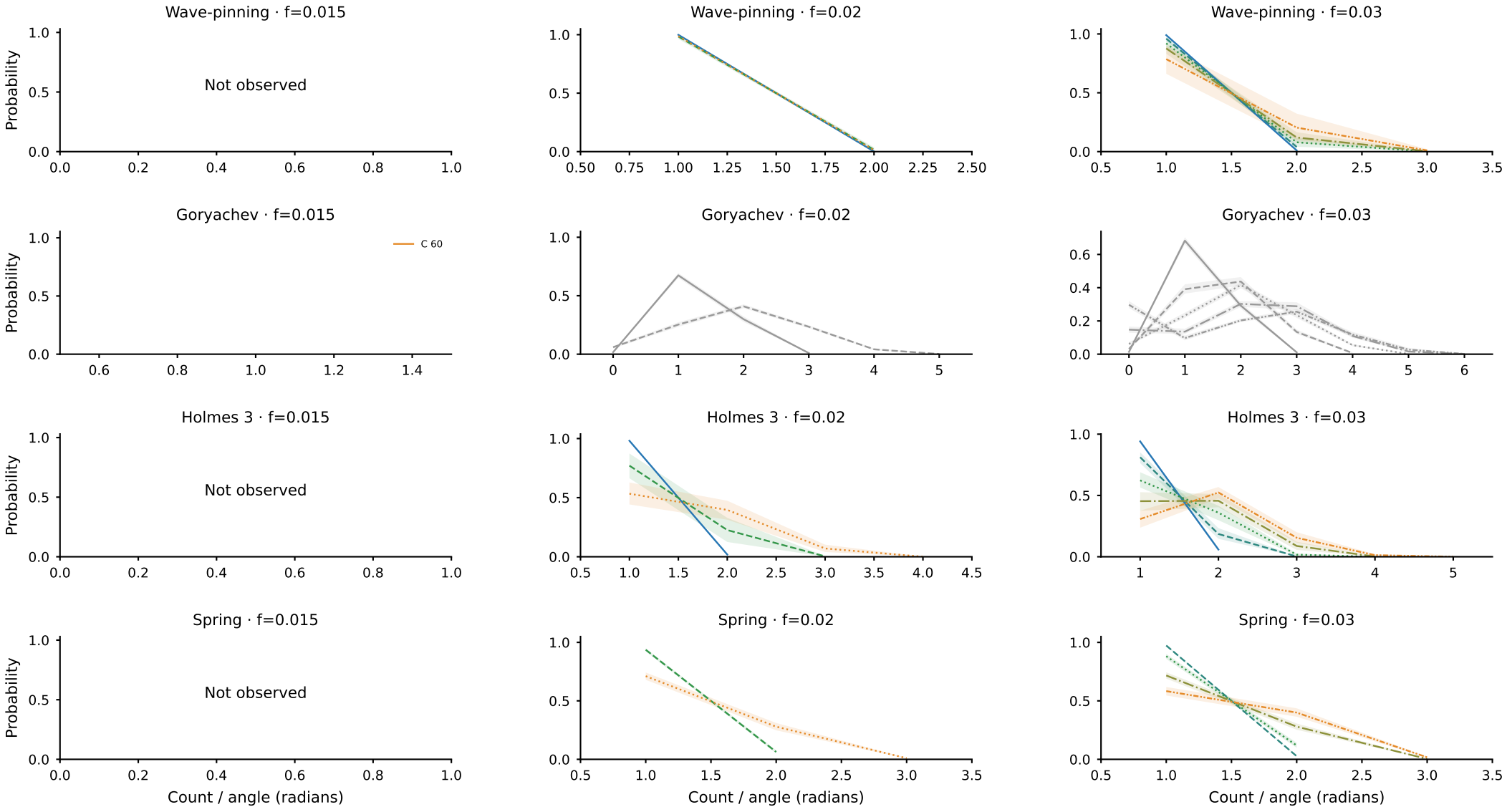
Common canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_098_S9_simultaneous_region_count_None_0.csv

S9 · Common canonical A5 observations · simultaneous region count



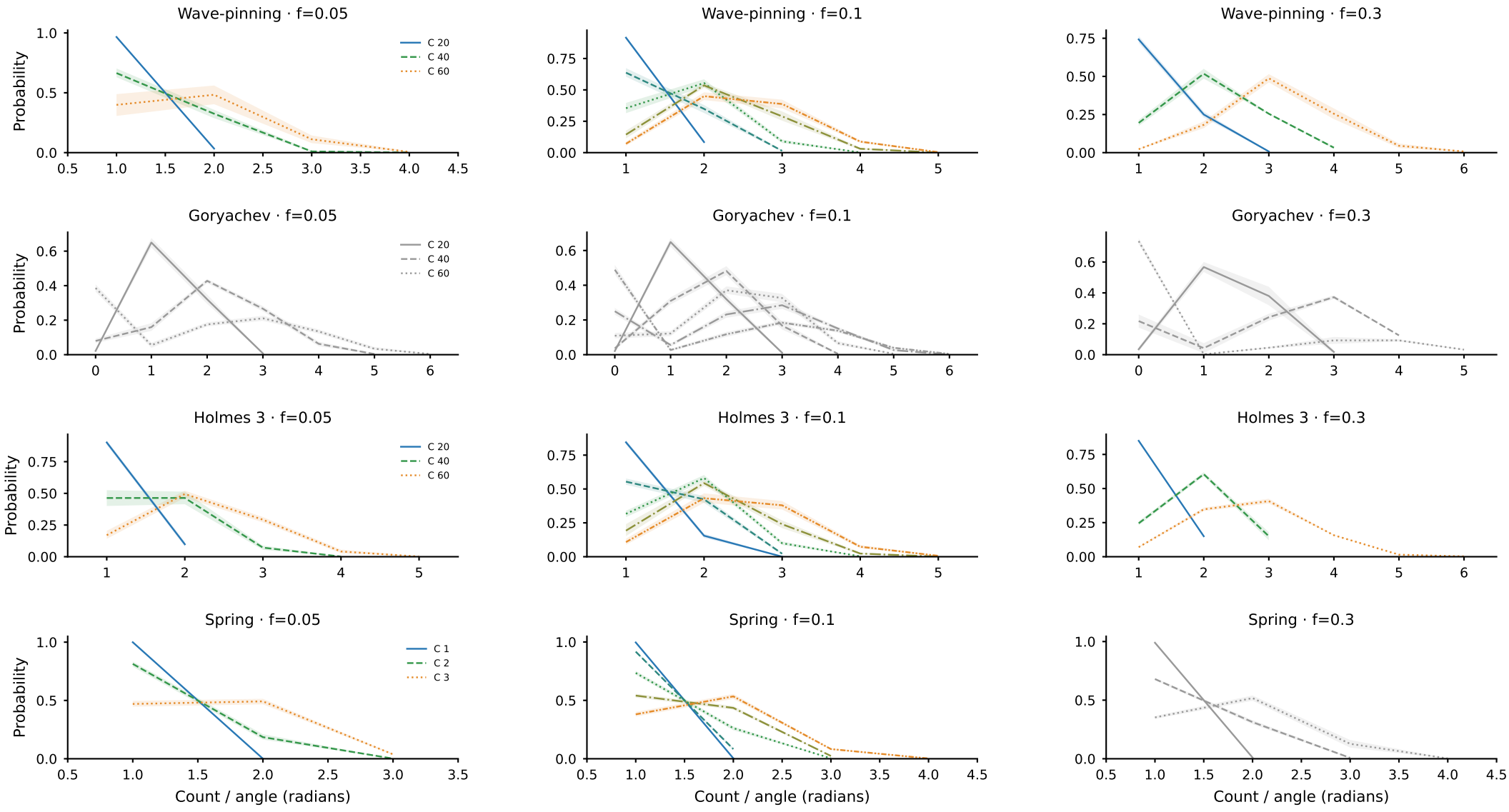
Common canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_099_S9_simultaneous_region_count_None_3.csv

S9 · Common canonical A5 observations · simultaneous region count



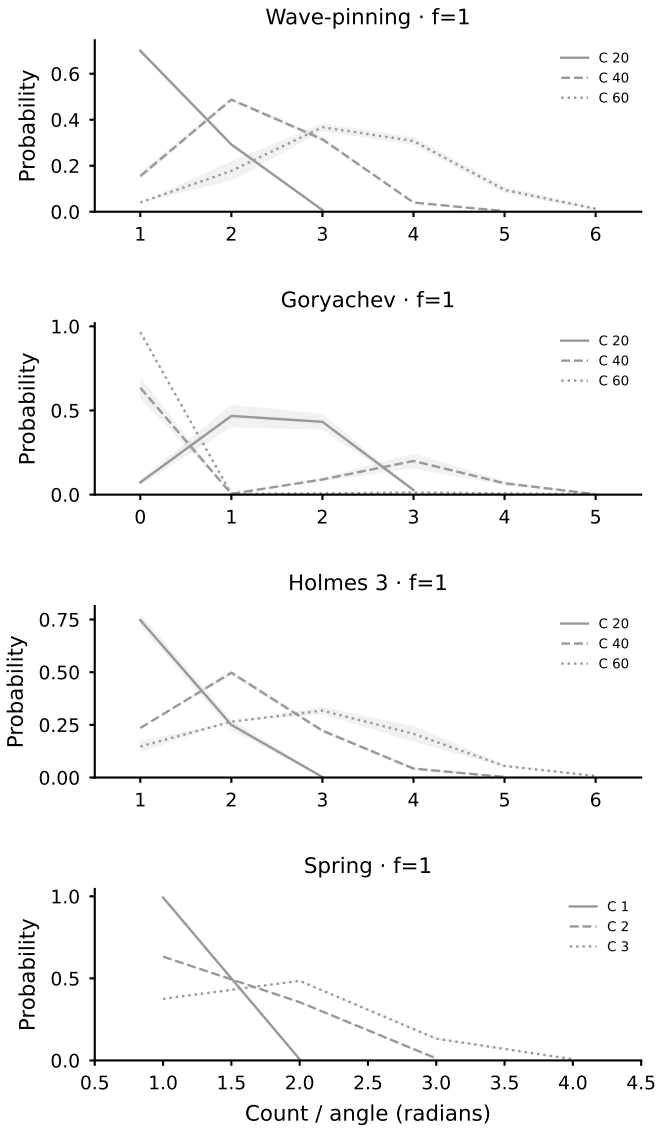
Common canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_100_S9_simultaneous_region_count_None_6.csv

S9 · Common canonical A5 observations · simultaneous region count



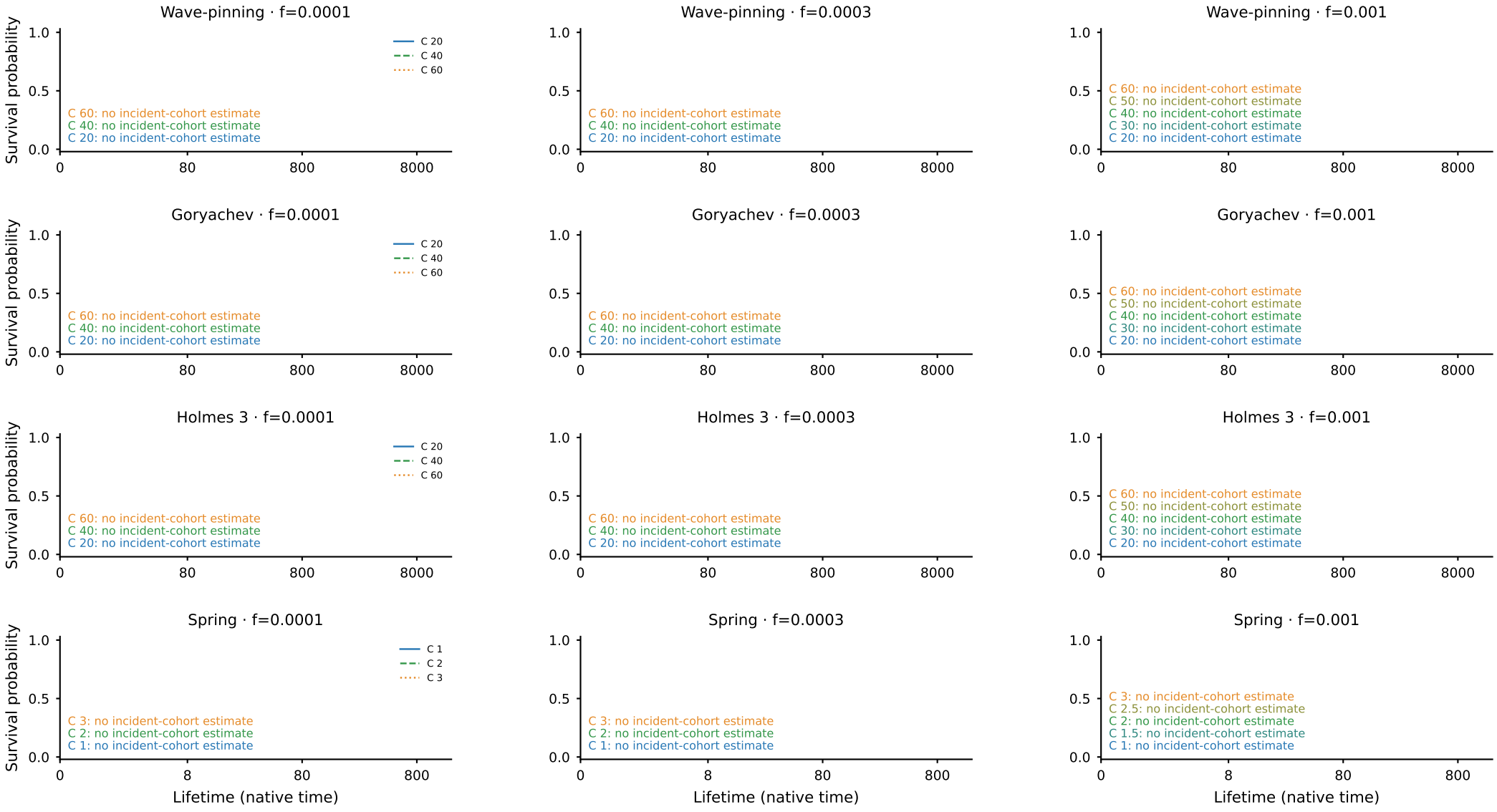
Common canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_101_S9_simultaneous_region_count_None_9.csv

S9 · Common canonical A5 observations · simultaneous region count



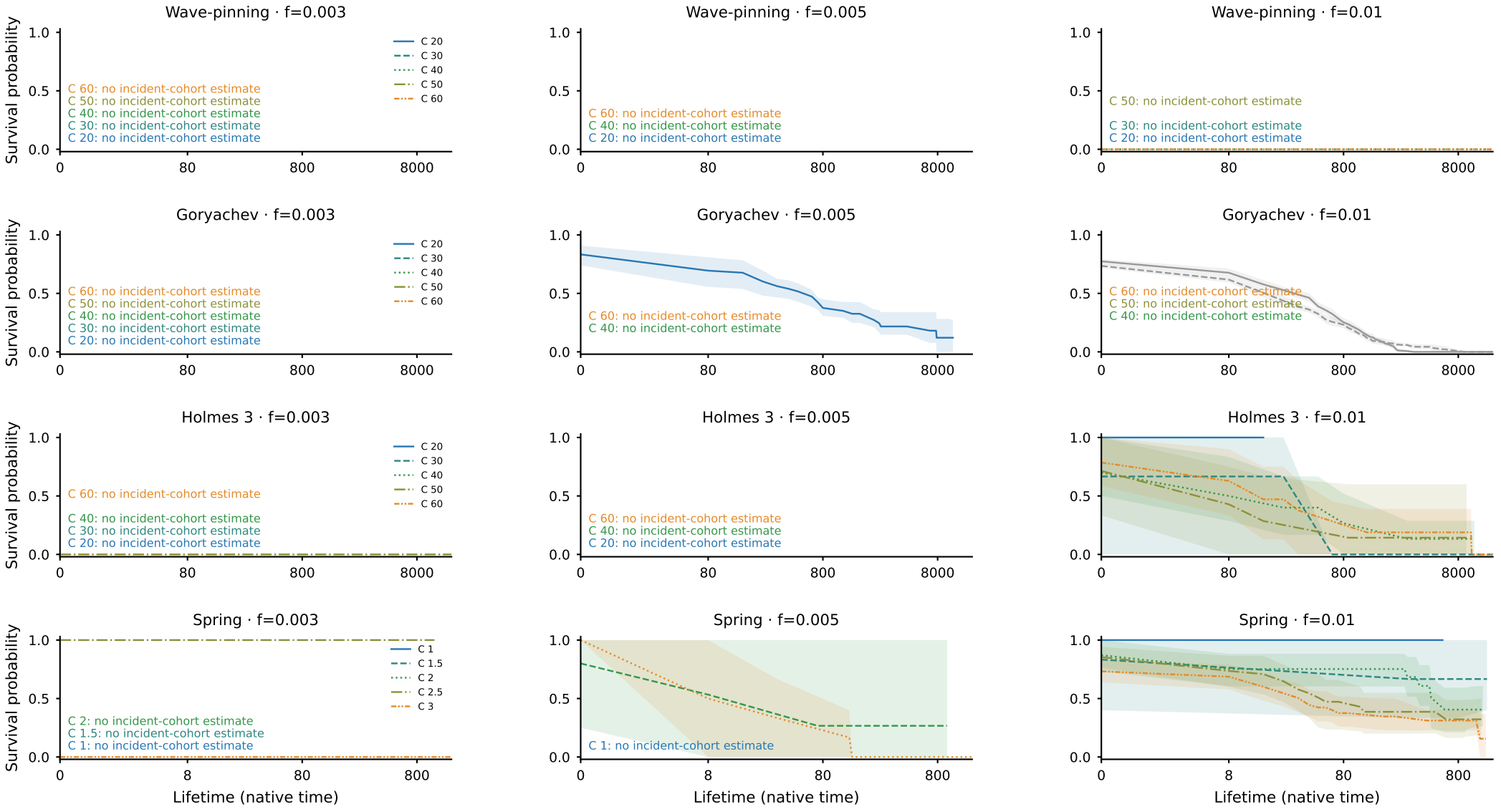
Common canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_102_S9_simultaneous_region_count_None_12.csv

S9 · Common canonical A5 observations · Incident-region lifetime survival · lower observation clock



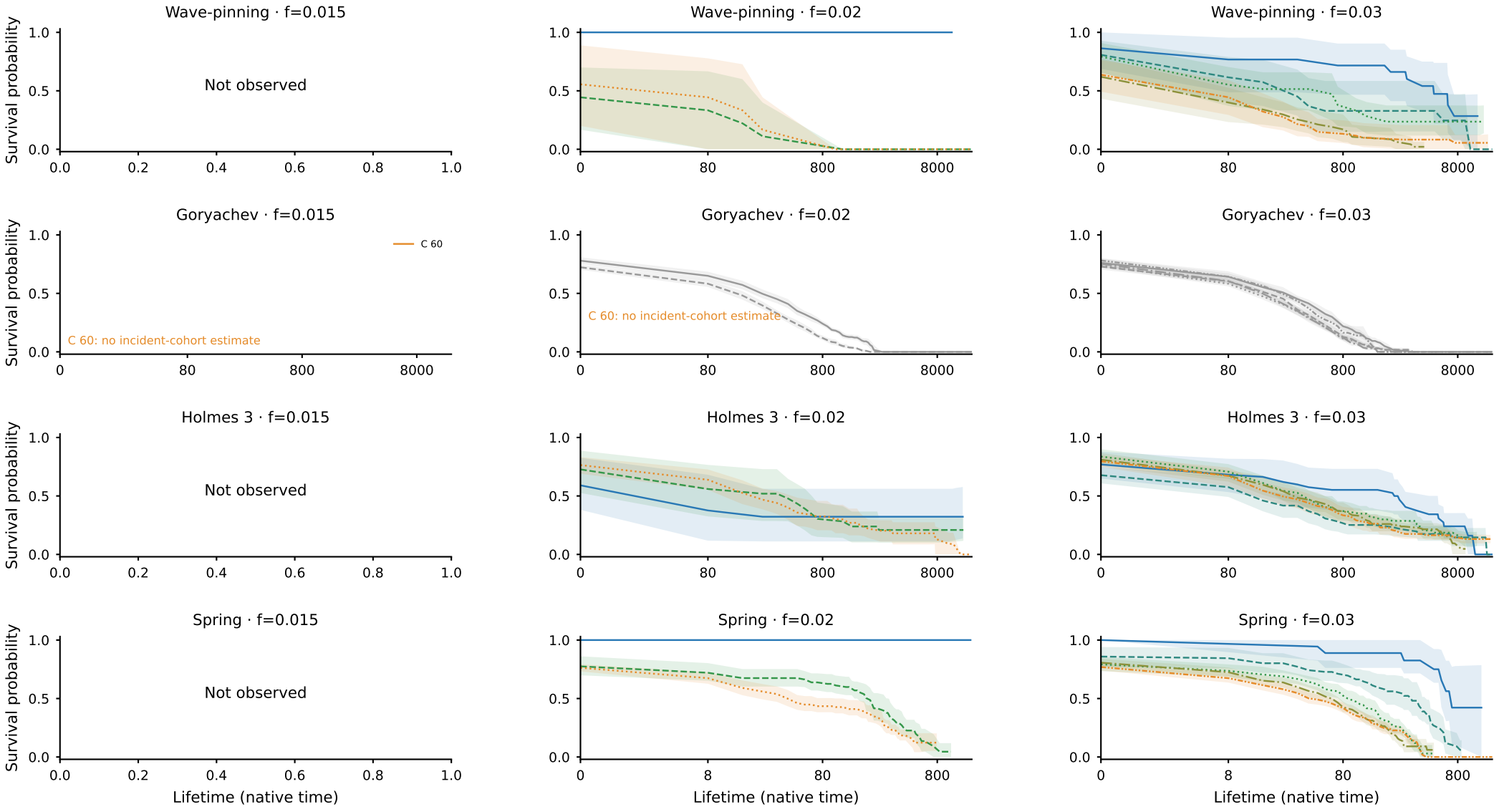
Common canonical A5 observations. Kaplan–Meier curves use incident tracks; preexisting left-censored regions are excluded. End-of-window and unresolved states retain right censoring. Lower/observed/upper clocks preserve cadence-limited times; symlog lifetime axes are linear through one observation cadence. Size colours and line patterns distinguish sizes; forced conditions are grey. Whole-seed CIs and at-risk/censoring counts remain in CSV. Unidentifiable tails are not extrapolated. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_103_S9_lifetime_lower_0.csv

S9 · Common canonical A5 observations · Incident-region lifetime survival · lower observation clock



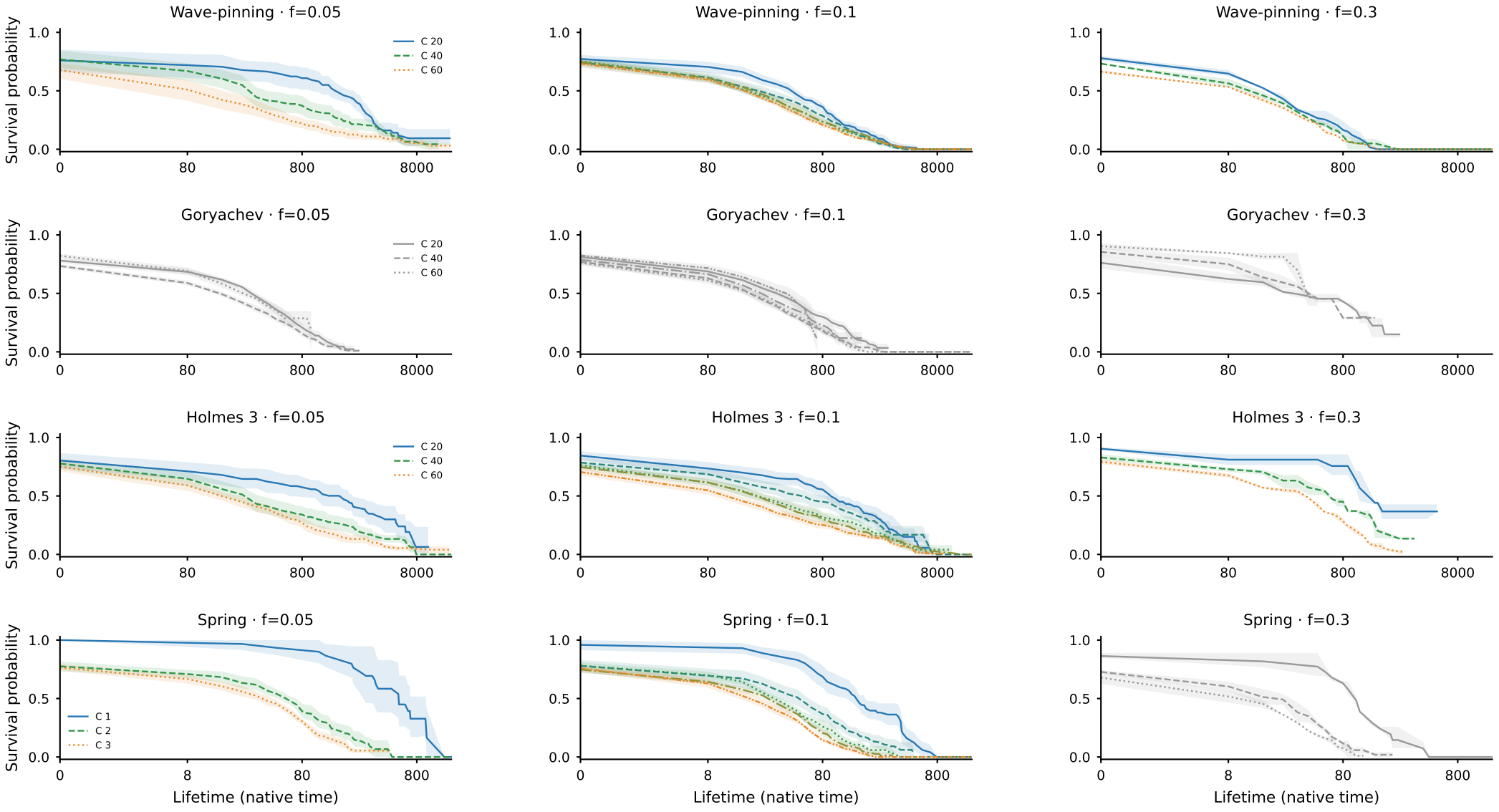
Common canonical A5 observations. Kaplan–Meier curves use incident tracks; preexisting left-censored regions are excluded. End-of-window and unresolved states retain right censoring. Lower/observed/upper clocks preserve cadence-limited times; symlog lifetime axes are linear through one observation cadence. Size colours and line patterns distinguish sizes; forced conditions are grey. Whole-seed CIs and at-risk/censoring counts remain in CSV. Unidentifiable tails are not extrapolated. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_104_S9_lifetime_lower_3.csv

S9 · Common canonical A5 observations · Incident-region lifetime survival · lower observation clock



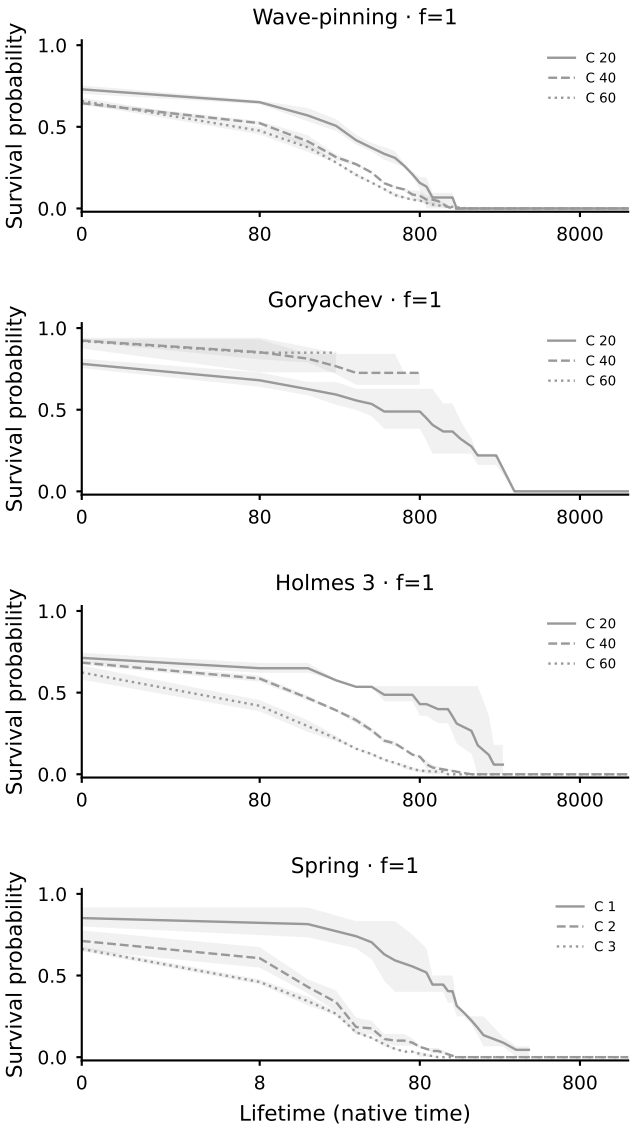
Common canonical A5 observations. Kaplan–Meier curves use incident tracks; preexisting left-censored regions are excluded. End-of-window and unresolved states retain right censoring. Lower/observed/upper clocks preserve cadence-limited times; symlog lifetime axes are linear through one observation cadence. Size colours and line patterns distinguish sizes; forced conditions are grey. Whole-seed CIs and at-risk/censoring counts remain in CSV. Unidentifiable tails are not extrapolated. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_105_S9_lifetime_lower_6.csv

S9 · Common canonical A5 observations · Incident-region lifetime survival · lower observation clock



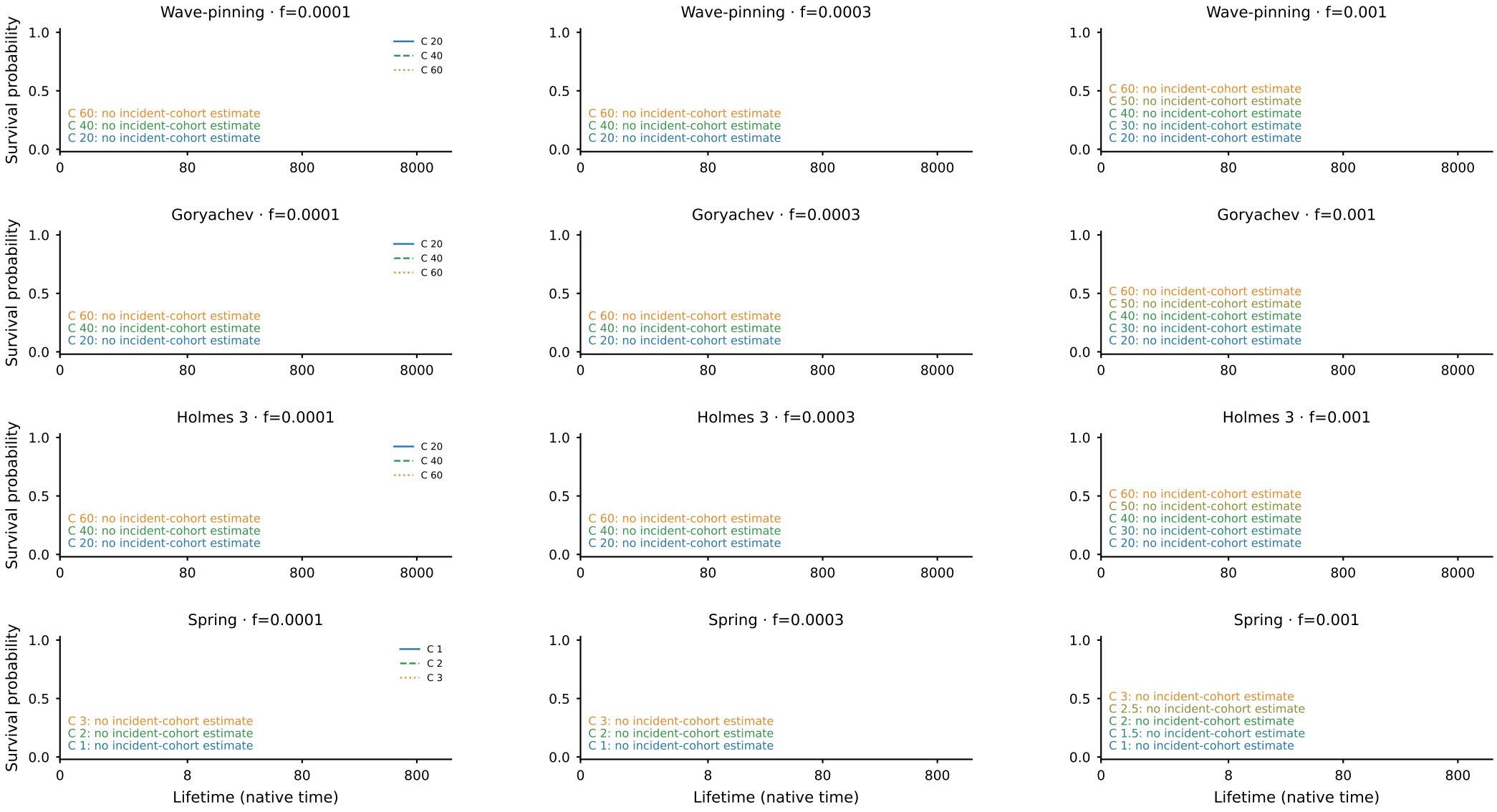
Common canonical A5 observations. Kaplan–Meier curves use incident tracks; preexisting left-censored regions are excluded. End-of-window and unresolved states retain right censoring. Lower/observed/upper clocks preserve cadence-limited times; symlog lifetime axes are linear through one observation cadence. Size colours and line patterns distinguish sizes; forced conditions are grey. Whole-seed CIs and at-risk/censoring counts remain in CSV. Unidentifiable tails are not extrapolated. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_106_S9_lifetime_lower_9.csv

S9 · Common canonical A5 observations · Incident-region lifetime survival · lower observation clock



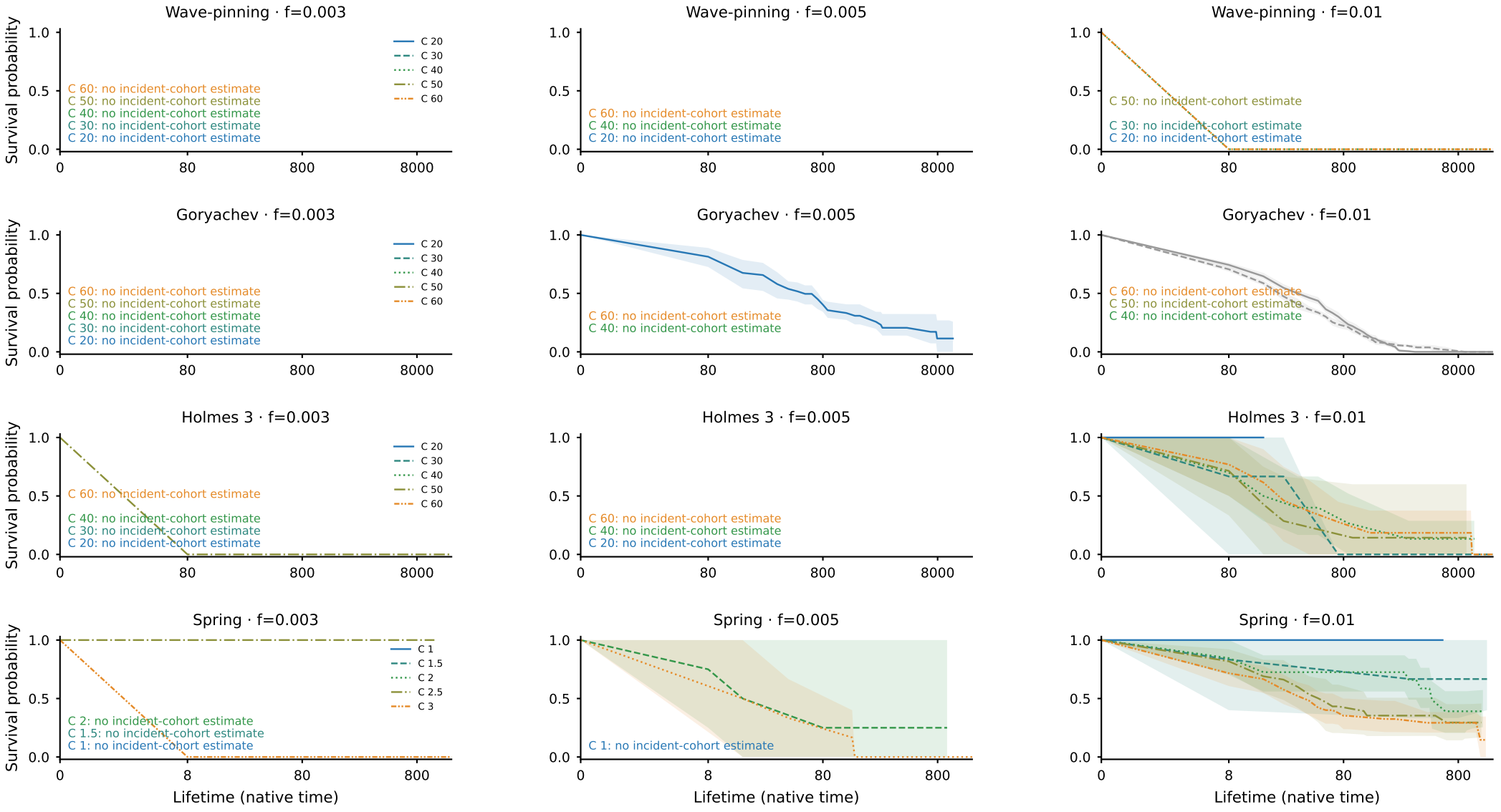
Common canonical A5 observations. Kaplan–Meier curves use incident tracks; preexisting left-censored regions are excluded. End-of-window and unresolved states retain right censoring. Lower/observed/upper clocks preserve cadence-limited times; symlog lifetime axes are linear through one observation cadence. Size colours and line patterns distinguish sizes; forced conditions are grey. Whole-seed CIs and at-risk/censoring counts remain in CSV. Unidentifiable tails are not extrapolated. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_107_S9_lifetime_lower_12.csv

S9 · Common canonical A5 observations · Incident-region lifetime survival · observed observation clock



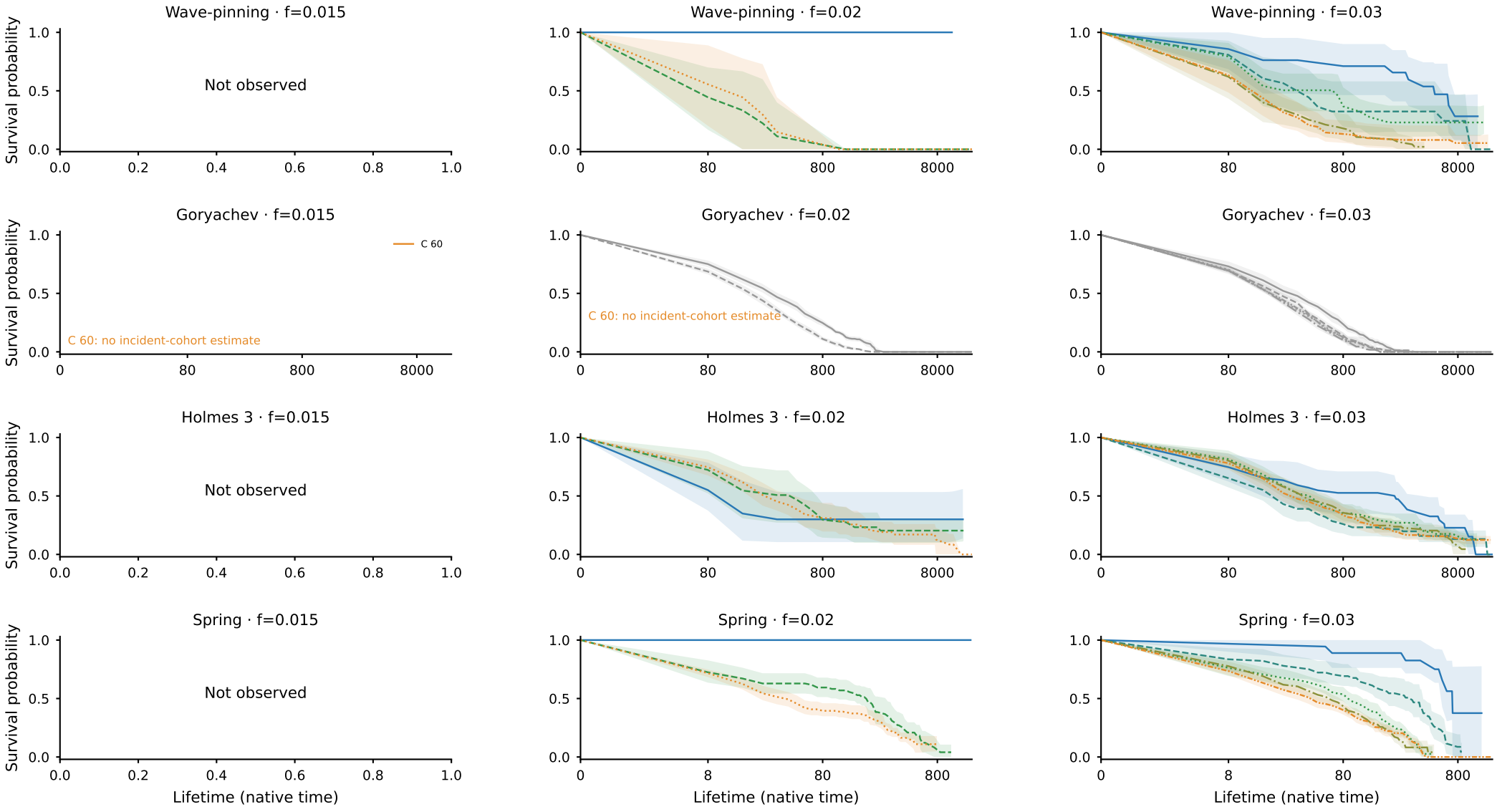
Common canonical A5 observations. Kaplan–Meier curves use incident tracks; preexisting left-censored regions are excluded. End-of-window and unresolved states retain right censoring. Lower/observed/upper clocks preserve cadence-limited times; symlog lifetime axes are linear through one observation cadence. Size colours and line patterns distinguish sizes; forced conditions are grey. Whole-seed CIs and at-risk/censoring counts remain in CSV. Unidentifiable tails are not extrapolated. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_108_S9_lifetime_observed_0.csv

S9 · Common canonical A5 observations · Incident-region lifetime survival · observed observation clock



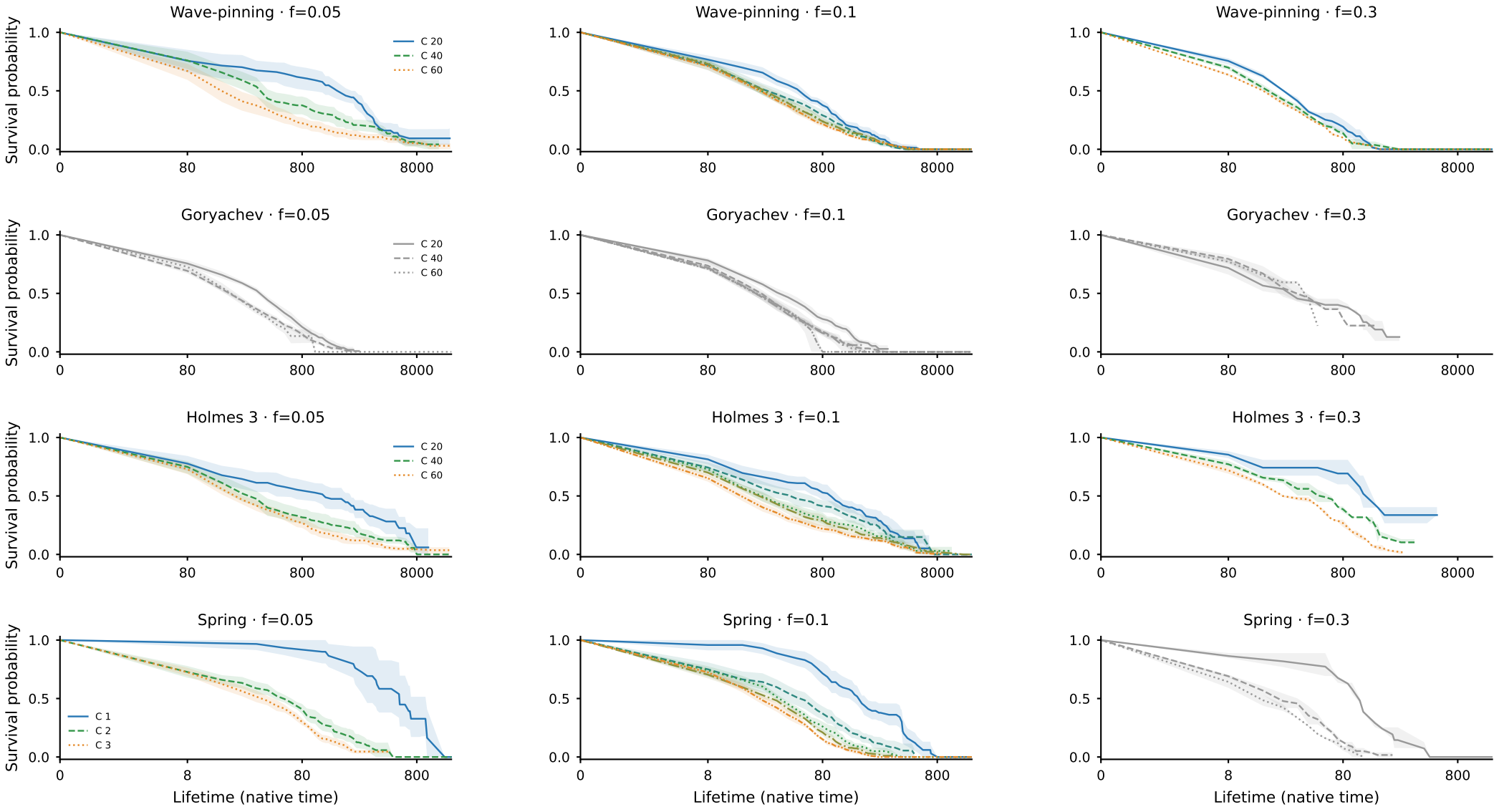
Common canonical A5 observations. Kaplan–Meier curves use incident tracks; preexisting left-censored regions are excluded. End-of-window and unresolved states retain right censoring. Lower/observed/upper clocks preserve cadence-limited times; symlog lifetime axes are linear through one observation cadence. Size colours and line patterns distinguish sizes; forced conditions are grey. Whole-seed CIs and at-risk/censoring counts remain in CSV. Unidentifiable tails are not extrapolated. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_109_S9_lifetime_observed_3.csv

S9 · Common canonical A5 observations · Incident-region lifetime survival · observed observation clock



Common canonical A5 observations. Kaplan–Meier curves use incident tracks; preexisting left-censored regions are excluded. End-of-window and unresolved states retain right censoring. Lower/observed/upper clocks preserve cadence-limited times; symlog lifetime axes are linear through one observation cadence. Size colours and line patterns distinguish sizes; forced conditions are grey. Whole-seed CIs and at-risk/censoring counts remain in CSV. Unidentifiable tails are not extrapolated. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_110_S9_lifetime_observed_6.csv

S9 · Common canonical A5 observations · Incident-region lifetime survival · observed observation clock



Common canonical A5 observations. Kaplan–Meier curves use incident tracks; preexisting left-censored regions are excluded. End-of-window and unresolved states retain right censoring. Lower/observed/upper clocks preserve cadence-limited times; symlog lifetime axes are linear through one observation cadence. Size colours and line patterns distinguish sizes; forced conditions are grey. Whole-seed CIs and at-risk/censoring counts remain in CSV. Unidentifiable tails are not extrapolated. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
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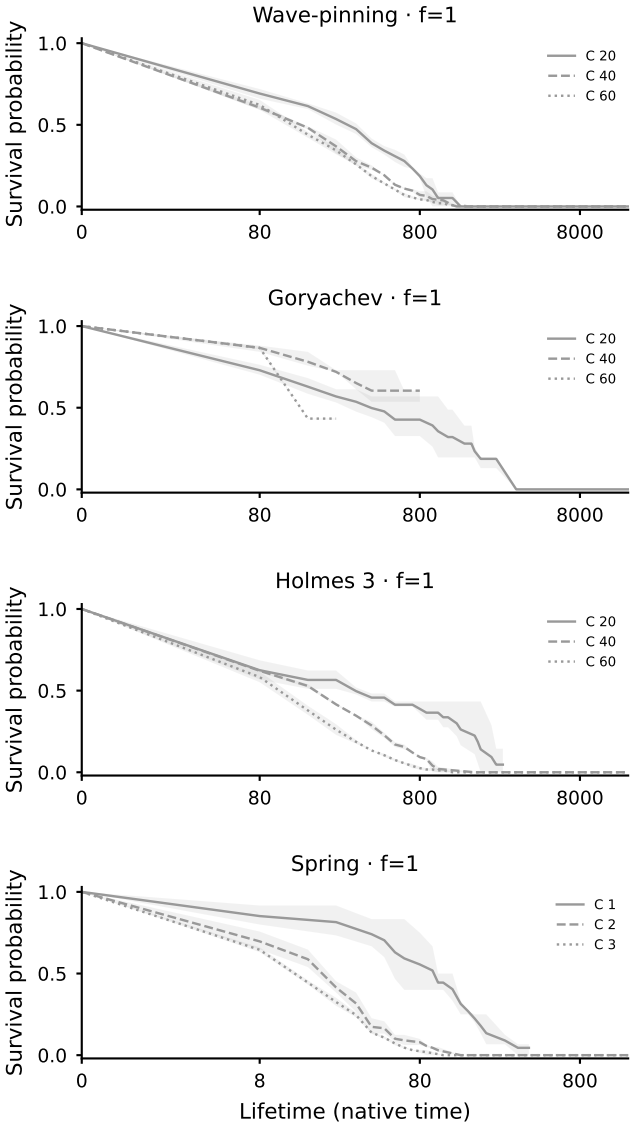
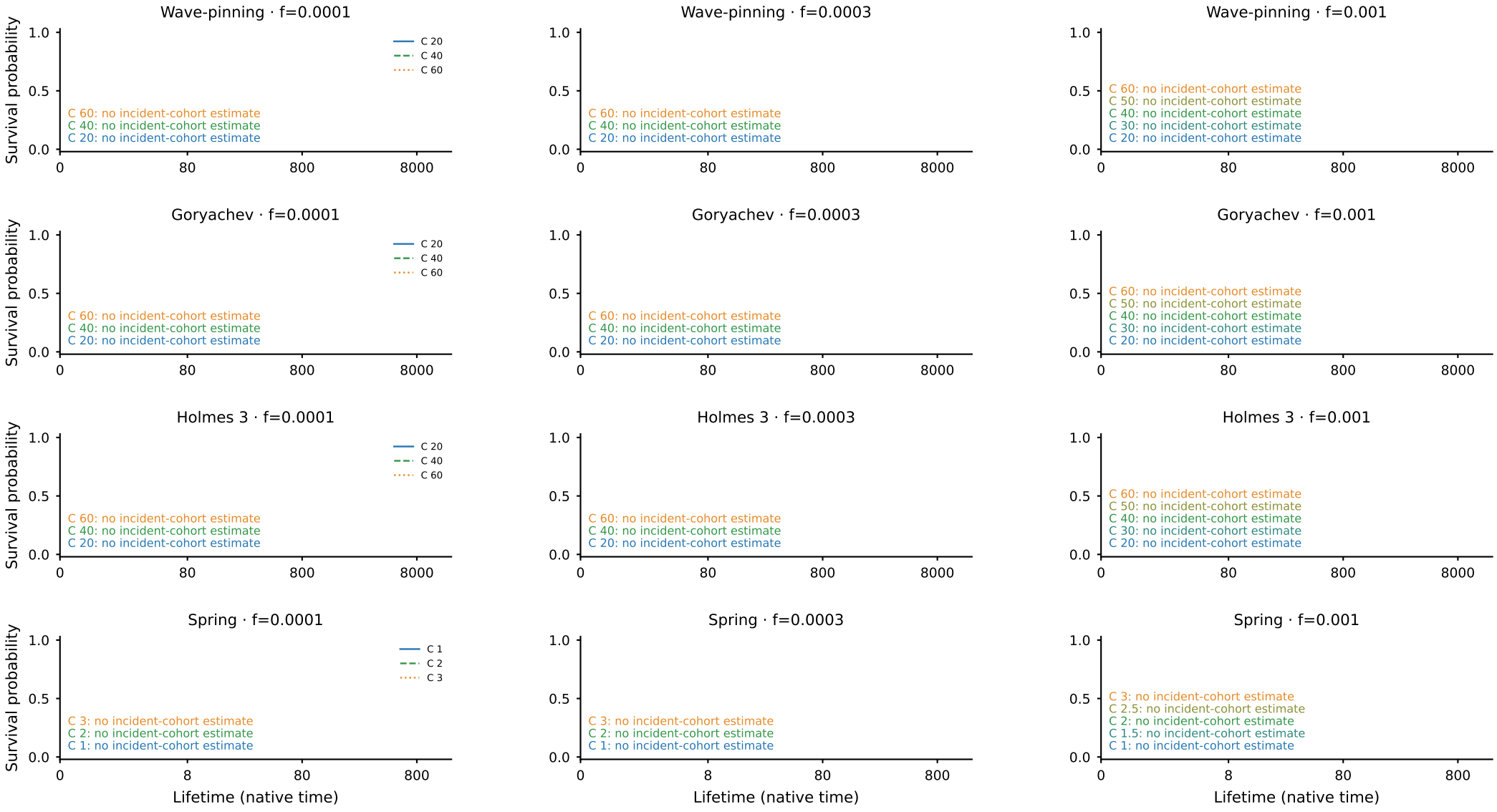
S9 · Common canonical A5 observations · Incident-region lifetime survival · observed observation clock



Common canonical A5 observations. Kaplan–Meier curves use incident tracks; preexisting left-censored regions are excluded. End-of-window and unresolved states retain right censoring. Lower/observed/upper clocks preserve cadence-limited times; symlog lifetime axes are linear through one observation cadence. Size colours and line patterns distinguish sizes; forced conditions are grey. Whole-seed CIs and at-risk/censoring counts remain in CSV. Unidentifiable tails are not extrapolated. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_112_S9_lifetime_observed_12.csv

S9 · Common canonical A5 observations · Incident-region lifetime survival · upper observation clock



Common canonical A5 observations. Kaplan–Meier curves use incident tracks; preexisting left-censored regions are excluded. End-of-window and unresolved states retain right censoring. Lower/observed/upper clocks preserve cadence-limited times; symlog lifetime axes are linear through one observation cadence. Size colours and line patterns distinguish sizes; forced conditions are grey. Whole-seed CIs and at-risk/censoring counts remain in CSV. Unidentifiable tails are not extrapolated. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
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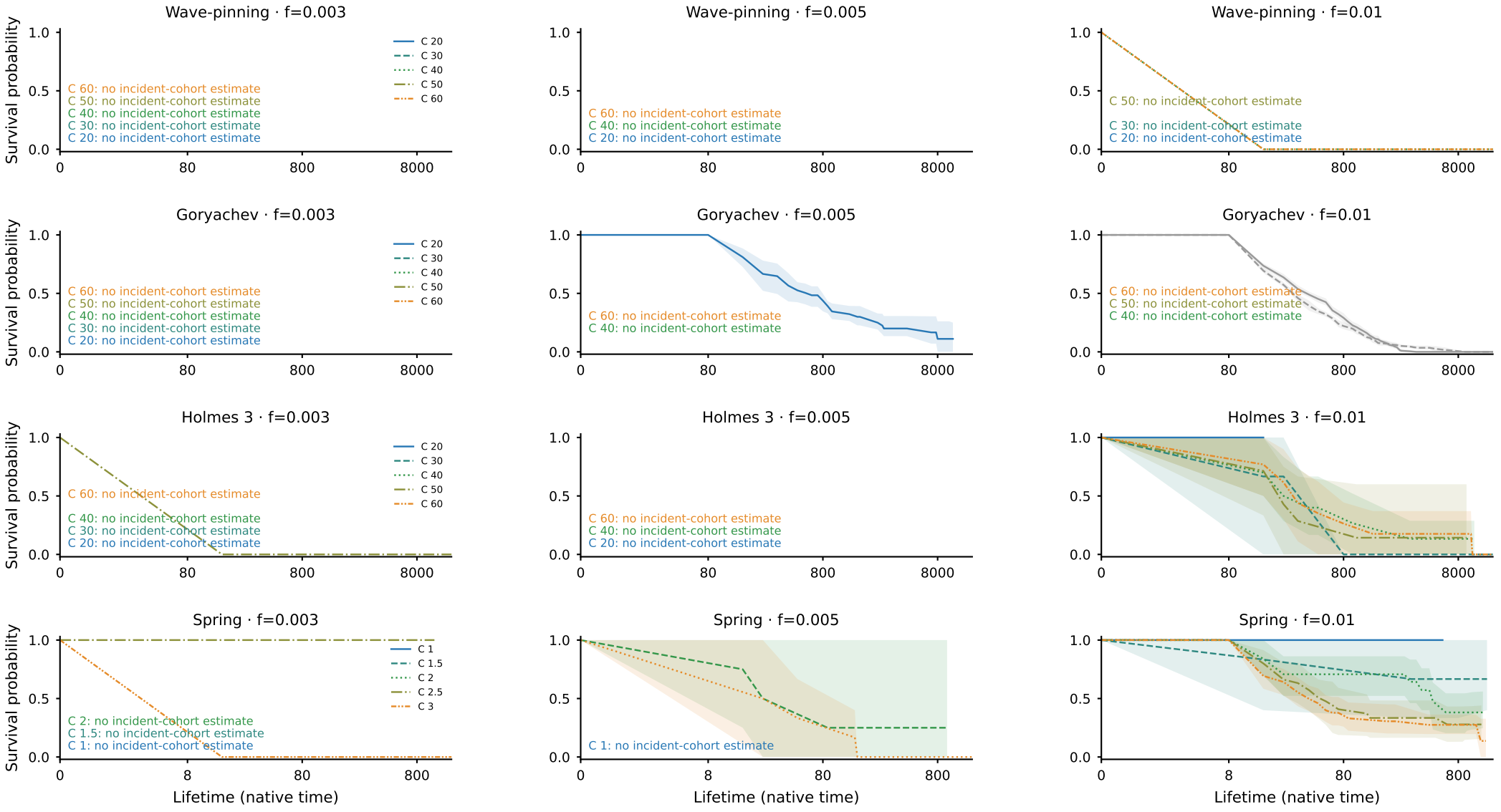
S9 · Common canonical A5 observations · Incident-region lifetime survival · upper observation clock



Common canonical A5 observations. Kaplan–Meier curves use incident tracks; preexisting left-censored regions are excluded. End-of-window and unresolved states retain right censoring. Lower/observed/upper clocks preserve cadence-limited times; symlog lifetime axes are linear through one observation cadence. Size colours and line patterns distinguish sizes; forced conditions are grey. Whole-seed CIs and at-risk/censoring counts remain in CSV. Unidentifiable tails are not extrapolated. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
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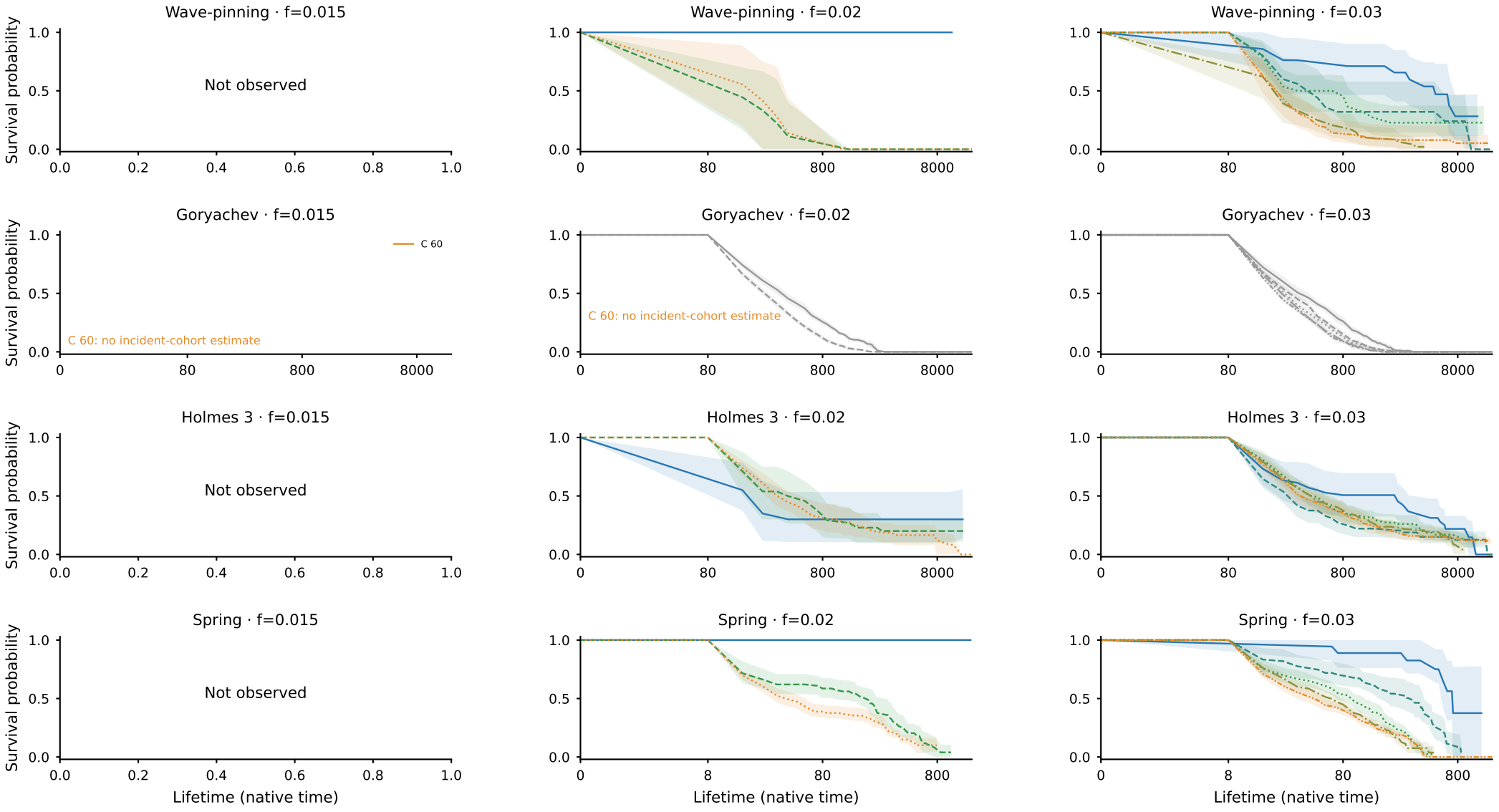
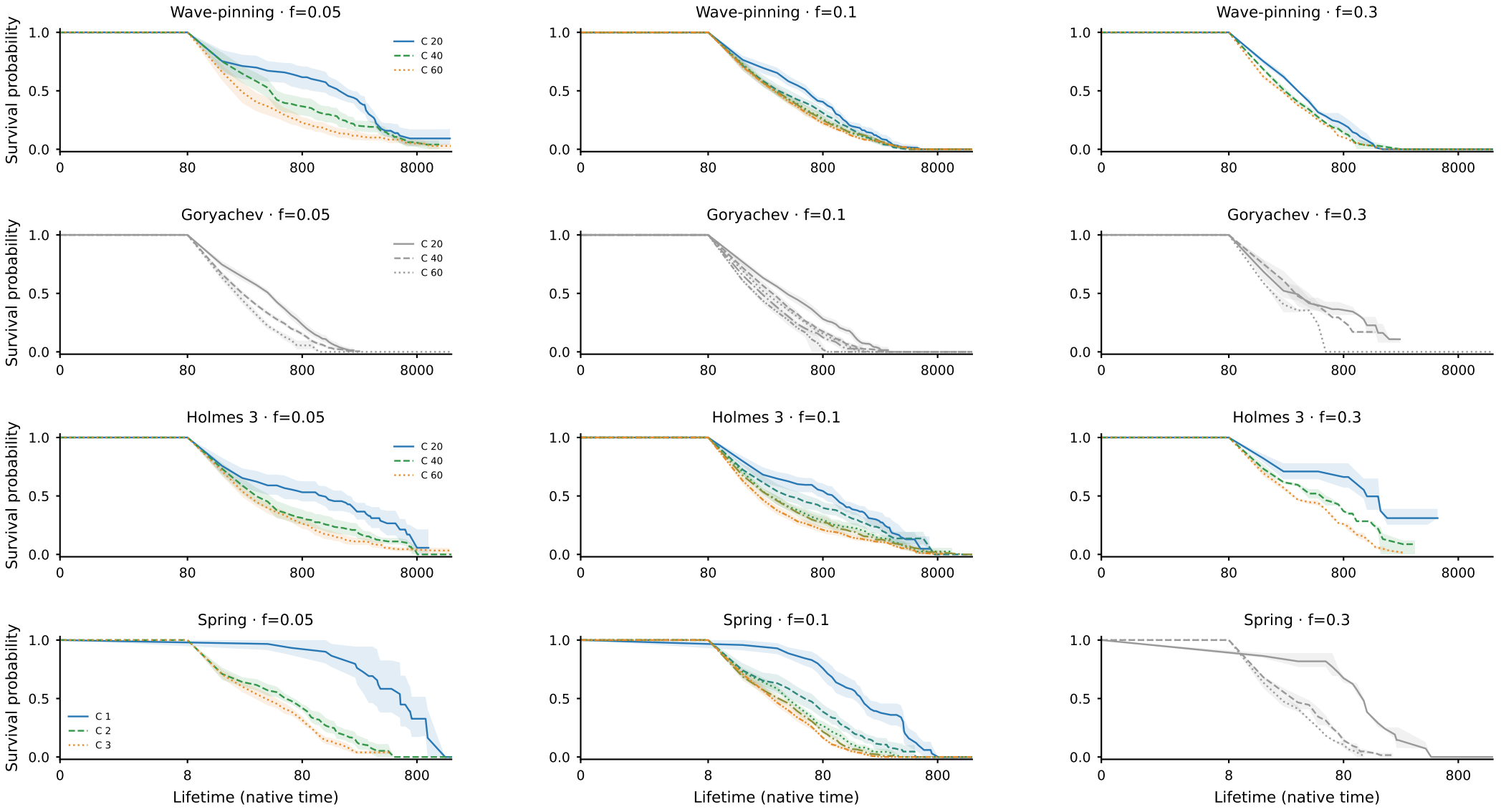
S9 · Common canonical A5 observations · Incident-region lifetime survival · upper observation clock



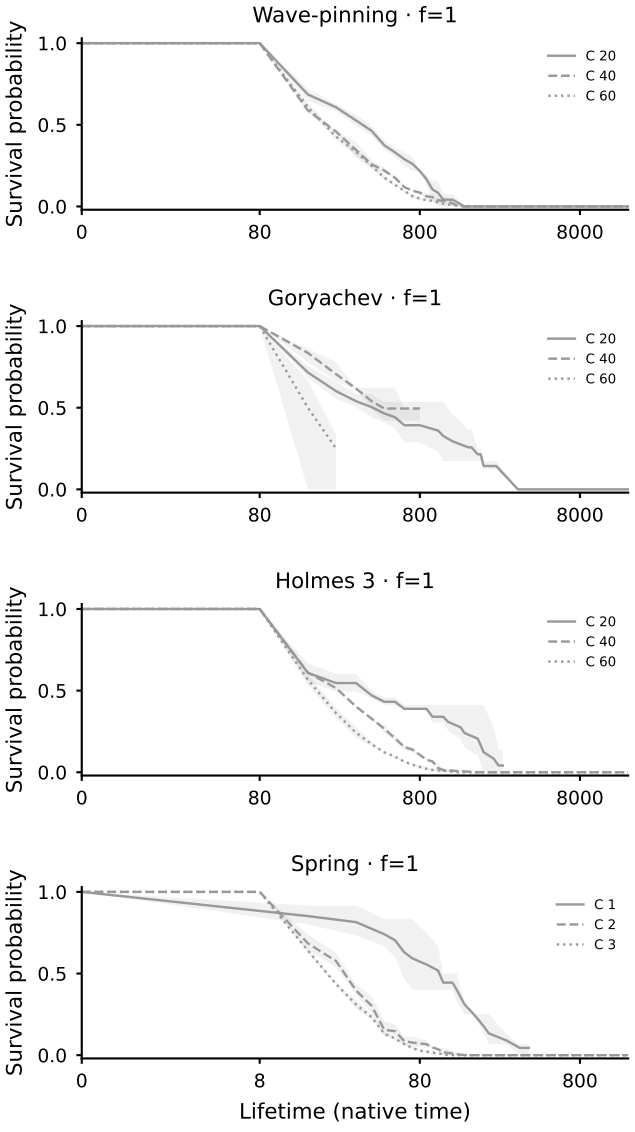
Common canonical A5 observations. Kaplan–Meier curves use incident tracks; preexisting left-censored regions are excluded. End-of-window and unresolved states retain right censoring. Lower/observed/upper clocks preserve cadence-limited times; symlog lifetime axes are linear through one observation cadence. Size colours and line patterns distinguish sizes; forced conditions are grey. Whole-seed CIs and at-risk/censoring counts remain in CSV. Unidentifiable tails are not extrapolated. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_115_S9_lifetime_upper_6.csv

S9 · Common canonical A5 observations · Incident-region lifetime survival · upper observation clock



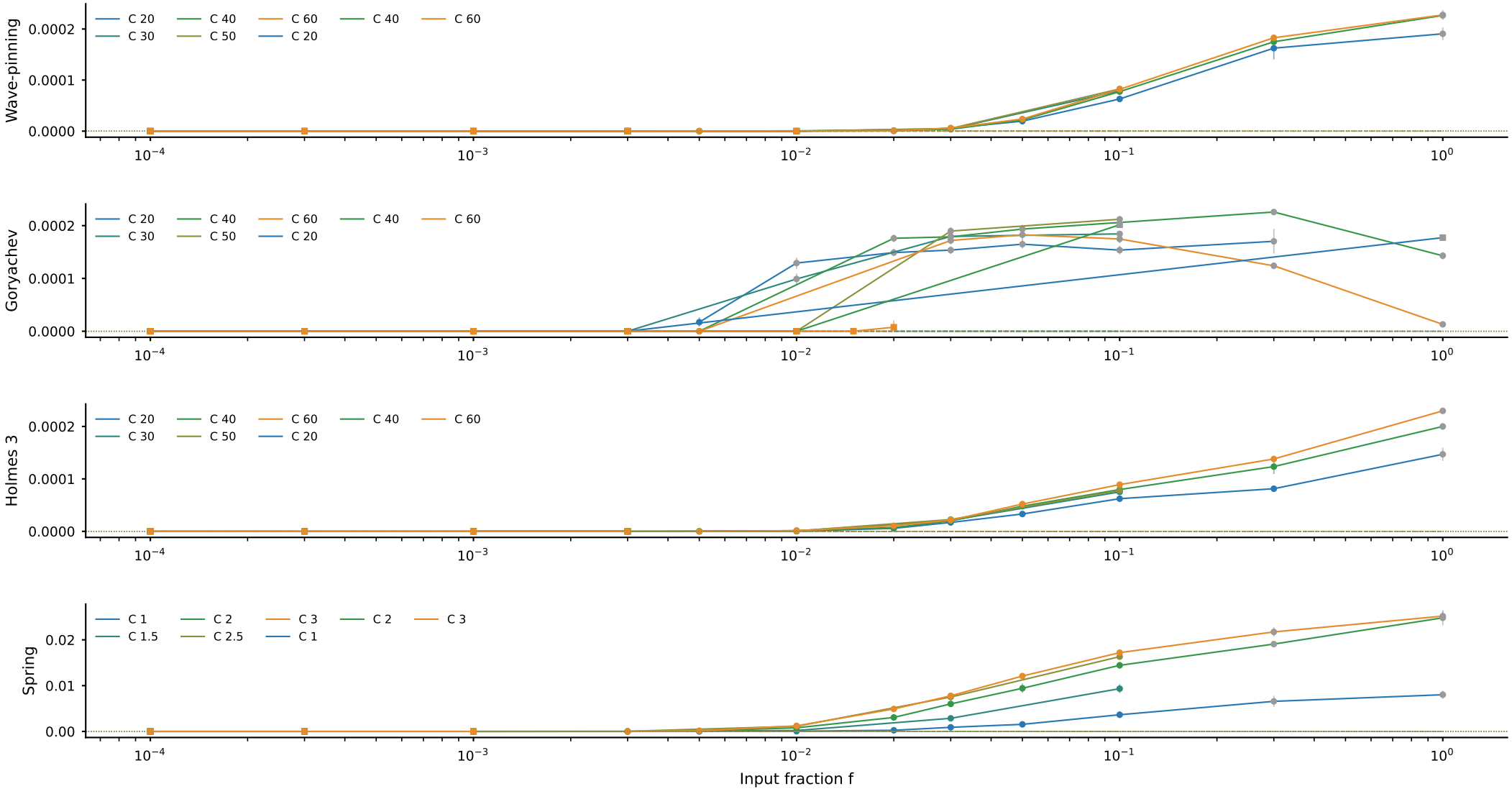
Common canonical A5 observations. Kaplan–Meier curves use incident tracks; preexisting left-censored regions are excluded. End-of-window and unresolved states retain right censoring. Lower/observed/upper clocks preserve cadence-limited times; symlog lifetime axes are linear through one observation cadence. Size colours and line patterns distinguish sizes; forced conditions are grey. Whole-seed CIs and at-risk/censoring counts remain in CSV. Unidentifiable tails are not extrapolated. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_116_S9_lifetime_upper_9.csv

S9 · Common canonical A5 observations · Incident-region lifetime survival · upper observation clock



Common canonical A5 observations. Kaplan–Meier curves use incident tracks; preexisting left-censored regions are excluded. End-of-window and unresolved states retain right censoring. Lower/observed/upper clocks preserve cadence-limited times; symlog lifetime axes are linear through one observation cadence. Size colours and line patterns distinguish sizes; forced conditions are grey. Whole-seed CIs and at-risk/censoring counts remain in CSV. Unidentifiable tails are not extrapolated. Native H: Wave-pinning 32000; Goryachev 32000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_common_canonical_20261020_117_S9_lifetime_upper_12.csv

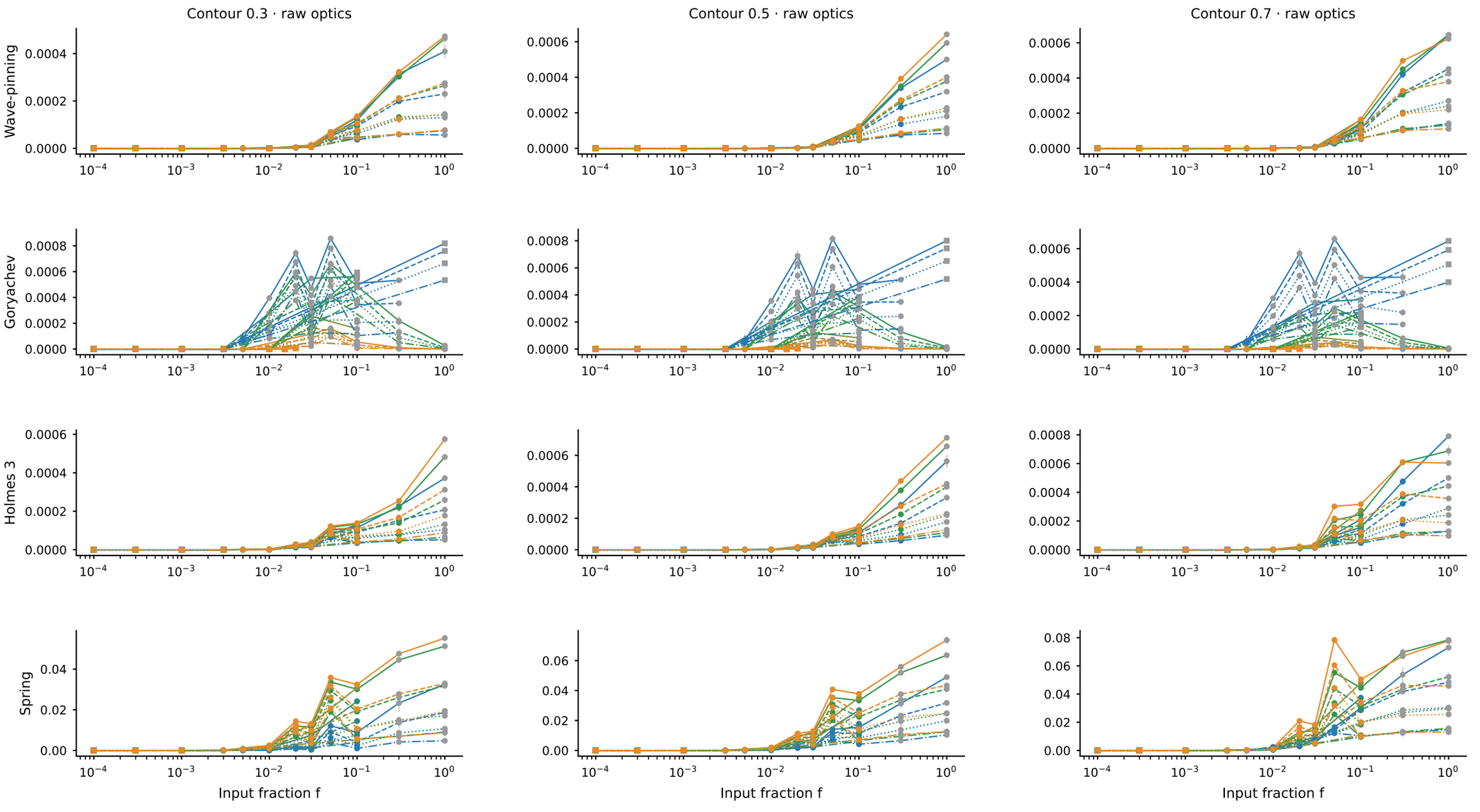
S9 · Longest canonical A5 observations · Region births per native time per physical perimeter



Longest canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.

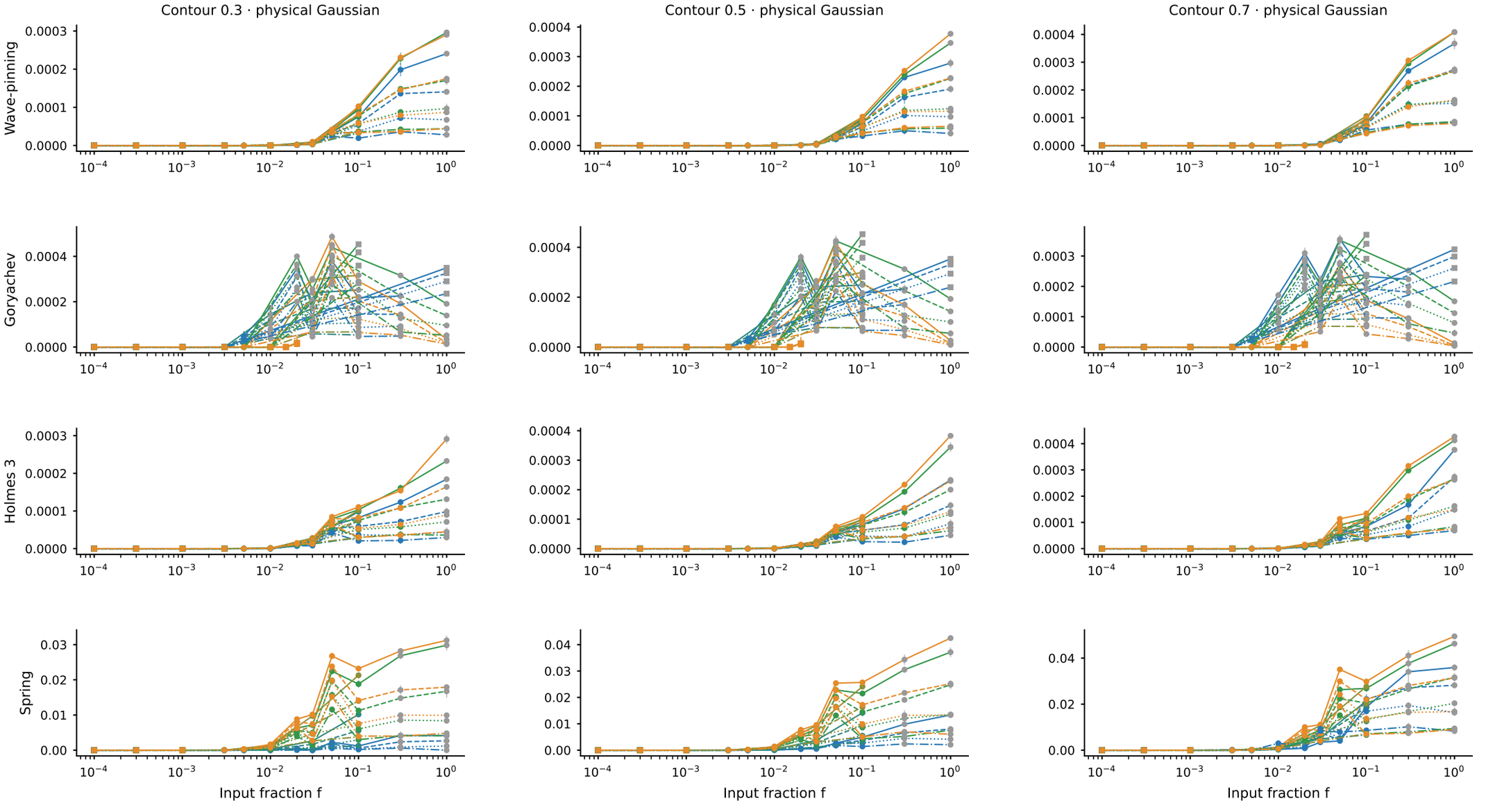
CSV: figure4_S9_longest_canonical_20261020_001_S9_births_per_time_per_perimeter.csv

S9 · Longest canonical A5 observations · Region births per native time per physical perimeter · full native-factor sensitivity



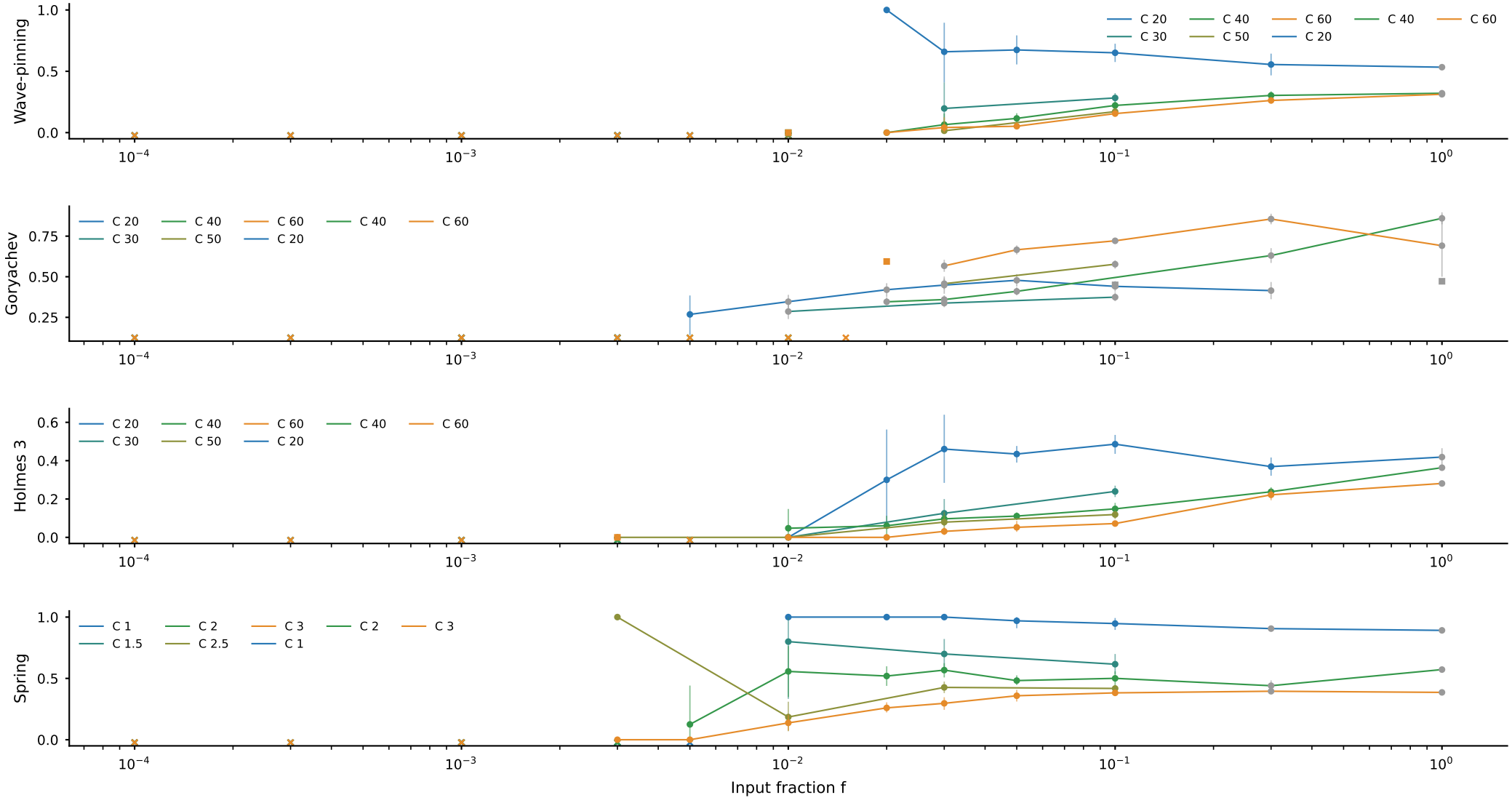
Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_002_S9_births_per_time_per_perimeter_raw_factorial.csv

S9 · Longest canonical A5 observations · Region births per native time per physical perimeter · full native-factor sensitivity



Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_003_S9_births_per_time_per_perimeter_smoothed_factorial.csv

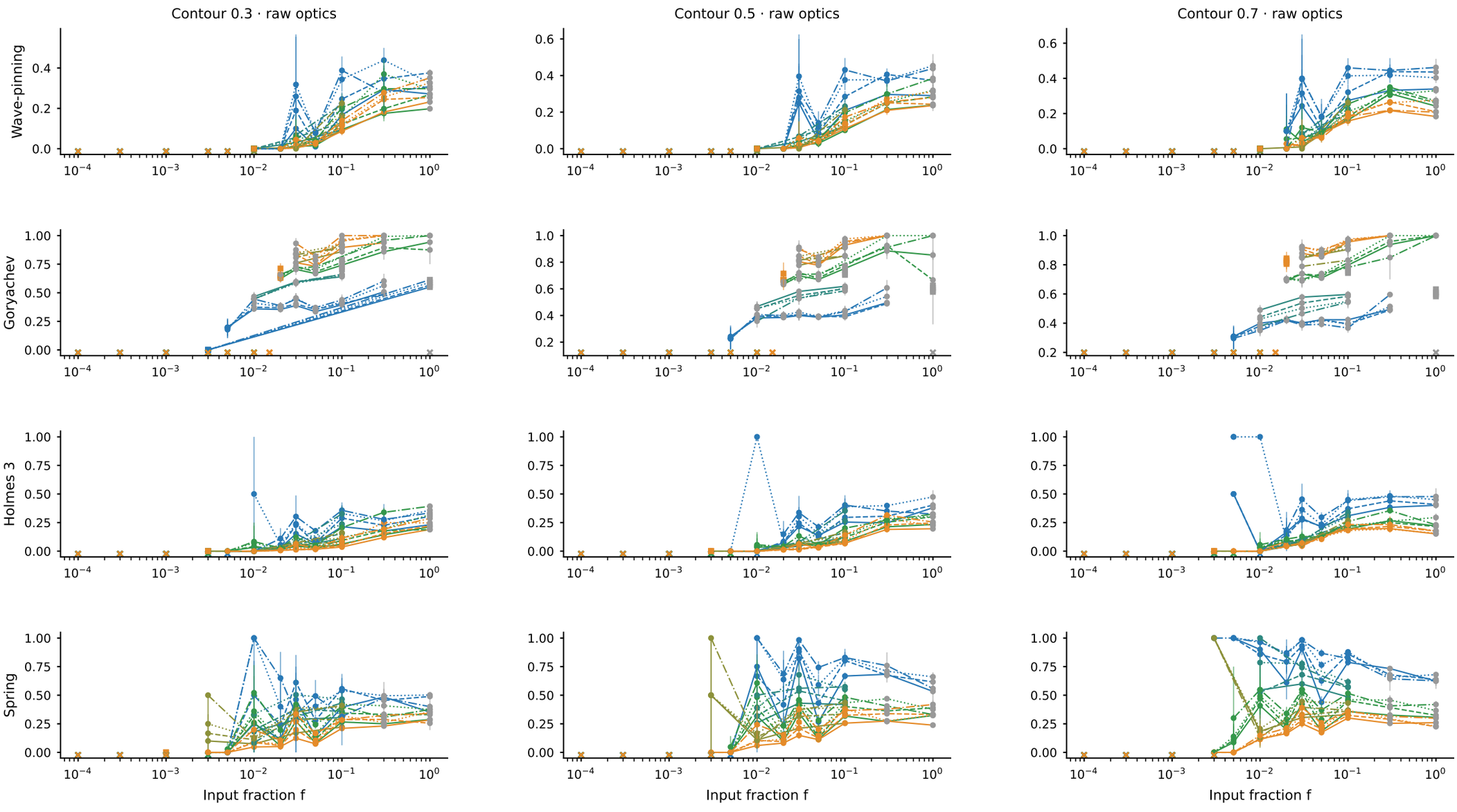
S9 · Longest canonical A5 observations · Fraction of trajectories with an unscorable takeover track



Longest canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.

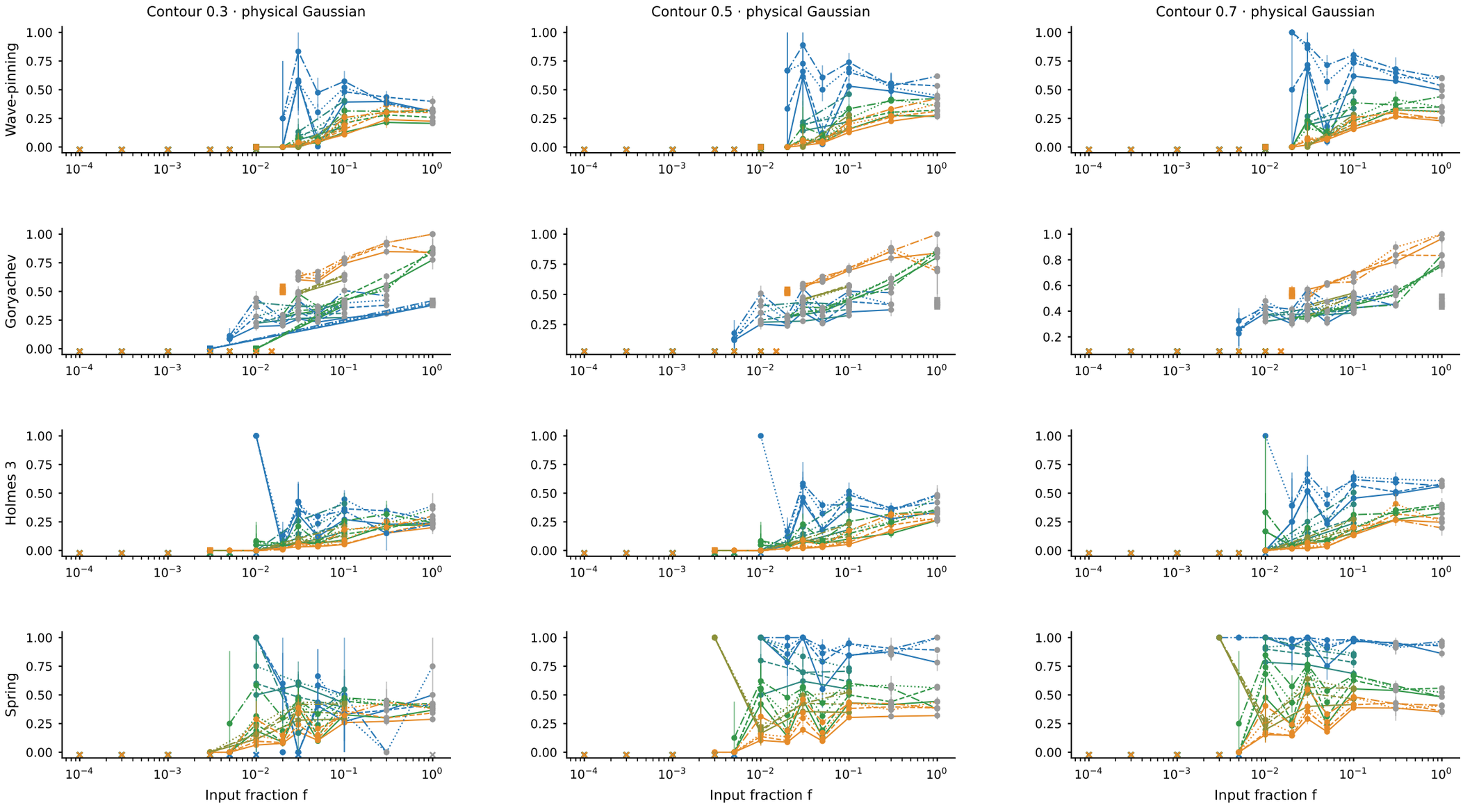
CSV: figure4_S9_longest_canonical_20261020_004_S9_ever_unscorable_takeover_fraction.csv

S9 · Longest canonical A5 observations · Fraction of trajectories with an unscorable takeover track · full native-factor sensitivity



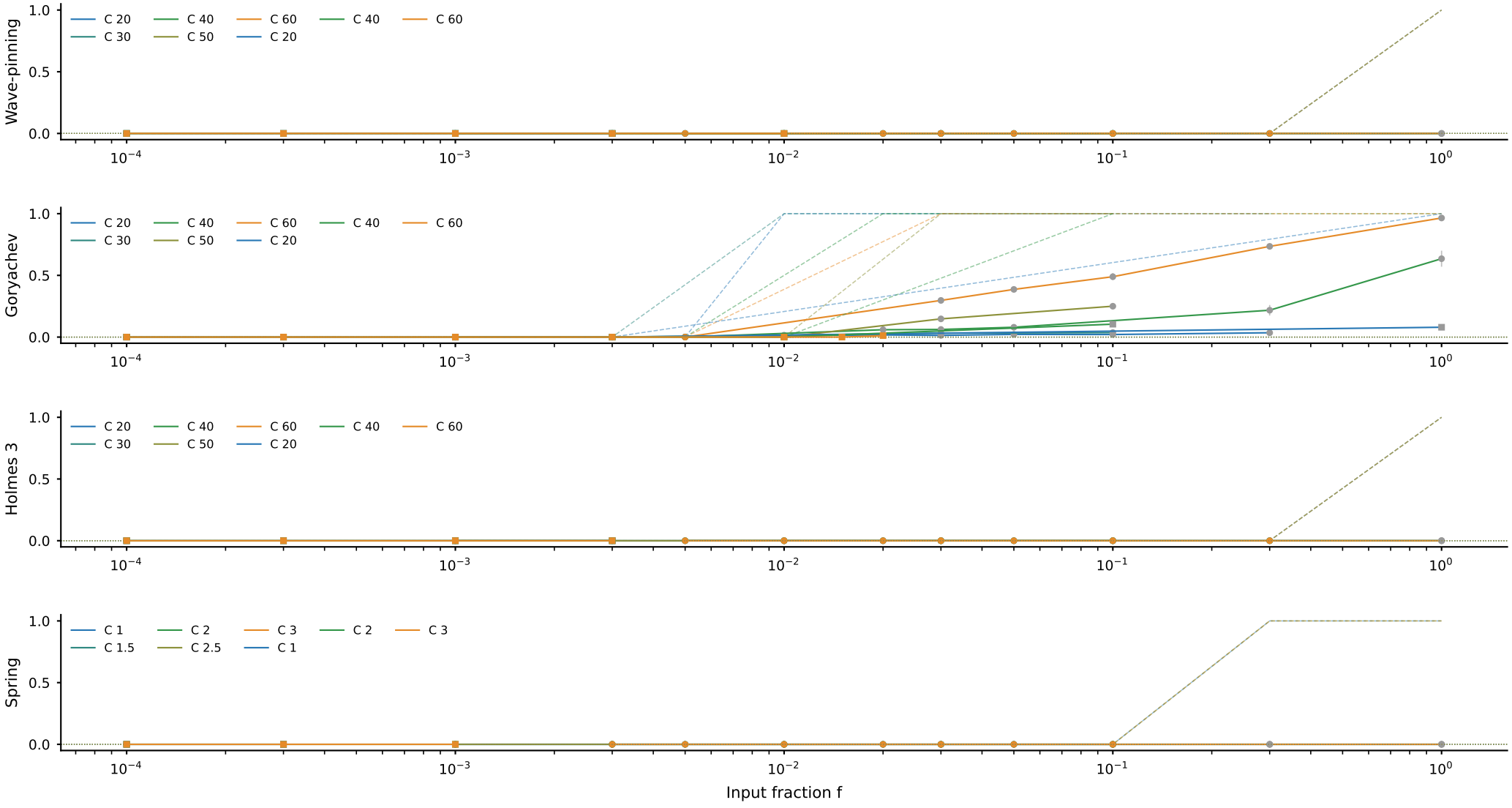
Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds.
× at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_005_S9_ever_unscorable_takeover_fraction_raw_factorial.csv

S9 · Longest canonical A5 observations · Fraction of trajectories with an unscorable takeover track · full native-factor sensitivity



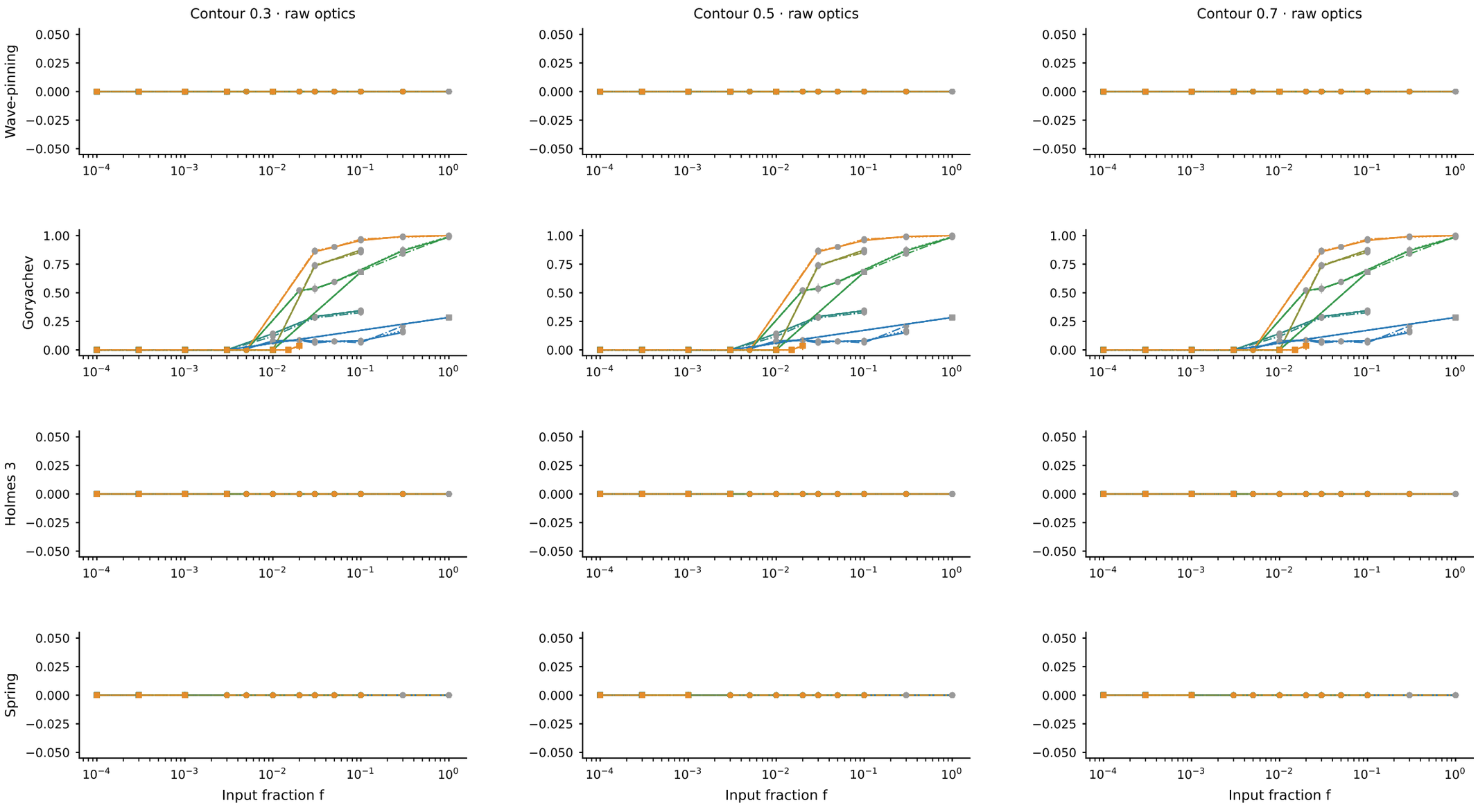
Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds.
× at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_006_S9_ever_unscorable_takeover_fraction_smoothed_factorial.csv

S9 · Longest canonical A5 observations · Time fraction below the contrast gate



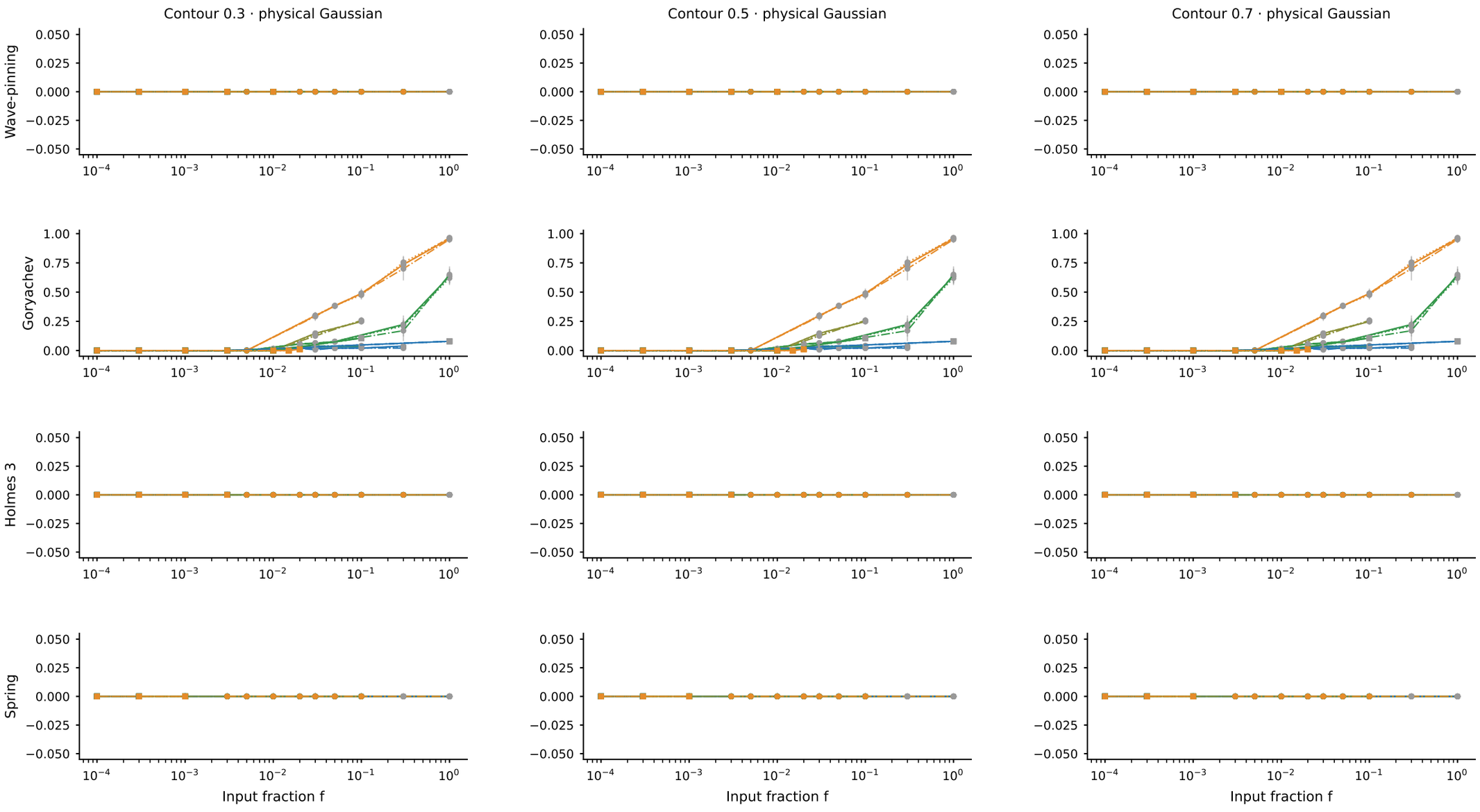
Longest canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_007_S9_low_contrast_time_fraction.csv

S9 · Longest canonical A5 observations · Time fraction below the contrast gate · full native-factor sensitivity



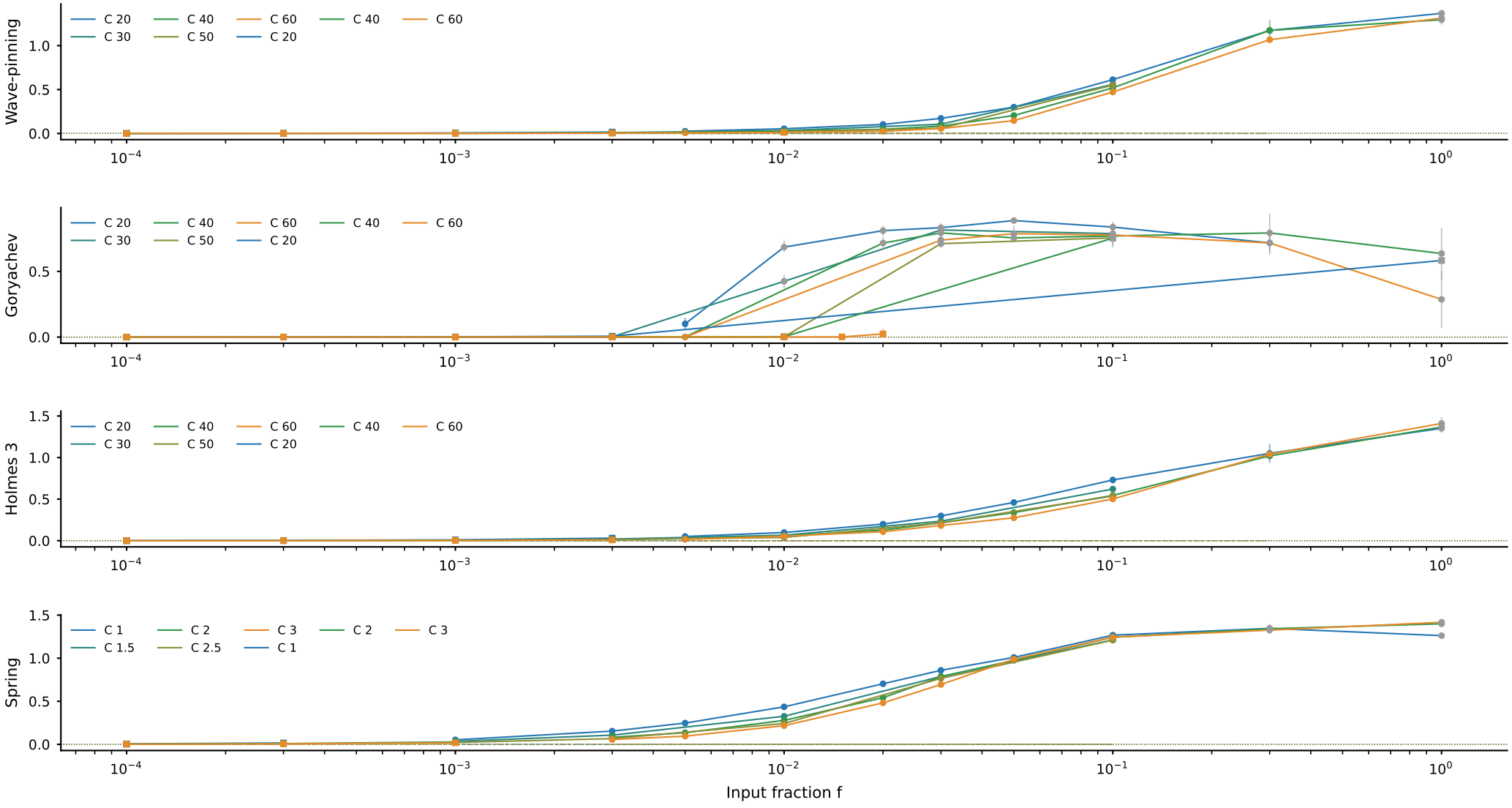
Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds.
× at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_008_S9_low_contrast_time_fraction_raw_factorial.csv

S9 · Longest canonical A5 observations · Time fraction below the contrast gate · full native-factor sensitivity



Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds.
× at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_009_S9_low_contrast_time_fraction_smoothed_factorial.csv

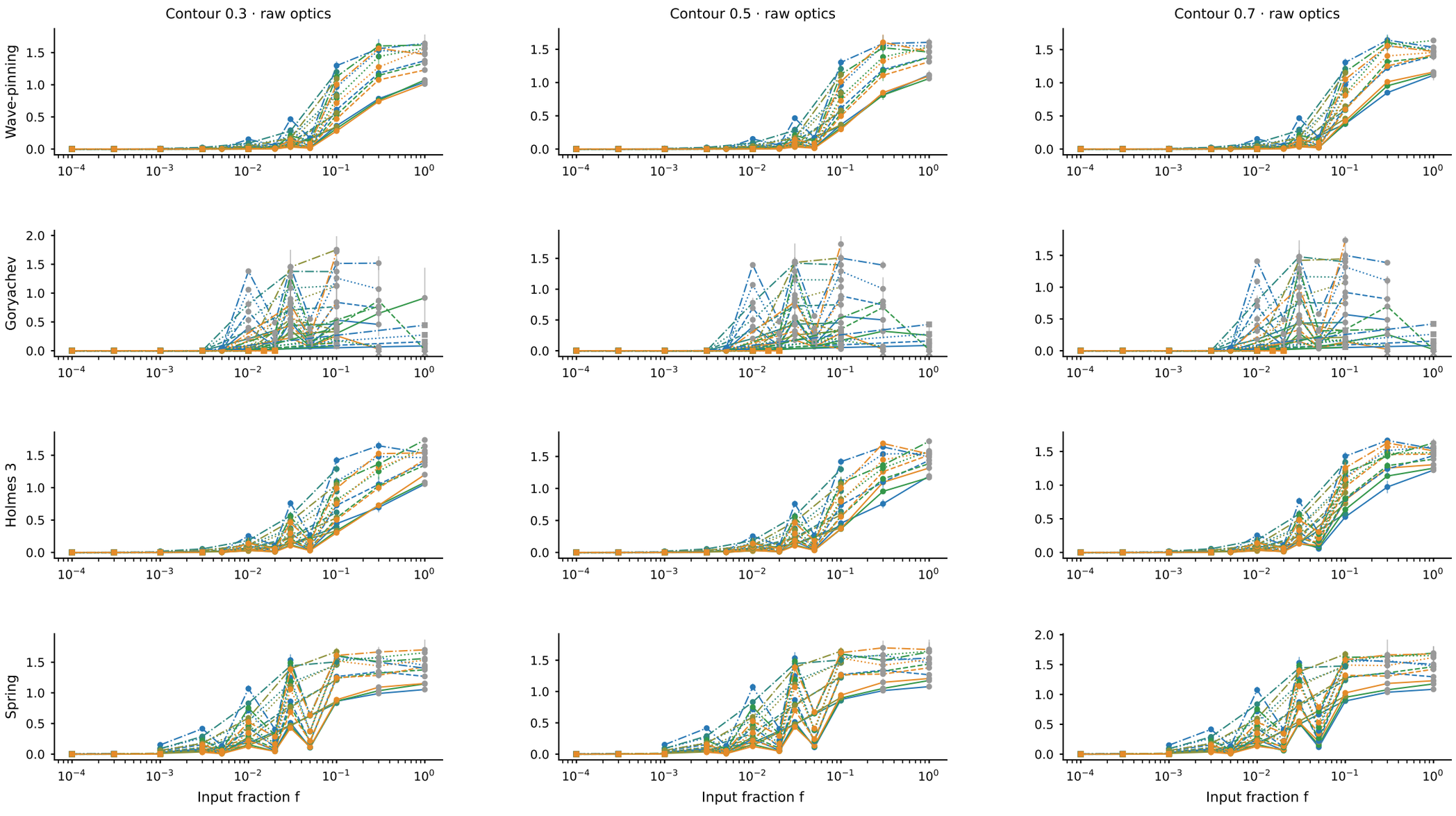
S9 · Longest canonical A5 observations · Mean absolute dominant-heading turn over 1 observation frames



Longest canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.

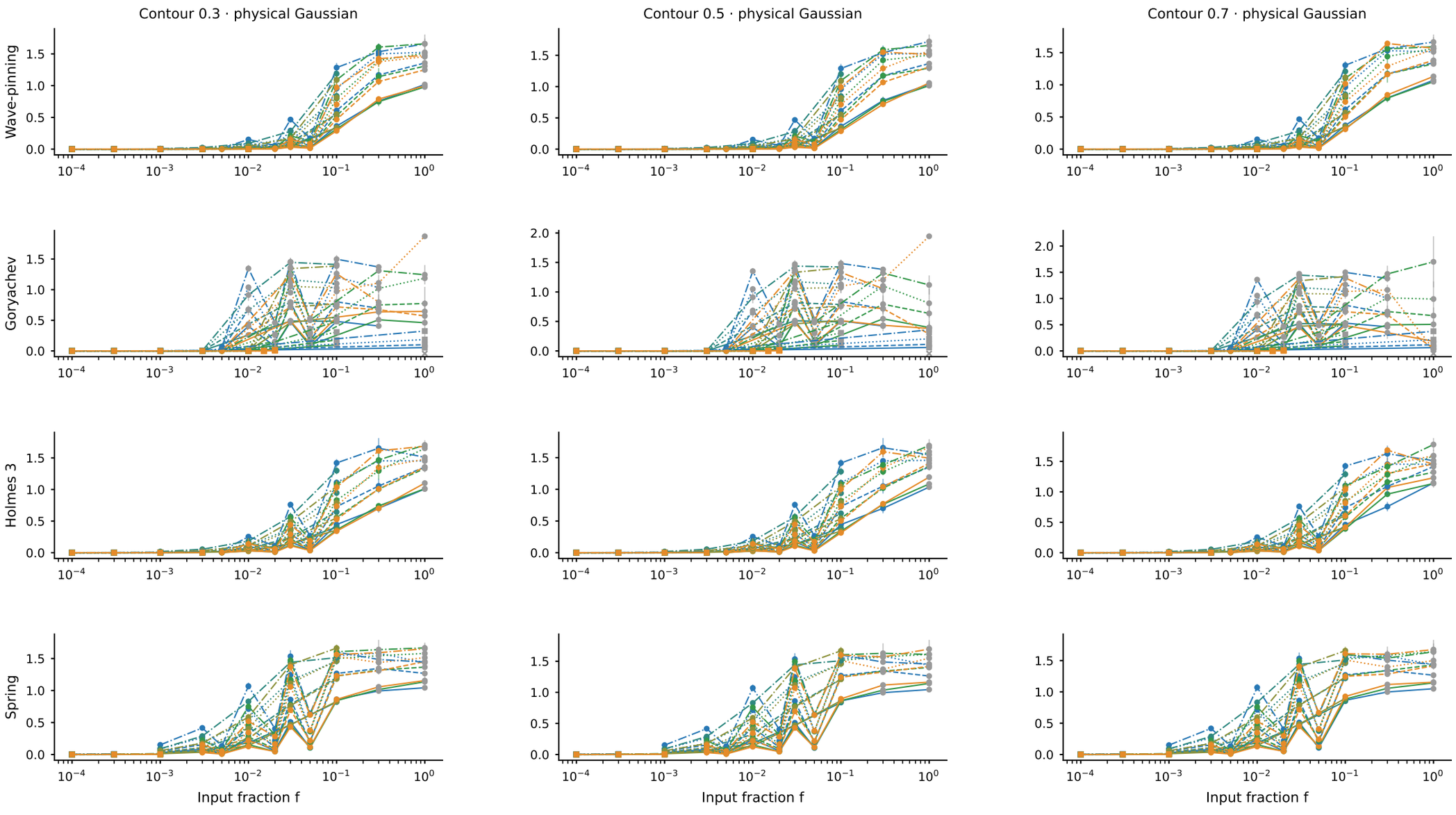
CSV: figure4_S9_longest_canonical_20261020_010_S9_mean_absolute_turn_1_frames.csv

S9 · Longest canonical A5 observations · Mean absolute dominant-heading turn over 1 observation frames · full native-factor sensitivity



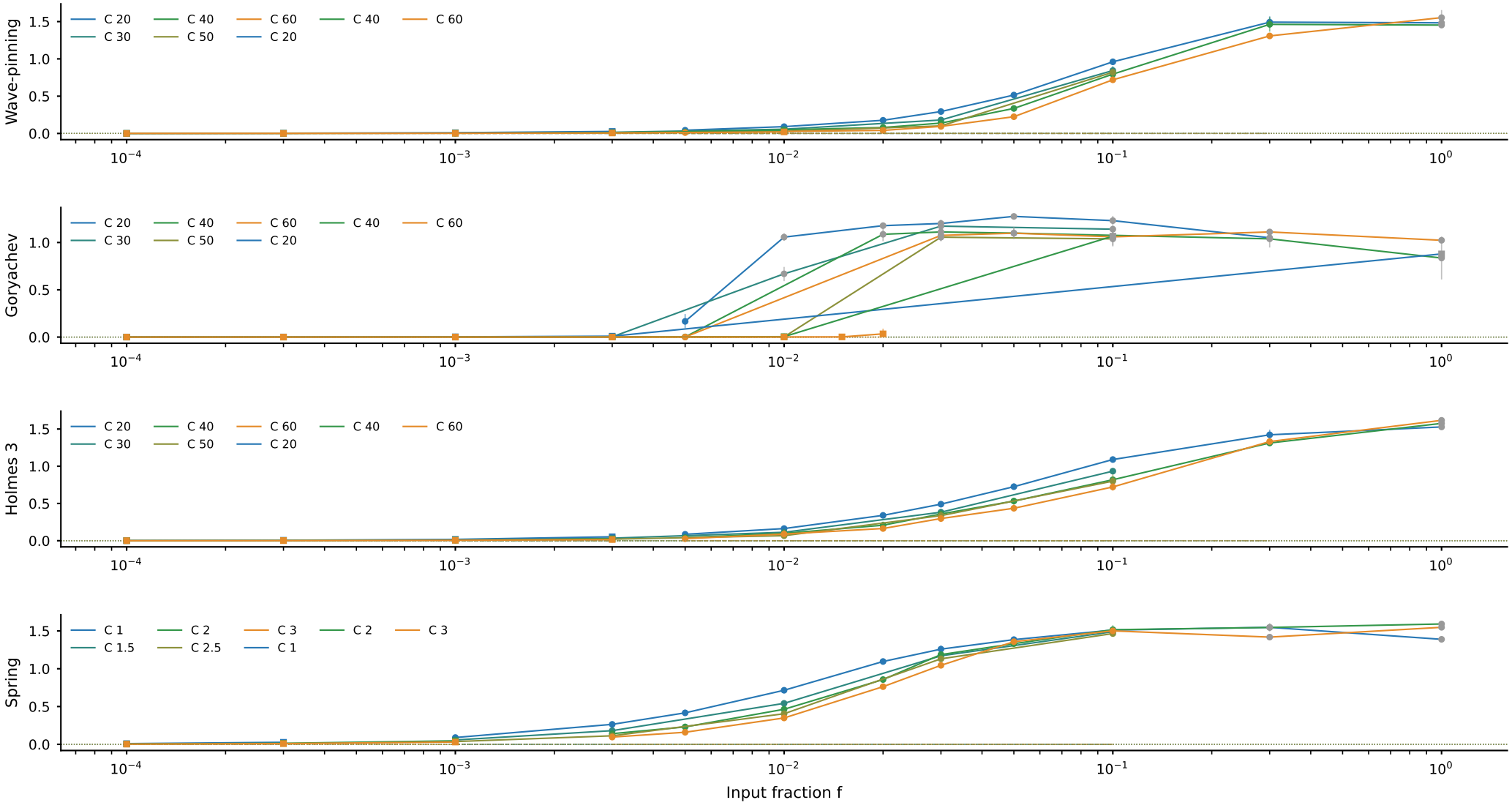
Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_011_S9_mean_absolute_turn_1_frames_raw_factorial.csv

S9 · Longest canonical A5 observations · Mean absolute dominant-heading turn over 1 observation frames · full native-factor sensitivity



Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_012_S9_mean_absolute_turn_1_frames_smoothed_factorial.csv

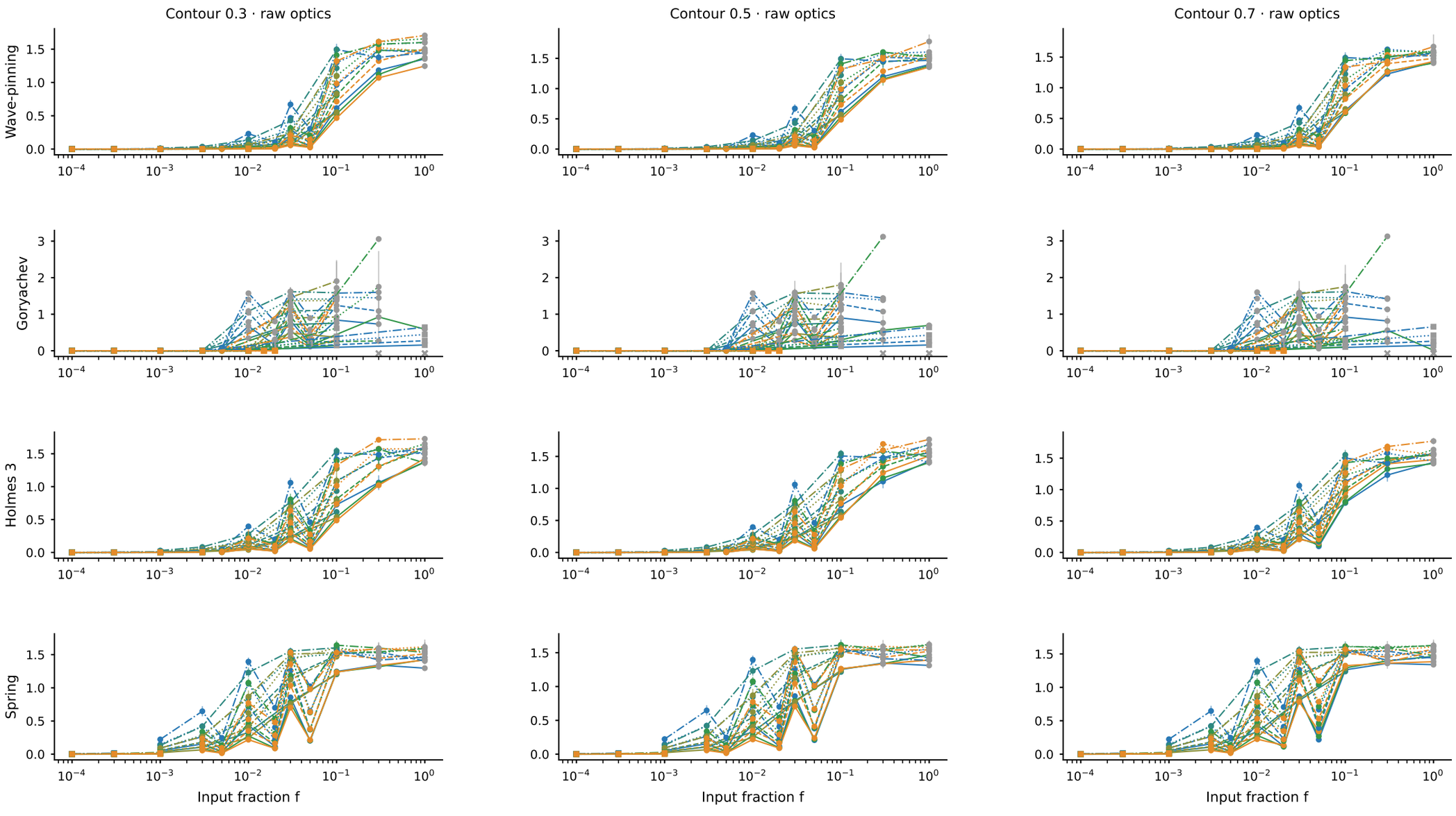
S9 · Longest canonical A5 observations · Mean absolute dominant-heading turn over 2 observation frames



Longest canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.

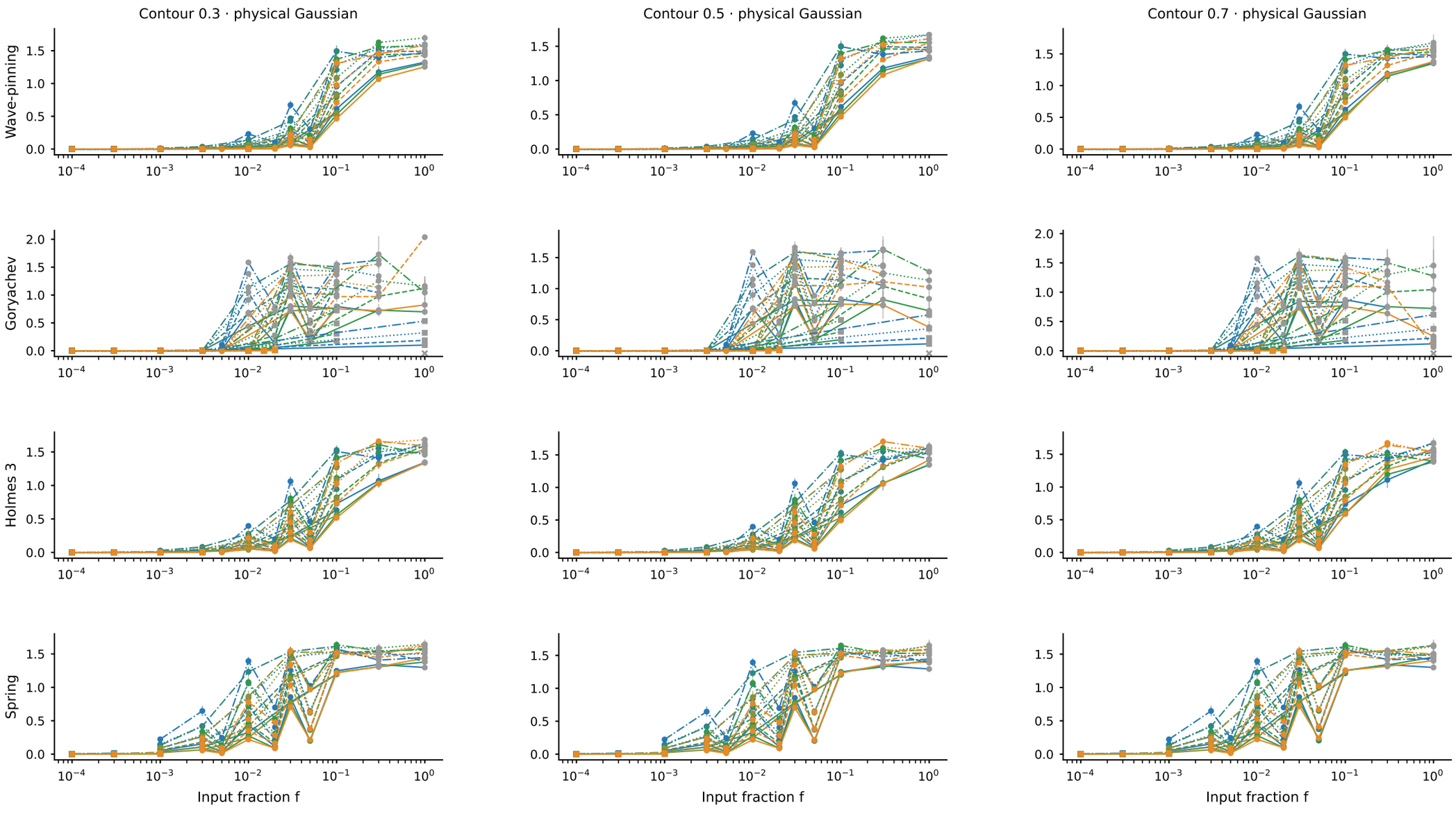
CSV: figure4_S9_longest_canonical_20261020_013_S9_mean_absolute_turn_2_frames.csv

S9 · Longest canonical A5 observations · Mean absolute dominant-heading turn over 2 observation frames · full native-factor sensitivity



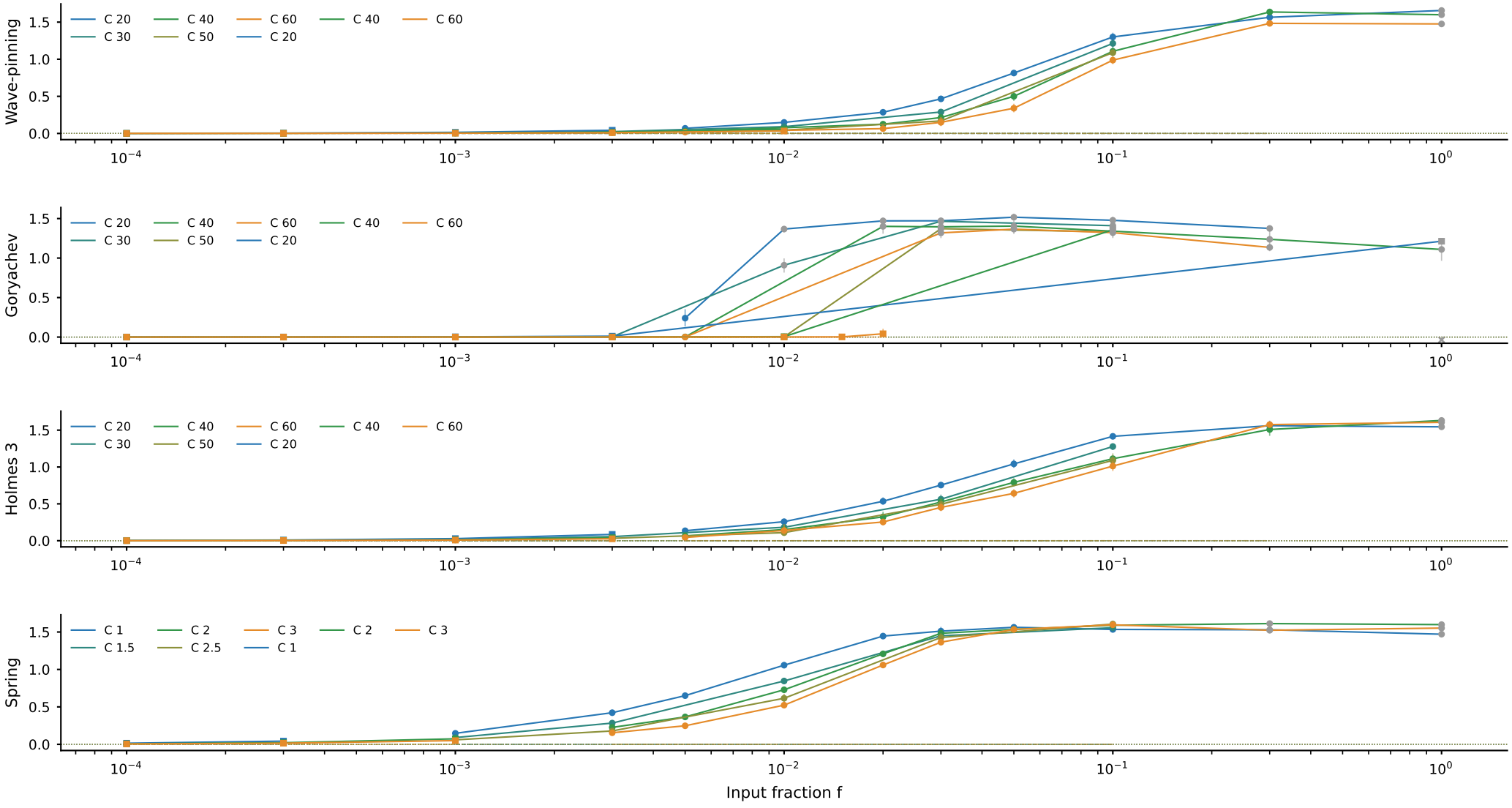
Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_014_S9_mean_absolute_turn_2_frames_raw_factorial.csv

S9 · Longest canonical A5 observations · Mean absolute dominant-heading turn over 2 observation frames · full native-factor sensitivity



Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_015_S9_mean_absolute_turn_2_frames_smoothed_factorial.csv

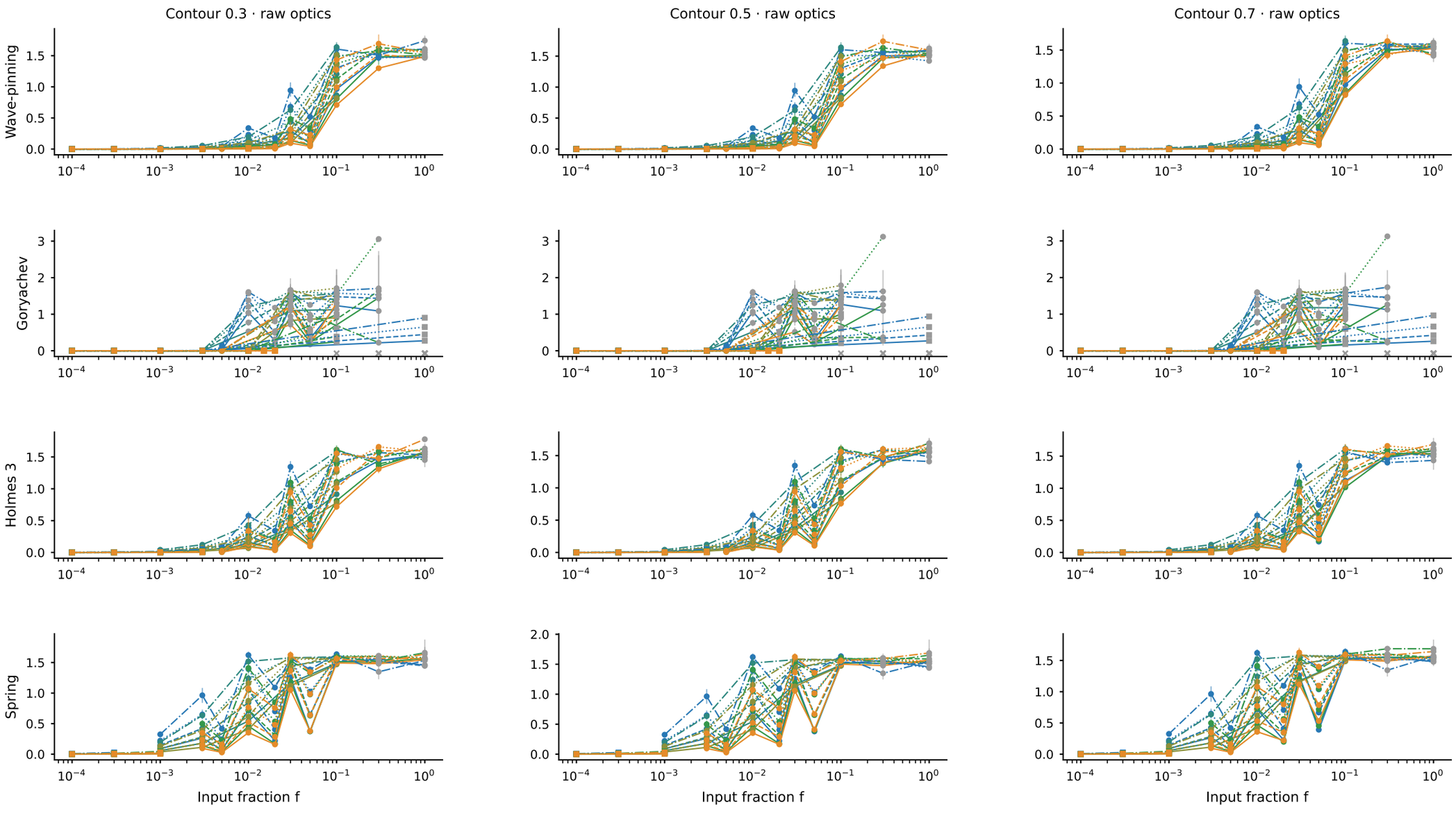
S9 · Longest canonical A5 observations · Mean absolute dominant-heading turn over 4 observation frames



Longest canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.

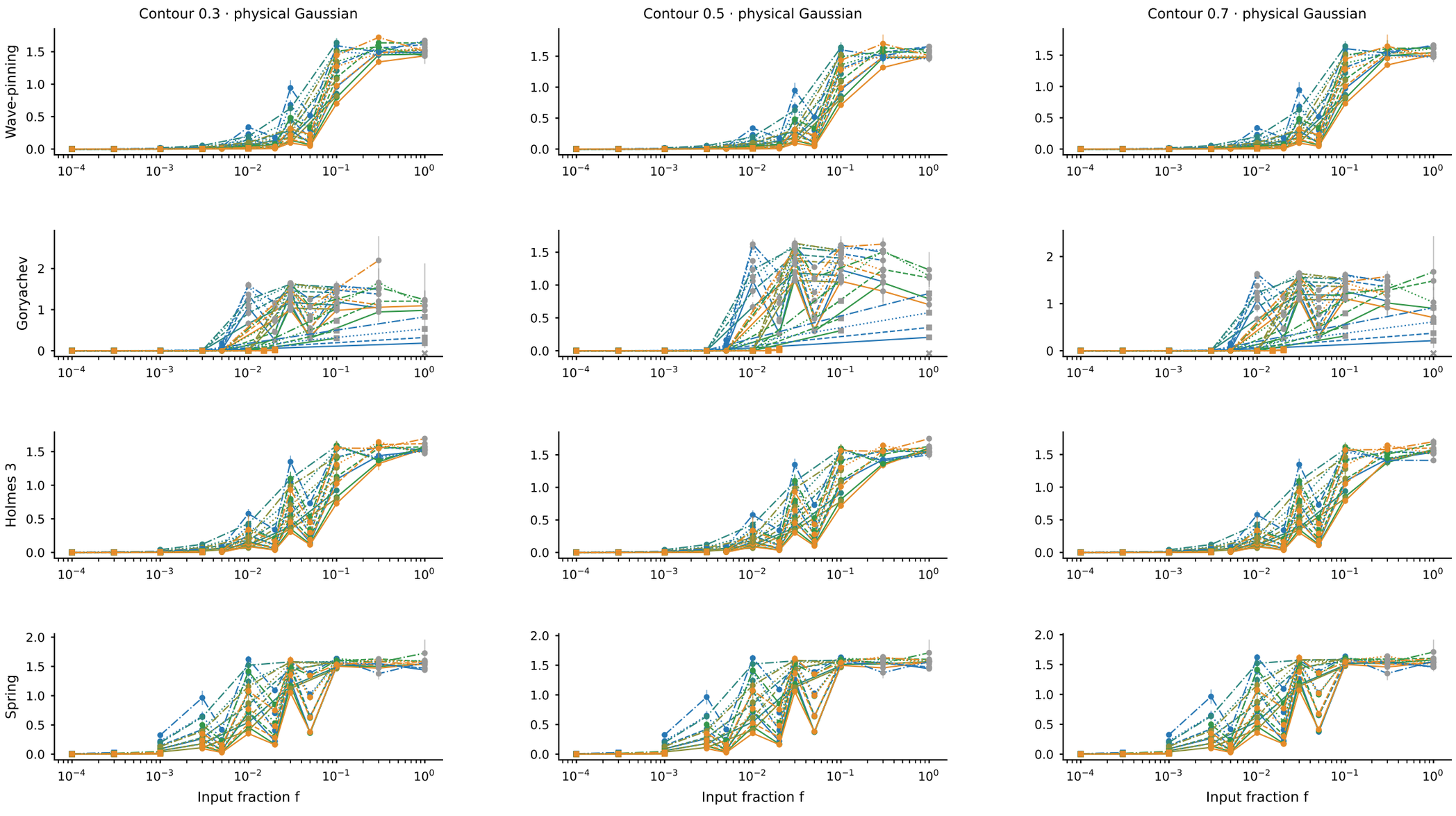
CSV: figure4_S9_longest_canonical_20261020_016_S9_mean_absolute_turn_4_frames.csv

S9 · Longest canonical A5 observations · Mean absolute dominant-heading turn over 4 observation frames · full native-factor sensitivity



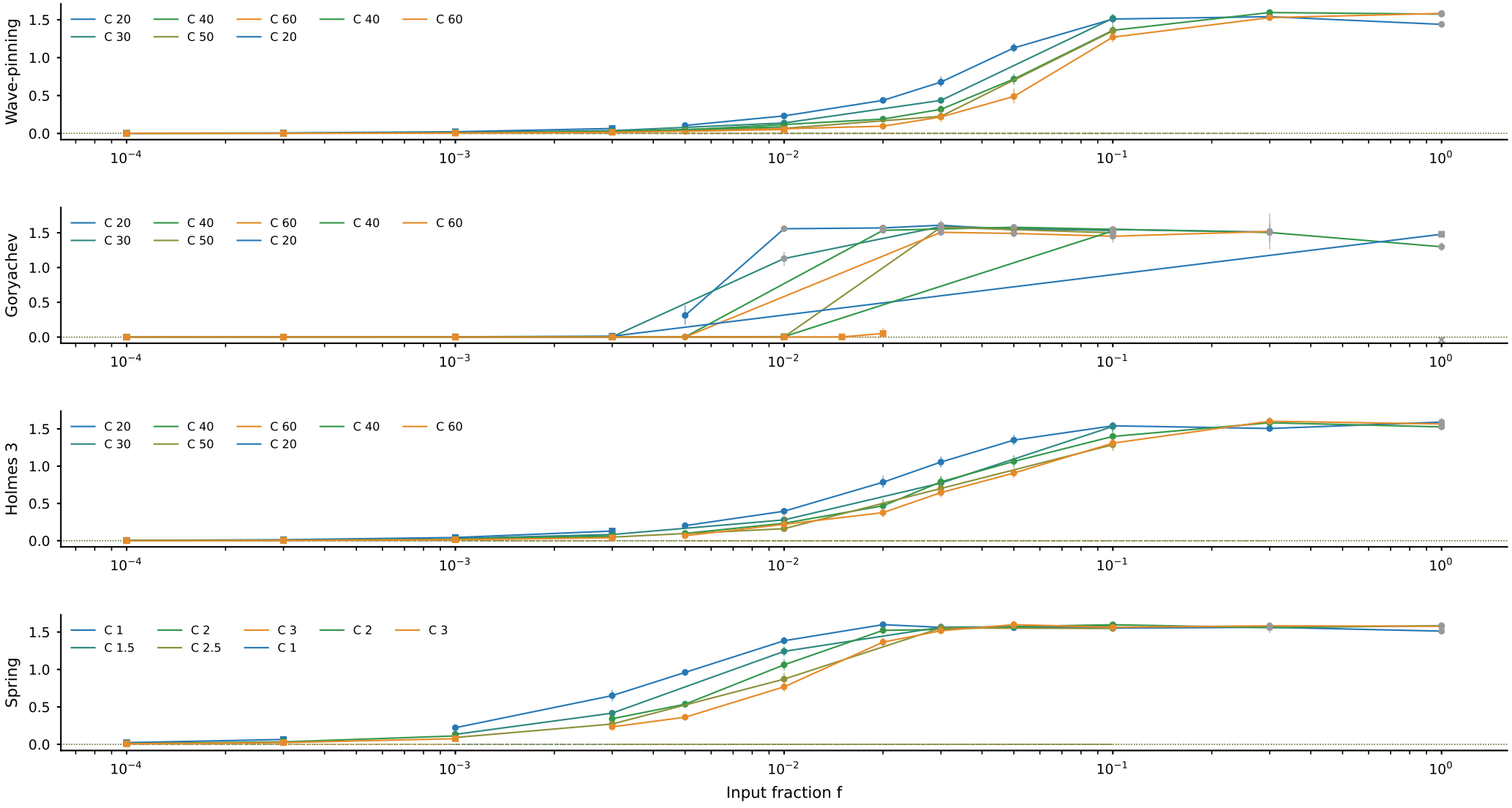
Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_017_S9_mean_absolute_turn_4_frames_raw_factorial.csv

S9 · Longest canonical A5 observations · Mean absolute dominant-heading turn over 4 observation frames · full native-factor sensitivity



Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_018_S9_mean_absolute_turn_4_frames_smoothed_factorial.csv

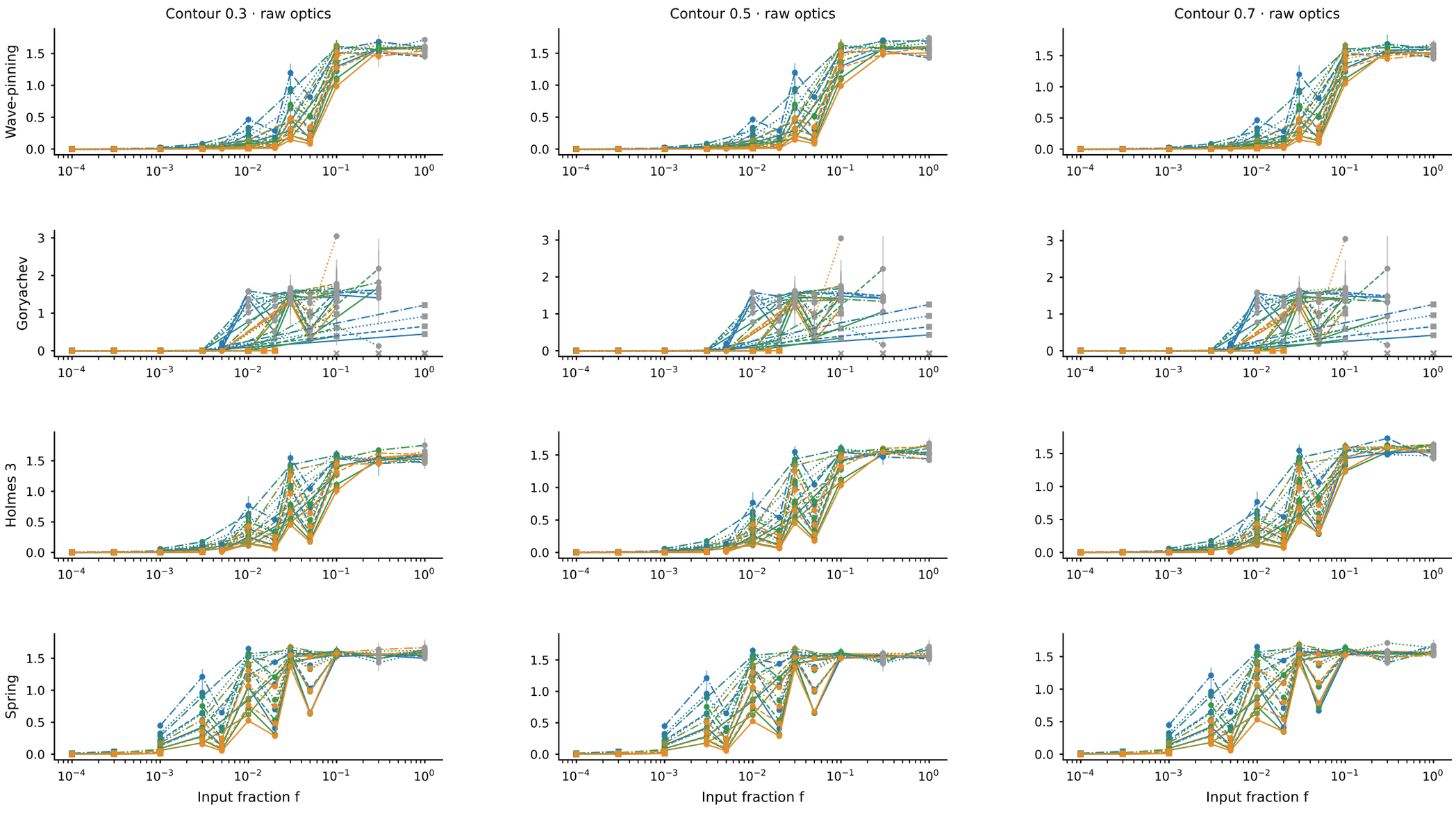
S9 · Longest canonical A5 observations · Mean absolute dominant-heading turn over 8 observation frames



Longest canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.

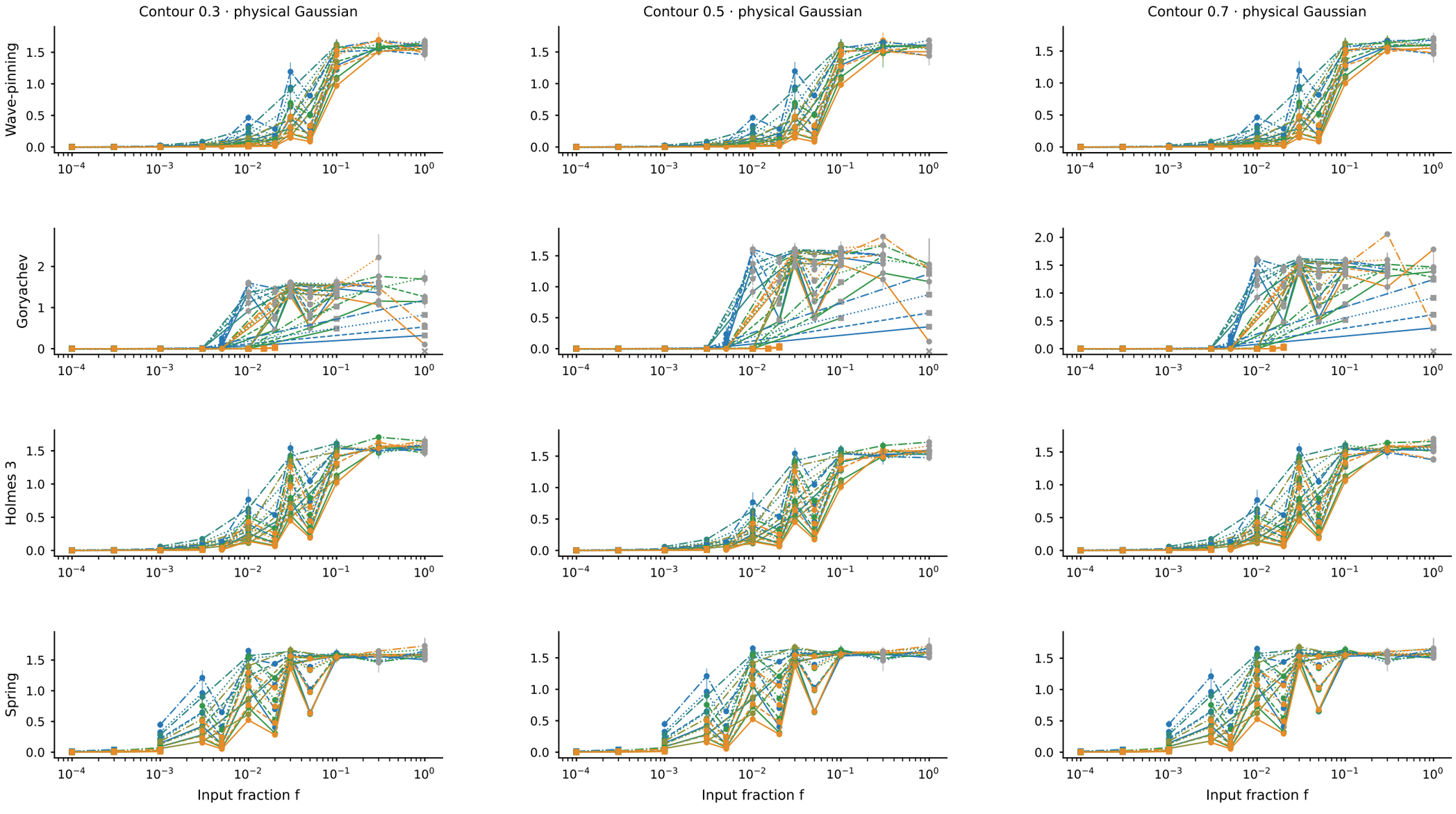
CSV: figure4_S9_longest_canonical_20261020_019_S9_mean_absolute_turn_8_frames.csv

S9 · Longest canonical A5 observations · Mean absolute dominant-heading turn over 8 observation frames · full native-factor sensitivity



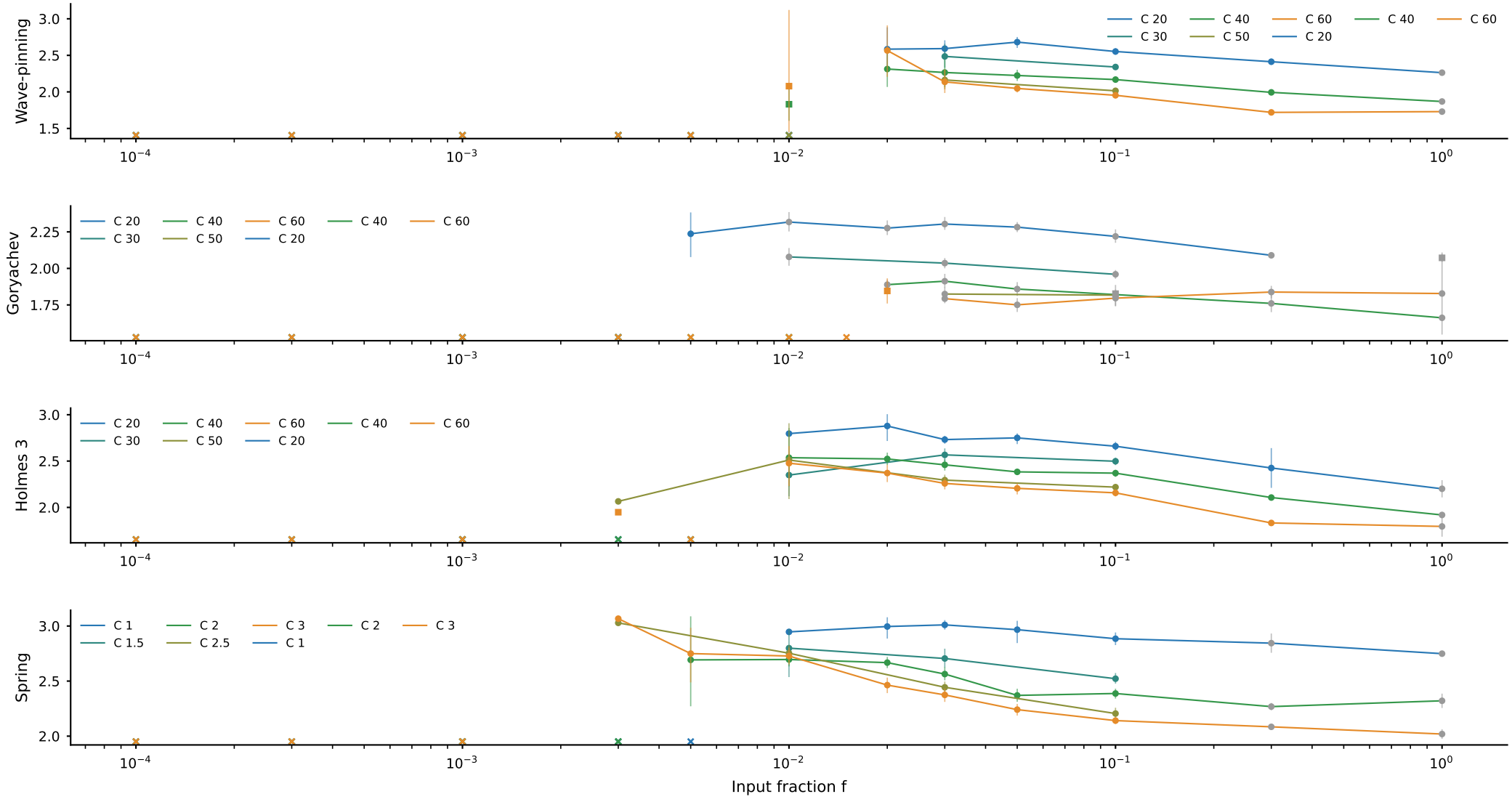
Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds.
× at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_020_S9_mean_absolute_turn_8_frames_raw_factorial.csv

S9 · Longest canonical A5 observations · Mean absolute dominant-heading turn over 8 observation frames · full native-factor sensitivity



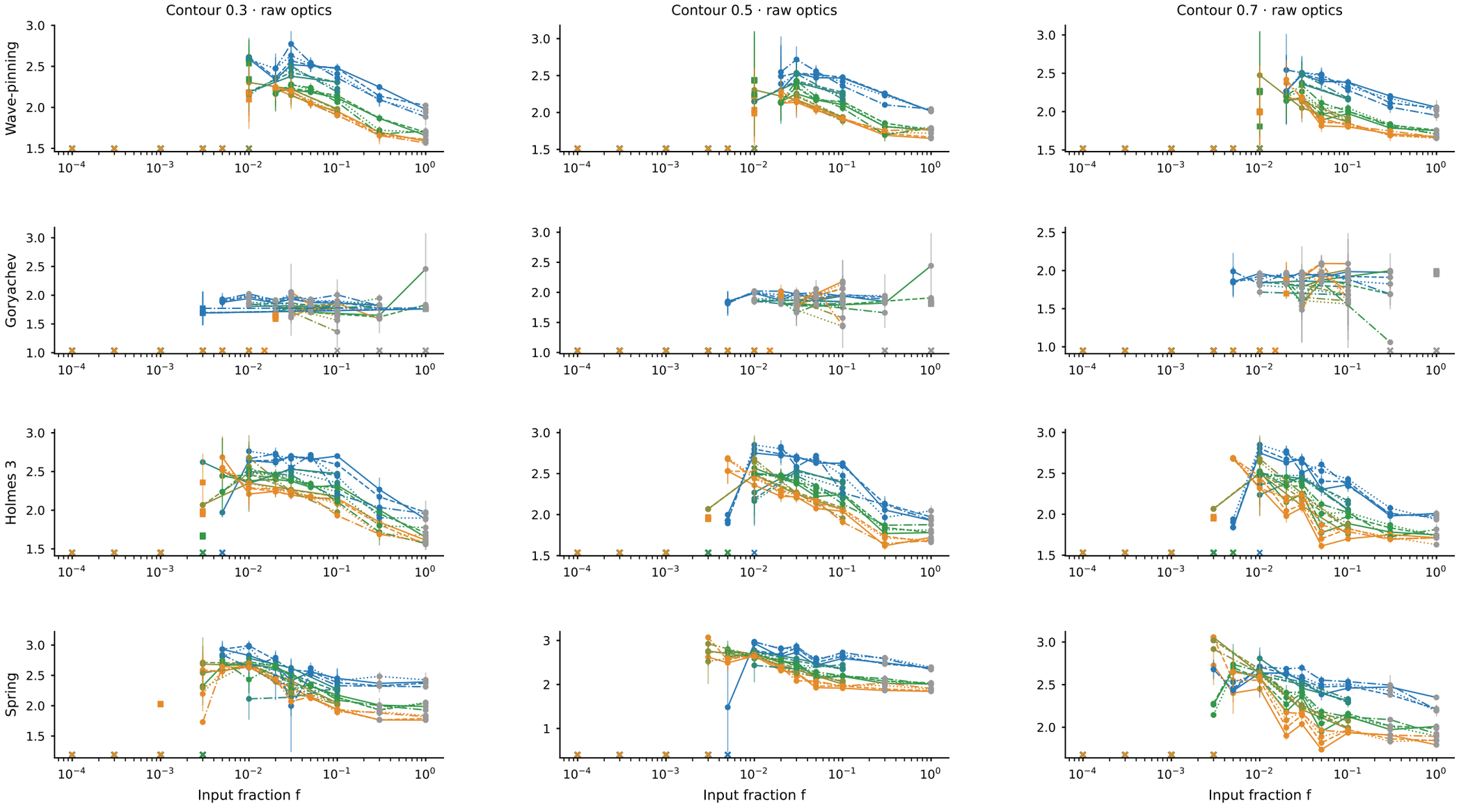
Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds.
× at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_021_S9_mean_absolute_turn_8_frames_smoothed_factorial.csv

S9 · Longest canonical A5 observations · Mean newborn angle from the previous dominant region



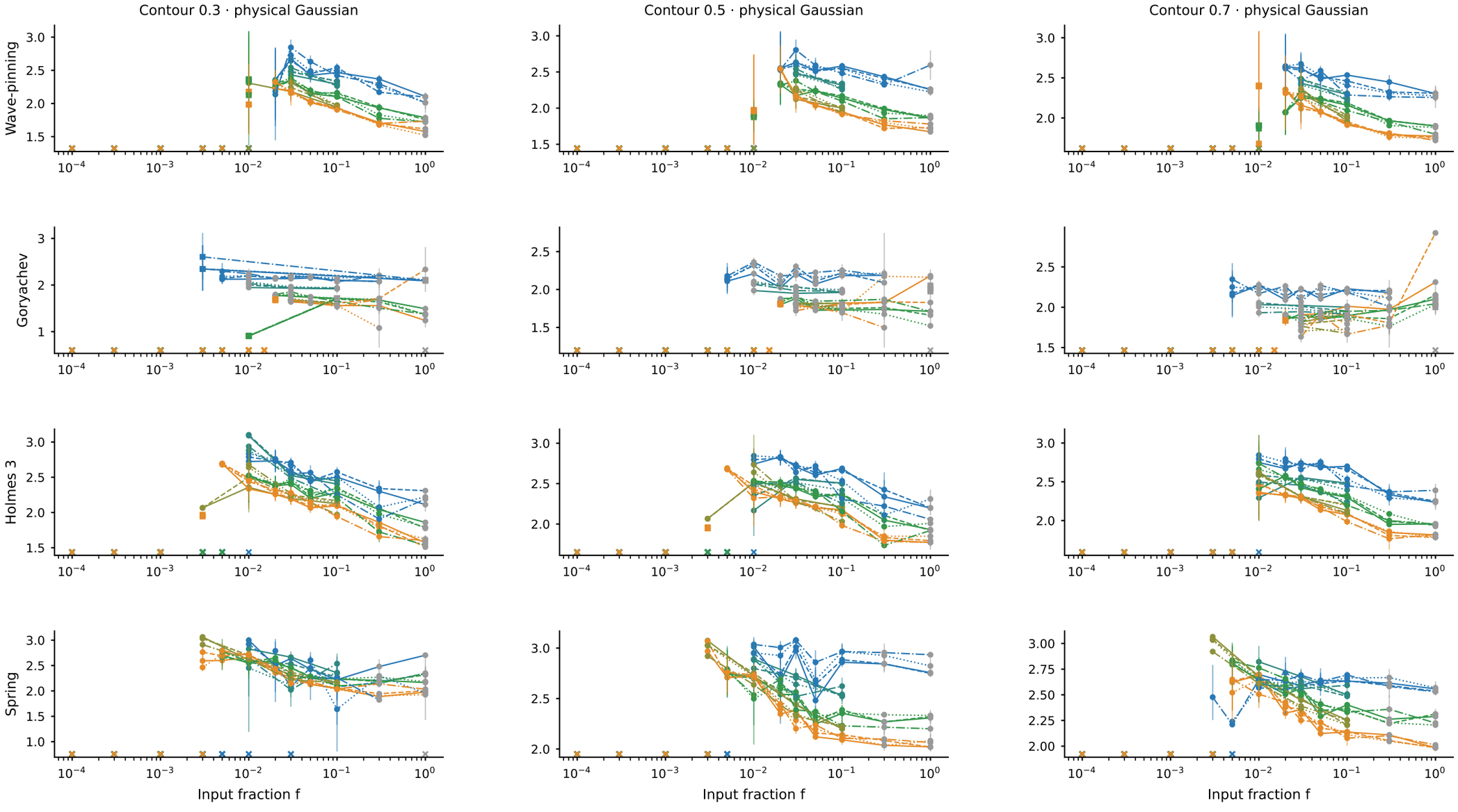
Longest canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_022_S9_mean_birth_angle.csv

S9 · Longest canonical A5 observations · Mean newborn angle from the previous dominant region · full native-factor sensitivity



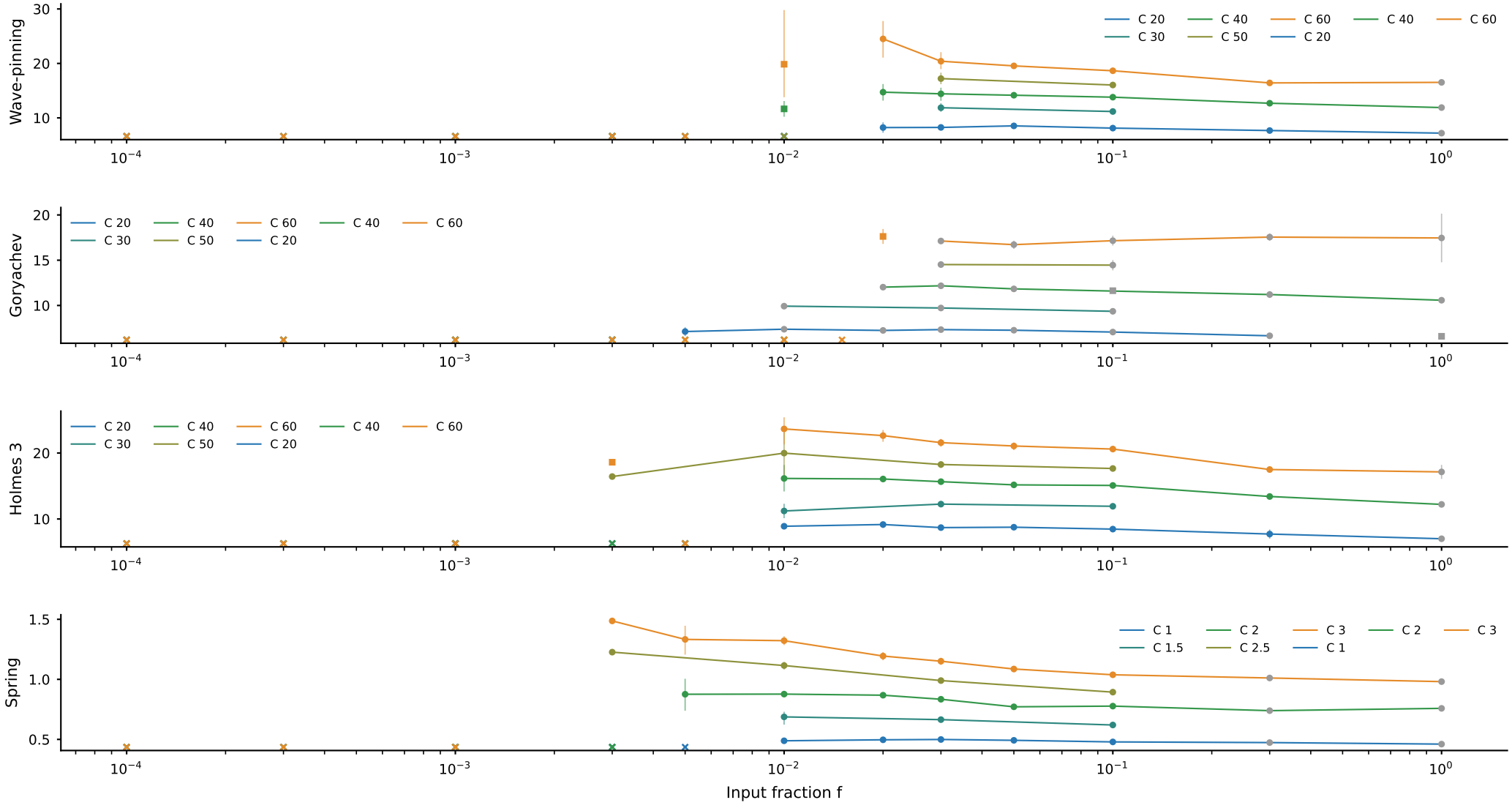
Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_023_S9_mean_birth_angle_raw_factorial.csv

S9 · Longest canonical A5 observations · Mean newborn angle from the previous dominant region · full native-factor sensitivity



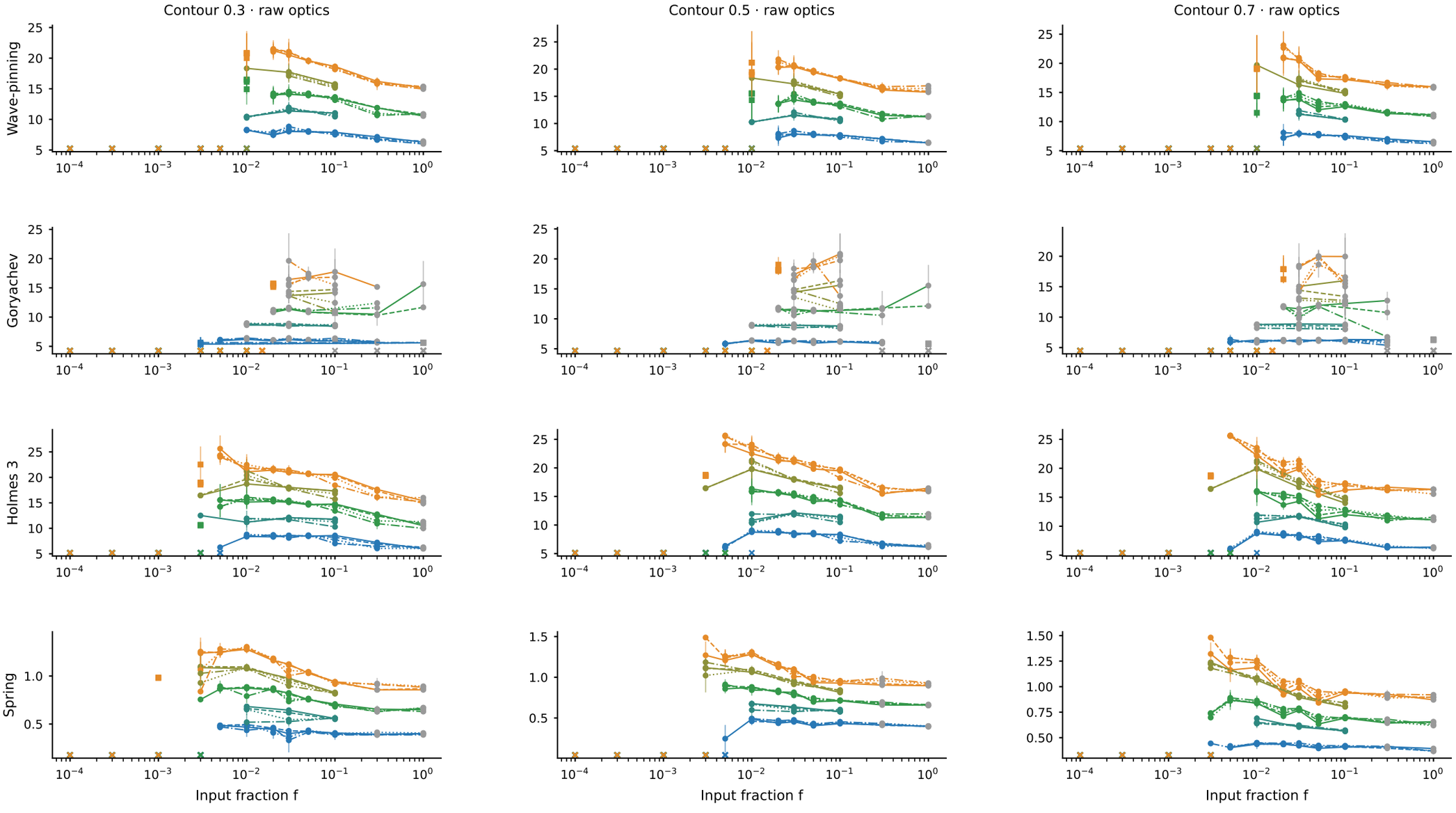
Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds.
× at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_024_S9_mean_birth_angle_smoothed_factorial.csv

S9 · Longest canonical A5 observations · Mean newborn physical arc distance



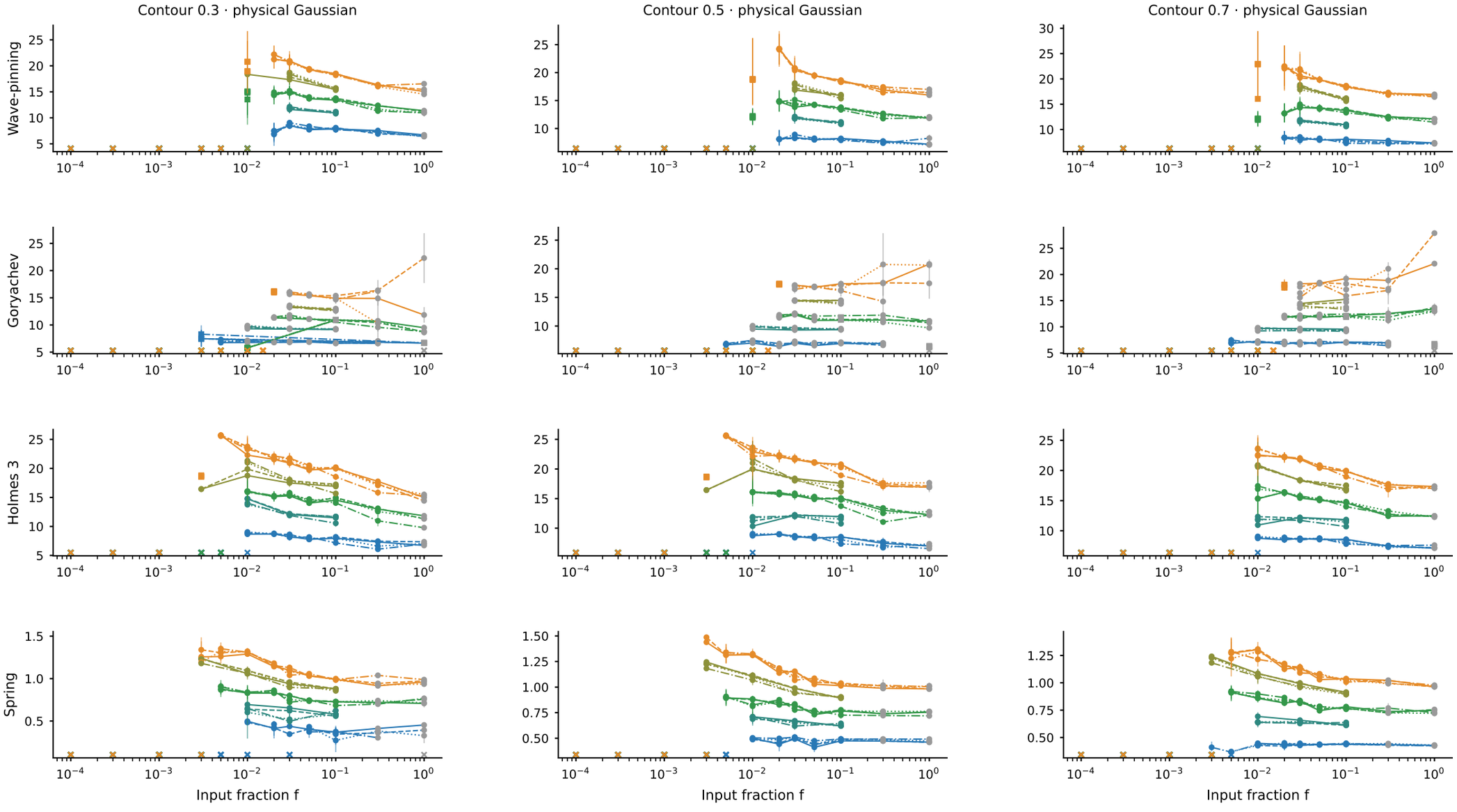
Longest canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_025_S9_mean_birth_physical_distance.csv

S9 · Longest canonical A5 observations · Mean newborn physical arc distance · full native-factor sensitivity



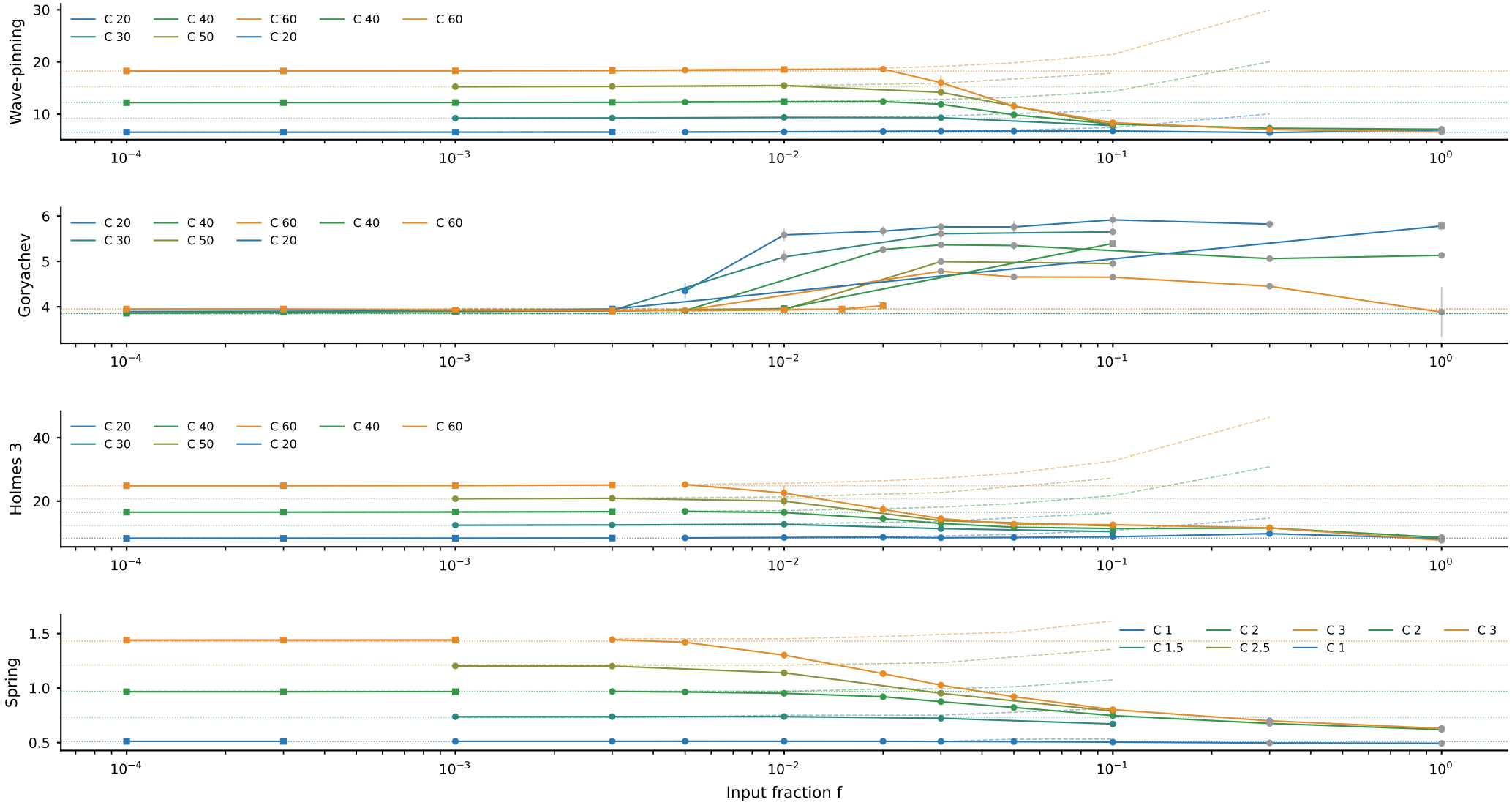
Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds.
× at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_026_S9_mean_birth_physical_distance_raw_factorial.csv

S9 · Longest canonical A5 observations · Mean newborn physical arc distance · full native-factor sensitivity



Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds.
× at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_027_S9_mean_birth_physical_distance_smoothed_factorial.csv

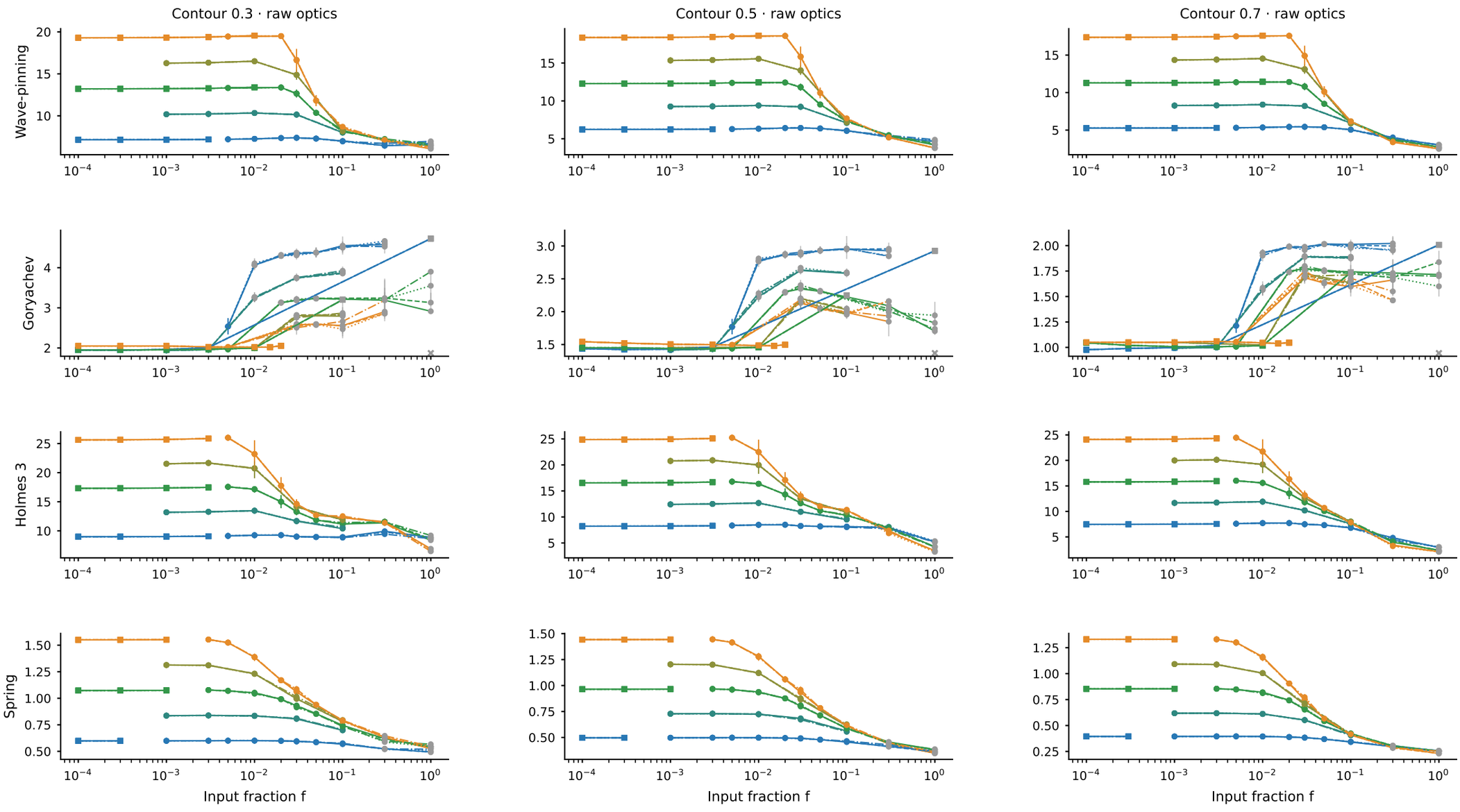
S9 · Longest canonical A5 observations · Mean physical region width



Longest canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.

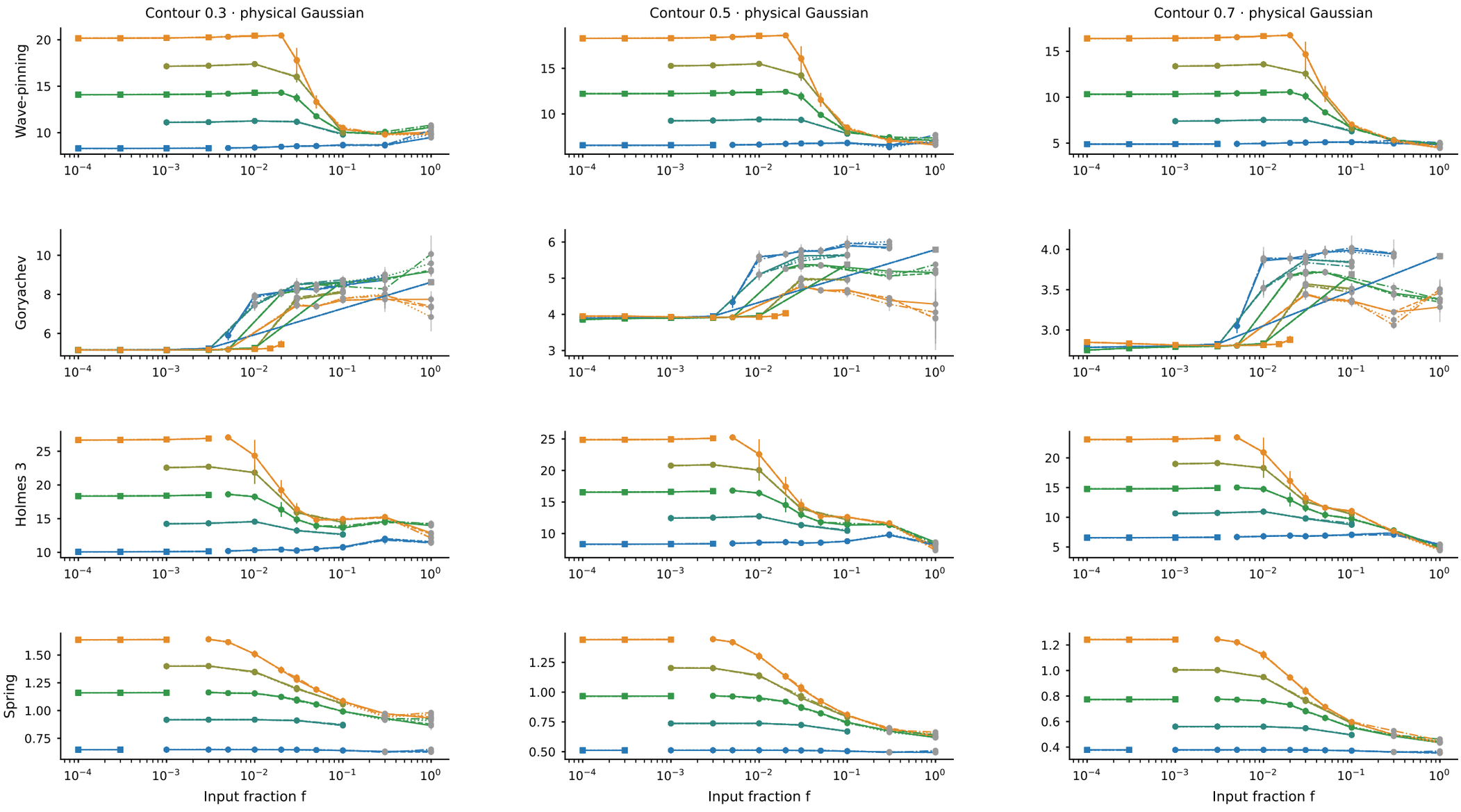
CSV: figure4_S9_longest_canonical_20261020_028_S9_mean_region_width.csv

S9 · Longest canonical A5 observations · Mean physical region width · full native-factor sensitivity



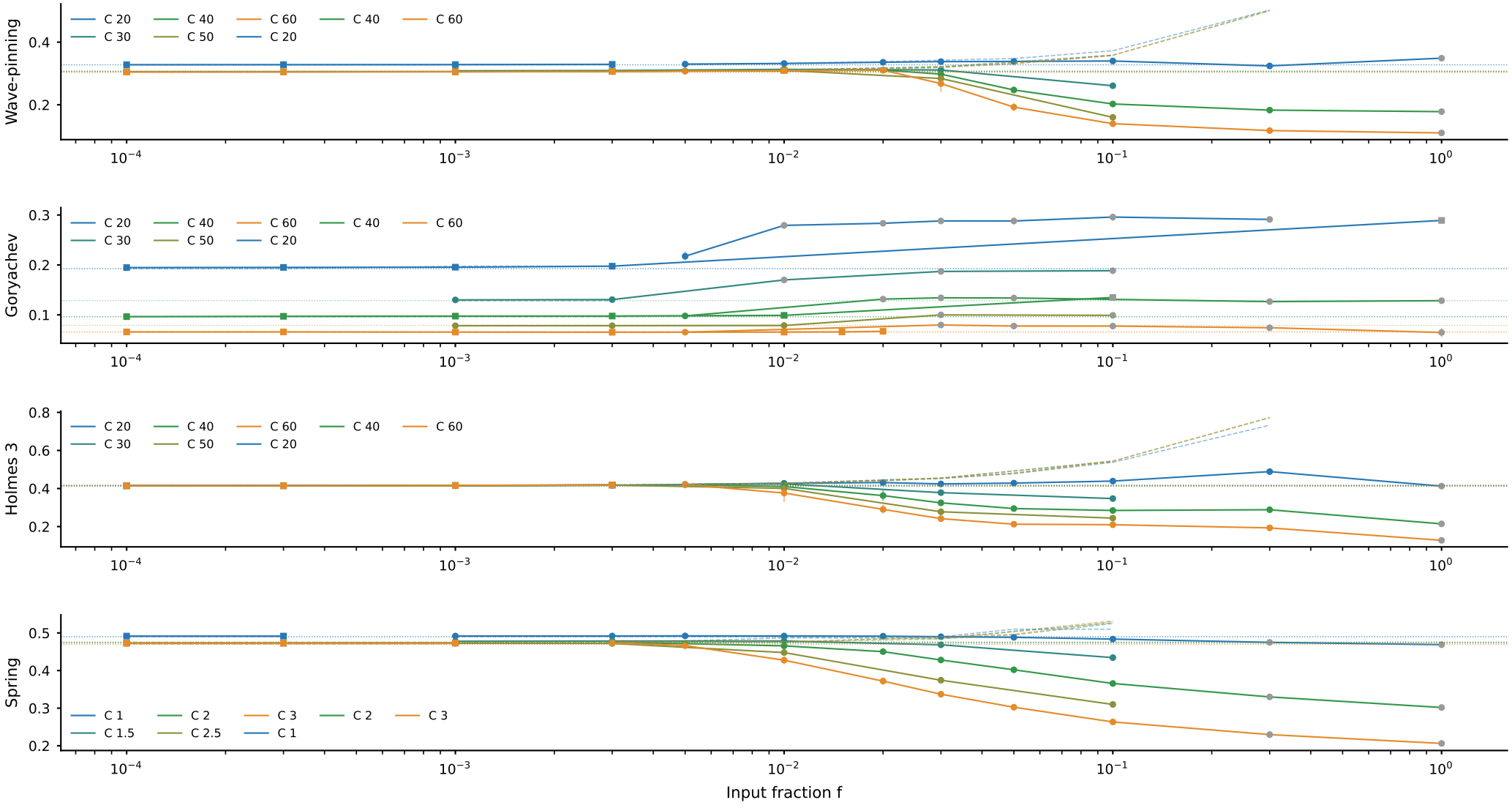
Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds.
× at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_029_S9_mean_region_width_raw_factorial.csv

S9 · Longest canonical A5 observations · Mean physical region width · full native-factor sensitivity



Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_030_S9_mean_region_width_smoothed_factorial.csv

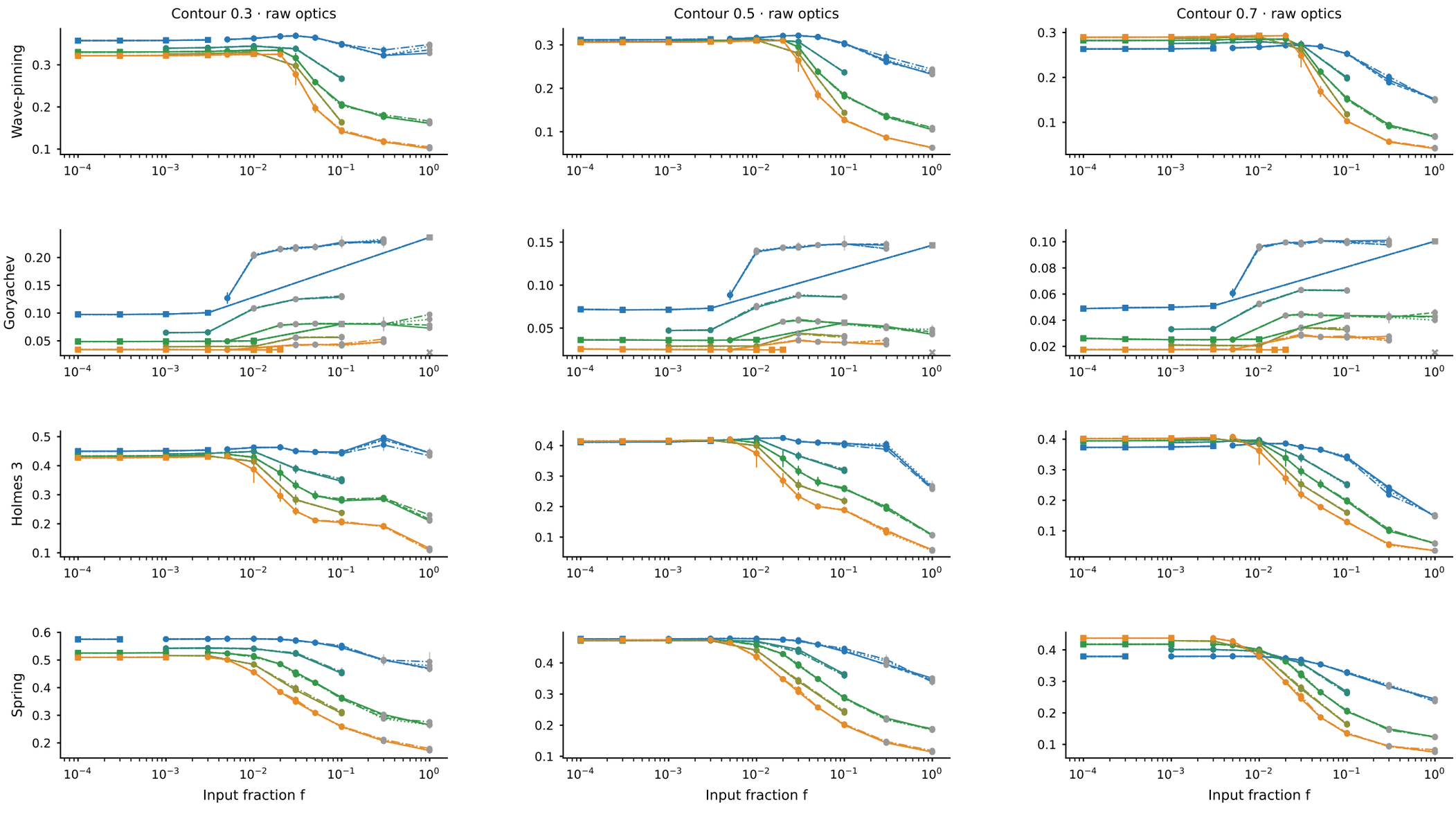
S9 · Longest canonical A5 observations · Mean region width / actual circumference



Longest canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.

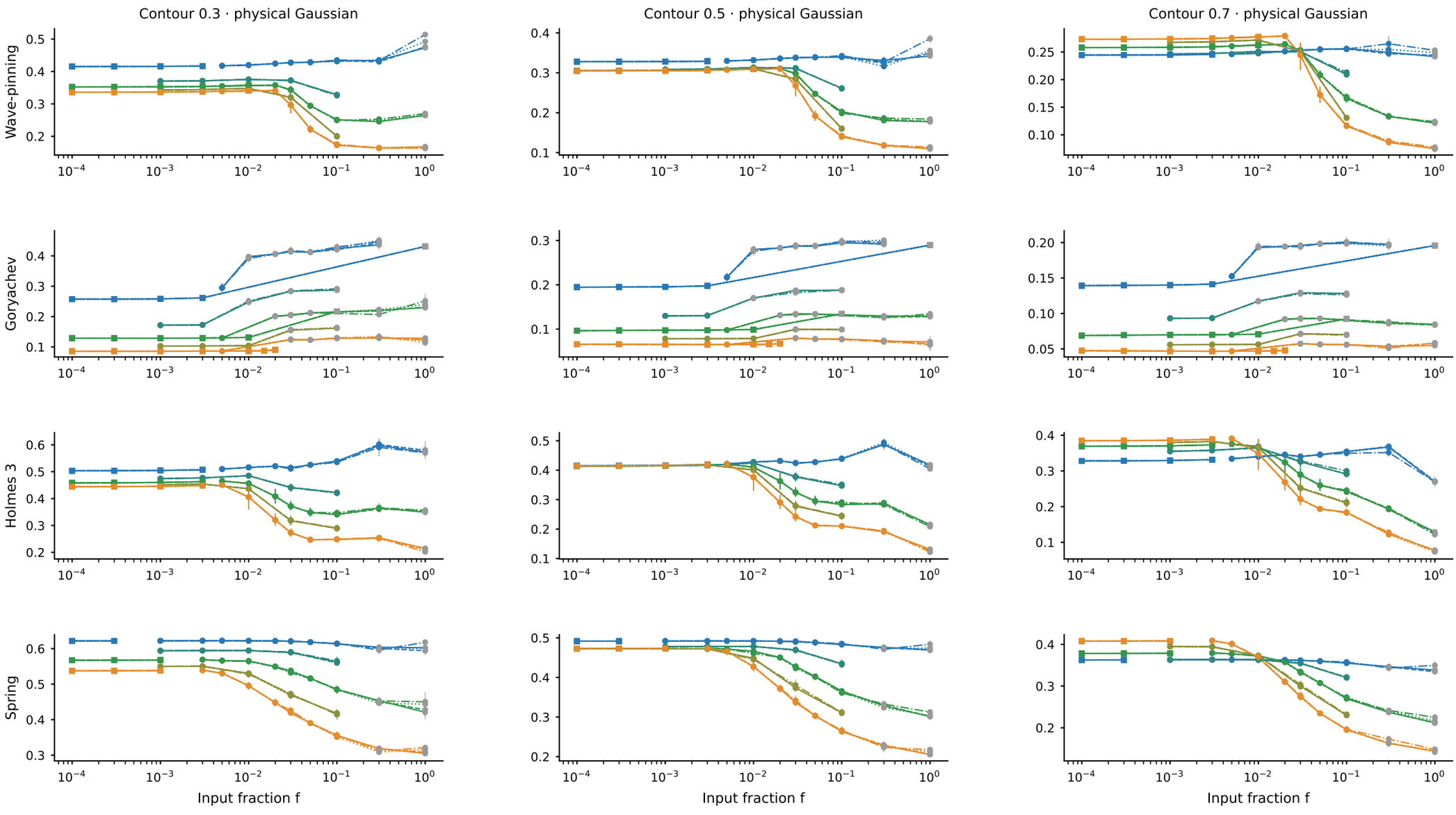
CSV: figure4_S9_longest_canonical_20261020_031_S9_mean_region_width_fraction.csv

S9 · Longest canonical A5 observations · Mean region width / actual circumference · full native-factor sensitivity



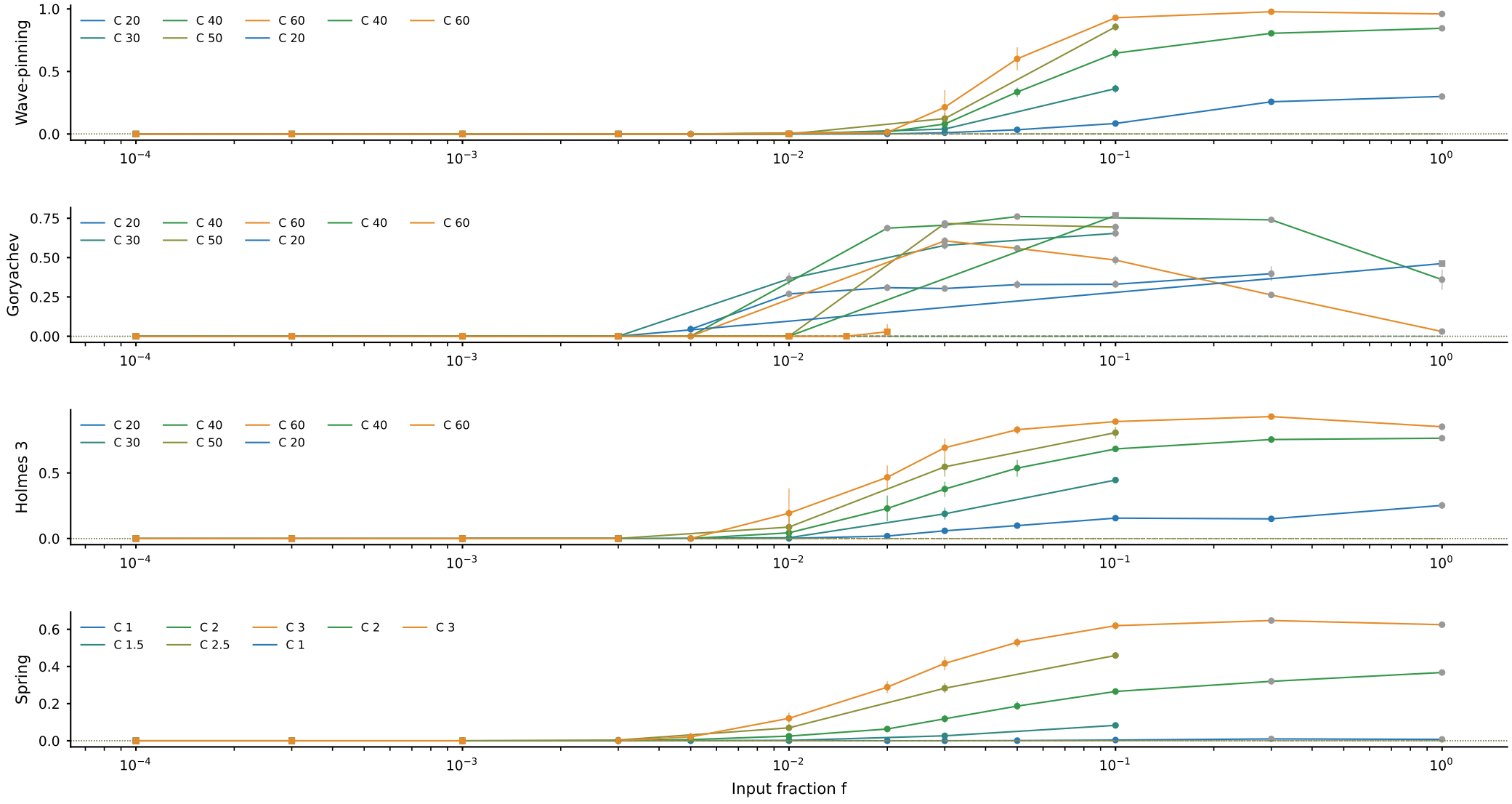
Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds.
× at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_032_S9_mean_region_width_fraction_raw_factorial.csv

S9 · Longest canonical A5 observations · Mean region width / actual circumference · full native-factor sensitivity



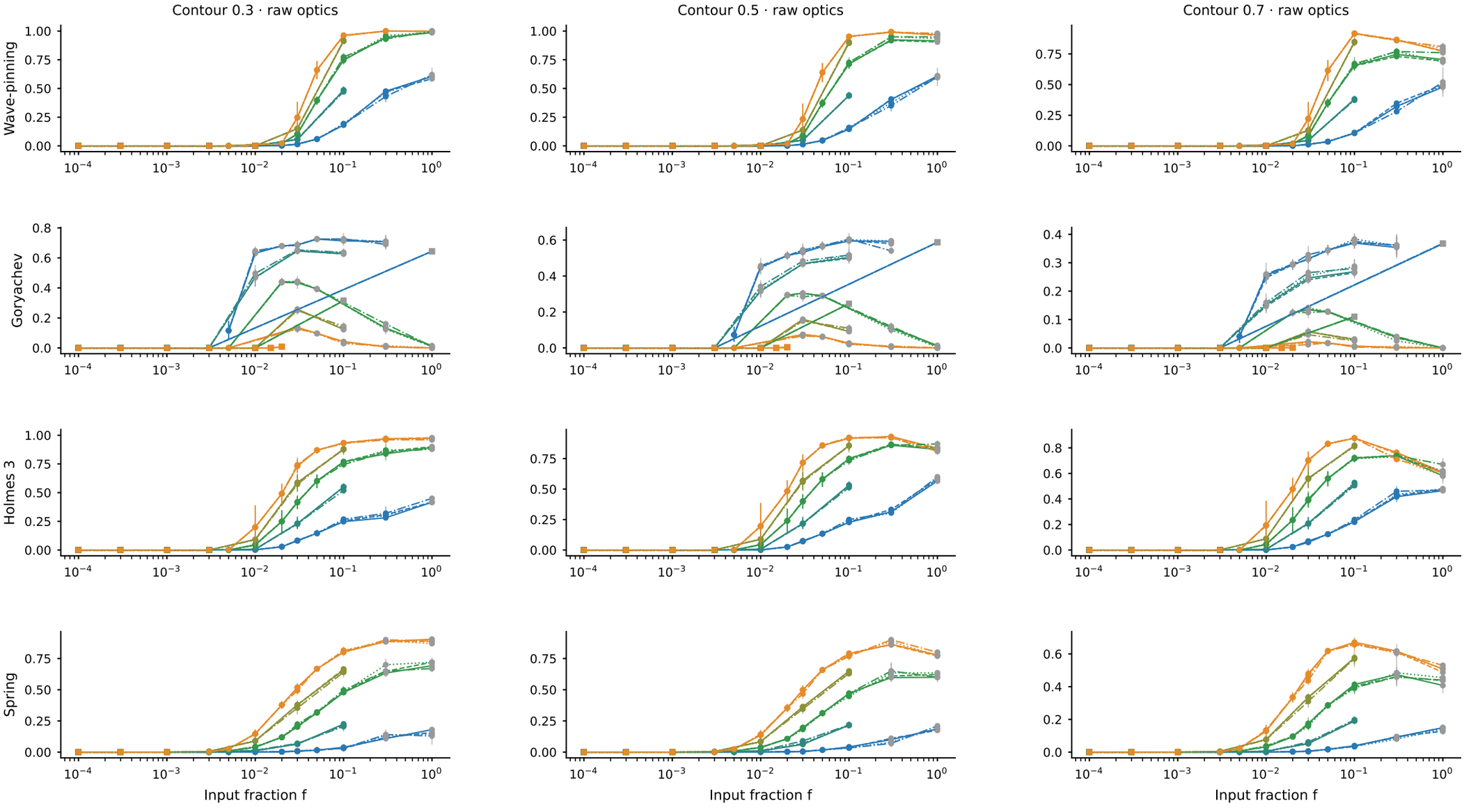
Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_033_S9_mean_region_width_fraction_smoothed_factorial.csv

S9 · Longest canonical A5 observations · Time fraction with multiple qualified regions



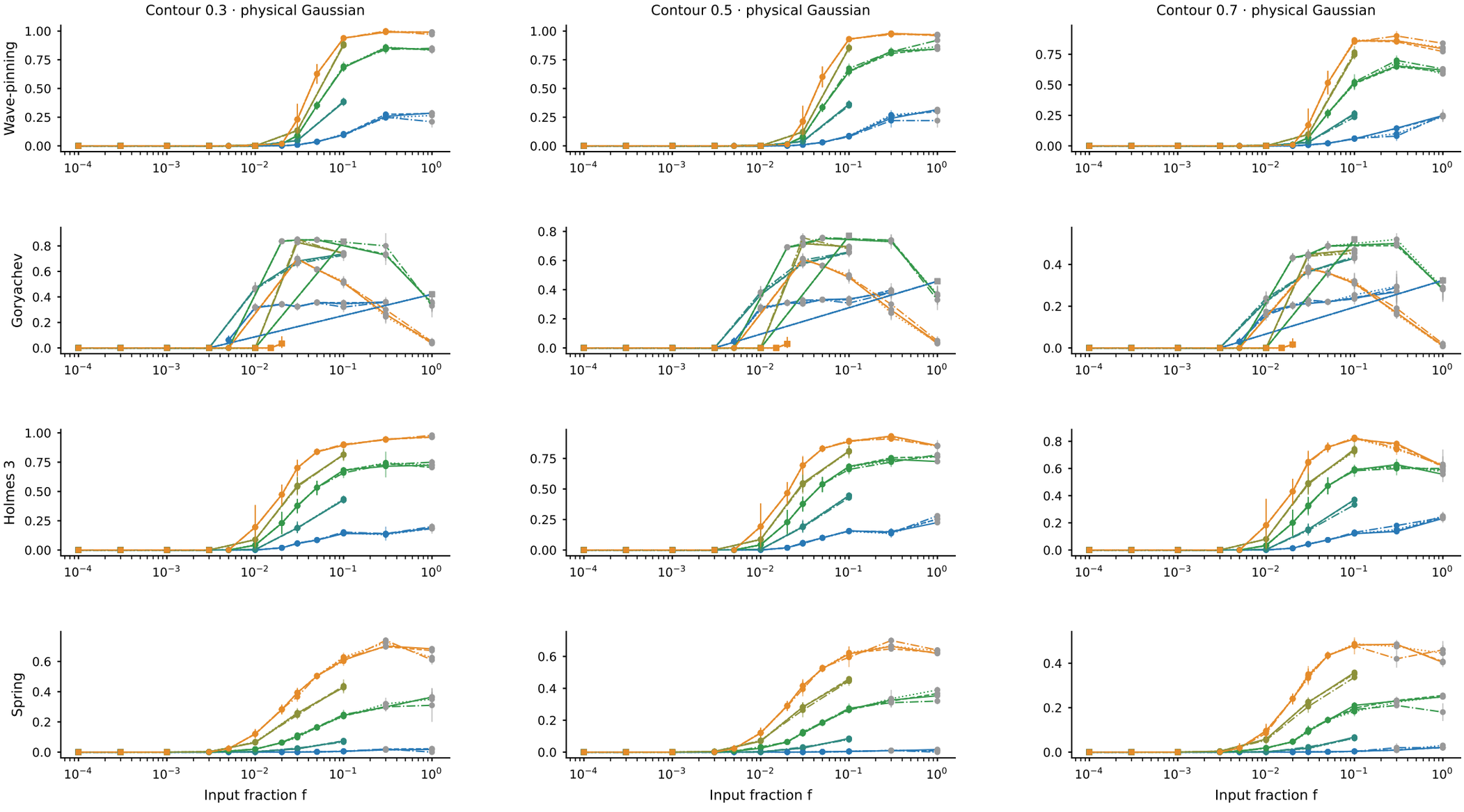
Longest canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_034_S9_multiple_front_fraction.csv

S9 · Longest canonical A5 observations · Time fraction with multiple qualified regions · full native-factor sensitivity



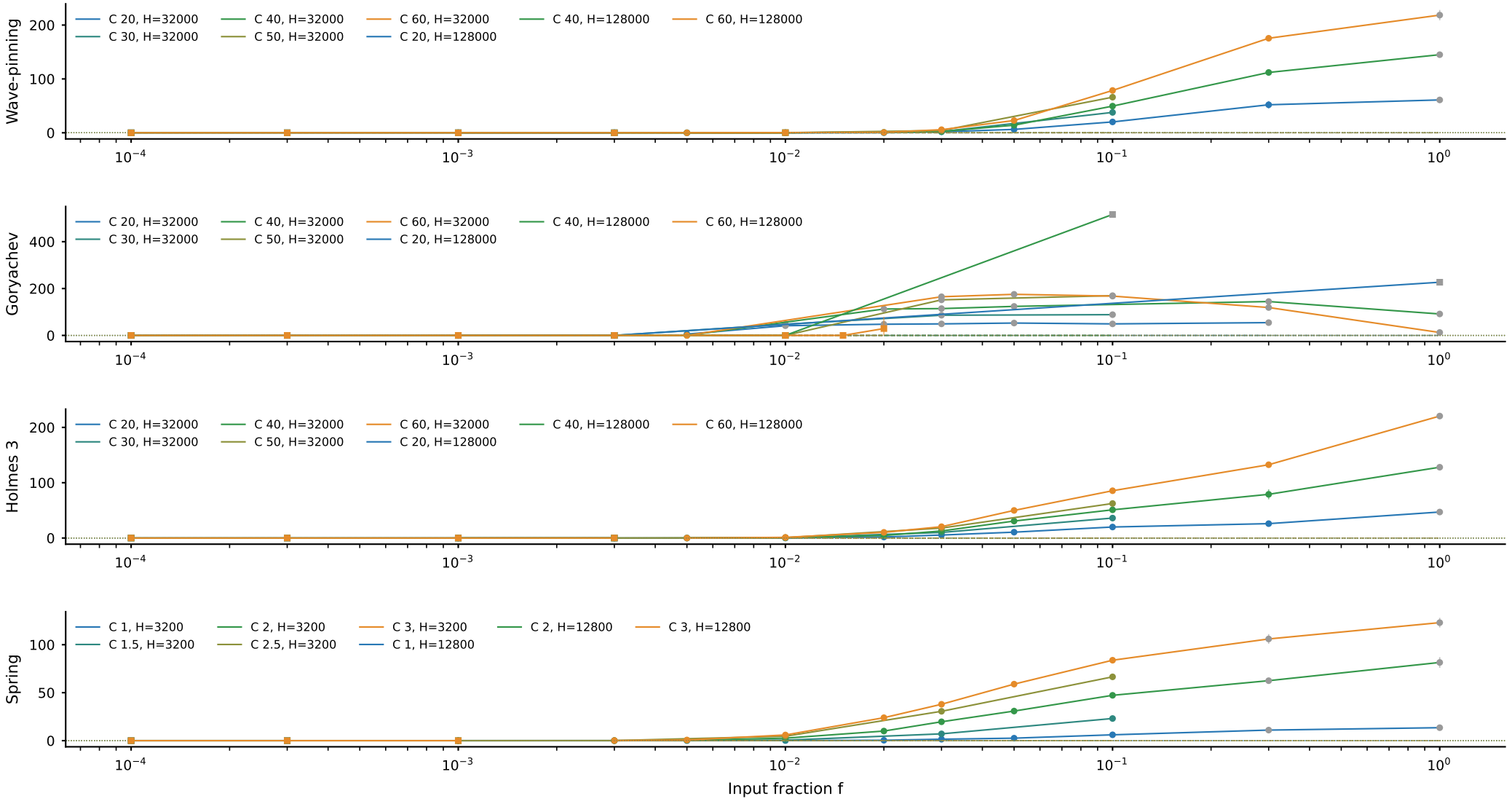
Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds.
× at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_035_S9_multiple_front_fraction_raw_factorial.csv

S9 · Longest canonical A5 observations · Time fraction with multiple qualified regions · full native-factor sensitivity



Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds.
× at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_036_S9_multiple_front_fraction_smoothed_factorial.csv

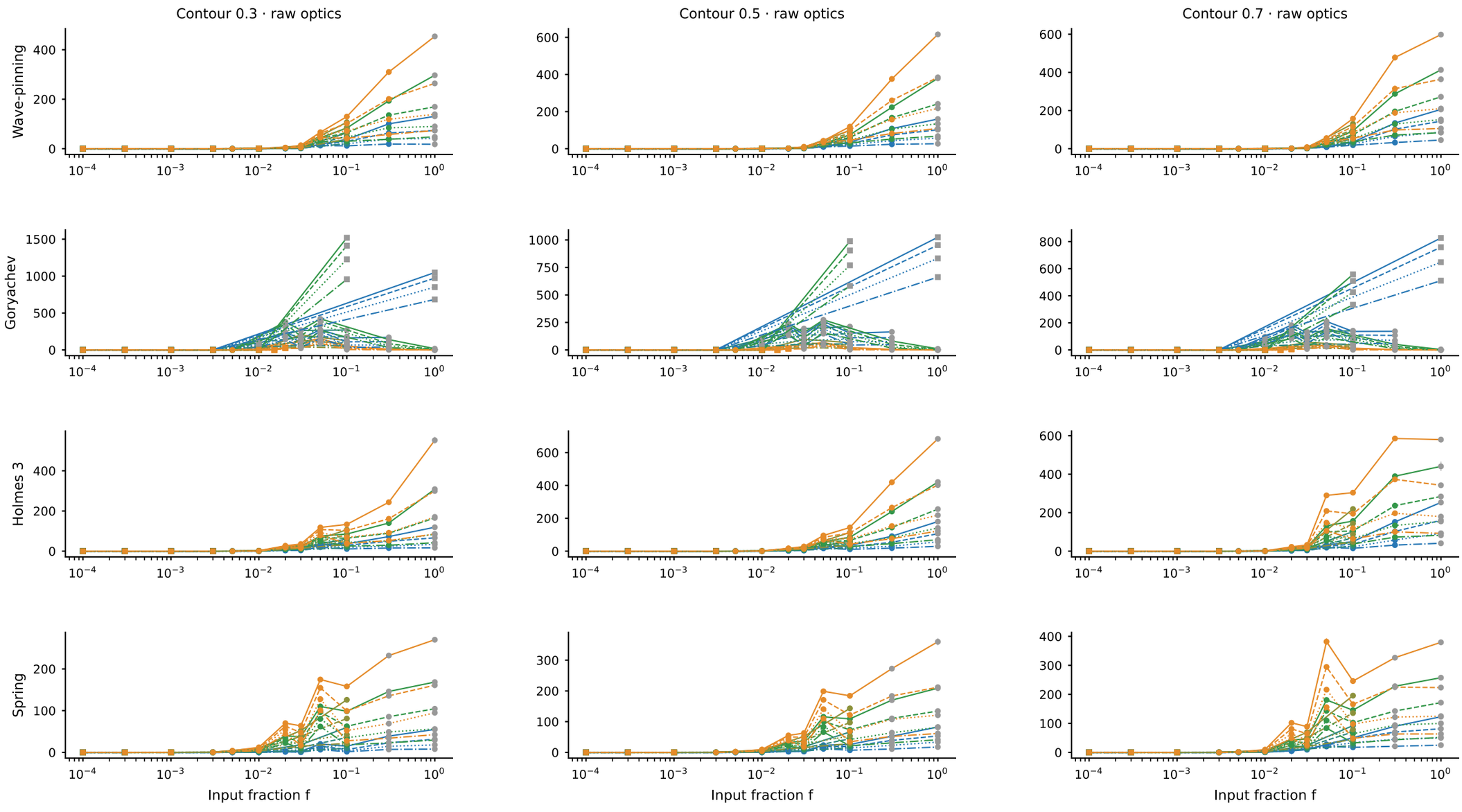
S9 · Longest canonical A5 observations · Observed region births per trajectory



Longest canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.

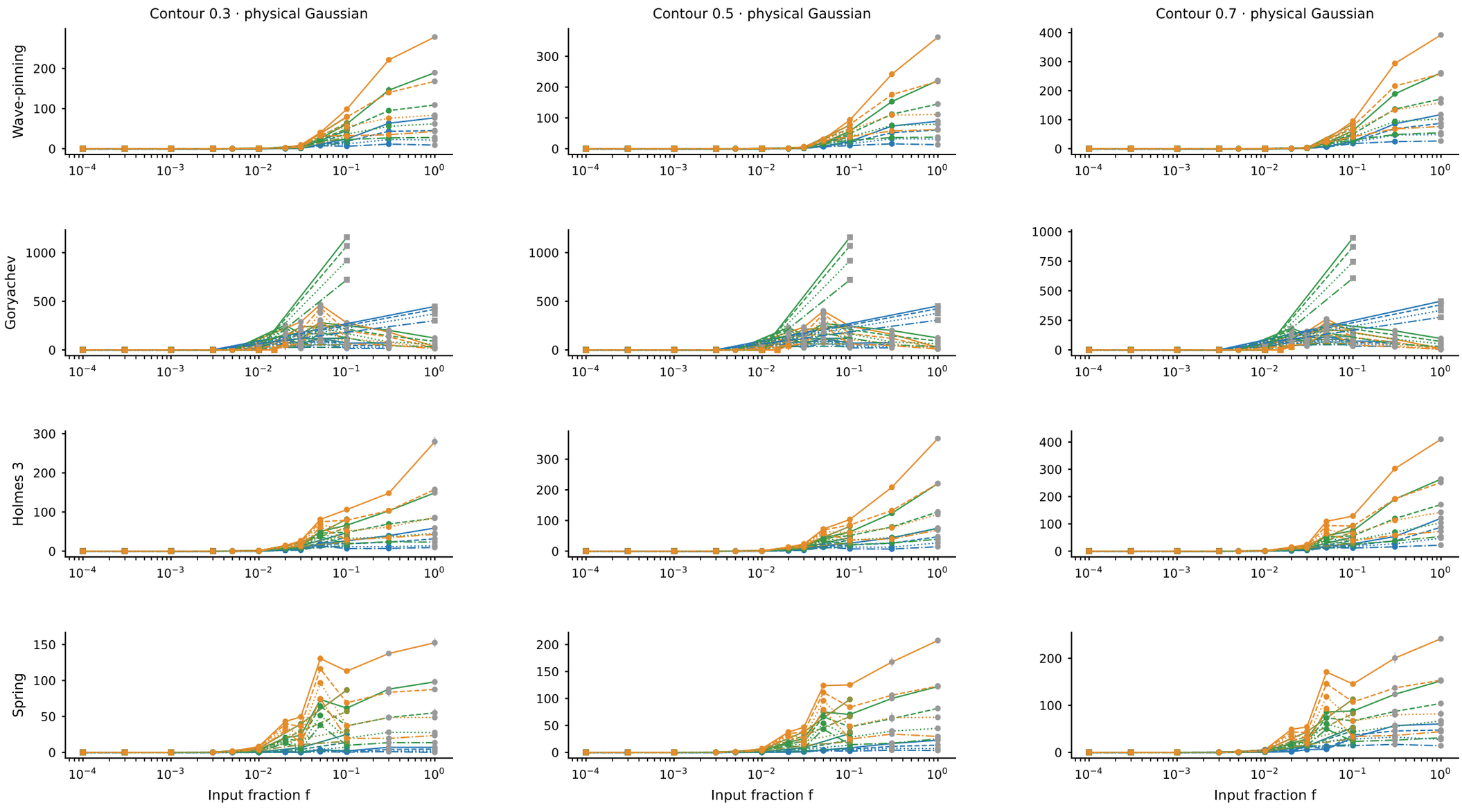
CSV: figure4_S9_longest_canonical_20261020_037_S9_observed_births.csv

S9 · Longest canonical A5 observations · Observed region births per trajectory · full native-factor sensitivity



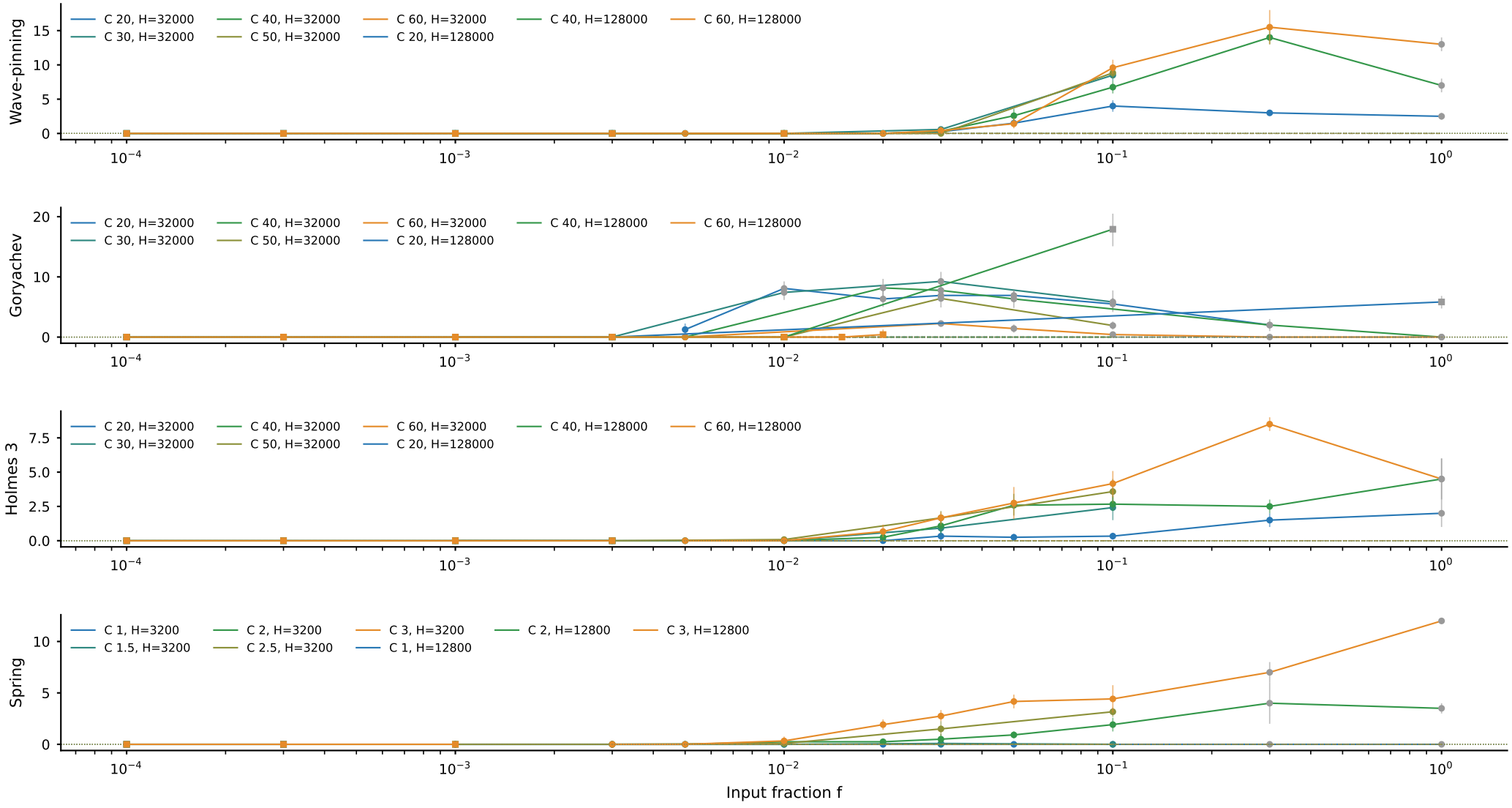
Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_038_S9_observed_births_raw_factorial.csv

S9 · Longest canonical A5 observations · Observed region births per trajectory · full native-factor sensitivity



Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_039_S9_observed_births_smoothed_factorial.csv

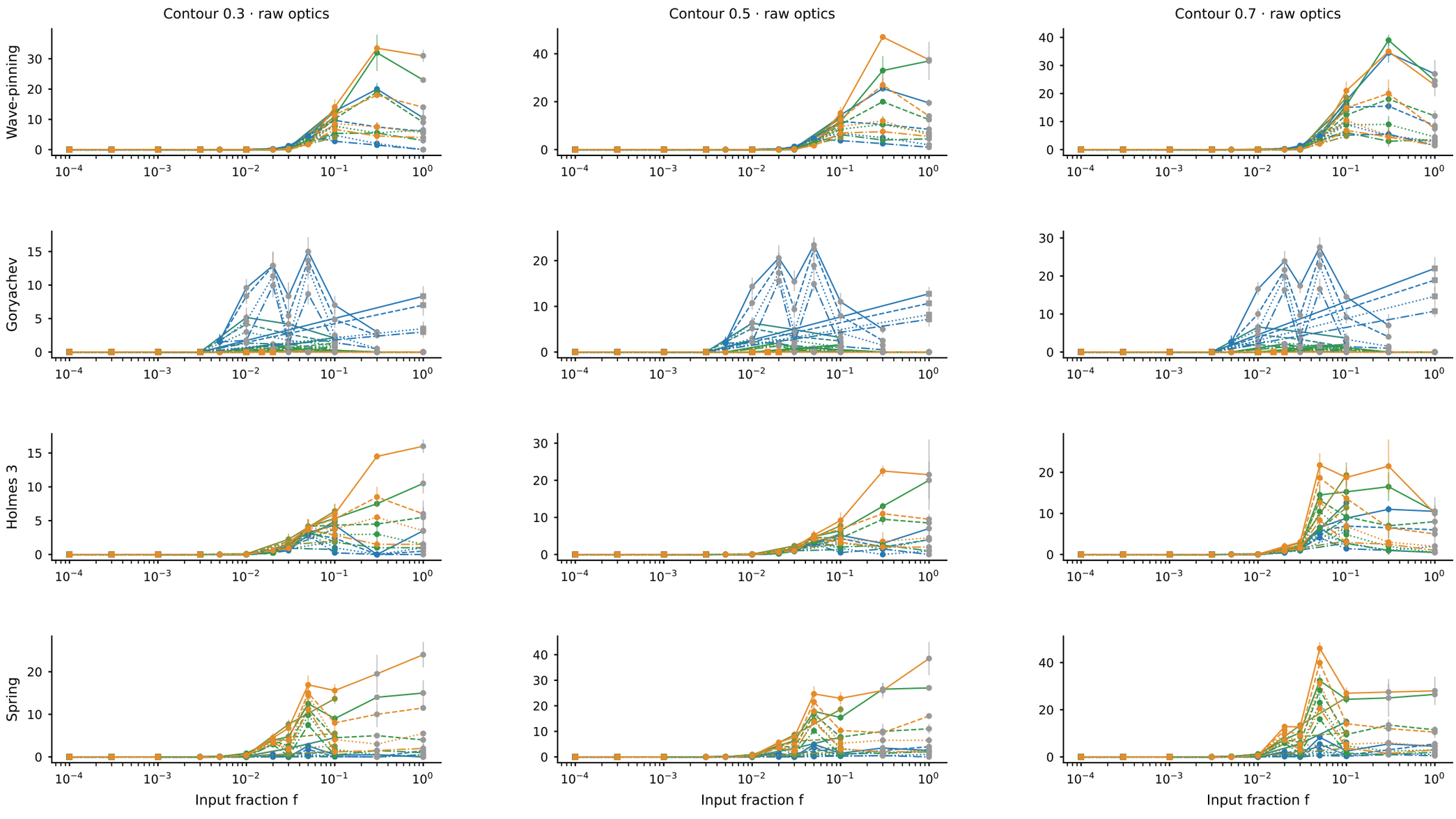
S9 · Longest canonical A5 observations · Literal pairwise takeover sensitivity per trajectory



Longest canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.

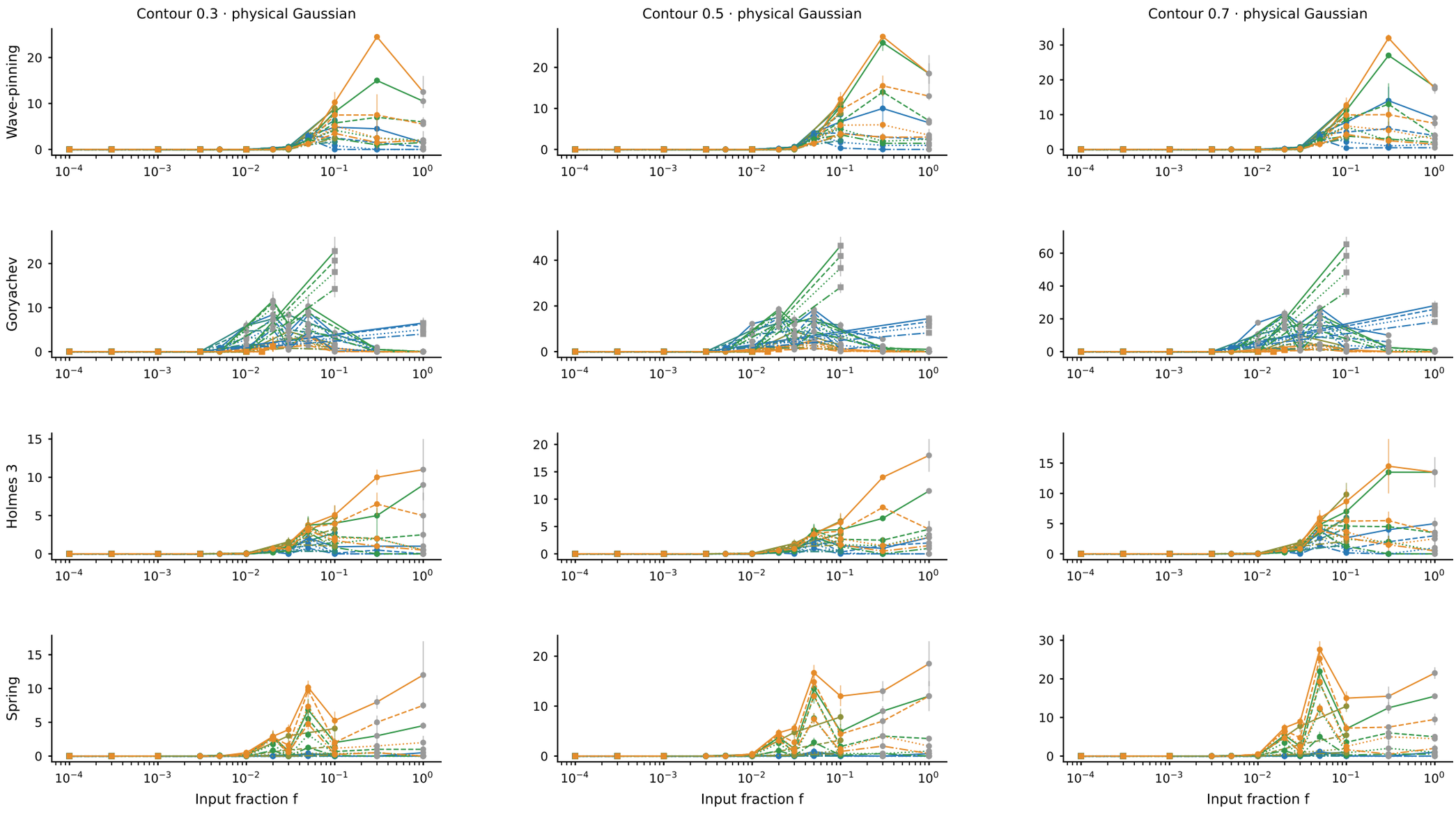
CSV: figure4_S9_longest_canonical_20261020_040_S9_observed_pairwise_takeovers.csv

S9 · Longest canonical A5 observations · Literal pairwise takeover sensitivity per trajectory · full native-factor sensitivity



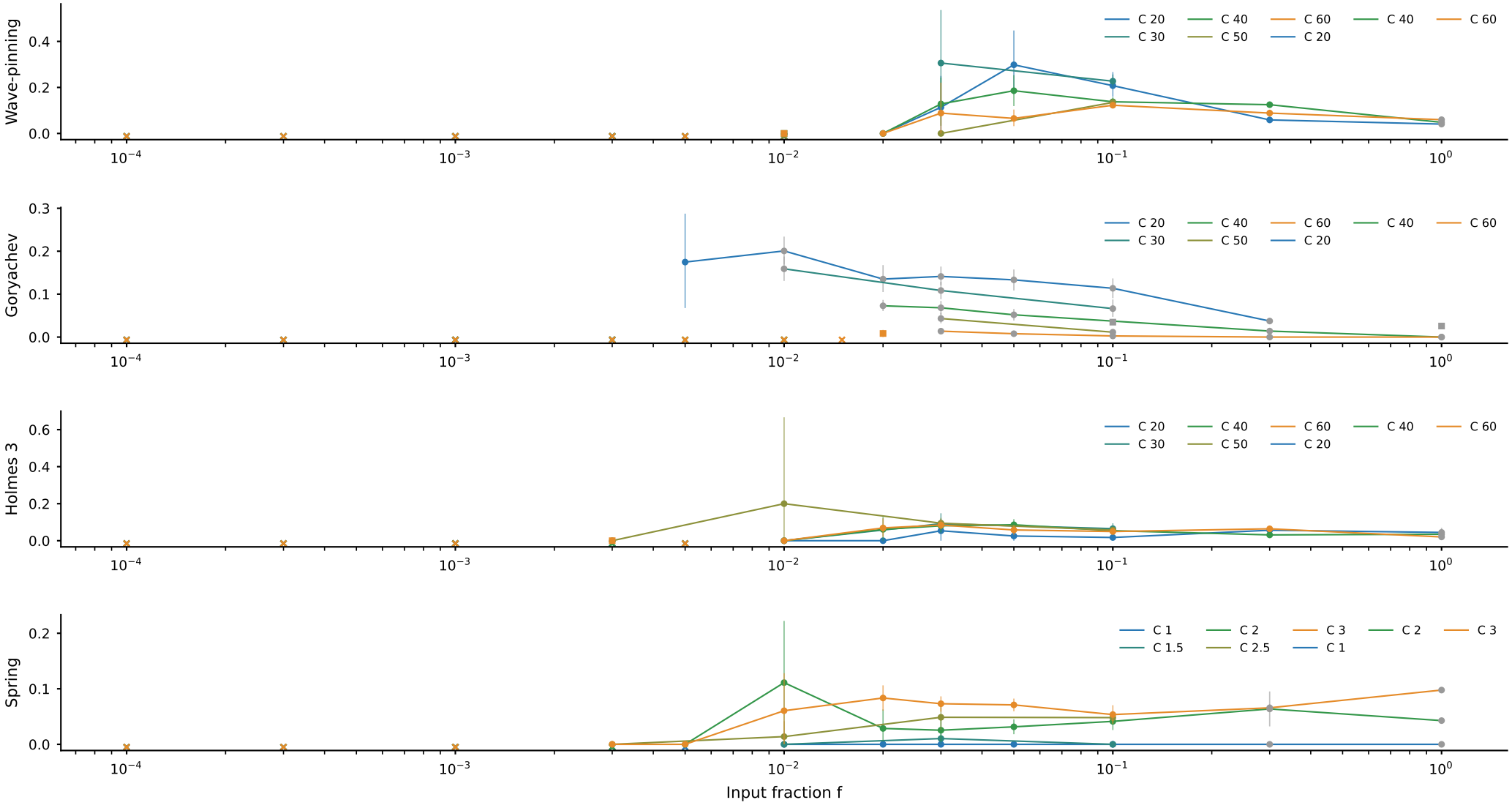
Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds.
× at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_041_S9_observed_pairwise_takeovers_raw_factorial.csv

S9 · Longest canonical A5 observations · Literal pairwise takeover sensitivity per trajectory · full native-factor sensitivity



Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_042_S9_observed_pairwise_takeovers_smoothed_factorial.csv

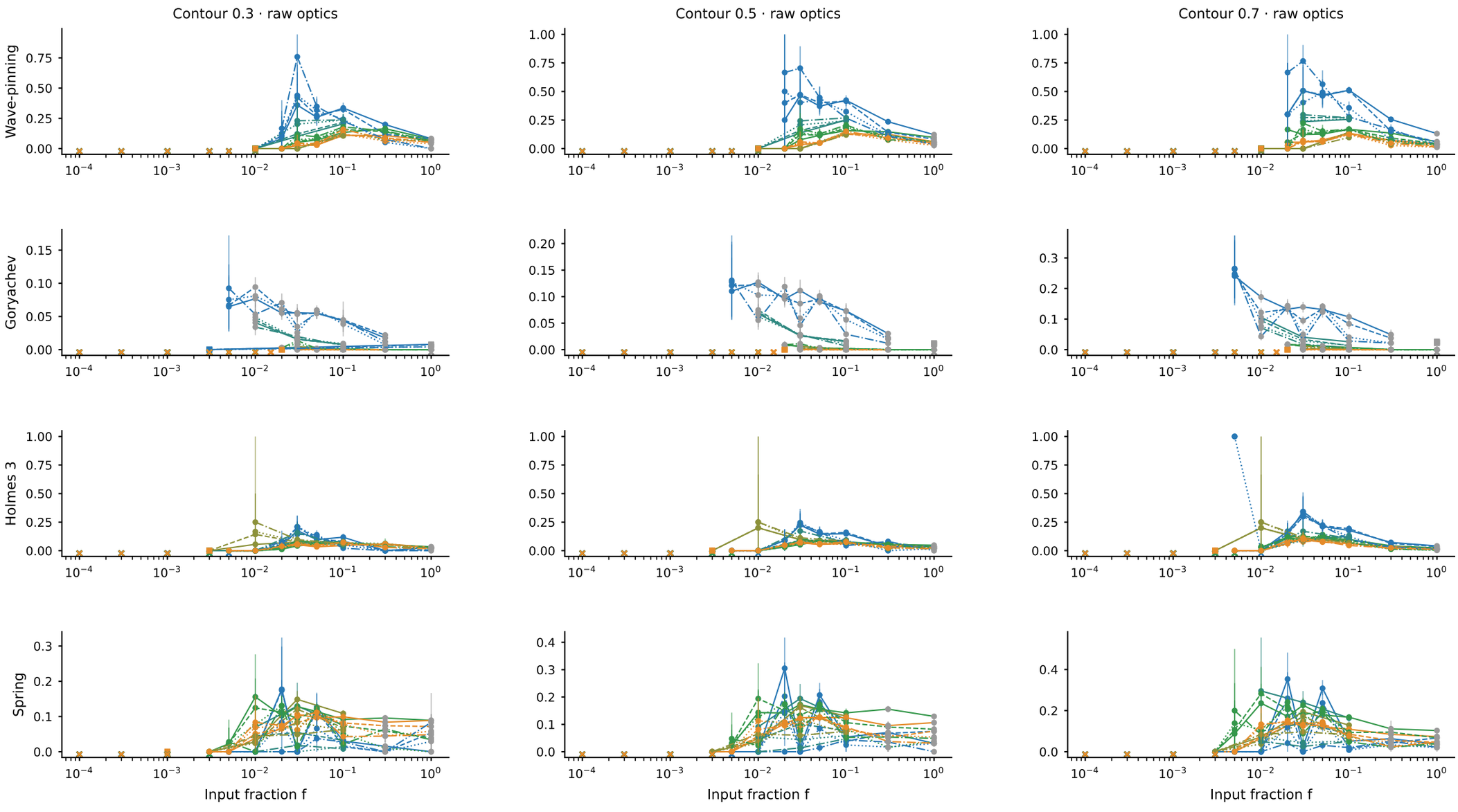
S9 · Longest canonical A5 observations · Literal pairwise takeover sensitivity per birth



Longest canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.

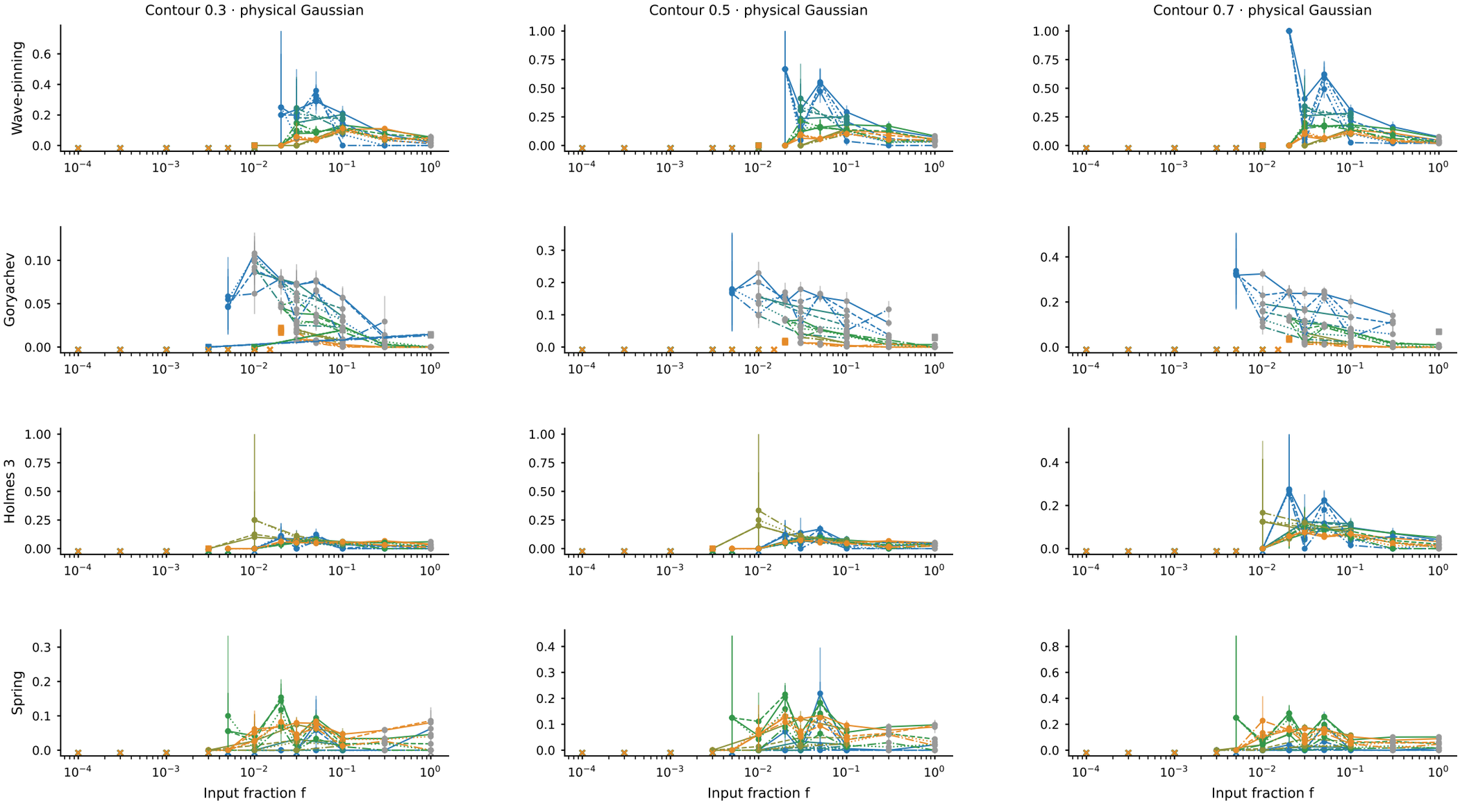
CSV: figure4_S9_longest_canonical_20261020_043_S9_observed_pairwise_takeovers_per_birth.csv

S9 · Longest canonical A5 observations · Literal pairwise takeover sensitivity per birth · full native-factor sensitivity



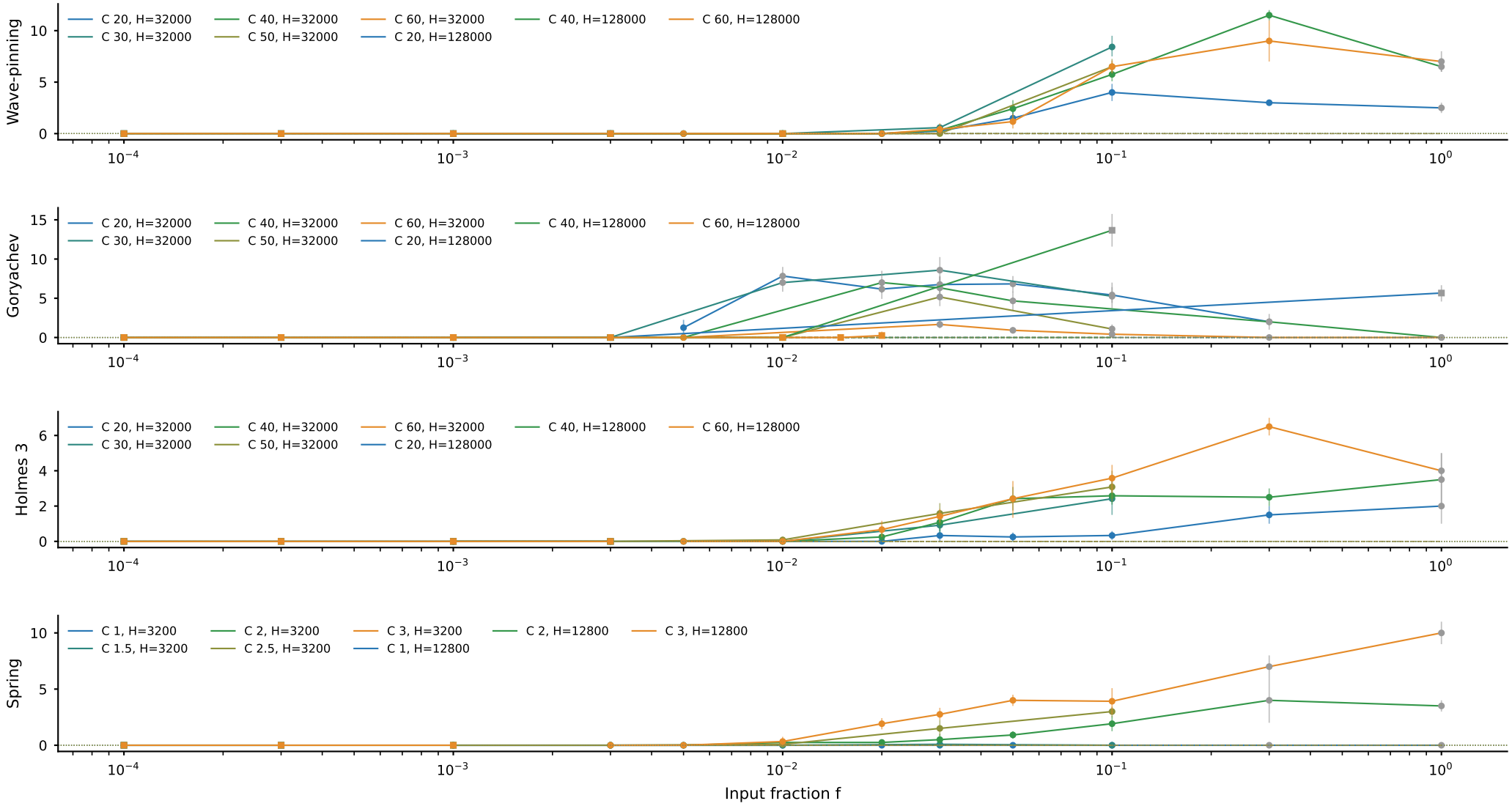
Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds.
× at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_044_S9_observed_pairwise_takeovers_per_birth_raw_factorial.csv

S9 · Longest canonical A5 observations · Literal pairwise takeover sensitivity per birth · full native-factor sensitivity



Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds.
× at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_045_S9_observed_pairwise_takeovers_per_birth_smoothed_factorial.csv

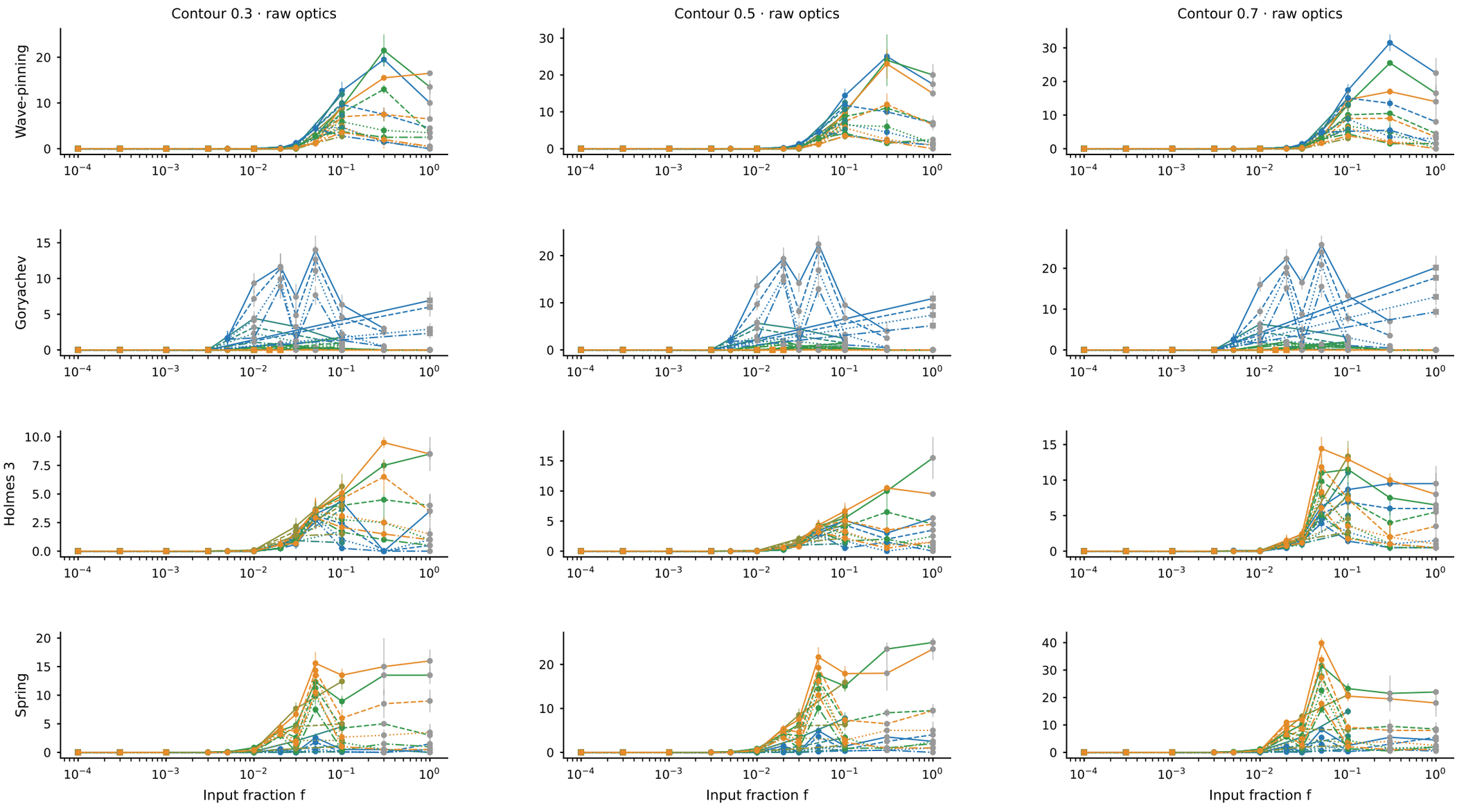
S9 · Longest canonical A5 observations · Observed globally dominant newborn takeovers per trajectory



Longest canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.

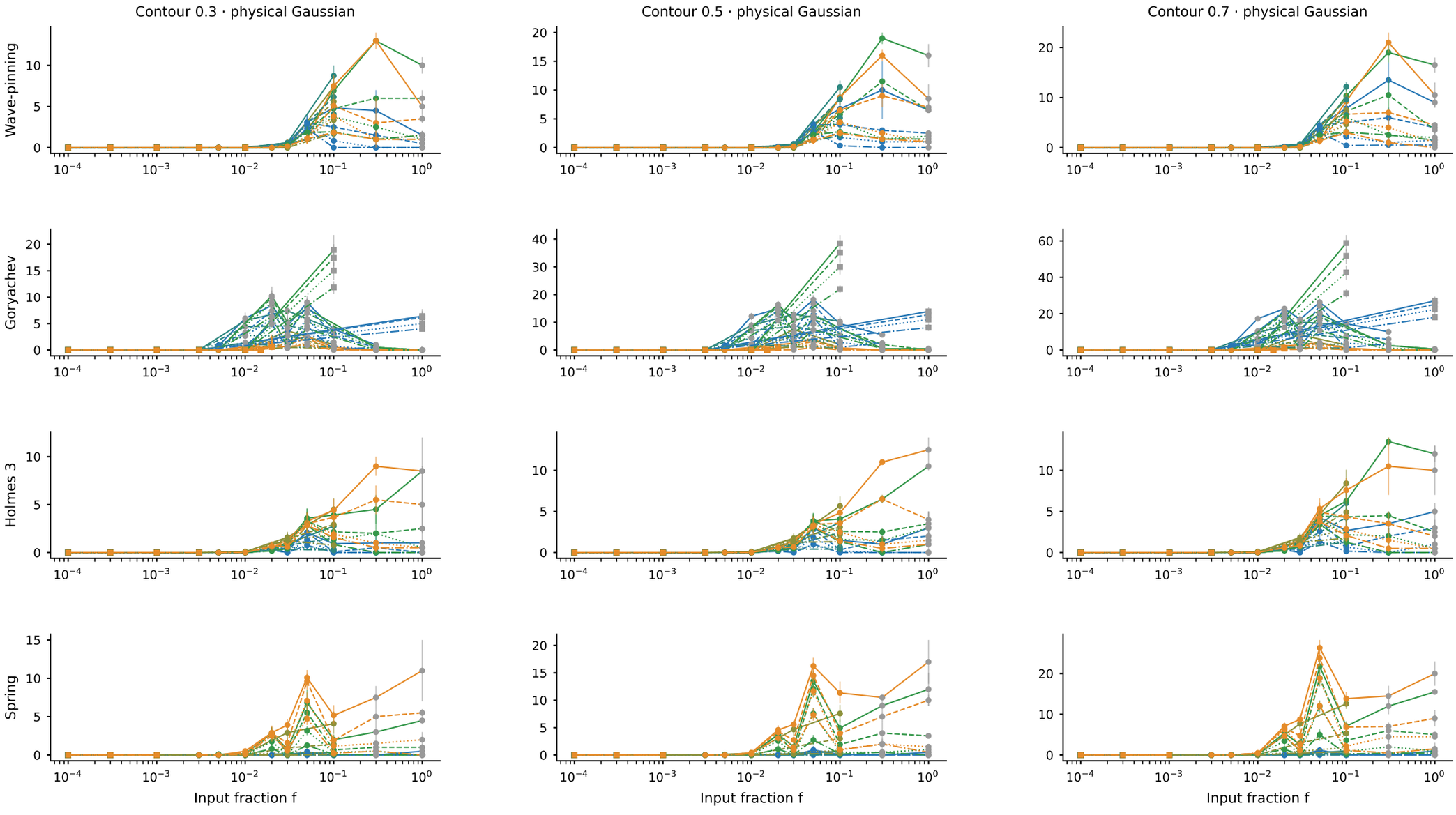
CSV: figure4_S9_longest_canonical_20261020_046_S9_observed_takeovers.csv

S9 · Longest canonical A5 observations · Observed globally dominant newborn takeovers per trajectory · full native-factor sensitivity



Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_047_S9_observed_takeovers_raw_factorial.csv

S9 · Longest canonical A5 observations · Observed globally dominant newborn takeovers per trajectory · full native-factor sensitivity



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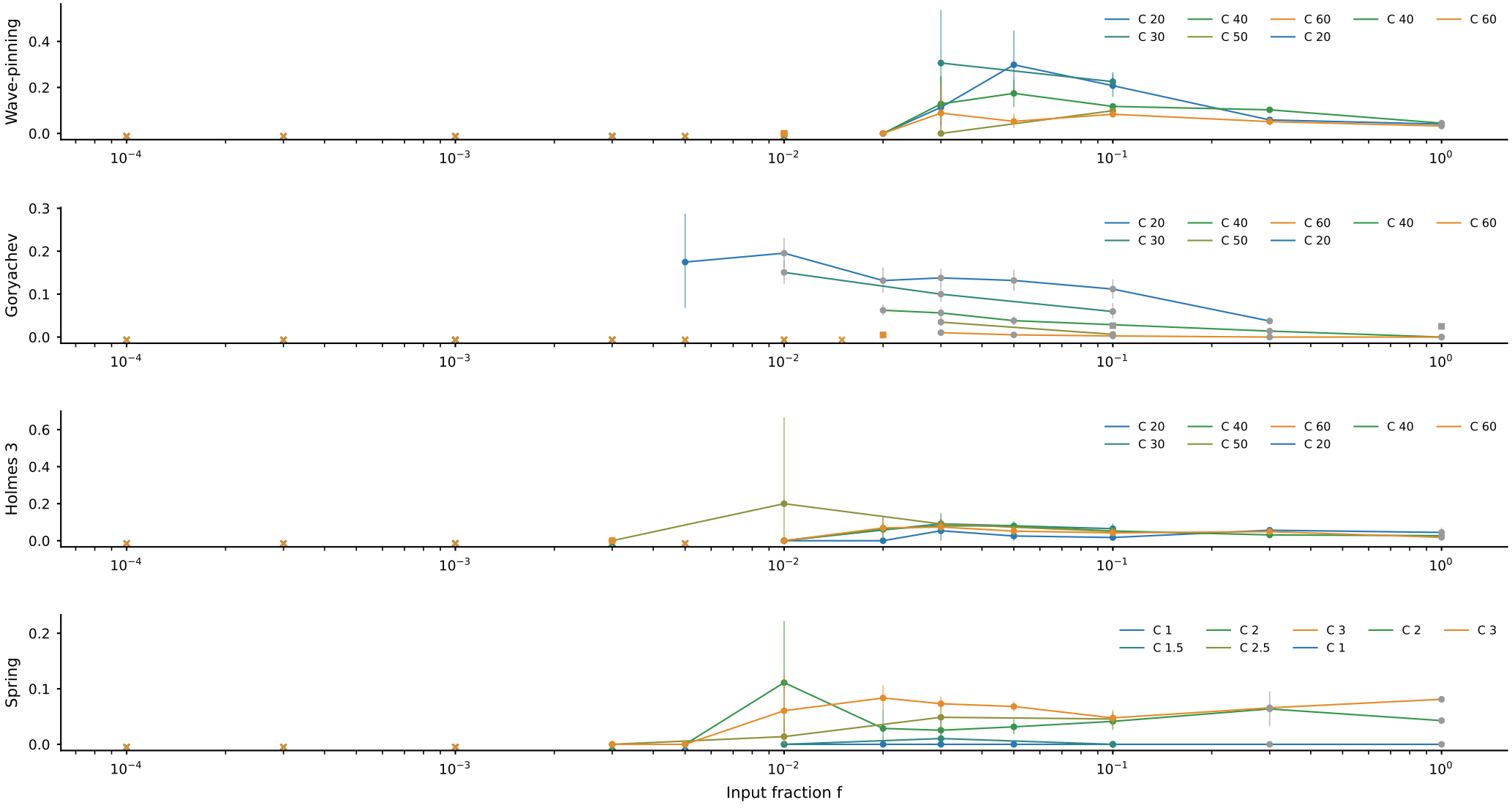
CSV: figure4_S9_longest_canonical_20261020_048_S9_observed_takeovers_smoothed_factorial.csv

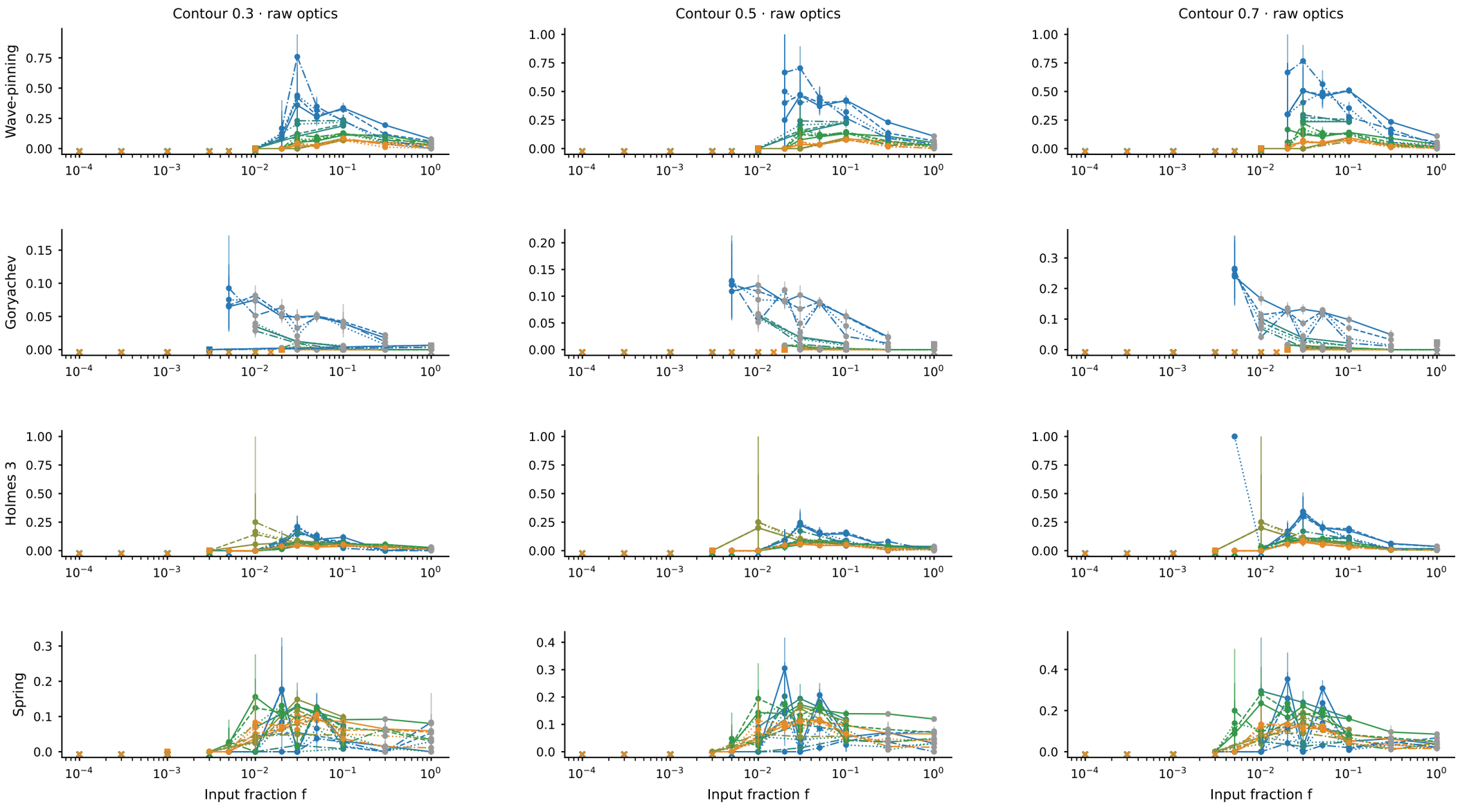
S9 · Longest canonical A5 observations · Globally dominant newborn takeovers per birth



Longest canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.

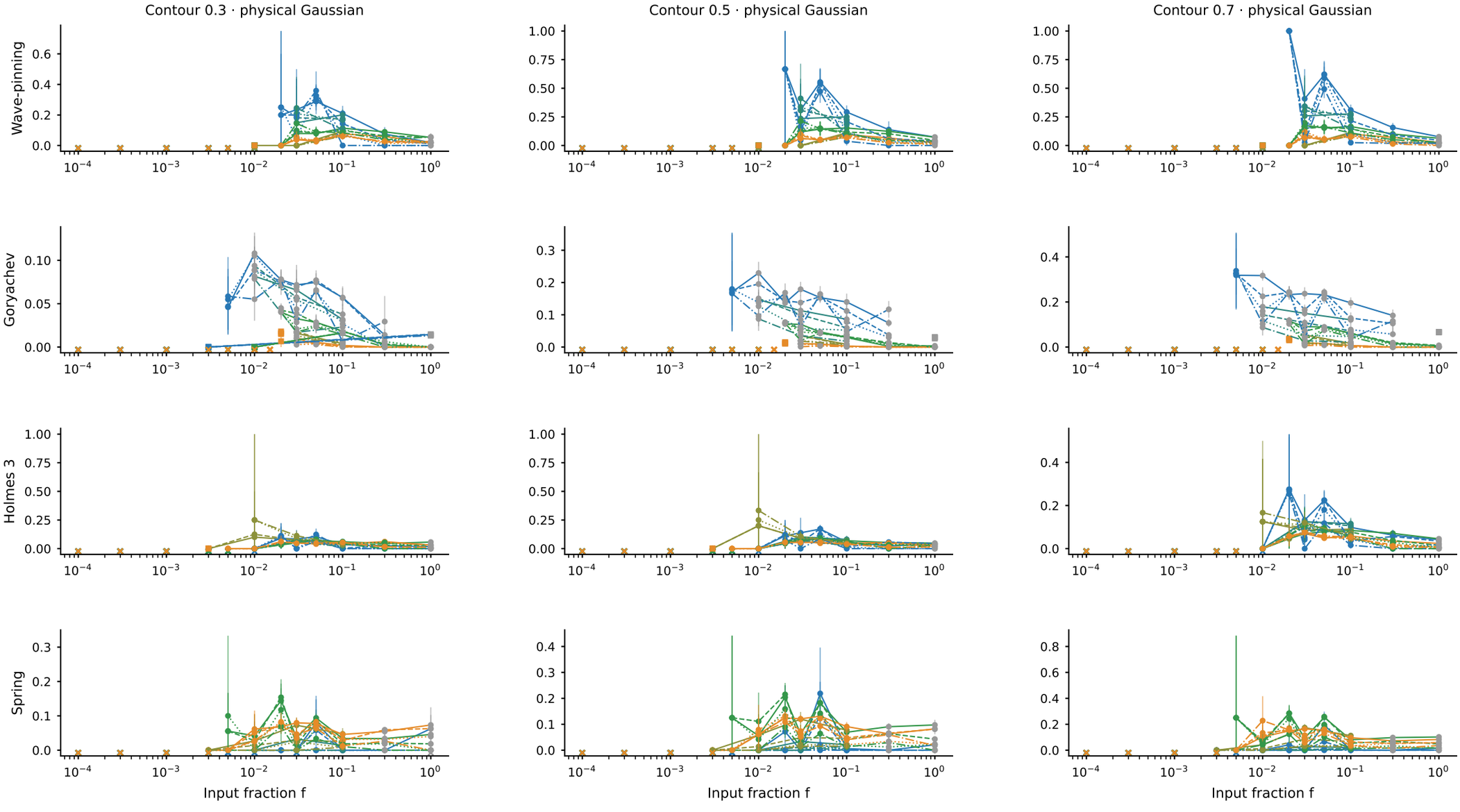
CSV: figure4_S9_longest_canonical_20261020_049_S9_observed_takeovers_per_birth.csv

S9 · Longest canonical A5 observations · Globally dominant newborn takeovers per birth · full native-factor sensitivity



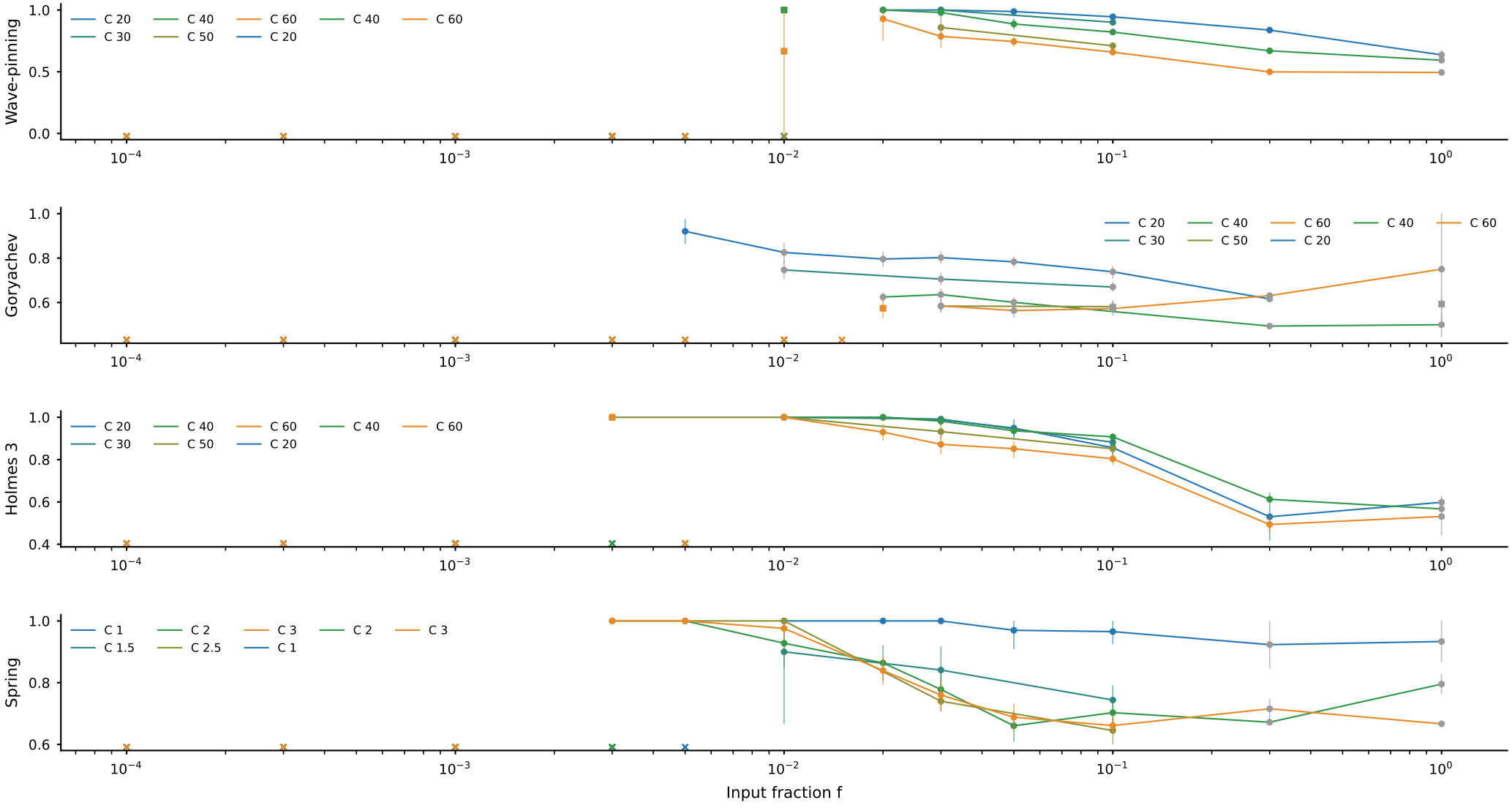
Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds.
× at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_050_S9_observed_takeovers_per_birth_raw_factorial.csv

S9 · Longest canonical A5 observations · Globally dominant newborn takeovers per birth · full native-factor sensitivity



Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds.
× at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_051_S9_observed_takeovers_per_birth_smoothed_factorial.csv

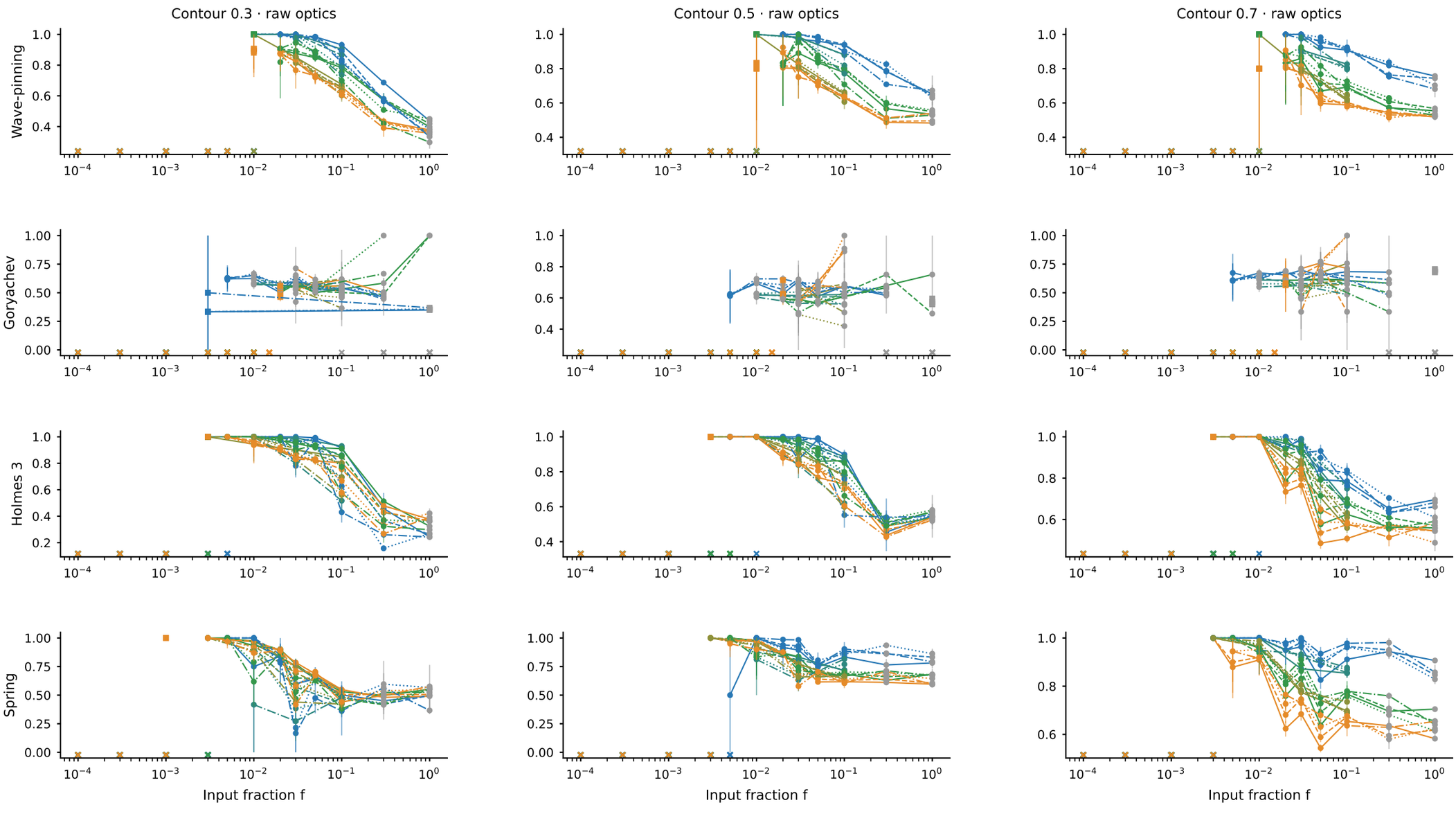
S9 · Longest canonical A5 observations · Fraction of births opposite the previous dominant region



Longest canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.

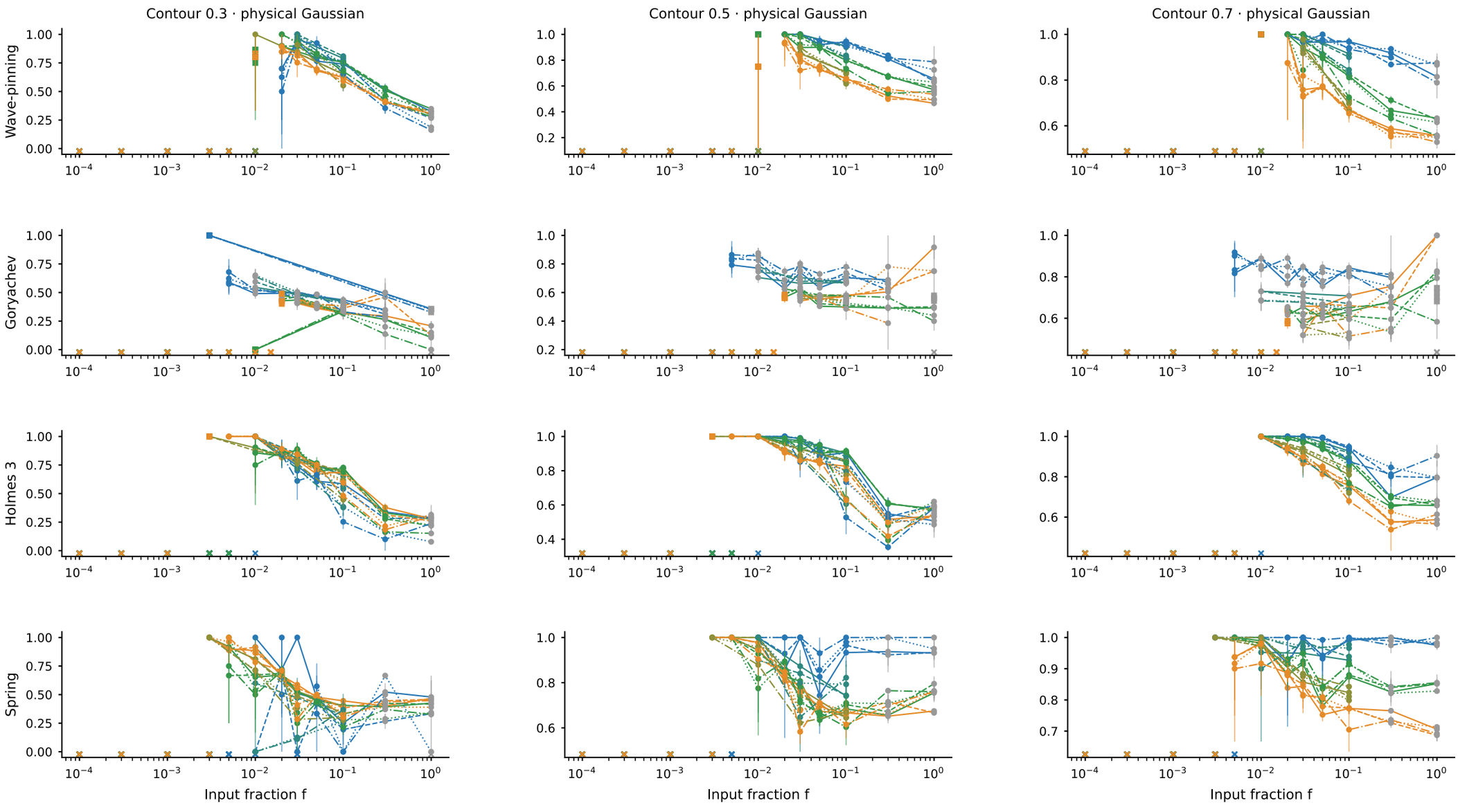
CSV: figure4_S9_longest_canonical_20261020_052_S9_opposite_birth_fraction.csv

S9 · Longest canonical A5 observations · Fraction of births opposite the previous dominant region · full native-factor sensitivity



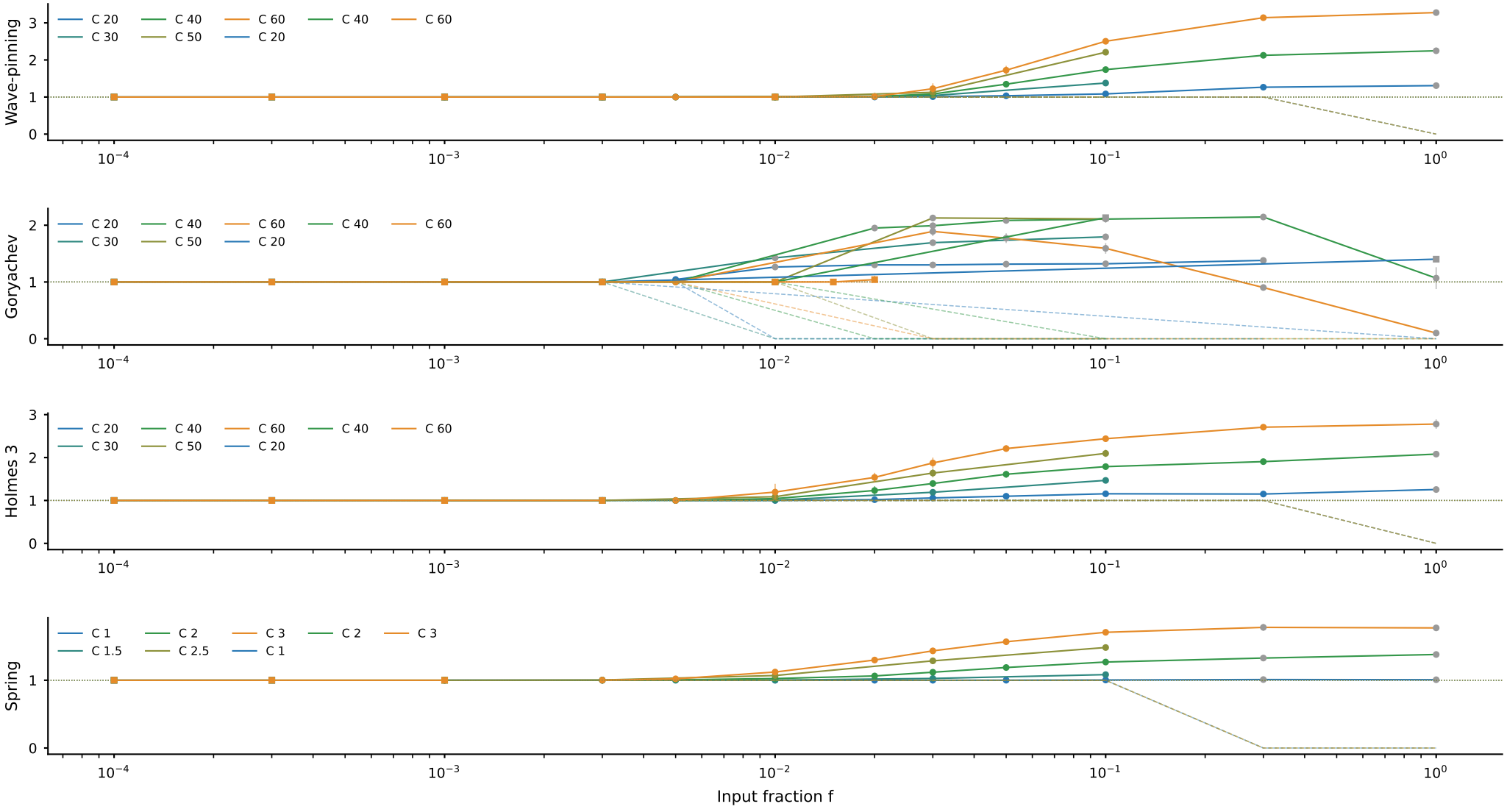
Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds.
× at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_053_S9_opposite_birth_fraction_raw_factorial.csv

S9 · Longest canonical A5 observations · Fraction of births opposite the previous dominant region · full native-factor sensitivity



Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds.
× at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_054_S9_opposite_birth_fraction_smoothed_factorial.csv

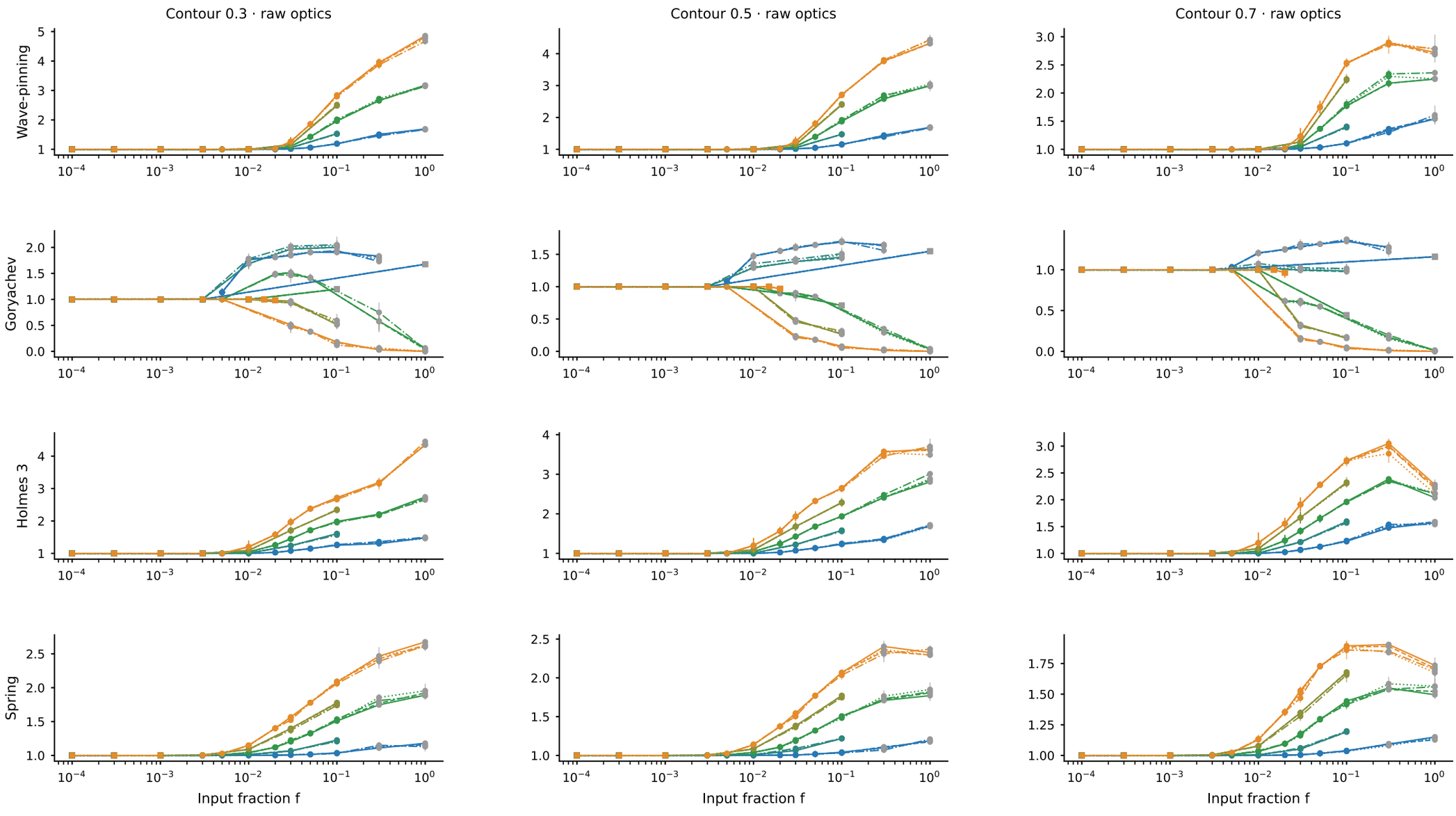
S9 · Longest canonical A5 observations · Mean simultaneous qualified-region count



Longest canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.

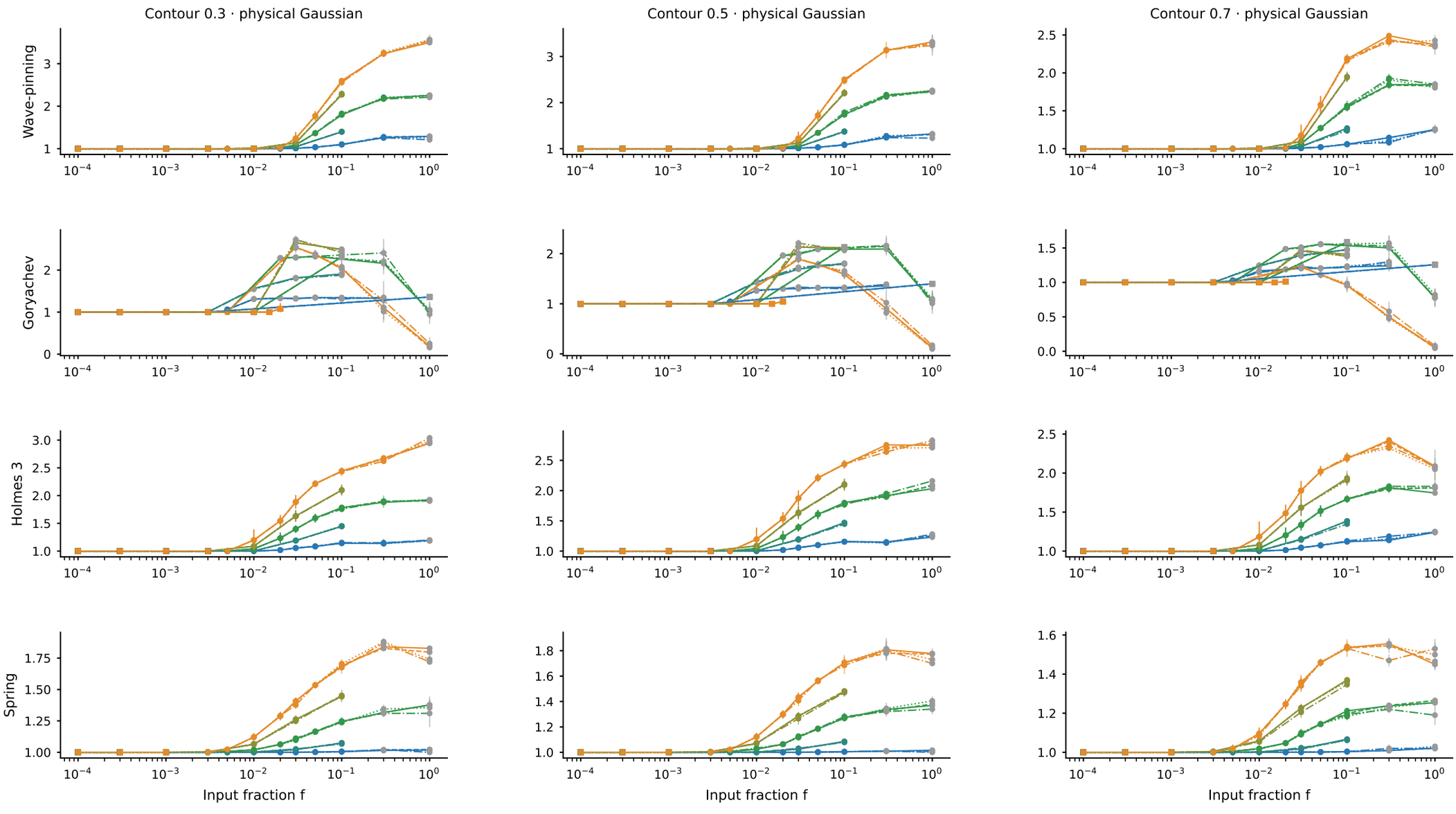
CSV: figure4_S9_longest_canonical_20261020_055_S9_region_count_mean.csv

S9 · Longest canonical A5 observations · Mean simultaneous qualified-region count · full native-factor sensitivity



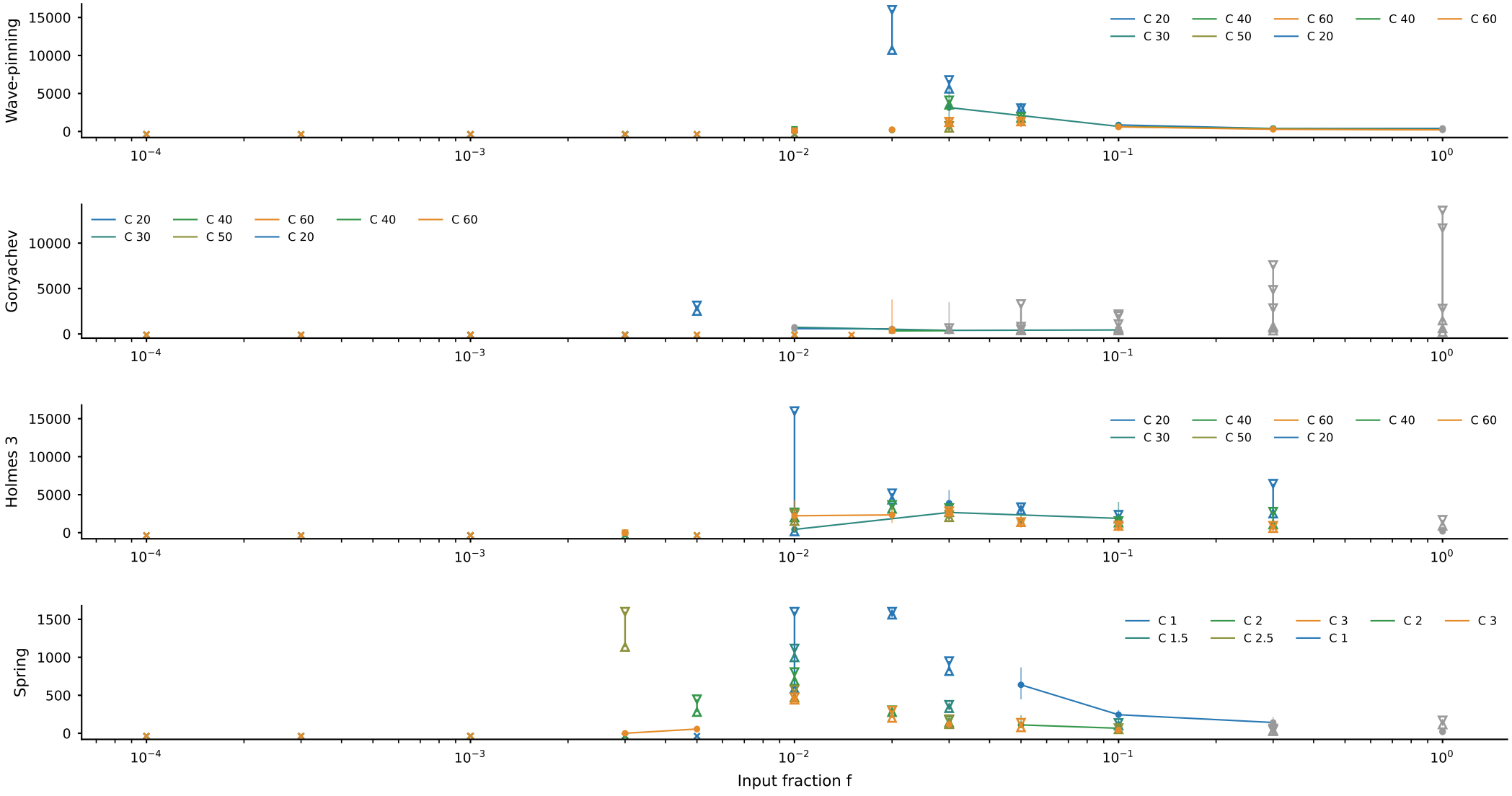
Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds.
× at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_056_S9_region_count_mean_raw_factorial.csv

S9 · Longest canonical A5 observations · Mean simultaneous qualified-region count · full native-factor sensitivity



Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds.
× at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_057_S9_region_count_mean_smoothed_factorial.csv

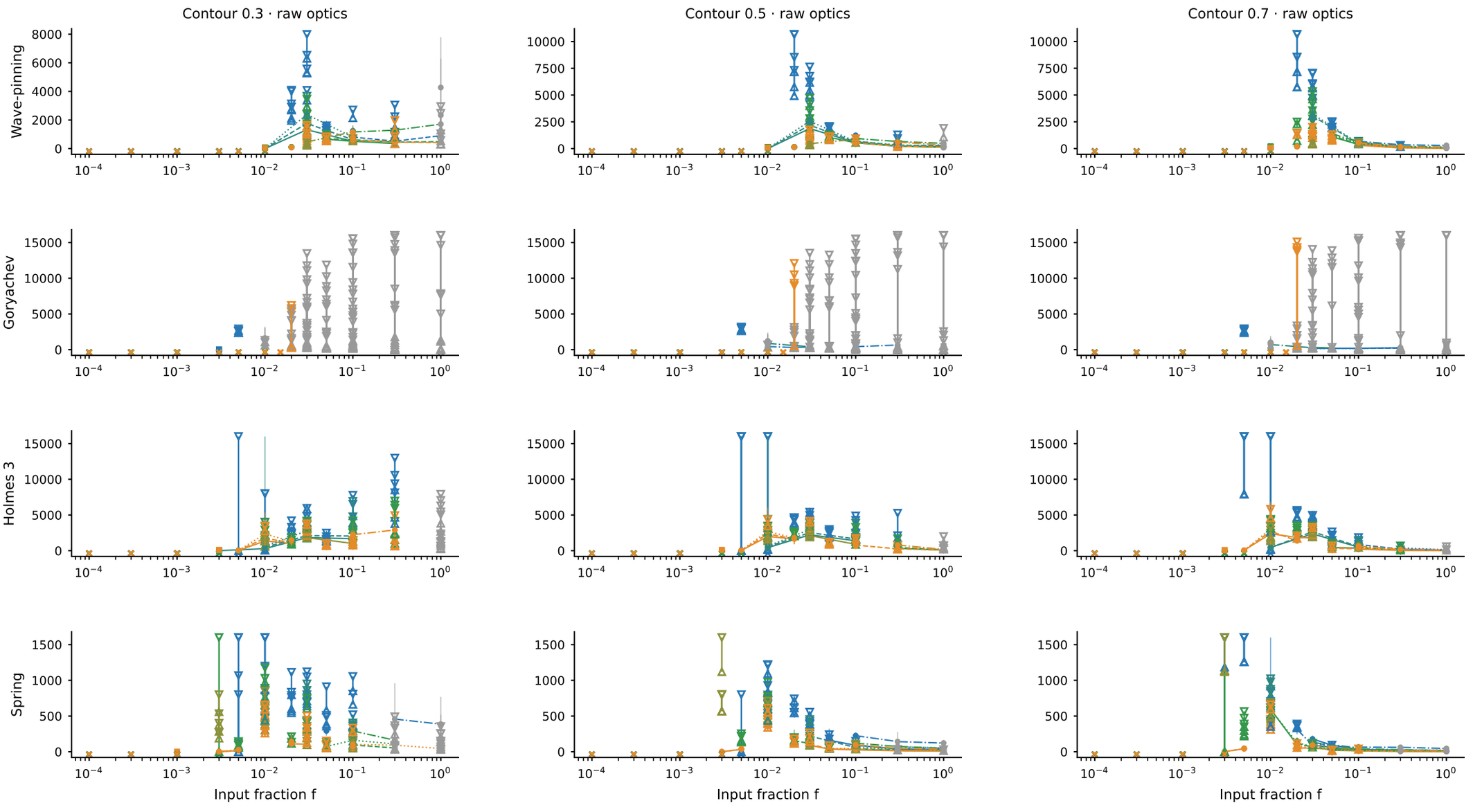
S9 · Longest canonical A5 observations · Lower-censoring restricted mean lifetime



Longest canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.

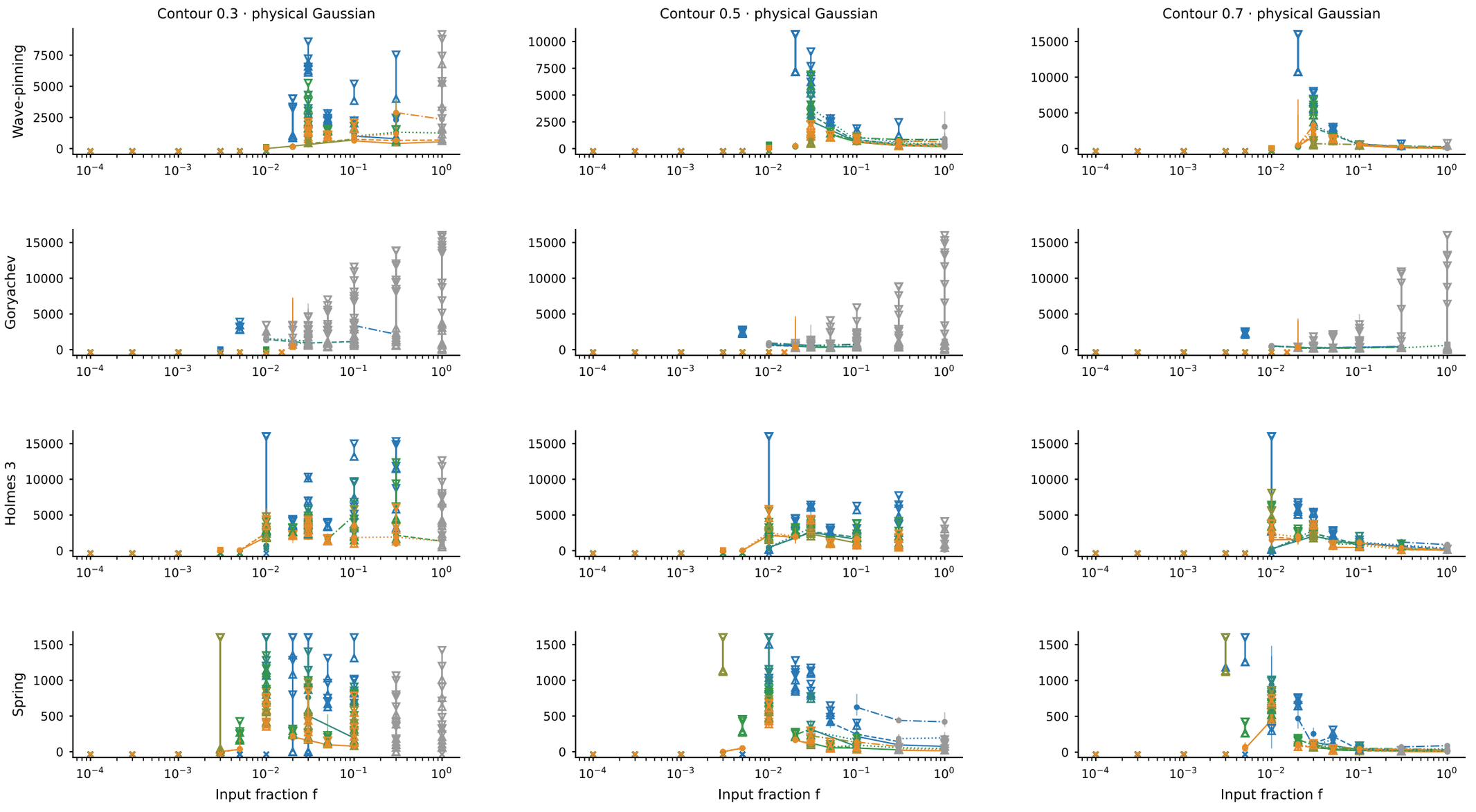
CSV: figure4_S9_longest_canonical_20261020_058_S9_restricted_mean_lifetime_lower.csv

S9 · Longest canonical A5 observations · Lower-censoring restricted mean lifetime · full native-factor sensitivity



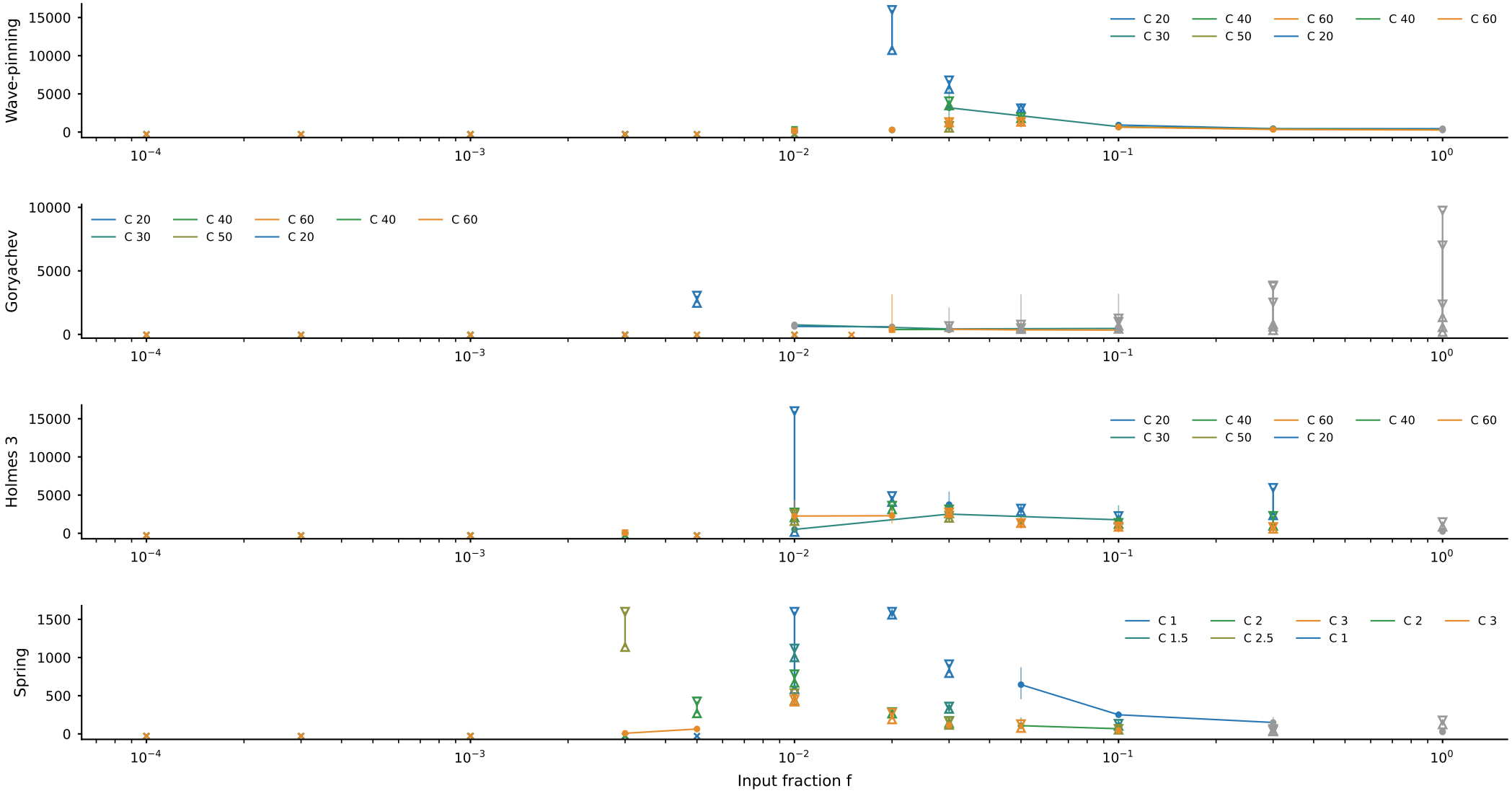
Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_059_S9_restricted_mean_lifetime_lower_raw_factorial.csv

S9 · Longest canonical A5 observations · Lower-censoring restricted mean lifetime · full native-factor sensitivity



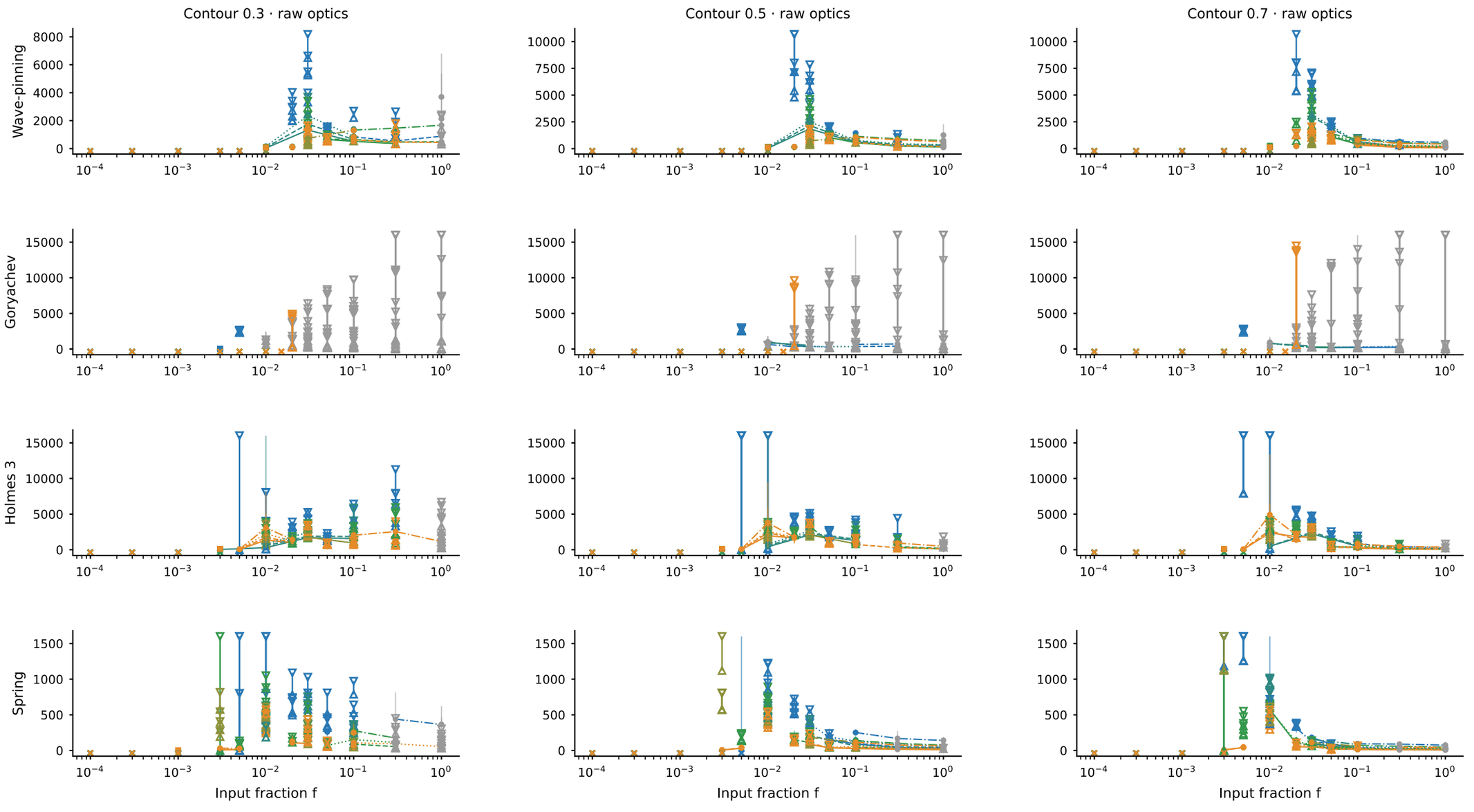
Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_060_S9_restricted_mean_lifetime_lower_smoothed_factorial.csv

S9 · Longest canonical A5 observations · Observed-frame restricted mean lifetime



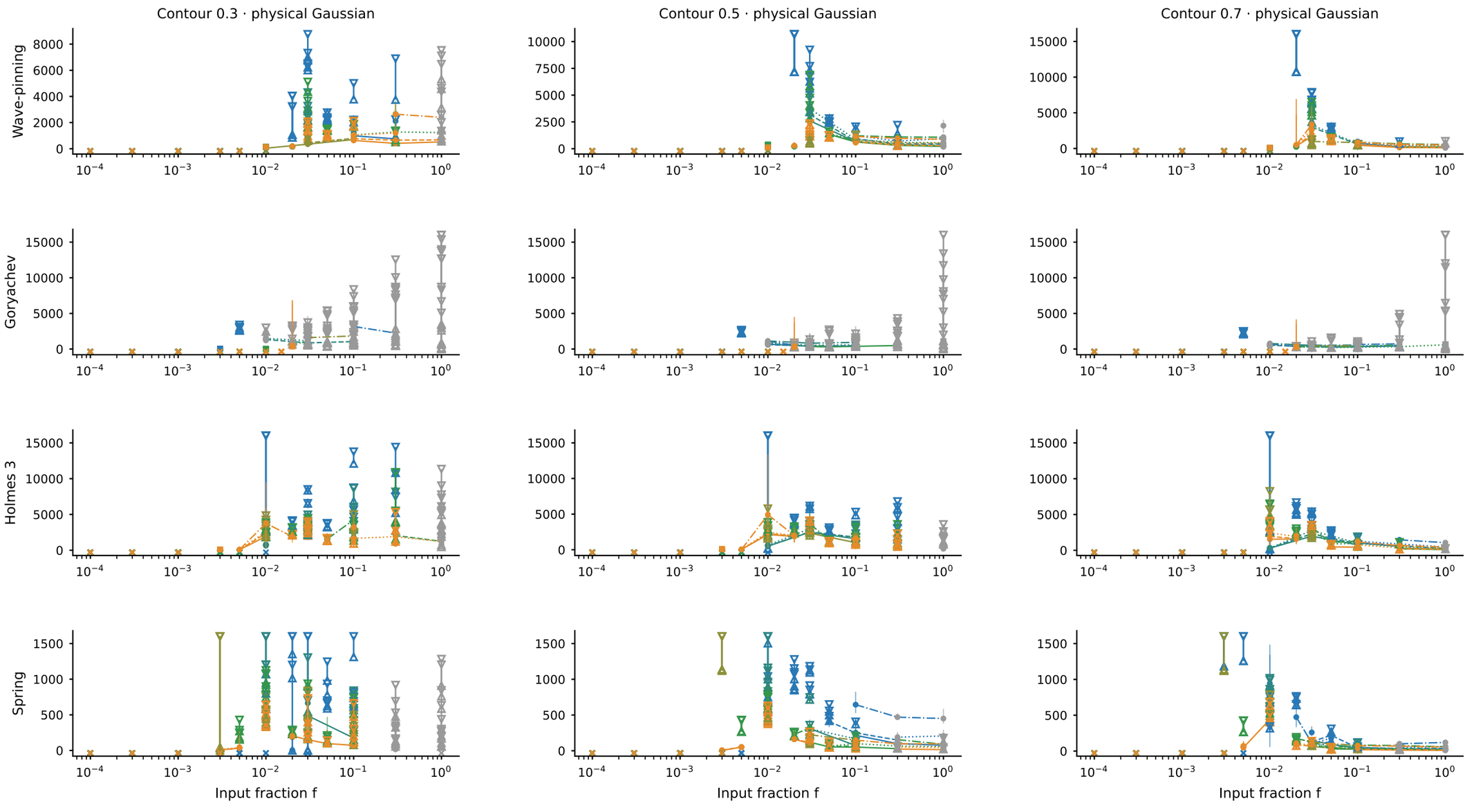
Longest canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_061_S9_restricted_mean_lifetime_observed.csv

S9 · Longest canonical A5 observations · Observed-frame restricted mean lifetime · full native-factor sensitivity



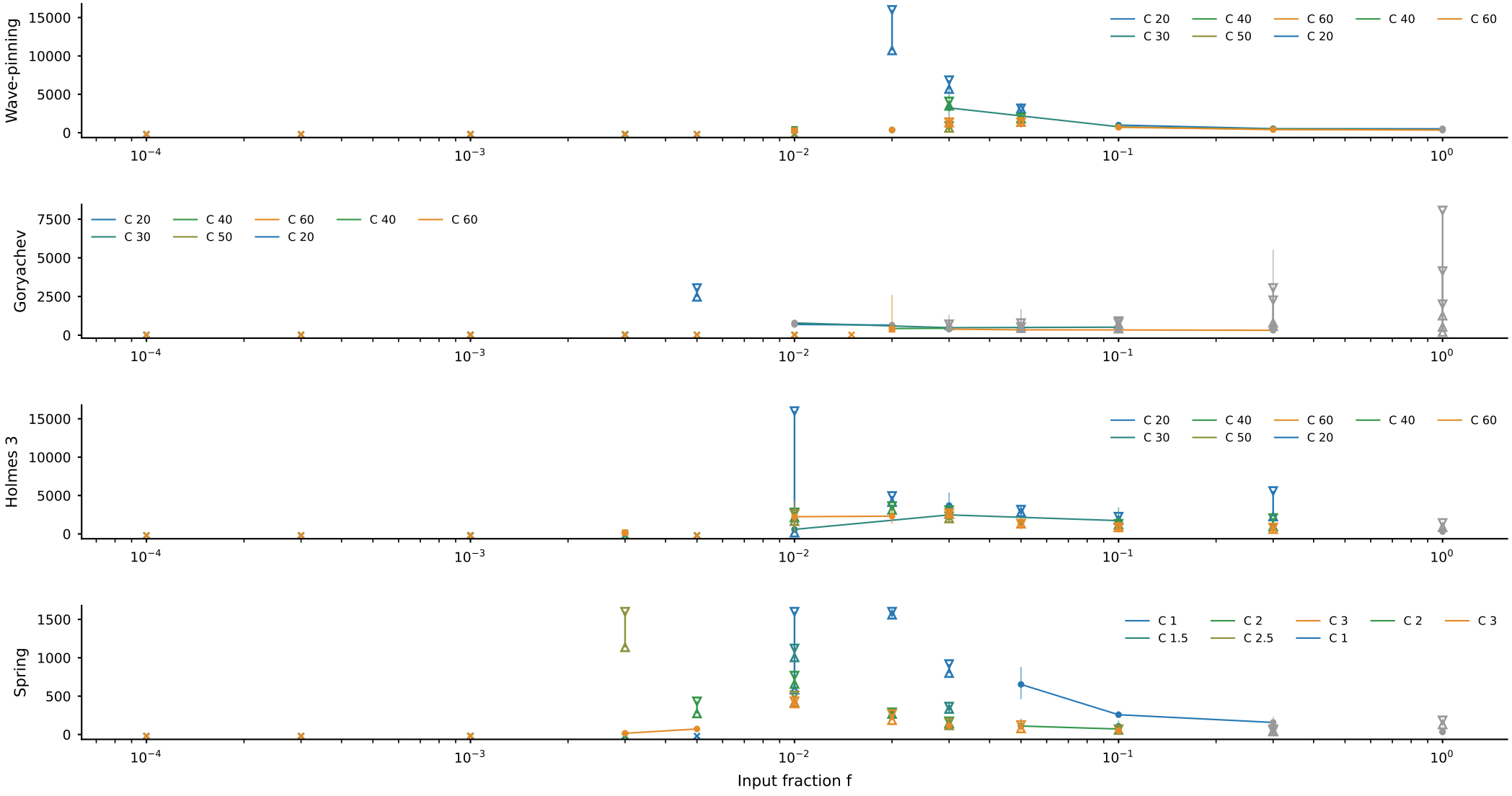
Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_062_S9_restricted_mean_lifetime_observed_raw_factorial.csv

S9 · Longest canonical A5 observations · Observed-frame restricted mean lifetime · full native-factor sensitivity



Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_063_S9_restricted_mean_lifetime_observed_smoothed_factorial.csv

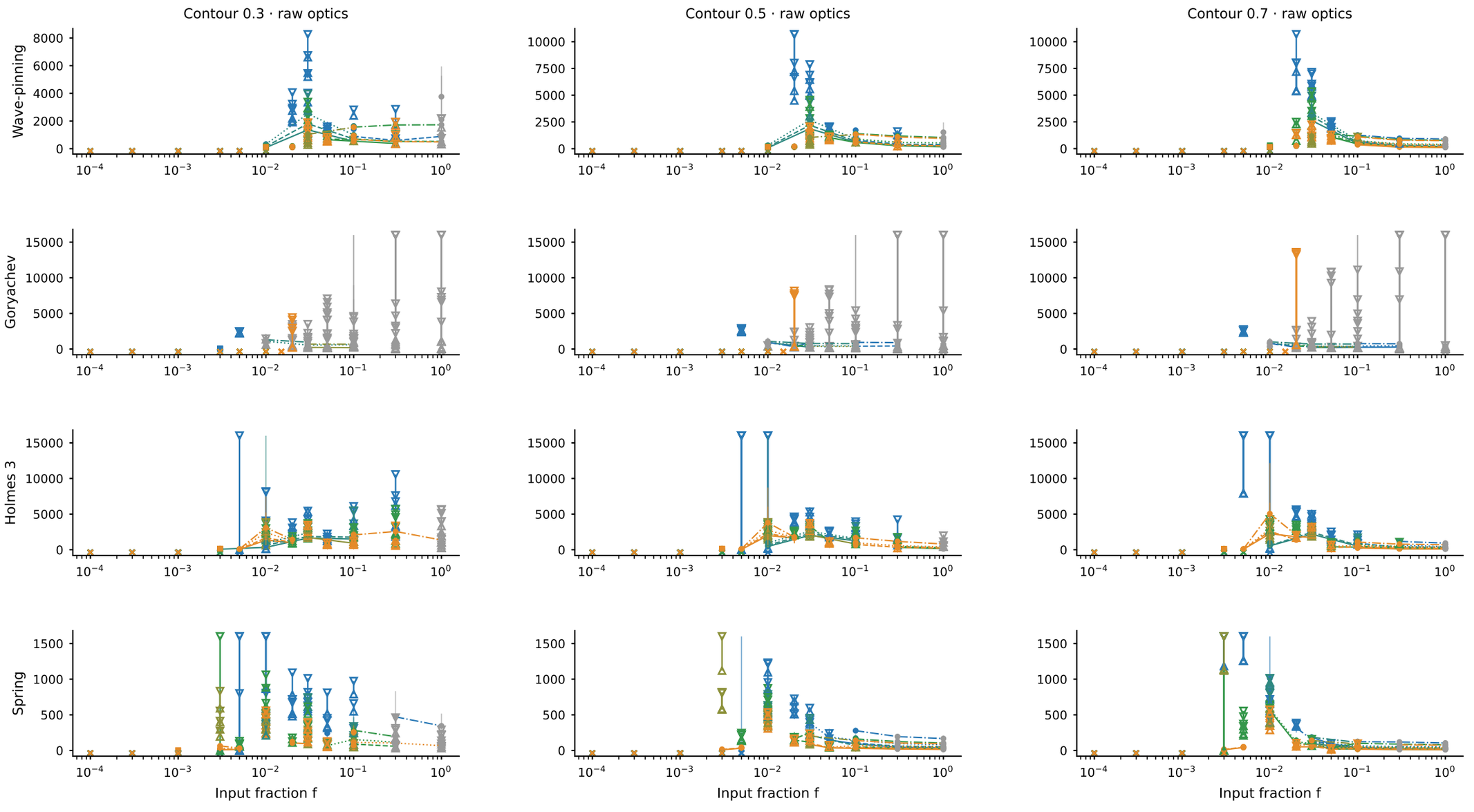
S9 · Longest canonical A5 observations · Upper-censoring restricted mean lifetime



Longest canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.

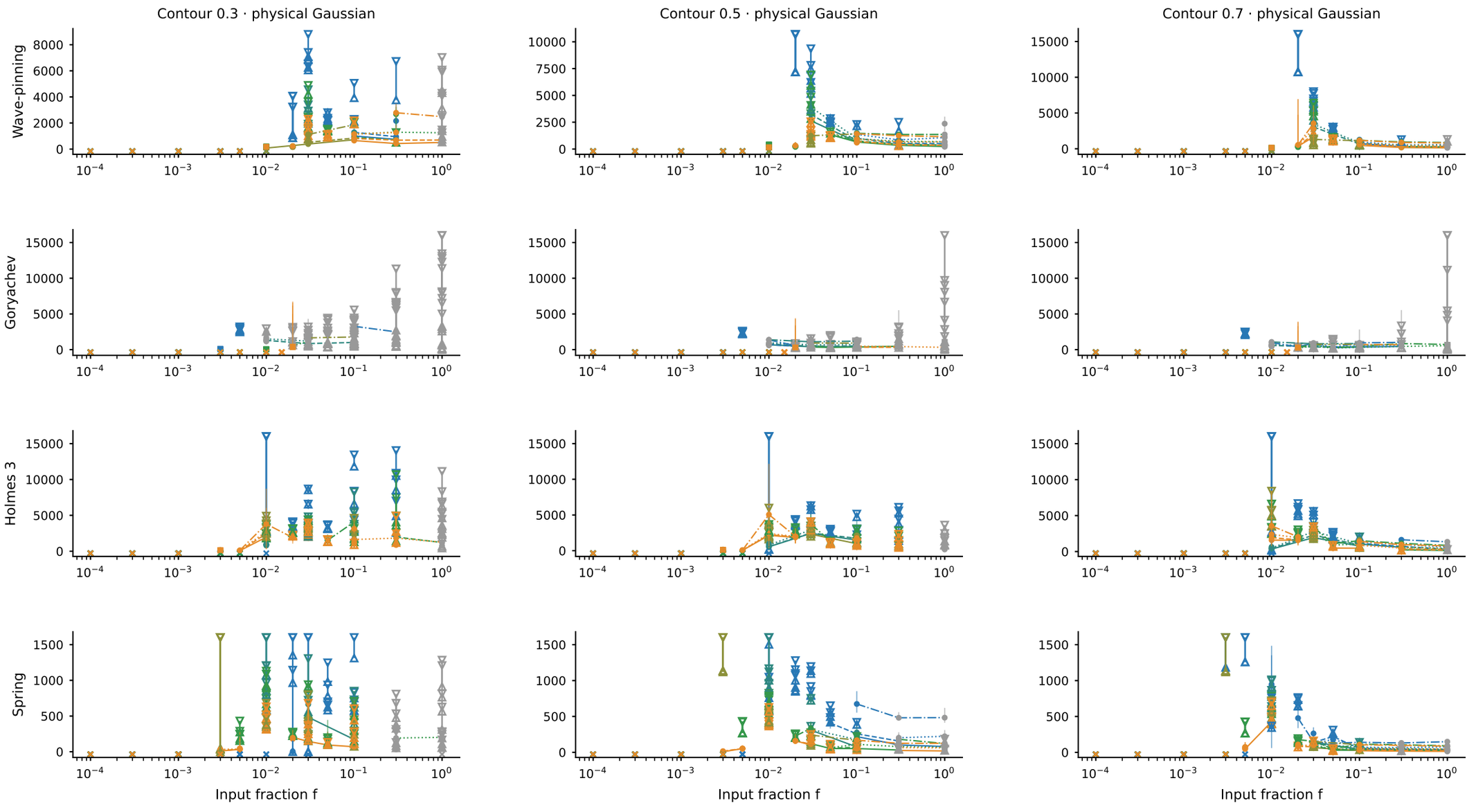
CSV: figure4_S9_longest_canonical_20261020_064_S9_restricted_mean_lifetime_upper.csv

S9 · Longest canonical A5 observations · Upper-censoring restricted mean lifetime · full native-factor sensitivity



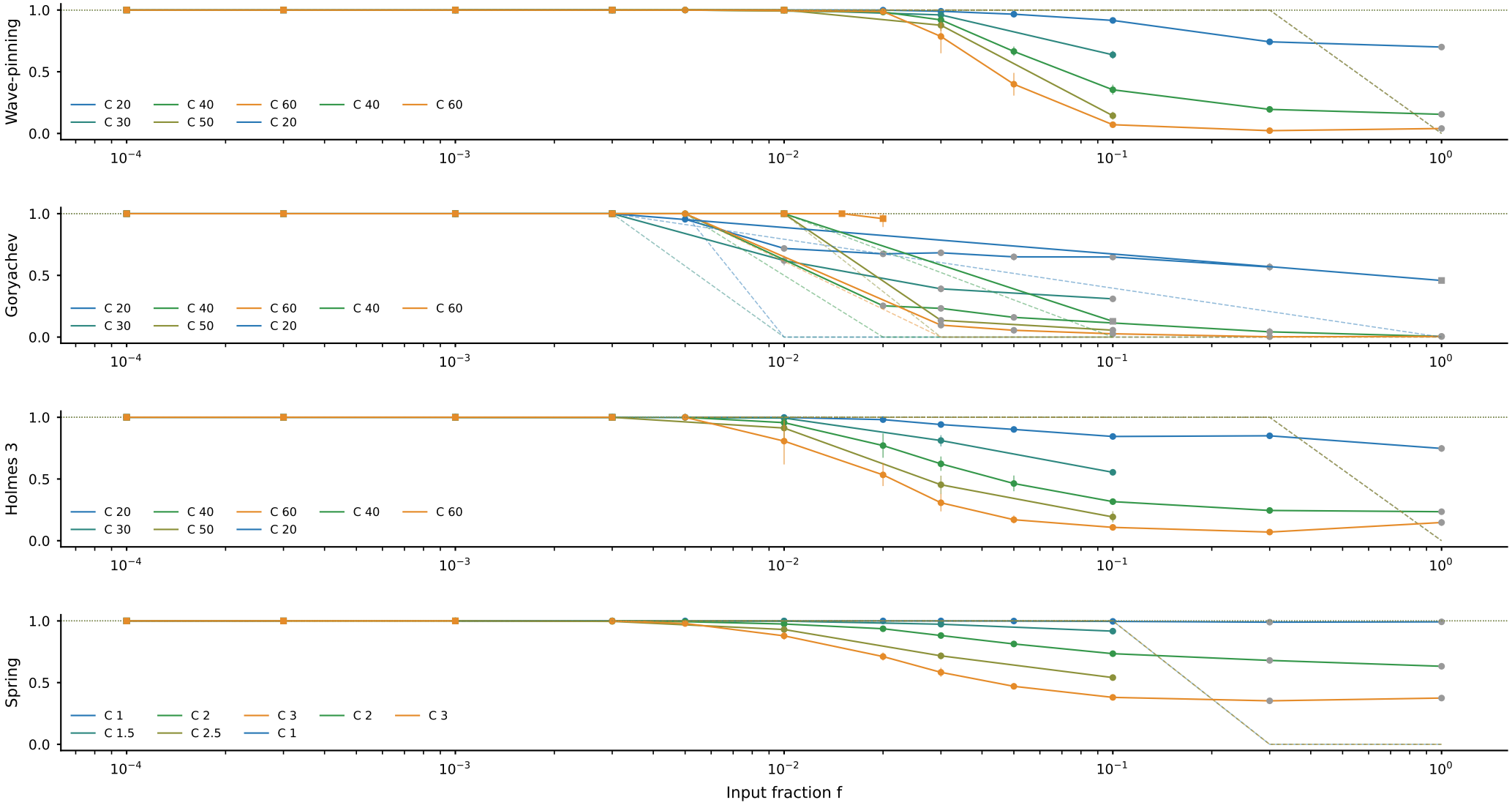
Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds.
× at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_065_S9_restricted_mean_lifetime_upper_raw_factorial.csv

S9 · Longest canonical A5 observations · Upper-censoring restricted mean lifetime · full native-factor sensitivity



Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_066_S9_restricted_mean_lifetime_upper_smoothed_factorial.csv

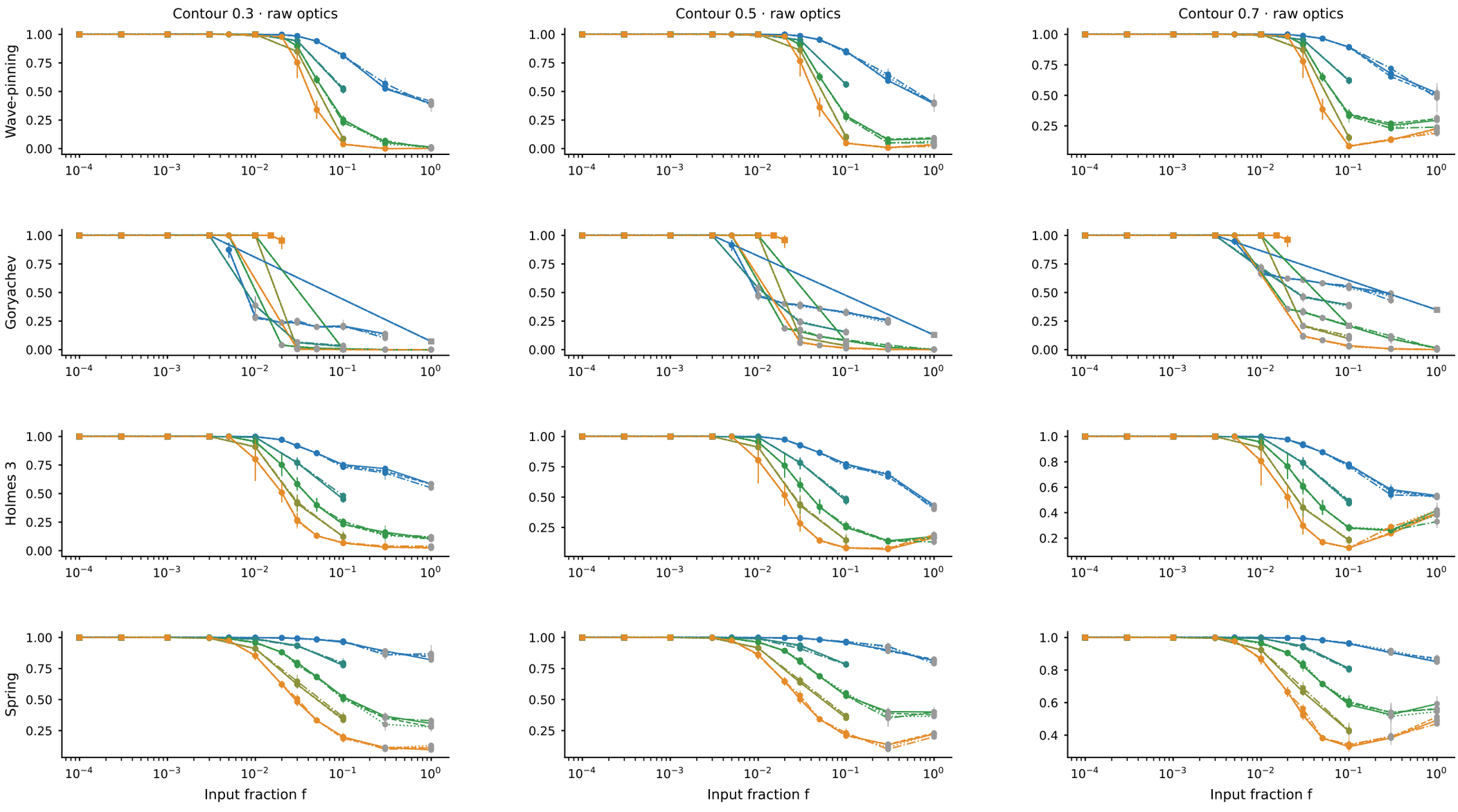
S9 · Longest canonical A5 observations · Time fraction with one qualified region



Longest canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.

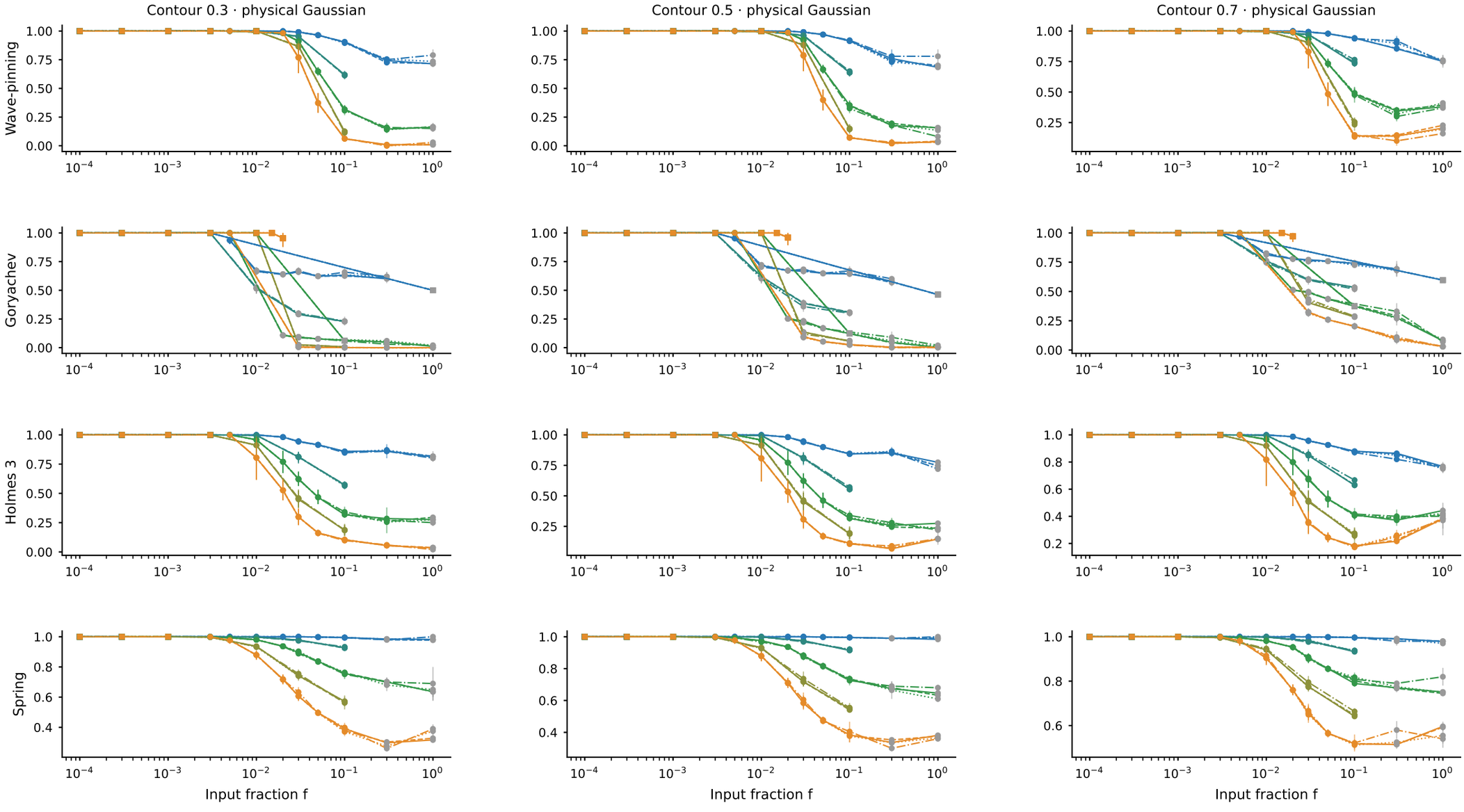
CSV: figure4_S9_longest_canonical_20261020_067_S9_single_front_fraction.csv

S9 · Longest canonical A5 observations · Time fraction with one qualified region · full native-factor sensitivity



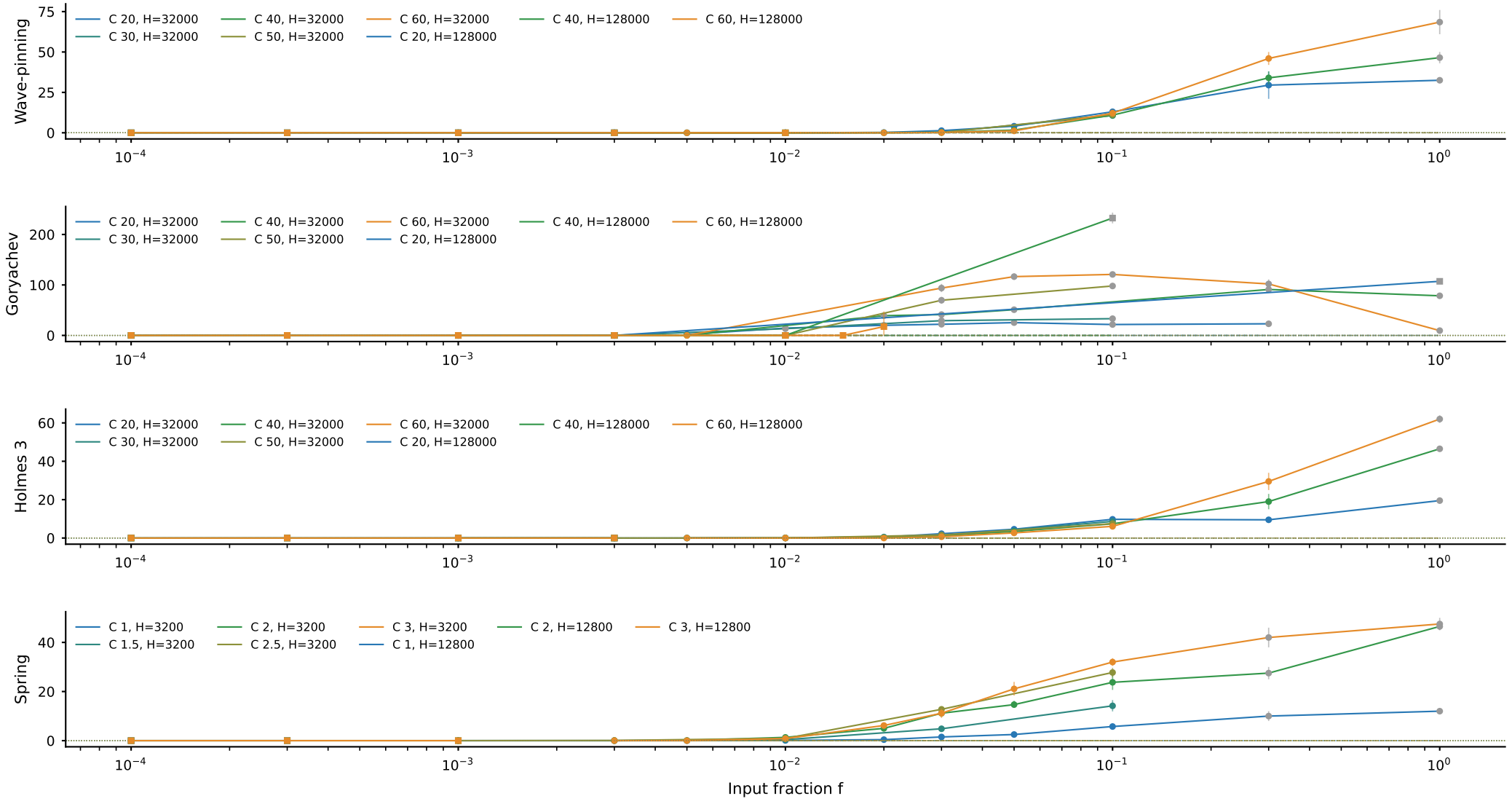
Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds.
× at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_068_S9_single_front_fraction_raw_factorial.csv

S9 · Longest canonical A5 observations · Time fraction with one qualified region · full native-factor sensitivity



Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds.
× at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_069_S9_single_front_fraction_smoothed_factorial.csv

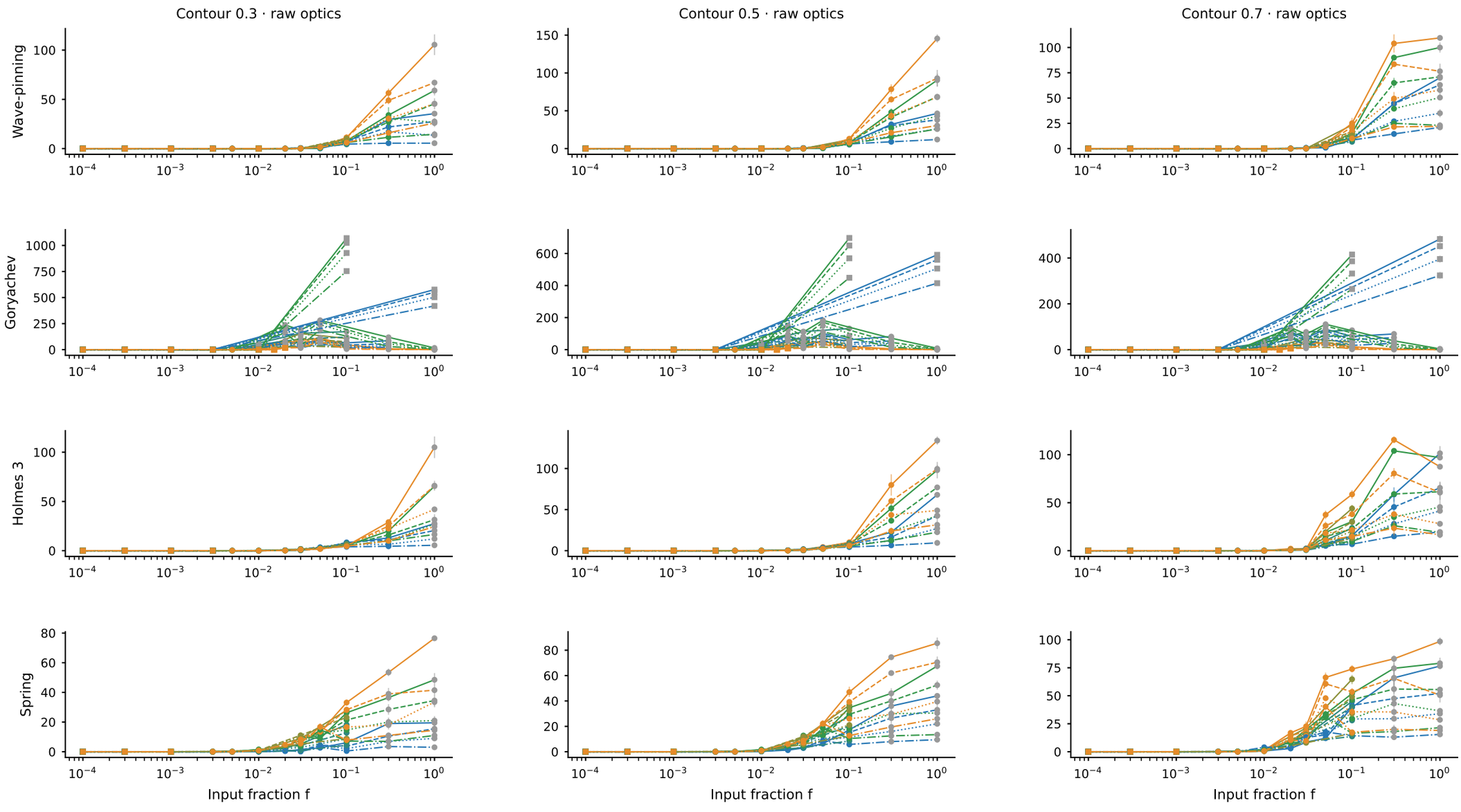
S9 · Longest canonical A5 observations · Tracks whose takeover cannot be scored



Longest canonical A5 observations. Imaging primary: half contour, fixed physical Gaussian 1.5 (Spring 0.15), physical cadence 80 (Spring 8), joined independently of native save factor. Grey marks retain operationally forced conditions; unknown flags stay explicit in CSV. Dashed matched-mean and dotted zero controls have no seed bootstrap. Complete native peak basins define positive active-signal excess, not molecule counts. Observed dominant-newborn takeover requires the newborn to be largest among qualified regions and pairwise D>0.5 against the previous dominant for two frames; this is not 75% of whole-cell activity with three or more regions. Lifetimes retain censoring; RMST restriction remains 16000 (Spring 1600) even when long survival curves extend farther. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.

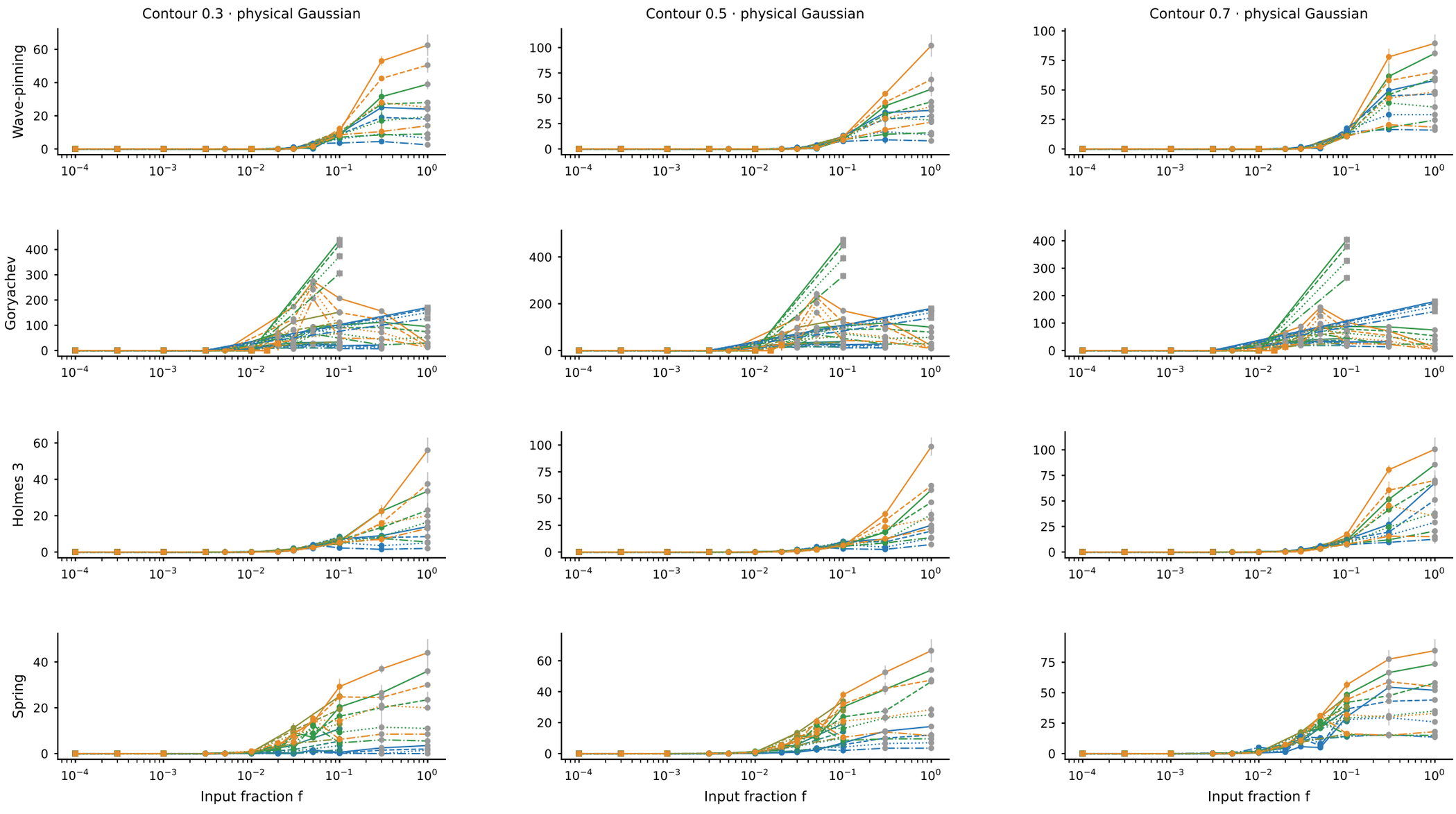
CSV: figure4_S9_longest_canonical_20261020_070_S9_unscorable_takeover_tracks.csv

S9 · Longest canonical A5 observations · Tracks whose takeover cannot be scored · full native-factor sensitivity



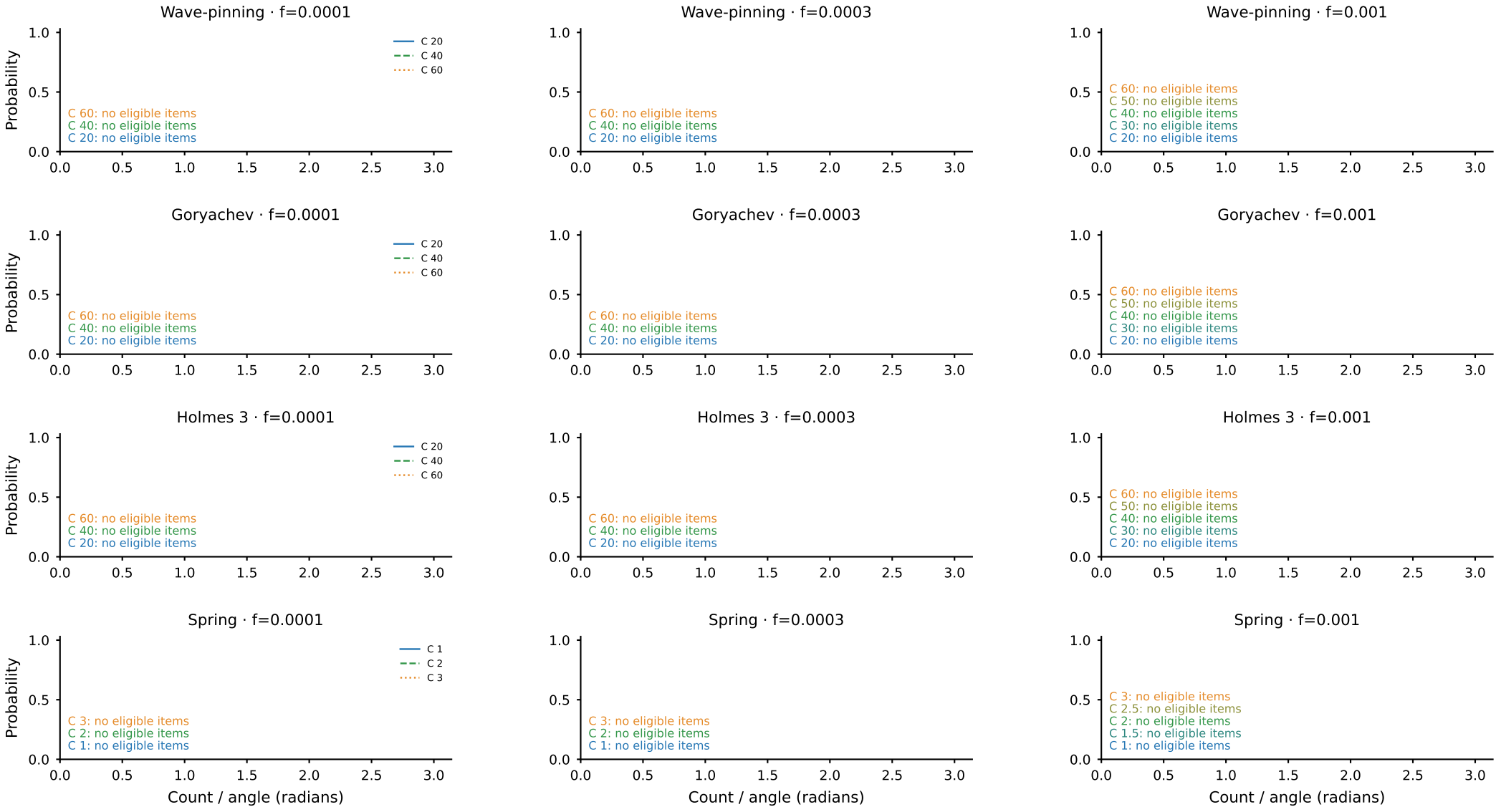
Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_071_S9_unscorable_takeover_tracks_raw_factorial.csv

S9 · Longest canonical A5 observations · Tracks whose takeover cannot be scored · full native-factor sensitivity



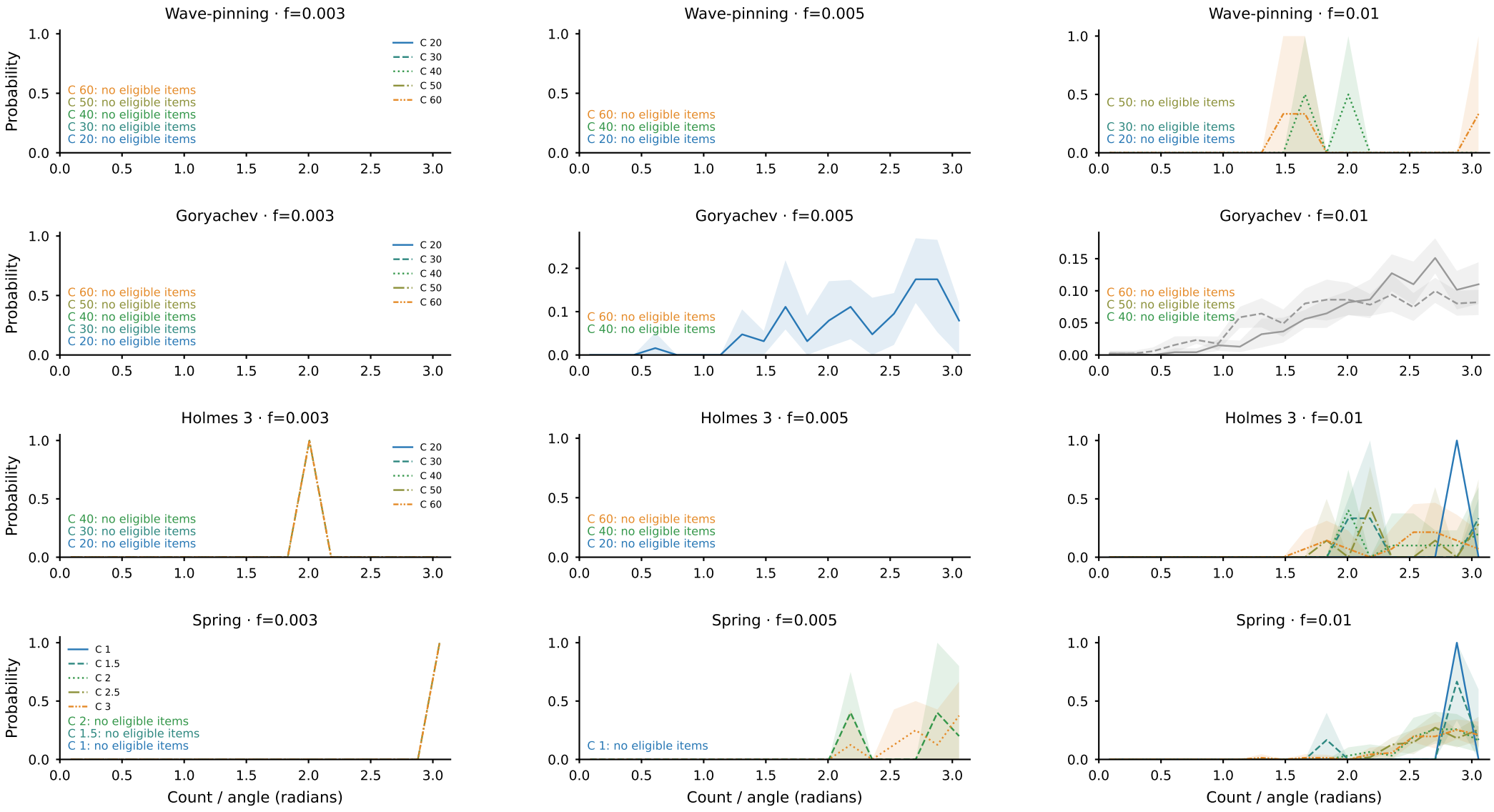
Longest canonical A5 observations. All three contours and both optics views are shown across these paired pages. Solid/dashed/dotted/dash-dot curves are native save factors 1/2/4/8; size colours encode actual C. Native profile cadences differ across source arms, so equal factors are not equal physical cadences. Every physical cadence is retained in the CSV. The separately displayed common-cadence primary may be a 25th variant, outside this 24-way factorial; no missing factor16 contour/optics sensitivity is invented. CIs resample whole independent seeds, not frames. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_072_S9_unscorable_takeover_tracks_smoothed_factorial.csv

S9 · Longest canonical A5 observations · birth angle from previous dominant



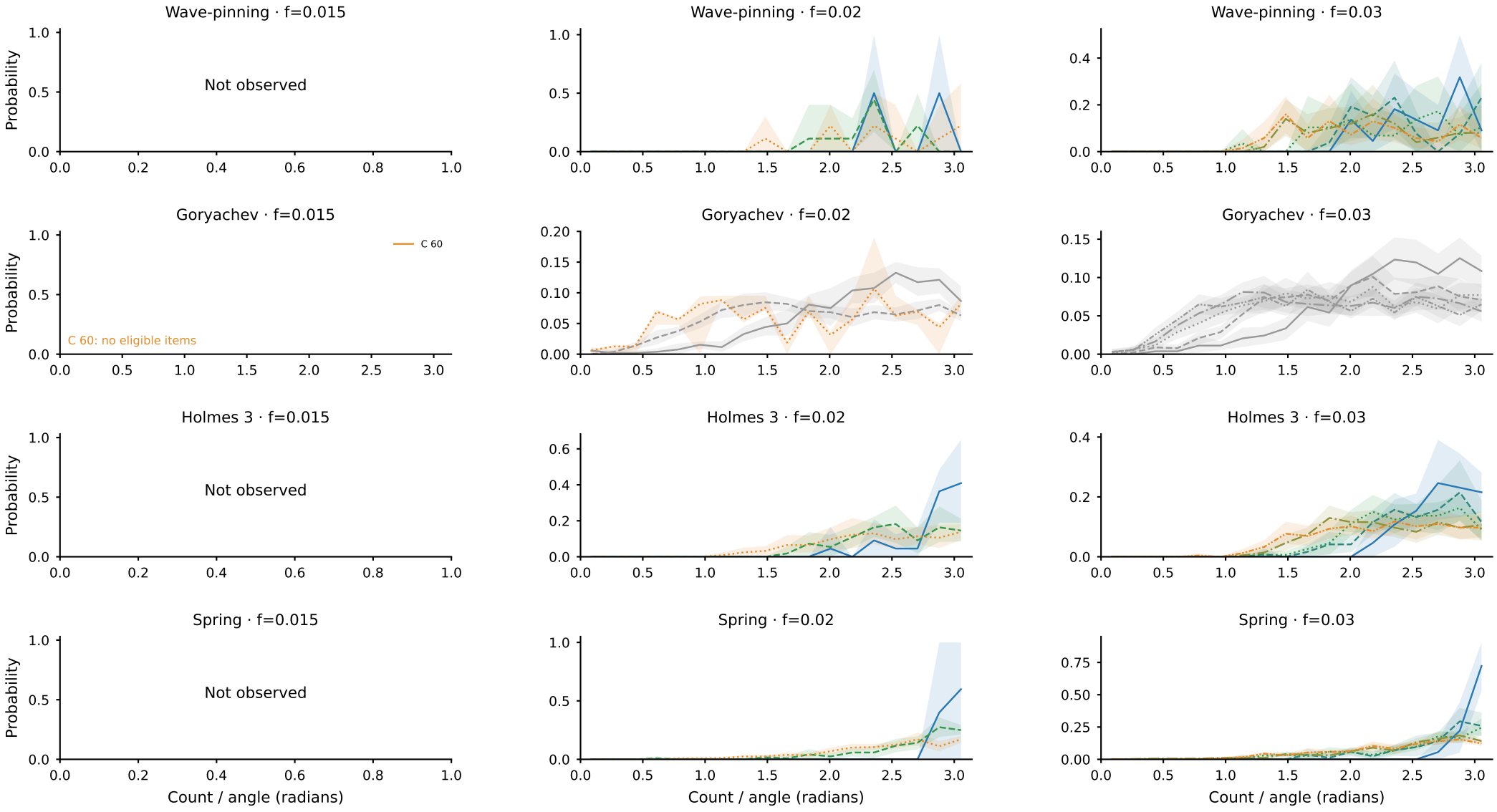
Longest canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_073_S9_birth_angle_from_previous_dominant_None_0.csv

S9 · Longest canonical A5 observations · birth angle from previous dominant



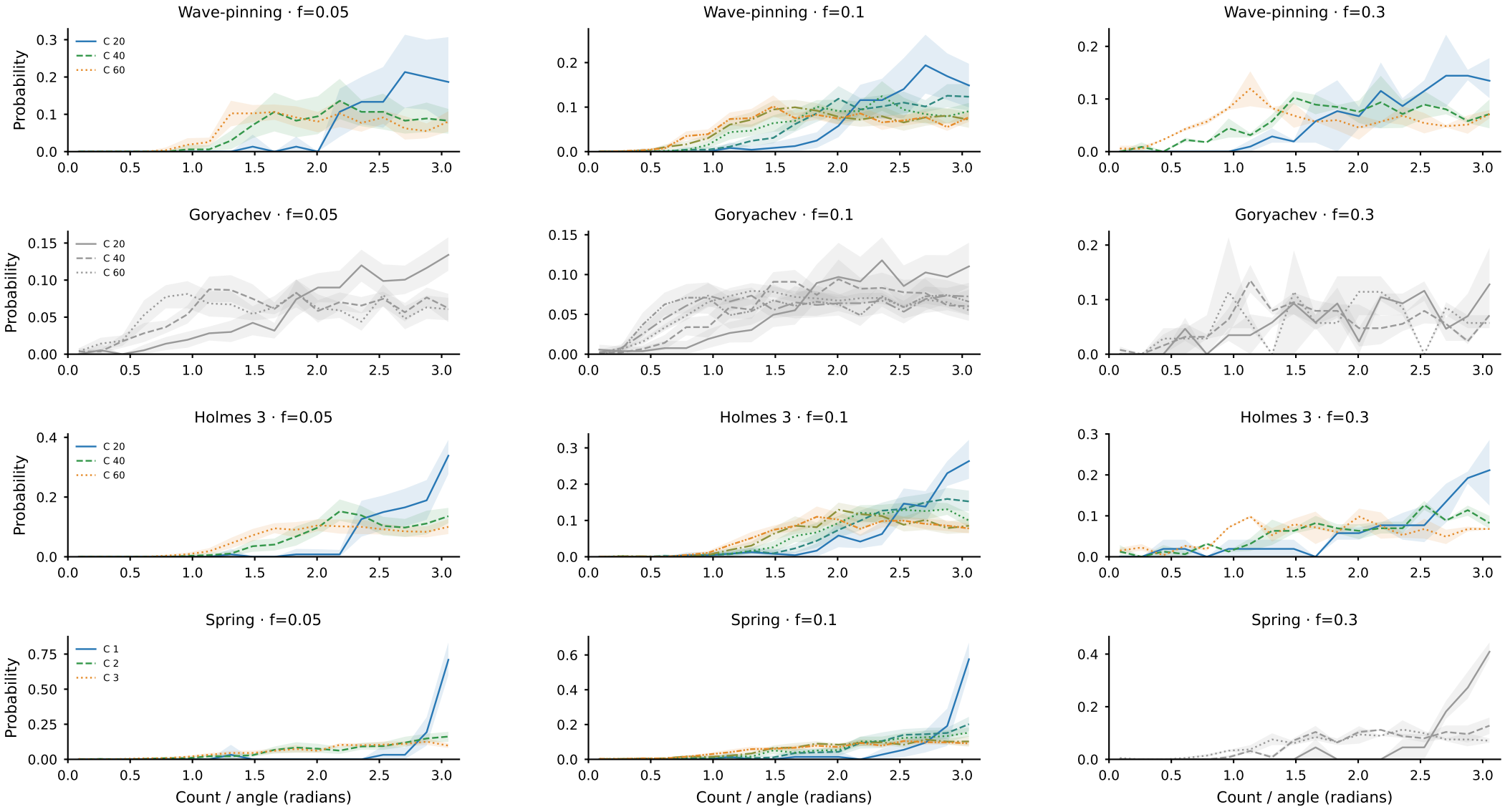
Longest canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_074_S9_birth_angle_from_previous_dominant_None_3.csv

S9 · Longest canonical A5 observations · birth angle from previous dominant



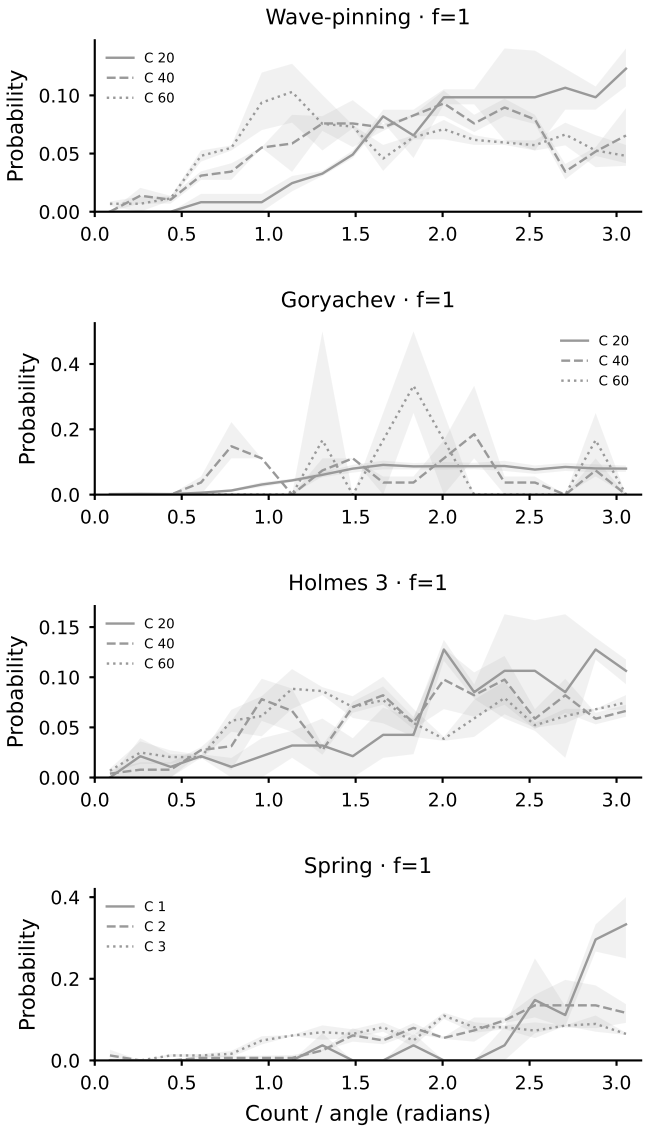
Longest canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000/128000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_075_S9_birth_angle_from_previous_dominant_None_6.csv

S9 · Longest canonical A5 observations · birth angle from previous dominant



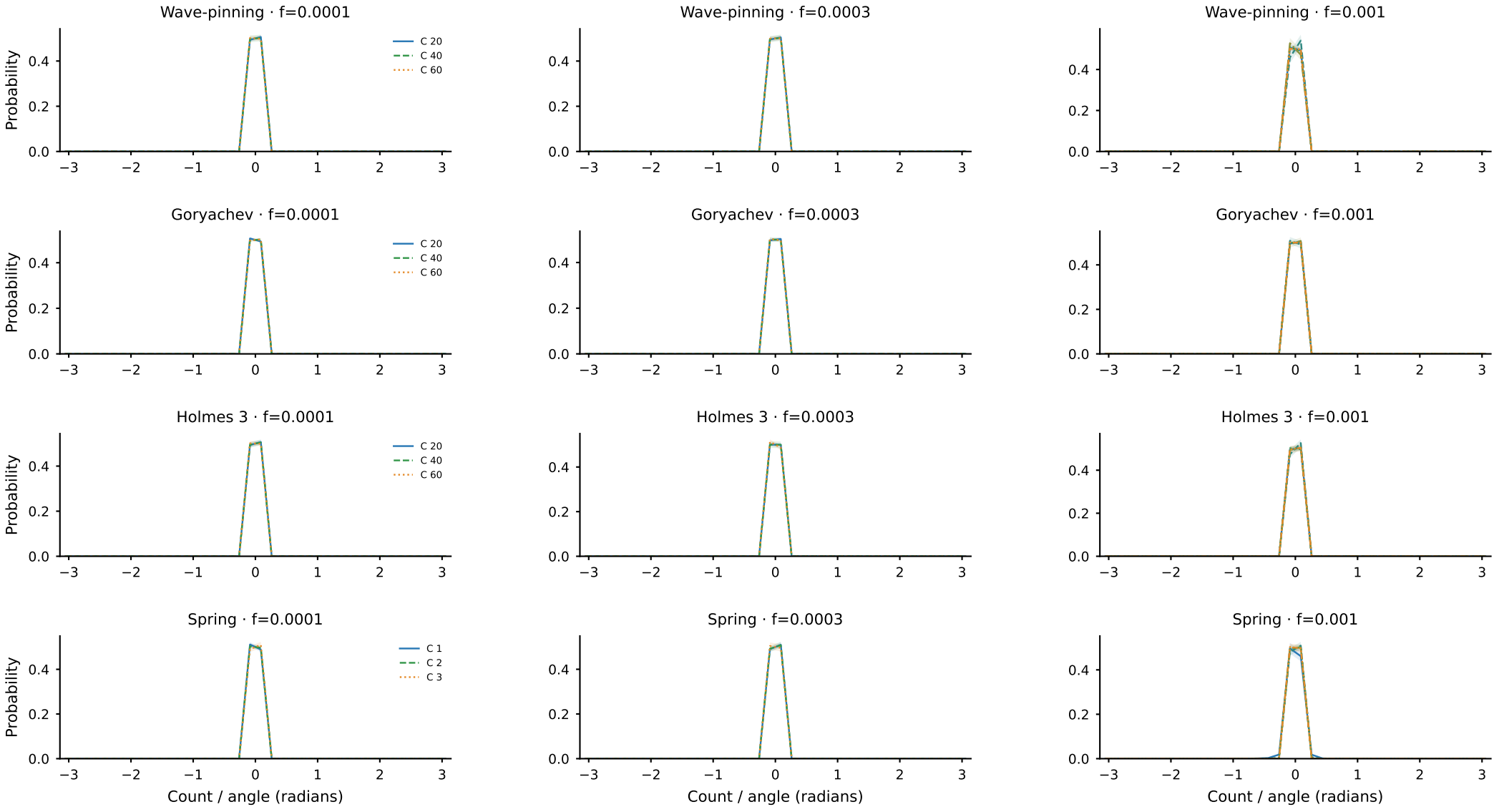
Longest canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000/128000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_076_S9_birth_angle_from_previous_dominant_None_9.csv

S9 · Longest canonical A5 observations · birth angle from previous dominant



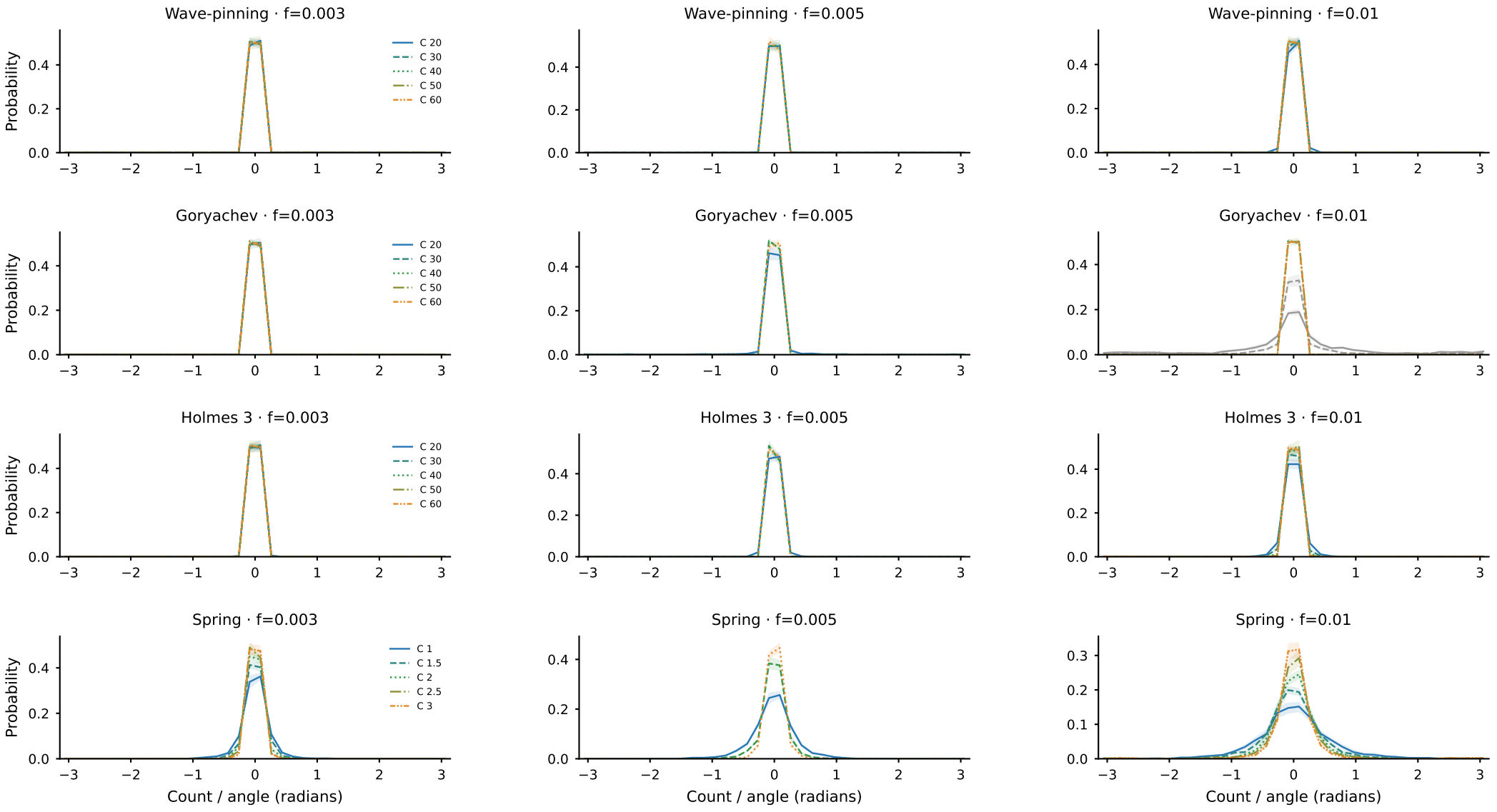
Longest canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000/128000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_077_S9_birth_angle_from_previous_dominant_None_12.csv

S9 · Longest canonical A5 observations · dominant heading turn · 1 observation frames



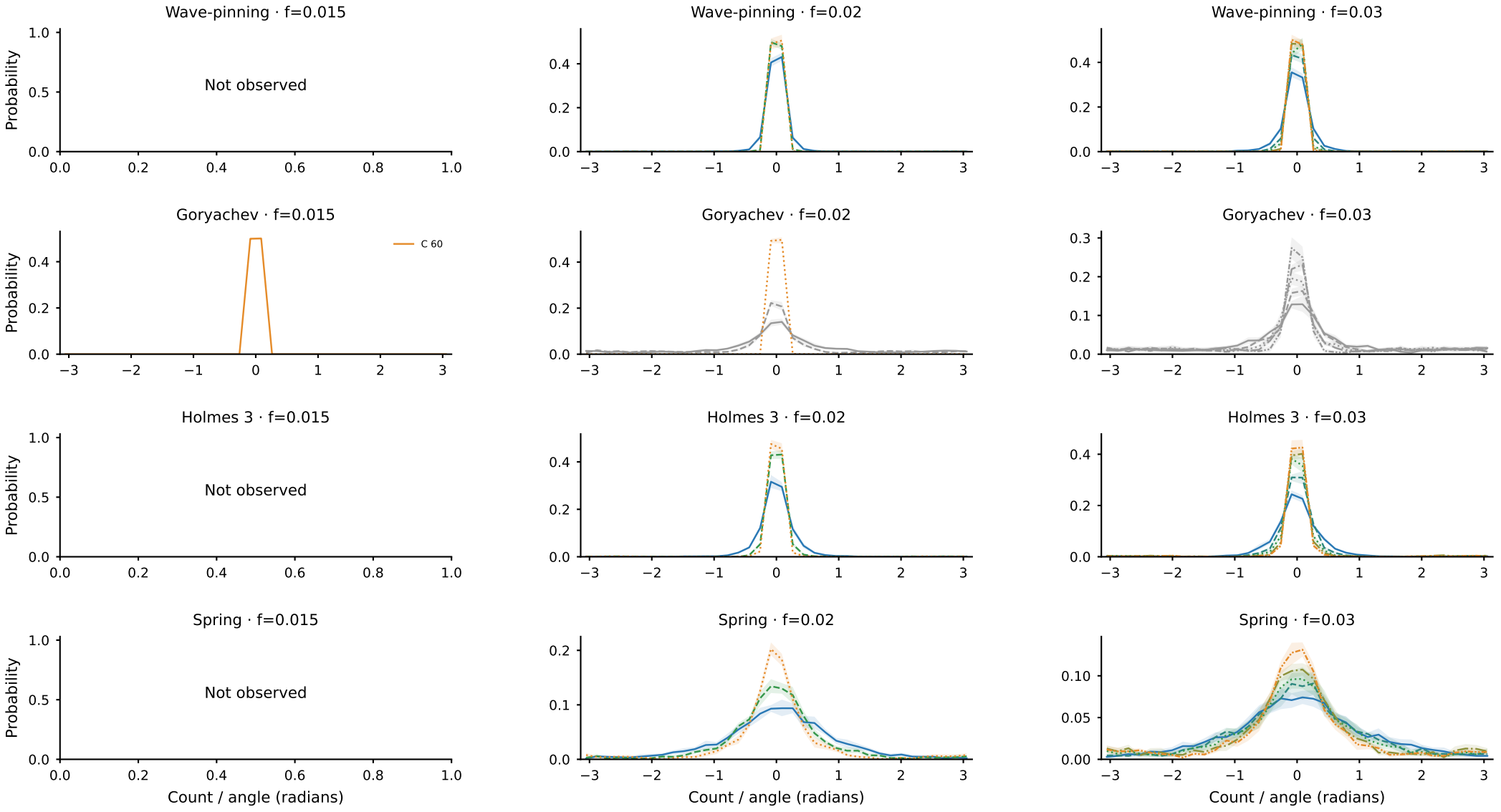
Longest canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_078_S9_dominant_heading_turn_1.0_0.csv

S9 · Longest canonical A5 observations · dominant heading turn · 1 observation frames



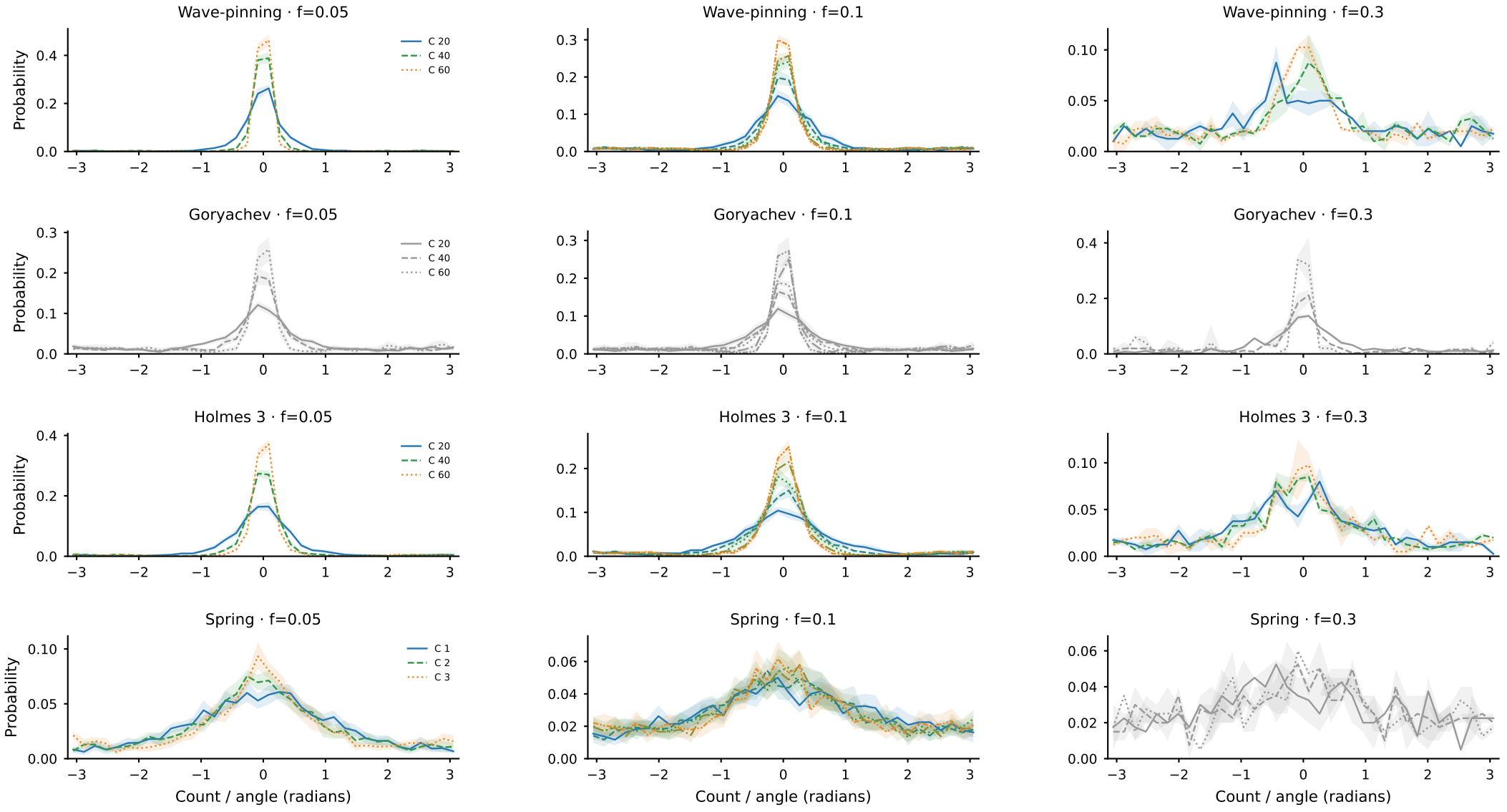
Longest canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_079_S9_dominant_heading_turn_1.0_3.csv

S9 · Longest canonical A5 observations · dominant heading turn · 1 observation frames



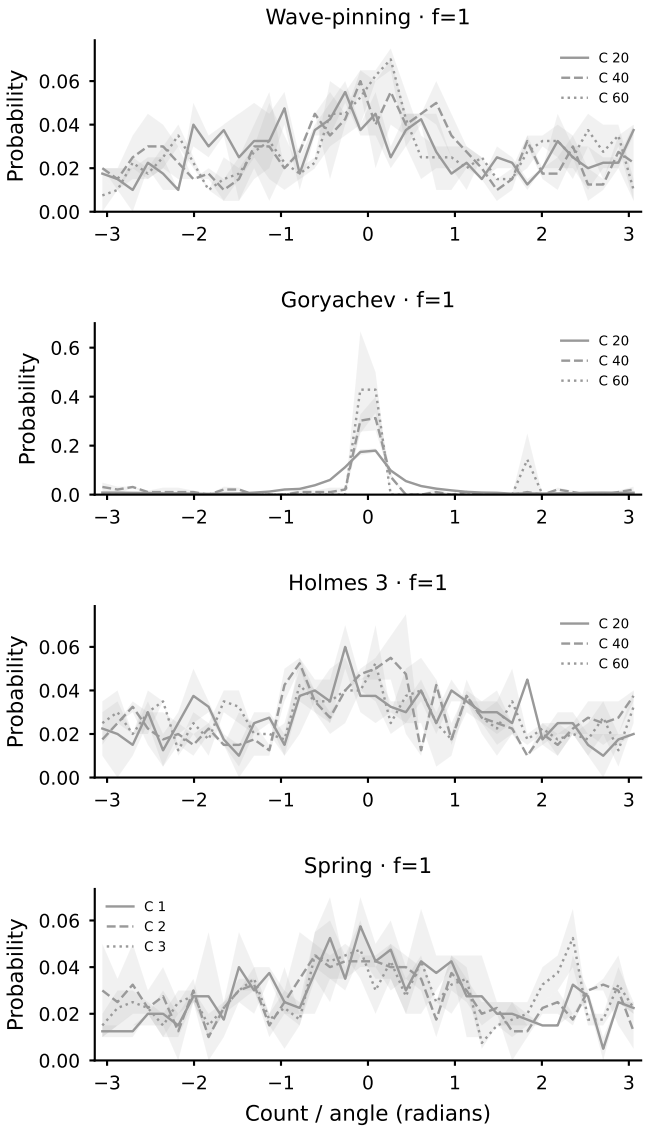
Longest canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000/128000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_080_S9_dominant_heading_turn_1.0_6.csv

S9 · Longest canonical A5 observations · dominant heading turn · 1 observation frames



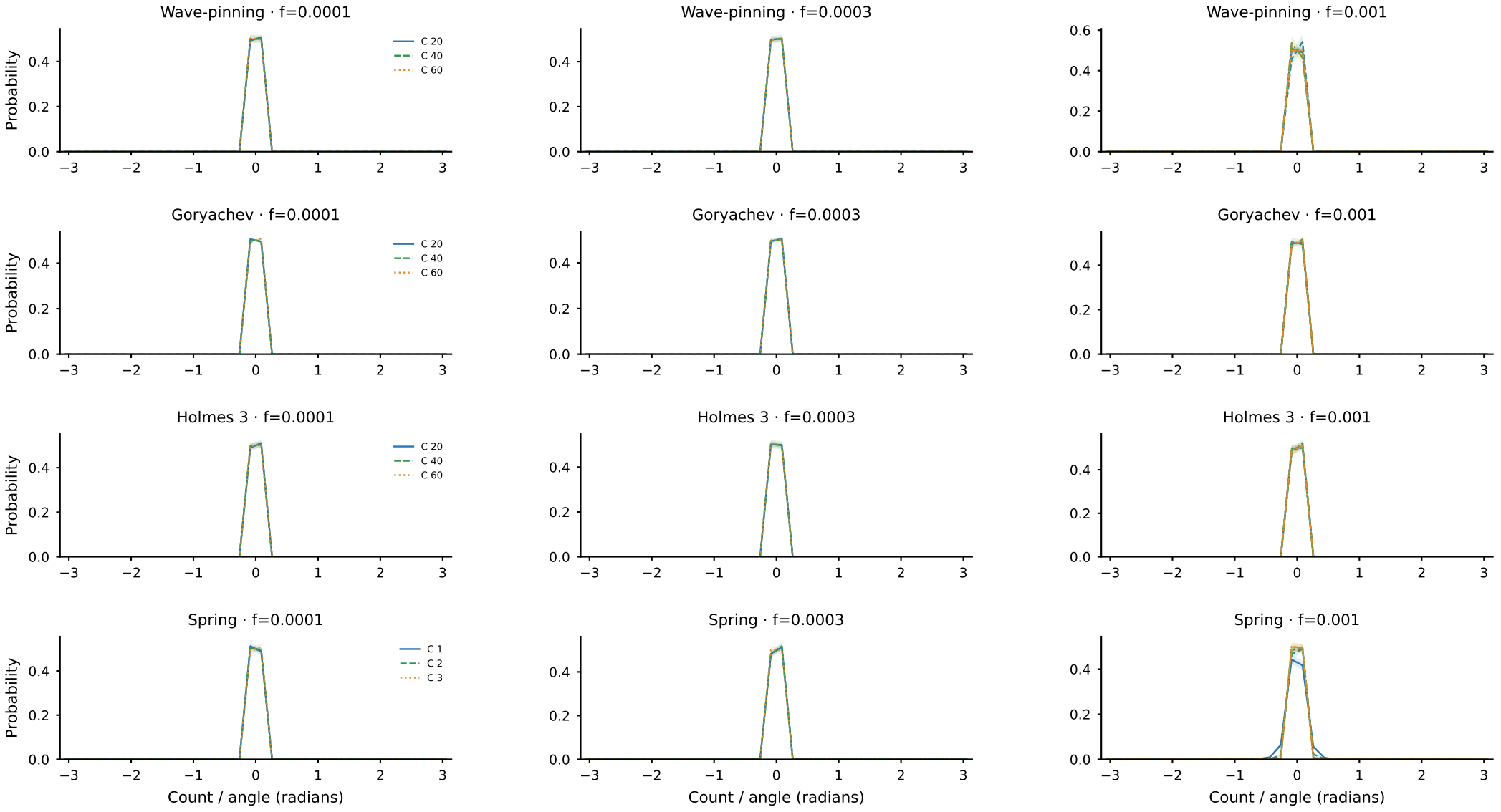
Longest canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000/128000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_081_S9_dominant_heading_turn_1.0_9.csv

S9 · Longest canonical A5 observations · dominant heading turn · 1 observation frames



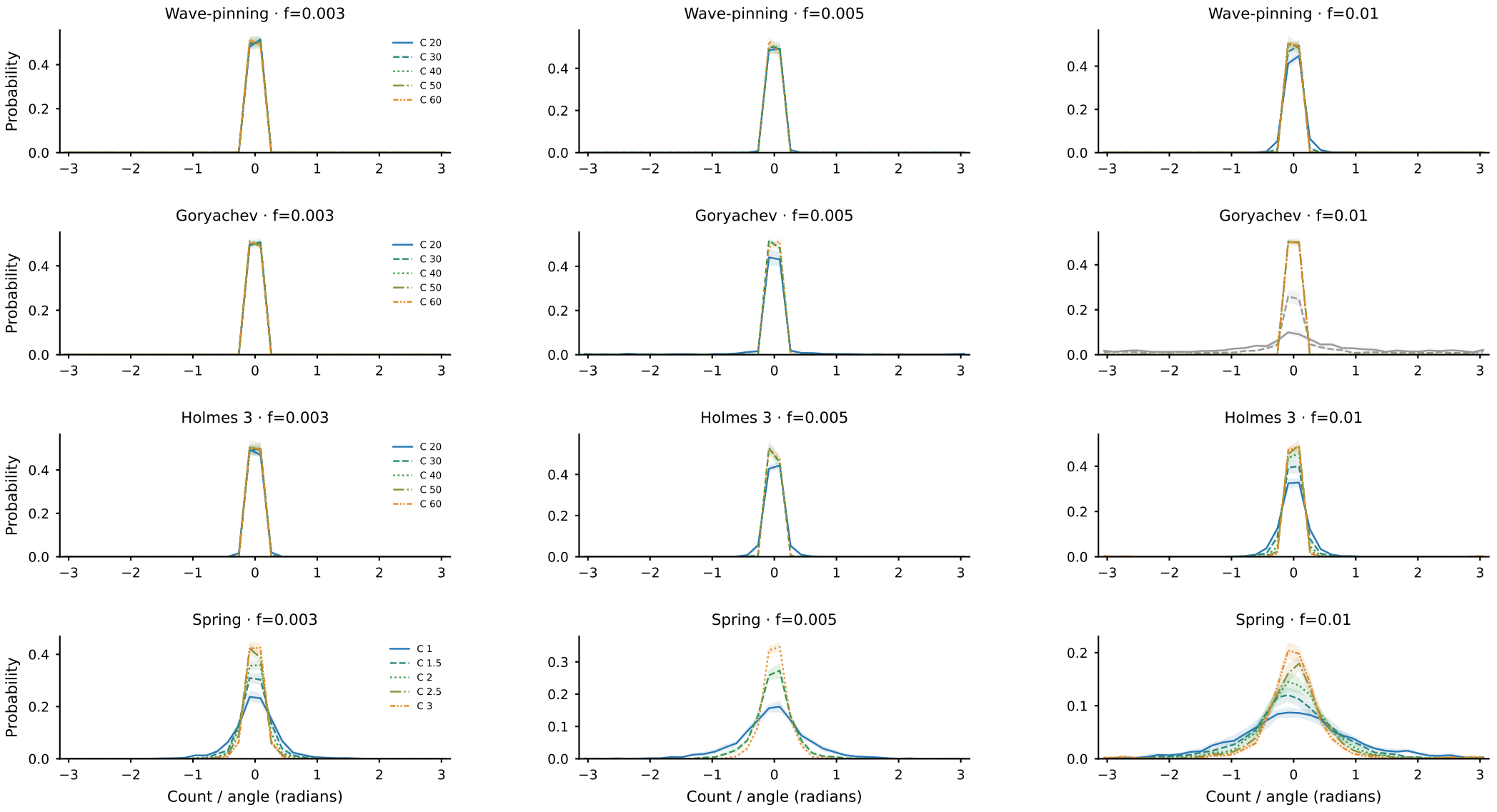
Longest canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000/128000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_082_S9_dominant_heading_turn_1.0_12.csv

S9 · Longest canonical A5 observations · dominant heading turn · 2 observation frames



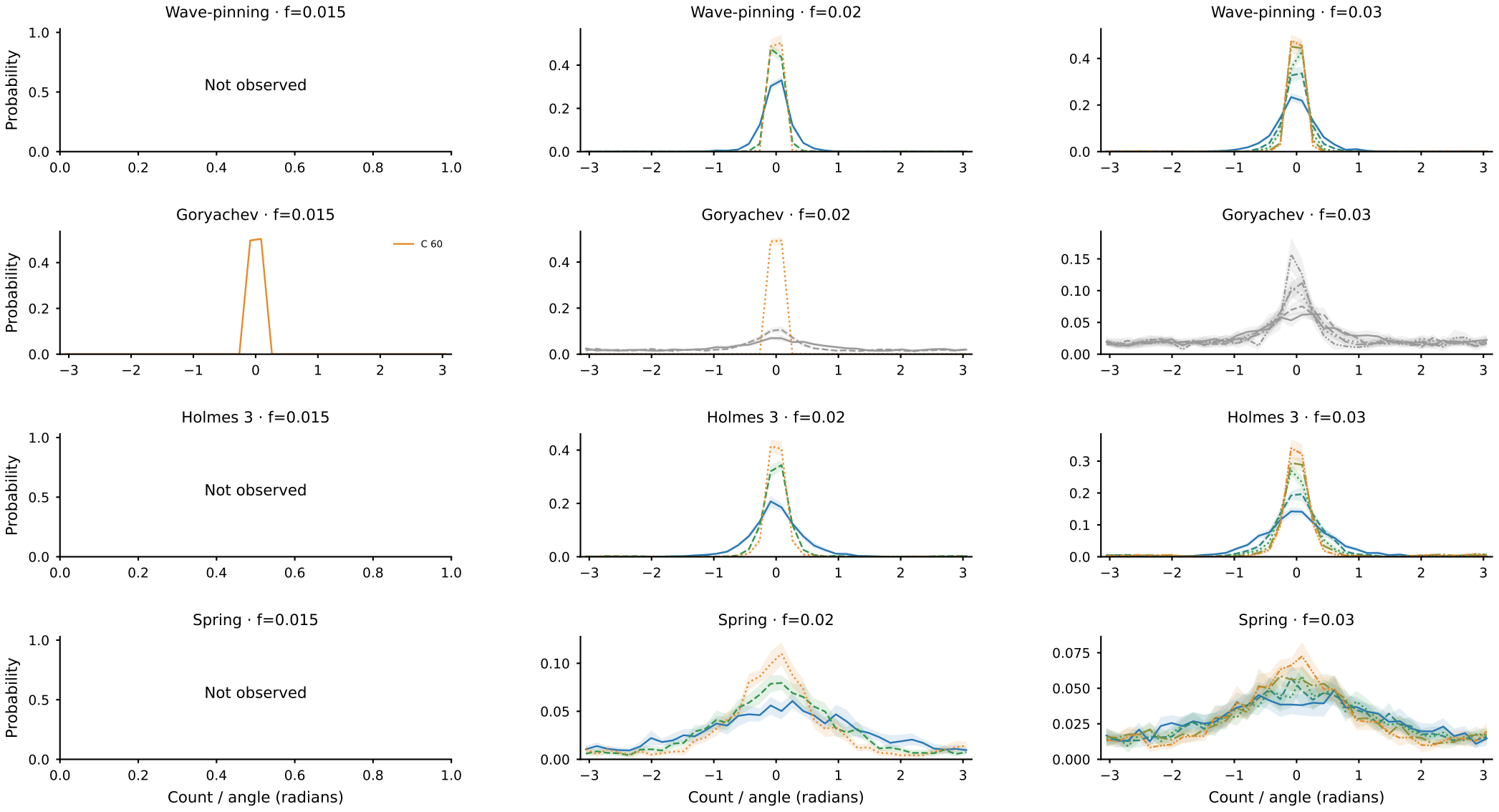
Longest canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_083_S9_dominant_heading_turn_2.0_0.csv

S9 · Longest canonical A5 observations · dominant heading turn · 2 observation frames



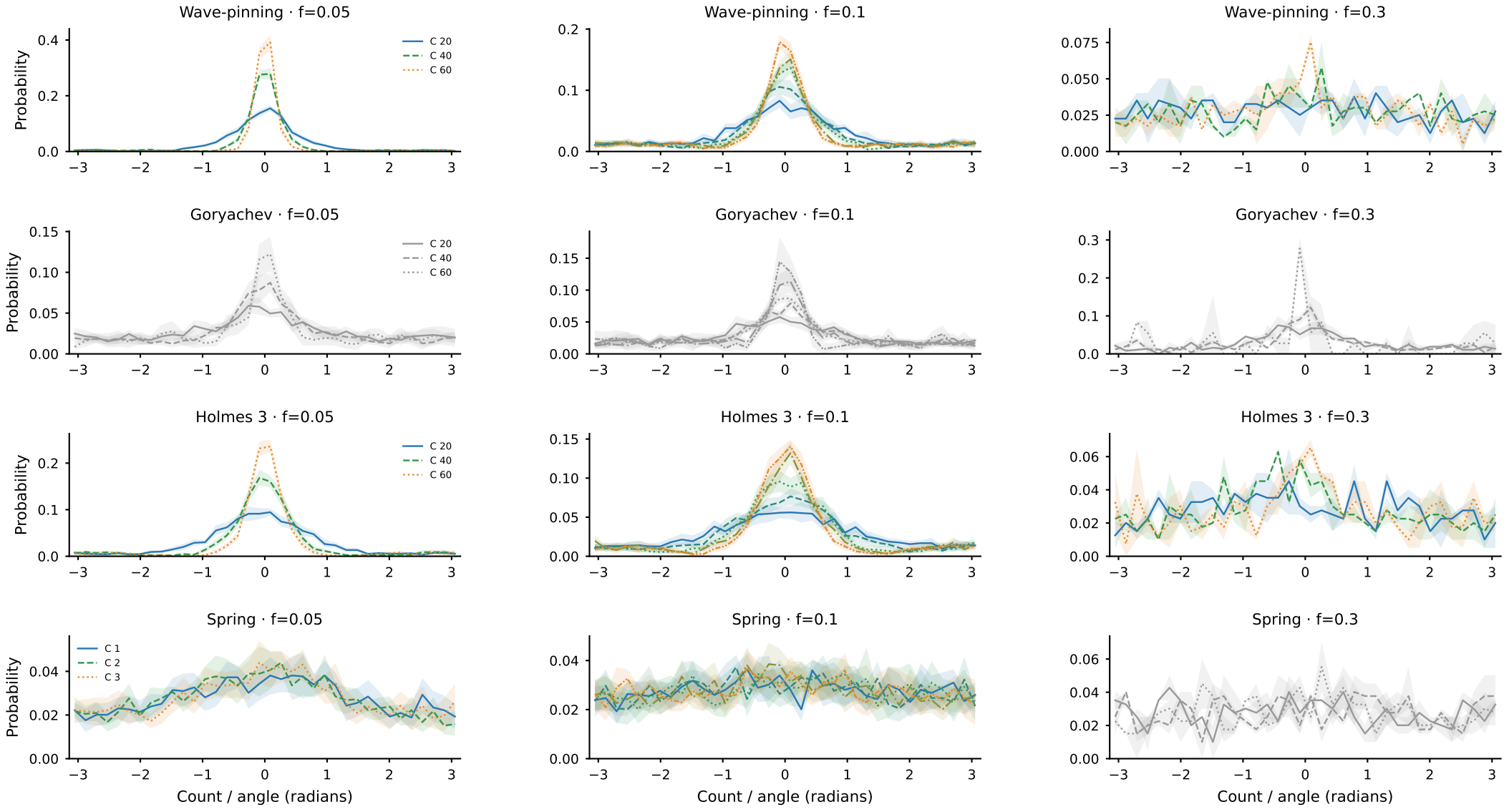
Longest canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_084_S9_dominant_heading_turn_2.0_3.csv

S9 · Longest canonical A5 observations · dominant heading turn · 2 observation frames



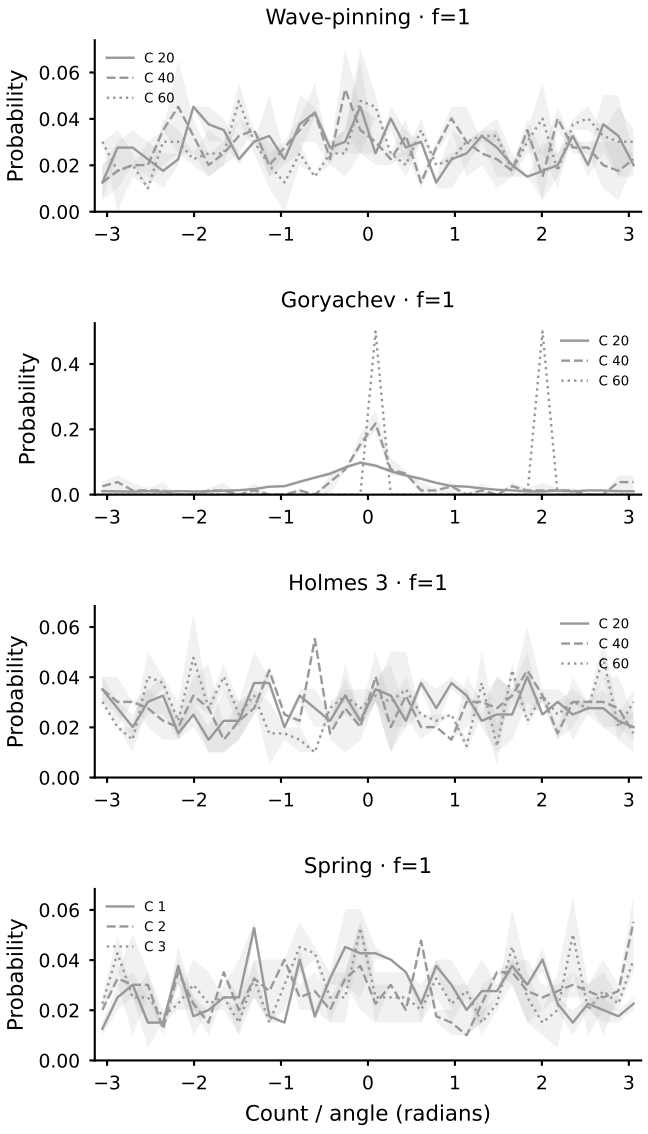
Longest canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000/128000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_085_S9_dominant_heading_turn_2.0_6.csv

S9 · Longest canonical A5 observations · dominant heading turn · 2 observation frames



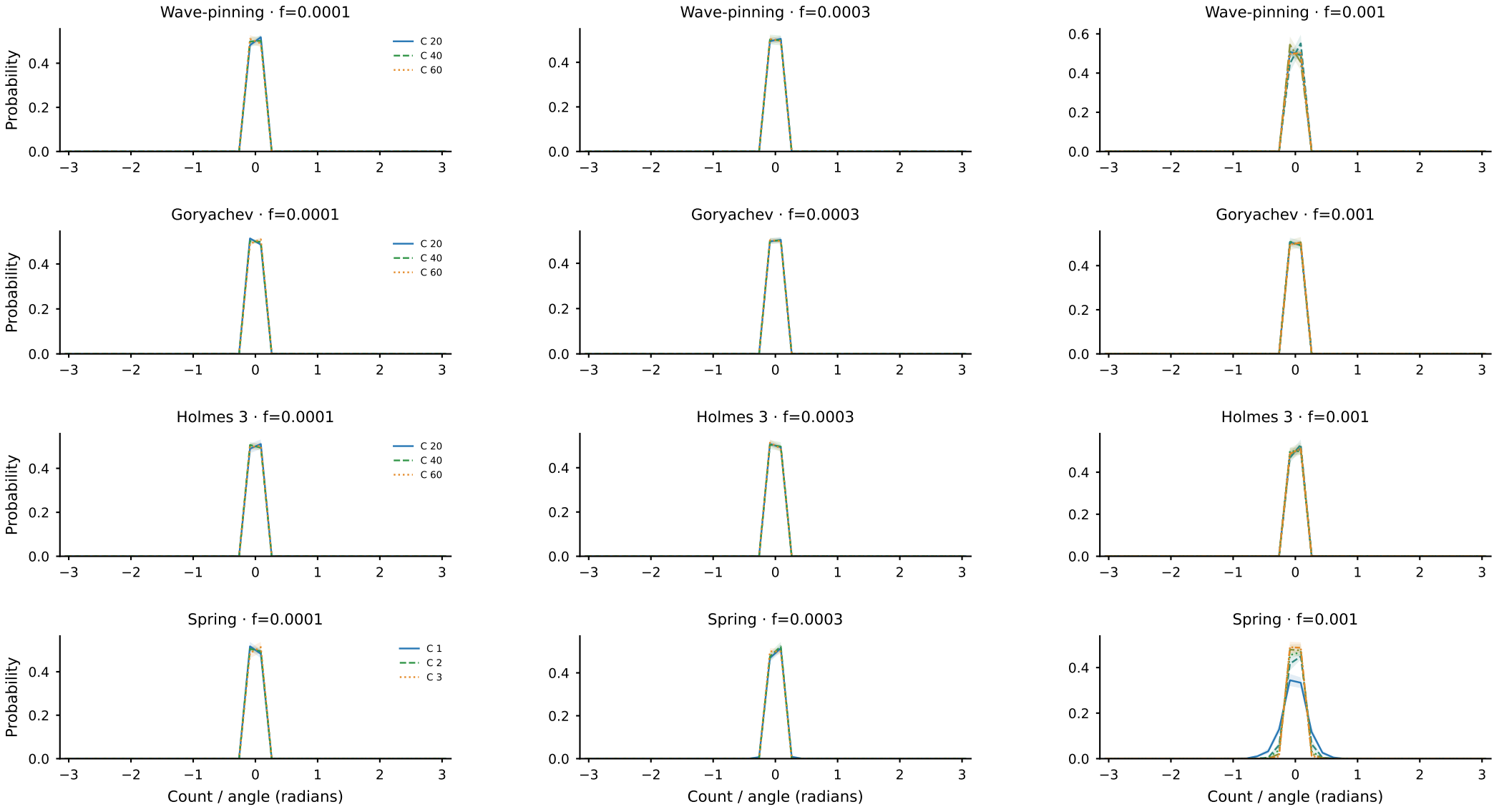
Longest canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000/128000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_086_S9_dominant_heading_turn_2.0_9.csv

S9 · Longest canonical A5 observations · dominant heading turn · 2 observation frames



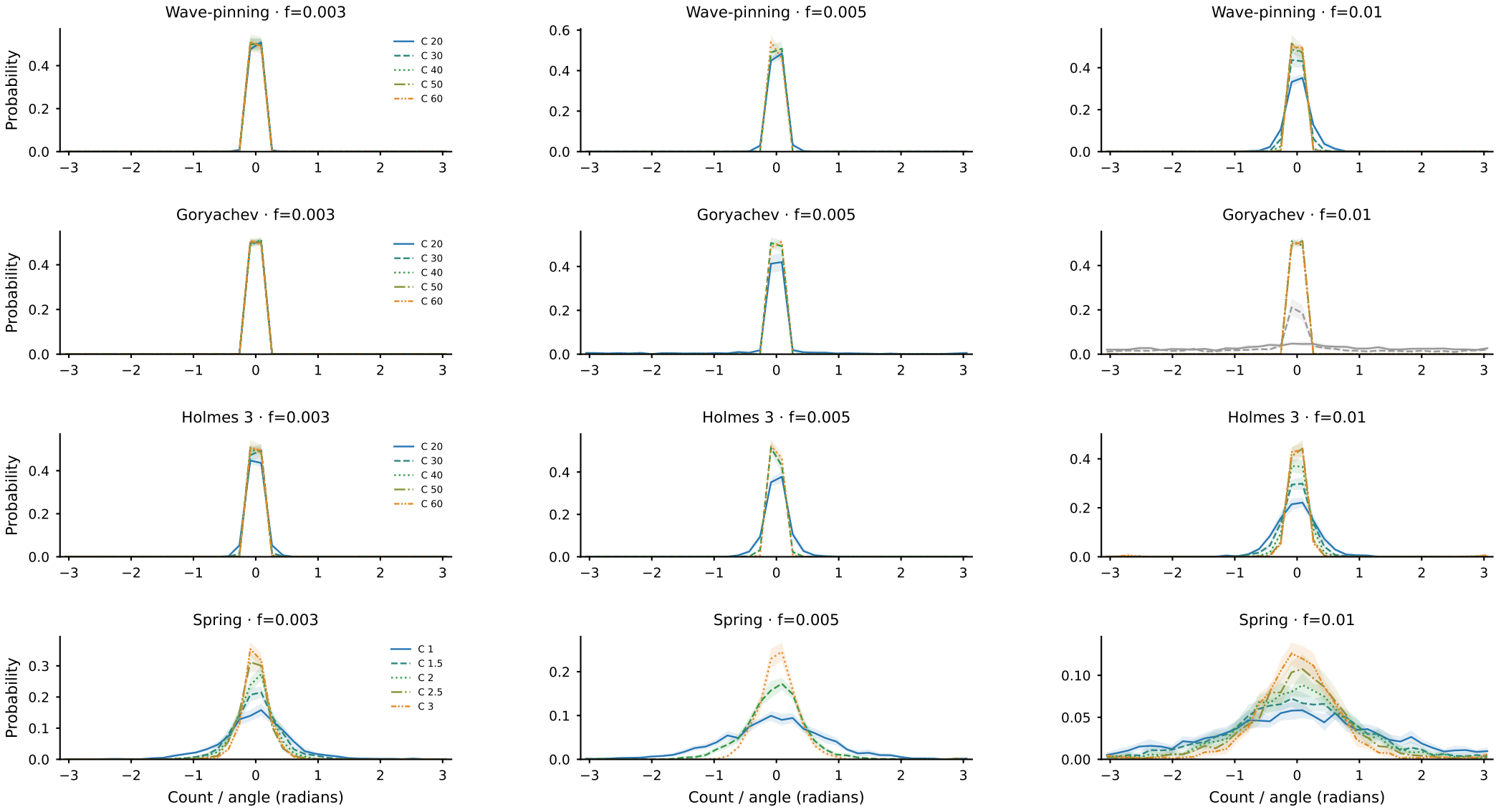
Longest canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000/128000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_087_S9_dominant_heading_turn_2.0_12.csv

S9 · Longest canonical A5 observations · dominant heading turn · 4 observation frames



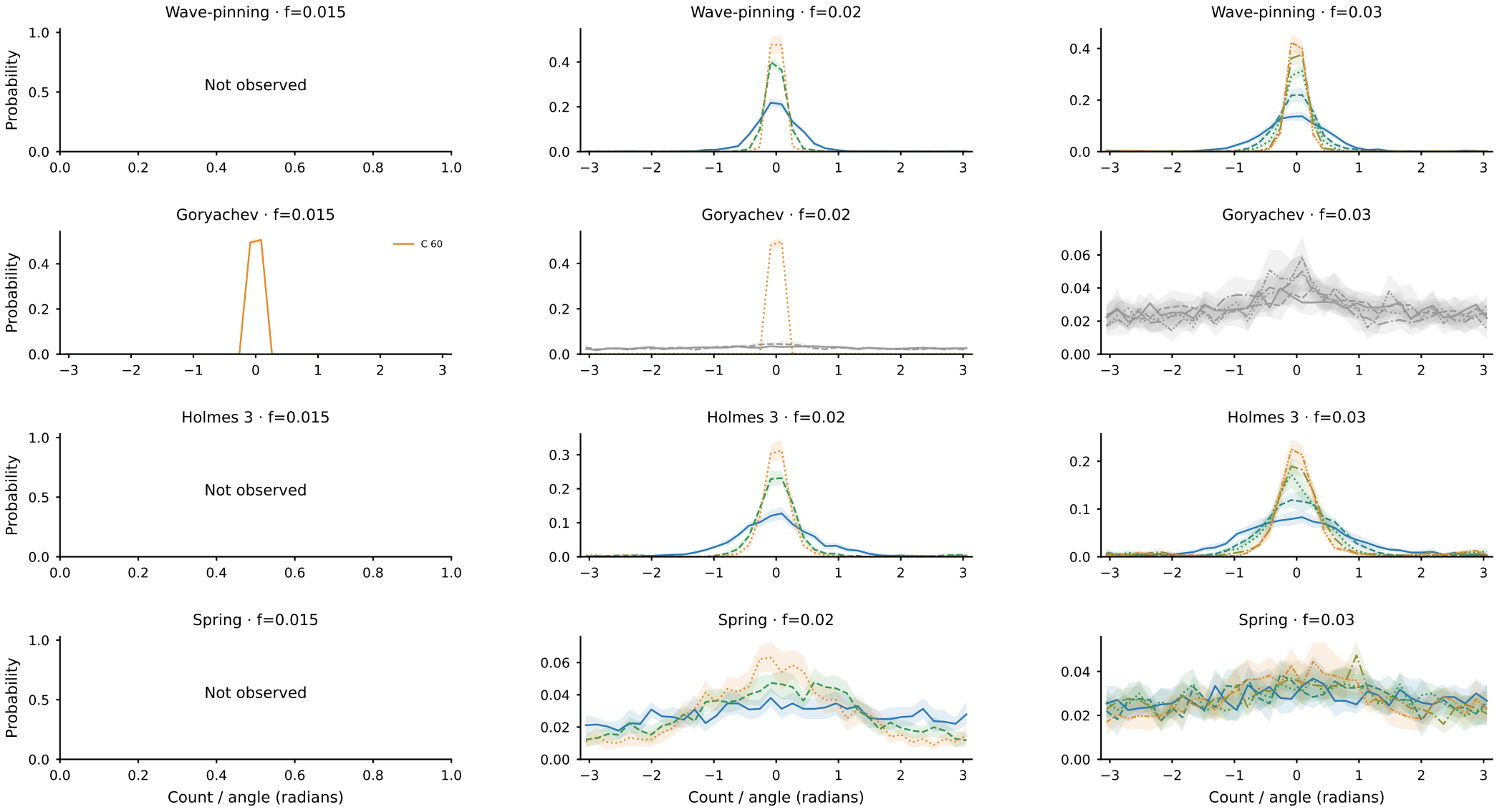
Longest canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_088_S9_dominant_heading_turn_4.0_0.csv

S9 · Longest canonical A5 observations · dominant heading turn · 4 observation frames



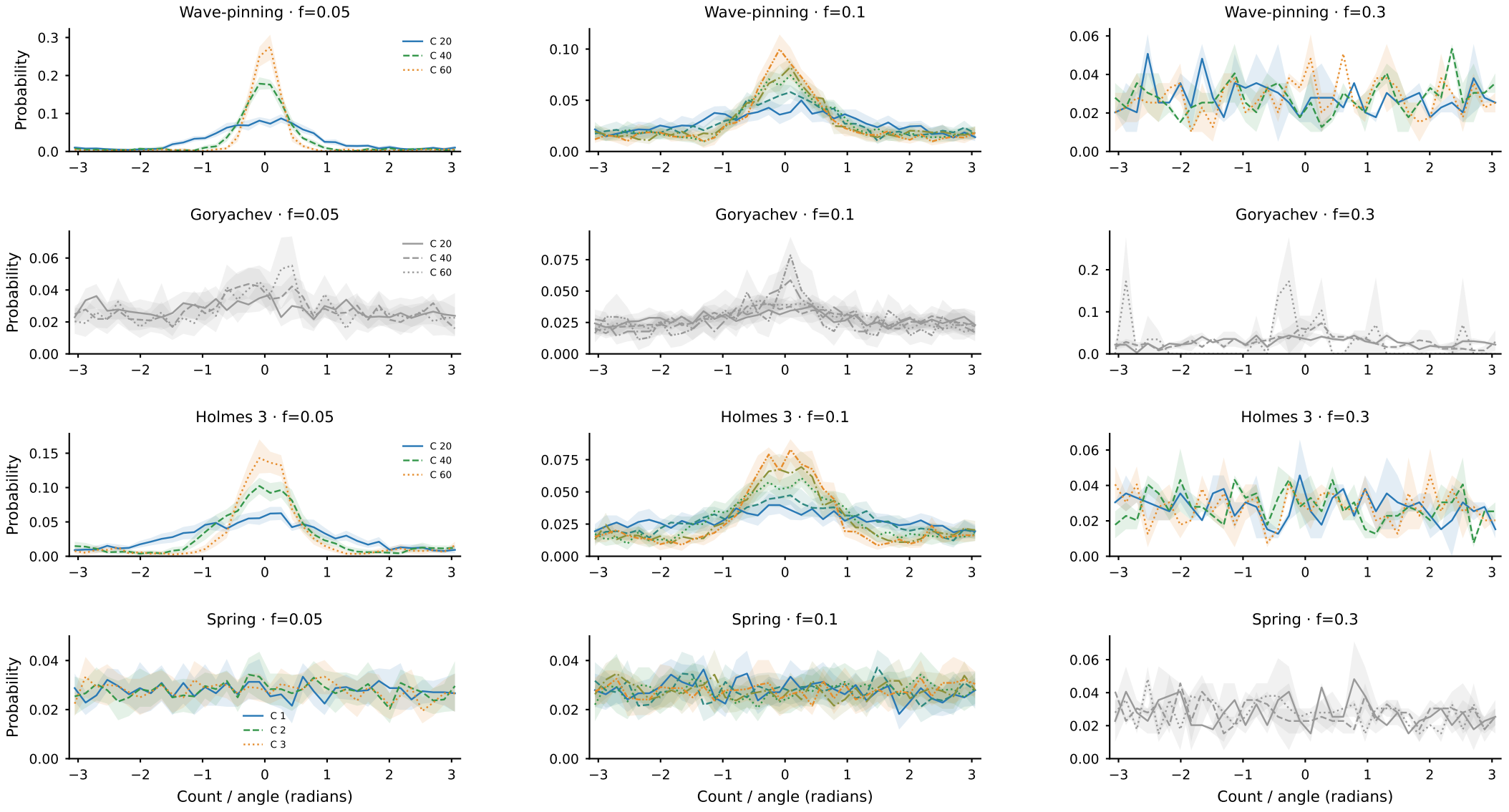
Longest canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_089_S9_dominant_heading_turn_4.0_3.csv

S9 · Longest canonical A5 observations · dominant heading turn · 4 observation frames



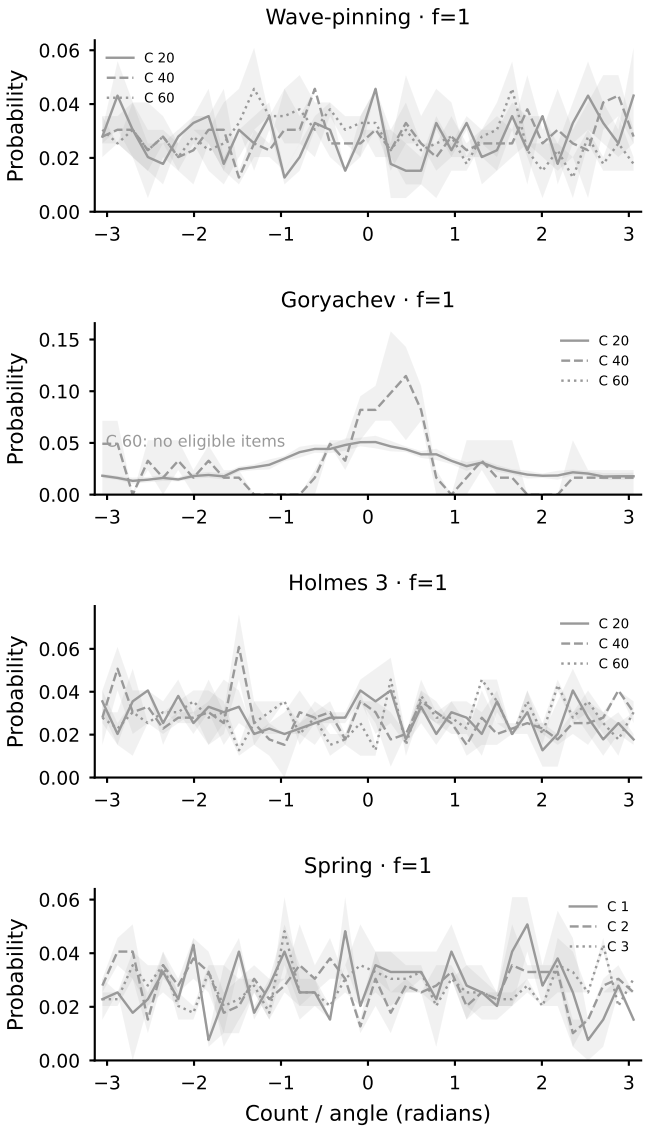
Longest canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000/128000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_090_S9_dominant_heading_turn_4.0_6.csv

S9 · Longest canonical A5 observations · dominant heading turn · 4 observation frames



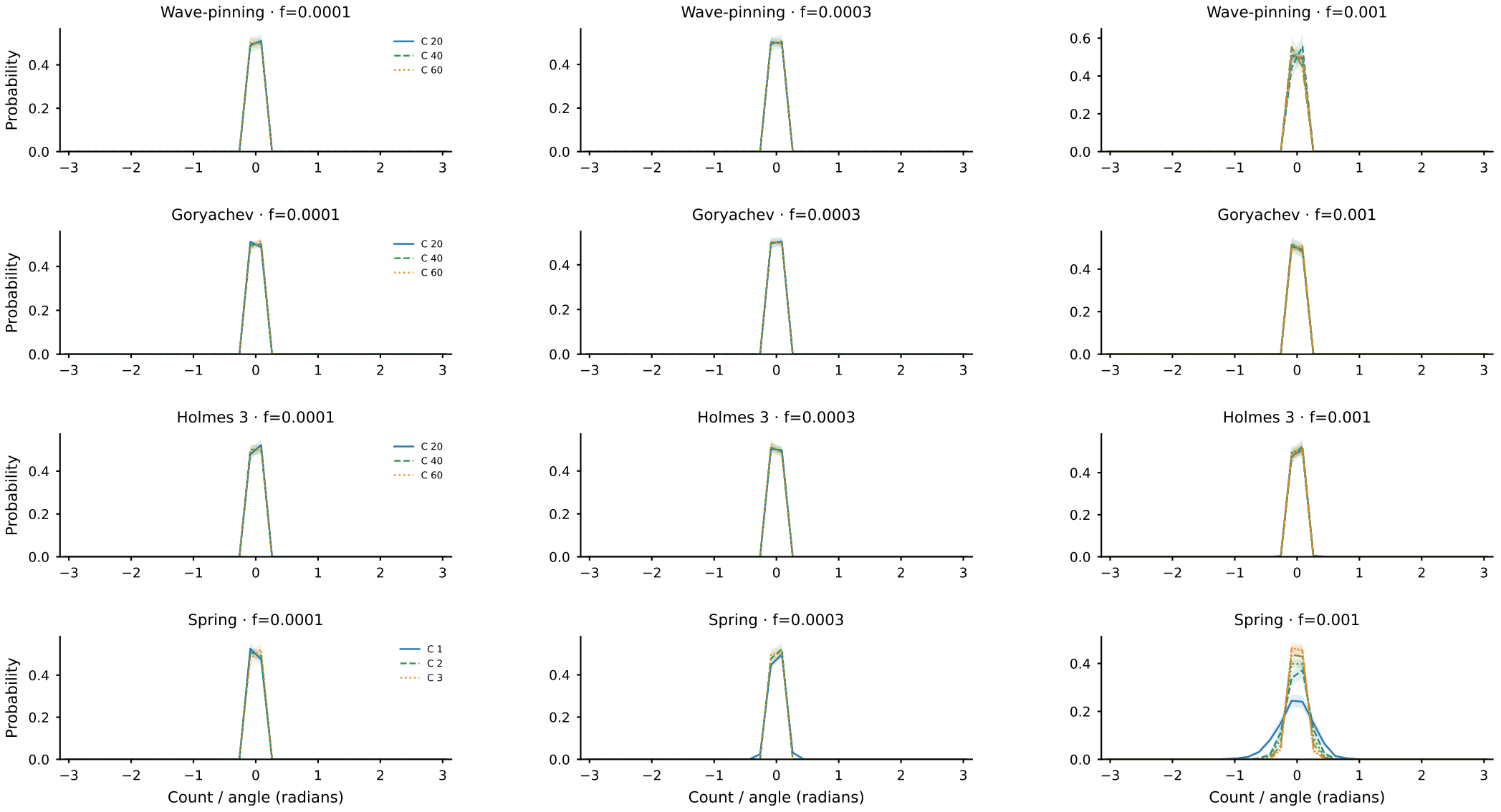
Longest canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000/128000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_091_S9_dominant_heading_turn_4.0_9.csv

S9 · Longest canonical A5 observations · dominant heading turn · 4 observation frames



Longest canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000/128000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_092_S9_dominant_heading_turn_4.0_12.csv

S9 · Longest canonical A5 observations · dominant heading turn · 8 observation frames



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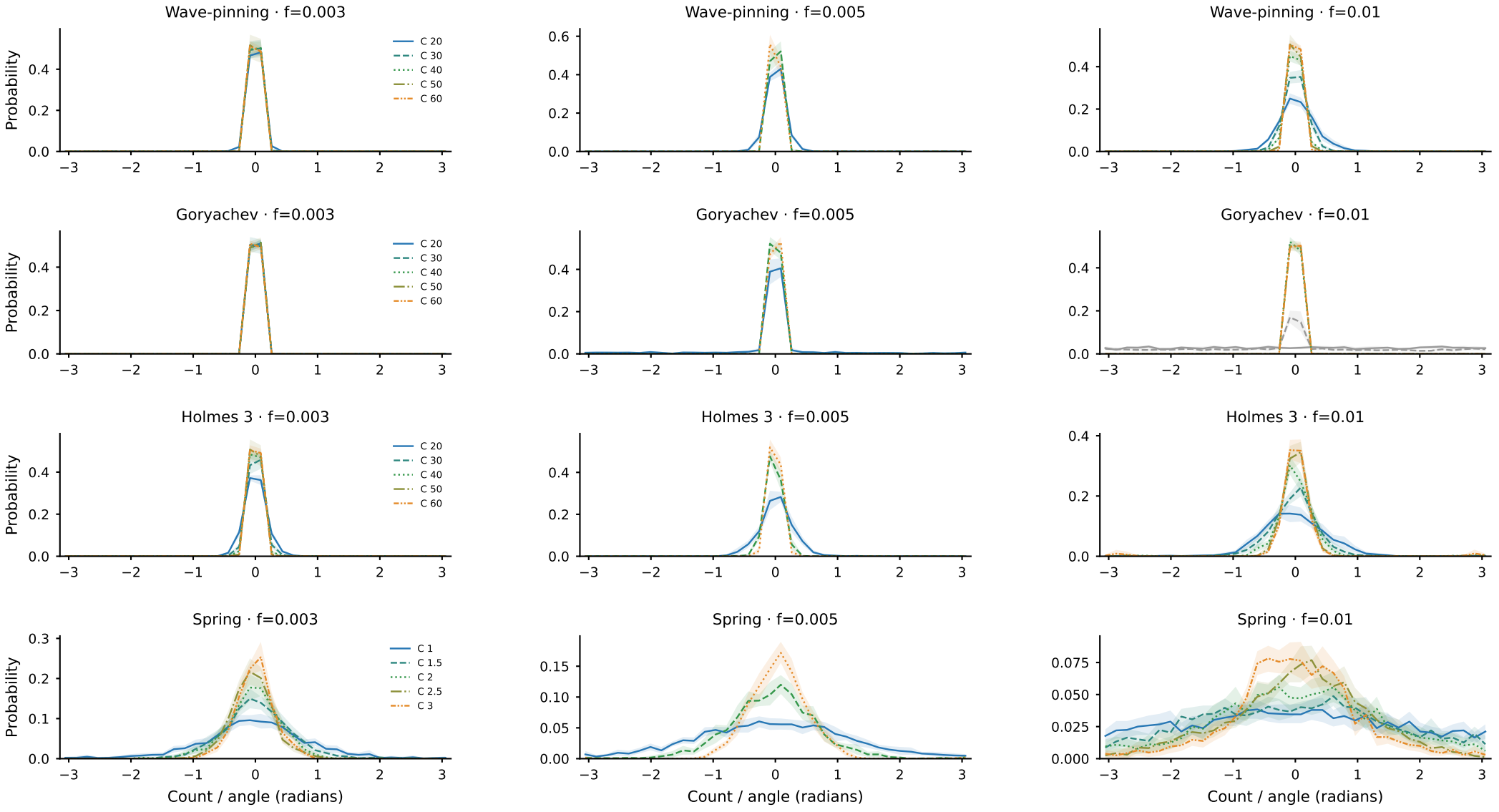
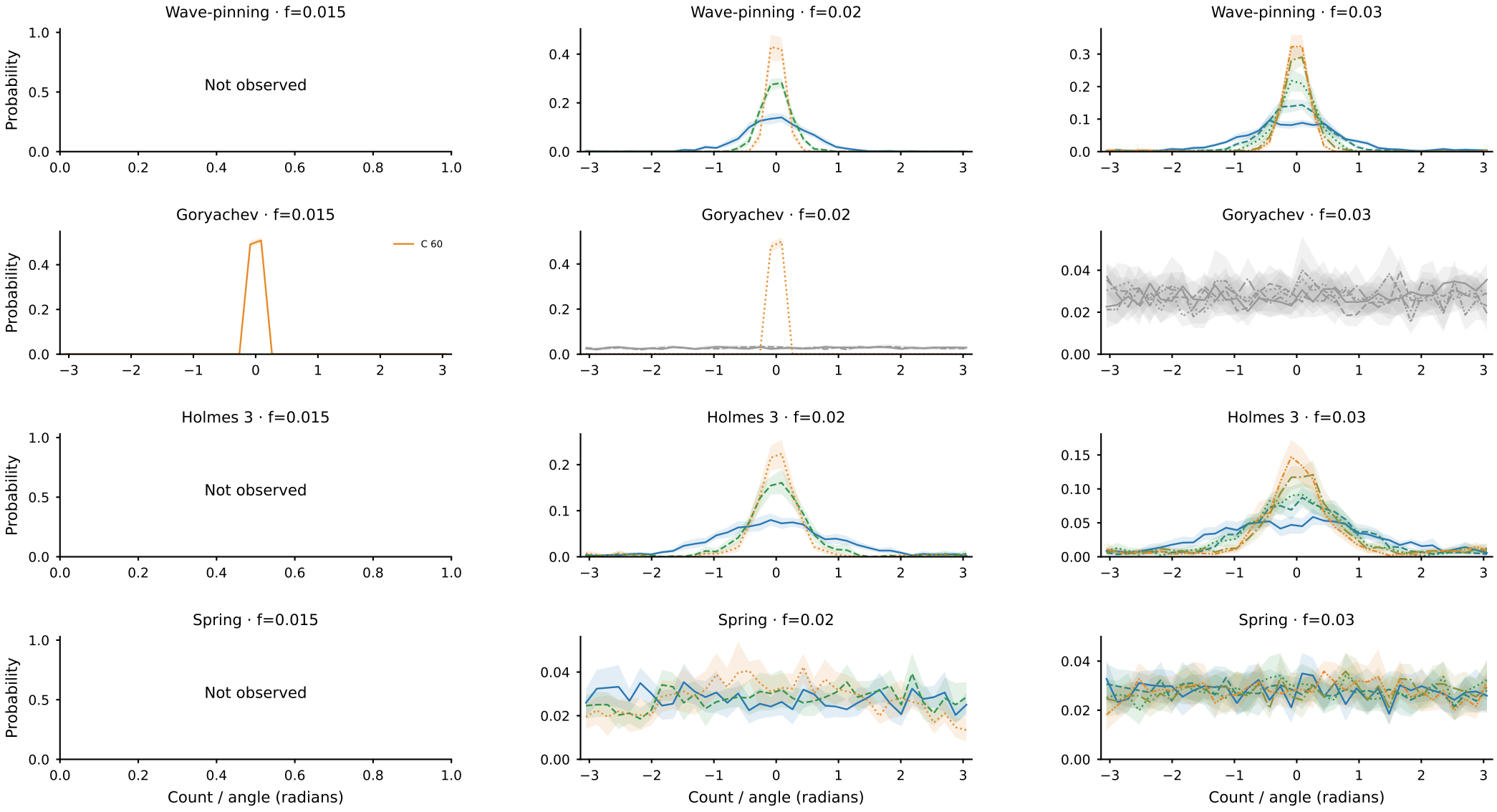
CSV: figure4_S9_longest_canonical_20261020_093_S9_dominant_heading_turn_8.0_0.csv

S9 · Longest canonical A5 observations · dominant heading turn · 8 observation frames



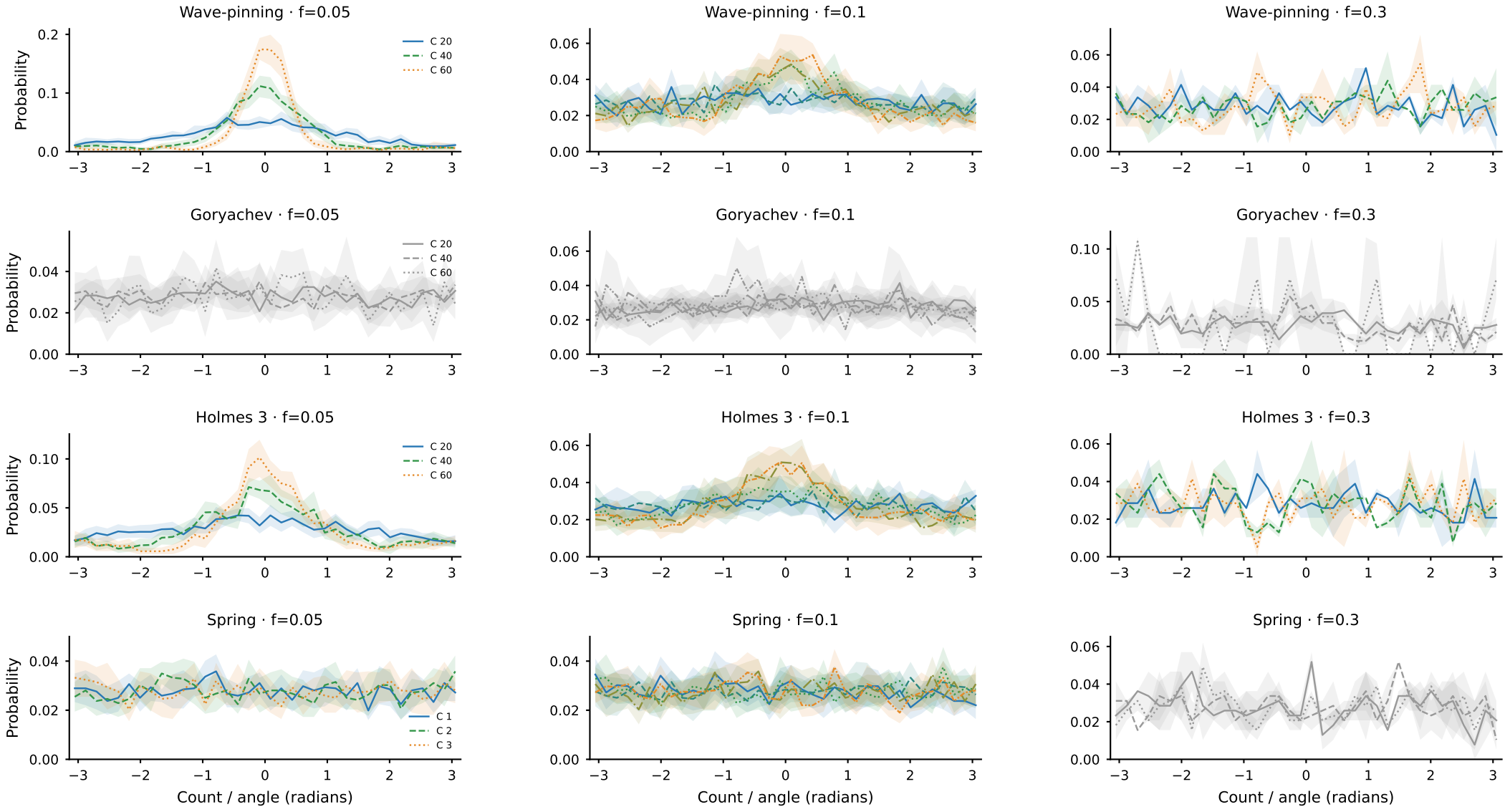
Longest canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_094_S9_dominant_heading_turn_8.0_3.csv

S9 · Longest canonical A5 observations · dominant heading turn · 8 observation frames



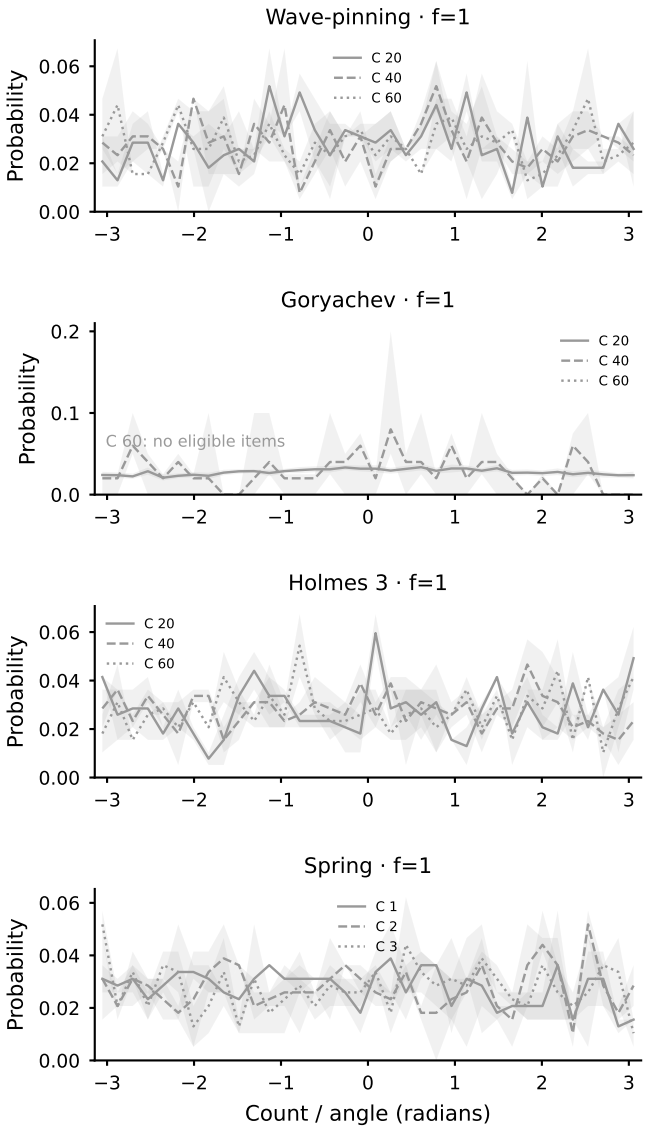
Longest canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000/128000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_095_S9_dominant_heading_turn_8.0_6.csv

S9 · Longest canonical A5 observations · dominant heading turn · 8 observation frames



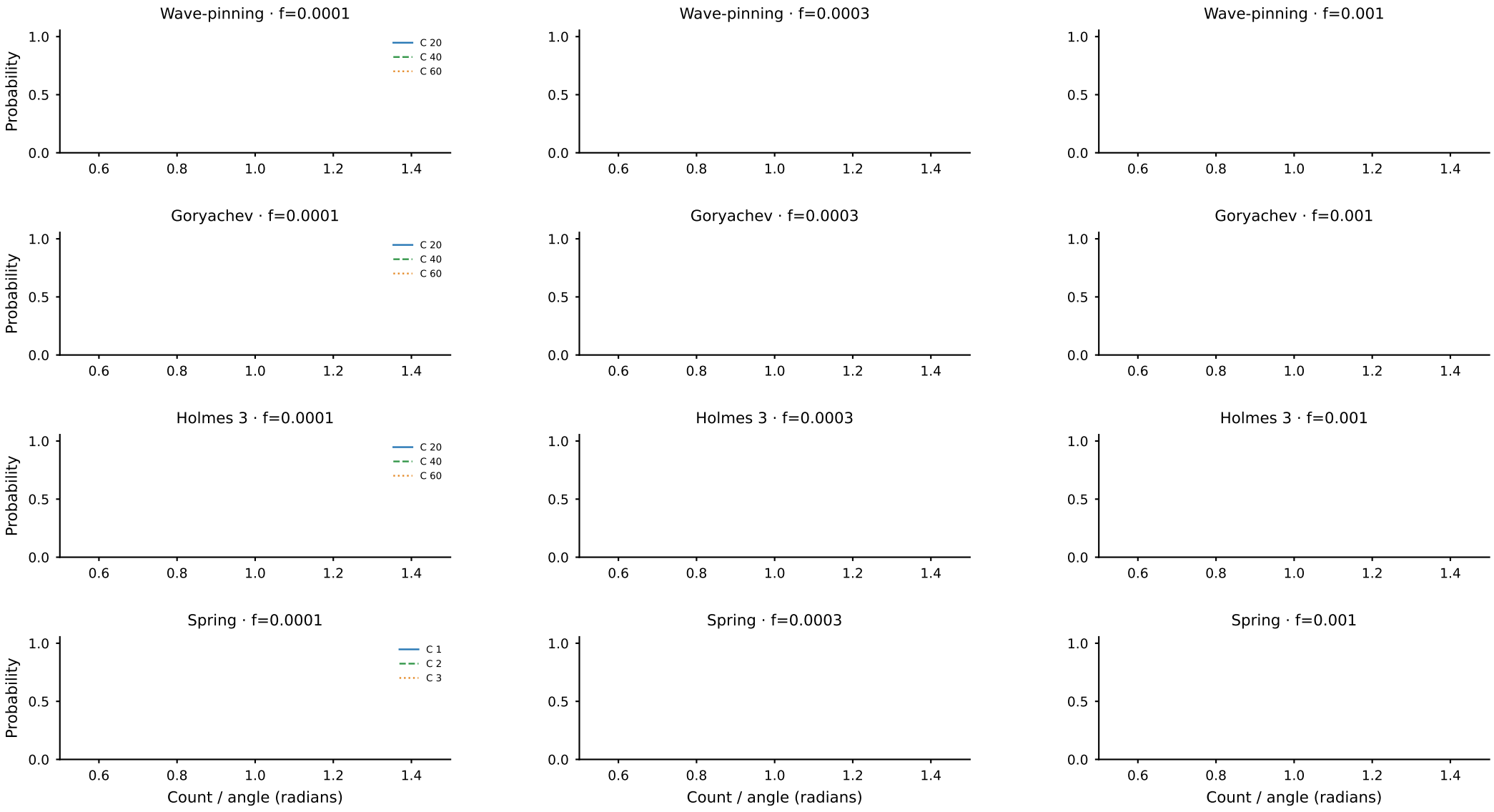
Longest canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000/128000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_096_S9_dominant_heading_turn_8.0_9.csv

S9 · Longest canonical A5 observations · dominant heading turn · 8 observation frames



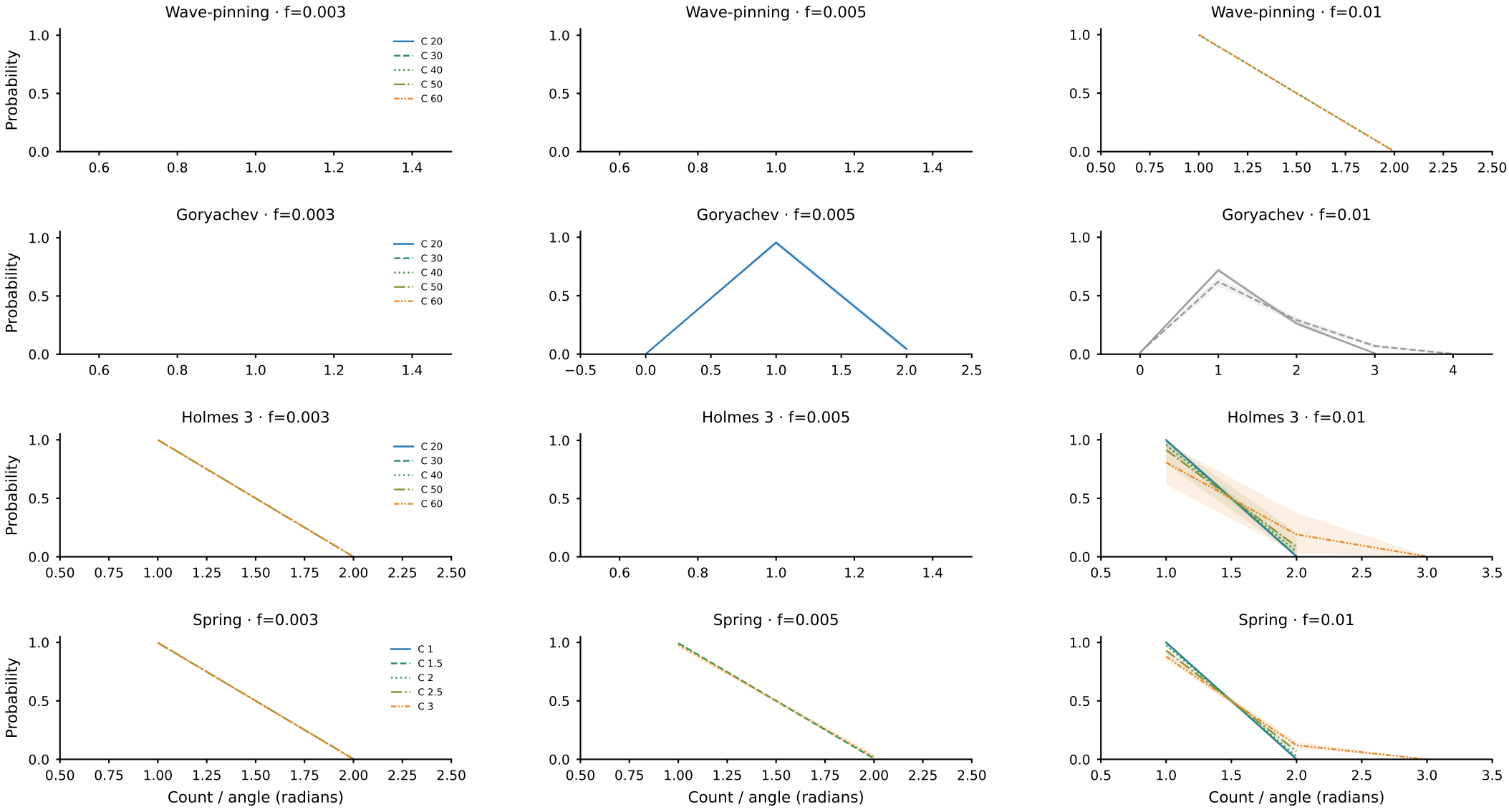
Longest canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000/128000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_097_S9_dominant_heading_turn_8.0_12.csv

S9 · Longest canonical A5 observations · simultaneous region count



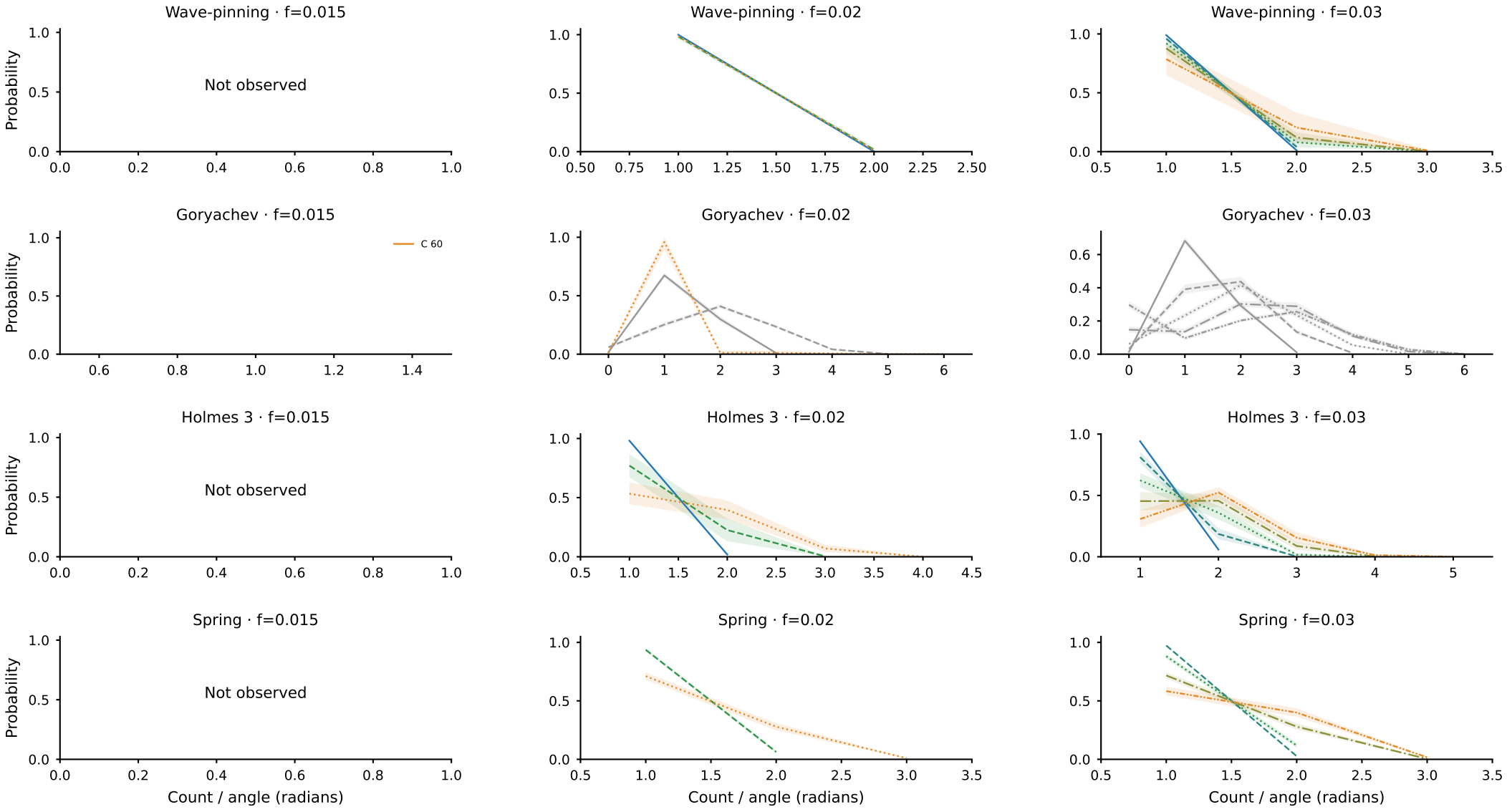
Longest canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_098_S9_simultaneous_region_count_None_0.csv

S9 · Longest canonical A5 observations · simultaneous region count



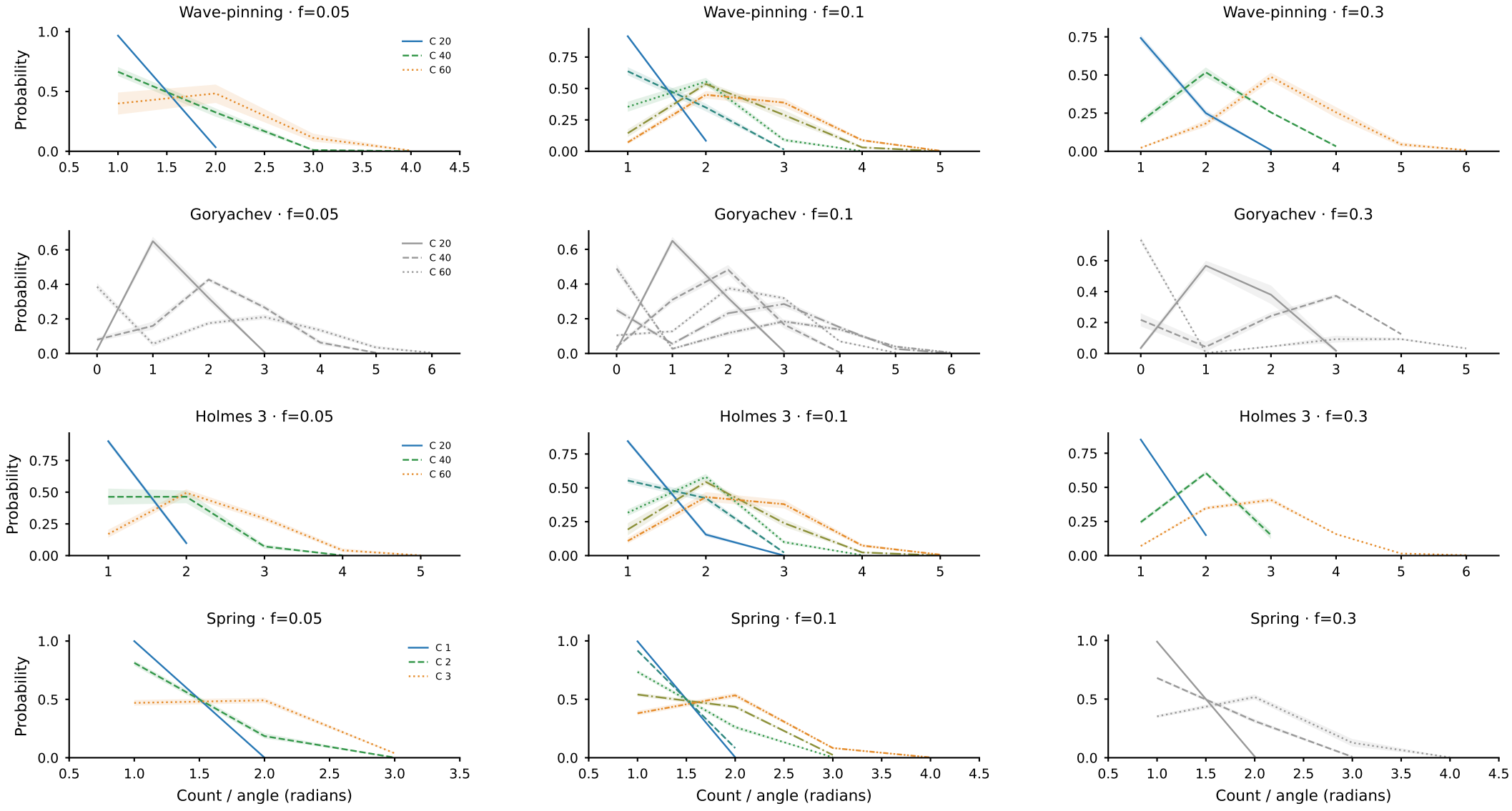
Longest canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_099_S9_simultaneous_region_count_None_3.csv

S9 · Longest canonical A5 observations · simultaneous region count



Longest canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000/128000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_100_S9_simultaneous_region_count_None_6.csv

S9 · Longest canonical A5 observations · simultaneous region count



Longest canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000/128000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
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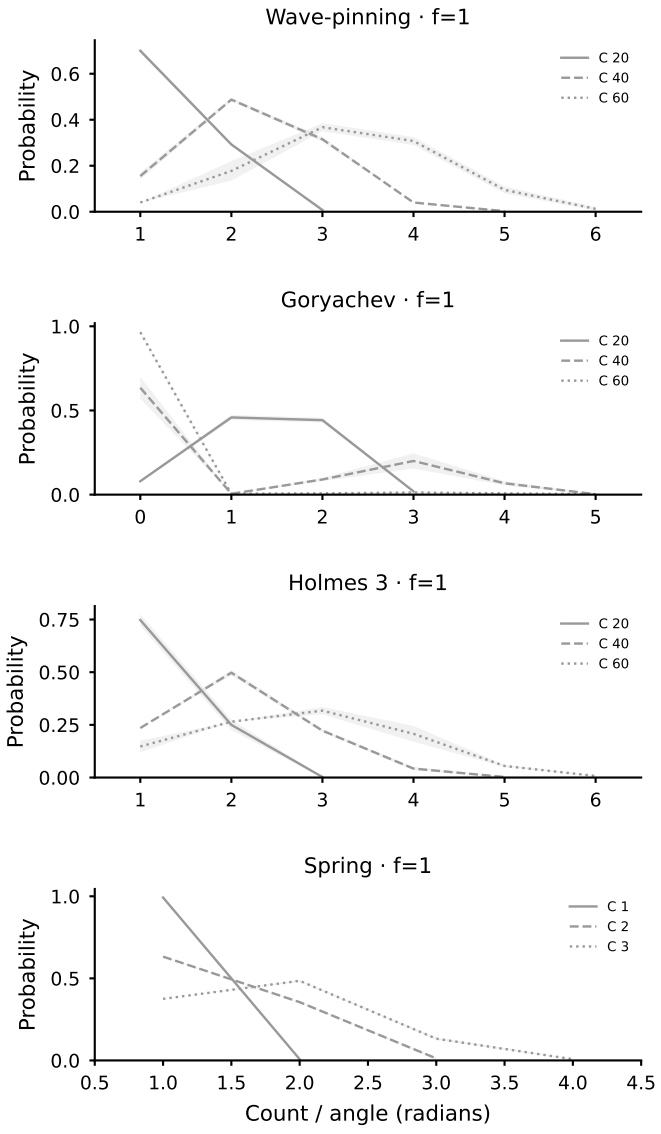
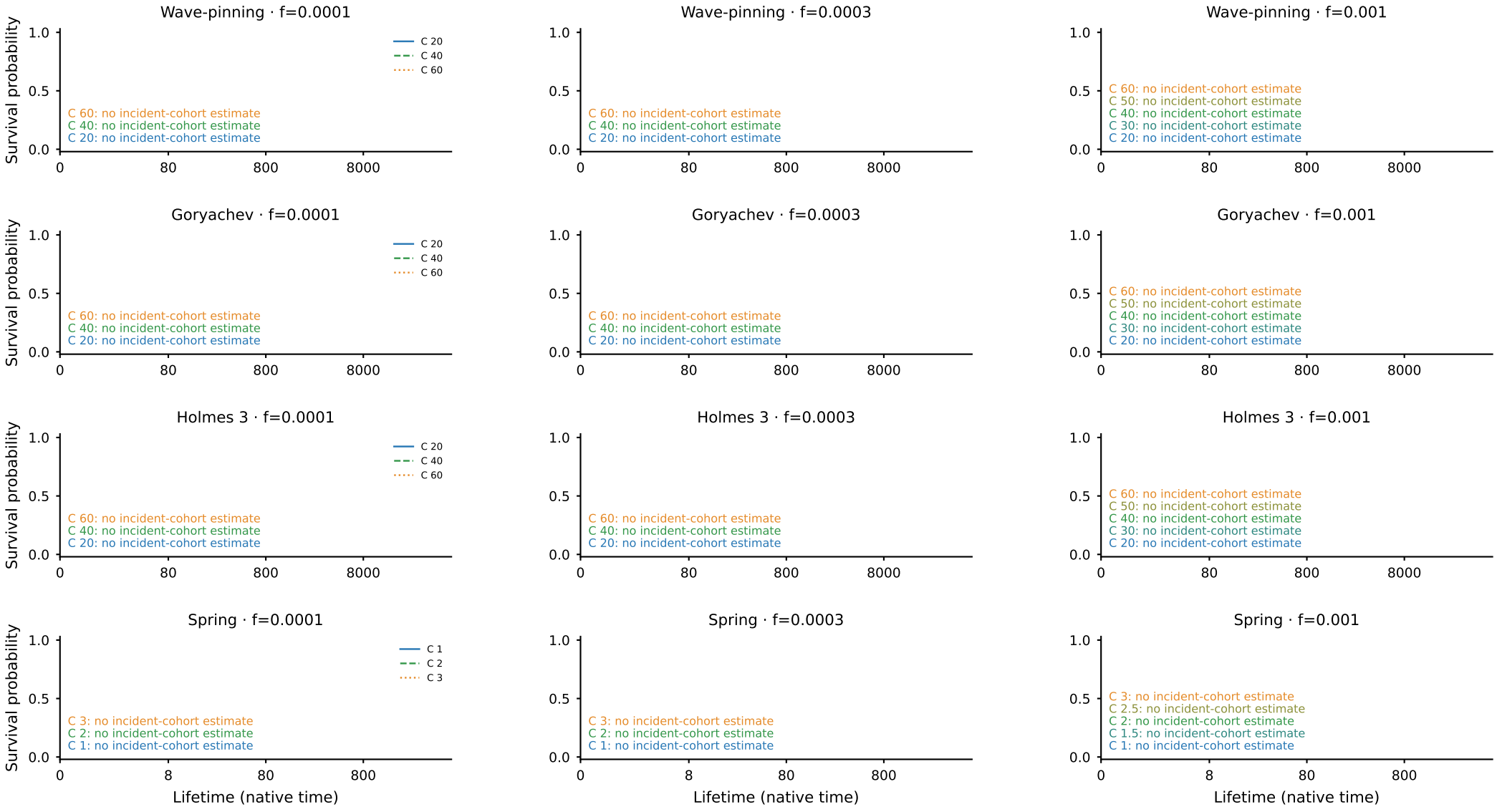
S9 · Longest canonical A5 observations · simultaneous region count



Longest canonical A5 observations. Imaging primary only; size colours match S1 and forced conditions are grey. Size colours and line patterns distinguish all declared sizes. Histogram bin centres are displayed; exact bins, finite-seed uncertainty, observed totals and frame intervals remain in CSV. Finite native and imaging cadences limit birth/turn observability; exact cadences remain in CSV. Missing densities remain missing, not zero. Native H: Wave-pinning 32000; Goryachev 32000/128000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_102_S9_simultaneous_region_count_None_12.csv

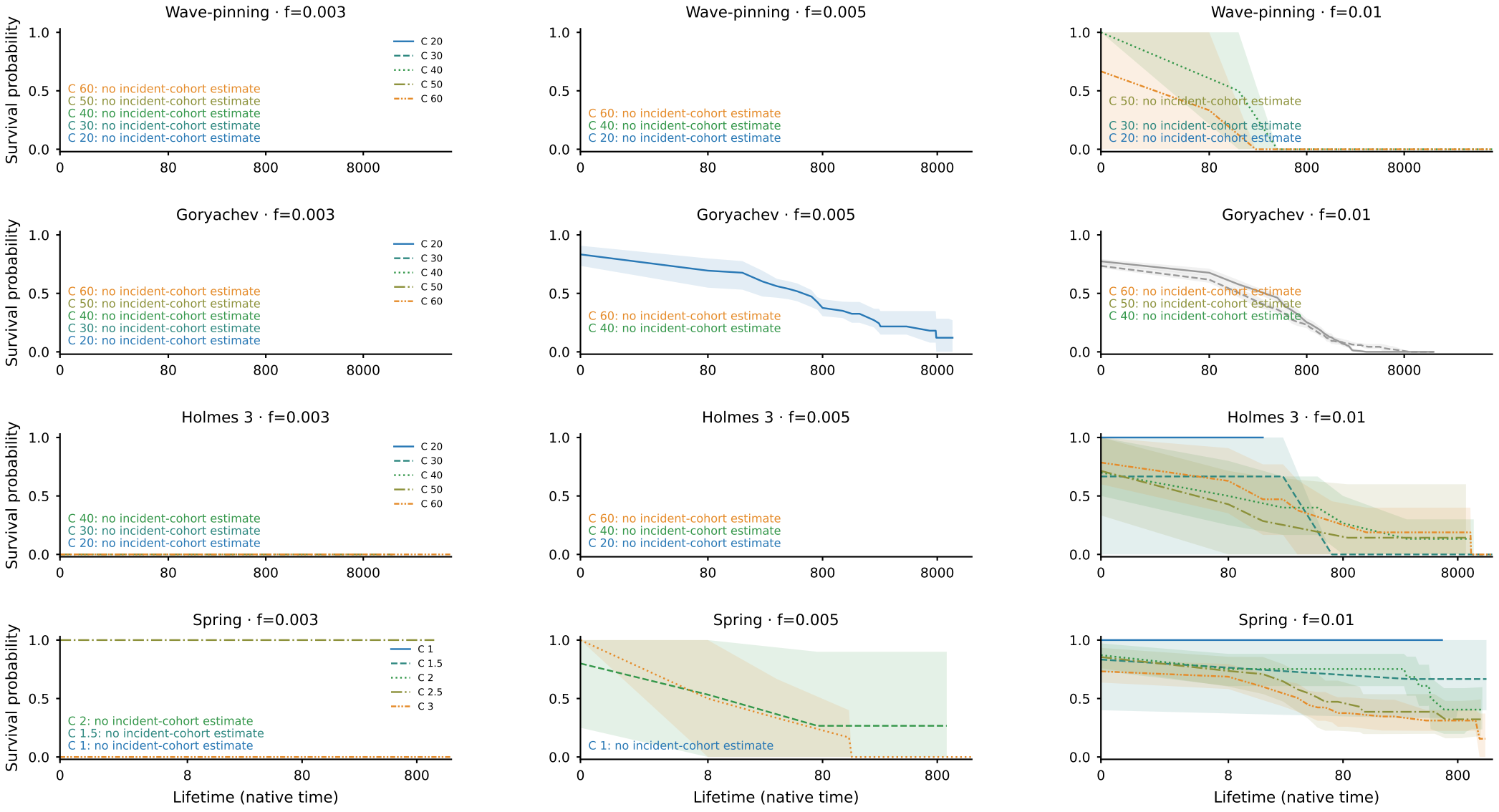
S9 · Longest canonical A5 observations · Incident-region lifetime survival · lower observation clock



Longest canonical A5 observations. Kaplan–Meier curves use incident tracks; preexisting left-censored regions are excluded. End-of-window and unresolved states retain right censoring. Lower/observed/upper clocks preserve cadence-limited times; symlog lifetime axes are linear through one observation cadence. Size colours and line patterns distinguish sizes; forced conditions are grey. Whole-seed CIs and at-risk/censoring counts remain in CSV. Unidentifiable tails are not extrapolated. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.

CSV: figure4_S9_longest_canonical_20261020_103_S9_lifetime_lower_0.csv

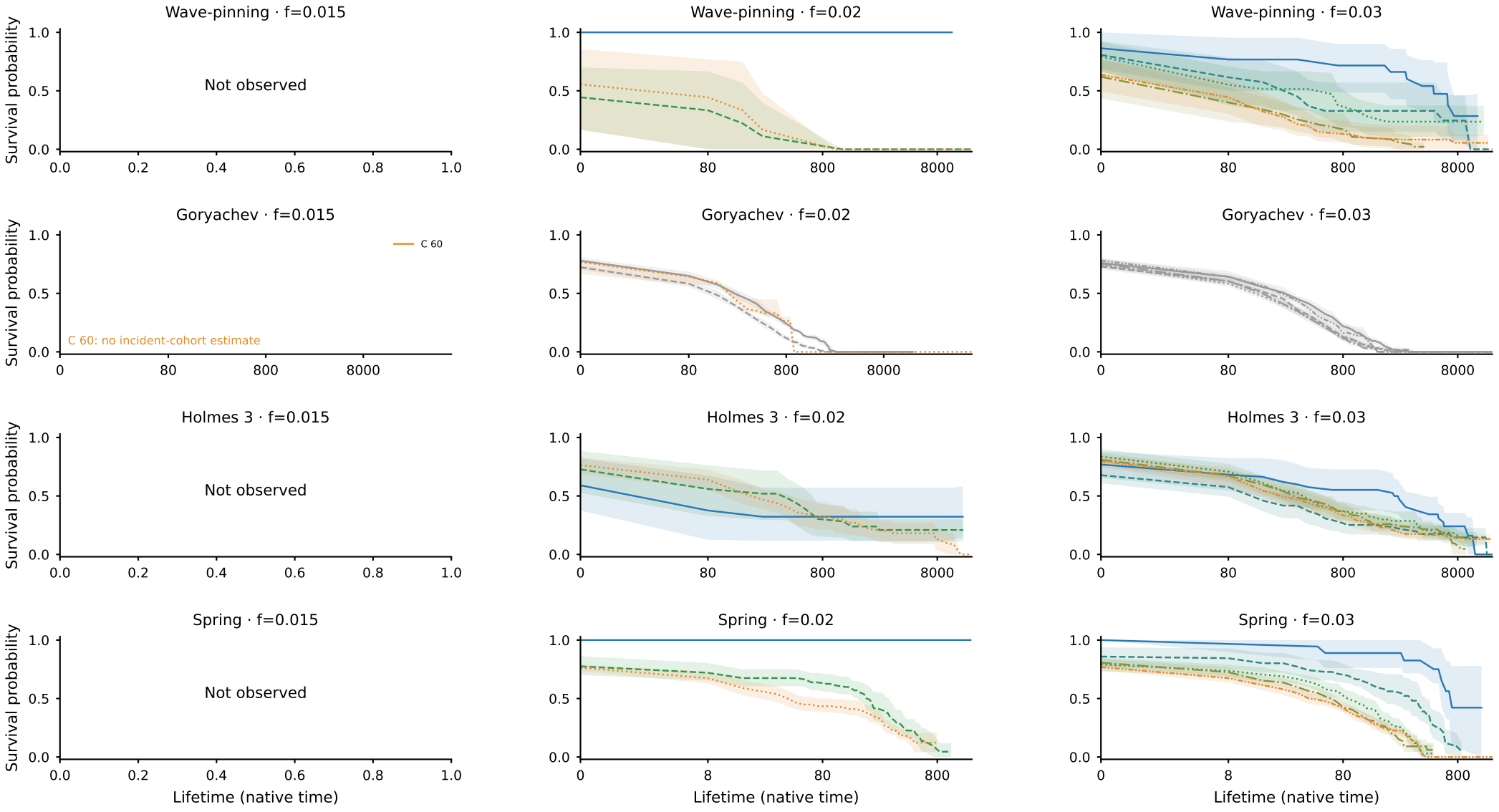
S9 · Longest canonical A5 observations · Incident-region lifetime survival · lower observation clock



Longest canonical A5 observations. Kaplan–Meier curves use incident tracks; preexisting left-censored regions are excluded. End-of-window and unresolved states retain right censoring. Lower/observed/upper clocks preserve cadence-limited times; symlog lifetime axes are linear through one observation cadence. Size colours and line patterns distinguish sizes; forced conditions are grey. Whole-seed CIs and at-risk/censoring counts remain in CSV. Unidentifiable tails are not extrapolated. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.

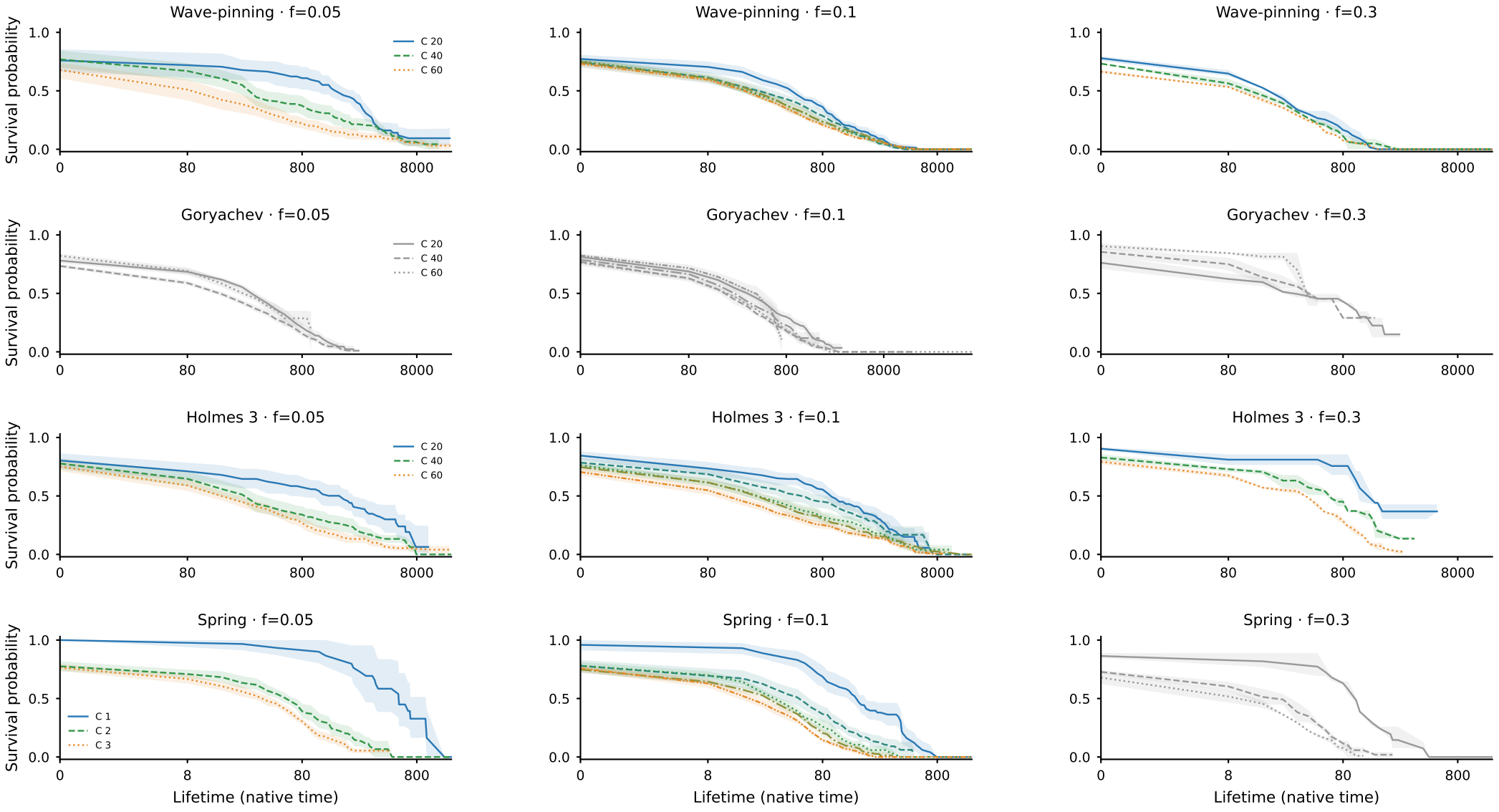
CSV: figure4_S9_longest_canonical_20261020_104_S9_lifetime_lower_3.csv

S9 · Longest canonical A5 observations · Incident-region lifetime survival · lower observation clock



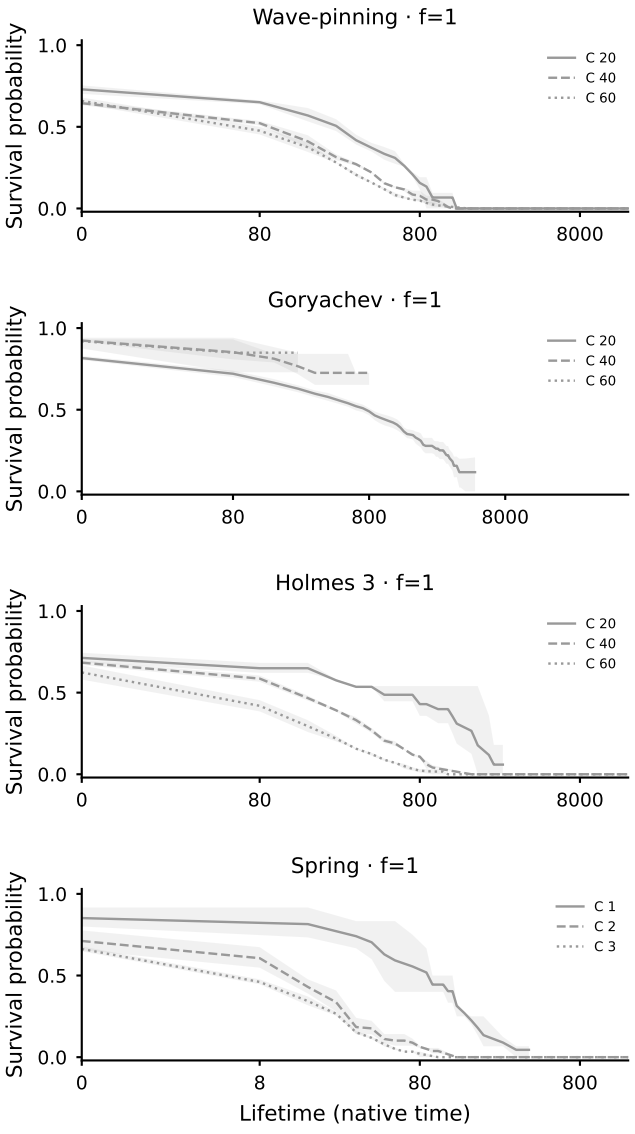
Longest canonical A5 observations. Kaplan–Meier curves use incident tracks; preexisting left-censored regions are excluded. End-of-window and unresolved states retain right censoring. Lower/observed/upper clocks preserve cadence-limited times; symlog lifetime axes are linear through one observation cadence. Size colours and line patterns distinguish sizes; forced conditions are grey. Whole-seed CIs and at-risk/censoring counts remain in CSV. Unidentifiable tails are not extrapolated. Native H: Wave-pinning 32000; Goryachev 32000/128000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_105_S9_lifetime_lower_6.csv

S9 · Longest canonical A5 observations · Incident-region lifetime survival · lower observation clock



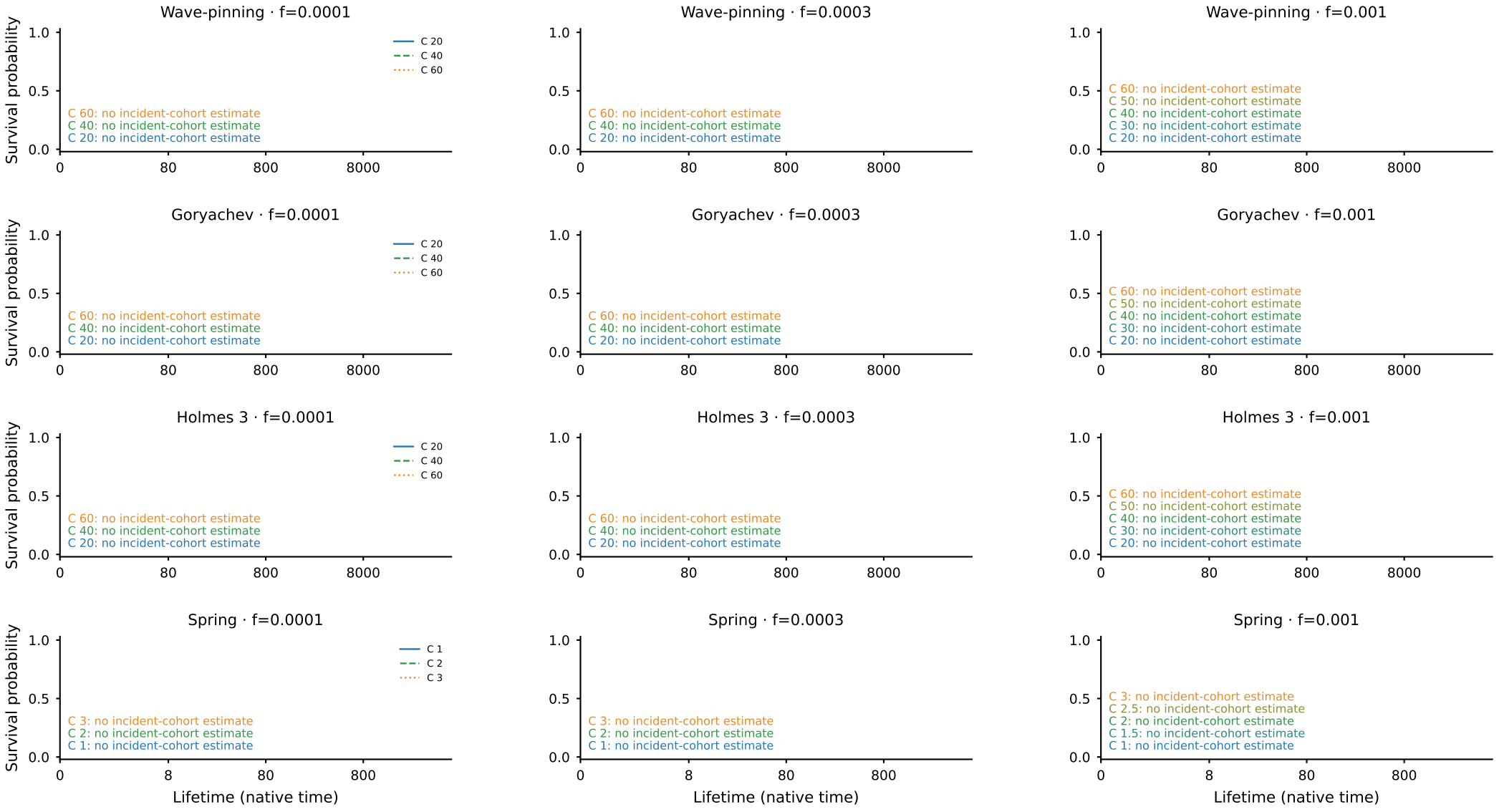
Longest canonical A5 observations. Kaplan–Meier curves use incident tracks; preexisting left-censored regions are excluded. End-of-window and unresolved states retain right censoring. Lower/observed/upper clocks preserve cadence-limited times; symlog lifetime axes are linear through one observation cadence. Size colours and line patterns distinguish sizes; forced conditions are grey. Whole-seed CIs and at-risk/censoring counts remain in CSV. Unidentifiable tails are not extrapolated. Native H: Wave-pinning 32000; Goryachev 32000/128000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_106_S9_lifetime_lower_9.csv

S9 · Longest canonical A5 observations · Incident-region lifetime survival · lower observation clock



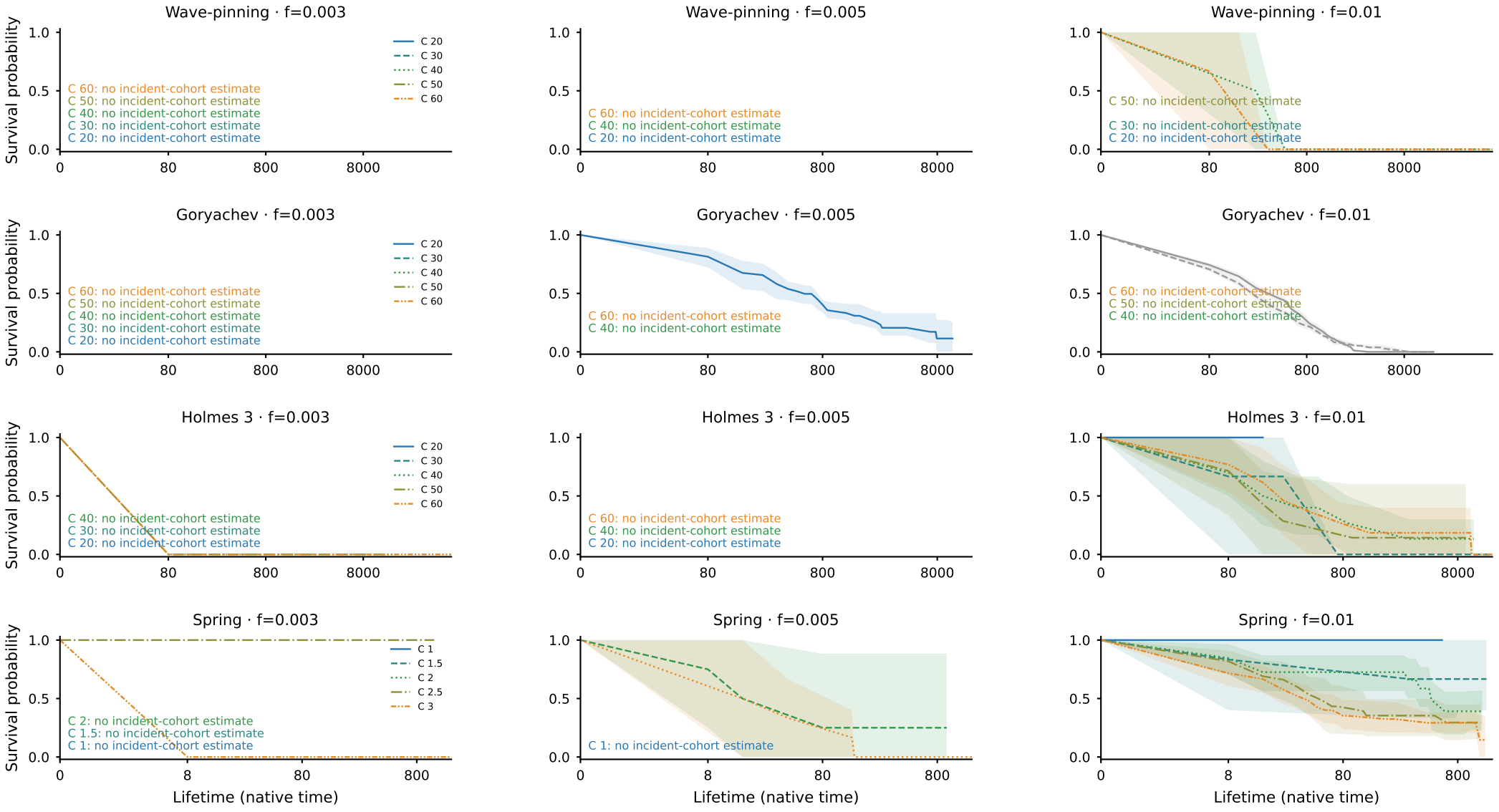
Longest canonical A5 observations. Kaplan–Meier curves use incident tracks; preexisting left-censored regions are excluded. End-of-window and unresolved states retain right censoring. Lower/observed/upper clocks preserve cadence-limited times; symlog lifetime axes are linear through one observation cadence. Size colours and line patterns distinguish sizes; forced conditions are grey. Whole-seed CIs and at-risk/censoring counts remain in CSV. Unidentifiable tails are not extrapolated. Native H: Wave-pinning 32000; Goryachev 32000/128000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_107_S9_lifetime_lower_12.csv

S9 · Longest canonical A5 observations · Incident-region lifetime survival · observed observation clock



Longest canonical A5 observations. Kaplan–Meier curves use incident tracks; preexisting left-censored regions are excluded. End-of-window and unresolved states retain right censoring. Lower/observed/upper clocks preserve cadence-limited times; symlog lifetime axes are linear through one observation cadence. Size colours and line patterns distinguish sizes; forced conditions are grey. Whole-seed CIs and at-risk/censoring counts remain in CSV. Unidentifiable tails are not extrapolated. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_108_S9_lifetime_observed_0.csv

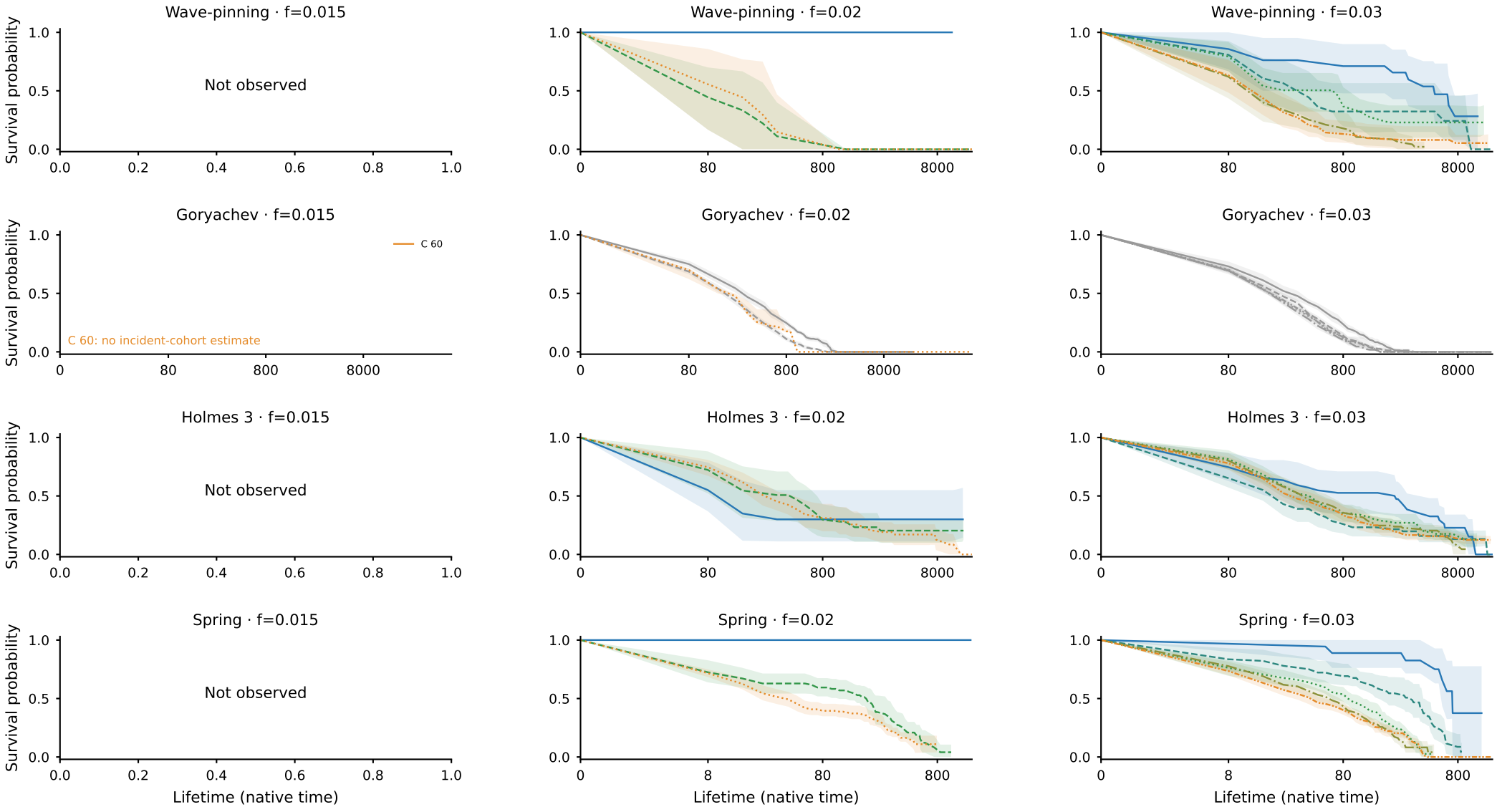
S9 · Longest canonical A5 observations · Incident-region lifetime survival · observed observation clock



Longest canonical A5 observations. Kaplan–Meier curves use incident tracks; preexisting left-censored regions are excluded. End-of-window and unresolved states retain right censoring. Lower/observed/upper clocks preserve cadence-limited times; symlog lifetime axes are linear through one observation cadence. Size colours and line patterns distinguish sizes; forced conditions are grey. Whole-seed CIs and at-risk/censoring counts remain in CSV. Unidentifiable tails are not extrapolated. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.

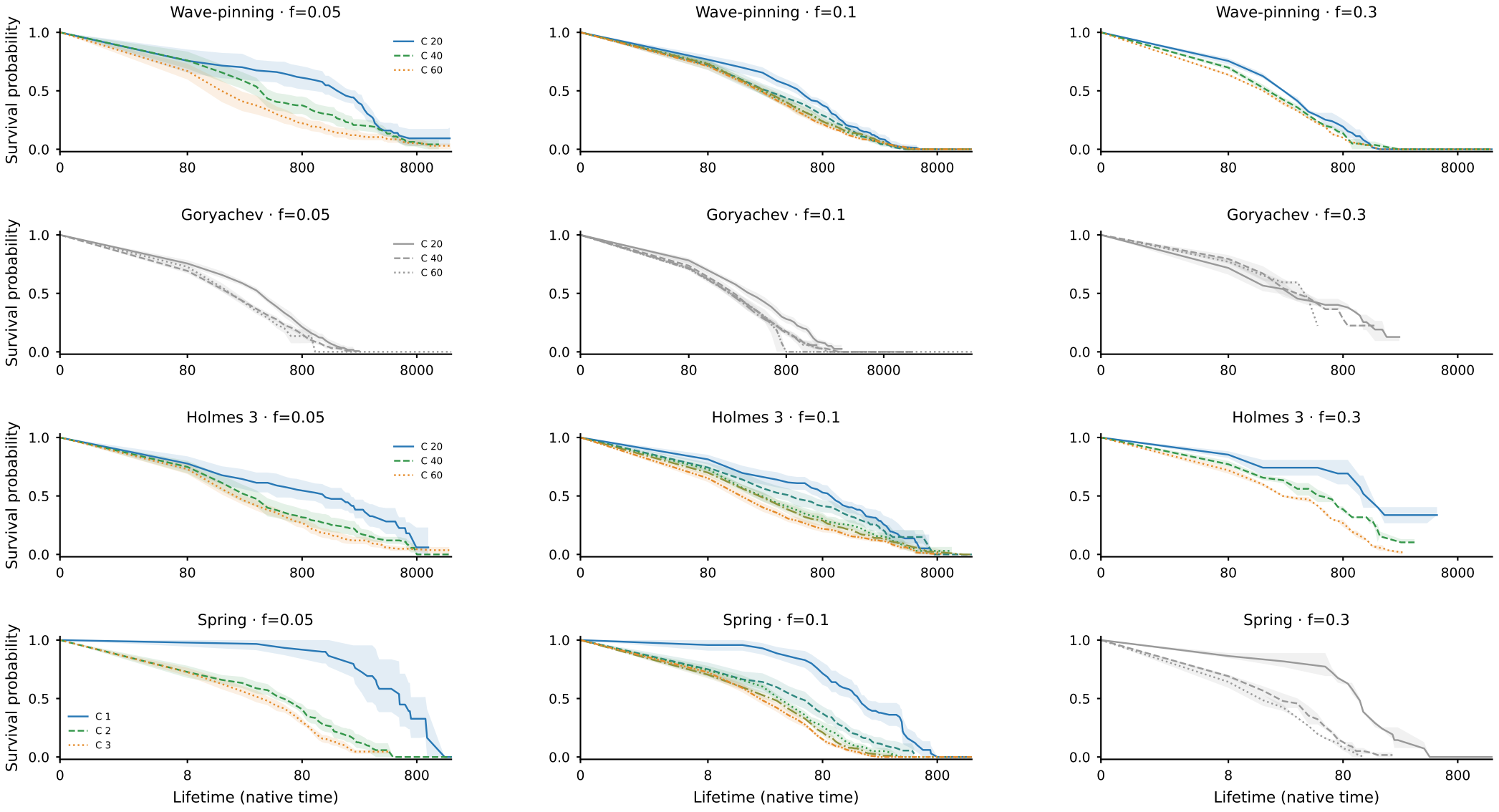
CSV: figure4_S9_longest_canonical_20261020_109_S9_lifetime_observed_3.csv

S9 · Longest canonical A5 observations · Incident-region lifetime survival · observed observation clock



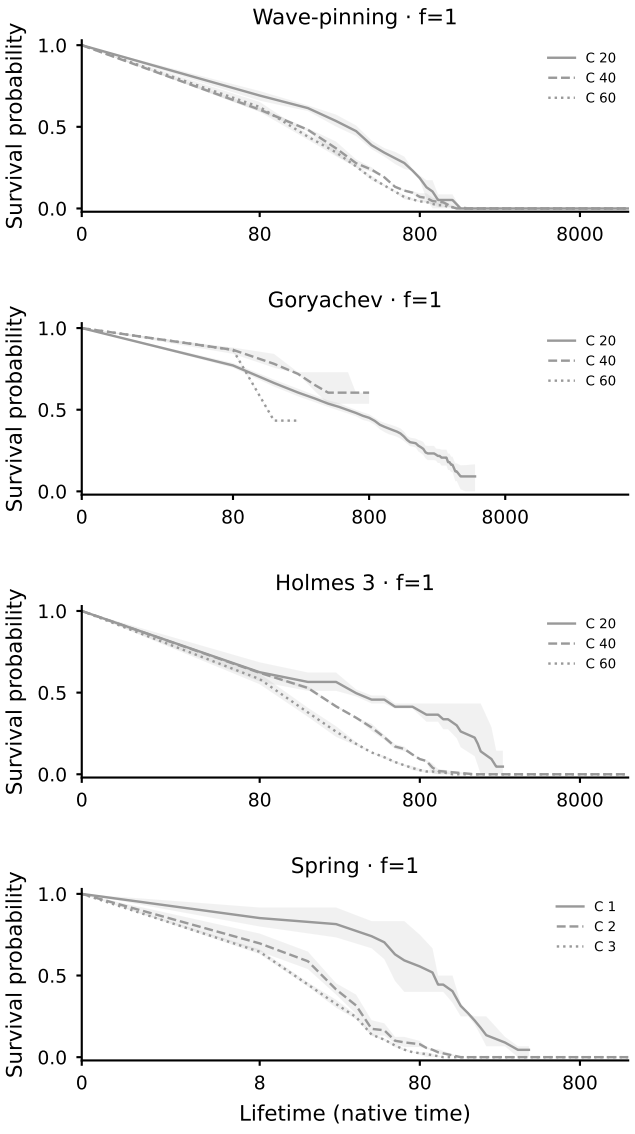
Longest canonical A5 observations. Kaplan–Meier curves use incident tracks; preexisting left-censored regions are excluded. End-of-window and unresolved states retain right censoring. Lower/observed/upper clocks preserve cadence-limited times; symlog lifetime axes are linear through one observation cadence. Size colours and line patterns distinguish sizes; forced conditions are grey. Whole-seed CIs and at-risk/censoring counts remain in CSV. Unidentifiable tails are not extrapolated. Native H: Wave-pinning 32000; Goryachev 32000/128000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_110_S9_lifetime_observed_6.csv

S9 · Longest canonical A5 observations · Incident-region lifetime survival · observed observation clock



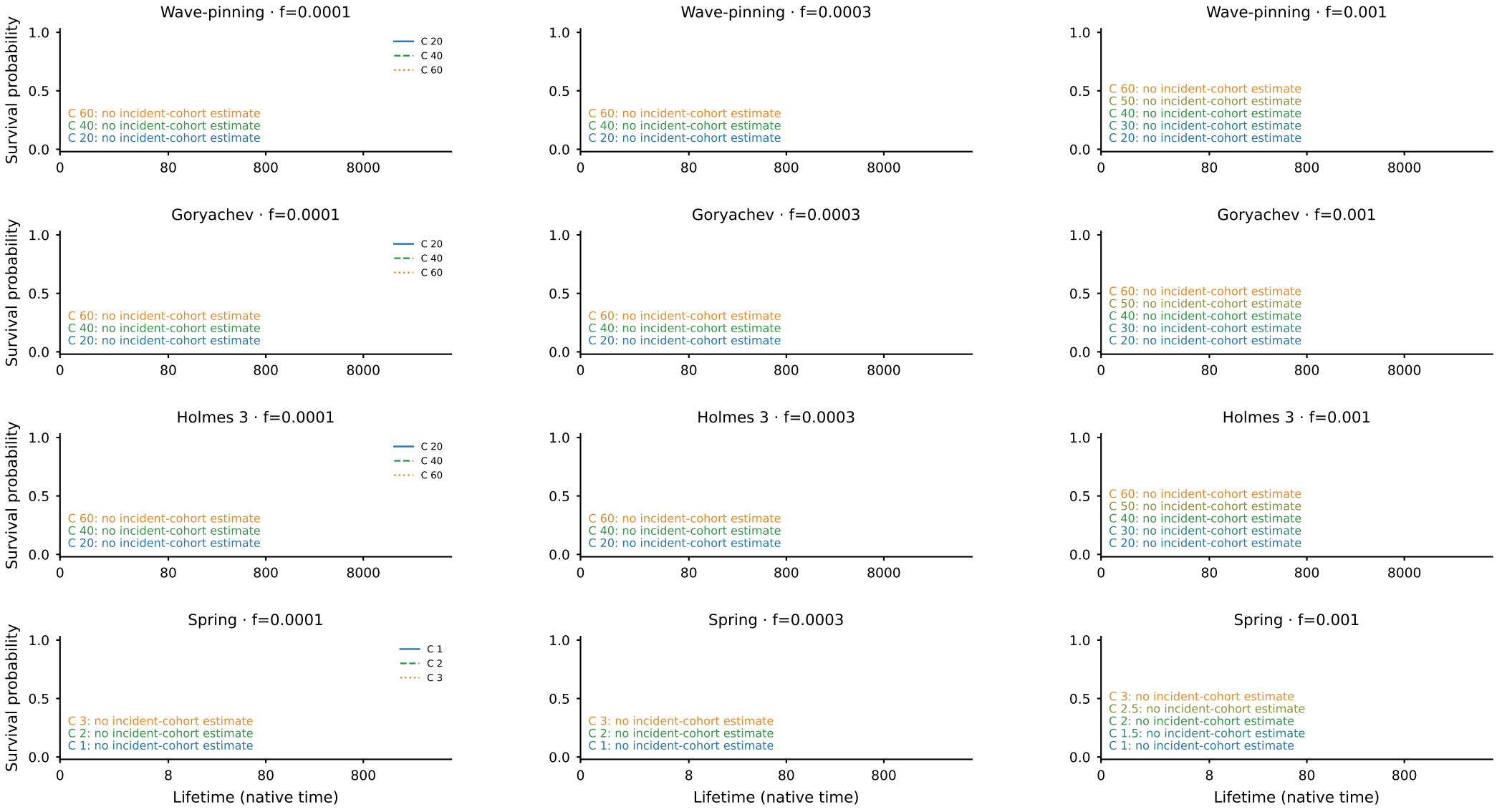
Longest canonical A5 observations. Kaplan–Meier curves use incident tracks; preexisting left-censored regions are excluded. End-of-window and unresolved states retain right censoring. Lower/observed/upper clocks preserve cadence-limited times; symlog lifetime axes are linear through one observation cadence. Size colours and line patterns distinguish sizes; forced conditions are grey. Whole-seed CIs and at-risk/censoring counts remain in CSV. Unidentifiable tails are not extrapolated. Native H: Wave-pinning 32000; Goryachev 32000/128000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_111_S9_lifetime_observed_9.csv

S9 · Longest canonical A5 observations · Incident-region lifetime survival · observed observation clock



Longest canonical A5 observations. Kaplan–Meier curves use incident tracks; preexisting left-censored regions are excluded. End-of-window and unresolved states retain right censoring. Lower/observed/upper clocks preserve cadence-limited times; symlog lifetime axes are linear through one observation cadence. Size colours and line patterns distinguish sizes; forced conditions are grey. Whole-seed CIs and at-risk/censoring counts remain in CSV. Unidentifiable tails are not extrapolated. Native H: Wave-pinning 32000; Goryachev 32000/128000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_112_S9_lifetime_observed_12.csv

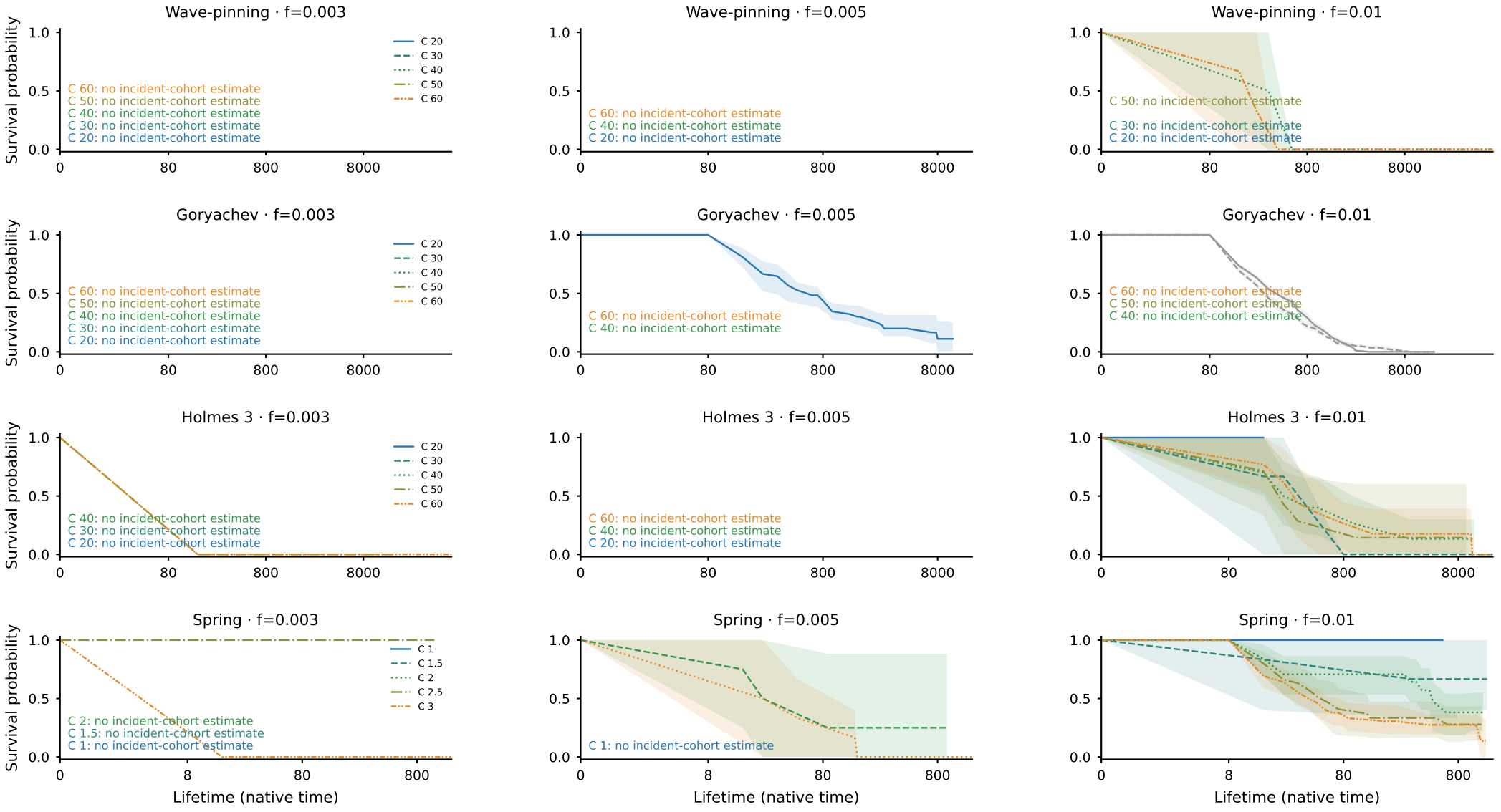
S9 · Longest canonical A5 observations · Incident-region lifetime survival · upper observation clock



Longest canonical A5 observations. Kaplan–Meier curves use incident tracks; preexisting left-censored regions are excluded. End-of-window and unresolved states retain right censoring. Lower/observed/upper clocks preserve cadence-limited times; symlog lifetime axes are linear through one observation cadence. Size colours and line patterns distinguish sizes; forced conditions are grey. Whole-seed CIs and at-risk/censoring counts remain in CSV. Unidentifiable tails are not extrapolated. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200/12800; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.

CSV: figure4_S9_longest_canonical_20261020_113_S9_lifetime_upper_0.csv

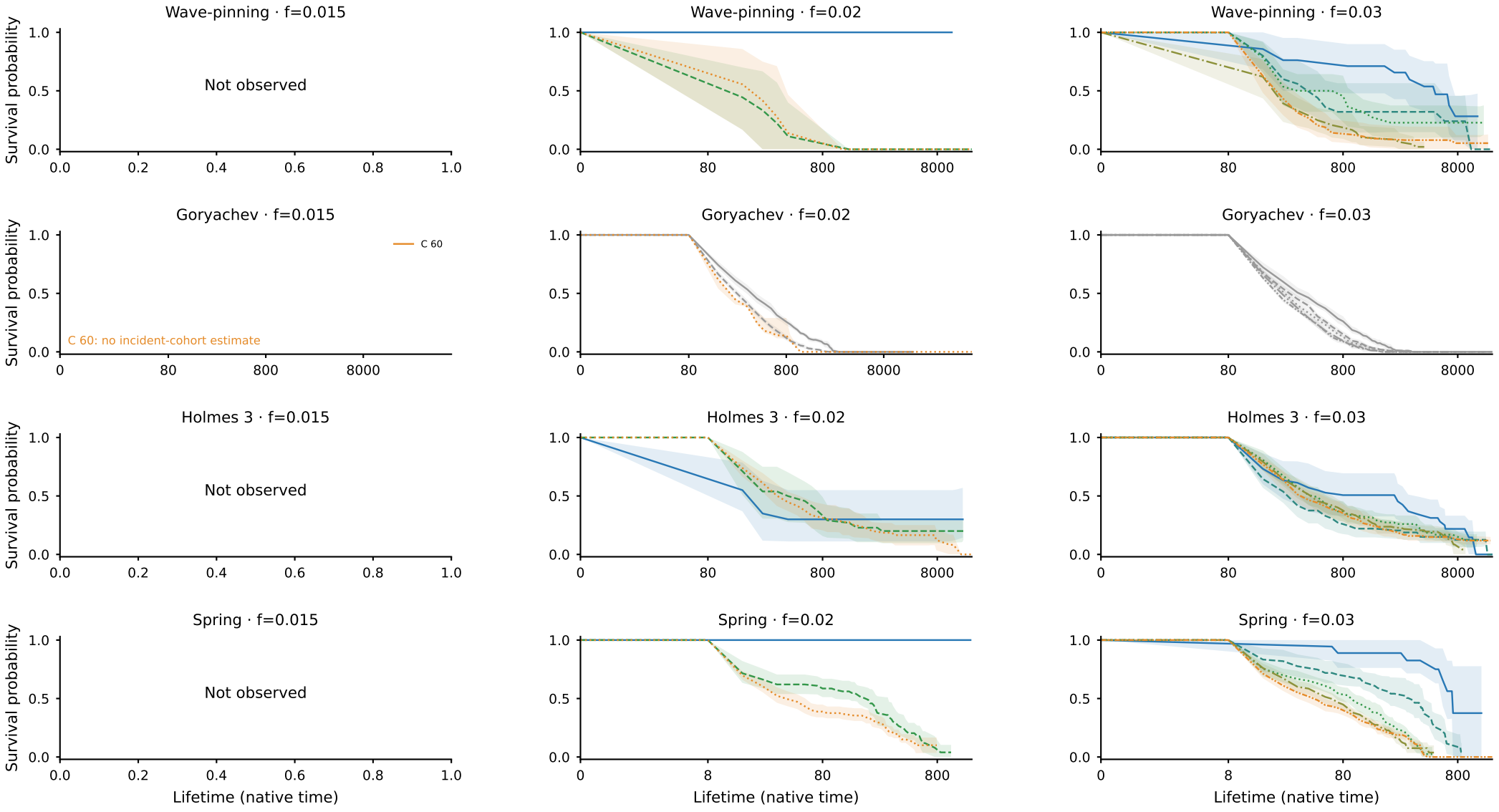
S9 · Longest canonical A5 observations · Incident-region lifetime survival · upper observation clock



Longest canonical A5 observations. Kaplan–Meier curves use incident tracks; preexisting left-censored regions are excluded. End-of-window and unresolved states retain right censoring. Lower/observed/upper clocks preserve cadence-limited times; symlog lifetime axes are linear through one observation cadence. Size colours and line patterns distinguish sizes; forced conditions are grey. Whole-seed CIs and at-risk/censoring counts remain in CSV. Unidentifiable tails are not extrapolated. Native H: Wave-pinning 32000/128000; Goryachev 32000/128000; Holmes 3 32000/128000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.

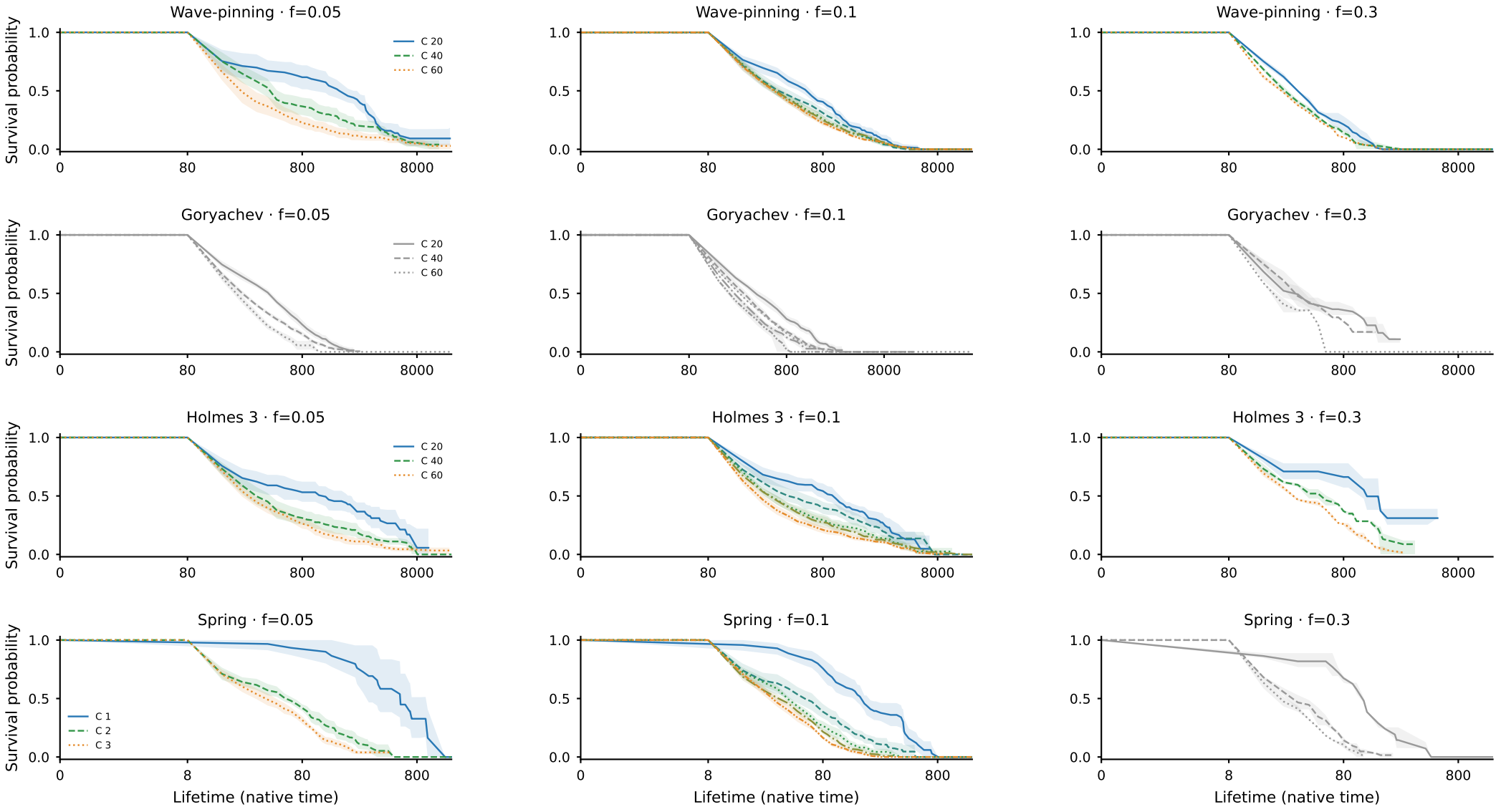
CSV: figure4_S9_longest_canonical_20261020_114_S9_lifetime_upper_3.csv

S9 · Longest canonical A5 observations · Incident-region lifetime survival · upper observation clock



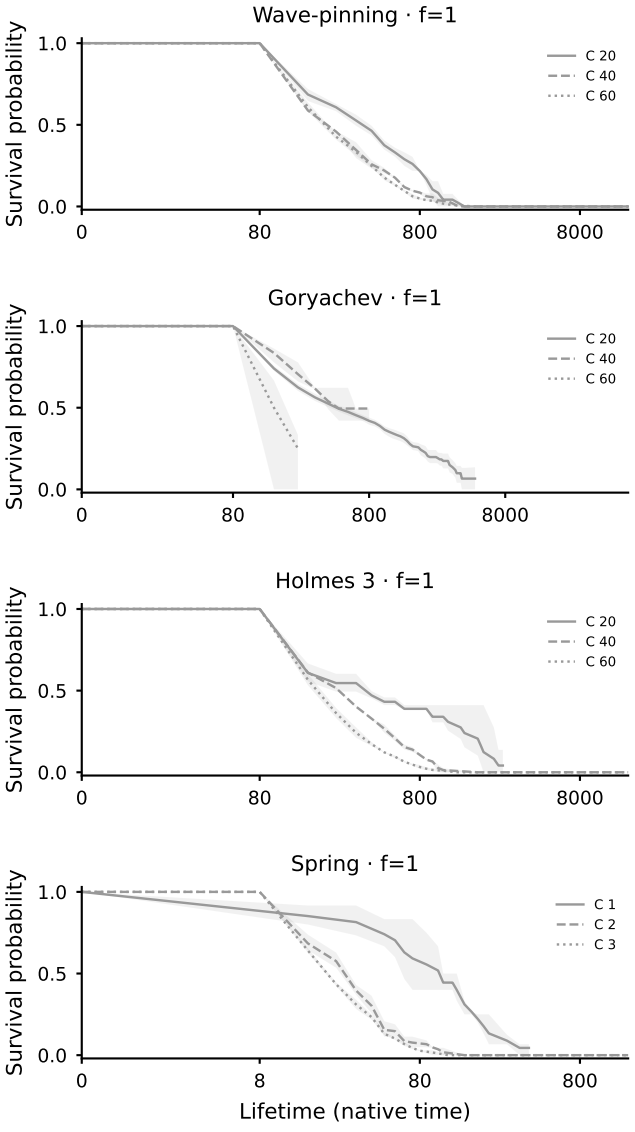
Longest canonical A5 observations. Kaplan–Meier curves use incident tracks; preexisting left-censored regions are excluded. End-of-window and unresolved states retain right censoring. Lower/observed/upper clocks preserve cadence-limited times; symlog lifetime axes are linear through one observation cadence. Size colours and line patterns distinguish sizes; forced conditions are grey. Whole-seed CIs and at-risk/censoring counts remain in CSV. Unidentifiable tails are not extrapolated. Native H: Wave-pinning 32000; Goryachev 32000/128000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_115_S9_lifetime_upper_6.csv

S9 · Longest canonical A5 observations · Incident-region lifetime survival · upper observation clock



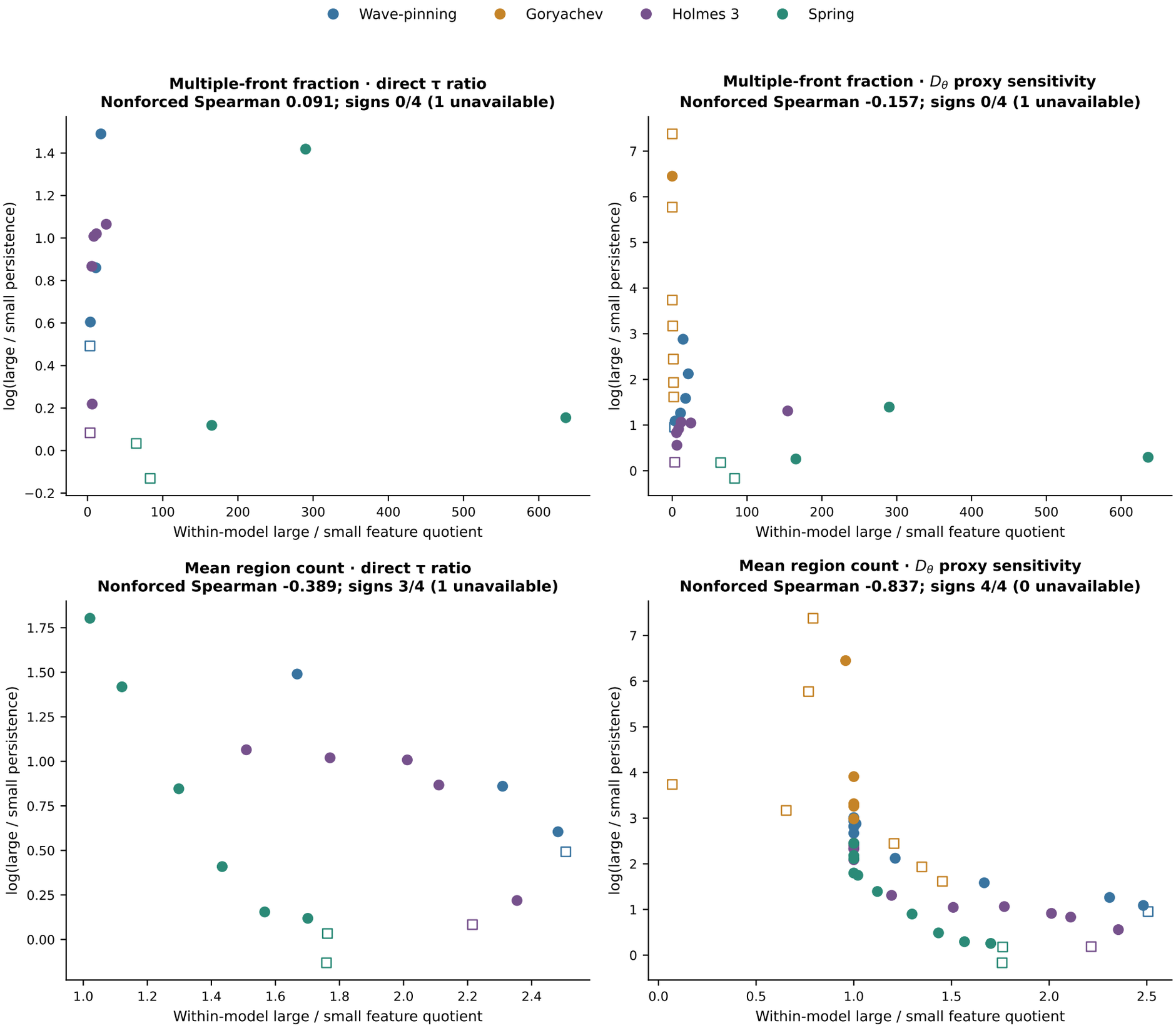
Longest canonical A5 observations. Kaplan–Meier curves use incident tracks; preexisting left-censored regions are excluded. End-of-window and unresolved states retain right censoring. Lower/observed/upper clocks preserve cadence-limited times; symlog lifetime axes are linear through one observation cadence. Size colours and line patterns distinguish sizes; forced conditions are grey. Whole-seed CIs and at-risk/censoring counts remain in CSV. Unidentifiable tails are not extrapolated. Native H: Wave-pinning 32000; Goryachev 32000/128000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_116_S9_lifetime_upper_9.csv

S9 · Longest canonical A5 observations · Incident-region lifetime survival · upper observation clock



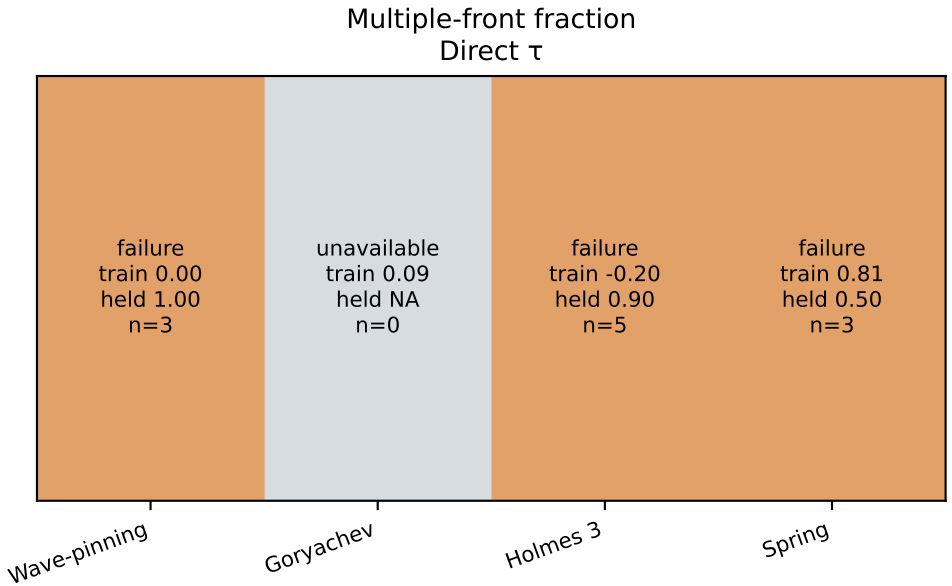
Longest canonical A5 observations. Kaplan–Meier curves use incident tracks; preexisting left-censored regions are excluded. End-of-window and unresolved states retain right censoring. Lower/observed/upper clocks preserve cadence-limited times; symlog lifetime axes are linear through one observation cadence. Size colours and line patterns distinguish sizes; forced conditions are grey. Whole-seed CIs and at-risk/censoring counts remain in CSV. Unidentifiable tails are not extrapolated. Native H: Wave-pinning 32000; Goryachev 32000/128000; Holmes 3 32000; Spring 3200; analysis window [H/2,H]. Event counts per trajectory cover that window; no cross-model rate ranking. Actual independent seed counts are condition-specific; some weakest-input groups have two, and controls/companions add no seeds. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S9_longest_canonical_20261020_117_S9_lifetime_upper_12.csv

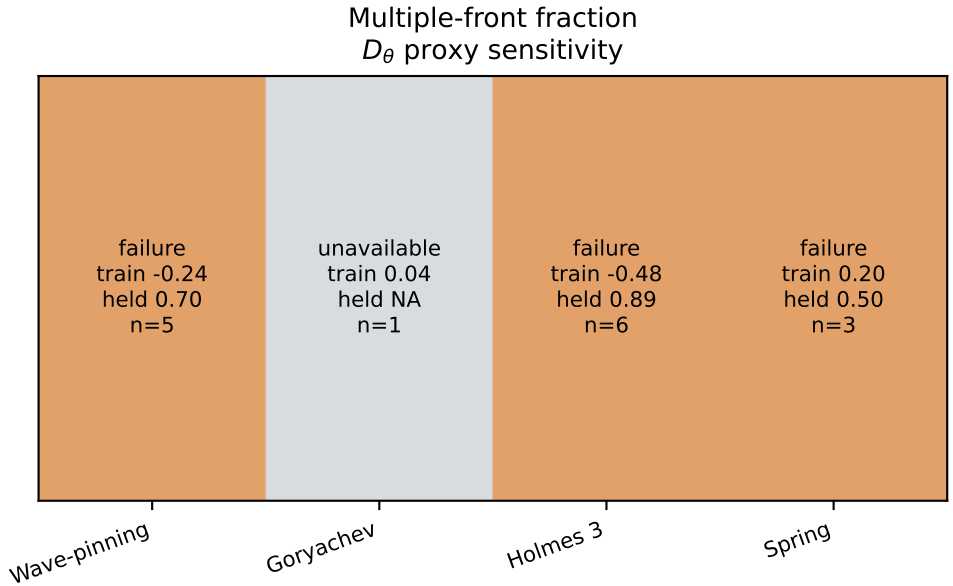
S10 · Direct-τ A6 prediction criterion is not met

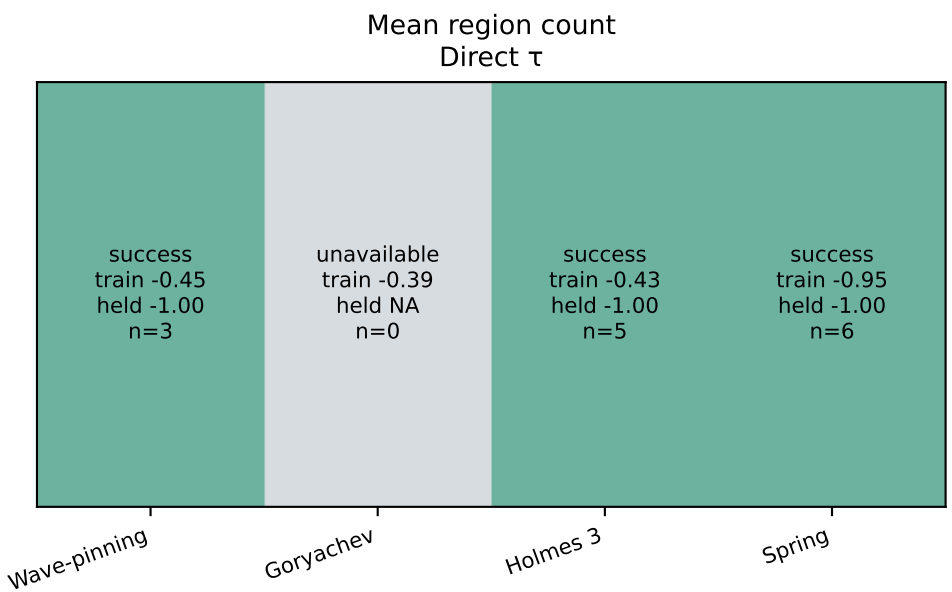


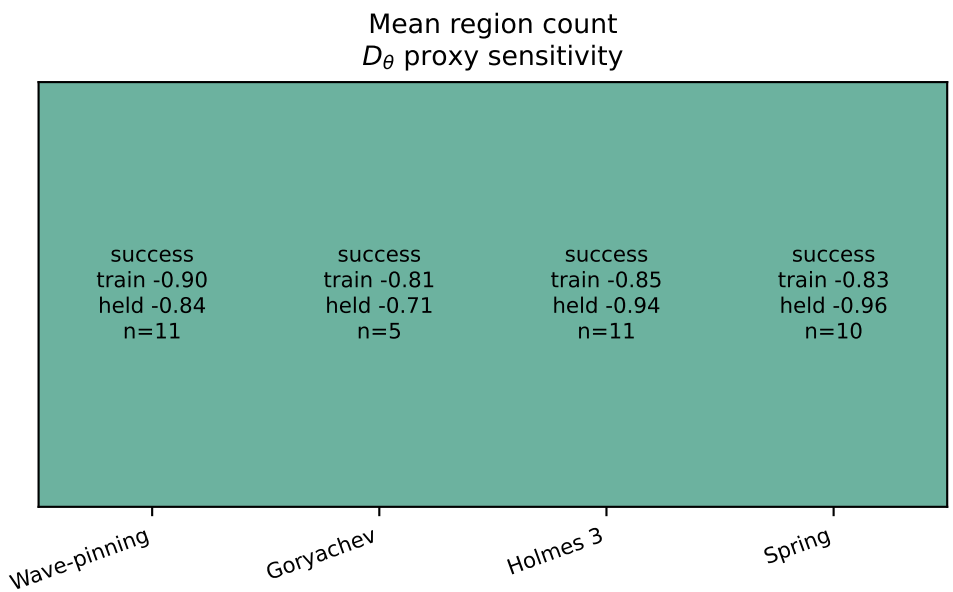
Arm common · σ=1 · actual per-size horizons retained in the source CSV. Crosses: unknown forcing, excluded from primary.
Filled circles: nonforced primary conditions; open squares: forced context. No absolute native quantity is pooled across models.
Direct resolved τ ratios and D_θ-derived proxy sensitivity are separate. Marginal whole-seed CIs remain in the source tables.
Primary support requires Spearman ≤ −0.6 and negative training/held-out sign in at least 3 of 4 planned folds.

S10 · Every planned held-out-model fold remains in the denominator



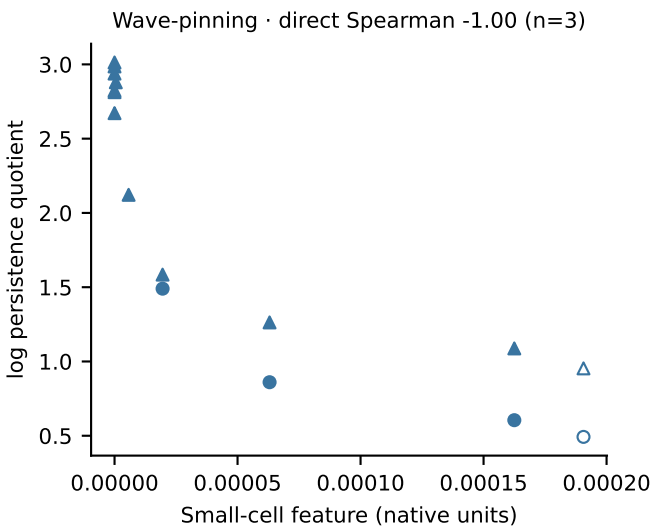


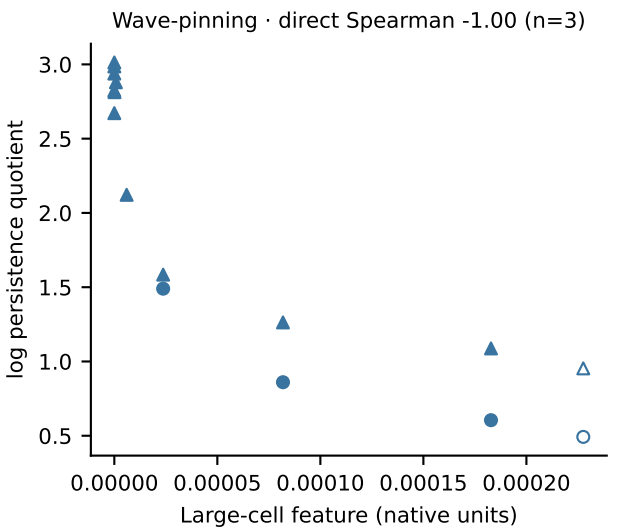


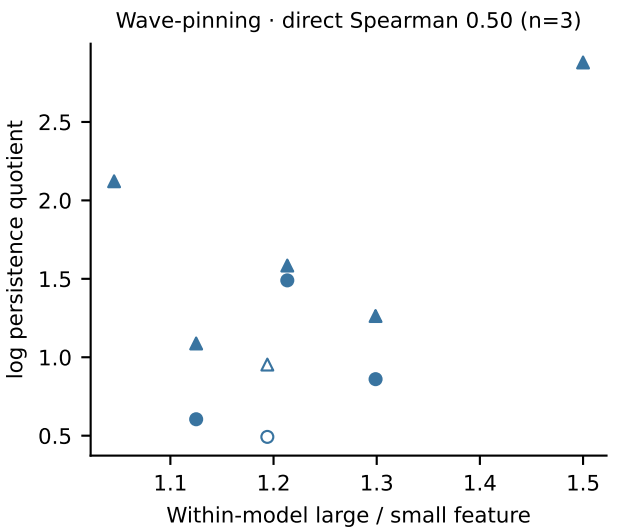


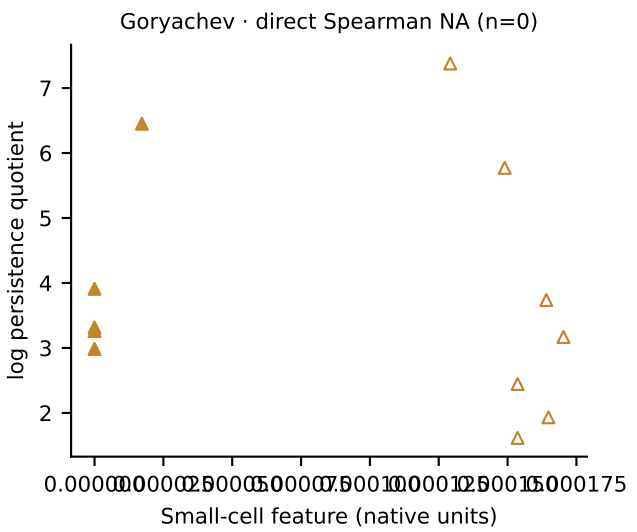
Arm common · σ=1 · actual per-size horizons retained in the source CSV.
Green: negative training and held-out signs. Orange: evaluated sign failure. Grey: insufficient/constant/censored data.
Each training and held-out sample needs at least 3 matched input fractions and nonconstant predictor/outcome.
Absence of a qualifying Goryachev or other model is explicit; no missing fold is counted as successful.

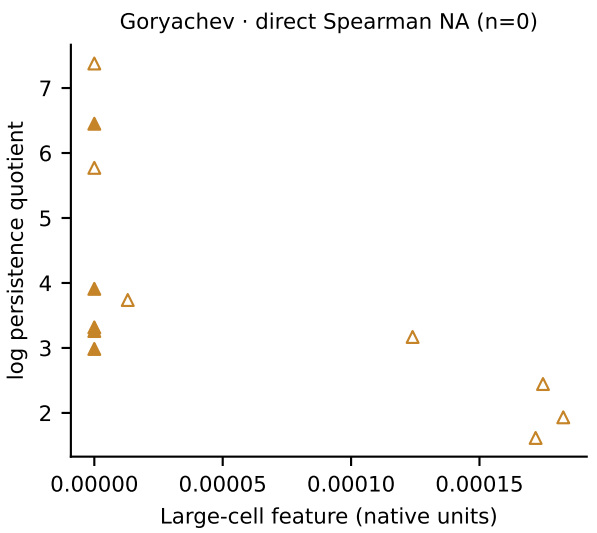
S10 · births per time per perimeter

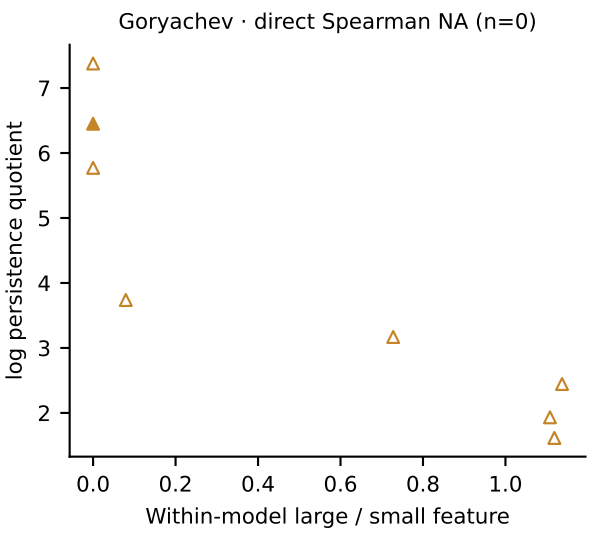


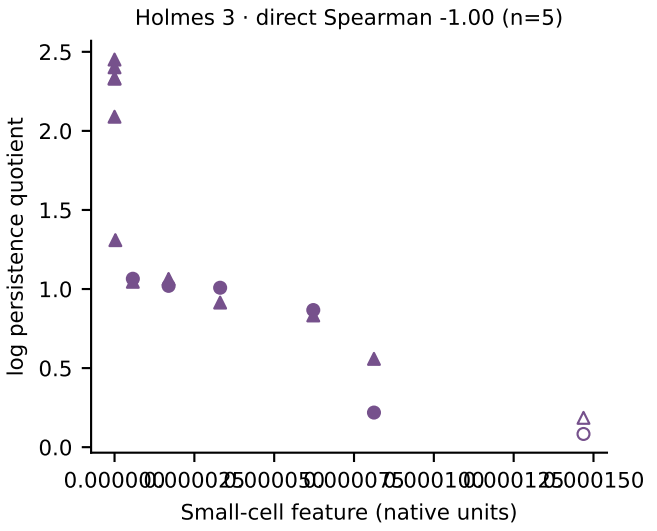


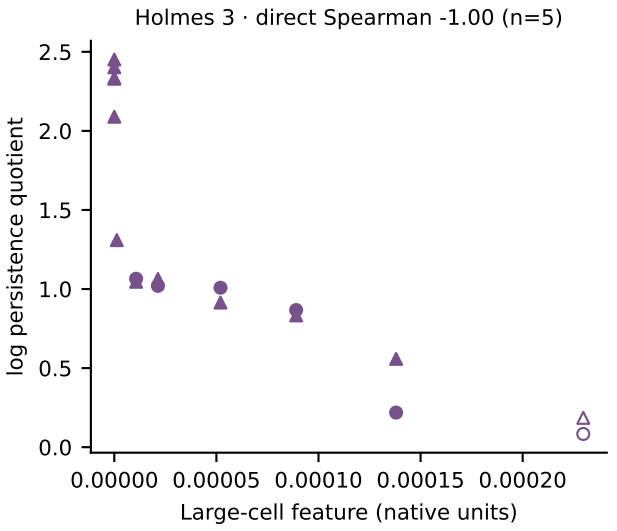


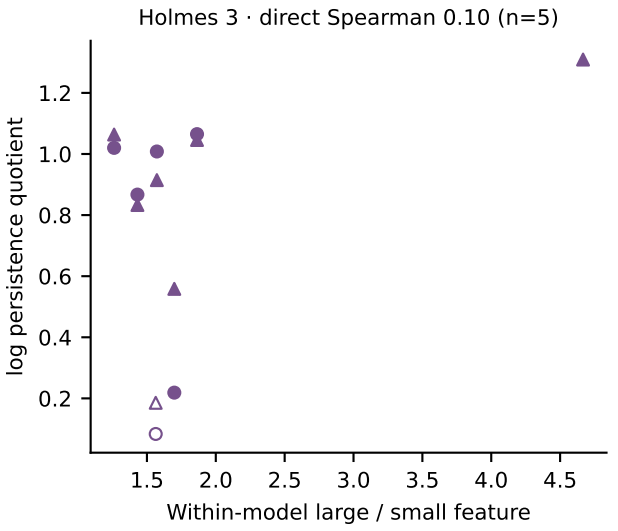


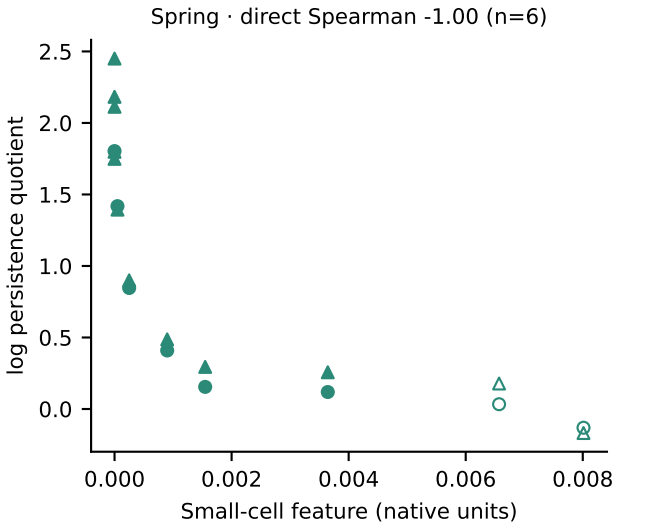


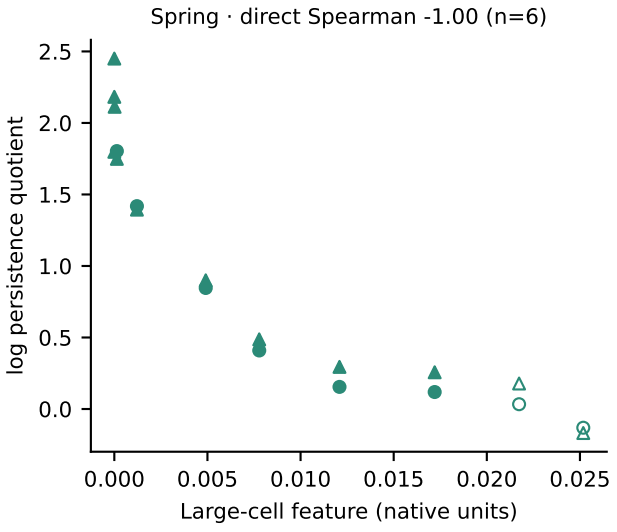


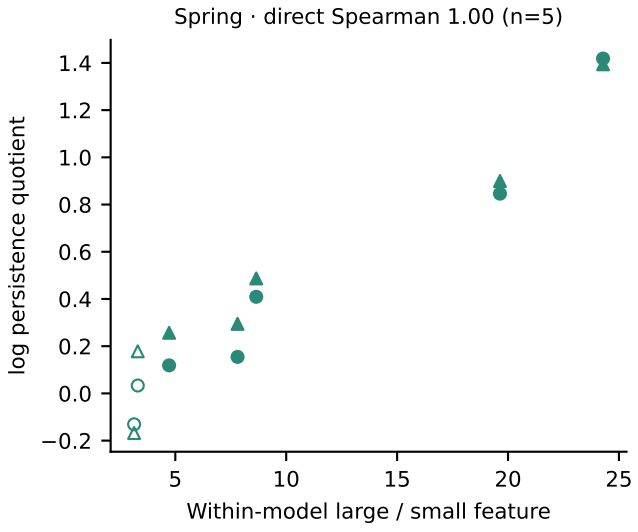






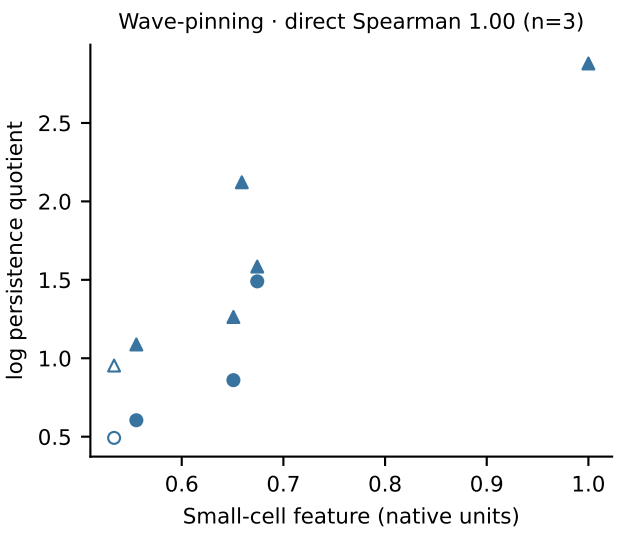


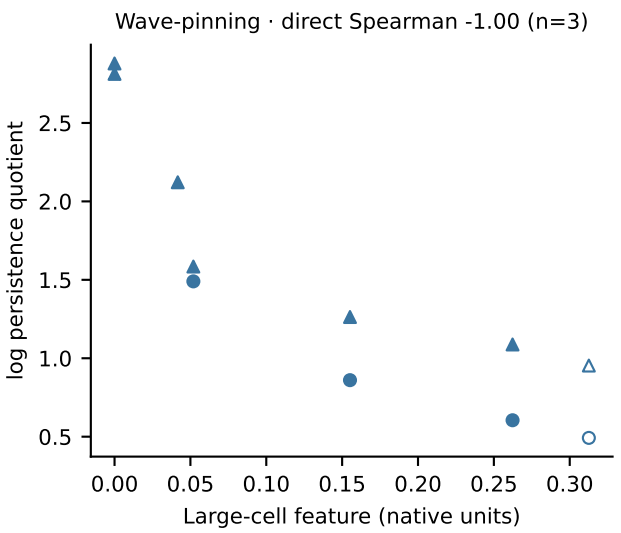


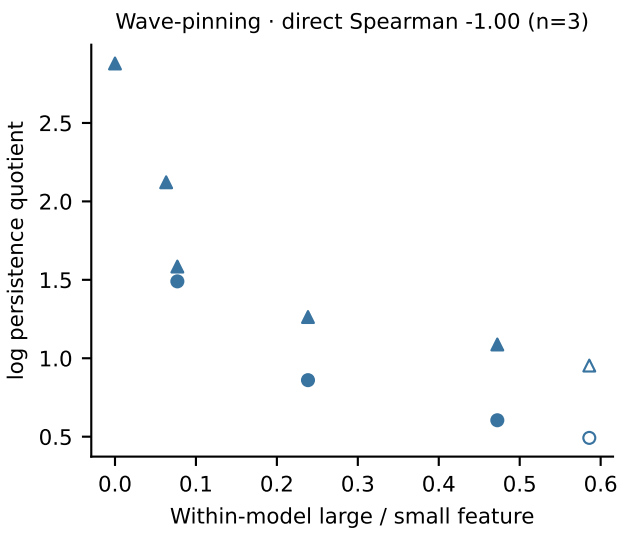


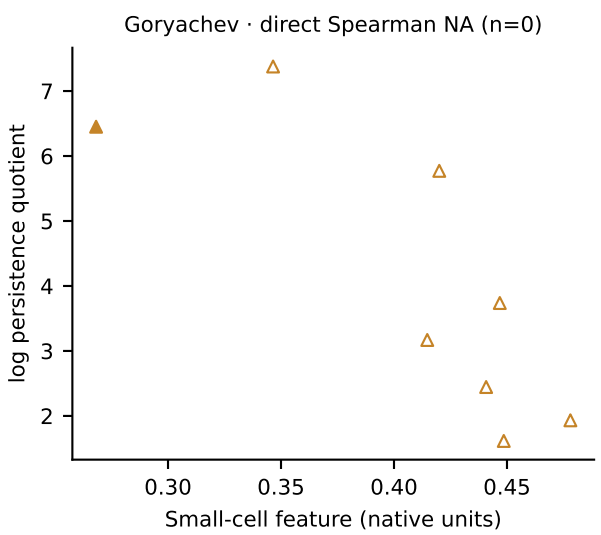
Arm common · σ=1 · actual per-size horizons retained in the source CSV. Crosses: unknown forcing, excluded from primary.
○ Direct resolved τ ratio; △ D_θ proxy sensitivity. Filled: nonforced; open: forced context.
All small/large native quantities remain within their own model. Cross-model tests use only size quotients.
NA retains insufficient, constant, censored or unscorable measurements; native caps are never substituted.

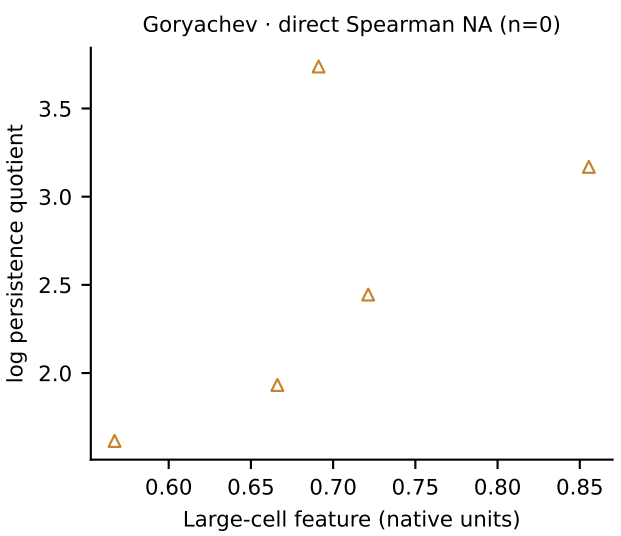
S10 · ever unscorable takeover fraction

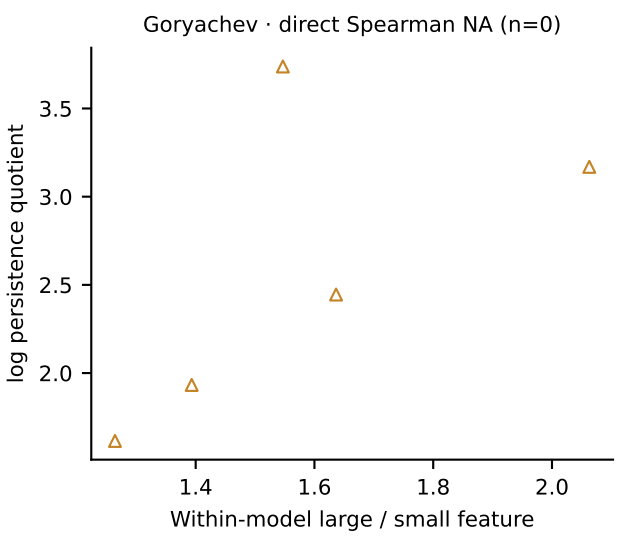


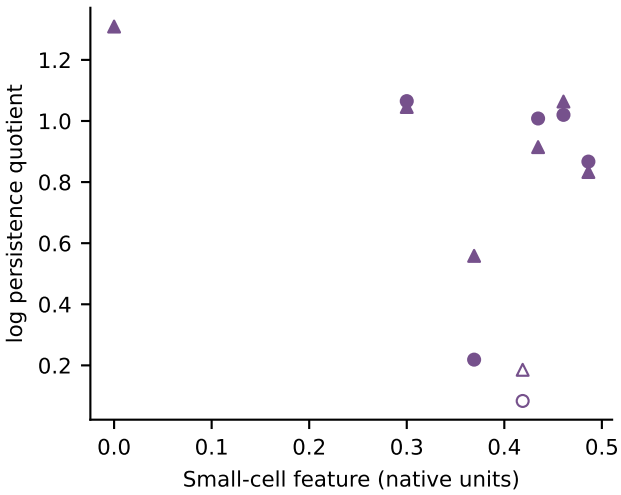


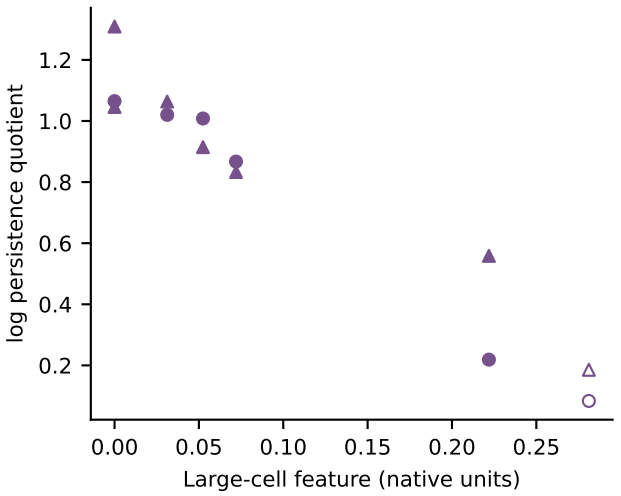


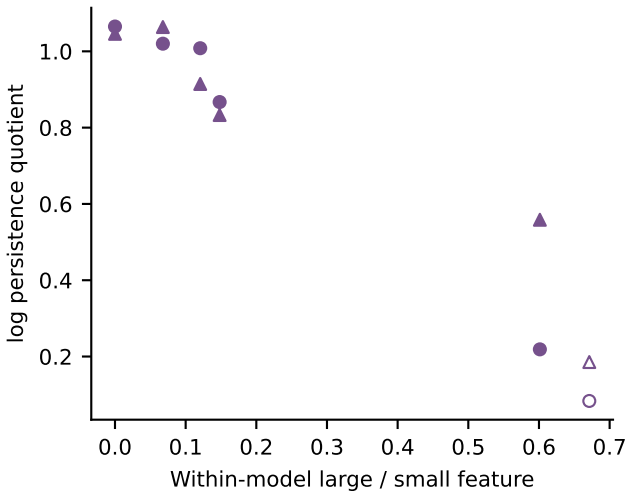


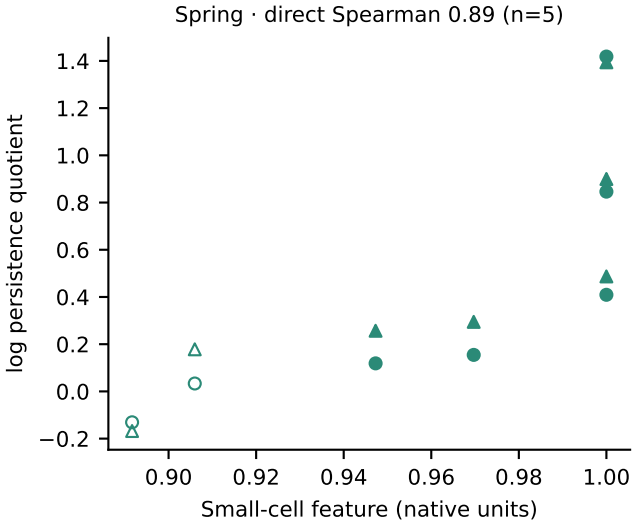


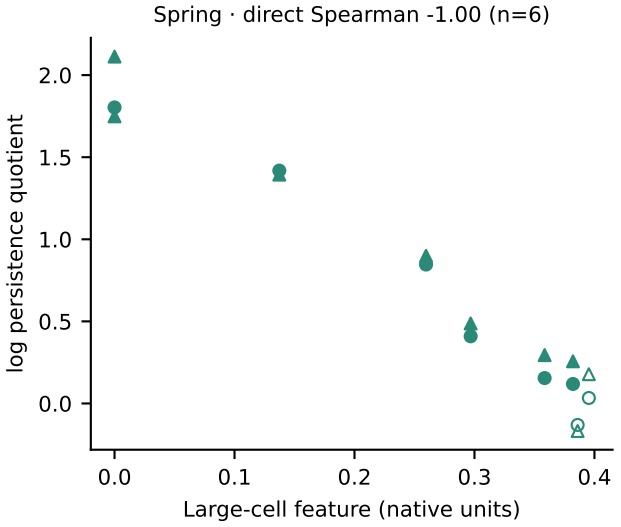


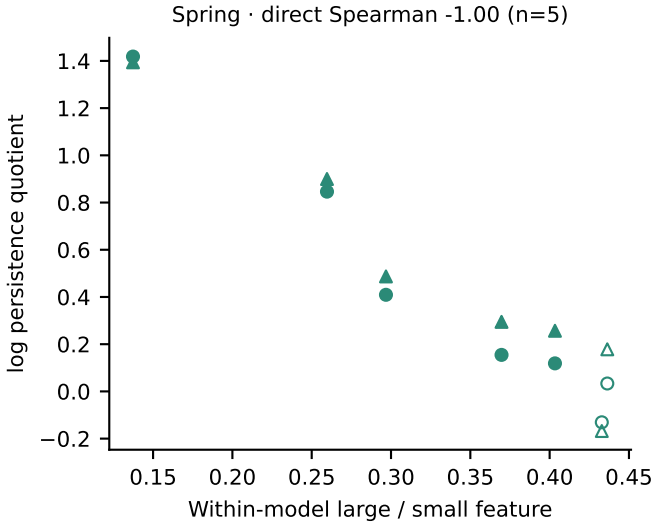












Arm common · σ=1 · actual per-size horizons retained in the source CSV. Crosses: unknown forcing, excluded from primary.
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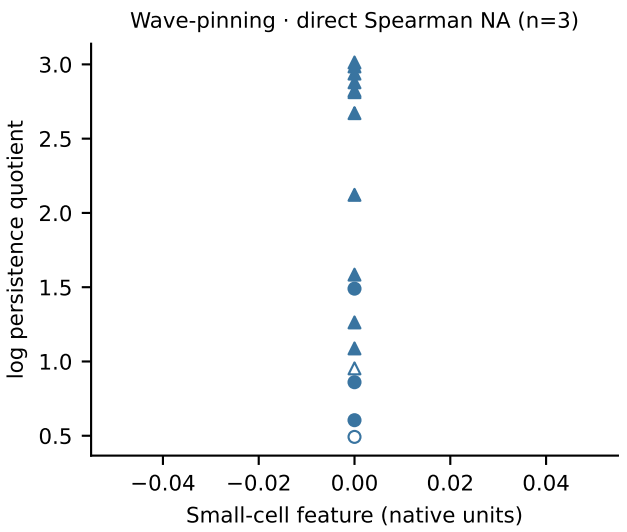
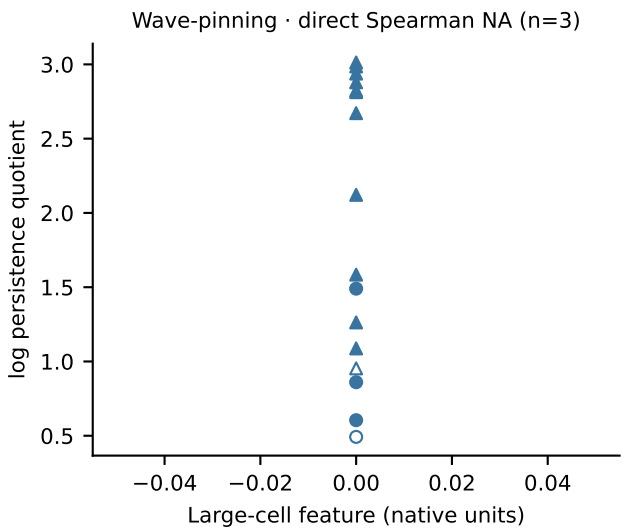
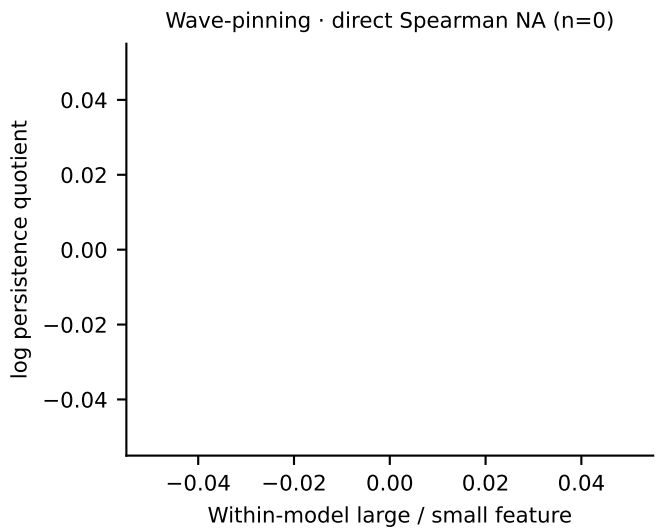
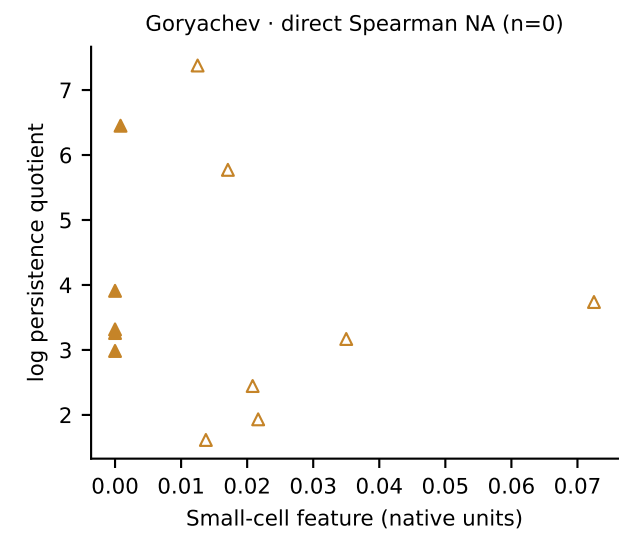
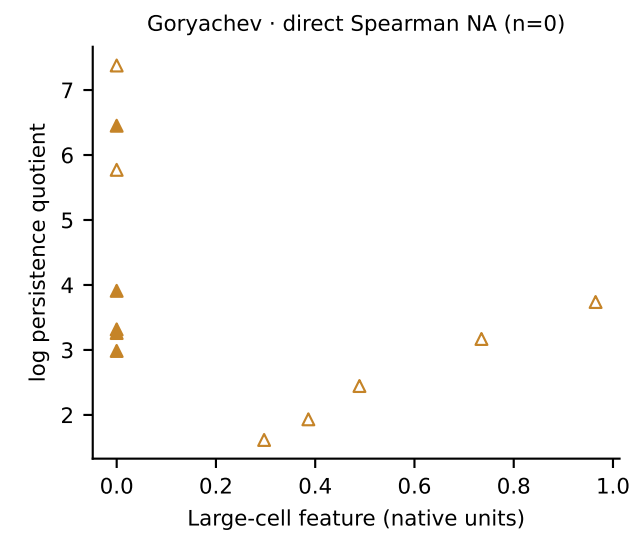
S10 · low contrast time fraction

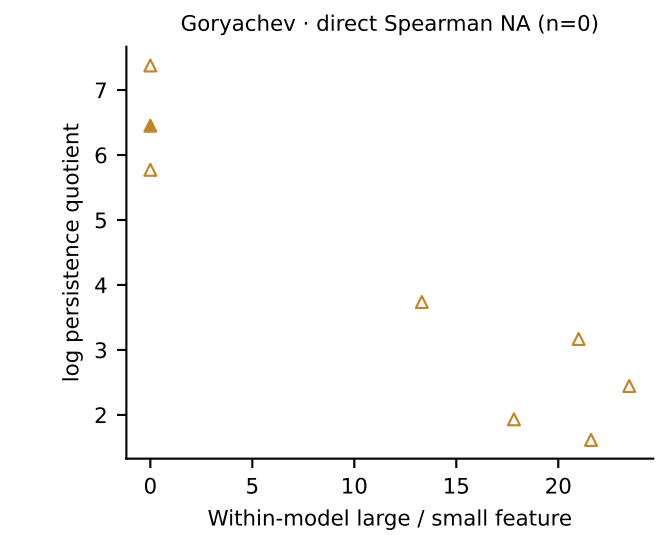


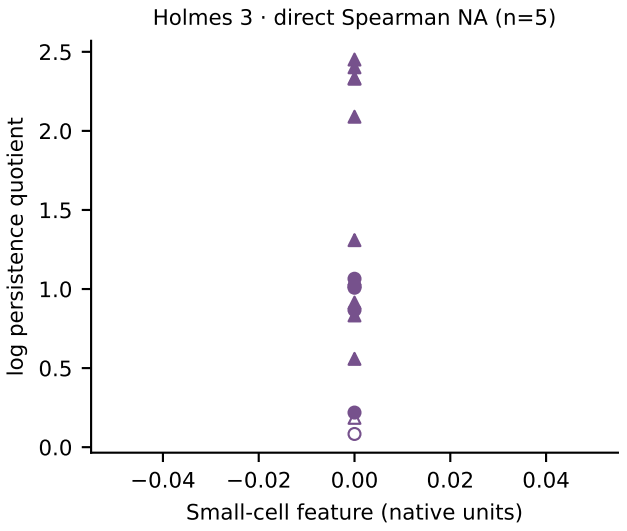


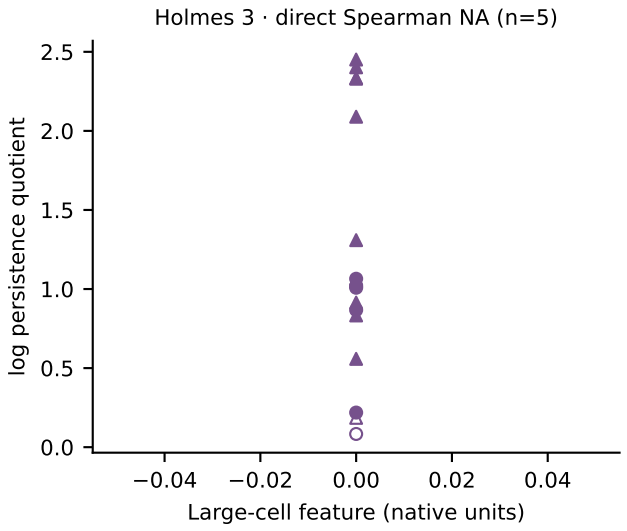


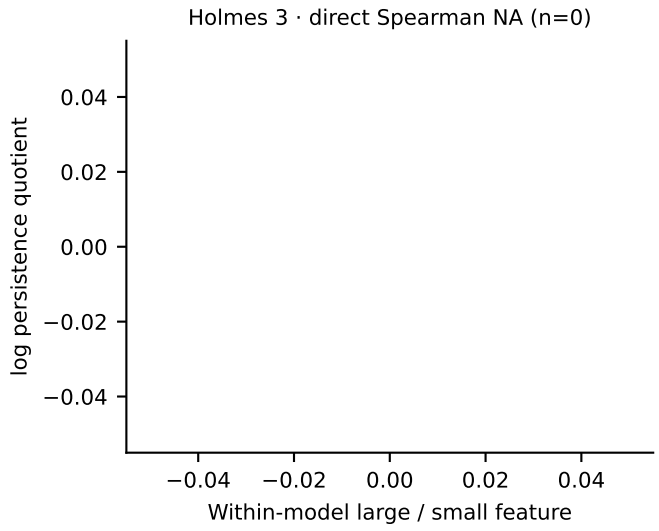


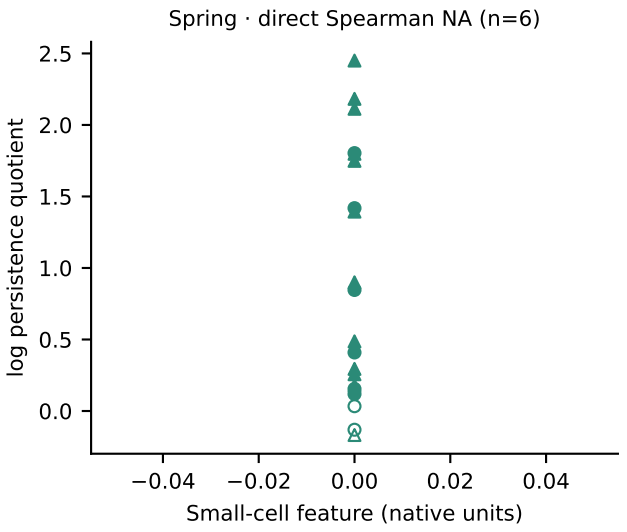


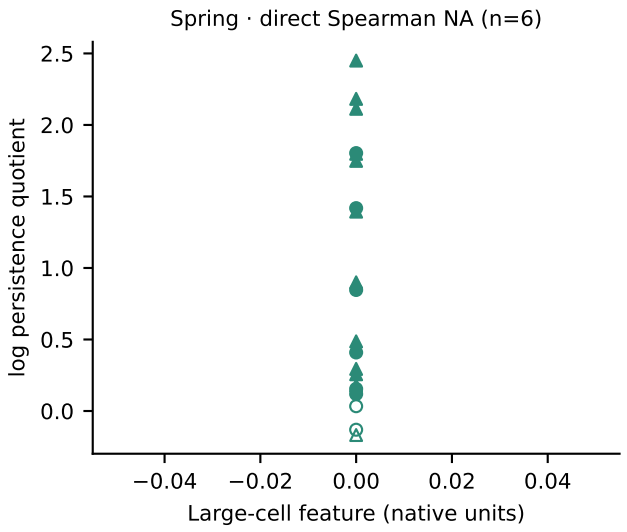


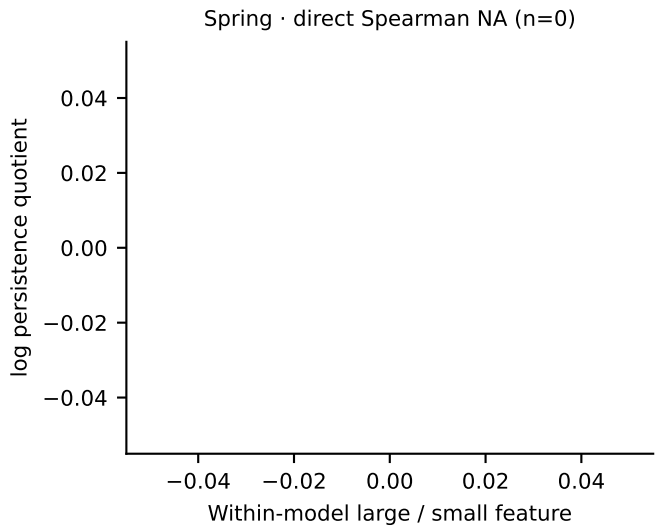












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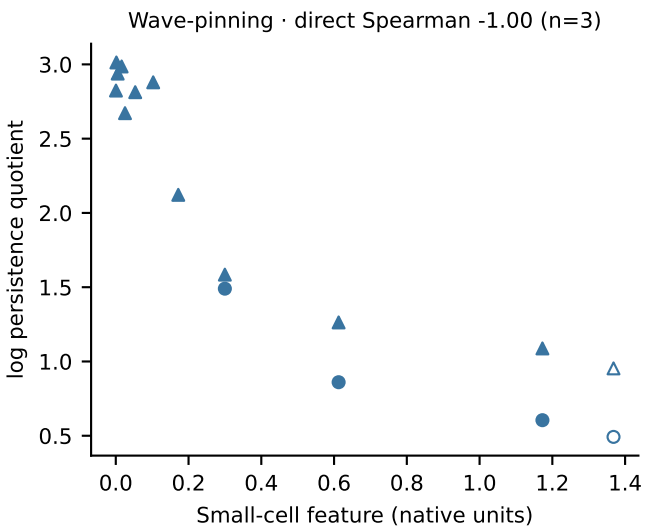
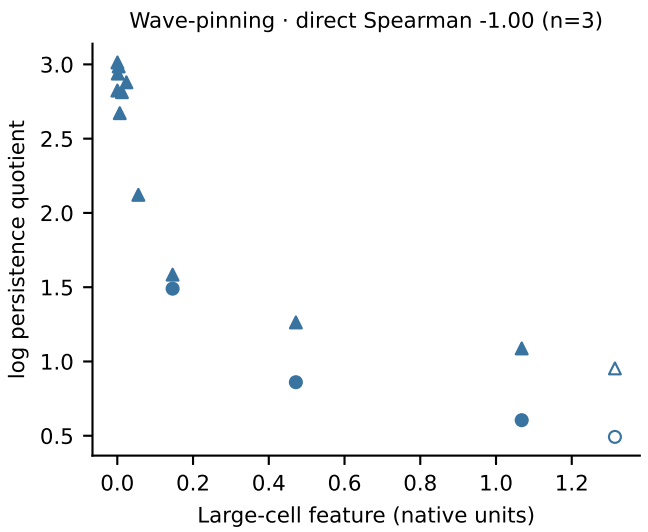
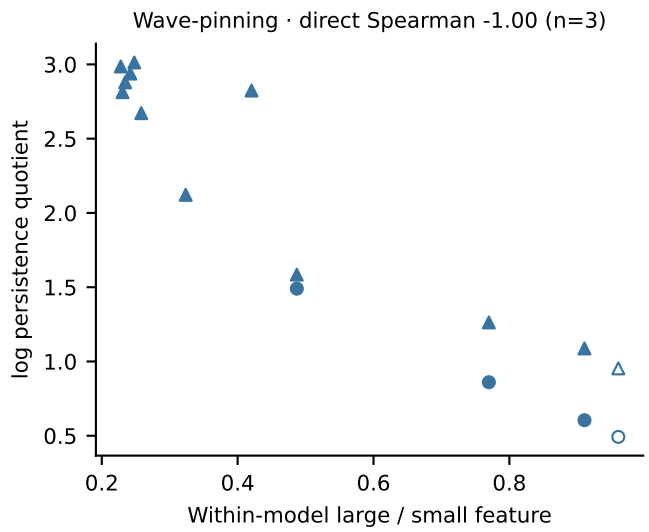
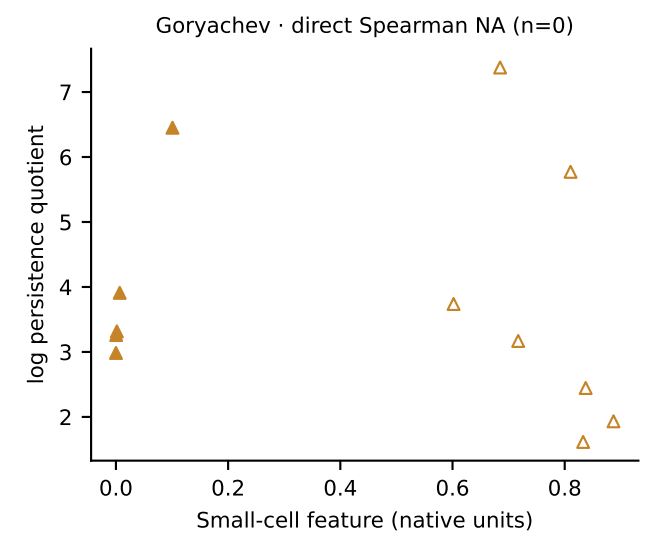
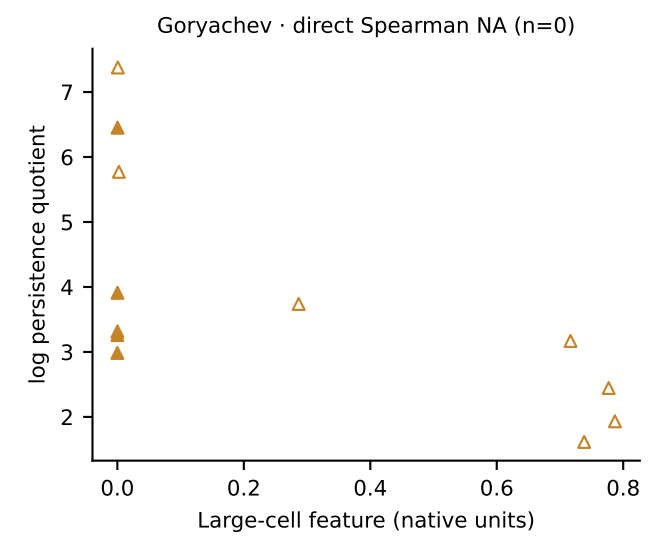
S10 · mean absolute turn 1 frames

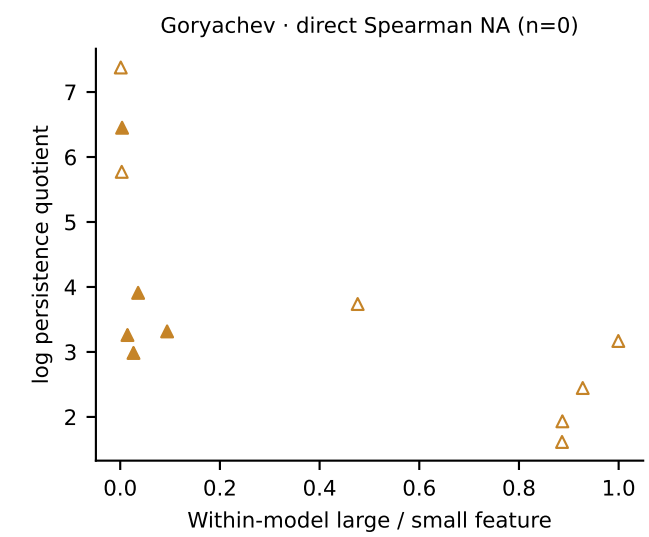


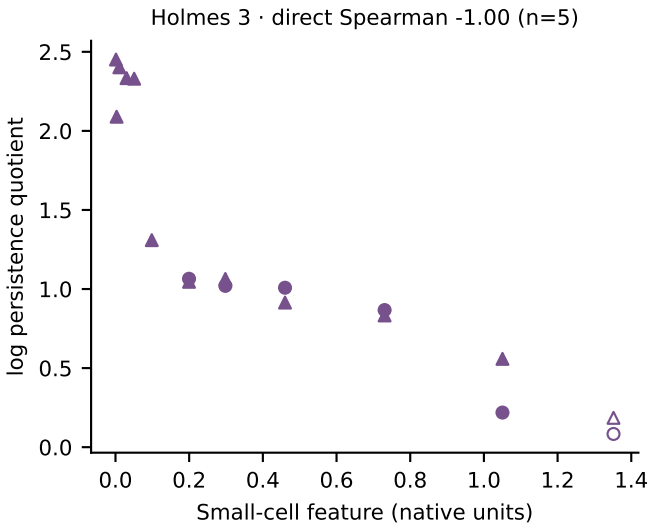


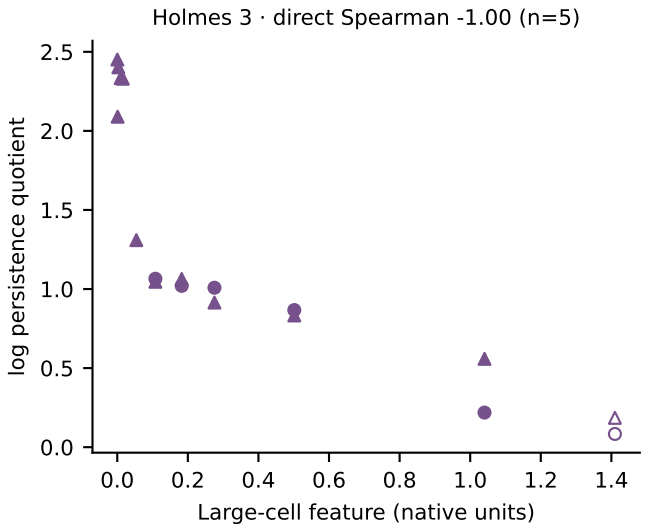


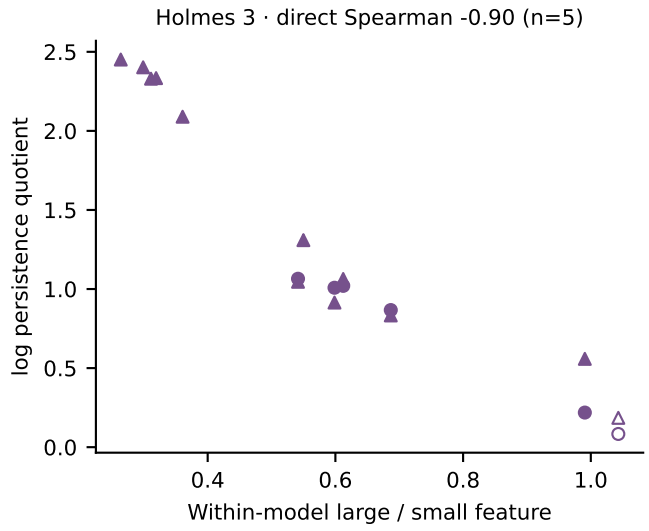


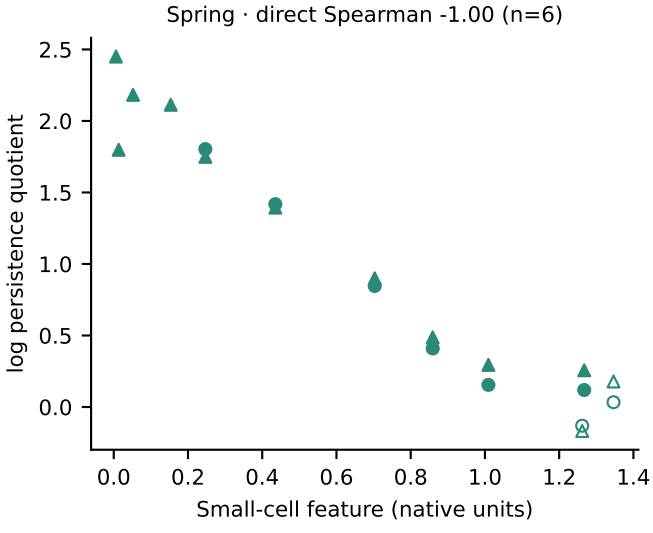


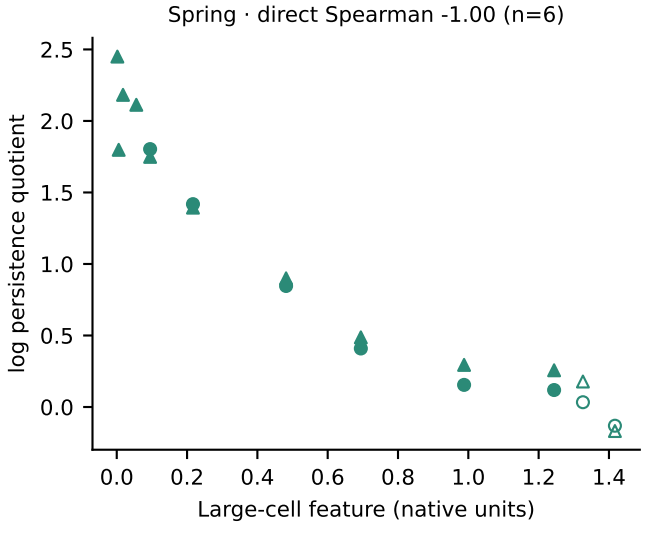


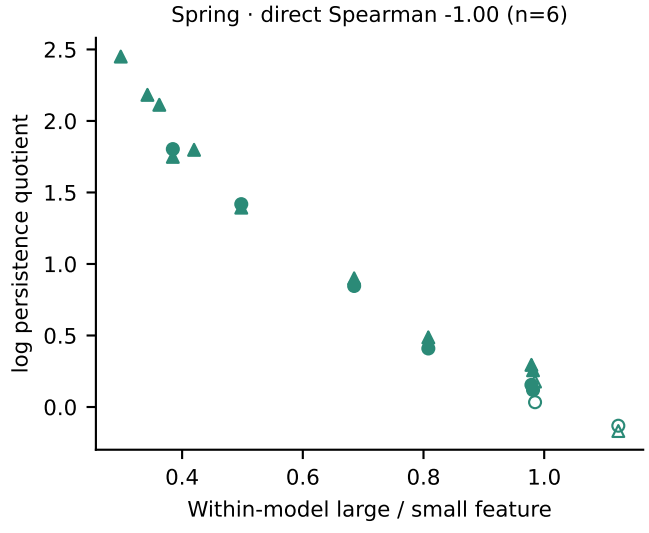












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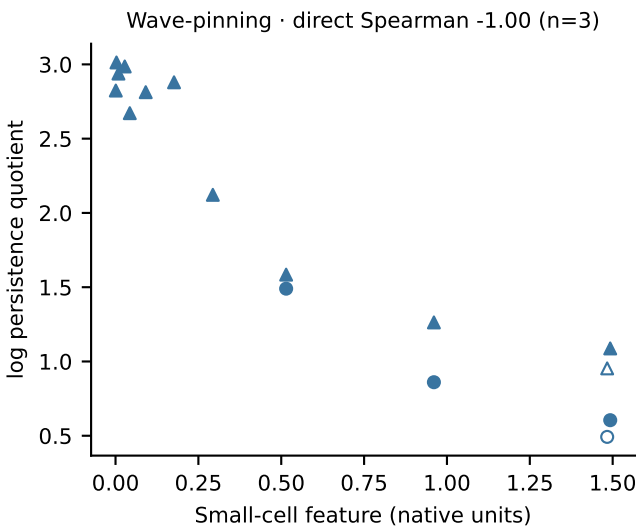
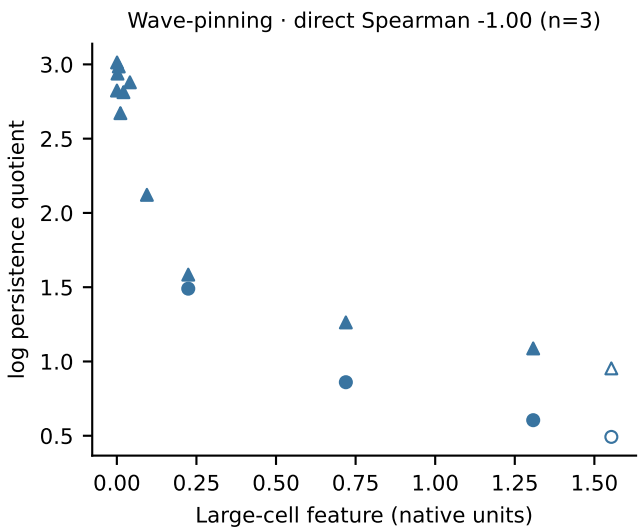
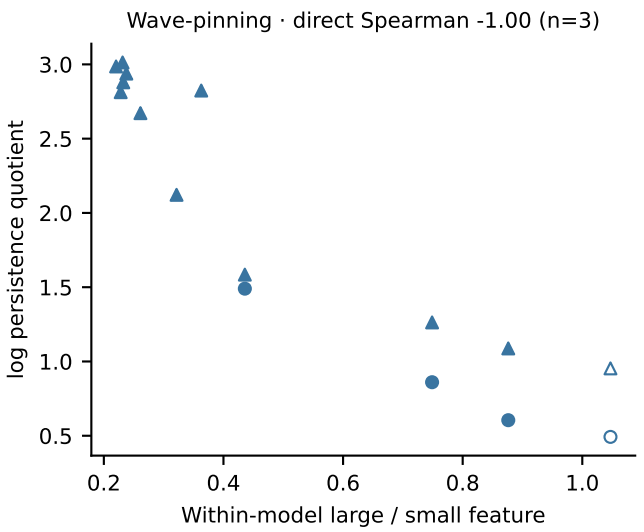
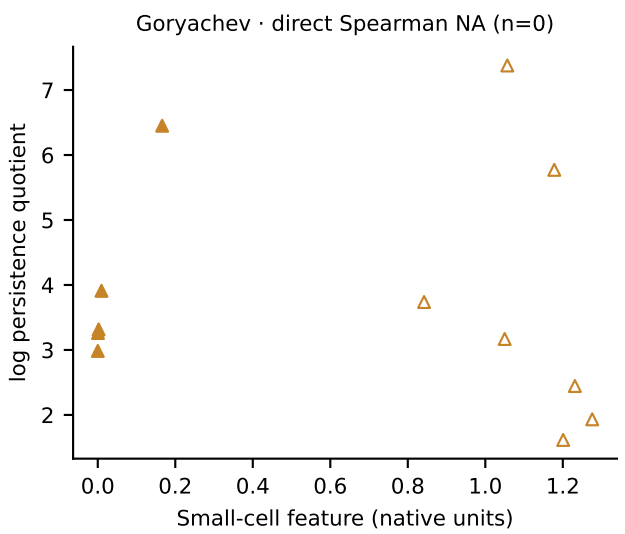
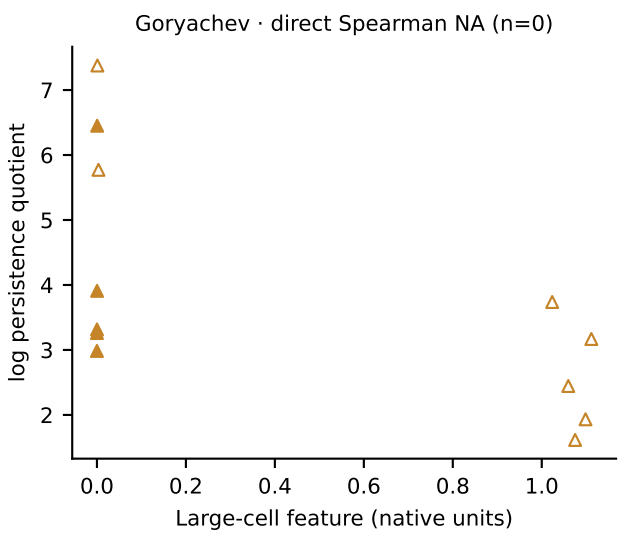
S10 · mean absolute turn 2 frames

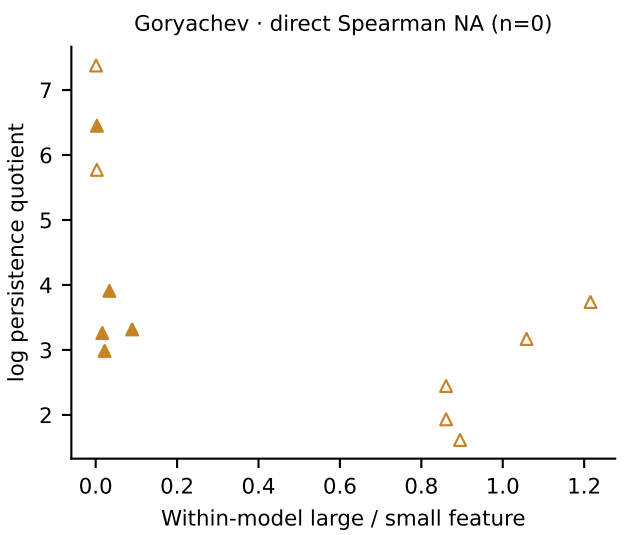


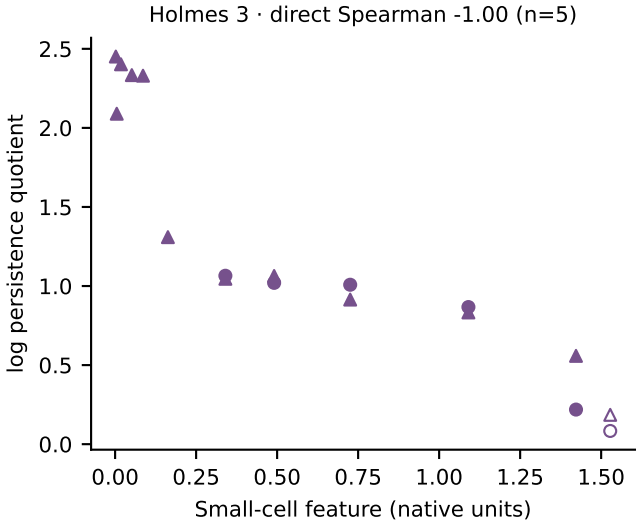


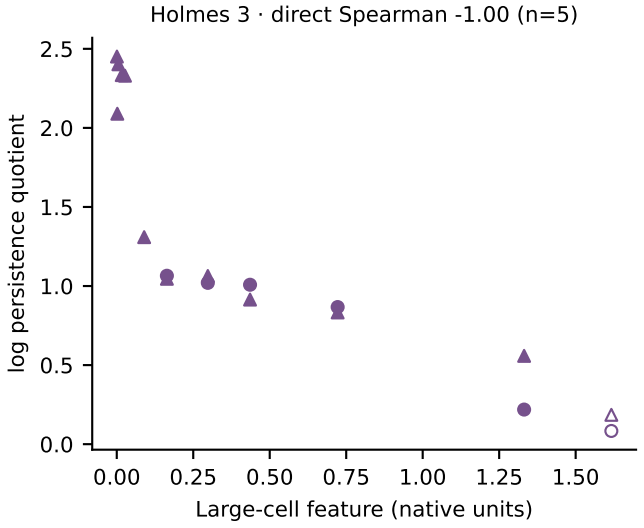


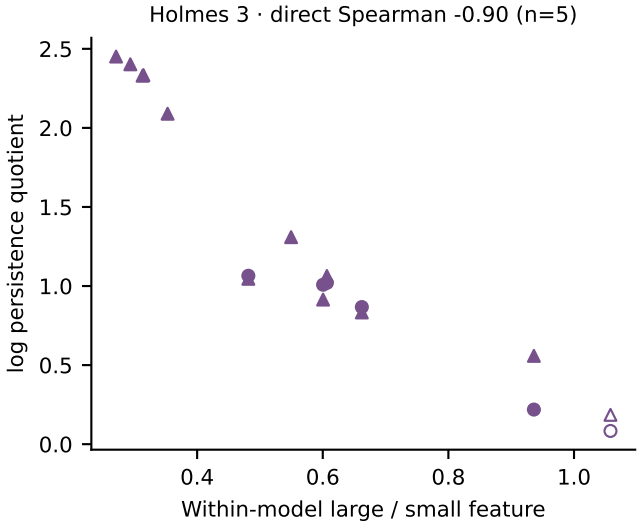


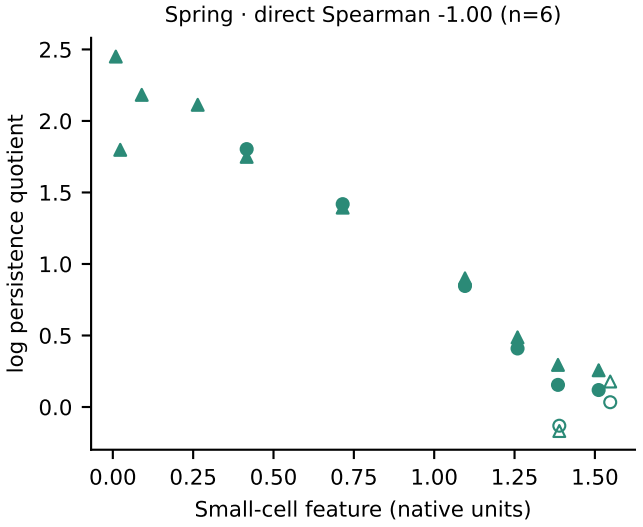


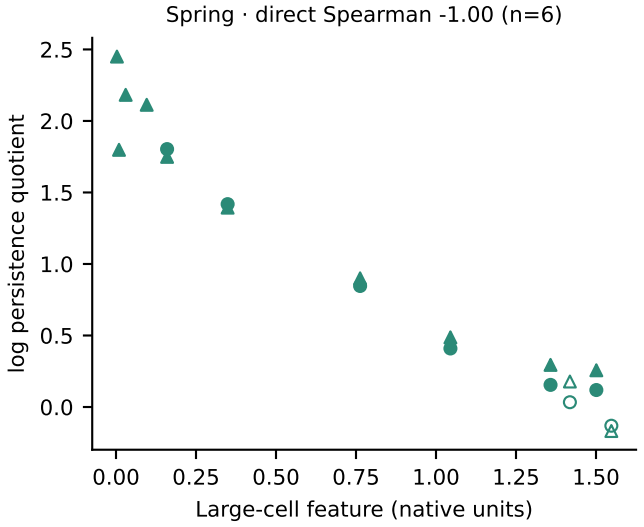


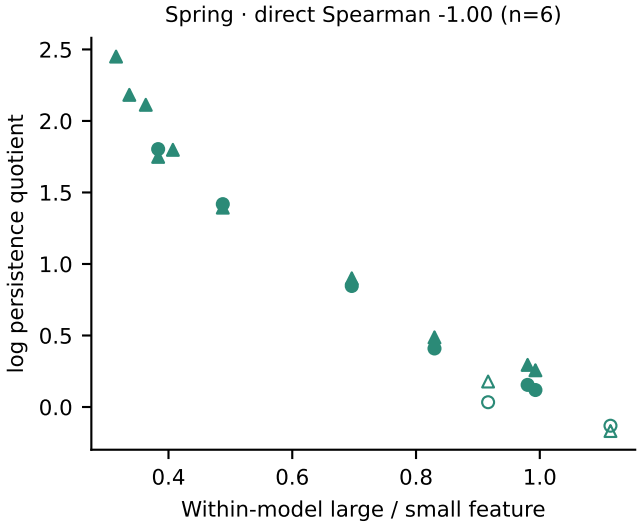












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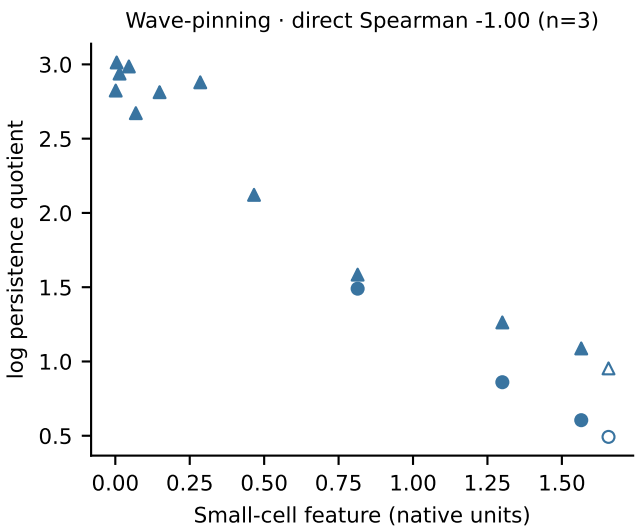
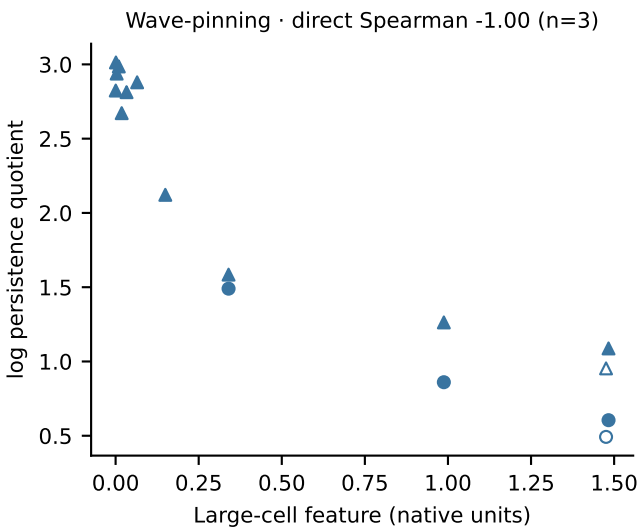
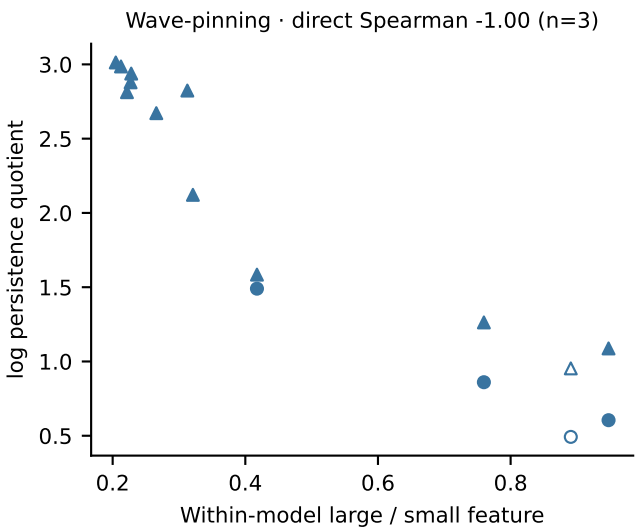
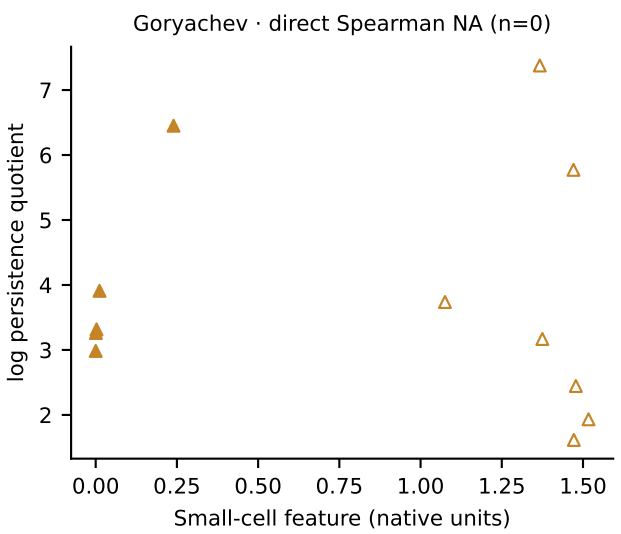
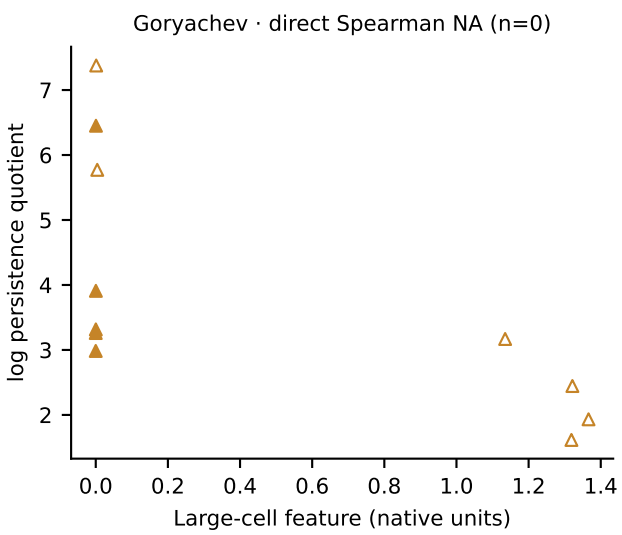
S10 · mean absolute turn 4 frames

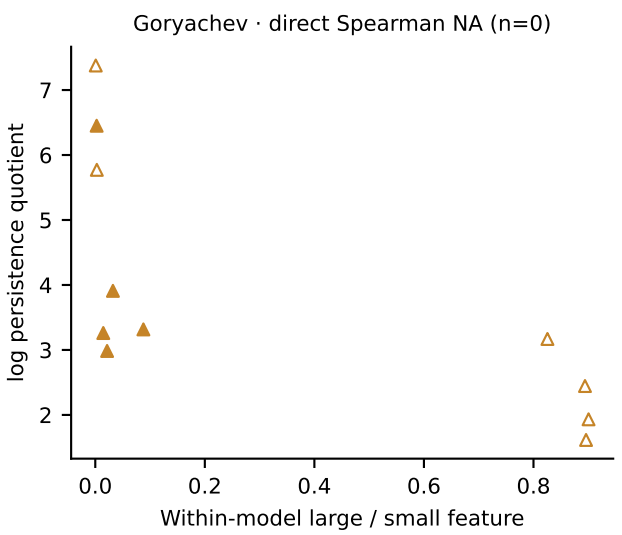


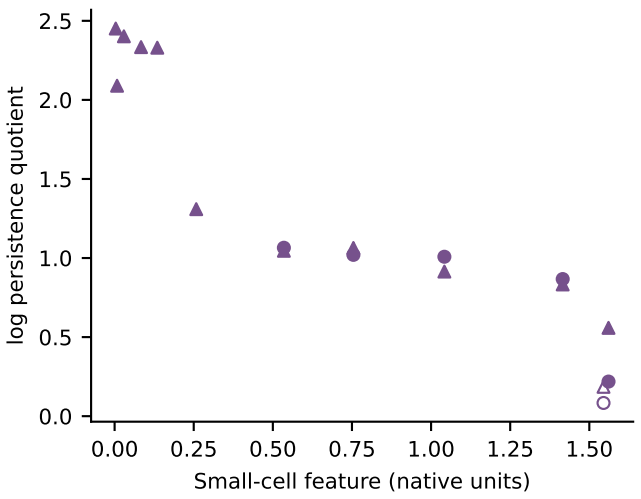


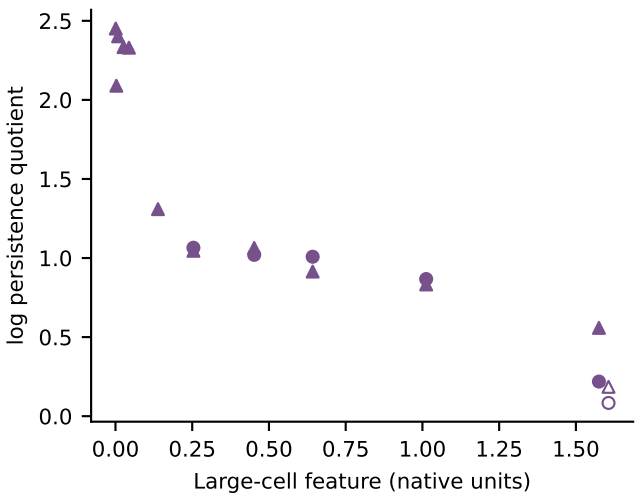


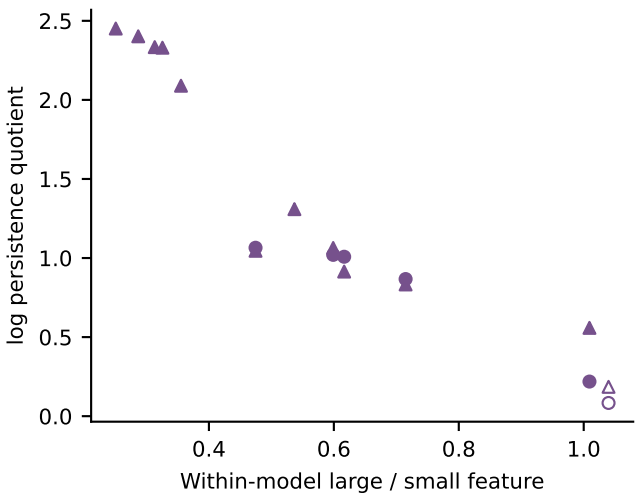


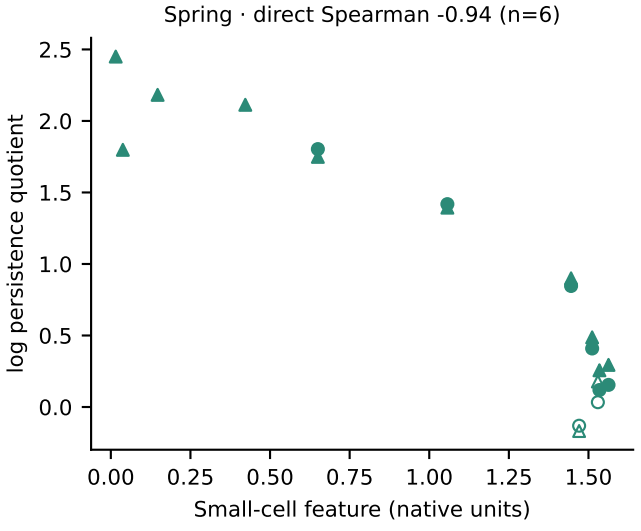


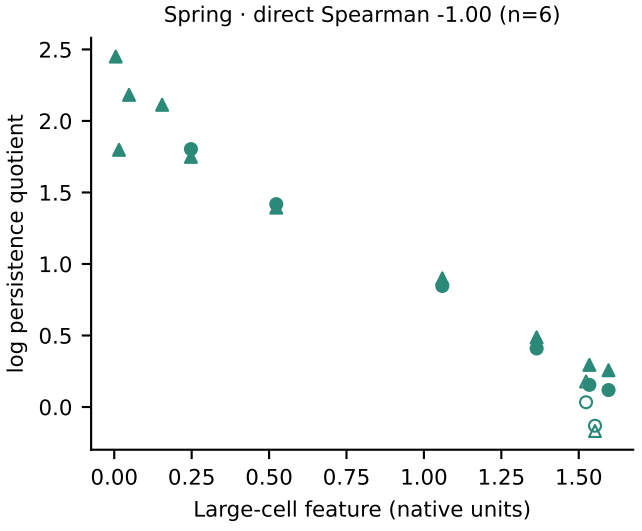


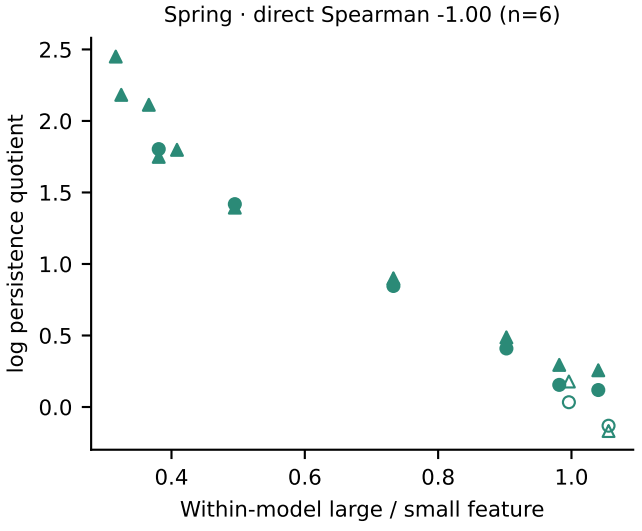












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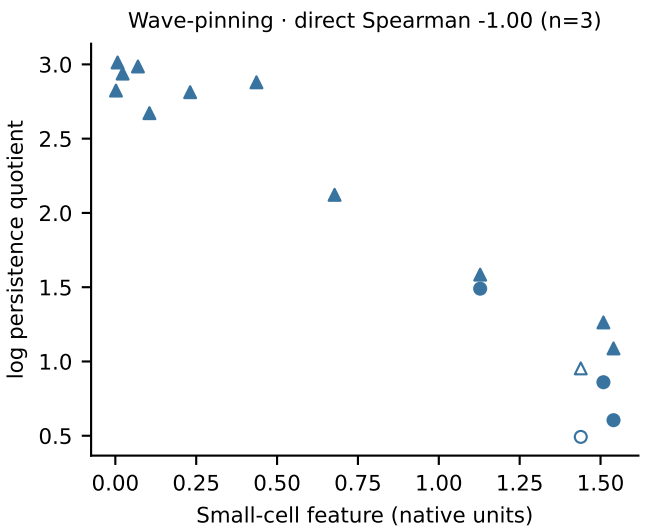
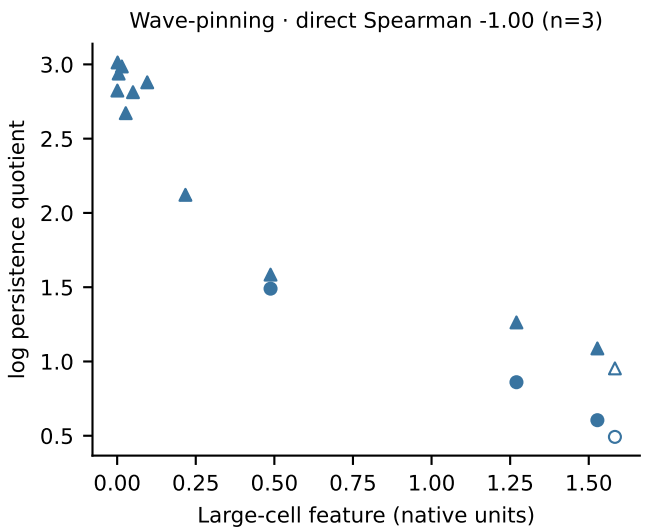
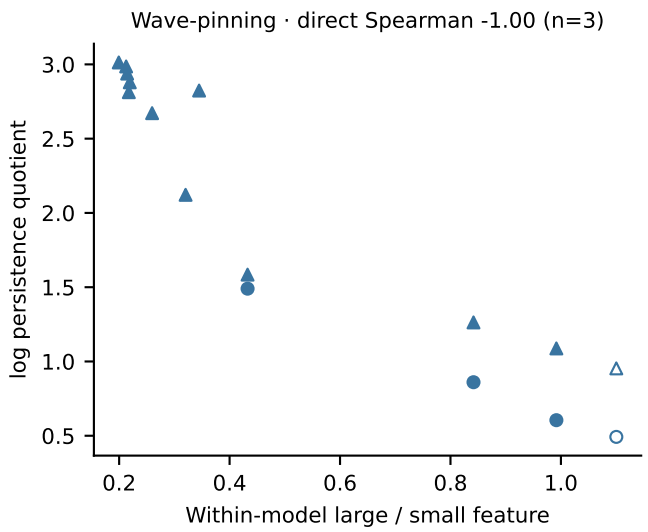
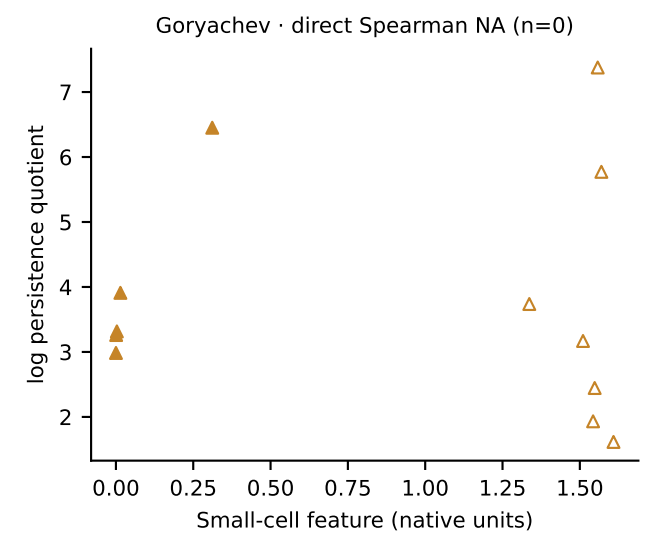
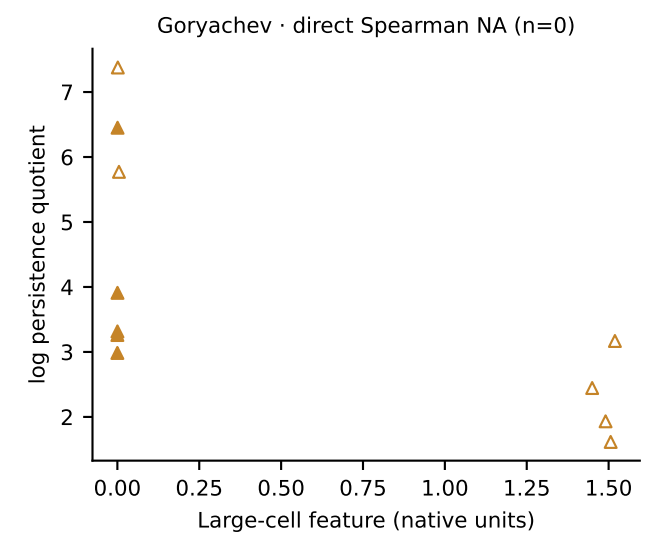
S10 · mean absolute turn 8 frames

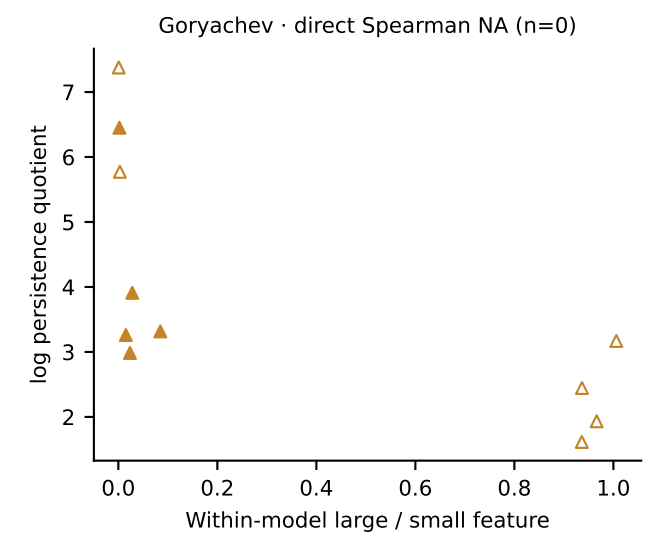


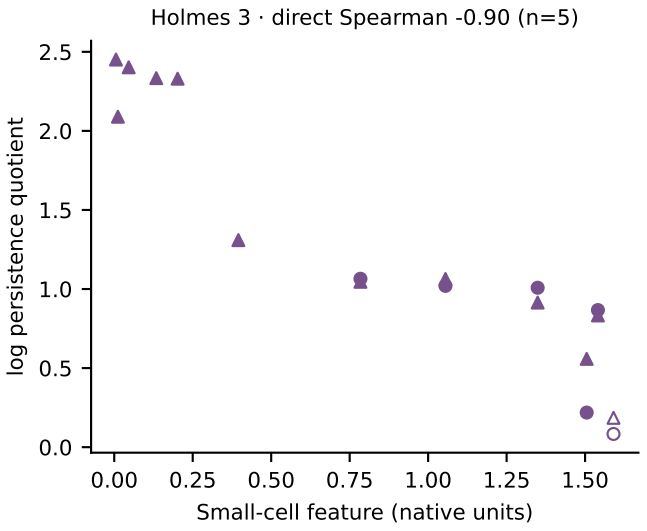


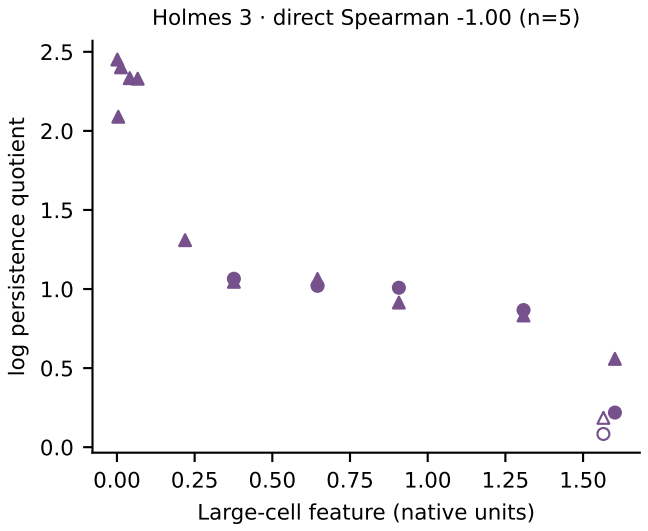


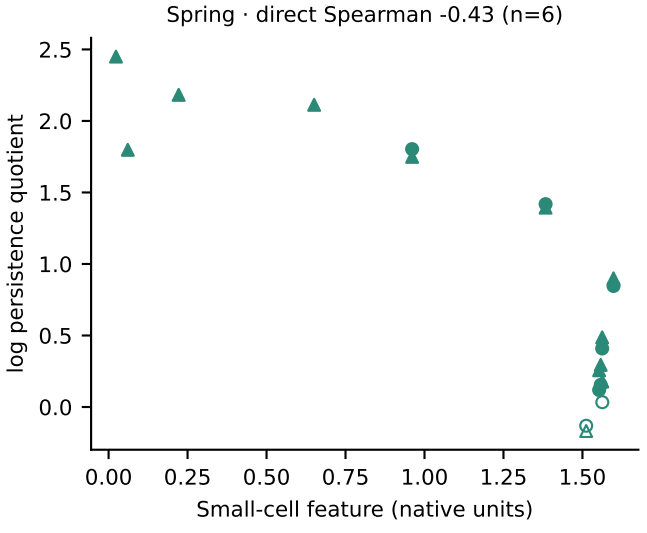


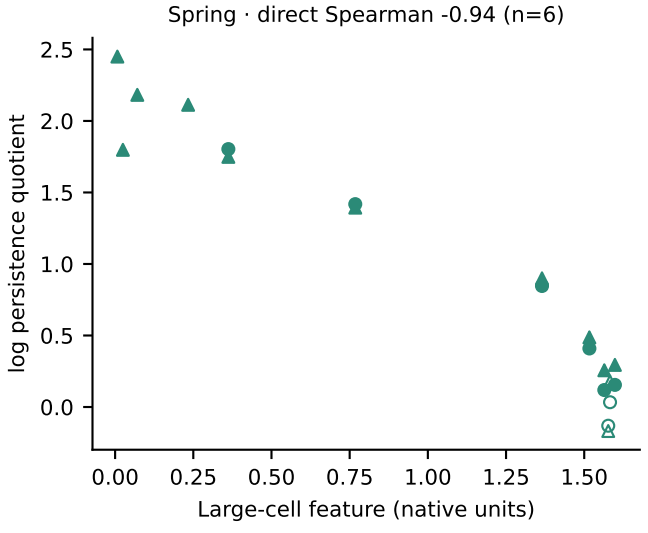


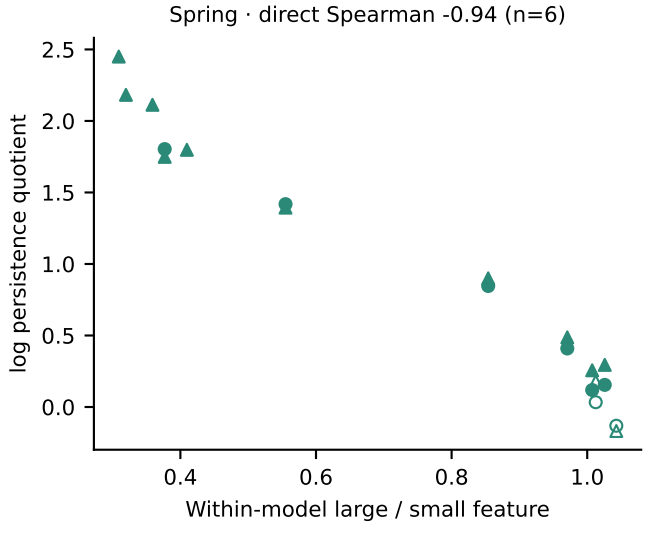












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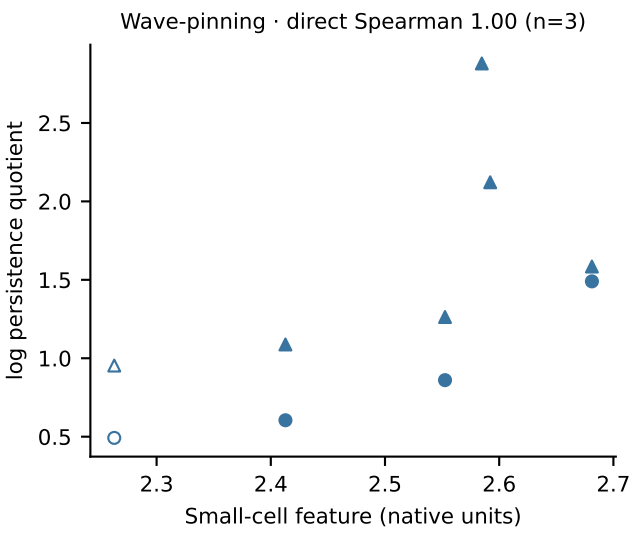
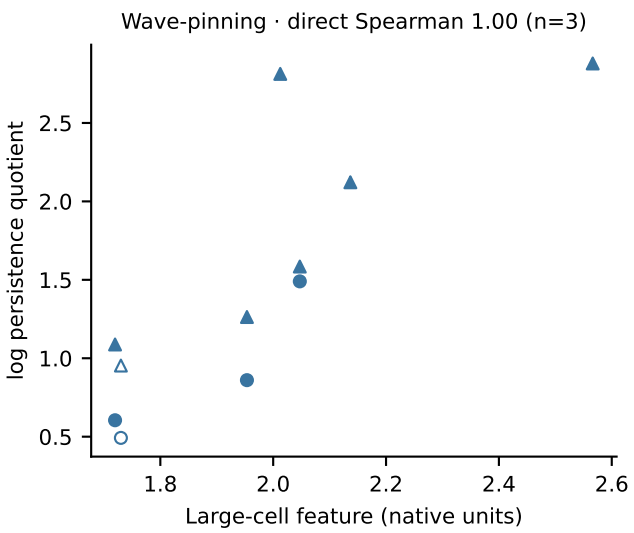
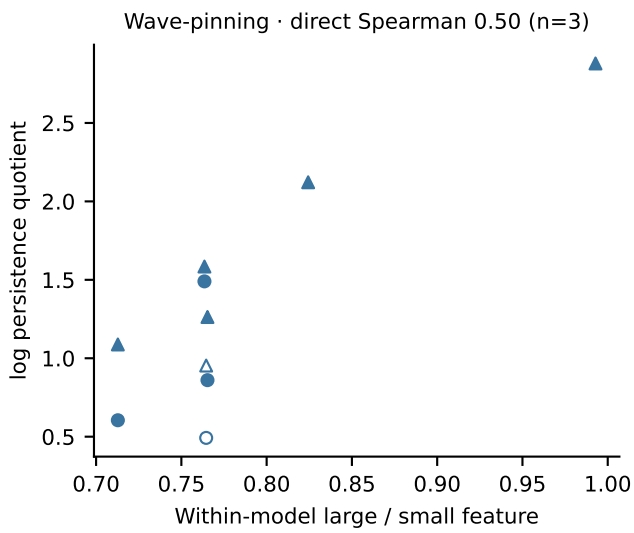
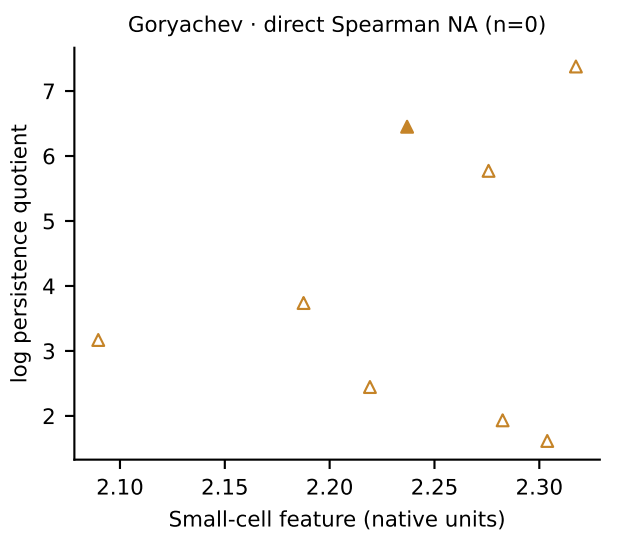
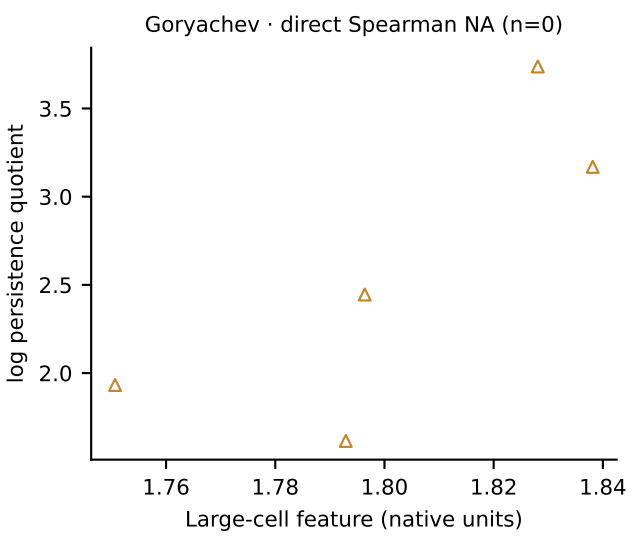
S10 · mean birth angle

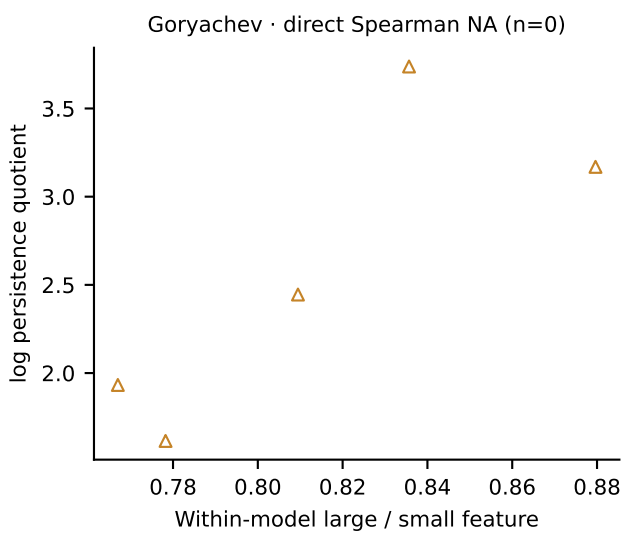


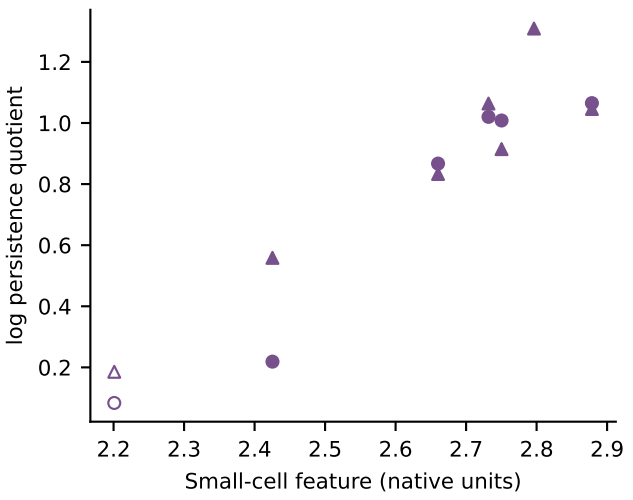


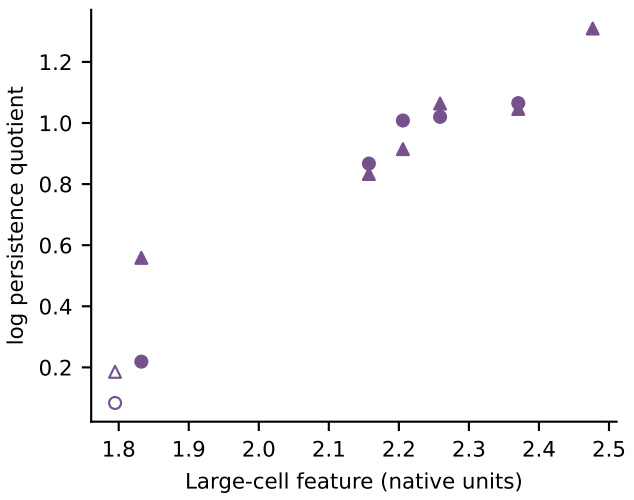


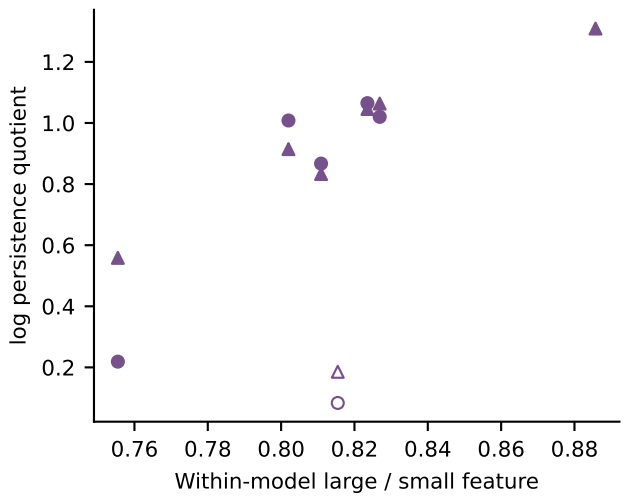


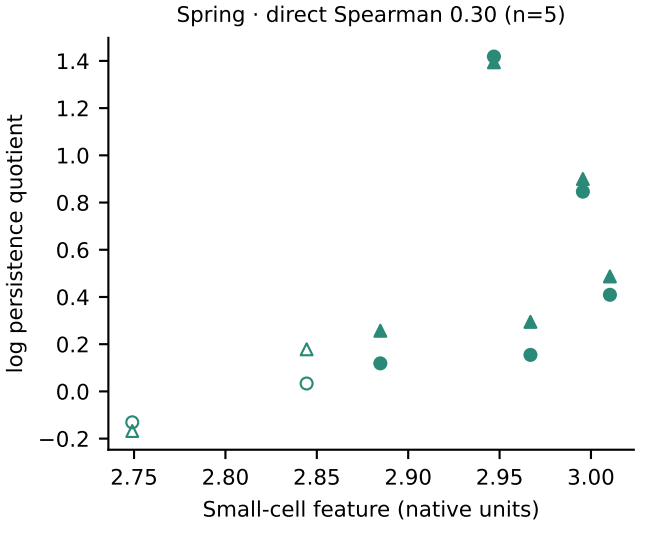


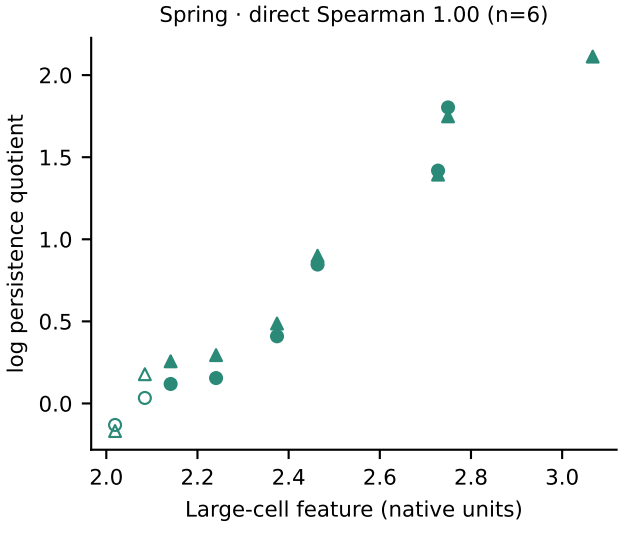


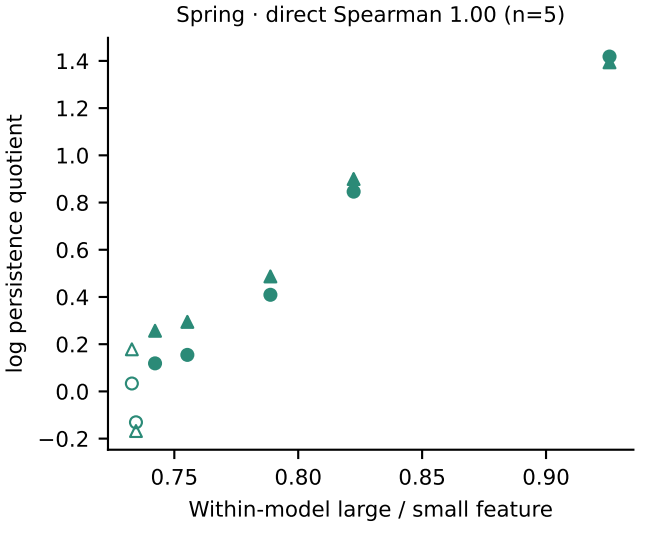












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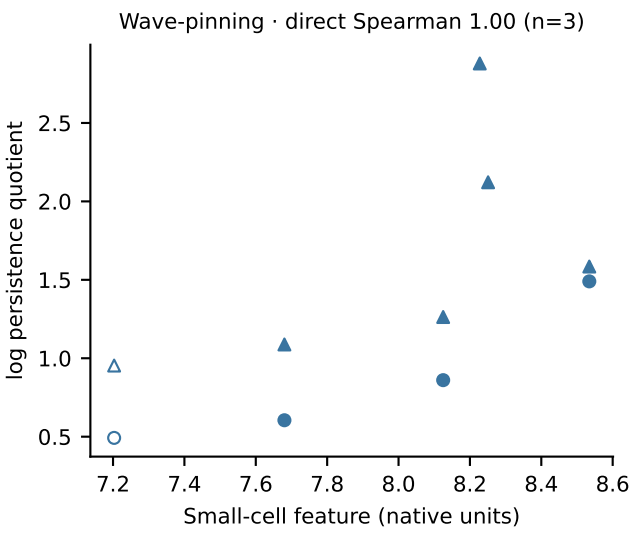
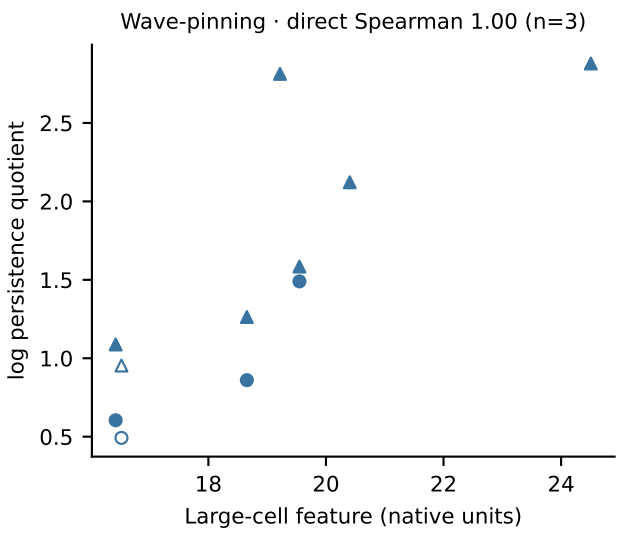
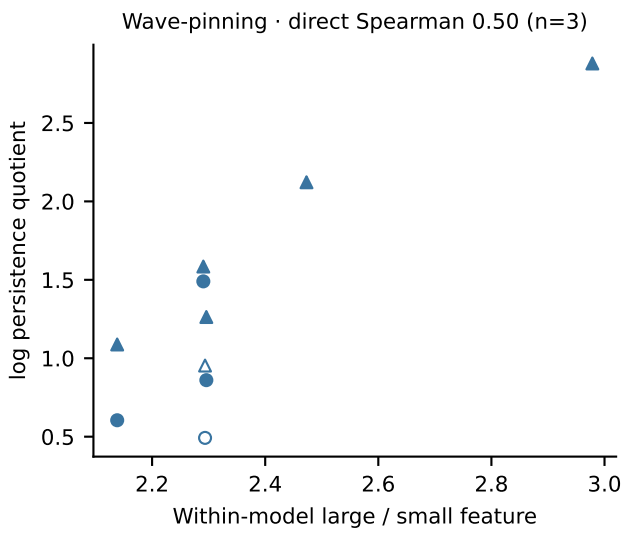
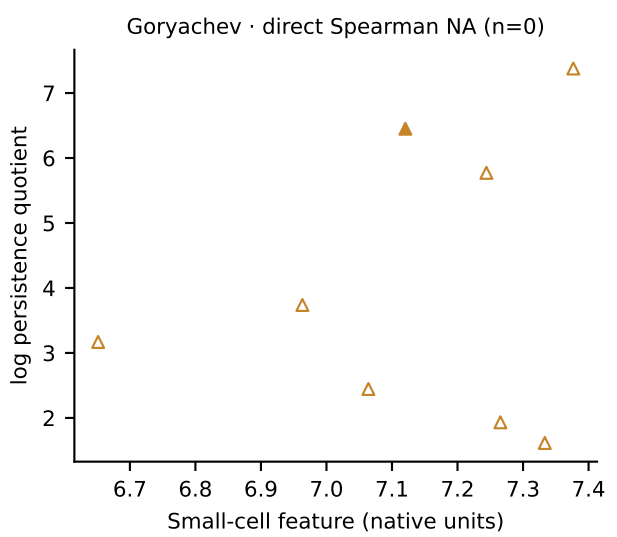
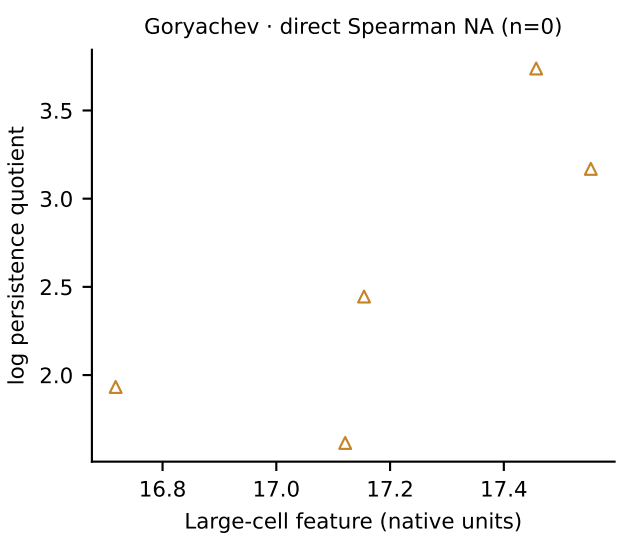
S10 · mean birth physical distance

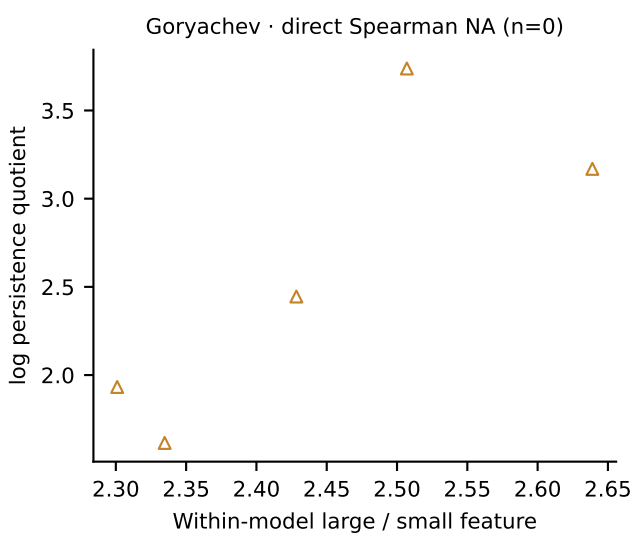


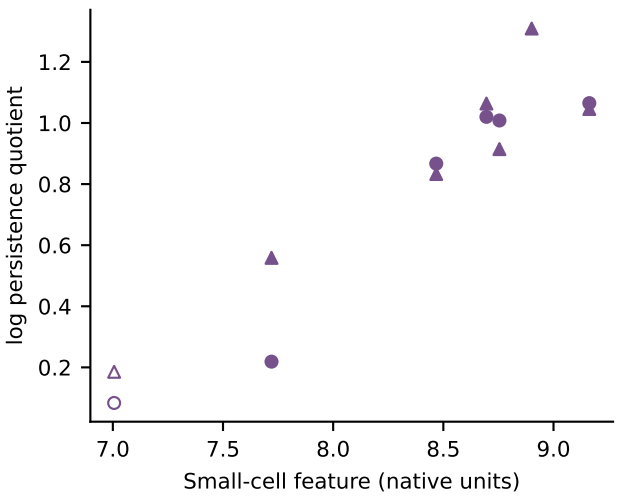


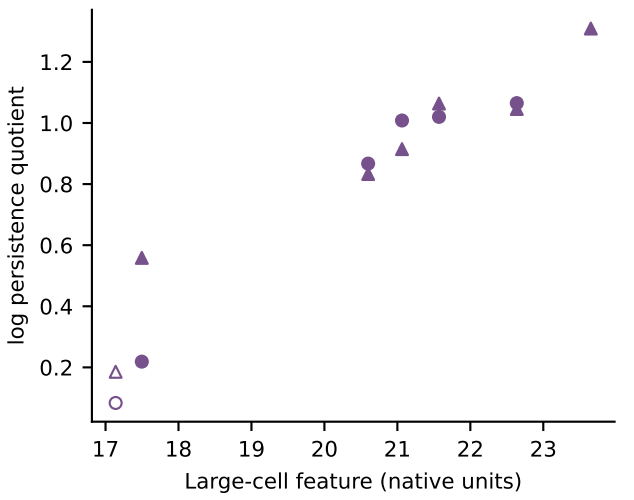


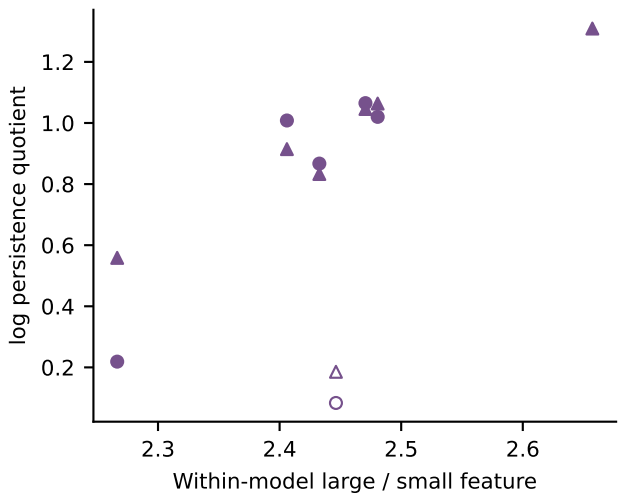


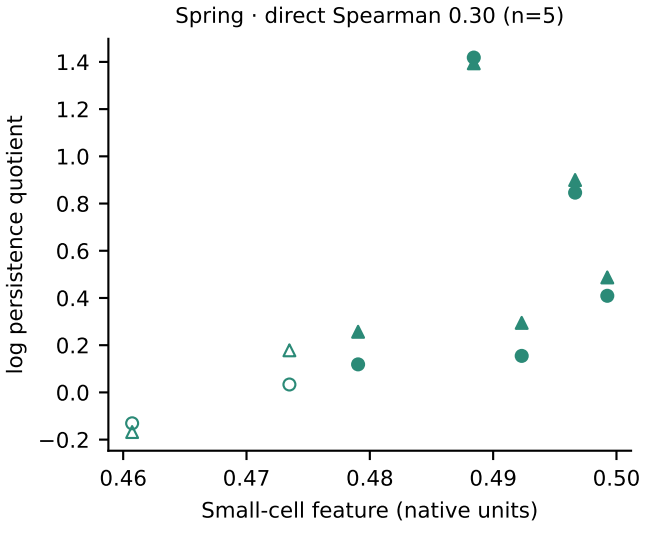


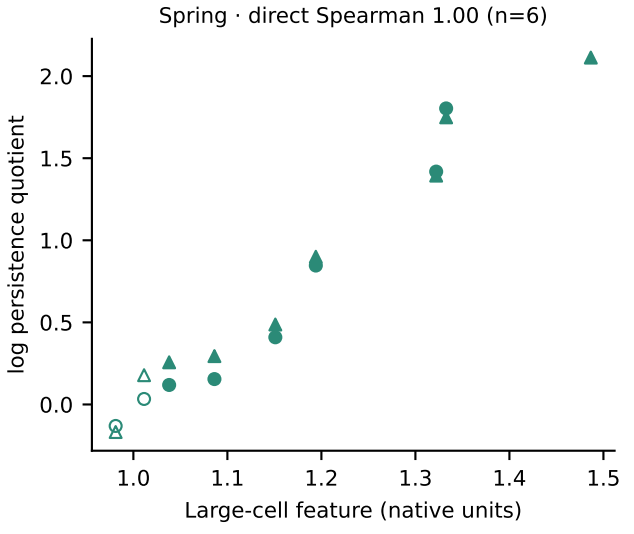


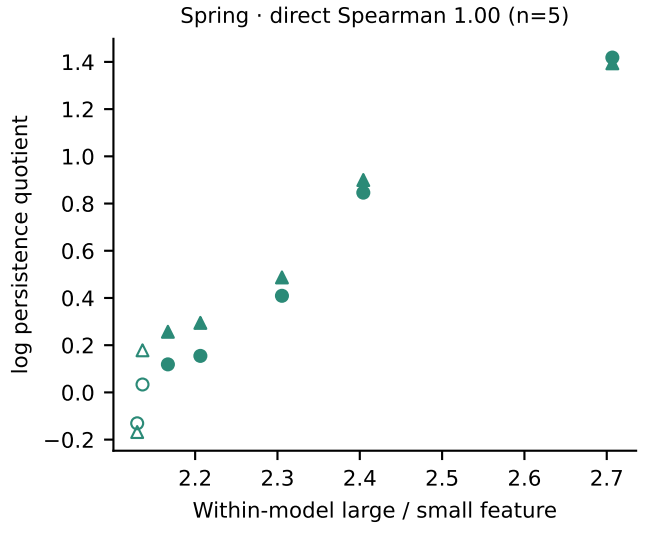












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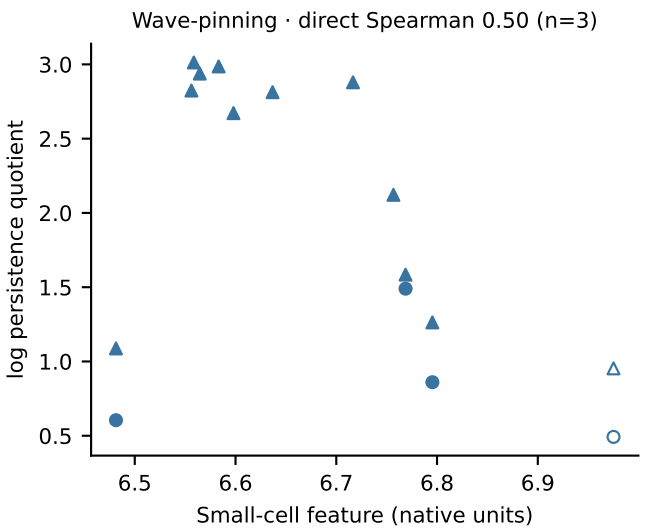
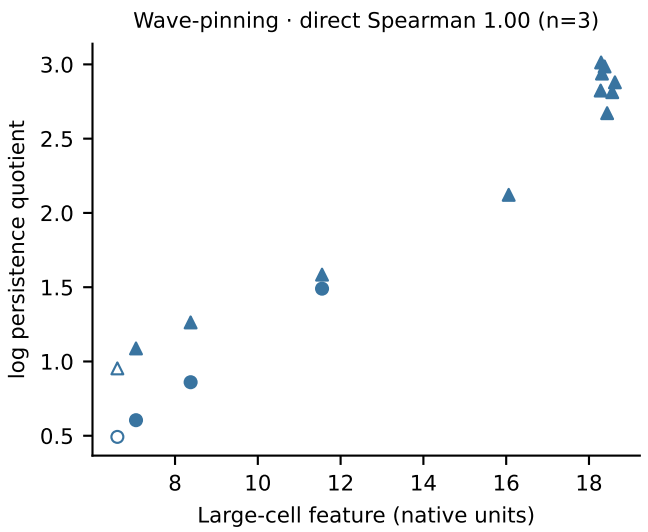
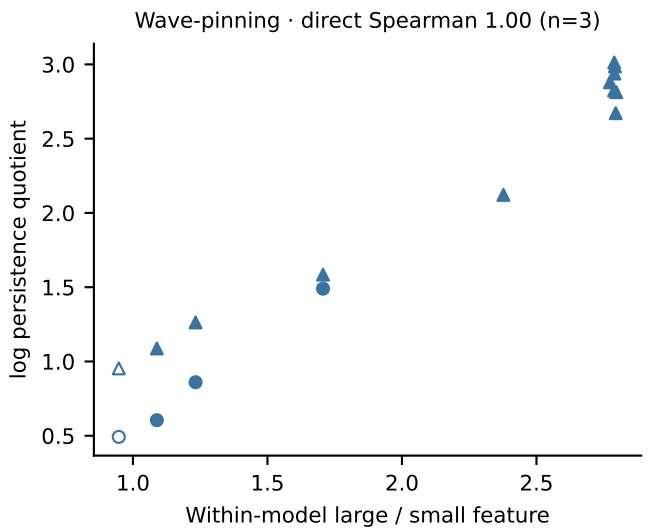
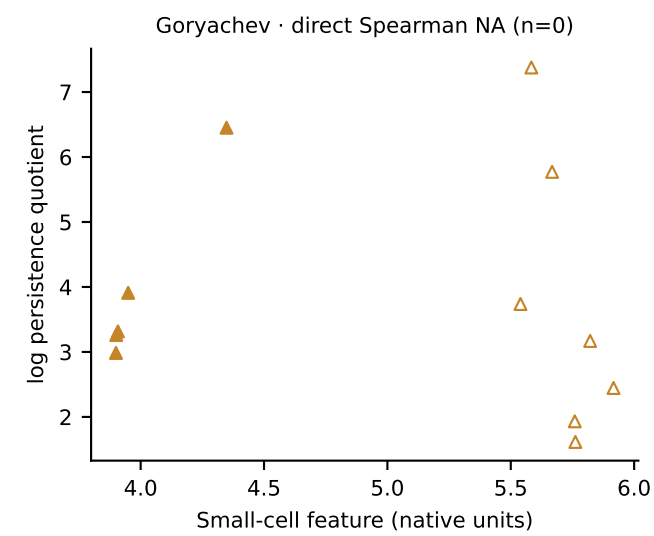
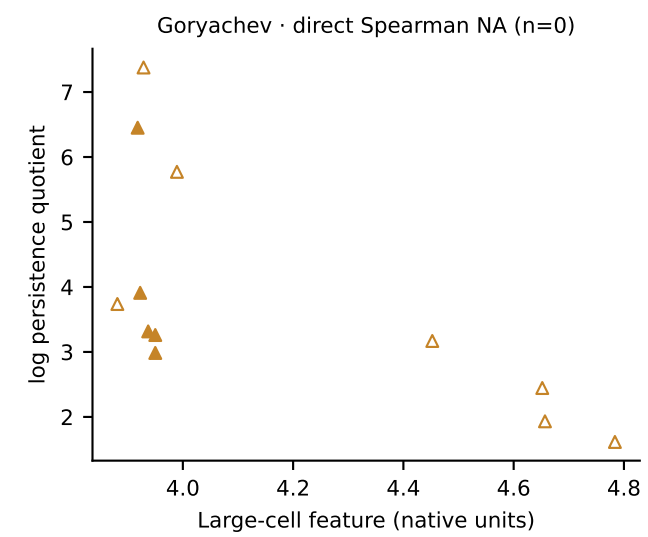
S10 · mean region width

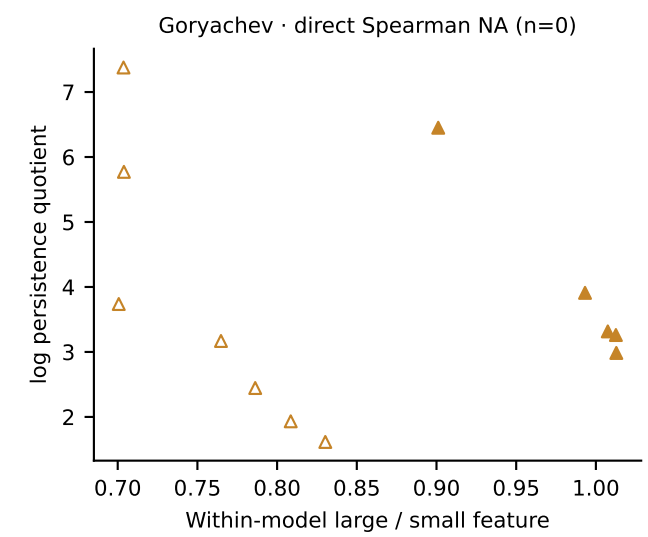


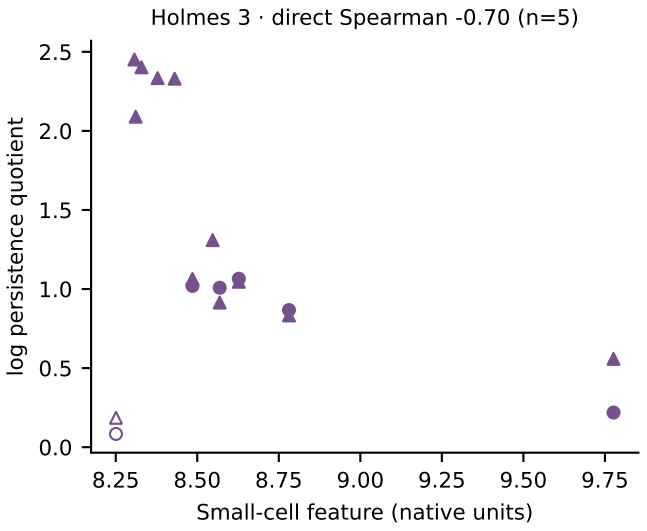


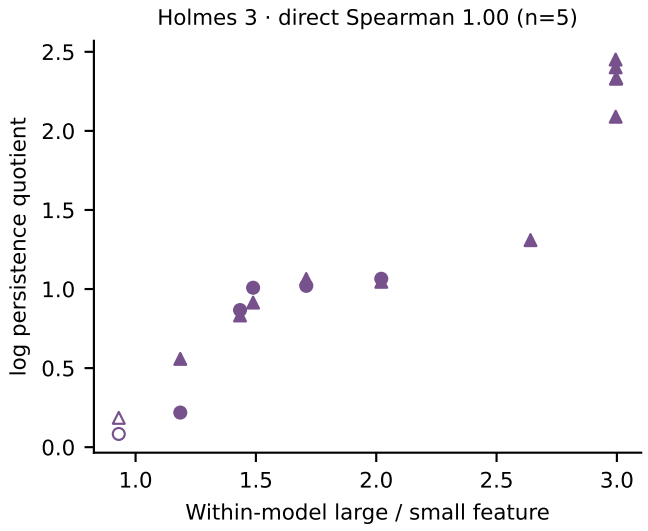


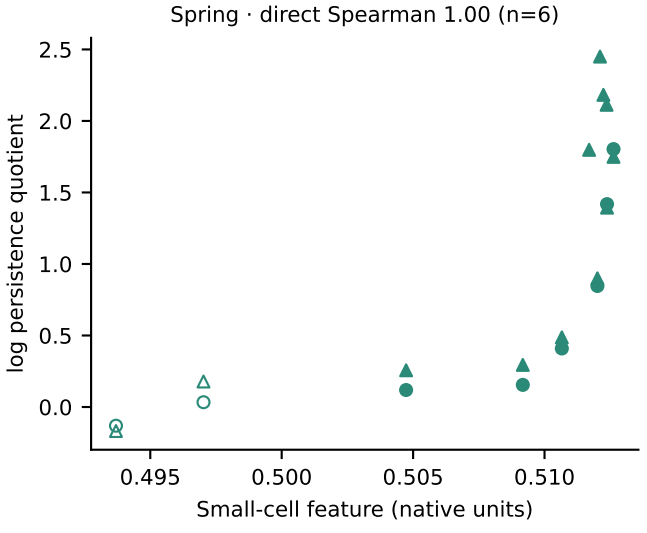


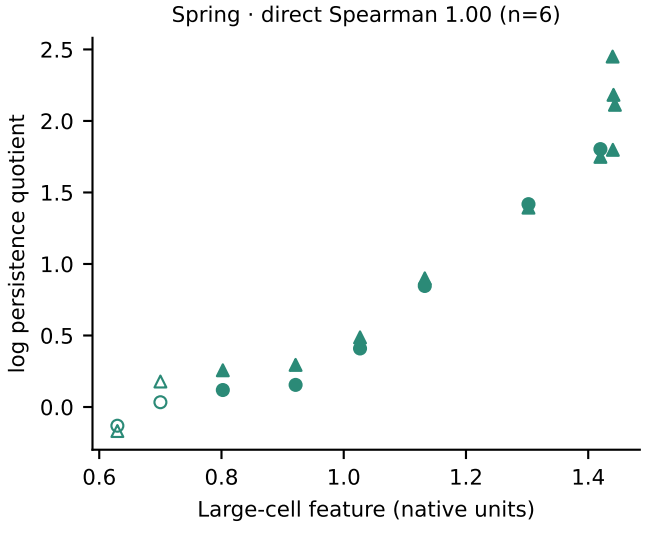


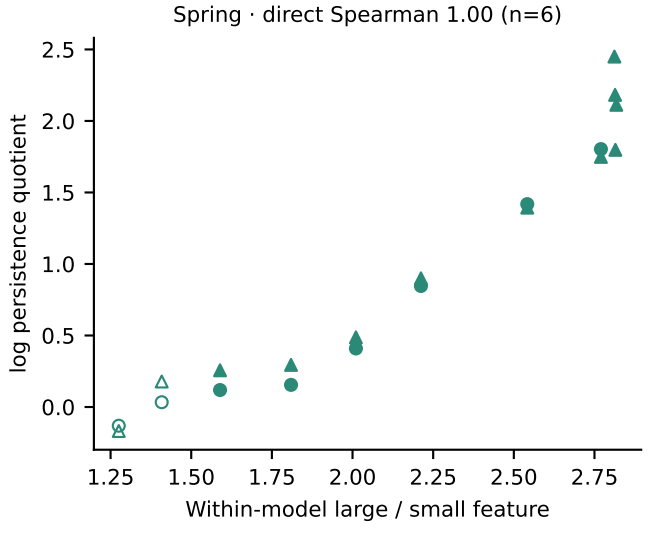












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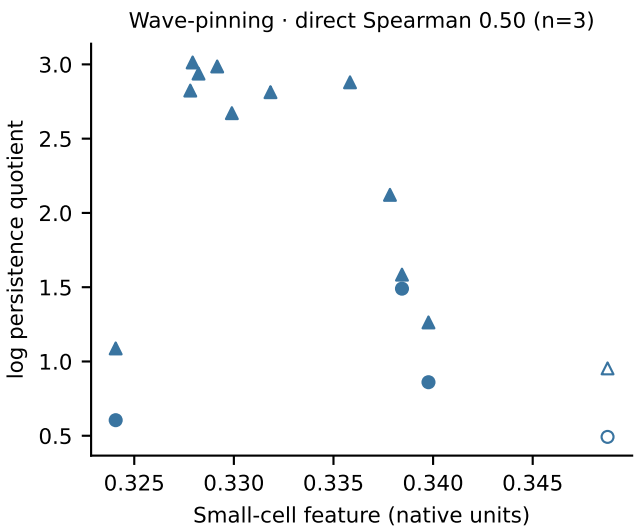
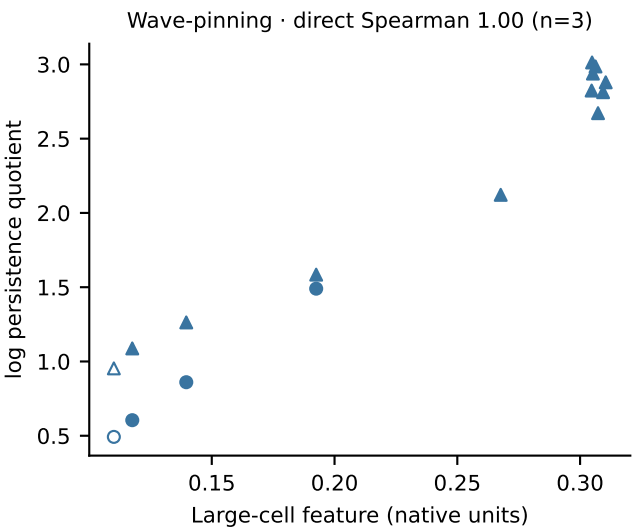
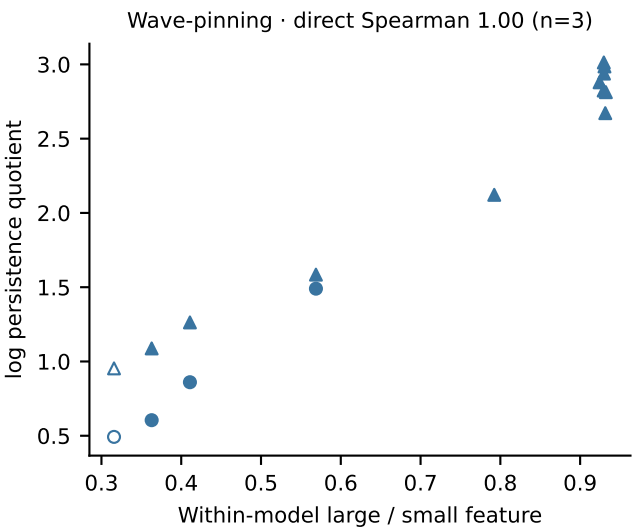
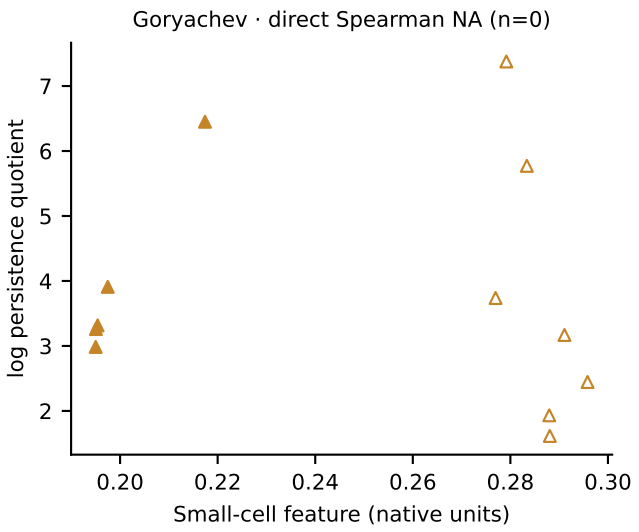
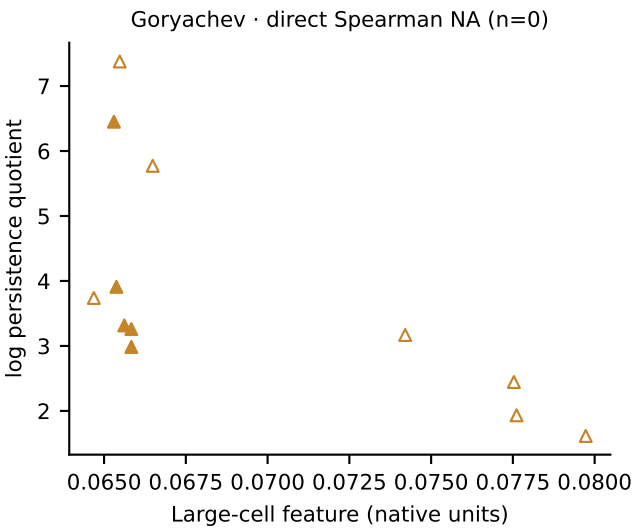
S10 · mean region width fraction

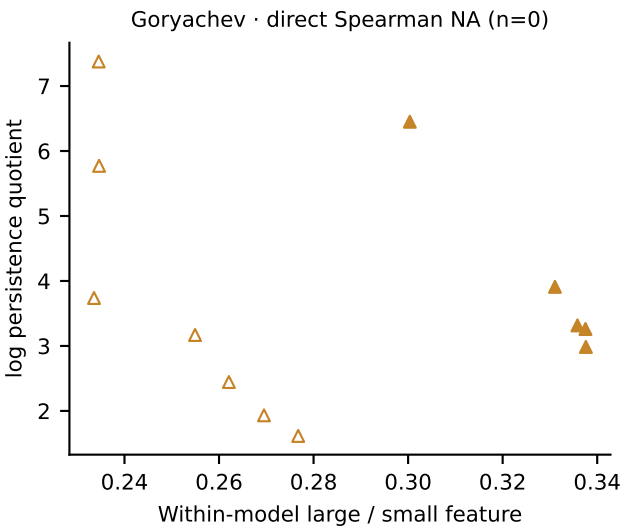


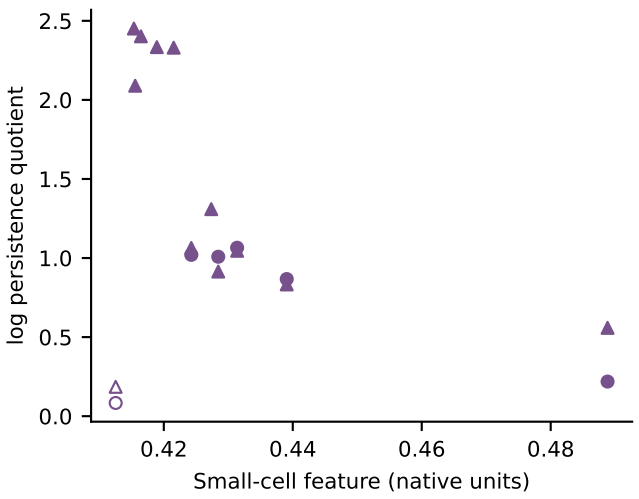


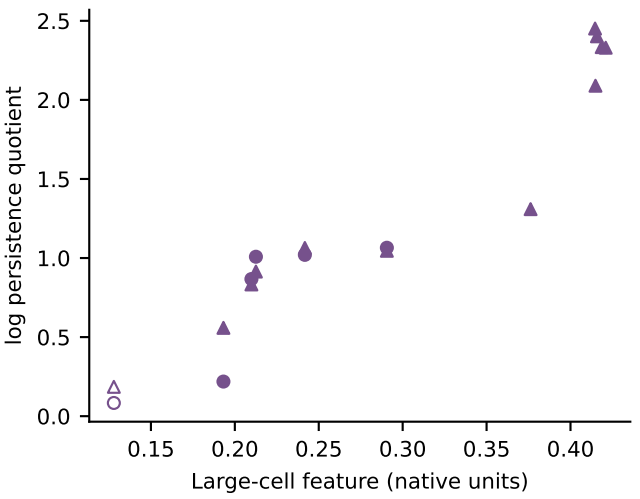


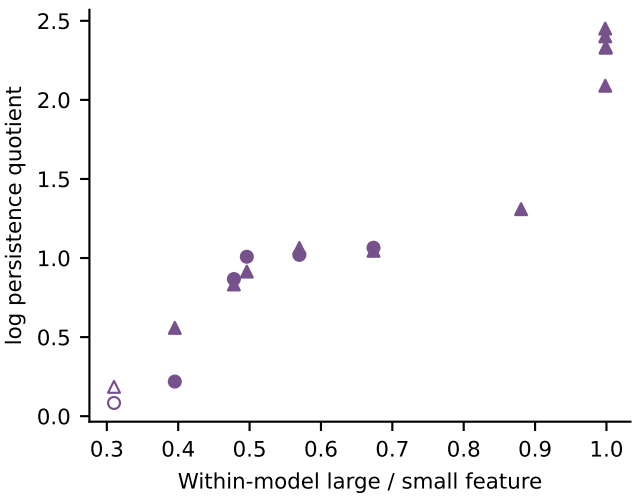


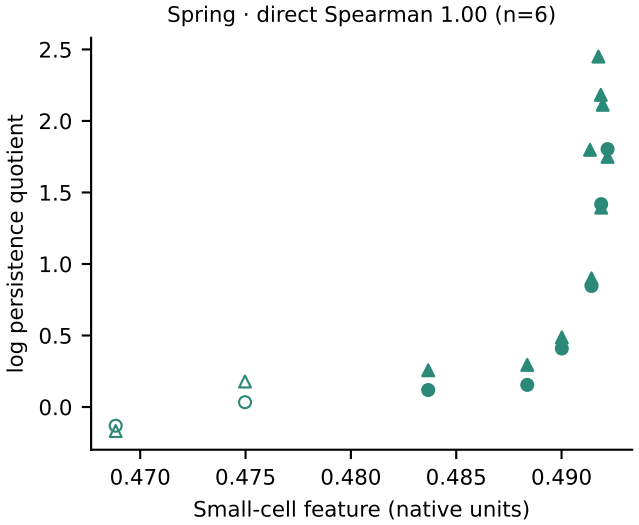


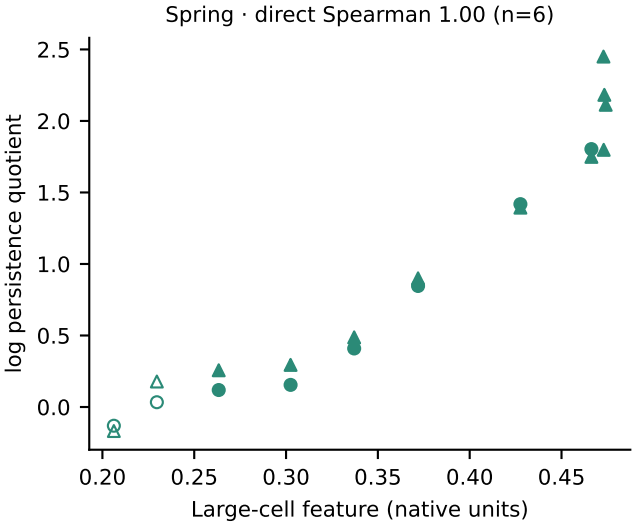


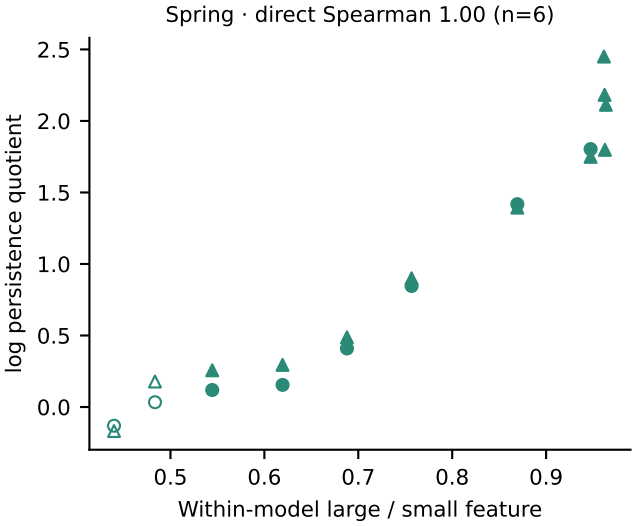












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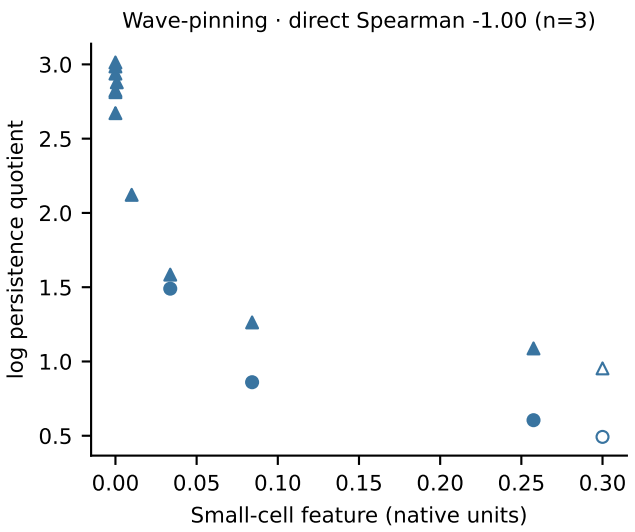
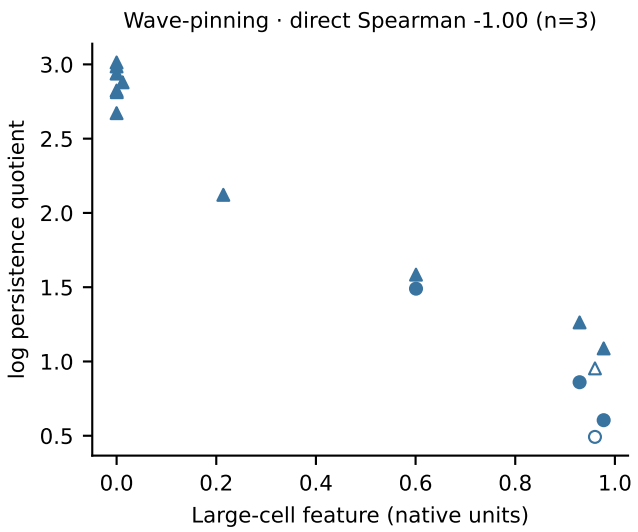
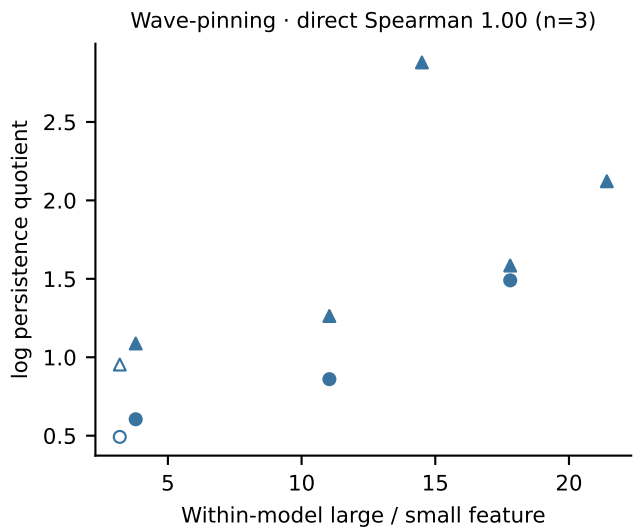
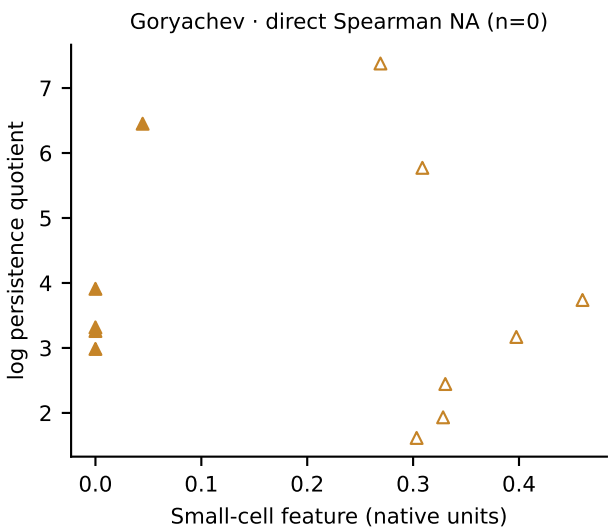
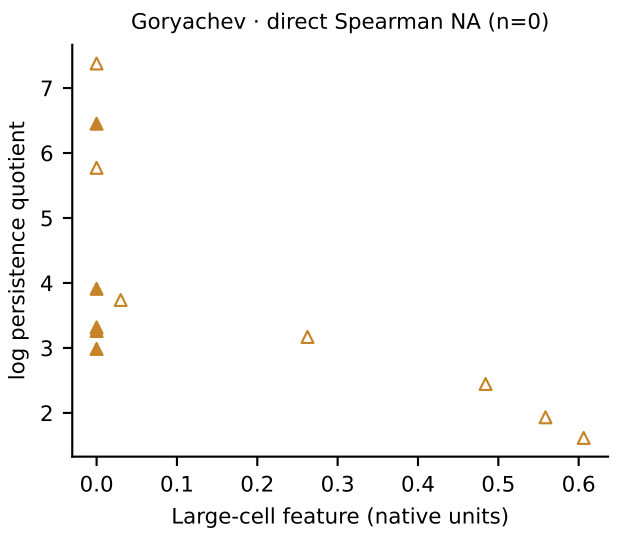
S10 · Multiple-front fraction

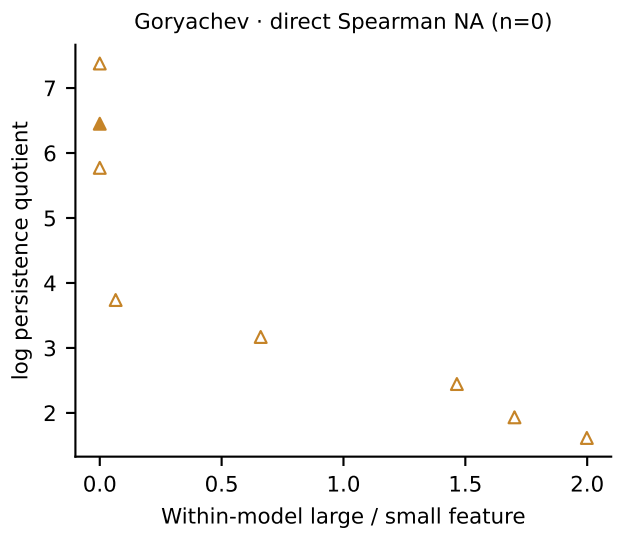


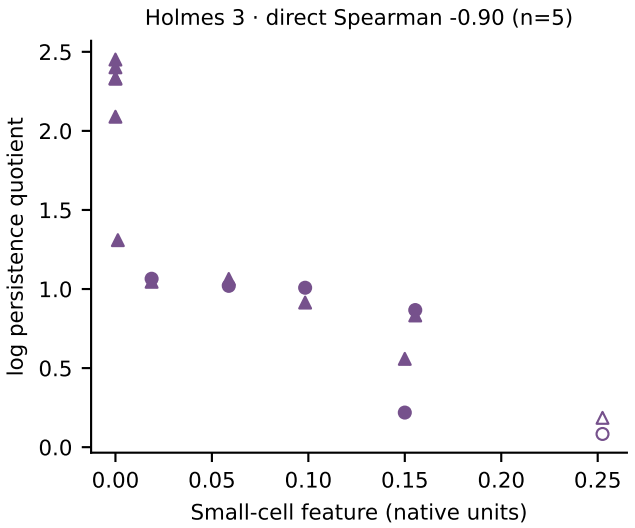


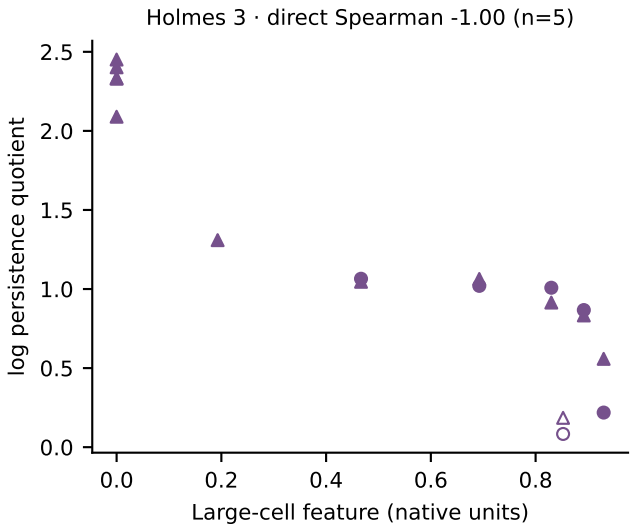


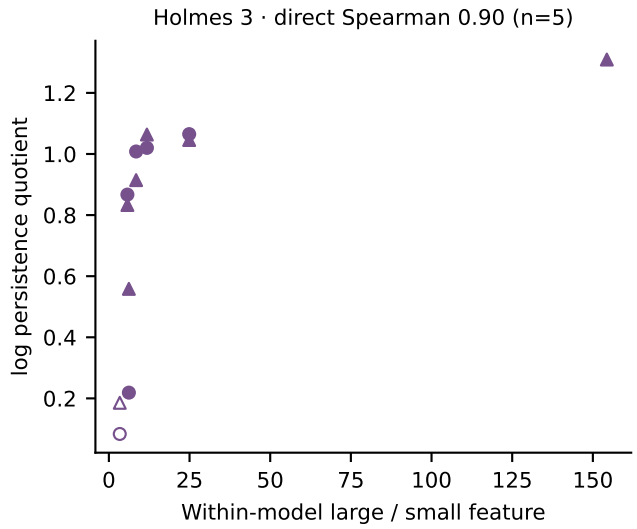


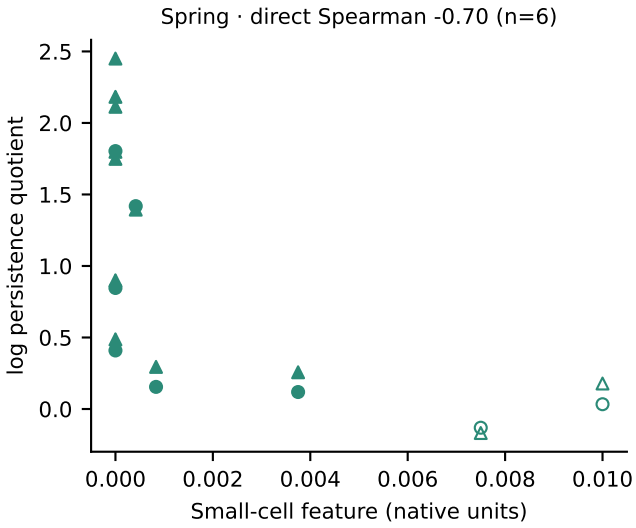


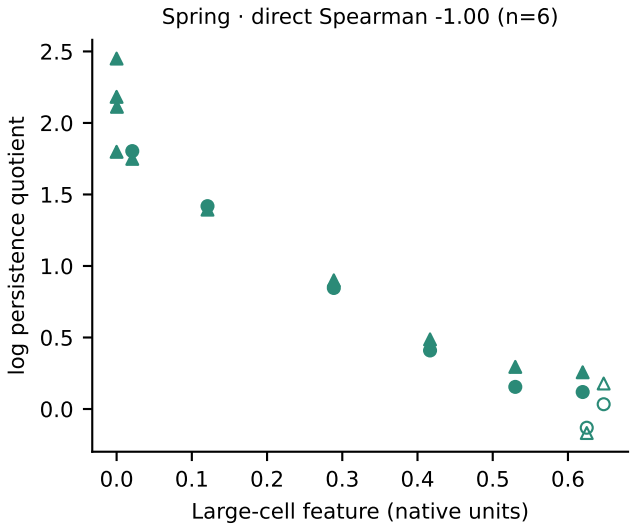


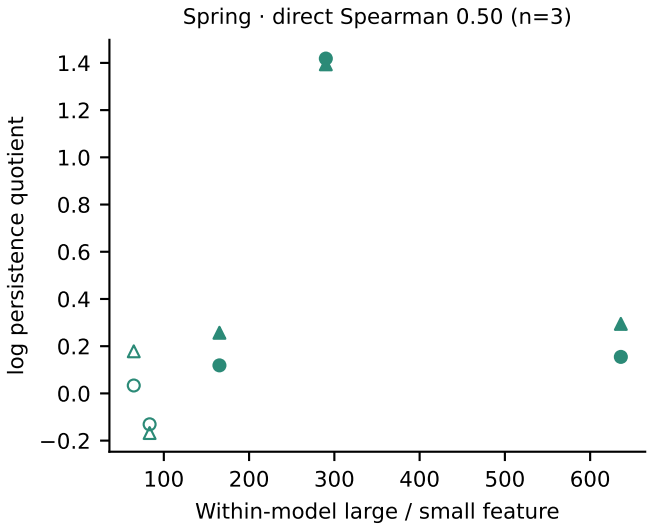












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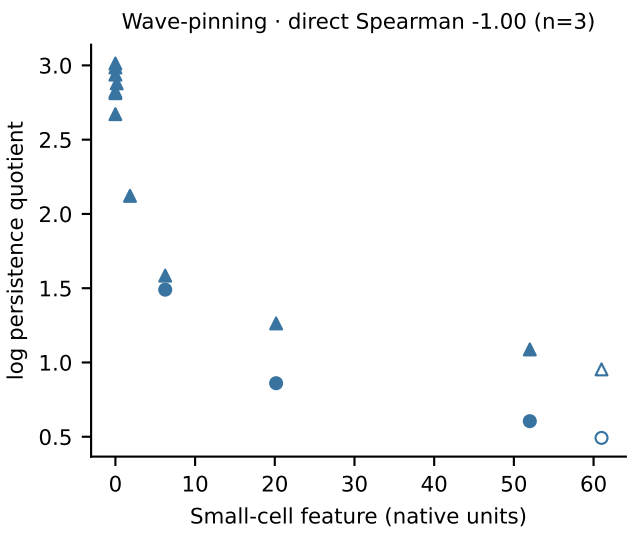
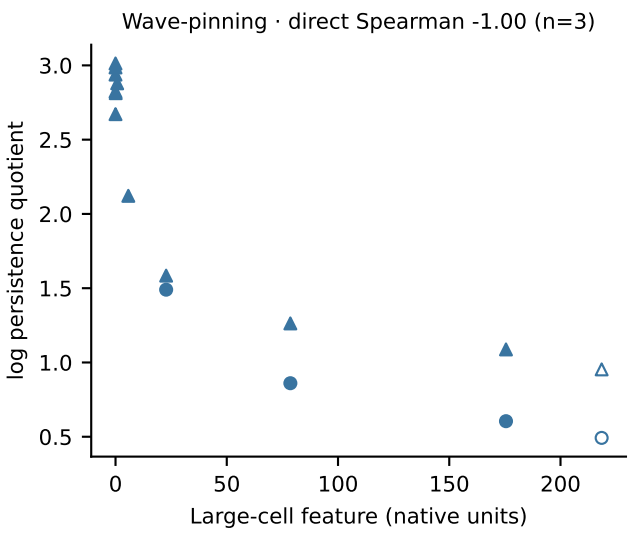
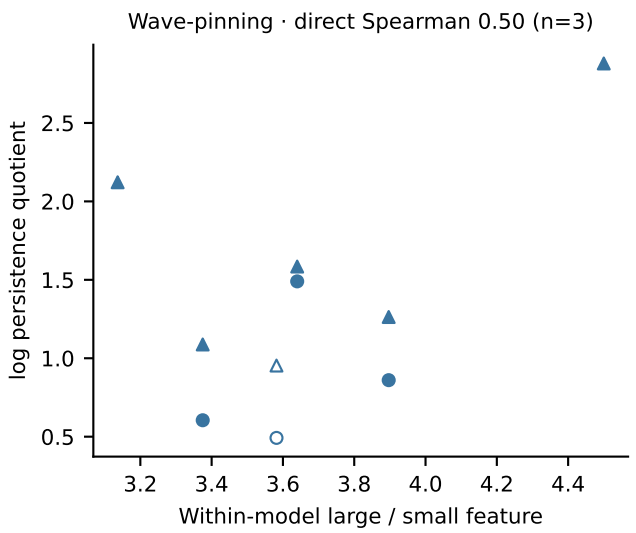
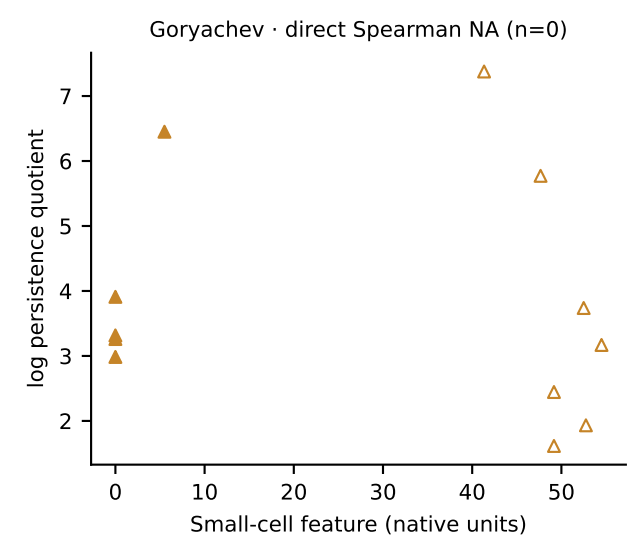
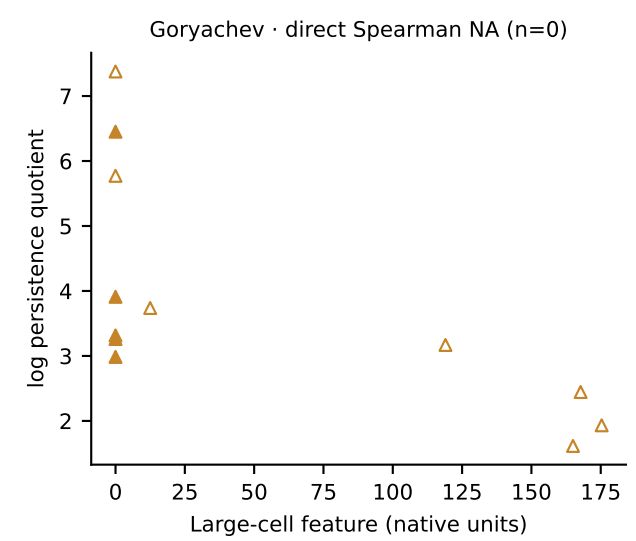
S10 · observed births

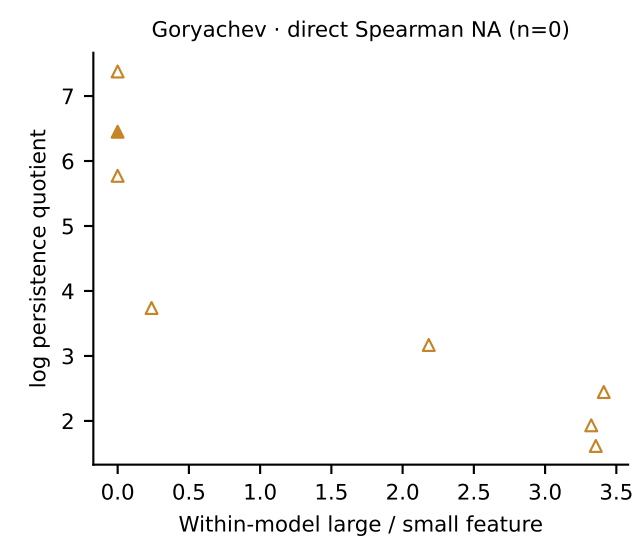


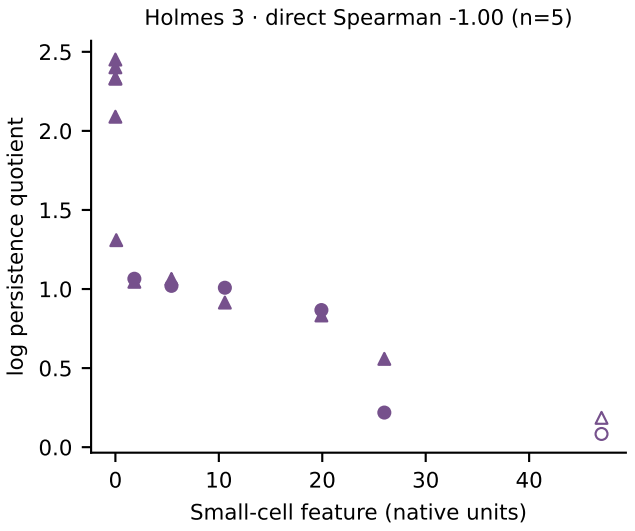


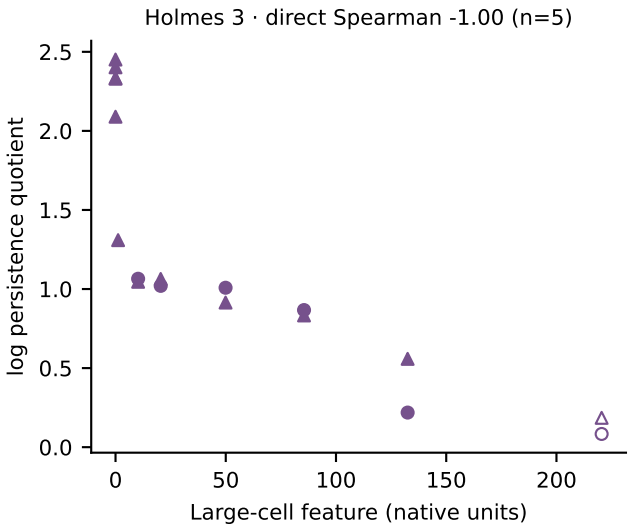


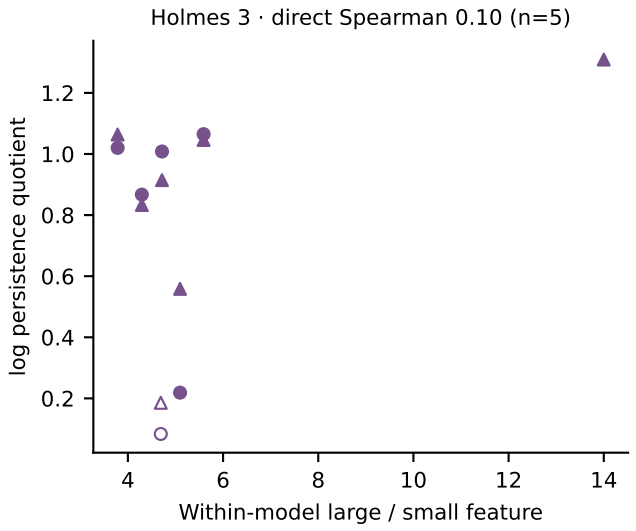


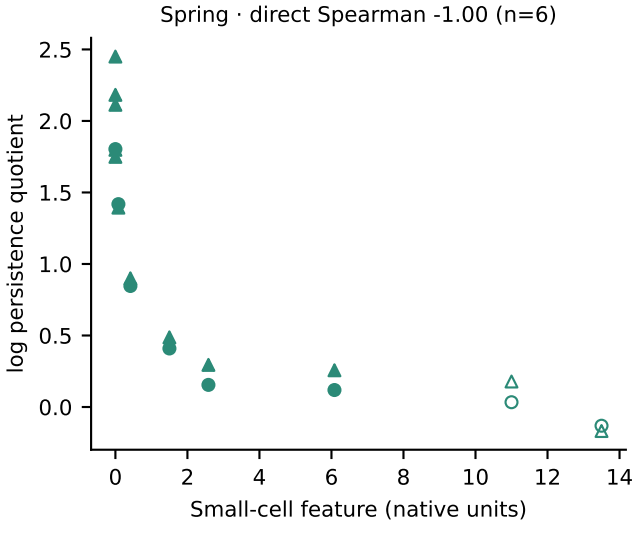


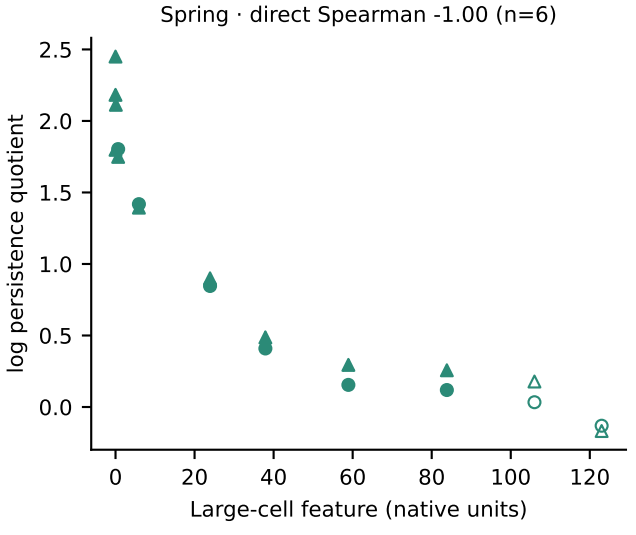


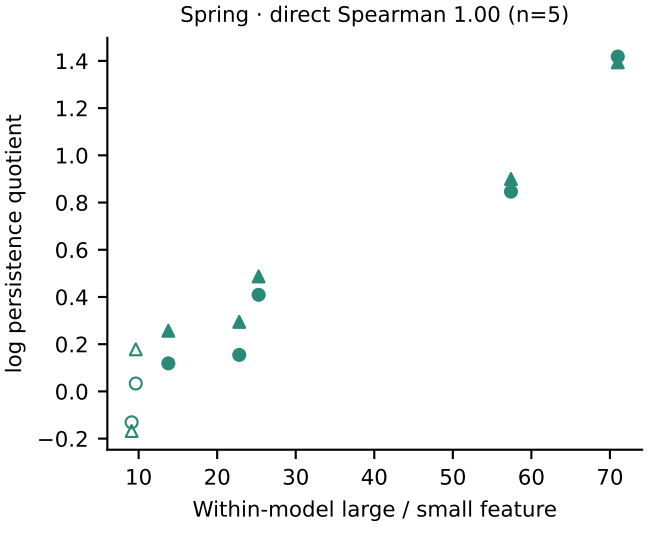












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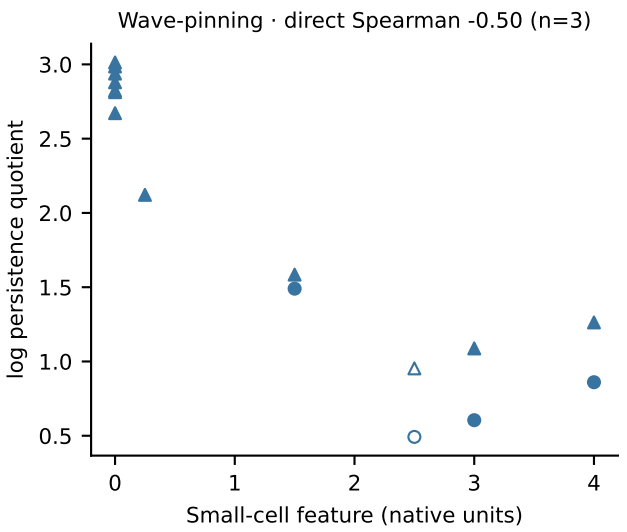
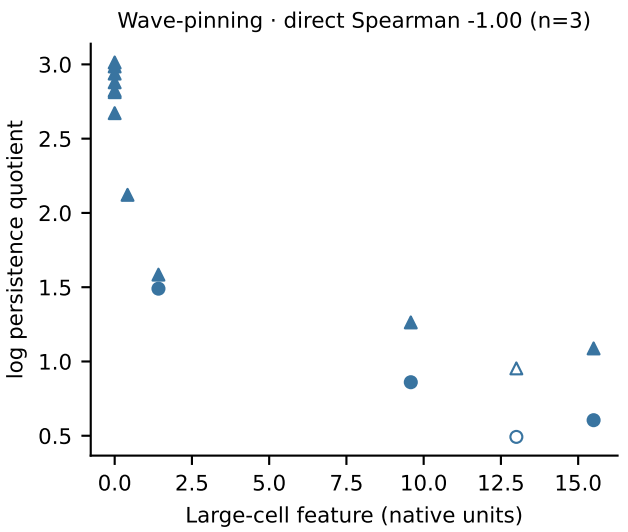
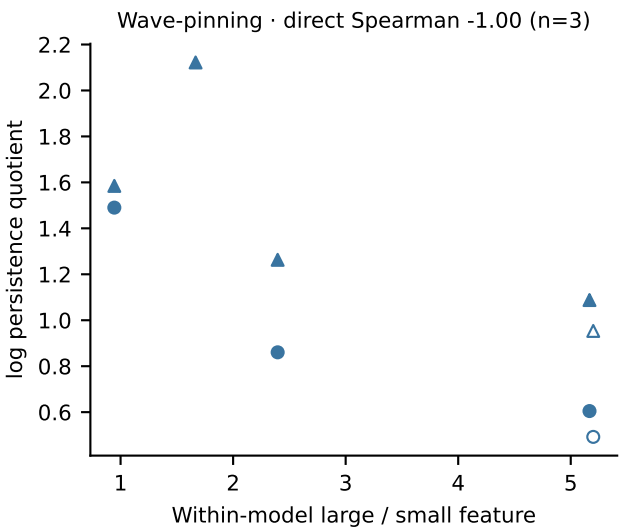
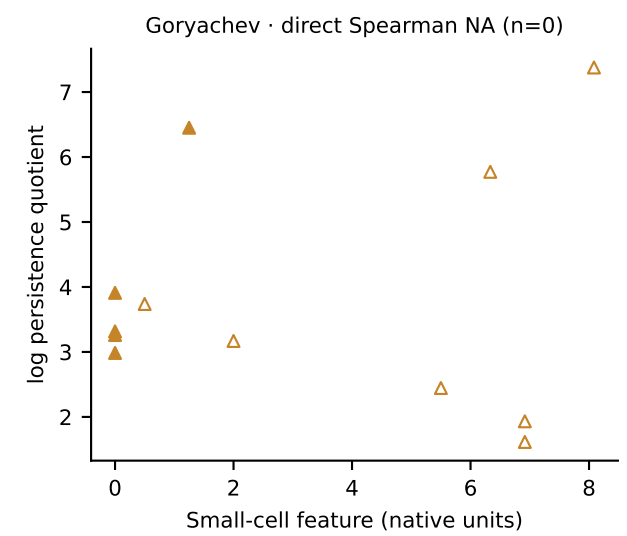
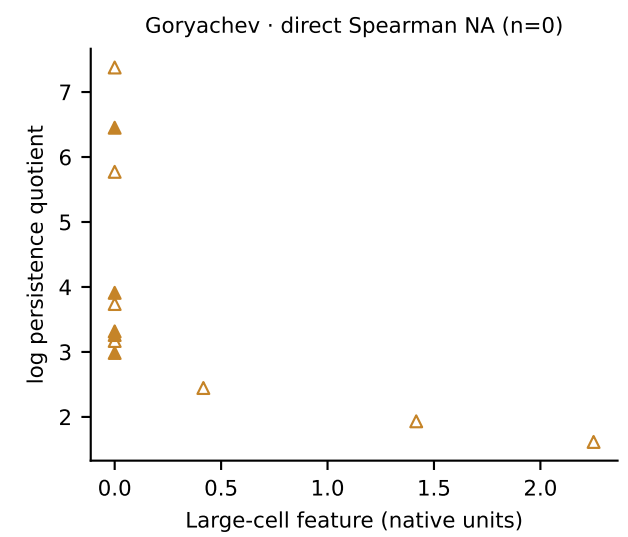
S10 · observed pairwise takeovers

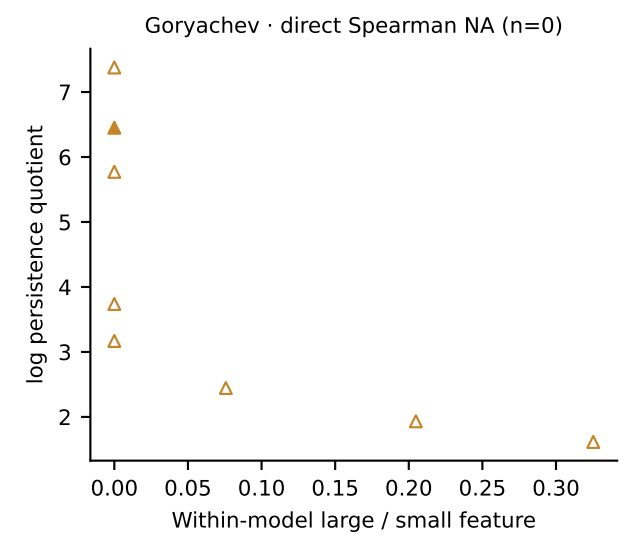


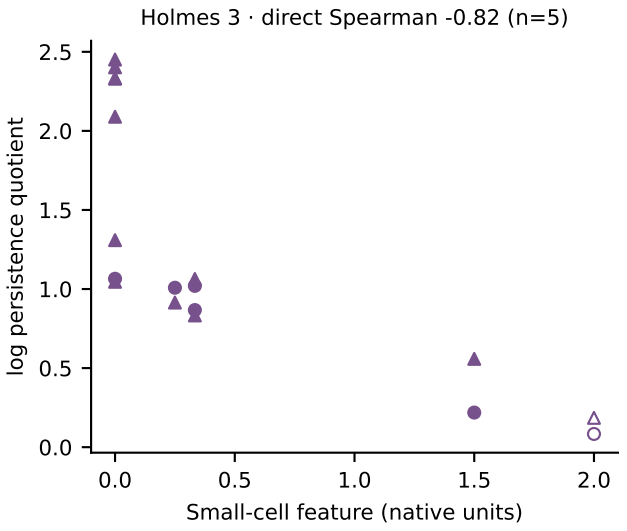


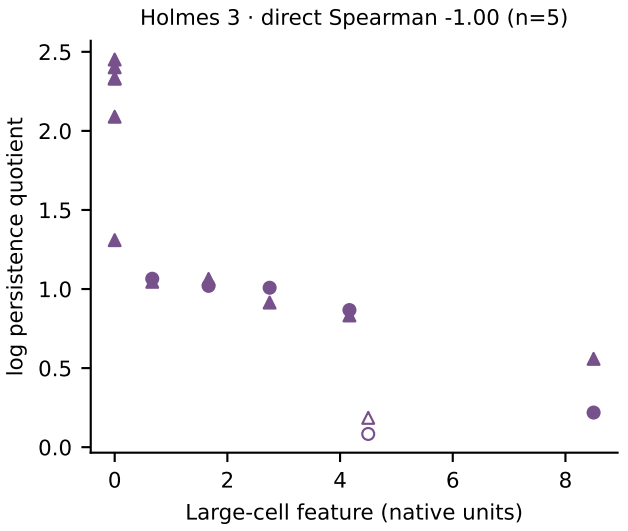


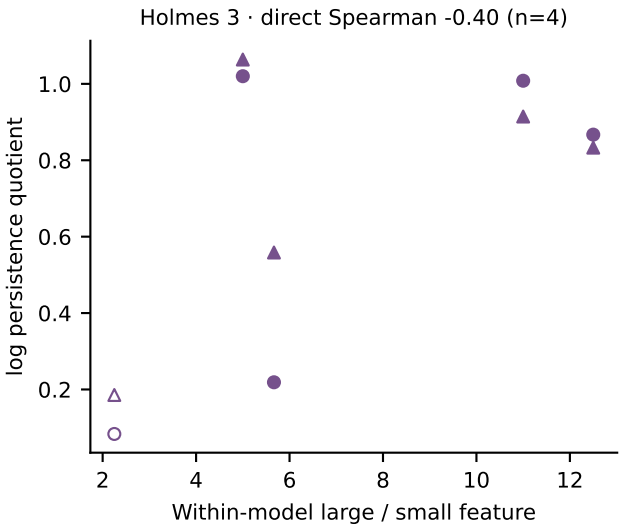


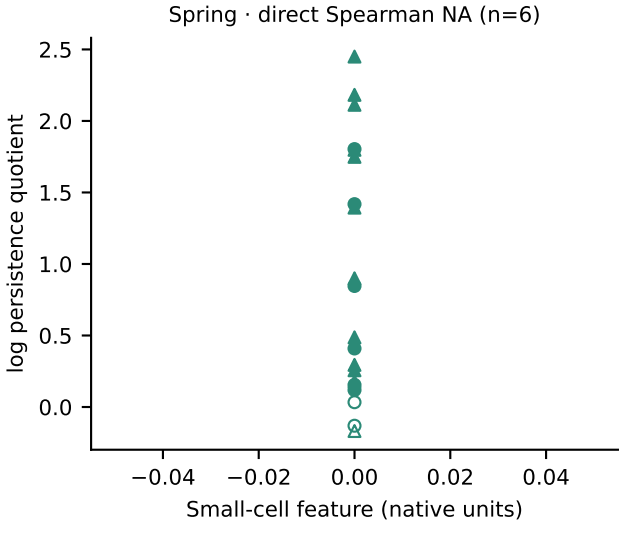


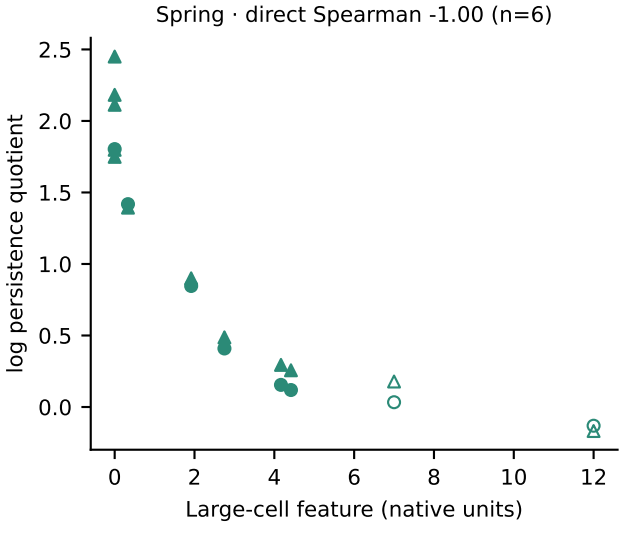


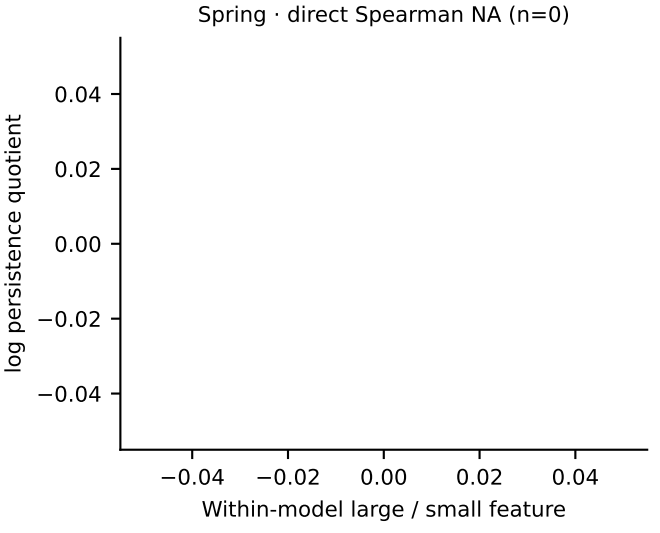












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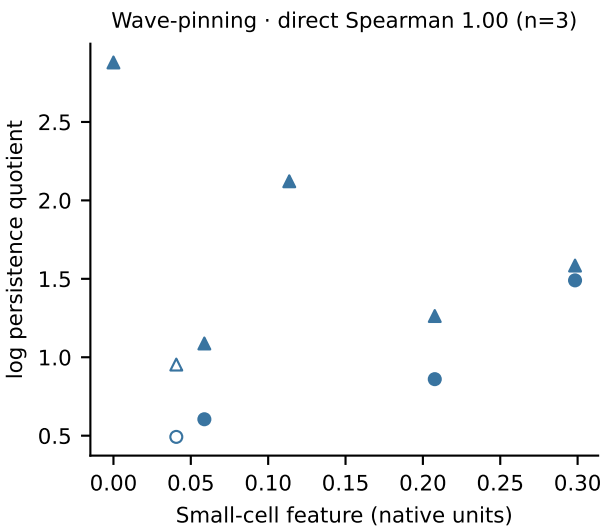
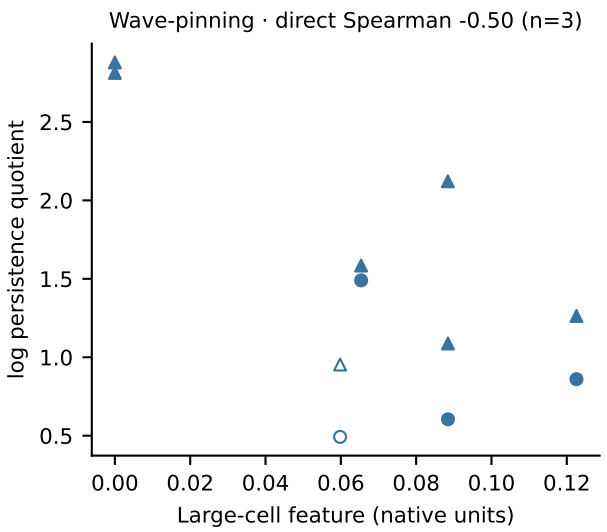
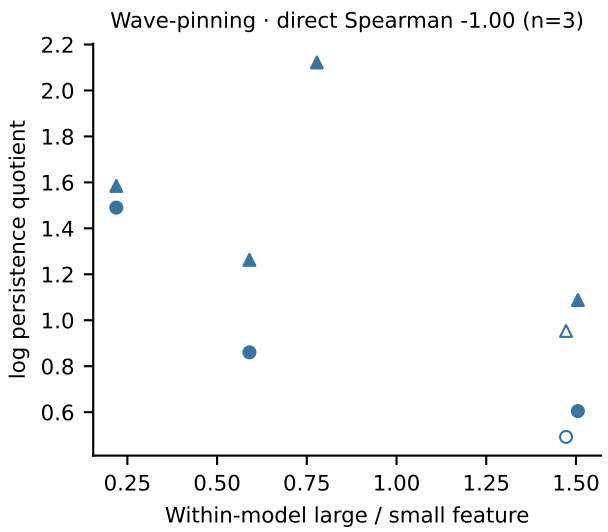
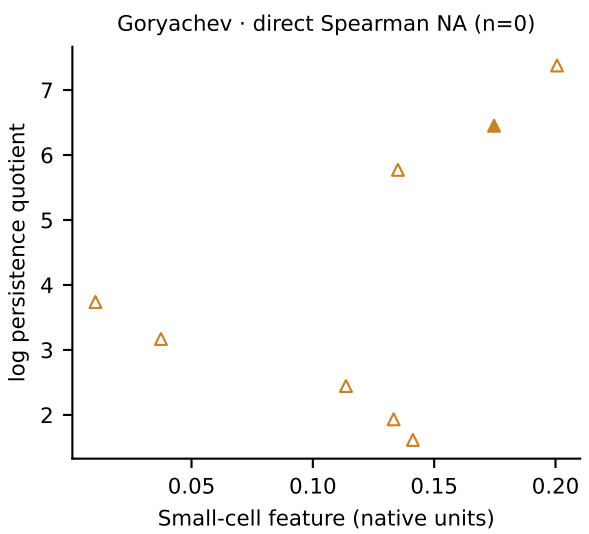
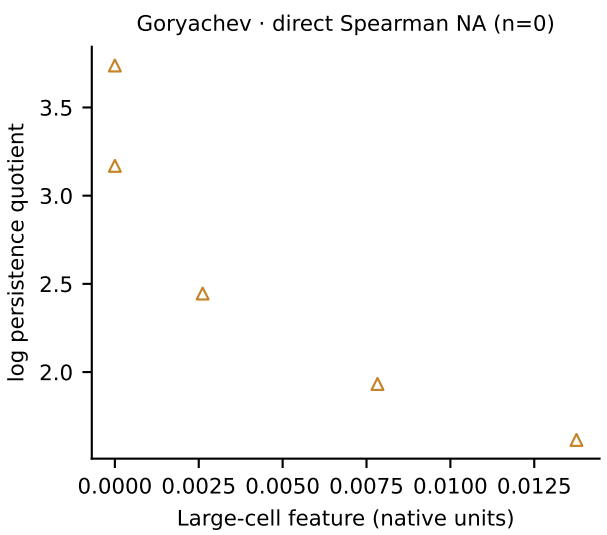
S10 · observed pairwise takeovers per birth

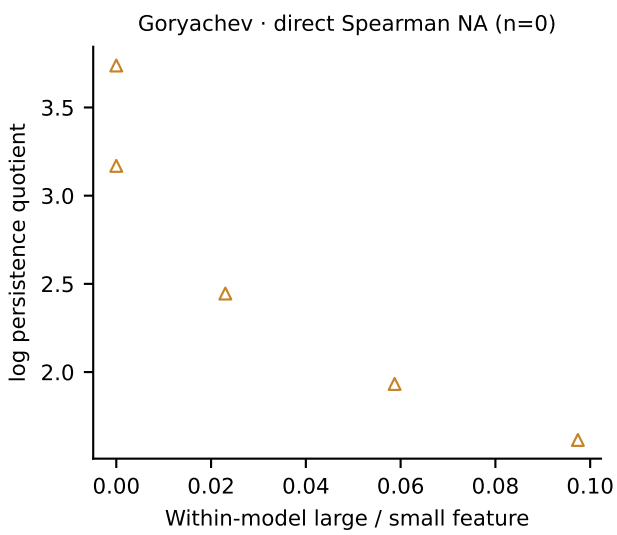


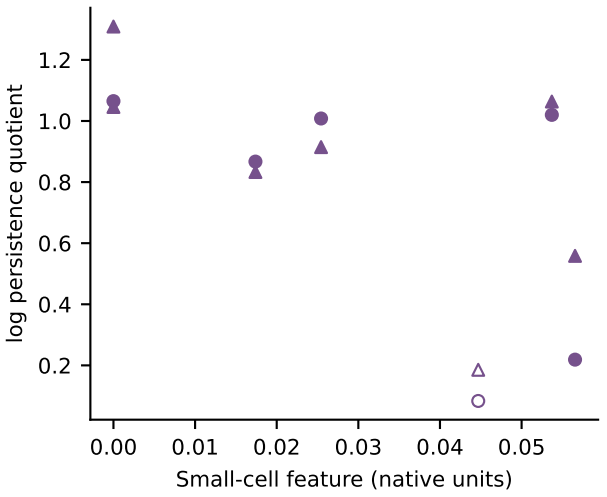


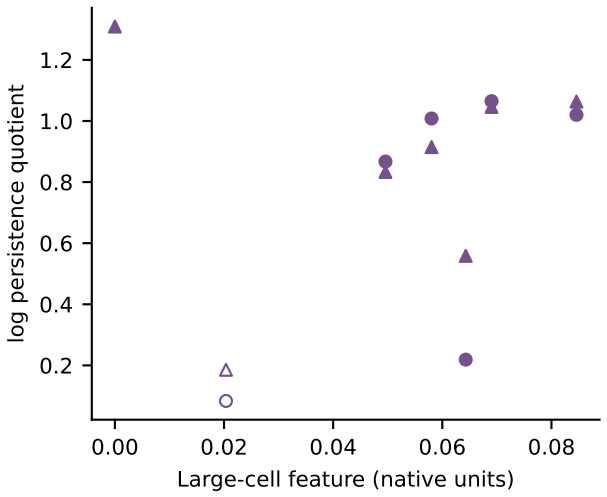


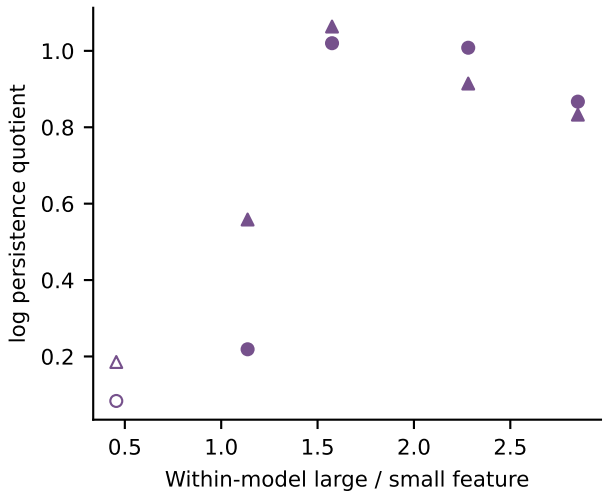


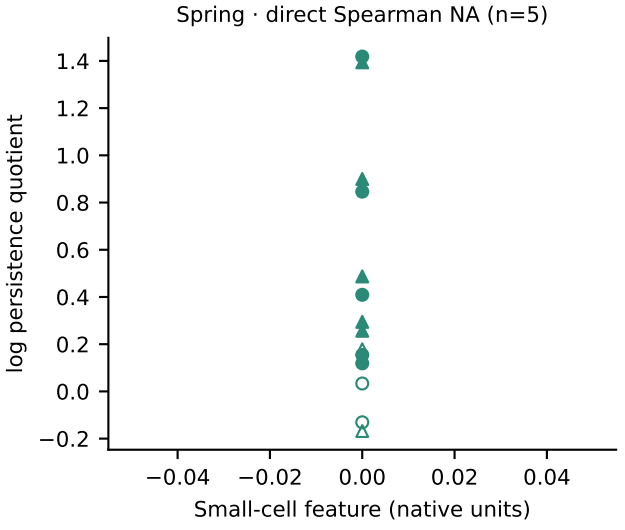


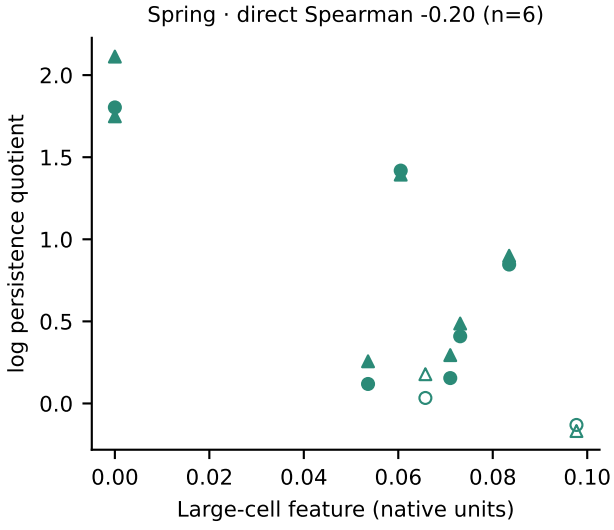


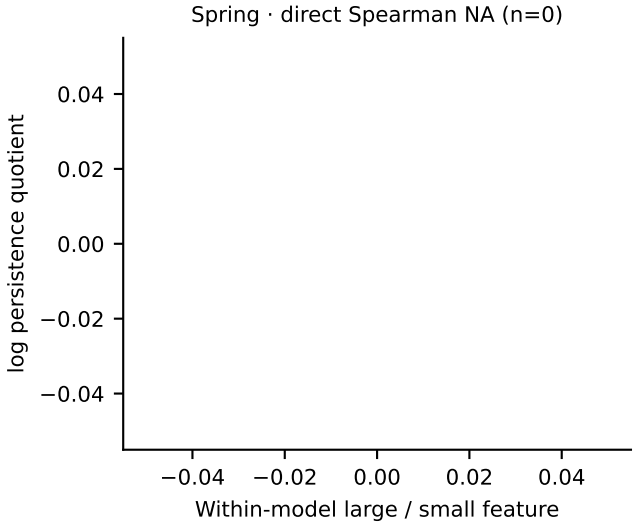












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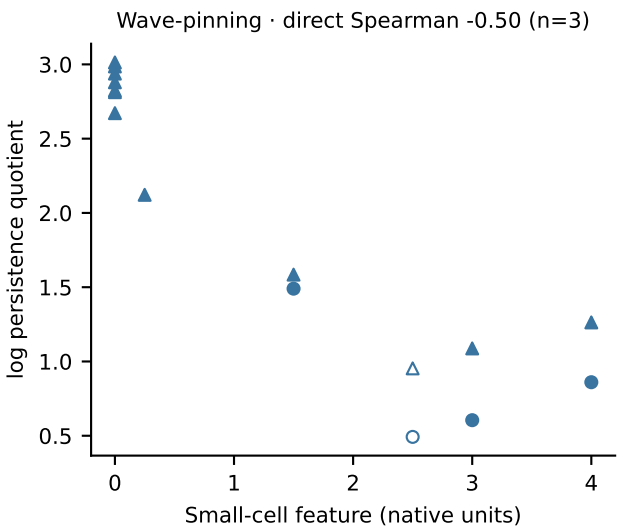
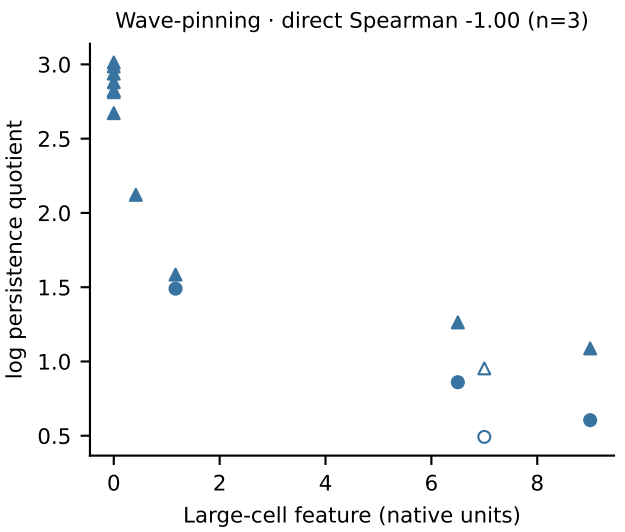
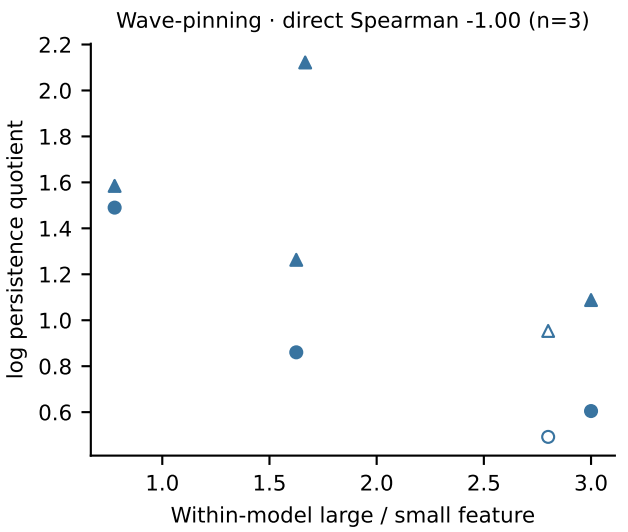
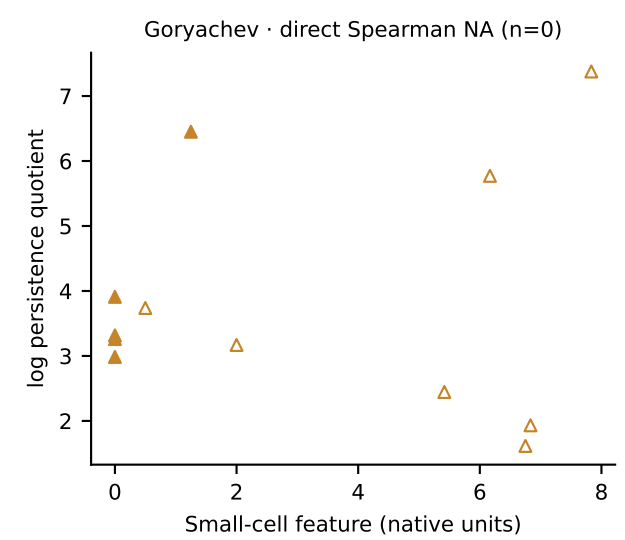
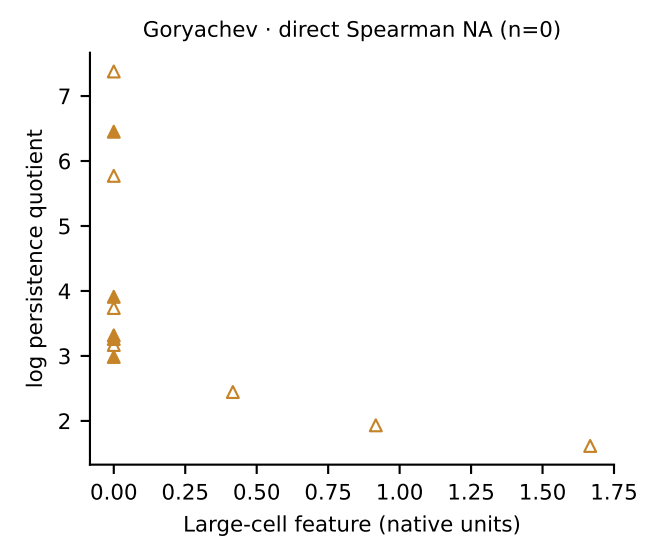
S10 · observed takeovers

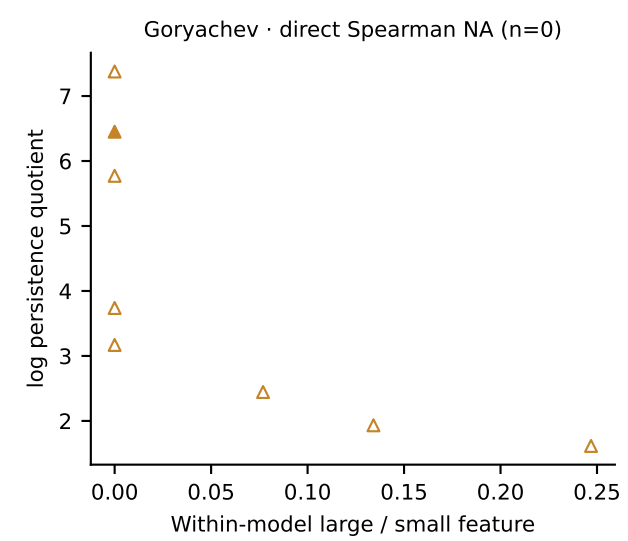


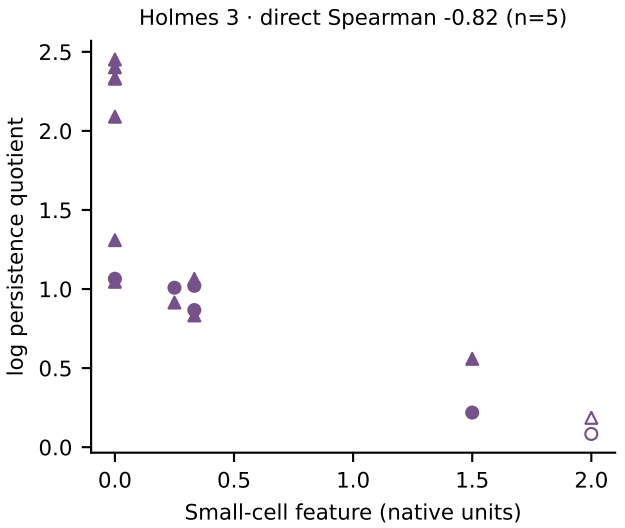


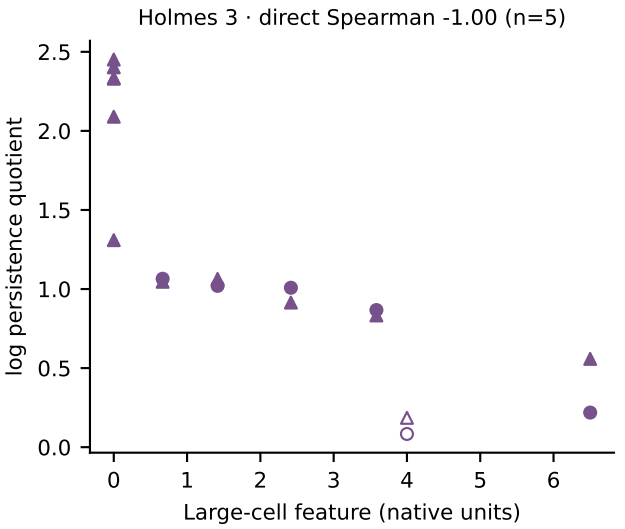


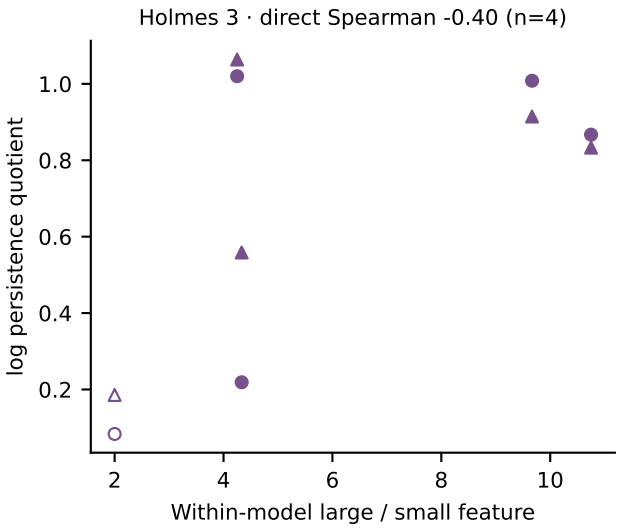


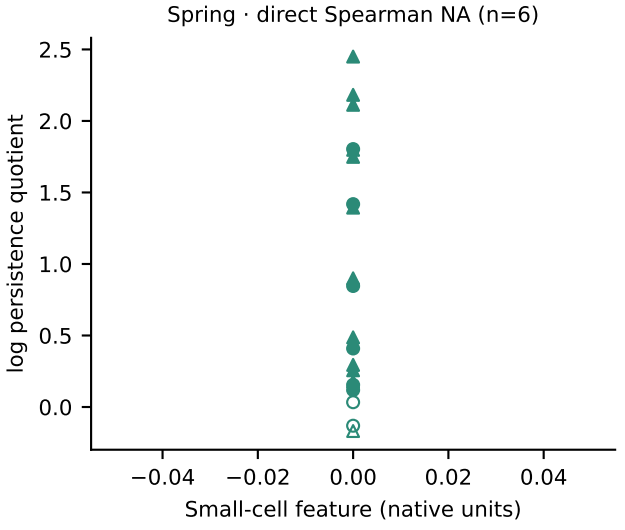


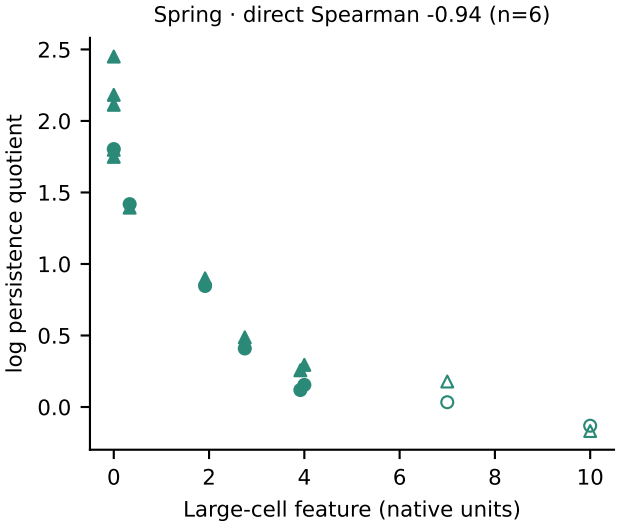


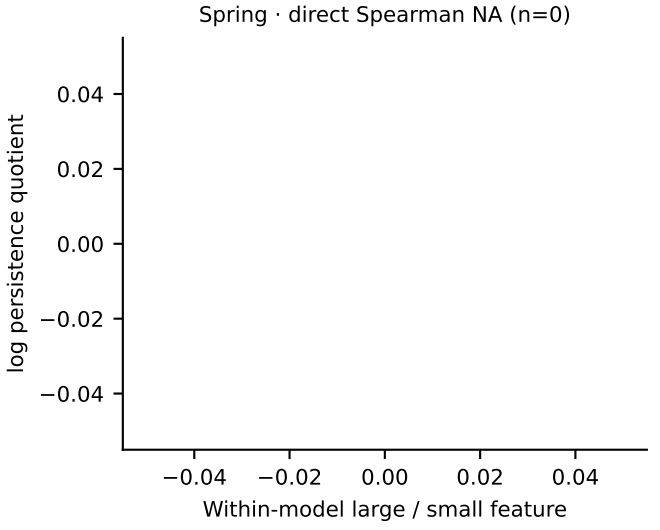












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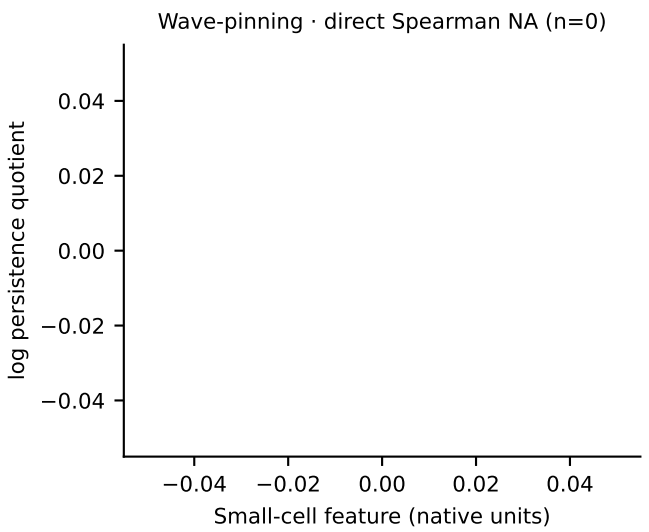
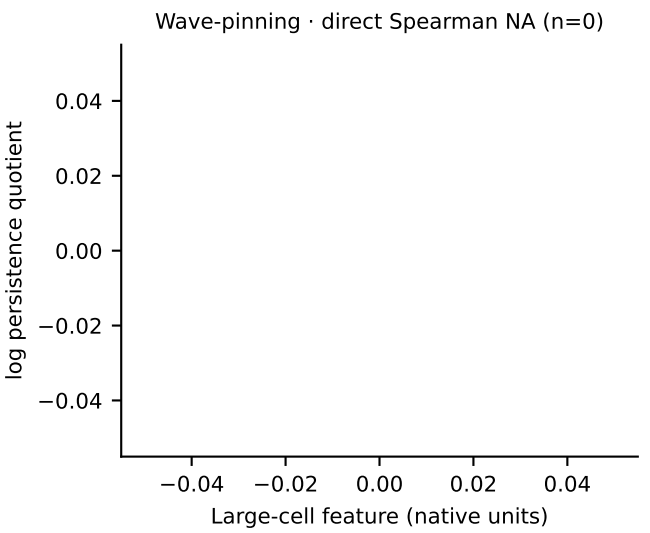
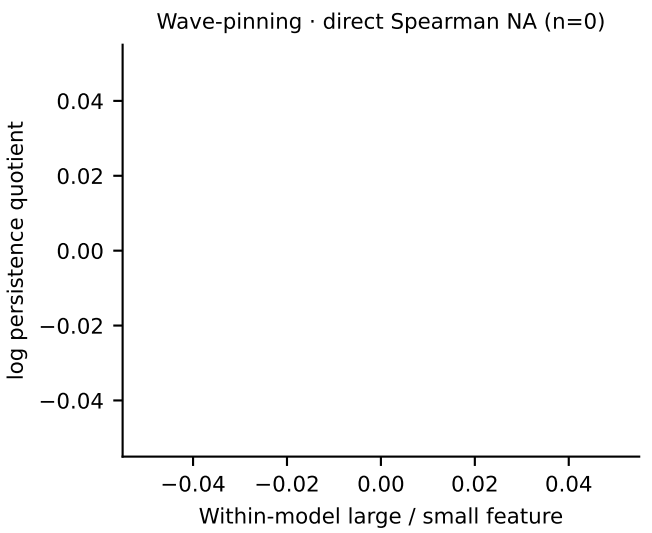
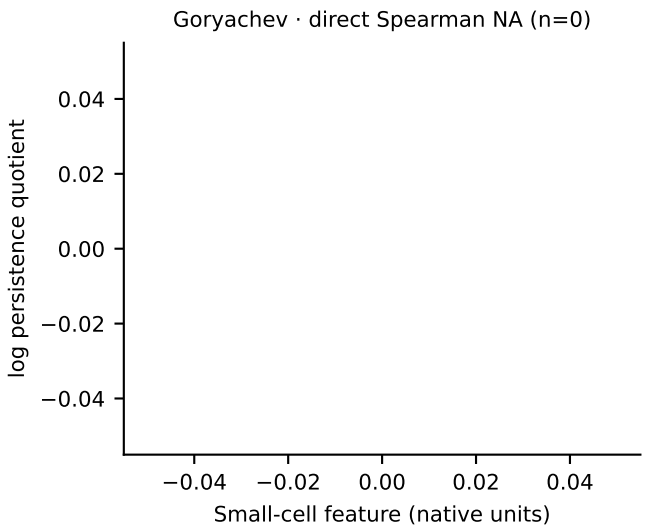
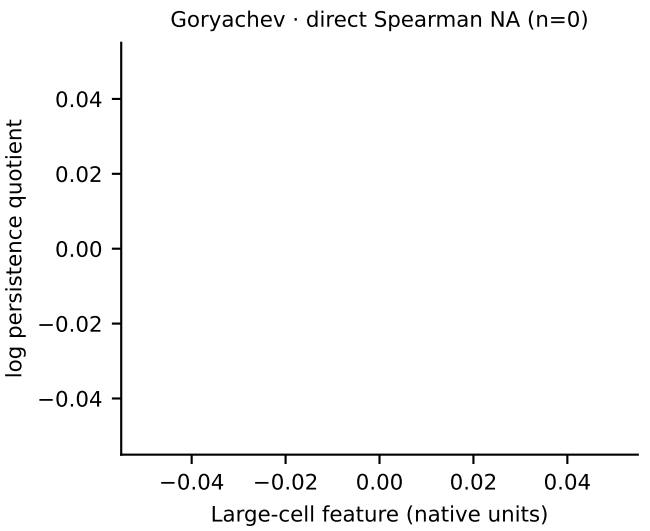
S10 · observed takeovers per birth

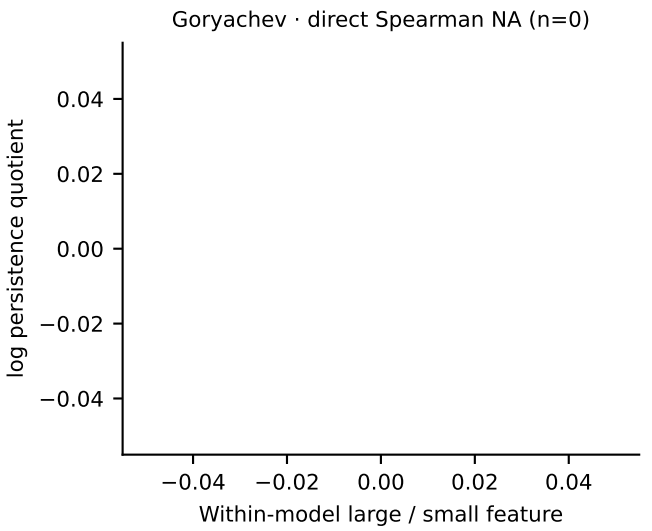


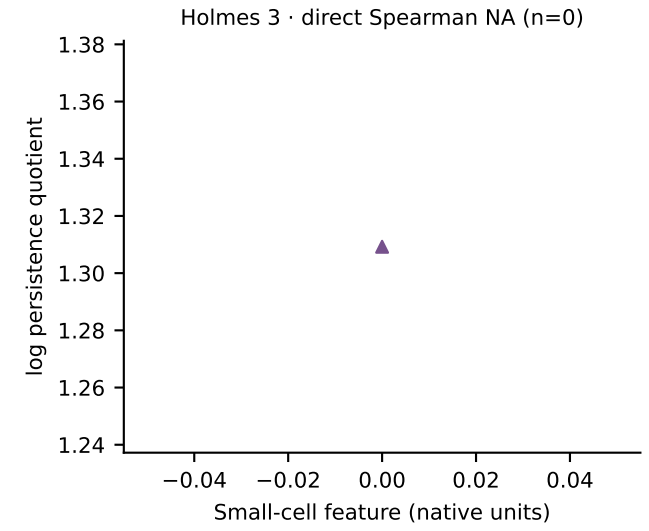


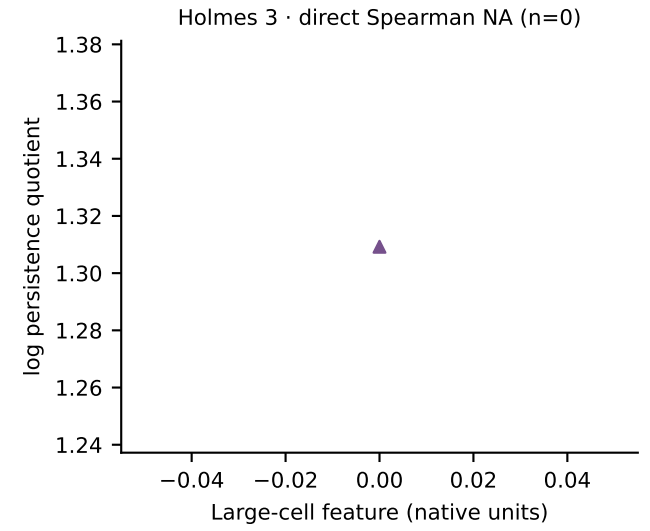


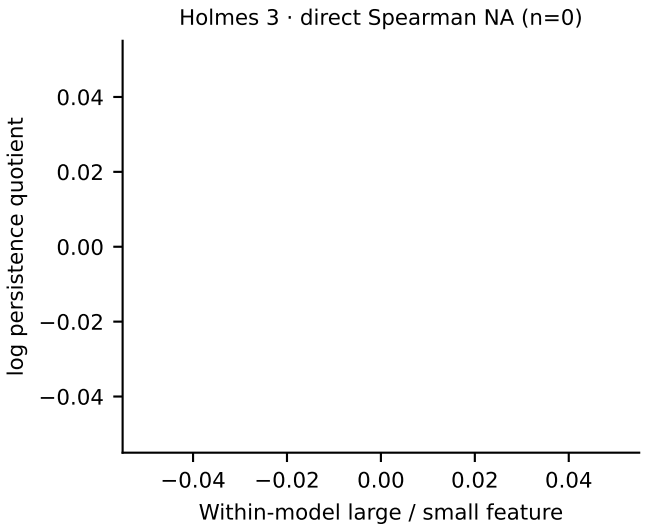


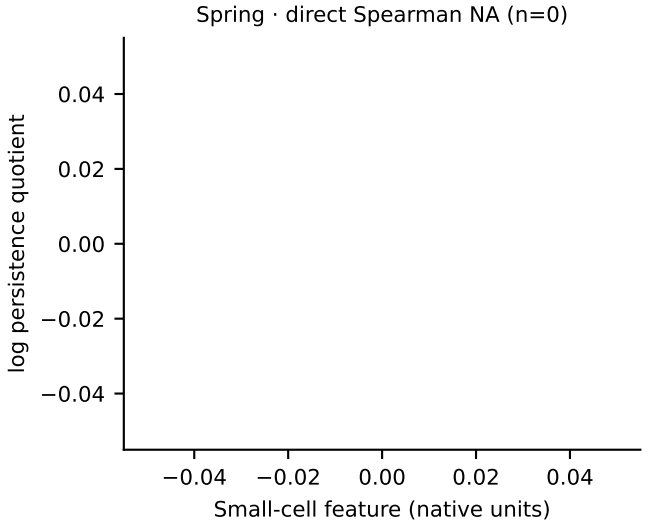


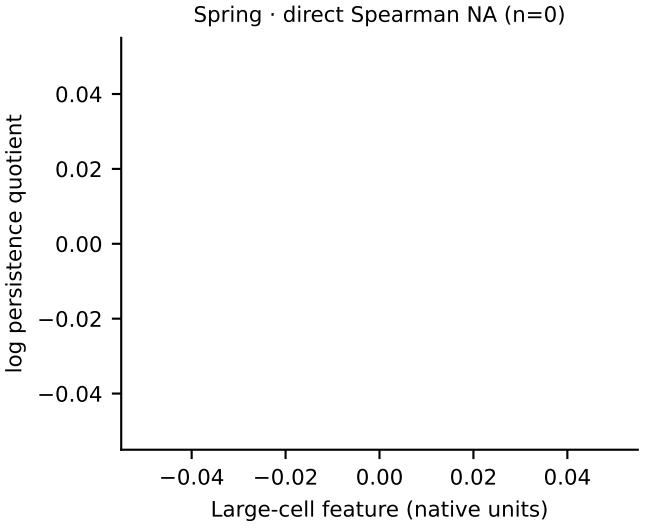


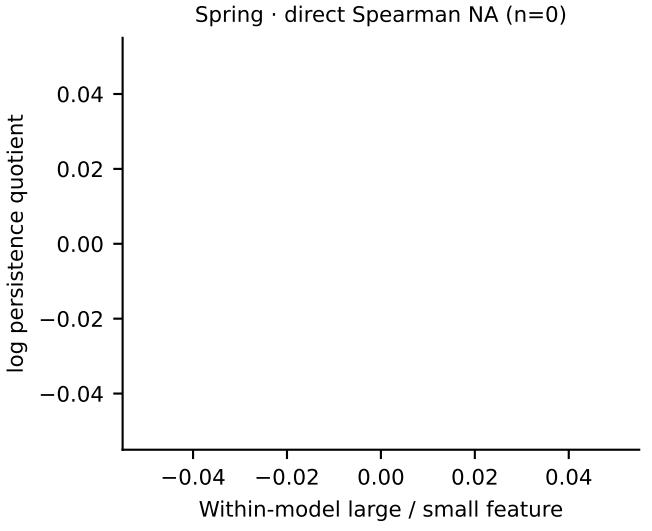












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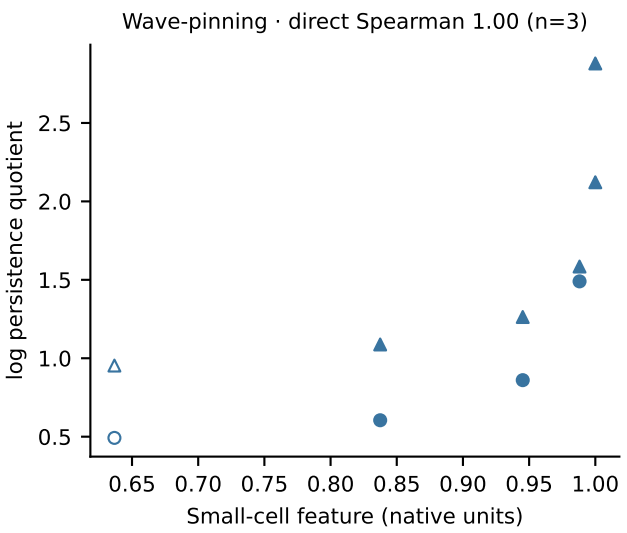
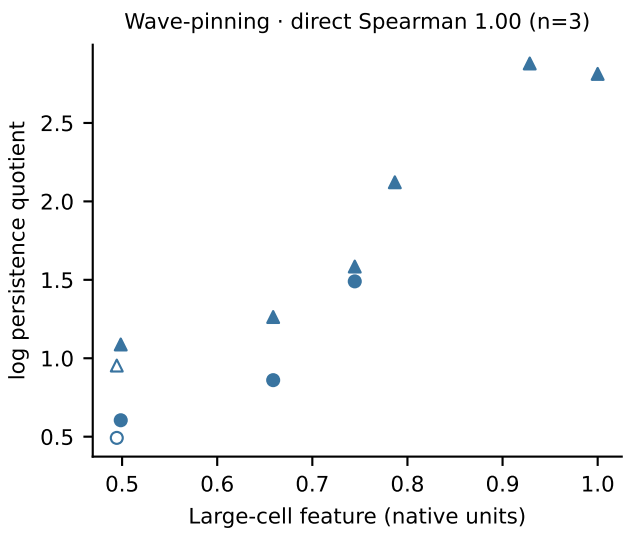
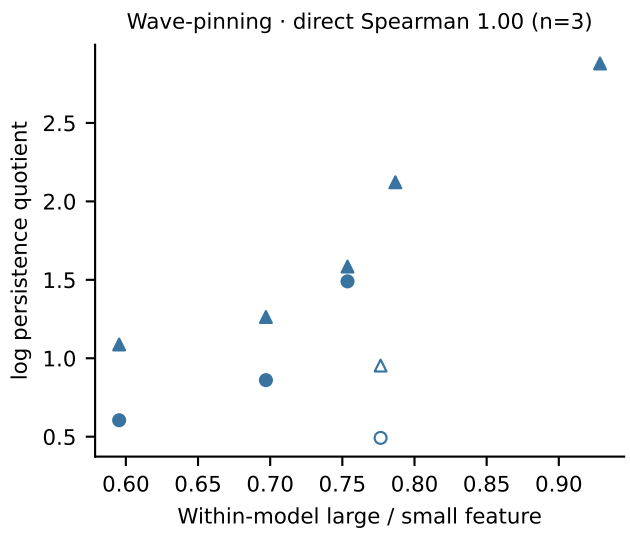
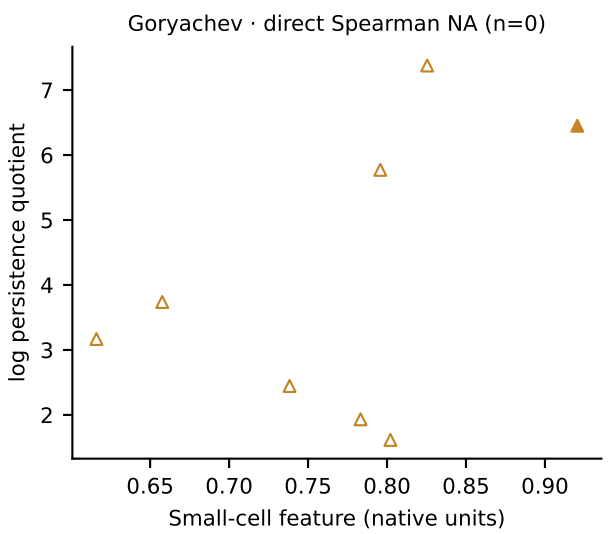
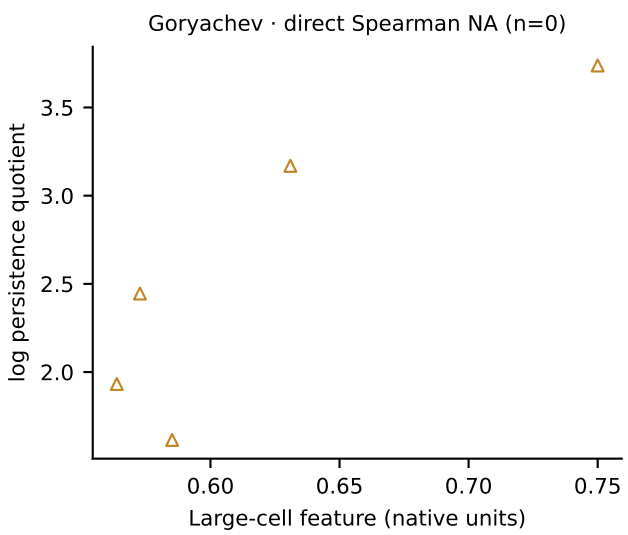
S10 · opposite birth fraction

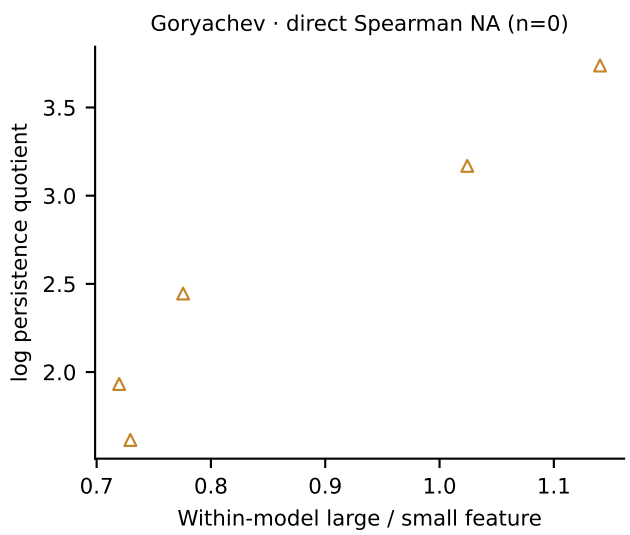


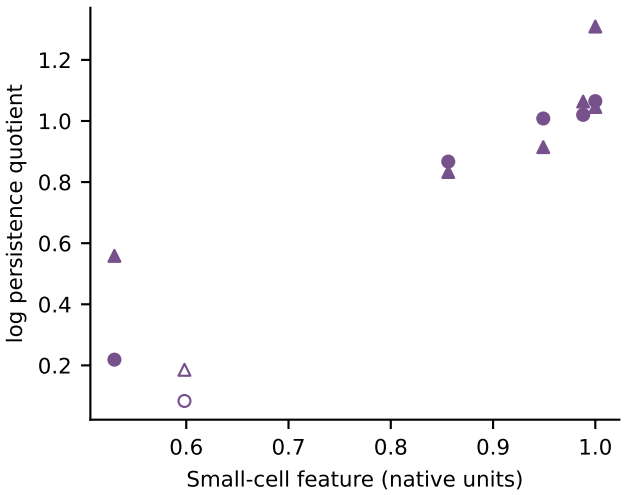


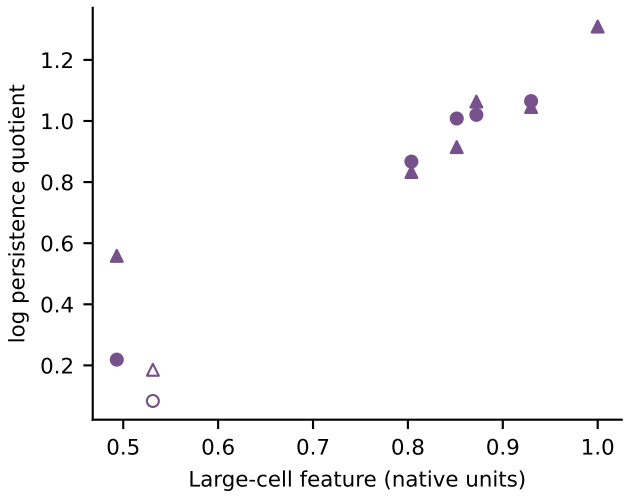


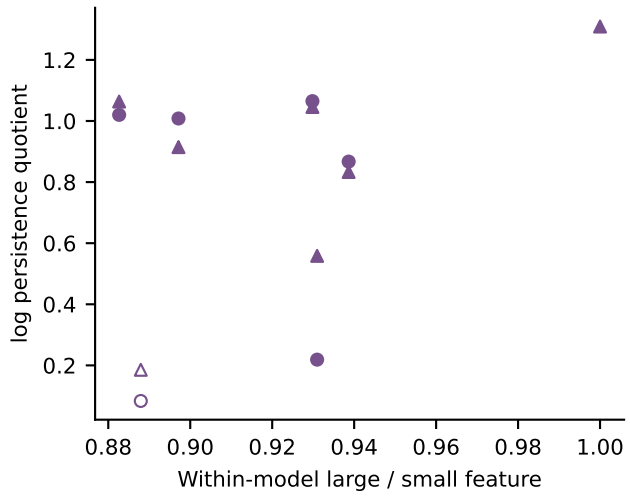


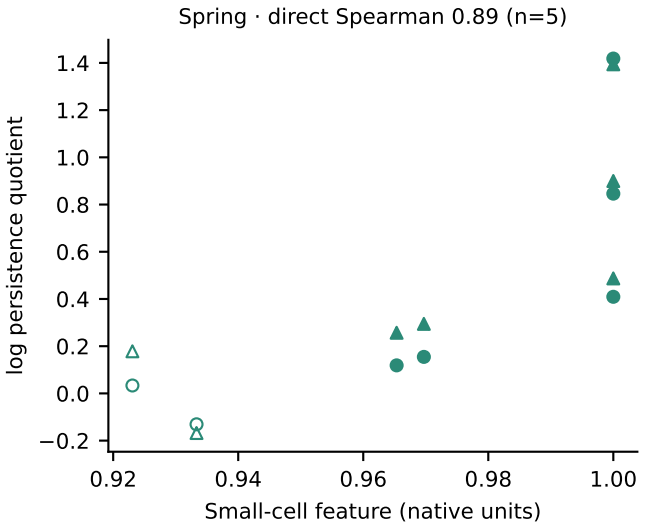


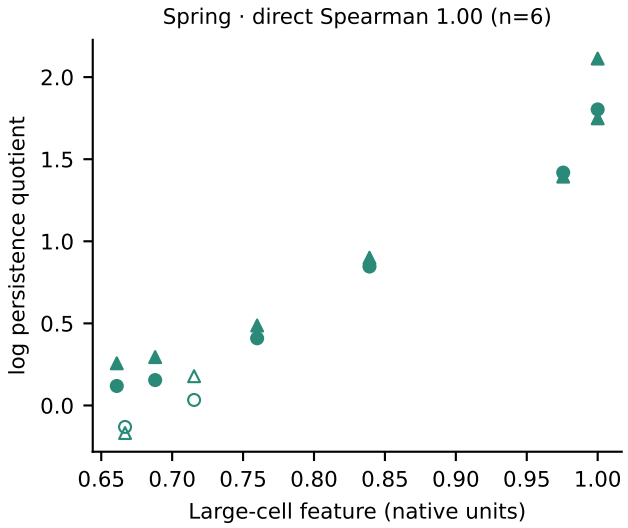


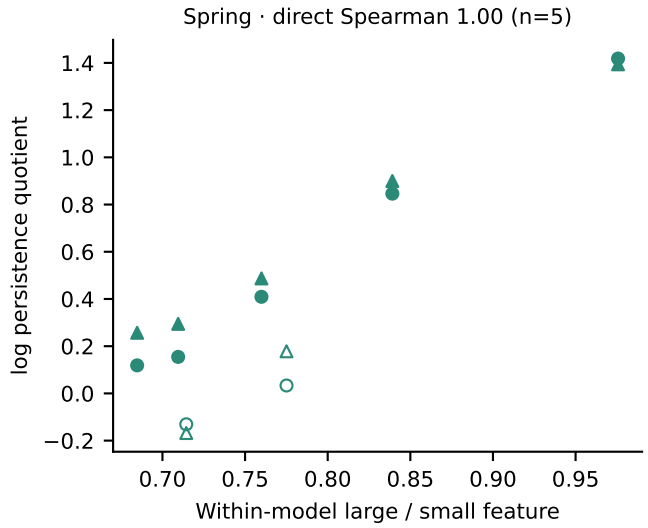












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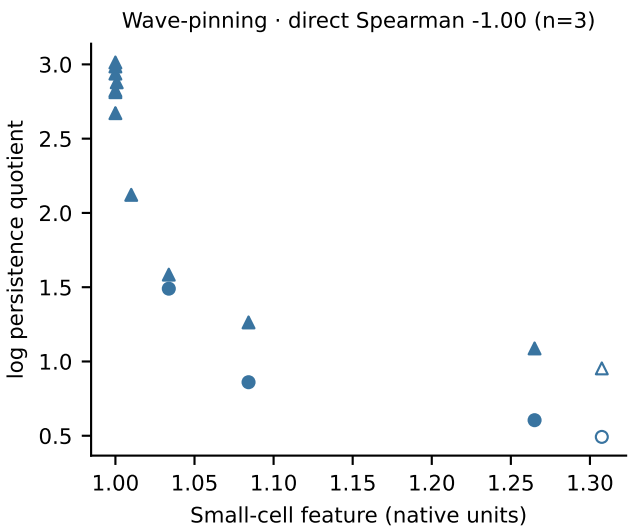
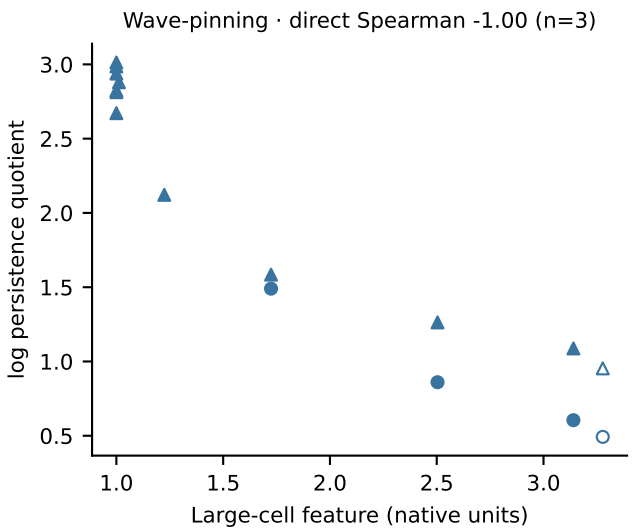
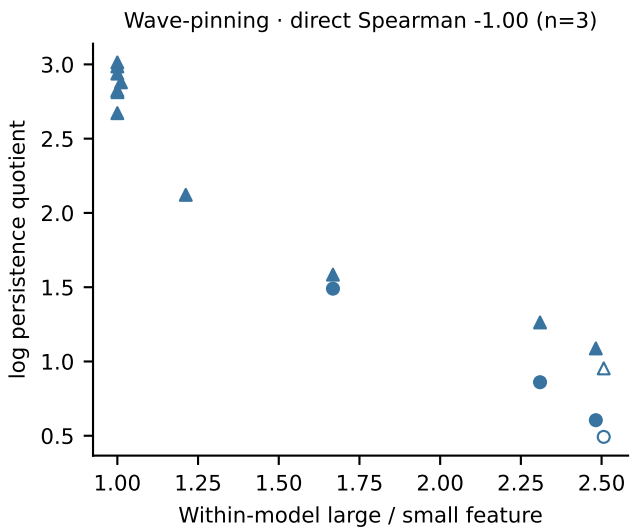
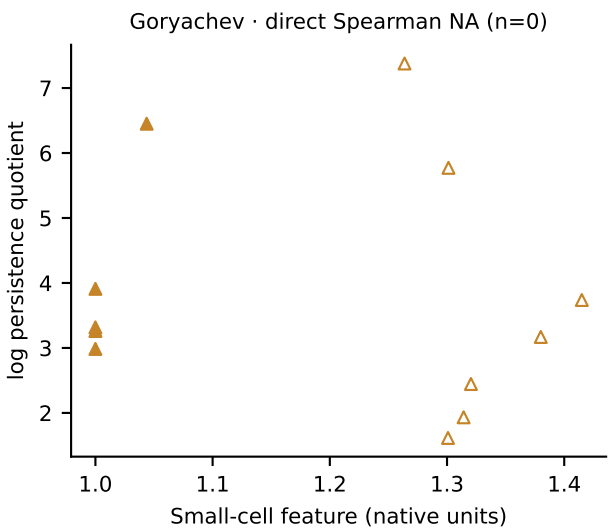
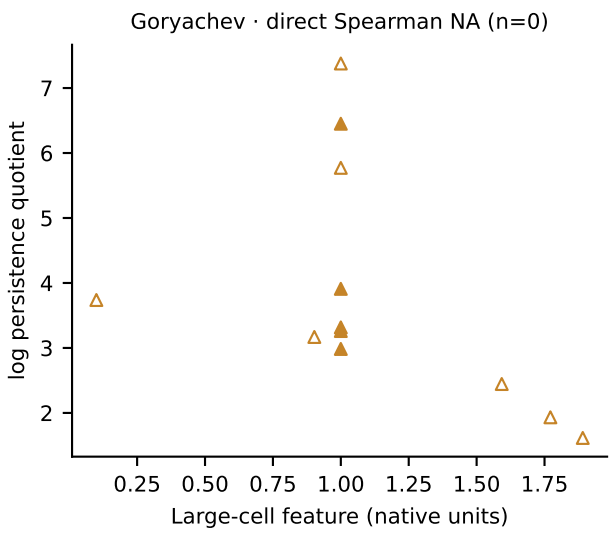
S10 · Mean region count

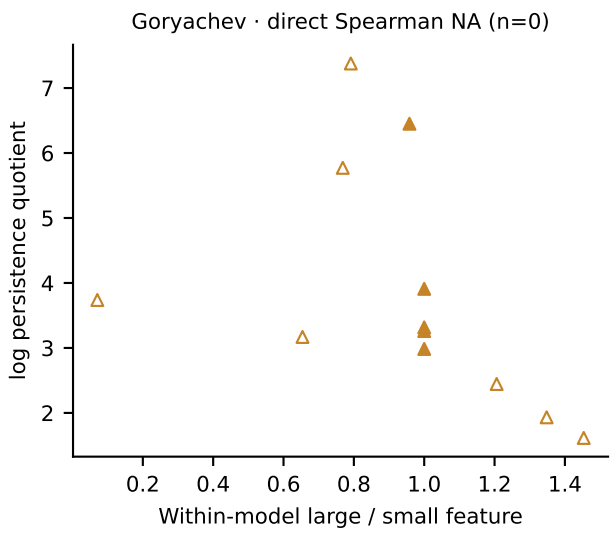


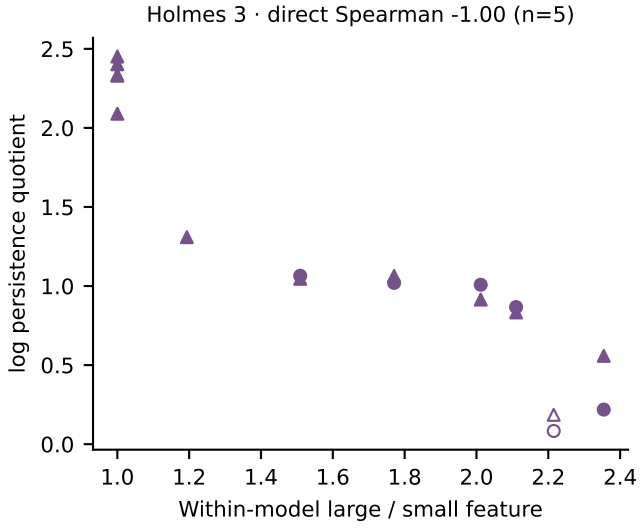


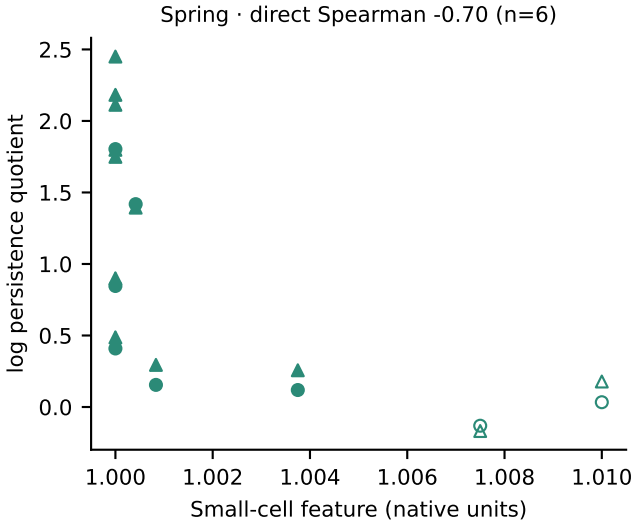


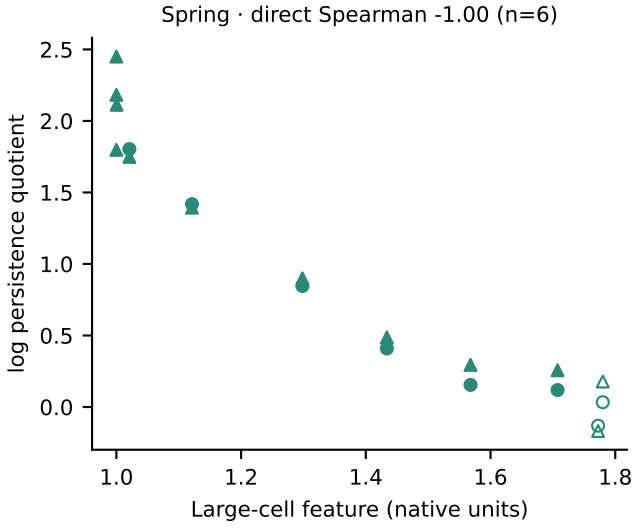


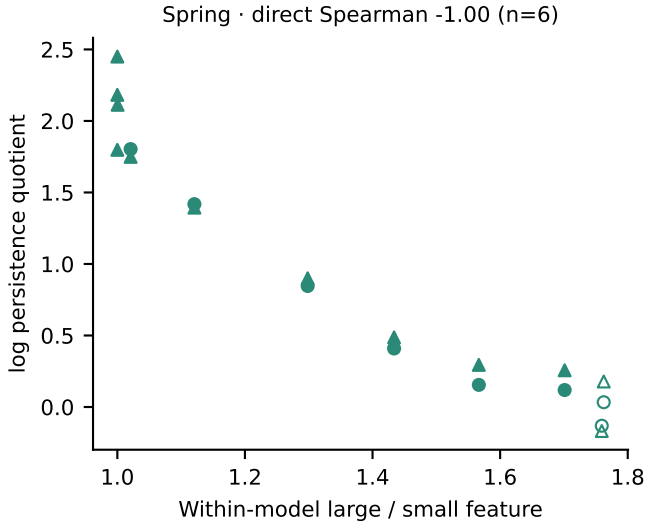












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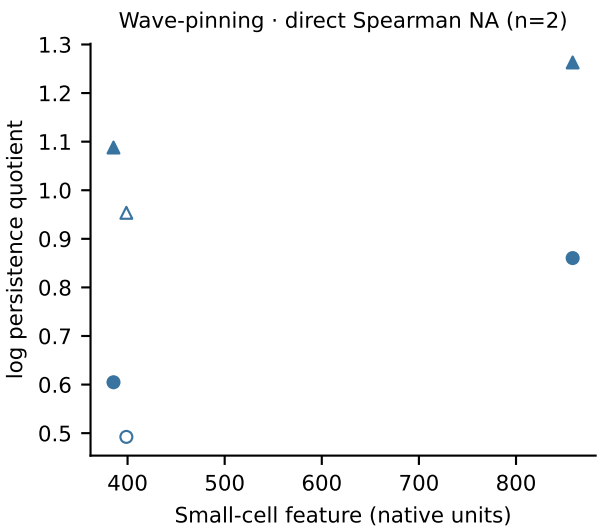
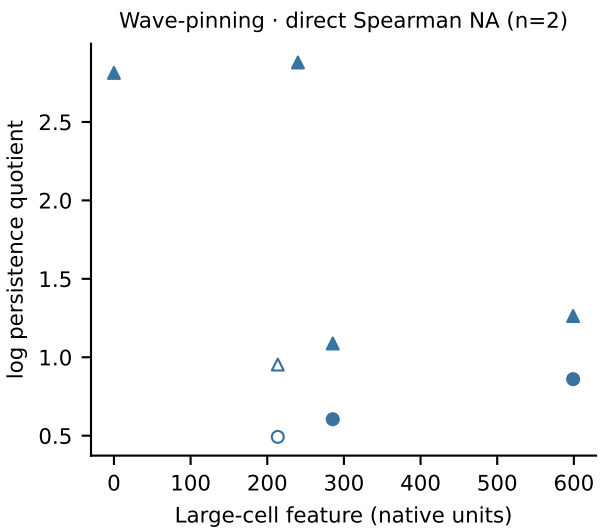
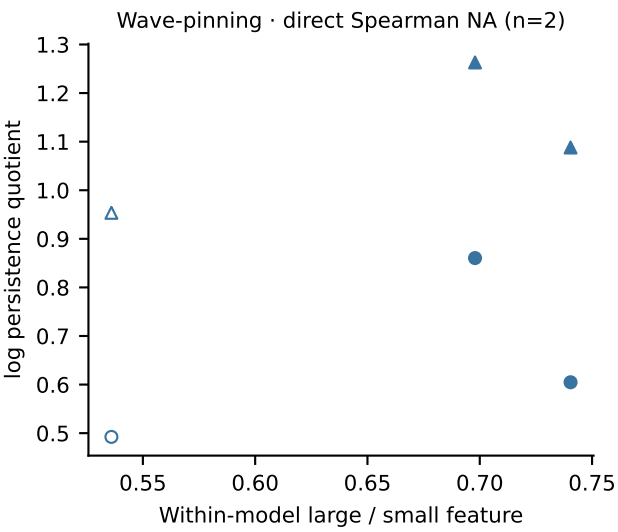
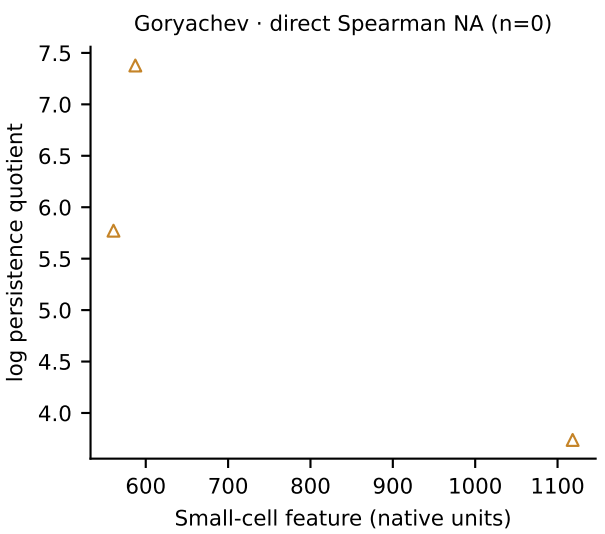
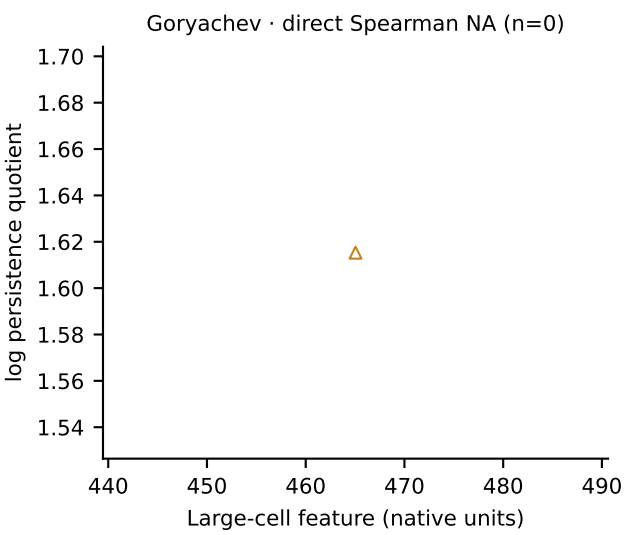
S10 · restricted mean lifetime lower

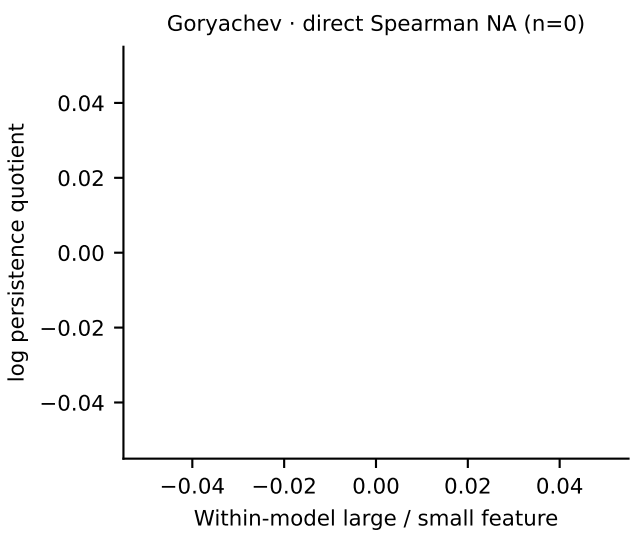


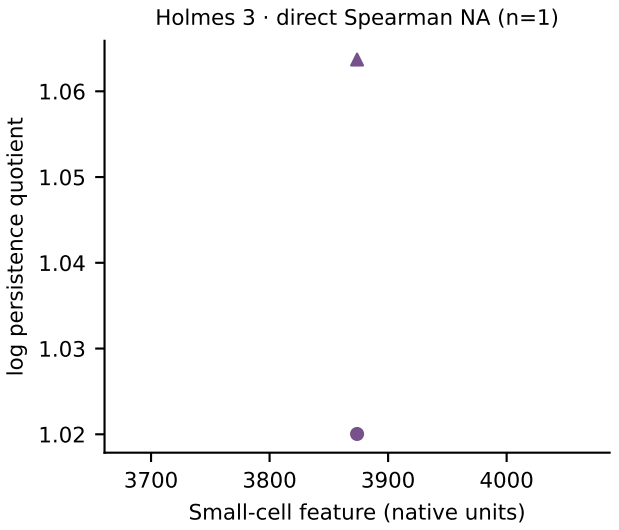


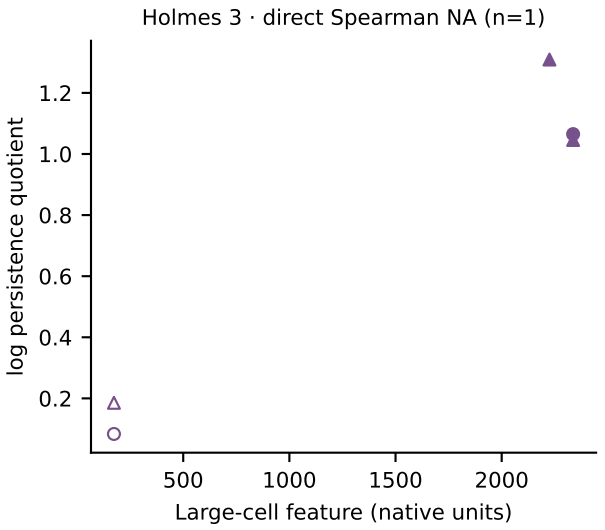


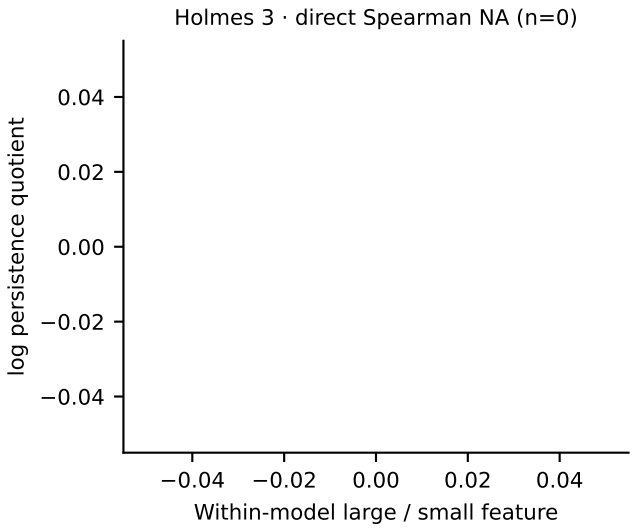


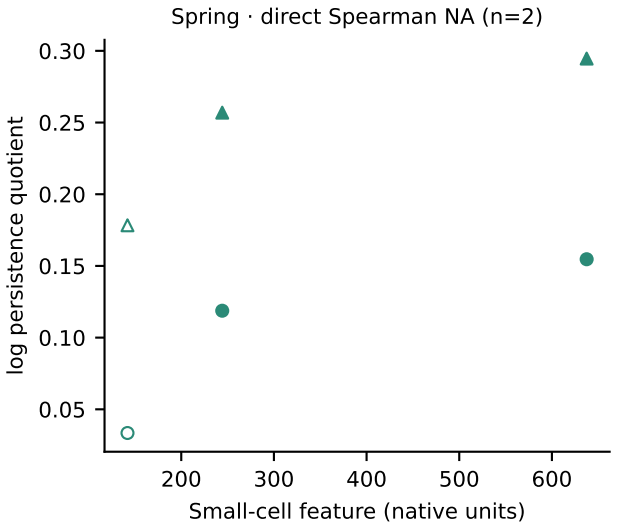


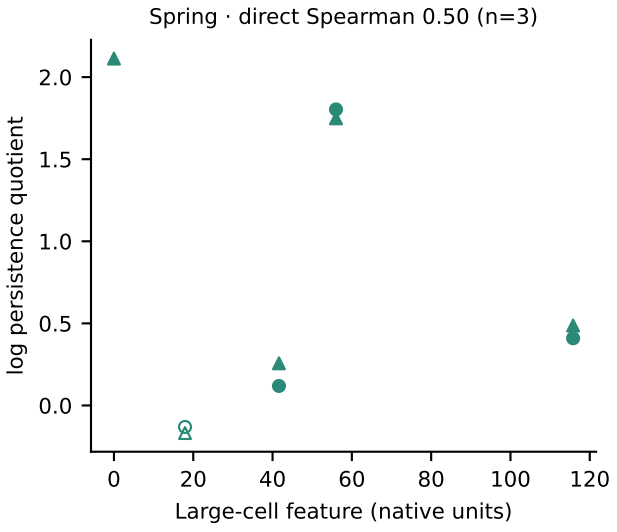


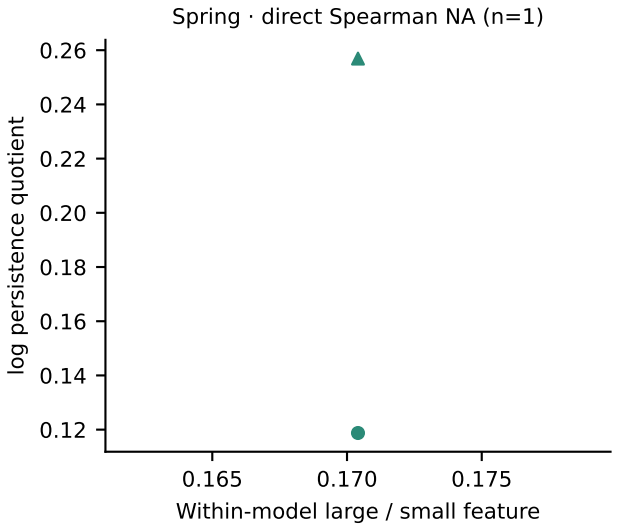












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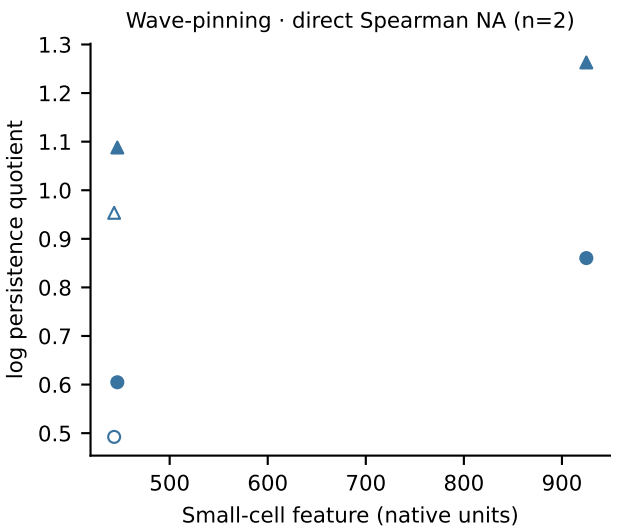
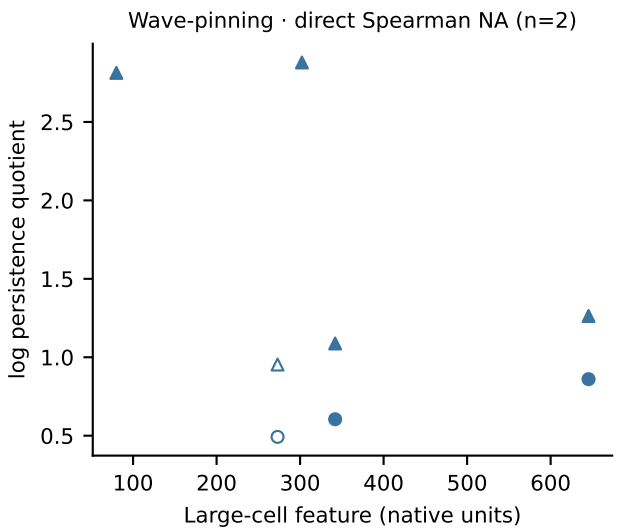
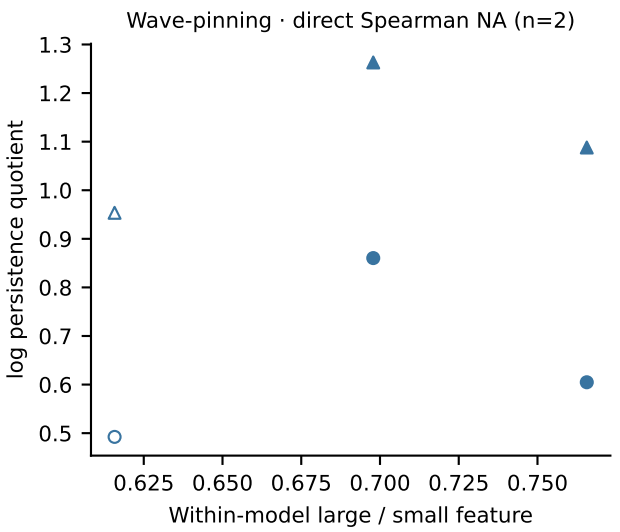
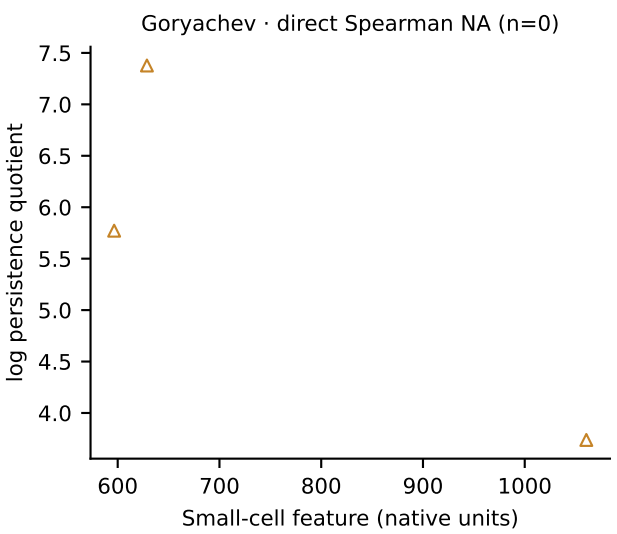
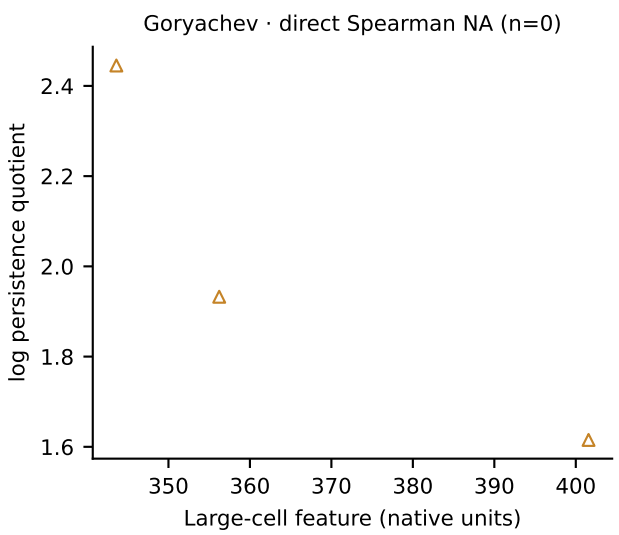
S10 · restricted mean lifetime observed

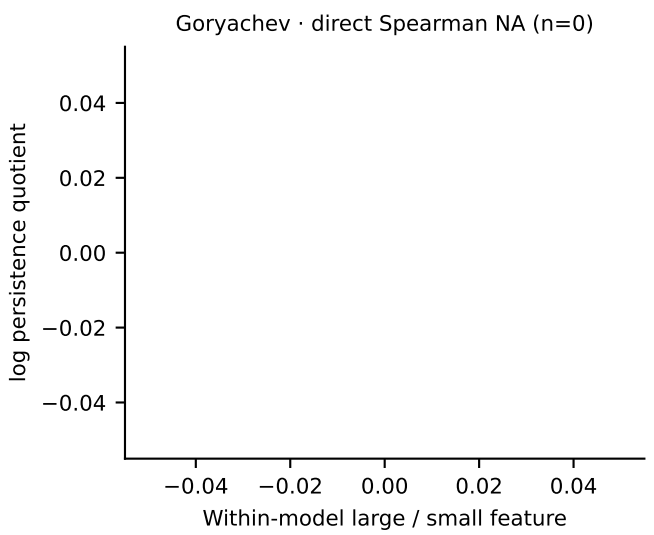


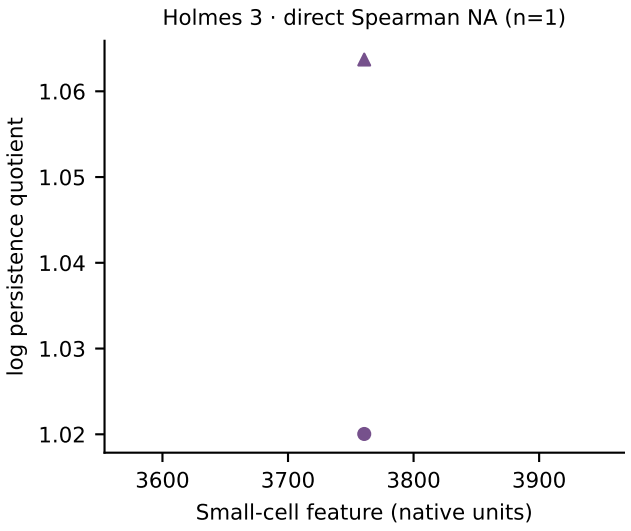


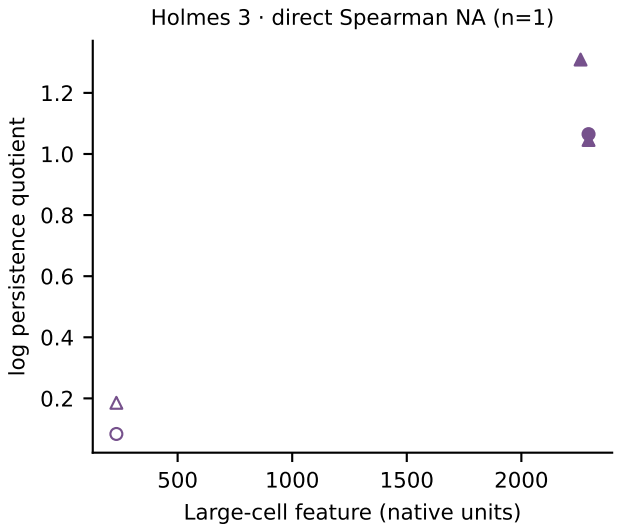


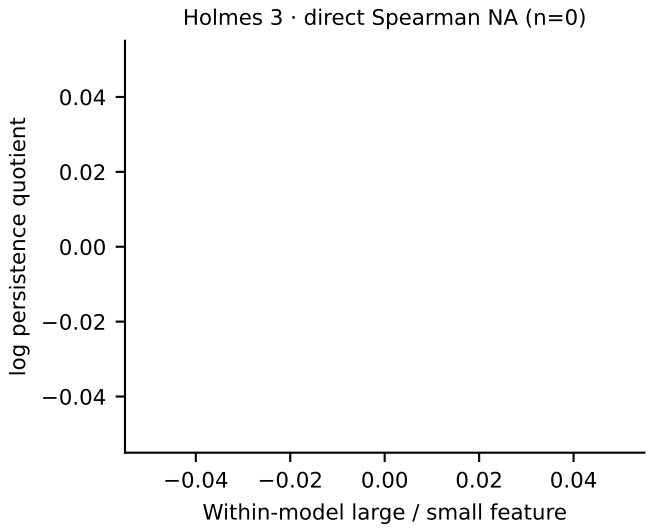


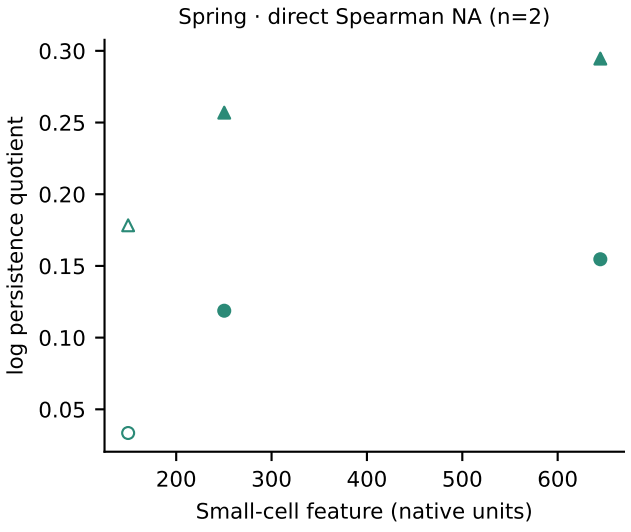


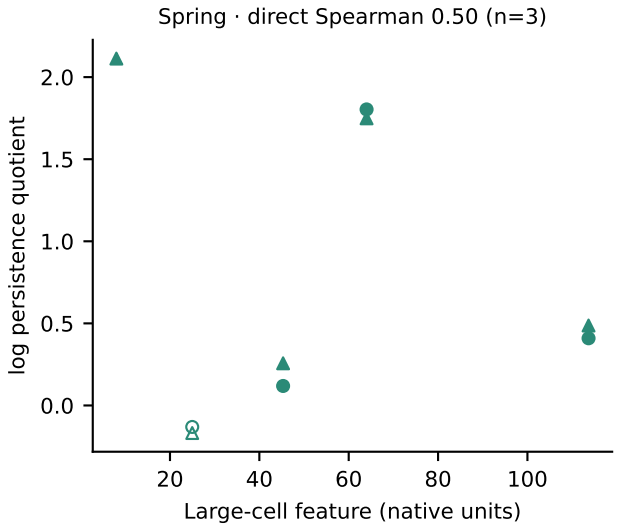


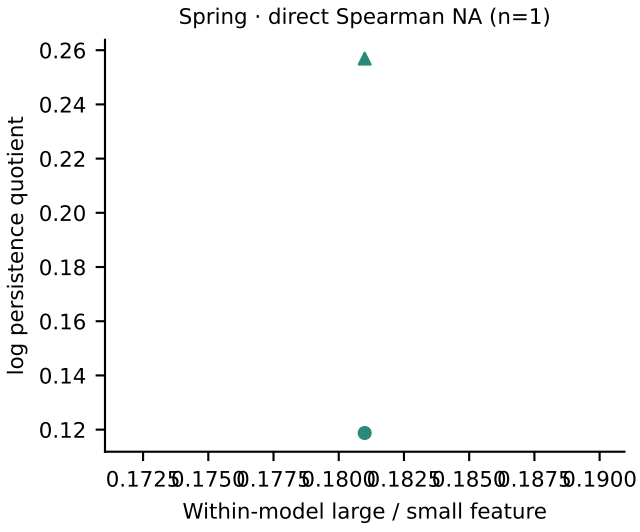












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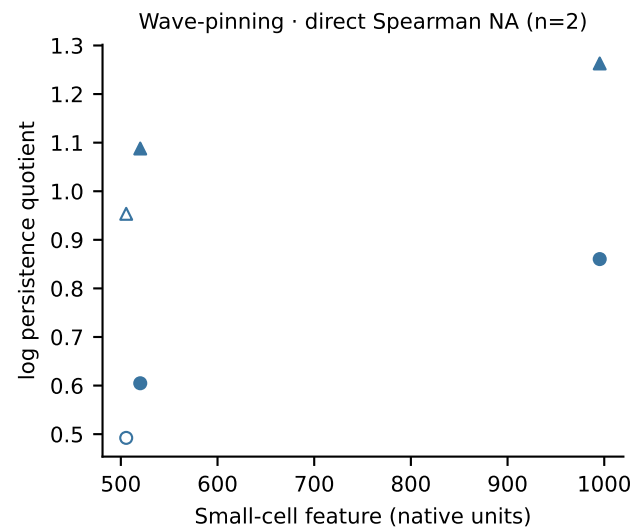
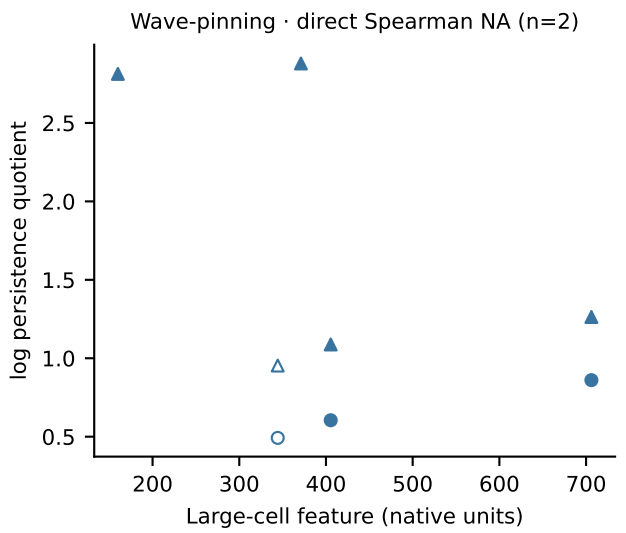
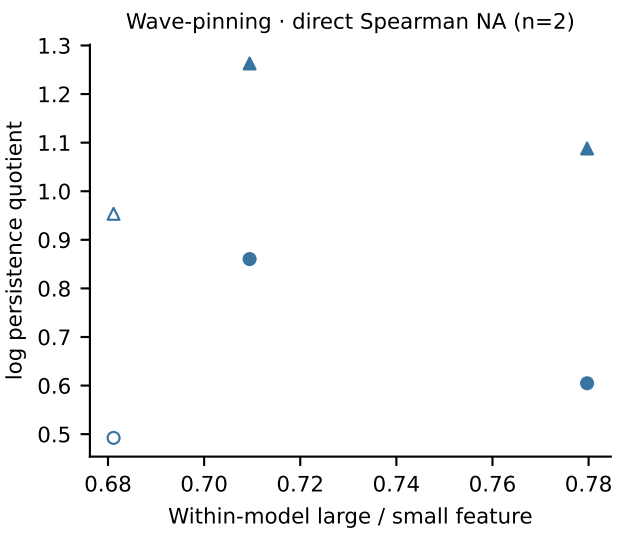
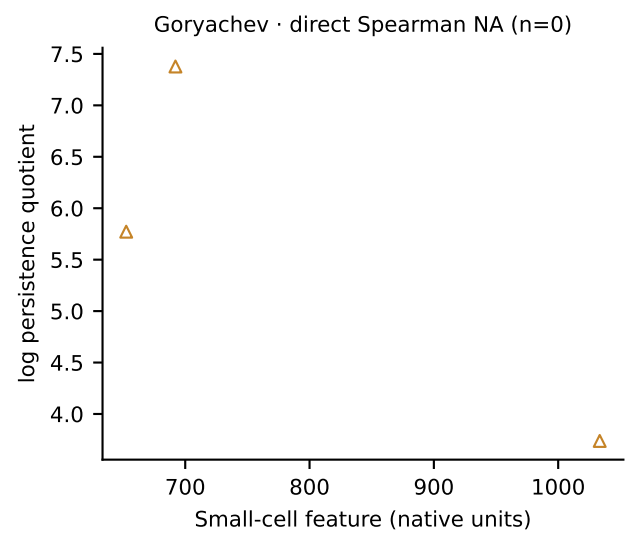
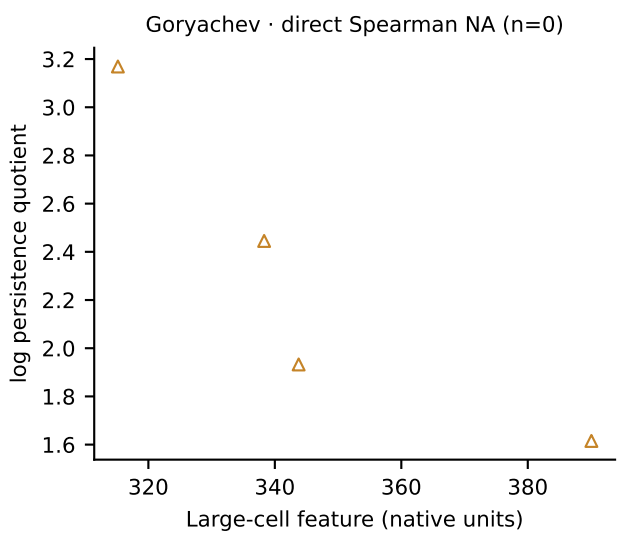
S10 · restricted mean lifetime upper

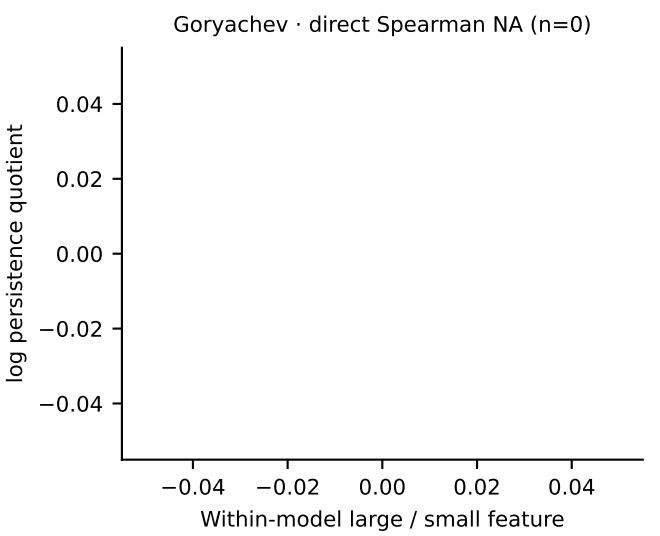


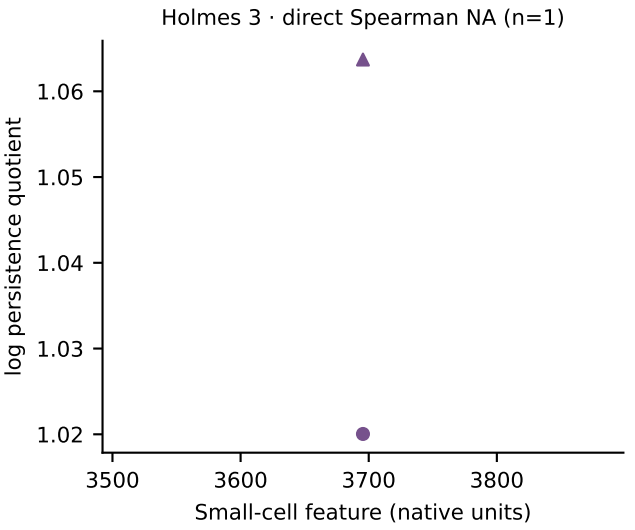


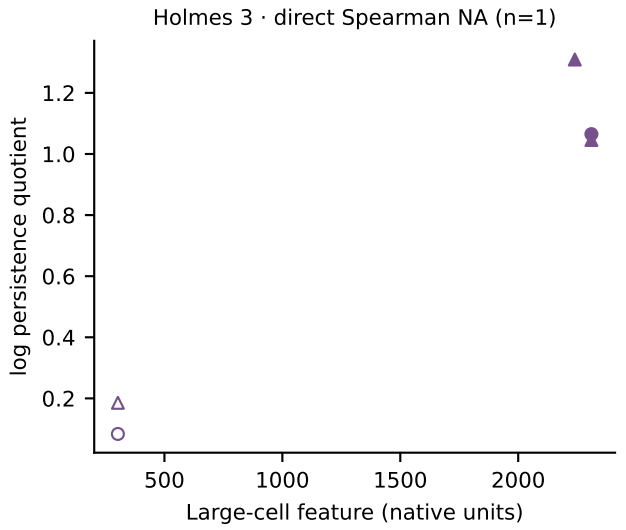


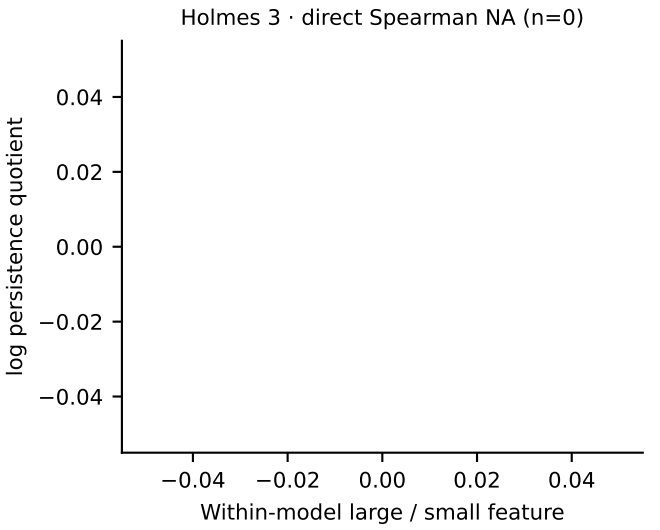


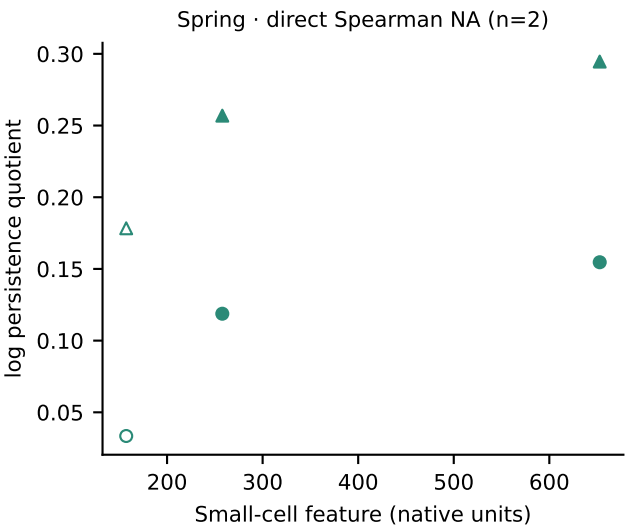


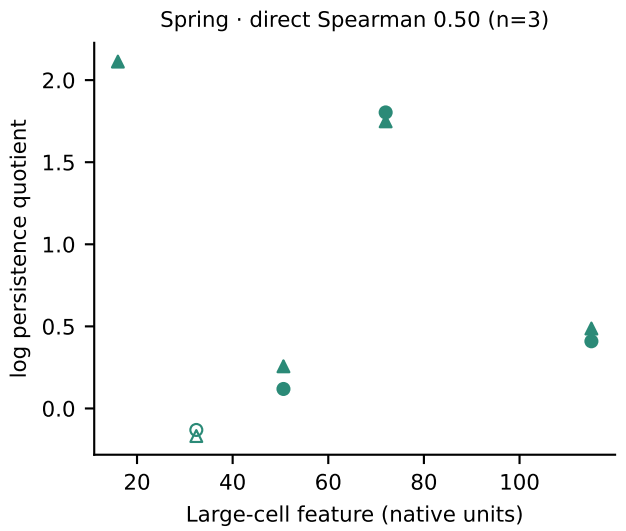


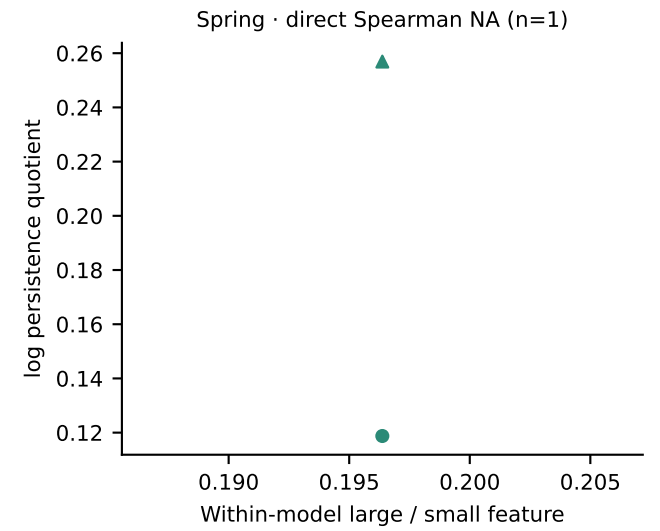






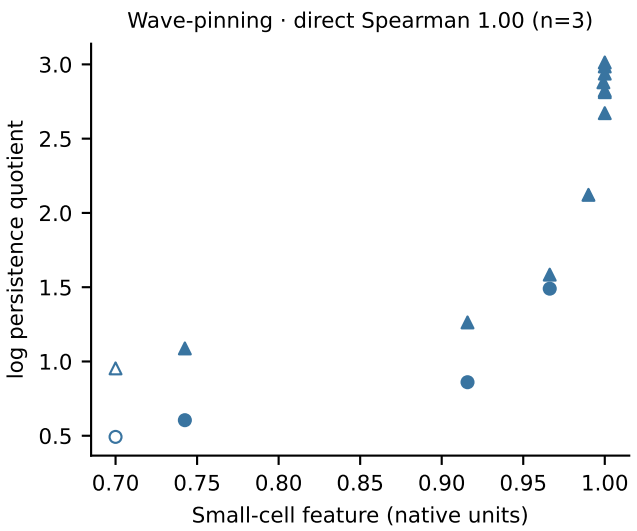


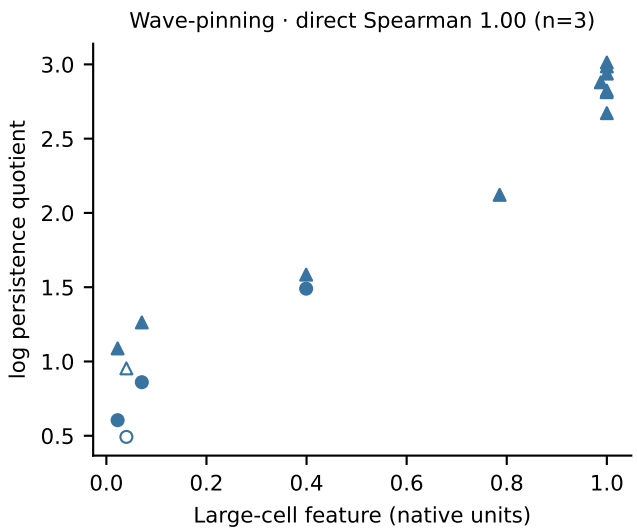


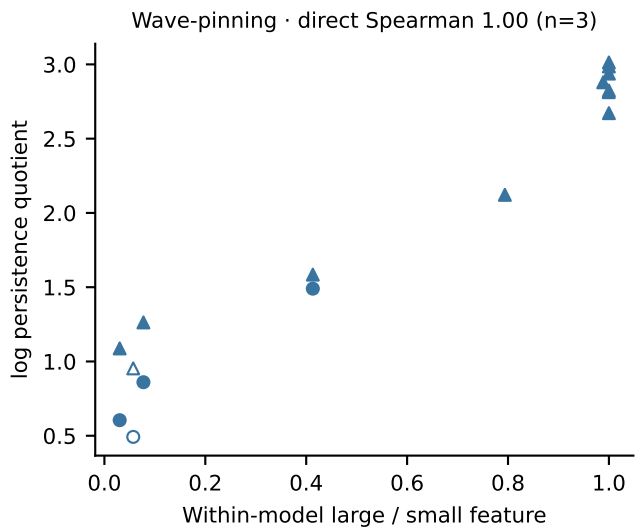


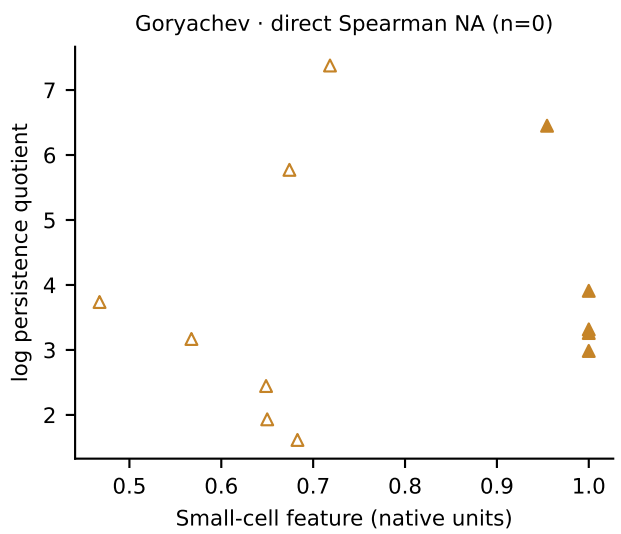
Arm common · σ=1 · actual per-size horizons retained in the source CSV. Crosses: unknown forcing, excluded from primary.
○ Direct resolved τ ratio; △ D_θ proxy sensitivity. Filled: nonforced; open: forced context.
All small/large native quantities remain within their own model. Cross-model tests use only size quotients.
NA retains insufficient, constant, censored or unscorable measurements; native caps are never substituted.

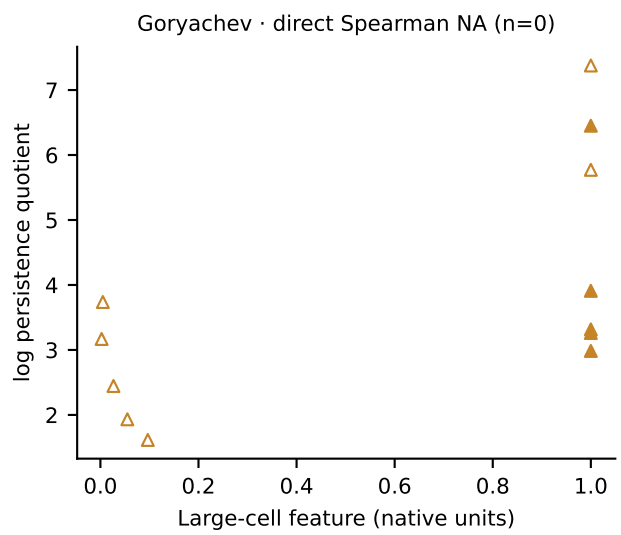
S10 · single front fraction

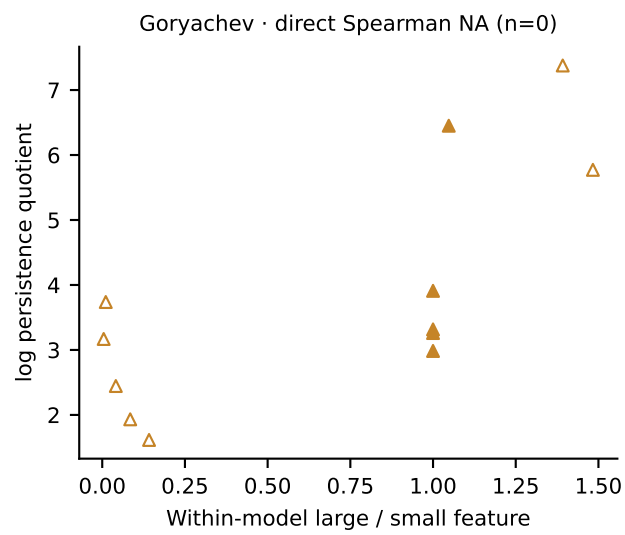


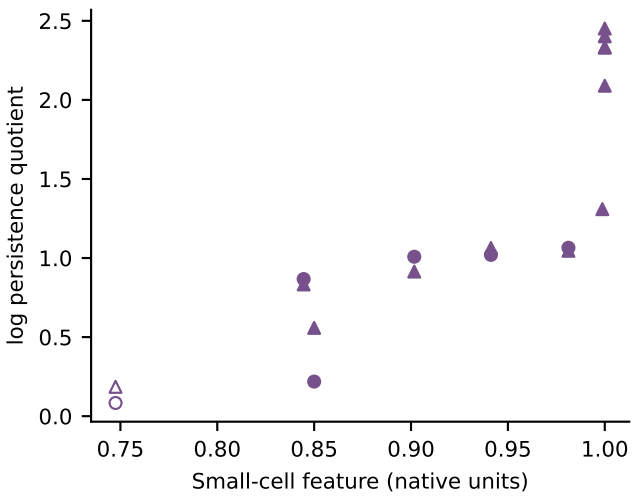


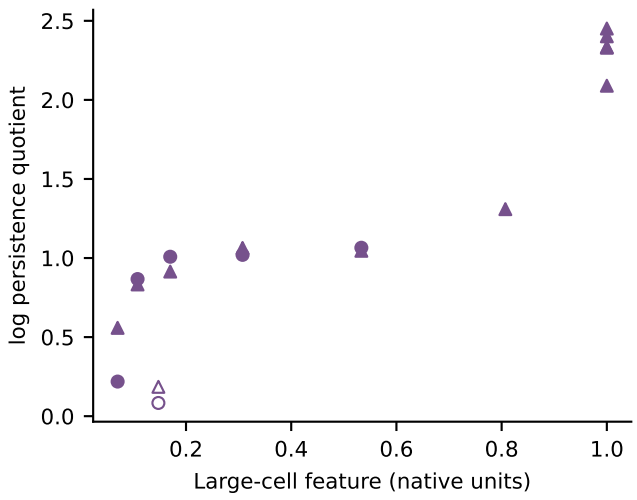


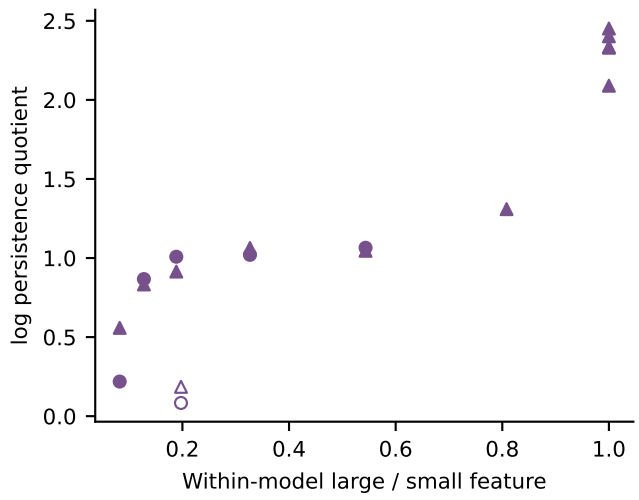


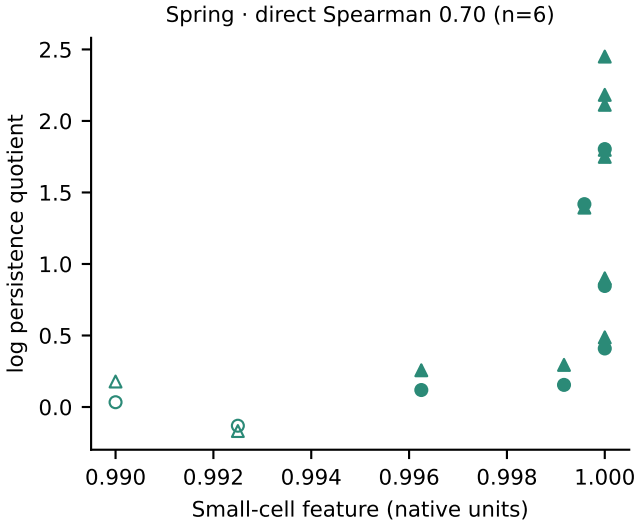


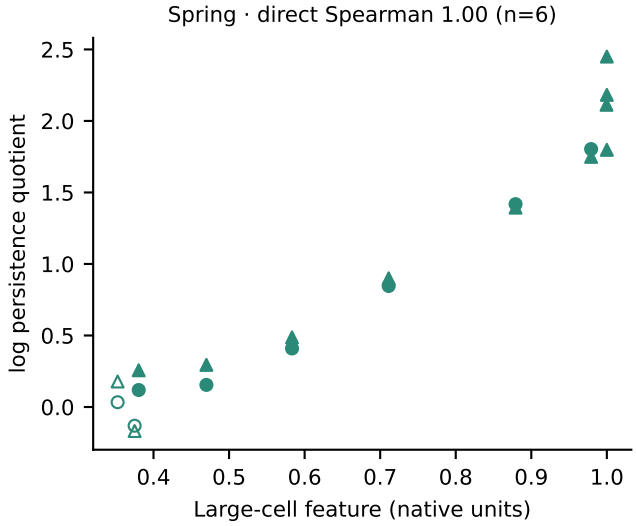


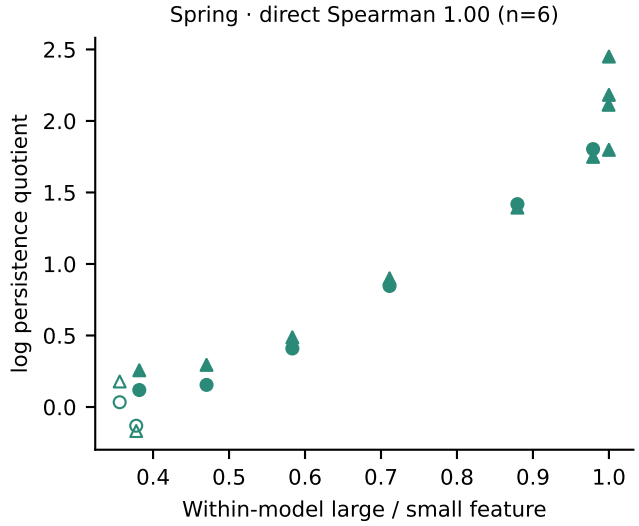






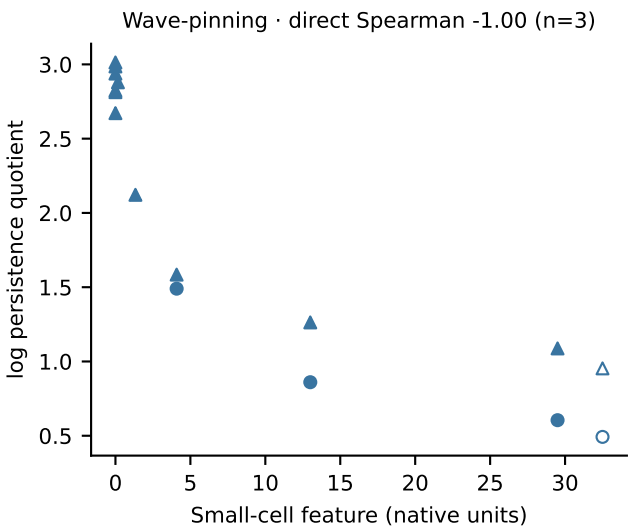


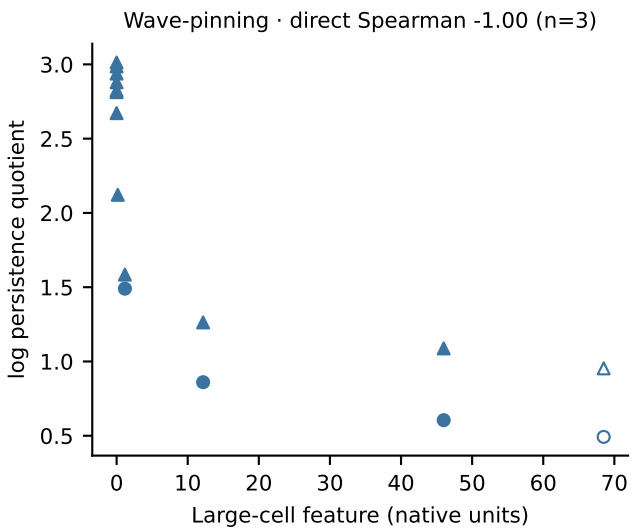


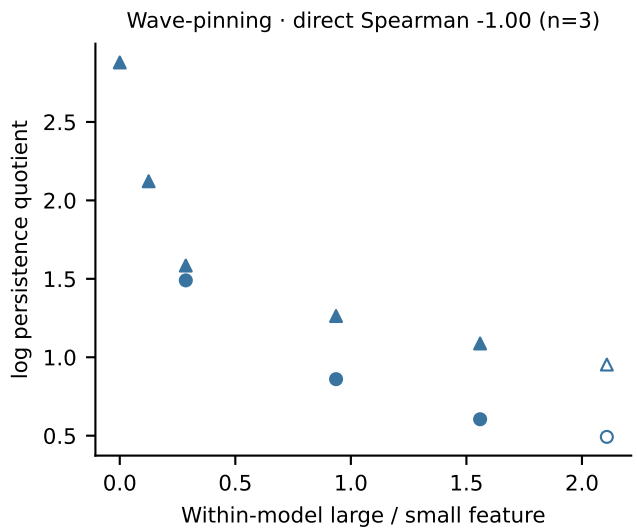


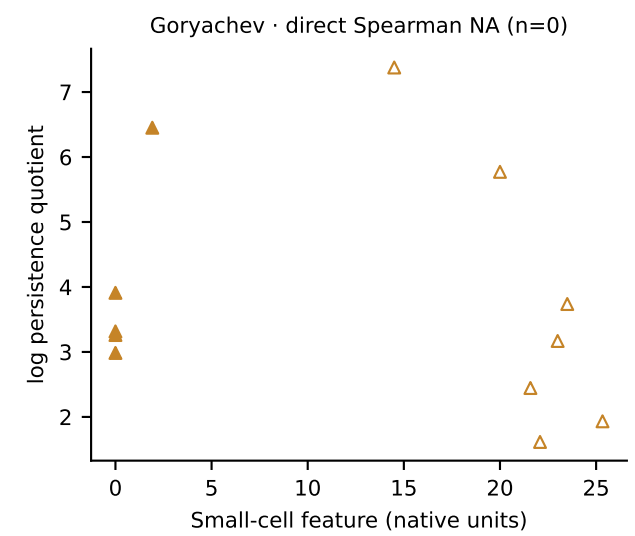
Arm common · σ=1 · actual per-size horizons retained in the source CSV. Crosses: unknown forcing, excluded from primary.
○ Direct resolved τ ratio; △ D_θ proxy sensitivity. Filled: nonforced; open: forced context.
All small/large native quantities remain within their own model. Cross-model tests use only size quotients.
NA retains insufficient, constant, censored or unscorable measurements; native caps are never substituted.

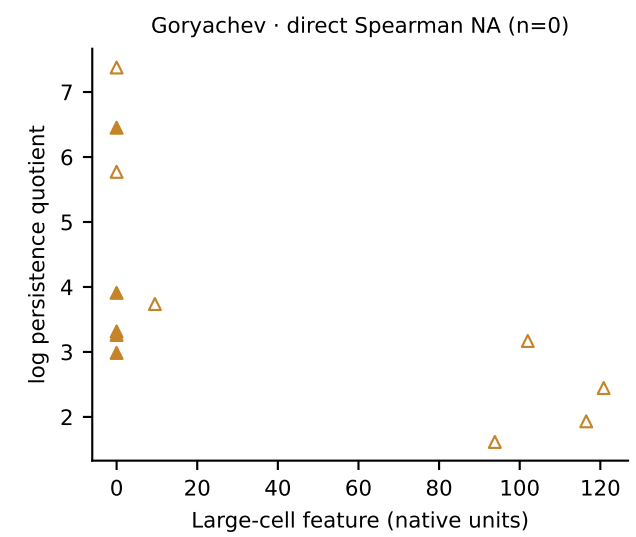
S10 · unscorable takeover tracks

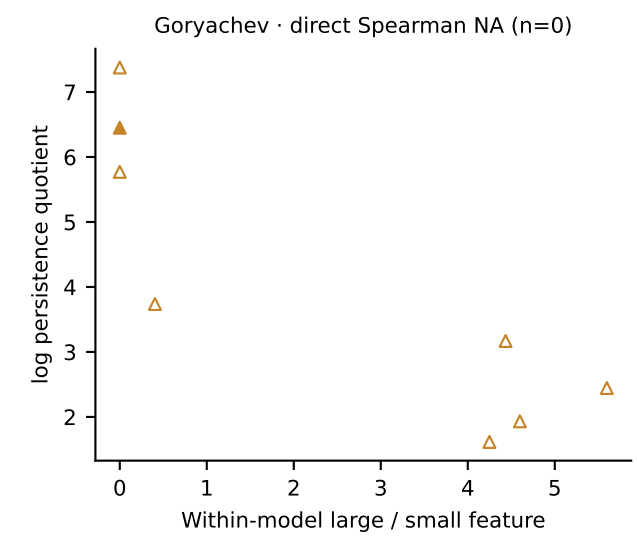


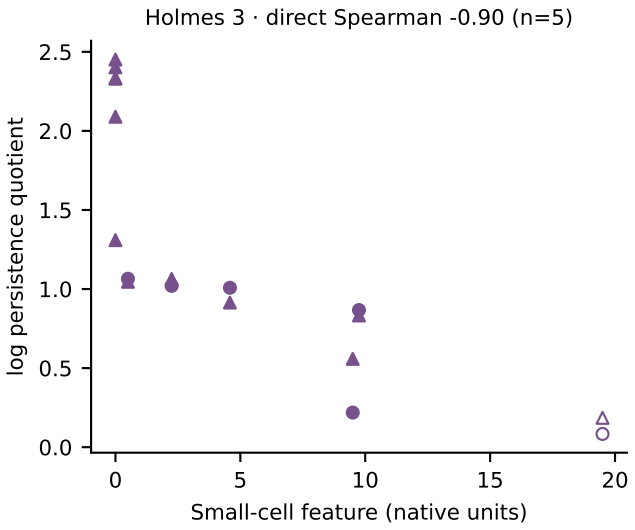


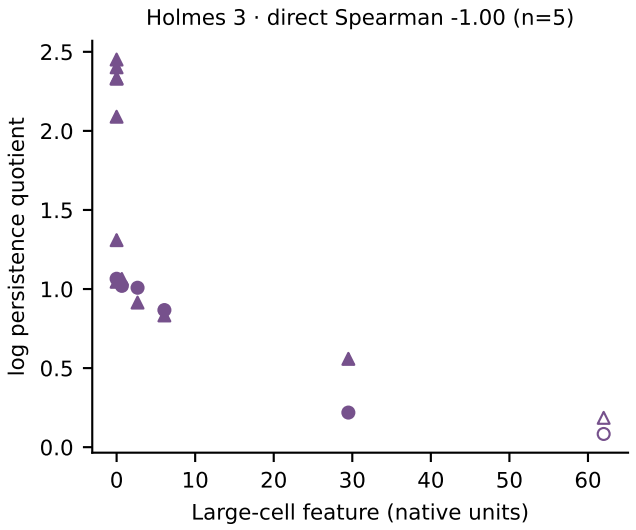


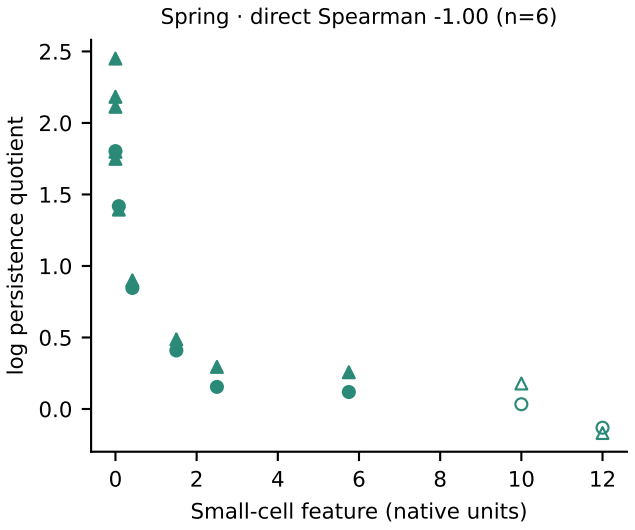


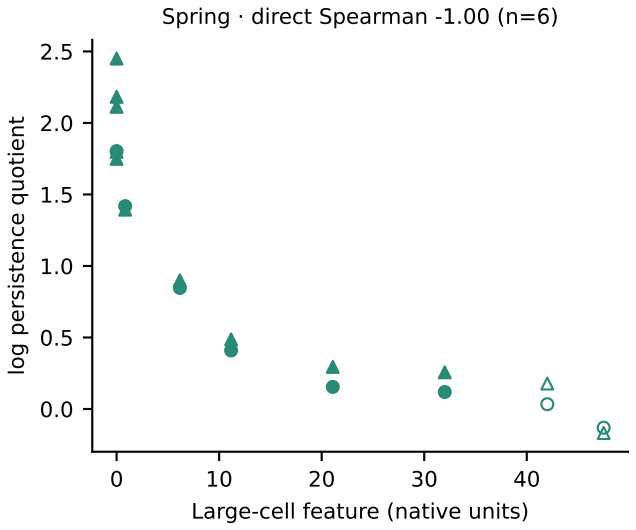


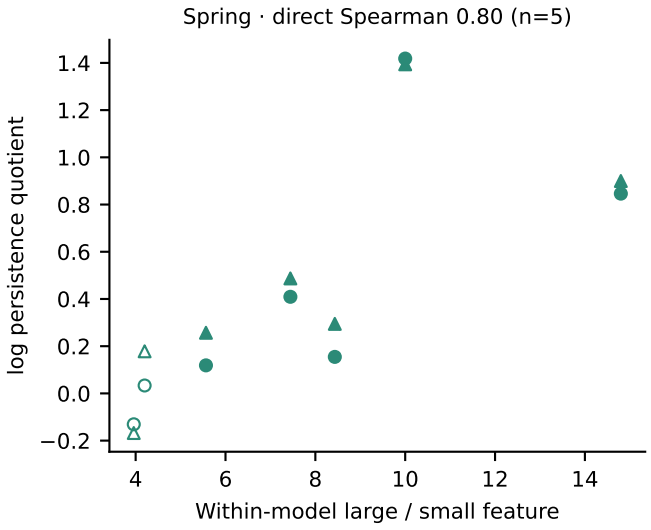












Raw event totals retain actual exposure; size quotients are unavailable for unequal horizons.
Arm common · σ=1 · actual per-size horizons retained in the source CSV. Crosses: unknown forcing, excluded from primary.
○ Direct resolved τ ratio; △ D_θ proxy sensitivity. Filled: nonforced; open: forced context.
All small/large native quantities remain within their own model. Cross-model tests use only size quotients.
NA retains insufficient, constant, censored or unscorable measurements; native caps are never substituted.

S10 · Primary predictor sensitivity · forced excluded

A5 imaging view	Multi-front τ rank	τ folds	Multi-front proxy rank	Proxy folds	Count τ rank	τ folds	Count proxy rank	Proxy folds
Primary imaging view	0.091	0/4 (1 NA)	-0.157	0/4 (1 NA)	-0.389	3/4 (1 NA)	-0.837	4/4 (0 NA)
q=0.3 / raw / Δt=0.0625×	0.086	0/4 (4 NA)	-0.450	0/4 (2 NA)	-0.486	0/4 (4 NA)	-0.855	3/4 (1 NA)
q=0.3 / raw / Δt=0.125×	-0.143	0/4 (4 NA)	-0.548	0/4 (3 NA)	-0.486	0/4 (4 NA)	-0.855	3/4 (1 NA)
q=0.3 / raw / Δt=0.25×	0.086	0/4 (4 NA)	-0.452	0/4 (3 NA)	-0.257	0/4 (4 NA)	-0.806	3/4 (1 NA)
q=0.3 / raw / Δt=0.5×	0.240	0/4 (1 NA)	-0.261	0/4 (1 NA)	-0.508	3/4 (1 NA)	-0.860	4/4 (0 NA)
q=0.3 / raw / Δt=1×	0.238	0/4 (2 NA)	0.145	0/4 (1 NA)	-0.405	2/4 (2 NA)	-0.813	3/4 (1 NA)
q=0.3 / raw / Δt=2×	0.238	0/4 (2 NA)	0.109	0/4 (1 NA)	-0.405	2/4 (2 NA)	-0.808	3/4 (1 NA)
q=0.3 / raw / Δt=4×	0.238	0/4 (2 NA)	0.224	0/4 (1 NA)	-0.333	2/4 (2 NA)	-0.808	3/4 (1 NA)
q=0.3 / optical / Δt=0.0625×	NA	0/4 (4 NA)	-0.750	0/4 (4 NA)	-0.257	0/4 (4 NA)	-0.806	3/4 (1 NA)
q=0.3 / optical / Δt=0.125×	NA	0/4 (4 NA)	-0.750	0/4 (4 NA)	-0.257	0/4 (4 NA)	-0.806	3/4 (1 NA)
q=0.3 / optical / Δt=0.25×	NA	0/4 (4 NA)	-0.750	0/4 (4 NA)	-0.257	0/4 (4 NA)	-0.806	3/4 (1 NA)
q=0.3 / optical / Δt=0.5×	-0.055	0/4 (1 NA)	-0.270	0/4 (1 NA)	-0.319	3/4 (1 NA)	-0.835	4/4 (0 NA)
q=0.3 / optical / Δt=1×	0.024	0/4 (2 NA)	0.067	0/4 (1 NA)	-0.190	2/4 (2 NA)	-0.803	3/4 (1 NA)
q=0.3 / optical / Δt=2×	-0.143	0/4 (4 NA)	0.238	0/4 (2 NA)	-0.262	2/4 (2 NA)	-0.805	3/4 (1 NA)
q=0.3 / optical / Δt=4×	-0.143	0/4 (4 NA)	-0.071	0/4 (2 NA)	-0.190	2/4 (2 NA)	-0.784	3/4 (1 NA)
q=0.5 / raw / Δt=0.0625×	0.086	0/4 (4 NA)	-0.517	0/4 (2 NA)	-0.486	0/4 (4 NA)	-0.855	3/4 (1 NA)
q=0.5 / raw / Δt=0.125×	NA	0/4 (4 NA)	-0.714	0/4 (3 NA)	-0.486	0/4 (4 NA)	-0.855	3/4 (1 NA)
q=0.5 / raw / Δt=0.25×	NA	0/4 (4 NA)	-0.786	1/4 (3 NA)	-0.486	0/4 (4 NA)	-0.855	3/4 (1 NA)
q=0.5 / raw / Δt=0.5×	0.038	0/4 (1 NA)	-0.286	0/4 (1 NA)	-0.516	3/4 (1 NA)	-0.848	4/4 (0 NA)
q=0.5 / raw / Δt=1×	0.238	0/4 (2 NA)	0.236	0/4 (1 NA)	-0.452	2/4 (2 NA)	-0.810	3/4 (1 NA)
q=0.5 / raw / Δt=2×	0.238	0/4 (2 NA)	0.224	0/4 (1 NA)	-0.405	2/4 (2 NA)	-0.789	3/4 (1 NA)
q=0.5 / raw / Δt=4×	0.119	0/4 (2 NA)	-0.083	0/4 (1 NA)	-0.405	2/4 (2 NA)	-0.789	3/4 (1 NA)
q=0.5 / optical / Δt=0.0625×	NA	0/4 (4 NA)	-0.750	0/4 (4 NA)	-0.257	0/4 (4 NA)	-0.806	3/4 (1 NA)
q=0.5 / optical / Δt=0.125×	NA	0/4 (4 NA)	-0.750	0/4 (4 NA)	-0.257	0/4 (4 NA)	-0.806	3/4 (1 NA)
q=0.5 / optical / Δt=0.25×	NA	0/4 (4 NA)	-0.750	0/4 (4 NA)	-0.257	0/4 (4 NA)	-0.806	3/4 (1 NA)
q=0.5 / optical / Δt=0.5×	-0.007	0/4 (1 NA)	-0.235	0/4 (1 NA)	-0.389	3/4 (1 NA)	-0.840	4/4 (0 NA)
q=0.5 / optical / Δt=2×	-0.143	0/4 (4 NA)	0.238	0/4 (2 NA)	-0.262	2/4 (2 NA)	-0.786	3/4 (1 NA)
q=0.5 / optical / Δt=4×	-0.143	0/4 (4 NA)	-0.071	0/4 (2 NA)	-0.262	2/4 (2 NA)	-0.809	3/4 (1 NA)
q=0.7 / raw / Δt=0.0625×	NA	0/4 (4 NA)	-0.786	1/4 (3 NA)	-0.486	0/4 (4 NA)	-0.855	3/4 (1 NA)
q=0.7 / raw / Δt=0.125×	NA	0/4 (4 NA)	-0.786	1/4 (3 NA)	-0.486	0/4 (4 NA)	-0.855	3/4 (1 NA)
q=0.7 / raw / Δt=0.25×	NA	0/4 (4 NA)	-0.750	0/4 (4 NA)	-0.486	0/4 (4 NA)	-0.855	3/4 (1 NA)
q=0.7 / raw / Δt=0.5×	0.181	0/4 (1 NA)	-0.017	0/4 (1 NA)	-0.446	3/4 (1 NA)	-0.848	4/4 (0 NA)
q=0.7 / raw / Δt=1×	0.286	0/4 (2 NA)	0.273	0/4 (1 NA)	-0.167	2/4 (2 NA)	-0.808	3/4 (1 NA)
q=0.7 / raw / Δt=2×	0.286	0/4 (2 NA)	0.261	0/4 (1 NA)	-0.167	2/4 (2 NA)	-0.786	3/4 (1 NA)
q=0.7 / raw / Δt=4×	-0.071	0/4 (3 NA)	-0.262	0/4 (2 NA)	-0.071	2/4 (2 NA)	-0.805	3/4 (1 NA)
q=0.7 / optical / Δt=0.0625×	NA	0/4 (4 NA)	-0.750	0/4 (4 NA)	-0.086	0/4 (4 NA)	-0.766	3/4 (1 NA)
q=0.7 / optical / Δt=0.125×	NA	0/4 (4 NA)	-0.750	0/4 (4 NA)	-0.086	0/4 (4 NA)	-0.766	3/4 (1 NA)
q=0.7 / optical / Δt=0.25×	NA	0/4 (4 NA)	-0.600	0/4 (4 NA)	-0.086	0/4 (4 NA)	-0.766	3/4 (1 NA)
q=0.7 / optical / Δt=0.5×	0.014	0/4 (1 NA)	-0.244	0/4 (1 NA)	-0.297	3/4 (1 NA)	-0.834	4/4 (0 NA)
q=0.7 / optical / Δt=1×	0.119	0/4 (2 NA)	0.139	0/4 (1 NA)	-0.262	2/4 (2 NA)	-0.812	3/4 (1 NA)
q=0.7 / optical / Δt=2×	-0.143	0/4 (3 NA)	0.000	0/4 (2 NA)	-0.190	2/4 (2 NA)	-0.809	3/4 (1 NA)
q=0.7 / optical / Δt=4×	-0.143	0/4 (3 NA)	-0.214	0/4 (2 NA)	-0.190	2/4 (2 NA)	-0.809	3/4 (1 NA)

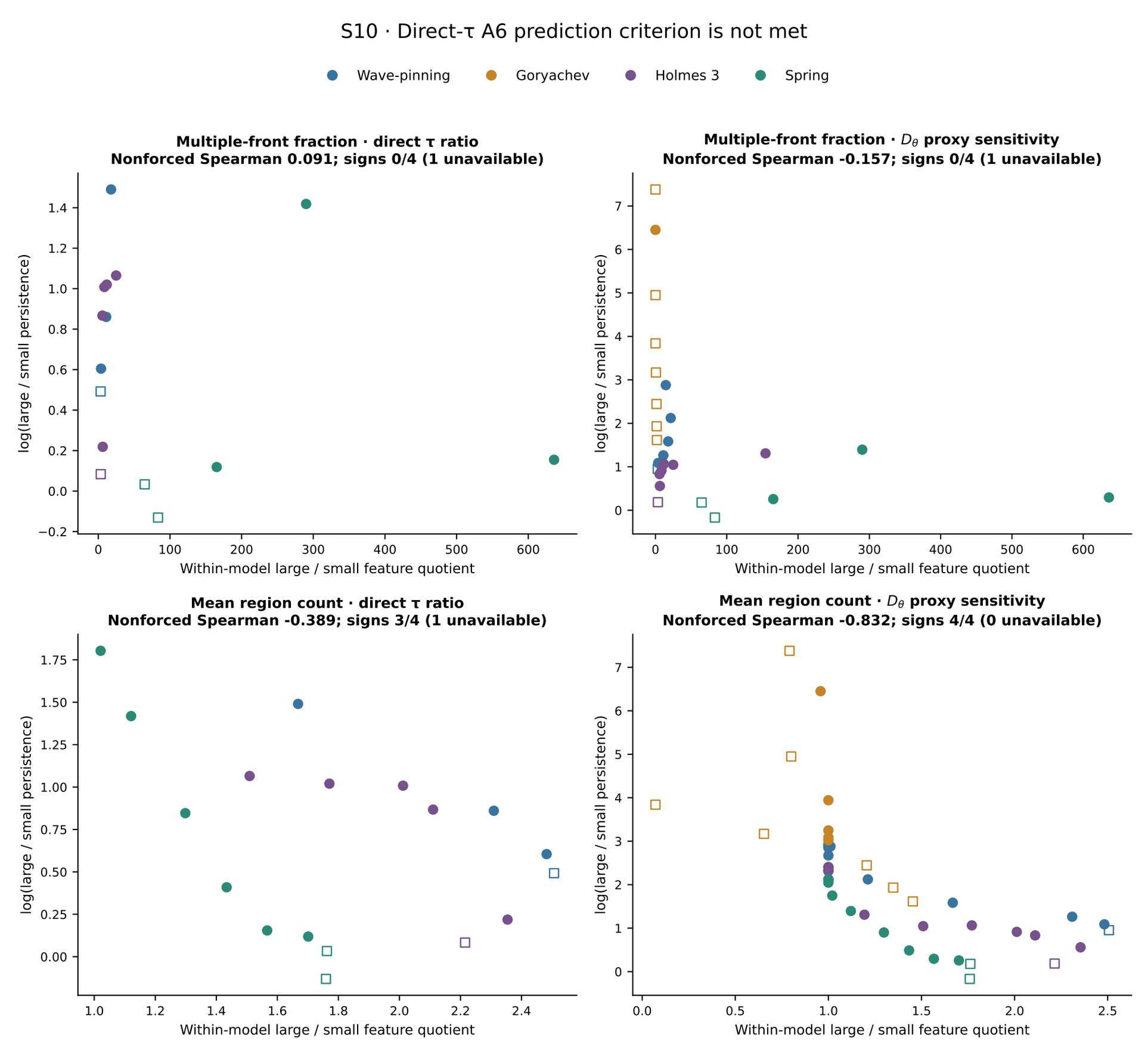
Arm common · σ=1 · actual per-size horizons retained in the source CSV.
Rank entries are Spearman coefficients; multi-front denotes the fraction of time with at least two qualified regions.
Every threshold, raw/physical-kernel and observation-cadence view is retained. Δt is relative to the declared common imaging cadence.
Direct resolved τ and D_θ-proxy associations remain separate. All four folds are planned; NA counts remain in the denominator.
Only the primary, forced-excluded, direct-τ rows enter the declared H2 A6 gate. Other rows are sensitivity checks.

S10 · Primary predictor sensitivity · forced included

A5 imaging view	Multi-front τ rank	τ folds	Multi-front proxy rank	Proxy folds	Count τ rank	τ folds	Count proxy rank	Proxy folds
Primary imaging view	-0.021	0/4 (1 NA)	-0.632	1/4 (0 NA)	-0.342	3/4 (1 NA)	-0.874	4/4 (0 NA)
q=0.3 / raw / Δt=0.0625×	0.086	0/4 (4 NA)	-0.636	1/4 (1 NA)	-0.486	0/4 (4 NA)	-0.804	3/4 (0 NA)
q=0.3 / raw / Δt=0.125×	-0.143	0/4 (4 NA)	-0.745	1/4 (2 NA)	-0.486	0/4 (4 NA)	-0.804	3/4 (0 NA)
q=0.3 / raw / Δt=0.25×	0.086	0/4 (4 NA)	-0.697	1/4 (2 NA)	-0.257	0/4 (4 NA)	-0.776	3/4 (0 NA)
q=0.3 / raw / Δt=0.5×	0.387	0/4 (1 NA)	-0.507	0/4 (0 NA)	-0.593	3/4 (1 NA)	-0.820	3/4 (0 NA)
q=0.3 / raw / Δt=1×	0.147	0/4 (1 NA)	-0.436	0/4 (0 NA)	-0.385	2/4 (1 NA)	-0.806	3/4 (0 NA)
q=0.3 / raw / Δt=2×	0.147	0/4 (1 NA)	-0.469	0/4 (0 NA)	-0.385	2/4 (1 NA)	-0.804	3/4 (0 NA)
q=0.3 / raw / Δt=4×	0.175	0/4 (1 NA)	-0.468	0/4 (0 NA)	-0.329	2/4 (1 NA)	-0.804	3/4 (0 NA)
q=0.3 / optical / Δt=0.0625×	NA	0/4 (4 NA)	-0.850	0/4 (4 NA)	-0.257	0/4 (4 NA)	-0.762	4/4 (0 NA)
q=0.3 / optical / Δt=0.125×	NA	0/4 (4 NA)	-0.850	0/4 (4 NA)	-0.257	0/4 (4 NA)	-0.762	4/4 (0 NA)
q=0.3 / optical / Δt=0.25×	NA	0/4 (4 NA)	-0.850	0/4 (4 NA)	-0.257	0/4 (4 NA)	-0.762	4/4 (0 NA)
q=0.3 / optical / Δt=0.5×	-0.039	0/4 (1 NA)	-0.640	1/4 (0 NA)	-0.379	3/4 (1 NA)	-0.856	4/4 (0 NA)
q=0.3 / optical / Δt=1×	-0.063	0/4 (1 NA)	-0.579	0/4 (0 NA)	-0.049	2/4 (1 NA)	-0.841	4/4 (0 NA)
q=0.3 / optical / Δt=2×	-0.345	0/4 (1 NA)	-0.647	1/4 (0 NA)	-0.077	2/4 (1 NA)	-0.841	4/4 (0 NA)
q=0.3 / optical / Δt=4×	-0.183	0/4 (2 NA)	-0.729	1/4 (1 NA)	-0.049	2/4 (1 NA)	-0.831	4/4 (0 NA)
q=0.5 / raw / Δt=0.0625×	0.086	0/4 (4 NA)	-0.688	1/4 (1 NA)	-0.486	0/4 (4 NA)	-0.804	3/4 (0 NA)
q=0.5 / raw / Δt=0.125×	NA	0/4 (4 NA)	-0.790	1/4 (2 NA)	-0.486	0/4 (4 NA)	-0.804	3/4 (0 NA)
q=0.5 / raw / Δt=0.25×	NA	0/4 (4 NA)	-0.839	2/4 (2 NA)	-0.486	0/4 (4 NA)	-0.804	3/4 (0 NA)
q=0.5 / raw / Δt=0.5×	0.292	0/4 (1 NA)	-0.529	0/4 (0 NA)	-0.484	3/4 (1 NA)	-0.806	3/4 (0 NA)
q=0.5 / raw / Δt=1×	0.266	0/4 (1 NA)	-0.438	0/4 (0 NA)	-0.098	2/4 (1 NA)	-0.794	3/4 (0 NA)
q=0.5 / raw / Δt=2×	0.203	0/4 (1 NA)	-0.452	0/4 (0 NA)	-0.140	2/4 (1 NA)	-0.784	3/4 (0 NA)
q=0.5 / raw / Δt=4×	0.161	0/4 (1 NA)	-0.547	0/4 (0 NA)	-0.175	2/4 (1 NA)	-0.785	3/4 (0 NA)
q=0.5 / optical / Δt=0.0625×	NA	0/4 (4 NA)	-0.845	0/4 (4 NA)	-0.257	0/4 (4 NA)	-0.839	4/4 (0 NA)
q=0.5 / optical / Δt=0.125×	NA	0/4 (4 NA)	-0.845	0/4 (4 NA)	-0.257	0/4 (4 NA)	-0.839	4/4 (0 NA)
q=0.5 / optical / Δt=0.25×	NA	0/4 (4 NA)	-0.845	0/4 (4 NA)	-0.257	0/4 (4 NA)	-0.839	4/4 (0 NA)
q=0.5 / optical / Δt=0.5×	-0.050	0/4 (1 NA)	-0.638	1/4 (0 NA)	-0.364	3/4 (1 NA)	-0.877	4/4 (0 NA)
q=0.5 / optical / Δt=2×	-0.297	0/4 (1 NA)	-0.632	1/4 (0 NA)	-0.021	2/4 (1 NA)	-0.844	4/4 (0 NA)
q=0.5 / optical / Δt=4×	-0.059	0/4 (2 NA)	-0.690	1/4 (1 NA)	-0.021	2/4 (1 NA)	-0.853	4/4 (0 NA)
q=0.7 / raw / Δt=0.0625×	NA	0/4 (4 NA)	-0.839	2/4 (2 NA)	-0.486	0/4 (4 NA)	-0.804	3/4 (0 NA)
q=0.7 / raw / Δt=0.125×	NA	0/4 (4 NA)	-0.839	2/4 (2 NA)	-0.486	0/4 (4 NA)	-0.804	3/4 (0 NA)
q=0.7 / raw / Δt=0.25×	NA	0/4 (4 NA)	-0.845	0/4 (4 NA)	-0.486	0/4 (4 NA)	-0.804	3/4 (0 NA)
q=0.7 / raw / Δt=0.5×	0.441	0/4 (1 NA)	-0.449	0/4 (0 NA)	-0.127	3/4 (1 NA)	-0.794	3/4 (0 NA)
q=0.7 / raw / Δt=1×	0.399	0/4 (1 NA)	-0.405	0/4 (0 NA)	0.259	0/4 (1 NA)	-0.787	3/4 (0 NA)
q=0.7 / raw / Δt=2×	0.399	0/4 (1 NA)	-0.405	0/4 (0 NA)	0.294	0/4 (1 NA)	-0.775	3/4 (0 NA)
q=0.7 / raw / Δt=4×	0.236	0/4 (1 NA)	-0.558	0/4 (0 NA)	0.294	0/4 (1 NA)	-0.782	3/4 (0 NA)
q=0.7 / optical / Δt=0.0625×	NA	0/4 (4 NA)	-0.845	0/4 (4 NA)	-0.086	0/4 (4 NA)	-0.781	3/4 (0 NA)
q=0.7 / optical / Δt=0.125×	NA	0/4 (4 NA)	-0.845	0/4 (4 NA)	-0.086	0/4 (4 NA)	-0.781	3/4 (0 NA)
q=0.7 / optical / Δt=0.25×	NA	0/4 (4 NA)	-0.755	0/4 (4 NA)	-0.086	0/4 (4 NA)	-0.781	3/4 (0 NA)
q=0.7 / optical / Δt=0.5×	0.062	0/4 (1 NA)	-0.603	0/4 (0 NA)	-0.112	3/4 (1 NA)	-0.832	4/4 (0 NA)
q=0.7 / optical / Δt=1×	0.042	0/4 (1 NA)	-0.521	0/4 (0 NA)	0.196	0/4 (1 NA)	-0.840	4/4 (0 NA)
q=0.7 / optical / Δt=2×	-0.191	0/4 (1 NA)	-0.595	0/4 (0 NA)	0.238	0/4 (1 NA)	-0.839	4/4 (0 NA)
q=0.7 / optical / Δt=4×	-0.200	0/4 (1 NA)	-0.716	1/4 (0 NA)	0.231	0/4 (1 NA)	-0.839	4/4 (0 NA)

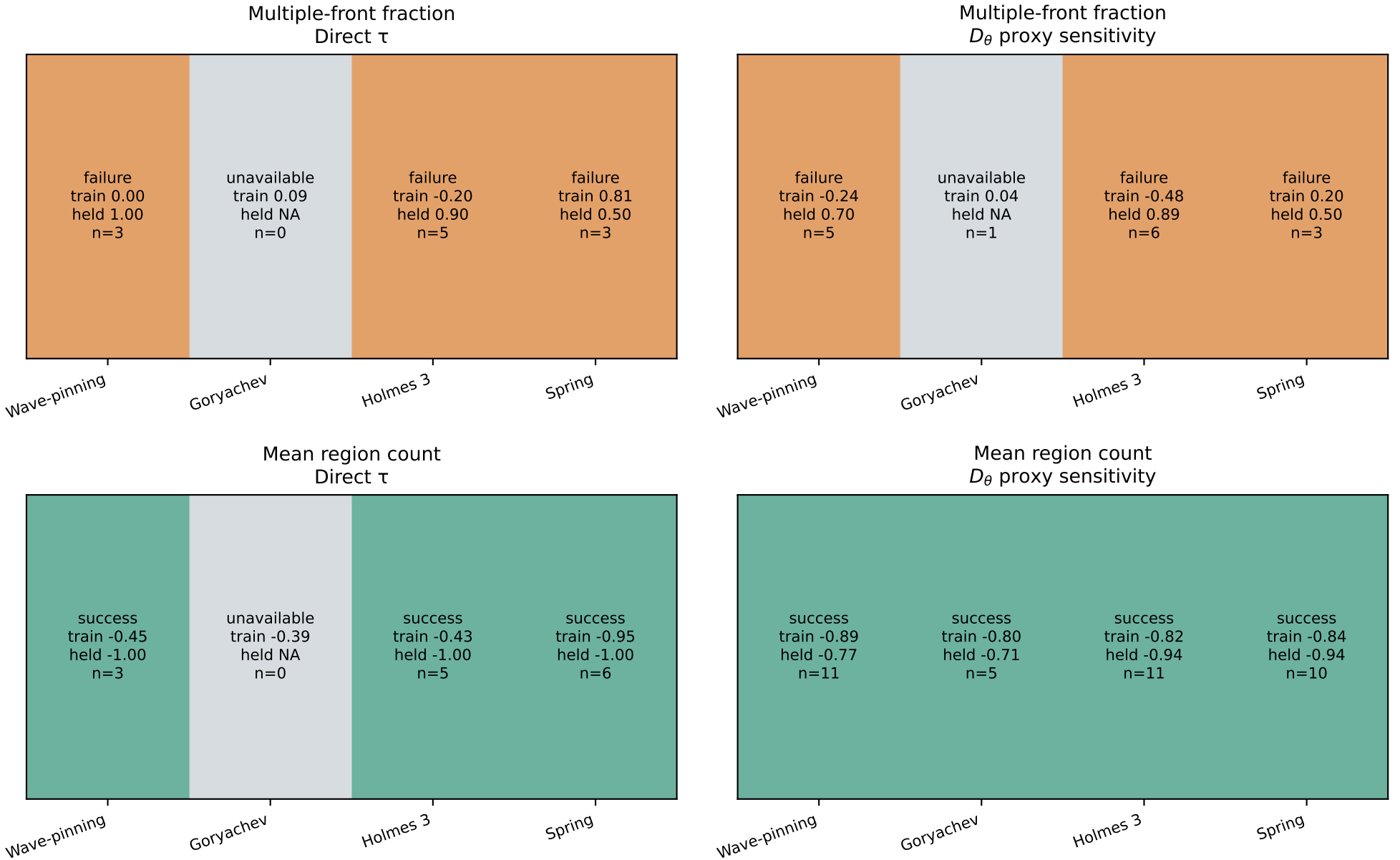
Arm common · σ=1 · actual per-size horizons retained in the source CSV.
Rank entries are Spearman coefficients; multi-front denotes the fraction of time with at least two qualified regions.
Every threshold, raw/physical-kernel and observation-cadence view is retained. Δt is relative to the declared common imaging cadence.
Direct resolved τ and D_θ-proxy associations remain separate. All four folds are planned; NA counts remain in the denominator.
Only the primary, forced-excluded, direct-τ rows enter the declared H2 A6 gate. Other rows are sensitivity checks.

S10 · Direct-τ A6 prediction criterion is not met



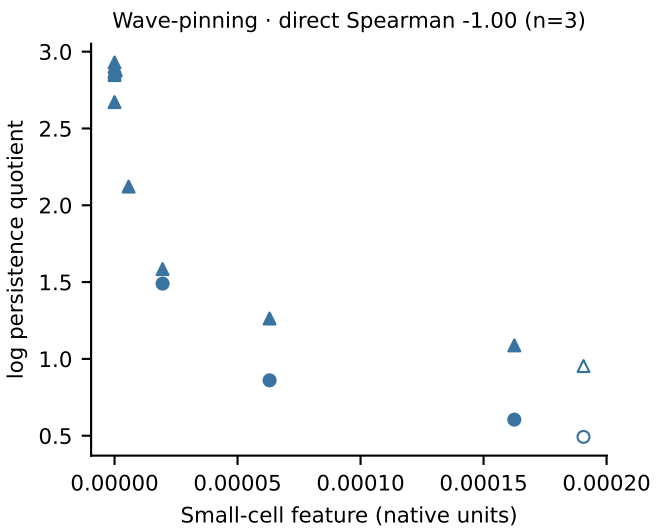
Arm longest · σ=1 · actual per-size horizons retained in the source CSV. Crosses: unknown forcing, excluded from primary.
Filled circles: nonforced primary conditions; open squares: forced context. No absolute native quantity is pooled across models.
Direct resolved τ ratios and D_θ-derived proxy sensitivity are separate. Marginal whole-seed CIs remain in the source tables.
Primary support requires Spearman ≤ −0.6 and negative training/held-out sign in at least 3 of 4 planned folds.

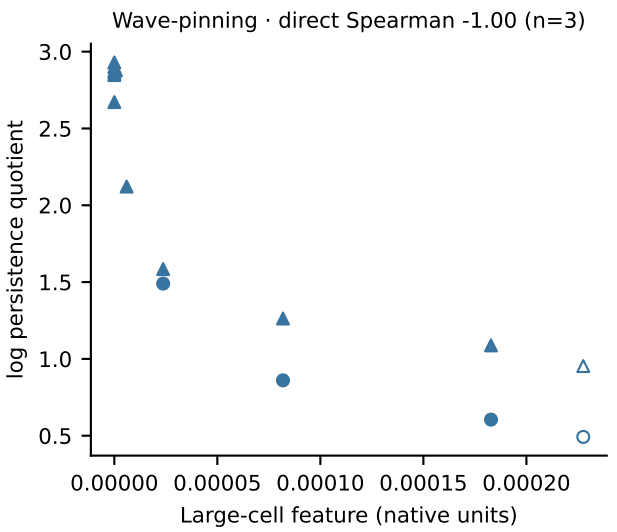
S10 · Every planned held-out-model fold remains in the denominator

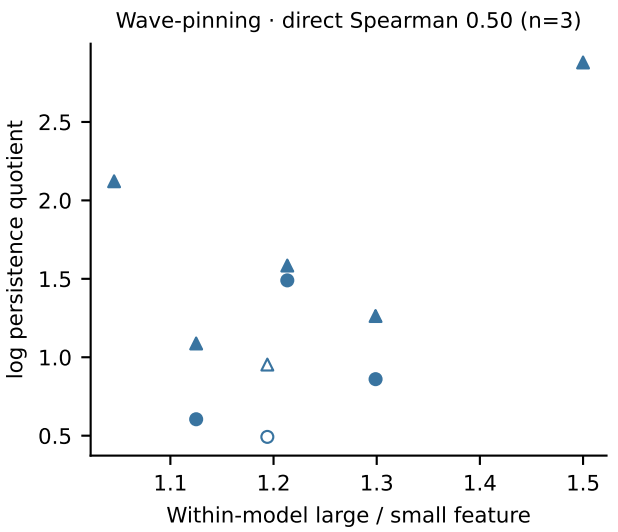


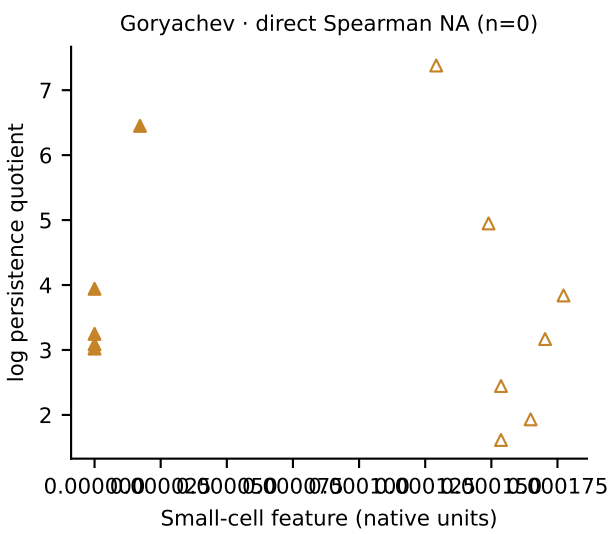
Arm longest · σ=1 · actual per-size horizons retained in the source CSV.
Green: negative training and held-out signs. Orange: evaluated sign failure. Grey: insufficient/constant/censored data.
Each training and held-out sample needs at least 3 matched input fractions and nonconstant predictor/outcome.
Absence of a qualifying Goryachev or other model is explicit; no missing fold is counted as successful.

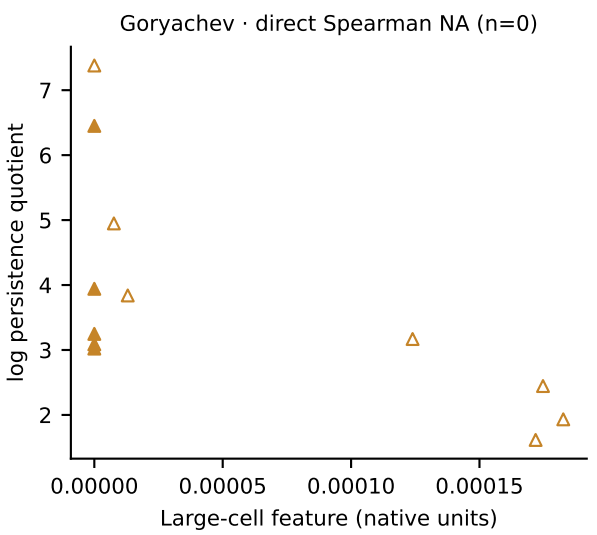
S10 · births per time per perimeter

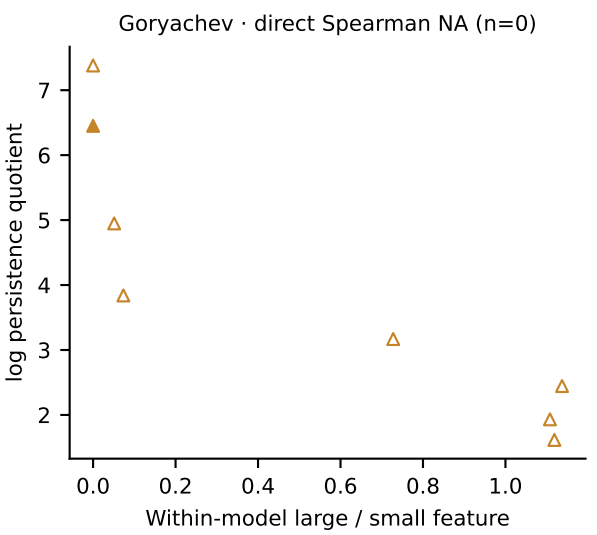


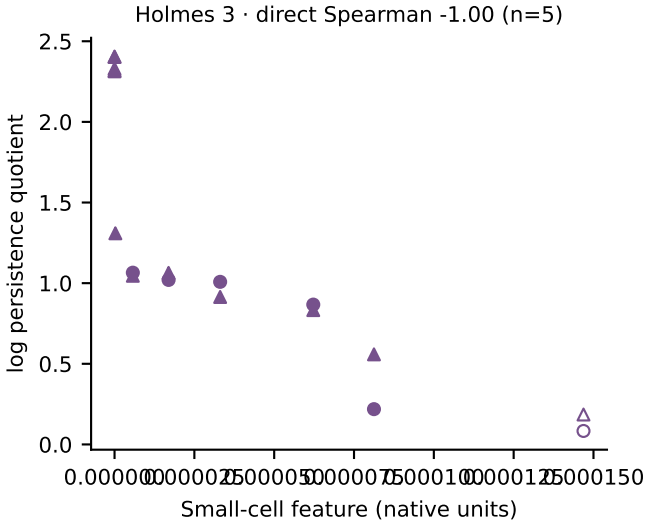


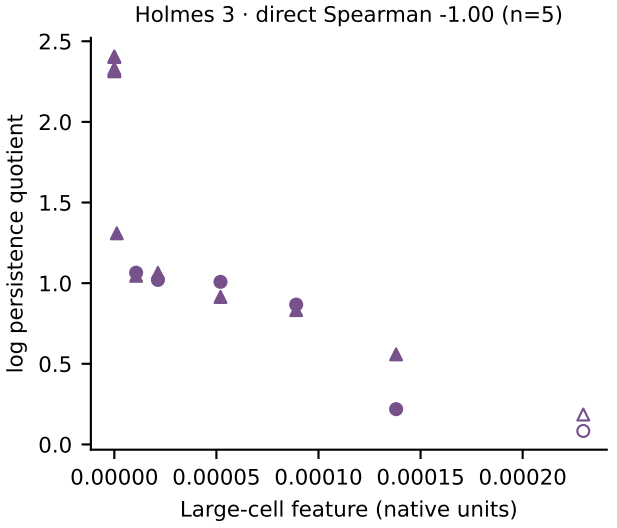


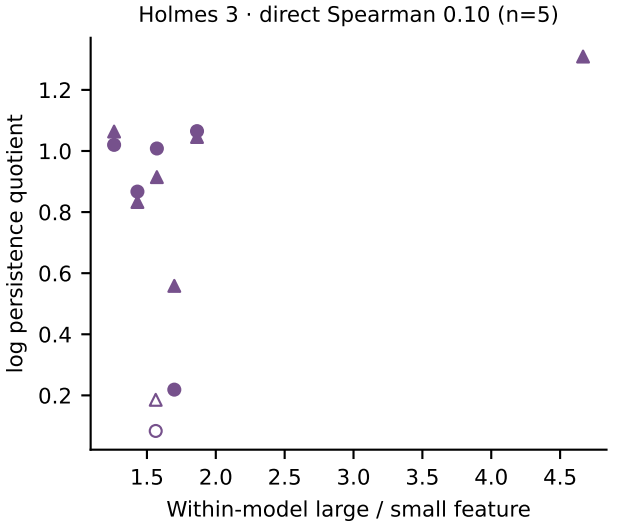


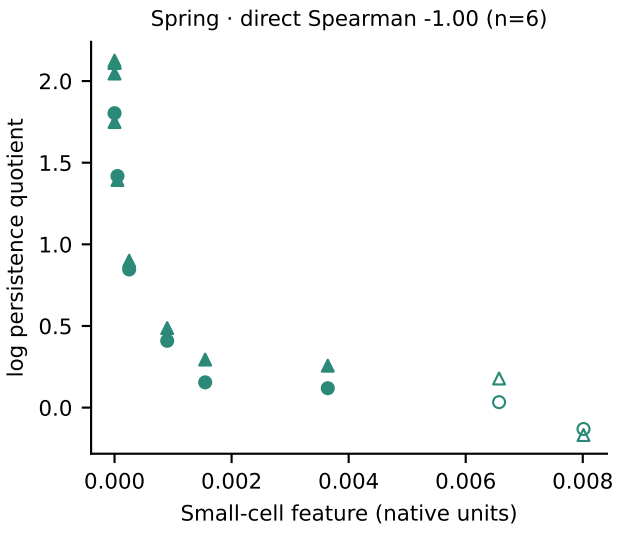


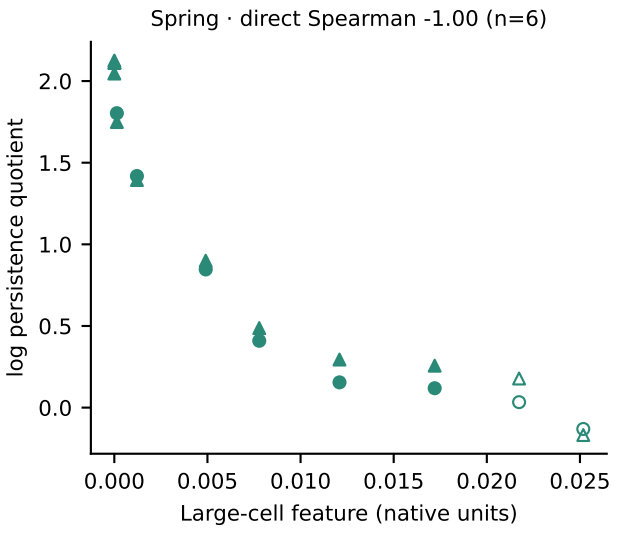


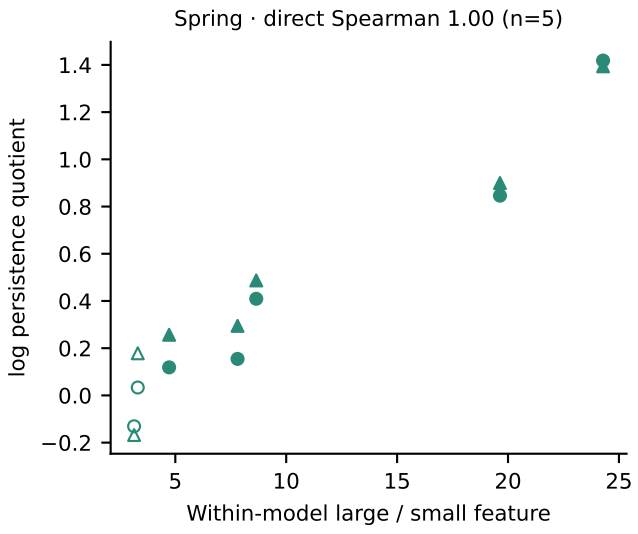






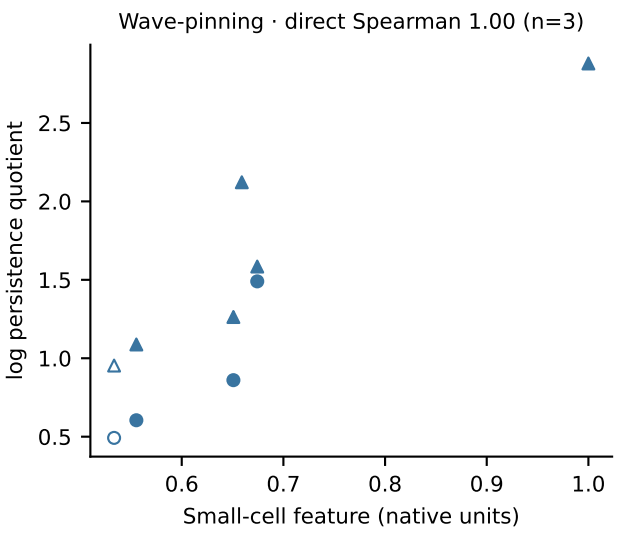


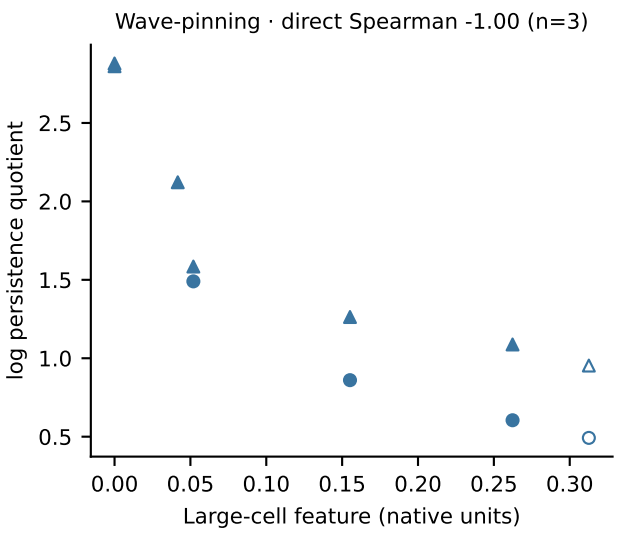


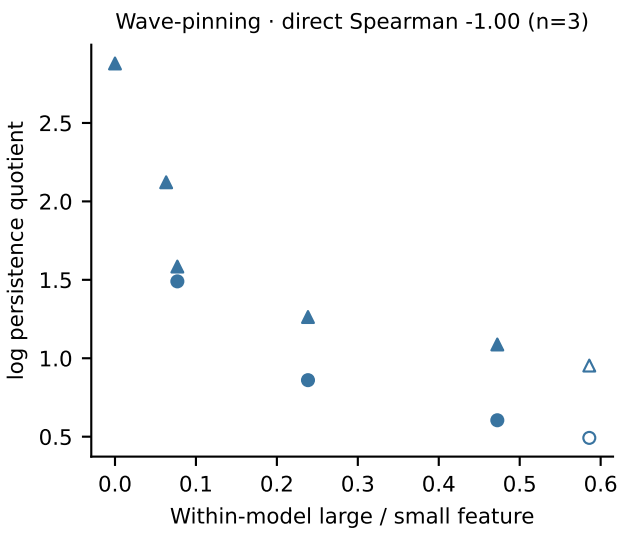


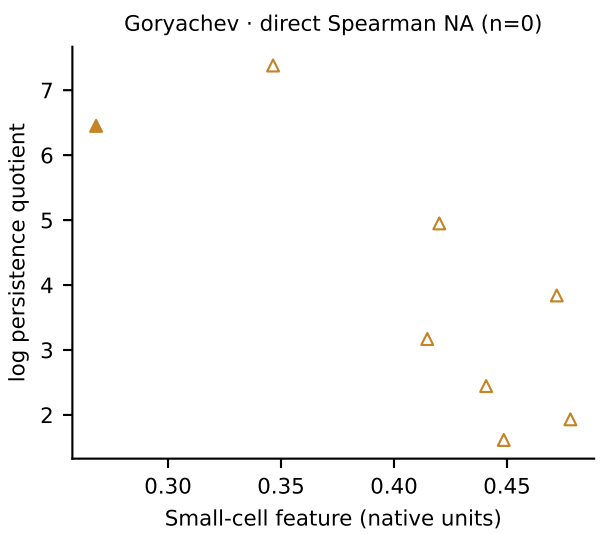
Arm longest · σ=1 · actual per-size horizons retained in the source CSV. Crosses: unknown forcing, excluded from primary.
○ Direct resolved τ ratio; △ D_θ proxy sensitivity. Filled: nonforced; open: forced context.
All small/large native quantities remain within their own model. Cross-model tests use only size quotients.
NA retains insufficient, constant, censored or unscorable measurements; native caps are never substituted.

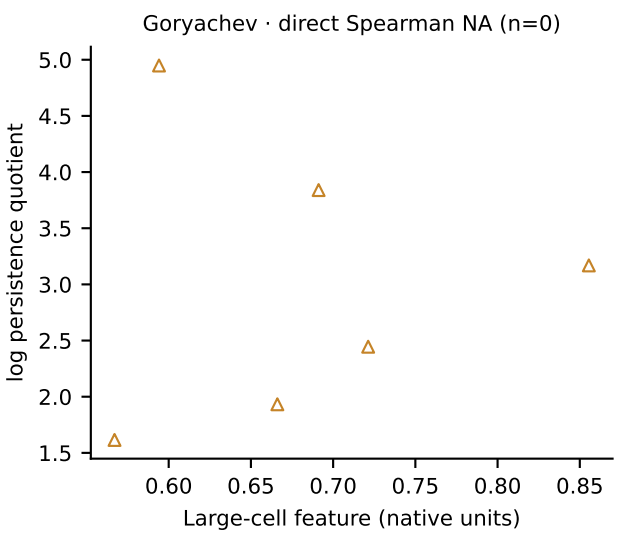
S10 · ever unscorable takeover fraction

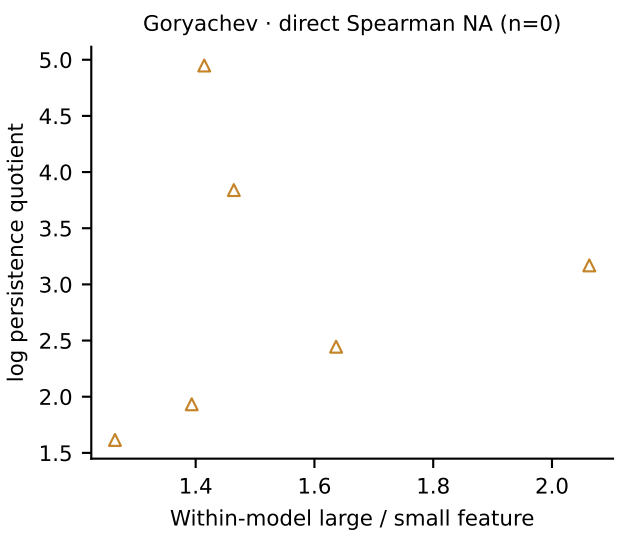


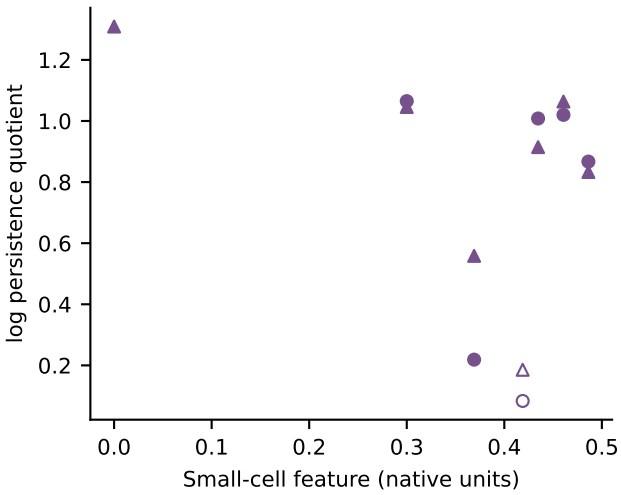


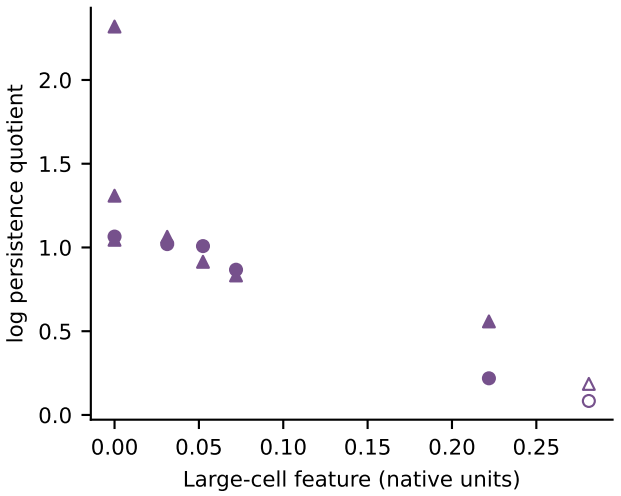


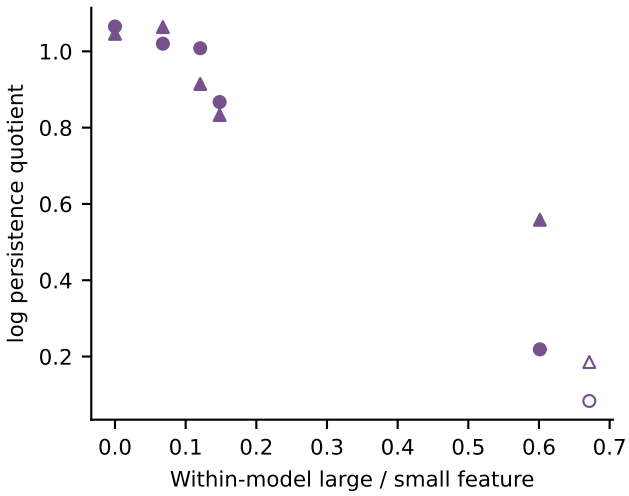


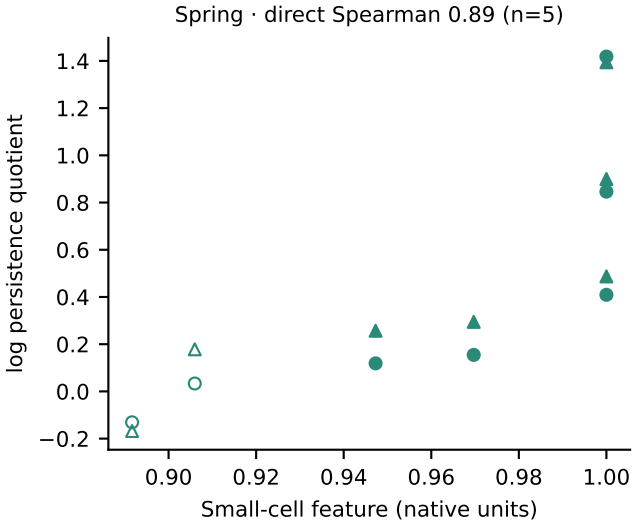


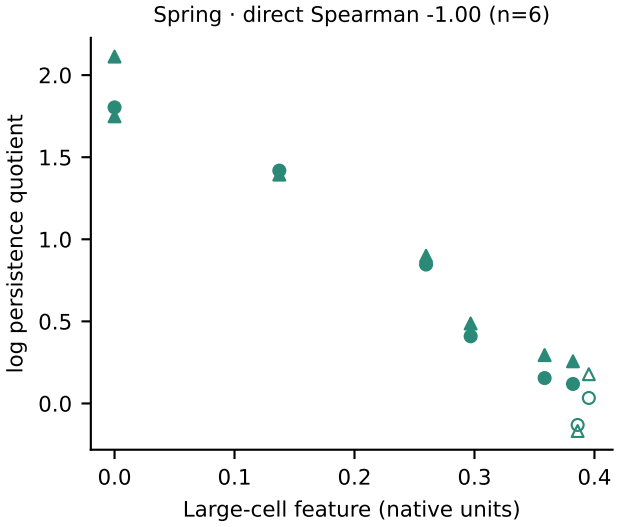


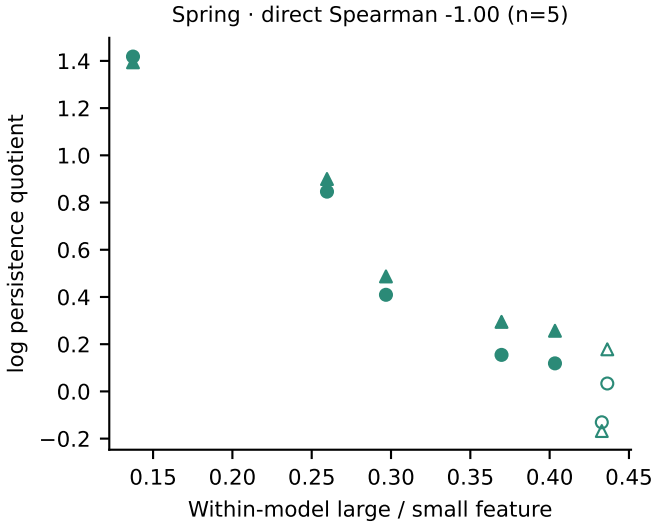






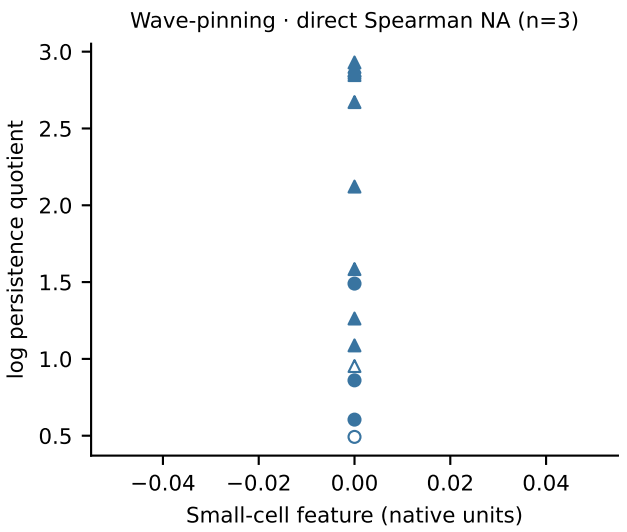


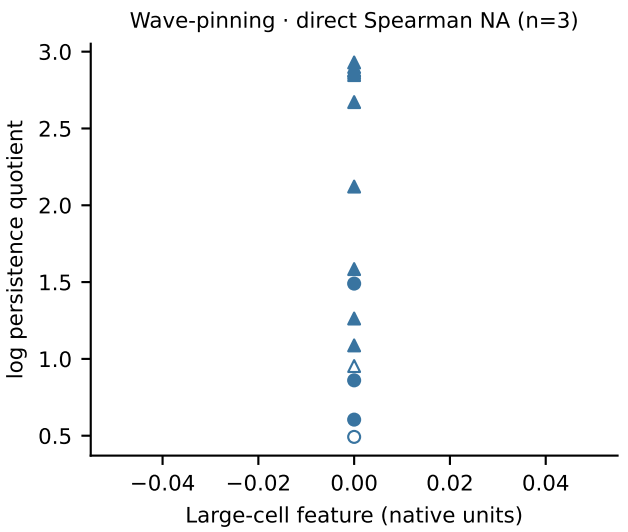


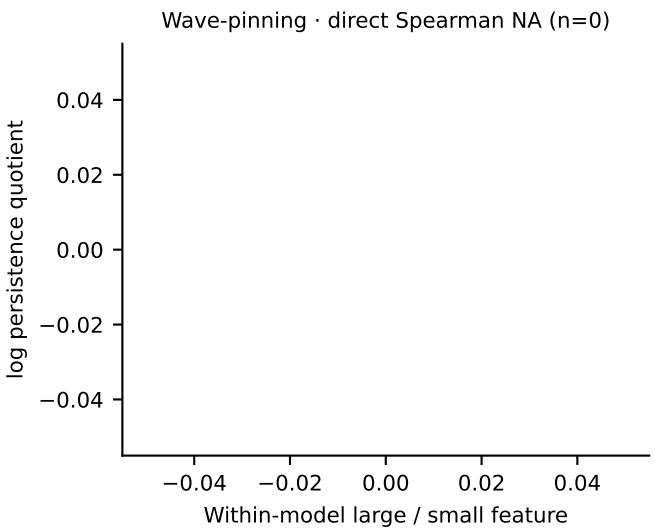


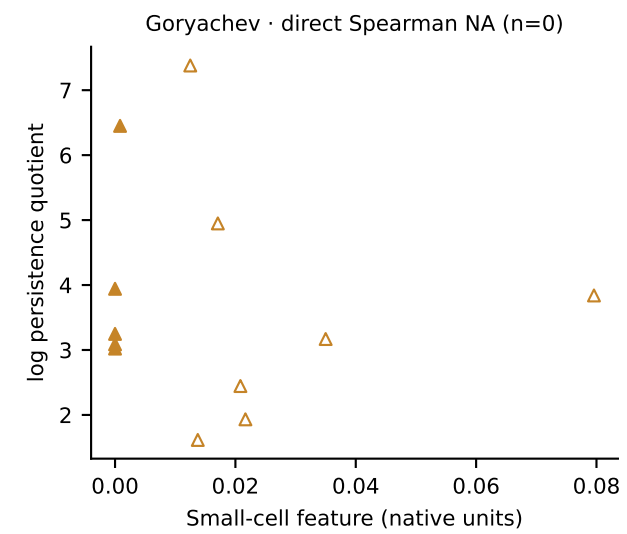
Arm longest · σ=1 · actual per-size horizons retained in the source CSV. Crosses: unknown forcing, excluded from primary.
○ Direct resolved τ ratio; △ D_θ proxy sensitivity. Filled: nonforced; open: forced context.
All small/large native quantities remain within their own model. Cross-model tests use only size quotients.
NA retains insufficient, constant, censored or unscorable measurements; native caps are never substituted.

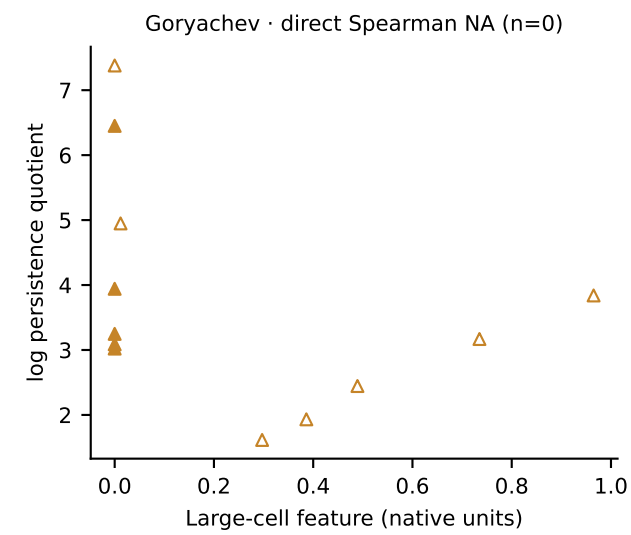
S10 · low contrast time fraction

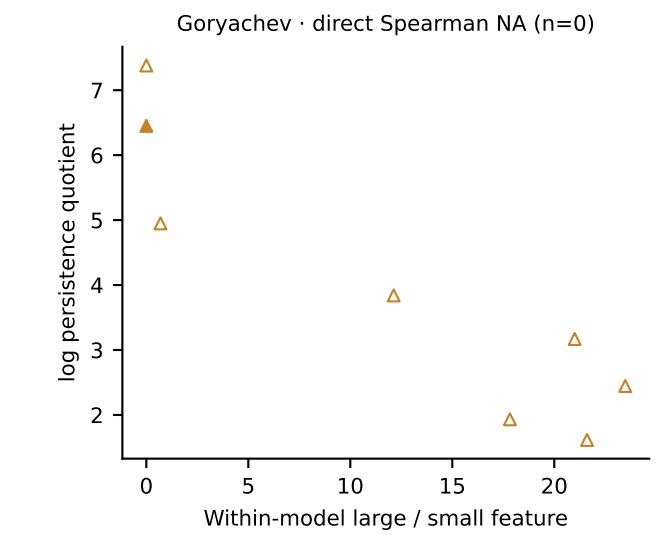


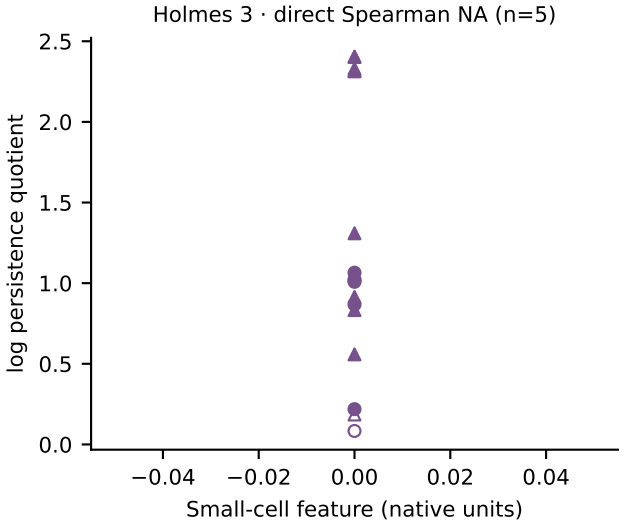


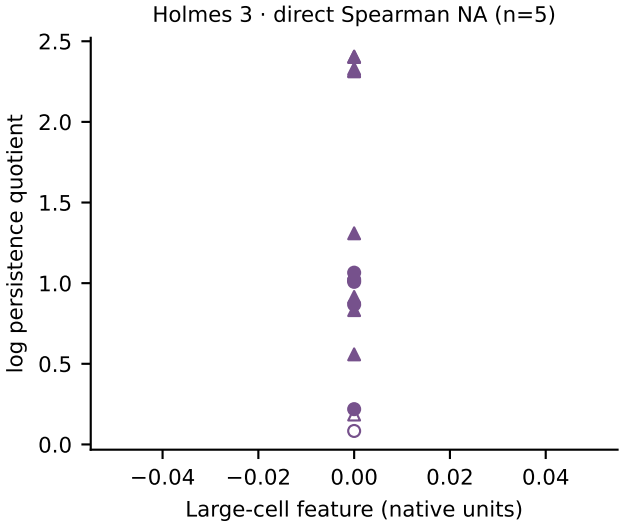


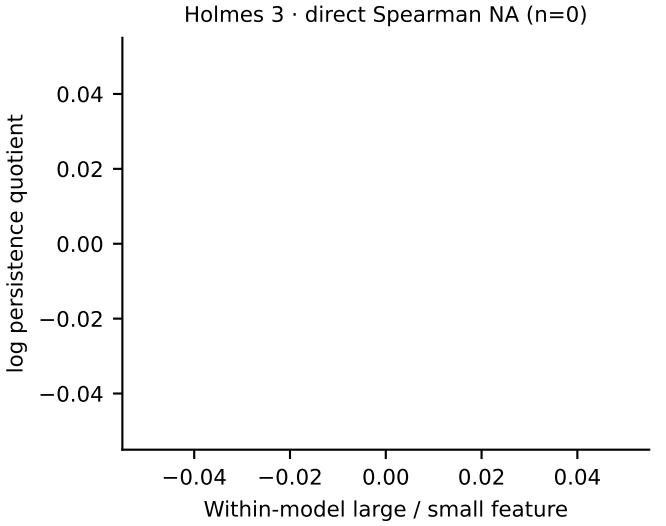


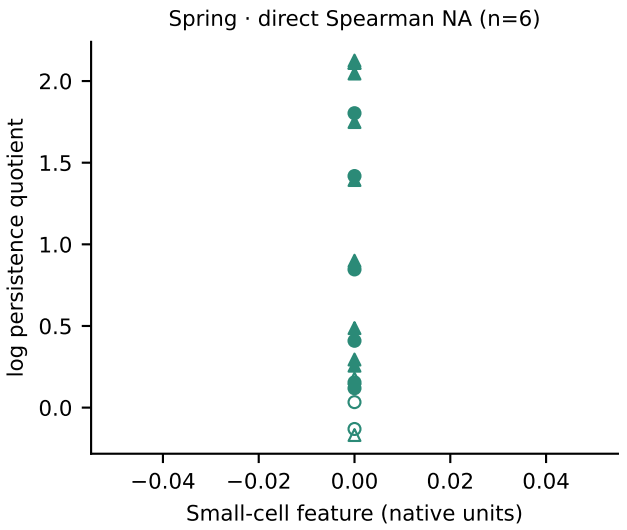


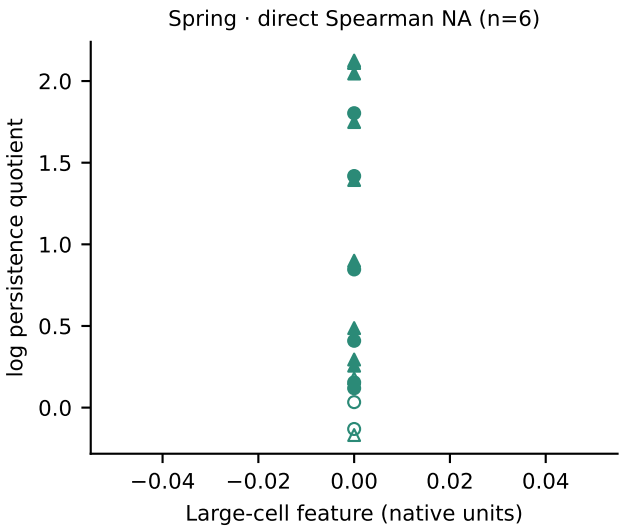


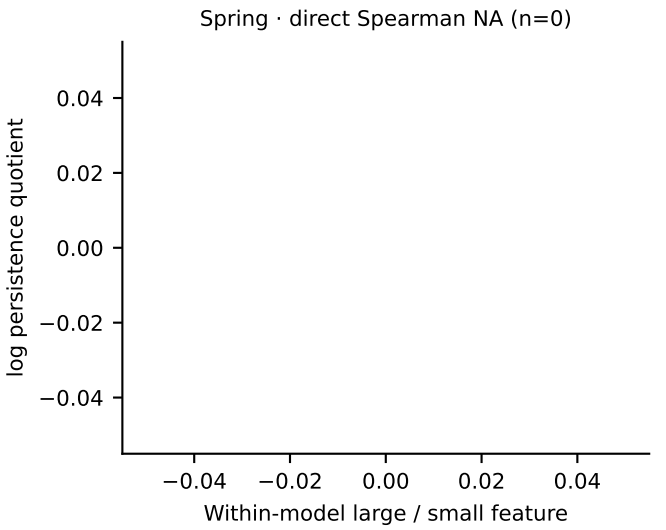












Arm longest · σ=1 · actual per-size horizons retained in the source CSV. Crosses: unknown forcing, excluded from primary.
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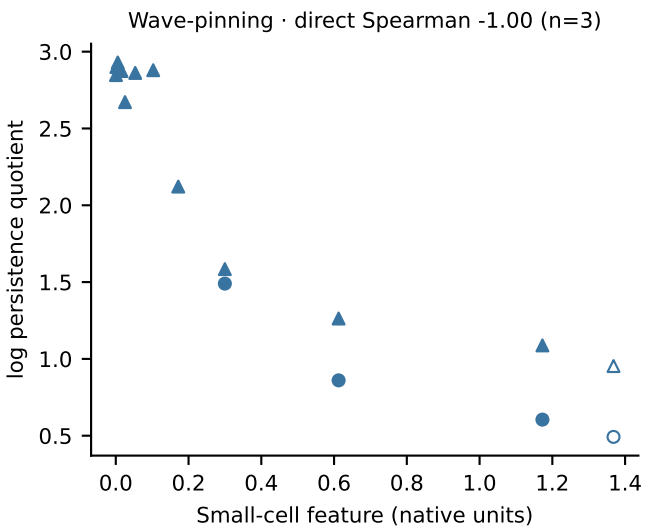
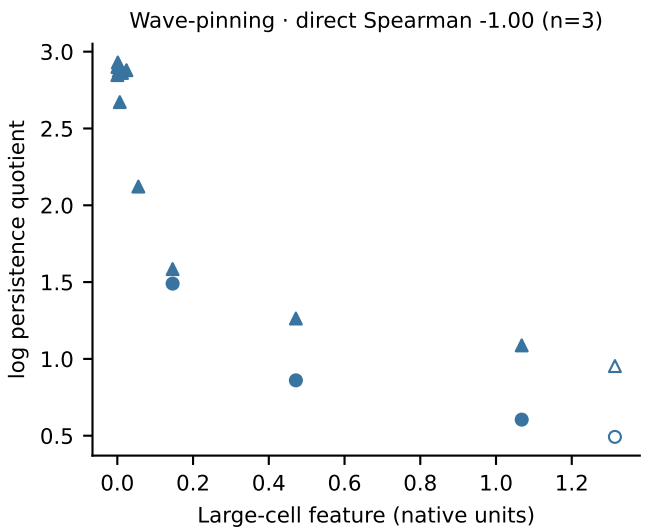
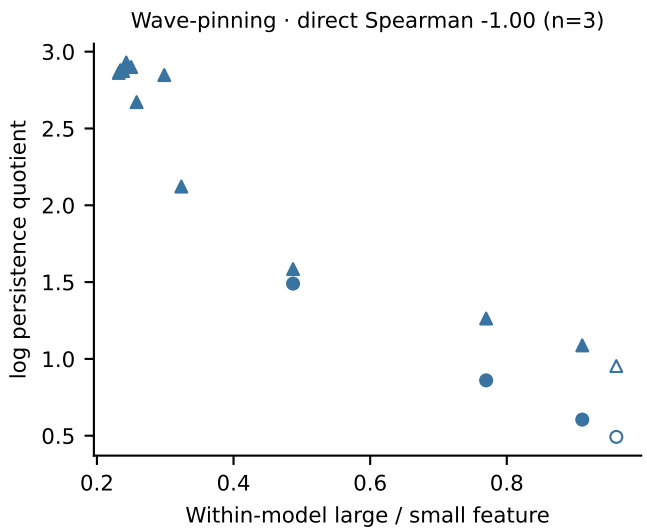
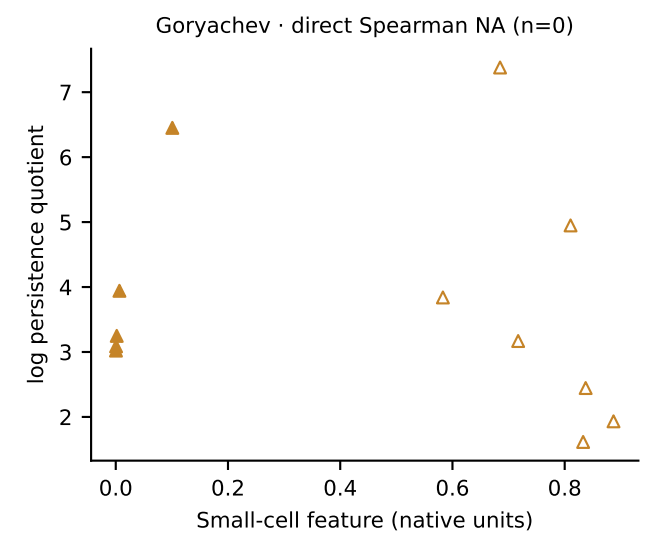
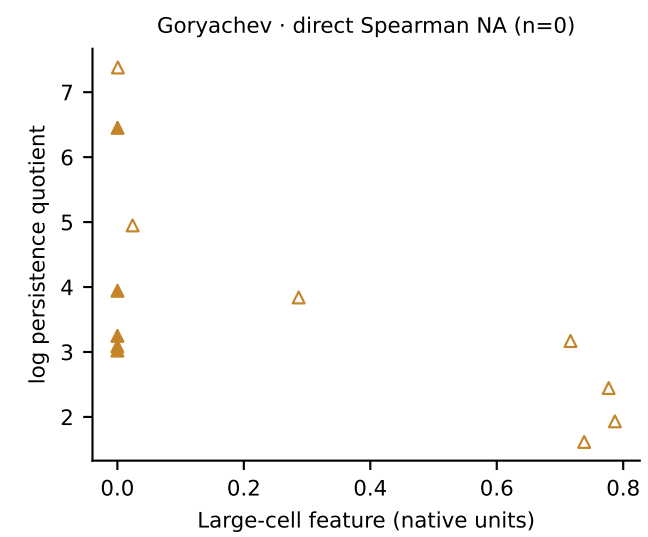
S10 · mean absolute turn 1 frames

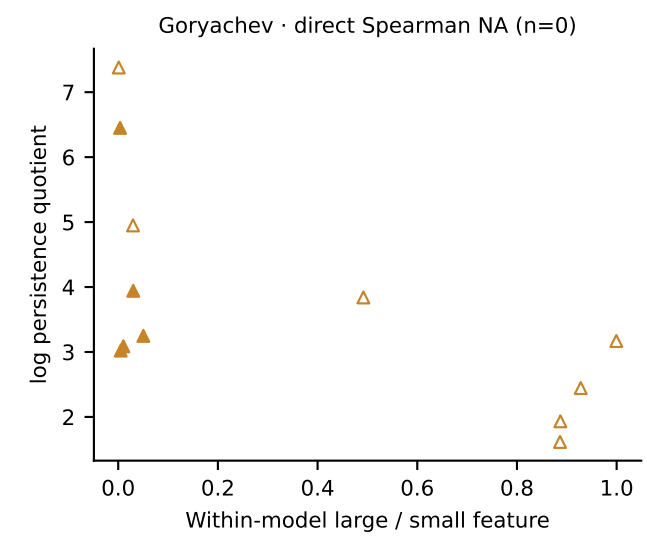


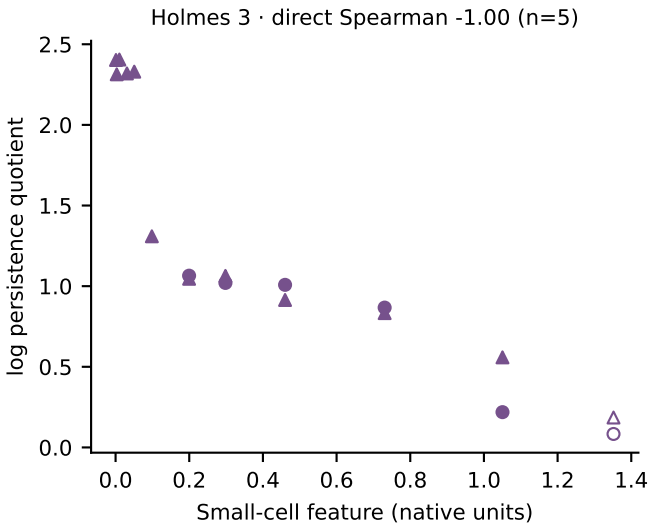


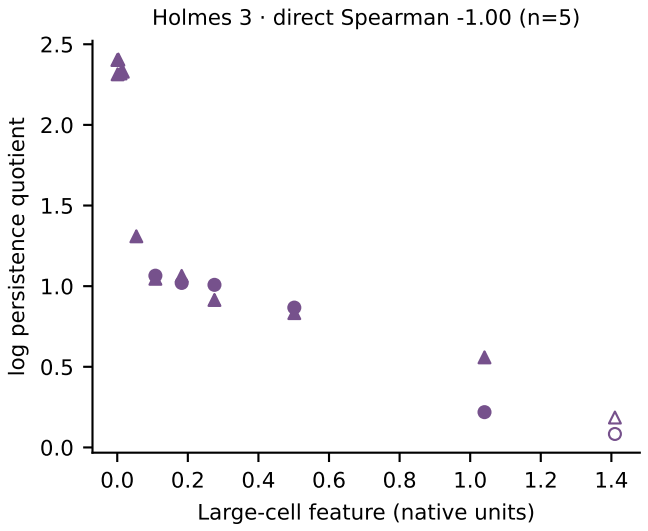


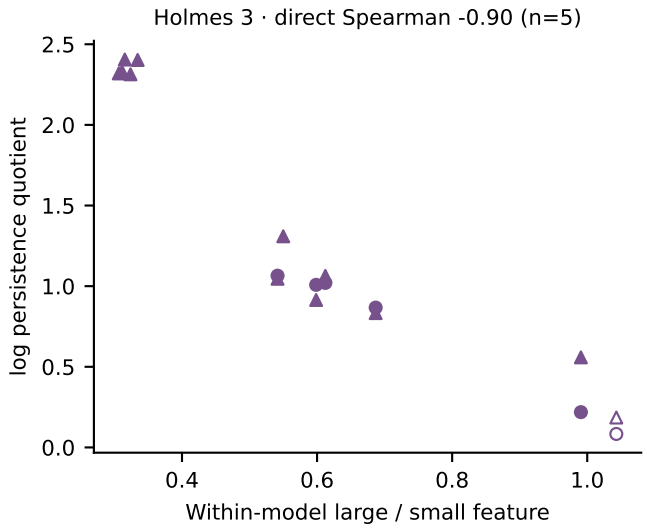


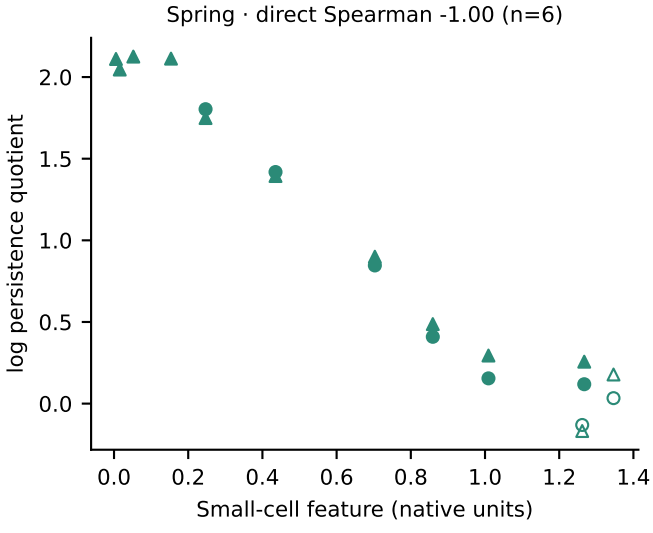


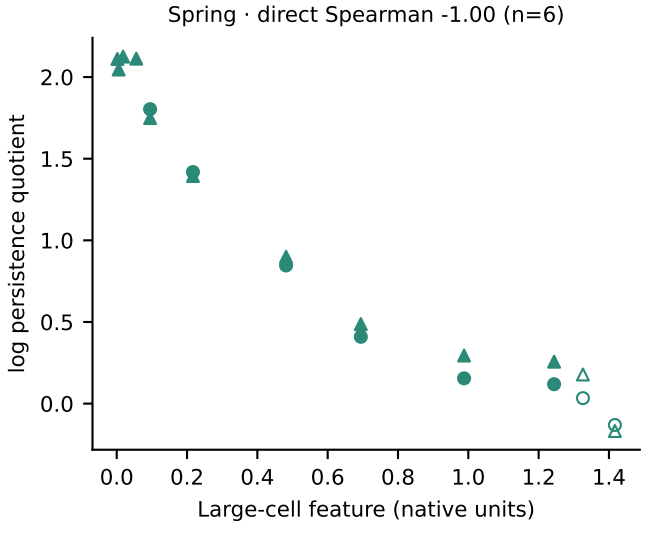


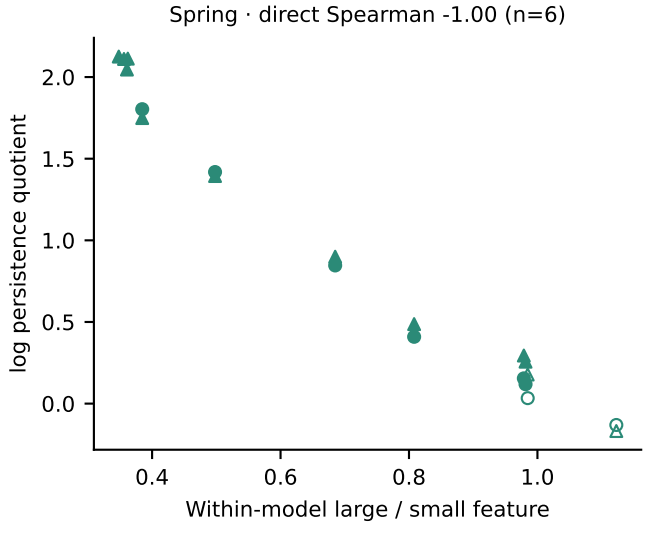












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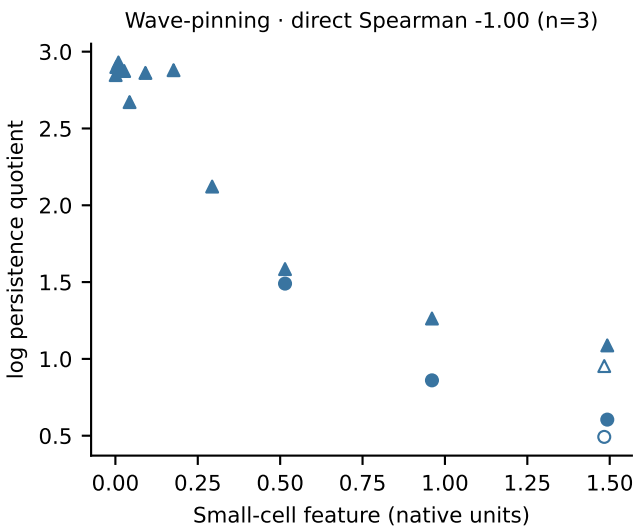
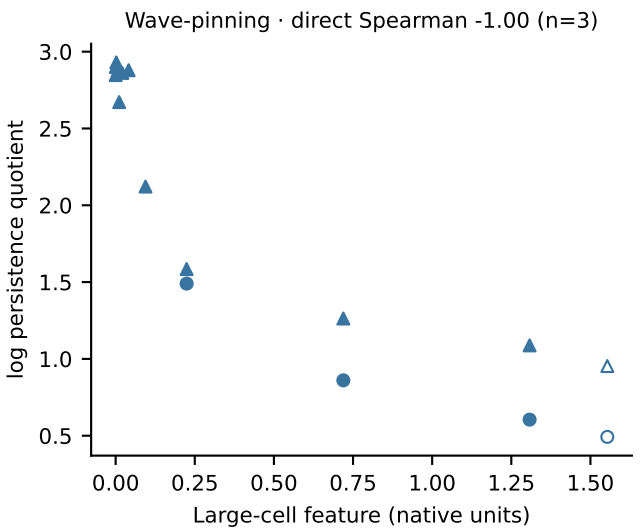
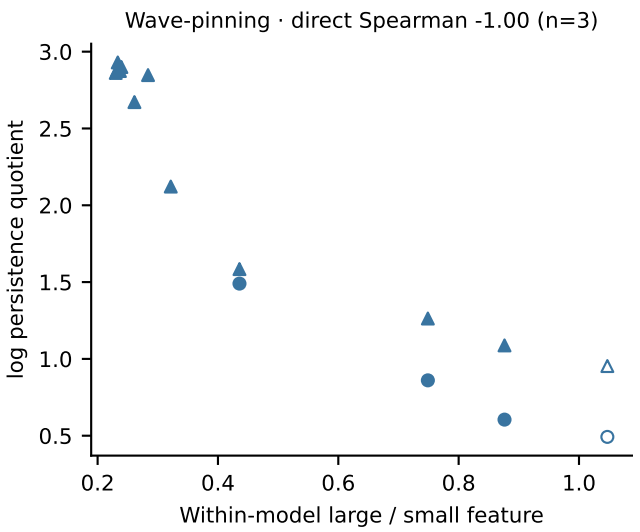
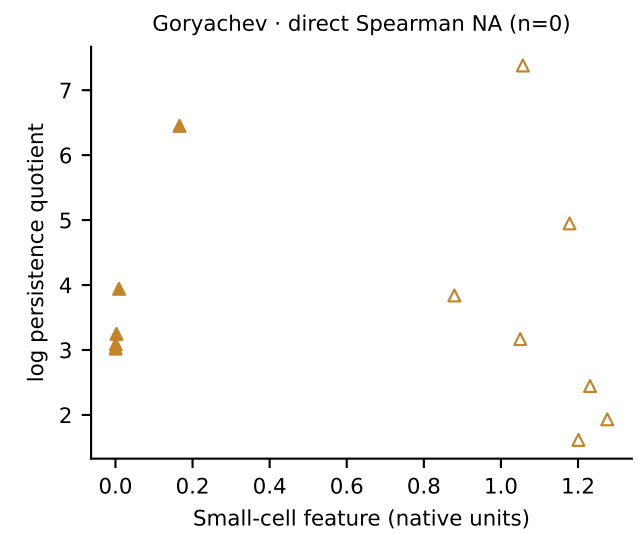
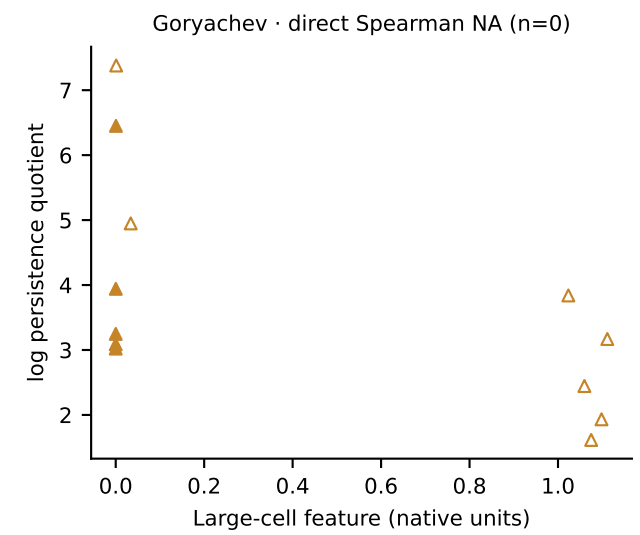
S10 · mean absolute turn 2 frames

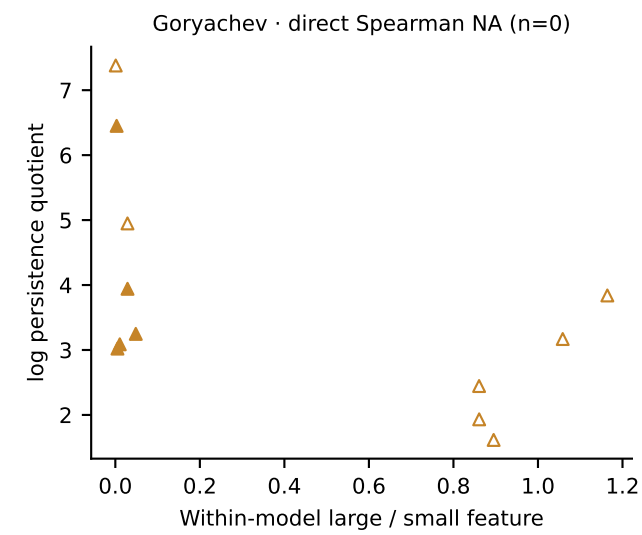


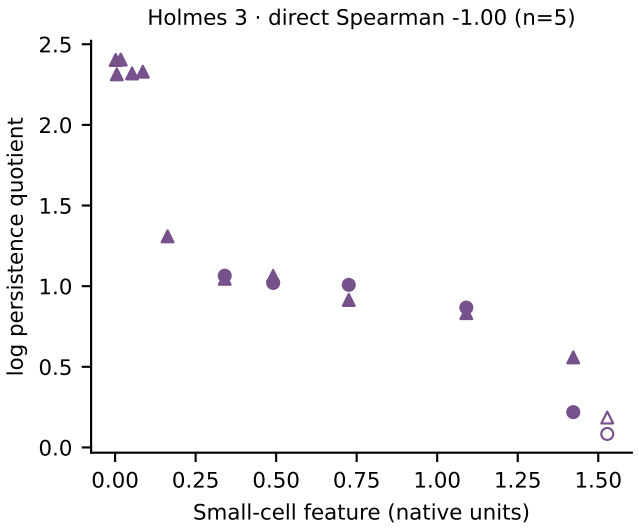


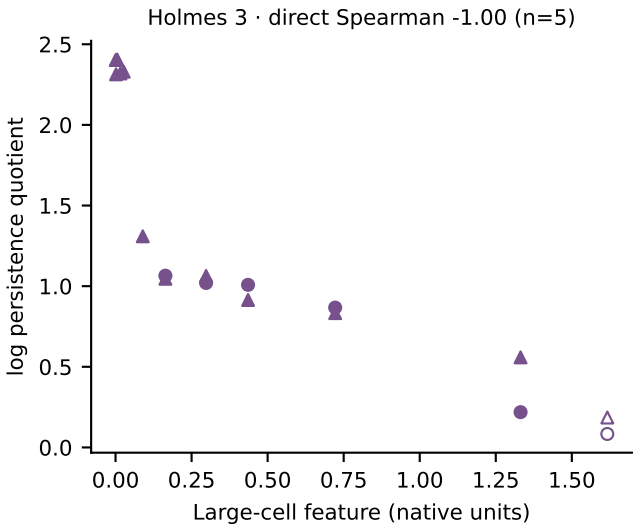


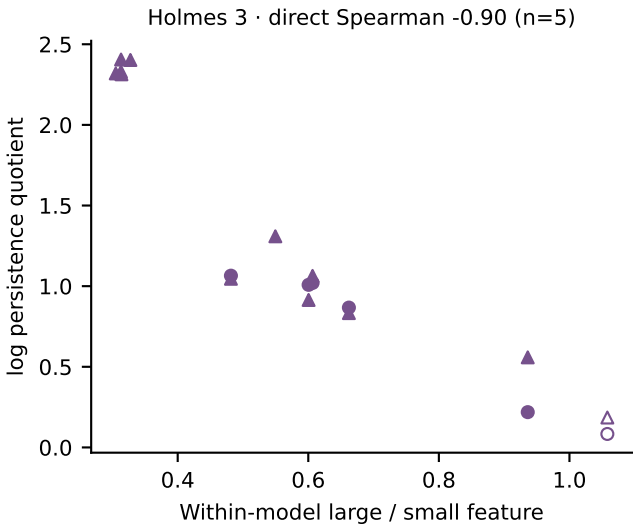


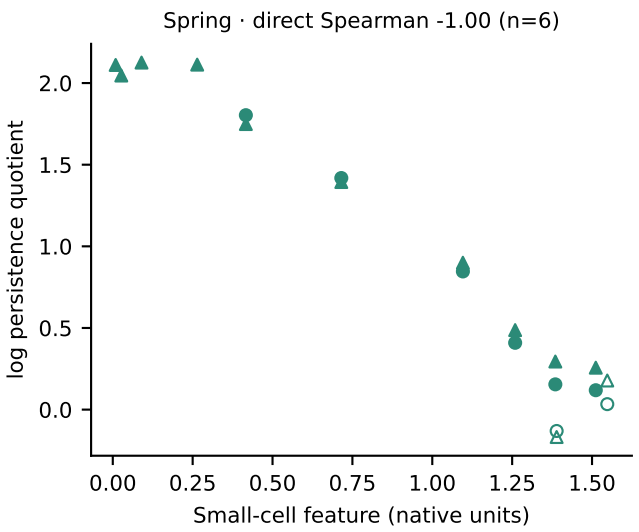


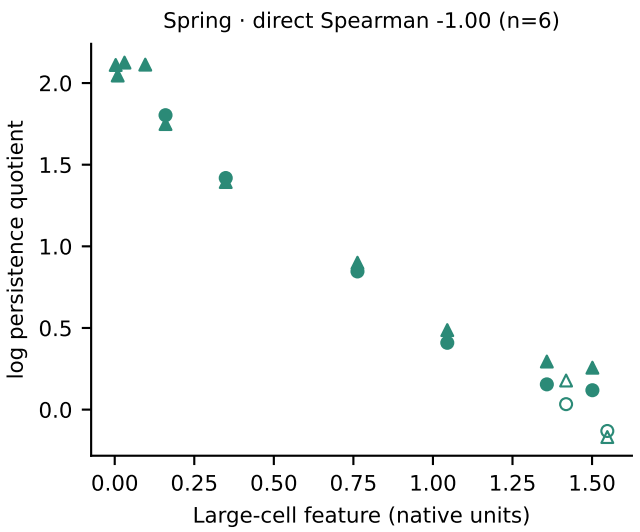


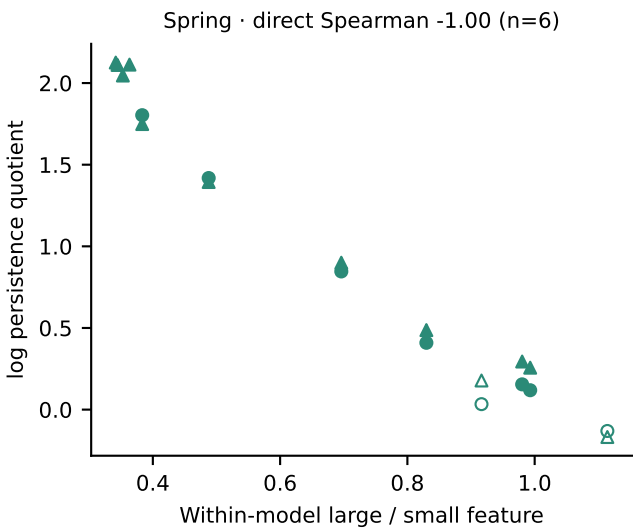












Arm longest · σ=1 · actual per-size horizons retained in the source CSV. Crosses: unknown forcing, excluded from primary.
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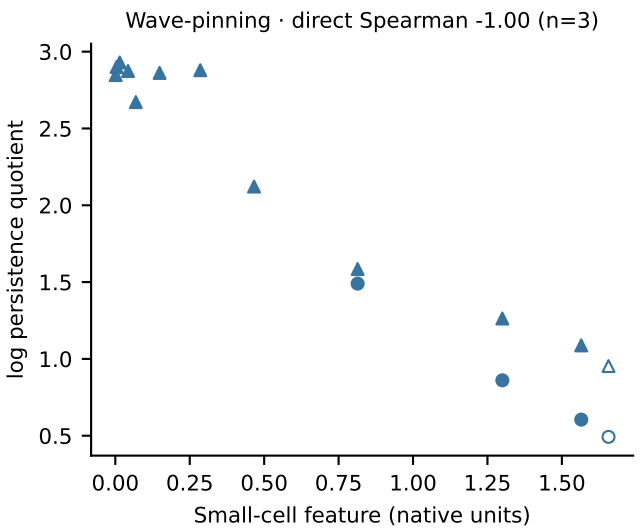
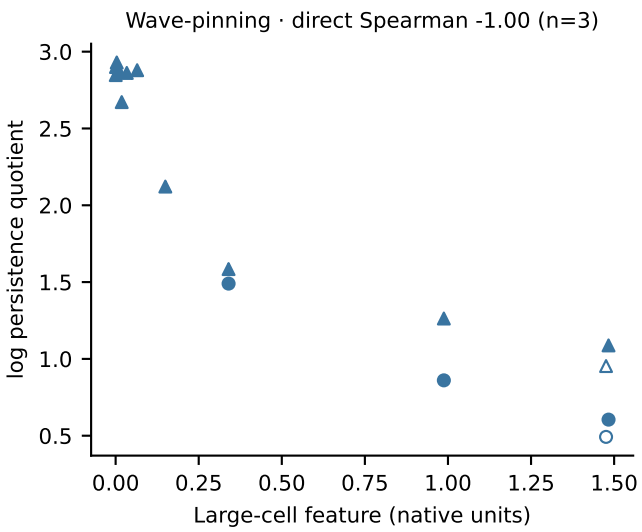
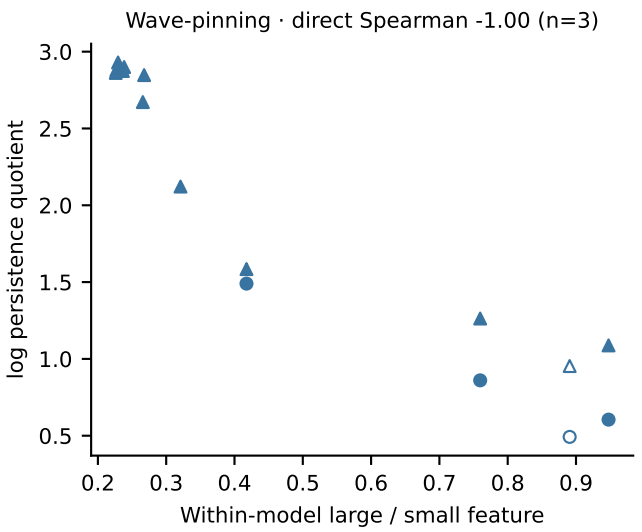
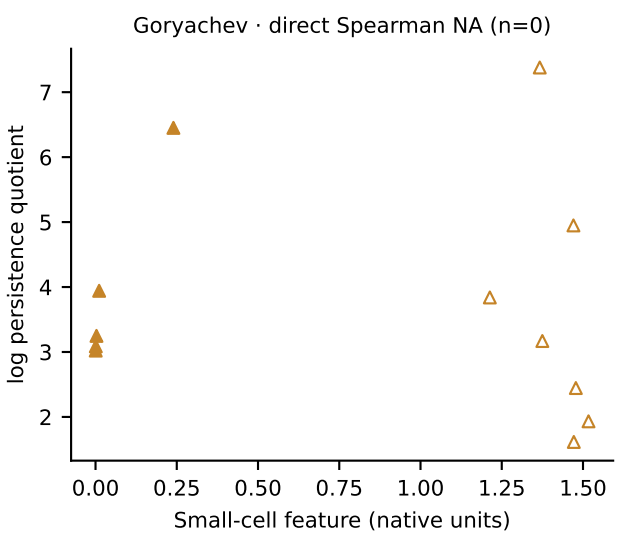
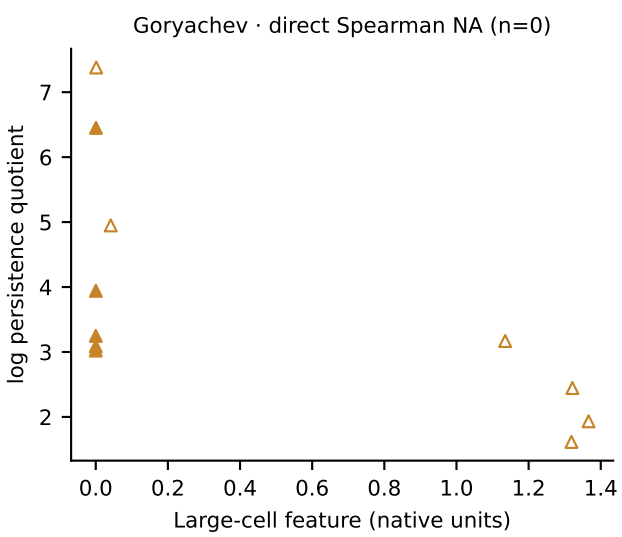
S10 · mean absolute turn 4 frames

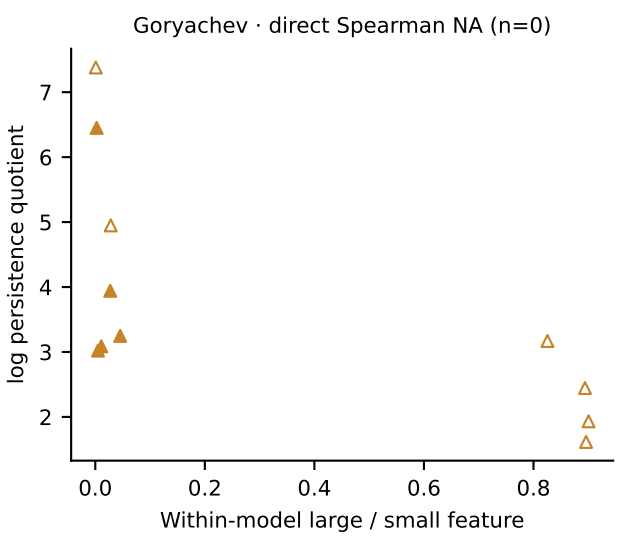


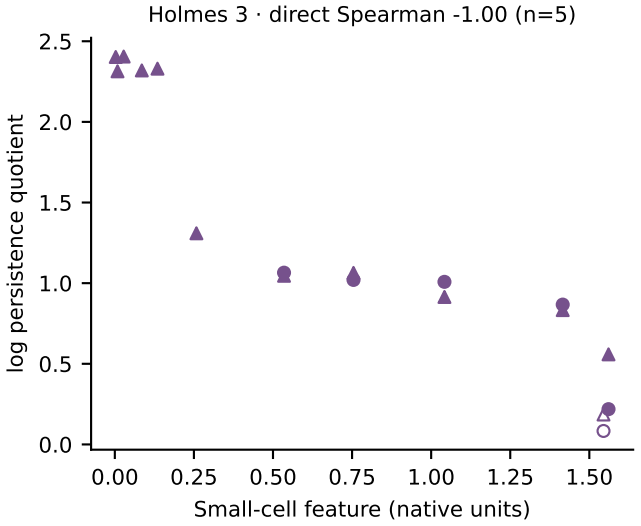


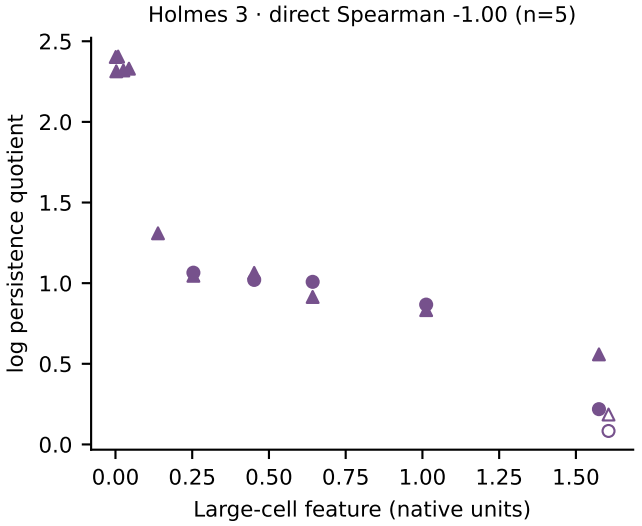


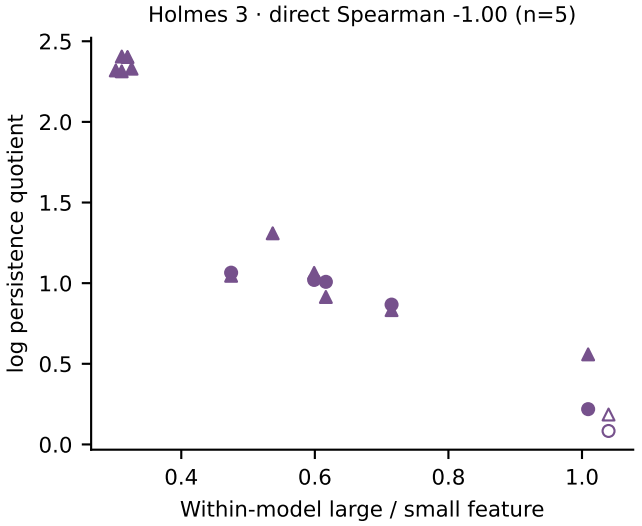


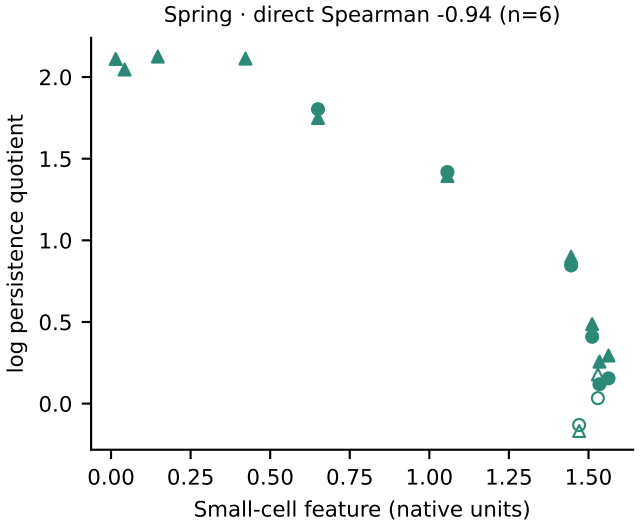


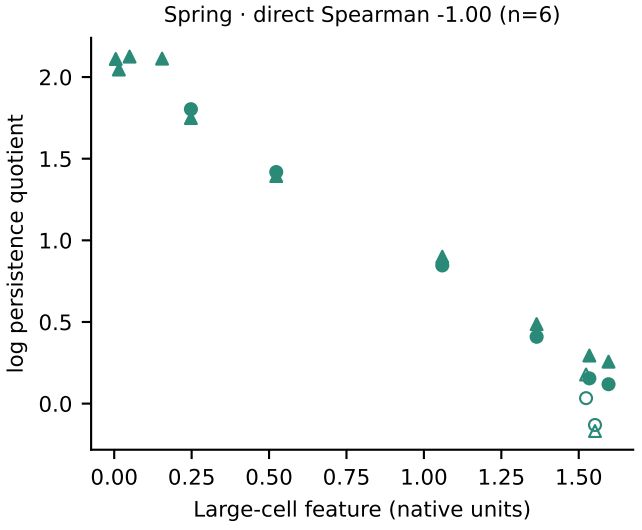


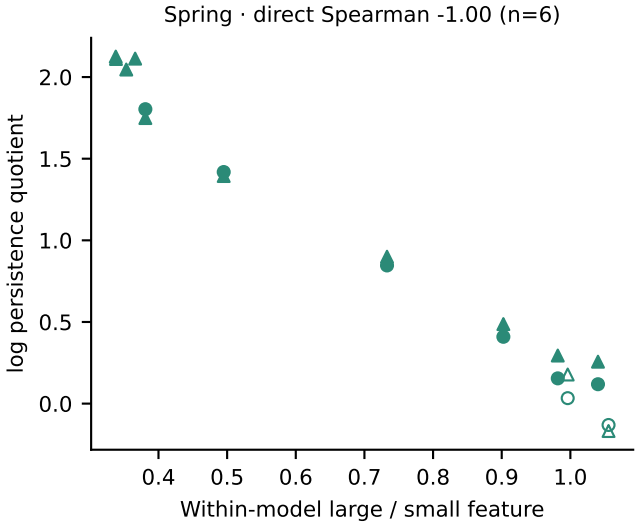












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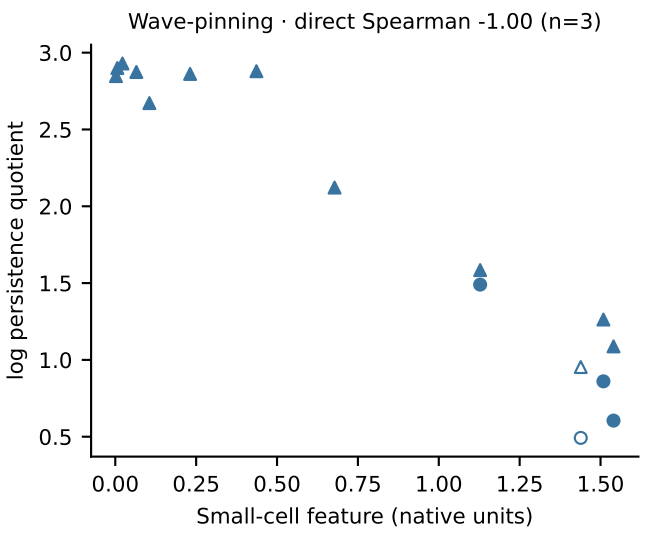
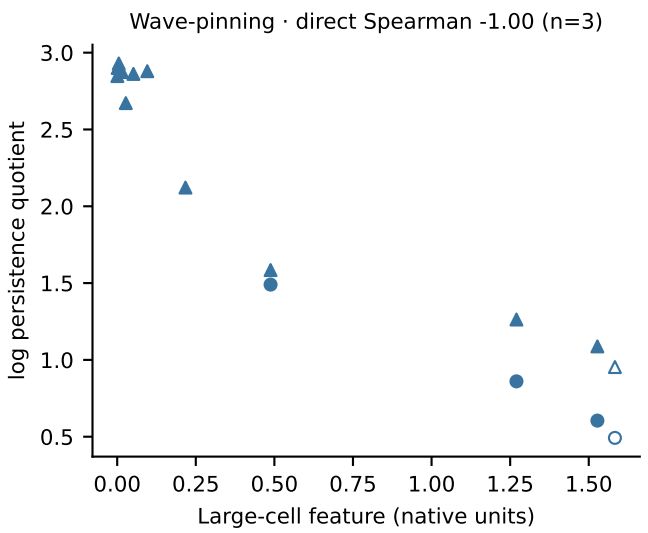
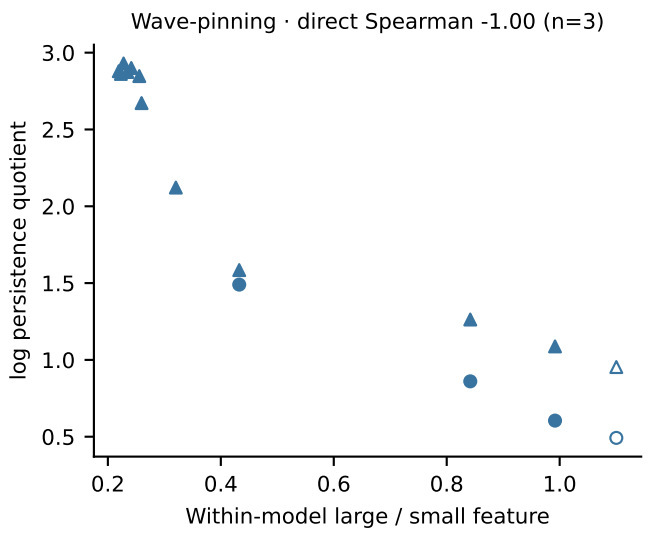
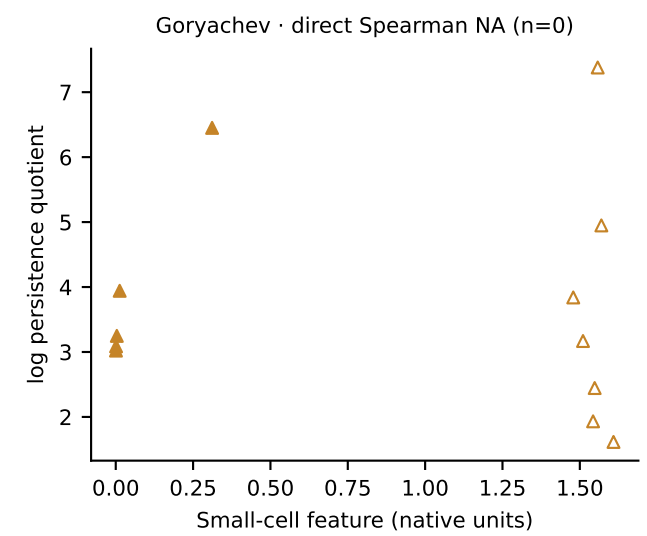
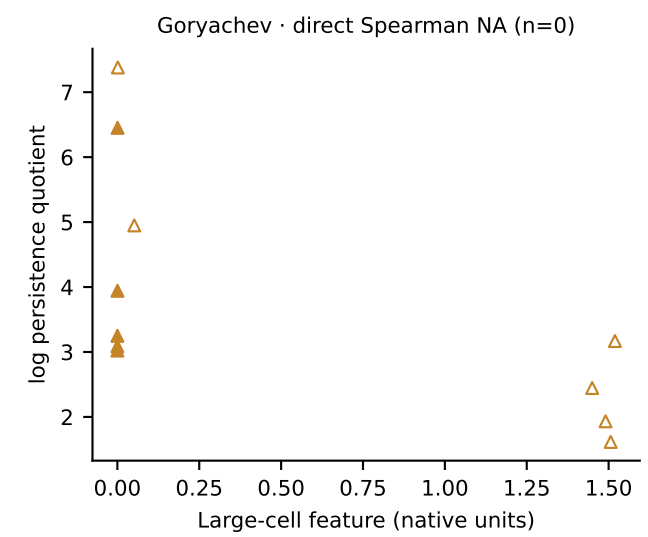
S10 · mean absolute turn 8 frames

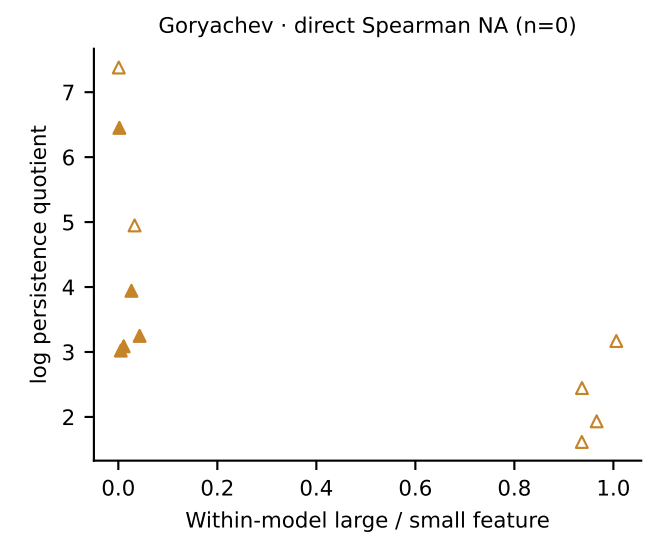


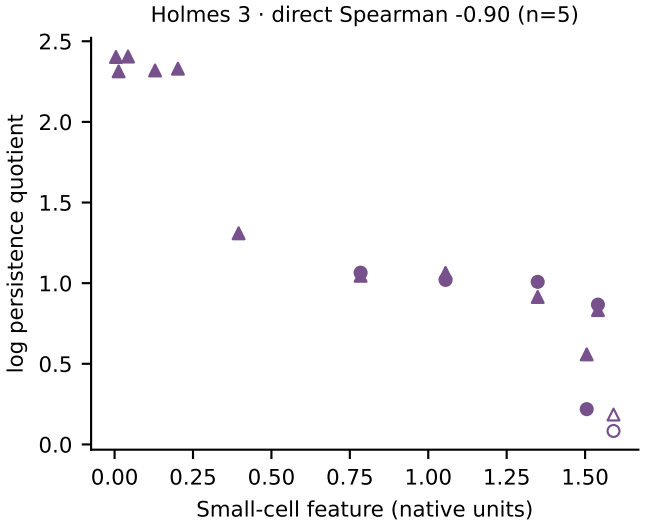


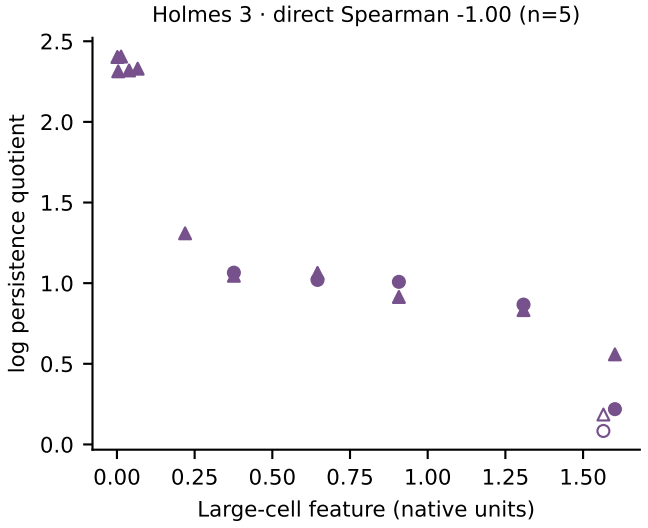


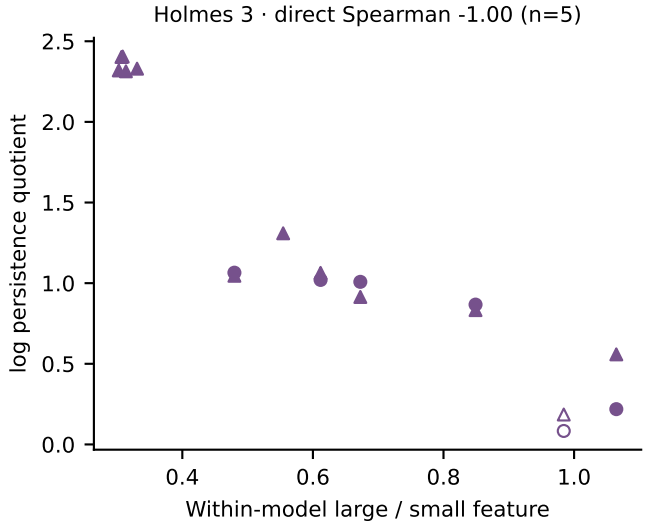


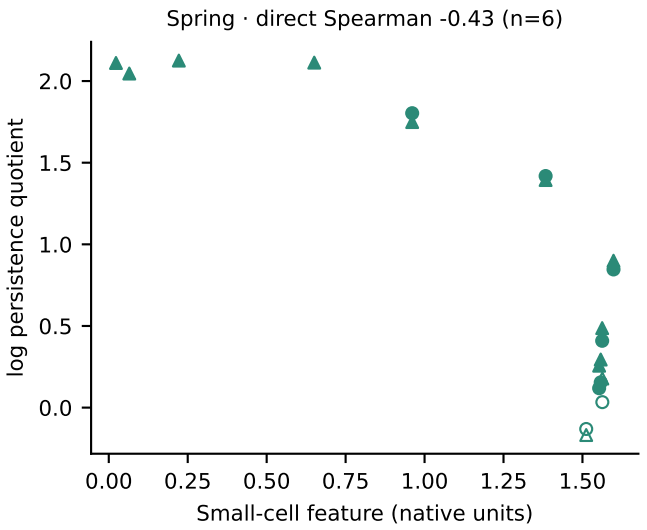


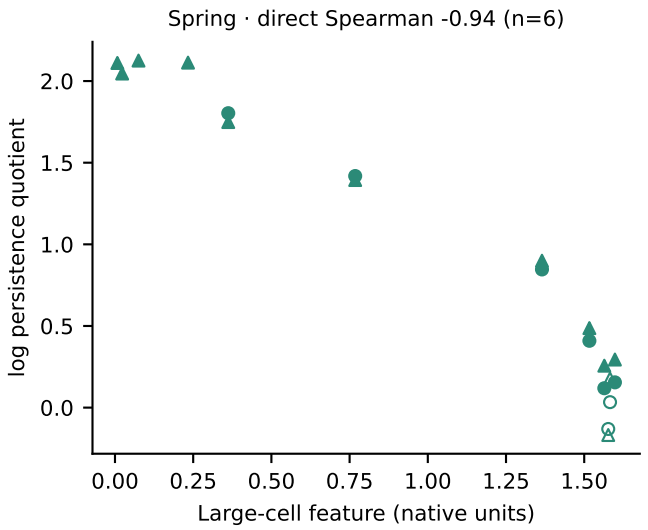


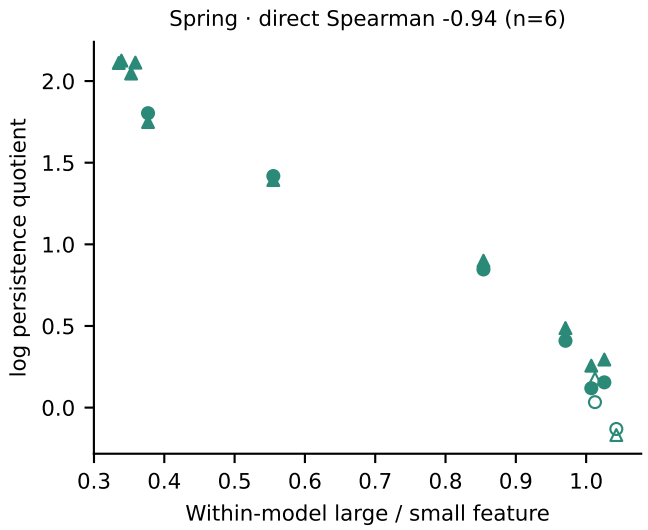












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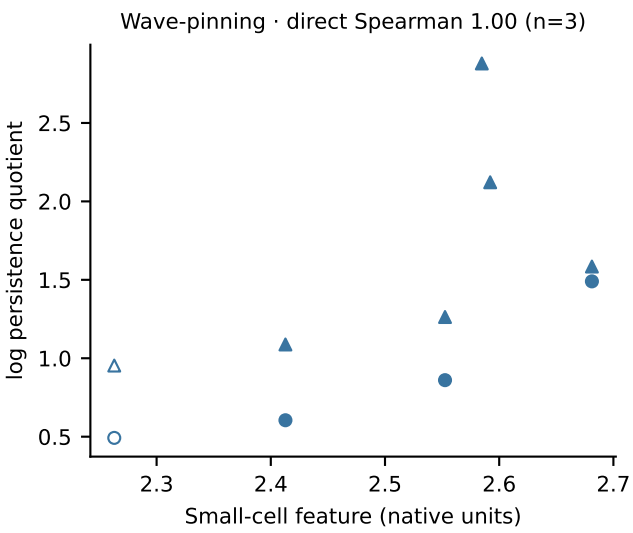
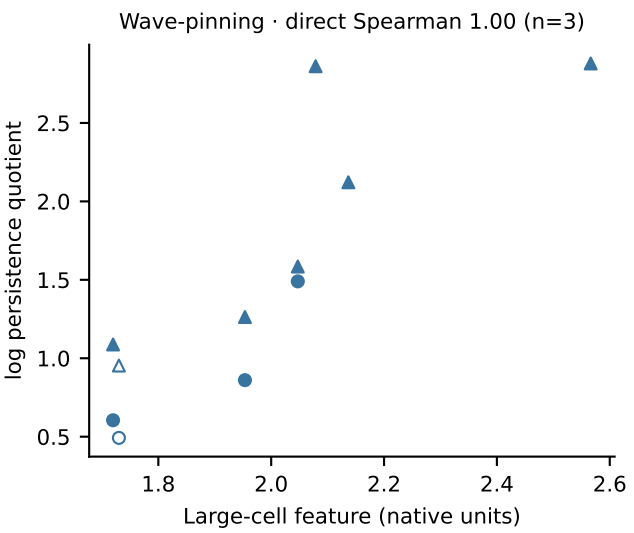
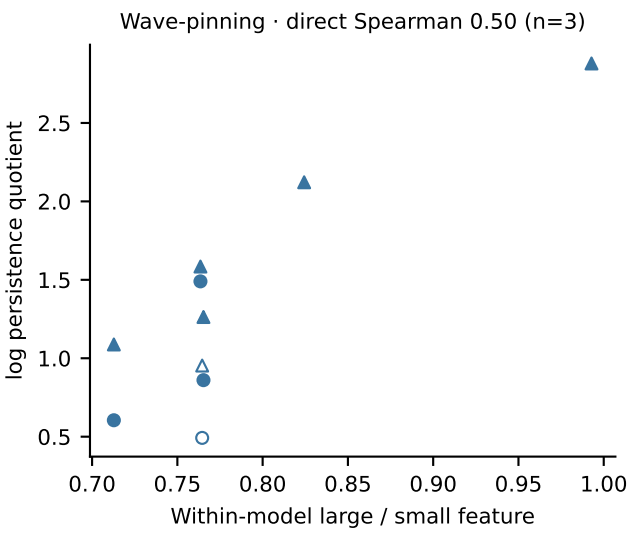
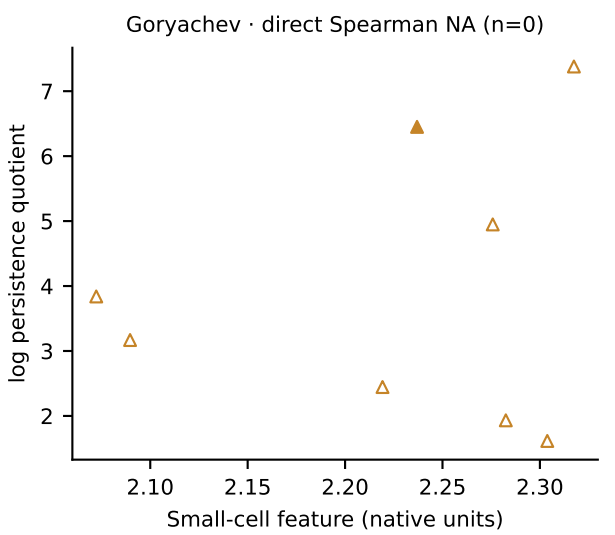
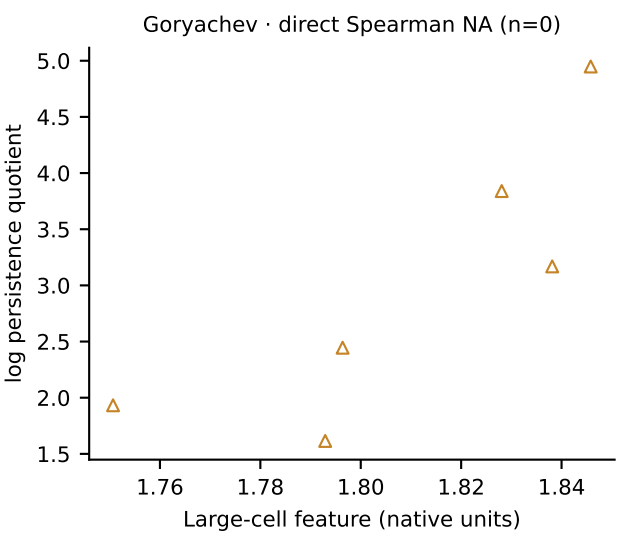
S10 · mean birth angle

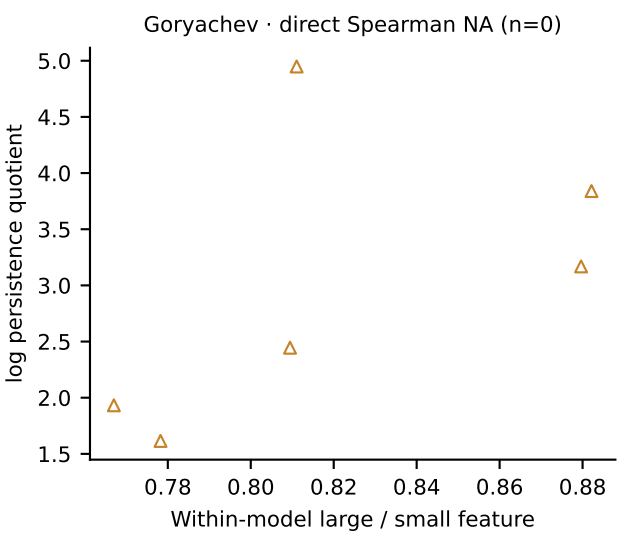


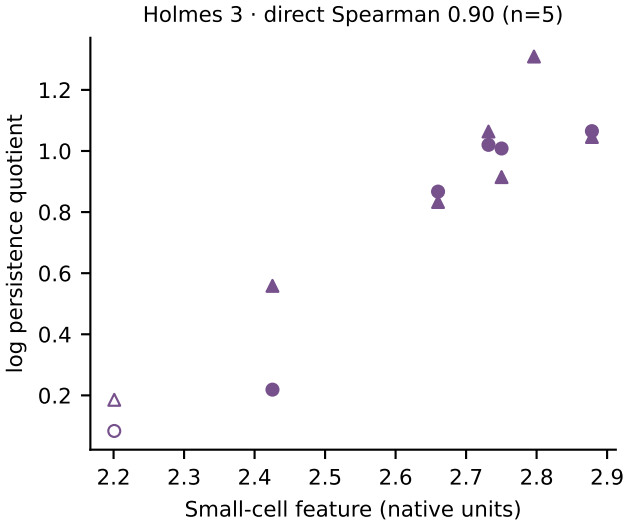


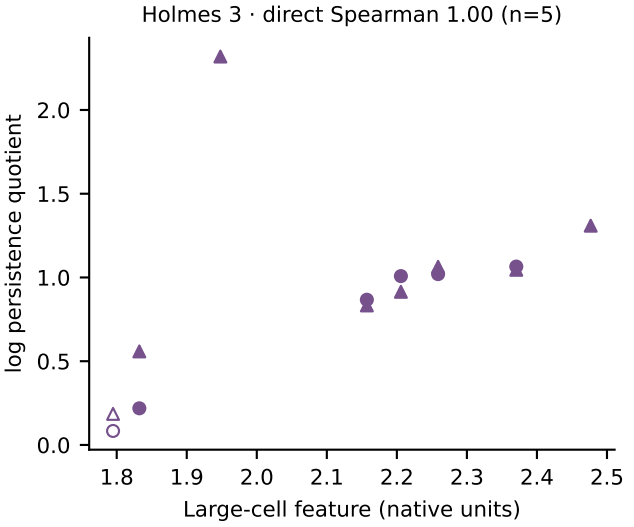


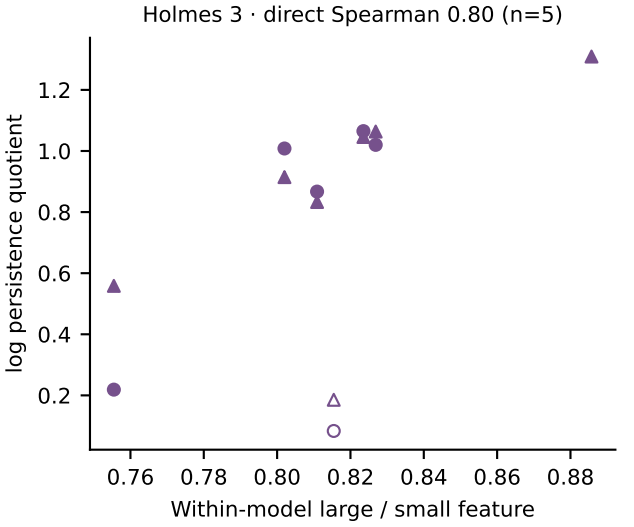


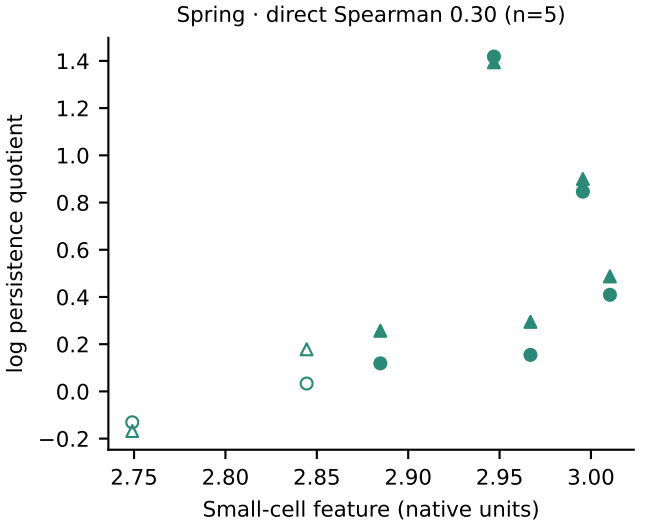


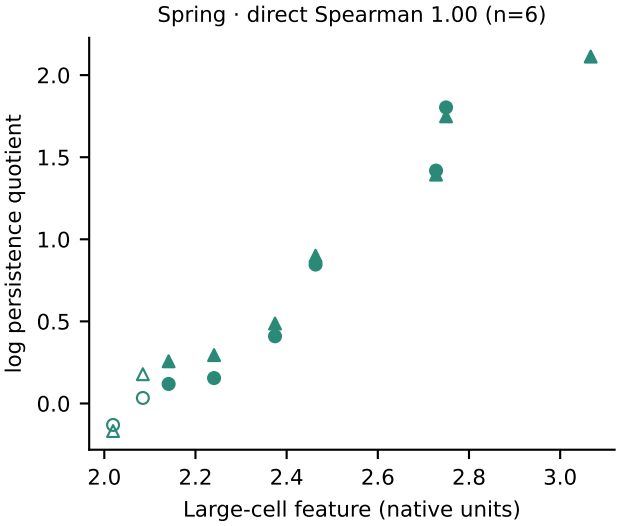


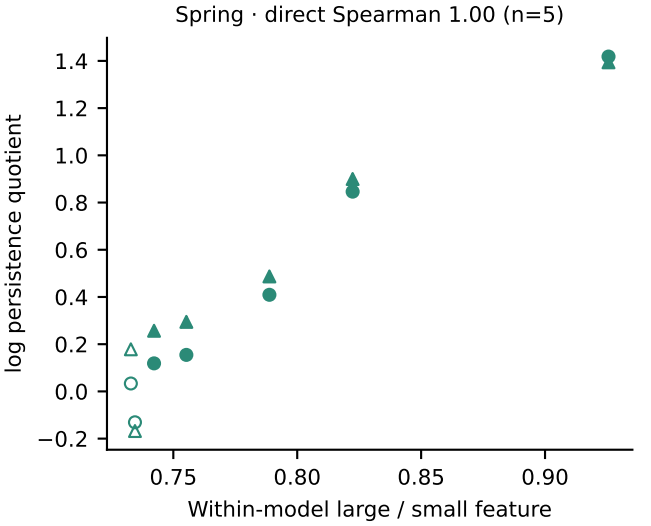












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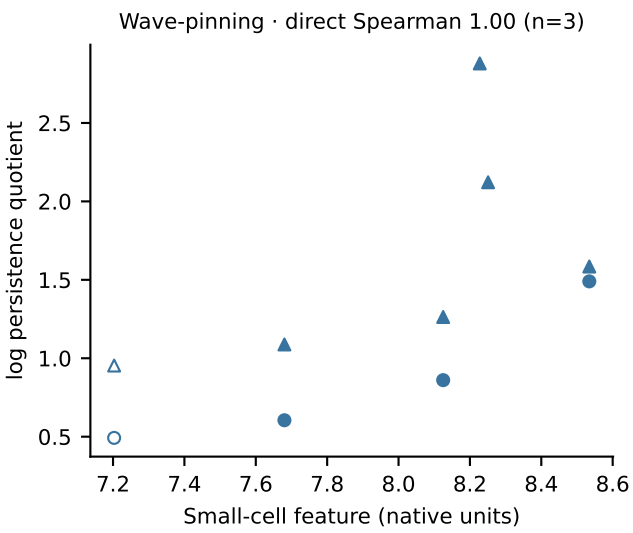
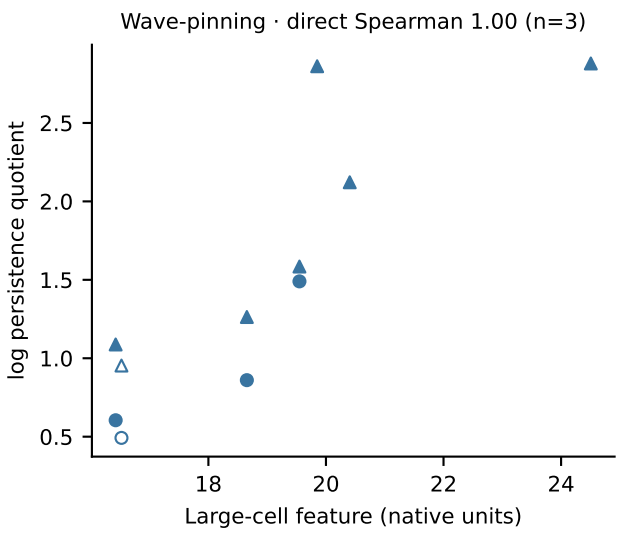
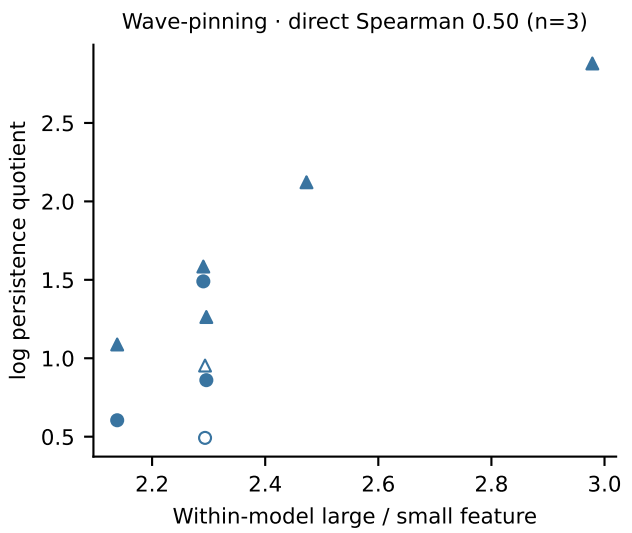
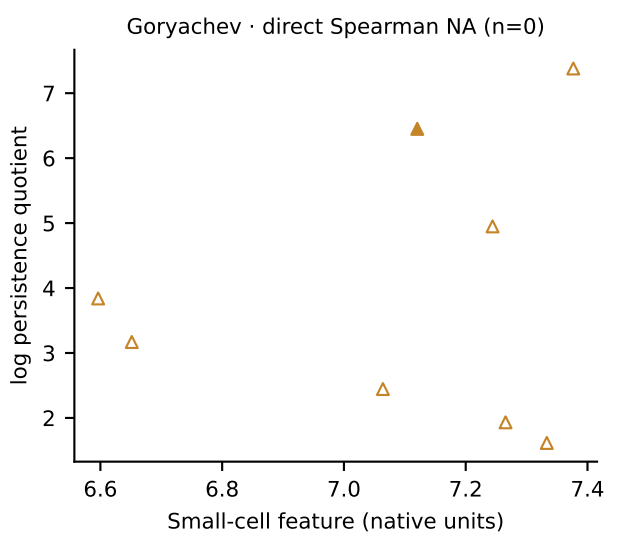
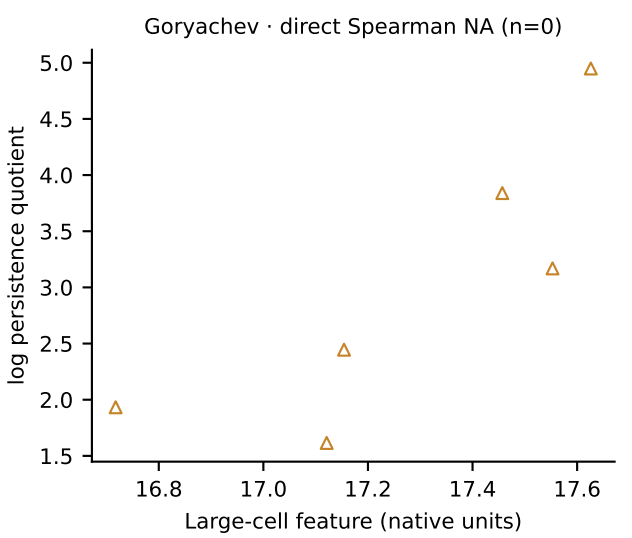
S10 · mean birth physical distance

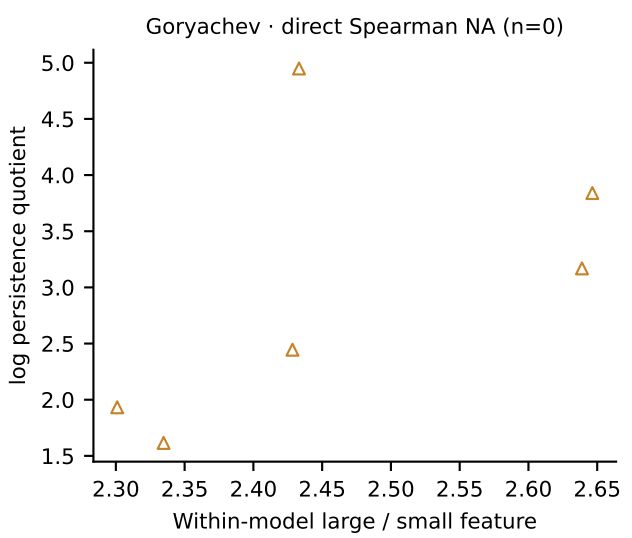


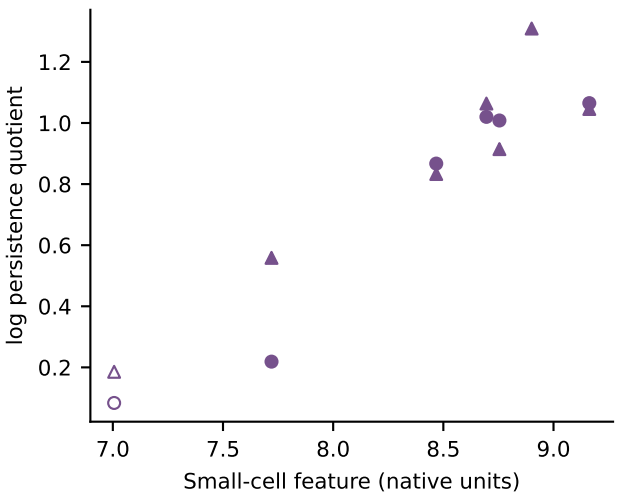


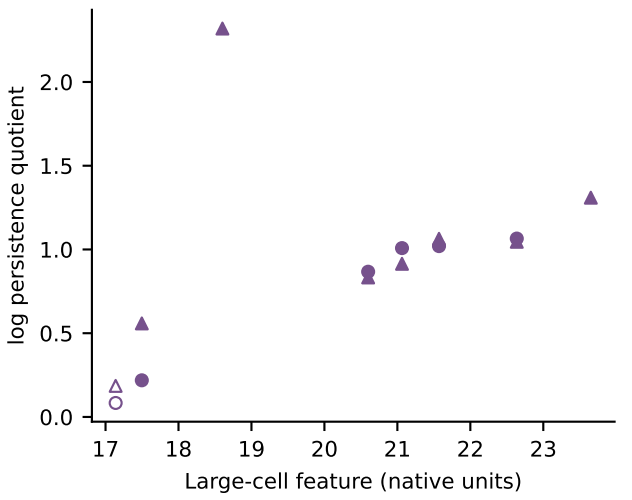


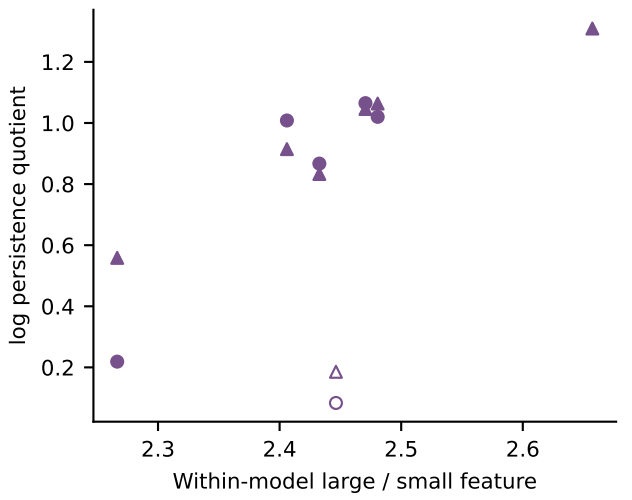


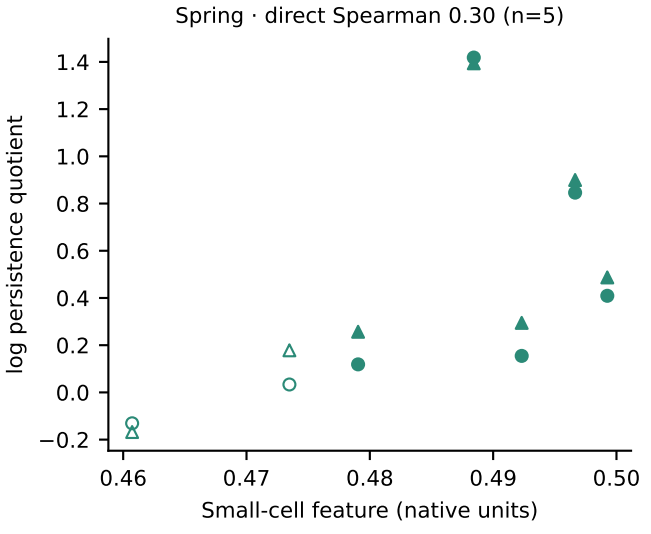


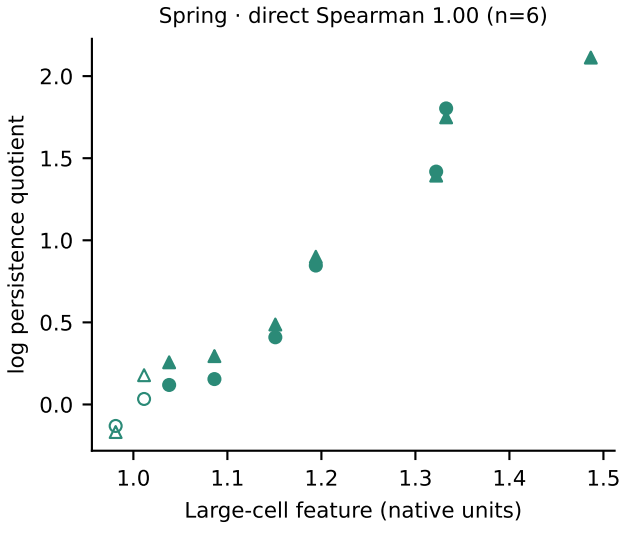


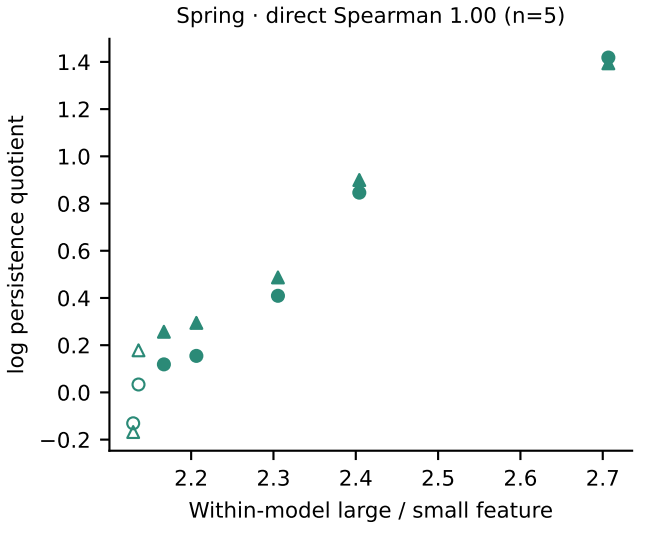












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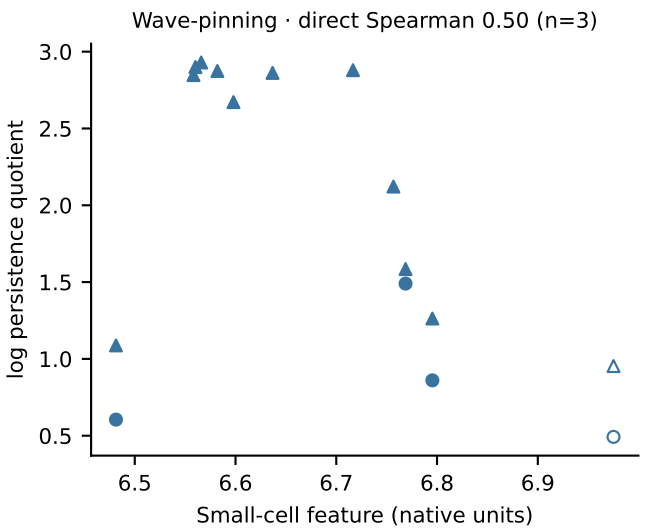
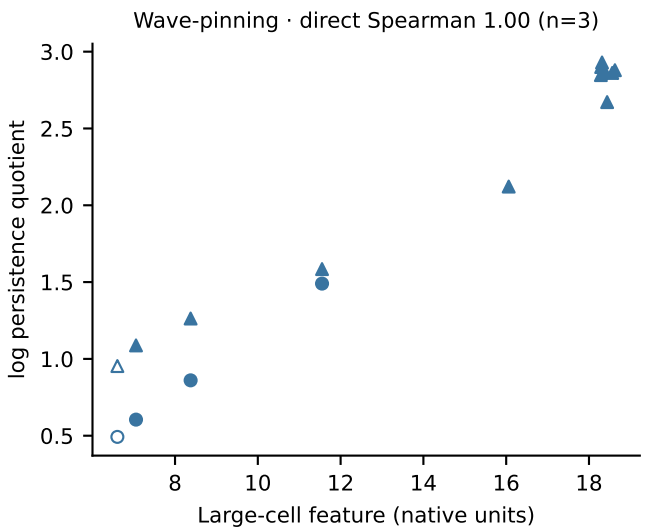
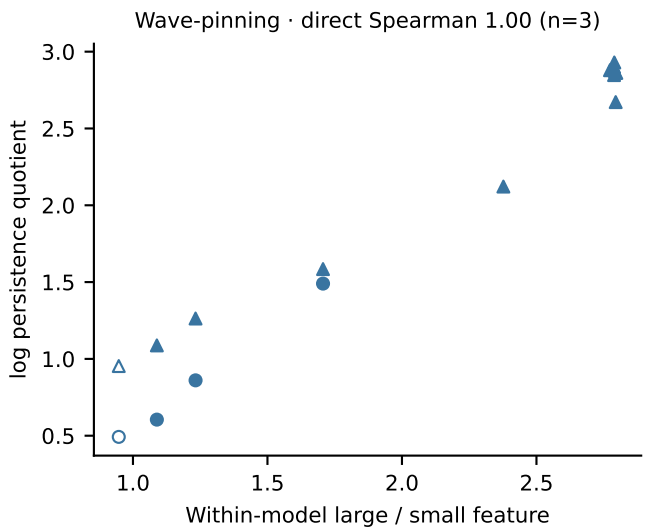
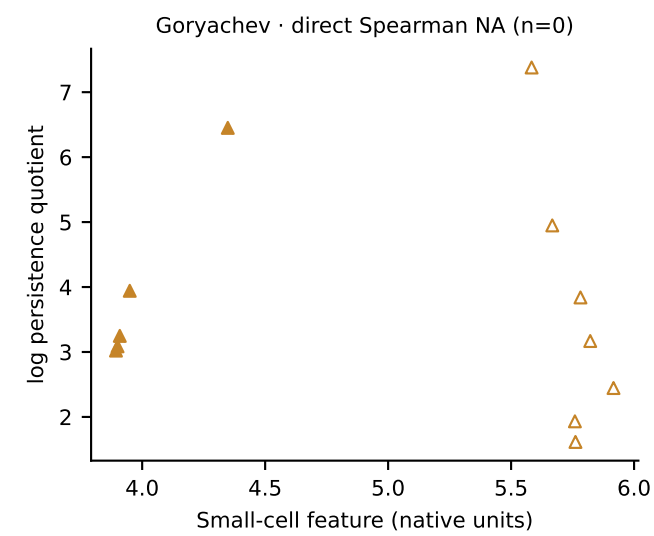
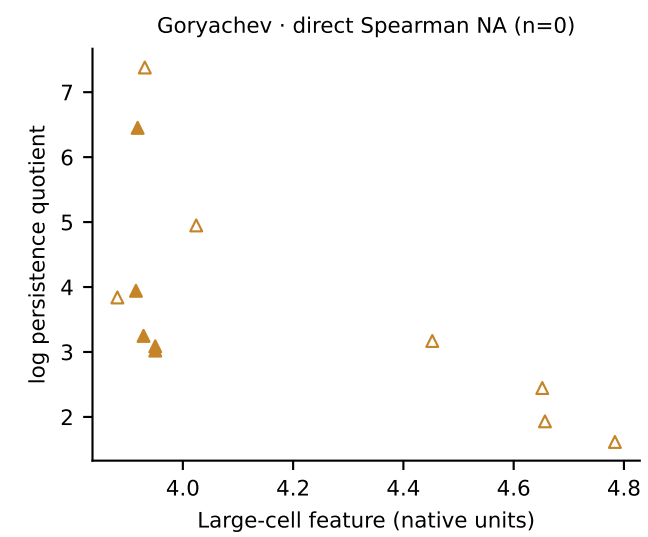
S10 · mean region width

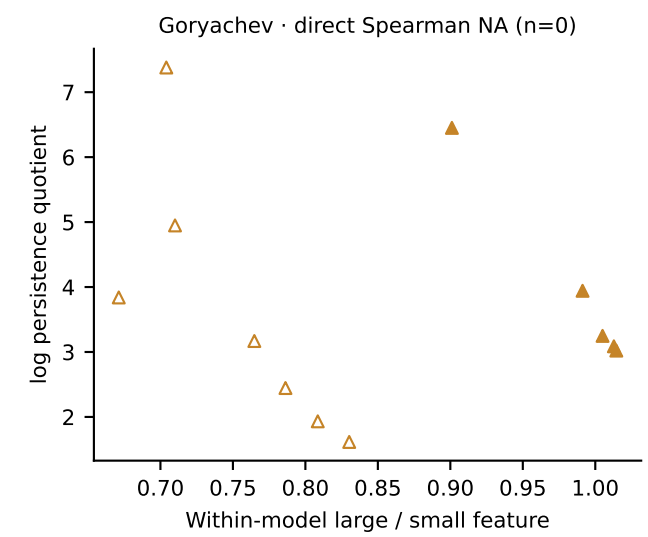


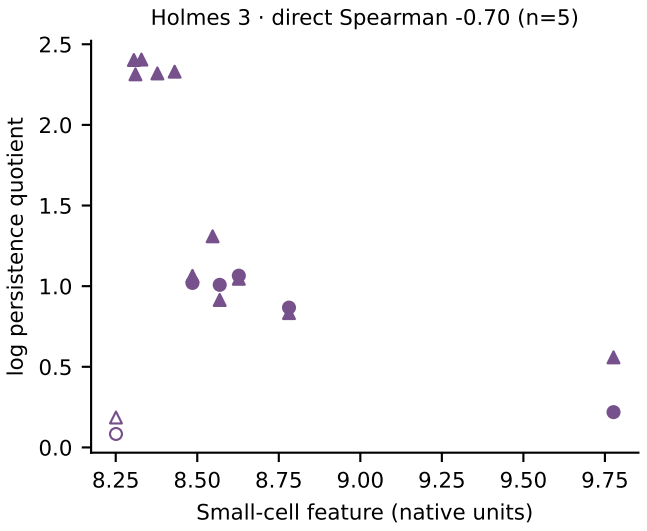


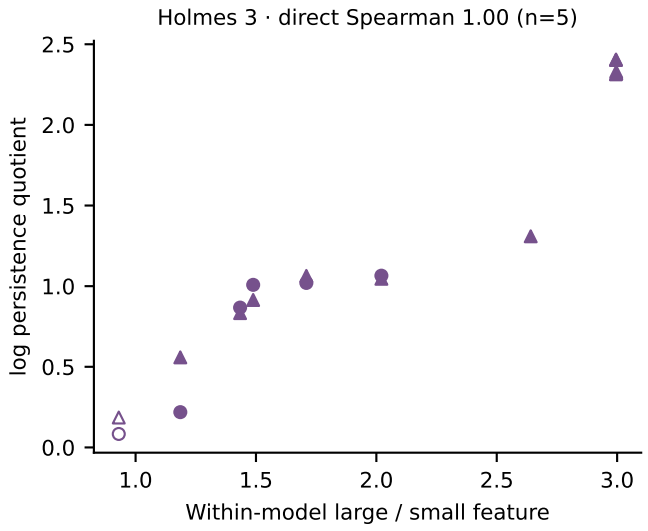


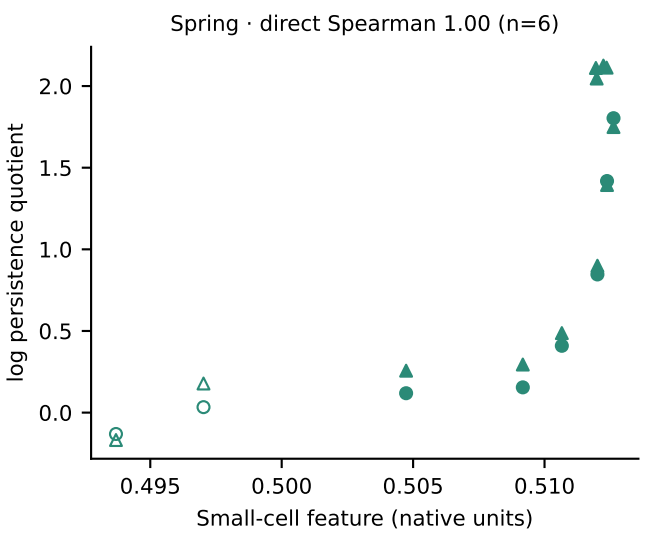


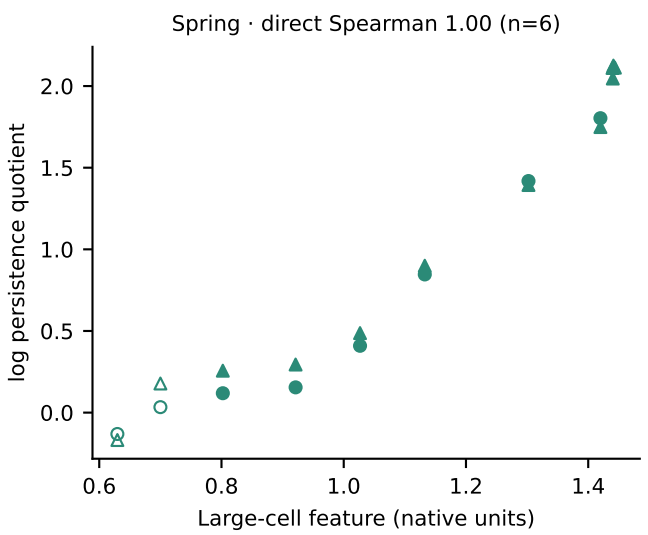


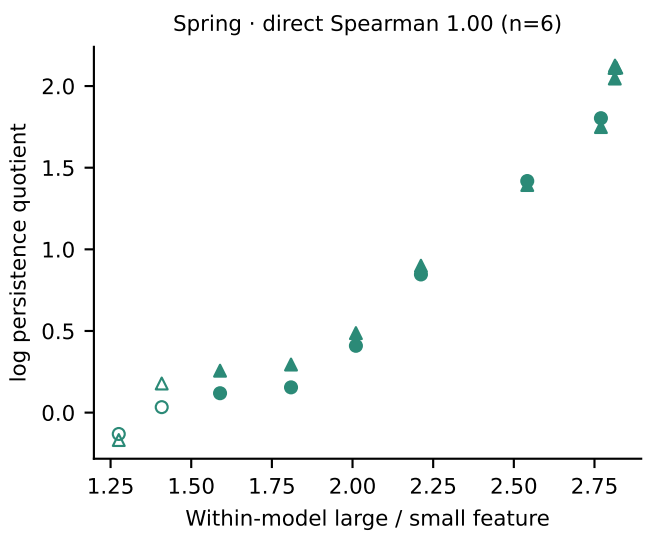












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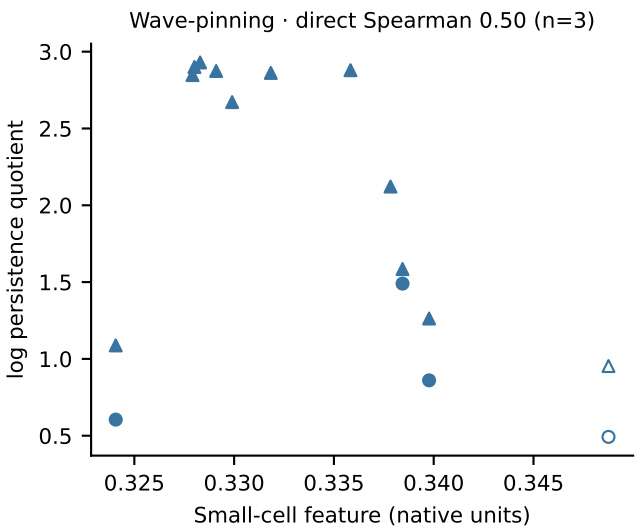
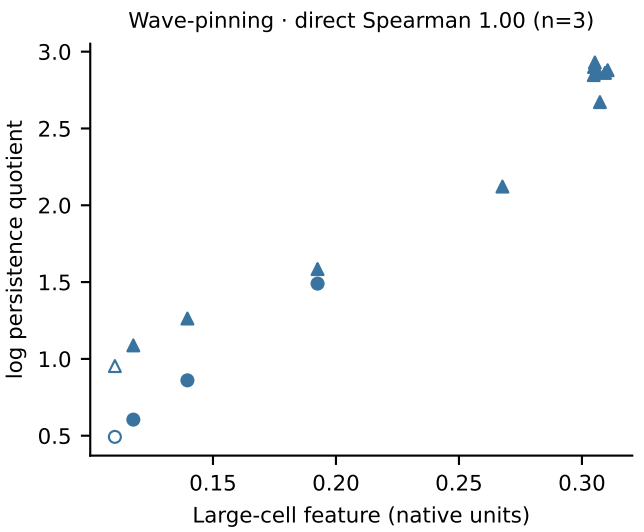
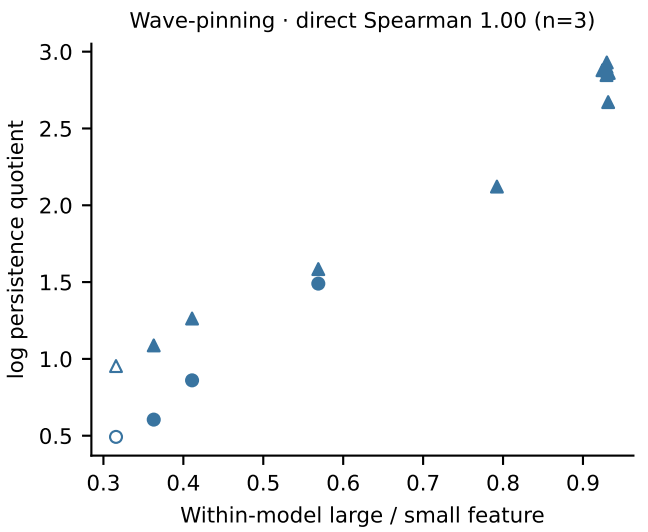
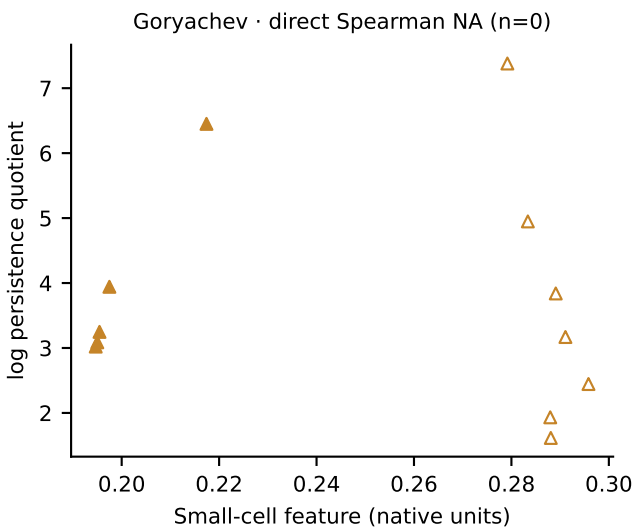
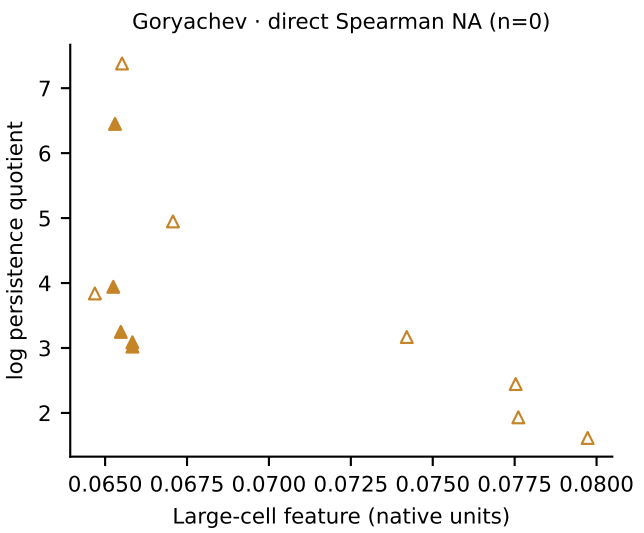
S10 · mean region width fraction

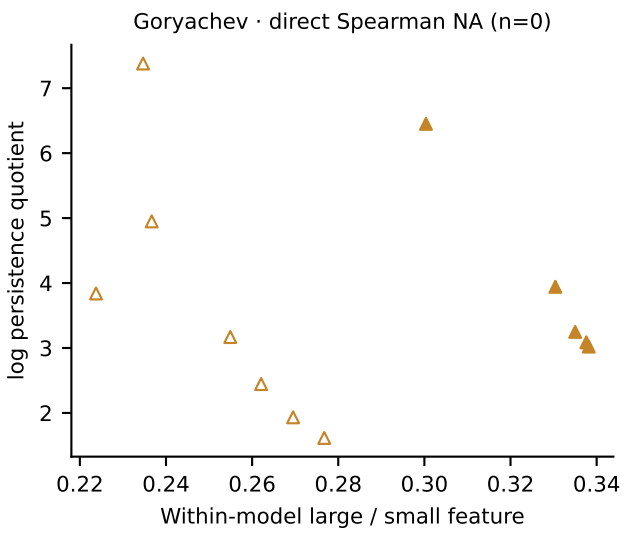


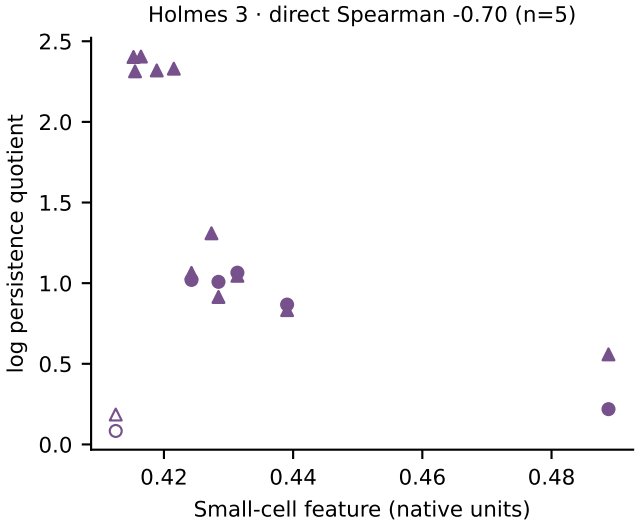


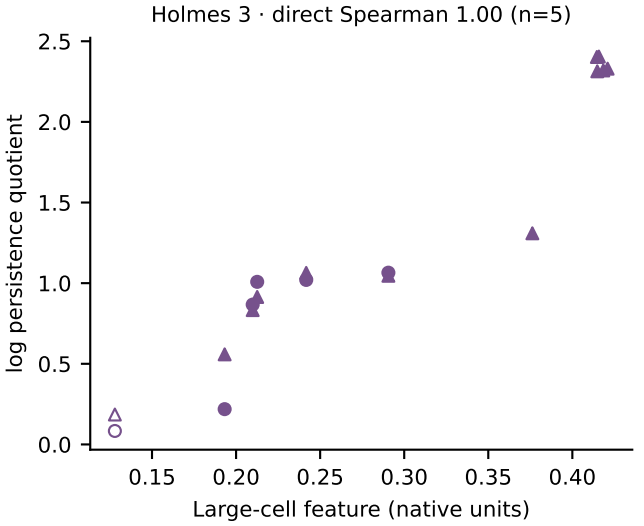


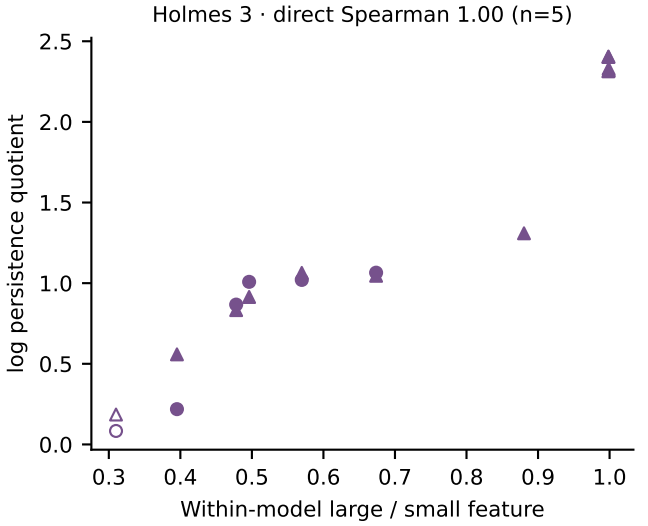


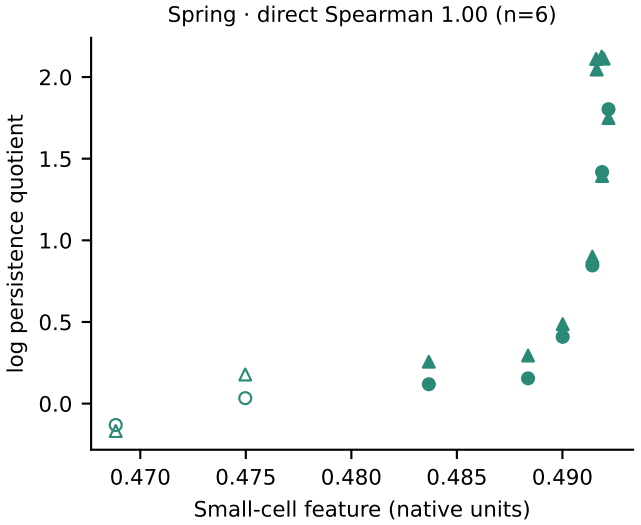


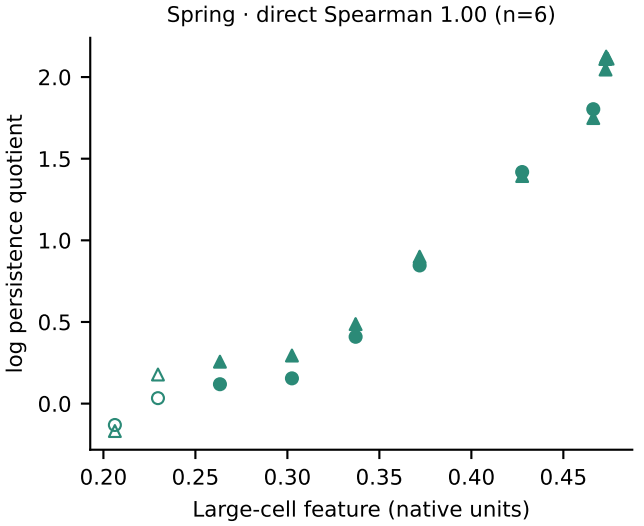


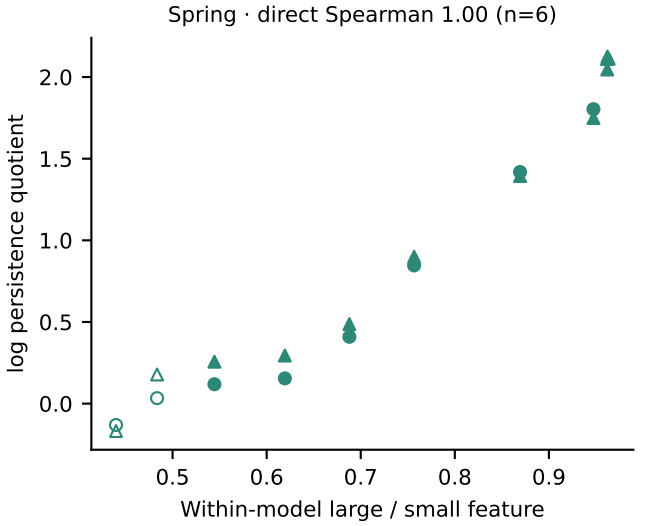












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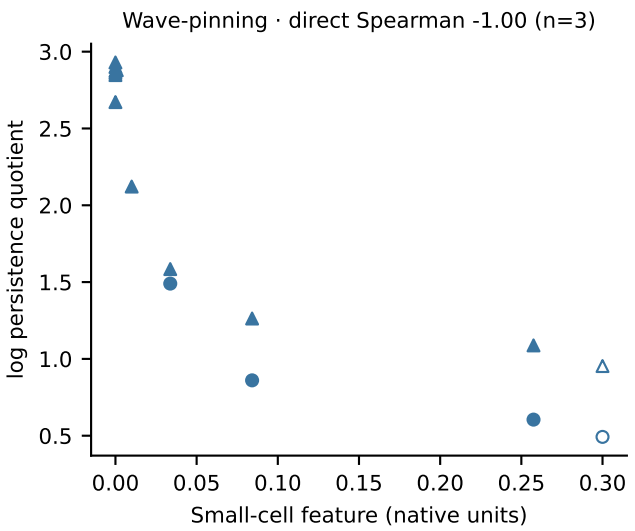
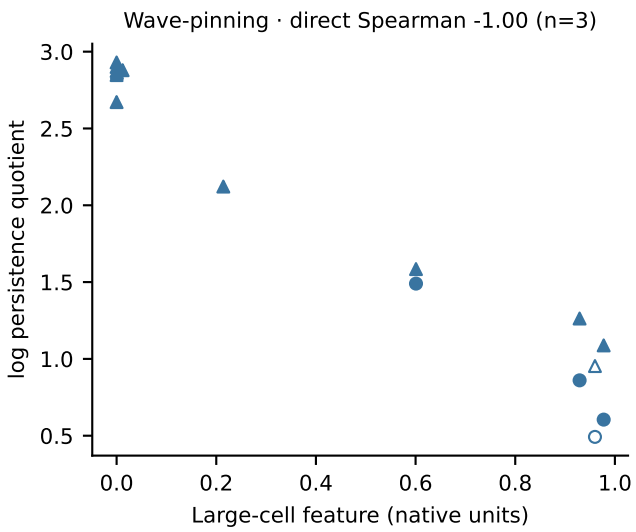
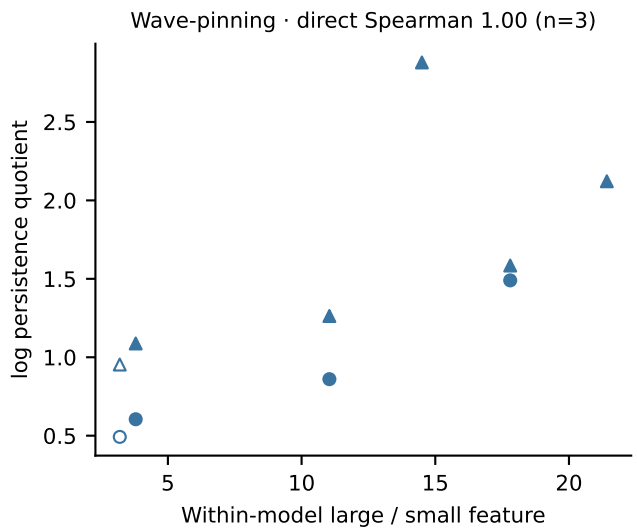
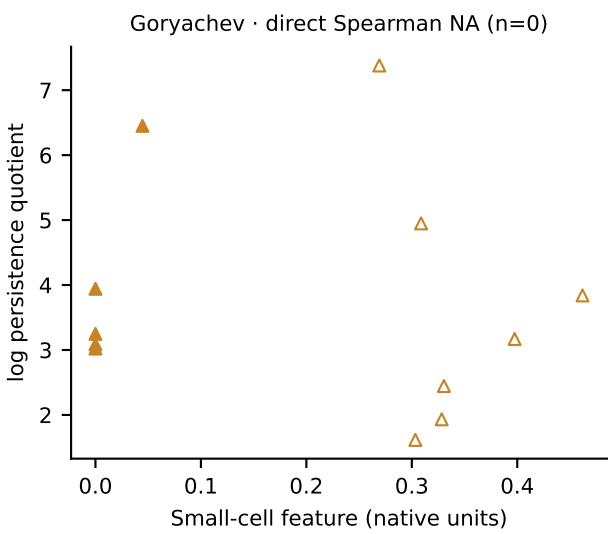
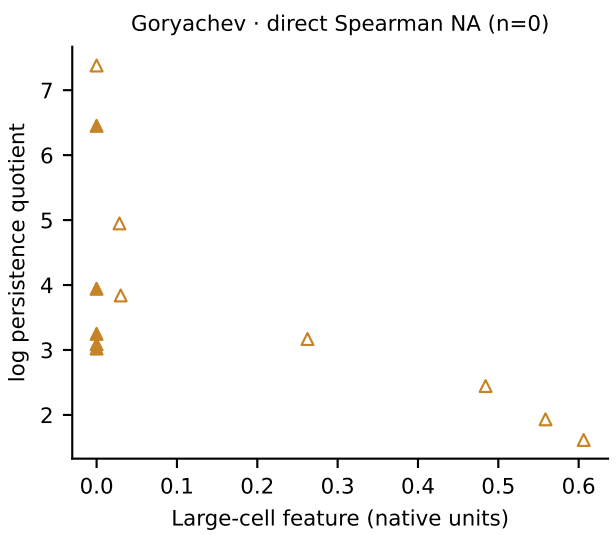
S10 · Multiple-front fraction

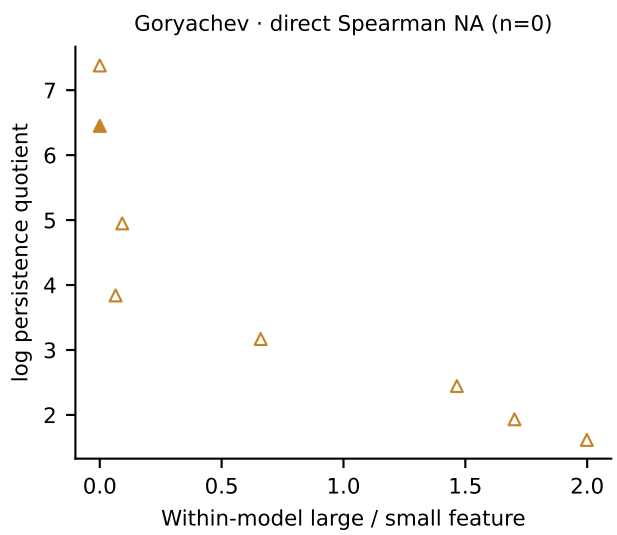


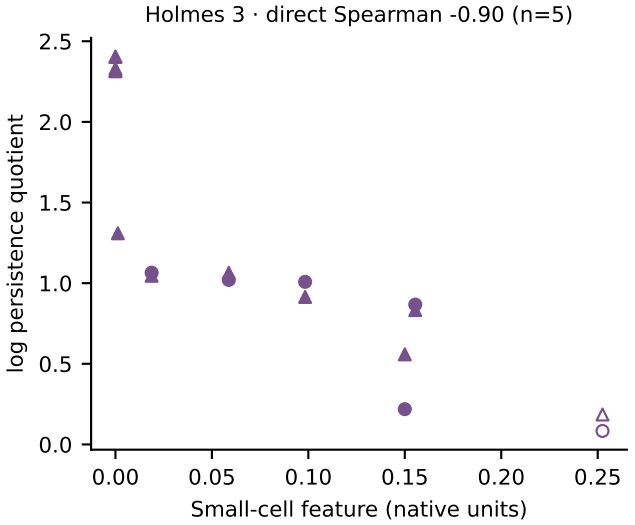


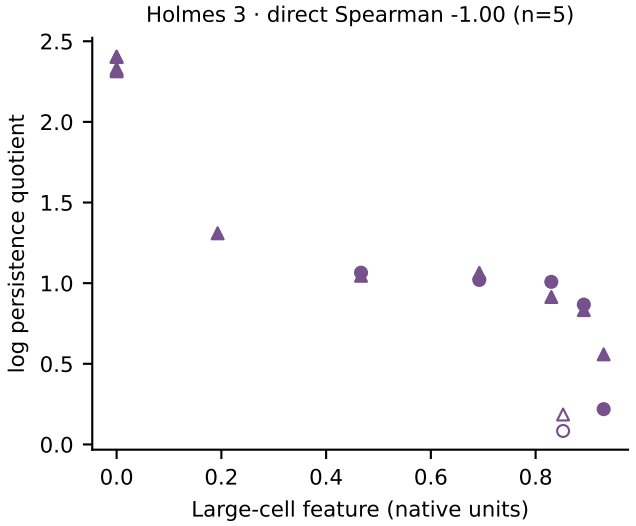


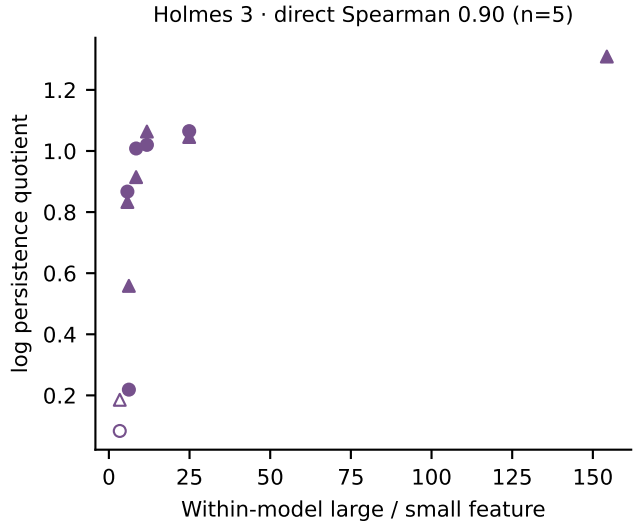


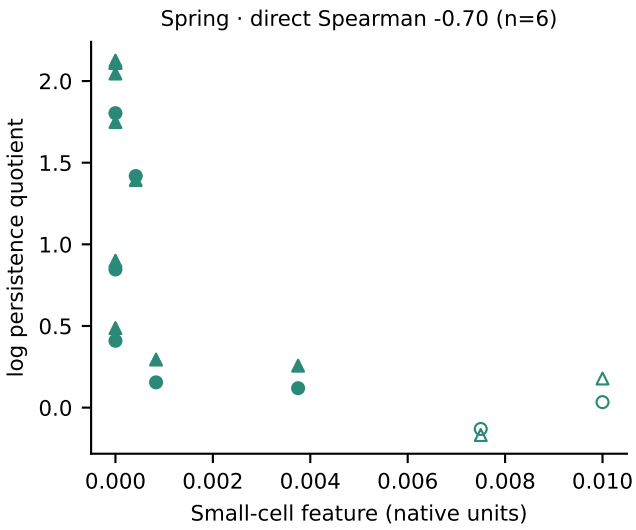


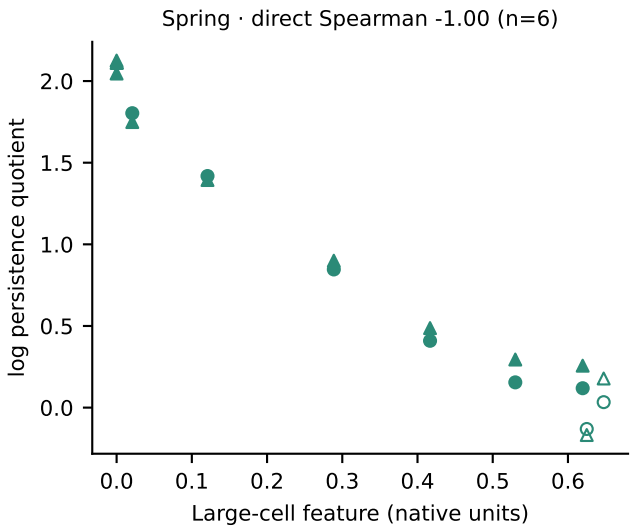


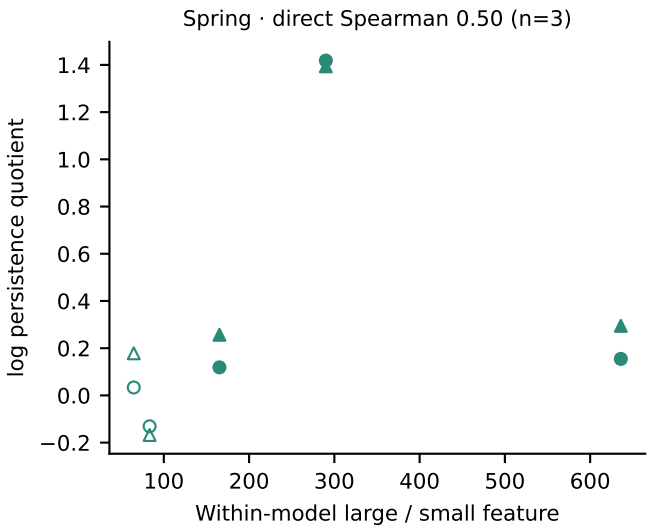












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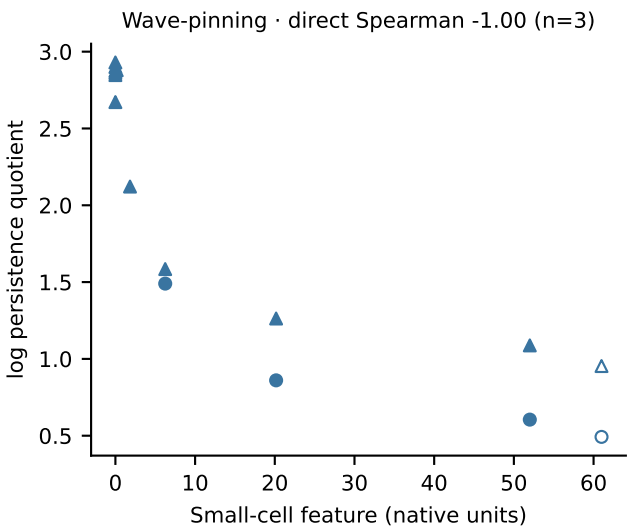
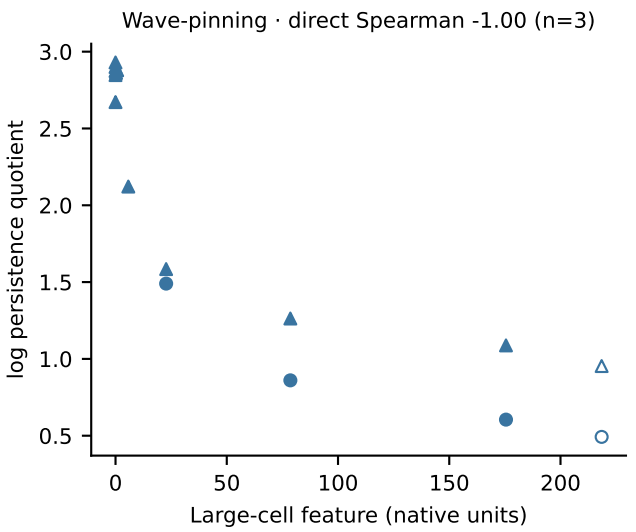
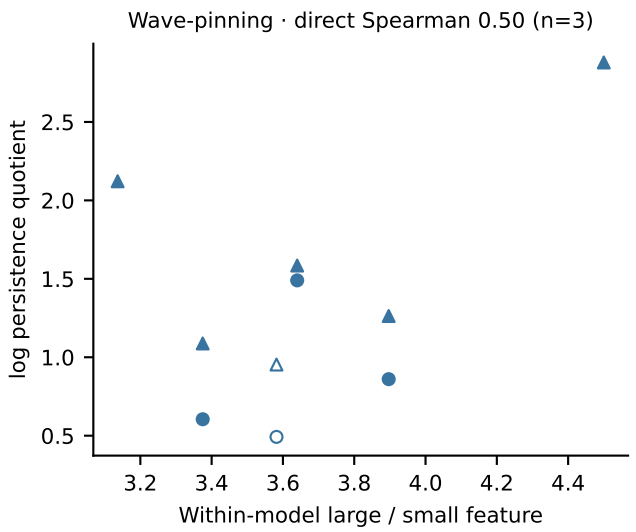
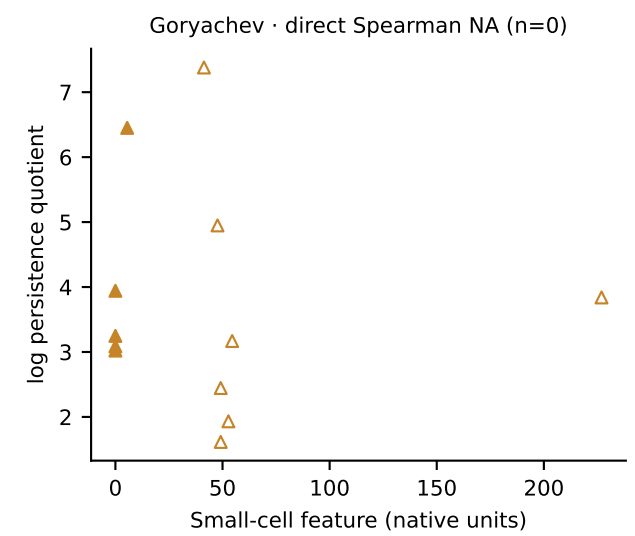
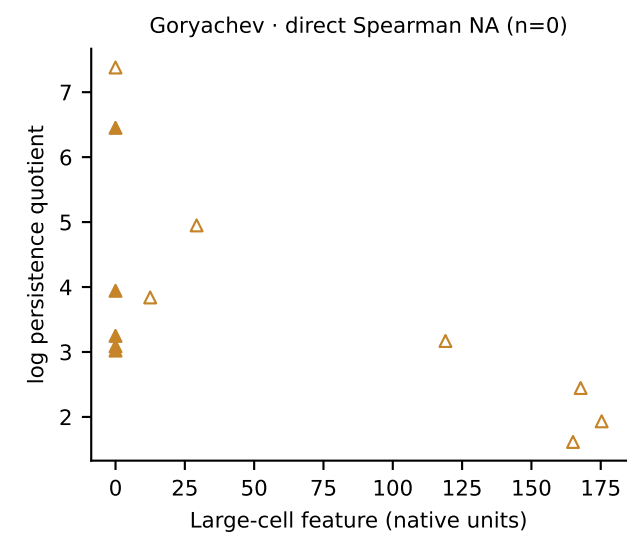
S10 · observed births

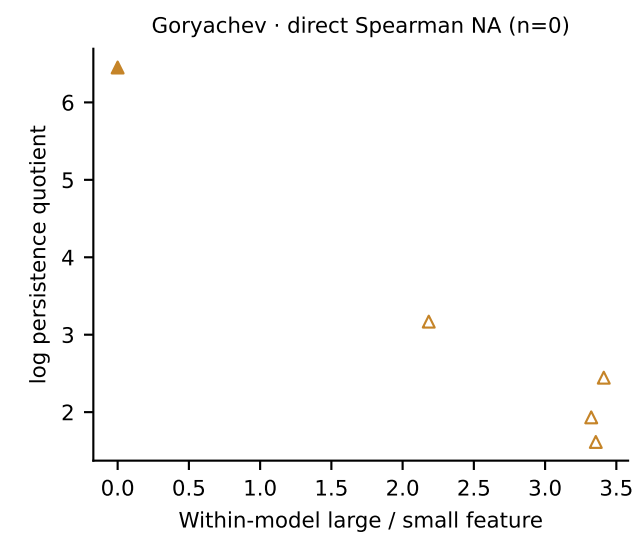


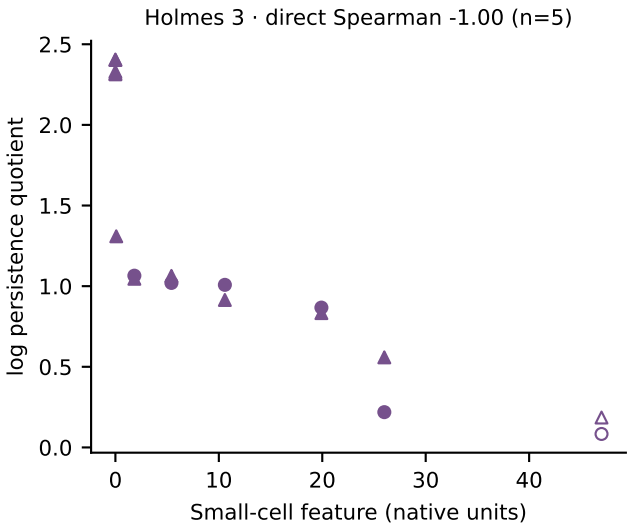


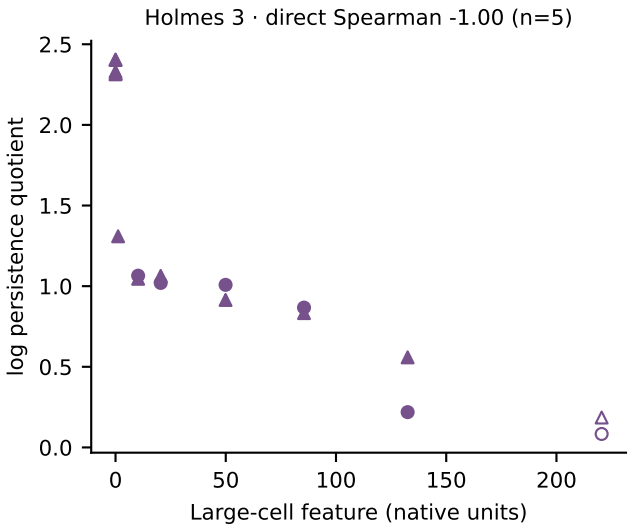


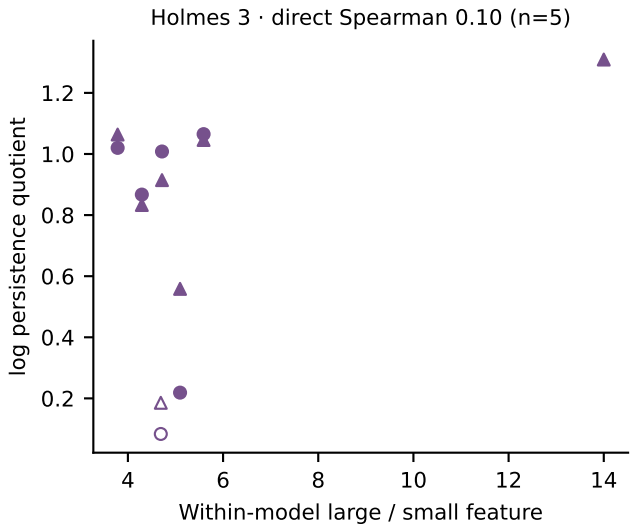


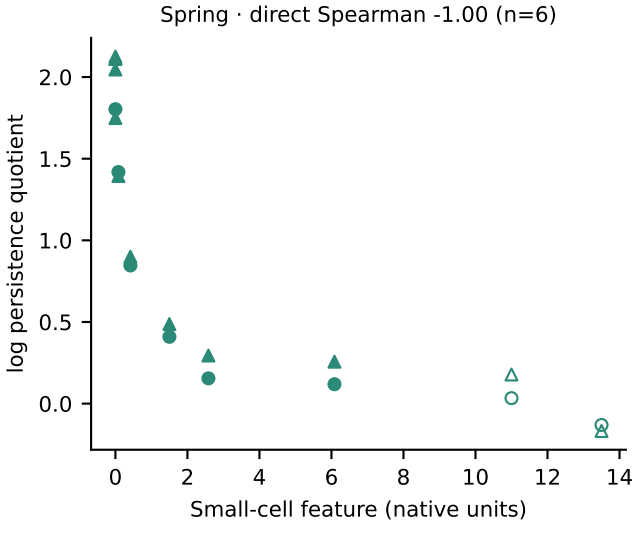


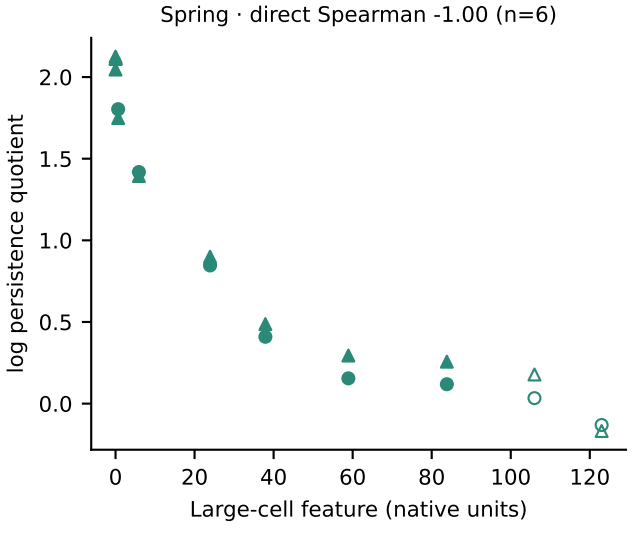


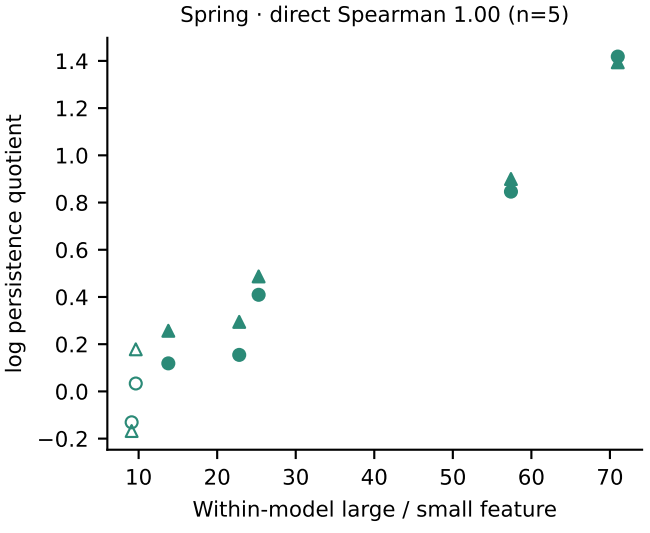












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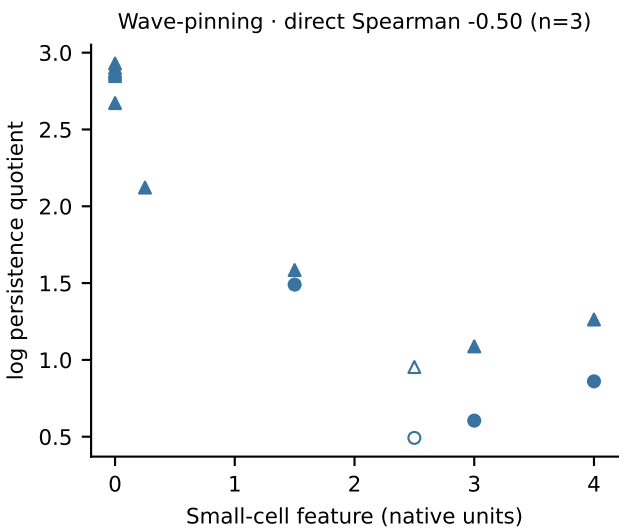
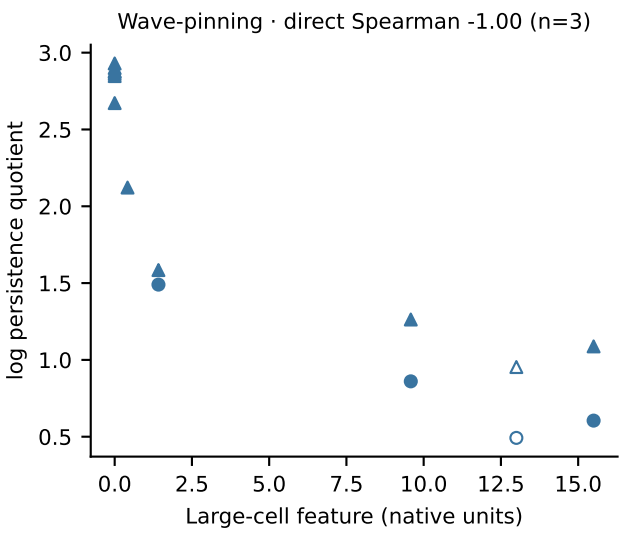
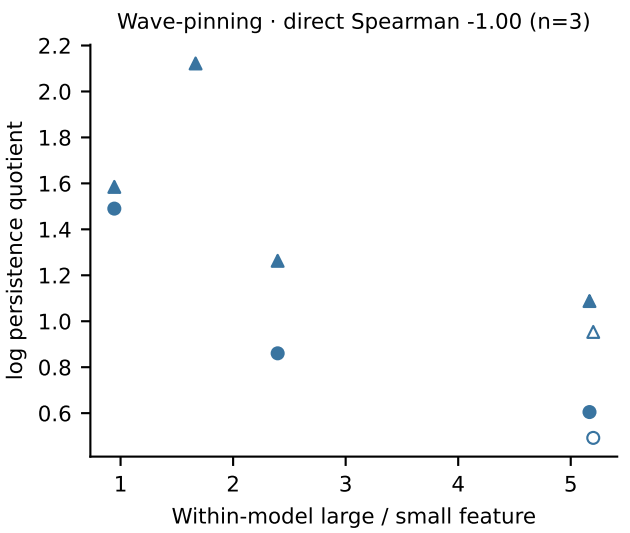
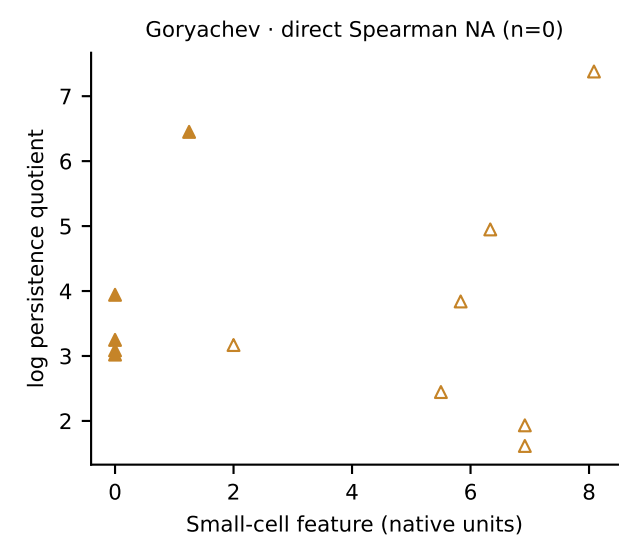
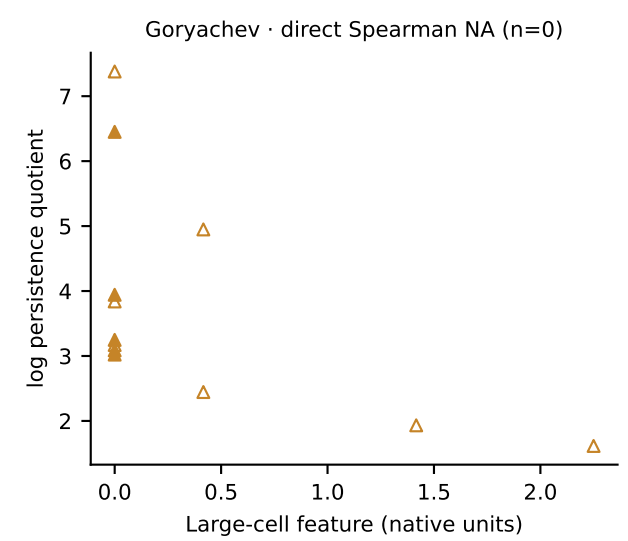
S10 · observed pairwise takeovers

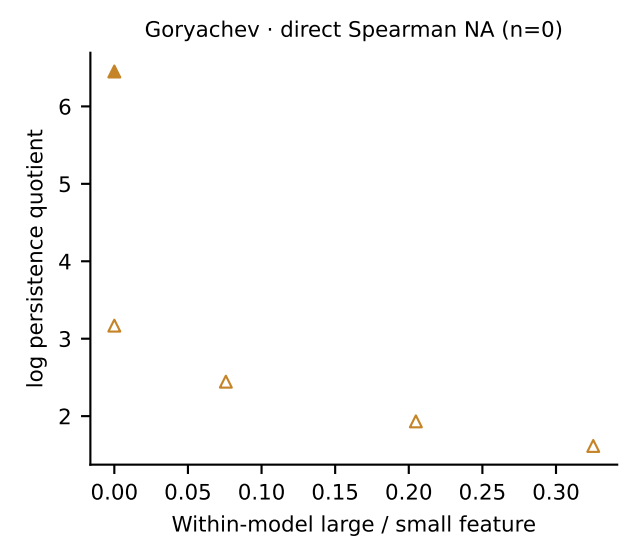


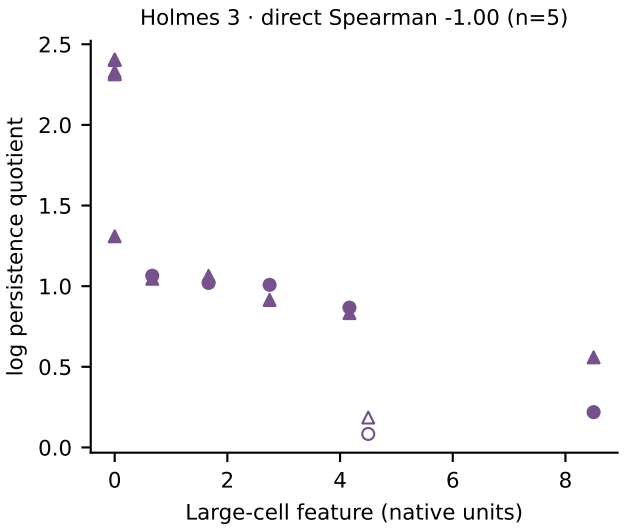


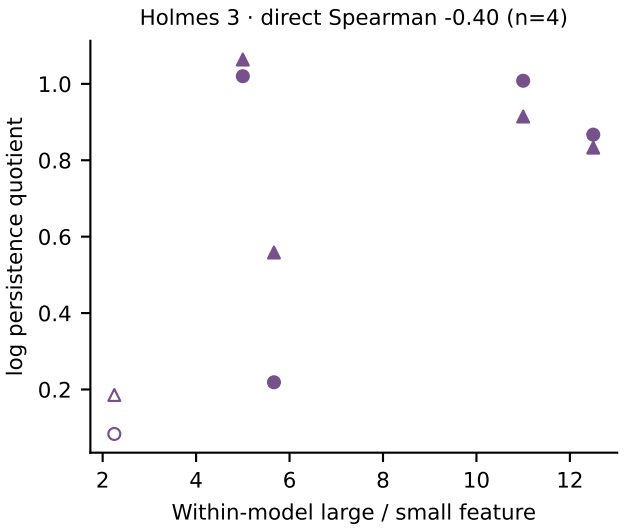


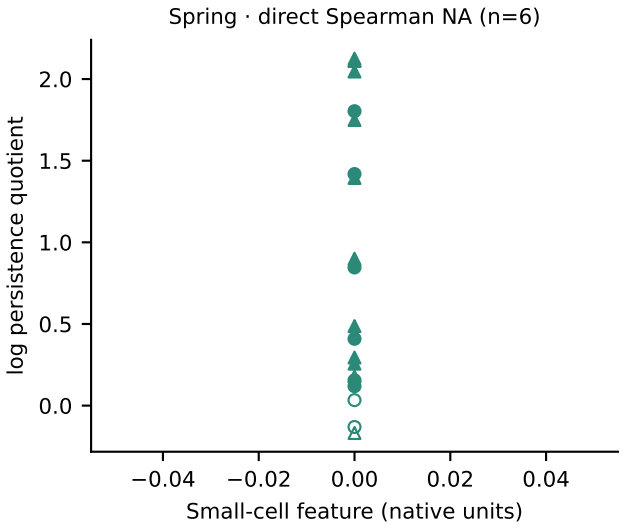


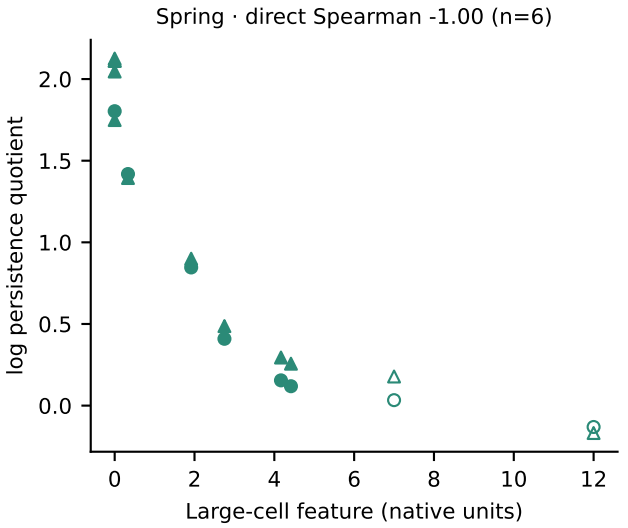


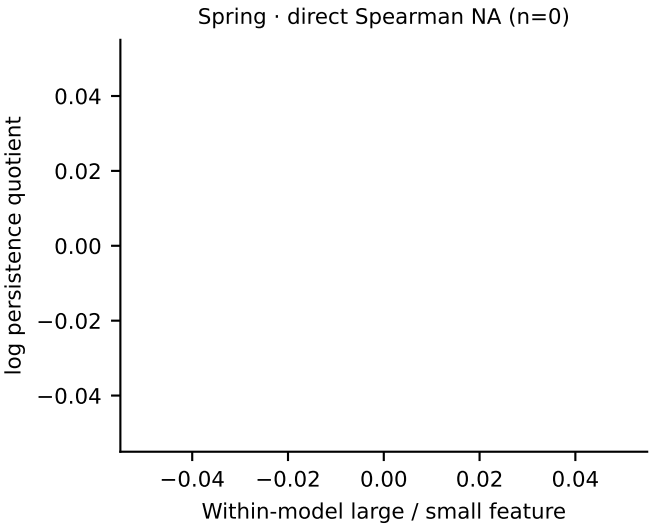












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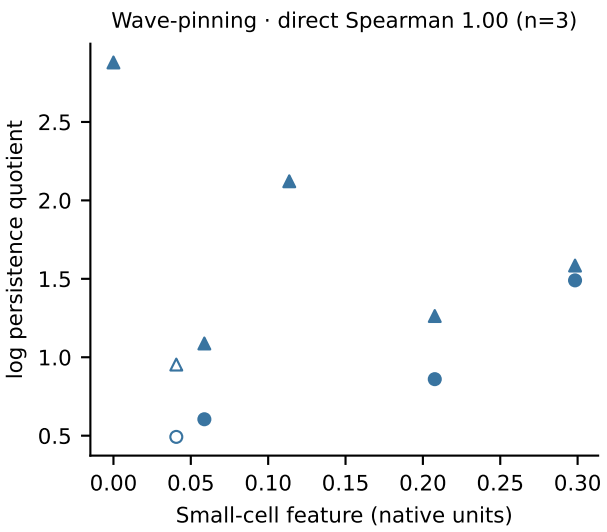
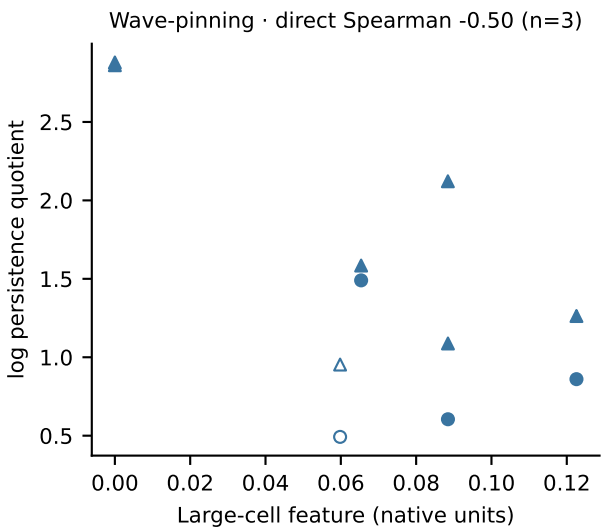
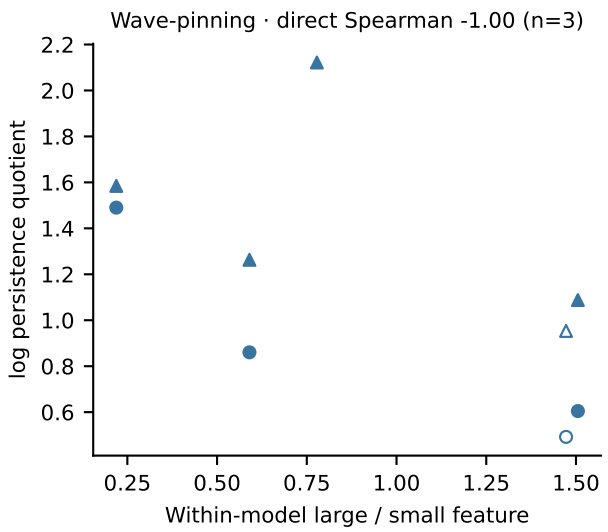
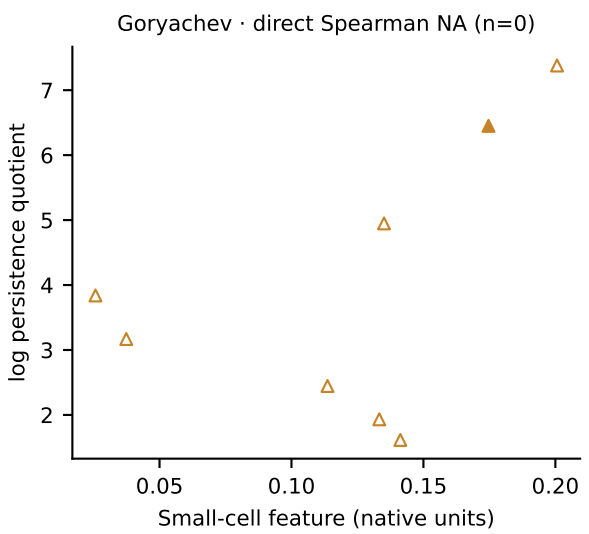
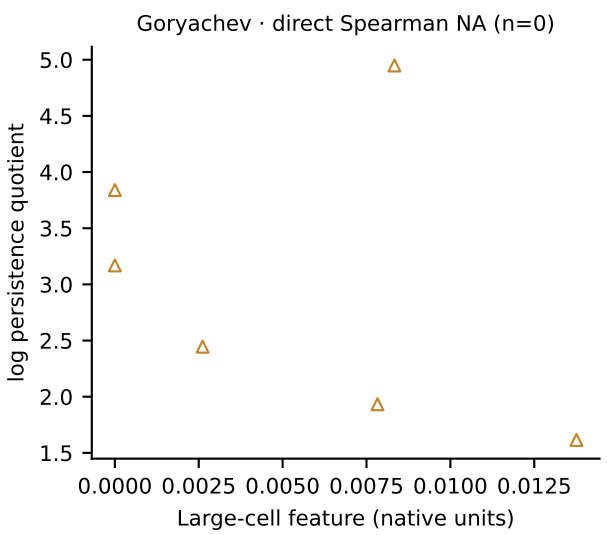
S10 · observed pairwise takeovers per birth

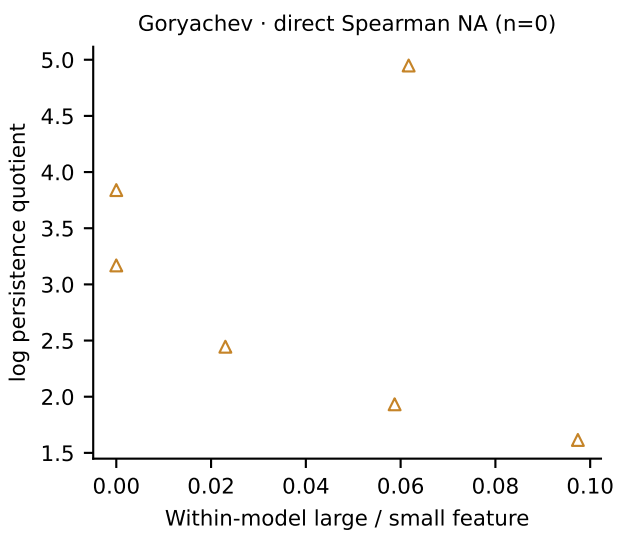


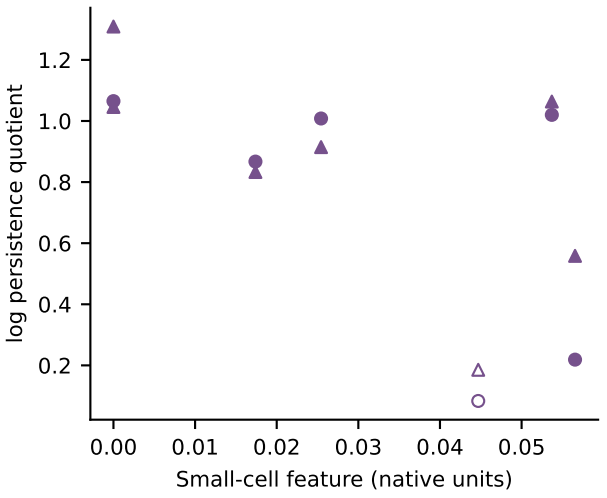


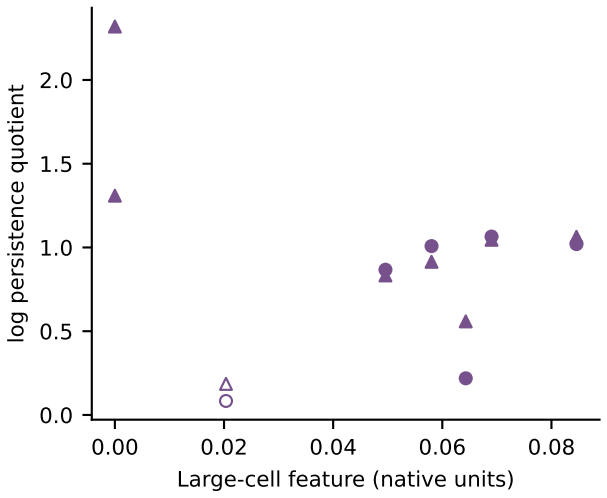


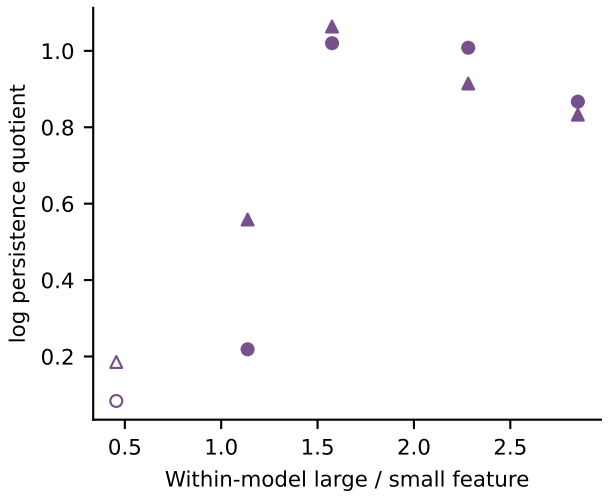


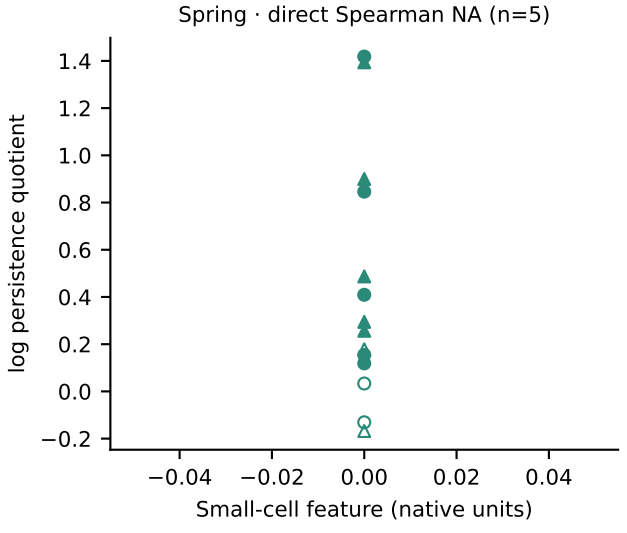


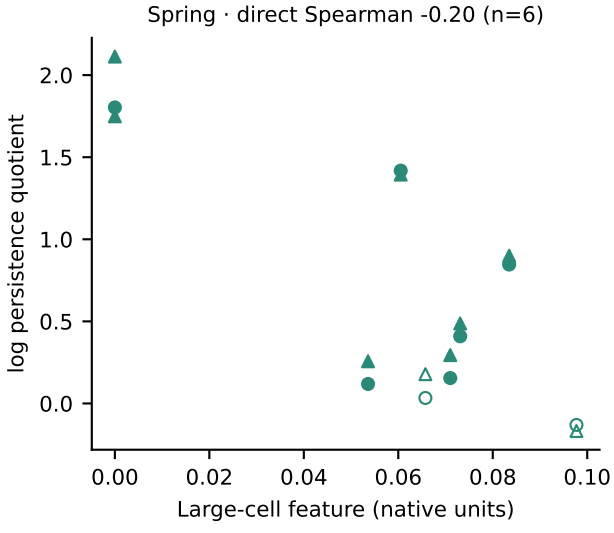


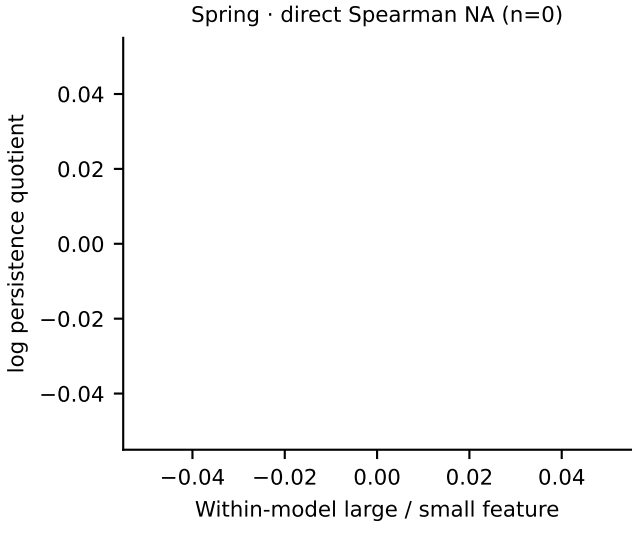












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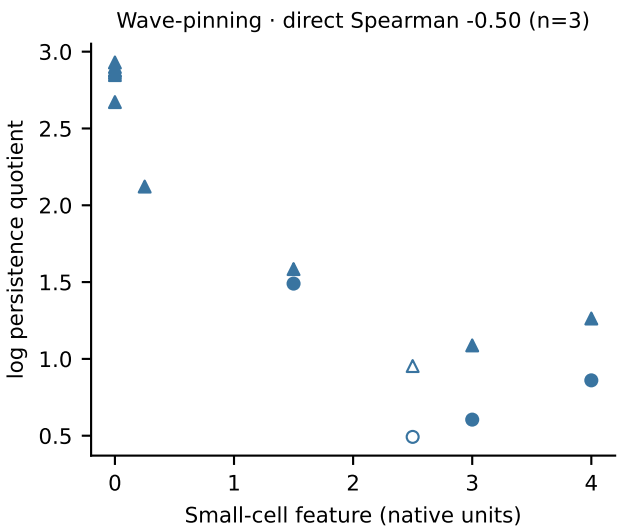
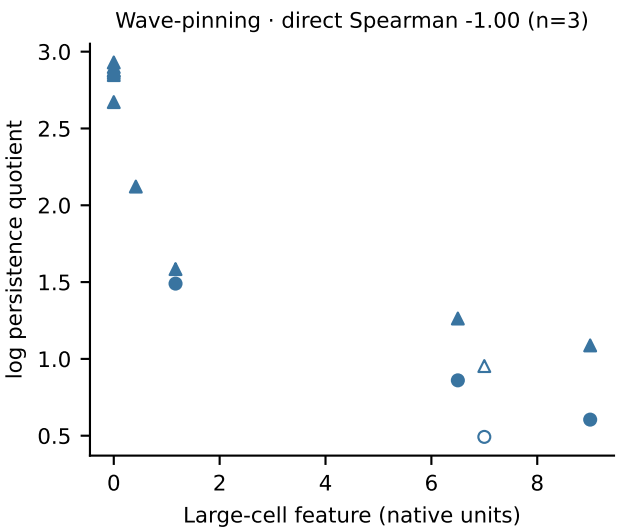
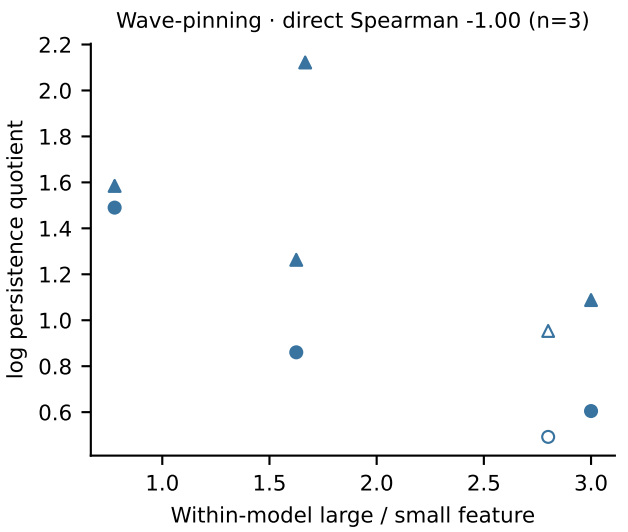
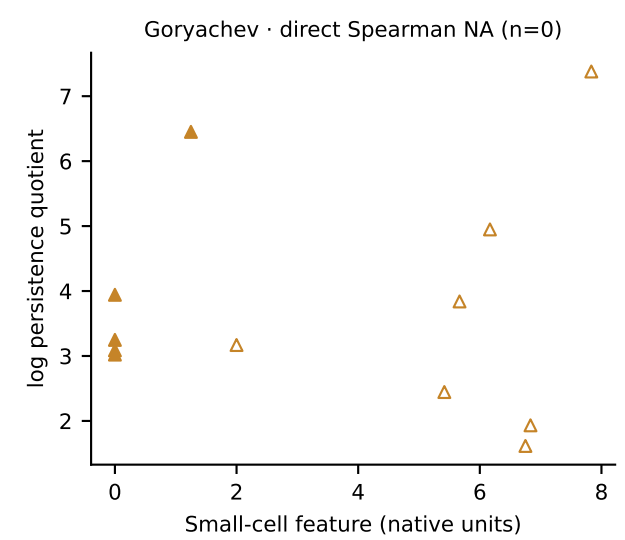
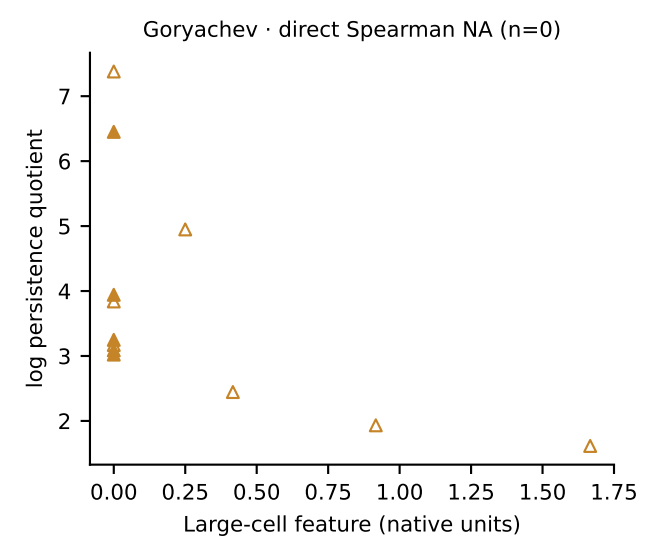
S10 · observed takeovers

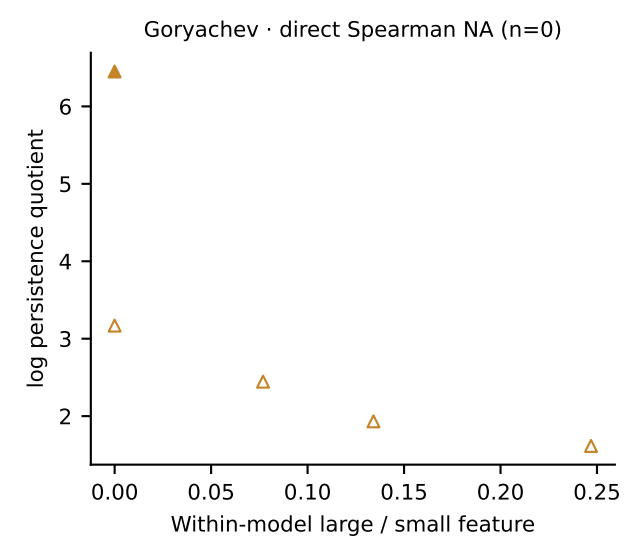


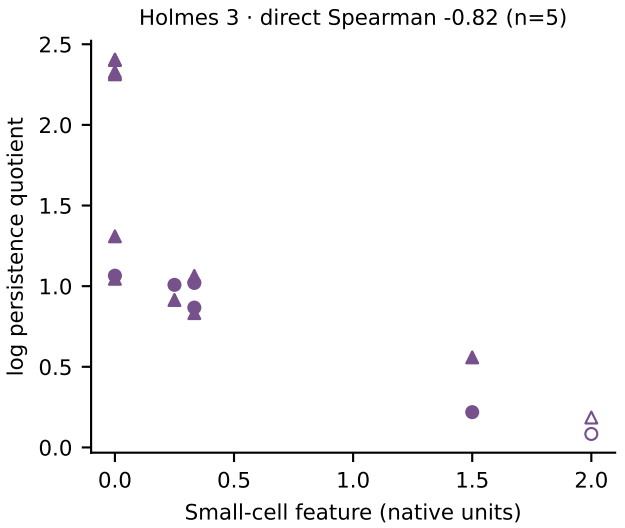


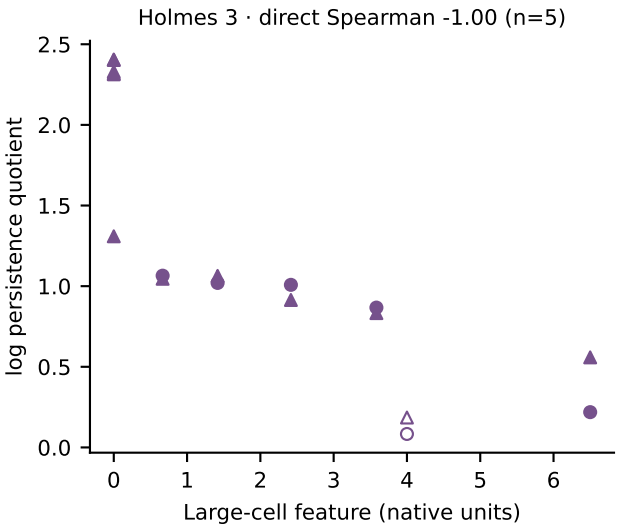


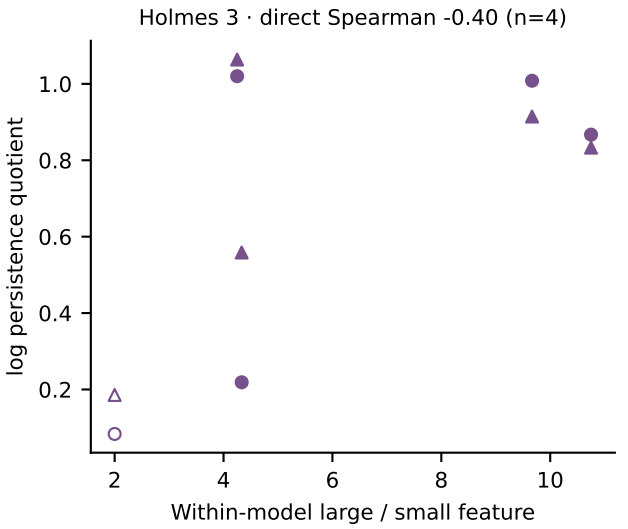


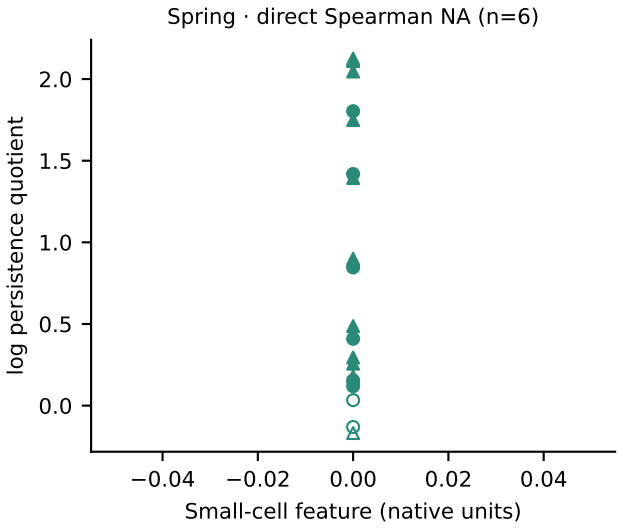


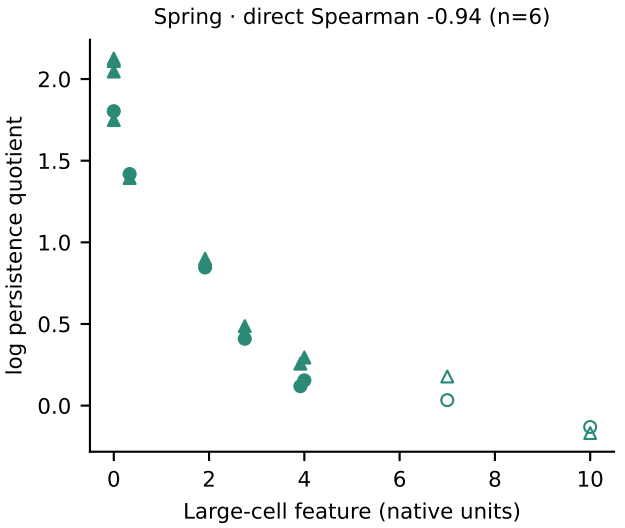


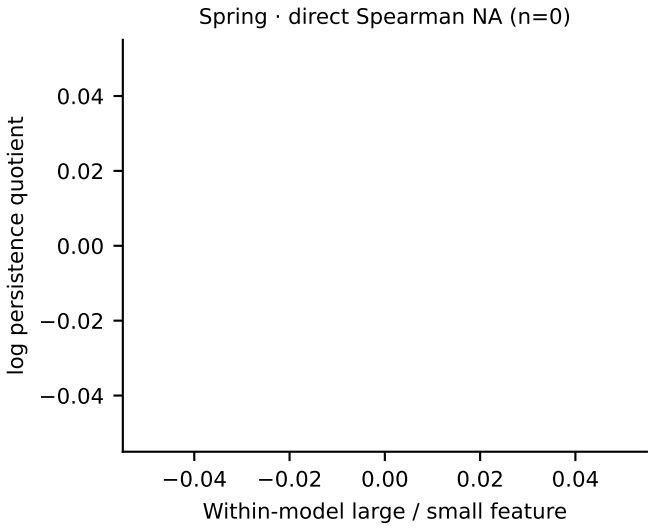












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S10 · observed takeovers per birth

Wave-pinning · direct Spearman NA (n=0)

Wave-pinning · direct Spearman NA (n=0)

Wave-pinning · direct Spearman NA (n=0)

Goryachev · direct Spearman NA (n=0)

Goryachev · direct Spearman NA (n=0)

Goryachev · direct Spearman NA (n=0)

Holmes 3 · direct Spearman NA (n=0)

Holmes 3 · direct Spearman NA (n=0)

Holmes 3 · direct Spearman NA (n=0)

Spring · direct Spearman NA (n=0)

Spring · direct Spearman NA (n=0)

Spring · direct Spearman NA (n=0)

log persistence quotient

Small-cell feature (native units)

Large-cell feature (native units)

Within-model large / small feature

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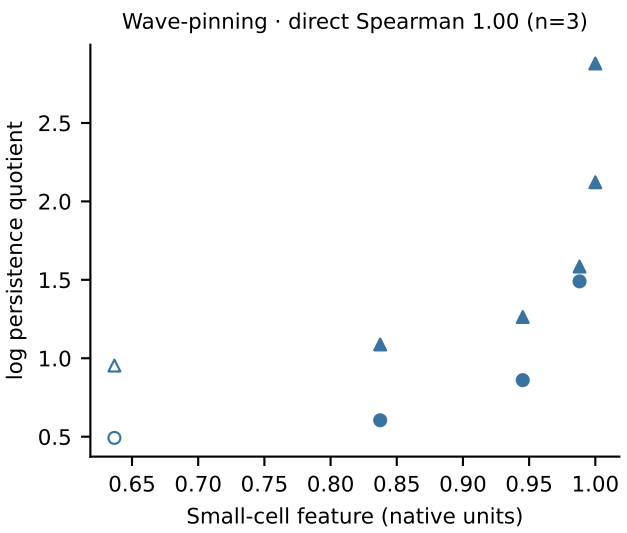
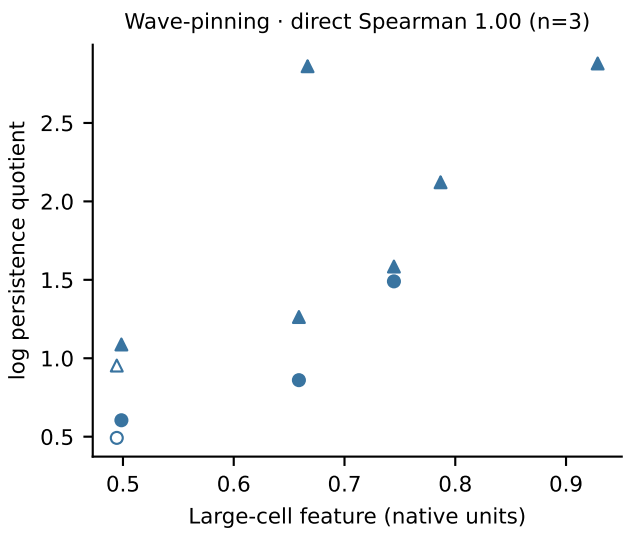
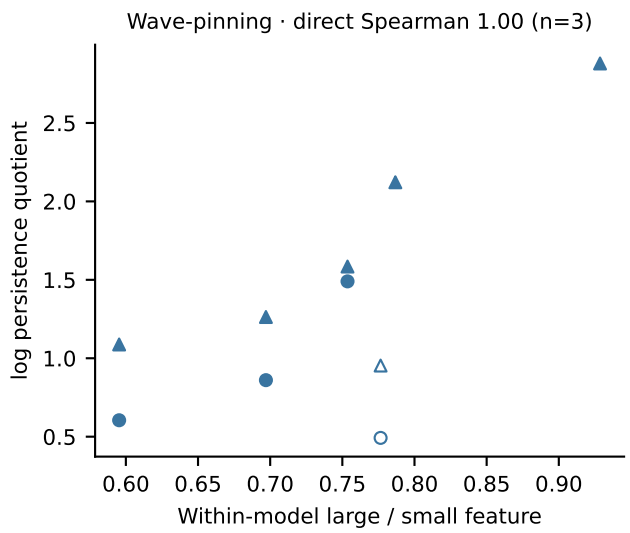
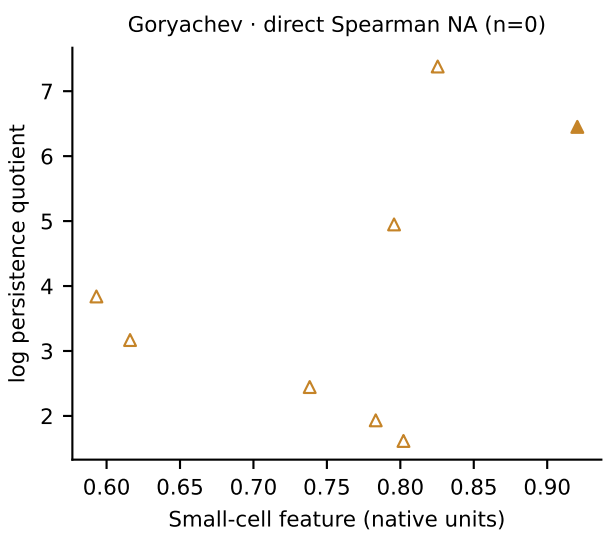
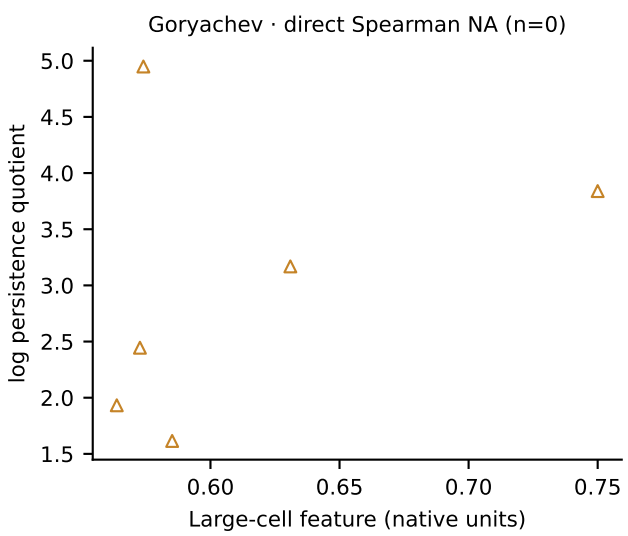
S10 · opposite birth fraction

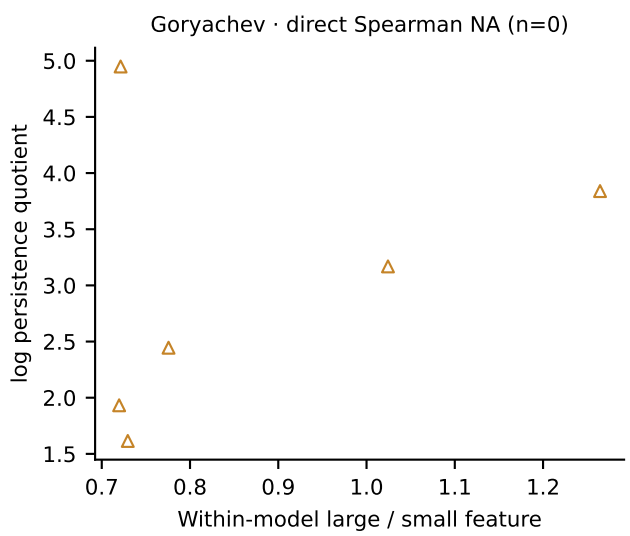


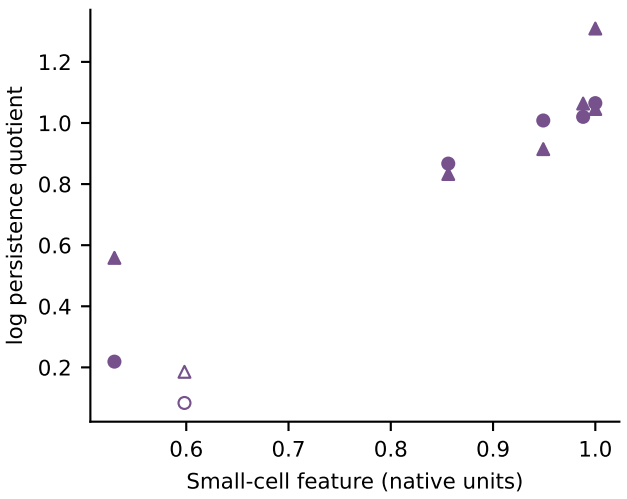


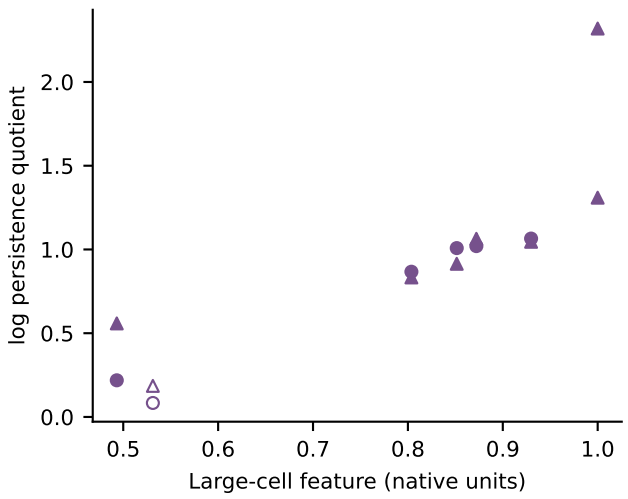


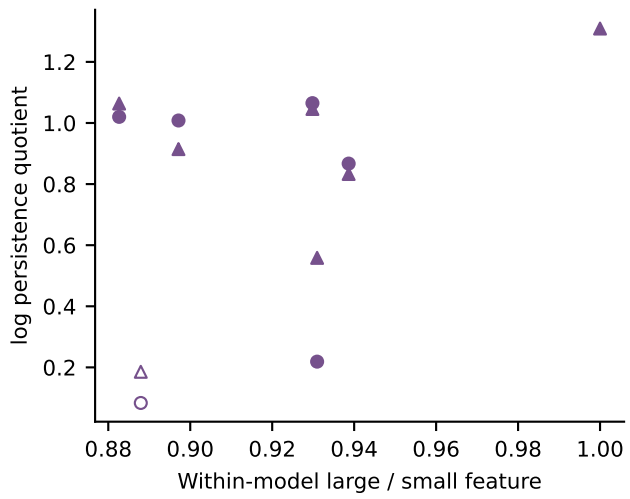


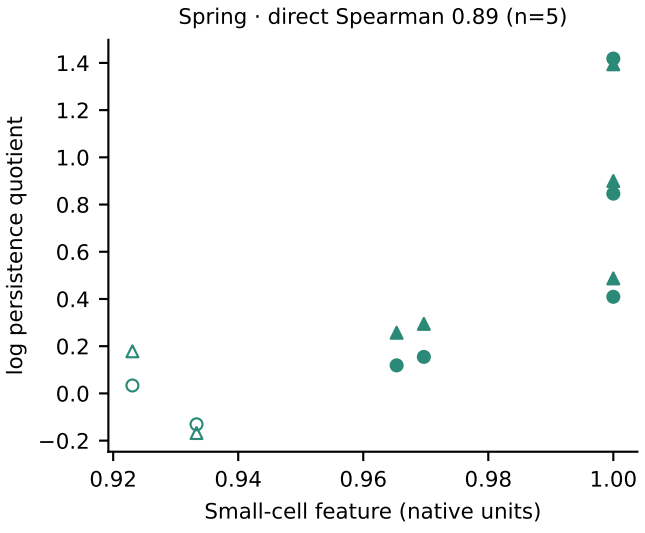


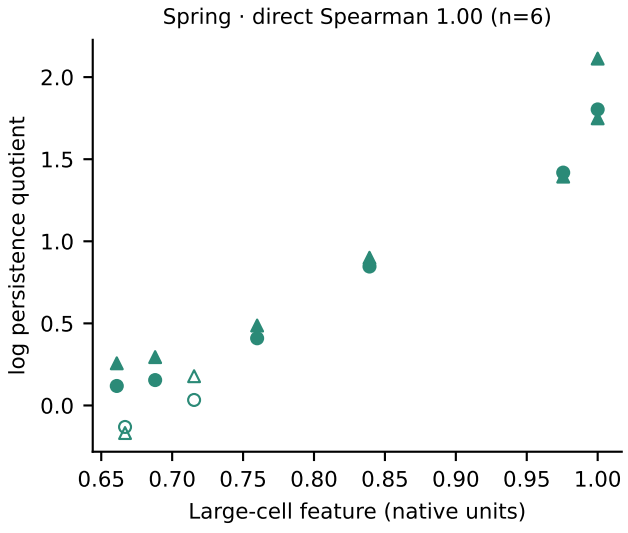


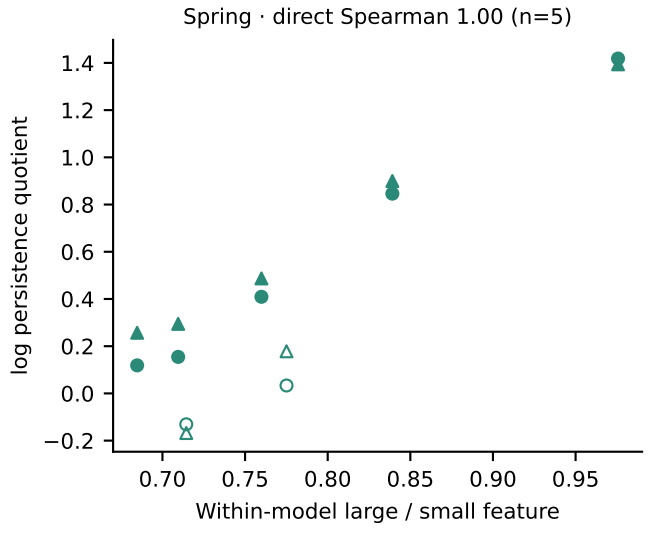












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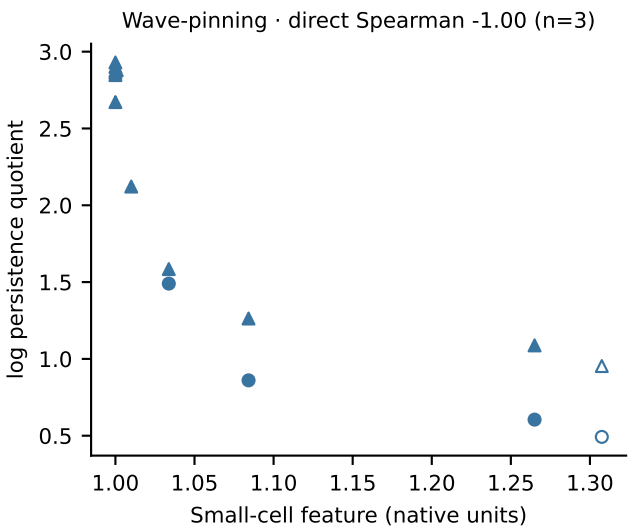
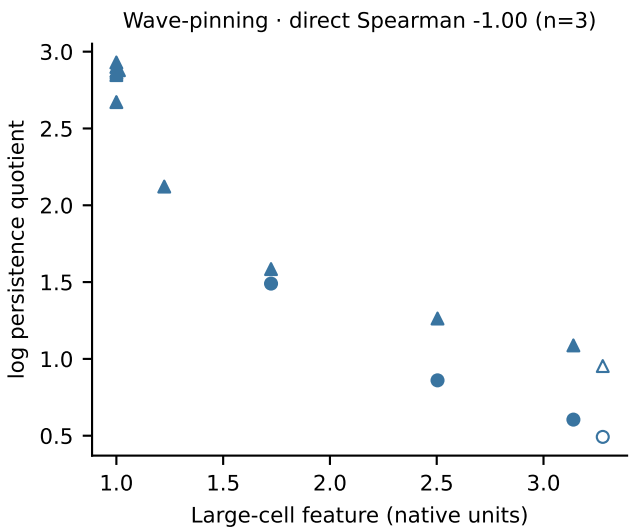
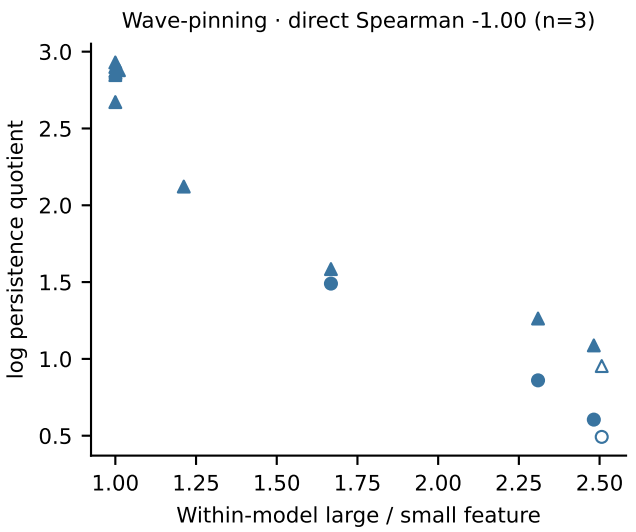
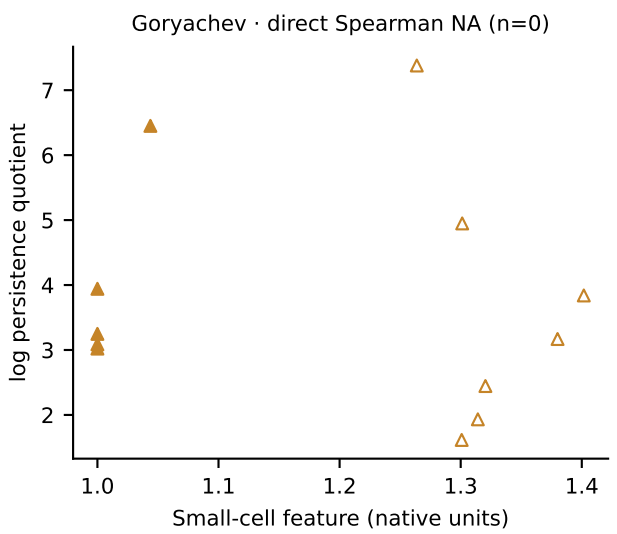
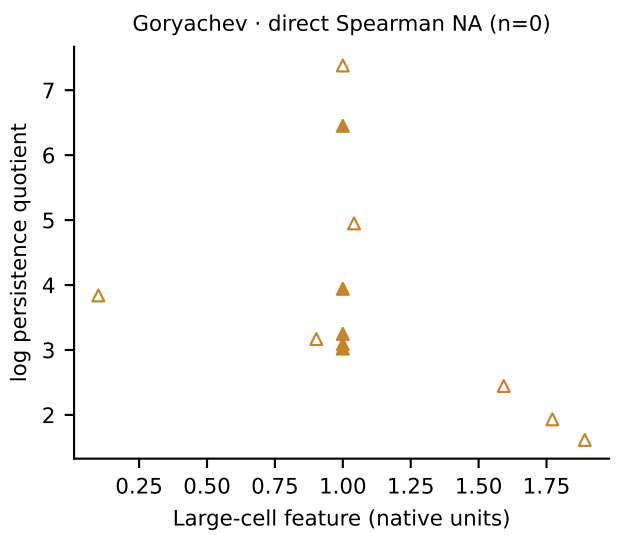
S10 · Mean region count

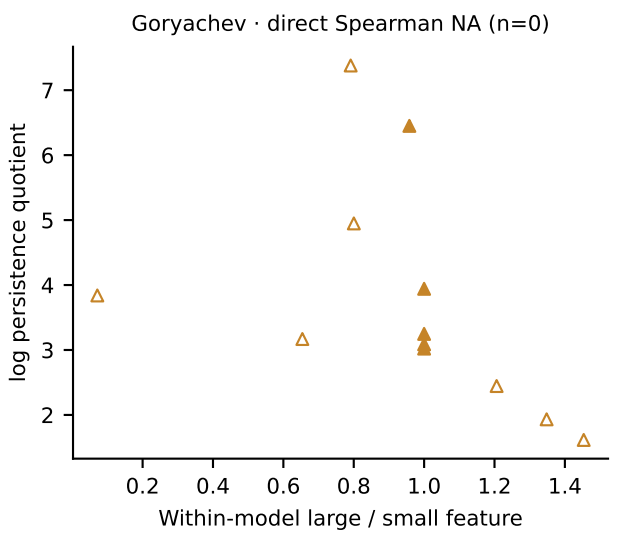


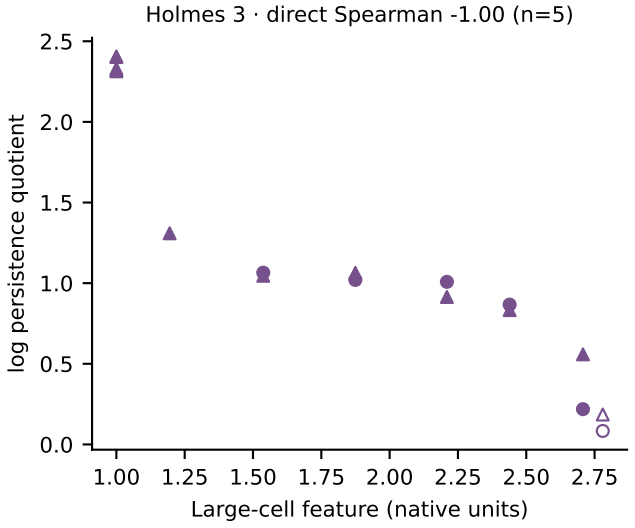


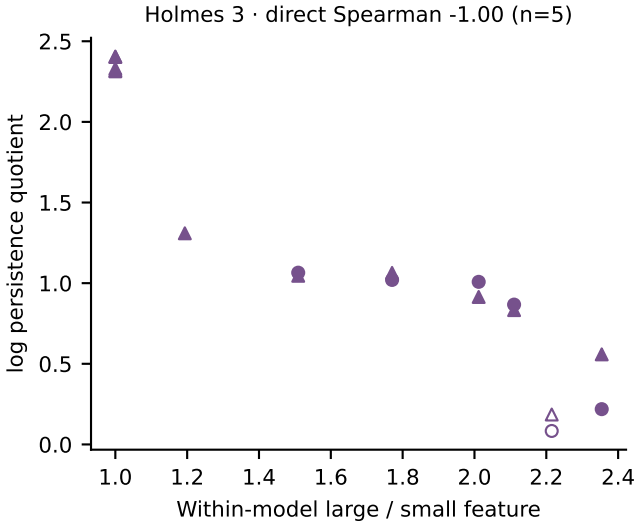


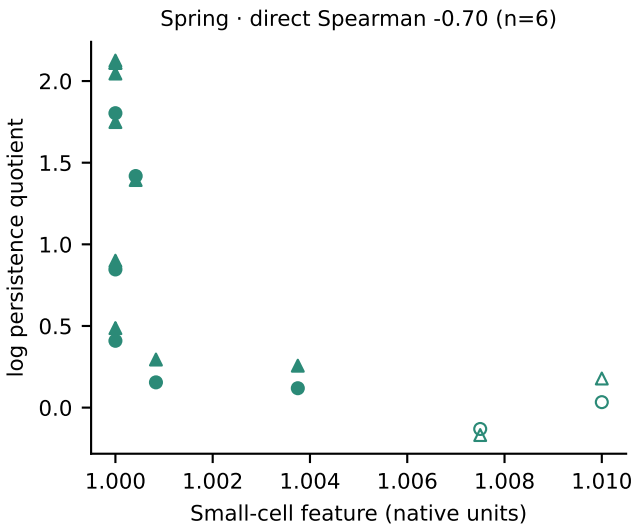


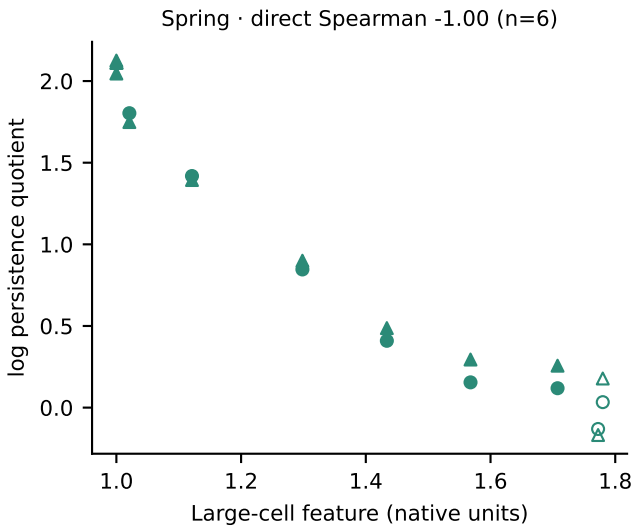


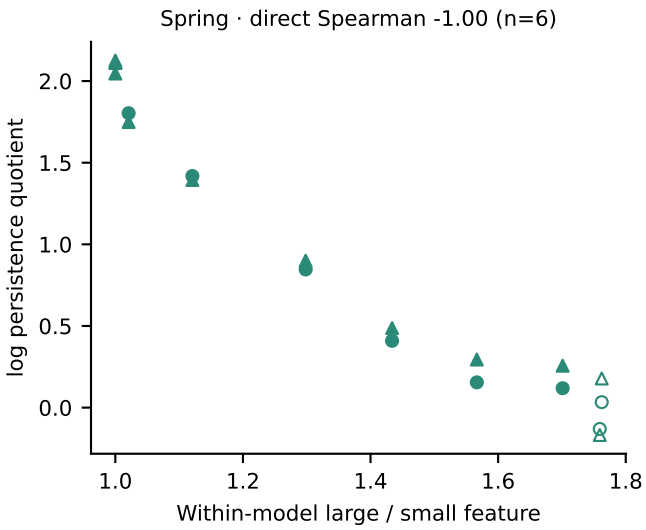












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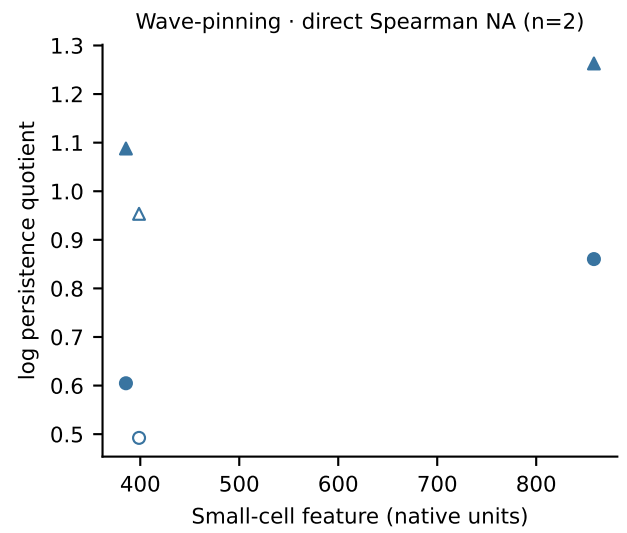
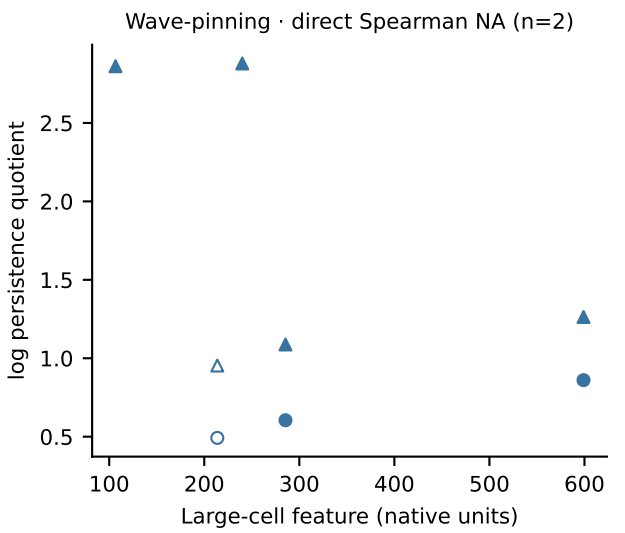
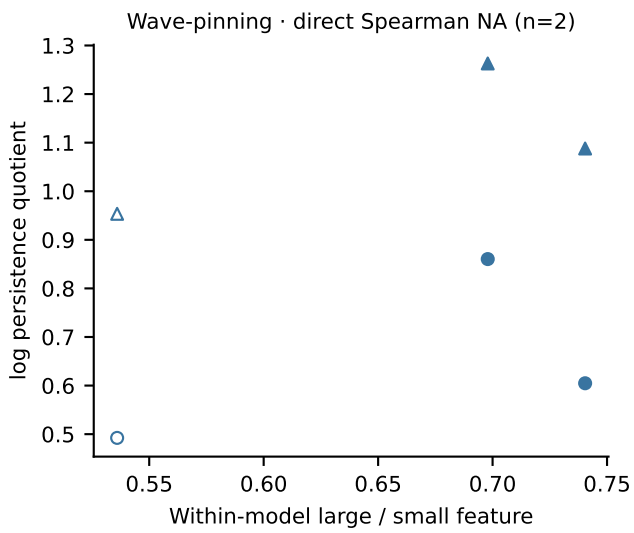
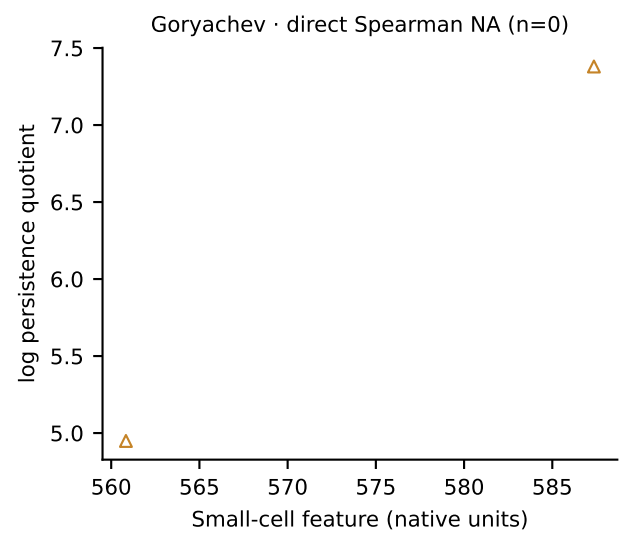
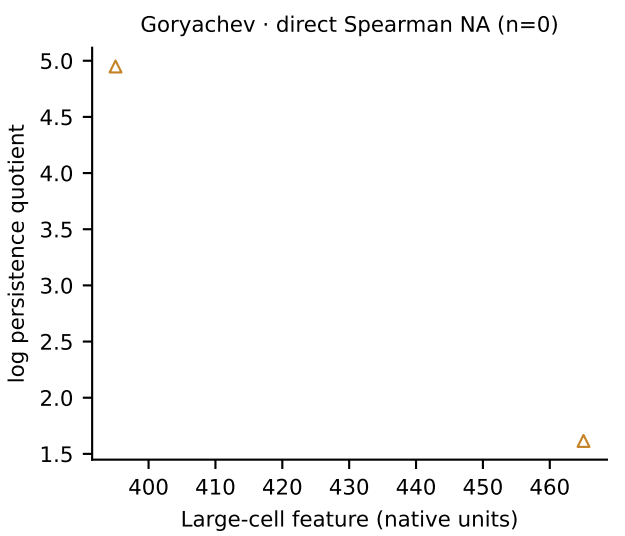
S10 · restricted mean lifetime lower

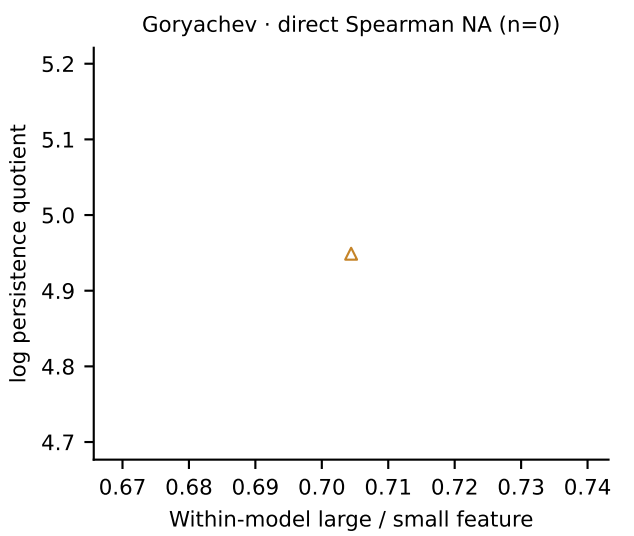


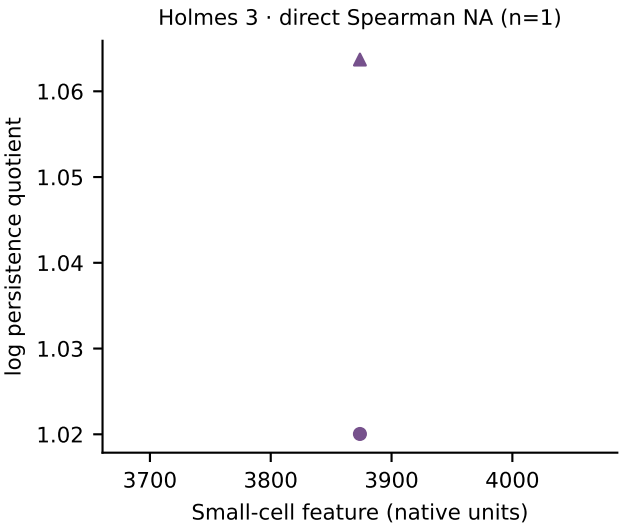


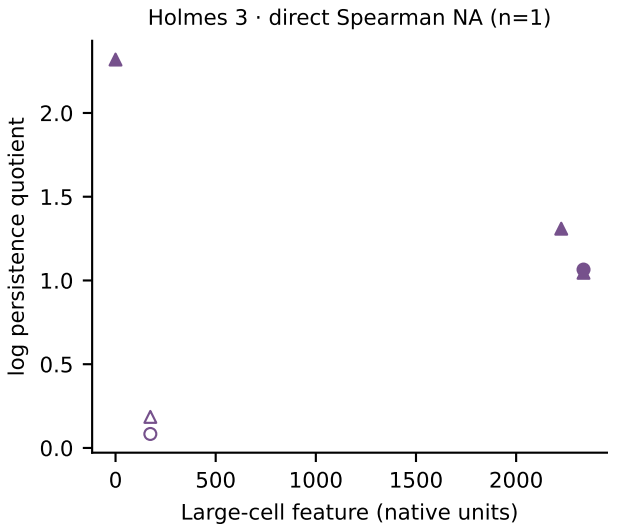


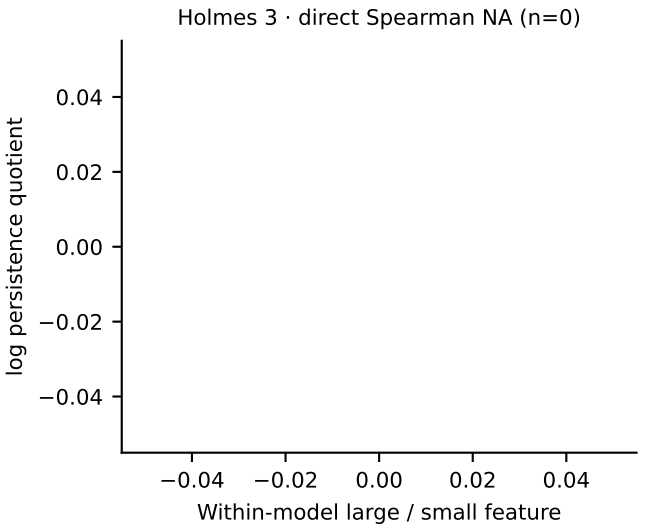


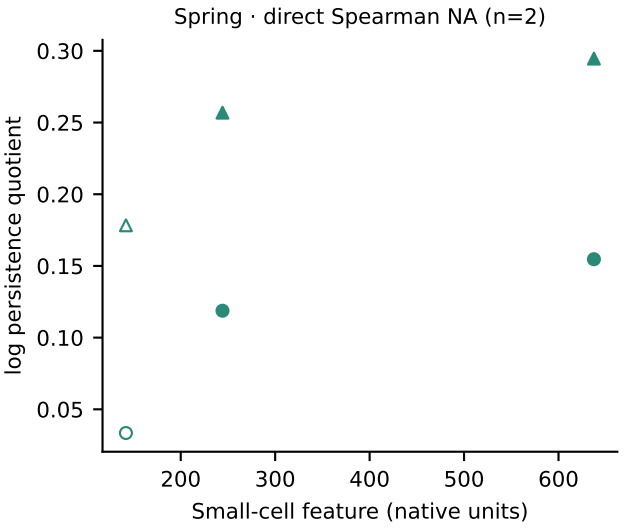


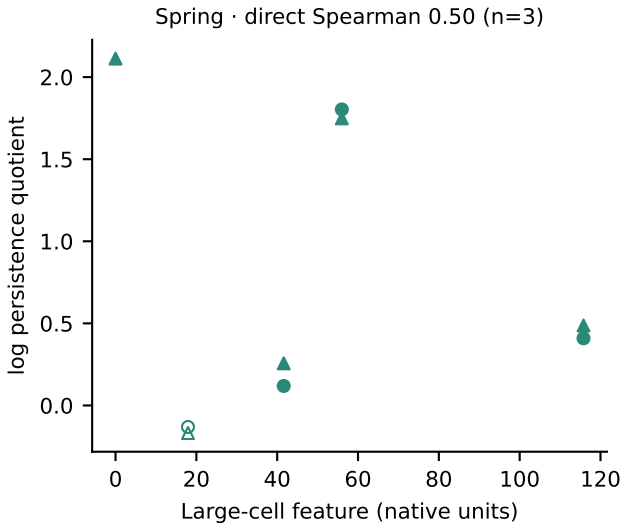


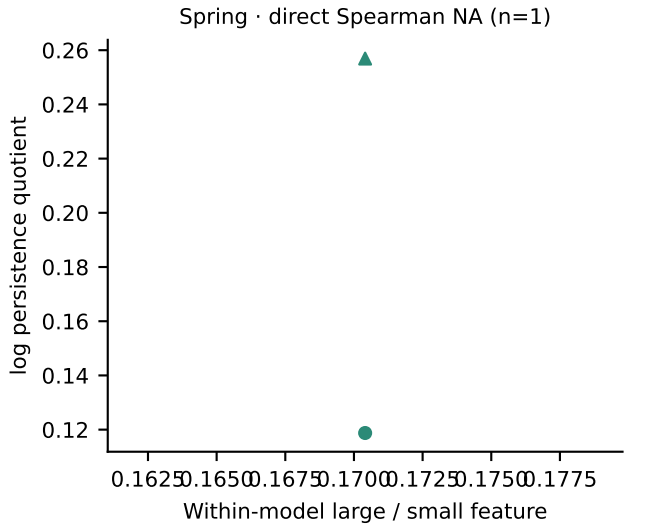












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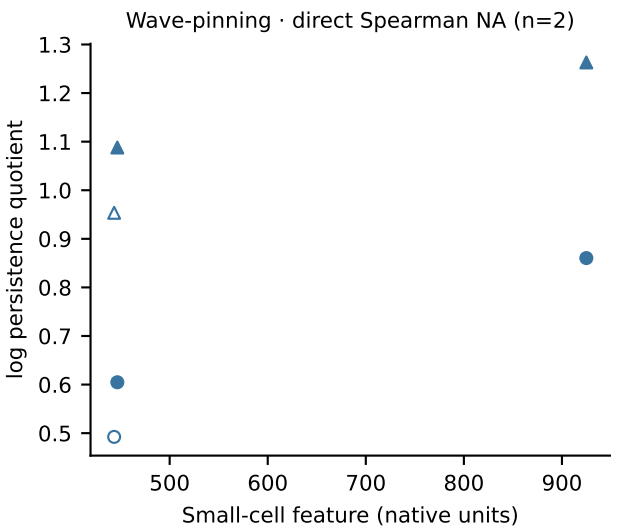
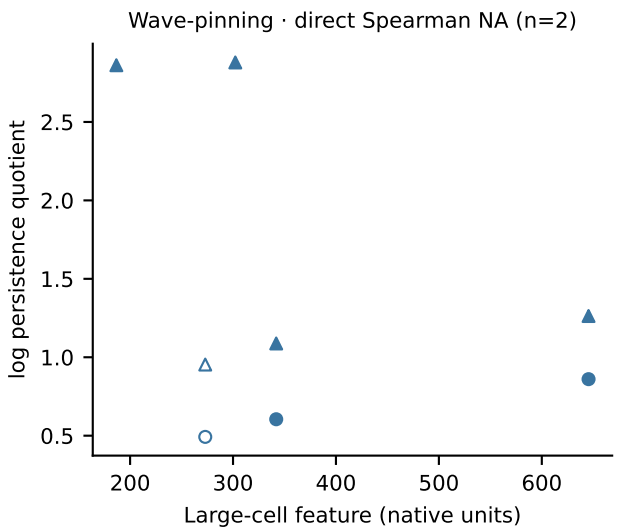
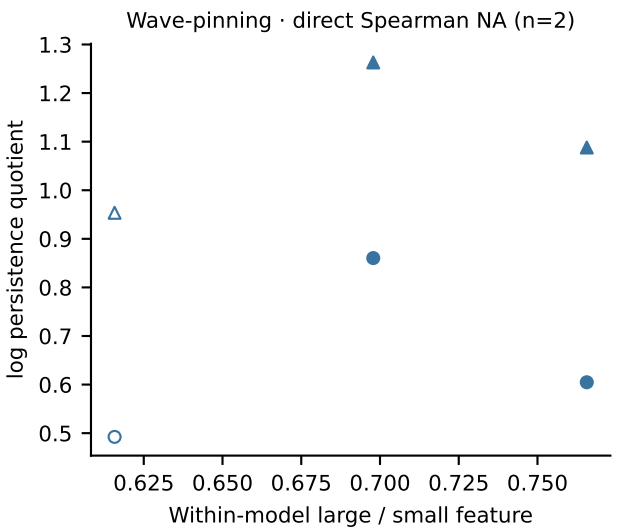
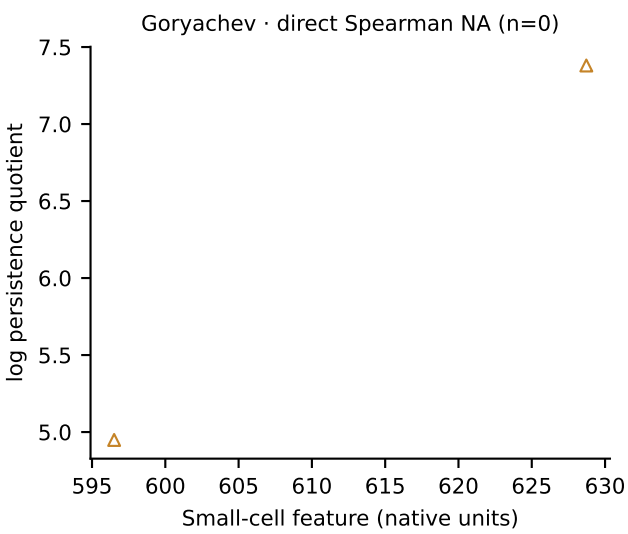
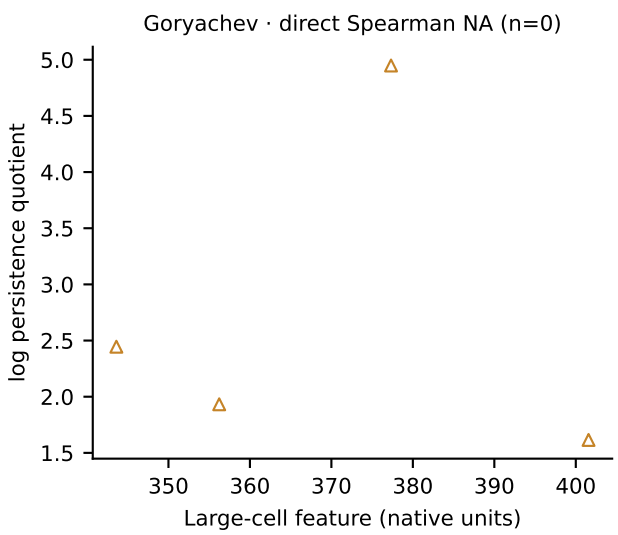
S10 · restricted mean lifetime observed

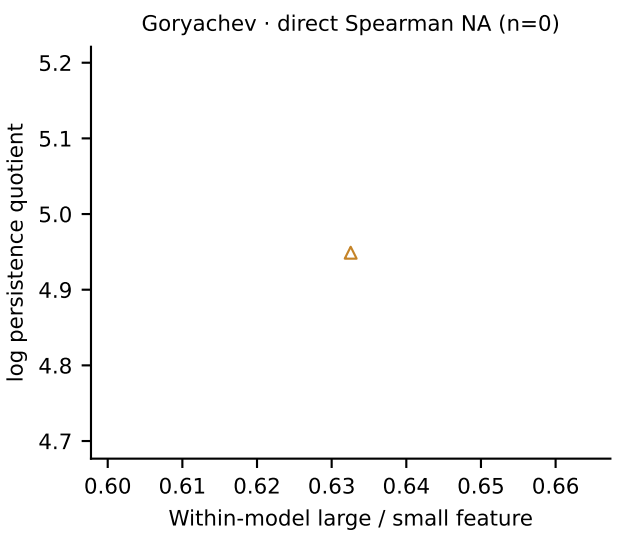


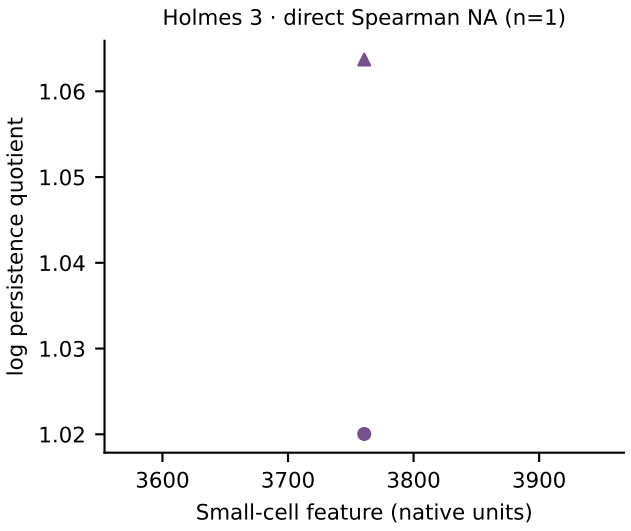


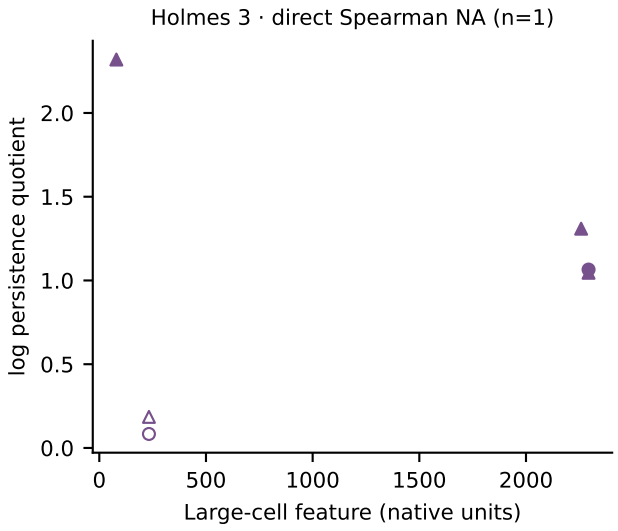


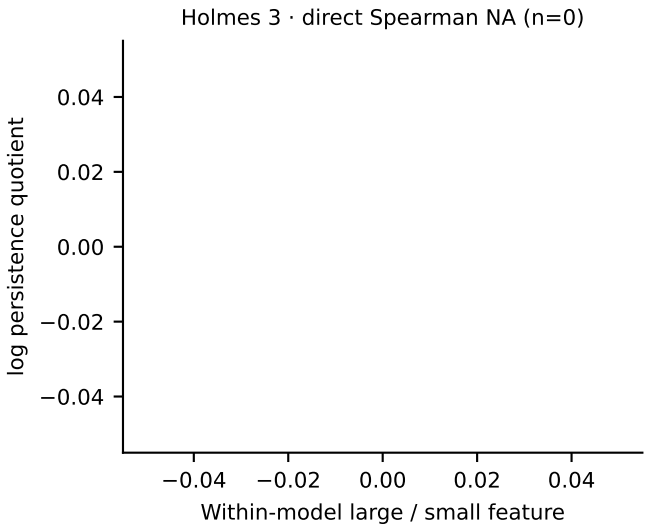


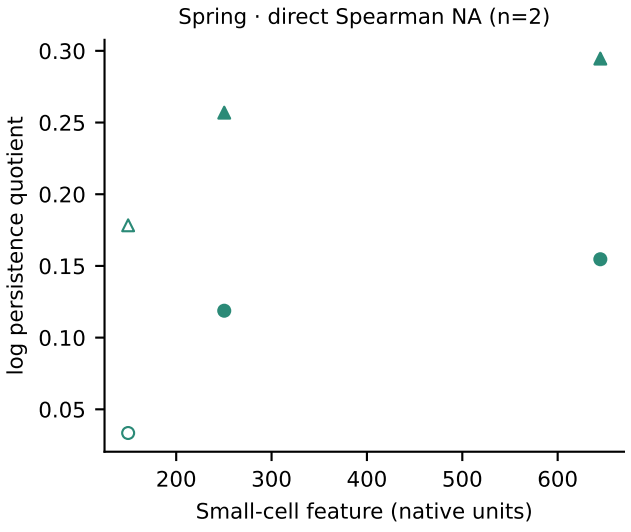


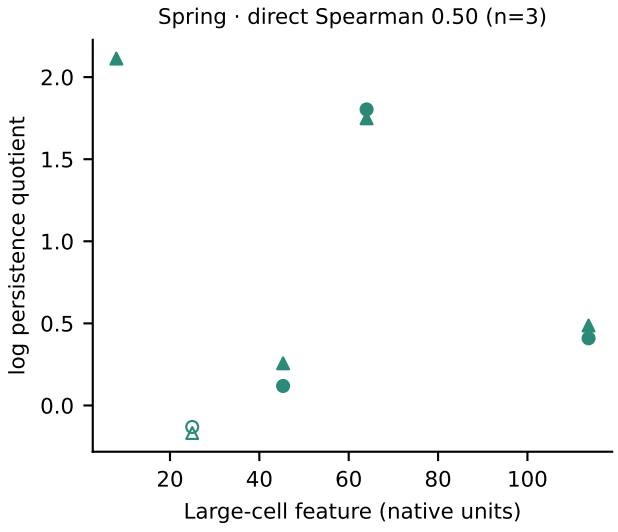


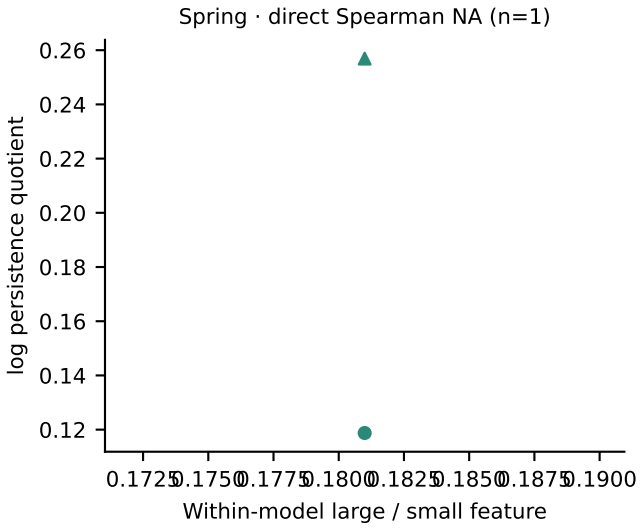












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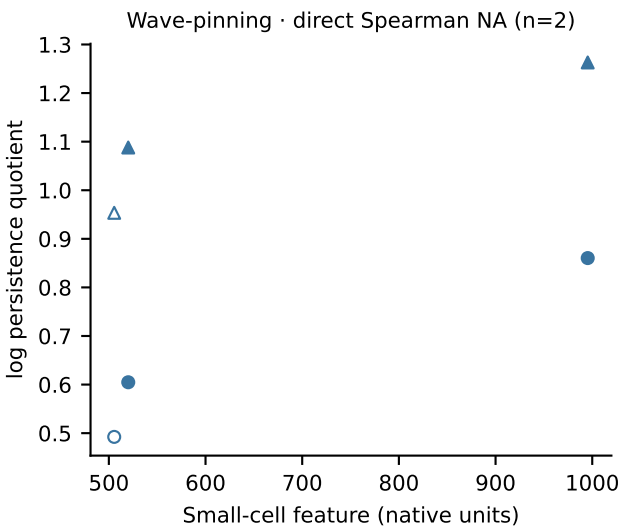
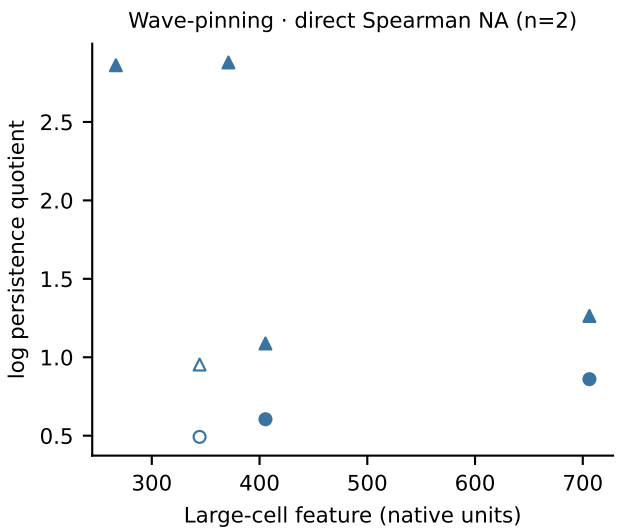
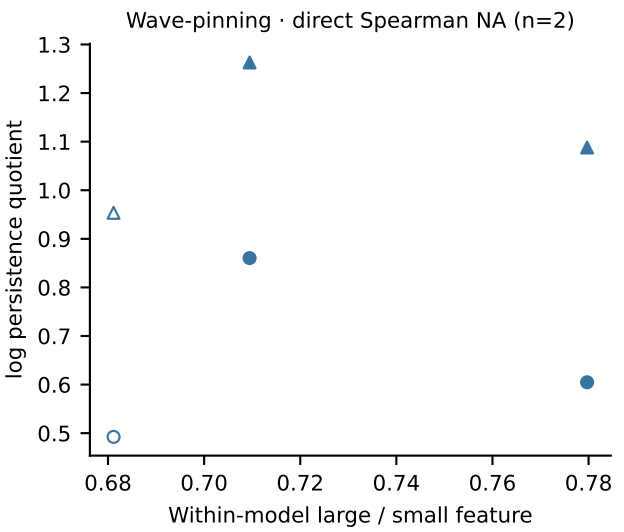
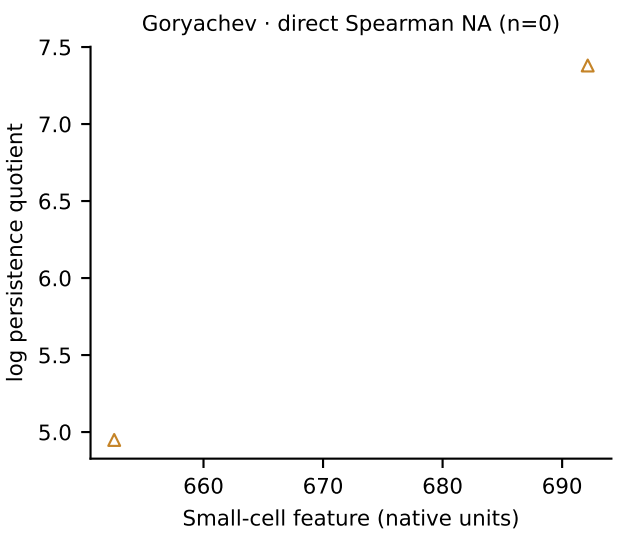
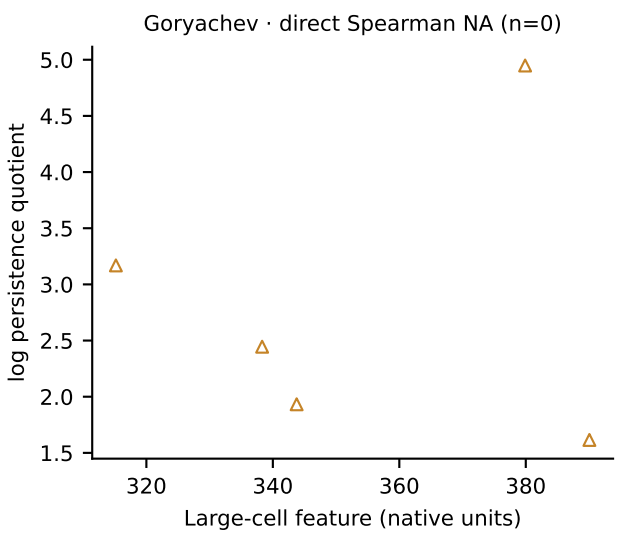
S10 · restricted mean lifetime upper

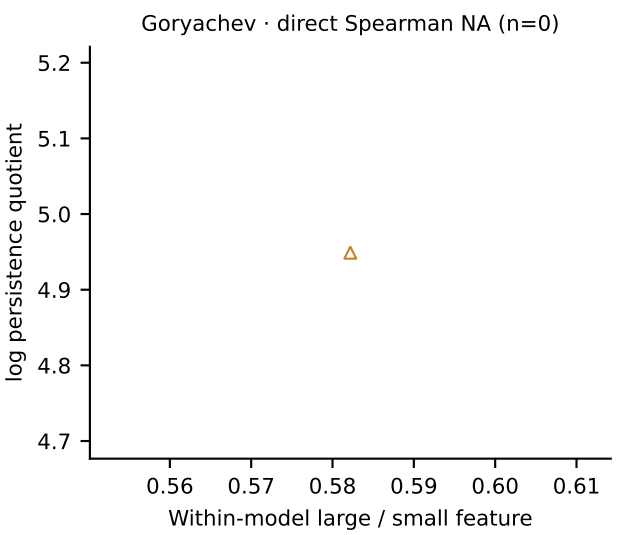


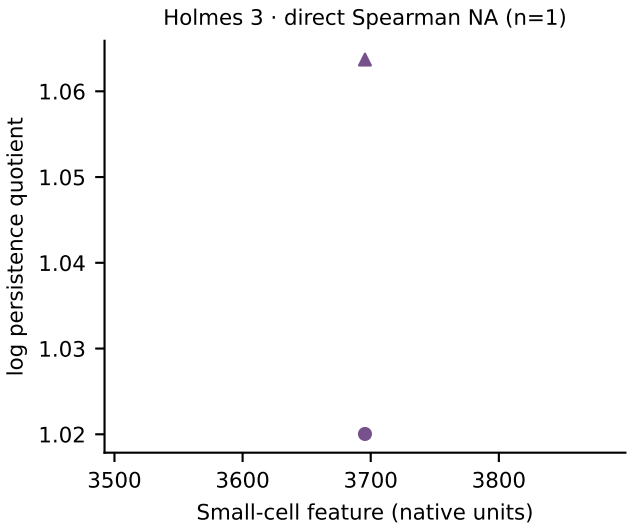


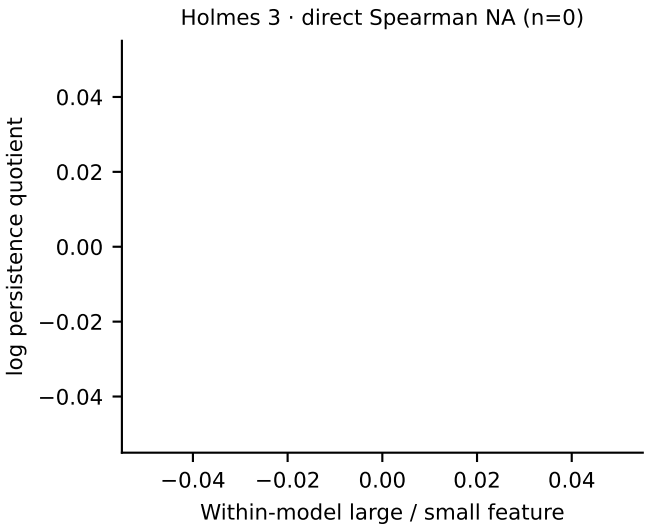


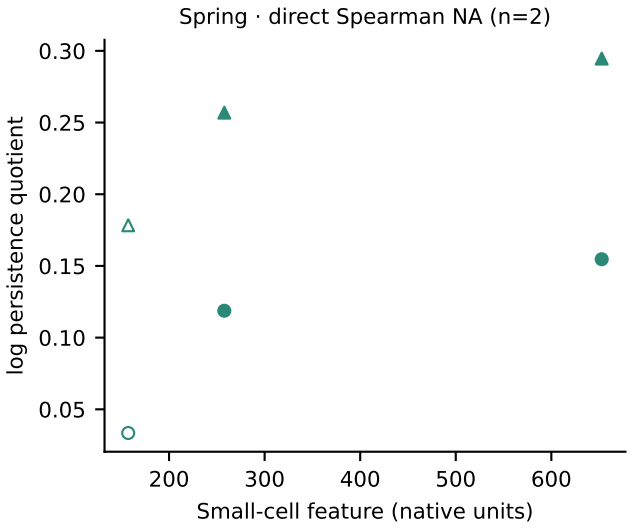


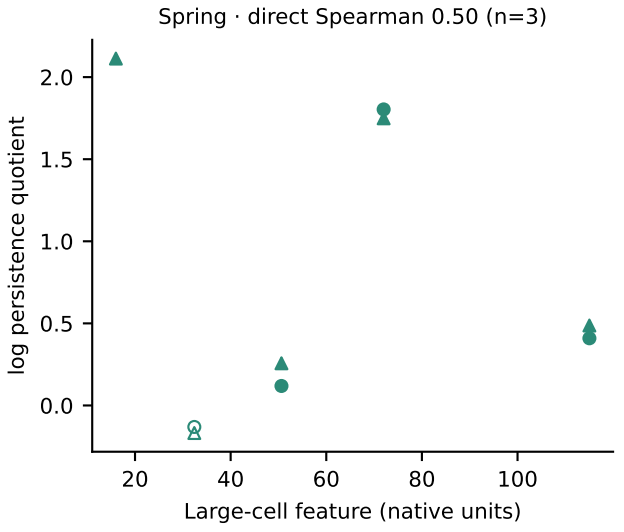


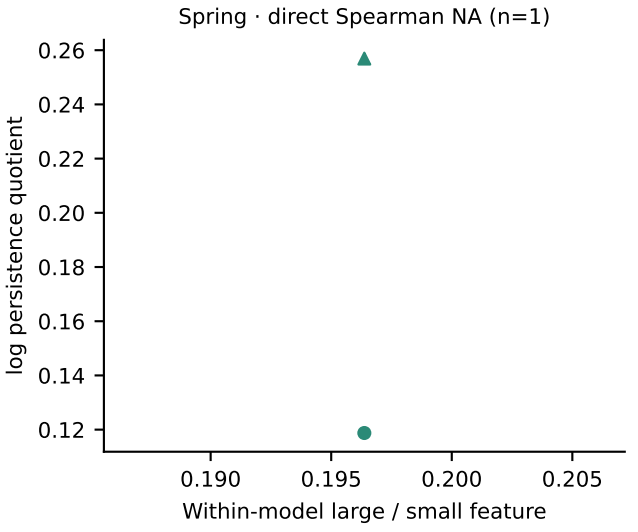






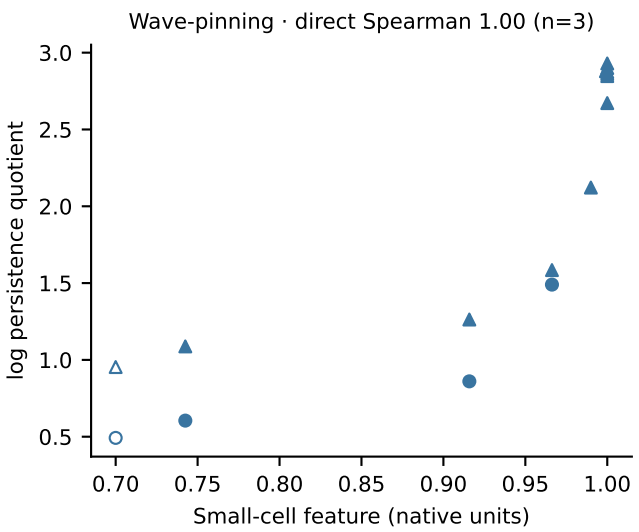


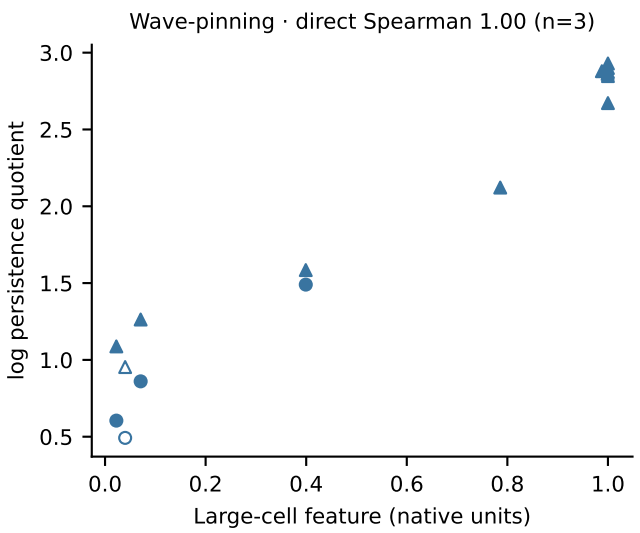


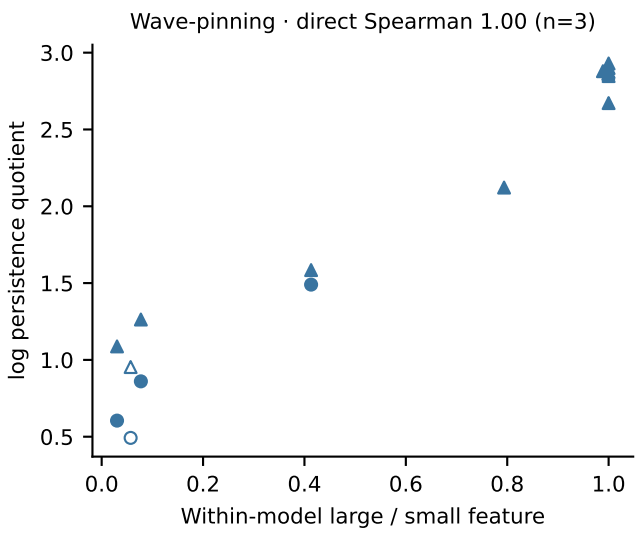


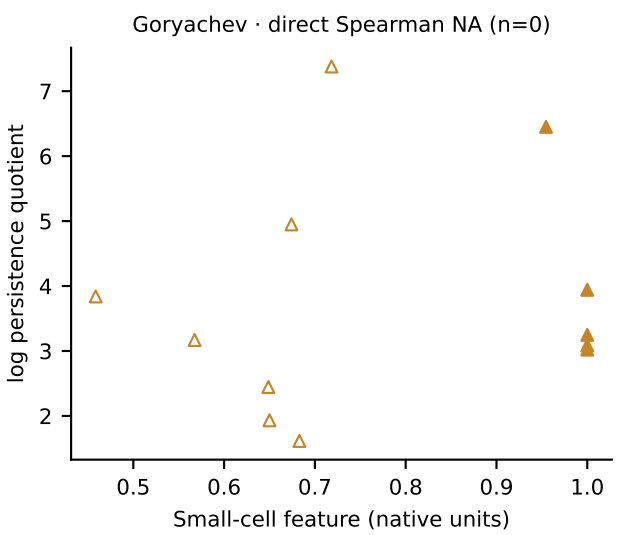
Arm longest · σ=1 · actual per-size horizons retained in the source CSV. Crosses: unknown forcing, excluded from primary.
○ Direct resolved τ ratio; △ D_θ proxy sensitivity. Filled: nonforced; open: forced context.
All small/large native quantities remain within their own model. Cross-model tests use only size quotients.
NA retains insufficient, constant, censored or unscorable measurements; native caps are never substituted.

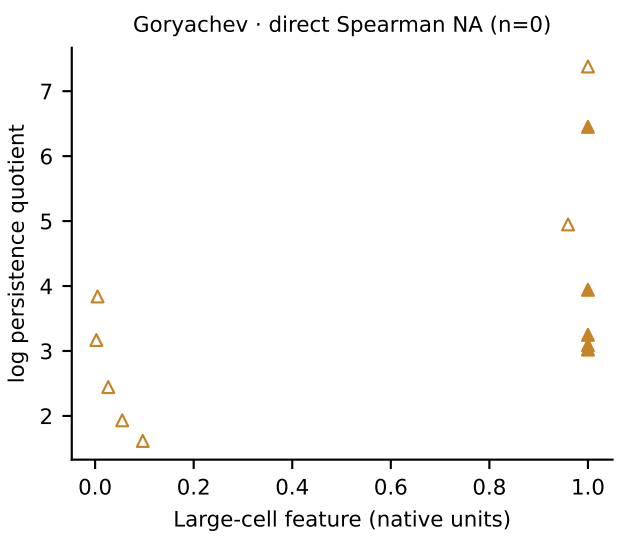
S10 · single front fraction

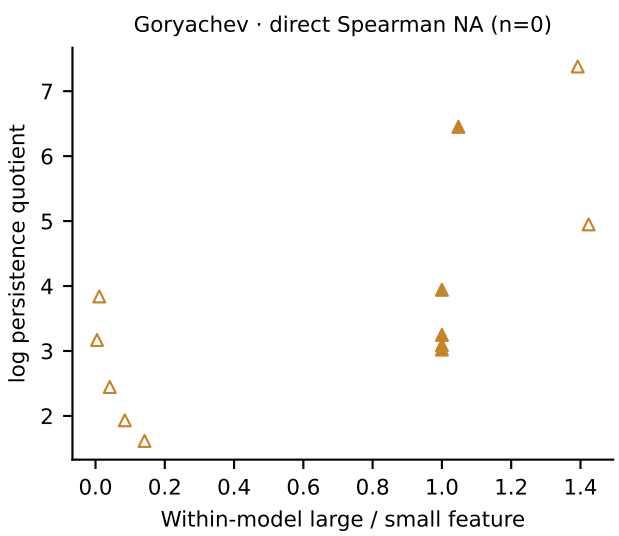


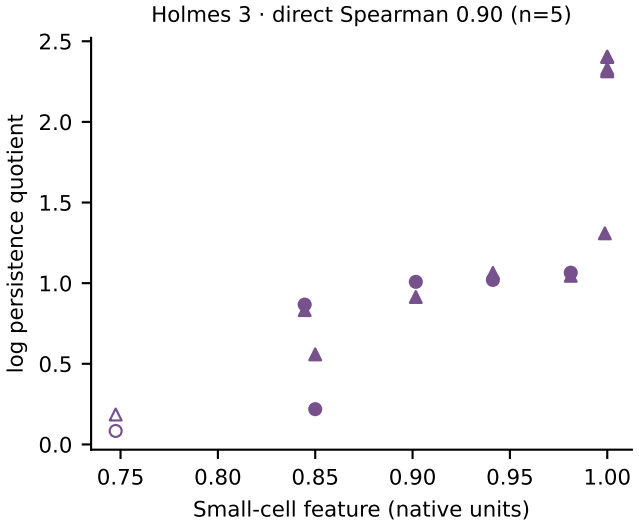


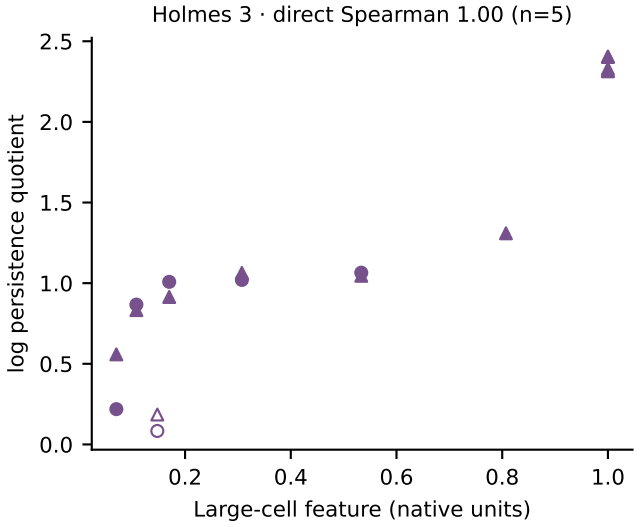


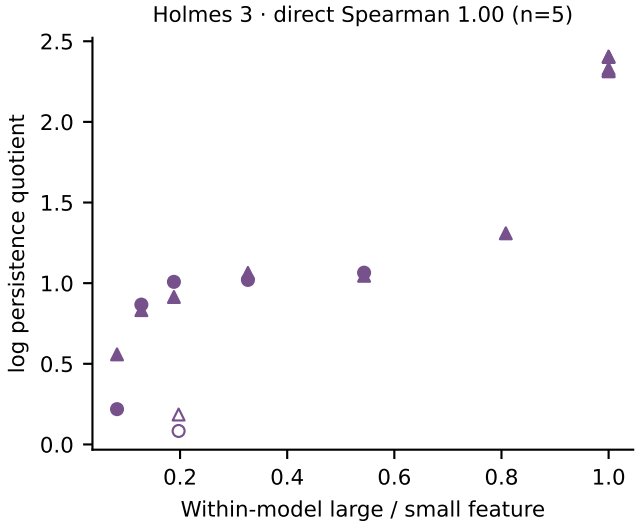


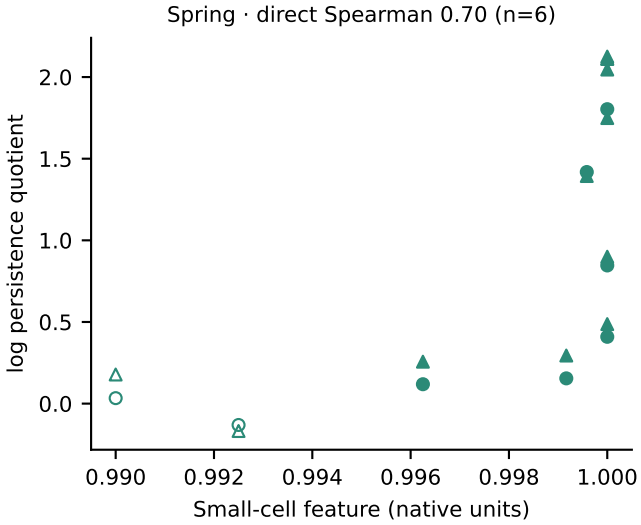


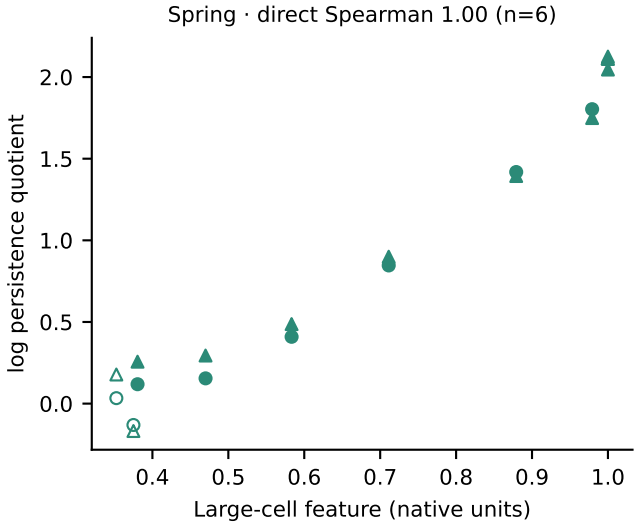


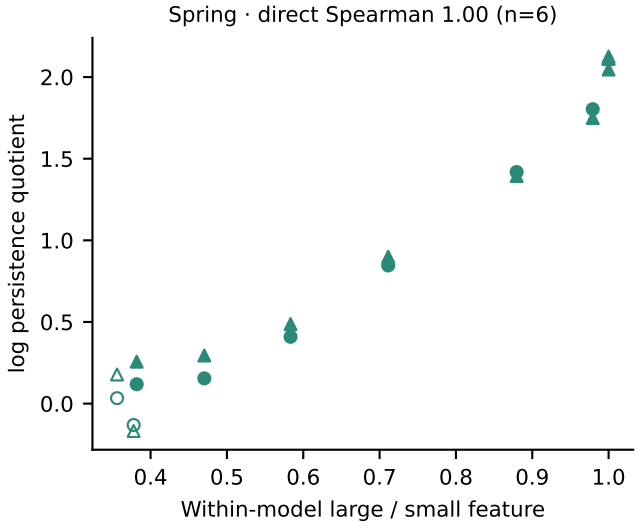






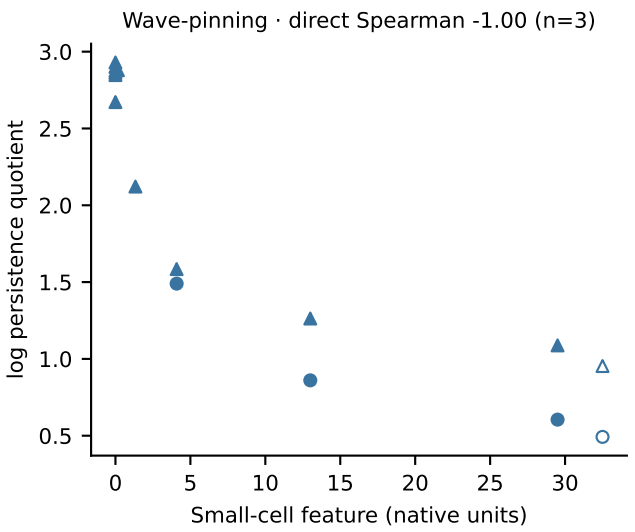


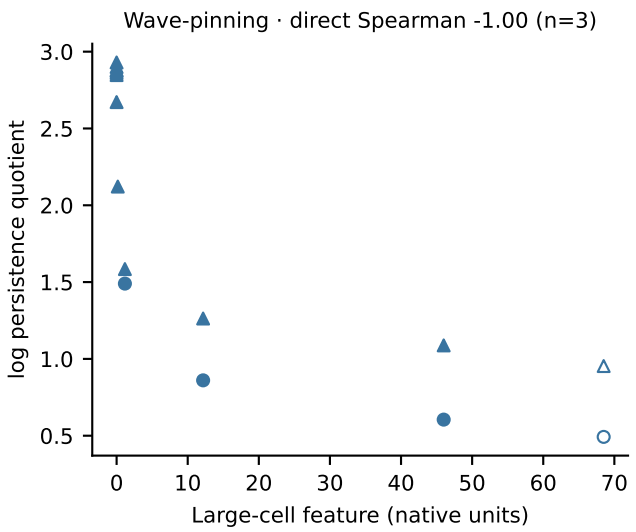


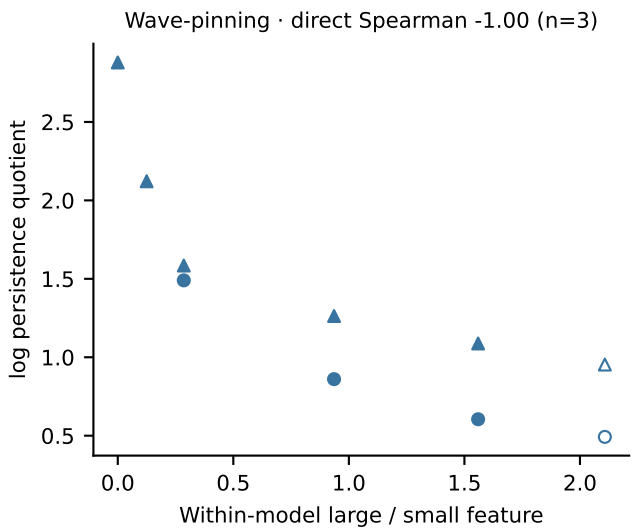


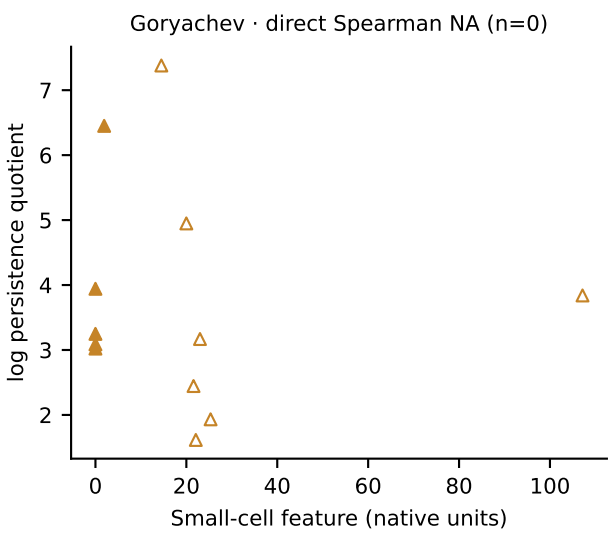
Arm longest · σ=1 · actual per-size horizons retained in the source CSV. Crosses: unknown forcing, excluded from primary.
○ Direct resolved τ ratio; △ D_θ proxy sensitivity. Filled: nonforced; open: forced context.
All small/large native quantities remain within their own model. Cross-model tests use only size quotients.
NA retains insufficient, constant, censored or unscorable measurements; native caps are never substituted.

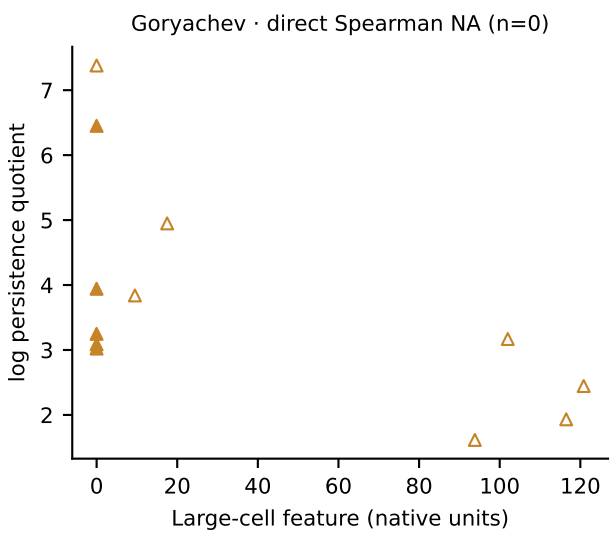
S10 · unscorable takeover tracks

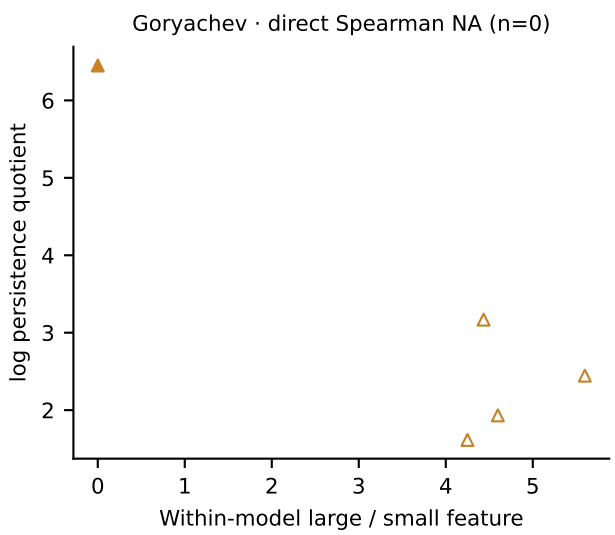


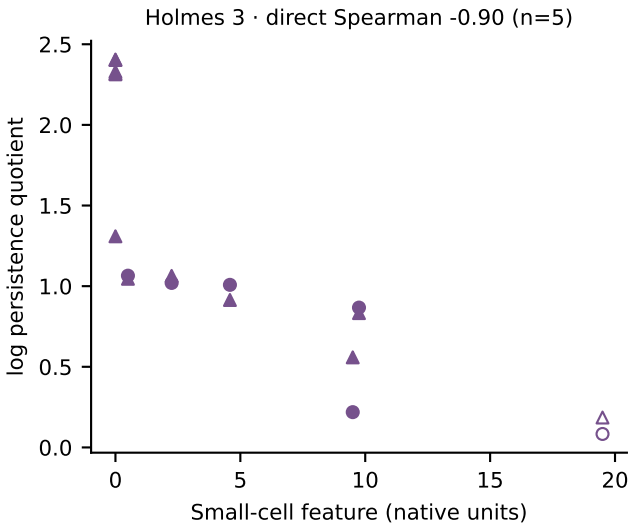


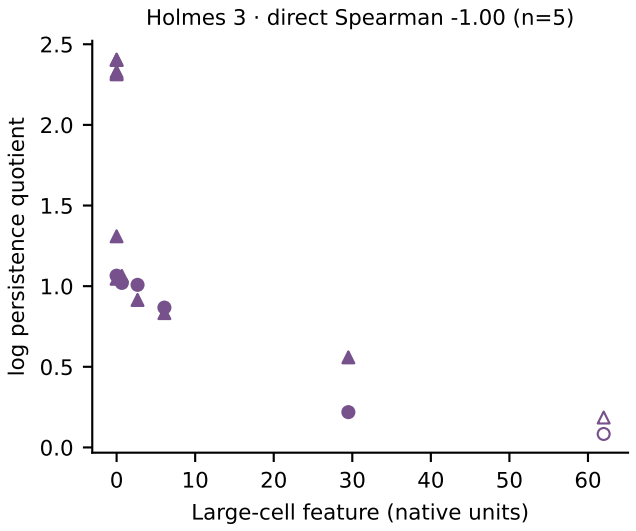


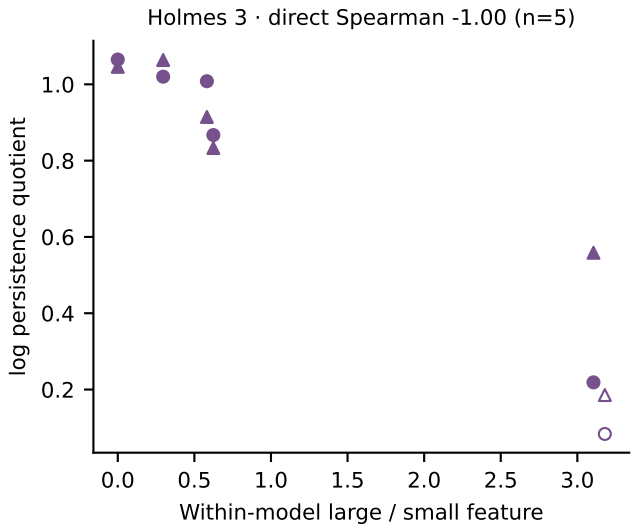


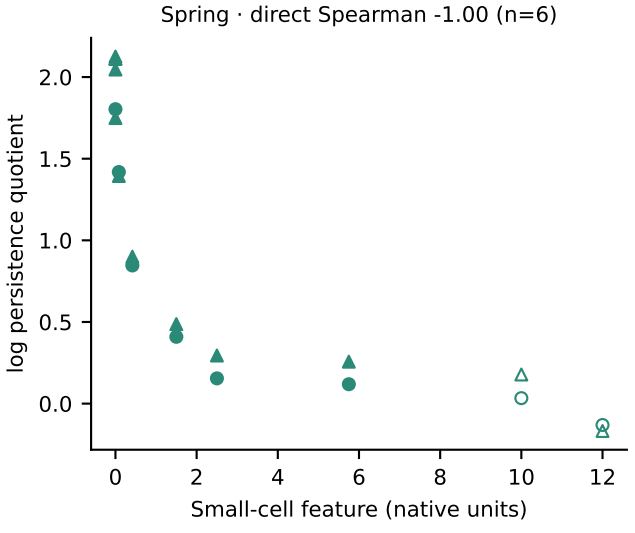


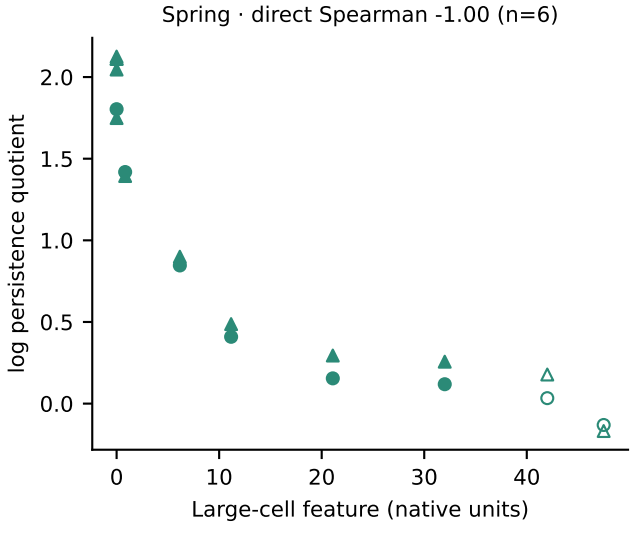


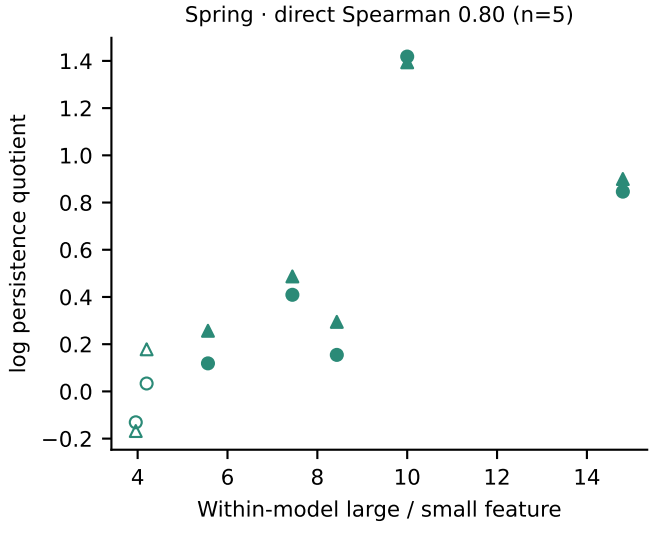












Raw event totals retain actual exposure; size quotients are unavailable for unequal horizons.
Arm longest · σ=1 · actual per-size horizons retained in the source CSV. Crosses: unknown forcing, excluded from primary.
○ Direct resolved τ ratio; △ D_θ proxy sensitivity. Filled: nonforced; open: forced context.
All small/large native quantities remain within their own model. Cross-model tests use only size quotients.
NA retains insufficient, constant, censored or unscorable measurements; native caps are never substituted.

S10 · Primary predictor sensitivity · forced excluded

A5 imaging view	Multi-front τ rank	τ folds	Multi-front proxy rank	Proxy folds	Count τ rank	τ folds	Count proxy rank	Proxy folds
Primary imaging view	0.091	0/4 (1 NA)	-0.157	0/4 (1 NA)	-0.389	3/4 (1 NA)	-0.832	4/4 (0 NA)
q=0.3 / raw / Δt=0.0625×	0.086	0/4 (4 NA)	-0.596	0/4 (2 NA)	-0.486	0/4 (4 NA)	-0.794	4/4 (0 NA)
q=0.3 / raw / Δt=0.125×	-0.143	0/4 (4 NA)	-0.678	0/4 (3 NA)	-0.486	0/4 (4 NA)	-0.794	4/4 (0 NA)
q=0.3 / raw / Δt=0.25×	0.086	0/4 (4 NA)	-0.611	0/4 (3 NA)	-0.257	0/4 (4 NA)	-0.790	4/4 (0 NA)
q=0.3 / raw / Δt=0.5×	0.240	0/4 (1 NA)	-0.261	0/4 (1 NA)	-0.508	3/4 (1 NA)	-0.853	4/4 (0 NA)
q=0.3 / raw / Δt=1×	0.238	0/4 (2 NA)	0.236	0/4 (1 NA)	-0.405	2/4 (2 NA)	-0.564	3/4 (1 NA)
q=0.3 / raw / Δt=2×	0.238	0/4 (2 NA)	0.236	0/4 (1 NA)	-0.405	2/4 (2 NA)	-0.564	3/4 (1 NA)
q=0.3 / raw / Δt=4×	0.238	0/4 (2 NA)	0.224	0/4 (1 NA)	-0.333	2/4 (2 NA)	-0.527	3/4 (1 NA)
q=0.3 / optical / Δt=0.0625×	NA	0/4 (4 NA)	-0.826	0/4 (4 NA)	-0.257	0/4 (4 NA)	-0.790	4/4 (0 NA)
q=0.3 / optical / Δt=0.125×	NA	0/4 (4 NA)	-0.826	0/4 (4 NA)	-0.257	0/4 (4 NA)	-0.790	4/4 (0 NA)
q=0.3 / optical / Δt=0.25×	NA	0/4 (4 NA)	-0.826	0/4 (4 NA)	-0.257	0/4 (4 NA)	-0.790	4/4 (0 NA)
q=0.3 / optical / Δt=0.5×	-0.055	0/4 (1 NA)	-0.384	0/4 (1 NA)	-0.319	3/4 (1 NA)	-0.839	4/4 (0 NA)
q=0.3 / optical / Δt=1×	0.024	0/4 (2 NA)	0.067	0/4 (1 NA)	-0.190	2/4 (2 NA)	-0.491	3/4 (1 NA)
q=0.3 / optical / Δt=2×	-0.143	0/4 (4 NA)	0.238	0/4 (2 NA)	-0.262	2/4 (2 NA)	-0.527	3/4 (1 NA)
q=0.3 / optical / Δt=4×	-0.143	0/4 (4 NA)	-0.071	0/4 (2 NA)	-0.190	2/4 (2 NA)	-0.491	3/4 (1 NA)
q=0.5 / raw / Δt=0.0625×	0.086	0/4 (4 NA)	-0.517	0/4 (2 NA)	-0.486	0/4 (4 NA)	-0.763	4/4 (0 NA)
q=0.5 / raw / Δt=0.125×	NA	0/4 (4 NA)	-0.714	0/4 (3 NA)	-0.486	0/4 (4 NA)	-0.763	4/4 (0 NA)
q=0.5 / raw / Δt=0.25×	NA	0/4 (4 NA)	-0.786	1/4 (3 NA)	-0.486	0/4 (4 NA)	-0.763	4/4 (0 NA)
q=0.5 / raw / Δt=0.5×	0.038	0/4 (1 NA)	-0.286	0/4 (1 NA)	-0.516	3/4 (1 NA)	-0.841	4/4 (0 NA)
q=0.5 / raw / Δt=1×	0.238	0/4 (2 NA)	0.236	0/4 (1 NA)	-0.452	2/4 (2 NA)	-0.591	3/4 (1 NA)
q=0.5 / raw / Δt=2×	0.238	0/4 (2 NA)	0.224	0/4 (1 NA)	-0.405	2/4 (2 NA)	-0.564	3/4 (1 NA)
q=0.5 / raw / Δt=4×	0.119	0/4 (2 NA)	-0.083	0/4 (1 NA)	-0.405	2/4 (2 NA)	-0.564	3/4 (1 NA)
q=0.5 / optical / Δt=0.0625×	NA	0/4 (4 NA)	-0.750	0/4 (4 NA)	-0.257	0/4 (4 NA)	-0.759	4/4 (0 NA)
q=0.5 / optical / Δt=0.125×	NA	0/4 (4 NA)	-0.750	0/4 (4 NA)	-0.257	0/4 (4 NA)	-0.759	4/4 (0 NA)
q=0.5 / optical / Δt=0.25×	NA	0/4 (4 NA)	-0.750	0/4 (4 NA)	-0.257	0/4 (4 NA)	-0.759	4/4 (0 NA)
q=0.5 / optical / Δt=0.5×	-0.007	0/4 (1 NA)	-0.235	0/4 (1 NA)	-0.389	3/4 (1 NA)	-0.833	4/4 (0 NA)
q=0.5 / optical / Δt=2×	-0.143	0/4 (4 NA)	0.238	0/4 (2 NA)	-0.262	2/4 (2 NA)	-0.527	3/4 (1 NA)
q=0.5 / optical / Δt=4×	-0.143	0/4 (4 NA)	-0.071	0/4 (2 NA)	-0.262	2/4 (2 NA)	-0.527	3/4 (1 NA)
q=0.7 / raw / Δt=0.0625×	NA	0/4 (4 NA)	-0.786	1/4 (3 NA)	-0.486	0/4 (4 NA)	-0.763	4/4 (0 NA)
q=0.7 / raw / Δt=0.125×	NA	0/4 (4 NA)	-0.786	1/4 (3 NA)	-0.486	0/4 (4 NA)	-0.763	4/4 (0 NA)
q=0.7 / raw / Δt=0.25×	NA	0/4 (4 NA)	-0.750	0/4 (4 NA)	-0.486	0/4 (4 NA)	-0.763	4/4 (0 NA)
q=0.7 / raw / Δt=0.5×	0.181	0/4 (1 NA)	-0.017	0/4 (1 NA)	-0.446	3/4 (1 NA)	-0.838	4/4 (0 NA)
q=0.7 / raw / Δt=1×	0.286	0/4 (2 NA)	0.273	0/4 (1 NA)	-0.167	2/4 (2 NA)	-0.527	3/4 (1 NA)
q=0.7 / raw / Δt=2×	0.286	0/4 (2 NA)	0.261	0/4 (1 NA)	-0.167	2/4 (2 NA)	-0.527	3/4 (1 NA)
q=0.7 / raw / Δt=4×	-0.071	0/4 (3 NA)	-0.262	0/4 (2 NA)	-0.071	2/4 (2 NA)	-0.473	3/4 (1 NA)
q=0.7 / optical / Δt=0.0625×	NA	0/4 (4 NA)	-0.750	0/4 (4 NA)	-0.086	0/4 (4 NA)	-0.735	4/4 (0 NA)
q=0.7 / optical / Δt=0.125×	NA	0/4 (4 NA)	-0.750	0/4 (4 NA)	-0.086	0/4 (4 NA)	-0.735	4/4 (0 NA)
q=0.7 / optical / Δt=0.25×	NA	0/4 (4 NA)	-0.600	0/4 (4 NA)	-0.086	0/4 (4 NA)	-0.735	4/4 (0 NA)
q=0.7 / optical / Δt=0.5×	0.014	0/4 (1 NA)	-0.244	0/4 (1 NA)	-0.297	3/4 (1 NA)	-0.817	4/4 (0 NA)
q=0.7 / optical / Δt=1×	0.119	0/4 (2 NA)	0.139	0/4 (1 NA)	-0.262	2/4 (2 NA)	-0.555	3/4 (1 NA)
q=0.7 / optical / Δt=2×	-0.143	0/4 (3 NA)	0.000	0/4 (2 NA)	-0.190	2/4 (2 NA)	-0.518	3/4 (1 NA)
q=0.7 / optical / Δt=4×	-0.143	0/4 (3 NA)	-0.214	0/4 (2 NA)	-0.190	2/4 (2 NA)	-0.518	3/4 (1 NA)

Arm longest · σ=1 · actual per-size horizons retained in the source CSV.
Rank entries are Spearman coefficients; multi-front denotes the fraction of time with at least two qualified regions.
Every threshold, raw/physical-kernel and observation-cadence view is retained. Δt is relative to the declared common imaging cadence.
Direct resolved τ and D_θ-proxy associations remain separate. All four folds are planned; NA counts remain in the denominator.
Only the primary, forced-excluded, direct-τ rows enter the declared H2 A6 gate. Other rows are sensitivity checks.

S10 · Primary predictor sensitivity · forced included

A5 imaging view	Multi-front τ rank	τ folds	Multi-front proxy rank	Proxy folds	Count τ rank	τ folds	Count proxy rank	Proxy folds
Primary imaging view	-0.021	0/4 (1 NA)	-0.632	1/4 (0 NA)	-0.342	3/4 (1 NA)	-0.873	4/4 (0 NA)
q=0.3 / raw / Δt=0.0625×	0.086	0/4 (4 NA)	-0.708	1/4 (1 NA)	-0.486	0/4 (4 NA)	-0.701	4/4 (0 NA)
q=0.3 / raw / Δt=0.125×	-0.143	0/4 (4 NA)	-0.793	1/4 (2 NA)	-0.486	0/4 (4 NA)	-0.701	4/4 (0 NA)
q=0.3 / raw / Δt=0.25×	0.086	0/4 (4 NA)	-0.756	1/4 (2 NA)	-0.257	0/4 (4 NA)	-0.698	4/4 (0 NA)
q=0.3 / raw / Δt=0.5×	0.387	0/4 (1 NA)	-0.506	0/4 (0 NA)	-0.593	3/4 (1 NA)	-0.821	3/4 (0 NA)
q=0.3 / raw / Δt=1×	0.147	0/4 (1 NA)	-0.270	0/4 (0 NA)	-0.385	2/4 (1 NA)	-0.728	4/4 (0 NA)
q=0.3 / raw / Δt=2×	0.147	0/4 (1 NA)	-0.270	0/4 (0 NA)	-0.385	2/4 (1 NA)	-0.728	4/4 (0 NA)
q=0.3 / raw / Δt=4×	0.175	0/4 (1 NA)	-0.257	0/4 (0 NA)	-0.329	2/4 (1 NA)	-0.717	4/4 (0 NA)
q=0.3 / optical / Δt=0.0625×	NA	0/4 (4 NA)	-0.869	0/4 (4 NA)	-0.257	0/4 (4 NA)	-0.825	4/4 (0 NA)
q=0.3 / optical / Δt=0.125×	NA	0/4 (4 NA)	-0.869	0/4 (4 NA)	-0.257	0/4 (4 NA)	-0.825	4/4 (0 NA)
q=0.3 / optical / Δt=0.25×	NA	0/4 (4 NA)	-0.869	0/4 (4 NA)	-0.257	0/4 (4 NA)	-0.825	4/4 (0 NA)
q=0.3 / optical / Δt=0.5×	-0.039	0/4 (1 NA)	-0.674	1/4 (0 NA)	-0.379	3/4 (1 NA)	-0.860	4/4 (0 NA)
q=0.3 / optical / Δt=1×	-0.063	0/4 (1 NA)	-0.412	0/4 (0 NA)	-0.049	2/4 (1 NA)	-0.490	4/4 (0 NA)
q=0.3 / optical / Δt=2×	-0.345	0/4 (1 NA)	-0.486	1/4 (0 NA)	-0.077	2/4 (1 NA)	-0.501	4/4 (0 NA)
q=0.3 / optical / Δt=4×	-0.183	0/4 (2 NA)	-0.582	1/4 (1 NA)	-0.049	2/4 (1 NA)	-0.490	4/4 (0 NA)
q=0.5 / raw / Δt=0.0625×	0.086	0/4 (4 NA)	-0.691	1/4 (1 NA)	-0.486	0/4 (4 NA)	-0.675	4/4 (0 NA)
q=0.5 / raw / Δt=0.125×	NA	0/4 (4 NA)	-0.794	1/4 (2 NA)	-0.486	0/4 (4 NA)	-0.675	4/4 (0 NA)
q=0.5 / raw / Δt=0.25×	NA	0/4 (4 NA)	-0.842	2/4 (2 NA)	-0.486	0/4 (4 NA)	-0.675	4/4 (0 NA)
q=0.5 / raw / Δt=0.5×	0.292	0/4 (1 NA)	-0.529	0/4 (0 NA)	-0.484	3/4 (1 NA)	-0.807	3/4 (0 NA)
q=0.5 / raw / Δt=1×	0.266	0/4 (1 NA)	-0.216	0/4 (0 NA)	-0.098	2/4 (1 NA)	-0.649	4/4 (0 NA)
q=0.5 / raw / Δt=2×	0.203	0/4 (1 NA)	-0.235	0/4 (0 NA)	-0.140	2/4 (1 NA)	-0.653	4/4 (0 NA)
q=0.5 / raw / Δt=4×	0.161	0/4 (1 NA)	-0.356	0/4 (0 NA)	-0.175	2/4 (1 NA)	-0.657	4/4 (0 NA)
q=0.5 / optical / Δt=0.0625×	NA	0/4 (4 NA)	-0.850	0/4 (4 NA)	-0.257	0/4 (4 NA)	-0.809	4/4 (0 NA)
q=0.5 / optical / Δt=0.125×	NA	0/4 (4 NA)	-0.850	0/4 (4 NA)	-0.257	0/4 (4 NA)	-0.809	4/4 (0 NA)
q=0.5 / optical / Δt=0.25×	NA	0/4 (4 NA)	-0.850	0/4 (4 NA)	-0.257	0/4 (4 NA)	-0.809	4/4 (0 NA)
q=0.5 / optical / Δt=0.5×	-0.050	0/4 (1 NA)	-0.638	1/4 (0 NA)	-0.364	3/4 (1 NA)	-0.874	4/4 (0 NA)
q=0.5 / optical / Δt=2×	-0.297	0/4 (1 NA)	-0.464	1/4 (0 NA)	-0.021	2/4 (1 NA)	-0.585	4/4 (0 NA)
q=0.5 / optical / Δt=4×	-0.059	0/4 (2 NA)	-0.523	1/4 (1 NA)	-0.021	2/4 (1 NA)	-0.585	4/4 (0 NA)
q=0.7 / raw / Δt=0.0625×	NA	0/4 (4 NA)	-0.842	2/4 (2 NA)	-0.486	0/4 (4 NA)	-0.675	4/4 (0 NA)
q=0.7 / raw / Δt=0.125×	NA	0/4 (4 NA)	-0.842	2/4 (2 NA)	-0.486	0/4 (4 NA)	-0.675	4/4 (0 NA)
q=0.7 / raw / Δt=0.25×	NA	0/4 (4 NA)	-0.850	0/4 (4 NA)	-0.486	0/4 (4 NA)	-0.675	4/4 (0 NA)
q=0.7 / raw / Δt=0.5×	0.441	0/4 (1 NA)	-0.449	0/4 (0 NA)	-0.127	3/4 (1 NA)	-0.793	3/4 (0 NA)
q=0.7 / raw / Δt=1×	0.399	0/4 (1 NA)	-0.169	0/4 (0 NA)	0.259	0/4 (1 NA)	-0.587	4/4 (0 NA)
q=0.7 / raw / Δt=2×	0.399	0/4 (1 NA)	-0.169	0/4 (0 NA)	0.294	0/4 (1 NA)	-0.579	4/4 (0 NA)
q=0.7 / raw / Δt=4×	0.236	0/4 (1 NA)	-0.361	0/4 (0 NA)	0.294	0/4 (1 NA)	-0.571	4/4 (0 NA)
q=0.7 / optical / Δt=0.0625×	NA	0/4 (4 NA)	-0.850	0/4 (4 NA)	-0.086	0/4 (4 NA)	-0.662	4/4 (0 NA)
q=0.7 / optical / Δt=0.125×	NA	0/4 (4 NA)	-0.850	0/4 (4 NA)	-0.086	0/4 (4 NA)	-0.662	4/4 (0 NA)
q=0.7 / optical / Δt=0.25×	NA	0/4 (4 NA)	-0.762	0/4 (4 NA)	-0.086	0/4 (4 NA)	-0.662	4/4 (0 NA)
q=0.7 / optical / Δt=0.5×	0.062	0/4 (1 NA)	-0.603	0/4 (0 NA)	-0.112	3/4 (1 NA)	-0.828	4/4 (0 NA)
q=0.7 / optical / Δt=1×	0.042	0/4 (1 NA)	-0.331	0/4 (0 NA)	0.196	0/4 (1 NA)	-0.593	4/4 (0 NA)
q=0.7 / optical / Δt=2×	-0.191	0/4 (1 NA)	-0.424	0/4 (0 NA)	0.238	0/4 (1 NA)	-0.579	4/4 (0 NA)
q=0.7 / optical / Δt=4×	-0.200	0/4 (1 NA)	-0.586	1/4 (0 NA)	0.231	0/4 (1 NA)	-0.581	4/4 (0 NA)

Arm longest · σ=1 · actual per-size horizons retained in the source CSV.
Rank entries are Spearman coefficients; multi-front denotes the fraction of time with at least two qualified regions.
Every threshold, raw/physical-kernel and observation-cadence view is retained. Δt is relative to the declared common imaging cadence.
Direct resolved τ and D_θ-proxy associations remain separate. All four folds are planned; NA counts remain in the denominator.
Only the primary, forced-excluded, direct-τ rows enter the declared H2 A6 gate. Other rows are sensitivity checks.

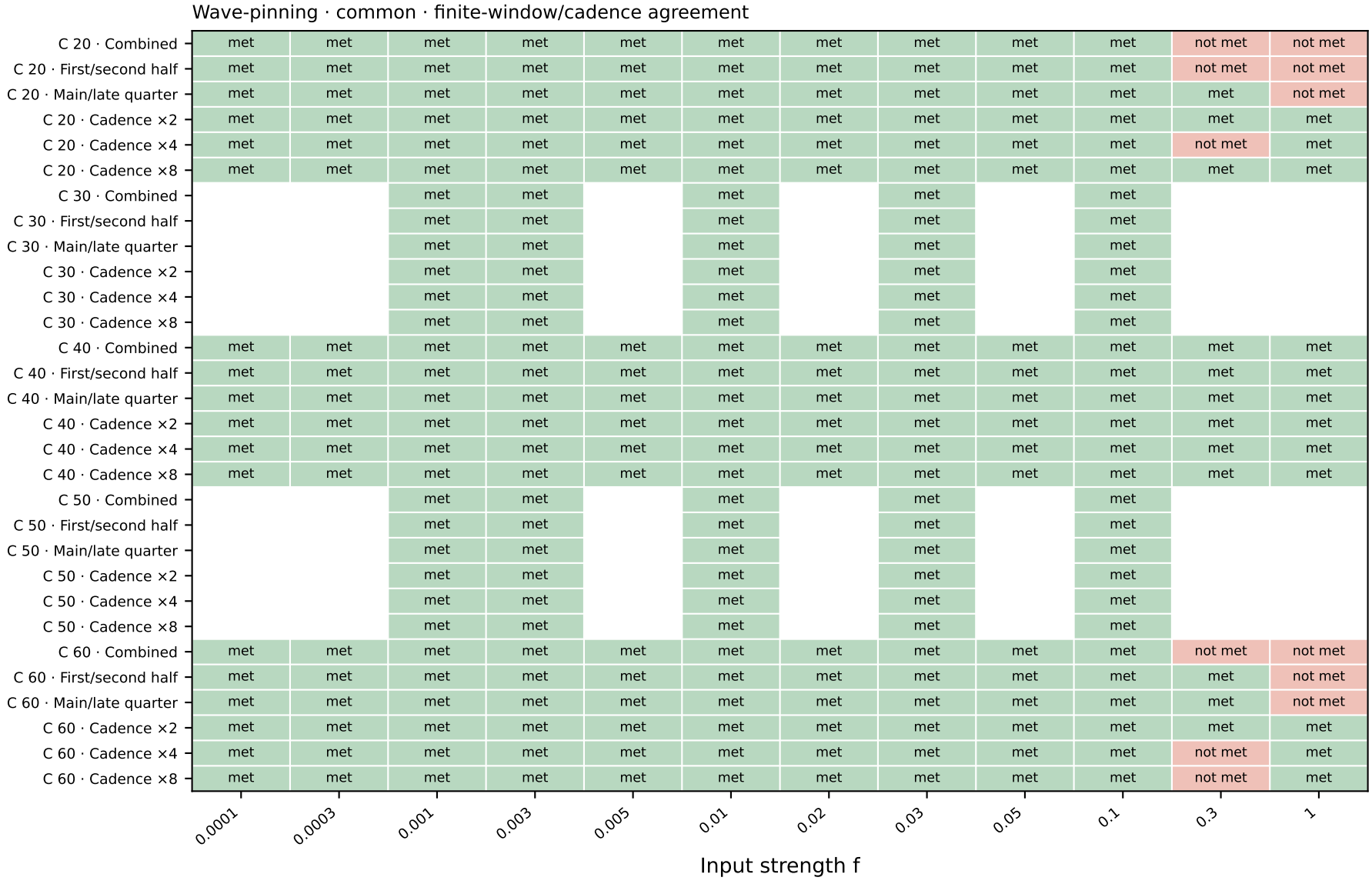
S11 · Native activity atlas appendices

All 578 native atlas pages are retained in 13 numbered PDF volumes. This core PDF indexes those appendices. Full plotted native fields remain in verified HPC wideCSV.gz archives; portable per-page inventories give exact paths, hashes, native grids and row counts. Statistical CSVs and selection/snapshot metadata are portable. Native profiles retain their saved grids and colour scales; no interpolation or quantization is introduced.

Volume	PDF filename	Native pages	MiB
1	figure4_S11_native_atlas_appendix_001.pdf	50	17.00
2	figure4_S11_native_atlas_appendix_002.pdf	50	16.74
3	figure4_S11_native_atlas_appendix_003.pdf	50	19.05
4	figure4_S11_native_atlas_appendix_004.pdf	50	20.43
5	figure4_S11_native_atlas_appendix_005.pdf	50	18.10
6	figure4_S11_native_atlas_appendix_006.pdf	50	19.45
7	figure4_S11_native_atlas_appendix_007.pdf	50	20.23
8	figure4_S11_native_atlas_appendix_008.pdf	40	17.11
9	figure4_S11_native_atlas_appendix_009.pdf	50	31.24
10	figure4_S11_native_atlas_appendix_010.pdf	22	15.51
11	figure4_S11_native_atlas_appendix_011.pdf	50	14.88
12	figure4_S11_native_atlas_appendix_012.pdf	50	15.26
13	figure4_S11_native_atlas_appendix_013.pdf	16	5.40

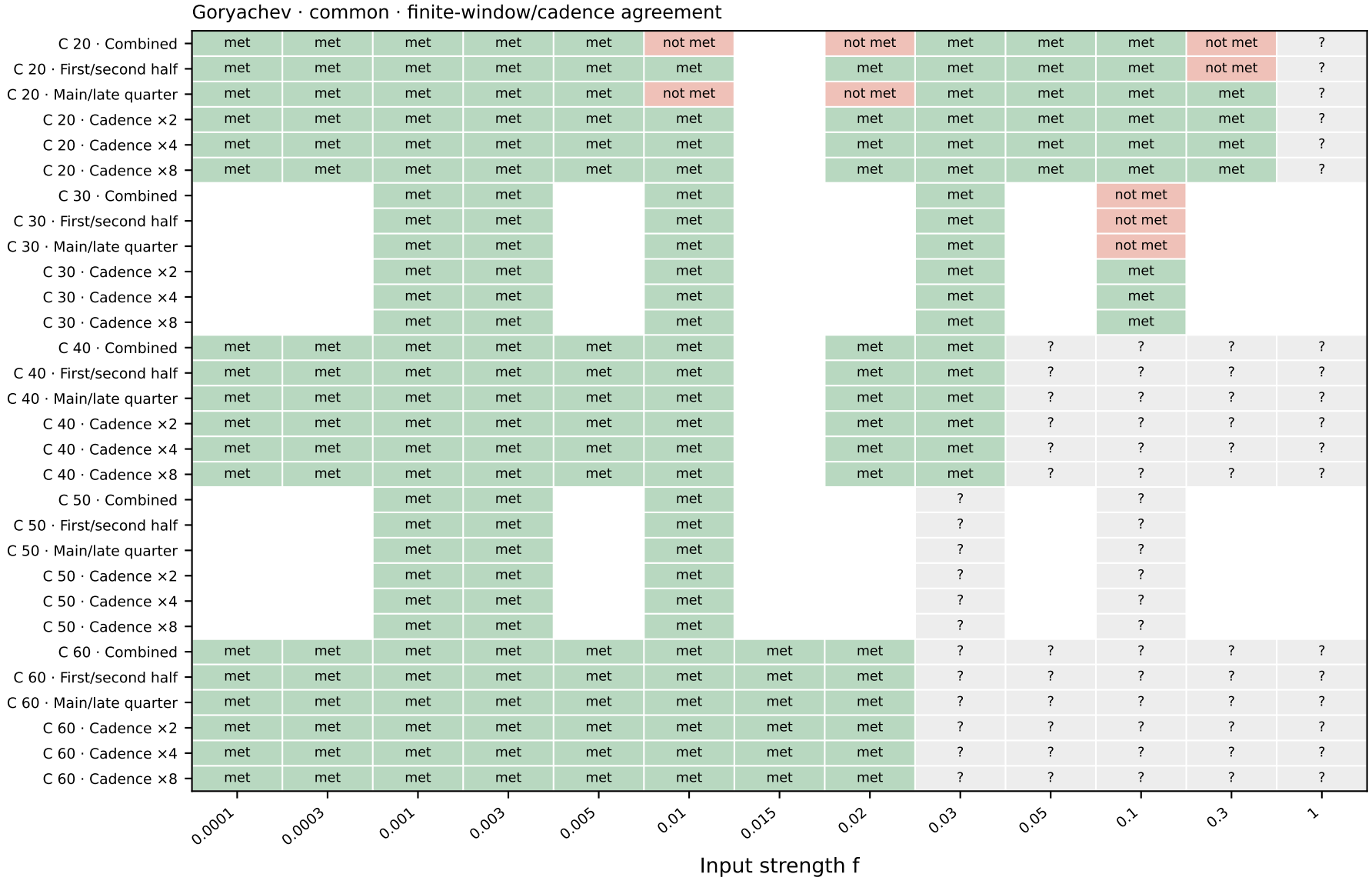
Complete per-page registry: figure4_S11_complete_appendix_page_registry.csv
Every appendix and native-field CSV is hash-verified. The reader links compact archive inventories; full-field files are available on HPC.

S12 · Completed audited Figure 4 criteria and qualifications · A7 qualification by condition and comparison



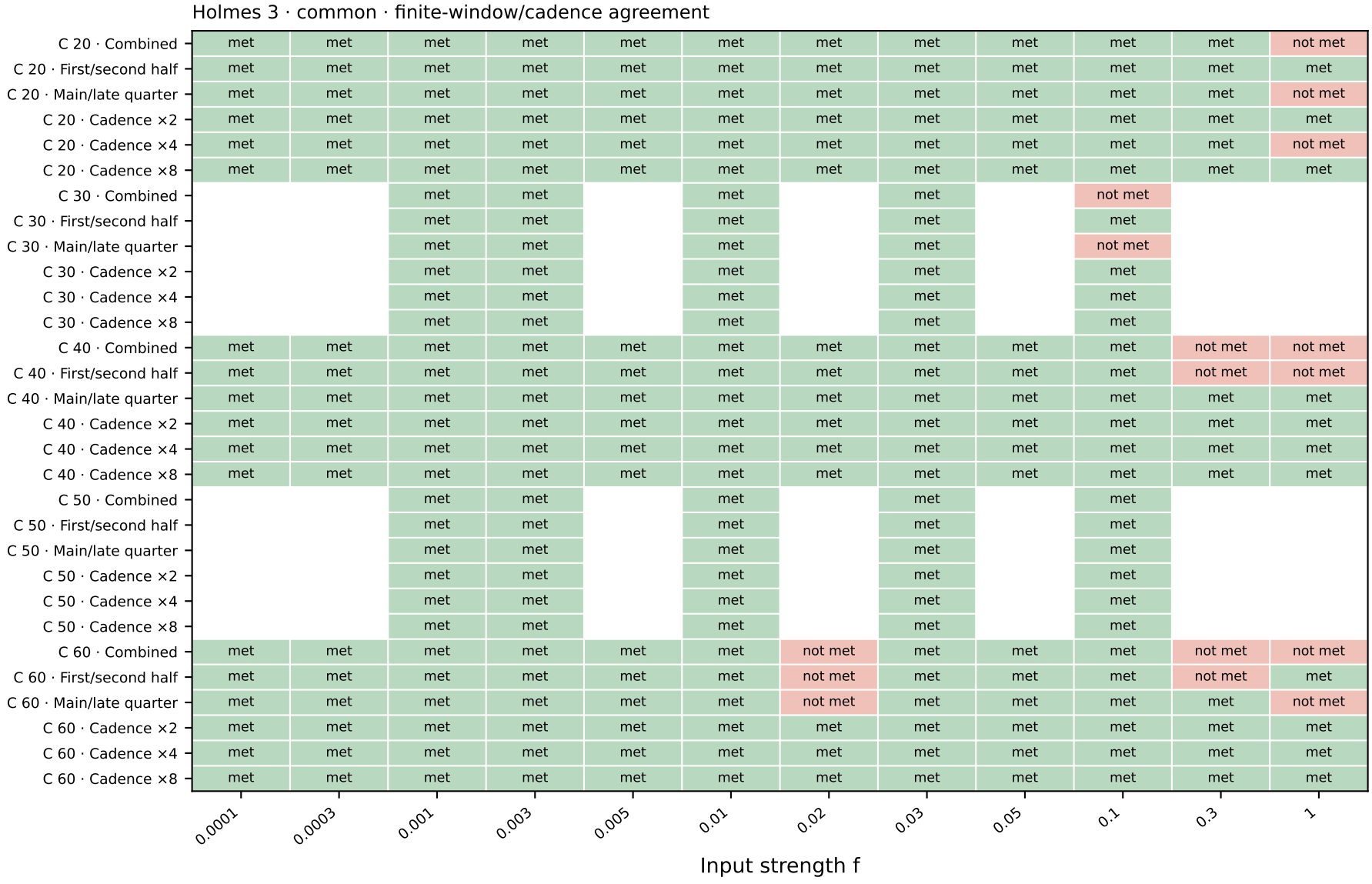
Completed audited Figure 4 criteria and qualifications. These are interval-and-native-regime agreement checks. Angular-mobility disagreement remains separately recorded in components. Blank cells are outside the declared grid. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_581_S12_windows_common_wave_pinning.csv

S12 · Completed audited Figure 4 criteria and qualifications · A7 qualification by condition and comparison



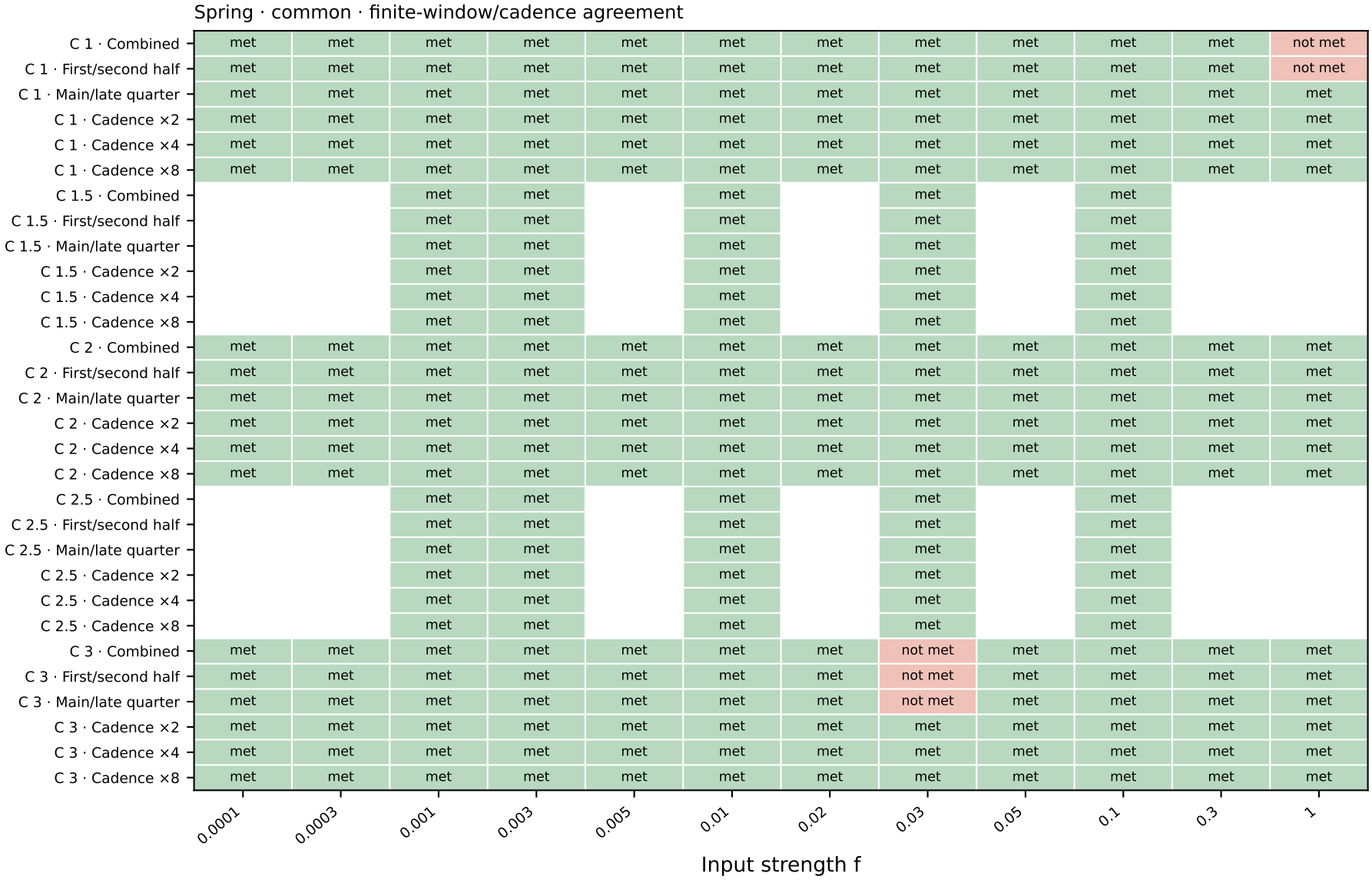
Completed audited Figure 4 criteria and qualifications. These are interval-and-native-regime agreement checks. Angular-mobility disagreement remains separately recorded in components. Blank cells are outside the declared grid. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_582_S12_windows_common_goryachev.csv

S12 · Completed audited Figure 4 criteria and qualifications · A7 qualification by condition and comparison



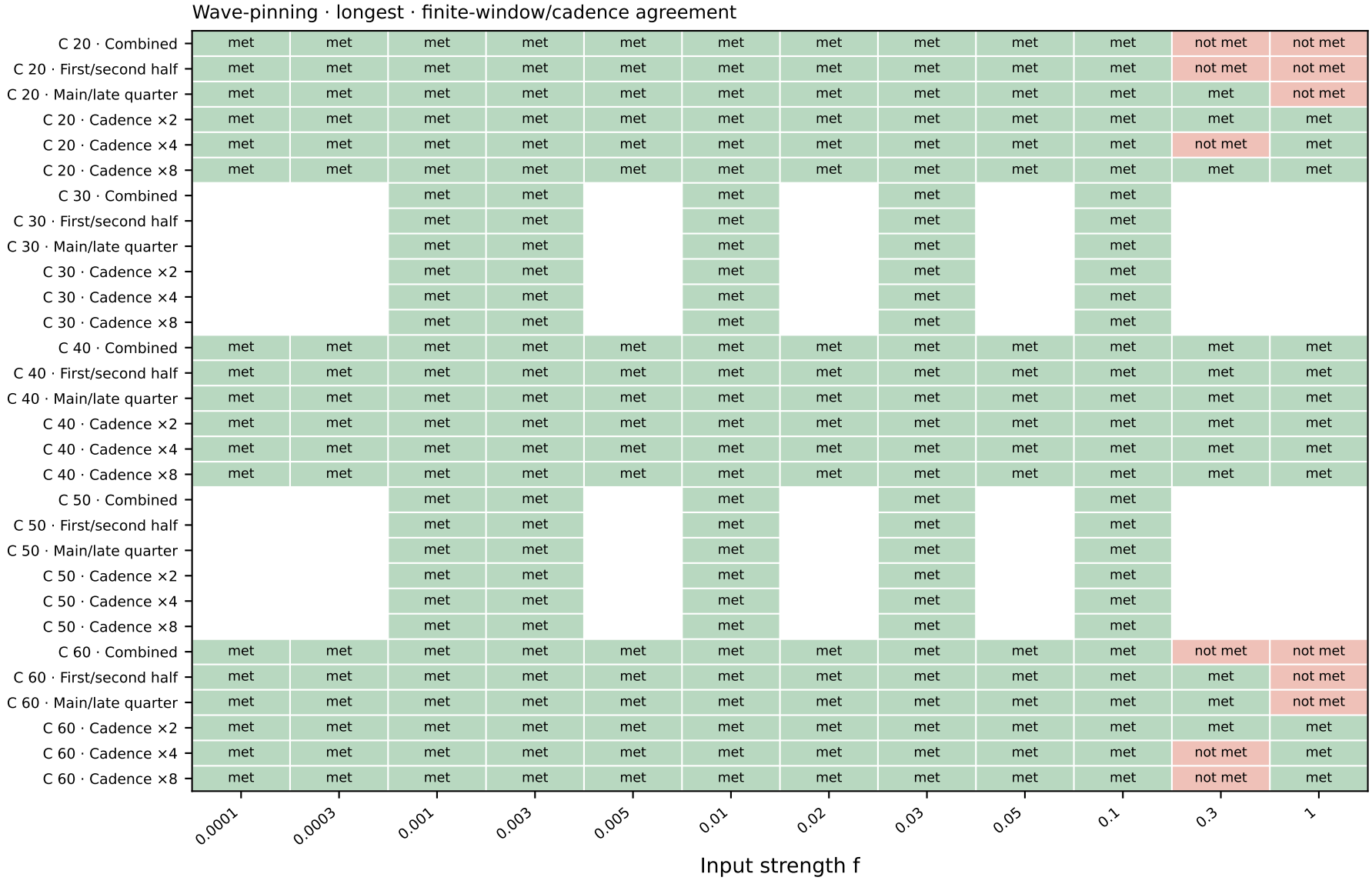
Completed audited Figure 4 criteria and qualifications. These are interval-and-native-regime agreement checks. Angular-mobility disagreement remains separately recorded in components. Blank cells are outside the declared grid. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_583_S12_windows_common_holmes_model3.csv

S12 · Completed audited Figure 4 criteria and qualifications · A7 qualification by condition and comparison



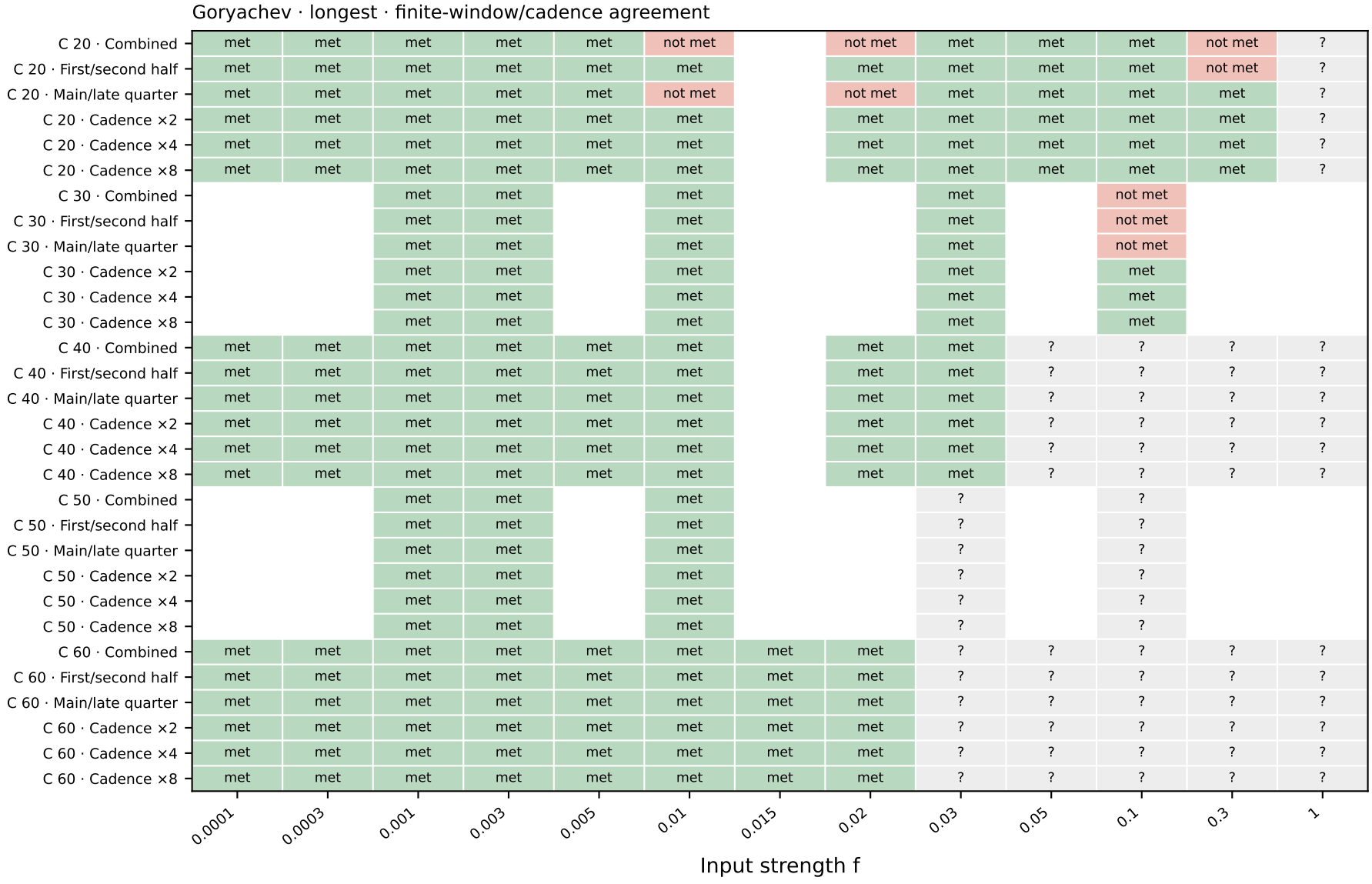
Completed audited Figure 4 criteria and qualifications. These are interval-and-native-regime agreement checks. Angular-mobility disagreement remains separately recorded in components. Blank cells are outside the declared grid. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_584_S12_windows_common_spring_ring.csv

S12 · Completed audited Figure 4 criteria and qualifications · A7 qualification by condition and comparison



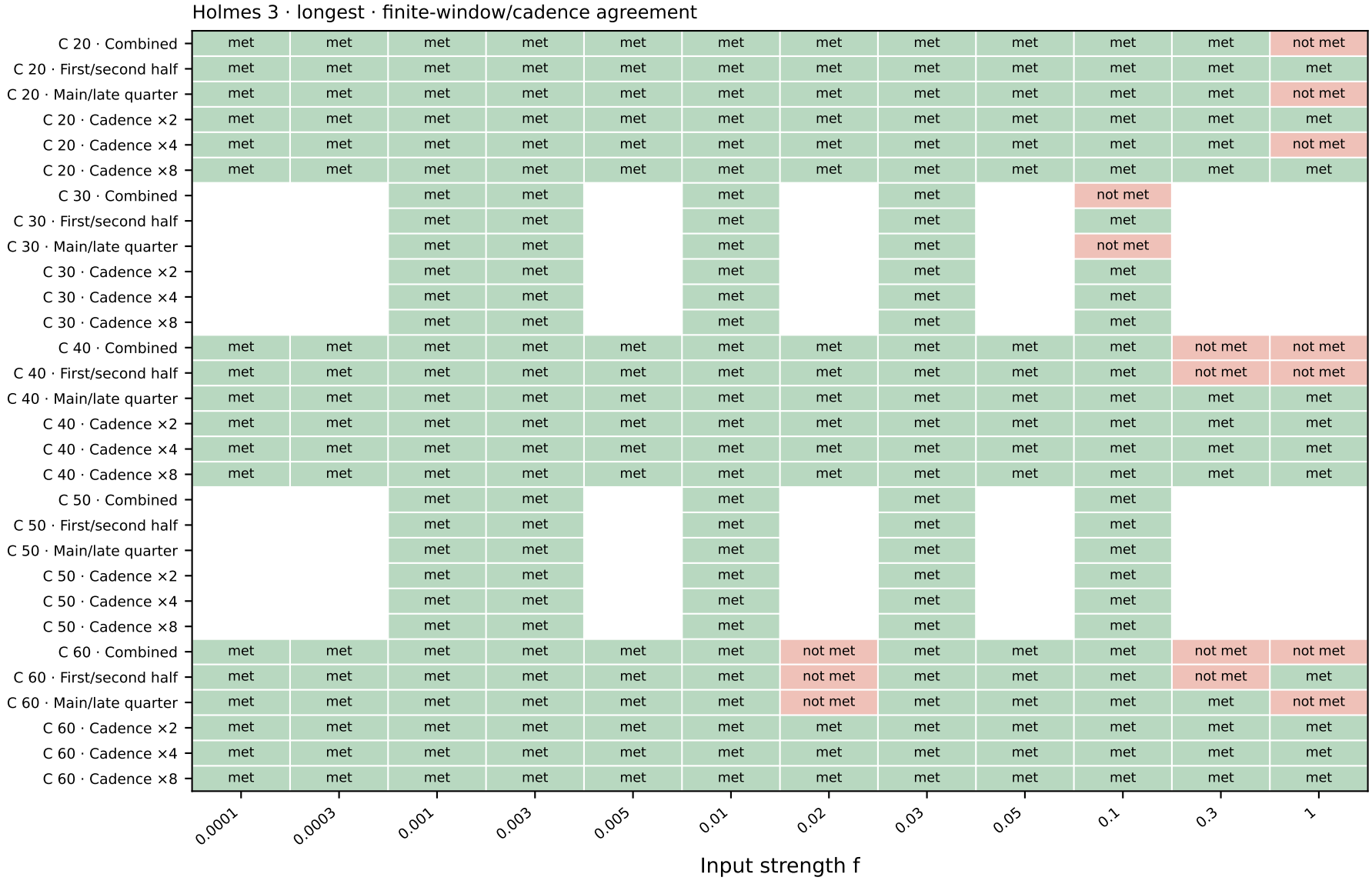
Completed audited Figure 4 criteria and qualifications. These are interval-and-native-regime agreement checks. Angular-mobility disagreement remains separately recorded in components. Blank cells are outside the declared grid. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_585_S12_windows_longest_wave_pinning.csv

S12 · Completed audited Figure 4 criteria and qualifications · A7 qualification by condition and comparison



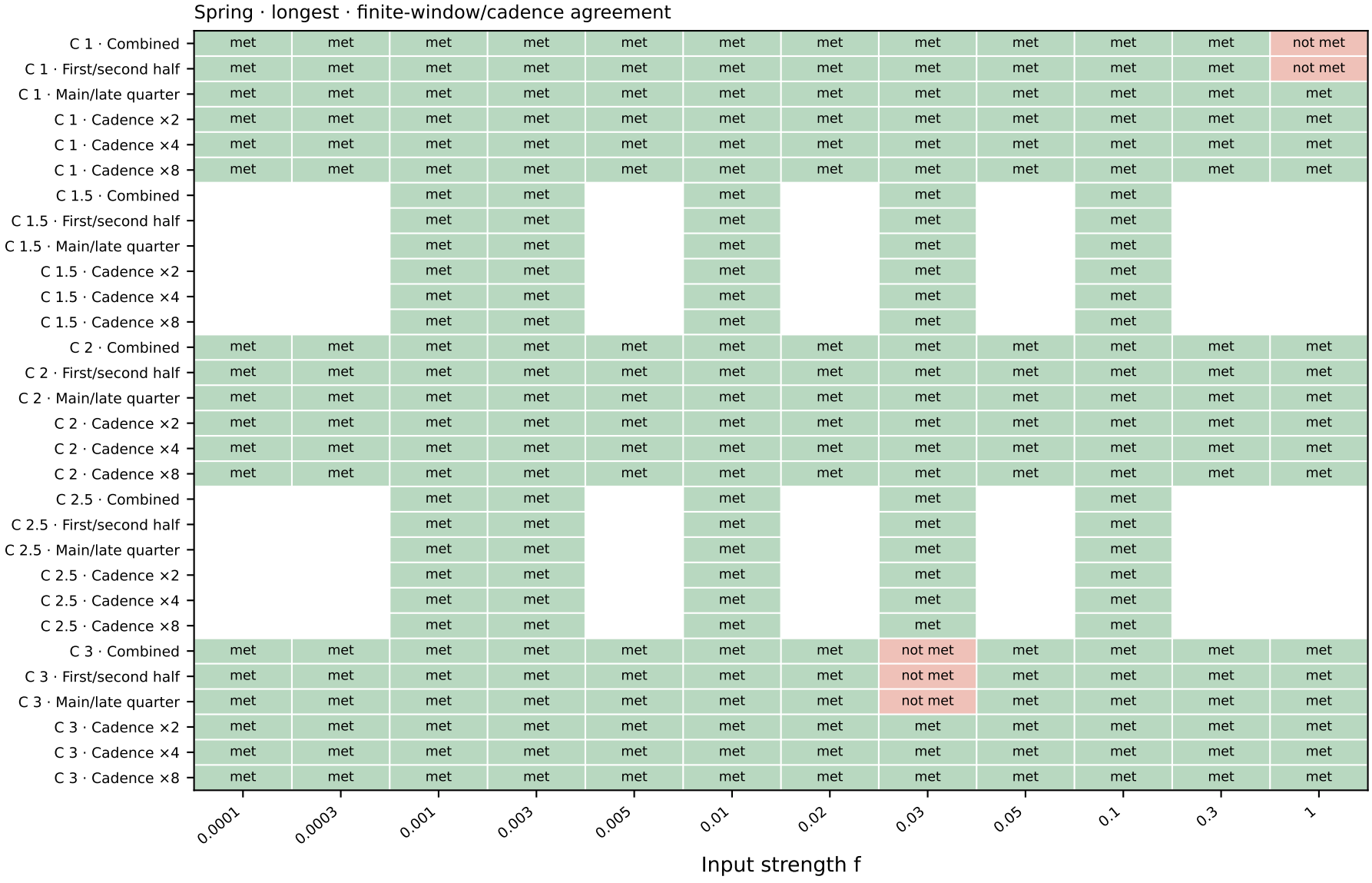
Completed audited Figure 4 criteria and qualifications. These are interval-and-native-regime agreement checks. Angular-mobility disagreement remains separately recorded in components. Blank cells are outside the declared grid. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_586_S12_windows_longest_goryachev.csv

S12 · Completed audited Figure 4 criteria and qualifications · A7 qualification by condition and comparison



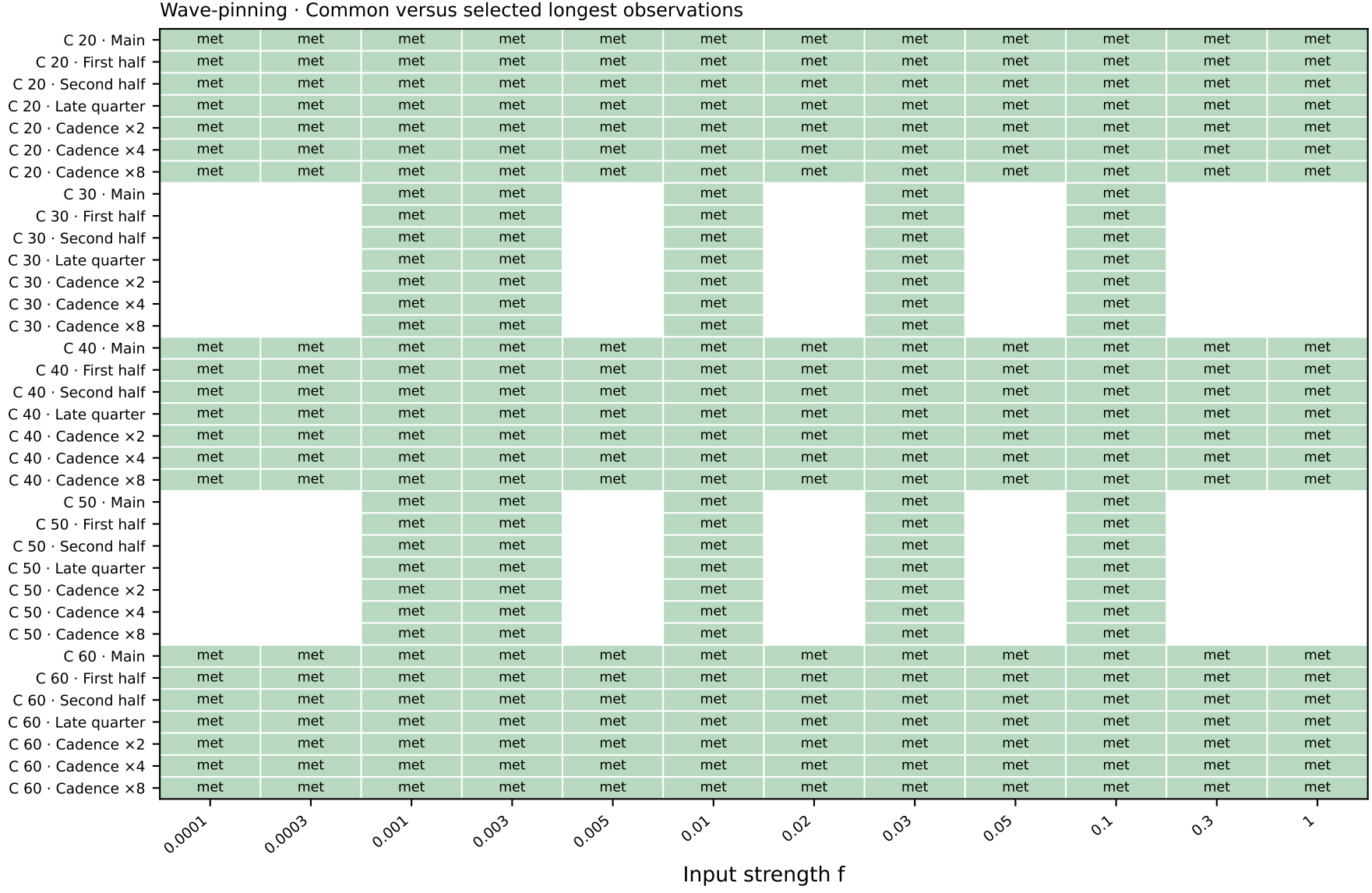
Completed audited Figure 4 criteria and qualifications. These are interval-and-native-regime agreement checks. Angular-mobility disagreement remains separately recorded in components. Blank cells are outside the declared grid. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_587_S12_windows_longest_holmes_model3.csv

S12 · Completed audited Figure 4 criteria and qualifications · A7 qualification by condition and comparison



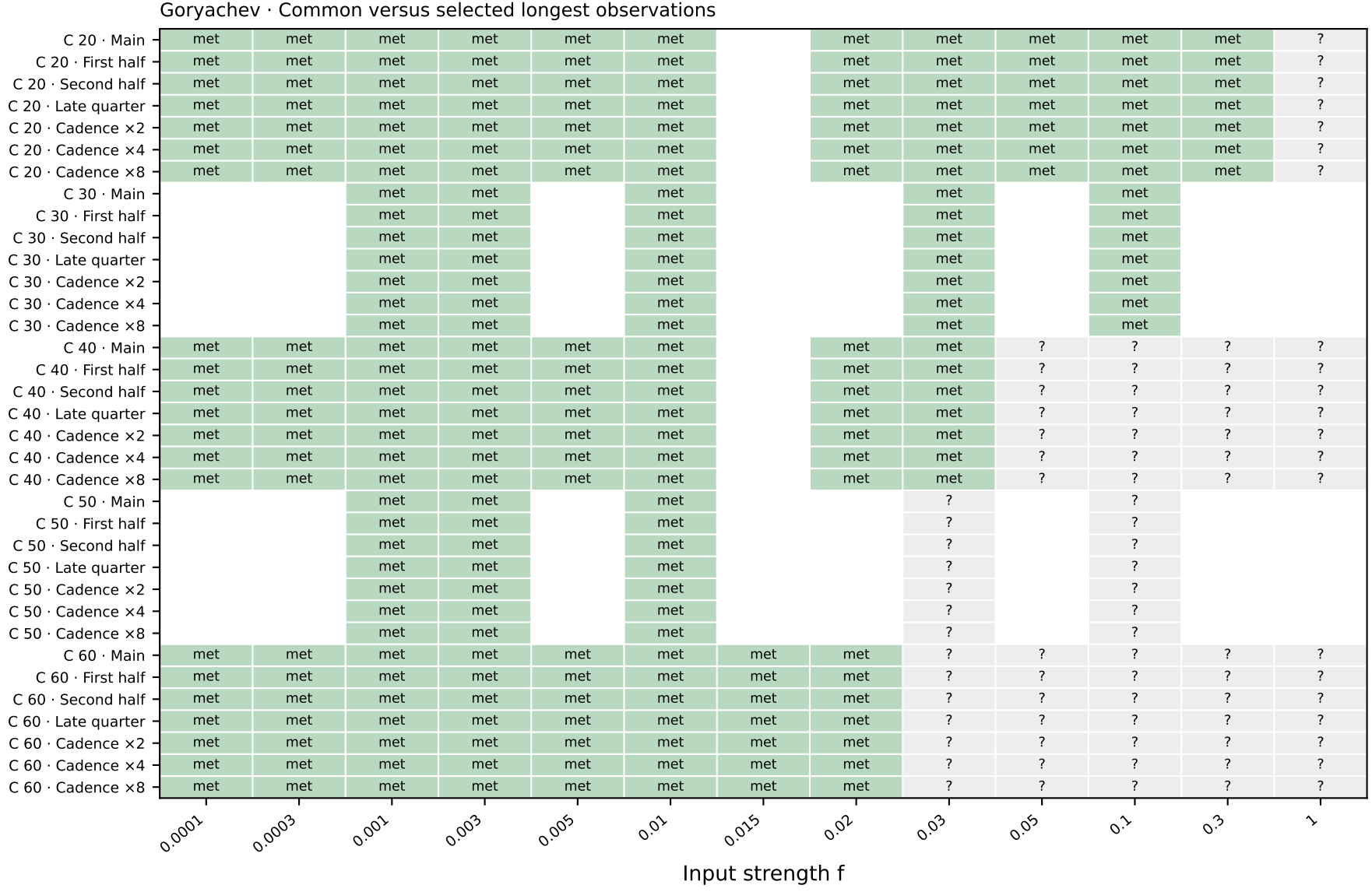
Completed audited Figure 4 criteria and qualifications. These are interval-and-native-regime agreement checks. Angular-mobility disagreement remains separately recorded in components. Blank cells are outside the declared grid. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_588_S12_windows_longest_spring_ring.csv

S12 · Completed audited Figure 4 criteria and qualifications · Common versus selected longest observations



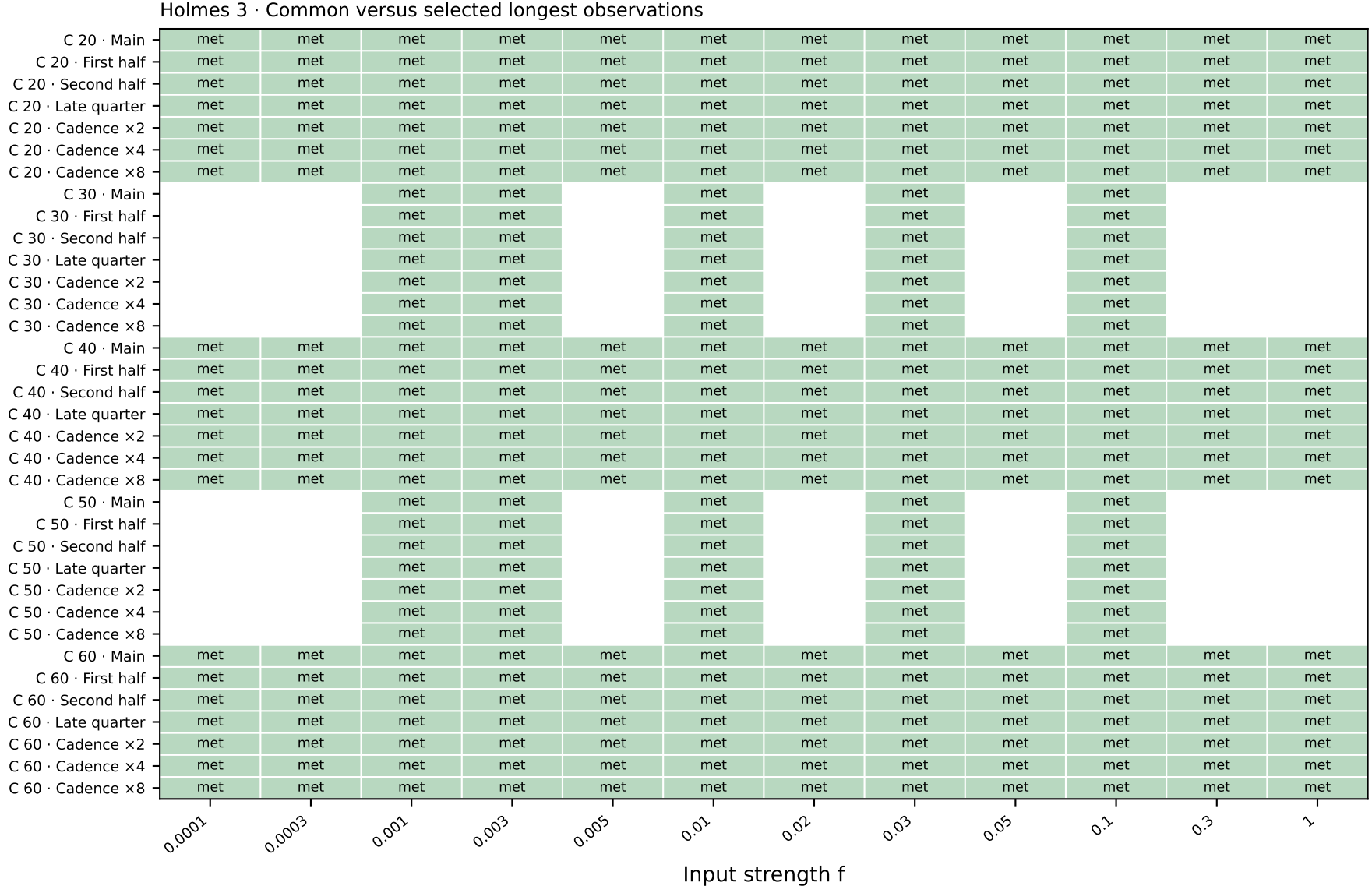
Completed audited Figure 4 criteria and qualifications. All seven views remain distinct. Exact original paths are a same-horizon subset, not additional independent replicates. A non-selected common observation is retained explicitly; no missing long observation is replaced. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_589_S12_horizon_qualification_wave_pinning.csv

S12 · Completed audited Figure 4 criteria and qualifications · Common versus selected longest observations



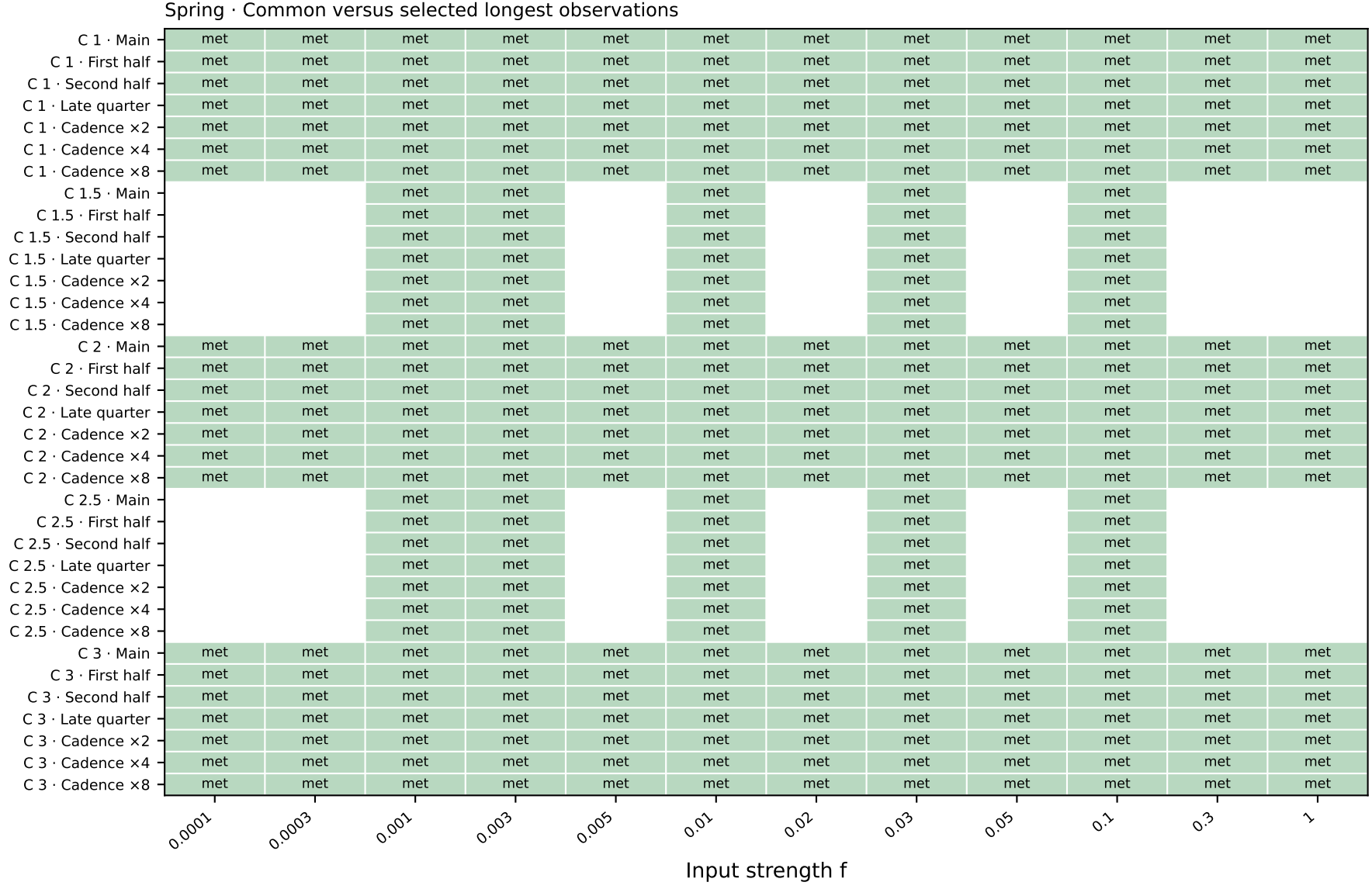
Completed audited Figure 4 criteria and qualifications. All seven views remain distinct. Exact original paths are a same-horizon subset, not additional independent replicates. A non-selected common observation is retained explicitly; no missing long observation is replaced. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_590_S12_horizon_qualification_goryachev.csv

S12 · Completed audited Figure 4 criteria and qualifications · Common versus selected longest observations



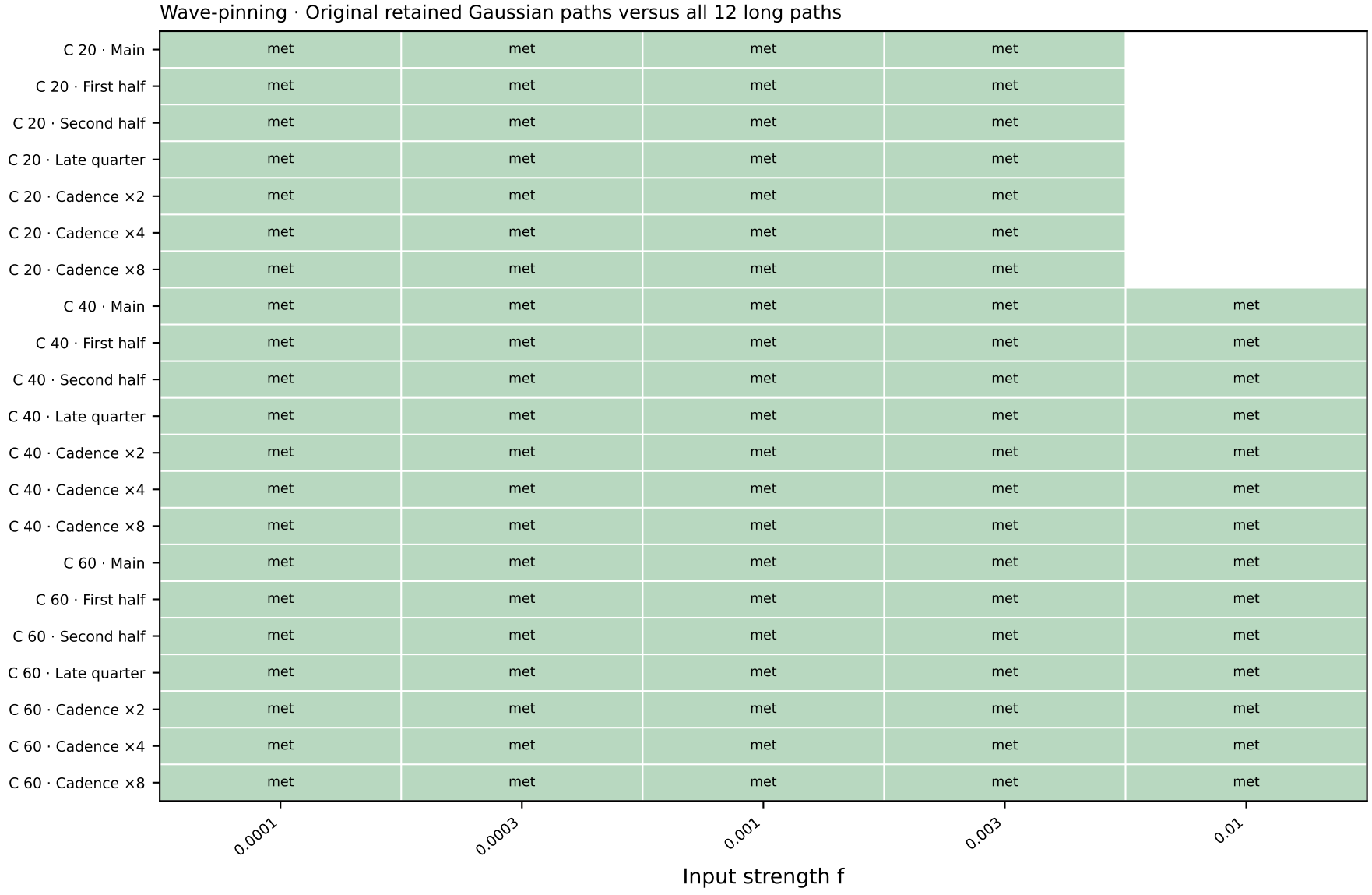
Completed audited Figure 4 criteria and qualifications. All seven views remain distinct. Exact original paths are a same-horizon subset, not additional independent replicates. A non-selected common observation is retained explicitly; no missing long observation is replaced. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_591_S12_horizon_qualification_holmes_model3.csv

S12 · Completed audited Figure 4 criteria and qualifications · Common versus selected longest observations



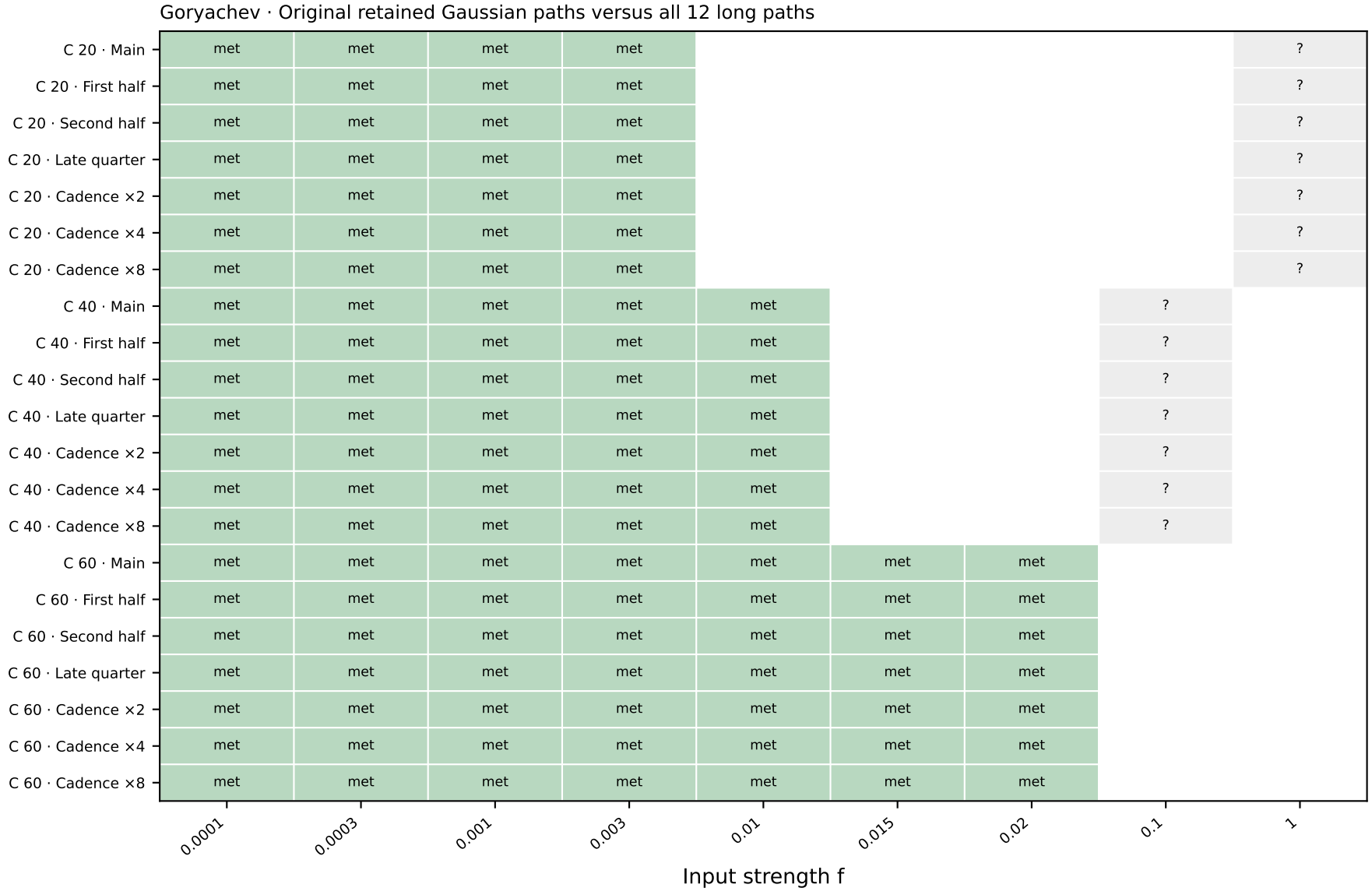
Completed audited Figure 4 criteria and qualifications. All seven views remain distinct. Exact original paths are a same-horizon subset, not additional independent replicates. A non-selected common observation is retained explicitly; no missing long observation is replaced. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_592_S12_horizon_qualification_spring_ring.csv

S12 · Completed audited Figure 4 criteria and qualifications · Original retained Gaussian paths versus all 12 long paths



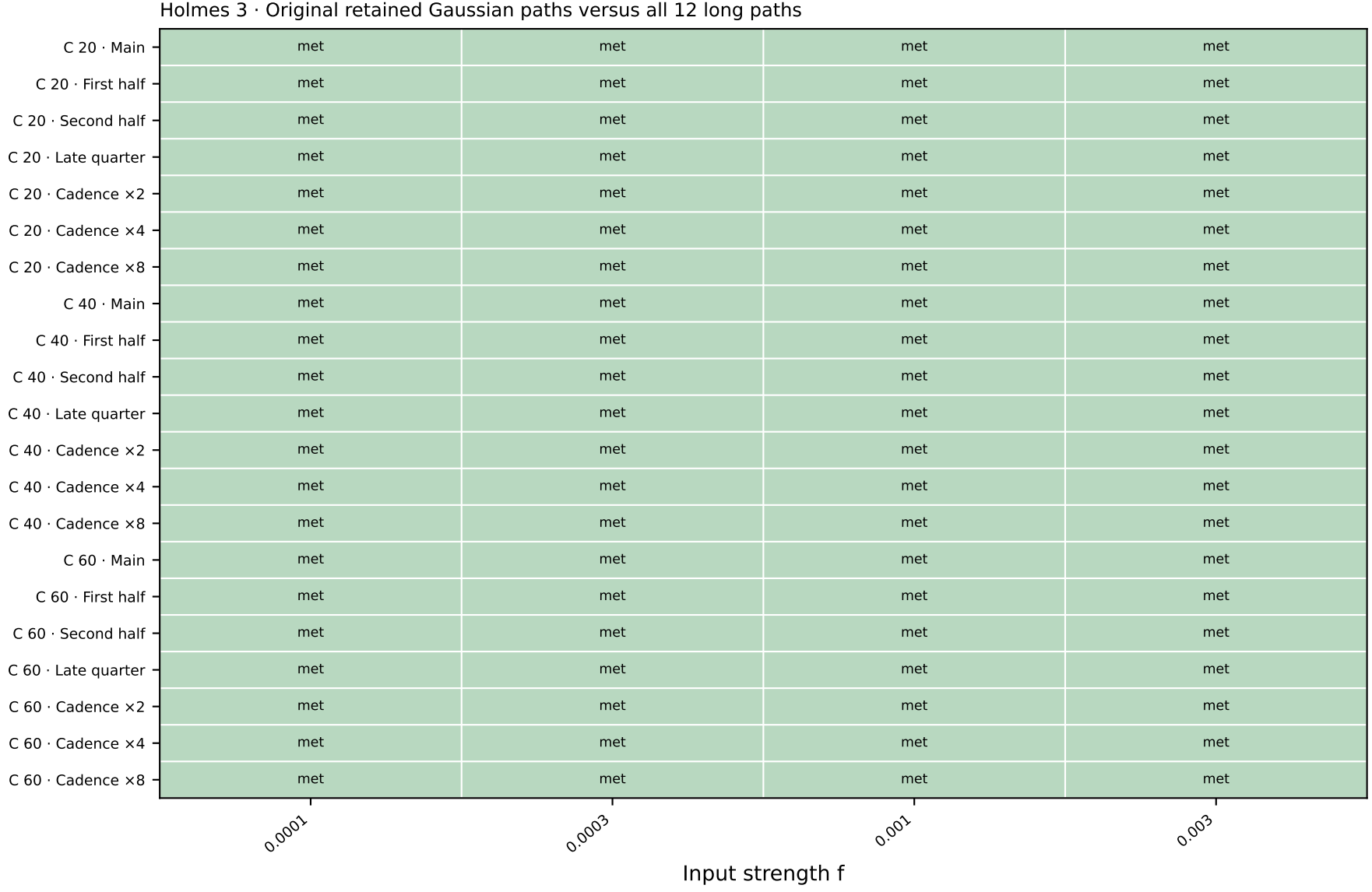
Completed audited Figure 4 criteria and qualifications. All seven views remain distinct. Exact original paths are a same-horizon subset, not additional independent replicates. A non-selected common observation is retained explicitly; no missing long observation is replaced. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_593_S12_original_seed_sensitivity_wave_pinning.csv

S12 · Completed audited Figure 4 criteria and qualifications · Original retained Gaussian paths versus all 12 long paths



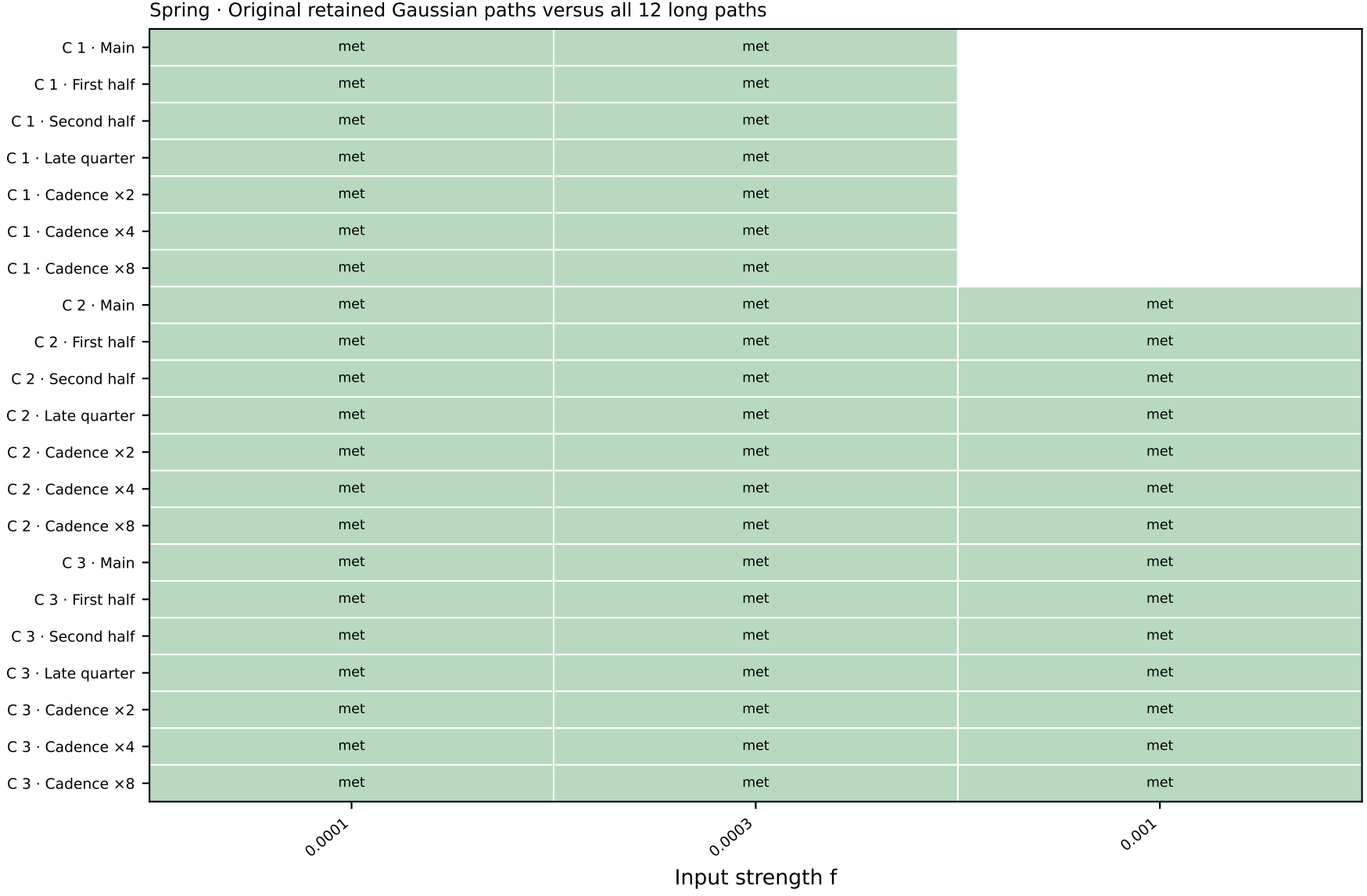
Completed audited Figure 4 criteria and qualifications. All seven views remain distinct. Exact original paths are a same-horizon subset, not additional independent replicates. A non-selected common observation is retained explicitly; no missing long observation is replaced. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_594_S12_original_seed_sensitivity_goryachev.csv

S12 · Completed audited Figure 4 criteria and qualifications · Original retained Gaussian paths versus all 12 long paths



Completed audited Figure 4 criteria and qualifications. All seven views remain distinct. Exact original paths are a same-horizon subset, not additional independent replicates. A non-selected common observation is retained explicitly; no missing long observation is replaced. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_595_S12_original_seed_sensitivity_holmes_model3.csv

S12 · Completed audited Figure 4 criteria and qualifications · Original retained Gaussian paths versus all 12 long paths



Completed audited Figure 4 criteria and qualifications. All seven views remain distinct. Exact original paths are a same-horizon subset, not additional independent replicates. A non-selected common observation is retained explicitly; no missing long observation is replaced. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_596_S12_original_seed_sensitivity_spring_ring.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

goryachev, 20, 0.0001, main: present in common arm yes; retained E3 reference.
goryachev, 20, 0.0001, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 20, 0.0001, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 20, 0.0001, late quarter: present in common arm yes; retained E3 reference.
goryachev, 20, 0.0001, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 20, 0.0001, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 20, 0.0001, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 20, 0.0003, main: present in common arm yes; retained E3 reference.
goryachev, 20, 0.0003, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 20, 0.0003, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 20, 0.0003, late quarter: present in common arm yes; retained E3 reference.
goryachev, 20, 0.0003, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 20, 0.0003, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 20, 0.0003, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 20, 0.001, main: present in common arm yes; retained E3 reference.
goryachev, 20, 0.001, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 20, 0.001, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 20, 0.001, late quarter: present in common arm yes; retained E3 reference.
goryachev, 20, 0.001, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 20, 0.001, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 20, 0.001, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 20, 0.003, main: present in common arm yes; retained E3 reference.
goryachev, 20, 0.003, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 20, 0.003, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 20, 0.003, late quarter: present in common arm yes; retained E3 reference.
goryachev, 20, 0.003, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 20, 0.003, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 20, 0.003, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 20, 0.01, main: present in common arm yes; retained E3 reference.
goryachev, 20, 0.01, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 20, 0.01, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 20, 0.01, late quarter: present in common arm yes; retained E3 reference.
goryachev, 20, 0.01, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 20, 0.01, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 20, 0.01, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_597_S12_E3_references_01.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

goryachev, 20, 0.03, main: present in common arm yes; retained E3 reference.
goryachev, 20, 0.03, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 20, 0.03, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 20, 0.03, late quarter: present in common arm yes; retained E3 reference.
goryachev, 20, 0.03, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 20, 0.03, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 20, 0.03, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 20, 0.1, main: present in common arm yes; retained E3 reference.
goryachev, 20, 0.1, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 20, 0.1, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 20, 0.1, late quarter: present in common arm yes; retained E3 reference.
goryachev, 20, 0.1, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 20, 0.1, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 20, 0.1, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 20, 0.3, main: present in common arm yes; retained E3 reference.
goryachev, 20, 0.3, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 20, 0.3, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 20, 0.3, late quarter: present in common arm yes; retained E3 reference.
goryachev, 20, 0.3, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 20, 0.3, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 20, 0.3, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 20, 1, main: present in common arm yes; retained E3 reference.
goryachev, 20, 1, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 20, 1, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 20, 1, late quarter: present in common arm yes; retained E3 reference.
goryachev, 20, 1, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 20, 1, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 20, 1, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 40, 0.0001, main: present in common arm yes; retained E3 reference.
goryachev, 40, 0.0001, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 40, 0.0001, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 40, 0.0001, late quarter: present in common arm yes; retained E3 reference.
goryachev, 40, 0.0001, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 40, 0.0001, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 40, 0.0001, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_598_S12_E3_references_02.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

goryachev, 40, 0.0003, main: present in common arm yes; retained E3 reference.
goryachev, 40, 0.0003, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 40, 0.0003, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 40, 0.0003, late quarter: present in common arm yes; retained E3 reference.
goryachev, 40, 0.0003, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 40, 0.0003, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 40, 0.0003, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 40, 0.001, main: present in common arm yes; retained E3 reference.
goryachev, 40, 0.001, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 40, 0.001, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 40, 0.001, late quarter: present in common arm yes; retained E3 reference.
goryachev, 40, 0.001, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 40, 0.001, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 40, 0.001, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 40, 0.003, main: present in common arm yes; retained E3 reference.
goryachev, 40, 0.003, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 40, 0.003, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 40, 0.003, late quarter: present in common arm yes; retained E3 reference.
goryachev, 40, 0.003, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 40, 0.003, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 40, 0.003, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 40, 0.01, main: present in common arm yes; retained E3 reference.
goryachev, 40, 0.01, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 40, 0.01, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 40, 0.01, late quarter: present in common arm yes; retained E3 reference.
goryachev, 40, 0.01, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 40, 0.01, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 40, 0.01, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 40, 0.03, main: present in common arm yes; retained E3 reference.
goryachev, 40, 0.03, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 40, 0.03, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 40, 0.03, late quarter: present in common arm yes; retained E3 reference.
goryachev, 40, 0.03, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 40, 0.03, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 40, 0.03, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_599_S12_E3_references_03.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

goryachev, 40, 0.1, main: present in common arm yes; retained E3 reference.
goryachev, 40, 0.1, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 40, 0.1, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 40, 0.1, late quarter: present in common arm yes; retained E3 reference.
goryachev, 40, 0.1, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 40, 0.1, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 40, 0.1, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 40, 0.3, main: present in common arm yes; retained E3 reference.
goryachev, 40, 0.3, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 40, 0.3, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 40, 0.3, late quarter: present in common arm yes; retained E3 reference.
goryachev, 40, 0.3, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 40, 0.3, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 40, 0.3, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 40, 1, main: present in common arm yes; retained E3 reference.
goryachev, 40, 1, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 40, 1, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 40, 1, late quarter: present in common arm yes; retained E3 reference.
goryachev, 40, 1, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 40, 1, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 40, 1, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 60, 0.0001, main: present in common arm yes; retained E3 reference.
goryachev, 60, 0.0001, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.0001, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.0001, late quarter: present in common arm yes; retained E3 reference.
goryachev, 60, 0.0001, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 60, 0.0001, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 60, 0.0001, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 60, 0.0003, main: present in common arm yes; retained E3 reference.
goryachev, 60, 0.0003, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.0003, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.0003, late quarter: present in common arm yes; retained E3 reference.
goryachev, 60, 0.0003, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 60, 0.0003, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 60, 0.0003, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_600_S12_E3_references_04.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

goryachev, 60, 0.001, main: present in common arm yes; retained E3 reference.
goryachev, 60, 0.001, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.001, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.001, late quarter: present in common arm yes; retained E3 reference.
goryachev, 60, 0.001, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 60, 0.001, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 60, 0.001, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 60, 0.003, main: present in common arm yes; retained E3 reference.
goryachev, 60, 0.003, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.003, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.003, late quarter: present in common arm yes; retained E3 reference.
goryachev, 60, 0.003, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 60, 0.003, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 60, 0.003, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 60, 0.01, main: present in common arm yes; retained E3 reference.
goryachev, 60, 0.01, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.01, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.01, late quarter: present in common arm yes; retained E3 reference.
goryachev, 60, 0.01, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 60, 0.01, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 60, 0.01, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 60, 0.015, main: present in common arm yes; retained E3 reference.
goryachev, 60, 0.015, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.015, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.015, late quarter: present in common arm yes; retained E3 reference.
goryachev, 60, 0.015, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 60, 0.015, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 60, 0.015, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 60, 0.02, main: present in common arm yes; retained E3 reference.
goryachev, 60, 0.02, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.02, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.02, late quarter: present in common arm yes; retained E3 reference.
goryachev, 60, 0.02, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 60, 0.02, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 60, 0.02, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_601_S12_E3_references_05.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

goryachev, 60, 0.03, main: present in common arm yes; retained E3 reference.
goryachev, 60, 0.03, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.03, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.03, late quarter: present in common arm yes; retained E3 reference.
goryachev, 60, 0.03, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 60, 0.03, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 60, 0.03, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 60, 0.1, main: present in common arm yes; retained E3 reference.
goryachev, 60, 0.1, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.1, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.1, late quarter: present in common arm yes; retained E3 reference.
goryachev, 60, 0.1, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 60, 0.1, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 60, 0.1, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 60, 0.3, main: present in common arm yes; retained E3 reference.
goryachev, 60, 0.3, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.3, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.3, late quarter: present in common arm yes; retained E3 reference.
goryachev, 60, 0.3, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 60, 0.3, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 60, 0.3, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 60, 1, main: present in common arm yes; retained E3 reference.
goryachev, 60, 1, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 60, 1, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 60, 1, late quarter: present in common arm yes; retained E3 reference.
goryachev, 60, 1, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 60, 1, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 60, 1, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.0001, main: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.0001, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.0001, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.0001, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.0001, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.0001, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.0001, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_602_S12_E3_references_06.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

holmes model3, 20, 0.0003, main: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.0003, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.0003, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.0003, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.0003, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.0003, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.0003, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.001, main: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.001, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.001, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.001, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.001, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.001, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.001, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.003, main: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.003, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.003, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.003, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.003, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.003, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.003, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.01, main: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.01, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.01, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.01, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.01, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.01, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.01, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.03, main: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.03, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.03, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.03, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.03, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.03, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.03, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_603_S12_E3_references_07.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

holmes model3, 20, 0.1, main: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.1, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.1, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.1, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.1, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.1, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.1, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.3, main: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.3, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.3, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.3, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.3, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.3, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.3, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 20, 1, main: present in common arm yes; retained E3 reference.
holmes model3, 20, 1, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 20, 1, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 20, 1, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 20, 1, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 20, 1, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 20, 1, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.0001, main: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.0001, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.0001, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.0001, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.0001, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.0001, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.0001, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.0003, main: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.0003, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.0003, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.0003, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.0003, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.0003, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.0003, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_604_S12_E3_references_08.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

holmes model3, 40, 0.001, main: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.001, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.001, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.001, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.001, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.001, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.001, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.003, main: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.003, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.003, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.003, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.003, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.003, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.003, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.01, main: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.01, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.01, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.01, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.01, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.01, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.01, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.03, main: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.03, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.03, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.03, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.03, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.03, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.03, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.1, main: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.1, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.1, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.1, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.1, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.1, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.1, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_605_S12_E3_references_09.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

holmes model3, 40, 0.3, main: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.3, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.3, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.3, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.3, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.3, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.3, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 40, 1, main: present in common arm yes; retained E3 reference.
holmes model3, 40, 1, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 40, 1, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 40, 1, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 40, 1, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 40, 1, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 40, 1, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.0001, main: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.0001, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.0001, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.0001, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.0001, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.0001, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.0001, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.0003, main: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.0003, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.0003, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.0003, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.0003, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.0003, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.0003, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.001, main: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.001, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.001, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.001, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.001, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.001, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.001, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_606_S12_E3_references_10.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

holmes model3, 60, 0.003, main: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.003, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.003, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.003, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.003, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.003, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.003, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.01, main: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.01, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.01, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.01, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.01, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.01, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.01, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.03, main: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.03, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.03, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.03, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.03, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.03, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.03, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.1, main: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.1, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.1, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.1, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.1, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.1, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.1, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.3, main: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.3, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.3, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.3, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.3, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.3, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.3, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_607_S12_E3_references_11.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

holmes model3, 60, 1, main: present in common arm yes; retained E3 reference.
holmes model3, 60, 1, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 60, 1, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 60, 1, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 60, 1, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 60, 1, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 60, 1, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 1, 0.0001, main: present in common arm yes; retained E3 reference.
spring ring, 1, 0.0001, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 1, 0.0001, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 1, 0.0001, late quarter: present in common arm yes; retained E3 reference.
spring ring, 1, 0.0001, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 1, 0.0001, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 1, 0.0001, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 1, 0.0003, main: present in common arm yes; retained E3 reference.
spring ring, 1, 0.0003, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 1, 0.0003, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 1, 0.0003, late quarter: present in common arm yes; retained E3 reference.
spring ring, 1, 0.0003, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 1, 0.0003, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 1, 0.0003, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 1, 0.001, main: present in common arm yes; retained E3 reference.
spring ring, 1, 0.001, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 1, 0.001, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 1, 0.001, late quarter: present in common arm yes; retained E3 reference.
spring ring, 1, 0.001, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 1, 0.001, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 1, 0.001, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 1, 0.003, main: present in common arm yes; retained E3 reference.
spring ring, 1, 0.003, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 1, 0.003, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 1, 0.003, late quarter: present in common arm yes; retained E3 reference.
spring ring, 1, 0.003, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 1, 0.003, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 1, 0.003, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_608_S12_E3_references_12.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

spring ring, 1, 0.01, main: present in common arm yes; retained E3 reference.
spring ring, 1, 0.01, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 1, 0.01, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 1, 0.01, late quarter: present in common arm yes; retained E3 reference.
spring ring, 1, 0.01, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 1, 0.01, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 1, 0.01, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 1, 0.03, main: present in common arm yes; retained E3 reference.
spring ring, 1, 0.03, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 1, 0.03, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 1, 0.03, late quarter: present in common arm yes; retained E3 reference.
spring ring, 1, 0.03, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 1, 0.03, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 1, 0.03, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 1, 0.1, main: present in common arm yes; retained E3 reference.
spring ring, 1, 0.1, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 1, 0.1, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 1, 0.1, late quarter: present in common arm yes; retained E3 reference.
spring ring, 1, 0.1, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 1, 0.1, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 1, 0.1, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 1, 0.3, main: present in common arm yes; retained E3 reference.
spring ring, 1, 0.3, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 1, 0.3, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 1, 0.3, late quarter: present in common arm yes; retained E3 reference.
spring ring, 1, 0.3, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 1, 0.3, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 1, 0.3, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 1, 1, main: present in common arm yes; retained E3 reference.
spring ring, 1, 1, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 1, 1, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 1, 1, late quarter: present in common arm yes; retained E3 reference.
spring ring, 1, 1, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 1, 1, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 1, 1, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_609_S12_E3_references_13.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

spring ring, 2, 0.0001, main: present in common arm yes; retained E3 reference.
spring ring, 2, 0.0001, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 2, 0.0001, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 2, 0.0001, late quarter: present in common arm yes; retained E3 reference.
spring ring, 2, 0.0001, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 2, 0.0001, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 2, 0.0001, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 2, 0.0003, main: present in common arm yes; retained E3 reference.
spring ring, 2, 0.0003, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 2, 0.0003, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 2, 0.0003, late quarter: present in common arm yes; retained E3 reference.
spring ring, 2, 0.0003, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 2, 0.0003, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 2, 0.0003, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 2, 0.001, main: present in common arm yes; retained E3 reference.
spring ring, 2, 0.001, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 2, 0.001, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 2, 0.001, late quarter: present in common arm yes; retained E3 reference.
spring ring, 2, 0.001, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 2, 0.001, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 2, 0.001, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 2, 0.003, main: present in common arm yes; retained E3 reference.
spring ring, 2, 0.003, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 2, 0.003, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 2, 0.003, late quarter: present in common arm yes; retained E3 reference.
spring ring, 2, 0.003, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 2, 0.003, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 2, 0.003, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 2, 0.01, main: present in common arm yes; retained E3 reference.
spring ring, 2, 0.01, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 2, 0.01, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 2, 0.01, late quarter: present in common arm yes; retained E3 reference.
spring ring, 2, 0.01, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 2, 0.01, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 2, 0.01, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_610_S12_E3_references_14.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

spring ring, 2, 0.03, main: present in common arm yes; retained E3 reference.
spring ring, 2, 0.03, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 2, 0.03, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 2, 0.03, late quarter: present in common arm yes; retained E3 reference.
spring ring, 2, 0.03, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 2, 0.03, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 2, 0.03, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 2, 0.1, main: present in common arm yes; retained E3 reference.
spring ring, 2, 0.1, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 2, 0.1, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 2, 0.1, late quarter: present in common arm yes; retained E3 reference.
spring ring, 2, 0.1, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 2, 0.1, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 2, 0.1, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 2, 0.3, main: present in common arm yes; retained E3 reference.
spring ring, 2, 0.3, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 2, 0.3, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 2, 0.3, late quarter: present in common arm yes; retained E3 reference.
spring ring, 2, 0.3, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 2, 0.3, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 2, 0.3, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 2, 1, main: present in common arm yes; retained E3 reference.
spring ring, 2, 1, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 2, 1, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 2, 1, late quarter: present in common arm yes; retained E3 reference.
spring ring, 2, 1, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 2, 1, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 2, 1, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 3, 0.0001, main: present in common arm yes; retained E3 reference.
spring ring, 3, 0.0001, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 3, 0.0001, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 3, 0.0001, late quarter: present in common arm yes; retained E3 reference.
spring ring, 3, 0.0001, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 3, 0.0001, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 3, 0.0001, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_611_S12_E3_references_15.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

spring ring, 3, 0.0003, main: present in common arm yes; retained E3 reference.
spring ring, 3, 0.0003, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 3, 0.0003, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 3, 0.0003, late quarter: present in common arm yes; retained E3 reference.
spring ring, 3, 0.0003, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 3, 0.0003, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 3, 0.0003, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 3, 0.001, main: present in common arm yes; retained E3 reference.
spring ring, 3, 0.001, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 3, 0.001, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 3, 0.001, late quarter: present in common arm yes; retained E3 reference.
spring ring, 3, 0.001, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 3, 0.001, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 3, 0.001, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 3, 0.003, main: present in common arm yes; retained E3 reference.
spring ring, 3, 0.003, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 3, 0.003, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 3, 0.003, late quarter: present in common arm yes; retained E3 reference.
spring ring, 3, 0.003, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 3, 0.003, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 3, 0.003, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 3, 0.01, main: present in common arm yes; retained E3 reference.
spring ring, 3, 0.01, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 3, 0.01, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 3, 0.01, late quarter: present in common arm yes; retained E3 reference.
spring ring, 3, 0.01, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 3, 0.01, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 3, 0.01, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 3, 0.03, main: present in common arm yes; retained E3 reference.
spring ring, 3, 0.03, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 3, 0.03, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 3, 0.03, late quarter: present in common arm yes; retained E3 reference.
spring ring, 3, 0.03, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 3, 0.03, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 3, 0.03, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_612_S12_E3_references_16.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

spring ring, 3, 0.1, main: present in common arm yes; retained E3 reference.
spring ring, 3, 0.1, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 3, 0.1, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 3, 0.1, late quarter: present in common arm yes; retained E3 reference.
spring ring, 3, 0.1, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 3, 0.1, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 3, 0.1, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 3, 0.3, main: present in common arm yes; retained E3 reference.
spring ring, 3, 0.3, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 3, 0.3, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 3, 0.3, late quarter: present in common arm yes; retained E3 reference.
spring ring, 3, 0.3, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 3, 0.3, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 3, 0.3, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 3, 1, main: present in common arm yes; retained E3 reference.
spring ring, 3, 1, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 3, 1, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 3, 1, late quarter: present in common arm yes; retained E3 reference.
spring ring, 3, 1, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 3, 1, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 3, 1, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.0001, main: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.0001, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.0001, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.0001, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.0001, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.0001, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.0001, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.0003, main: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.0003, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.0003, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.0003, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.0003, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.0003, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.0003, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_613_S12_E3_references_17.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

wave pinning, 20, 0.001, main: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.001, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.001, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.001, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.001, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.001, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.001, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.003, main: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.003, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.003, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.003, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.003, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.003, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.003, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.01, main: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.01, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.01, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.01, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.01, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.01, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.01, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.03, main: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.03, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.03, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.03, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.03, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.03, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.03, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.1, main: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.1, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.1, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.1, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.1, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.1, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.1, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_614_S12_E3_references_18.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

wave pinning, 20, 0.3, main: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.3, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.3, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.3, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.3, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.3, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.3, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 20, 1, main: present in common arm yes; retained E3 reference.
wave pinning, 20, 1, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 20, 1, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 20, 1, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 20, 1, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 20, 1, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 20, 1, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.0001, main: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.0001, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.0001, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.0001, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.0001, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.0001, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.0001, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.0003, main: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.0003, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.0003, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.0003, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.0003, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.0003, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.0003, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.001, main: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.001, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.001, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.001, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.001, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.001, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.001, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_615_S12_E3_references_19.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

wave pinning, 40, 0.003, main: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.003, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.003, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.003, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.003, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.003, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.003, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.01, main: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.01, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.01, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.01, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.01, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.01, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.01, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.03, main: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.03, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.03, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.03, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.03, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.03, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.03, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.1, main: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.1, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.1, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.1, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.1, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.1, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.1, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.3, main: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.3, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.3, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.3, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.3, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.3, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.3, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_616_S12_E3_references_20.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

wave pinning, 40, 1, main: present in common arm yes; retained E3 reference.
wave pinning, 40, 1, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 40, 1, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 40, 1, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 40, 1, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 40, 1, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 40, 1, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.0001, main: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.0001, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.0001, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.0001, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.0001, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.0001, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.0001, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.0003, main: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.0003, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.0003, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.0003, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.0003, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.0003, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.0003, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.001, main: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.001, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.001, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.001, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.001, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.001, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.001, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.003, main: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.003, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.003, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.003, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.003, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.003, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.003, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_617_S12_E3_references_21.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

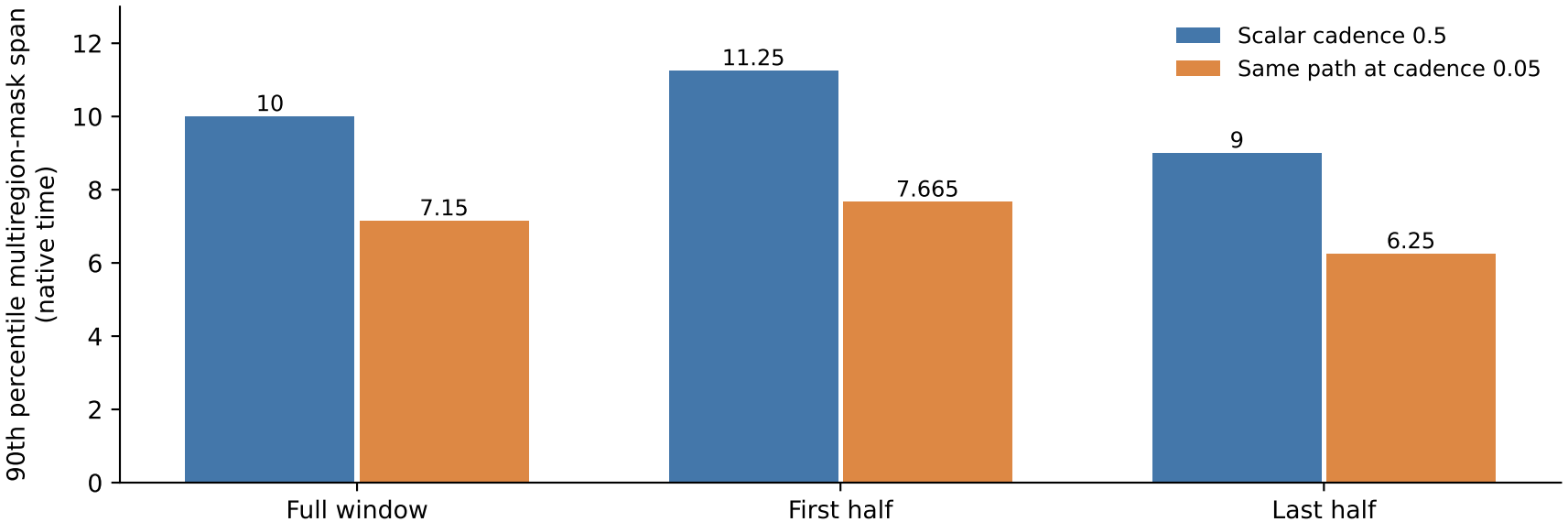
Retained E3 observations remain references; they do not replace canonical common observations.

wave pinning, 60, 0.01, main: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.01, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.01, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.01, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.01, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.01, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.01, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.03, main: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.03, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.03, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.03, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.03, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.03, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.03, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.1, main: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.1, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.1, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.1, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.1, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.1, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.1, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.3, main: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.3, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.3, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.3, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.3, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.3, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.3, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 60, 1, main: present in common arm yes; retained E3 reference.
wave pinning, 60, 1, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 60, 1, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 60, 1, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 60, 1, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 60, 1, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 60, 1, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_618_S12_E3_references_22.csv

S12 · Retained Spring numerical and observation qualification

Spring $C_0 = 3$ · input fraction f = 1 · seed 202610192320800
Original full gate: failed. Accepted stable scope: 180/180 rows; all 18 q90 rows excluded.



The 90th percentile of consecutive saved multiregion-mask spans (last true time minus first true time); this is not the lifetime of a tracked individual region.

All 12 failures are scalar-observation cadence comparisons (standard and tighter ODE, raw and contrast-qualified masks, three windows). All six ODE-only q90 comparisons pass.

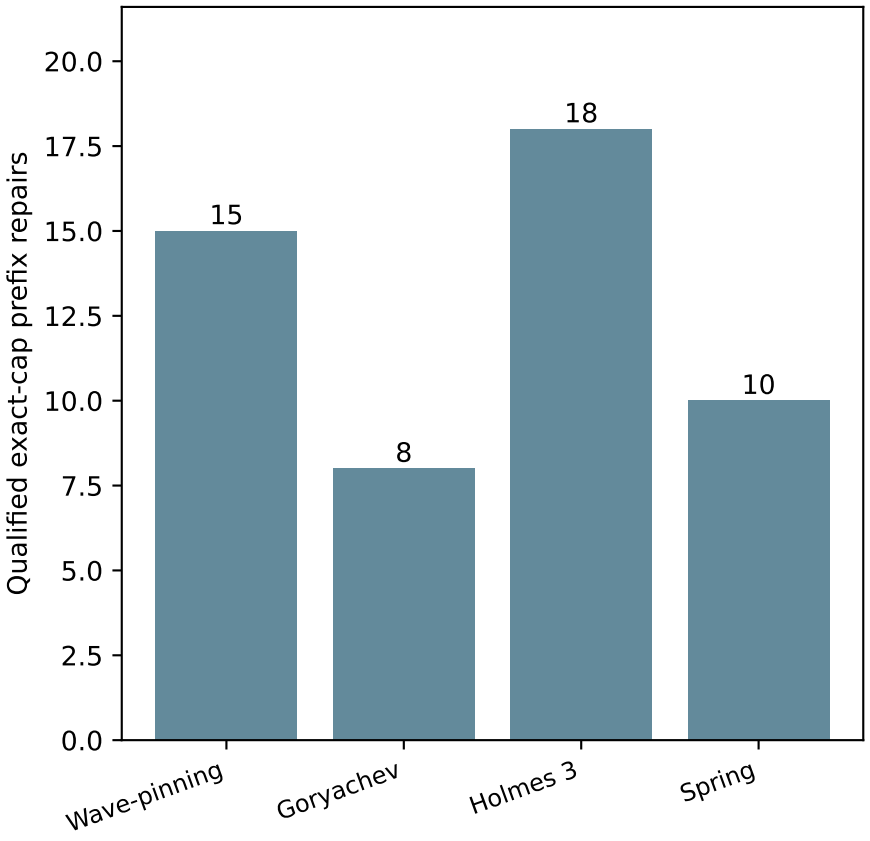
The original full gate remains failed. The separate accepted stable scope excludes all 18 q90 rows. It does not establish convergence of sampled durations, event rates, continuous region lifetimes, every condition, stationarity, or migration.

Exact sigma-one replay establishes preservation of an existing path and forcing. It does not establish numerical or observation convergence. New optical tracking and cadence diagnostics require their own separately retained qualification.

Dense output samples the same native path. Tight-ODE companions are separate numerical controls, never extra independent noise realizations in the 936+26 E0 scientific inventory.

CSV retains all 198 rows, including all 12 failures and all 18 excluded q90 comparisons.
figure4_S12_historical_numerics.csv

S12 · Accepted E4 numerical prefix repair; scientific outcomes assessed separately · E4 exact-cap repair retains the declared native prefix rule



All 688 original integrations reached the full horizon.
637 passed the original prefix gate and were retained unchanged.
51 failed the prefix gate; precisely these were replayed to the exact old cap and continued.
All 688 selected histories pass the independent native/source/forcing/prefix gates.

1911 native numeric prefix checks; 1870 bitwise equal.
Maximum absolute prefix difference: 2.8422e-14.
Unchanged gate: relative tolerance 2×10^{-7}; absolute tolerance 2×10^{-9}.

One original normalization and global-time forcing closure; native kinetics, mesh, fields, observation clocks and seeds are unchanged.
Failed attempts remain archived. No biological response was used to select replacements.

This is engineering equivalence, not a stationarity or persistence result.
The separate historical Spring scalar-cadence q90 sensitivity remains unresolved.

Accepted E4 numerical prefix repair; scientific outcomes assessed separately. All 51 repair records, original failure reasons and exact source/proof identities remain in the page CSV. Counts are engineering integration attempts, not additional independent scientific seeds. Successful replay neither resolves the historical Spring observation-cadence limitation nor qualifies long-horizon biological claims. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_E4_prefix_recovery_20261020_001_S12_E4_prefix_recovery.csv